

Lectures on Symplectic Geometry

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Foreword

These notes approximately transcribe a 15-week course on symplectic geometry I taught at UC Berkeley in the Fall of 1997.

The course at Berkeley was greatly inspired in content and style by Victor Guillemin, whose masterly teaching of beautiful courses on topics related to symplectic geometry at MIT, I was lucky enough to experience as a graduate student. I am very thankful to him!

That course also borrowed from the 1997 Park City summer courses on symplectic geometry and topology, and from many talks and discussions of the symplectic geometry group at MIT. Among the regular participants in the MIT informal symplectic seminar 93-96, I would like to acknowledge the contributions of Allen Knutson, Chris Woodward, David Metzler, Eckhard Meinrenken, Elisa Prato, Eugene Lerman, Jonathan Weitsman, Lisa Jeffrey, Reyer Sjamaar, Shaun Martin, Stephanie Singer, Sue Tolman and, last but not least, Yael Karshon.

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Introduction

The goal of these notes is to provide a fast introduction to symplectic geometry.

A symplectic form is a closed nondegenerate 2-form. A symplectic manifold is a manifold equipped with a symplectic form. Symplectic geometry is the geometry of symplectic manifolds. Symplectic manifolds are necessarily even-dimensional and orientable, since nondegeneracy says that the top exterior power of a symplectic form is a volume form. The closedness condition is a natural differential equation, which forces all symplectic manifolds to be locally indistinguishable. (These assertions will be explained in Lecture 1 and Homework 2.)

The list of questions on symplectic forms begins with those of existence and uniqueness on a given manifold. For specific symplectic manifolds, one would like to understand the geometry and the topology of special submanifolds, the dynamics of certain vector fields or systems of differential equations, the symmetries and extra structure, etc.

Two centuries ago, symplectic geometry provided a language for classical mechanics. Through its recent huge development, it conquered an independent and rich territory, as a central branch of differential geometry and topology. To mention just a few key landmarks, one may say that symplectic geometry began to take its modern shape with the formulation of the Arnold conjectures in the 60's and with the foundational work of Weinstein in the 70's. A paper of Gromov [49] in the 80's gave the subject a whole new set of tools: pseudo-holomorphic curves. Gromov also first showed that important results from complex Kähler geometry remain true in the more general symplectic category, and this direction was continued rather dramatically in the 90's in the work of Donaldson on the topology of symplectic manifolds and their symplectic submanifolds, and in the work of Taubes in the context of the Seiberg-Witten invariants. Symplectic geometry is significantly stimulated by important interactions with global analysis, mathematical physics, low-dimensional topology, dynamical systems, algebraic geometry, integrable systems, microlocal analysis, partial differential equations, representation theory, quantization, equivariant cohomology, geometric combinatorics, etc.

As a curiosity, note that two centuries ago the name *symplectic geometry* did not exist. If you consult a major English dictionary, you are likely to find that *symplectic* is the name for a bone in a fish's head. However, as clarified in [105], the word *symplectic* in mathematics was coined by Weyl [110, p.165] who substituted the Latin root in *complex* by the corresponding Greek root, in order to label the symplectic group. Weyl thus avoided that this group connote the complex numbers, and also spared us from much confusion that would have arisen, had the name remained the former one in honor of Abel: *abelian linear group*.

This text is essentially the set of notes of a 15-week course on symplectic geometry with 2 hour-and-a-half lectures per week. The course targeted second-year graduate students in mathematics, though the audience was more diverse, including advanced undergraduates, post-docs and graduate students from other departments. The present text should hence still be appropriate for a second-year graduate course or for an independent study project.

There are scattered short exercises throughout the text. At the end of most lectures, some longer guided problems, called homework, were designed to complement the exposition or extend the reader's understanding.

Geometry of manifolds was the basic prerequisite for the original course, so the same holds now for the notes. In particular, some familiarity with de Rham theory and classical Lie groups is expected.

As for conventions: unless otherwise indicated, all vector spaces are real and finite-dimensional, all maps are smooth (i.e., C^∞) and all manifolds are smooth, Hausdorff and second countable.

Here is a brief summary of the contents of this book. Parts I-III explain classical topics, including cotangent bundles, symplectomorphisms, lagrangian submanifolds and local forms. Parts IV-VI concentrate on important related areas, such as contact geometry and Kähler geometry. Classical hamiltonian theory enters in Parts VII-VIII, starting the second half of this book, which is devoted to a selection of themes from hamiltonian dynamical systems and symmetry. Parts IX-XI discuss the moment map whose preponderance has been growing steadily for the past twenty years.

There are by now excellent references on symplectic geometry, a subset of which is in the bibliography. However, the most efficient introduction to a subject is often a short elementary treatment, and these notes attempt to serve that purpose. The author hopes that these notes provide a taste of areas of current research, and will prepare the reader to explore recent papers and extensive books in symplectic geometry, where the pace is much faster.

Part I

Symplectic Manifolds

A symplectic form is a 2-form satisfying an algebraic condition – nondegeneracy – and an analytical condition – closedness. In Lectures 1 and 2 we define symplectic forms, describe some of their basic properties, introduce the first examples, namely even-dimensional euclidean spaces and cotangent bundles.

1 Symplectic Forms

1.1 Skew-Symmetric Bilinear Maps

Let V be an m -dimensional vector space over $\mathbb{R}$, and let $\Omega : V \times V \rightarrow \mathbb{R}$ be a bilinear map. The map Ω is **skew-symmetric** if $\Omega(u, v) = -\Omega(v, u)$, for all $u, v \in V$.

Theorem 1.1 (Standard Form for Skew-symmetric Bilinear Maps)

Let Ω be a skew-symmetric bilinear map on V . Then there is a basis $u_1, \dots, u_k, e_1, \dots, e_n, f_1, \dots, f_n$ of V such that

$$\begin{aligned} \Omega(u_i, v) &= 0, & \text{for all } i \text{ and all } v \in V, \\ \Omega(e_i, e_j) &= 0 = \Omega(f_i, f_j), & \text{for all } i, j, \text{ and} \\ \Omega(e_i, f_j) &= \delta_{ij}, & \text{for all } i, j. \end{aligned}$$

Remarks.

1. The basis in Theorem 1.1 is not unique, though it is traditionally also called a “canonical” basis.
2. In matrix notation with respect to such basis, we have

$$\Omega(u, v) = [-u] \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & \text{Id} \\ 0 & -\text{Id} & 0 \end{bmatrix} \begin{bmatrix} | \\ v \\ | \end{bmatrix}.$$

◇

Proof. This induction proof is a skew-symmetric version of the Gram-Schmidt process.

Let $U := \{u \in V \mid \Omega(u, v) = 0 \text{ for all } v \in V\}$. Choose a basis $u_1, \dots, u_k$ of U , and choose a complementary space W to U in V ,

$$V = U \oplus W.$$

Take any nonzero $e_1 \in W$. Then there is $f_1 \in W$ such that $\Omega(e_1, f_1) \neq 0$. Assume that $\Omega(e_1, f_1) = 1$. Let

$$\begin{aligned} W_1 &= \text{span of } e_1, f_1 \\ W_1^\Omega &= \{w \in W \mid \Omega(w, v) = 0 \text{ for all } v \in W_1\} . \end{aligned}$$

Claim. $W_1 \cap W_1^\Omega = \{0\}$.

Suppose that $v = ae_1 + bf_1 \in W_1 \cap W_1^\Omega$.

$$\left. \begin{aligned} 0 &= \Omega(v, e_1) = -b \\ 0 &= \Omega(v, f_1) = a \end{aligned} \right\} \implies v = 0 .$$

Claim. $W = W_1 \oplus W_1^\Omega$.

Suppose that $v \in W$ has $\Omega(v, e_1) = c$ and $\Omega(v, f_1) = d$. Then

$$v = \underbrace{(-cf_1 + de_1)}_{\in W_1} + \underbrace{(v + cf_1 - de_1)}_{\in W_1^\Omega} .$$

Go on: let $e_2 \in W_1^\Omega$, $e_2 \neq 0$. There is $f_2 \in W_1^\Omega$ such that $\Omega(e_2, f_2) \neq 0$. Assume that $\Omega(e_2, f_2) = 1$. Let $W_2 = \text{span of } e_2, f_2$. Etc.

This process eventually stops because $\dim V < \infty$. We hence obtain

$$V = U \oplus W_1 \oplus W_2 \oplus \dots \oplus W_n$$

where all summands are orthogonal with respect to Ω , and where W_i has basis e_i, f_i with $\Omega(e_i, f_i) = 1$. $\square$

The dimension of the subspace $U = \{u \in V \mid \Omega(u, v) = 0, \text{ for all } v \in V\}$ does not depend on the choice of basis.

$\implies k := \dim U$ is an invariant of (V, Ω) .

Since $k + 2n = m = \dim V$,

$\implies n$ is an invariant of (V, Ω) ; $2n$ is called the **rank** of Ω .

1.2 Symplectic Vector Spaces

Let V be an m -dimensional vector space over $\mathbb{R}$, and let $\Omega : V \times V \rightarrow \mathbb{R}$ be a bilinear map.

Definition 1.2 The map $\tilde{\Omega} : V \rightarrow V^*$ is the linear map defined by $\tilde{\Omega}(v)(u) = \Omega(v, u)$.

The kernel of $\tilde{\Omega}$ is the subspace U above.

Definition 1.3 A skew-symmetric bilinear map Ω is **symplectic** (or **nondegenerate**) if $\tilde{\Omega}$ is bijective, i.e., $U = \{0\}$. The map Ω is then called a **linear symplectic structure** on V , and (V, Ω) is called a **symplectic vector space**.

The following are immediate properties of a linear symplectic structure Ω :

- **Duality**: the map $\tilde{\Omega} : V \xrightarrow{\sim} V^*$ is a bijection.
- By the standard form theorem, $k = \dim U = 0$, so $\dim V = 2n$ is **even**.
- By Theorem 1.1, a symplectic vector space (V, Ω) has a basis $e_1, \dots, e_n, f_1, \dots, f_n$ satisfying

$$\Omega(e_i, f_j) = \delta_{ij} \quad \text{and} \quad \Omega(e_i, e_j) = 0 = \Omega(f_i, f_j) .$$

Such a basis is called a **symplectic basis** of (V, Ω) . We have

$$\Omega(u, v) = [-u-] \begin{bmatrix} 0 & \text{Id} \\ -\text{Id} & 0 \end{bmatrix} \begin{bmatrix} | \\ v \\ | \end{bmatrix} ,$$

where the symbol $\begin{bmatrix} | \\ v \\ | \end{bmatrix}$ represents the column of coordinates of the vector v with respect to a symplectic basis $e_1, \dots, e_n, f_1, \dots, f_n$ whereas $[-v-]$ represents its transpose line.

Not all subspaces W of a symplectic vector space (V, Ω) look the same:

- A subspace W is called **symplectic** if $\Omega|_W$ is nondegenerate. For instance, the span of e_1, f_1 is symplectic.
- A subspace W is called **isotropic** if $\Omega|_W \equiv 0$. For instance, the span of e_1, e_2 is isotropic.

Homework 1 describes subspaces W of (V, Ω) in terms of the relation between W and W^Ω .

The **prototype of a symplectic vector space** is $(\mathbb{R}^{2n}, \Omega_0)$ with Ω_0 such that the basis

$$\begin{aligned} e_1 &= (1, 0, \dots, 0), & \dots, & \quad e_n = (0, \dots, 0, \overbrace{1}^n, 0, \dots, 0), \\ f_1 &= (0, \dots, 0, \underbrace{1}_{n+1}, 0, \dots, 0), & \dots, & \quad f_n = (0, \dots, 0, 1), \end{aligned}$$

is a symplectic basis. The map Ω_0 on other vectors is determined by its values on a basis and bilinearity.

Definition 1.4 A **symplectomorphism** φ between symplectic vector spaces (V, Ω) and (V', Ω') is a linear isomorphism $\varphi : V \xrightarrow{\sim} V'$ such that $\varphi^* \Omega' = \Omega$. (By definition, $(\varphi^* \Omega')(u, v) = \Omega'(\varphi(u), \varphi(v))$.) If a symplectomorphism exists, (V, Ω) and (V', Ω') are said to be **symplectomorphic**.

The relation of being symplectomorphic is clearly an equivalence relation in the set of all even-dimensional vector spaces. Furthermore, by Theorem 1.1, every $2n$ -dimensional symplectic vector space (V, Ω) is symplectomorphic to the prototype $(\mathbb{R}^{2n}, \Omega_0)$; a choice of a symplectic basis for (V, Ω) yields a symplectomorphism to $(\mathbb{R}^{2n}, \Omega_0)$. Hence, nonnegative even integers classify equivalence classes for the relation of being symplectomorphic.

1.3 Symplectic Manifolds

Let ω be a de Rham 2-form on a manifold M , that is, for each $p \in M$, the map $\omega_p : T_p M \times T_p M \rightarrow \mathbb{R}$ is skew-symmetric bilinear on the tangent space to M at p , and ω_p varies smoothly in p . We say that ω is closed if it satisfies the differential equation $d\omega = 0$, where d is the de Rham differential (i.e., exterior derivative).

Definition 1.5 The 2-form ω is **symplectic** if ω is closed and ω_p is symplectic for all $p \in M$.

If ω is symplectic, then $\dim T_p M = \dim M$ must be even.

Definition 1.6 A **symplectic manifold** is a pair (M, ω) where M is a manifold and ω is a symplectic form.

Example. Let $M = \mathbb{R}^{2n}$ with linear coordinates $x_1, \dots, x_n, y_1, \dots, y_n$. The form

$$\omega_0 = \sum_{i=1}^n dx_i \wedge dy_i$$

is symplectic as can be easily checked, and the set

$$\left\{ \left(\frac{\partial}{\partial x_1} \right)_p, \dots, \left(\frac{\partial}{\partial x_n} \right)_p, \left(\frac{\partial}{\partial y_1} \right)_p, \dots, \left(\frac{\partial}{\partial y_n} \right)_p \right\}$$

is a symplectic basis of $T_p M$. ◇

Example. Let $M = \mathbb{C}^n$ with linear coordinates $z_1, \dots, z_n$. The form

$$\omega_0 = \frac{i}{2} \sum_{k=1}^n dz_k \wedge d\bar{z}_k$$

is symplectic. In fact, this form equals that of the previous example under the identification $\mathbb{C}^n \simeq \mathbb{R}^{2n}$, $z_k = x_k + iy_k$. ◇

Example. Let $M = S^2$ regarded as the set of unit vectors in $\mathbb{R}^3$. Tangent vectors to S^2 at p may then be identified with vectors orthogonal to p . The standard symplectic form on S^2 is induced by the inner and exterior products:

$$\omega_p(u, v) := \langle p, u \times v \rangle, \quad \text{for } u, v \in T_p S^2 = \{p\}^\perp.$$

This form is closed because it is of top degree; it is nondegenerate because $\langle p, u \times v \rangle \neq 0$ when $u \neq 0$ and we take, for instance, $v = u \times p$. ◇

1.4 Symplectomorphisms

Definition 1.7 Let (M_1, ω_1) and (M_2, ω_2) be $2n$ -dimensional symplectic manifolds, and let $\varphi : M_1 \rightarrow M_2$ be a diffeomorphism. Then φ is a **symplectomorphism** if $\varphi^*\omega_2 = \omega_1$.¹

We would like to classify symplectic manifolds up to symplectomorphism. The Darboux theorem (proved in Lecture 8 and stated below) takes care of this classification locally: the dimension is the only local invariant of symplectic manifolds up to symplectomorphisms. Just as any n -dimensional manifold looks locally like $\mathbb{R}^n$, any $2n$ -dimensional *symplectic* manifold looks locally like $(\mathbb{R}^{2n}, \omega_0)$. More precisely, any symplectic manifold (M^{2n}, ω) is locally symplectomorphic to $(\mathbb{R}^{2n}, \omega_0)$.

Theorem 8.1 (Darboux) Let (M, ω) be a $2n$ -dimensional symplectic manifold, and let p be any point in M .

Then there is a coordinate chart $(\mathcal{U}, x_1, \dots, x_n, y_1, \dots, y_n)$ centered at p such that on $\mathcal{U}$

$$\omega = \sum_{i=1}^n dx_i \wedge dy_i .$$

A chart $(\mathcal{U}, x_1, \dots, x_n, y_1, \dots, y_n)$ as in Theorem 8.1 is called a **Darboux chart**.

By Theorem 8.1, the **prototype of a local piece of a $2n$ -dimensional symplectic manifold** is $M = \mathbb{R}^{2n}$, with linear coordinates $(x_1, \dots, x_n, y_1, \dots, y_n)$, and with symplectic form

$$\omega_0 = \sum_{i=1}^n dx_i \wedge dy_i .$$

¹Recall that, by definition of pullback, at tangent vectors $u, v \in T_p M_1$, we have $(\varphi^*\omega_2)_p(u, v) = (\omega_2)_{\varphi(p)}(d\varphi_p(u), d\varphi_p(v))$.

Homework 1: Symplectic Linear Algebra

Given a linear subspace Y of a symplectic vector space (V, Ω) , its **symplectic orthogonal** Y^Ω is the linear subspace defined by

$$Y^\Omega := \{v \in V \mid \Omega(v, u) = 0 \text{ for all } u \in Y\} .$$

1. Show that $\dim Y + \dim Y^\Omega = \dim V$.

Hint: What is the kernel and image of the map

$$\begin{aligned} V &\longrightarrow Y^* = \text{Hom}(Y, \mathbb{R}) \quad ? \\ v &\longmapsto \Omega(v, \cdot)|_Y \end{aligned}$$

2. Show that $(Y^\Omega)^\Omega = Y$.
3. Show that, if Y and W are subspaces, then

$$Y \subseteq W \iff W^\Omega \subseteq Y^\Omega .$$

4. Show that:

$$Y \text{ is } \mathbf{symplectic} \text{ (i.e., } \Omega|_{Y \times Y} \text{ is nondegenerate)} \iff Y \cap Y^\Omega = \{0\} \iff V = Y \oplus Y^\Omega .$$

5. We call Y **isotropic** when $Y \subseteq Y^\Omega$ (i.e., $\Omega|_{Y \times Y} \equiv 0$).

Show that, if Y is isotropic, then $\dim Y \leq \frac{1}{2} \dim V$.

6. We call Y **coisotropic** when $Y^\Omega \subseteq Y$.

Check that every codimension 1 subspace Y is coisotropic.

7. An isotropic subspace Y of (V, Ω) is called **lagrangian** when $\dim Y = \frac{1}{2} \dim V$.

Check that:

$$Y \text{ is lagrangian} \iff Y \text{ is isotropic and coisotropic} \iff Y = Y^\Omega .$$

8. Show that, if Y is a lagrangian subspace of (V, Ω) , then any basis $e_1, \dots, e_n$ of Y can be extended to a symplectic basis $e_1, \dots, e_n, f_1, \dots, f_n$ of (V, Ω) .

Hint: Choose f_1 in W^Ω , where W is the linear span of $\{e_2, \dots, e_n\}$.

9. Show that, if Y is a lagrangian subspace, (V, Ω) is symplectomorphic to the space $(Y \oplus Y^*, \Omega_0)$, where Ω_0 is determined by the formula

$$\Omega_0(u \oplus \alpha, v \oplus \beta) = \beta(u) - \alpha(v) .$$

In fact, for any vector space E , the direct sum $V = E \oplus E^*$ has a canonical symplectic structure determined by the formula above. If $e_1, \dots, e_n$ is a basis of E , and $f_1, \dots, f_n$ is the dual basis, then $e_1 \oplus 0, \dots, e_n \oplus 0, 0 \oplus f_1, \dots, 0 \oplus f_n$ is a symplectic basis for V .

2 Symplectic Form on the Cotangent Bundle

2.1 Cotangent Bundle

Let X be any n -dimensional manifold and $M = T^*X$ its cotangent bundle. If the manifold structure on X is described by coordinate charts $(\mathcal{U}, x_1, \dots, x_n)$ with $x_i : \mathcal{U} \rightarrow \mathbb{R}$, then at any $x \in \mathcal{U}$, the differentials $(dx_1)_x, \dots, (dx_n)_x$ form a basis of T_x^*X . Namely, if $\xi \in T_x^*X$, then $\xi = \sum_{i=1}^n \xi_i (dx_i)_x$ for some real coefficients $\xi_1, \dots, \xi_n$. This induces a map

$$\begin{aligned} T^*\mathcal{U} &\longrightarrow \mathbb{R}^{2n} \\ (x, \xi) &\longmapsto (x_1, \dots, x_n, \xi_1, \dots, \xi_n) . \end{aligned}$$

The chart $(T^*\mathcal{U}, x_1, \dots, x_n, \xi_1, \dots, \xi_n)$ is a coordinate chart for T^*X ; the coordinates $x_1, \dots, x_n, \xi_1, \dots, \xi_n$ are the **cotangent coordinates** associated to the coordinates $x_1, \dots, x_n$ on $\mathcal{U}$. The transition functions on the overlaps are smooth: given two charts $(\mathcal{U}, x_1, \dots, x_n)$, $(\mathcal{U}', x'_1, \dots, x'_n)$, and $x \in \mathcal{U} \cap \mathcal{U}'$, if $\xi \in T_x^*X$, then

$$\xi = \sum_{i=1}^n \xi_i (dx_i)_x = \sum_{i,j} \xi_i \left(\frac{\partial x_i}{\partial x'_j} \right) (dx'_j)_x = \sum_{j=1}^n \xi'_j (dx'_j)_x$$

where $\xi'_j = \sum_i \xi_i \left(\frac{\partial x_i}{\partial x'_j} \right)$ is smooth. Hence, T^*X is a $2n$ -dimensional manifold.

We will now construct a major class of examples of symplectic forms. The *canonical forms* on cotangent bundles are relevant for several branches, including analysis of differential operators, dynamical systems and classical mechanics.

2.2 Tautological and Canonical Forms in Coordinates

Let $(\mathcal{U}, x_1, \dots, x_n)$ be a coordinate chart for X , with associated cotangent coordinates $(T^*\mathcal{U}, x_1, \dots, x_n, \xi_1, \dots, \xi_n)$. Define a 2-form ω on $T^*\mathcal{U}$ by

$$\omega = \sum_{i=1}^n dx_i \wedge d\xi_i .$$

In order to check that this definition is coordinate-independent, consider the 1-form on $T^*\mathcal{U}$

$$\alpha = \sum_{i=1}^n \xi_i dx_i .$$

Clearly, $\omega = -d\alpha$.

Claim. The form α is intrinsically defined (and hence the form ω is also intrinsically defined) .

Proof. Let $(\mathcal{U}, x_1, \dots, x_n, \xi_1, \dots, \xi_n)$ and $(\mathcal{U}', x'_1, \dots, x'_n, \xi'_1, \dots, \xi'_n)$ be two cotangent coordinate charts. On $\mathcal{U} \cap \mathcal{U}'$, the two sets of coordinates are related by $\xi'_j = \sum_i \xi_i \left(\frac{\partial x_i}{\partial x'_j} \right)$. Since $dx'_j = \sum_i \left(\frac{\partial x'_j}{\partial x_i} \right) dx_i$, we have

$$\alpha = \sum_i \xi_i dx_i = \sum_j \xi'_j dx'_j = \alpha' .$$

□

The 1-form α is the **tautological form** or **Liouville 1-form** and the 2-form ω is the **canonical symplectic form**. The following section provides an alternative proof of the intrinsic character of these forms.

2.3 Coordinate-Free Definitions

Let

$$\begin{array}{ccc} M = T^*X & p = (x, \xi) & \xi \in T_x^*X \\ \downarrow \pi & \downarrow & \\ X & x & \end{array}$$

be the natural projection. The **tautological 1-form** α may be defined pointwise by

$$\alpha_p = (d\pi_p)^* \xi \in T_p^*M ,$$

where $(d\pi_p)^*$ is the transpose of $d\pi_p$, that is, $(d\pi_p)^* \xi = \xi \circ d\pi_p$:

$$\begin{array}{ccc} p = (x, \xi) & T_p M & T_p^* M \\ \downarrow \pi & \downarrow d\pi_p & \uparrow (d\pi_p)^* \\ x & T_x X & T_x^* X \end{array}$$

Equivalently,

$$\alpha_p(v) = \xi \left((d\pi_p)v \right) , \quad \text{for } v \in T_p M .$$

Exercise. Let $(\mathcal{U}, x_1, \dots, x_n)$ be a chart on X with associated cotangent coordinates $x_1, \dots, x_n, \xi_1, \dots, \xi_n$. Show that on $T^*\mathcal{U}$, $\alpha = \sum_{i=1}^n \xi_i dx_i$. ◇

The **canonical symplectic 2-form** ω on T^*X is defined as

$$\omega = -d\alpha .$$

Locally, $\omega = \sum_{i=1}^n dx_i \wedge d\xi_i$.

Exercise. Show that the tautological 1-form α is uniquely characterized by the property that, for every 1-form $\mu : X \rightarrow T^*X$, $\mu^* \alpha = \mu$. (See Lecture 3.) ◇

2.4 Naturality of the Tautological and Canonical Forms

Let X_1 and X_2 be n -dimensional manifolds with cotangent bundles $M_1 = T^*X_1$ and $M_2 = T^*X_2$, and tautological 1-forms α_1 and α_2 . Suppose that $f : X_1 \rightarrow X_2$ is a diffeomorphism. Then there is a natural diffeomorphism

$$f_{\sharp} : M_1 \rightarrow M_2$$

which **lifts** f ; namely, if $p_1 = (x_1, \xi_1) \in M_1$ for $x_1 \in X_1$ and $\xi_1 \in T_{x_1}^*X_1$, then we define

$$f_{\sharp}(p_1) = p_2 = (x_2, \xi_2), \quad \text{with } \begin{cases} x_2 = f(x_1) \in X_2 \text{ and} \\ \xi_2 = (df_{x_1})^* \xi_1, \end{cases}$$

where $(df_{x_1})^* : T_{x_2}^*X_2 \xrightarrow{\sim} T_{x_1}^*X_1$, so $f_{\sharp}|_{T_{x_1}^*X_1}$ is the inverse map of $(df_{x_1})^*$.

Exercise. Check that $f_{\sharp}$ is a diffeomorphism. Here are some hints:

1.
$$\begin{array}{ccc} M_1 & \xrightarrow{f_{\sharp}} & M_2 \\ \pi_1 \downarrow & & \downarrow \pi_2 \\ X_1 & \xrightarrow{f} & X_2 \end{array} \quad \text{commutes;}$$
2. $f_{\sharp} : M_1 \rightarrow M_2$ is bijective;
3. $f_{\sharp}$ and $f_{\sharp}^{-1}$ are smooth.

◇

Proposition 2.1 *The lift $f_{\sharp}$ of a diffeomorphism $f : X_1 \rightarrow X_2$ pulls the tautological form on T^*X_2 back to the tautological form on T^*X_1 , i.e.,*

$$(f_{\sharp})^* \alpha_2 = \alpha_1.$$

Proof. At $p_1 = (x_1, \xi_1) \in M_1$, this identity says

$$(df_{\sharp})_{p_1}^* (\alpha_2)_{p_2} = (\alpha_1)_{p_1} \quad (\star)$$

where $p_2 = f_{\sharp}(p_1)$.

Using the following facts,

- Definition of $f_{\sharp}$:

$$p_2 = f_{\sharp}(p_1) \iff p_2 = (x_2, \xi_2) \text{ where } x_2 = f(x_1) \text{ and } (df_{x_1})^* \xi_2 = \xi_1.$$

- Definition of tautological 1-form:

$$(\alpha_1)_{p_1} = (d\pi_1)_{p_1}^* \xi_1 \quad \text{and} \quad (\alpha_2)_{p_2} = (d\pi_2)_{p_2}^* \xi_2.$$

- The diagram
$$\begin{array}{ccc} M_1 & \xrightarrow{f_{\sharp}} & M_2 \\ \pi_1 \downarrow & & \downarrow \pi_2 \\ X_1 & \xrightarrow{f} & X_2 \end{array} \quad \text{commutes.}$$

the proof of $(\star)$ is:

$$\begin{aligned}
 (df_{\sharp})_{p_1}^* (\alpha_2)_{p_2} &= (df_{\sharp})_{p_1}^* (d\pi_2)_{p_2}^* \xi_2 = (d(\pi_2 \circ f_{\sharp}))_{p_1}^* \xi_2 \\
 &= (d(f \circ \pi_1))_{p_1}^* \xi_2 = (d\pi_1)_{p_1}^* (df)_{x_1}^* \xi_2 \\
 &= (d\pi_1)_{p_1}^* \xi_1 = (\alpha_1)_{p_1} .
 \end{aligned}$$

□

Corollary 2.2 *The lift $f_{\sharp}$ of a diffeomorphism $f : X_1 \rightarrow X_2$ is a symplectomorphism, i.e.,*

$$(f_{\sharp})^* \omega_2 = \omega_1 ,$$

where ω_1, ω_2 are the canonical symplectic forms.

In summary, a diffeomorphism of manifolds induces a canonical symplectomorphism of cotangent bundles:

$$\begin{array}{ccc}
 f_{\sharp} : T^*X_1 & \longrightarrow & T^*X_2 \\
 & \uparrow & \\
 f : X_1 & \longrightarrow & X_2
 \end{array}$$

Example. Let $X_1 = X_2 = S^1$. Then T^*S^1 is an infinite cylinder $S^1 \times \mathbb{R}$. The canonical 2-form ω is the area form $\omega = d\theta \wedge d\xi$. If $f : S^1 \rightarrow S^1$ is any diffeomorphism, then $f_{\sharp} : S^1 \times \mathbb{R} \rightarrow S^1 \times \mathbb{R}$ is a symplectomorphism, i.e., is an area-preserving diffeomorphism of the cylinder. ◇

If $f : X_1 \rightarrow X_2$ and $g : X_2 \rightarrow X_3$ are diffeomorphisms, then $(g \circ f)_{\sharp} = g_{\sharp} \circ f_{\sharp}$. In terms of the group $\text{Diff}(X)$ of diffeomorphisms of X and the group $\text{Symp}(M, \omega)$ of symplectomorphisms of (M, ω) , we say that the map

$$\begin{array}{ccc}
 \text{Diff}(X) & \longrightarrow & \text{Symp}(M, \omega) \\
 f & \longmapsto & f_{\sharp}
 \end{array}$$

is a group homomorphism. This map is clearly injective. Is it surjective? Do all symplectomorphisms $T^*X \rightarrow T^*X$ come from diffeomorphisms $X \rightarrow X$? No: for instance, translation along cotangent fibers is not induced by a diffeomorphism of the base manifold. A criterion for which symplectomorphisms arise as lifts of diffeomorphisms is discussed in Homework 3.

Homework 2: Symplectic Volume

1. Given a vector space V , the exterior algebra of its dual space is

$$\wedge^*(V^*) = \bigoplus_{k=0}^{\dim V} \wedge^k(V^*),$$

where $\wedge^k(V^*)$ is the set of maps $\alpha : \overbrace{V \times \cdots \times V}^k \rightarrow \mathbb{R}$ which are linear in each entry, and for any permutation π , $\alpha(v_{\pi_1}, \dots, v_{\pi_k}) = (\text{sign } \pi) \cdot \alpha(v_1, \dots, v_k)$. The elements of $\wedge^k(V^*)$ are known as **skew-symmetric k -linear maps** or **k -forms** on V .

- (a) Show that any $\Omega \in \wedge^2(V^*)$ is of the form $\Omega = e_1^* \wedge f_1^* + \dots + e_n^* \wedge f_n^*$, where $u_1^*, \dots, u_k^*, e_1^*, \dots, e_n^*, f_1^*, \dots, f_n^*$ is a basis of V^* dual to the standard basis ($k + 2n = \dim V$).
- (b) In this language, a symplectic map $\Omega : V \times V \rightarrow \mathbb{R}$ is just a nondegenerate 2-form $\Omega \in \wedge^2(V^*)$, called a **symplectic form** on V .
Show that, if Ω is any symplectic form on a vector space V of dimension $2n$, then the n th exterior power $\Omega^n = \underbrace{\Omega \wedge \dots \wedge \Omega}_n$ does not vanish.
- (c) Deduce that the n th exterior power ω^n of any symplectic form ω on a $2n$ -dimensional manifold M is a volume form.²
Hence, any symplectic manifold (M, ω) is canonically oriented by the symplectic structure. The form $\frac{\omega^n}{n!}$ is called the **symplectic volume** or the **Liouville volume** of (M, ω) .
Does the Möbius strip support a symplectic structure?
- (d) Conversely, given a 2-form $\Omega \in \wedge^2(V^*)$, show that, if $\Omega^n \neq 0$, then Ω is symplectic.

Hint: Standard form.

2. Let (M, ω) be a $2n$ -dimensional symplectic manifold, and let ω^n be the volume form obtained by wedging ω with itself n times.

- (a) Show that, if M is compact, the de Rham cohomology class $[\omega^n] \in H^{2n}(M; \mathbb{R})$ is non-zero.
Hint: Stokes' theorem.
- (b) Conclude that $[\omega]$ itself is non-zero (in other words, that ω is not exact).
- (c) Show that if $n > 1$ there are no symplectic structures on the sphere S^{2n} .

²A **volume form** is a nonvanishing form of top degree.

Part II

Symplectomorphisms

Equivalence between symplectic manifolds is expressed by a *symplectomorphism*. By Weinstein's lagrangian creed [105], everything is a lagrangian manifold! We will study symplectomorphisms according to the creed.

3 Lagrangian Submanifolds

3.1 Submanifolds

Let M and X be manifolds with $\dim X < \dim M$.

Definition 3.1 A map $i : X \rightarrow M$ is an **immersion** if $di_p : T_p X \rightarrow T_{i(p)} M$ is injective for any point $p \in X$.

An **embedding** is an immersion which is a homeomorphism onto its image.³

A **closed embedding** is a proper⁴ injective immersion.

Exercise. Show that a map $i : X \rightarrow M$ is a closed embedding if and only if i is an embedding and its image $i(X)$ is closed in M .

Hints:

- If i is injective and proper, then for any neighborhood $\mathcal{U}$ of $p \in X$, there is a neighborhood $\mathcal{V}$ of $i(p)$ such that $f^{-1}(\mathcal{V}) \subseteq \mathcal{U}$.
- On a Hausdorff space, any compact set is closed. On any topological space, a closed subset of a compact set is compact.
- An embedding is proper if and only if its image is closed.

◇

Definition 3.2 A **submanifold** of M is a manifold X with a closed embedding $i : X \hookrightarrow M$.⁵

Notation. Given a submanifold, we regard the embedding $i : X \hookrightarrow M$ as an inclusion, in order to identify points and tangent vectors:

$$p = i(p) \quad \text{and} \quad T_p X = di_p(T_p X) \subset T_p M .$$

³The image has the topology induced by the target manifold.

⁴A map is **proper** if the preimage of any compact set is compact.

⁵When X is an open subset of a manifold M , we refer to it as an *open* submanifold.

3.2 Lagrangian Submanifolds of T^*X

Definition 3.3 Let (M, ω) be a $2n$ -dimensional symplectic manifold. A submanifold Y of M is a **lagrangian submanifold** if, at each $p \in Y$, $T_p Y$ is a lagrangian subspace of $T_p M$, i.e., $\omega_p|_{T_p Y} \equiv 0$ and $\dim T_p Y = \frac{1}{2} \dim T_p M$. Equivalently, if $i : Y \hookrightarrow M$ is the inclusion map, then Y is **lagrangian** if and only if $i^* \omega = 0$ and $\dim Y = \frac{1}{2} \dim M$.

Let X be an n -dimensional manifold, with $M = T^*X$ its cotangent bundle. If $x_1, \dots, x_n$ are coordinates on $U \subseteq X$, with associated cotangent coordinates $x_1, \dots, x_n, \xi_1, \dots, \xi_n$ on T^*U , then the tautological 1-form on T^*X is

$$\alpha = \sum \xi_i dx_i$$

and the canonical 2-form on T^*X is

$$\omega = -d\alpha = \sum dx_i \wedge d\xi_i .$$

The **zero section** of T^*X

$$X_0 := \{(x, \xi) \in T^*X \mid \xi = 0 \text{ in } T_x^*X\}$$

is an n -dimensional submanifold of T^*X whose intersection with T^*U is given by the equations $\xi_1 = \dots = \xi_n = 0$. Clearly $\alpha = \sum \xi_i dx_i$ vanishes on $X_0 \cap T^*U$. In particular, if $i_0 : X_0 \hookrightarrow T^*X$ is the inclusion map, we have $i_0^* \alpha = 0$. Hence, $i_0^* \omega = i_0^* d\alpha = 0$, and X_0 is lagrangian.

What are all the lagrangian submanifolds of T^*X which are “ C^1 -close to X_0 ”?

Let X_μ be (the image of) another section, that is, an n -dimensional submanifold of T^*X of the form

$$X_\mu = \{(x, \mu_x) \mid x \in X, \mu_x \in T_x^*X\} \quad (\star)$$

where the covector μ_x depends smoothly on x , and $\mu : X \rightarrow T^*X$ is a de Rham 1-form. Relative to the inclusion $i : X_\mu \hookrightarrow T^*X$ and the cotangent projection $\pi : T^*X \rightarrow X$, X_μ is of the form $(\star)$ if and only if $\pi \circ i : X_\mu \rightarrow X$ is a diffeomorphism.

When is such an X_μ lagrangian?

Proposition 3.4 Let X_μ be of the form $(\star)$, and let μ be the associated de Rham 1-form. Denote by $s_\mu : X \rightarrow T^*X$, $x \mapsto (x, \mu_x)$, the 1-form μ regarded exclusively as a map. Notice that the image of s_μ is X_μ . Let α be the tautological 1-form on T^*X . Then

$$s_\mu^* \alpha = \mu .$$

Proof. By definition of α (previous lecture), $\alpha_p = (d\pi_p)^* \xi$ at $p = (x, \xi) \in M$. For $p = s_\mu(x) = (x, \mu_x)$, we have $\alpha_p = (d\pi_p)^* \mu_x$. Then

$$(s_\mu^* \alpha)_x = (ds_\mu)_x^* \alpha_p = (ds_\mu)_x^* (d\pi_p)^* \mu_x = (d(\underbrace{\pi \circ s_\mu}_{\text{id}_X}))_x^* \mu_x = \mu_x .$$

□

Suppose that X_μ is an n -dimensional submanifold of T^*X of the form $(\star)$, with associated de Rham 1-form μ . Then $s_\mu : X \rightarrow T^*X$ is an embedding with image X_μ , and there is a diffeomorphism $\tau : X \rightarrow X_\mu$, $\tau(x) := (x, \mu_x)$, such that the following diagram commutes.

$$\begin{array}{ccc} X & \xrightarrow{s_\mu} & T^*X \\ & \searrow \tau & \nearrow i \\ & X_\mu & \end{array}$$

We want to express the condition of X_μ being lagrangian in terms of the form μ :

$$\begin{aligned} X_\mu \text{ is lagrangian} &\iff i^*d\alpha = 0 \\ &\iff \tau^*i^*d\alpha = 0 \\ &\iff (i \circ \tau)^*d\alpha = 0 \\ &\iff s_\mu^*d\alpha = 0 \\ &\iff ds_\mu^*\alpha = 0 \\ &\iff d\mu = 0 \\ &\iff \mu \text{ is closed .} \end{aligned}$$

Therefore, there is a one-to-one correspondence between the set of lagrangian submanifolds of T^*X of the form $(\star)$ and the set of closed 1-forms on X .

When X is simply connected, $H_{\text{deRham}}^1(X) = 0$, so every closed 1-form μ is equal to df for some $f \in C^\infty(X)$. Any such primitive f is then called a **generating function** for the lagrangian submanifold X_μ associated to μ . (Two functions generate the same lagrangian submanifold if and only if they differ by a locally constant function.) On arbitrary manifolds X , functions $f \in C^\infty(X)$ originate lagrangian submanifolds as images of df .

Exercise. Check that, if X is compact (and not just one point) and $f \in C^\infty(X)$, then $\#(X_{df} \cap X_0) \geq 2$. $\diamond$

There are lots of lagrangian submanifolds of T^*X not covered by the description in terms of closed 1-forms, starting with the cotangent fibers.

3.3 Conormal Bundles

Let S be any k -dimensional submanifold of an n -dimensional manifold X .

Definition 3.5 *The conormal space at $x \in S$ is*

$$N_x^*S = \{\xi \in T_x^*X \mid \xi(v) = 0, \text{ for all } v \in T_x S\} .$$

The conormal bundle of S is

$$N^*S = \{(x, \xi) \in T^*X \mid x \in S, \xi \in N_x^*S\} .$$

Exercise. The conormal bundle N^*S is an n -dimensional submanifold of T^*X .

Hint: Use coordinates on X adapted⁶ to S . $\diamond$

Proposition 3.6 *Let $i : N^*S \hookrightarrow T^*X$ be the inclusion, and let α be the tautological 1-form on T^*X . Then*

$$i^*\alpha = 0 .$$

Proof. Let $(\mathcal{U}, x_1, \dots, x_n)$ be a coordinate system on X centered at $x \in S$ and adapted to S , so that $\mathcal{U} \cap S$ is described by $x_{k+1} = \dots = x_n = 0$. Let $(T^*\mathcal{U}, x_1, \dots, x_n, \xi_1, \dots, \xi_n)$ be the associated cotangent coordinate system. The submanifold $N^*S \cap T^*\mathcal{U}$ is then described by

$$x_{k+1} = \dots = x_n = 0 \quad \text{and} \quad \xi_1 = \dots = \xi_k = 0 .$$

Since $\alpha = \sum \xi_i dx_i$ on $T^*\mathcal{U}$, we conclude that, at $p \in N^*S$,

$$(i^*\alpha)_p = \alpha_p|_{T_p(N^*S)} = \sum_{i>k} \xi_i dx_i \Big|_{\text{span}\{\frac{\partial}{\partial x_i}, i \leq k\}} = 0 .$$

□

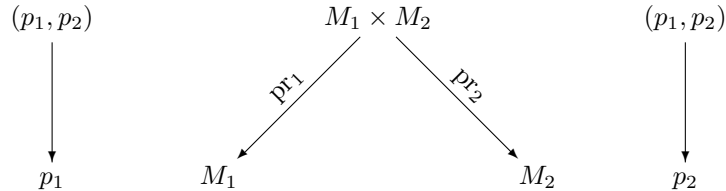
Corollary 3.7 *For any submanifold $S \subset X$, the conormal bundle N^*S is a Lagrangian submanifold of T^*X .*

Taking $S = \{x\}$ to be one point, the conormal bundle $L = N^*S = T_x^*X$ is a cotangent fiber. Taking $S = X$, the conormal bundle $L = X_0$ is the zero section of T^*X .

3.4 Application to Symplectomorphisms

Let (M_1, ω_1) and (M_2, ω_2) be two $2n$ -dimensional symplectic manifolds. Given a diffeomorphism $\varphi : M_1 \xrightarrow{\sim} M_2$, when is it a symplectomorphism? (i.e., when is $\varphi^*\omega_2 = \omega_1$?)

Consider the two projection maps



⁶A coordinate chart $(\mathcal{U}, x_1, \dots, x_n)$ on X is adapted to a k -dimensional submanifold S if $S \cap \mathcal{U}$ is described by $x_{k+1} = \dots = x_n = 0$.

Then $\omega = (\text{pr}_1)^*\omega_1 + (\text{pr}_2)^*\omega_2$ is a 2-form on $M_1 \times M_2$ which is closed,

$$d\omega = (\text{pr}_1)^* \underbrace{d\omega_1}_0 + (\text{pr}_2)^* \underbrace{d\omega_2}_0 = 0 ,$$

and symplectic,

$$\omega^{2n} = \binom{2n}{n} \left((\text{pr}_1)^*\omega_1 \right)^n \wedge \left((\text{pr}_2)^*\omega_2 \right)^n \neq 0 .$$

More generally, if $\lambda_1, \lambda_2 \in \mathbb{R} \setminus \{0\}$, then $\lambda_1(\text{pr}_1)^*\omega_1 + \lambda_2(\text{pr}_2)^*\omega_2$ is also a symplectic form on $M_1 \times M_2$. Take $\lambda_1 = 1$, $\lambda_2 = -1$ to obtain the **twisted product form** on $M_1 \times M_2$:

$$\tilde{\omega} = (\text{pr}_1)^*\omega_1 - (\text{pr}_2)^*\omega_2 .$$

The graph of a diffeomorphism $\varphi : M_1 \xrightarrow{\cong} M_2$ is the $2n$ -dimensional submanifold of $M_1 \times M_2$:

$$\Gamma_\varphi := \text{Graph } \varphi = \{ (p, \varphi(p)) \mid p \in M_1 \} .$$

The submanifold Γ_φ is an embedded image of M_1 in $M_1 \times M_2$, the embedding being the map

$$\begin{aligned} \gamma : M_1 &\longrightarrow M_1 \times M_2 \\ p &\longmapsto (p, \varphi(p)) . \end{aligned}$$

Proposition 3.8 *A diffeomorphism φ is a symplectomorphism if and only if Γ_φ is a lagrangian submanifold of $(M_1 \times M_2, \tilde{\omega})$.*

Proof. The graph Γ_φ is lagrangian if and only if $\gamma^*\tilde{\omega} = 0$. But

$$\begin{aligned} \gamma^*\tilde{\omega} &= \gamma^* \text{pr}_1^* \omega_1 - \gamma^* \text{pr}_2^* \omega_2 \\ &= (\text{pr}_1 \circ \gamma)^* \omega_1 - (\text{pr}_2 \circ \gamma)^* \omega_2 \end{aligned}$$

and $\text{pr}_1 \circ \gamma$ is the identity map on M_1 whereas $\text{pr}_2 \circ \gamma = \varphi$. Therefore,

$$\gamma^*\tilde{\omega} = 0 \iff \varphi^*\omega_2 = \omega_1 .$$

□

Homework 3: Tautological Form and Symplectomorphisms

This set of problems is from [53].

1. Let (M, ω) be a symplectic manifold, and let α be a 1-form such that

$$\omega = -d\alpha .$$

Show that there exists a unique vector field v such that its interior product with ω is α , i.e., $\iota_v \omega = -\alpha$.

Prove that, if g is a symplectomorphism which preserves α (that is, $g^* \alpha = \alpha$), then g commutes with the one-parameter group of diffeomorphisms generated by v , i.e.,

$$(\exp tv) \circ g = g \circ (\exp tv) .$$

Hint: Recall that, for $p \in M$, $(\exp tv)(p)$ is the *unique* curve in M solving the ordinary differential equation

$$\begin{cases} \frac{d}{dt}(\exp tv(p)) = v(\exp tv(p)) \\ (\exp tv)(p)|_{t=0} = p \end{cases}$$

for t in some neighborhood of 0. Show that $g \circ (\exp tv) \circ g^{-1}$ is the one-parameter group of diffeomorphisms generated by $g_* v$. (The push-forward of v by g is defined by $(g_* v)_{g(p)} = dg_p(v_p)$.) Finally check that g preserves v (that is, $g_* v = v$).

2. Let X be an arbitrary n -dimensional manifold, and let $M = T^*X$. Let $(\mathcal{U}, x_1, \dots, x_n)$ be a coordinate system on X , and let $x_1, \dots, x_n, \xi_1, \dots, \xi_n$ be the corresponding coordinates on $T^*\mathcal{U}$.

Show that, when α is the tautological 1-form on M (which, in these coordinates, is $\sum \xi_i dx_i$), the vector field v in the previous exercise is just the vector field $\sum \xi_i \frac{\partial}{\partial \xi_i}$.

Let $\exp tv$, $-\infty < t < \infty$, be the one-parameter group of diffeomorphisms generated by v .

Show that, for every point $p = (x, \xi)$ in M ,

$$(\exp tv)(p) = p_t \quad \text{where} \quad p_t = (x, e^t \xi) .$$

3. Let M be as in exercise 2.

Show that, if g is a symplectomorphism of M which preserves α , then

$$g(x, \xi) = (y, \eta) \implies g(x, \lambda\xi) = (y, \lambda\eta)$$

for all $(x, \xi) \in M$ and $\lambda \in \mathbb{R}$.

Conclude that g has to preserve the cotangent fibration, i.e., show that there exists a diffeomorphism $f : X \rightarrow X$ such that $\pi \circ g = f \circ \pi$, where $\pi : M \rightarrow X$ is the projection map $\pi(x, \xi) = x$.

Finally prove that $g = f_\#$, the map $f_\#$ being the symplectomorphism of M lifting f .

Hint: Suppose that $g(p) = q$ where $p = (x, \xi)$ and $q = (y, \eta)$.
Combine the identity

$$(dg_p)^* \alpha_q = \alpha_p$$

with the identity

$$d\pi_q \circ dg_p = df_x \circ d\pi_p .$$

(The first identity expresses the fact that $g^* \alpha = \alpha$, and the second identity is obtained by differentiating both sides of the equation $\pi \circ g = f \circ \pi$ at p .)

4. Let M be as in exercise 2, and let h be a smooth function on X . Define $\tau_h : M \rightarrow M$ by setting

$$\tau_h(x, \xi) = (x, \xi + dh_x) .$$

Prove that

$$\tau_h^* \alpha = \alpha + \pi^* dh$$

where π is the projection map

$$\begin{array}{ccc} M & & (x, \xi) \\ \downarrow \pi & & \downarrow \\ X & & x \end{array}$$

Deduce that

$$\tau_h^* \omega = \omega ,$$

i.e., that τ_h is a symplectomorphism.

4 Generating Functions

4.1 Constructing Symplectomorphisms

Let X_1, X_2 be n -dimensional manifolds, with cotangent bundles $M_1 = T^*X_1$, $M_2 = T^*X_2$, tautological 1-forms α_1, α_2 , and canonical 2-forms ω_1, ω_2 .

Under the natural identification

$$M_1 \times M_2 = T^*X_1 \times T^*X_2 \simeq T^*(X_1 \times X_2),$$

the tautological 1-form on $T^*(X_1 \times X_2)$ is

$$\alpha = (\text{pr}_1)^*\alpha_1 + (\text{pr}_2)^*\alpha_2,$$

where $\text{pr}_i : M_1 \times M_2 \rightarrow M_i$, $i = 1, 2$ are the two projections. The canonical 2-form on $T^*(X_1 \times X_2)$ is

$$\omega = -d\alpha = -d\text{pr}_1^*\alpha_1 - d\text{pr}_2^*\alpha_2 = \text{pr}_1^*\omega_1 + \text{pr}_2^*\omega_2.$$

In order to describe the twisted form $\tilde{\omega} = \text{pr}_1^*\omega_1 - \text{pr}_2^*\omega_2$, we define an involution of $M_2 = T^*X_2$ by

$$\begin{aligned} \sigma_2 : M_2 &\longrightarrow M_2 \\ (x_2, \xi_2) &\longmapsto (x_2, -\xi_2) \end{aligned}$$

which yields $\sigma_2^*\alpha_2 = -\alpha_2$. Let $\sigma = \text{id}_{M_1} \times \sigma_2 : M_1 \times M_2 \rightarrow M_1 \times M_2$. Then

$$\sigma^*\tilde{\omega} = \text{pr}_1^*\omega_1 + \text{pr}_2^*\omega_2 = \omega.$$

If Y is a lagrangian submanifold of $(M_1 \times M_2, \omega)$, then its “twist” $Y^\sigma := \sigma(Y)$ is a lagrangian submanifold of $(M_1 \times M_2, \tilde{\omega})$.

Recipe for producing symplectomorphisms $M_1 = T^*X_1 \rightarrow M_2 = T^*X_2$:

1. Start with a lagrangian submanifold Y of $(M_1 \times M_2, \omega)$.
2. Twist it to obtain a lagrangian submanifold Y^σ of $(M_1 \times M_2, \tilde{\omega})$.
3. Check whether Y^σ is the graph of some diffeomorphism $\varphi : M_1 \rightarrow M_2$.
4. If it is, then φ is a symplectomorphism (by Proposition 3.8).

Let $i : Y^\sigma \hookrightarrow M_1 \times M_2$ be the inclusion map

$$\begin{array}{ccc} & Y^\sigma & \\ \text{pr}_1 \circ i \swarrow & & \searrow \text{pr}_2 \circ i \\ M_1 & \xrightarrow{\varphi?} & M_2 \end{array}$$

Step 3 amounts to checking whether $\text{pr}_1 \circ i$ and $\text{pr}_2 \circ i$ are diffeomorphisms. If yes, then $\varphi := (\text{pr}_2 \circ i) \circ (\text{pr}_1 \circ i)^{-1}$ is a diffeomorphism.

In order to obtain lagrangian submanifolds of $M_1 \times M_2 \simeq T^*(X_1 \times X_2)$, we can use the *method of generating functions*.

4.2 Method of Generating Functions

For any $f \in C^\infty(X_1 \times X_2)$, df is a closed 1-form on $X_1 \times X_2$. The **lagrangian submanifold generated by f** is

$$Y_f := \{((x, y), (df)_{(x, y)}) \mid (x, y) \in X_1 \times X_2\} .$$

We adopt the notation

$$\begin{aligned} d_x f &:= (df)_{(x, y)} \text{ projected to } T_x^* X_1 \times \{0\}, \\ d_y f &:= (df)_{(x, y)} \text{ projected to } \{0\} \times T_y^* X_2, \end{aligned}$$

which enables us to write

$$Y_f = \{(x, y, d_x f, d_y f) \mid (x, y) \in X_1 \times X_2\}$$

and

$$Y_f^\sigma = \{(x, y, d_x f, -d_y f) \mid (x, y) \in X_1 \times X_2\} .$$

When Y_f^σ is in fact the graph of a diffeomorphism $\varphi : M_1 \rightarrow M_2$, we call φ the **symplectomorphism generated by f** , and call f the **generating function**, of $\varphi : M_1 \rightarrow M_2$.

So when is Y_f^σ the graph of a diffeomorphism $\varphi : M_1 \rightarrow M_2$?

Let $(\mathcal{U}_1, x_1, \dots, x_n), (\mathcal{U}_2, y_1, \dots, y_n)$ be coordinate charts for X_1, X_2 , with associated charts $(T^*\mathcal{U}_1, x_1, \dots, x_n, \xi_1, \dots, \xi_n), (T^*\mathcal{U}_2, y_1, \dots, y_n, \eta_1, \dots, \eta_n)$ for M_1, M_2 . The set Y_f^σ is the graph of $\varphi : M_1 \rightarrow M_2$ if and only if, for any $(x, \xi) \in M_1$ and $(y, \eta) \in M_2$, we have

$$\varphi(x, \xi) = (y, \eta) \iff \xi = d_x f \text{ and } \eta = -d_y f .$$

Therefore, given a point $(x, \xi) \in M_1$, to find its image $(y, \eta) = \varphi(x, \xi)$ we must solve the ‘‘Hamilton’’ equations

$$\begin{cases} \xi_i &= \frac{\partial f}{\partial x_i}(x, y) & (\star) \\ \eta_i &= -\frac{\partial f}{\partial y_i}(x, y) . & (\star\star) \end{cases}$$

If there is a solution $y = \varphi_1(x, \xi)$ of $(\star)$, we may feed it to $(\star\star)$ thus obtaining $\eta = \varphi_2(x, \xi)$, so that $\varphi(x, \xi) = (\varphi_1(x, \xi), \varphi_2(x, \xi))$. Now by the implicit function theorem, in order to solve $(\star)$ locally for y in terms of x and ξ , we need the condition

$$\det \left[\frac{\partial}{\partial y_j} \left(\frac{\partial f}{\partial x_i} \right) \right]_{i, j=1}^n \neq 0 .$$

This is a necessary local condition for f to generate a symplectomorphism φ . Locally this is also sufficient, but globally there is the usual bijectivity issue.

Example. Let $X_1 = \mathcal{U}_1 \simeq \mathbb{R}^n$, $X_2 = \mathcal{U}_2 \simeq \mathbb{R}^n$, and $f(x, y) = -\frac{|x-y|^2}{2}$, the square of euclidean distance up to a constant.

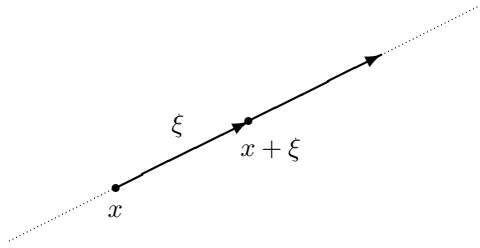
The “Hamilton” equations are

$$\begin{cases} \xi_i &= \frac{\partial f}{\partial x_i} &= y_i - x_i \\ \eta_i &= -\frac{\partial f}{\partial y_i} &= y_i - x_i \end{cases} \iff \begin{cases} y_i &= x_i + \xi_i \\ \eta_i &= \xi_i . \end{cases}$$

The symplectomorphism generated by f is

$$\varphi(x, \xi) = (x + \xi, \xi) .$$

If we use the euclidean inner product to identify $T^*\mathbb{R}^n$ with $T\mathbb{R}^n$, and hence regard φ as $\tilde{\varphi} : T\mathbb{R}^n \rightarrow T\mathbb{R}^n$ and interpret ξ as the velocity vector, then the symplectomorphism φ corresponds to free translational motion in euclidean space.



◇

4.3 Application to Geodesic Flow

Let V be an n -dimensional vector space. A **positive inner product** G on V is a bilinear map $G : V \times V \rightarrow \mathbb{R}$ which is

$$\begin{array}{ll} \text{symmetric :} & G(v, w) = G(w, v) , \quad \text{and} \\ \text{positive-definite :} & G(v, v) > 0 \quad \text{when} \quad v \neq 0 . \end{array}$$

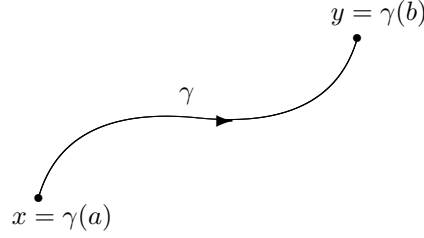
Definition 4.1 A **riemannian metric** on a manifold X is a function g which assigns to each point $x \in X$ a positive inner product g_x on $T_x X$.

A **riemannian metric** g is **smooth** if for every smooth vector field $v : X \rightarrow TX$ the real-valued function $x \mapsto g_x(v_x, v_x)$ is a smooth function on X .

Definition 4.2 A **riemannian manifold** (X, g) is a manifold X equipped with a smooth riemannian metric g .

The **arc-length** of a piecewise smooth curve $\gamma : [a, b] \rightarrow X$ on a riemannian manifold (X, g) is

$$\int_a^b \sqrt{g_{\gamma(t)} \left(\frac{d\gamma}{dt}, \frac{d\gamma}{dt} \right)} dt .$$



Definition 4.3 The **riemannian distance** between two points x and y of a connected riemannian manifold (X, g) is the infimum $d(x, y)$ of the set of all arc-lengths for piecewise smooth curves joining x to y .

A smooth curve joining x to y is a **minimizing geodesic**⁷ if its arc-length is the riemannian distance $d(x, y)$.

A riemannian manifold (X, g) is **geodesically convex** if every point x is joined to every other point y by a unique minimizing geodesic.

Example. On $X = \mathbb{R}^n$ with $TX \simeq \mathbb{R}^n \times \mathbb{R}^n$, let $g_x(v, w) = \langle v, w \rangle$, $g_x(v, v) = |v|^2$, where $\langle \cdot, \cdot \rangle$ is the euclidean inner product, and $|\cdot|$ is the euclidean norm. Then $(\mathbb{R}^n, \langle \cdot, \cdot \rangle)$ is a geodesically convex riemannian manifold, and the riemannian distance is the usual euclidean distance $d(x, y) = |x - y|$. $\diamond$

Suppose that (X, g) is a geodesically convex riemannian manifold. Consider the function

$$f : X \times X \longrightarrow \mathbb{R}, \quad f(x, y) = -\frac{d(x, y)^2}{2} .$$

What is the symplectomorphism $\varphi : T^*X \rightarrow T^*X$ generated by f ?

The metric $g_x : T_x X \times T_x X \rightarrow \mathbb{R}$ induces an identification

$$\begin{array}{ccc} \tilde{g}_x : T_x X & \xrightarrow{\simeq} & T_x^* X \\ v & \longmapsto & g_x(v, \cdot) \end{array}$$

Use $\tilde{g}$ to translate φ into a map $\tilde{\varphi} : TX \rightarrow TX$.

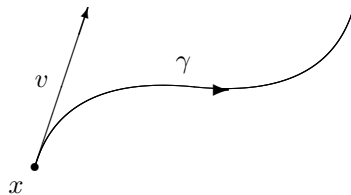
We need to solve

$$\begin{cases} \tilde{g}_x(v) &= \xi_i &= d_x f(x, y) \\ \tilde{g}_y(w) &= \eta_i &= -d_y f(x, y) \end{cases}$$

⁷In riemannian geometry, a **geodesic** is a curve which locally minimizes distance and whose velocity is constant.

for (y, η) in terms of (x, ξ) in order to find φ , or, equivalently, for (y, w) in terms (x, v) in order to find $\tilde{\varphi}$.

Let γ be the geodesic with initial conditions $\gamma(0) = x$ and $\frac{d\gamma}{dt}(0) = v$.



Then the symplectomorphism φ corresponds to the map

$$\begin{aligned} \tilde{\varphi}: \quad TX &\longrightarrow TX \\ (x, v) &\longmapsto (\gamma(1), \frac{d\gamma}{dt}(1)) . \end{aligned}$$

This is called the **geodesic flow** on X (see Homework 4).

Homework 4: Geodesic Flow

This set of problems is adapted from [53].

Let (X, g) be a riemannian manifold. The arc-length of a smooth curve $\gamma : [a, b] \rightarrow X$ is

$$\text{arc-length of } \gamma := \int_a^b \left| \frac{d\gamma}{dt} \right| dt, \quad \text{where} \quad \left| \frac{d\gamma}{dt} \right| := \sqrt{g_{\gamma(t)} \left(\frac{d\gamma}{dt}, \frac{d\gamma}{dt} \right)}.$$

1. Show that the arc-length of γ is independent of the parametrization of γ , i.e., show that, if we reparametrize γ by $\tau : [a', b'] \rightarrow [a, b]$, the new curve $\gamma' = \gamma \circ \tau : [a', b'] \rightarrow X$ has the same arc-length. A curve γ is called a curve of *constant velocity* when $\left| \frac{d\gamma}{dt} \right|$ is independent of t . Show that, given any curve $\gamma : [a, b] \rightarrow X$ (with $\frac{d\gamma}{dt}$ never vanishing), there is a reparametrization $\tau : [a, b] \rightarrow [a, b]$ such that $\gamma \circ \tau : [a, b] \rightarrow X$ is of constant velocity.

2. Given a smooth curve $\gamma : [a, b] \rightarrow X$, the *action* of γ is $\mathcal{A}(\gamma) := \int_a^b \left| \frac{d\gamma}{dt} \right|^2 dt$.

Show that, among all curves joining x to y , γ minimizes the action if and only if γ is of constant velocity and γ minimizes arc-length.

Hint: Suppose that γ is of constant velocity, and let $\tau : [a, b] \rightarrow [a, b]$ be a reparametrization. Show that $\mathcal{A}(\gamma \circ \tau) \geq \mathcal{A}(\gamma)$, with equality only when $\tau =$ identity.

3. Assume that (X, g) is geodesically convex, that is, any two points $x, y \in X$ are joined by a unique (up to reparametrization) minimizing geodesic; its arc-length $d(x, y)$ is called the riemannian distance between x and y .

Assume also that (X, g) is *geodesically complete*, that is, every geodesic can be extended indefinitely. Given $(x, v) \in TX$, let $\exp(x, v) : \mathbb{R} \rightarrow X$ be the unique minimizing geodesic of constant velocity with initial conditions $\exp(x, v)(0) = x$ and $\frac{d\exp(x, v)}{dt}(0) = v$.

Consider the function $f : X \times X \rightarrow \mathbb{R}$ given by $f(x, y) = -\frac{1}{2} \cdot d(x, y)^2$. Let $d_x f$ and $d_y f$ be the components of $df_{(x, y)}$ with respect to $T_{(x, y)}^*(X \times X) \simeq T_x^*X \times T_y^*X$. Recall that, if

$$\Gamma_f^\sigma = \{(x, y, d_x f, -d_y f) \mid (x, y) \in X \times X\}$$

is the graph of a diffeomorphism $f : T^*X \rightarrow T^*X$, then f is the symplectomorphism generated by f . In this case, $f(x, \xi) = (y, \eta)$ if and only if $\xi = d_x f$ and $\eta = -d_y f$.

Show that, under the identification of TX with T^*X by g , the symplectomorphism generated by f coincides with the map $TX \rightarrow TX$, $(x, v) \mapsto \exp(x, v)(1)$.

Hint: The metric g provides the identifications $T_x X v \simeq \xi(\cdot) = g_x(v, \cdot) \in T_x^* X$. We need to show that, given $(x, v) \in TX$, the unique solution of

$$(*) \begin{cases} g_x(v, \cdot) = d_x f(\cdot) \\ g_y(w, \cdot) = -d_y f(\cdot) \end{cases} \text{ is } (y, w) = (\exp(x, v)(1), d \frac{\exp(x, v)}{dt}(1)).$$

Look up the Gauss lemma in a book on riemannian geometry. It asserts that geodesics are orthogonal to the level sets of the distance function.

To solve the first line in $(*)$ for y , evaluate both sides at $v = \frac{d \exp(x, v)}{dt}(0)$. Conclude that $y = \exp(x, v)(1)$. Check that $d_x f(v') = 0$ for vectors $v' \in T_x X$ orthogonal to v (that is, $g_x(v, v') = 0$); this is a consequence of $f(x, y)$ being the arc-length of a *minimizing* geodesic, and it suffices to check locally.

The vector w is obtained from the second line of $(*)$. Compute $-d_y f(\frac{d \exp(x, v)}{dt}(1))$. Then evaluate $-d_y f$ at vectors $w' \in T_y X$ orthogonal to $\frac{d \exp(x, v)}{dt}(1)$; this pairing is again 0 because $f(x, y)$ is the arc-length of a minimizing geodesic. Conclude, using the nondegeneracy of g , that $w = \frac{d \exp(x, v)}{dt}(1)$. For both steps, it might be useful to recall that, given a function $\varphi : X \rightarrow \mathbb{R}$ and a tangent vector $v \in T_x X$, we have $d\varphi_x(v) = \frac{d}{du} [\varphi(\exp(x, v)(u))]_{u=0}$.

5 Recurrence

5.1 Periodic Points

Let X be an n -dimensional manifold. Let $M = T^*X$ be its cotangent bundle with canonical symplectic form ω .

Suppose that we are given a smooth function $f : X \times X \rightarrow \mathbb{R}$ which generates a symplectomorphism $\varphi : M \rightarrow M$, $\varphi(x, d_x f) = (y, -d_y f)$, by the recipe of the previous lecture.

What are the fixed points of φ ?

Define $\psi : X \rightarrow \mathbb{R}$ by $\psi(x) = f(x, x)$.

Proposition 5.1 *There is a one-to-one correspondence between the fixed points of φ and the critical points of ψ .*

Proof. At $x_0 \in X$, $d_{x_0} \psi = (d_x f + d_y f)|_{(x,y)=(x_0,x_0)}$. Let $\xi = d_x f|_{(x,y)=(x_0,x_0)}$.

$$x_0 \text{ is a critical point of } \psi \iff d_{x_0} \psi = 0 \iff d_y f|_{(x,y)=(x_0,x_0)} = -\xi .$$

Hence, the point in Γ_f^σ corresponding to $(x, y) = (x_0, x_0)$ is (x_0, x_0, ξ, ξ) . But Γ_f^σ is the graph of φ , so $\varphi(x_0, \xi) = (x_0, \xi)$ is a fixed point. This argument also works backwards. $\square$

Consider the iterates of φ ,

$$\varphi^{(N)} = \underbrace{\varphi \circ \varphi \circ \dots \circ \varphi}_N : M \longrightarrow M , \quad N = 1, 2, \dots ,$$

each of which is a symplectomorphism of M . According to the previous proposition, if $\varphi^{(N)} : M \rightarrow M$ is generated by $f^{(N)}$, then there is a correspondence

$$\left\{ \text{fixed points of } \varphi^{(N)} \right\} \xrightarrow{1-1} \left\{ \begin{array}{c} \text{critical points of} \\ \psi^{(N)} : X \rightarrow \mathbb{R} , \ \psi^{(N)}(x) = f^{(N)}(x, x) \end{array} \right\}$$

Knowing that φ is generated by f , does $\varphi^{(2)}$ have a generating function? The answer is a partial yes:

Fix $x, y \in X$. Define a map

$$\begin{aligned} X &\longrightarrow \mathbb{R} \\ z &\longmapsto f(x, z) + f(z, y) . \end{aligned}$$

Suppose that this map has a unique critical point z_0 , and that z_0 is nondegenerate. Let

$$f^{(2)}(x, y) := f(x, z_0) + f(z_0, y) .$$

Proposition 5.2 *The function $f^{(2)} : X \times X \rightarrow \mathbb{R}$ is smooth and is a generating function for $\varphi^{(2)}$ if we assume that, for each $\xi \in T_x^*X$, there is a unique $y \in X$ for which $d_x f^{(2)} = \xi$.*

Proof. The point z_0 is given implicitly by $d_y f(x, z_0) + d_x f(z_0, y) = 0$. The nondegeneracy condition is

$$\det \left[\frac{\partial}{\partial z_i} \left(\frac{\partial f}{\partial y_j}(x, z) + \frac{\partial f}{\partial x_j}(z, y) \right) \right] \neq 0.$$

By the implicit function theorem, $z_0 = z_0(x, y)$ is smooth.

As for the second assertion, $f^{(2)}(x, y)$ is a generating function for $\varphi^{(2)}$ if and only if

$$\varphi^{(2)}(x, d_x f^{(2)}) = (y, -d_y f^{(2)})$$

(assuming that, for each $\xi \in T_x^*X$, there is a unique $y \in X$ for which $d_x f^{(2)} = \xi$). Since φ is generated by f , and z_0 is critical, we obtain

$$\begin{aligned} \varphi^{(2)}(x, d_x f^{(2)}(x, y)) &= \varphi(\varphi(x, \underbrace{d_x f^{(2)}(x, y)}_{=d_x f(x, z_0)})) = \varphi(z_0, -d_y f(x, z_0)) \\ &= \varphi(z_0, d_x f(z_0, y)) = (y, \underbrace{-d_y f(z_0, y)}_{=-d_y f^{(2)}(x, y)}). \end{aligned}$$

□

Exercise. What is a generating function for $\varphi^{(3)}$?

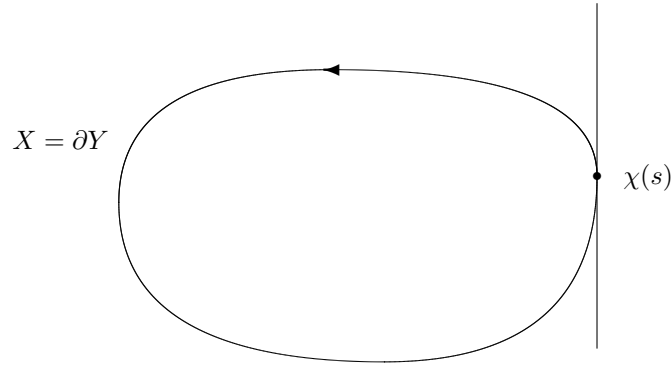
Hint: Suppose that the function

$$\begin{aligned} X \times X &\longrightarrow \mathbb{R} \\ (z, u) &\longmapsto f(x, z) + f(z, u) + f(u, y) \end{aligned}$$

has a unique critical point (z_0, u_0) , and that it is a nondegenerate critical point. Let $f^{(3)}(x, y) = f(x, z_0) + f(z_0, u_0) + f(u_0, y)$. ◇

5.2 Billiards

Let $\chi : \mathbb{R} \rightarrow \mathbb{R}^2$ be a smooth plane curve which is 1-periodic, i.e., $\chi(s+1) = \chi(s)$, and parametrized by arc-length, i.e., $\left| \frac{d\chi}{ds} \right| = 1$. Assume that the region Y enclosed by χ is convex, i.e., for any $s \in \mathbb{R}$, the tangent line $\{\chi(s) + t \frac{d\chi}{ds} \mid t \in \mathbb{R}\}$ intersects $X := \partial Y$ (= the image of χ) at only the point $\chi(s)$.

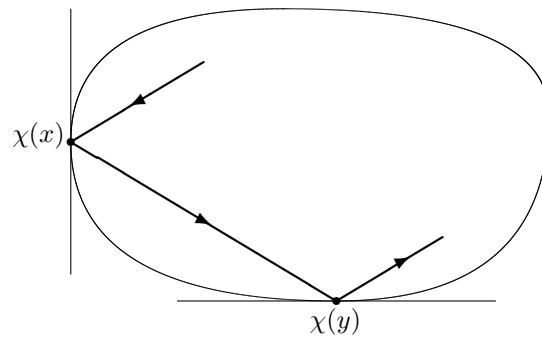


Suppose that we throw a ball into Y rolling with constant velocity and bouncing off the boundary with the usual law of reflection. This determines a map

$$\begin{aligned} \varphi : \mathbb{R}/\mathbb{Z} \times (-1, 1) &\longrightarrow \mathbb{R}/\mathbb{Z} \times (-1, 1) \\ (x, v) &\longmapsto (y, w) \end{aligned}$$

by the rule

when the ball bounces off $\chi(x)$ with angle $\theta = \arccos v$, it will next collide with $\chi(y)$ and bounce off with angle $\nu = \arccos w$.



Let $f : \mathbb{R}/\mathbb{Z} \times \mathbb{R}/\mathbb{Z} \rightarrow \mathbb{R}$ be defined by $f(x, y) = -|\chi(x) - \chi(y)|$; f is smooth off the diagonal. Use χ to identify $\mathbb{R}/\mathbb{Z}$ with the image curve X .

Suppose that $\varphi(x, v) = (y, w)$, i.e., (x, v) and (y, w) are successive points on

the orbit described by the ball. Then

$$\begin{cases} \frac{df}{dx} = -\frac{x-y}{|x-y|} \text{ projected onto } T_x X & = v \\ \frac{df}{dy} = -\frac{y-x}{|x-y|} \text{ projected onto } T_y X & = -w \end{cases}$$

or, equivalently,

$$\begin{cases} \frac{d}{ds} f(\chi(s), y) = \frac{y-x}{|x-y|} \cdot \frac{d\chi}{ds} = \cos \theta = v \\ \frac{d}{ds} f(x, \chi(s)) = \frac{x-y}{|x-y|} \cdot \frac{d\chi}{ds} = -\cos \nu = -w. \end{cases}$$

We conclude that f is a generating function for φ . Similar approaches work for higher dimensional billiards problems.

Periodic points are obtained by finding critical points of

$$\underbrace{X \times \dots \times X}_N \longrightarrow \mathbb{R}, \quad N > 1$$

$$(x_1, \dots, x_N) \longmapsto f(x_1, x_2) + f(x_2, x_3) + \dots + f(x_{N-1}, x_N) + f(x_N, x_1)$$

$$= |x_1 - x_2| + \dots + |x_{N-1} - x_N| + |x_N - x_1|,$$

that is, by finding the N -sided (generalized) polygons inscribed in X of critical perimeter.

Notice that

$$\mathbb{R}/\mathbb{Z} \times (-1, 1) \simeq \{(x, v) \mid x \in X, v \in T_x X, |v| < 1\} \simeq A$$

is the open unit tangent ball bundle of a circle X , that is, an open annulus A . The map $\varphi : A \rightarrow A$ is area-preserving.

5.3 Poincaré Recurrence

Theorem 5.3 (Poincaré Recurrence Theorem) *Suppose that $\varphi : A \rightarrow A$ is an area-preserving diffeomorphism of a finite-area manifold A . Let $p \in A$, and let $\mathcal{U}$ be a neighborhood of p . Then there is $q \in \mathcal{U}$ and a positive integer N such that $\varphi^{(N)}(q) \in \mathcal{U}$.*

Proof. Let $\mathcal{U}_0 = \mathcal{U}, \mathcal{U}_1 = \varphi(\mathcal{U}), \mathcal{U}_2 = \varphi^{(2)}(\mathcal{U}), \dots$. If all of these sets were disjoint, then, since $\text{Area}(\mathcal{U}_i) = \text{Area}(\mathcal{U}) > 0$ for all i , we would have

$$\text{Area } A \geq \text{Area}(\mathcal{U}_0 \cup \mathcal{U}_1 \cup \mathcal{U}_2 \cup \dots) = \sum_i \text{Area}(\mathcal{U}_i) = \infty.$$

To avoid this contradiction we must have $\varphi^{(k)}(\mathcal{U}) \cap \varphi^{(l)}(\mathcal{U}) \neq \emptyset$ for some $k > l$, which implies $\varphi^{(k-l)}(\mathcal{U}) \cap \mathcal{U} \neq \emptyset$. $\square$

Hence, eternal return applies to billiards...

Remark. Theorem 5.3 clearly generalizes to volume-preserving diffeomorphisms in higher dimensions. $\diamond$

Theorem 5.4 (Poincaré's Last Geometric Theorem) *Suppose $\varphi : A \rightarrow A$ is an area-preserving diffeomorphism of the closed annulus $A = \mathbb{R}/\mathbb{Z} \times [-1, 1]$ which preserves the two components of the boundary, and twists them in opposite directions. Then φ has at least two fixed points.*

This theorem was proved in 1913 by Birkhoff, and hence is also called the **Poincaré-Birkhoff theorem**. It has important applications to dynamical systems and celestial mechanics. The Arnold conjecture (1966) on the existence of fixed points for symplectomorphisms of compact manifolds (see Lecture 9) may be regarded as a generalization of the Poincaré-Birkhoff theorem. This conjecture has motivated a significant amount of recent research involving a more general notion of generating function; see, for instance, [34, 45].

Part III

Local Forms

Inspired by the elementary normal form in symplectic linear algebra (Theorem 1.1), we will go on to describe normal neighborhoods of a point (the Darboux theorem) and of a lagrangian submanifold (the Weinstein theorems), inside a symplectic manifold. The main tool is the Moser trick, explained in Lecture 7, which leads to the crucial Moser theorems and which is at the heart of many arguments in symplectic geometry.

In order to prove the normal forms, we need the (non-symplectic) ingredients discussed in Lecture 6; for more on these topics, see, for instance, [18, 55, 96].

6 Preparation for the Local Theory

6.1 Isotopies and Vector Fields

Let M be a manifold, and $\rho : M \times \mathbb{R} \rightarrow M$ a map, where we set $\rho_t(p) := \rho(p, t)$.

Definition 6.1 *The map ρ is an **isotopy** if each $\rho_t : M \rightarrow M$ is a diffeomorphism, and $\rho_0 = \text{id}_M$.*

Given an isotopy ρ , we obtain a **time-dependent vector field**, that is, a family of vector fields v_t , $t \in \mathbb{R}$, which at $p \in M$ satisfy

$$v_t(p) = \left. \frac{d}{ds} \rho_s(q) \right|_{s=t} \quad \text{where} \quad q = \rho_t^{-1}(p) ,$$

i.e.,

$$\frac{d\rho_t}{dt} = v_t \circ \rho_t .$$

Conversely, given a time-dependent vector field v_t , if M is compact or if the v_t 's are compactly supported, there exists an isotopy ρ satisfying the previous ordinary differential equation.

Suppose that M is compact. Then we have a one-to-one correspondence

$$\begin{array}{ccc} \{\text{isotopies of } M\} & \xleftrightarrow{1-1} & \{\text{time-dependent vector fields on } M\} \\ \rho_t, t \in \mathbb{R} & \longleftrightarrow & v_t, t \in \mathbb{R} \end{array}$$

Definition 6.2 *When $v_t = v$ is independent of t , the associated isotopy is called the **exponential map** or the **flow** of v and is denoted $\exp tv$; i.e., $\{\exp tv : M \rightarrow M \mid t \in \mathbb{R}\}$ is the unique smooth family of diffeomorphisms satisfying*

$$\exp tv|_{t=0} = \text{id}_M \quad \text{and} \quad \frac{d}{dt}(\exp tv)(p) = v(\exp tv(p)) .$$

Definition 6.3 *The Lie derivative is the operator*

$$\mathcal{L}_v : \Omega^k(M) \longrightarrow \Omega^k(M) \quad \text{defined by} \quad \mathcal{L}_v \omega := \frac{d}{dt}(\exp tv)^* \omega|_{t=0} .$$

When a vector field v_t is time-dependent, its flow, that is, the corresponding isotopy ρ , still locally exists by Picard's theorem. More precisely, in the neighborhood of any point p and for sufficiently small time t , there is a one-parameter family of local diffeomorphisms ρ_t satisfying

$$\frac{d\rho_t}{dt} = v_t \circ \rho_t \quad \text{and} \quad \rho_0 = \text{id} .$$

Hence, we say that the **Lie derivative** by v_t is

$$\mathcal{L}_{v_t} : \Omega^k(M) \longrightarrow \Omega^k(M) \quad \text{defined by} \quad \mathcal{L}_{v_t} \omega := \frac{d}{dt}(\rho_t)^* \omega|_{t=0} .$$

Exercise. Prove the **Cartan magic formula**,

$$\mathcal{L}_v \omega = \iota_v d\omega + d\iota_v \omega ,$$

and the formula

$$\frac{d}{dt} \rho_t^* \omega = \rho_t^* \mathcal{L}_{v_t} \omega , \quad (\star)$$

where ρ is the (local) isotopy generated by v_t . A good strategy for each formula is to follow the steps:

1. Check the formula for 0-forms $\omega \in \Omega^0(M) = C^\infty(M)$.
2. Check that both sides commute with d .
3. Check that both sides are derivations of the algebra $(\Omega^*(M), \wedge)$. For instance, check that

$$\mathcal{L}_v(\omega \wedge \alpha) = (\mathcal{L}_v \omega) \wedge \alpha + \omega \wedge (\mathcal{L}_v \alpha) .$$

4. Notice that, if $\mathcal{U}$ is the domain of a coordinate system, then $\Omega^\bullet(\mathcal{U})$ is generated as an algebra by $\Omega^0(\mathcal{U})$ and $d\Omega^0(\mathcal{U})$, i.e., every element in $\Omega^\bullet(\mathcal{U})$ is a linear combination of wedge products of elements in $\Omega^0(\mathcal{U})$ and elements in $d\Omega^0(\mathcal{U})$.

◇

We will need the following improved version of formula $(\star)$.

Proposition 6.4 *For a smooth family ω_t , $t \in \mathbb{R}$, of d -forms, we have*

$$\frac{d}{dt} \rho_t^* \omega_t = \rho_t^* \left(\mathcal{L}_{v_t} \omega_t + \frac{d\omega_t}{dt} \right) .$$

Proof. If $f(x, y)$ is a real function of two variables, by the chain rule we have

$$\frac{d}{dt}f(t, t) = \left. \frac{d}{dx}f(x, t) \right|_{x=t} + \left. \frac{d}{dy}f(t, y) \right|_{y=t} .$$

Therefore,

$$\begin{aligned} \frac{d}{dt}\rho_t^*\omega_t &= \underbrace{\left. \frac{d}{dx}\rho_x^*\omega_t \right|_{x=t}}_{\rho_x^*\mathcal{L}_{v_x}\omega_t \big|_{x=t} \text{ by } (\star)} + \underbrace{\left. \frac{d}{dy}\rho_t^*\omega_y \right|_{y=t}}_{\rho_t^*\frac{d\omega_y}{dy} \big|_{y=t}} \\ &= \rho_t^*\left(\mathcal{L}_{v_t}\omega_t + \frac{d\omega_t}{dt}\right) . \end{aligned}$$

□

6.2 Tubular Neighborhood Theorem

Let M be an n -dimensional manifold, and let X be a k -dimensional submanifold where $k < n$ and with inclusion map

$$i : X \hookrightarrow M .$$

At each $x \in X$, the tangent space to X is viewed as a subspace of the tangent space to M via the linear inclusion $di_x : T_x X \hookrightarrow T_x M$, where we denote $x = i(x)$. The quotient $N_x X := T_x M / T_x X$ is an $(n - k)$ -dimensional vector space, known as the **normal space** to X at x . The **normal bundle** of X is

$$NX = \{(x, v) \mid x \in X, v \in N_x X\} .$$

The set NX has the structure of a vector bundle over X of rank $n - k$ under the natural projection, hence as a manifold NX is n -dimensional. The zero section of NX ,

$$i_0 : X \hookrightarrow NX, \quad x \mapsto (x, 0) ,$$

embeds X as a closed submanifold of NX . A neighborhood $\mathcal{U}_0$ of the zero section X in NX is called **convex** if the intersection $\mathcal{U}_0 \cap N_x X$ with each fiber is convex.

Theorem 6.5 (Tubular Neighborhood Theorem) *There exist a convex neighborhood $\mathcal{U}_0$ of X in NX , a neighborhood $\mathcal{U}$ of X in M , and a diffeomorphism $\varphi : \mathcal{U}_0 \rightarrow \mathcal{U}$ such that*

$$\begin{array}{ccc} NX \supseteq \mathcal{U}_0 & \xrightarrow[\simeq]{\varphi} & \mathcal{U} \subseteq M \\ & \swarrow i_0 \quad \searrow i & \\ & X & \end{array} \quad \text{commutes.}$$

Outline of the proof.

- Case of $M = \mathbb{R}^n$, and X is a compact submanifold of $\mathbb{R}^n$.

Theorem 6.6 (ε -Neighborhood Theorem)

Let $\mathcal{U}^\varepsilon = \{p \in \mathbb{R}^n : |p - q| < \varepsilon \text{ for some } q \in X\}$ be the set of points at a distance less than ε from X . Then, for ε sufficiently small, each $p \in \mathcal{U}^\varepsilon$ has a unique nearest point $q \in X$ (i.e., a unique $q \in X$ minimizing $|q - p|$).

Moreover, setting $q = \pi(p)$, the map $\mathcal{U}^\varepsilon \xrightarrow{\pi} X$ is a (smooth) submersion with the property that, for all $p \in \mathcal{U}^\varepsilon$, the line segment $(1 - t)p + tq$, $0 \leq t \leq 1$, is in $\mathcal{U}^\varepsilon$.

The proof is part of Homework 5. Here are some hints.

At any $x \in X$, the *normal* space $N_x X$ may be regarded as an $(n - k)$ -dimensional subspace of $\mathbb{R}^n$, namely the orthogonal complement in $\mathbb{R}^n$ of the tangent space to X at x :

$$N_x X \simeq \{v \in \mathbb{R}^n : v \perp w, \text{ for all } w \in T_x X\}.$$

We define the following open neighborhood of X in NX :

$$NX^\varepsilon = \{(x, v) \in NX : |v| < \varepsilon\}.$$

Let

$$\begin{aligned} \exp : NX &\longrightarrow \mathbb{R}^n \\ (x, v) &\longmapsto x + v. \end{aligned}$$

Restricted to the zero section, $\exp$ is the identity map on X .

Prove that, for ε sufficiently small, $\exp$ maps NX^ε diffeomorphically onto $\mathcal{U}^\varepsilon$, and show also that the diagram

$$\begin{array}{ccc} NX^\varepsilon & \xrightarrow{\exp} & \mathcal{U}^\varepsilon \\ & \searrow \pi_0 & \swarrow \pi \\ & X & \end{array} \quad \text{commutes.}$$

- Case where X is a compact submanifold of an arbitrary manifold M .

Put a riemannian metric g on M , and let $d(p, q)$ be the riemannian distance between $p, q \in M$. The ε -neighborhood of a compact submanifold X is

$$\mathcal{U}^\varepsilon = \{p \in M \mid d(p, q) < \varepsilon \text{ for some } q \in X\}.$$

Prove the ε -neighborhood theorem in this setting: for ε small enough, the following assertions hold.

- Any $p \in \mathcal{U}^\varepsilon$ has a unique point $q \in X$ with minimal $d(p, q)$. Set $q = \pi(p)$.
- The map $\mathcal{U}^\varepsilon \xrightarrow{\pi} X$ is a submersion and, for all $p \in \mathcal{U}^\varepsilon$, there is a unique geodesic curve γ joining p to $q = \pi(p)$.
- The normal space to X at $x \in X$ is naturally identified with a subspace of $T_x M$:

$$N_x X \simeq \{v \in T_x M \mid g_x(v, w) = 0, \text{ for any } w \in T_x X\}.$$

Let $NX^\varepsilon = \{(x, v) \in NX \mid \sqrt{g_x(v, v)} < \varepsilon\}$.

- Define $\exp : NX^\varepsilon \rightarrow M$ by $\exp(x, v) = \gamma(1)$, where $\gamma : [0, 1] \rightarrow M$ is the geodesic with $\gamma(0) = x$ and $\frac{d\gamma}{dt}(0) = v$. Then $\exp$ maps NX^ε diffeomorphically to $\mathcal{U}^\varepsilon$.

- *General case.*

When X is not compact, adapt the previous argument by replacing ε by an appropriate continuous function $\varepsilon : X \rightarrow \mathbb{R}^+$ which tends to zero fast enough as x tends to infinity.

□

Restricting to the subset $\mathcal{U}^0 \subseteq NX$ from the tubular neighborhood theorem, we obtain a submersion $\mathcal{U}_0 \xrightarrow{\pi_0} X$ with all fibers $\pi_0^{-1}(x)$ convex. We can carry this fibration to $\mathcal{U}$ by setting $\pi = \pi_0 \circ \varphi^{-1}$:

$$\begin{array}{ccc} \mathcal{U}_0 & \subseteq NX & \text{is a fibration} \\ \pi_0 \downarrow & \implies & \mathcal{U} \subseteq M \text{ is a fibration} \\ X & & \pi \downarrow \\ & & X \end{array}$$

This is called the **tubular neighborhood fibration**.

6.3 Homotopy Formula

Let $\mathcal{U}$ be a tubular neighborhood of a submanifold X in M . The restriction $i^* : H_{\text{deRham}}^d(\mathcal{U}) \rightarrow H_{\text{deRham}}^d(X)$ by the inclusion map is surjective. As a corollary of the tubular neighborhood fibration, i^* is also injective: this follows from the homotopy-invariance of de Rham cohomology.

Corollary 6.7 *For any degree ℓ , $H_{\text{deRham}}^\ell(\mathcal{U}) \simeq H_{\text{deRham}}^\ell(X)$.*

At the level of forms, this means that, if ω is a closed ℓ -form on $\mathcal{U}$ and $i^*\omega$ is exact on X , then ω is exact. We will need the following related result.

Proposition 6.8 *If a closed ℓ -form ω on $\mathcal{U}$ has restriction $i^*\omega = 0$, then ω is exact, i.e., $\omega = d\mu$ for some $\mu \in \Omega^{d-1}(\mathcal{U})$. Moreover, we can choose μ such that $\mu_x = 0$ at all $x \in X$.*

Proof. Via $\varphi : \mathcal{U}_0 \xrightarrow{\sim} \mathcal{U}$, it is equivalent to work over $\mathcal{U}_0$. Define for every $0 \leq t \leq 1$ a map

$$\begin{aligned} \rho_t : \mathcal{U}_0 &\longrightarrow \mathcal{U}_0 \\ (x, v) &\longmapsto (x, tv) . \end{aligned}$$

This is well-defined since $\mathcal{U}_0$ is convex. The map ρ_1 is the identity, $\rho_0 = i_0 \circ \pi_0$, and each ρ_t fixes X , that is, $\rho_t \circ i_0 = i_0$. We hence say that the family $\{\rho_t \mid 0 \leq t \leq 1\}$ is a **homotopy** from $i_0 \circ \pi_0$ to the identity fixing X . The map $\pi_0 : \mathcal{U}_0 \rightarrow X$ is called a **retraction** because $\pi_0 \circ i_0$ is the identity. The submanifold X is then called a **deformation retract** of $\mathcal{U}$.

A (de Rham) **homotopy operator** between $\rho_0 = i_0 \circ \pi_0$ and $\rho_1 = \text{id}$ is a linear map

$$Q : \Omega^d(\mathcal{U}_0) \longrightarrow \Omega^{d-1}(\mathcal{U}_0)$$

satisfying the **homotopy formula**

$$\text{Id} - (i_0 \circ \pi_0)^* = dQ + Qd .$$

When $d\omega = 0$ and $i_0^*\omega = 0$, the operator Q gives $\omega = dQ\omega$, so that we can take $\mu = Q\omega$. A concrete operator Q is given by the formula:

$$Q\omega = \int_0^1 \rho_t^*(\iota_{v_t}\omega) dt ,$$

where v_t , at the point $q = \rho_t(p)$, is the vector tangent to the curve $\rho_s(p)$ at $s = t$. The proof that Q satisfies the homotopy formula is below.

In our case, for $x \in X$, $\rho_t(x) = x$ (all t) is the constant curve, so v_t vanishes at all x for all t , hence $\mu_x = 0$. $\square$

To check that Q above satisfies the homotopy formula, we compute

$$\begin{aligned} Qd\omega + dQ\omega &= \int_0^1 \rho_t^*(\iota_{v_t}d\omega)dt + d \int_0^1 \rho_t^*(\iota_{v_t}\omega)dt \\ &= \int_0^1 \rho_t^* \underbrace{(\iota_{v_t}d\omega + d\iota_{v_t}\omega)}_{\mathcal{L}_{v_t}\omega} dt , \end{aligned}$$

where $\mathcal{L}_v$ denotes the Lie derivative along v (reviewed in the next section), and we used the Cartan magic formula: $\mathcal{L}_v\omega = \iota_v d\omega + d\iota_v\omega$. The result now follows from

$$\frac{d}{dt}\rho_t^*\omega = \rho_t^*\mathcal{L}_{v_t}\omega$$

and from the fundamental theorem of calculus:

$$Qd\omega + dQ\omega = \int_0^1 \frac{d}{dt}\rho_t^*\omega dt = \rho_1^*\omega - \rho_0^*\omega .$$

Homework 5: Tubular Neighborhoods in $\mathbb{R}^n$

1. Let X be a k -dimensional submanifold of an n -dimensional manifold M . Let x be a point in X . The **normal space** to X at x is the quotient space

$$N_x X = T_x M / T_x X ,$$

and the **normal bundle** of X in M is the vector bundle NX over X whose fiber at x is $N_x X$.

- (a) Prove that NX is indeed a vector bundle.
 - (b) If M is $\mathbb{R}^n$, show that $N_x X$ can be identified with the usual “normal space” to X in $\mathbb{R}^n$, that is, the orthogonal complement in $\mathbb{R}^n$ of the tangent space to X at x .
2. Let X be a k -dimensional compact submanifold of $\mathbb{R}^n$. Prove the **tubular neighborhood theorem** in the following form.
- (a) Given $\varepsilon > 0$ let $\mathcal{U}_\varepsilon$ be the set of all points in $\mathbb{R}^n$ which are at a distance less than ε from X . Show that, for ε sufficiently small, every point $p \in \mathcal{U}_\varepsilon$ has a *unique* nearest point $\pi(p) \in X$.
 - (b) Let $\pi : \mathcal{U}_\varepsilon \rightarrow X$ be the map defined in (a) for ε sufficiently small. Show that, if $p \in \mathcal{U}_\varepsilon$, then the line segment $(1 - t) \cdot p + t \cdot \pi(p)$, $0 \leq t \leq 1$, joining p to $\pi(p)$ lies in $\mathcal{U}_\varepsilon$.
 - (c) Let $NX_\varepsilon = \{(x, v) \in NX \text{ such that } |v| < \varepsilon\}$. Let $\exp : NX \rightarrow \mathbb{R}^n$ be the map $(x, v) \mapsto x + v$, and let $\nu : NX_\varepsilon \rightarrow X$ be the map $(x, v) \mapsto x$. Show that, for ε sufficiently small, $\exp$ maps NX_ε diffeomorphically onto $\mathcal{U}_\varepsilon$, and show also that the following diagram commutes:

$$\begin{array}{ccc}
 NX_\varepsilon & \xrightarrow{\exp} & \mathcal{U}_\varepsilon \\
 \searrow \wr & & \nearrow \wr \\
 & X &
 \end{array}$$

3. Suppose that the manifold X in the previous exercise is not compact. Prove that the assertion about $\exp$ is still true provided we replace ε by a continuous function

$$\varepsilon : X \rightarrow \mathbb{R}^+$$

which tends to zero fast enough as x tends to infinity.

7 Moser Theorems

7.1 Notions of Equivalence for Symplectic Structures

Let M be a $2n$ -dimensional manifold with two symplectic forms ω_0 and ω_1 , so that (M, ω_0) and (M, ω_1) are two symplectic manifolds.

Definition 7.1 *We say that*

- (M, ω_0) and (M, ω_1) are **symplectomorphic** if there is a diffeomorphism $\varphi : M \rightarrow M$ with $\varphi^*\omega_1 = \omega_0$;
- (M, ω_0) and (M, ω_1) are **strongly isotopic** if there is an isotopy $\rho_t : M \rightarrow M$ such that $\rho_1^*\omega_1 = \omega_0$;
- (M, ω_0) and (M, ω_1) are **deformation-equivalent** if there is a smooth family ω_t of symplectic forms joining ω_0 to ω_1 ;
- (M, ω_0) and (M, ω_1) are **isotopic** if they are deformation-equivalent with $[\omega_t]$ independent of t .

Clearly, we have

$$\begin{aligned} \text{strongly isotopic} &\implies \text{symplectomorphic} , & \text{and} \\ \text{isotopic} &\implies \text{deformation-equivalent} . \end{aligned}$$

We also have

$$\text{strongly isotopic} \implies \text{isotopic}$$

because, if $\rho_t : M \rightarrow M$ is an isotopy such that $\rho_1^*\omega_1 = \omega_0$, then the set $\omega_t := \rho_t^*\omega_1$ is a smooth family of symplectic forms joining ω_1 to ω_0 and $[\omega_t] = [\omega_1]$, $\forall t$, by the homotopy invariance of de Rham cohomology. As we will see below, the Moser theorem states that, on a compact manifold,

$$\text{isotopic} \implies \text{strongly isotopic} .$$

7.2 Moser Trick

Problem. Given a $2n$ -dimensional manifold M , a k -dimensional submanifold X , neighborhoods $\mathcal{U}_0, \mathcal{U}_1$ of X , and symplectic forms ω_0, ω_1 on $\mathcal{U}_0, \mathcal{U}_1$, does there exist a symplectomorphism preserving X ? More precisely, does there exist a diffeomorphism $\varphi : \mathcal{U}_0 \rightarrow \mathcal{U}_1$ with $\varphi^*\omega_1 = \omega_0$ and $\varphi(X) = X$?

At the two extremes, we have:

- Case $X = \text{point}$: Darboux theorem – see Lecture 8.
Case $X = M$: Moser theorem – discussed here:

Let M be a *compact* manifold with symplectic forms ω_0 and ω_1 .

– Are (M, ω_0) and (M, ω_1) symplectomorphic?

I.e., does there exist a diffeomorphism $\varphi : M \rightarrow M$ such that $\varphi_1^* \omega_0 = \omega_1$?

Moser asked whether we can find such an φ which is homotopic to id_M . A necessary condition is $[\omega_0] = [\omega_1] \in H^2(M; \mathbb{R})$ because: if $\varphi \sim \text{id}_M$, then, by the homotopy formula, there exists a homotopy operator Q such that

$$\begin{aligned} \text{id}_M^* \omega_1 - \varphi^* \omega_1 &= dQ\omega_1 + Q \underbrace{d\omega_1}_0 \\ \implies \omega_1 &= \varphi^* \omega_1 + d(Q\omega_1) \\ \implies [\omega_1] &= [\varphi^* \omega_1] = [\omega_0] . \end{aligned}$$

– If $[\omega_0] = [\omega_1]$, does there exist a diffeomorphism φ homotopic to id_M such that $\varphi^* \omega_1 = \omega_0$?

Moser [87] proved that the answer is yes, with a further hypothesis as in Theorem 7.2. McDuff showed that, in general, the answer is no; for a counterexample, see Example 7.23 in [83].

Theorem 7.2 (Moser Theorem – Version I) *Suppose that M is compact, $[\omega_0] = [\omega_1]$ and that the 2-form $\omega_t = (1-t)\omega_0 + t\omega_1$ is symplectic for each $t \in [0, 1]$. Then there exists an isotopy $\rho : M \times \mathbb{R} \rightarrow M$ such that $\rho_t^* \omega_t = \omega_0$ for all $t \in [0, 1]$.*

In particular, $\varphi = \rho_1 : M \xrightarrow{\sim} M$, satisfies $\varphi^* \omega_1 = \omega_0$.

The following argument, due to Moser, is extremely useful; it is known as the **Moser trick**.

Proof. Suppose that there exists an isotopy $\rho : M \times \mathbb{R} \rightarrow M$ such that $\rho_t^* \omega_t = \omega_0$, $0 \leq t \leq 1$. Let

$$v_t = \frac{d\rho_t}{dt} \circ \rho_t^{-1} , \quad t \in \mathbb{R} .$$

Then

$$\begin{aligned} 0 &= \frac{d}{dt}(\rho_t^* \omega_t) = \rho_t^* \left(\mathcal{L}_{v_t} \omega_t + \frac{d\omega_t}{dt} \right) \\ \iff \mathcal{L}_{v_t} \omega_t + \frac{d\omega_t}{dt} &= 0 . \quad (\star) \end{aligned}$$

Suppose conversely that we can find a smooth time-dependent vector field v_t , $t \in \mathbb{R}$, such that $(\star)$ holds for $0 \leq t \leq 1$. Since M is compact, we can integrate v_t to an isotopy $\rho : M \times \mathbb{R} \rightarrow M$ with

$$\frac{d}{dt}(\rho_t^* \omega_t) = 0 \implies \rho_t^* \omega_t = \rho_0^* \omega_0 = \omega_0 .$$

So everything boils down to solving $(\star)$ for v_t .

First, from $\omega_t = (1-t)\omega_0 + t\omega_1$, we conclude that

$$\frac{d\omega_t}{dt} = \omega_1 - \omega_0 .$$

Second, since $[\omega_0] = [\omega_1]$, there exists a 1-form μ such that

$$\omega_1 - \omega_0 = d\mu .$$

Third, by the Cartan magic formula, we have

$$\mathcal{L}_{v_t}\omega_t = d\iota_{v_t}\omega_t + \underbrace{\iota_{v_t}d\omega_t}_0 .$$

Putting everything together, we must find v_t such that

$$d\iota_{v_t}\omega_t + d\mu = 0 .$$

It is sufficient to solve $\iota_{v_t}\omega_t + \mu = 0$. By the nondegeneracy of ω_t , we can solve this pointwise, to obtain a unique (smooth) v_t . $\square$

Theorem 7.3 (Moser Theorem – Version II) *Let M be a compact manifold with symplectic forms ω_0 and ω_1 . Suppose that ω_t , $0 \leq t \leq 1$, is a smooth family of closed 2-forms joining ω_0 to ω_1 and satisfying:*

- (1) cohomology assumption: $[\omega_t]$ is independent of t , i.e., $\frac{d}{dt}[\omega_t] = [\frac{d}{dt}\omega_t] = 0$,
- (2) nondegeneracy assumption: ω_t is nondegenerate for $0 \leq t \leq 1$.

Then there exists an isotopy $\rho : M \times \mathbb{R} \rightarrow M$ such that $\rho_t^\omega_t = \omega_0$, $0 \leq t \leq 1$.*

Proof. (*Moser trick*) We have the following implications from the hypotheses:

- (1) $\implies \exists$ family of 1-forms μ_t such that

$$\frac{d\omega_t}{dt} = d\mu_t , \quad 0 \leq t \leq 1 .$$

We can indeed find a *smooth* family of 1-forms μ_t such that $\frac{d\omega_t}{dt} = d\mu_t$. The argument involves the Poincaré lemma for compactly-supported forms, together with the Mayer-Vietoris sequence in order to use induction on the number of charts in a good cover of M . For a sketch of the argument, see page 95 in [83].

- (2) $\implies \exists$ unique family of vector fields v_t such that

$$\iota_{v_t}\omega_t + \mu_t = 0 \quad \textbf{(Moser equation)} .$$

Extend v_t to all $t \in \mathbb{R}$. Let ρ be the isotopy generated by v_t (ρ exists by compactness of M). Then we indeed have

$$\frac{d}{dt}(\rho_t^*\omega_t) = \rho_t^*(\mathcal{L}_{v_t}\omega_t + \frac{d\omega_t}{dt}) = \rho_t^*(d\iota_{v_t}\omega_t + d\mu_t) = 0 .$$

$\square$

The compactness of M was used to be able to integrate v_t for all $t \in \mathbb{R}$. If M is *not* compact, we need to check the existence of a solution ρ_t for the differential equation $\frac{d\rho_t}{dt} = v_t \circ \rho_t$ for $0 \leq t \leq 1$.

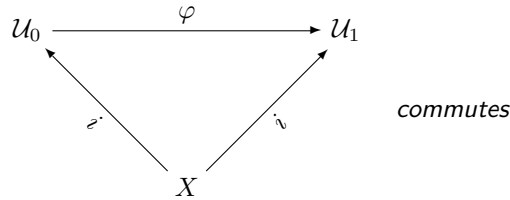
Picture. Fix $c \in H^2(M)$. Define $S_c = \{\text{symplectic forms } \omega \text{ in } M \text{ with } [\omega] = c\}$. The Moser theorem implies that, on a compact manifold, all symplectic forms on the same path-connected component of S_c are symplectomorphic.

7.3 Moser Relative Theorem

Theorem 7.4 (Moser Theorem – Relative Version) *Let M be a manifold, X a compact submanifold of M , $i : X \hookrightarrow M$ the inclusion map, ω_0 and ω_1 symplectic forms in M .*

Hypothesis: $\omega_0|_p = \omega_1|_p$, $\forall p \in X$.

Conclusion: *There exist neighborhoods $\mathcal{U}_0, \mathcal{U}_1$ of X in M , and a diffeomorphism $\varphi : \mathcal{U}_0 \rightarrow \mathcal{U}_1$ such that*



commutes

and $\varphi^\omega_1 = \omega_0$.*

Proof.

1. Pick a tubular neighborhood $\mathcal{U}_0$ of X . The 2-form $\omega_1 - \omega_0$ is closed on $\mathcal{U}_0$, and $(\omega_1 - \omega_0)_p = 0$ at all $p \in X$. By the homotopy formula on the tubular neighborhood, there exists a 1-form μ on $\mathcal{U}_0$ such that $\omega_1 - \omega_0 = d\mu$ and $\mu_p = 0$ at all $p \in X$.
2. Consider the family $\omega_t = (1-t)\omega_0 + t\omega_1 = \omega_0 + td\mu$ of closed 2-forms on $\mathcal{U}_0$. Shrinking $\mathcal{U}_0$ if necessary, we can assume that ω_t is symplectic for $0 \leq t \leq 1$.
3. Solve the Moser equation: $v_t \omega_t = -\mu$. Notice that $v_t = 0$ on X .
4. Integrate v_t . Shrinking $\mathcal{U}_0$ again if necessary, there exists an isotopy $\rho : \mathcal{U}_0 \times [0, 1] \rightarrow M$ with $\rho_t^* \omega_t = \omega_0$, for all $t \in [0, 1]$. Since $v_t|_X = 0$, we have $\rho_t|_X = \text{id}_X$.

Set $\varphi = \rho_1$, $\mathcal{U}_1 = \rho_1(\mathcal{U}_0)$. □

Exercise. Prove the Darboux theorem. (Hint: apply the relative version of the Moser theorem to $X = \{p\}$, as in the next lecture.) ◇

8 Darboux-Moser-Weinstein Theory

8.1 Darboux Theorem

Theorem 8.1 (Darboux) *Let (M, ω) be a symplectic manifold, and let p be any point in M . Then we can find a coordinate system $(\mathcal{U}, x_1, \dots, x_n, y_1, \dots, y_n)$ centered at p such that on $\mathcal{U}$*

$$\omega = \sum_{i=1}^n dx_i \wedge dy_i .$$

As a consequence of Theorem 8.1, if we prove for $(\mathbb{R}^{2n}, \sum dx_i \wedge dy_i)$ a local assertion which is invariant under symplectomorphisms, then that assertion holds for any symplectic manifold.

Proof. Apply the Moser relative theorem (Theorem 7.4) to $X = \{p\}$:

Use any symplectic basis for $T_p M$ to construct coordinates $(x'_1, \dots, x'_n, y'_1, \dots, y'_n)$ centered at p and valid on some neighborhood $\mathcal{U}'$, so that

$$\omega_p = \sum dx'_i \wedge dy'_i \Big|_p .$$

There are two symplectic forms on $\mathcal{U}'$: the given $\omega_0 = \omega$ and $\omega_1 = \sum dx'_i \wedge dy'_i$. By the Moser theorem, there are neighborhoods $\mathcal{U}_0$ and $\mathcal{U}_1$ of p , and a diffeomorphism $\varphi : \mathcal{U}_0 \rightarrow \mathcal{U}_1$ such that

$$\varphi(p) = p \quad \text{and} \quad \varphi^*(\sum dx'_i \wedge dy'_i) = \omega .$$

Since $\varphi^*(\sum dx'_i \wedge dy'_i) = \sum d(x'_i \circ \varphi) \wedge d(y'_i \circ \varphi)$, we only need to set new coordinates $x_i = x'_i \circ \varphi$ and $y_i = y'_i \circ \varphi$. $\square$

If in the Moser relative theorem (Theorem 7.4) we assume instead

Hypothesis: X is an n -dimensional submanifold with
 $i^* \omega_0 = i^* \omega_1 = 0$ where $i : X \hookrightarrow M$ is inclusion, i.e.,
 X is a submanifold lagrangian for ω_0 and ω_1 ,

then Weinstein [104] proved that the conclusion still holds. We need some algebra for the Weinstein theorem.

8.2 Lagrangian Subspaces

Suppose that U, W are n -dimensional vector spaces, and $\Omega : U \times W \rightarrow \mathbb{R}$ is a bilinear pairing; the map Ω gives rise to a linear map $\tilde{\Omega} : U \rightarrow W^*$, $\tilde{\Omega}(u) = \Omega(u, \cdot)$. Then Ω is nondegenerate if and only if $\tilde{\Omega}$ is bijective.

Proposition 8.2 Suppose that (V, Ω) is a $2n$ -dimensional symplectic vector space. Let U be a lagrangian subspace of (V, Ω) (i.e., $\Omega|_{U \times U} = 0$ and U is n -dimensional). Let W be any vector space complement to U , not necessarily lagrangian. Then from W we can canonically build a lagrangian complement to U .

Proof. The pairing Ω gives a nondegenerate pairing $U \times W \xrightarrow{\Omega'} \mathbb{R}$. Therefore, $\tilde{\Omega}' : U \rightarrow W^*$ is bijective. We look for a lagrangian complement to U of the form

$$W' = \{w + Aw \mid w \in W\},$$

the map $A : W \rightarrow U$ being linear. For W' to be lagrangian we need

$$\begin{aligned} \forall w_1, w_2 \in W, \quad \Omega(w_1 + Aw_1, w_2 + Aw_2) &= 0 \\ \implies \Omega(w_1, w_2) + \Omega(w_1, Aw_2) + \Omega(Aw_1, w_2) + \underbrace{\Omega(Aw_1, Aw_2)}_{\substack{\in U \\ 0}} &= 0 \\ \implies \Omega(w_1, w_2) &= \Omega(Aw_2, w_1) - \Omega(Aw_1, w_2) \\ &= \tilde{\Omega}'(Aw_2)(w_1) - \tilde{\Omega}'(Aw_1)(w_2). \end{aligned}$$

Let $A' = \tilde{\Omega}' \circ A : W \rightarrow W^*$, and look for A' such that

$$\forall w_1, w_2 \in W, \quad \Omega(w_1, w_2) = A'(w_2)(w_1) - A'(w_1)(w_2).$$

The canonical choice is $A'(w) = -\frac{1}{2}\Omega(w, \cdot)$. Then set $A = (\tilde{\Omega}')^{-1} \circ A'$. $\square$

Proposition 8.3 Let V be a $2n$ -dimensional vector space, let Ω_0 and Ω_1 be symplectic forms in V , let U be a subspace of V lagrangian for Ω_0 and Ω_1 , and let W be any complement to U in V . Then from W we can canonically construct a linear isomorphism $L : V \xrightarrow{\sim} V$ such that $L|_U = \text{Id}_U$ and $L^*\Omega_1 = \Omega_0$.

Proof. From W we canonically obtain complements W_0 and W_1 to U in V such that W_0 is lagrangian for Ω_0 and W_1 is lagrangian for Ω_1 . The nondegenerate bilinear pairings

$$\begin{array}{ccc} W_0 \times U & \xrightarrow{\Omega_0} & \mathbb{R} \\ W_1 \times U & \xrightarrow{\Omega_1} & \mathbb{R} \end{array} \quad \text{give isomorphisms} \quad \begin{array}{ccc} \tilde{\Omega}_0 : W_0 & \xrightarrow{\sim} & U^* \\ \tilde{\Omega}_1 : W_1 & \xrightarrow{\sim} & U^* \end{array}.$$

Consider the diagram

$$\begin{array}{ccc} W_0 & \xrightarrow{\tilde{\Omega}_0} & U^* \\ B \downarrow & & \downarrow \text{id} \\ W_1 & \xrightarrow{\tilde{\Omega}_1} & U^* \end{array}$$

where the linear map B satisfies $\tilde{\Omega}_1 \circ B = \tilde{\Omega}_0$, i.e., $\Omega_0(w_0, u) = \Omega_1(Bw_0, u)$, $\forall w_0 \in W_0, \forall u \in U$. Extend B to the rest of V by setting it to be the identity on U :

$$L := \text{Id}_U \oplus B : U \oplus W_0 \longrightarrow U \oplus W_1.$$

Finally, we check that $L^*\Omega_1 = \Omega_0$.

$$\begin{aligned}
 (L^*\Omega_1)(u \oplus w_0, u' \oplus w'_0) &= \Omega_1(u \oplus Bw_0, u' \oplus Bw'_0) \\
 &= \Omega_1(u, Bw'_0) + \Omega_1(Bw_0, u') \\
 &= \Omega_0(u, w'_0) + \Omega_0(w_0, u') \\
 &= \Omega_0(u \oplus w_0, u' \oplus w'_0) .
 \end{aligned}$$

□

8.3 Weinstein Lagrangian Neighborhood Theorem

Theorem 8.4 (Weinstein Lagrangian Neighborhood Theorem [104]) *Let M be a $2n$ -dimensional manifold, X a compact n -dimensional submanifold, $i : X \hookrightarrow M$ the inclusion map, and ω_0 and ω_1 symplectic forms on M such that $i^*\omega_0 = i^*\omega_1 = 0$, i.e., X is a lagrangian submanifold of both (M, ω_0) and (M, ω_1) . Then there exist neighborhoods $\mathcal{U}_0$ and $\mathcal{U}_1$ of X in M and a diffeomorphism $\varphi : \mathcal{U}_0 \rightarrow \mathcal{U}_1$ such that*

$$\begin{array}{ccc}
 \mathcal{U}_0 & \xrightarrow{\varphi} & \mathcal{U}_1 \\
 & \searrow i & \nearrow i \\
 & X &
 \end{array}
 \quad \text{commutes} \quad \text{and} \quad \varphi^*\omega_1 = \omega_0 .$$

The proof of the Weinstein theorem uses the Whitney extension theorem.

Theorem 8.5 (Whitney Extension Theorem) *Let M be an n -dimensional manifold and X a k -dimensional submanifold with $k < n$. Suppose that at each $p \in X$ we are given a linear isomorphism $L_p : T_pM \xrightarrow{\cong} T_pM$ such that $L_p|_{T_pX} = \text{Id}_{T_pX}$ and L_p depends smoothly on p . Then there exists an embedding $h : \mathcal{N} \rightarrow M$ of some neighborhood $\mathcal{N}$ of X in M such that $h|_X = \text{id}_X$ and $dh_p = L_p$ for all $p \in X$.*

The linear maps L serve as “germs” for the embedding.

Proof of the Weinstein theorem. Put a riemannian metric g on M ; at each $p \in M$, $g_p(\cdot, \cdot)$ is a positive-definite inner product. Fix $p \in X$, and let $V = T_pM$, $U = T_pX$ and $W = U^\perp$ the orthocomplement of U in V relative to $g_p(\cdot, \cdot)$.

Since $i^*\omega_0 = i^*\omega_1 = 0$, the space U is a lagrangian subspace of both $(V, \omega_0|_p)$ and $(V, \omega_1|_p)$. By symplectic linear algebra, we canonically get from $U^\perp$ a linear isomorphism $L_p : T_pM \rightarrow T_pM$, such that $L_p|_{T_pX} = \text{Id}_{T_pX}$ and $L_p^*\omega_1|_p = \omega_0|_p$. L_p varies smoothly with respect to p since our recipe is canonical!

By the Whitney theorem, there are a neighborhood $\mathcal{N}$ of X and an embedding $h : \mathcal{N} \hookrightarrow M$ with $h|_X = \text{id}_X$ and $dh_p = L_p$ for $p \in X$. Hence, at any $p \in X$,

$$(h^*\omega_1)_p = (dh_p)^*\omega_1|_p = L_p^*\omega_1|_p = \omega_0|_p .$$

Applying the Moser relative theorem (Theorem 7.4) to ω_0 and $h^*\omega_1$, we find a neighborhood $\mathcal{U}_0$ of X and an embedding $f : \mathcal{U}_0 \rightarrow \mathcal{N}$ such that $f|_X = \text{id}_X$ and $f^*(h^*\omega_1) = \omega_0$ on $\mathcal{U}_0$. Set $\varphi = h \circ f$. $\square$

Sketch of proof for the Whitney theorem.

Case $M = \mathbb{R}^n$:

For a compact k -dimensional submanifold X , take a neighborhood of the form

$$\mathcal{U}^\varepsilon = \{p \in M \mid \text{distance}(p, X) \leq \varepsilon\}.$$

For ε sufficiently small so that any $p \in \mathcal{U}^\varepsilon$ has a unique nearest point in X , define a projection $\pi : \mathcal{U}^\varepsilon \rightarrow X$, $p \mapsto \text{point on } X \text{ closest to } p$. If $\pi(p) = q$, then $p = q + v$ for some $v \in N_q X$ where $N_q X = (T_q X)^\perp$ is the normal space at q ; see Homework 5. Let

$$\begin{aligned} h : \mathcal{U}^\varepsilon &\longrightarrow \mathbb{R}^n \\ p &\longmapsto q + L_q v, \end{aligned}$$

where $q = \pi(p)$ and $v = p - \pi(p) \in N_q X$. Then $h_X = \text{id}_X$ and $dh_p = L_p$ for $p \in X$. If X is not compact, replace ε by a continuous function $\varepsilon : X \rightarrow \mathbb{R}^+$ which tends to zero fast enough as x tends to infinity.

General case:

Choose a riemannian metric on M . Replace distance by riemannian distance, replace straight lines $q + tv$ by geodesics $\exp(q, v)(t)$ and replace $q + L_q v$ by the value at $t = 1$ of the geodesic with initial value q and initial velocity $L_q v$. $\square$

In Lecture 30 we will need the following generalization of Theorem 8.4. For a proof see, for instance, either of [47, 58, 107].

Theorem 8.6 (Coisotropic Embedding Theorem) *Let M be a manifold of dimension $2n$, X a submanifold of dimension $k \geq n$, $i : X \hookrightarrow M$ the inclusion map, and ω_0 and ω_1 symplectic forms on M , such that $i^*\omega_0 = i^*\omega_1$ and X is coisotropic for both (M, ω_0) and (M, ω_1) . Then there exist neighborhoods $\mathcal{U}_0$ and $\mathcal{U}_1$ of X in M and a diffeomorphism $\varphi : \mathcal{U}_0 \rightarrow \mathcal{U}_1$ such that*

$$\begin{array}{ccc} \mathcal{U}_0 & \xrightarrow{\varphi} & \mathcal{U}_1 \\ & \nwarrow i & \nearrow i \\ & X & \end{array} \quad \text{commutes} \quad \text{and} \quad \varphi^*\omega_1 = \omega_0.$$

Homework 6: Oriented Surfaces

1. The standard symplectic form on the 2-sphere is the standard area form:

If we think of S^2 as the unit sphere in 3-space

$$S^2 = \{u \in \mathbb{R}^3 \text{ such that } |u| = 1\},$$

then the induced area form is given by

$$\omega_u(v, w) = \langle u, v \times w \rangle$$

where $u \in S^2$, $v, w \in T_u S^2$ are vectors in $\mathbb{R}^3$, $\times$ is the exterior product, and $\langle \cdot, \cdot \rangle$ is the standard inner product. With this form, the total area of S^2 is 4π .

Consider cylindrical polar coordinates (θ, z) on S^2 away from its poles, where $0 \leq \theta < 2\pi$ and $-1 \leq z \leq 1$.

Show that, in these coordinates,

$$\omega = d\theta \wedge dz.$$

2. Prove the Darboux theorem in the 2-dimensional case, using the fact that every nonvanishing 1-form on a surface can be written locally as $f dg$ for suitable functions f, g .

Hint: $\omega = df \wedge dg$ is nondegenerate $\iff (f, g)$ is a local diffeomorphism.

3. Any oriented 2-dimensional manifold with an area form is a symplectic manifold.

- (a) Show that convex combinations of two area forms ω_0, ω_1 that induce the same orientation are symplectic.

This is wrong in dimension 4: find two symplectic forms on the vector space $\mathbb{R}^4$ that induce the same orientation, yet some convex combination of which is degenerate. Find a path of symplectic forms that connect them.

- (b) Suppose that we have two area forms ω_0, ω_1 on a compact 2-dimensional manifold M representing the same de Rham cohomology class, i.e., $[\omega_0] = [\omega_1] \in H_{\text{deRham}}^2(M)$.

Prove that there is a 1-parameter family of diffeomorphisms $\varphi_t : M \rightarrow M$ such that $\varphi_1^* \omega_0 = \omega_1$, $\varphi_0 = \text{id}$, and $\varphi_t^* \omega_0$ is symplectic for all $t \in [0, 1]$.

Hint: Exercise (a) and the Moser trick.

Such a 1-parameter family φ_t is called a *strong isotopy* between ω_0 and ω_1 . In this language, this exercise shows that, up to strong isotopy, there is a unique symplectic representative in each non-zero 2-cohomology class of M .

9 Weinstein Tubular Neighborhood Theorem

9.1 Observation from Linear Algebra

Let (V, Ω) be a symplectic linear space, and let U be a lagrangian subspace.

Claim. There is a canonical nondegenerate bilinear pairing $\Omega' : V/U \times U \rightarrow \mathbb{R}$.

Proof. Define $\Omega'([v], u) = \Omega(v, u)$ where $[v]$ is the equivalence class of v in V/U .

Exercise. Check that Ω' is well-defined and nondegenerate. $\diamond$ $\square$

Consequently, we get

$\implies \tilde{\Omega}' : V/U \rightarrow U^*$ defined by $\tilde{\Omega}'([v]) = \Omega'([v], \cdot)$ is an isomorphism.

$\implies V/U \simeq U^*$ are canonically identified.

In particular, if (M, ω) is a symplectic manifold, and X is a lagrangian submanifold, then $T_x X$ is a lagrangian subspace of $(T_x M, \omega_x)$ for each $x \in X$.

The space $N_x X := T_x M / T_x X$ is called the **normal space** of X at x .

$\implies$ There is a canonical identification $N_x X \simeq T_x^* X$.

$\implies$

Theorem 9.1 *The vector bundles NX and T^*X are canonically identified.*

9.2 Tubular Neighborhoods

Theorem 9.2 (Standard Tubular Neighborhood Theorem) *Let M be an n -dimensional manifold, X a k -dimensional submanifold, NX the normal bundle of X in M , $i_0 : X \hookrightarrow NX$ the zero section, and $i : X \hookrightarrow M$ inclusion. Then there are neighborhoods $\mathcal{U}_0$ of X in NX , $\mathcal{U}$ of X in M and a diffeomorphism $\psi : \mathcal{U}_0 \rightarrow \mathcal{U}$ such that*

$$\begin{array}{ccc}
 \mathcal{U}_0 & \xrightarrow{\psi} & \mathcal{U} \\
 \swarrow i_0 & & \searrow i \\
 & X &
 \end{array}
 \quad \text{commutes.}$$

For the proof, see Lecture 6.

Theorem 9.3 (Weinstein Tubular Neighborhood Theorem) *Let (M, ω) be a symplectic manifold, X a compact lagrangian submanifold, ω_0 the canonical symplectic form on T^*X , $i_0 : X \hookrightarrow T^*X$ the lagrangian embedding as the zero section, and $i : X \hookrightarrow M$ the lagrangian embedding given by inclusion.*

Then there are neighborhoods $\mathcal{U}_0$ of X in T^*X , $\mathcal{U}$ of X in M , and a diffeomorphism $\varphi : \mathcal{U}_0 \rightarrow \mathcal{U}$ such that

$$\begin{array}{ccc}
 \mathcal{U}_0 & \xrightarrow{\varphi} & \mathcal{U} \\
 \searrow i_0 & & \nearrow i \\
 & X &
 \end{array}
 \quad \text{commutes} \quad \text{and} \quad \varphi^* \omega = \omega_0 .$$

Proof. This proof relies on (1) the standard tubular neighborhood theorem, and (2) the Weinstein lagrangian neighborhood theorem.

- (1) Since $NX \simeq T^*X$, we can find a neighborhood $\mathcal{N}_0$ of X in T^*X , a neighborhood $\mathcal{N}$ of X in M , and a diffeomorphism $\psi : \mathcal{N}_0 \rightarrow \mathcal{N}$ such that

$$\begin{array}{ccc}
 \mathcal{N}_0 & \xrightarrow{\psi} & \mathcal{N} \\
 \searrow i_0 & & \nearrow i \\
 & X &
 \end{array}
 \quad \text{commutes} .$$

Let $\left. \begin{array}{l} \omega_0 = \text{canonical form on } T^*X \\ \omega_1 = \psi^* \omega \end{array} \right\}$ symplectic forms on $\mathcal{N}_0$.

The submanifold X is lagrangian for both ω_0 and ω_1 .

- (2) There exist neighborhoods $\mathcal{U}_0$ and $\mathcal{U}_1$ of X in $\mathcal{N}_0$ and a diffeomorphism $\theta : \mathcal{U}_0 \rightarrow \mathcal{U}_1$ such that

$$\begin{array}{ccc}
 \mathcal{U}_0 & \xrightarrow{\theta} & \mathcal{U}_1 \\
 \searrow i_0 & & \nearrow i_0 \\
 & X &
 \end{array}
 \quad \text{commutes} \quad \text{and} \quad \theta^* \omega_1 = \omega_0 .$$

Take $\varphi = \psi \circ \theta$ and $\mathcal{U} = \varphi(\mathcal{U}_0)$. Check that $\varphi^* \omega = \underbrace{\theta^* \psi^* \omega}_{\omega_1} = \omega_0$.

□

Remark. Theorem 9.3 classifies lagrangian embeddings: up to local symplectomorphism, the set of lagrangian embeddings is the set of embeddings of manifolds into their cotangent bundles as zero sections.

The classification of *isotropic* embeddings was also carried out by Weinstein in [105, 107]. An **isotropic embedding** of a manifold X into a symplectic manifold (M, ω) is a closed embedding $i : X \hookrightarrow M$ such that $i^*\omega = 0$. Weinstein showed that neighbourhood equivalence of isotropic embeddings is in one-to-one correspondence with isomorphism classes of symplectic vector bundles.

The classification of *coisotropic embeddings* is due to Gotay [47]. A **coisotropic embedding** of a manifold X carrying a closed 2-form α of constant rank into a symplectic manifold (M, ω) is an embedding $i : X \hookrightarrow M$ such that $i^*\omega = \alpha$ and $i(X)$ is coisotropic as a submanifold of M . Let E be the **characteristic distribution** of a closed form α of constant rank on X , i.e., E_p is the kernel of α_p at $p \in X$. Gotay showed that then E^* carries a symplectic structure in a neighbourhood of the zero section, such that X embeds coisotropically onto this zero section, and, moreover every coisotropic embedding is equivalent to this in some neighbourhood of the zero section. $\diamond$

9.3 Application 1: Tangent Space to the Group of Symplectomorphisms

The symplectomorphisms of a symplectic manifold (M, ω) form the group

$$\text{Symp}(M, \omega) = \{f : M \xrightarrow{\simeq} M \mid f^*\omega = \omega\}.$$

- What is $T_{\text{id}}(\text{Symp}(M, \omega))$?
- (What is the “Lie algebra” of the group of symplectomorphisms?)
- What does a neighborhood of id in $\text{Symp}(M, \omega)$ look like?

We use notions from the C^1 -topology:

C^1 -topology.

Let X and Y be manifolds.

Definition 9.4 A sequence of maps $f_i : X \rightarrow Y$ **converges in the C^0 -topology** to $f : X \rightarrow Y$ if and only if f_i converges uniformly on compact sets.

Definition 9.5 A sequence of C^1 maps $f_i : X \rightarrow Y$ **converges in the C^1 -topology** to $f : X \rightarrow Y$ if and only if it and the sequence of derivatives $df_i : TX \rightarrow TY$ converge uniformly on compact sets.

Let (M, ω) be a compact symplectic manifold and $f \in \text{Symp}(M, \omega)$. Then

$\left. \begin{array}{l} \text{Graph } f \\ \text{Graph } \text{id} = \Delta \end{array} \right\}$ are lagrangian submanifolds of $(M \times M, \text{pr}_1^*\omega - \text{pr}_2^*\omega)$.
($\text{pr}_i : M \times M \rightarrow M$, $i = 1, 2$, are the projections to each factor.)

By the Weinstein tubular neighborhood theorem, there exists a neighborhood $\mathcal{U}$ of $\Delta (\simeq M)$ in $(M \times M, \text{pr}_1^*\omega - \text{pr}_2^*\omega)$ which is symplectomorphic to a neighborhood $\mathcal{U}_0$ of M in (T^*M, ω_0) . Let $\varphi : \mathcal{U} \rightarrow \mathcal{U}_0$ be the symplectomorphism satisfying $\varphi(p, p) = (p, 0)$, $\forall p \in M$.

Suppose that f is sufficiently C^1 -close to id , i.e., f is in some sufficiently small neighborhood of id in the C^1 -topology. Then:

1. We can assume that $\text{Graph } f \subseteq \mathcal{U}$.

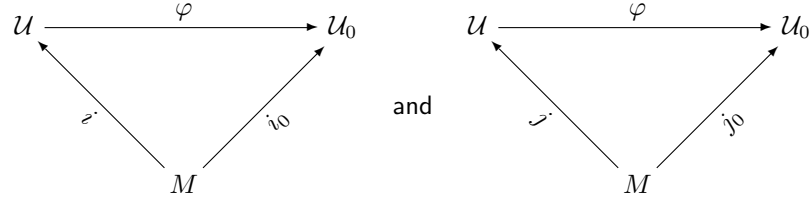
Let $j : M \hookrightarrow \mathcal{U}$ be the embedding as $\text{Graph } f$,
 $i : M \hookrightarrow \mathcal{U}$ be the embedding as $\text{Graph } \text{id} = \Delta$.

2. The map j is sufficiently C^1 -close to i .

3. By the Weinstein theorem, $\mathcal{U} \simeq \mathcal{U}_0 \subseteq T^*M$, so the above j and i induce

$j_0 : M \hookrightarrow \mathcal{U}_0$ embedding, where $j_0 = \varphi \circ j$,
 $i_0 : M \hookrightarrow \mathcal{U}_0$ embedding as 0-section .

Hence, we have



where $i(p) = (p, p)$, $i_0(p) = (p, 0)$, $j(p) = (p, f(p))$ and $j_0(p) = \varphi(p, f(p))$ for $p \in M$.

4. The map j_0 is sufficiently C^1 -close to i_0 .

$\Downarrow$

The image set $j_0(M)$ intersects each T_p^*M at one point μ_p depending smoothly on p .

5. The image of j_0 is the image of a smooth section $\mu : M \rightarrow T^*M$, that is, a 1-form $\mu = j_0 \circ (\pi \circ j_0)^{-1}$.

Therefore, $\text{Graph } f \simeq \{(p, \mu_p) \mid p \in M, \mu_p \in T_p^*M\}$.

Exercise. Vice-versa: if μ is a 1-form sufficiently C^1 -close to the zero 1-form, then

$$\{(p, \mu_p) \mid p \in M, \mu_p \in T_p^*M\} \simeq \text{Graph } f ,$$

for some diffeomorphism $f : M \rightarrow M$. By Lecture 3, we have

$\text{Graph } f \text{ is lagrangian} \iff \mu \text{ is closed.}$

◇

Conclusion. A small C^1 -neighborhood of id in $\text{Symp}(M, \omega)$ is homeomorphic to a C^1 -neighborhood of zero in the vector space of closed 1-forms on M . So:

$$T_{\text{id}}(\text{Symp}(M, \omega)) \simeq \{\mu \in \Omega^1(M) \mid d\mu = 0\} .$$

In particular, $T_{\text{id}}(\text{Symp}(M, \omega))$ contains the space of exact 1-forms

$$\{\mu = dh \mid h \in C^\infty(M)\} \simeq C^\infty(M) / \text{locally constant functions} .$$

9.4 Application 2: Fixed Points of Symplectomorphisms

Theorem 9.6 *Let (M, ω) be a compact symplectic manifold with $H_{\text{deRham}}^1(M) = 0$. Then any symplectomorphism of M which is sufficiently C^1 -close to the identity has at least two fixed points.*

Proof. Suppose that $f \in \text{Symp}(M, \omega)$ is sufficiently C^1 -close to id .

Then $\text{Graph } f \simeq \text{closed 1-form } \mu \text{ on } M$.

$$\left. \begin{array}{l} d\mu = 0 \\ H_{\text{deRham}}^1(M) = 0 \end{array} \right\} \implies \mu = dh \text{ for some } h \in C^\infty(M) .$$

Since M is compact, h has at least 2 critical points.

$$\begin{array}{ccc} \text{Fixed points of } f & = & \text{critical points of } h \\ \parallel & & \parallel \\ \text{Graph } f \cap \Delta & = & \{p : \mu_p = dh_p = 0\} . \end{array}$$

□

Lagrangian intersection problem:

A submanifold Y of M is C^1 -close to X when there is a diffeomorphism $X \rightarrow Y$ which is, as a map into M , C^1 -close to the inclusion $X \hookrightarrow M$.

Theorem 9.7 *Let (M, ω) be a symplectic manifold. Suppose that X is a compact lagrangian submanifold of M with $H_{\text{deRham}}^1(X) = 0$. Then every lagrangian submanifold of M which is C^1 -close to X intersects X in at least two points.*

Proof. Exercise. □

Arnold conjecture:

Let (M, ω) be a compact symplectic manifold, and $f : M \rightarrow M$ a symplectomorphism which is “exactly homotopic to the identity” (see below). Then

$$\#\{\text{fixed points of } f\} \geq \begin{array}{l} \text{minimal \# of critical points} \\ \text{a smooth function on } M \text{ can have} . \end{array}$$

Together with Morse theory,⁸ we obtain⁹

$$\begin{aligned} \#\{\text{nondegenerate fixed points of } f\} &\geq \begin{array}{l} \text{minimal \# of critical points} \\ \text{a Morse function on } M \text{ can have} \end{array} \\ &\geq \sum_{i=0}^{2n} \dim H^i(M; \mathbb{R}) . \end{aligned}$$

⁸A **Morse function** on M is a function $h : M \rightarrow \mathbb{R}$ whose critical points (i.e., points p where $dh_p = 0$) are all nondegenerate (i.e., the hessian at those points is nonsingular: $\det \left(\frac{\partial^2 h}{\partial x_i \partial x_j} \right)_p \neq 0$).

⁹A fixed point p of $f : M \rightarrow M$ is **nondegenerate** if $df_p : T_p M \rightarrow T_p M$ is nonsingular.

The Arnold conjecture was proved by Conley-Zehnder, Floer, Hofer-Salamon, Ono, Fukaya-Ono, Liu-Tian using Floer homology (which is an ∞ -dimensional analogue of Morse theory). There are open conjectures for sharper bounds on the number of fixed points.

Meaning of “ f is exactly homotopic to the identity:”

Suppose that $h_t : M \rightarrow \mathbb{R}$ is a smooth family of functions which is 1-periodic, i.e., $h_t = h_{t+1}$. Let $\rho : M \times \mathbb{R} \rightarrow M$ be the isotopy generated by the time-dependent vector field v_t defined by $\omega(v_t, \cdot) = dh_t$. Then “ f being exactly homotopic to the identity” means $f = \rho_1$ for some such h_t .

In other words, f is **exactly homotopic to the identity** when f is the time-1 map of an isotopy generated by some smooth time-dependent 1-periodic hamiltonian function.

There is a one-to-one correspondence

$$\text{fixed points of } f \xleftrightarrow{1-1} \text{period-1 orbits of } \rho : M \times \mathbb{R} \rightarrow M$$

because $f(p) = p$ if and only if $\{\rho(t, p) \mid t \in [0, 1]\}$ is a closed orbit.

Proof of the Arnold conjecture in the case when $h : M \rightarrow \mathbb{R}$ is independent of t :

$$p \text{ is a critical point of } h \iff dh_p = 0 \iff v_p = 0$$

$$\implies \rho(t, p) = p, \forall t \in \mathbb{R} \implies p \text{ is a fixed point of } \rho_1.$$

□

Exercise. Compute these estimates for the number of fixed points on some compact symplectic manifolds (for instance, S^2 , $S^2 \times S^2$ and $T^2 = S^1 \times S^1$). ◇

Part IV

Contact Manifolds

Contact geometry is also known as “the odd-dimensional analogue of symplectic geometry.” We will browse through the basics of contact manifolds and their relation to symplectic manifolds.

10 Contact Forms

10.1 Contact Structures

Definition 10.1 A **contact element** on a manifold M is a point $p \in M$, called the **contact point**, together with a tangent hyperplane at p , $H_p \subset T_p M$, that is, a codimension-1 subspace of $T_p M$.

A hyperplane $H_p \subset T_p M$ determines a covector $\alpha_p \in T_p^* M \setminus \{0\}$, up to multiplication by a nonzero scalar:

$$(p, H_p) \text{ is a contact element} \iff H_p = \ker \alpha_p \text{ with } \alpha_p : T_p M \longrightarrow \mathbb{R} \text{ linear, } \neq 0$$

$$\ker \alpha_p = \ker \alpha'_p \iff \alpha_p = \lambda \alpha'_p \text{ for some } \lambda \in \mathbb{R} \setminus \{0\}.$$

Suppose that H is a smooth field of contact elements (i.e., of tangent hyperplanes) on M :

$$H : p \longmapsto H_p \subset T_p M.$$

Locally, $H = \ker \alpha$ for some 1-form α , called a **locally defining 1-form** for H . (α is not unique: $\ker \alpha = \ker(f\alpha)$, for any nowhere vanishing $f : M \rightarrow \mathbb{R}$.)

Definition 10.2 A **contact structure** on M is a smooth field of tangent hyperplanes $H \subset TM$, such that, for any locally defining 1-form α , we have $d\alpha|_H$ nondegenerate (i.e., symplectic). The pair (M, H) is then called a **contact manifold** and α is called a **local contact form**.

At each $p \in M$,

$$T_p M = \underbrace{\ker \alpha_p}_{H_p} \oplus \underbrace{\ker d\alpha_p}_{1\text{-dimensional}}.$$

The $\ker d\alpha_p$ summand in this splitting depends on the choice of α .

$$\begin{aligned} d\alpha_p|_{H_p} \text{ nondegenerate} &\implies \begin{cases} \dim H_p = 2n & \text{is even} \\ (d\alpha_p)^n|_{H_p} \neq 0 & \text{is a volume form on } H_p \end{cases} \\ \alpha_p|_{\ker d\alpha_p} \text{ nondegenerate} & \end{aligned}$$

Therefore,

- any contact manifold (M, H) has $\dim M = 2n + 1$ odd, and

- if α is a (global) contact form, then $\alpha \wedge (d\alpha)^n$ is a volume form on M .

Remark. Let (M, H) be a contact manifold. A *global* contact form exists if and only if the quotient line bundle TM/H is orientable. Since H is also orientable, this implies that M is orientable. $\diamond$

Proposition 10.3 *Let H be a field of tangent hyperplanes on M . Then*

H is a contact structure $\iff \alpha \wedge (d\alpha)^n \neq 0$ for every locally defining 1-form α .

Proof.

$\implies$ Done above.

$\impliedby$ Suppose that $H = \ker \alpha$ locally. We need to show:

$$d\alpha|_H \text{ nondegenerate} \iff \alpha \wedge (d\alpha)^n \neq 0.$$

Take a local trivialization $\{e_1, f_1, \dots, e_n, f_n, r\}$ of $TM = \ker \alpha \oplus \text{rest}$, such that $\ker \alpha = \text{span}\{e_1, f_1, \dots, e_n, f_n\}$ and $\text{rest} = \text{span}\{r\}$.

$$(\alpha \wedge (d\alpha)^n)(e_1, f_1, \dots, e_n, f_n, r) = \underbrace{\alpha(r)}_{\neq 0} \cdot (d\alpha)^n(e_1, f_1, \dots, e_n, f_n)$$

and hence $\alpha \wedge (d\alpha)^n \neq 0 \iff (d\alpha)^n|_H \neq 0 \iff d\alpha|_H$ is nondegenerate. $\square$

10.2 Examples

1. On $\mathbb{R}^3$ with coordinates (x, y, z) , consider $\alpha = xdy + dz$. Since

$$\alpha \wedge d\alpha = (xdy + dz) \wedge (dx \wedge dy) = dx \wedge dy \wedge dz \neq 0,$$

α is a contact form on $\mathbb{R}^3$.

The corresponding field of hyperplanes $H = \ker \alpha$ at $(x, y, z) \in \mathbb{R}^3$ is

$$H_{(x,y,z)} = \left\{ v = a \frac{\partial}{\partial x} + b \frac{\partial}{\partial y} + c \frac{\partial}{\partial z} \mid \alpha(v) = bx + c = 0 \right\}.$$

Exercise. Picture these hyperplanes. $\diamond$

2. **(Martinet [80], 1971)** Any compact orientable 3-manifold admits a contact structure.

Open Problem, 2000. The classification of compact orientable contact 3-manifolds is still not known. There is by now a huge collection of results in contact topology related to the classification of contact manifolds. For a review of the state of the knowledge and interesting questions on contact 3-manifolds, see [33, 43, 100].

3. Let X be a manifold and T^*X its cotangent bundle. There are two canonical contact manifolds associated to X (see Homework 7):

$$\begin{aligned}\mathbb{P}(T^*X) &= \text{the projectivization of } T^*X, \text{ and} \\ S(T^*X) &= \text{the cotangent sphere bundle.}\end{aligned}$$

4. On $\mathbb{R}^{2n+1}$ with coordinates $(x_1, y_1, \dots, x_n, y_n, z)$, $\alpha = \sum_i x_i dy_i + dz$ is contact.

10.3 First Properties

There is a local normal form theorem for contact manifolds analogous to the Darboux theorem for symplectic manifolds.

Theorem 10.4 *Let (M, H) be a contact manifold and $p \in M$. Then there exists a coordinate system $(\mathcal{U}, x_1, y_1, \dots, x_n, y_n, z)$ centered at p such that on $\mathcal{U}$*

$$\alpha = \sum x_i dy_i + dz \text{ is a local contact form for } H.$$

The idea behind the proof is sketched in the next lecture.

There is also a Moser-type theorem for contact forms.

Theorem 10.5 (Gray) *Let M be a compact manifold. Suppose that $\alpha_t, t \in [0, 1]$, is a smooth family of (global) contact forms on M . Let $H_t = \ker \alpha_t$. Then there exists an isotopy $\rho : M \times \mathbb{R} \rightarrow M$ such that $H_t = \rho_{t*} H_0$, for all $0 \leq t \leq 1$.*

Exercise. Show that $H_t = \rho_{t*} H_0 \iff \rho_t^* \alpha_t = u_t \cdot \alpha_0$ for some family $u_t : M \rightarrow \mathbb{R}$, $0 \leq t \leq 1$, of nowhere vanishing functions. $\diamond$

Proof. (*À la Moser*)

We need to find ρ_t such that $\begin{cases} \rho_0 = \text{id} \\ \frac{d}{dt}(\rho_t^* \alpha_t) = \frac{d}{dt}(u_t \alpha_0) \end{cases}$. For any isotopy ρ ,

$$\frac{d}{dt}(\rho_t^* \alpha_t) = \rho_t^* \left(\mathcal{L}_{v_t} \alpha_t + \frac{d\alpha_t}{dt} \right),$$

where $v_t = \frac{d\rho_t}{dt} \circ \rho_t^{-1}$ is the vector field generated by ρ_t . By the Moser trick, it suffices to find v_t and then integrate it to ρ_t . We will search for v_t in $H_t = \ker \alpha_t$; this unnecessary assumption simplifies the proof.

We need to solve

$$\begin{aligned}
 \rho_t^* \left(\underbrace{\mathcal{L}_{v_t} \alpha_t}_{d\iota_{v_t} \alpha_t + \iota_{v_t} d\alpha_t} + \frac{d\alpha_t}{dt} \right) &= \frac{du_t}{dt} \underbrace{\alpha_0}_{\frac{1}{u_t} \rho_t^* \alpha_t} \\
 \Rightarrow \quad \rho_t^* \left(\iota_{v_t} d\alpha_t + \frac{d\alpha_t}{dt} \right) &= \frac{du_t}{dt} \cdot \frac{1}{u_t} \cdot \rho_t^* \alpha_t \\
 \Leftrightarrow \quad \iota_{v_t} d\alpha_t + \frac{d\alpha_t}{dt} &= (\rho_t^*)^{-1} \left(\frac{du_t}{dt} \cdot \frac{1}{u_t} \right) \alpha_t . \quad (\star)
 \end{aligned}$$

Restricting to the hyperplane $H_t = \ker \alpha_t$, equation $(\star)$ reads

$$\iota_{v_t} d\alpha_t|_{H_t} = - \frac{d\alpha_t}{dt} \Big|_{H_t}$$

which determines v_t uniquely, since $d\alpha_t|_{H_t}$ is nondegenerate. After integrating v_t to ρ_t , the factor u_t is determined by the relation $\rho_t^* \alpha_t = u_t \cdot \alpha_0$. Check that this indeed gives a solution. $\square$

Homework 7: Manifolds of Contact Elements

Given any manifold X of dimension n , there is a canonical symplectic manifold of dimension $2n$ attached to it, namely its cotangent bundle with the standard symplectic structure. The exercises below show that there is also a canonical *contact* manifold of dimension $2n - 1$ attached to X .

The **manifold of contact elements** of an n -dimensional manifold X is

$$\mathcal{C} = \{(x, \chi_x) \mid x \in X \text{ and } \chi_x \text{ is a hyperplane in } T_x X\}.$$

On the other hand, the projectivization of the cotangent bundle of X is

$$\mathbb{P}^* X = (T^* X \setminus \text{zero section}) / \sim$$

where $(x, \xi) \sim (x, \xi')$ whenever $\xi = \lambda \xi'$ for some $\lambda \in \mathbb{R} \setminus \{0\}$ (here $x \in X$ and $\xi, \xi' \in T_x^* X \setminus \{0\}$). We will denote elements of $\mathbb{P}^* X$ by $(x, [\xi])$, $[\xi]$ being the $\sim$ equivalence class of ξ .

1. Show that $\mathcal{C}$ is naturally isomorphic to $\mathbb{P}^* X$ as a bundle over X , i.e., exhibit a diffeomorphism $\varphi : \mathcal{C} \rightarrow \mathbb{P}^* X$ such that the following diagram commutes:

$$\begin{array}{ccc} \mathcal{C} & \xrightarrow{\varphi} & \mathbb{P}^* X \\ \pi \downarrow & & \downarrow \pi \\ X & = & X \end{array}$$

where the vertical maps are the natural projections $(x, \chi_x) \mapsto x$ and $(x, \xi) \mapsto x$.

Hint: The kernel of a non-zero $\xi \in T_x^* X$ is a hyperplane $\chi_x \subset T_x X$. What is the relation between ξ and ξ' if $\ker \xi = \ker \xi'$?

2. There is on $\mathcal{C}$ a canonical field of hyperplanes $\mathcal{H}$ (that is, a smooth map attaching to each point in $\mathcal{C}$ a hyperplane in the tangent space to $\mathcal{C}$ at that point): $\mathcal{H}$ at the point $p = (x, \chi_x) \in \mathcal{C}$ is the hyperplane

$$\mathcal{H}_p = (d\pi_p)^{-1} \chi_x \subset T_p \mathcal{C},$$

where

$$\begin{array}{ccc} \mathcal{C} & p = (x, \chi_x) & T_p \mathcal{C} \\ \downarrow \pi & \downarrow & \downarrow d\pi_p \\ X & x & T_x X \end{array}$$

are the natural projections, and $(d\pi_p)^{-1} \chi_x$ is the preimage of $\chi_x \subset T_x X$ by $d\pi_p$.

Under the isomorphism $\mathcal{C} \simeq \mathbb{P}^* X$ from exercise 1, $\mathcal{H}$ induces a field of hyperplanes $\mathbb{H}$ on $\mathbb{P}^* X$. Describe $\mathbb{H}$.

Hint: If $\xi \in T_x^* X \setminus \{0\}$ has kernel χ_x , what is the kernel of the canonical 1-form $\alpha_{(x, \xi)} = (d\pi_{(x, \xi)})^* \xi$?

3. Check that $(\mathbb{P}^*X, \mathbb{H})$ is a contact manifold, and therefore $(\mathcal{C}, \mathcal{H})$ is a contact manifold.

Hint: Let $(x, [\xi]) \in \mathbb{P}^*X$. For any ξ representing the class $[\xi]$, we have

$$\mathbb{H}_{(x, [\xi])} = \ker((d\pi_{(x, [\xi])})^* \xi) .$$

Let $x_1, \dots, x_n$ be local coordinates on X , and let $x_1, \dots, x_n, \xi_1, \dots, \xi_n$ be the associated local coordinates on T^*X . In these coordinates, $(x, [\xi])$ is given by $(x_1, \dots, x_n, [\xi_1, \dots, \xi_n])$. Since at least one of the ξ_i 's is nonzero, without loss of generality we may assume that $\xi_1 \neq 0$ so that we may divide ξ by ξ_1 to obtain a representative with coordinates $(1, \xi_2, \dots, \xi_n)$. Hence, by choosing always the representative of $[\xi]$ with $\xi_1 = 1$, the set $x_1, \dots, x_n, \xi_2, \dots, \xi_n$ defines coordinates on some neighborhood $\mathcal{U}$ of $(x, [\xi])$ in $\mathbb{P}^*X$. On $\mathcal{U}$, consider the 1-form

$$\alpha = dx_1 + \sum_{i \geq 2} \xi_i dx_i .$$

Show that α is a contact form on $\mathcal{U}$, i.e., show that $\ker \alpha_{(x, [\xi])} = \mathbb{H}_{(x, [\xi])}$, and that $d\alpha_{(x, [\xi])}$ is nondegenerate on $\mathbb{H}_{(x, [\xi])}$.

4. What is the symplectization of $\mathcal{C}$?

What is the manifold $\mathcal{C}$ when $X = \mathbb{R}^3$ and when $X = S^1 \times S^1$?

Remark. Similarly, we could have defined the **manifold of oriented contact elements** of X to be

$$\mathcal{C}^o = \left\{ (x, \chi_x^o) \mid x \in X \text{ and } \chi_x^o \text{ is a hyperplane in } T_x X \text{ equipped with an orientation} \right\} .$$

The manifold $\mathcal{C}^o$ is isomorphic to the cotangent sphere bundle of X

$$S^*X := (T^*X \setminus \text{zero section}) / \approx$$

where $(x, \xi) \approx (x, \xi')$ whenever $\xi = \lambda \xi'$ for some $\lambda \in \mathbb{R}^+$.

A construction analogous to the above produces a canonical contact structure on $\mathcal{C}^o$. See [3, Appendix 4].

◇

11 Contact Dynamics

11.1 Reeb Vector Fields

Let (M, H) be a contact manifold with a contact form α .

Claim. There exists a unique vector field R on M such that $\begin{cases} \iota_R d\alpha = 0 \\ \iota_R \alpha = 1 \end{cases}$

Proof. $\begin{cases} \iota_R d\alpha = 0 & \implies R \in \ker d\alpha, \text{ which is a line bundle, and} \\ \iota_R \alpha = 1 & \implies \text{normalizes } R. \end{cases} \quad \square$

The vector field R is called the **Reeb vector field** determined by α .

Claim. The flow of R preserves the contact form, i.e., if $\rho_t = \exp tR$ is the isotopy generated by R , then $\rho_t^* \alpha = \alpha, \forall t \in \mathbb{R}$.

Proof. We have $\frac{d}{dt}(\rho_t^* \alpha) = \rho_t^*(\mathcal{L}_R \alpha) = \rho_t^*(\underbrace{d \iota_R \alpha}_1 + \underbrace{\iota_R d\alpha}_0) = 0$.

Hence, $\rho_t^* \alpha = \rho_0^* \alpha = \alpha, \forall t \in \mathbb{R}$. $\square$

Definition 11.1 A **contactomorphism** is a diffeomorphism f of a contact manifold (M, H) which preserves the contact structure (i.e., $f_* H = H$).

Examples.

1. Euclidean space $\mathbb{R}^{2n+1}$ with $\alpha = \sum_i x_i dy_i + dz$.

$$\left. \begin{aligned} \iota_R \sum dx_i \wedge dy_i &= 0 \\ \iota_R \sum x_i dy_i + dz &= 1 \end{aligned} \right\} \implies R = \frac{\partial}{\partial z} \text{ is the Reeb vector field.}$$

The contactomorphisms generated by R are translations

$$\rho_t(x_1, y_1, \dots, x_n, y_n, z) = (x_1, y_1, \dots, x_n, y_n, z + t).$$

2. Regard the odd sphere $S^{2n-1} \xrightarrow{i} \mathbb{R}^{2n}$ as the set of unit vectors

$$\{(x_1, y_1, \dots, x_n, y_n) \mid \sum (x_i^2 + y_i^2) = 1\}.$$

Consider the 1-form on $\mathbb{R}^{2n}$, $\sigma = \frac{1}{2} \sum (x_i dy_i - y_i dx_i)$.

Claim. The form $\alpha = i^* \sigma$ is a contact form on S^{2n-1} .

Proof. We need to show that $\alpha \wedge (d\alpha)^{n-1} \neq 0$. The 1-form on $\mathbb{R}^{2n}$ $\nu = d \sum (x_i^2 + y_i^2) = 2 \sum (x_i dx_i + y_i dy_i)$ satisfies $T_p S^{2n-1} = \ker \nu_p$, at $p \in S^{2n-1}$. Check that $\nu \wedge \sigma \wedge (d\sigma)^{n-1} \neq 0$. $\square$

The distribution $H = \ker \alpha$ is called the **standard contact structure** on S^{2n-1} . The Reeb vector field is $R = 2 \sum \left(x_i \frac{\partial}{\partial y_i} - y_i \frac{\partial}{\partial x_i} \right)$, and is also known as the **Hopf vector field** on S^{2n-1} , as the orbits of its flow are the circles of the Hopf fibration.

◇

11.2 Symplectization

Example. Let $\widetilde{M} = S^{2n-1} \times \mathbb{R}$, with coordinate τ in the $\mathbb{R}$ -factor, and projection $\pi : \widetilde{M} \rightarrow S^{2n-1}$, $(p, \tau) \mapsto p$. Under the identification $\widetilde{M} \simeq \mathbb{R}^{2n} \setminus \{0\}$, where the $\mathbb{R}$ -factor represents the logarithm of the square of the radius, the projection π becomes

$$\begin{aligned} \pi : \quad \mathbb{R}^{2n} \setminus \{0\} &\longrightarrow S^{2n-1} \\ (X_1, Y_1, \dots, X_n, Y_n) &\longmapsto \left(\frac{X_1}{\sqrt{e^\tau}}, \frac{Y_1}{\sqrt{e^\tau}}, \dots, \frac{X_n}{\sqrt{e^\tau}}, \frac{Y_n}{\sqrt{e^\tau}} \right) \end{aligned}$$

where $e^\tau = \sum (X_i^2 + Y_i^2)$. Let $\alpha = i^* \sigma$ be the standard contact form on S^{2n-1} (see the previous example). Then $\omega = d(e^\tau \pi^* \alpha)$ is a closed 2-form on $\mathbb{R}^{2n} \setminus \{0\}$. Since $\pi^* i^* x_i = \frac{X_i}{\sqrt{e^\tau}}$, $\pi^* i^* y_i = \frac{Y_i}{\sqrt{e^\tau}}$, we have

$$\begin{aligned} \pi^* \alpha &= \pi^* i^* \sigma = \frac{1}{2} \sum \left(\frac{X_i}{\sqrt{e^\tau}} d\left(\frac{Y_i}{\sqrt{e^\tau}}\right) - \frac{Y_i}{\sqrt{e^\tau}} d\left(\frac{X_i}{\sqrt{e^\tau}}\right) \right) \\ &= \frac{1}{2e^\tau} \sum (X_i dY_i - Y_i dX_i). \end{aligned}$$

Therefore, $\omega = \sum dX_i \wedge dY_i$ is the standard symplectic form on $\mathbb{R}^{2n} \setminus \{0\} \subset \mathbb{R}^{2n}$. $(\widetilde{M}, \omega)$ is called the *symplectization* of (S^{2n-1}, α) . ◇

Proposition 11.2 *Let (M, H) be a contact manifold with a contact form α . Let $\widetilde{M} = M \times \mathbb{R}$, and let $\pi : \widetilde{M} \rightarrow M$, $(p, \tau) \mapsto p$, be the projection. Then $\omega = d(e^\tau \pi^* \alpha)$ is a symplectic form on $\widetilde{M}$, where τ is a coordinate on $\mathbb{R}$.*

Proof. Exercise. □

Hence, $\widetilde{M}$ has a symplectic form ω canonically determined by a contact form α on M and a coordinate function on $\mathbb{R}$; $(\widetilde{M}, \omega)$ is called the **symplectization** of (M, α) .

Remarks.

1. The contact version of the Darboux theorem can now be derived by applying the symplectic theorem to the symplectization of the contact manifold (with appropriate choice of coordinates); see [3, Appendix 4].

2. There is a coordinate-free description of $\widetilde{M}$ as

$$\widetilde{M} = \{(p, \xi) \mid p \in M, \xi \in T_p^*M, \text{ such that } \ker \xi = H_p\} .$$

The group $\mathbb{R} \setminus \{0\}$ acts on $\widetilde{M}$ by multiplication on the cotangent vector:

$$\lambda \cdot (p, \xi) = (p, \lambda \xi) , \quad \lambda \in \mathbb{R} \setminus \{0\} .$$

The quotient $\widetilde{M}/(\mathbb{R} \setminus \{0\})$ is diffeomorphic to M . $\widetilde{M}$ has a canonical 1-form $\tilde{\alpha}$ defined at $v \in T_{(p, \xi)}\widetilde{M}$ by

$$\tilde{\alpha}_{(p, \xi)}(v) = \xi((d \operatorname{pr})_{(p, \xi)} v) ,$$

where $\operatorname{pr} : \widetilde{M} \rightarrow M$ is the bundle projection.

◇

11.3 Conjectures of Seifert and Weinstein

Question. (Seifert, 1948) Let v be a nowhere vanishing vector field on the 3-sphere. Does the flow of v have any periodic orbits?

Counterexamples.

- **(Schweitzer, 1974)** $\exists C^1$ vector field without periodic orbits.
- **(Kristina Kuperberg, 1994)** $\exists C^\infty$ vector field without periodic orbits.

Question. How about volume-preserving vector fields?

- **(Greg Kuperberg, 1997)** $\exists C^1$ counterexample.
- C^∞ counterexamples are not known.

Natural generalization of this problem:

Let $M = S^3$ be the 3-sphere, and let γ be a volume form on M . Suppose that v is a nowhere vanishing vector field, and suppose that v is volume-preserving, i.e.,

$$\mathcal{L}_v \gamma = 0 \iff d\iota_v \gamma = 0 \iff \iota_v \gamma = d\alpha$$

for some 1-form α , since $H^2(S^3) = 0$.

Given a 1-form α , we would like to study vector fields v such that

$$\begin{cases} \iota_v \gamma = d\alpha \\ \iota_v \alpha > 0 . \end{cases}$$

A vector field v satisfying $\iota_v \alpha > 0$ is called **positive**. For instance, vector fields in a neighborhood of the Hopf vector field are positive relative to the standard contact form on S^3 .

Renormalizing as $R := \frac{v}{\iota_v \alpha}$, we should study instead

$$\begin{cases} \iota_R d\alpha = 0 \\ \iota_\alpha = 1 \\ \alpha \wedge d\alpha \text{ is a volume form,} \end{cases}$$

that is, study pairs (α, R) where

$$\begin{cases} \alpha \text{ is a **contact** form, and} \\ R \text{ is its **Reeb** vector field.} \end{cases}$$

Conjecture. (Weinstein, 1978 [106]) *Suppose that M is a 3-dimensional manifold with a (global) contact form α . Let v be the Reeb vector field for α . Then v has a periodic orbit.*

Theorem 11.3 (Viterbo and Hofer, 1993 [63, 64, 103]) *The Weinstein conjecture is true when*

- 1) $M = S^3$, or
- 2) $\pi_2(M) \neq 0$, or
- 3) *the contact structure is overtwisted.*¹⁰

Open questions.

- How many periodic orbits are there?
- What do they look like?
- Is there always an unknotted one?
- What about the linking behavior?

¹⁰A surface S inside a contact 3-manifold determines a singular foliation on S , called the **characteristic foliation** of S , by the intersection of the contact planes with the tangent spaces to S . A contact structure on a 3-manifold M is called **overtwisted** if there exists an embedded 2-disk whose characteristic foliation contains one closed leaf C and exactly one singular point inside C ; otherwise, the contact structure is called **tight**. Eliashberg [32] showed that the isotopy classification of overtwisted contact structures on closed 3-manifolds coincides with their homotopy classification as tangent plane fields. The classification of tight contact structures is still open.

Part V

Compatible Almost Complex Structures

The fact that any symplectic manifold possesses almost complex structures, and even so in a *compatible* sense, establishes a link from symplectic geometry to complex geometry, and is the point of departure for the modern technique of counting pseudo-holomorphic curves, as first proposed by Gromov [49].

12 Almost Complex Structures

12.1 Three Geometries

1. Symplectic geometry:
geometry of a closed nondegenerate skew-symmetric bilinear form.
2. Riemannian geometry:
geometry of a positive-definite symmetric bilinear map.
3. Complex geometry:
geometry of a linear map with square -1 .

Example. The euclidean space $\mathbb{R}^{2n}$ with the standard linear coordinates $(x_1, \dots, x_n, y_1, \dots, y_n)$ has standard structures:

$$\begin{aligned}\omega_0 &= \sum dx_j \wedge dy_j, & \text{standard symplectic structure;} \\ g_0 &= \langle \cdot, \cdot \rangle, & \text{standard inner product; and}\end{aligned}$$

if we identify $\mathbb{R}^{2n}$ with $\mathbb{C}^n$ with coordinates $z_j = x_j + \sqrt{-1} y_j$, then multiplication by $\sqrt{-1}$ induces a constant linear map J_0 on the tangent spaces of $\mathbb{R}^{2n}$:

$$J_0\left(\frac{\partial}{\partial x_j}\right) = \frac{\partial}{\partial y_j}, \quad J_0\left(\frac{\partial}{\partial y_j}\right) = -\frac{\partial}{\partial x_j},$$

with $J_0^2 = -\text{Id}$. Relative to the basis $\frac{\partial}{\partial x_1}, \dots, \frac{\partial}{\partial x_n}, \frac{\partial}{\partial y_1}, \dots, \frac{\partial}{\partial y_n}$, the maps J_0 , ω_0 and g_0 are represented by

$$\begin{aligned}J_0(u) &= \begin{pmatrix} 0 & -\text{Id} \\ \text{Id} & 0 \end{pmatrix} u \\ \omega_0(u, v) &= v^t \begin{pmatrix} 0 & -\text{Id} \\ \text{Id} & 0 \end{pmatrix} u \\ g_0(u, v) &= v^t u\end{aligned}$$

where $u, v \in \mathbb{R}^{2n}$ and v^t is the transpose of v . The following compatibility relation holds:

$$\omega_0(u, v) = g_0(J_0(u), v) .$$

◇

12.2 Complex Structures on Vector Spaces

Definition 12.1 Let V be a vector space. A **complex structure** on V is a linear map:

$$J : V \rightarrow V \quad \text{with} \quad J^2 = -\text{Id} .$$

The pair (V, J) is called a **complex vector space**.

A complex structure J is equivalent to a structure of vector space over $\mathbb{C}$ if we identify the map J with multiplication by $\sqrt{-1}$.

Definition 12.2 Let (V, Ω) be a symplectic vector space. A complex structure J on V is said to be **compatible** (with Ω , or Ω -compatible) if

$$G_J(u, v) := \Omega(u, Jv) , \quad \forall u, v \in V , \text{ is a positive inner product on } V .$$

That is,

$$J \text{ is } \Omega\text{-compatible} \iff \begin{cases} \Omega(Ju, Jv) = \Omega(u, v) & [\text{symplectomorphism}] \\ \Omega(u, Ju) > 0, \quad \forall u \neq 0 & [\text{taming condition}] \end{cases}$$

Compatible complex structures always exist on symplectic vector spaces:

Proposition 12.3 Let (V, Ω) be a symplectic vector space. Then there is a compatible complex structure J on V .

Proof. Choose a positive inner product G on V . Since Ω and G are nondegenerate,

$$\left. \begin{array}{l} u \in V \mapsto \Omega(u, \cdot) \in V^* \\ w \in V \mapsto G(w, \cdot) \in V^* \end{array} \right\} \text{ are isomorphisms between } V \text{ and } V^* .$$

Hence, $\Omega(u, v) = G(Au, v)$ for some linear map $A : V \rightarrow V$. This map A is skew-symmetric because

$$\begin{aligned} G(A^*u, v) &= G(u, Av) = G(Av, u) \\ &= \Omega(v, u) = -\Omega(u, v) = G(-Au, v) . \end{aligned}$$

Also:

- AA^* is symmetric: $(AA^*)^* = AA^*$.
- AA^* is positive: $G(AA^*u, u) = G(A^*u, A^*u) > 0$, for $u \neq 0$.

These properties imply that AA^* diagonalizes with positive eigenvalues λ_i ,

$$AA^* = B \operatorname{diag}(\lambda_1, \dots, \lambda_{2n}) B^{-1}.$$

We may hence define an arbitrary real power of AA^* by rescaling the eigenspaces, in particular,

$$\sqrt{AA^*} := B \operatorname{diag}(\sqrt{\lambda_1}, \dots, \sqrt{\lambda_{2n}}) B^{-1}.$$

Then $\sqrt{AA^*}$ is symmetric and positive-definite. Let

$$J = (\sqrt{AA^*})^{-1} A.$$

The factorization $A = \sqrt{AA^*} J$ is called the **polar decomposition** of A . Since A commutes with $\sqrt{AA^*}$, J commutes with $\sqrt{AA^*}$. Check that J is orthogonal, $JJ^* = \operatorname{Id}$, as well as skew-adjoint, $J^* = -J$, and hence it is a complex structure on V :

$$J^2 = -JJ^* = -\operatorname{Id}.$$

Compatibility:

$$\begin{aligned} \Omega(Ju, Jv) &= G(AJu, Jv) = G(JAu, Jv) = G(Au, v) \\ &= \Omega(u, v) \\ \Omega(u, Ju) &= G(Au, Ju) = G(-JAu, u) \\ &= G(\sqrt{AA^*}u, u) > 0, \quad \text{for } u \neq 0. \end{aligned}$$

Therefore, J is a compatible complex structure on V . □

As indicated in the proof, in general, the positive inner product defined by

$$\Omega(u, Jv) = G(\sqrt{AA^*}u, v) \text{ is different from } G(u, v).$$

Remarks.

1. This construction is canonical after an initial choice of G . To see this, notice that $\sqrt{AA^*}$ does not depend on the choice of B nor of the ordering of the eigenvalues in $\operatorname{diag}(\sqrt{\lambda_1}, \dots, \sqrt{\lambda_{2n}})$. The linear transformation $\sqrt{AA^*}$ is completely determined by its effect on each eigenspace of AA^* : on the eigenspace corresponding to the eigenvalue λ_k , the map $\sqrt{AA^*}$ is defined to be multiplication by $\sqrt{\lambda_k}$.
2. If (V_t, Ω_t) is a family of symplectic vector spaces with a family G_t of positive inner products, all depending smoothly on a real parameter t , then, adapting the proof of the previous proposition, we can show that there is a smooth family J_t of compatible complex structures on V_t .
3. To check just the existence of compatible complex structures on a symplectic vector space (V, Ω) , we could also proceed as follows. Given a symplectic basis $e_1, \dots, e_n, f_1, \dots, f_n$ (i.e., $\Omega(e_i, e_j) = \Omega(f_i, f_j) = 0$ and $\Omega(e_i, f_j) = \delta_{ij}$), one can define $Je_j = f_j$ and $Jf_j = -e_j$. This is a compatible complex

structure on (V, Ω) . Moreover, given Ω and J compatible on V , there exists a symplectic basis of V of the form:

$$e_1, \dots, e_n, f_1 = Je_1, \dots, f_n = Je_n.$$

The proof is part of Homework 8.

4. Conversely, given (V, J) , there is always a symplectic structure Ω such that J is Ω -compatible: pick any positive inner product G such that $J^* = -J$ and take $\Omega(u, v) = G(Ju, v)$.

◇

12.3 Compatible Structures

Definition 12.4 An almost complex structure on a manifold M is a smooth field of complex structures on the tangent spaces:

$$x \longmapsto J_x : T_x M \rightarrow T_x M \quad \text{linear,} \quad \text{and} \quad J_x^2 = -\text{Id}.$$

The pair (M, J) is then called an **almost complex manifold**.

Definition 12.5 Let (M, ω) be a symplectic manifold. An almost complex structure J on M is called **compatible** (with ω or ω -compatible) if the assignment

$$\begin{aligned} x \longmapsto g_x : T_x M \times T_x M &\rightarrow \mathbb{R} \\ g_x(u, v) &:= \omega_x(u, J_x v) \end{aligned}$$

is a riemannian metric on M .

For a manifold M ,

$$\begin{array}{lll} \omega \text{ is a symplectic form} & \implies & x \longmapsto \omega_x : T_x M \times T_x M \rightarrow \mathbb{R} \text{ is bilinear,} \\ & & \text{nondegenerate, skew-symmetric;} \\ g \text{ is a riemannian metric} & \implies & x \longmapsto g_x : T_x M \times T_x M \rightarrow \mathbb{R} \\ & & \text{is a positive inner product;} \\ J \text{ almost complex structure} & \implies & x \longmapsto J_x : T_x M \rightarrow T_x M \\ & & \text{is linear and } J^2 = -\text{Id}. \end{array}$$

The triple (ω, g, J) is called a **compatible triple** when $g(\cdot, \cdot) = \omega(\cdot, J\cdot)$.

Proposition 12.6 Let (M, ω) be a symplectic manifold, and g a riemannian metric on M . Then there exists a canonical almost complex structure J on M which is compatible.

Proof. The polar decomposition is *canonical* (after a choice of metric), hence this construction of J on M is *smooth*; cf. Remark 2 of the previous section. □

Remark. In general, $g_J(\cdot, \cdot) := \omega(\cdot, J\cdot) \neq g(\cdot, \cdot)$. ◇

Since riemannian metrics always exist, we conclude:

Corollary 12.7 *Any symplectic manifold has compatible almost complex structures.*

– How different can compatible almost complex structures be?

Proposition 12.8 *Let (M, ω) be a symplectic manifold, and J_0, J_1 two almost complex structures compatible with ω . Then there is a smooth family $J_t, 0 \leq t \leq 1$, of compatible almost complex structures joining J_0 to J_1 .*

Proof. By compatibility, we get

$$\left. \begin{array}{l} \omega, J_0 \rightsquigarrow g_0(\cdot, \cdot) = \omega(\cdot, J_0 \cdot) \\ \omega, J_1 \rightsquigarrow g_1(\cdot, \cdot) = \omega(\cdot, J_1 \cdot) \end{array} \right\} \quad \text{two riemannian metrics on } M .$$

Their convex combinations

$$g_t(\cdot, \cdot) = (1 - t)g_0(\cdot, \cdot) + tg_1(\cdot, \cdot) , \quad 0 \leq t \leq 1 ,$$

form a smooth family of riemannian metrics. Apply the polar decomposition to (ω, g_t) to obtain a smooth family of J_t 's joining J_0 to J_1 . $\square$

Corollary 12.9 *The set of all compatible almost complex structures on a symplectic manifold is path-connected.*

Homework 8: Compatible Linear Structures

1. Let $\Omega(V)$ and $J(V)$ be the spaces of symplectic forms and complex structures on the vector space V , respectively. Take $\Omega \in \Omega(V)$ and $J \in J(V)$. Let $\text{GL}(V)$ be the group of all isomorphisms of V , let $\text{Sp}(V, \Omega)$ be the group of symplectomorphisms of (V, Ω) , and let $\text{GL}(V, J)$ be the group of complex isomorphisms of (V, J) .

Show that

$$\Omega(V) \simeq \text{GL}(V)/\text{Sp}(V, \Omega) \quad \text{and} \quad J(V) \simeq \text{GL}(V)/\text{GL}(V, J) .$$

Hint: The group $\text{GL}(V)$ acts on $\Omega(V)$ by pullback. What is the stabilizer of a given Ω ?

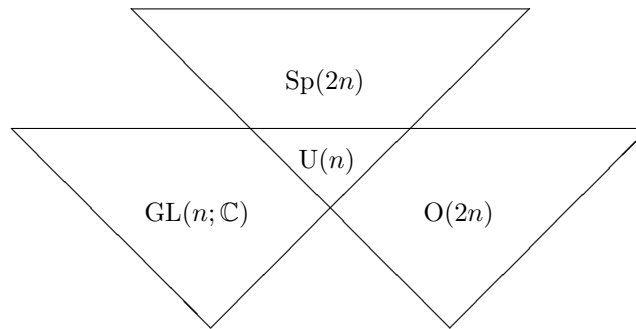
2. Let $(\mathbb{R}^{2n}, \Omega_0)$ be the standard $2n$ -dimensional symplectic euclidean space. The **symplectic linear group** is the group of all linear transformations of $\mathbb{R}^{2n}$ which preserve the symplectic structure:

$$\text{Sp}(2n) := \{A \in \text{GL}(2n; \mathbb{R}) \mid \Omega_0(Au, Av) = \Omega_0(u, v) \text{ for all } u, v \in \mathbb{R}^{2n}\} .$$

Identifying the complex $n \times n$ matrix $X + iY$ with the real $2n \times 2n$ matrix $\begin{pmatrix} X & -Y \\ Y & X \end{pmatrix}$, consider the following subgroups of $\text{GL}(2n; \mathbb{R})$:

$$\text{Sp}(2n) , \text{O}(2n) , \text{GL}(n; \mathbb{C}) \text{ and } \text{U}(n) .$$

Show that the intersection of any two of them is $\text{U}(n)$. (From [83, p.41].)



3. Let (V, Ω) be a symplectic vector space of dimension $2n$, and let $J : V \rightarrow V$, $J^2 = -\text{Id}$, be a complex structure on V .
- (a) Prove that, if J is Ω -compatible and L is a lagrangian subspace of (V, Ω) , then JL is also lagrangian and $JL = L^\perp$, where $\perp$ denotes orthogonality with respect to the positive inner product $G_J(u, v) = \Omega(u, Jv)$.
 - (b) Deduce that J is Ω -compatible if and only if there exists a symplectic basis for V of the form

$$e_1, e_2, \dots, e_n, f_1 = Je_1, f_2 = Je_2, \dots, f_n = Je_n$$

where $\Omega(e_i, e_j) = \Omega(f_i, f_j) = 0$ and $\Omega(e_i, f_j) = \delta_{ij}$.

13 Compatible Triples

13.1 Compatibility

Let (M, ω) be a symplectic manifold. As shown in the previous lecture, compatible almost complex structures always exist on (M, ω) . We also showed that the set of all compatible almost complex structures on (M, ω) is path-connected. In fact, the set of all compatible almost complex structures is even contractible. (This is important for defining invariants.) Let $\mathcal{J}(T_x M, \omega_x)$ be the set of all compatible complex structures on $(T_x M, \omega_x)$ for $x \in M$.

Proposition 13.1 *The set $\mathcal{J}(T_x M, \omega_x)$ is contractible, i.e., there exists a homotopy*

$$h_t : \mathcal{J}(T_x M, \omega_x) \longrightarrow \mathcal{J}(T_x M, \omega_x) , \quad 0 \leq t \leq 1 ,$$

starting at the identity $h_0 = \text{Id}$,

finishing at a trivial map $h_1 : \mathcal{J}(T_x M, \omega_x) \rightarrow \{J_0\}$,

and fixing J_0 (i.e., $h_t(J_0) = J_0, \forall t$) for some $J_0 \in \mathcal{J}(T_x M, \omega_x)$.

Proof. Homework 9. □

Consider the fiber bundle $\mathcal{J} \rightarrow M$ with fiber

$$\mathcal{J}_x := \mathcal{J}(T_x M, \omega_x) \quad \text{over } x \in M .$$

A compatible almost complex structure J on (M, ω) is a section of $\mathcal{J}$. The space of sections of $\mathcal{J}$ is contractible because the fibers are contractible.

Remarks.

- We never used the closedness of ω to construct compatible almost complex structures. The construction holds for an **almost symplectic manifold** (M, ω) , that is, a pair of a manifold M and a nondegenerate 2-form ω , not necessarily closed.
- Similarly, we could define a **symplectic vector bundle** to be a vector bundle $E \rightarrow M$ over a manifold M equipped with a smooth field ω of fiberwise nondegenerate skew-symmetric bilinear maps

$$\omega_x : E_x \times E_x \longrightarrow \mathbb{R} .$$

The existence of such a field ω is equivalent to being able to reduce the structure group of the bundle from the general linear group to the linear symplectic group. As a consequence of our discussion, a symplectic vector bundle is always a complex vector bundle, and vice-versa. ◇

13.2 Triple of Structures

If (ω, J, g) is a **compatible triple**, then any one of ω , J or g can be written in terms of the other two:

$$\begin{aligned} g(u, v) &= \omega(u, Jv) \\ \omega(u, v) &= g(Ju, v) \\ J(u) &= \tilde{g}^{-1}(\tilde{\omega}(u)) \end{aligned}$$

where

$$\begin{aligned} \tilde{\omega} : TM &\longrightarrow T^*M & u &\longmapsto \omega(u, \cdot) \\ \tilde{g} : TM &\longrightarrow T^*M & u &\longmapsto g(u, \cdot) \end{aligned}$$

are the linear isomorphisms induced by the bilinear forms ω and g .

The relations among ω , J and g can be summarized in the following table. The last column lists differential equations these structures are usually asked to satisfy.

Data	Condition/Technique	Consequence	Question
ω, J	$\omega(Ju, Jv) = \omega(u, v)$ $\omega(u, Ju) > 0, u \neq 0$	$g(u, v) := \omega(u, Jv)$ is positive inner product	$(g \text{ flat?})$
g, J	$g(Ju, Jv) = g(u, v)$ (i.e., J is orthogonal)	$\omega(u, v) := g(Ju, v)$ is nondeg., skew-symm.	ω closed?
ω, g	polar decomposition $\rightsquigarrow$	J almost complex str.	J integrable?

An almost complex structure J on a manifold M is called **integrable** if and only if J is induced by a structure of complex manifold on M . In Lecture 15 we will discuss tests to check whether a given J is integrable.

13.3 First Consequences

Proposition 13.2 *Let (M, J) be an almost complex manifold. Suppose that J is compatible with two symplectic structures ω_0, ω_1 . Then ω_0, ω_1 are deformation-equivalent, that is, there exists a smooth family ω_t , $0 \leq t \leq 1$, of symplectic forms joining ω_0 to ω_1 .*

Proof. Take $\omega_t = (1 - t)\omega_0 + t\omega_1$, $0 \leq t \leq 1$. Then:

- ω_t is closed.
- ω_t is nondegenerate, since

$$g_t(\cdot, \cdot) := \omega_t(\cdot, J\cdot) = (1 - t)g_0(\cdot, \cdot) + tg_1(\cdot, \cdot)$$

is positive, hence nondegenerate.

□

Remark. The converse of this proposition is not true. A counterexample is provided by the following family in $\mathbb{R}^4$:

$$\omega_t = \cos \pi t \, dx_1 dy_1 + \sin \pi t \, dx_1 dy_2 + \sin \pi t \, dy_1 dx_2 + \cos \pi t \, dx_2 dy_2, \quad 0 \leq t \leq 1.$$

There is no J in $\mathbb{R}^4$ compatible with both ω_0 and ω_1 . ◇

Definition 13.3 *A submanifold X of an almost complex manifold (M, J) is an **almost complex submanifold** when $J(TX) \subseteq TX$, i.e., for all $x \in X, v \in T_x X$, we have $J_x v \in T_x X$.*

Proposition 13.4 *Let (M, ω) be a symplectic manifold equipped with a compatible almost complex structure J . Then any almost complex submanifold X of (M, J) is a symplectic submanifold of (M, ω) .*

Proof. Let $i : X \hookrightarrow M$ be the inclusion. Then $i^* \omega$ is a closed 2-form on X .
Nondegeneracy:

$$\omega_x(u, v) = g_x(J_x u, v), \quad \forall x \in X, \forall u, v \in T_x X.$$

Since $g_x|_{T_x X}$ is nondegenerate, so is $\omega_x|_{T_x X}$. Hence, $i^* \omega$ is symplectic. □

– When is an almost complex manifold a complex manifold? See Lecture 15.

Examples.

S^2 is an almost complex manifold and it is a complex manifold.

S^4 is not an almost complex manifold (proved by Ehresmann and Hopf).

S^6 is almost complex and it is not yet known whether it is complex.

S^8 and higher spheres are not almost complex manifolds.

◇

Homework 9: Contractibility

The following proof illustrates in a geometric way the relation between lagrangian subspaces, complex structures and inner products; from [11, p.45].

Let (V, Ω) be a symplectic vector space, and let $\mathcal{J}(V, \Omega)$ be the set of all complex structures on (V, Ω) which are Ω -compatible; i.e., given a complex structure J on V we have

$$J \in \mathcal{J}(V, \Omega) \iff G_J(\cdot, \cdot) := \Omega(\cdot, J\cdot) \text{ is a positive inner product on } V.$$

Fix a lagrangian subspace L_0 of (V, Ω) . Let $\mathcal{L}(V, \Omega, L_0)$ be the space of all lagrangian subspaces of (V, Ω) which intersect L_0 transversally. Let $\mathcal{G}(L_0)$ be the space of all positive inner products on L_0 .

Consider the map

$$\begin{aligned} \Psi : \mathcal{J}(V, \Omega) &\rightarrow \mathcal{L}(V, \Omega, L_0) \times \mathcal{G}(L_0) \\ J &\mapsto (JL_0, G_J|_{L_0}) \end{aligned}$$

Show that:

1. Ψ is well-defined.
2. Ψ is a bijection.

Hint: Given $(L, G) \in \mathcal{L}(V, \Omega, L_0) \times \mathcal{G}(L_0)$, define J in the following manner: For $v \in L_0$, $v^\perp = \{u \in L_0 \mid G(u, v) = 0\}$ is a $(n-1)$ -dimensional space of L_0 ; its symplectic orthogonal $(v^\perp)^\Omega$ is $(n+1)$ -dimensional. Check that $(v^\perp)^\Omega \cap L$ is 1-dimensional. Let Jv be the unique vector in this line such that $\Omega(v, Jv) = 1$. Check that, if we take v 's in some G -orthonormal basis of L_0 , this defines the required element of $\mathcal{J}(V, \Omega)$.

3. $\mathcal{L}(V, \Omega, L_0)$ is contractible.

Hint: Prove that $\mathcal{L}(V, \Omega, L_0)$ can be identified with the vector space of all symmetric $n \times n$ matrices. Notice that any n -dimensional subspace L of V which is transversal to L_0 is the graph of a linear map $S : JL_0 \rightarrow L_0$, i.e.,

$$\begin{aligned} L &= \text{span of } \{Je_1 + SJe_1, \dots, Je_n + SJe_n\} \\ \text{when } L_0 &= \text{span of } \{e_1, \dots, e_n\}. \end{aligned}$$

4. $\mathcal{G}(L_0)$ is contractible.

Hint: $\mathcal{G}(L_0)$ is even convex.

Conclude that $\mathcal{J}(V, \Omega)$ is contractible.

14 Dolbeault Theory

14.1 Splittings

Let (M, J) be an almost complex manifold. The complexified tangent bundle of M is the bundle

$$\begin{array}{c} TM \otimes \mathbb{C} \\ \downarrow \\ M \end{array}$$

with fiber $(TM \otimes \mathbb{C})_p = T_p M \otimes \mathbb{C}$ at $p \in M$. If

$$\begin{array}{ll} T_p M & \text{is a } 2n\text{-dimensional vector space over } \mathbb{R}, \text{ then} \\ T_p M \otimes \mathbb{C} & \text{is a } 2n\text{-dimensional vector space over } \mathbb{C}. \end{array}$$

We may extend J linearly to $TM \otimes \mathbb{C}$:

$$J(v \otimes c) = Jv \otimes c, \quad v \in TM, \quad c \in \mathbb{C}.$$

Since $J^2 = -\text{Id}$, on the complex vector space $(TM \otimes \mathbb{C})_p$, the linear map J_p has eigenvalues $\pm i$. Let

$$\begin{aligned} T_{1,0} &= \{v \in TM \otimes \mathbb{C} \mid Jv = +iv\} = (+i)\text{-eigenspace of } J \\ &= \{v \otimes 1 - Jv \otimes i \mid v \in TM\} \\ &= \mathbf{(J)\text{-holomorphic tangent vectors}}; \\ T_{0,1} &= \{v \in TM \otimes \mathbb{C} \mid Jv = -iv\} = (-i)\text{-eigenspace of } J \\ &= \{v \otimes 1 + Jv \otimes i \mid v \in TM\} \\ &= \mathbf{(J)\text{-anti-holomorphic tangent vectors}}. \end{aligned}$$

Since

$$\begin{array}{ccc} \pi_{1,0} : TM & \longrightarrow & T_{1,0} \\ v & \longmapsto & \frac{1}{2}(v \otimes 1 - Jv \otimes i) \end{array}$$

is a (real) bundle isomorphism such that $\pi_{1,0} \circ J = i\pi_{1,0}$, and

$$\begin{array}{ccc} \pi_{0,1} : TM & \longrightarrow & T_{0,1} \\ v & \longmapsto & \frac{1}{2}(v \otimes 1 + Jv \otimes i) \end{array}$$

is also a (real) bundle isomorphism such that $\pi_{0,1} \circ J = -i\pi_{0,1}$, we conclude that we have isomorphisms of complex vector bundles

$$(TM, J) \simeq T_{1,0} \simeq \overline{T_{0,1}},$$

where $\overline{T_{0,1}}$ denotes the complex conjugate bundle of $T_{0,1}$. Extending $\pi_{1,0}$ and $\pi_{0,1}$ to projections of $TM \otimes \mathbb{C}$, we obtain an isomorphism

$$(\pi_{1,0}, \pi_{0,1}) : TM \otimes \mathbb{C} \xrightarrow{\simeq} T_{1,0} \oplus T_{0,1}.$$

Similarly, the complexified cotangent bundle splits as

$$(\pi^{1,0}, \pi^{0,1}) : T^*M \otimes \mathbb{C} \xrightarrow{\cong} T^{1,0} \oplus T^{0,1}$$

where

$$\begin{aligned} T^{1,0} &= (T_{1,0})^* = \{\eta \in T^* \otimes \mathbb{C} \mid \eta(J\omega) = i\eta(\omega), \forall \omega \in TM \otimes \mathbb{C}\} \\ &= \{\xi \otimes 1 - (\xi \circ J) \otimes i \mid \xi \in T^*M\} \\ &= \text{complex-linear cotangent vectors} , \\ T^{0,1} &= (T_{0,1})^* = \{\eta \in T^* \otimes \mathbb{C} \mid \eta(J\omega) = -i\eta(\omega), \forall \omega \in TM \otimes \mathbb{C}\} \\ &= \{\xi \otimes 1 + (\xi \circ J) \otimes i \mid \xi \in T^*M\} \\ &= \text{complex-antilinear cotangent vectors} , \end{aligned}$$

and $\pi^{1,0}, \pi^{0,1}$ are the two natural projections

$$\begin{aligned} \pi^{1,0} : T^*M \otimes \mathbb{C} &\longrightarrow T^{1,0} \\ \eta &\longmapsto \eta^{1,0} := \frac{1}{2}(\eta - i\eta \circ J) ; \\ \pi^{0,1} : T^*M \otimes \mathbb{C} &\longrightarrow T^{0,1} \\ \eta &\longmapsto \eta^{0,1} := \frac{1}{2}(\eta + i\eta \circ J) . \end{aligned}$$

14.2 Forms of Type (ℓ, m)

For an almost complex manifold (M, J) , let

$$\begin{aligned} \Omega^k(M; \mathbb{C}) &:= \text{sections of } \Lambda^k(T^*M \otimes \mathbb{C}) \\ &= \text{complex-valued k-forms on } M, \text{ where} \\ \Lambda^k(T^*M \otimes \mathbb{C}) &:= \Lambda^k(T^{1,0} \oplus T^{0,1}) \\ &= \bigoplus_{\ell+m=k} \underbrace{(\Lambda^\ell T^{1,0}) \wedge (\Lambda^m T^{0,1})}_{\Lambda^{\ell,m}(\text{definition})} \\ &= \bigoplus_{\ell+m=k} \Lambda^{\ell,m} . \end{aligned}$$

In particular, $\Lambda^{1,0} = T^{1,0}$ and $\Lambda^{0,1} = T^{0,1}$.

Definition 14.1 *The differential forms of type (ℓ, m) on (M, J) are the sections of $\Lambda^{\ell,m}$:*

$$\Omega^{\ell,m} := \text{sections of } \Lambda^{\ell,m} .$$

Then

$$\Omega^k(M; \mathbb{C}) = \bigoplus_{\ell+m=k} \Omega^{\ell,m} .$$

Let $\pi^{\ell,m} : \Lambda^k(T^*M \otimes \mathbb{C}) \rightarrow \Lambda^{\ell,m}$ be the projection map, where $\ell + m = k$. The usual exterior derivative d composed with two of these projections induces differential operators ∂ and $\bar{\partial}$ on forms of type (ℓ, m) :

$$\begin{aligned} \partial &:= \pi^{\ell+1,m} \circ d : \Omega^{\ell,m}(M) \longrightarrow \Omega^{\ell+1,m}(M) \\ \bar{\partial} &:= \pi^{\ell,m+1} \circ d : \Omega^{\ell,m}(M) \longrightarrow \Omega^{\ell,m+1}(M) . \end{aligned}$$

If $\beta \in \Omega^{\ell,m}(M)$, with $k = \ell + m$, then $d\beta \in \Omega^{k+1}(M; \mathbb{C})$:

$$d\beta = \sum_{r+s=k+1} \pi^{r,s} d\beta = \pi^{k+1,0} d\beta + \cdots + \partial\beta + \bar{\partial}\beta + \cdots + \pi^{0,k+1} d\beta .$$

14.3 J -Holomorphic Functions

Let $f : M \rightarrow \mathbb{C}$ be a smooth complex-valued function on M . The exterior derivative d extends linearly to $\mathbb{C}$ -valued functions as $df = d(\operatorname{Re} f) + i d(\operatorname{Im} f)$.

Definition 14.2 A function f is **(J -)holomorphic at $x \in M$** if df_p is complex linear, i.e., $df_p \circ J = i df_p$. A function f is **(J -)holomorphic** if it is holomorphic at all $p \in M$.

Exercise. Show that

$$df_p \circ J = i df_p \iff df_p \in T_p^{1,0} \iff \pi_p^{0,1} df_p = 0 .$$

◇

Definition 14.3 A function f is **(J -)anti-holomorphic at $p \in M$** if df_p is complex antilinear, i.e., $df_p \circ J = -i df_p$.

Exercise.

$$\begin{aligned} df_p \circ J = -i df_p &\iff df_p \in T_p^{0,1} \iff \pi_p^{1,0} df_p = 0 \\ &\iff d\bar{f}_p \in T_p^{1,0} \iff \pi_p^{0,1} d\bar{f}_p = 0 \\ &\iff \bar{f} \text{ is holomorphic at } p \in M . \end{aligned}$$

◇

Definition 14.4 On functions, $d = \partial + \bar{\partial}$, where

$$\partial := \pi^{1,0} \circ d \quad \text{and} \quad \bar{\partial} := \pi^{0,1} \circ d .$$

Then

$$\begin{aligned} f \text{ is holomorphic} &\iff \bar{\partial} f = 0 , \\ f \text{ is anti-holomorphic} &\iff \partial f = 0 . \end{aligned}$$

– What about higher differential forms?

14.4 Dolbeault Cohomology

Suppose that $d = \partial + \bar{\partial}$, i.e.,

$$d\beta = \underbrace{\partial\beta}_{\in \Omega^{\ell+1,m}} + \underbrace{\bar{\partial}\beta}_{\in \Omega^{\ell,m+1}}, \quad \forall \beta \in \Omega^{\ell,m}.$$

Then, for any form $\beta \in \Omega^{\ell,m}$,

$$0 = d^2\beta = \underbrace{\partial^2\beta}_{\in \Omega^{\ell+2,m}} + \underbrace{\partial\bar{\partial}\beta + \bar{\partial}\partial\beta}_{\in \Omega^{\ell+1,m+1}} + \underbrace{\bar{\partial}^2\beta}_{\in \Omega^{\ell,m+2}},$$

which implies

$$\begin{cases} \bar{\partial}^2 = 0 \\ \partial\bar{\partial} + \bar{\partial}\partial = 0 \\ \partial^2 = 0 \end{cases}$$

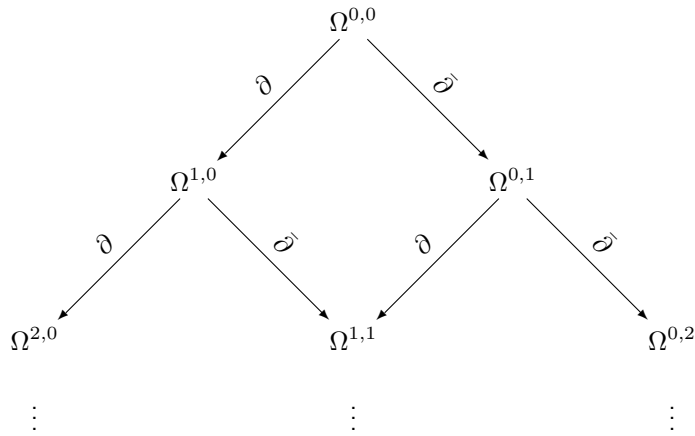
Since $\bar{\partial}^2 = 0$, the chain

$$0 \longrightarrow \Omega^{\ell,0} \xrightarrow{\bar{\partial}} \Omega^{\ell,1} \xrightarrow{\bar{\partial}} \Omega^{\ell,2} \xrightarrow{\bar{\partial}} \dots$$

is a differential complex; its cohomology groups

$$H_{\text{Dolbeault}}^{\ell,m}(M) := \frac{\ker \bar{\partial} : \Omega^{\ell,m} \longrightarrow \Omega^{\ell,m+1}}{\text{im } \bar{\partial} : \Omega^{\ell,m-1} \longrightarrow \Omega^{\ell,m}}$$

are called the **Dolbeault cohomology** groups.



– When is $d = \partial + \bar{\partial}$? See the next lecture.

Homework 10: Integrability

This set of problems is from [11, p.46-47].

1. Let (M, J) be an almost complex manifold. Its **Nijenhuis tensor** $\mathcal{N}$ is:

$$\mathcal{N}(v, w) := [Jv, Jw] - J[v, Jw] - J[Jv, w] - [v, w] ,$$

where v and w are vector fields on M , $[\cdot, \cdot]$ is the usual bracket

$$[v, w] \cdot f := v \cdot (w \cdot f) - w \cdot (v \cdot f) , \text{ for } f \in C^\infty(M) ,$$

and $v \cdot f = df(v)$.

- (a) Check that, if the map $v \mapsto [v, w]$ is complex linear (in the sense that it commutes with J), then $\mathcal{N} \equiv 0$.
- (b) Show that $\mathcal{N}$ is actually a tensor, that is: $\mathcal{N}(v, w)$ at $x \in M$ depends only on the values $v_x, w_x \in T_x M$ and not really on the vector fields v and w .
- (c) Compute $\mathcal{N}(v, Jv)$. Deduce that, if M is a surface, then $\mathcal{N} \equiv 0$.

A theorem of Newlander and Nirenberg [89] states that an almost complex manifold (M, J) is a complex (analytic) manifold if and only if $\mathcal{N} \equiv 0$. Combining (c) with the fact that any orientable surface is symplectic, we conclude that any orientable surface is a complex manifold, a result already known to Gauss.

2. Let $\mathcal{N}$ be as above. For any map $f : \mathbb{R}^{2n} \rightarrow \mathbb{C}$ and any vector field v on $\mathbb{R}^{2n}$, we have $v \cdot f = v \cdot (f_1 + if_2) = v \cdot f_1 + i v \cdot f_2$, so that $f \mapsto v \cdot f$ is a complex linear map.

- (a) Let $\mathbb{R}^{2n}$ be endowed with an almost complex structure J , and suppose that f is a **J -holomorphic function**, that is,

$$df \circ J = i df .$$

Show that $df(\mathcal{N}(v, w)) = 0$ for all vector fields v, w .

- (b) Suppose that there exist n J -holomorphic functions, $f_1, \dots, f_n$, on $\mathbb{R}^{2n}$, which are independent at some point p , i.e., the real and imaginary parts of $(df_1)_p, \dots, (df_n)_p$ form a basis of $T_p^* \mathbb{R}^{2n}$. Show that $\mathcal{N}$ vanishes identically at p .
- (c) Assume that M is a complex manifold and J is its complex structure. Show that $\mathcal{N}$ vanishes identically everywhere on M .

In general, an almost complex manifold has *no* J -holomorphic functions at all. On the other hand, it has *plenty* of **J -holomorphic curves**: maps $f : \mathbb{C} \rightarrow M$ such that $df \circ i = J \circ df$. J -holomorphic curves, also known as **pseudo-holomorphic curves**, provide a main tool in symplectic topology, as first realized by Gromov [49].

Part VI

Kähler Manifolds

Kähler geometry lies at the intersection of complex, riemannian and symplectic geometries, and plays a central role in all of these fields. We will start by reviewing complex manifolds. After describing the local normal form for Kähler manifolds (Lecture 16), we conclude with a summary of Hodge theory for compact Kähler manifolds (Lecture 17).

15 Complex Manifolds

15.1 Complex Charts

Definition 15.1 A complex manifold of (complex) dimension n is a set M with a complete complex atlas

$$\mathcal{A} = \{(\mathcal{U}_\alpha, \mathcal{V}_\alpha, \varphi_\alpha), \alpha \in \text{index set } I\}$$

where $M = \cup_\alpha \mathcal{U}_\alpha$, the $\mathcal{V}_\alpha$'s are open subsets of $\mathbb{C}^n$, and the maps $\varphi_\alpha : \mathcal{U}_\alpha \rightarrow \mathcal{V}_\alpha$ are such that the transition maps $\psi_{\alpha\beta}$ are biholomorphic as maps on open subsets of $\mathbb{C}^n$:

$$\begin{array}{ccc} & \mathcal{U}_\alpha \cap \mathcal{U}_\beta & \\ \varphi_\alpha \swarrow & & \searrow \varphi_\beta \\ \mathcal{V}_{\alpha\beta} & \xrightarrow{\psi_{\alpha\beta} = \varphi_\beta \circ \varphi_\alpha^{-1}} & \mathcal{V}_{\beta\alpha} \end{array}$$

where $\mathcal{V}_{\alpha\beta} = \varphi_\alpha(\mathcal{U}_\alpha \cap \mathcal{U}_\beta) \subseteq \mathbb{C}^n$ and $\mathcal{V}_{\beta\alpha} = \varphi_\beta(\mathcal{U}_\alpha \cap \mathcal{U}_\beta) \subseteq \mathbb{C}^n$. $\psi_{\alpha\beta}$ being biholomorphic means that $\psi_{\alpha\beta}$ is a bijection and that $\psi_{\alpha\beta}$ and $\psi_{\alpha\beta}^{-1}$ are both holomorphic.

Proposition 15.2 Any complex manifold has a canonical almost complex structure.

Proof.

1) Local definition of J :

Let $(\mathcal{U}, \mathcal{V}, \varphi : \mathcal{U} \rightarrow \mathcal{V})$ be a complex chart for a complex manifold M with $\varphi = (z_1, \dots, z_n)$ written in components relative to complex coordinates $z_j = x_j + iy_j$. At $p \in \mathcal{U}$

$$T_p M = \mathbb{R}\text{-span of } \left\{ \frac{\partial}{\partial x_j} \Big|_p, \frac{\partial}{\partial y_j} \Big|_p : j = 1, \dots, n \right\} .$$

Define J over $\mathcal{U}$ by

$$\begin{aligned} J_p \left(\frac{\partial}{\partial x_j} \Big|_p \right) &= \frac{\partial}{\partial y_j} \Big|_p \\ J_p \left(\frac{\partial}{\partial y_j} \Big|_p \right) &= -\frac{\partial}{\partial x_j} \Big|_p \end{aligned} \quad j = 1, \dots, n .$$

2) *This J is well-defined globally.*

If $(\mathcal{U}, \mathcal{V}, \varphi)$ and $(\mathcal{U}', \mathcal{V}', \varphi')$ are two charts, we need to show that $J = J'$ on their overlap.

On $\mathcal{U} \cap \mathcal{U}'$, $\psi \circ \varphi = \varphi'$. If $z_j = x_j + iy_j$ and $w_j = u_j + iv_j$ are coordinates on $\mathcal{U}$ and $\mathcal{U}'$, respectively, so that φ and φ' can be written in components $\varphi = (z_1, \dots, z_n)$, $\varphi' = (w_1, \dots, w_n)$, then $\psi(z_1, \dots, z_n) = (w_1, \dots, w_n)$. Taking the derivative of a composition

$$\begin{cases} \frac{\partial}{\partial x_k} = \sum_j \left(\frac{\partial u_j}{\partial x_k} \frac{\partial}{\partial u_j} + \frac{\partial v_j}{\partial x_k} \frac{\partial}{\partial v_j} \right) \\ \frac{\partial}{\partial y_k} = \sum_j \left(\frac{\partial u_j}{\partial y_k} \frac{\partial}{\partial u_j} + \frac{\partial v_j}{\partial y_k} \frac{\partial}{\partial v_j} \right) \end{cases}$$

Since ψ is biholomorphic, each component of ψ satisfies the **Cauchy-Riemann equations**:

$$\begin{cases} \frac{\partial u_j}{\partial x_k} = \frac{\partial v_j}{\partial y_k} \\ \frac{\partial u_j}{\partial y_k} = -\frac{\partial v_j}{\partial x_k} \end{cases} \quad j, k = 1, \dots, n .$$

These equations imply

$$\begin{aligned} J' \sum_j \left(\frac{\partial u_j}{\partial x_k} \frac{\partial}{\partial u_j} + \frac{\partial v_j}{\partial x_k} \frac{\partial}{\partial v_j} \right) &= \sum_j \left(\frac{\partial u_j}{\partial y_k} \frac{\partial}{\partial u_j} + \frac{\partial v_j}{\partial y_k} \frac{\partial}{\partial v_j} \right) \\ &= \sum_j \left(\underbrace{\frac{\partial u_j}{\partial x_k} \frac{\partial}{\partial v_j}}_{\frac{\partial v_j}{\partial y_k}} - \underbrace{\frac{\partial v_j}{\partial x_k} \frac{\partial}{\partial u_j}}_{-\frac{\partial u_j}{\partial y_k}} \right) \end{aligned}$$

which matches the equation

$$J \frac{\partial}{\partial x_k} = \frac{\partial}{\partial y_k} .$$

□

15.2 Forms on Complex Manifolds

Suppose that M is a complex manifold and J is its canonical almost complex structure. What does the splitting $\Omega^k(M; \mathbb{C}) = \oplus_{\ell+m=k} \Omega^{\ell,m}$ look like? ([22, 48, 66, 109] are good references for this material.)

Let $\mathcal{U} \subseteq M$ be a coordinate neighborhood with complex coordinates $z_1, \dots, z_n$, $z_j = x_j + iy_j$, and real coordinates $x_1, y_1, \dots, x_n, y_n$. At $p \in \mathcal{U}$,

$$\begin{aligned} T_p M &= \mathbb{R}\text{-span} \left\{ \frac{\partial}{\partial x_j} \Big|_p, \frac{\partial}{\partial y_j} \Big|_p \right\} \\ T_p M \otimes \mathbb{C} &= \mathbb{C}\text{-span} \left\{ \frac{\partial}{\partial x_j} \Big|_p, \frac{\partial}{\partial y_j} \Big|_p \right\} \\ &= \underbrace{\mathbb{C}\text{-span} \left\{ \frac{1}{2} \left(\frac{\partial}{\partial x_j} \Big|_p - i \frac{\partial}{\partial y_j} \Big|_p \right) \right\}}_{\substack{T_{1,0} = (+i)\text{-eigenspace of } J \\ J \left(\frac{\partial}{\partial x_j} - i \frac{\partial}{\partial y_j} \right) = i \left(\frac{\partial}{\partial x_j} - i \frac{\partial}{\partial y_j} \right)}} \oplus \underbrace{\mathbb{C}\text{-span} \left\{ \frac{1}{2} \left(\frac{\partial}{\partial x_j} \Big|_p + i \frac{\partial}{\partial y_j} \Big|_p \right) \right\}}_{\substack{T_{0,1} = (-i)\text{-eigenspace of } J \\ J \left(\frac{\partial}{\partial x_j} + i \frac{\partial}{\partial y_j} \right) = -i \left(\frac{\partial}{\partial x_j} + i \frac{\partial}{\partial y_j} \right)}} \end{aligned}$$

This can be written more concisely using:

Definition 15.3

$$\frac{\partial}{\partial z_j} := \frac{1}{2} \left(\frac{\partial}{\partial x_j} - i \frac{\partial}{\partial y_j} \right) \quad \text{and} \quad \frac{\partial}{\partial \bar{z}_j} := \frac{1}{2} \left(\frac{\partial}{\partial x_j} + i \frac{\partial}{\partial y_j} \right).$$

Hence,

$$(T_{1,0})_p = \mathbb{C}\text{-span} \left\{ \frac{\partial}{\partial z_j} \Big|_p : j = 1, \dots, n \right\}, \quad (T_{0,1})_p = \mathbb{C}\text{-span} \left\{ \frac{\partial}{\partial \bar{z}_j} \Big|_p : j = 1, \dots, n \right\}.$$

Similarly,

$$\begin{aligned} T^* M \otimes \mathbb{C} &= \mathbb{C}\text{-span} \{ dx_j, dy_j : j = 1, \dots, n \} \\ &= \underbrace{\mathbb{C}\text{-span} \{ dx_j + i dy_j : j = 1, \dots, n \}}_{T^{1,0}} \oplus \underbrace{\mathbb{C}\text{-span} \{ dx_j - i dy_j : j = 1, \dots, n \}}_{T^{0,1}} \\ &\quad (dx_j + i dy_j) \circ J = i(dx_j + i dy_j) \quad (dx_j - i dy_j) \circ J = -i(dx_j - i dy_j) \end{aligned}$$

Putting

$$dz_j = dx_j + i dy_j \quad \text{and} \quad d\bar{z}_j = dx_j - i dy_j,$$

we obtain

$$T^{1,0} = \mathbb{C}\text{-span} \{ dz_j : j = 1, \dots, n \}, \quad T^{0,1} = \mathbb{C}\text{-span} \{ d\bar{z}_j : j = 1, \dots, n \}.$$

On the coordinate neighborhood $\mathcal{U}$,

$$\begin{aligned} (1,0)\text{-forms} &= \left\{ \sum_j b_j dz_j \mid b_j \in C^\infty(\mathcal{U}; \mathbb{C}) \right\} \\ (0,1)\text{-forms} &= \left\{ \sum_j b_j d\bar{z}_j \mid b_j \in C^\infty(\mathcal{U}; \mathbb{C}) \right\} \\ (2,0)\text{-forms} &= \left\{ \sum_{j_1 < j_2} b_{j_1, j_2} dz_{j_1} \wedge dz_{j_2} \mid b_{j_1, j_2} \in C^\infty(\mathcal{U}; \mathbb{C}) \right\} \\ (1,1)\text{-forms} &= \left\{ \sum_{j_1, j_2} b_{j_1, j_2} dz_{j_1} \wedge d\bar{z}_{j_2} \mid b_{j_1, j_2} \in C^\infty(\mathcal{U}; \mathbb{C}) \right\} \\ (0,2)\text{-forms} &= \left\{ \sum_{j_1 < j_2} b_{j_1, j_2} d\bar{z}_{j_1} \wedge d\bar{z}_{j_2} \mid b_{j_1, j_2} \in C^\infty(\mathcal{U}; \mathbb{C}) \right\} \end{aligned}$$

If we use multi-index notation:

$$\begin{aligned} J &= (j_1, \dots, j_m) & 1 \leq j_1 < \dots < j_m \leq n \\ |J| &= m \\ dz_J &= dz_{j_1} \wedge dz_{j_2} \wedge \dots \wedge dz_{j_m} \end{aligned}$$

then

$$\Omega^{\ell, m} = (\ell, m)\text{-forms} = \left\{ \sum_{|J|=\ell, |K|=m} b_{J, K} dz_J \wedge d\bar{z}_K \mid b_{J, K} \in C^\infty(\mathcal{U}; \mathbb{C}) \right\} .$$

15.3 Differentials

On a coordinate neighborhood $\mathcal{U}$, a form $\beta \in \Omega^k(M; \mathbb{C})$ may be written as

$$\beta = \sum_{|J|+|K|=k} a_{J, K} dx_J \wedge dy_K, \quad \text{with } a_{J, K} \in C^\infty(\mathcal{U}; \mathbb{C}) .$$

We would like to know whether the following equality holds:

$$d\beta = \sum (\partial a_{J, K} + \bar{\partial} a_{J, K}) dx_J \wedge dy_K \stackrel{?}{=} (\partial + \bar{\partial}) \sum a_{J, K} dx_J \wedge dy_K .$$

If we use the identities

$$\begin{cases} dx_j + i dy_j = dz_j \\ dx_j - i dy_j = d\bar{z}_j \end{cases} \iff \begin{cases} dx_j = \frac{1}{2}(dz_j + d\bar{z}_j) \\ dy_j = \frac{1}{2i}(dz_j - d\bar{z}_j) \end{cases}$$

after substituting and reshuffling, we obtain

$$\begin{aligned}
\beta &= \sum_{|J|+|K|=k} b_{J,K} dz_J \wedge d\bar{z}_K \\
&= \sum_{\ell+m=k} \underbrace{\left(\sum_{|J|=\ell, |K|=m} b_{J,K} dz_J \wedge d\bar{z}_K \right)}_{\in \Omega^{\ell,m}}, \\
d\beta &= \sum_{\ell+m=k} \left(\sum_{|J|=\ell, |K|=m} db_{J,K} \wedge dz_J \wedge d\bar{z}_K \right) \\
&= \sum_{\ell+m=k} \sum_{|J|=\ell, |K|=m} (\partial b_{J,K} + \bar{\partial} b_{J,K}) \wedge dz_J \wedge d\bar{z}_K \\
&\quad \text{(because } d = \partial + \bar{\partial} \text{ on functions)} \\
&= \sum_{\ell+m=k} \left(\underbrace{\sum_{|J|=\ell, |K|=m} \partial b_{J,K} \wedge dz_J \wedge d\bar{z}_K}_{\in \Omega^{\ell+1,m}} + \underbrace{\sum_{|J|=\ell, |K|=m} \bar{\partial} b_{J,K} \wedge dz_J \wedge d\bar{z}_K}_{\in \Omega^{\ell,m+1}} \right) \\
&= \partial\beta + \bar{\partial}\beta.
\end{aligned}$$

Therefore, $d = \partial + \bar{\partial}$ on forms of any degree for a *complex* manifold.

Conclusion. If M is a complex manifold, then $d = \partial + \bar{\partial}$. (For an almost complex manifold this fails because there are no coordinate functions z_j to give a suitable basis of 1-forms.)

Remark. If $b \in C^\infty(\mathcal{U}; \mathbb{C})$, in terms of z and $\bar{z}$, we obtain the following formulas:

$$\begin{aligned}
db &= \sum_j \left(\frac{\partial b}{\partial x_j} dx_j + \frac{\partial b}{\partial y_j} dy_j \right) \\
&= \sum_j \left[\frac{1}{2} \left(\frac{\partial b}{\partial x_j} - i \frac{\partial b}{\partial y_j} \right) (dx_j + i dy_j) + \frac{1}{2} \left(\frac{\partial b}{\partial x_j} + i \frac{\partial b}{\partial y_j} \right) (dx_j - i dy_j) \right] \\
&= \sum_j \left(\frac{\partial b}{\partial z_j} dz_j + \frac{\partial b}{\partial \bar{z}_j} d\bar{z}_j \right).
\end{aligned}$$

Hence:

$$\begin{cases} \partial b &= \pi^{1,0} db = \sum_j \frac{\partial b}{\partial z_j} dz_j \\ \bar{\partial} b &= \pi^{0,1} db = \sum_j \frac{\partial b}{\partial \bar{z}_j} d\bar{z}_j \end{cases}$$

◇

In the case where $\beta \in \Omega^{\ell,m}$, we have

$$\begin{aligned}
 d\beta &= \partial\beta + \bar{\partial}\beta = (\ell+1, m)\text{-form} + (\ell, m+1)\text{-form} \\
 0 = d^2\beta &= (\ell+2, m)\text{-form} + (\ell+1, m+1)\text{-form} + (\ell, m+2)\text{-form} \\
 &= \underbrace{\partial^2\beta}_0 + \underbrace{(\partial\bar{\partial} + \bar{\partial}\partial)\beta}_0 + \underbrace{\bar{\partial}^2\beta}_0 .
 \end{aligned}$$

Hence, $\bar{\partial}^2 = 0$.

The Dolbeault theorem states that for complex manifolds

$$H_{\text{Dolbeault}}^{\ell,m}(M) = H^m(M; \mathcal{O}(\Omega^{(\ell,0)})) ,$$

where $\mathcal{O}(\Omega^{(\ell,0)})$ is the sheaf of forms of type $(\ell, 0)$ over M .

Theorem 15.4 (Newlander-Nirenberg, 1957 [89])

Let (M, J) be an almost complex manifold. Let $\mathcal{N}$ be the **Nijenhuis tensor** (defined in Homework 10). Then:

$$\begin{aligned}
 M \text{ is a complex manifold} &\iff J \text{ is integrable} \\
 &\iff \mathcal{N} \equiv 0 \\
 &\iff d = \partial + \bar{\partial} \\
 &\iff \bar{\partial}^2 = 0 \\
 &\iff \pi^{2,0}d|_{\Omega^{0,1}} = 0 .
 \end{aligned}$$

For the proof of this theorem, besides the original reference, see also [22, 30, 48, 66, 109]. Naturally most almost complex manifolds have $d \neq \partial + \bar{\partial}$.

Homework 11: Complex Projective Space

The complex projective space $\mathbb{CP}^n$ is the space of complex lines in $\mathbb{C}^{n+1}$:

$\mathbb{CP}^n$ is obtained from $\mathbb{C}^{n+1} \setminus \{0\}$ by making the identifications $(z_0, \dots, z_n) \sim (\lambda z_0, \dots, \lambda z_n)$ for all $\lambda \in \mathbb{C} \setminus \{0\}$. One denotes by $[z_0, \dots, z_n]$ the equivalence class of $(z_0, \dots, z_n)$, and calls $z_0, \dots, z_n$ the homogeneous coordinates of the point $p = [z_0, \dots, z_n]$. (The homogeneous coordinates are, of course, only determined up to multiplication by a non-zero complex number λ .)

Let $\mathcal{U}_i$ be the subset of $\mathbb{CP}^n$ consisting of all points $p = [z_0, \dots, z_n]$ for which $z_i \neq 0$. Let $\varphi_i : \mathcal{U}_i \rightarrow \mathbb{C}^n$ be the map

$$\varphi_i([z_0, \dots, z_n]) = \left(\frac{z_0}{z_i}, \dots, \frac{z_{i-1}}{z_i}, \frac{z_{i+1}}{z_i}, \dots, \frac{z_n}{z_i} \right).$$

1. Show that the collection

$$\{(\mathcal{U}_i, \mathbb{C}^n, \varphi_i), i = 0, \dots, n\}$$

is an atlas in the *complex* sense, i.e., the transition maps are biholomorphic. Conclude that $\mathbb{CP}^n$ is a complex manifold.

Hint: Work out the transition maps associated with $(\mathcal{U}_0, \mathbb{C}^n, \varphi_0)$ and $(\mathcal{U}_1, \mathbb{C}^n, \varphi_1)$. Show that the transition diagram has the form

$$\begin{array}{ccc} & \mathcal{U}_0 \cap \mathcal{U}_1 & \\ \varphi_0 \swarrow & & \searrow \varphi_1 \\ \mathcal{V}_{0,1} & \xrightarrow{\varphi_{0,1}} & \mathcal{V}_{1,0} \end{array}$$

where $\mathcal{V}_{0,1} = \mathcal{V}_{1,0} = \{(z_1, \dots, z_n) \in \mathbb{C}^n \mid z_1 \neq 0\}$ and

$$\varphi_{0,1}(z_1, \dots, z_n) = \left(\frac{1}{z_1}, \frac{z_2}{z_1}, \dots, \frac{z_n}{z_1} \right).$$

2. Show that the 1-dimensional complex manifold $\mathbb{CP}^1$ is diffeomorphic, as a real 2-dimensional manifold, to S^2 .

Hint: Stereographic projection.

16 Kähler Forms

16.1 Kähler Forms

Definition 16.1 A **Kähler manifold** is a symplectic manifold (M, ω) equipped with an integrable compatible almost complex structure. The symplectic form ω is then called a **Kähler form**.

It follows immediately from the previous definition that

$$\begin{aligned} (M, \omega) \text{ is Kähler} &\implies M \text{ is a complex manifold} \\ &\implies \begin{cases} \Omega^k(M; \mathbb{C}) = \oplus_{\ell+m=k} \Omega^{\ell, m} \\ d = \partial + \bar{\partial} \end{cases} \end{aligned}$$

where

$$\begin{aligned} \partial &= \pi^{\ell+1, m} \circ d : \Omega^{\ell, m} \rightarrow \Omega^{\ell+1, m} \\ \bar{\partial} &= \pi^{\ell, m+1} \circ d : \Omega^{\ell, m} \rightarrow \Omega^{\ell, m+1} . \end{aligned}$$

On a complex chart $(\mathcal{U}, z_1, \dots, z_n)$, $n = \dim_{\mathbb{C}} M$,

$$\Omega^{\ell, m} = \left\{ \sum_{|J|=\ell, |K|=m} b_{JK} dz_J \wedge d\bar{z}_K \mid b_{JK} \in C^\infty(\mathcal{U}; \mathbb{C}) \right\} ,$$

where

$$\begin{aligned} J &= (j_1, \dots, j_\ell) , & j_1 < \dots < j_\ell , & & dz_J &= dz_{j_1} \wedge \dots \wedge dz_{j_\ell} , \\ K &= (k_1, \dots, k_m) , & k_1 < \dots < k_m , & & d\bar{z}_K &= d\bar{z}_{k_1} \wedge \dots \wedge d\bar{z}_{k_m} . \end{aligned}$$

On the other hand,

$$(M, \omega) \text{ is Kähler} \implies \omega \text{ is a symplectic form .}$$

– Where does ω fit with respect to the above decomposition?

A Kähler form ω is

1. a 2-form,
2. compatible with the complex structure,
3. closed,
4. real-valued, and
5. nondegenerate.

These properties translate into:

$$1. \Omega^2(M; \mathbb{C}) = \Omega^{2,0} \oplus \Omega^{1,1} \oplus \Omega^{0,2}.$$

On a local complex chart $(\mathcal{U}, z_1, \dots, z_n)$,

$$\omega = \sum a_{jk} dz_j \wedge dz_k + \sum b_{jk} dz_j \wedge d\bar{z}_k + \sum c_{jk} d\bar{z}_j \wedge d\bar{z}_k$$

for some $a_{jk}, b_{jk}, c_{jk} \in C^\infty(\mathcal{U}; \mathbb{C})$.

$$2. J \text{ is a symplectomorphism, that is, } J^*\omega = \omega \text{ where } (J^*\omega)(u, v) := \omega(Ju, Jv).$$

$$J^*dz_j = dz_j \circ J = idz_j$$

$$J^*d\bar{z}_j = d\bar{z}_j \circ J = -id\bar{z}_j$$

$$J^*\omega = \sum (i \cdot i) a_{jk} dz_j \wedge dz_k + i(-i) \sum b_{jk} dz_j \wedge d\bar{z}_k + (-i)^2 \sum c_{jk} d\bar{z}_j \wedge d\bar{z}_k$$

$$J^*\omega = \omega \iff a_{jk} = 0 = c_{jk}, \text{ all } j, k \iff \omega \in \Omega^{1,1}.$$

$$3. 0 = d\omega = \underbrace{\partial\omega}_{(2,1)\text{-form}} + \underbrace{\bar{\partial}\omega}_{(1,2)\text{-form}} \implies \begin{cases} \partial\omega = 0 & \omega \text{ is } \partial\text{-closed} \\ \bar{\partial}\omega = 0 & \omega \text{ is } \bar{\partial}\text{-closed} \end{cases}$$

Hence, ω defines a Dolbeault $(1, 1)$ cohomology class,

$$[\omega] \in H_{\text{Dolbeault}}^{1,1}(M).$$

Putting $b_{jk} = \frac{i}{2} h_{jk}$,

$$\omega = \frac{i}{2} \sum_{j,k=1}^n h_{jk} dz_j \wedge d\bar{z}_k, \quad h_{jk} \in C^\infty(\mathcal{U}; \mathbb{C}).$$

$$4. \omega \text{ real-valued} \iff \omega = \bar{\omega}.$$

$$\bar{\omega} = -\frac{i}{2} \sum \overline{h_{jk}} d\bar{z}_j \wedge dz_k = \frac{i}{2} \sum \overline{h_{jk}} dz_k \wedge d\bar{z}_j = \frac{i}{2} \sum \overline{h_{kj}} dz_j \wedge d\bar{z}_k$$

$$\omega \text{ real} \iff h_{jk} = \overline{h_{kj}},$$

i.e., at every point $p \in \mathcal{U}$, the $n \times n$ matrix $(h_{jk}(p))$ is hermitian.

$$5. \text{nondegeneracy: } \omega^n = \underbrace{\omega \wedge \dots \wedge \omega}_n \neq 0.$$

Exercise. Check that

$$\omega^n = n! \left(\frac{i}{2}\right)^n \det(h_{jk}) dz_1 \wedge d\bar{z}_1 \wedge \dots \wedge dz_n \wedge d\bar{z}_n.$$

◇

Now

$$\omega \text{ nondegenerate} \iff \det_{\mathbb{C}}(h_{jk}) \neq 0,$$

i.e., at every $p \in M$, $(h_{jk}(p))$ is a nonsingular matrix.

2. Again the positivity condition: $\omega(v, Jv) > 0, \forall v \neq 0$.

Exercise. Show that $(h_{jk}(p))$ is positive-definite. $\diamond$

$$\omega \text{ positive} \iff (h_{jk}) \gg 0 ,$$

i.e., at each $p \in \mathcal{U}$, $(h_{jk}(p))$ is positive-definite.

Conclusion. Kähler forms are ∂ - and $\bar{\partial}$ -closed $(1,1)$ -forms, which are given on a local chart $(\mathcal{U}, z_1, \dots, z_n)$ by

$$\omega = \frac{i}{2} \sum_{j,k=1}^n h_{jk} dz_j \wedge d\bar{z}_k$$

where, at every point $p \in \mathcal{U}$, $(h_{jk}(p))$ is a positive-definite hermitian matrix.

16.2 An Application

Theorem 16.2 (Banyaga) *Let M be a compact complex manifold. Let ω_0 and ω_1 be Kähler forms on M . If $[\omega_0] = [\omega_1] \in H_{\text{deRham}}^2(M)$, then (M, ω_0) and (M, ω_1) are symplectomorphic.*

Proof. Any combination $\omega_t = (1-t)\omega_0 + t\omega_1$ is symplectic for $0 \leq t \leq 1$, because, on a complex chart $(\mathcal{U}, z_1, \dots, z_n)$, where $n = \dim_{\mathbb{C}} M$, we have

$$\omega_0 = \frac{i}{2} \sum h_{jk}^0 dz_j \wedge d\bar{z}_k$$

$$\omega_1 = \frac{i}{2} \sum h_{jk}^1 dz_j \wedge d\bar{z}_k$$

$$\omega_t = \frac{i}{2} \sum h_{jk}^t dz_j \wedge d\bar{z}_k, \quad \text{where } h_{jk}^t = (1-t)h_{jk}^0 + th_{jk}^1.$$

$$(h_{jk}^0) \gg 0, (h_{jk}^1) \gg 0 \implies (h_{jk}^t) \gg 0.$$

Apply the Moser theorem (Theorem 7.2). $\square$

16.3 Recipe to Obtain Kähler Forms

Definition 16.3 *Let M be a complex manifold. A function $\rho \in C^\infty(M; \mathbb{R})$ is strictly plurisubharmonic (s.p.s.h.) if, on each local complex chart $(\mathcal{U}, z_1, \dots, z_n)$, where $n = \dim_{\mathbb{C}} M$, the matrix $\left(\frac{\partial^2 \rho}{\partial z_j \partial \bar{z}_k}(p) \right)$ is positive-definite at all $p \in \mathcal{U}$.*

Proposition 16.4 *Let M be a complex manifold and let $\rho \in C^\infty(M; \mathbb{R})$ be s.p.s.h.. Then*

$$\omega = \frac{i}{2} \partial \bar{\partial} \rho \quad \text{is Kähler.}$$

A function ρ as in the previous proposition is called a (global) **Kähler potential**.

Proof. Simply observe that:

$$\begin{cases} \partial\omega &= \frac{i}{2} \overbrace{\partial^2}^0 \bar{\partial}\rho = 0 \\ \bar{\partial}\omega &= \frac{i}{2} \underbrace{\bar{\partial}\partial}_{-\partial\bar{\partial}} \bar{\partial}\rho = -\frac{i}{2} \partial \underbrace{\bar{\partial}^2}_0 \rho = 0 \end{cases}$$

$$d\omega = \partial\omega + \bar{\partial}\omega = 0 \implies \omega \text{ is closed .}$$

$$\bar{\omega} = -\frac{i}{2} \bar{\partial}\partial\rho = \frac{i}{2} \partial\bar{\partial}\rho = \omega \implies \omega \text{ is real .}$$

$$\omega \in \Omega^{1,1} \implies J^*\omega = \omega \implies \omega(\cdot, J\cdot) \text{ is symmetric .}$$

Exercise. Show that, for $f \in C^\infty(\mathcal{U}; \mathbb{C})$,

$$\partial f = \sum \frac{\partial f}{\partial z_j} dz_j \quad \text{and} \quad \bar{\partial} f = \sum \frac{\partial f}{\partial \bar{z}_j} d\bar{z}_j .$$

Since the right-hand sides are in $\Omega^{1,0}$ and $\Omega^{0,1}$, respectively, it suffices to show that the sum of the two expressions is df . $\diamond$

$$\omega = \frac{i}{2} \partial\bar{\partial}\rho = \frac{i}{2} \sum \frac{\partial}{\partial z_j} \left(\frac{\partial \rho}{\partial \bar{z}_k} \right) dz_j \wedge d\bar{z}_k = \frac{i}{2} \sum \underbrace{\left(\frac{\partial^2 \rho}{\partial z_j \partial \bar{z}_k} \right)}_{h_{jk}} dz_j \wedge d\bar{z}_k .$$

$$\rho \text{ is s.p.s.h} \implies (h_{jk}) \gg 0 \implies \omega(\cdot, J\cdot) \text{ is positive .}$$

In particular, ω is nondegenerate. $\square$

Example. Let $M = \mathbb{C}^n \simeq \mathbb{R}^{2n}$, with complex coordinates $(z_1, \dots, z_n)$ and corresponding real coordinates $(x_1, y_1, \dots, x_n, y_n)$ via $z_j = x_j + iy_j$. Let

$$\rho(x_1, y_1, \dots, x_n, y_n) = \sum_{j=1}^n (x_j^2 + y_j^2) = \sum |z_j|^2 = \sum z_j \bar{z}_j .$$

Then

$$\frac{\partial}{\partial z_j} \frac{\partial \rho}{\partial \bar{z}_k} = \frac{\partial}{\partial z_j} z_k = \delta_{jk} ,$$

so

$$(h_{jk}) = \left(\frac{\partial^2 \rho}{\partial z_j \partial \bar{z}_k} \right) = (\delta_{jk}) = \text{Id} \gg 0 \implies \rho \text{ is s.p.s.h. .}$$

The corresponding Kähler form

$$\begin{aligned}\omega &= \frac{i}{2} \partial \bar{\partial} \rho = \frac{i}{2} \sum_{j,k} \delta_{jk} dz_j \wedge d\bar{z}_k \\ &= \frac{i}{2} \sum_j dz_j \wedge d\bar{z}_j = \sum_j dx_j \wedge dy_j \quad \text{is the standard form .}\end{aligned}$$

◇

16.4 Local Canonical Form for Kähler Forms

There is a local converse to the previous construction of Kähler forms.

Theorem 16.5 *Let ω be a closed real-valued $(1,1)$ -form on a complex manifold M and let $p \in M$. Then there exist a neighborhood $\mathcal{U}$ of p and $\rho \in C^\infty(\mathcal{U}; \mathbb{R})$ such that, on $\mathcal{U}$,*

$$\omega = \frac{i}{2} \partial \bar{\partial} \rho .$$

The function ρ is then called a (local) **Kähler potential**.

The proof requires holomorphic versions of Poincaré's lemma, namely, the local triviality of Dolbeault groups:

$$\forall p \in M \quad \exists \text{ neighborhood } \mathcal{U} \text{ of } p \text{ such that } H_{\text{Dolbeault}}^{\ell,m}(\mathcal{U}) = 0, \quad m > 0 ,$$

and the local triviality of the holomorphic de Rham groups; see [48].

Proposition 16.6 *Let M be a complex manifold, $\rho \in C^\infty(M; \mathbb{R})$ s.p.s.h., X a complex submanifold, and $i : X \hookrightarrow M$ the inclusion map. Then $i^* \rho$ is s.p.s.h..*

Proof. Let $\dim_{\mathbb{C}} M = n$ and $\dim_{\mathbb{C}} X = n - m$. For $p \in X$, choose a chart $(\mathcal{U}, z_1, \dots, z_n)$ for M centered at p and adapted to X , i.e., $X \cap \mathcal{U}$ is given by $z_1 = \dots = z_m = 0$. In this chart, $i^* \rho = \rho(0, 0, \dots, 0, z_{m+1}, \dots, z_n)$.

$$i^* \rho \text{ is s.p.s.h.} \iff \left(\frac{\partial^2 \rho}{\partial z_{m+j} \partial \bar{z}_{m+k}}(0, \dots, 0, z_{m+1}, \dots, z_n) \right) \text{ is positive-definite ,}$$

which holds since this is a minor of $\left(\frac{\partial^2}{\partial z_j \partial \bar{z}_k}(0, \dots, 0, z_{m+1}, \dots, z_n) \right)$. □

Corollary 16.7 *Any complex submanifold of a Kähler manifold is also Kähler.*

Definition 16.8 *Let (M, ω) be a Kähler manifold, X a complex submanifold, and $i : X \hookrightarrow M$ the inclusion. Then $(X, i^* \omega)$ is called a **Kähler submanifold**.*

Example. Complex vector space $(\mathbb{C}^n, \omega_0)$ where $\omega_0 = \frac{i}{2} \sum dz_j \wedge d\bar{z}_j$ is Kähler. Every complex submanifold of $\mathbb{C}^n$ is Kähler. $\diamond$

Example. The complex projective space is

$$\mathbb{CP}^n = \mathbb{C}^{n+1} \setminus \{0\} / \sim$$

where

$$(z_0, \dots, z_n) \sim (\lambda z_0, \dots, \lambda z_n), \quad \lambda \in \mathbb{C} \setminus \{0\}.$$

The Fubini-Study form (see Homework 12) is Kähler. Therefore, every **non-singular projective variety** is a Kähler submanifold. Here we mean

$$\begin{aligned} \text{non-singular} &= \text{smooth} \\ \text{projective variety} &= \text{zero locus of a collection} \\ &\quad \text{of homogeneous polynomials.} \end{aligned}$$

$\diamond$

Homework 12: The Fubini-Study Structure

The purpose of the following exercises is to describe the natural Kähler structure on complex projective space, $\mathbb{CP}^n$.

1. Show that the function on $\mathbb{C}^n$

$$z \mapsto \log(|z|^2 + 1)$$

is strictly plurisubharmonic. Conclude that the 2-form

$$\omega_{\text{FS}} = \frac{i}{2} \partial \bar{\partial} \log(|z|^2 + 1)$$

is a Kähler form. (It is usually called the **Fubini-Study form** on $\mathbb{C}^n$.)

Hint: A hermitian $n \times n$ matrix H is positive definite if and only if $v^* H v > 0$ for any $v \in \mathbb{C}^n \setminus \{0\}$, where v^* is the transpose of the vector $\bar{v}$. To prove positive-definiteness, either apply the Cauchy-Schwarz inequality, or use the following symmetry observation: $U(n)$ acts transitively on S^{2n-1} and ω_{FS} is $U(n)$ -invariant, thus it suffices to show positive-definiteness along *one* direction.

2. Let $\mathcal{U}$ be the open subset of $\mathbb{C}^n$ defined by the inequality $z_1 \neq 0$, and let $\varphi : \mathcal{U} \rightarrow \mathcal{U}$ be the map

$$\varphi(z_1, \dots, z_n) = \frac{1}{z_1} (1, z_2, \dots, z_n) .$$

Show that φ maps $\mathcal{U}$ biholomorphically onto $\mathcal{U}$ and that

$$\varphi^* \log(|z|^2 + 1) = \log(|z|^2 + 1) + \log \frac{1}{|z_1|^2} . \quad (\star)$$

3. Notice that, for every point $p \in \mathcal{U}$, we can write the second term in $(\star)$ as the sum of a holomorphic and an anti-holomorphic function:

$$-\log z_1 - \log \bar{z}_1$$

on a neighborhood of p . Conclude that

$$\partial \bar{\partial} \varphi^* \log(|z|^2 + 1) = \partial \bar{\partial} \log(|z|^2 + 1)$$

and hence that $\varphi^* \omega_{\text{FS}} = \omega_{\text{FS}}$.

Hint: You need to use the fact that the pullback by a holomorphic map φ^* commutes with the ∂ and $\bar{\partial}$ operators. This is a consequence of φ^* preserving form type, $\varphi^*(\Omega^{p,q}) \subseteq \Omega^{p,q}$, which in turn is implied by $\varphi^* dz_j = \partial \varphi_j \subseteq \Omega^{1,0}$ and $\varphi^* d\bar{z}_j = \bar{\partial} \bar{\varphi}_j \subseteq \Omega^{0,1}$, where φ_j is the j th component of φ with respect to local complex coordinates $(z_1, \dots, z_n)$.

4. Recall that $\mathbb{CP}^n$ is obtained from $\mathbb{C}^{n+1} \setminus \{0\}$ by making the identifications $(z_0, \dots, z_n) \sim (\lambda z_0, \dots, \lambda z_n)$ for all $\lambda \in \mathbb{C} \setminus \{0\}$; $[z_0, \dots, z_n]$ is the equivalence class of $(z_0, \dots, z_n)$.

For $i = 0, 1, \dots, n$, let

$$\mathcal{U}_i = \{[z_0, \dots, z_n] \in \mathbb{CP}^n \mid z_i \neq 0\}$$

$$\varphi_i : \mathcal{U}_i \rightarrow \mathbb{C}^n \quad \varphi_i([z_0, \dots, z_n]) = \left(\frac{z_0}{z_i}, \dots, \frac{z_{i-1}}{z_i}, \frac{z_{i+1}}{z_i}, \dots, \frac{z_n}{z_i} \right).$$

Homework 11 showed that the collection $\{(\mathcal{U}_i, \mathbb{C}^n, \varphi_i), i = 0, \dots, n\}$ is a complex atlas (i.e., the transition maps are biholomorphic). In particular, it was shown that the transition diagram associated with $(\mathcal{U}_0, \mathbb{C}^n, \varphi_0)$ and $(\mathcal{U}_1, \mathbb{C}^n, \varphi_1)$ has the form

$$\begin{array}{ccc} & \mathcal{U}_0 \cap \mathcal{U}_1 & \\ \varphi_0 \swarrow & & \searrow \varphi_1 \\ \mathcal{V}_{0,1} & \xrightarrow{\varphi_{0,1}} & \mathcal{V}_{1,0} \end{array}$$

where $\mathcal{V}_{0,1} = \mathcal{V}_{1,0} = \{(z_1, \dots, z_n) \in \mathbb{C}^n \mid z_1 \neq 0\}$ and $\varphi_{0,1}(z_1, \dots, z_n) = (\frac{1}{z_1}, \frac{z_2}{z_1}, \dots, \frac{z_n}{z_1})$. Now the set $\mathcal{U}$ in exercise 2 is equal to the sets $\mathcal{V}_{0,1}$ and $\mathcal{V}_{1,0}$, and the map φ coincides with $\varphi_{0,1}$.

Show that $\varphi_0^* \omega_{\text{FS}}$ and $\varphi_1^* \omega_{\text{FS}}$ are identical on the overlap $\mathcal{U}_0 \cap \mathcal{U}_1$.

More generally, show that the Kähler forms $\varphi_i^* \omega_{\text{FS}}$ “glue together” to define a Kähler structure on $\mathbb{CP}^n$. This is called the **Fubini-Study form** on complex projective space.

5. Prove that for $\mathbb{CP}^1$ the Fubini-Study form on the chart $\mathcal{U}_0 = \{[z_0, z_1] \in \mathbb{CP}^1 \mid z_0 \neq 0\}$ is given by the formula

$$\omega_{\text{FS}} = \frac{dx \wedge dy}{(x^2 + y^2 + 1)^2}$$

where $\frac{z_1}{z_0} = z = x + iy$ is the usual coordinate on $\mathbb{C}$.

6. Compute the total area of $\mathbb{CP}^1 = \mathbb{C} \cup \{\infty\}$ with respect to ω_{FS} :

$$\int_{\mathbb{CP}^1} \omega_{\text{FS}} = \int_{\mathbb{R}^2} \frac{dx \wedge dy}{(x^2 + y^2 + 1)^2}.$$

7. Recall that $\mathbb{CP}^1 \simeq S^2$ as real 2-dimensional manifolds (Homework 11). On S^2 there is the standard area form ω_{std} induced by regarding it as the unit sphere in $\mathbb{R}^3$ (Homework 6): in cylindrical polar coordinates (θ, h) on S^2 away from its poles ($0 \leq \theta < 2\pi$ and $-1 \leq h \leq 1$), we have

$$\omega_{\text{std}} = d\theta \wedge dh.$$

Using stereographic projection, show that

$$\omega_{\text{FS}} = \frac{1}{4} \omega_{\text{std}}.$$

17 Compact Kähler Manifolds

17.1 Hodge Theory

Let M be a complex manifold. A Kähler form ω on M is a symplectic form which is compatible with the complex structure. Equivalently, a Kähler form ω is a ∂ - and $\bar{\partial}$ -closed form of type $(1,1)$ which, on a local chart $(\mathcal{U}, z_1, \dots, z_n)$ is given by $\omega = \frac{i}{2} \sum_{j,k=1}^n h_{j\bar{k}} dz_j \wedge d\bar{z}_k$, where, at each $x \in \mathcal{U}$, $(h_{j\bar{k}}(x))$ is a positive-definite hermitian matrix. The pair (M, ω) is then called a Kähler manifold.

Theorem 17.1 (Hodge) *On a compact Kähler manifold (M, ω) the Dolbeault cohomology groups satisfy*

$$H_{\text{deRham}}^k(M; \mathbb{C}) \simeq \bigoplus_{\ell+m=k} H_{\text{Dolbeault}}^{\ell, m}(M) \quad (\text{Hodge decomposition})$$

with $H^{\ell, m} \simeq \overline{H^{m, \ell}}$. In particular, the spaces $H_{\text{Dolbeault}}^{\ell, m}$ are finite-dimensional.

Hodge identified the spaces of cohomology classes of forms with spaces of actual forms, by picking *the* representative from each class which solves a certain differential equation, namely the *harmonic* representative.

(1) The **Hodge *-operator**.

Each tangent space $V = T_x M$ has a positive inner product $\langle \cdot, \cdot \rangle$, part of the riemannian metric in a compatible triple; we forget about the complex and symplectic structures until part (4).

Let $e_1, \dots, e_n$ be a positively oriented orthonormal basis of V .

The star operator is a linear operator $*$: $\Lambda(V) \rightarrow \Lambda(V)$ defined by

$$\begin{aligned} *(1) &= e_1 \wedge \dots \wedge e_n \\ *(e_1 \wedge \dots \wedge e_n) &= 1 \\ *(e_1 \wedge \dots \wedge e_k) &= e_{k+1} \wedge \dots \wedge e_n. \end{aligned}$$

We see that $*$: $\Lambda^k(V) \rightarrow \Lambda^{n-k}(V)$ and satisfies $** = (-1)^{k(n-k)}$.

(2) The **codifferential** and the **laplacian** are the operators defined by:

$$\begin{aligned} \delta &= (-1)^{n(k+1)+1} * d * : \Omega^k(M) \rightarrow \Omega^{k-1}(M) \\ \Delta &= d\delta + \delta d : \Omega^k(M) \rightarrow \Omega^k(M). \end{aligned}$$

The operator Δ is also called the **Laplace-Beltrami operator**.

Exercise. Check that, on $\Omega^0(\mathbb{R}^n) = C^\infty(\mathbb{R}^n)$, $\Delta = -\sum_{i=1}^n \frac{\partial^2}{\partial x_i^2}$. ◇

Exercise. Check that $\Delta * = * \Delta$. ◇

Suppose that M is compact. Define an inner product on forms by

$$\langle \cdot, \cdot \rangle : \Omega^k \times \Omega^k \rightarrow \mathbb{R}, \quad \langle \alpha, \beta \rangle = \int_M \alpha \wedge * \beta.$$

Exercise. Check that this is symmetric, positive-definite and satisfies $\langle d\alpha, \beta \rangle = \langle \alpha, \delta\beta \rangle$. $\diamond$

Therefore, δ is often denoted by d^* and called the adjoint of d . (When M is not compact, we still have a formal adjoint of d with respect to the nondegenerate bilinear pairing $\langle \cdot, \cdot \rangle : \Omega^k \times \Omega_c^k \rightarrow \mathbb{R}$ defined by a similar formula, where Ω_c^k is the space of compactly supported k -forms.) Also, Δ is self-adjoint:

Exercise. Check that $\langle \Delta\alpha, \beta \rangle = \langle \alpha, \Delta\beta \rangle$, and that $\langle \Delta\alpha, \alpha \rangle = |d\alpha|^2 + |\delta\alpha|^2 \geq 0$, where $|\cdot|$ is the norm with respect to this inner product. $\diamond$

(3) The **harmonic k -forms** are the elements of $\mathcal{H}^k := \{\alpha \in \Omega^k \mid \Delta\alpha = 0\}$.

Note that $\Delta\alpha = 0 \iff d\alpha = \delta\alpha = 0$. Since a harmonic form is d -closed, it defines a de Rham cohomology class.

Theorem 17.2 (Hodge) *Every de Rham cohomology class on a compact oriented riemannian manifold (M, g) possesses a unique harmonic representative, i.e.,*

$$\mathcal{H}^k \simeq H_{\text{deRham}}^k(M; \mathbb{R}) .$$

In particular, the spaces $\mathcal{H}^k$ are finite-dimensional. We also have the following orthogonal decomposition with respect to $\langle \cdot, \cdot \rangle$:

$$\begin{aligned} \Omega^k &\simeq \mathcal{H}^k \oplus \Delta(\Omega^k(M)) \\ &\simeq \mathcal{H}^k \oplus d\Omega^{k-1} \oplus \delta\Omega^{k+1} \end{aligned} \quad \textbf{(Hodge decomposition on forms)} .$$

The proof involves functional analysis, elliptic differential operators, pseudodifferential operators and Fourier analysis; see [48, 109].

So far, this was ordinary Hodge theory, considering only the metric and not the complex structure.

(4) **Complex Hodge Theory.**

When M is Kähler, the laplacian satisfies $\Delta = 2(\bar{\partial}\bar{\partial}^* + \bar{\partial}^*\bar{\partial})$ (see, for example, [48]) and preserves the decomposition according to type, $\Delta : \Omega^{\ell, m} \rightarrow \Omega^{\ell, m}$. Hence, harmonic forms are also bigraded

$$\mathcal{H}^k = \bigoplus_{\ell+m=k} \mathcal{H}^{\ell, m} .$$

Theorem 17.3 (Hodge) *Every Dolbeault cohomology class on a compact Kähler manifold (M, ω) possesses a unique harmonic representative, i.e.,*

$$\mathcal{H}^{\ell, m} \simeq H_{\text{Dolbeault}}^{\ell, m}(M)$$

and the spaces $\mathcal{H}^{\ell,m}$ are finite-dimensional. Hence, we have the following isomorphisms:

$$H_{\text{deRham}}^k(M) \simeq \mathcal{H}^k \simeq \bigoplus_{\ell+m=k} \mathcal{H}^{\ell,m} \simeq \bigoplus_{\ell+m=k} H_{\text{Dolbeault}}^{\ell,m}(M) .$$

For the proof, see for instance [48, 109].

17.2 Immediate Topological Consequences

Let $b^k(M) := \dim H_{\text{deRham}}^k(M)$ be the usual **Betti numbers** of M , and let $h^{\ell,m}(M) := \dim H_{\text{Dolbeault}}^{\ell,m}(M)$ be the so-called **Hodge numbers** of M .

$$\text{Hodge Theorem} \implies \begin{cases} b^k &= \sum_{\ell+m=k} h^{\ell,m} \\ h^{\ell,m} &= h^{m,\ell} \end{cases}$$

Some immediate topological consequences are:

1. On compact Kähler manifolds “the odd Betti numbers are even:”

$$b^{2k+1} = \sum_{\ell+m=2k+1} h^{\ell,m} = 2 \sum_{\ell=0}^k h^{\ell, (2k+1)-\ell} \quad \text{is even} .$$

2. On compact Kähler manifolds, $h^{1,0} = \frac{1}{2}b^1$ is a topological invariant.
3. On compact symplectic manifolds, “even Betti numbers are positive,” because ω^k is closed but not exact ($k = 0, 1, \dots, n$).

Proof. If $\omega^k = d\alpha$, by Stokes’ theorem, $\int_M \omega^n = \int_M d(\alpha \wedge \omega^{n-k}) = 0$. This cannot happen since ω^n is a volume form. $\square$

4. On compact Kähler manifolds, the $h^{\ell,\ell}$ are positive.

Claim. $0 \neq [\omega^\ell] \in H_{\text{Dolbeault}}^{\ell,\ell}(M)$.

Proof.

$$\begin{aligned} \omega \in \Omega^{1,1} &\implies \omega^\ell \in \Omega^{\ell,\ell} \\ d\omega = 0 &\implies 0 = d\omega^\ell = \underbrace{\partial\omega^\ell}_{(\ell+1,\ell)} + \underbrace{\bar{\partial}\omega^\ell}_{(\ell,\ell+1)} \\ &\implies \bar{\partial}\omega^\ell = 0 , \end{aligned}$$

so $[\omega^\ell]$ defines an element of $H_{\text{Dolbeault}}^{\ell,\ell}$. Why is ω^ℓ not $\bar{\partial}$ -exact? If $\omega^\ell = \bar{\partial}\beta$ for some $\beta \in \Omega^{\ell-1,\ell}$, then

$$\omega^n = \omega^\ell \wedge \omega^{n-\ell} = \bar{\partial}(\beta \wedge \omega^{n-\ell}) \implies 0 = [\omega^n] \in H_{\text{Dolbeault}}^{n,n}(M) .$$

But $[\omega^n] \neq 0$ in $H_{\text{deRham}}^{2n}(M; \mathbb{C}) \simeq H_{\text{Dolbeault}}^{n,n}(M)$ since it is a volume form. $\square$

There are other constraints on the Hodge numbers of compact Kähler manifolds, and ongoing research on how to compute $H_{\text{Dolbeault}}^{\ell,m}$. A popular picture to describe the relations is the **Hodge diamond**:

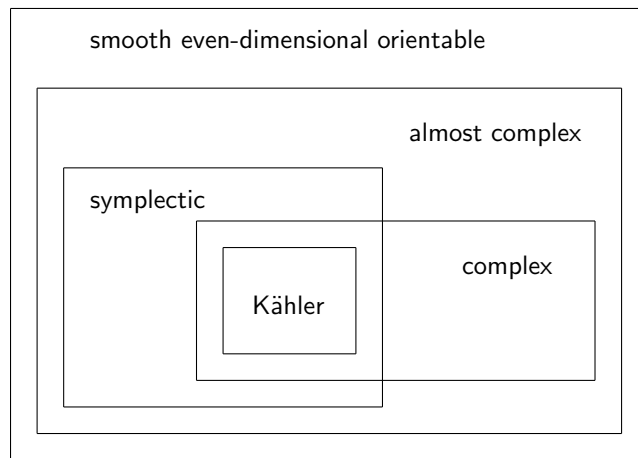
$$\begin{array}{ccccc}
 & & h^{n,n} & & \\
 & h^{n,n-1} & & h^{n-1,n} & \\
 h^{n,n-2} & & h^{n-1,n-1} & & h^{n-2,n} \\
 & & \vdots & & \\
 & h^{2,0} & & h^{1,1} & h^{0,2} \\
 & h^{1,0} & & h^{0,1} & \\
 & & h^{0,0} & &
 \end{array}$$

Complex conjugation gives symmetry with respect to the middle vertical, whereas the Hodge $*$ induces symmetry about the center of the diamond. The middle vertical axis is all non-zero. There are further symmetries induced by isomorphisms given by wedging with ω .

The **Hodge conjecture** relates $H_{\text{Dolbeault}}^{\ell,\ell}(M) \cap H^{2\ell}(M; \mathbb{Z})$ for projective manifolds M (i.e., submanifolds of complex projective space) to $\text{codim}_{\mathbb{C}} = \ell$ complex submanifolds of M .

17.3 Compact Examples and Counterexamples

$$\begin{array}{ccc}
 \text{symplectic} & \Leftarrow & \text{Kähler} \\
 \Downarrow & & \Downarrow \\
 \text{almost complex} & \Leftarrow & \text{complex}
 \end{array}$$



Is each of these regions nonempty? Can we even find representatives of each region which are simply connected or have any specified fundamental group?

- Not all smooth even-dimensional manifolds are almost complex. For example, S^4 , S^8 , S^{10} , etc., are not almost complex.
- If M is both symplectic and complex, is it necessarily Kähler?

No. For some time, it had been suspected that every compact symplectic manifold might have an underlying Kähler structure, or, at least, that a compact symplectic manifold might have to satisfy the Hodge relations on its Betti numbers [52]. The following example first demonstrated otherwise.

The Kodaira-Thurston example (Thurston, 1976 [101]):

Take $\mathbb{R}^4$ with $dx_1 \wedge dy_1 + dx_2 \wedge dy_2$, and Γ the discrete group generated by the following symplectomorphisms:

$$\begin{aligned} \gamma_1 &: (x_1, x_2, y_1, y_2) \mapsto (x_1, x_2 + 1, y_1, y_2) \\ \gamma_2 &: (x_1, x_2, y_1, y_2) \mapsto (x_1, x_2, y_1, y_2 + 1) \\ \gamma_3 &: (x_1, x_2, y_1, y_2) \mapsto (x_1 + 1, x_2, y_1, y_2) \\ \gamma_4 &: (x_1, x_2, y_1, y_2) \mapsto (x_1, x_2 + y_2, y_1 + 1, y_2) \end{aligned}$$

Then $M = \mathbb{R}^4/\Gamma$ is a flat 2-torus bundle over a 2-torus. Kodaira [70] had shown that M has a complex structure. However, $\pi_1(M) = \Gamma$, hence $H^1(\mathbb{R}^4/\Gamma; \mathbb{Z}) = \Gamma/[\Gamma, \Gamma]$ has rank 3, $b^1 = 3$ is *odd*, so M is *not* Kähler [101].

- Does any symplectic manifold admit some complex structure (not necessarily compatible)?

No.

(Fernández-Gotay-Gray, 1988 [37]): There are symplectic manifolds which do *not* admit any complex structure [37]. Their examples are circle bundles over circle bundles over a 2-torus.

$$\begin{array}{ccc} S^1 & \hookrightarrow & M \\ & & \downarrow \\ S^1 & \hookrightarrow & P \\ & & \downarrow \\ & & \mathbb{T}^2 \end{array} \quad \text{tower of circle fibrations}$$

- Given a complex structure on M , is there always a symplectic structure (not necessarily compatible)?

No.

The **Hopf surface** $S^1 \times S^3$ is not symplectic because $H^2(S^1 \times S^3) = 0$. But it is complex since $S^1 \times S^3 \simeq \mathbb{C}^2 \setminus \{0\}/\Gamma$ where $\Gamma = \{2^n \text{Id} \mid n \in \mathbb{Z}\}$ is a group of *complex* transformations, i.e., we factor $\mathbb{C}^2 \setminus \{0\}$ by the equivalence relation $(z_1, z_2) \sim (2z_1, 2z_2)$.

- Is any almost complex manifold either complex or symplectic?

No.

$\mathbb{CP}^2 \# \mathbb{CP}^2 \# \mathbb{CP}^2$ is almost complex (proved by a computation with characteristic classes), but is neither complex (since it does not fit Kodaira's classification of complex surfaces), nor symplectic (as shown by **Taubes** [97] in 1995 using Seiberg-Witten invariants).

- In 1993 **Gompf** [46] provided a construction that yields a compact symplectic 4-manifold with fundamental group equal to any given finitely-presented group. In particular, we can find simply connected examples. His construction can be adapted to produce *nonKähler* examples.

17.4 Main Kähler Manifolds

- **Compact Riemann surfaces**

As real manifolds, these are the 2-dimensional compact orientable manifolds classified by genus. An area form is a symplectic form. Any compatible almost complex structure is always integrable for dimension reasons (see Homework 10).

- **Stein manifolds**

Definition 17.4 A **Stein manifold** is a Kähler manifold (M, ω) which admits a global proper Kähler potential, i.e., $\omega = \frac{i}{2} \partial \bar{\partial} \rho$ for some proper function $\rho : M \rightarrow \mathbb{R}$.

Proper means that the preimage by ρ of a compact set is compact, i.e., " $\rho(p) \rightarrow \infty$ as $p \rightarrow \infty$."

Stein manifolds can be also characterized as the properly embedded analytic submanifolds of $\mathbb{C}^n$.

- **Complex tori**

Complex tori look like $M = \mathbb{C}^n / \mathbb{Z}^n$ where $\mathbb{Z}^n$ is a lattice in $\mathbb{C}^n$. The form $\omega = \sum dz_j \wedge d\bar{z}_j$ induced by the euclidean structure is Kähler.

- **Complex projective spaces**

The standard Kähler form on $\mathbb{CP}^n$ is the Fubini-Study form (see Homework 12). (In 1995, Taubes showed that $\mathbb{CP}^2$ has a unique symplectic structure up to symplectomorphism.)

- **Products of Kähler manifolds**
- **Complex submanifolds of Kähler manifolds**

Part VII

Hamiltonian Mechanics

The equations of motion in classical mechanics arise as solutions of variational problems. For a general mechanical system of n particles in $\mathbb{R}^3$, the physical path satisfies Newton's second law. On the other hand, the physical path minimizes the mean value of kinetic minus potential energy. This quantity is called the action. For a system with constraints, the physical path is the path which minimizes the action among all paths satisfying the constraint.

The Legendre transform (Lecture 20) gives the relation between the variational (Euler-Lagrange) and the symplectic (Hamilton-Jacobi) formulations of the equations of motion.

18 Hamiltonian Vector Fields

18.1 Hamiltonian and Symplectic Vector Fields

– What does a symplectic geometer do with a real function?...

Let (M, ω) be a symplectic manifold and let $H : M \rightarrow \mathbb{R}$ be a smooth function. Its differential dH is a 1-form. By nondegeneracy, there is a unique vector field X_H on M such that $\iota_{X_H} \omega = dH$. Integrate X_H . Supposing that M is compact, or at least that X_H is complete, let $\rho_t : M \rightarrow M$, $t \in \mathbb{R}$, be the one-parameter family of diffeomorphisms generated by X_H :

$$\begin{cases} \rho_0 = \text{id}_M \\ \frac{d\rho_t}{dt} \circ \rho_t^{-1} = X_H \end{cases} .$$

Claim. Each diffeomorphism ρ_t preserves ω , i.e., $\rho_t^* \omega = \omega$, $\forall t$.

Proof. We have $\frac{d}{dt} \rho_t^* \omega = \rho_t^* \mathcal{L}_{X_H} \omega = \rho_t^* (\underbrace{d\iota_{X_H} \omega}_{dH} + \underbrace{\iota_{X_H} d\omega}_0) = 0$. □

Therefore, every function on (M, ω) gives a family of symplectomorphisms. Notice how the proof involved both the *nondegeneracy* and the *closedness* of ω .

Definition 18.1 A vector field X_H as above is called the **hamiltonian vector field** with **hamiltonian function** H .

Example. The height function $H(\theta, h) = h$ on the sphere $(M, \omega) = (S^2, d\theta \wedge dh)$ has

$$\iota_{X_H} (d\theta \wedge dh) = dh \iff X_H = \frac{\partial}{\partial \theta} .$$

Thus, $\rho_t(\theta, h) = (\theta + t, h)$, which is rotation about the vertical axis; the height function H is preserved by this motion. $\diamond$

Exercise. Let X be a vector field on an abstract manifold W . There is a unique vector field $X_\sharp$ on the cotangent bundle T^*W , whose flow is the lift of the flow of X ; cf. Lecture 2. Let α be the tautological 1-form on T^*W and let $\omega = -d\alpha$ be the canonical symplectic form on T^*W . Show that $X_\sharp$ is a hamiltonian vector field with hamiltonian function $H := \iota_{X_\sharp}\alpha$. $\diamond$

Remark. If X_H is hamiltonian, then

$$\mathcal{L}_{X_H}H = \iota_{X_H}dH = \iota_{X_H}\iota_{X_H}\omega = 0 .$$

Therefore, hamiltonian vector fields preserve their hamiltonian functions, and each integral curve $\{\rho_t(x) \mid t \in \mathbb{R}\}$ of X_H must be contained in a level set of H :

$$H(x) = (\rho_t^*H)(x) = H(\rho_t(x)) , \quad \forall t .$$

$\diamond$

Definition 18.2 A vector field X on M preserving ω (i.e., such that $\mathcal{L}_X\omega = 0$) is called a **symplectic vector field**.

$$\begin{cases} X \text{ is symplectic} & \iff \iota_X\omega \text{ is closed} , \\ X \text{ is hamiltonian} & \iff \iota_X\omega \text{ is exact} . \end{cases}$$

Locally, on every contractible open set, every symplectic vector field is hamiltonian. If $H_{\text{deRham}}^1(M) = 0$, then globally every symplectic vector field is hamiltonian. In general, $H_{\text{deRham}}^1(M)$ measures the obstruction for symplectic vector fields to be hamiltonian.

Example. On the 2-torus $(M, \omega) = (\mathbb{T}^2, d\theta_1 \wedge d\theta_2)$, the vector fields $X_1 = \frac{\partial}{\partial \theta_1}$ and $X_2 = \frac{\partial}{\partial \theta_2}$ are symplectic but not hamiltonian. $\diamond$

To summarize, vector fields on a symplectic manifold (M, ω) which preserve ω are called **symplectic**. The following are equivalent:

- X is a symplectic vector field;
- the flow ρ_t of X preserves ω , i.e., $\rho_t^*\omega = \omega$, for all t ;
- $\mathcal{L}_X\omega = 0$;
- $\iota_X\omega$ is closed.

A **hamiltonian** vector field is a vector field X for which

- $\iota_X\omega$ is exact,

i.e., $\iota_X\omega = dH$ for some $H \in C^\infty(M)$. A primitive H of $\iota_X\omega$ is then called a **hamiltonian function** of X .

18.2 Classical Mechanics

Consider euclidean space $\mathbb{R}^{2n}$ with coordinates $(q_1, \dots, q_n, p_1, \dots, p_n)$ and $\omega_0 = \sum dq_j \wedge dp_j$. The curve $\rho_t = (q(t), p(t))$ is an integral curve for X_H exactly if

$$\begin{cases} \frac{dq_i}{dt}(t) = \frac{\partial H}{\partial p_i} \\ \frac{dp_i}{dt}(t) = -\frac{\partial H}{\partial q_i} \end{cases} \quad \text{(Hamilton equations)}$$

Indeed, let $X_H = \sum_{i=1}^n \left(\frac{\partial H}{\partial p_i} \frac{\partial}{\partial q_i} - \frac{\partial H}{\partial q_i} \frac{\partial}{\partial p_i} \right)$. Then,

$$\begin{aligned} \iota_{X_H} \omega &= \sum_{j=1}^n \iota_{X_H} (dq_j \wedge dp_j) = \sum_{j=1}^n [(\iota_{X_H} dq_j) \wedge dp_j - dq_j \wedge (\iota_{X_H} dp_j)] \\ &= \sum_{j=1}^n \left(\frac{\partial H}{\partial p_j} dp_j + \frac{\partial H}{\partial q_j} dq_j \right) = dH. \end{aligned}$$

Remark. The gradient vector field of H relative to the euclidean metric is

$$\nabla H := \sum_{i=1}^n \left(\frac{\partial H}{\partial q_i} \frac{\partial}{\partial q_i} + \frac{\partial H}{\partial p_i} \frac{\partial}{\partial p_i} \right).$$

If J is the standard (almost) complex structure so that $J(\frac{\partial}{\partial q_i}) = \frac{\partial}{\partial p_i}$ and $J(\frac{\partial}{\partial p_i}) = -\frac{\partial}{\partial q_i}$, we have $JX_H = \nabla H$. $\diamond$

The case where $n = 3$ has a simple physical illustration. Newton's second law states that a particle of mass m moving in **configuration space** $\mathbb{R}^3$ with coordinates $q = (q_1, q_2, q_3)$ under a potential $V(q)$ moves along a curve $q(t)$ such that

$$m \frac{d^2 q}{dt^2} = -\nabla V(q).$$

Introduce the **momenta** $p_i = m \frac{dq_i}{dt}$ for $i = 1, 2, 3$, and **energy** function $H(p, q) = \frac{1}{2m} |p|^2 + V(q)$. Let $\mathbb{R}^6 = T^*\mathbb{R}^3$ be the corresponding **phase space**, with coordinates $(q_1, q_2, q_3, p_1, p_2, p_3)$. Newton's second law in $\mathbb{R}^3$ is equivalent to the Hamilton equations in $\mathbb{R}^6$:

$$\begin{cases} \frac{dq_i}{dt} = \frac{1}{m} p_i = \frac{\partial H}{\partial p_i} \\ \frac{dp_i}{dt} = m \frac{d^2 q_i}{dt^2} = -\frac{\partial V}{\partial q_i} = -\frac{\partial H}{\partial q_i} \end{cases}.$$

The energy H is conserved by the physical motion.

18.3 Brackets

Vector fields are differential operators on functions: if X is a vector field and $f \in C^\infty(M)$, df being the corresponding 1-form, then

$$X \cdot f := df(X) = \mathcal{L}_X f .$$

Given two vector fields X, Y , there is a unique vector field W such that

$$\mathcal{L}_W f = \mathcal{L}_X(\mathcal{L}_Y f) - \mathcal{L}_Y(\mathcal{L}_X f) .$$

The vector field W is called the **Lie bracket** of the vector fields X and Y and denoted $W = [X, Y]$, since $\mathcal{L}_W = [\mathcal{L}_X, \mathcal{L}_Y]$ is the commutator.

Exercise. Check that, for any form α ,

$$\iota_{[X,Y]}\alpha = \mathcal{L}_X \iota_Y \alpha - \iota_Y \mathcal{L}_X \alpha = [\mathcal{L}_X, \iota_Y] \alpha .$$

Since each side is an anti-derivation with respect to the wedge product, it suffices to check this formula on local generators of the exterior algebra of forms, namely functions and exact 1-forms. $\diamond$

Proposition 18.3 *If X and Y are symplectic vector fields on a symplectic manifold (M, ω) , then $[X, Y]$ is hamiltonian with hamiltonian function $\omega(Y, X)$.*

Proof.

$$\begin{aligned} \iota_{[X,Y]}\omega &= \mathcal{L}_X \iota_Y \omega - \iota_Y \mathcal{L}_X \omega \\ &= d\iota_X \iota_Y \omega + \underbrace{\iota_X d\iota_Y \omega}_0 - \iota_Y \underbrace{d\iota_X \omega}_0 - \iota_Y \iota_X \underbrace{d\omega}_0 \\ &= d(\omega(Y, X)) . \end{aligned}$$

□

A (real) **Lie algebra** is a (real) vector space $\mathfrak{g}$ together with a **Lie bracket** $[\cdot, \cdot]$, i.e., a bilinear map $[\cdot, \cdot] : \mathfrak{g} \times \mathfrak{g} \rightarrow \mathfrak{g}$ satisfying:

$$(a) \quad [x, y] = -[y, x] , \quad \forall x, y \in \mathfrak{g} , \quad \textbf{(antisymmetry)}$$

$$(b) \quad [x, [y, z]] + [y, [z, x]] + [z, [x, y]] = 0 , \quad \forall x, y, z \in \mathfrak{g} . \quad \textbf{(Jacobi identity)}$$

Let

$$\begin{aligned} \chi(M) &= \{ \text{vector fields on } M \} \\ \chi^{\text{symp}}(M) &= \{ \text{symplectic vector fields on } M \} \\ \chi^{\text{ham}}(M) &= \{ \text{hamiltonian vector fields on } M \} . \end{aligned}$$

Corollary 18.4 *The inclusions $(\chi^{\text{ham}}(M), [\cdot, \cdot]) \subseteq (\chi^{\text{symp}}(M), [\cdot, \cdot]) \subseteq (\chi(M), [\cdot, \cdot])$ are inclusions of Lie algebras.*

Definition 18.5 *The **Poisson bracket** of two functions $f, g \in C^\infty(M; \mathbb{R})$ is*

$$\{f, g\} := \omega(X_f, X_g) .$$

We have $X_{\{f,g\}} = -[X_f, X_g]$ because $X_{\omega(X_f, X_g)} = [X_g, X_f]$.

Theorem 18.6 *The bracket $\{\cdot, \cdot\}$ satisfies the Jacobi identity, i.e.,*

$$\{f, \{g, h\}\} + \{g, \{h, f\}\} + \{h, \{f, g\}\} = 0 .$$

Proof. Exercise. □

Definition 18.7 *A Poisson algebra $(\mathcal{P}, \{\cdot, \cdot\})$ is a commutative associative algebra $\mathcal{P}$ with a Lie bracket $\{\cdot, \cdot\}$ satisfying the **Leibniz rule**:*

$$\{f, gh\} = \{f, g\}h + g\{f, h\} .$$

Exercise. Check that the Poisson bracket $\{\cdot, \cdot\}$ defined above satisfies the Leibniz rule. ◇

We conclude that, if (M, ω) is a symplectic manifold, then $(C^\infty(M), \{\cdot, \cdot\})$ is a Poisson algebra. Furthermore, we have a Lie algebra anti-homomorphism

$$\begin{array}{ccc} C^\infty(M) & \longrightarrow & \chi(M) \\ H & \longmapsto & X_H \\ \{\cdot, \cdot\} & \rightsquigarrow & -[\cdot, \cdot] . \end{array}$$

18.4 Integrable Systems

Definition 18.8 *A hamiltonian system is a triple (M, ω, H) , where (M, ω) is a symplectic manifold and $H \in C^\infty(M; \mathbb{R})$ is a function, called the **hamiltonian function**.*

Theorem 18.9 *We have $\{f, H\} = 0$ if and only if f is constant along integral curves of X_H .*

Proof. Let ρ_t be the flow of X_H . Then

$$\begin{aligned} \frac{d}{dt}(f \circ \rho_t) &= \rho_t^* \mathcal{L}_{X_H} f = \rho_t^* \iota_{X_H} df = \rho_t^* \iota_{X_H} \iota_{X_f} \omega \\ &= \rho_t^* \omega(X_f, X_H) = \rho_t^* \{f, H\} . \end{aligned}$$

□

A function f as in Theorem 18.9 is called an **integral of motion** (or a **first integral** or a **constant of motion**). In general, hamiltonian systems do not admit integrals of motion which are *independent* of the hamiltonian function. Functions $f_1, \dots, f_n$ on M are said to be **independent** if their differentials $(df_1)_p, \dots, (df_n)_p$ are linearly independent at all points p in some open dense subset of M . Loosely speaking, a hamiltonian system is (*completely*) *integrable* if it has as many commuting integrals of motion as possible. Commutativity is with respect to the Poisson

bracket. Notice that, if $f_1, \dots, f_n$ are commuting integrals of motion for a hamiltonian system (M, ω, H) , then, at each $p \in M$, their hamiltonian vector fields generate an isotropic subspace of $T_p M$:

$$\omega(X_{f_i}, X_{f_j}) = \{f_i, f_j\} = 0.$$

If $f_1, \dots, f_n$ are independent, then, by symplectic linear algebra, n can be at most half the dimension of M .

Definition 18.10 *A hamiltonian system (M, ω, H) is **(completely) integrable** if it possesses $n = \frac{1}{2} \dim M$ independent integrals of motion, $f_1 = H, f_2, \dots, f_n$, which are pairwise in involution with respect to the Poisson bracket, i.e., $\{f_i, f_j\} = 0$, for all i, j .*

Example. The simple pendulum (Homework 13) and the harmonic oscillator are trivially integrable systems – any 2-dimensional hamiltonian system (where the set of non-fixed points is dense) is integrable. $\diamond$

Example. A hamiltonian system (M, ω, H) where M is 4-dimensional is integrable if there is an integral of motion independent of H (the commutativity condition is automatically satisfied). Homework 18 shows that the spherical pendulum is integrable. $\diamond$

For sophisticated examples of integrable systems, see [10, 62].

Let (M, ω, H) be an integrable system of dimension $2n$ with integrals of motion $f_1 = H, f_2, \dots, f_n$. Let $c \in \mathbb{R}^n$ be a regular value of $f := (f_1, \dots, f_n)$. The corresponding level set, $f^{-1}(c)$, is a lagrangian submanifold, because it is n -dimensional and its tangent bundle is isotropic.

Lemma 18.11 *If the hamiltonian vector fields $X_{f_1}, \dots, X_{f_n}$ are complete on the level $f^{-1}(c)$, then the connected components of $f^{-1}(c)$ are homogeneous spaces for $\mathbb{R}^n$, i.e., are of the form $\mathbb{R}^{n-k} \times \mathbb{T}^k$ for some k , $0 \leq k \leq n$, where $\mathbb{T}^k$ is a k -dimensional torus.*

Proof. Exercise (just follow the flows to obtain coordinates). $\square$

Any compact component of $f^{-1}(c)$ must hence be a torus. These components, when they exist, are called **Liouville tori**. (The easiest way to ensure that compact components exist is to have one of the f_i 's proper.)

Theorem 18.12 (Arnold-Liouville [3]) *Let (M, ω, H) be an integrable system of dimension $2n$ with integrals of motion $f_1 = H, f_2, \dots, f_n$. Let $c \in \mathbb{R}^n$ be a regular value of $f := (f_1, \dots, f_n)$. The corresponding level $f^{-1}(c)$ is a lagrangian submanifold of M .*

- (a) *If the flows of $X_{f_1}, \dots, X_{f_n}$ starting at a point $p \in f^{-1}(c)$ are complete, then the connected component of $f^{-1}(c)$ containing p is a homogeneous space for $\mathbb{R}^n$. With respect to this affine structure, that component has coordinates $\varphi_1, \dots, \varphi_n$, known as **angle coordinates**, in which the flows of the vector fields $X_{f_1}, \dots, X_{f_n}$ are linear.*

- (b) *There are coordinates $\psi_1, \dots, \psi_n$, known as **action coordinates**, complementary to the angle coordinates such that the ψ_i 's are integrals of motion and $\varphi_1, \dots, \varphi_n, \psi_1, \dots, \psi_n$ form a Darboux chart.*

Therefore, the dynamics of an integrable system is extremely simple and the system has an explicit solution in action-angle coordinates. The proof of part (a) – the easy part – of the Arnold-Liouville theorem is sketched above. For the proof of part (b), see [3, 28].

Geometrically, regular levels being lagrangian submanifolds implies that, in a neighborhood of a regular value, the map $f : M \rightarrow \mathbb{R}^n$ collecting the given integrals of motion is a **lagrangian fibration**, i.e., it is locally trivial and its fibers are lagrangian submanifolds. Part (a) of the Arnold-Liouville theorem states that there are coordinates along the fibers, the angle coordinates φ_i ,¹¹ in which the flows of $X_{f_1}, \dots, X_{f_n}$ are linear. Part (b) of the theorem guarantees the existence of coordinates on $\mathbb{R}^n$, the action coordinates ψ_i , which (Poisson) commute among themselves and satisfy $\{\varphi_i, \psi_j\} = \delta_{ij}$ with respect to the angle coordinates. Notice that, in general, the action coordinates are not the given integrals of motion because $\varphi_1, \dots, \varphi_n, f_1, \dots, f_n$ do not form a Darboux chart.

¹¹The name “angle coordinates” is used even if the fibers are not tori.

Homework 13: Simple Pendulum

This problem is adapted from [53].

The **simple pendulum** is a mechanical system consisting of a massless rigid rod of length l , fixed at one end, whereas the other end has a plumb bob of mass m , which may oscillate in the vertical plane. Assume that the force of gravity is constant pointing vertically downwards, and that this is the only external force acting on this system.

- (a) Let θ be the oriented angle between the rod (regarded as a line segment) and the vertical direction. Let ξ be the coordinate along the fibers of T^*S^1 induced by the standard angle coordinate on S^1 . Show that the function $H : T^*S^1 \rightarrow \mathbb{R}$ given by

$$H(\theta, \xi) = \underbrace{\frac{\xi^2}{2ml^2}}_K + \underbrace{ml(1 - \cos \theta)}_V,$$

is an appropriate hamiltonian function to describe the simple pendulum. More precisely, check that gravity corresponds to the potential energy $V(\theta) = ml(1 - \cos \theta)$ (we omit universal constants), and that the kinetic energy is given by $K(\theta, \xi) = \frac{1}{2ml^2}\xi^2$.

- (b) For simplicity assume that $m = l = 1$.

Plot the level curves of H in the (θ, ξ) plane.

Show that there exists a number c such that for $0 < h < c$ the level curve $H = h$ is a disjoint union of closed curves. Show that the projection of each of these curves onto the θ -axis is an interval of length less than π .

Show that neither of these assertions is true if $h > c$.

What types of motion are described by these two types of curves?

What about the case $H = c$?

- (c) Compute the critical points of the function H . Show that, modulo 2π in θ , there are exactly two critical points: a critical point s where H vanishes, and a critical point u where H equals c . These points are called the **stable** and **unstable** points of H , respectively. Justify this terminology, i.e., show that a trajectory of the hamiltonian vector field of H whose initial point is close to s stays close to s forever, and show that this is not the case for u . What is happening physically?

19 Variational Principles

19.1 Equations of Motion

The equations of motion in classical mechanics arise as solutions of variational problems:

A general mechanical system possesses both kinetic and potential energy. The quantity that is minimized is the mean value of kinetic minus potential energy.

Example. Suppose that a point-particle of mass m moves in $\mathbb{R}^3$ under a force field F ; let $x(t)$, $a \leq t \leq b$, be its path of motion in $\mathbb{R}^3$. Newton's second law states that

$$m \frac{d^2 x}{dt^2}(t) = F(x(t)) .$$

Define the **work** of a path $\gamma : [a, b] \longrightarrow \mathbb{R}^3$, with $\gamma(a) = p$ and $\gamma(b) = q$, to be

$$W_\gamma = \int_a^b F(\gamma(t)) \cdot \frac{d\gamma}{dt}(t) dt .$$

Suppose that F is **conservative**, i.e., W_γ depends only on p and q . Then we can define the **potential energy** $V : \mathbb{R}^3 \longrightarrow \mathbb{R}$ of the system as

$$V(q) := W_\gamma$$

where γ is a path joining a fixed base point $p_0 \in \mathbb{R}^3$ (the "origin") to q . Newton's second law can now be written

$$m \frac{d^2 x}{dt^2}(t) = - \frac{\partial V}{\partial x}(x(t)) .$$

In the previous lecture we saw that

$$\begin{array}{ccc} \text{Newton's second law} & \Longleftrightarrow & \text{Hamilton equations} \\ \text{in } \mathbb{R}^3 = \{(q_1, q_2, q_3)\} & & \text{in } T^*\mathbb{R}^3 = \{(q_1, q_2, q_3, p_1, p_2, p_3)\} \end{array}$$

where $p_i = m \frac{dq_i}{dt}$ and the hamiltonian is $H(p, q) = \frac{1}{2m}|p|^2 + V(q)$. Hence, solving Newton's second law in **configuration space** $\mathbb{R}^3$ is equivalent to solving in **phase space** $T^*\mathbb{R}^3$ for the integral curve of the hamiltonian vector field with hamiltonian function H . $\diamond$

Example. The motion of earth about the sun, both regarded as point-masses and assuming that the sun to be stationary at the origin, obeys the **inverse square law**

$$m \frac{d^2 x}{dt^2} = - \frac{\partial V}{\partial x} ,$$

where $x(t)$ is the position of earth at time t , and $V(x) = \frac{\text{const.}}{|x|}$ is the **gravitational potential**. $\diamond$

19.2 Principle of Least Action

When we need to deal with systems with constraints, such as the simple pendulum, or two point masses attached by a rigid rod, or a rigid body, the language of variational principles becomes more appropriate than the explicit analogues of Newton's second laws. Variational principles are due mostly to D'Alembert, Maupertius, Euler and Lagrange.

Example. (The n -particle system.) Suppose that we have n point-particles of masses $m_1, \dots, m_n$ moving in 3-space. At any time t , the configuration of this system is described by a vector in configuration space $\mathbb{R}^{3n}$

$$x = (x_1, \dots, x_n) \in \mathbb{R}^{3n}$$

with $x_i \in \mathbb{R}^3$ describing the position of the i th particle. If $V \in C^\infty(\mathbb{R}^{3n})$ is the potential energy, then a path of motion $x(t)$, $a \leq t \leq b$, satisfies

$$m_i \frac{d^2 x_i}{dt^2}(t) = - \frac{\partial V}{\partial x_i}(x_1(t), \dots, x_n(t)) .$$

Consider this path in configuration space as a map $\gamma_0 : [a, b] \rightarrow \mathbb{R}^{3n}$ with $\gamma_0(a) = p$ and $\gamma_0(b) = q$, and let

$$\mathcal{P} = \{\gamma : [a, b] \rightarrow \mathbb{R}^{3n} \mid \gamma(a) = p \text{ and } \gamma(b) = q\}$$

be the set of all paths going from p to q over time $t \in [a, b]$. ◇

Definition 19.1 *The action of a path $\gamma \in \mathcal{P}$ is*

$$\mathcal{A}_\gamma := \int_a^b \left(\sum_{i=1}^n \frac{m_i}{2} \left| \frac{d\gamma_i}{dt}(t) \right|^2 - V(\gamma(t)) \right) dt .$$

Principle of least action.

The physical path γ_0 is the path for which $\mathcal{A}_\gamma$ is minimal.

Newton's second law for a constrained system.

Suppose that the n point-masses are restricted to move on a submanifold M of $\mathbb{R}^{3n}$ called the **constraint set**. We can now single out the actual physical path $\gamma_0 : [a, b] \rightarrow M$, with $\gamma_0(a) = p$ and $\gamma_0(b) = q$, as being "the" path which minimizes $\mathcal{A}_\gamma$ among all those hypothetical paths $\gamma : [a, b] \rightarrow \mathbb{R}^{3n}$ with $\gamma(a) = p$, $\gamma(b) = q$ and satisfying the rigid constraints $\gamma(t) \in M$ for all t .

19.3 Variational Problems

Let M be an n -dimensional manifold. Its tangent bundle TM is a $2n$ -dimensional manifold. Let $F : TM \rightarrow \mathbb{R}$ be a smooth function.

If $\gamma : [a, b] \rightarrow M$ is a smooth curve on M , define the **lift of γ to TM** to be the smooth curve on TM given by

$$\begin{aligned} \tilde{\gamma} : [a, b] &\longrightarrow TM \\ t &\longmapsto \left(\gamma(t), \frac{d\gamma}{dt}(t) \right) . \end{aligned}$$

The **action** of γ is

$$\mathcal{A}_\gamma := \int_a^b (\tilde{\gamma}^* F)(t) dt = \int_a^b F \left(\gamma(t), \frac{d\gamma}{dt}(t) \right) dt .$$

For fixed $p, q \in M$, let

$$\mathcal{P}(a, b, p, q) := \{ \gamma : [a, b] \longrightarrow M \mid \gamma(a) = p, \gamma(b) = q \} .$$

Problem.

Find, among all $\gamma \in \mathcal{P}(a, b, p, q)$, the curve γ_0 which “minimizes” $\mathcal{A}_\gamma$.

First observe that minimizing curves are always locally minimizing:

Lemma 19.2 *Suppose that $\gamma_0 : [a, b] \rightarrow M$ is minimizing. Let $[a_1, b_1]$ be a subinterval of $[a, b]$ and let $p_1 = \gamma_0(a_1)$, $q_1 = \gamma_0(b_1)$. Then $\gamma_0|_{[a_1, b_1]}$ is minimizing among the curves in $\mathcal{P}(a_1, b_1, p_1, q_1)$.*

Proof. Exercise:

Argue by contradiction. Suppose that there were $\gamma_1 \in \mathcal{P}(a_1, b_1, p_1, q_1)$ for which $\mathcal{A}_{\gamma_1} < \mathcal{A}_{\gamma_0|_{[a_1, b_1]}}$. Consider a broken path obtained from γ_0 by replacing the segment $\gamma_0|_{[a_1, b_1]}$ by γ_1 . Construct a smooth curve $\gamma_2 \in \mathcal{P}(a, b, p, q)$ for which $\mathcal{A}_{\gamma_2} < \mathcal{A}_{\gamma_0}$ by rounding off the corners of the broken path. $\square$

We will now assume that p, q and γ_0 lie in a coordinate neighborhood $(\mathcal{U}, x_1, \dots, x_n)$. On $T\mathcal{U}$ we have coordinates $(x_1, \dots, x_n, v_1, \dots, v_n)$ associated with a trivialization of $T\mathcal{U}$ by $\frac{\partial}{\partial x_1}, \dots, \frac{\partial}{\partial x_n}$. Using this trivialization, the curve

$$\gamma : [a, b] \longrightarrow \mathcal{U} , \quad \gamma(t) = (\gamma_1(t), \dots, \gamma_n(t))$$

lifts to

$$\tilde{\gamma} : [a, b] \longrightarrow T\mathcal{U} , \quad \tilde{\gamma}(t) = \left(\gamma_1(t), \dots, \gamma_n(t), \frac{d\gamma_1}{dt}(t), \dots, \frac{d\gamma_n}{dt}(t) \right) .$$

Necessary condition for $\gamma_0 \in \mathcal{P}(a, b, p, q)$ to minimize the action.

Let $c_1, \dots, c_n \in C^\infty([a, b])$ be such that $c_i(a) = c_i(b) = 0$. Let $\gamma_\varepsilon : [a, b] \longrightarrow \mathcal{U}$ be the curve

$$\gamma_\varepsilon(t) = (\gamma_1(t) + \varepsilon c_1(t), \dots, \gamma_n(t) + \varepsilon c_n(t)) .$$

For ε small, γ_ε is well-defined and in $\mathcal{P}(a, b, p, q)$.

Let $\mathcal{A}_\varepsilon = \mathcal{A}_{\gamma_\varepsilon} = \int_a^b F\left(\gamma_\varepsilon(t), \frac{d\gamma_\varepsilon}{dt}(t)\right) dt$. If γ_0 minimizes $\mathcal{A}$, then

$$\frac{d\mathcal{A}_\varepsilon}{d\varepsilon}(0) = 0 .$$

$$\begin{aligned} \frac{d\mathcal{A}_\varepsilon}{d\varepsilon}(0) &= \int_a^b \sum_i \left[\frac{\partial F}{\partial x_i} \left(\gamma_0(t), \frac{d\gamma_0}{dt}(t) \right) c_i(t) + \frac{\partial F}{\partial v_i} \left(\gamma_0(t), \frac{d\gamma_0}{dt}(t) \right) \frac{dc_i}{dt}(t) \right] dt \\ &= \int_a^b \sum_i \left[\frac{\partial F}{\partial x_i}(\dots) - \frac{d}{dt} \frac{\partial F}{\partial v_i}(\dots) \right] c_i(t) dt = 0 \end{aligned}$$

where the first equality follows from the Leibniz rule and the second equality follows from integration by parts. Since this is true for all c_i 's satisfying the boundary conditions $c_i(a) = c_i(b) = 0$, we conclude that

$$\frac{\partial F}{\partial x_i} \left(\gamma_0(t), \frac{d\gamma_0}{dt}(t) \right) = \frac{d}{dt} \frac{\partial F}{\partial v_i} \left(\gamma_0(t), \frac{d\gamma_0}{dt}(t) \right) . \quad \mathbf{E-L}$$

These are the **Euler-Lagrange equations**.

19.4 Solving the Euler-Lagrange Equations

Case 1: Suppose that $F(x, v)$ does not depend on v .

The Euler-Lagrange equations become

$$\frac{\partial F}{\partial x_i} \left(\gamma_0(t), \frac{d\gamma_0}{dt}(t) \right) = 0 \iff \text{the curve } \gamma_0 \text{ sits on the critical set of } F .$$

For generic F , the critical points are isolated, hence $\gamma_0(t)$ must be a constant curve.

Case 2: Suppose that $F(x, v)$ depends affinely on v :

$$F(x, v) = F_0(x) + \sum_{j=1}^n F_j(x) v_j .$$

$$\text{LHS of } \mathbf{E-L} : \quad \frac{\partial F_0}{\partial x_i}(\gamma(t)) + \sum_{j=1}^n \frac{\partial F_j}{\partial x_i}(\gamma(t)) \frac{d\gamma_j}{dt}(t)$$

$$\text{RHS of } \mathbf{E-L} : \quad \frac{d}{dt} F_i(\gamma(t)) = \sum_{j=1}^n \frac{\partial F_i}{\partial x_j}(\gamma(t)) \frac{d\gamma_j}{dt}(t)$$

The Euler-Lagrange equations become

$$\frac{\partial F_0}{\partial x_i}(\gamma(t)) = \sum_{j=1}^n \underbrace{\left(\frac{\partial F_i}{\partial x_j} - \frac{\partial F_j}{\partial x_i} \right)}_{n \times n \text{ matrix}}(\gamma(t)) \frac{d\gamma_j}{dt}(t) .$$

If the $n \times n$ matrix $\left(\frac{\partial F_i}{\partial x_j} - \frac{\partial F_j}{\partial x_i} \right)$ has an inverse $G_{ij}(x)$, then

$$\frac{d\gamma_j}{dt}(t) = \sum_{i=1}^n G_{ji}(\gamma(t)) \frac{\partial F_0}{\partial x_i}(\gamma(t))$$

is a system of first order ordinary differential equations. Locally it has a unique solution through each point p . If q is not on this curve, there is no solution at all to the Euler-Lagrange equations belonging to $\mathcal{P}(a, b, p, q)$.

Therefore, we need non-linear dependence of F on the v variables in order to have appropriate solutions. From now on, assume that the

Legendre condition: $\det \left(\frac{\partial^2 F}{\partial v_i \partial v_j} \right) \neq 0$.

Letting $G_{ij}(x, v) = \left(\frac{\partial^2 F}{\partial v_i \partial v_j}(x, v) \right)^{-1}$, the Euler-Lagrange equations become

$$\frac{d^2 \gamma_j}{dt^2} = \sum_i G_{ji} \frac{\partial F}{\partial x_i} \left(\gamma, \frac{d\gamma}{dt} \right) - \sum_{i,k} G_{ji} \frac{\partial^2 F}{\partial v_i \partial x_k} \left(\gamma, \frac{d\gamma}{dt} \right) \frac{d\gamma_k}{dt}.$$

This second order ordinary differential equation has a unique solution given initial conditions

$$\gamma(a) = p \quad \text{and} \quad \frac{d\gamma}{dt}(a) = v.$$

19.5 Minimizing Properties

Is the above solution locally minimizing?

Assume that $\left(\frac{\partial^2 F}{\partial v_i \partial v_j}(x, v) \right) \gg 0$, $\forall (x, v)$, i.e., with the x variable frozen, the function $v \mapsto F(x, v)$ is **strictly convex**.

Suppose that $\gamma_0 \in \mathcal{P}(a, b, p, q)$ satisfies **E-L**. Does γ_0 minimize $\mathcal{A}_\gamma$? Locally, yes, according to the following theorem. (Globally it is only critical.)

Theorem 19.3 *For every sufficiently small subinterval $[a_1, b_1]$ of $[a, b]$, $\gamma_0|_{[a_1, b_1]}$ is locally minimizing in $\mathcal{P}(a_1, b_1, p_1, q_1)$ where $p_1 = \gamma_0(a_1)$, $q_1 = \gamma_0(b_1)$.*

Proof. As an exercise in Fourier series, show the **Wirtinger inequality**: for $f \in C^1([a, b])$ with $f(a) = f(b) = 0$, we have

$$\int_a^b \left| \frac{df}{dt} \right|^2 dt \geq \frac{\pi^2}{(b-a)^2} \int_a^b |f|^2 dt.$$

Suppose that $\gamma_0 : [a, b] \rightarrow \mathcal{U}$ satisfies **E-L**. Take $c_i \in C^\infty([a, b])$, $c_i(a) = c_i(b) = 0$. Let $c = (c_1, \dots, c_n)$. Let $\gamma_\varepsilon = \gamma_0 + \varepsilon c \in \mathcal{P}(a, b, p, q)$, and let $\mathcal{A}_\varepsilon = \mathcal{A}_{\gamma_\varepsilon}$.

$$\mathbf{E-L} \iff \frac{d\mathcal{A}_\varepsilon}{d\varepsilon}(0) = 0.$$

$$\frac{d^2\mathcal{A}_\varepsilon}{d\varepsilon^2}(0) = \int_a^b \sum_{i,j} \frac{\partial^2 F}{\partial x_i \partial x_j} \left(\gamma_0, \frac{d\gamma_0}{dt} \right) c_i c_j dt \quad (\text{I})$$

$$+ 2 \int_a^b \sum_{i,j} \frac{\partial^2 F}{\partial x_i \partial v_j} \left(\gamma_0, \frac{d\gamma_0}{dt} \right) c_i \frac{dc_j}{dt} dt \quad (\text{II})$$

$$+ \int_a^b \sum_{i,j} \frac{\partial^2 F}{\partial v_i \partial v_j} \left(\gamma_0, \frac{d\gamma_0}{dt} \right) \frac{dc_i}{dt} \frac{dc_j}{dt} dt \quad (\text{III}) .$$

Since $\left(\frac{\partial^2 F}{\partial v_i \partial v_j}(x, v) \right) \gg 0$ at all x, v ,

$$\text{III} \geq K_{\text{III}} \left| \frac{dc}{dt} \right|_{L^2[a,b]}^2$$

$$|\text{I}| \leq K_{\text{I}} |c|_{L^2[a,b]}^2$$

$$|\text{II}| \leq K_{\text{II}} |c|_{L^2[a,b]} \left| \frac{dc}{dt} \right|_{L^2[a,b]}$$

where $K_{\text{I}}, K_{\text{II}}, K_{\text{III}} > 0$. By the Wirtinger inequality, if $b - a$ is very small, then $\text{III} > |\text{I}| + |\text{II}|$ when $c \neq 0$. Hence, γ_0 is a local minimum. $\square$

Homework 14: Minimizing Geodesics

This set of problems is adapted from [53].

Let (M, g) be a riemannian manifold. From the riemannian metric, we get a function $F : TM \rightarrow \mathbb{R}$, whose restriction to each tangent space $T_p M$ is the quadratic form defined by the metric.

Let p and q be points on M , and let $\gamma : [a, b] \rightarrow M$ be a smooth curve joining p to q . Let $\tilde{\gamma} : [a, b] \rightarrow TM$, $\tilde{\gamma}(t) = (\gamma(t), \frac{d\gamma}{dt}(t))$ be the lift of γ to TM . The **action** of γ is

$$\mathcal{A}(\gamma) = \int_a^b (\tilde{\gamma}^* F) dt = \int_a^b \left| \frac{d\gamma}{dt} \right|^2 dt .$$

1. Let $\gamma : [a, b] \rightarrow M$ be a smooth curve joining p to q . Show that the arc-length of γ is independent of the parametrization of γ , i.e., show that if we reparametrize γ by $\tau : [a', b'] \rightarrow [a, b]$, the new curve $\gamma' = \gamma \circ \tau : [a', b'] \rightarrow M$ has the same arc-length.
2. Show that, given any curve $\gamma : [a, b] \rightarrow M$ (with $\frac{d\gamma}{dt}$ never vanishing), there is a reparametrization $\tau : [a, b] \rightarrow [a, b]$ such that $\gamma \circ \tau : [a, b] \rightarrow M$ is of constant velocity, that is, $|\frac{d\gamma}{dt}|$ is independent of t .
3. Let $\tau : [a, b] \rightarrow [a, b]$ be a smooth monotone map taking the endpoints of $[a, b]$ to the endpoints of $[a, b]$. Prove that

$$\int_a^b \left(\frac{d\tau}{dt} \right)^2 dt \geq b - a ,$$

with equality holding if and only if $\frac{d\tau}{dt} = 1$.

4. Let $\gamma : [a, b] \rightarrow M$ be a smooth curve joining p to q . Suppose that, as s goes from a to b , its image $\gamma(s)$ moves at constant velocity, i.e., suppose that $|\frac{d\gamma}{ds}|$ is constant as a function of s . Let $\gamma' = \gamma \circ \tau : [a, b] \rightarrow M$ be a reparametrization of γ . Show that $\mathcal{A}(\gamma') \geq \mathcal{A}(\gamma)$, with equality holding if and only if $\tau(t) \equiv t$.

5. Let $\gamma_0 : [a, b] \rightarrow M$ be a curve joining p to q . Suppose that γ_0 is **action-minimizing**, i.e., suppose that

$$\mathcal{A}(\gamma_0) \leq \mathcal{A}(\gamma)$$

for any other curve $\gamma : [a, b] \rightarrow M$ joining p to q . Prove that γ_0 is also **arc-length-minimizing**, i.e., show that γ_0 is the shortest geodesic joining p to q .

6. Show that, among all curves joining p to q , γ_0 minimizes the action if and only if γ_0 is of constant velocity and γ_0 minimizes arc-length.

7. On a coordinate chart $(\mathcal{U}, x^1, \dots, x^n)$ on M , we have

$$F(x, v) = \sum g_{ij}(x) v^i v^j .$$

Show that the Euler-Lagrange equations associated to the action reduce to the **Christoffel equations** for a geodesic

$$\frac{d^2 \gamma^k}{dt^2} + \sum (\Gamma_{ij}^k \circ \gamma) \frac{d\gamma^i}{dt} \frac{d\gamma^j}{dt} = 0 ,$$

where the Γ_{ij}^k 's (called the **Christoffel symbols**) are defined in terms of the coefficients of the riemannian metric by

$$\Gamma_{ij}^k = \frac{1}{2} \sum_{\ell} g^{\ell k} \left(\frac{\partial g_{\ell i}}{\partial x_j} + \frac{\partial g_{\ell j}}{\partial x_i} - \frac{\partial g_{ij}}{\partial x_{\ell}} \right) ,$$

(g^{ij}) being the matrix inverse to (g_{ij}) .

8. Let p and q be two non-antipodal points on S^n . Show that the geodesic joining p to q is an arc of a great circle, the great circle in question being the intersection of S^n with the two-dimensional subspace of $\mathbb{R}^{n+1}$ spanned by p and q .

Hint: No calculations are needed: Show that an isometry of a riemannian manifold has to carry geodesics into geodesics, and show that there is an isometry of $\mathbb{R}^{n+1}$ whose fixed point set is the plane spanned by p and q , and show that this isometry induces on S^n an isometry whose fixed point set is the great circle containing p and q .

20 Legendre Transform

20.1 Strict Convexity

Let V be an n -dimensional vector space, with $e_1, \dots, e_n$ a basis of V and $v_1, \dots, v_n$ the associated coordinates. Let $F : V \rightarrow \mathbb{R}$, $F = F(v_1, \dots, v_n)$, be a smooth function. Let $p \in V$, $u = \sum_{i=1}^n u_i e_i \in V$. The **hessian** of F is the quadratic function on V defined by

$$(d^2 F)_p(u) := \sum_{i,j} \frac{\partial^2 F}{\partial v_i \partial v_j}(p) u_i u_j .$$

Exercise. Show that $(d^2 F)_p(u) = \frac{d^2}{dt^2} F(p + tu)|_{t=0}$. ◇

Definition 20.1 The function F is **strictly convex** if $(d^2 F)_p \gg 0$, $\forall p \in V$.

Proposition 20.2 For a strictly convex function F on V , the following are equivalent:

- (a) F has a critical point, i.e., a point where $dF_p = 0$;
- (b) F has a local minimum at some point;
- (c) F has a unique critical point (global minimum); and
- (d) F is proper, that is, $F(p) \rightarrow +\infty$ as $p \rightarrow \infty$ in V .

Proof. Homework 15. □

Definition 20.3 A strictly convex function F is **stable** when it satisfies conditions (a)-(d) in Proposition 20.2.

Example. The function $e^x + ax$ is strictly convex for any $a \in \mathbb{R}$, but it is stable only for $a < 0$. The function $x^2 + ax$ is strictly convex and stable for any $a \in \mathbb{R}$. ◇

20.2 Legendre Transform

Let F be any strictly convex function on V . Given $\ell \in V^*$, let

$$F_\ell : V \longrightarrow \mathbb{R} , \quad F_\ell(v) = F(v) - \ell(v) .$$

Since $(d^2 F)_p = (d^2 F_\ell)_p$,

$$F \text{ is strictly convex} \iff F_\ell \text{ is strictly convex.}$$

Definition 20.4 The **stability set** of a strictly convex function F is

$$S_F = \{\ell \in V^* \mid F_\ell \text{ is stable}\} .$$

Proposition 20.5 The set S_F is an open and convex subset of V^* .

Proof. Homework 15. □

Homework 15 also describes a sufficient condition for $S_F = V^*$.

Definition 20.6 The **Legendre transform** associated to $F \in C^\infty(V; \mathbb{R})$ is the map

$$\begin{aligned} L_F : V &\longrightarrow V^* \\ p &\longmapsto dF_p \in T_p^*V \simeq V^* . \end{aligned}$$

Proposition 20.7 Suppose that F is strictly convex. Then

$$L_F : V \xrightarrow{\simeq} S_F ,$$

i.e., L_F is a diffeomorphism onto S_F .

The inverse map $L_F^{-1} : S_F \rightarrow V$ is described as follows: for $\ell \in S_F$, the value $L_F^{-1}(\ell)$ is the unique minimum point $p_\ell \in V$ of $F_\ell = F - \ell$.

Exercise. Check that p is the minimum of $F(v) - dF_p(v)$. ◇

Definition 20.8 The **dual function** F^* to F is

$$F^* : S_F \longrightarrow \mathbb{R} , \quad F^*(\ell) = - \min_{p \in V} F_\ell(p) .$$

Theorem 20.9 We have that $L_F^{-1} = L_{F^*}$.

Proof. Homework 15. □

20.3 Application to Variational Problems

Let M be a manifold and $F : TM \rightarrow \mathbb{R}$ a function on TM .

Problem. Minimize $\mathcal{A}_\gamma = \int \tilde{\gamma}^* F$.

At $p \in M$, let

$$F_p := F|_{T_p M} : T_p M \longrightarrow \mathbb{R} .$$

Assume that F_p is strictly convex for all $p \in M$. To simplify notation, assume also that $S_{F_p} = T_p^* M$. The Legendre transform on each tangent space

$$L_{F_p} : T_p M \xrightarrow{\simeq} T_p^* M$$

is essentially given by the first derivatives of F in the v directions. The dual function to F_p is $F_p^* : T_p^*M \rightarrow \mathbb{R}$. Collect these fiberwise maps into

$$\begin{aligned} \mathcal{L} : TM &\longrightarrow T^*M, & \mathcal{L}|_{T_p M} &= L_{F_p}, & \text{and} \\ H : T^*M &\longrightarrow \mathbb{R}, & H|_{T_p^* M} &= F_p^*. \end{aligned}$$

Exercise. The maps H and $\mathcal{L}$ are smooth, and $\mathcal{L}$ is a diffeomorphism. $\diamond$

Let

$$\begin{aligned} \gamma : [a, b] &\longrightarrow M & \text{be a curve,} & \text{and} \\ \tilde{\gamma} : [a, b] &\longrightarrow TM & \text{its lift.} \end{aligned}$$

Theorem 20.10 *The curve γ satisfies the Euler-Lagrange equations on every coordinate chart if and only if $\mathcal{L} \circ \tilde{\gamma} : [a, b] \rightarrow T^*M$ is an integral curve of the hamiltonian vector field X_H .*

Proof. Let

$$\begin{aligned} (\mathcal{U}, x_1, \dots, x_n) &\quad \text{coordinate neighborhood in } M, \\ (T\mathcal{U}, x_1, \dots, x_n, v_1, \dots, v_n) &\quad \text{coordinates in } TM, \\ (T^*\mathcal{U}, x_1, \dots, x_n, \xi_1, \dots, \xi_n) &\quad \text{coordinates in } T^*M. \end{aligned}$$

On $T\mathcal{U}$ we have $F = F(x, v)$.

On $T^*\mathcal{U}$ we have $H = H(u, \xi)$.

$$\begin{aligned} \mathcal{L} : T\mathcal{U} &\longrightarrow T^*\mathcal{U} \\ (x, v) &\longmapsto (x, \xi) \quad \text{where} \quad \xi = L_{F_x}(v) = \frac{\partial F}{\partial v}(x, v). \end{aligned}$$

(This is the definition of **momentum** ξ .)

$$H(x, \xi) = F_x^*(\xi) = \xi \cdot v - F(x, v) \quad \text{where} \quad \mathcal{L}(x, v) = (x, \xi).$$

Integral curves $(x(t), \xi(t))$ of X_H satisfy the Hamilton equations:

$$\mathbf{H} \quad \begin{cases} \frac{dx}{dt} = \frac{\partial H}{\partial \xi}(x, \xi) \\ \frac{d\xi}{dt} = -\frac{\partial H}{\partial x}(x, \xi), \end{cases}$$

whereas the physical path $x(t)$ satisfies the Euler-Lagrange equations:

$$\mathbf{E-L} \quad \frac{\partial F}{\partial x} \left(x, \frac{dx}{dt} \right) = \frac{d}{dt} \frac{\partial F}{\partial v} \left(x, \frac{dx}{dt} \right).$$

Let $(x(t), \xi(t)) = \mathcal{L} \left(x(t), \frac{dx}{dt}(t) \right)$. We want to prove:

$$t \mapsto (x(t), \xi(t)) \text{ satisfies } \mathbf{H} \quad \Longleftrightarrow \quad t \mapsto \left(x(t), \frac{dx}{dt}(t) \right) \text{ satisfies } \mathbf{E-L}.$$

The first line of **H** is automatically satisfied:

$$\frac{dx}{dt} = \frac{\partial H}{\partial \xi}(x, \xi) = L_{F_x^*}(\xi) = L_{F_x}^{-1}(\xi) \quad \Longleftrightarrow \quad \xi = L_{F_x} \left(\frac{dx}{dt} \right)$$

Claim. If $(x, \xi) = \mathcal{L}(x, v)$, then $\frac{\partial F}{\partial x}(x, v) = -\frac{\partial H}{\partial x}(x, \xi)$.

This follows from differentiating both sides of $H(x, \xi) = \xi \cdot v - F(x, v)$ with respect to x , where $\xi = L_{F_x}(v) = \xi(x, v)$.

$$\frac{\partial H}{\partial x} + \underbrace{\frac{\partial H}{\partial \xi} \frac{\partial \xi}{\partial x}}_v = \frac{\partial \xi}{\partial x} \cdot v - \frac{\partial F}{\partial x}.$$

Now the second line of **H** becomes

$$\underbrace{\frac{d}{dt} \frac{\partial F}{\partial v}(x, v)}_{\text{since } \xi = L_{F_x}(v)} = \frac{d\xi}{dt} = \underbrace{-\frac{\partial H}{\partial x}(x, \xi)}_{\text{by the claim}} = \frac{\partial F}{\partial x}(x, v) \quad \Longleftrightarrow \quad \mathbf{E-L}.$$

□

Homework 15: Legendre Transform

This set of problems is adapted from [54].

- Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a smooth function. f is called **strictly convex** if $f''(x) > 0$ for all $x \in \mathbb{R}$. Assuming that f is strictly convex, prove that the following four conditions are equivalent:
 - $f'(x) = 0$ for some point x_0 ,
 - f has a local minimum at some point x_0 ,
 - f has a unique (global) minimum at some point x_0 ,
 - $f(x) \rightarrow +\infty$ as $x \rightarrow \pm\infty$.

The function f is **stable** if it satisfies one (and hence all) of these conditions.

For what values of a is the function $e^x + ax$ stable? For those values of a for which it is not stable, what does the graph look like?

- Let V be an n -dimensional vector space and $F : V \rightarrow \mathbb{R}$ a smooth function. The function F is said to be **strictly convex** if for every pair of elements $p, v \in V$, $v \neq 0$, the restriction of F to the line $\{p + xv \mid x \in \mathbb{R}\}$ is strictly convex.

The **hessian** of F at p is the quadratic form

$$d^2 F_p : v \mapsto \frac{d^2}{dx^2} F(p + xv)|_{x=0} .$$

Show that F is strictly convex if and only if $d^2 F_p$ is positive definite for all $p \in V$.

Prove the n -dimensional analogue of the result you proved in (1). Namely, assuming that F is strictly convex, show that the four following assertions are equivalent:

- $dF_p = 0$ at some point p_0 ,
 - F has a local minimum at some point p_0 ,
 - F has a unique (global) minimum at some point p_0 ,
 - $F(p) \rightarrow +\infty$ as $p \rightarrow \infty$.
- As in exercise 2, let V be an n -dimensional vector space and $F : V \rightarrow \mathbb{R}$ a smooth function. Since V is a vector space, there is a canonical identification $T_p^* V \simeq V^*$, for every $p \in V$. Therefore, we can define a map

$$L_F : V \longrightarrow V^* \quad \textbf{(Legendre transform)}$$

by setting

$$L_F(p) = dF_p \in T_p^* V \simeq V^* .$$

Show that, if F is strictly convex, then, for every point $p \in V$, L_F maps a neighborhood of p diffeomorphically onto a neighborhood of $L_F(p)$.

4. A strictly convex function $F : V \rightarrow \mathbb{R}$ is **stable** if it satisfies the four equivalent conditions of exercise 2. Given any strictly convex function F , we will denote by S_F the set of $\ell \in V^*$ for which the function $F_\ell : V \rightarrow \mathbb{R}, p \mapsto F(p) - \ell(p)$, is stable. Prove that:

- (a) The set S_F is open and convex.
- (b) L_F maps V diffeomorphically onto S_F .
- (c) If $\ell \in S_F$ and $p_0 = L_F^{-1}(\ell)$, then p_0 is the unique minimum point of the function F_ℓ .

Let $F^* : S_F \rightarrow \mathbb{R}$ be the function whose value at ℓ is the quantity $-\min_{p \in V} F_\ell(p)$.

Show that F^* is a smooth function.

The function F^* is called the **dual** of the function F .

5. Let F be a strictly convex function. F is said to have **quadratic growth at infinity** if there exists a positive-definite quadratic form Q on V and a constant K such that $F(p) \geq Q(p) - K$, for all p . Show that, if F has quadratic growth at infinity, then $S_F = V^*$ and hence L_F maps V diffeomorphically onto V^* .
6. Let $F : V \rightarrow \mathbb{R}$ be strictly convex and let $F^* : S_F \rightarrow \mathbb{R}$ be the dual function. Prove that for all $p \in V$ and all $\ell \in S_F$,

$$F(p) + F^*(\ell) \geq \ell(p) \quad (\text{Young inequality}) .$$

7. On one hand we have $V \times V^* \simeq T^*V$, and on the other hand, since $V = V^{**}$, we have $V \times V^* \simeq V^* \times V \simeq T^*V^*$.

Let α_1 be the canonical 1-form on T^*V and α_2 be the canonical 1-form on T^*V^* . Via the identifications above, we can think of both of these forms as living on $V \times V^*$. Show that $\alpha_1 = d\beta - \alpha_2$, where $\beta : V \times V^* \rightarrow \mathbb{R}$ is the function $\beta(p, \ell) = \ell(p)$.

Conclude that the forms $\omega_1 = d\alpha_1$ and $\omega_2 = d\alpha_2$ satisfy $\omega_1 = -\omega_2$.

8. Let $F : V \rightarrow \mathbb{R}$ be strictly convex. Assume that F has quadratic growth at infinity so that $S_F = V^*$. Let Λ_F be the graph of the Legendre transform L_F . The graph Λ_F is a lagrangian submanifold of $V \times V^*$ with respect to the symplectic form ω_1 ; why? Hence, Λ_F is also lagrangian for ω_2 .

Let $\text{pr}_1 : \Lambda_F \rightarrow V$ and $\text{pr}_2 : \Lambda_F \rightarrow V^*$ be the restrictions of the projection maps $V \times V^* \rightarrow V$ and $V \times V^* \rightarrow V^*$, and let $i : \Lambda_F \hookrightarrow V \times V^*$ be the inclusion map. Show that

$$i^* \alpha_1 = d(\text{pr}_1)^* F .$$

Conclude that

$$i^* \alpha_2 = d(i^* \beta - (\text{pr}_1)^* F) = d(\text{pr}_2)^* F^* ,$$

and from this conclude that the inverse of the Legendre transform associated with F is the Legendre transform associated with F^* .

Part VIII

Moment Maps

The concept of a *moment map*¹² is a generalization of that of a hamiltonian function. The notion of a moment map associated to a group action on a symplectic manifold formalizes the Noether principle, which states that to every symmetry (such as a group action) in a mechanical system, there corresponds a conserved quantity.

21 Actions

21.1 One-Parameter Groups of Diffeomorphisms

Let M be a manifold and X a complete vector field on M . Let $\rho_t : M \rightarrow M$, $t \in \mathbb{R}$, be the family of diffeomorphisms generated by X . For each $p \in M$, $\rho_t(p)$, $t \in \mathbb{R}$, is by definition the unique integral curve of X passing through p at time 0, i.e., $\rho_t(p)$ satisfies

$$\begin{cases} \rho_0(p) &= p \\ \frac{d\rho_t(p)}{dt} &= X(\rho_t(p)) . \end{cases}$$

Claim. We have that $\rho_t \circ \rho_s = \rho_{t+s}$.

Proof. Let $\rho_s(q) = p$. We need to show that $(\rho_t \circ \rho_s)(q) = \rho_{t+s}(q)$, for all $t \in \mathbb{R}$. Reparametrize as $\tilde{\rho}_t(q) := \rho_{t+s}(q)$. Then

$$\begin{cases} \tilde{\rho}_0(q) &= \rho_s(q) = p \\ \frac{d\tilde{\rho}_t(q)}{dt} &= \frac{d\rho_{t+s}(q)}{dt} = X(\rho_{t+s}(q)) = X(\tilde{\rho}_t(q)) , \end{cases}$$

i.e., $\tilde{\rho}_t(q)$ is an integral curve of X through p . By uniqueness we must have $\tilde{\rho}_t(q) = \rho_t(p)$, that is, $\rho_{t+s}(q) = \rho_t(\rho_s(q))$. $\square$

Consequence. We have that $\rho_t^{-1} = \rho_{-t}$.

In terms of the group $(\mathbb{R}, +)$ and the group $(\text{Diff}(M), \circ)$ of all diffeomorphisms of M , these results can be summarized as:

Corollary 21.1 *The map $\mathbb{R} \rightarrow \text{Diff}(M)$, $t \mapsto \rho_t$, is a group homomorphism.*

The family $\{\rho_t \mid t \in \mathbb{R}\}$ is then called a **one-parameter group of diffeomorphisms** of M and denoted

$$\rho_t = \exp tX .$$

¹²Souriau [95] invented the french name “application moment.” In the US, East and West coasts could be distinguished by the choice of translation: *moment map* and *momentum map*, respectively. We will stick to the more economical version.

21.2 Lie Groups

Definition 21.2 A **Lie group** is a manifold G equipped with a group structure where the group operations

$$\begin{array}{ccc} G \times G & \longrightarrow & G \\ (a, b) & \longmapsto & a \cdot b \end{array} \quad \text{and} \quad \begin{array}{ccc} G & \longrightarrow & G \\ a & \longmapsto & a^{-1} \end{array}$$

are smooth maps.

Examples.

- $\mathbb{R}$ (with addition¹³).
- S^1 regarded as unit complex numbers with multiplication, represents rotations of the plane: $S^1 = \mathrm{U}(1) = \mathrm{SO}(2)$.
- $\mathrm{U}(n)$, unitary linear transformations of $\mathbb{C}^n$.
- $\mathrm{SU}(n)$, unitary linear transformations of $\mathbb{C}^n$ with $\det = 1$.
- $\mathrm{O}(n)$, orthogonal linear transformations of $\mathbb{R}^n$.
- $\mathrm{SO}(n)$, elements of $\mathrm{O}(n)$ with $\det = 1$.
- $\mathrm{GL}(V)$, invertible linear transformations of a vector space V .

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Definition 21.3 A **representation** of a Lie group G on a vector space V is a group homomorphism $G \rightarrow \mathrm{GL}(V)$.

21.3 Smooth Actions

Let M be a manifold.

Definition 21.4 An **action** of a Lie group G on M is a group homomorphism

$$\begin{array}{ccc} \psi : G & \longrightarrow & \mathrm{Diff}(M) \\ g & \longmapsto & \psi_g . \end{array}$$

(We will only consider left actions where ψ is a homomorphism. A **right action** is defined with ψ being an anti-homomorphism.) The **evaluation map** associated with an action $\psi : G \rightarrow \mathrm{Diff}(M)$ is

$$\begin{array}{ccc} \mathrm{ev}_\psi : M \times G & \longrightarrow & M \\ (p, g) & \longmapsto & \psi_g(p) . \end{array}$$

The action ψ is **smooth** if ev_ψ is a smooth map.

¹³The operation will be omitted when it is clear from the context.

Example. If X is a complete vector field on M , then

$$\begin{aligned} \rho : \mathbb{R} &\longrightarrow \text{Diff}(M) \\ t &\longmapsto \rho_t = \exp tX \end{aligned}$$

is a smooth action of $\mathbb{R}$ on M . $\diamond$

Every complete vector field gives rise to a smooth action of $\mathbb{R}$ on M . Conversely, every smooth action of $\mathbb{R}$ on M is defined by a complete vector field.

$$\{\text{complete vector fields on } M\} \xrightarrow{1-1} \{\text{smooth actions of } \mathbb{R} \text{ on } M\}$$

$$X \longmapsto \exp tX$$

$$X_p = \left. \frac{d\psi_t(p)}{dt} \right|_{t=0} \longleftarrow \psi$$

21.4 Symplectic and Hamiltonian Actions

Let (M, ω) be a symplectic manifold, and G a Lie group. Let $\psi : G \longrightarrow \text{Diff}(M)$ be a (smooth) action.

Definition 21.5 *The action ψ is a symplectic action if*

$$\psi : G \longrightarrow \text{Symp}(M, \omega) \subset \text{Diff}(M) ,$$

i.e., G “acts by symplectomorphisms.”

$$\{\text{complete symplectic vector fields on } M\} \xrightarrow{1-1} \{\text{symplectic actions of } \mathbb{R} \text{ on } M\}$$

Example. On $\mathbb{R}^{2n}$ with $\omega = \sum dx_i \wedge dy_i$, let $X = -\frac{\partial}{\partial y_1}$. The orbits of the action generated by X are lines parallel to the y_1 -axis,

$$\{(x_1, y_1 - t, x_2, y_2, \dots, x_n, y_n) \mid t \in \mathbb{R}\} .$$

Since $X = X_{x_1}$ is hamiltonian (with hamiltonian function $H = x_1$), this is actually an example of a *hamiltonian action* of $\mathbb{R}$. $\diamond$

Example. On S^2 with $\omega = d\theta \wedge dh$ (cylindrical coordinates), let $X = \frac{\partial}{\partial \theta}$. Each orbit is a horizontal circle (called a “parallel”) $\{(\theta + t, h) \mid t \in \mathbb{R}\}$. Notice that all orbits of this $\mathbb{R}$ -action close up after time 2π , so that this is an action of S^1 :

$$\begin{aligned} \psi : S^1 &\longrightarrow \text{Symp}(S^2, \omega) \\ t &\longmapsto \text{rotation by angle } t \text{ around } h\text{-axis} . \end{aligned}$$

Since $X = X_h$ is hamiltonian (with hamiltonian function $H = h$), this is an example of a *hamiltonian action* of S^1 . $\diamond$

Definition 21.6 A symplectic action ψ of S^1 or $\mathbb{R}$ on (M, ω) is **hamiltonian** if the vector field generated by ψ is hamiltonian. Equivalently, an action ψ of S^1 or $\mathbb{R}$ on (M, ω) is **hamiltonian** if there is $H : M \rightarrow \mathbb{R}$ with $dH = \iota_X \omega$, where X is the vector field generated by ψ .

What is a “hamiltonian action” of an arbitrary Lie group?

For the case where $G = \mathbb{T}^n = S^1 \times \dots \times S^1$ is an n -torus, an action $\psi : G \rightarrow \text{Symp}(M, \omega)$ should be called *hamiltonian* when each restriction

$$\psi^i := \psi|_{\text{ith } S^1 \text{ factor}} : S^1 \longrightarrow \text{Symp}(M, \omega)$$

is hamiltonian in the previous sense with hamiltonian function preserved by the action of the rest of G .

When G is not a product of S^1 's or $\mathbb{R}$'s, the solution is to use an upgraded hamiltonian function, known as a *moment map*. Before its definition though (in Lecture 22), we need a little Lie theory.

21.5 Adjoint and Coadjoint Representations

Let G be a Lie group. Given $g \in G$ let

$$\begin{aligned} L_g : G &\longrightarrow G \\ a &\longmapsto g \cdot a \end{aligned}$$

be **left multiplication** by g . A vector field X on G is called **left-invariant** if $(L_g)_* X = X$ for every $g \in G$. (There are similar *right* notions.)

Let $\mathfrak{g}$ be the vector space of all left-invariant vector fields on G . Together with the Lie bracket $[\cdot, \cdot]$ of vector fields, $\mathfrak{g}$ forms a Lie algebra, called the **Lie algebra of the Lie group G** .

Exercise. Show that the map

$$\begin{aligned} \mathfrak{g} &\longrightarrow T_e G \\ X &\longmapsto X_e \end{aligned}$$

where e is the identity element in G , is an isomorphism of vector spaces. $\diamond$

Any Lie group G acts on itself by **conjugation**:

$$\begin{aligned} G &\longrightarrow \text{Diff}(G) \\ g &\longmapsto \psi_g, \quad \psi_g(a) = g \cdot a \cdot g^{-1}. \end{aligned}$$

The derivative at the identity of

$$\begin{aligned} \psi_g : G &\longrightarrow G \\ a &\longmapsto g \cdot a \cdot g^{-1} \end{aligned}$$

is an invertible linear map $\text{Ad}_g : \mathfrak{g} \longrightarrow \mathfrak{g}$. Here we identify the Lie algebra $\mathfrak{g}$ with the tangent space $T_e G$. Letting g vary, we obtain the **adjoint representation** (or **adjoint action**) of G on $\mathfrak{g}$:

$$\begin{aligned} \text{Ad} : G &\longrightarrow \text{GL}(\mathfrak{g}) \\ g &\longmapsto \text{Ad}_g . \end{aligned}$$

Exercise. Check for matrix groups that

$$\left. \frac{d}{dt} \text{Ad}_{\exp tX} Y \right|_{t=0} = [X, Y] , \quad \forall X, Y \in \mathfrak{g} .$$

Hint: For a matrix group G (i.e., a subgroup of $\text{GL}(n; \mathbb{R})$ for some n), we have

$$\text{Ad}_g(Y) = gYg^{-1} , \quad \forall g \in G , \forall Y \in \mathfrak{g}$$

and

$$[X, Y] = XY - YX , \quad \forall X, Y \in \mathfrak{g} .$$

◇

Let $\langle \cdot, \cdot \rangle$ be the natural pairing between $\mathfrak{g}^*$ and $\mathfrak{g}$:

$$\begin{aligned} \langle \cdot, \cdot \rangle : \mathfrak{g}^* \times \mathfrak{g} &\longrightarrow \mathbb{R} \\ (\xi, X) &\longmapsto \langle \xi, X \rangle = \xi(X) . \end{aligned}$$

Given $\xi \in \mathfrak{g}^*$, we define $\text{Ad}_g^* \xi$ by

$$\langle \text{Ad}_g^* \xi, X \rangle = \langle \xi, \text{Ad}_{g^{-1}} X \rangle , \quad \text{for any } X \in \mathfrak{g} .$$

The collection of maps Ad_g^* forms the **coadjoint representation** (or **coadjoint action**) of G on $\mathfrak{g}^*$:

$$\begin{aligned} \text{Ad}^* : G &\longrightarrow \text{GL}(\mathfrak{g}^*) \\ g &\longmapsto \text{Ad}_g^* . \end{aligned}$$

We take g^{-1} in the definition of $\text{Ad}_g^* \xi$ in order to obtain a (left) representation, i.e., a group homomorphism, instead of a “right” representation, i.e., a group anti-homomorphism.

Exercise. Show that $\text{Ad}_g \circ \text{Ad}_h = \text{Ad}_{gh}$ and $\text{Ad}_g^* \circ \text{Ad}_h^* = \text{Ad}_{gh}^*$. ◇

Homework 16: Hermitian Matrices

Let $\mathcal{H}$ be the vector space of $n \times n$ complex hermitian matrices.

The unitary group $U(n)$ acts on $\mathcal{H}$ by conjugation: $A \cdot \xi = A\xi A^{-1}$, for $A \in U(n)$, $\xi \in \mathcal{H}$.

For each $\lambda = (\lambda_1, \dots, \lambda_n) \in \mathbb{R}^n$, let $\mathcal{H}_\lambda$ be the set of all $n \times n$ complex hermitian matrices whose spectrum is λ .

1. Show that the orbits of the $U(n)$ -action are the manifolds $\mathcal{H}_\lambda$.

For a fixed $\lambda \in \mathbb{R}^n$, what is the stabilizer of a point in $\mathcal{H}_\lambda$?

Hint: If $\lambda_1, \dots, \lambda_n$ are all distinct, the stabilizer of the diagonal matrix is the torus $\mathbb{T}^n$ of all diagonal unitary matrices.

2. Show that the symmetric bilinear form on $\mathcal{H}$, $(X, Y) \mapsto \text{trace}(XY)$, is nondegenerate.

For $\xi \in \mathcal{H}$, define a skew-symmetric bilinear form ω_ξ on $\mathfrak{u}(n) = T_1 U(n) = i\mathcal{H}$ (space of skew-hermitian matrices) by

$$\omega_\xi(X, Y) = i \text{trace}([X, Y]\xi), \quad X, Y \in i\mathcal{H}.$$

Check that $\omega_\xi(X, Y) = i \text{trace}(X(Y\xi - \xi Y))$ and $Y\xi - \xi Y \in \mathcal{H}$.

Show that the kernel of ω_ξ is $K_\xi := \{Y \in \mathfrak{u}(n) \mid [Y, \xi] = 0\}$.

3. Show that K_ξ is the Lie algebra of the stabilizer of ξ .

Hint: Differentiate the relation $A\xi A^{-1} = \xi$.

Show that the ω_ξ 's induce nondegenerate 2-forms on the orbits $\mathcal{H}_\lambda$.

Show that these 2-forms are closed.

Conclude that all the orbits $\mathcal{H}_\lambda$ are compact symplectic manifolds.

4. Describe the manifolds $\mathcal{H}_\lambda$.

When all eigenvalues are equal, there is only one point in the orbit.

Suppose that $\lambda_1 \neq \lambda_2 = \dots = \lambda_n$. Then the eigenspace associated with λ_1 is a line, and the one associated with λ_2 is the orthogonal hyperplane. Show that there is a diffeomorphism $\mathcal{H}_\lambda \simeq \mathbb{CP}^{n-1}$. We have thus exhibited a lot of symplectic forms on $\mathbb{CP}^{n-1}$, one for each pair of distinct real numbers.

What about the other cases?

Hint: When the eigenvalues $\lambda_1 < \dots < \lambda_n$ are all distinct, any element in $\mathcal{H}_\lambda$ defines a family of pairwise orthogonal lines in $\mathbb{C}^n$: its eigenspaces.

5. Show that, for any skew-hermitian matrix $X \in \mathfrak{u}(n)$, the vector field on $\mathcal{H}$ generated by $X \in \mathfrak{u}(n)$ for the $U(n)$ -action by conjugation is $X_\xi^\# = [X, \xi]$.

22 Hamiltonian Actions

22.1 Moment and Comoment Maps

Let

(M, ω) be a symplectic manifold,
 G a Lie group, and
 $\psi : G \rightarrow \text{Symp}(M, \omega)$ a (smooth) symplectic action, i.e., a group homomorphism
 such that the evaluation map $\text{ev}_\psi(g, p) := \psi_g(p)$ is smooth.

Case $G = \mathbb{R}$:

We have the following bijective correspondence:

$$\{\text{symplectic actions of } \mathbb{R} \text{ on } M\} \xleftrightarrow{1-1} \{\text{complete symplectic vector fields on } M\}$$

$$\psi \longmapsto X_p = \frac{d\psi_t(p)}{dt}$$

$$\psi = \exp tX \longleftarrow X$$

$$\text{"flow of } X" \qquad \text{"vector field generated by } \psi"$$

The action ψ is **hamiltonian** if there exists a function $H : M \rightarrow \mathbb{R}$ such that $dH = \iota_X \omega$ where X is the vector field on M generated by ψ .

Case $G = S^1$:

An action of S^1 is an action of $\mathbb{R}$ which is 2π -periodic: $\psi_{2\pi} = \psi_0$. The S^1 -action is called **hamiltonian** if the underlying $\mathbb{R}$ -action is hamiltonian.

General case:

Let

(M, ω) be a symplectic manifold,
 G a Lie group,
 $\mathfrak{g}$ the Lie algebra of G ,
 $\mathfrak{g}^*$ the dual vector space of $\mathfrak{g}$, and

$\psi : G \longrightarrow \text{Symp}(M, \omega)$ a symplectic action.

Definition 22.1 *The action ψ is a **hamiltonian action** if there exists a map*

$$\mu : M \longrightarrow \mathfrak{g}^*$$

satisfying:

1. For each $X \in \mathfrak{g}$, let

- $\mu^X : M \rightarrow \mathbb{R}$, $\mu^X(p) := \langle \mu(p), X \rangle$, be the component of μ along X ,
- $X^\#$ be the vector field on M generated by the one-parameter subgroup $\{\exp tX \mid t \in \mathbb{R}\} \subseteq G$.

Then

$$d\mu^X = \iota_{X^\#}\omega$$

i.e., μ^X is a hamiltonian function for the vector field $X^\#$.

2. μ is equivariant with respect to the given action ψ of G on M and the coadjoint action Ad^* of G on $\mathfrak{g}^*$:

$$\mu \circ \psi_g = \text{Ad}_g^* \circ \mu, \quad \text{for all } g \in G.$$

The vector (M, ω, G, μ) is then called a **hamiltonian G -space** and μ is a **moment map**.

For connected Lie groups, hamiltonian actions can be equivalently defined in terms of a **comoment map**

$$\mu^* : \mathfrak{g} \longrightarrow C^\infty(M),$$

with the two conditions rephrased as:

1. $\mu^*(X) := \mu^X$ is a hamiltonian function for the vector field $X^\#$,
2. μ^* is a Lie algebra homomorphism:

$$\mu^*[X, Y] = \{\mu^*(X), \mu^*(Y)\}$$

where $\{\cdot, \cdot\}$ is the Poisson bracket on $C^\infty(M)$.

These definitions match the previous ones for the cases $G = \mathbb{R}$, S^1 , torus, where equivariance becomes invariance since the coadjoint action is trivial.

Case $G = S^1$ (or $\mathbb{R}$):

Here $\mathfrak{g} \simeq \mathbb{R}$, $\mathfrak{g}^* \simeq \mathbb{R}$. A moment map $\mu : M \longrightarrow \mathbb{R}$ satisfies:

1. For the generator $X = 1$ of $\mathfrak{g}$, we have $\mu^X(p) = \mu(p) \cdot 1$, i.e., $\mu^X = \mu$, and $X^\#$ is the standard vector field on M generated by S^1 . Then $d\mu = \iota_{X^\#}\omega$.
2. μ is invariant: $\mathcal{L}_{X^\#}\mu = \iota_{X^\#}d\mu = 0$.

Case $G = \mathbb{T}^n = n$ -torus:

Here $\mathfrak{g} \simeq \mathbb{R}^n$, $\mathfrak{g}^* \simeq \mathbb{R}^n$. A moment map $\mu : M \longrightarrow \mathbb{R}^n$ satisfies:

1. For each basis vector X_i of $\mathbb{R}^n$, μ^{X_i} is a hamiltonian function for $X_i^\#$.
2. μ is invariant.

22.2 Orbit Spaces

Let $\psi : G \rightarrow \text{Diff}(M)$ be any action.

Definition 22.2 The **orbit** of G through $p \in M$ is $\{\psi_g(p) \mid g \in G\}$.

The **stabilizer** (or **isotropy**) of $p \in M$ is the subgroup $G_p := \{g \in G \mid \psi_g(p) = p\}$.

Exercise. If q is in the orbit of p , then G_q and G_p are conjugate subgroups. $\diamond$

Definition 22.3 We say that the action of G on M is ...

- **transitive** if there is just one orbit,
- **free** if all stabilizers are trivial $\{e\}$,
- **locally free** if all stabilizers are discrete.

Let $\sim$ be the orbit equivalence relation; for $p, q \in M$,

$$p \sim q \iff p \text{ and } q \text{ are on the same orbit.}$$

The space of orbits $M/\sim = M/G$ is called the **orbit space**. Let

$$\begin{aligned} \pi : M &\longrightarrow M/G \\ p &\longmapsto \text{orbit through } p \end{aligned}$$

be the **point-orbit projection**.

Topology of the orbit space:

We equip M/G with the weakest topology for which π is continuous, i.e., $\mathcal{U} \subseteq M/G$ is open if and only if $\pi^{-1}(\mathcal{U})$ is open in M . This is called the **quotient topology**. This topology can be “bad.” For instance:

Example. Let $G = \mathbb{R}$ act on $M = \mathbb{R}$ by

$$t \longmapsto \psi_t = \text{multiplication by } e^t.$$

There are three orbits $\mathbb{R}^+$, $\mathbb{R}^-$ and $\{0\}$. The point in the three-point orbit space corresponding to the orbit $\{0\}$ is not open, so the orbit space with the quotient topology is *not* Hausdorff. $\diamond$

Example. Let $G = \mathbb{C} \setminus \{0\}$ act on $M = \mathbb{C}^n$ by

$$\lambda \longmapsto \psi_\lambda = \text{multiplication by } \lambda.$$

The orbits are the punctured complex lines (through non-zero vectors $z \in \mathbb{C}^n$), plus one “unstable” orbit through 0, which has a single point. The orbit space is

$$M/G = \mathbb{CP}^{n-1} \sqcup \{\text{point}\}.$$

The quotient topology restricts to the usual topology on $\mathbb{CP}^{n-1}$. The only open set containing $\{\text{point}\}$ in the quotient topology is the full space. Again the quotient topology in M/G is *not* Hausdorff.

However, it suffices to remove 0 from $\mathbb{C}^n$ to obtain a Hausdorff orbit space: $\mathbb{CP}^{n-1}$. Then there is also a compact (yet not complex) description of the orbit space by taking only unit vectors:

$$\mathbb{CP}^{n-1} = (\mathbb{C}^n \setminus \{0\}) / (\mathbb{C} \setminus \{0\}) = S^{2n-1} / S^1 .$$

◇

22.3 Preview of Reduction

Let $\omega = \frac{i}{2} \sum dz_i \wedge d\bar{z}_i = \sum dx_i \wedge dy_i = \sum r_i dr_i \wedge d\theta_i$ be the standard symplectic form on $\mathbb{C}^n$. Consider the following S^1 -action on $(\mathbb{C}^n, \omega)$:

$$t \in S^1 \mapsto \psi_t = \text{multiplication by } t .$$

The action ψ is hamiltonian with moment map

$$\begin{aligned} \mu : \mathbb{C}^n &\longrightarrow \mathbb{R} \\ z &\longmapsto -\frac{|z|^2}{2} + \text{constant} \end{aligned}$$

since

$$\begin{aligned} d\mu &= -\frac{1}{2}d(\sum r_i^2) \\ X^\# &= \frac{\partial}{\partial \theta_1} + \frac{\partial}{\partial \theta_2} + \dots + \frac{\partial}{\partial \theta_n} \\ \iota_{X^\#} \omega &= -\sum r_i dr_i = -\frac{1}{2} \sum dr_i^2 . \end{aligned}$$

If we choose the constant to be $\frac{1}{2}$, then $\mu^{-1}(0) = S^{2n-1}$ is the unit sphere. The orbit space of the zero level of the moment map is

$$\mu^{-1}(0)/S^1 = S^{2n-1}/S^1 = \mathbb{CP}^{n-1} .$$

$\mathbb{CP}^{n-1}$ is thus called a **reduced space**. Notice also that the image of the moment map is half-space.

These particular observations are related to major theorems:
Under assumptions (explained in Lectures 23-29),

- [Marsden-Weinstein-Meyer] reduced spaces are symplectic manifolds;
- [Atiyah-Guillemin-Sternberg] the image of the moment map is a convex polytope;
- [Delzant] hamiltonian $\mathbb{T}^n$ -spaces are classified by the image of the moment map.

22.4 Classical Examples

Example.

Let $G = \mathrm{SO}(3) = \{A \in \mathrm{GL}(3; \mathbb{R}) \mid A^t A = \mathrm{Id} \text{ and } \det A = 1\}$. Then $\mathfrak{g} = \{A \in \mathfrak{gl}(3; \mathbb{R}) \mid A + A^t = 0\}$ is the space of 3×3 skew-symmetric matrices and can be identified with $\mathbb{R}^3$. The Lie bracket on $\mathfrak{g}$ can be identified with the exterior product via

$$A = \begin{bmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{bmatrix} \longmapsto \vec{a} = (a_1, a_2, a_3)$$

$$[A, B] = AB - BA \longmapsto \vec{a} \times \vec{b}.$$

Exercise. Under the identifications $\mathfrak{g}, \mathfrak{g}^* \simeq \mathbb{R}^3$, the adjoint and coadjoint actions are the usual $\mathrm{SO}(3)$ -action on $\mathbb{R}^3$ by rotations. $\diamond$

Therefore, the coadjoint orbits are the spheres in $\mathbb{R}^3$ centered at the origin. Homework 17 shows that coadjoint orbits are symplectic. $\diamond$

The name “moment map” comes from being the generalization of linear and angular momenta in classical mechanics.

Translation: Consider $\mathbb{R}^6$ with coordinates $x_1, x_2, x_3, y_1, y_2, y_3$ and symplectic form $\omega = \sum dx_i \wedge dy_i$. Let $\mathbb{R}^3$ act on $\mathbb{R}^6$ by translations:

$$\vec{a} \in \mathbb{R}^3 \longmapsto \psi_{\vec{a}} \in \mathrm{Symp}(\mathbb{R}^6, \omega)$$

$$\psi_{\vec{a}}(\vec{x}, \vec{y}) = (\vec{x} + \vec{a}, \vec{y}).$$

Then $X^\# = a_1 \frac{\partial}{\partial x_1} + a_2 \frac{\partial}{\partial x_2} + a_3 \frac{\partial}{\partial x_3}$ for $X = \vec{a}$, and

$$\mu : \mathbb{R}^6 \longrightarrow \mathbb{R}^3, \quad \mu(\vec{x}, \vec{y}) = \vec{y}$$

is a moment map, with

$$\mu_{\vec{a}}(\vec{x}, \vec{y}) = \langle \mu(\vec{x}, \vec{y}), \vec{a} \rangle = \vec{y} \cdot \vec{a}.$$

Classically, $\vec{y}$ is called the **momentum vector** corresponding to the **position vector** $\vec{x}$, and the map μ is called the **linear momentum**.

Rotation: The $\mathrm{SO}(3)$ -action on $\mathbb{R}^3$ by rotations lifts to a symplectic action ψ on the cotangent bundle $\mathbb{R}^6$. The infinitesimal version of this action is

$$\vec{a} \in \mathbb{R}^3 \longmapsto d\psi(\vec{a}) \in \chi^{\mathrm{symp}}(\mathbb{R}^6)$$

$$d\psi(\vec{a})(\vec{x}, \vec{y}) = (\vec{a} \times \vec{x}, \vec{a} \times \vec{y}).$$

Then

$$\mu : \mathbb{R}^6 \longrightarrow \mathbb{R}^3, \quad \mu(\vec{x}, \vec{y}) = \vec{x} \times \vec{y}$$

is a moment map, with

$$\mu^{\vec{a}}(\vec{x}, \vec{y}) = \langle \mu(\vec{x}, \vec{y}), \vec{a} \rangle = (\vec{x} \times \vec{y}) \cdot \vec{a}.$$

The map μ is called the **angular momentum**.

Homework 17: Coadjoint Orbits

Let G be a Lie group, $\mathfrak{g}$ its Lie algebra and $\mathfrak{g}^*$ the dual vector space of $\mathfrak{g}$.

1. Let ${}^{\mathfrak{g}}X^{\#}$ be the vector field generated by $X \in \mathfrak{g}$ for the adjoint representation of G on $\mathfrak{g}$. Show that

$${}^{\mathfrak{g}}X_Y^{\#} = [X, Y] \quad \forall Y \in \mathfrak{g} .$$

2. Let $X^{\#}$ be the vector field generated by $X \in \mathfrak{g}$ for the coadjoint representation of G on $\mathfrak{g}^*$. Show that

$$\langle X_{\xi}^{\#}, Y \rangle = \langle \xi, [Y, X] \rangle \quad \forall Y \in \mathfrak{g} .$$

3. For any $\xi \in \mathfrak{g}^*$, define a skew-symmetric bilinear form on $\mathfrak{g}$ by

$$\omega_{\xi}(X, Y) := \langle \xi, [X, Y] \rangle .$$

Show that the kernel of ω_{ξ} is the Lie algebra $\mathfrak{g}_{\xi}$ of the stabilizer of ξ for the coadjoint representation.

4. Show that ω_{ξ} defines a nondegenerate 2-form on the tangent space at ξ to the coadjoint orbit through ξ .
5. Show that ω_{ξ} defines a closed 2-form on the orbit of ξ in $\mathfrak{g}^*$.

Hint: The tangent space to the orbit being generated by the vector fields $X^{\#}$, this is a consequence of the Jacobi identity in $\mathfrak{g}$.

This **canonical symplectic form** on the coadjoint orbits in $\mathfrak{g}^*$ is also known as the **Lie-Poisson** or **Kostant-Kirillov symplectic structure**.

6. The Lie algebra structure of $\mathfrak{g}$ defines a canonical Poisson structure on $\mathfrak{g}^*$:

$$\{f, g\}(\xi) := \langle \xi, [df_{\xi}, dg_{\xi}] \rangle$$

for $f, g \in C^{\infty}(\mathfrak{g}^*)$ and $\xi \in \mathfrak{g}^*$. Notice that $df_{\xi} : T_{\xi}\mathfrak{g}^* \simeq \mathfrak{g}^* \rightarrow \mathbb{R}$ is identified with an element of $\mathfrak{g} \simeq \mathfrak{g}^{**}$.

Check that $\{\cdot, \cdot\}$ satisfies the Leibniz rule:

$$\{f, gh\} = g\{f, h\} + h\{f, g\} .$$

7. Show that the **jacobiator**

$$J(f, g, h) := \{\{f, g\}, h\} + \{\{g, h\}, f\} + \{\{h, f\}, g\}$$

is a trivector field, i.e., J is a skew-symmetric trilinear map $C^{\infty}(\mathfrak{g}^*) \times C^{\infty}(\mathfrak{g}^*) \times C^{\infty}(\mathfrak{g}^*) \rightarrow C^{\infty}(\mathfrak{g}^*)$, which is a derivation in each argument.

Hint: Being a derivation amounts to the Leibniz rule from exercise 6.

8. Show that $J \equiv 0$, i.e., $\{\cdot, \cdot\}$ satisfies the Jacobi identity.

Hint: Follows from the Jacobi identity for $[\cdot, \cdot]$ in $\mathfrak{g}$. It is enough to check on coordinate functions.

Part IX

Symplectic Reduction

The phase space of a system of n particles is the space parametrizing the position and momenta of the particles. The mathematical model for the phase space is a symplectic manifold. Classical physicists realized that, whenever there is a symmetry group of dimension k acting on a mechanical system, then the number of degrees of freedom for the position and momenta of the particles may be reduced by $2k$. Symplectic reduction formulates this feature mathematically.

23 The Marsden-Weinstein-Meyer Theorem

23.1 Statement

Theorem 23.1 (Marsden-Weinstein-Meyer [77, 85]) *Let (M, ω, G, μ) be a hamiltonian G -space for a compact Lie group G . Let $i : \mu^{-1}(0) \hookrightarrow M$ be the inclusion map. Assume that G acts freely on $\mu^{-1}(0)$. Then*

- the orbit space $M_{\text{red}} = \mu^{-1}(0)/G$ is a manifold,
- $\pi : \mu^{-1}(0) \rightarrow M_{\text{red}}$ is a principal G -bundle, and
- there is a symplectic form ω_{red} on M_{red} satisfying $i^*\omega = \pi^*\omega_{\text{red}}$.

Definition 23.2 *The pair $(M_{\text{red}}, \omega_{\text{red}})$ is called the **reduction** of (M, ω) with respect to G, μ , or the **reduced space**, or the **symplectic quotient**, or the **Marsden-Weinstein-Meyer quotient**, etc.*

Low-brow proof for the case $G = S^1$ and $\dim M = 4$.

In this case the moment map is $\mu : M \rightarrow \mathbb{R}$. Let $p \in \mu^{-1}(0)$. Choose local coordinates:

- θ along the orbit through p ,
- μ given by the moment map, and
- η_1, η_2 pullback of coordinates on $\mu^{-1}(0)/S^1$.

Then the symplectic form can be written

$$\omega = A d\theta \wedge d\mu + B_j d\theta \wedge d\eta_j + C_j d\mu \wedge d\eta_j + D d\eta_1 \wedge d\eta_2 .$$

Since $d\mu = \iota\left(\frac{\partial}{\partial \theta}\right)\omega$, we must have $A = 1$, $B_j = 0$. Hence,

$$\omega = d\theta \wedge d\mu + C_j d\mu \wedge d\eta_j + D d\eta_1 \wedge d\eta_2 .$$

Since ω is symplectic, we must have $D \neq 0$. Therefore, $i^*\omega = D d\eta_1 \wedge d\eta_2$ is the pullback of a symplectic form on M_{red} . $\square$

The actual proof of the Marsden-Weinstein-Meyer theorem requires the following ingredients.

23.2 Ingredients

1. Let $\mathfrak{g}_p$ be the Lie algebra of the stabilizer of $p \in M$. Then $d\mu_p : T_p M \rightarrow \mathfrak{g}^*$ has

$$\begin{aligned} \ker d\mu_p &= (T_p \mathcal{O}_p)^{\omega_p} \\ \text{im } d\mu_p &= \mathfrak{g}_p^0 \end{aligned}$$

where $\mathcal{O}_p$ is the G -orbit through p , and $\mathfrak{g}_p^0 = \{\xi \in \mathfrak{g}^* \mid \langle \xi, X \rangle = 0, \forall X \in \mathfrak{g}_p\}$ is the annihilator of $\mathfrak{g}_p$.

Proof. Stare at the expression $\omega_p(X_p^\#, v) = \langle d\mu_p(v), X \rangle$, for all $v \in T_p M$ and all $X \in \mathfrak{g}$, and count dimensions. $\square$

Consequences:

- The action is locally free at p
 - $\iff \mathfrak{g}_p = \{0\}$
 - $\iff d\mu_p$ is surjective
 - $\iff p$ is a regular point of μ .
 - G acts freely on $\mu^{-1}(0)$
 - $\implies 0$ is a regular value of μ
 - $\implies \mu^{-1}(0)$ is a closed submanifold of M of codimension equal to $\dim G$.
 - G acts freely on $\mu^{-1}(0)$
 - $\implies T_p \mu^{-1}(0) = \ker d\mu_p$ (for $p \in \mu^{-1}(0)$)
 - $\implies T_p \mu^{-1}(0)$ and $T_p \mathcal{O}_p$ are symplectic orthocomplements in $T_p M$.
- In particular, the tangent space to the orbit through $p \in \mu^{-1}(0)$ is an isotropic subspace of $T_p M$. Hence, orbits in $\mu^{-1}(0)$ are isotropic.

Since any tangent vector to the orbit is the value of a vector field generated by the group, we can confirm that orbits are isotropic directly by computing, for any $X, Y \in \mathfrak{g}$ and any $p \in \mu^{-1}(0)$,

$$\begin{aligned} \omega_p(X_p^\#, Y_p^\#) &= \text{hamiltonian function for } [Y^\#, X^\#] \text{ at } p \\ &= \text{hamiltonian function for } [Y, X]^\# \text{ at } p \\ &= \mu^{[Y, X]}(p) = 0. \end{aligned}$$

2. **Lemma 23.3** *Let (V, ω) be a symplectic vector space. Suppose that I is an isotropic subspace, that is, $\omega|_I \equiv 0$. Then ω induces a canonical symplectic form Ω on I^ω/I .*

Proof. Let $u, v \in I^\omega$, and $[u], [v] \in I^\omega/I$. Define $\Omega([u], [v]) = \omega(u, v)$.

- Ω is well-defined:

$$\omega(u + i, v + j) = \omega(u, v) + \underbrace{\omega(u, j)}_0 + \underbrace{\omega(i, v)}_0 + \underbrace{\omega(i, j)}_0, \quad \forall i, j \in I.$$

- Ω is nondegenerate:

Suppose that $u \in I^\omega$ has $\omega(u, v) = 0$, for all $v \in I^\omega$.

Then $u \in (I^\omega)^\omega = I$, i.e., $[u] = 0$.

□

3. **Theorem 23.4** *If a compact Lie group G acts freely on a manifold M , then M/G is a manifold and the map $\pi : M \rightarrow M/G$ is a principal G -bundle.*

Proof. We will first show that, for any $p \in M$, the G -orbit through p is a compact embedded submanifold of M diffeomorphic to G .

Since the action is smooth, the evaluation map $\text{ev} : G \times M \rightarrow M$, $\text{ev}(g, p) = g \cdot p$, is smooth. Let $\text{ev}_p : G \rightarrow M$ be defined by $\text{ev}_p(g) = g \cdot p$. The map ev_p provides the embedding we seek:

The image of ev_p is the G -orbit through p . Injectivity of ev_p follows from the action of G being free. The map ev_p is proper because, if A is a compact, hence closed, subset of M , then its inverse image $(\text{ev}_p)^{-1}(A)$, being a closed subset of the compact Lie group G , is also compact. It remains to show that ev_p is an immersion. For $X \in \mathfrak{g} \simeq T_e G$, we have

$$d(\text{ev}_p)_e(X) = 0 \iff X_p^\# = 0 \iff X = 0 ,$$

as the action is free. We conclude that $d(\text{ev}_p)_e$ is injective. At any other point $g \in G$, for $X \in T_g G$, we have

$$d(\text{ev}_p)_g(X) = 0 \iff d(\text{ev}_p \circ R_g)_e \circ (dR_{g^{-1}})_g(X) = 0 ,$$

where $R_g : G \rightarrow G$ is right multiplication by g . But $\text{ev}_p \circ R_g = \text{ev}_{g \cdot p}$ has an injective differential at e , and $(dR_{g^{-1}})_g$ is an isomorphism. It follows that $d(\text{ev}_p)_g$ is always injective.

Exercise. Show that, even if the action is not free, the G -orbit through p is a compact embedded submanifold of M . In that case, the orbit is diffeomorphic to the quotient of G by the isotropy of p : $\mathcal{O}_p \simeq G/G_p$. ◇

Let S be a transverse section to $\mathcal{O}_p$ at p ; this is called a **slice**. Choose a coordinate system $x_1, \dots, x_n$ centered at p such that

$$\begin{array}{llll} \mathcal{O}_p \simeq G & : & x_1 & = \dots = x_k = 0 \\ S & : & x_{k+1} & = \dots = x_n = 0 . \end{array}$$

Let $S_\varepsilon = S \cap B_\varepsilon(0, \mathbb{R}^n)$ where $B_\varepsilon(0, \mathbb{R}^n)$ is the ball of radius ε centered at 0 in $\mathbb{R}^n$. Let $\eta : G \times S \rightarrow M$, $\eta(g, s) = g \cdot s$. Apply the following equivariant tubular neighborhood theorem.

Theorem 23.5 (Slice Theorem) *Let G be a compact Lie group acting on a manifold M such that G acts freely at $p \in M$. For sufficiently small ε , $\eta : G \times S_\varepsilon \rightarrow M$ maps $G \times S_\varepsilon$ diffeomorphically onto a G -invariant neighborhood $\mathcal{U}$ of the G -orbit through p .*

The proof of this slice theorem is sketched further below.

Corollary 23.6 *If the action of G is free at p , then the action is free on $\mathcal{U}$.*

Corollary 23.7 *The set of points where G acts freely is open.*

Corollary 23.8 *The set $G \times S_\varepsilon \simeq \mathcal{U}$ is G -invariant. Hence, the quotient $\mathcal{U}/G \simeq S_\varepsilon$ is smooth.*

Conclusion of the proof that M/G is a manifold and $\pi : M \rightarrow M/G$ is a smooth fiber map.

For $p \in M$, let $q = \pi(p) \in M/G$. Choose a G -invariant neighborhood $\mathcal{U}$ of p as in the slice theorem: $\mathcal{U} \simeq G \times S$ (where $S = S_\varepsilon$ for an appropriate ε). Then $\pi(\mathcal{U}) = \mathcal{U}/G =: \mathcal{V}$ is an open neighborhood of q in M/G . By the slice theorem, $S \xrightarrow{\cong} \mathcal{V}$ is a homeomorphism. We will use such neighborhoods $\mathcal{V}$ as charts on M/G . To show that the transition functions associated with these charts are smooth, consider two G -invariant open sets $\mathcal{U}_1, \mathcal{U}_2$ in M and corresponding slices S_1, S_2 of the G -action. Then $S_{12} = S_1 \cap \mathcal{U}_2$, $S_{21} = S_2 \cap \mathcal{U}_1$ are both slices for the G -action on $\mathcal{U}_1 \cap \mathcal{U}_2$. To compute the transition map $S_{12} \rightarrow S_{21}$, consider the diagram

$$\begin{array}{ccccc}
 S_{12} & \xrightarrow{\cong} & \text{id} \times S_{12} & \hookrightarrow & G \times S_{12} \\
 & & & & \searrow \cong \\
 & & & & \mathcal{U}_1 \cap \mathcal{U}_2 \\
 & & & & \nearrow \cong \\
 S_{21} & \xrightarrow{\cong} & \text{id} \times S_{21} & \hookrightarrow & G \times S_{21}
 \end{array}$$

Then the composition

$$S_{12} \hookrightarrow \mathcal{U}_1 \cap \mathcal{U}_2 \xrightarrow{\cong} G \times S_{21} \xrightarrow{pr} S_{21}$$

is smooth.

Finally, we need to show that $\pi : M \rightarrow M/G$ is a smooth fiber map. For $p \in M$, $q = \pi(p)$, choose a G -invariant neighborhood $\mathcal{U}$ of the G -orbit through p of the form $\eta : G \times S \xrightarrow{\cong} \mathcal{U}$. Then $\mathcal{V} = \mathcal{U}/G \simeq S$ is the corresponding neighborhood of q in M/G :

$$\begin{array}{ccccc}
 M \supseteq & \mathcal{U} & \xrightarrow{\eta} & G \times S & \simeq & G \times \mathcal{V} \\
 & \downarrow \pi & & & & \downarrow \\
 M/G \supseteq & \mathcal{V} & & = & & \mathcal{V}
 \end{array}$$

Since the projection on the right is smooth, π is smooth.

Exercise. Check that the transition functions for the bundle defined by π are smooth. $\diamond$

□

Sketch for the proof of the slice theorem. We need to show that, for ε sufficiently small, $\eta : G \times S_\varepsilon \rightarrow \mathcal{U}$ is a diffeomorphism where $\mathcal{U} \subseteq M$ is a G -invariant neighborhood of the G -orbit through p . Show that:

- (a) $d\eta_{(\text{id}, p)}$ is bijective.
- (b) Let G act on $G \times S$ by the product of its left action on G and trivial action on S . Then $\eta : G \times S \rightarrow M$ is G -equivariant.
- (c) $d\eta$ is bijective at all points of $G \times \{p\}$. This follows from (a) and (b).
- (d) The set $G \times \{p\}$ is compact, and $\eta : G \times S \rightarrow M$ is injective on $G \times \{p\}$ with $d\eta$ bijective at all these points. By the implicit function theorem, there is a neighborhood $\mathcal{U}_0$ of $G \times \{p\}$ in $G \times S$ such that η maps $\mathcal{U}_0$ diffeomorphically onto a neighborhood $\mathcal{U}$ of the G -orbit through p .
- (e) The sets $G \times S_\varepsilon$, varying ε , form a neighborhood base for $G \times \{p\}$ in $G \times S$. So in (d) we may take $\mathcal{U}_0 = G \times S_\varepsilon$.

□

23.3 Proof of the Marsden-Weinstein-Meyer Theorem

Since

$$\begin{aligned} G \text{ acts freely on } \mu^{-1}(0) &\implies d\mu_p \text{ is surjective for all } p \in \mu^{-1}(0) \\ &\implies 0 \text{ is a regular value} \\ &\implies \mu^{-1}(0) \text{ is a submanifold of codimension} = \dim G \end{aligned}$$

for the first two parts of the Marsden-Weinstein-Meyer theorem it is enough to apply the third ingredient from Section 23.2 to the free action of G on $\mu^{-1}(0)$.

At $p \in \mu^{-1}(0)$ the tangent space to the orbit $T_p\mathcal{O}_p$ is an isotropic subspace of the symplectic vector space (T_pM, ω_p) , i.e., $T_p\mathcal{O}_p \subseteq (T_p\mathcal{O}_p)^\omega$.

$$(T_p\mathcal{O}_p)^\omega = \ker d\mu_p = T_p\mu^{-1}(0).$$

The lemma (second ingredient) gives a canonical symplectic structure on the quotient $T_p\mu^{-1}(0)/T_p\mathcal{O}_p$. The point $[p] \in M_{\text{red}} = \mu^{-1}(0)/G$ has tangent space $T_{[p]}M_{\text{red}} \simeq T_p\mu^{-1}(0)/T_p\mathcal{O}_p$. Thus the lemma defines a nondegenerate 2-form ω_{red} on M_{red} . This is well-defined because ω is G -invariant.

By construction $i^*\omega = \pi^*\omega_{\text{red}}$ where

$$\begin{array}{ccc} \mu^{-1}(0) & \xhookrightarrow{i} & M \\ \downarrow \pi & & \\ M_{\text{red}} & & \end{array}$$

Hence, $\pi^* d\omega_{\text{red}} = d\pi^* \omega_{\text{red}} = di^* \omega = i^* d\omega = 0$. The closedness of ω_{red} follows from the injectivity of π^* . $\square$

Remark. Suppose that another Lie group H acts on (M, ω) in a hamiltonian way with moment map $\phi : M \rightarrow \mathfrak{h}^*$. If the H -action commutes with the G -action, and if ϕ is G -invariant, then M_{red} inherits a hamiltonian action of H , with moment map $\phi_{\text{red}} : M_{\text{red}} \rightarrow \mathfrak{h}^*$ satisfying $\phi_{\text{red}} \circ \pi = \phi \circ i$. $\diamond$

24 Reduction

24.1 Noether Principle

Let (M, ω, G, μ) be a hamiltonian G -space.

Theorem 24.1 (Noether) *A function $f : M \rightarrow \mathbb{R}$ is G -invariant if and only if μ is constant on the trajectories of the hamiltonian vector field of f .*

Proof. Let v_f be the hamiltonian vector field of f . Let $X \in \mathfrak{g}$ and $\mu^X = \langle \mu, X \rangle : M \rightarrow \mathbb{R}$. We have

$$\begin{aligned} \mathcal{L}_{v_f} \mu^X &= \iota_{v_f} d\mu^X = \iota_{v_f} \iota_{X\#} \omega \\ &= -\iota_{X\#} \iota_{v_f} \omega = -\iota_{X\#} df \\ &= -\mathcal{L}_{X\#} f = 0 \end{aligned}$$

because f is G -invariant. □

Definition 24.2 *A G -invariant function $f : M \rightarrow \mathbb{R}$ is called an **integral of motion** of (M, ω, G, μ) . If μ is constant on the trajectories of a hamiltonian vector field v_f , then the corresponding one-parameter group of diffeomorphisms $\{\exp tv_f \mid t \in \mathbb{R}\}$ is called a **symmetry** of (M, ω, G, μ) .*

The **Noether principle** asserts that there is a one-to-one correspondence between symmetries and integrals of motion.

24.2 Elementary Theory of Reduction

Finding a symmetry for a $2n$ -dimensional mechanical problem may reduce it to a $(2n - 2)$ -dimensional problem as follows: an integral of motion f for a $2n$ -dimensional hamiltonian system (M, ω, H) may enable us to understand the trajectories of this system in terms of the trajectories of a $(2n - 2)$ -dimensional hamiltonian system $(M_{\text{red}}, \omega_{\text{red}}, H_{\text{red}})$. To make this precise, we will describe this process locally. Suppose that $\mathcal{U}$ is an open set in M with Darboux coordinates $x_1, \dots, x_n, \xi_1, \dots, \xi_n$ such that $f = \xi_n$ for this chart, and write H in these coordinates: $H = H(x_1, \dots, x_n, \xi_1, \dots, \xi_n)$. Then

$$\xi_n \text{ is an integral of motion} \implies \begin{cases} \text{the trajectories of } v_H \text{ lie on the} \\ \text{hyperplane } \xi_n = \text{constant} \\ \{\xi_n, H\} = 0 = -\frac{\partial H}{\partial x_n} \\ \implies H = H(x_1, \dots, x_{n-1}, \xi_1, \dots, \xi_n) . \end{cases}$$

If we set $\xi_n = c$, the motion of the system on this hyperplane is described by the following Hamilton equations:

$$\left\{ \begin{array}{l} \frac{dx_1}{dt} = \frac{\partial H}{\partial \xi_1} (x_1, \dots, x_{n-1}, \xi_1, \dots, \xi_{n-1}, c) \\ \vdots \\ \frac{dx_{n-1}}{dt} = \frac{\partial H}{\partial \xi_{n-1}} (x_1, \dots, x_{n-1}, \xi_1, \dots, \xi_{n-1}, c) \\ \frac{d\xi_1}{dt} = -\frac{\partial H}{\partial x_1} (x_1, \dots, x_{n-1}, \xi_1, \dots, \xi_{n-1}, c) \\ \vdots \\ \frac{d\xi_{n-1}}{dt} = -\frac{\partial H}{\partial x_{n-1}} (x_1, \dots, x_{n-1}, \xi_1, \dots, \xi_{n-1}, c) \end{array} \right.$$

$$\frac{dx_n}{dt} = \frac{\partial H}{\partial \xi_n}$$

$$\frac{d\xi_n}{dt} = -\frac{\partial H}{\partial x_n} = 0 .$$

The **reduced phase space** is

$$\mathcal{U}_{\text{red}} = \{ (x_1, \dots, x_{n-1}, \xi_1, \dots, \xi_{n-1}) \in \mathbb{R}^{2n-2} \mid (x_1, \dots, x_{n-1}, a, \xi_1, \dots, \xi_{n-1}, c) \in \mathcal{U} \text{ for some } a \} .$$

The **reduced hamiltonian** is

$$\begin{aligned} H_{\text{red}} : \mathcal{U}_{\text{red}} &\longrightarrow \mathbb{R} , \\ H_{\text{red}}(x_1, \dots, x_{n-1}, \xi_1, \dots, \xi_{n-1}) &= H(x_1, \dots, x_{n-1}, \xi_1, \dots, \xi_{n-1}, c) . \end{aligned}$$

In order to find the trajectories of the original system on the hypersurface $\xi_n = c$, we look for the trajectories

$$x_1(t), \dots, x_{n-1}(t), \xi_1(t), \dots, \xi_{n-1}(t)$$

of the reduced system on $\mathcal{U}_{\text{red}}$. We integrate the equation

$$\frac{dx_n}{dt}(t) = \frac{\partial H}{\partial \xi_n}(x_1(t), \dots, x_{n-1}(t), \xi_1(t), \dots, \xi_{n-1}(t), c)$$

to obtain the original trajectories

$$\left\{ \begin{array}{l} x_n(t) = x_n(0) + \int_0^t \frac{\partial H}{\partial \xi_n}(\dots) dt \\ \xi_n(t) = c . \end{array} \right.$$

24.3 Reduction for Product Groups

Let G_1 and G_2 be compact connected Lie groups and let $G = G_1 \times G_2$. Then

$$\mathfrak{g} = \mathfrak{g}_1 \oplus \mathfrak{g}_2 \quad \text{and} \quad \mathfrak{g}^* = \mathfrak{g}_1^* \oplus \mathfrak{g}_2^* .$$

Suppose that (M, ω, G, ψ) is a hamiltonian G -space with moment map

$$\psi : M \longrightarrow \mathfrak{g}_1^* \oplus \mathfrak{g}_2^* .$$

Write $\psi = (\psi_1, \psi_2)$ where $\psi_i : M \rightarrow \mathfrak{g}_i^*$ for $i = 1, 2$. The fact that ψ is equivariant implies that ψ_1 is invariant under G_2 and ψ_2 is invariant under G_1 . Now reduce (M, ω) with respect to the G_1 -action. Let

$$Z_1 = \psi_1^{-1}(0) .$$

Assume that G_1 acts freely on Z_1 . Let $M_1 = Z_1/G_1$ be the reduced space and let ω_1 be the corresponding reduced symplectic form. The action of G_2 on Z_1 commutes with the G_1 -action. Since G_2 preserves ω , it follows that G_2 acts symplectically on (M_1, ω_1) . Since G_1 preserves ψ_2 , G_1 also preserves $\psi_2 \circ \iota_1 : Z_1 \rightarrow \mathfrak{g}_2^*$, where $\iota_1 : Z_1 \hookrightarrow M$ is inclusion. Thus $\psi_2 \circ \iota_1$ is constant on fibers of $Z_1 \xrightarrow{p_1} M_1$. We conclude that there exists a smooth map $\mu_2 : M_1 \rightarrow \mathfrak{g}_2^*$ such that $\mu_2 \circ p_1 = \psi_2 \circ \iota_1$.

Exercise. Show that:

- (a) the map μ_2 is a moment map for the action of G_2 on (M_1, ω_1) , and
- (b) if G acts freely on $\psi^{-1}(0, 0)$, then G_2 acts freely on $\mu_2^{-1}(0)$, and there is a natural symplectomorphism

$$\mu_2^{-1}(0)/G_2 \simeq \psi^{-1}(0, 0)/G .$$

◇

This technique of performing reduction with respect to one factor of a product group at a time is called **reduction in stages**. It may be extended to reduction by a normal subgroup $H \subset G$ and by the corresponding quotient group G/H .

24.4 Reduction at Other Levels

Suppose that a compact Lie group G acts on a symplectic manifold (M, ω) in a hamiltonian way with moment map $\mu : M \rightarrow \mathfrak{g}^*$. Let $\xi \in \mathfrak{g}^*$.

To reduce at the level ξ of μ , we need $\mu^{-1}(\xi)$ to be preserved by G , or else take the G -orbit of $\mu^{-1}(\xi)$, or else take the quotient by the maximal subgroup of G which preserves $\mu^{-1}(\xi)$.

Since μ is equivariant,

$$\begin{aligned} G \text{ preserves } \mu^{-1}(\xi) &\iff G \text{ preserves } \xi \\ &\iff \text{Ad}_g^* \xi = \xi, \forall g \in G . \end{aligned}$$

Of course the level 0 is always preserved. Also, when G is a torus, any level is preserved and reduction at ξ for the moment map μ , is equivalent to reduction at 0 for a shifted moment map $\phi : M \rightarrow \mathfrak{g}^*$, $\phi(p) := \mu(p) - \xi$.

Let $\mathcal{O}$ be a coadjoint orbit in $\mathfrak{g}^*$ equipped with the **canonical symplectic form** (also known as the **Kostant-Kirillov symplectic form** or the **Lie-Poisson symplectic form**) $\omega_{\mathcal{O}}$ defined in Homework 17. Let $\mathcal{O}^-$ be the orbit $\mathcal{O}$ equipped with $-\omega_{\mathcal{O}}$. The natural product action of G on $M \times \mathcal{O}^-$ is hamiltonian with moment map $\mu_{\mathcal{O}}(p, \xi) = \mu(p) - \xi$. If the Marsden-Weinstein-Meyer hypothesis is satisfied for $M \times \mathcal{O}^-$, then one obtains a **reduced space with respect to the coadjoint orbit $\mathcal{O}$** .

24.5 Orbifolds

Example. Let $G = \mathbb{T}^n$ be an n -torus. For any $\xi \in (\mathfrak{t}^n)^*$, $\mu^{-1}(\xi)$ is preserved by the $\mathbb{T}^n$ -action. Suppose that ξ is a regular value of μ . (By Sard's theorem, the singular values of μ form a set of measure zero.) Then $\mu^{-1}(\xi)$ is a submanifold of codimension n . Note that

$$\begin{aligned} \xi \text{ regular} &\implies d\mu_p \text{ is surjective at all } p \in \mu^{-1}(\xi) \\ &\implies \mathfrak{g}_p = 0 \text{ for all } p \in \mu^{-1}(\xi) \\ &\implies \text{the stabilizers on } \mu^{-1}(\xi) \text{ are finite} \\ &\implies \mu^{-1}(\xi)/G \text{ is an } \mathbf{orbifold} \text{ [91, 92]}. \end{aligned}$$

Let G_p be the stabilizer of p . By the slice theorem (Lecture 23), $\mu^{-1}(\xi)/G$ is modeled by S/G_p , where S is a G_p -invariant disk in $\mu^{-1}(\xi)$ through p and transverse to $\mathcal{O}_p$. Hence, locally $\mu^{-1}(\xi)/G$ looks indeed like $\mathbb{R}^n$ divided by a finite group action. $\diamond$

Example. Consider the S^1 -action on $\mathbb{C}^2$ given by $e^{i\theta} \cdot (z_1, z_2) = (e^{ik\theta} z_1, e^{i\ell\theta} z_2)$ for some fixed integer $k \geq 2$. This is hamiltonian with moment map

$$\begin{aligned} \mu : \quad \mathbb{C}^2 &\longrightarrow \mathbb{R} \\ (z_1, z_2) &\longmapsto -\frac{1}{2}(k|z_1|^2 + |z_2|^2). \end{aligned}$$

Any $\xi < 0$ is a regular value and $\mu^{-1}(\xi)$ is a 3-dimensional ellipsoid. The stabilizer of $(z_1, z_2) \in \mu^{-1}(\xi)$ is $\{1\}$ if $z_2 \neq 0$, and is $\mathbb{Z}_k = \left\{ e^{i\frac{2\pi\ell}{k}} \mid \ell = 0, 1, \dots, k-1 \right\}$ if $z_2 = 0$. The reduced space $\mu^{-1}(\xi)/S^1$ is called a **teardrop orbifold** or **conehead**; it has one **cone** (also known as a **dunce cap**) singularity of type k (with cone angle $\frac{2\pi}{k}$). $\diamond$

Example. Let S^1 act on $\mathbb{C}^2$ by $e^{i\theta} \cdot (z_1, z_2) = (e^{ik\theta} z_1, e^{i\ell\theta} z_2)$ for some integers $k, \ell \geq 2$. Suppose that k and ℓ are relatively prime. Then

$$\begin{aligned} (z_1, 0) &\text{ has stabilizer } \mathbb{Z}_k & (\text{for } z_1 \neq 0), \\ (0, z_2) &\text{ has stabilizer } \mathbb{Z}_\ell & (\text{for } z_2 \neq 0), \\ (z_1, z_2) &\text{ has stabilizer } \{1\} & (\text{for } z_1, z_2 \neq 0). \end{aligned}$$

The quotient $\mu^{-1}(\xi)/S^1$ is called a **football** orbifold. It has two cone singularities, one of type k and another of type ℓ . $\diamond$

Example. More generally, the reduced spaces of S^1 acting on $\mathbb{C}^n$ by

$$e^{i\theta} \cdot (z_1, \dots, z_n) = (e^{ik_1\theta} z_1, \dots, e^{ik_n\theta} z_n) ,$$

are called **weighted** (or **twisted**) **projective spaces**. $\diamond$

Homework 18: Spherical Pendulum

This set of problems is from [53].

The **spherical pendulum** is a mechanical system consisting of a massless rigid rod of length l , fixed at one end, whereas the other end has a plumb bob of mass m , which may oscillate freely in all directions. Assume that the force of gravity is constant pointing vertically downwards, and that this is the only external force acting on this system.

Let φ, θ ($0 < \varphi < \pi$, $0 < \theta < 2\pi$) be spherical coordinates for the bob. For simplicity assume that $m = l = 1$.

1. Let η, ξ be the coordinates along the fibers of T^*S^2 induced by the spherical coordinates φ, θ on S^2 . Show that the function $H : T^*S^2 \rightarrow \mathbb{R}$ given by

$$H(\varphi, \theta, \eta, \xi) = \frac{1}{2} \left(\eta^2 + \frac{\xi^2}{(\sin \varphi)^2} \right) + \cos \varphi ,$$

is an appropriate hamiltonian function to describe the spherical pendulum.

2. Compute the critical points of the function H . Show that, on S^2 , there are exactly two critical points: s (where H has a minimum) and u . These points are called the **stable** and **unstable** points of H , respectively. Justify this terminology, i.e., show that a trajectory whose initial point is close to s stays close to s forever, and show that this is not the case for u . What is happening physically?
3. Show that the group of rotations about the vertical axis is a group of symmetries of the spherical pendulum.

Show that, in the coordinates above, the integral of motion associated with these symmetries is the function

$$J(\varphi, \theta, \eta, \xi) = \xi .$$

Give a more coordinate-independent description of J , one that makes sense also on the cotangent fibers above the North and South poles.

4. Locate all points $p \in T^*S^2$ where dH_p and dJ_p are linearly dependent:
- Clearly, the two critical points s and u belong to this set. Show that these are the only two points where $dH_p = dJ_p = 0$.
 - Show that, if $x \in S^2$ is in the southern hemisphere ($x_3 < 0$), then there exist exactly two points, $p_+ = (x, \eta, \xi)$ and $p_- = (x, -\eta, -\xi)$, in the cotangent fiber above x where dH_p and dJ_p are linearly dependent.
 - Show that dH_p and dJ_p are linearly dependent along the trajectory of the hamiltonian vector field of H through p_+ .
Conclude that this trajectory is also a trajectory of the hamiltonian vector field of J , and, hence, that its projection onto S^2 is a latitudinal circle (of the form $x_3 = \text{constant}$).
Show that the projection of the trajectory through p_- is the same latitudinal circle traced in the opposite direction.
5. Show that any nonzero value j is a regular value of J , and that S^1 acts freely on the level set $J = j$. What happens on the cotangent fibers above the North and South poles?
6. For $j \neq 0$ describe the reduced system and sketch the level curves of the reduced hamiltonian.
7. Show that the integral curves of the original system on the level set $J = j$ can be obtained from those of the reduced system by “quadrature”, in other words, by a simple integration.
8. Show that the reduced system for $j \neq 0$ has exactly one equilibrium point. Show that the corresponding relative equilibrium for the original system is one of the horizontal curves in exercise 4.
9. The **energy-momentum map** is the map $(H, J) : T^*S^2 \rightarrow \mathbb{R}^2$. Show that, if $j \neq 0$, the level set $(H, J) = (h, j)$ of the energy-momentum map is either a circle (in which case it is one of the horizontal curves in exercise 4), or a two-torus. Show that the projection onto the configuration space of the two-torus is an annular region on S^2 .

Part X

Moment Maps Revisited

Moment maps and symplectic reduction have been finding infinite-dimensional incarnations with amazing consequences for differential geometry. Lecture 25 sketches the symplectic approach of Atiyah and Bott to Yang-Mills theory.

Lecture 27 describes the convexity of the image of a torus moment map, one of the most striking geometric characteristics of moment maps.

25 Moment Map in Gauge Theory

25.1 Connections on a Principal Bundle

Let G be a Lie group and B a manifold.

Definition 25.1 A **principal G -bundle over B** is a manifold P with a smooth map $\pi : P \rightarrow B$ satisfying the following conditions:

- (a) G acts freely on P (on the left),
- (b) B is the orbit space for this action and π is the point-orbit projection, and
- (c) there is an open covering of B , such that, to each set $\mathcal{U}$ in that covering corresponds a map $\varphi_{\mathcal{U}} : \pi^{-1}(\mathcal{U}) \rightarrow \mathcal{U} \times G$ with

$$\varphi_{\mathcal{U}}(p) = (\pi(p), s_{\mathcal{U}}(p)) \quad \text{and} \quad s_{\mathcal{U}}(g \cdot p) = g \cdot s_{\mathcal{U}}(p), \quad \forall p \in \pi^{-1}(\mathcal{U}).$$

The G -valued maps $s_{\mathcal{U}}$ are determined by the corresponding $\varphi_{\mathcal{U}}$. Condition (c) is called the property of being **locally trivial**.

If P with map $\pi : P \rightarrow B$ is a principal G -bundle over B , then the manifold B is called the **base**, the manifold P is called the **total space**, the Lie group G is called the **structure group**, and the map π is called the **projection**. This principal bundle is also represented by the following diagram:

$$\begin{array}{ccc} G & \hookrightarrow & P \\ & & \downarrow \pi \\ & & B \end{array}$$

Example. Let P be the 3-sphere regarded as unit vectors in $\mathbb{C}^2$:

$$P = S^3 = \{(z_1, z_2) \in \mathbb{C}^2 : |z_1|^2 + |z_2|^2 = 1\}.$$

Let G be the circle group, where $e^{i\theta} \in S^1$ acts on S^3 by complex multiplication,

$$(z_1, z_2) \mapsto (e^{i\theta} z_1, e^{i\theta} z_2) .$$

Then the quotient space B is the first complex projective space, that is, the two-sphere. This data forms a principal S^1 -bundle, known as the **Hopf fibration**:

$$\begin{array}{ccc} S^1 & \hookrightarrow & S^3 \\ & & \downarrow \pi \\ & & S^2 \end{array}$$

◇

An action $\psi : G \rightarrow \text{Diff}(P)$ induces an infinitesimal action

$$\begin{array}{ccc} d\psi : & \mathfrak{g} & \longrightarrow \chi(P) \\ X & \longmapsto & X^\# = \text{vector field generated by the} \\ & & \text{one-parameter group } \{\exp tX(e) \mid t \in \mathbb{R}\} . \end{array}$$

From now on, fix a basis $X_1, \dots, X_k$ of $\mathfrak{g}$.

Let P be a principal G -bundle over B . Since the G -action is free, the vector fields $X_1^\#, \dots, X_k^\#$ are linearly independent at each $p \in P$. The **vertical bundle** V is the rank k subbundle of TP generated by $X_1^\#, \dots, X_k^\#$.

Exercise. Check that the vertical bundle V is the set of vectors tangent to P which lie in the kernel of the derivative of the bundle projection π . (This shows that V is independent of the choice of basis for $\mathfrak{g}$.) ◇

Definition 25.2 A (**Ehresmann**) **connection** on a principal bundle P is a choice of a splitting

$$TP = V \oplus H ,$$

where H is a G -invariant subbundle of TP complementary to the vertical bundle V . The bundle H is called the **horizontal bundle**.

25.2 Connection and Curvature Forms

A connection on a principal bundle P may be equivalently described in terms of 1-forms.

Definition 25.3 A **connection form** on a principal bundle P is a Lie-algebra-valued 1-form

$$A = \sum_{i=1}^k A_i \otimes X_i \quad \in \Omega^1(P) \otimes \mathfrak{g}$$

such that:

- (a) A is G -invariant, with respect to the product action of G on $\Omega^1(P)$ (induced by the action on P) and on $\mathfrak{g}$ (the adjoint representation), and
- (b) A is vertical, in the sense that $\iota_{X\#} A = X$ for any $X \in \mathfrak{g}$.

Exercise. Show that a connection $TP = V \oplus H$ determines a connection form A and vice-versa by the formula

$$H = \ker A = \{v \in TP \mid \iota_v A = 0\} .$$

◇

Given a connection on P , the splitting $TP = V \oplus H$ induces the following splittings for bundles:

$$\begin{aligned} T^*P &= V^* \oplus H^* \\ \wedge^2 T^*P &= (\wedge^2 V^*) \oplus (V^* \wedge H^*) \oplus (\wedge^2 H^*) \\ &\vdots \end{aligned}$$

and for their sections:

$$\begin{aligned} \Omega^1(P) &= \Omega_{\text{vert}}^1(P) \oplus \Omega_{\text{horiz}}^1(P) \\ \Omega^2(P) &= \Omega_{\text{vert}}^2(P) \oplus \Omega_{\text{mix}}^2(P) \oplus \Omega_{\text{horiz}}^2(P) \\ &\vdots \end{aligned}$$

The corresponding connection form A is in $\Omega_{\text{vert}}^1 \otimes \mathfrak{g}$. Its exterior derivative dA is in

$$\Omega^2(P) \otimes \mathfrak{g} = (\Omega_{\text{vert}}^2 \oplus \Omega_{\text{mix}}^2 \oplus \Omega_{\text{horiz}}^2) \otimes \mathfrak{g} ,$$

and thus decomposes into three components,

$$dA = (dA)_{\text{vert}} + (dA)_{\text{mix}} + (dA)_{\text{horiz}} .$$

Exercise. Check that:

- (a) $(dA)_{\text{vert}}(X, Y) = [X, Y]$, i.e., $(dA)_{\text{vert}} = \frac{1}{2} \sum_{i, \ell, m} c_{\ell m}^i A_\ell \wedge A_m \otimes X_i$, where

the $c_{\ell m}^i$'s are the **structure constants** of the Lie algebra with respect to the chosen basis, and defined by $[X_\ell, X_m] = \sum_{i, \ell, m} c_{\ell m}^i X_i$;

- (b) $(dA)_{\text{mix}} = 0$.

◇

According to the previous exercise, the relevance of dA may come only from its horizontal component.

Definition 25.4 The **curvature form** of a connection is the horizontal component of its connection form. I.e., if A is the connection form, then

$$\text{curv } A = (dA)_{\text{horiz}} \in \Omega_{\text{horiz}}^2 \otimes \mathfrak{g} .$$

Definition 25.5 A connection is called **flat** if its curvature is zero.

25.3 Symplectic Structure on the Space of Connections

Let P be a principal G -bundle over B . If A is a connection form on P , and if $a \in \Omega_{\text{horiz}}^1 \otimes \mathfrak{g}$ is G -invariant for the product action, then it is easy to check that $A + a$ is also a connection form on P . Reciprocally, any two connection forms on P differ by an $a \in (\Omega_{\text{horiz}}^1 \otimes \mathfrak{g})^G$. We conclude that the set $\mathcal{A}$ of all connections on the principal G -bundle P is an affine space modeled on the linear space

$$\mathfrak{a} = (\Omega_{\text{horiz}}^1 \otimes \mathfrak{g})^G .$$

Now let P be a principal G -bundle over a compact oriented 2-dimensional riemannian manifold B (for instance, B is a Riemann surface). Suppose that the group G is compact or semisimple. Atiyah and Bott [7] noticed that the corresponding space $\mathcal{A}$ of all connections may be treated as an *infinite-dimensional symplectic manifold*. This will require choosing a G -invariant inner product $\langle \cdot, \cdot \rangle$ on $\mathfrak{g}$, which always exists, either by averaging any inner product when G is compact, or by using the Killing form on semisimple groups.

Since $\mathcal{A}$ is an affine space, its tangent space at any point A is identified with the model linear space $\mathfrak{a}$. With respect to a basis $X_1, \dots, X_k$ for the Lie algebra $\mathfrak{g}$, elements $a, b \in \mathfrak{a}^{14}$ are written

$$a = \sum a_i \otimes X_i \quad \text{and} \quad b = \sum b_i \otimes X_i .$$

If we wedge a and b , and then integrate over B using the riemannian volume, we obtain a real number:

$$\begin{aligned} \omega : \mathfrak{a} \times \mathfrak{a} &\longrightarrow (\Omega_{\text{horiz}}^2(P))^G \simeq \Omega^2(B) \longrightarrow \mathbb{R} \\ (a, b) &\longmapsto \sum_{i,j} a_i \wedge b_j \langle X_i, X_j \rangle \longmapsto \int_B \sum_{i,j} a_i \wedge b_j \langle X_i, X_j \rangle . \end{aligned}$$

We have used that the pullback $\pi^* : \Omega^2(B) \rightarrow \Omega^2(P)$ is an isomorphism onto its image $(\Omega_{\text{horiz}}^2(P))^G$.

Exercise. Show that if $\omega(a, b) = 0$ for all $b \in \mathfrak{a}$, then a must be zero. $\diamond$

The map ω is nondegenerate, skew-symmetric, bilinear and constant in the sense that it does not depend on the base point A . Therefore, it has the right to be called a symplectic form on $\mathcal{A}$, so the pair $(\mathcal{A}, \omega)$ is an infinite-dimensional symplectic manifold.

25.4 Action of the Gauge Group

Let P be a principal G -bundle over B . A diffeomorphism $f : P \rightarrow P$ commuting with the G -action determines a diffeomorphism $f_{\text{basic}} : B \rightarrow B$ by projection.

¹⁴The choice of symbols is in honor of Atiyah and Bott!

Definition 25.6 A diffeomorphism $f : P \rightarrow P$ commuting with the G -action is a **gauge transformation** if the induced f_{basic} is the identity. The **gauge group** of P is the group $\mathcal{G}$ of all gauge transformations of P .

The derivative of an $f \in \mathcal{G}$ takes a connection $TP = V \oplus H$ to another connection $TP = V \oplus H_f$, and thus induces an action of $\mathcal{G}$ in the space $\mathcal{A}$ of all connections. Recall that $\mathcal{A}$ has a symplectic form ω . Atiyah and Bott [7] noticed that the action of $\mathcal{G}$ on $(\mathcal{A}, \omega)$ is hamiltonian, where the moment map (appropriately interpreted) is the map

$$\begin{aligned} \mu : \mathcal{A} &\longrightarrow (\Omega^2(P) \otimes \mathfrak{g})^G \\ A &\longmapsto \text{curv } A, \end{aligned}$$

i.e., the moment map “is” the curvature! We will describe this construction in detail for the case of circle bundles in the next section.

Remark. The reduced space at level zero

$$\mathcal{M} = \mu^{-1}(0)/\mathcal{G}$$

is the space of flat connections modulo gauge equivalence, known as the **moduli space of flat connections**. It turns out that $\mathcal{M}$ is a finite-dimensional symplectic orbifold. $\diamond$

25.5 Case of Circle Bundles

What does the Atiyah-Bott construction of the previous section look like for the case when $G = S^1$?

$$\begin{array}{ccc} S^1 & \hookrightarrow & P \\ & & \downarrow \pi \\ & & B \end{array}$$

Let v be the generator of the S^1 -action on P , corresponding to the basis 1 of $\mathfrak{g} \simeq \mathbb{R}$. A connection form on P is a usual 1-form $A \in \Omega^1(P)$ such that

$$\mathcal{L}_v A = 0 \quad \text{and} \quad \iota_v A = 1.$$

If we fix one particular connection A_0 , then any other connection is of the form $A = A_0 + a$ for some $a \in \mathfrak{a} = (\Omega^1_{\text{horiz}}(P))^G = \Omega^1(B)$. The symplectic form on $\mathfrak{a} = \Omega^1(B)$ is simply

$$\begin{aligned} \omega : \mathfrak{a} \times \mathfrak{a} &\longrightarrow \mathbb{R} \\ (a, b) &\longmapsto \int_B \underbrace{a \wedge b}_{\in \Omega^2(B)}. \end{aligned}$$

The gauge group is $\mathcal{G} = \text{Maps}(B, S^1)$, because a gauge transformation is multiplication by some element of S^1 over each point in B :

$$\begin{aligned} \phi : \quad \mathcal{G} &\longrightarrow \text{Diff}(P) \\ h : B \rightarrow S^1 &\longmapsto \phi_h : \begin{array}{l} P \rightarrow P \\ p \rightarrow h(\pi(p)) \cdot p \end{array} \end{aligned}$$

The Lie algebra of $\mathcal{G}$ is

$$\text{Lie } \mathcal{G} = \text{Maps}(B, \mathbb{R}) = C^\infty(B) .$$

Its dual space is

$$(\text{Lie } \mathcal{G})^* = \Omega^2(B) ,$$

where the duality is provided by integration over B

$$\begin{aligned} C^\infty(B) \times \Omega^2(B) &\longrightarrow \mathbb{R} \\ (h, \beta) &\longmapsto \int_B h\beta . \end{aligned}$$

(it is topological or smooth duality, as opposed to algebraic duality) .

The gauge group acts on the space of all connections by

$$\begin{aligned} \mathcal{G} &\longrightarrow \text{Diff}(\mathcal{A}) \\ h(x) = e^{i\theta(x)} &\longmapsto (A \mapsto \underbrace{A - \pi^* d\theta}_{\in \mathfrak{a}}) \end{aligned}$$

Exercise. Check the previous assertion about the action on connections.

Hint: First deal with the case where $P = S^1 \times B$ is a trivial bundle, in which case $h \in \mathcal{G}$ acts on P by

$$\phi_h : (t, x) \longmapsto (t + \theta(x), x) ,$$

and where every connection can be written $A = dt + \beta$, with $\beta \in \Omega^1(B)$. A gauge transformation $h \in \mathcal{G}$ acts on $\mathcal{A}$ by

$$A \longmapsto \phi_{h^{-1}}^*(A) .$$

◇

The infinitesimal action of $\mathcal{G}$ on $\mathcal{A}$ is

$$\begin{aligned} d\phi : \text{Lie } \mathcal{G} &\longrightarrow \chi(\mathcal{A}) \\ X &\longmapsto X^\# = \text{vector field described by the transformation} \\ &\quad (A \mapsto \underbrace{A - dX}_{\in \Omega^1(B) = \mathfrak{a}}) \end{aligned}$$

so that $X^\# = -dX$.

Finally, we will check that

$$\begin{aligned}\mu : \mathcal{A} &\longrightarrow (\text{Lie } \mathcal{G})^* = \Omega^2(B) \\ A &\longmapsto \text{curv } A\end{aligned}$$

is indeed a moment map for the action of the gauge group on $\mathcal{A}$.

Exercise. Check that in this case:

- (a) $\text{curv } A = dA \in (\Omega_{\text{horiz}}^2(P))^G = \Omega^2(B)$,
- (b) μ is $\mathcal{G}$ -invariant.

◇

The previous exercise takes care of the equivariance condition, since the action of $\mathcal{G}$ on $\Omega^2(B)$ is trivial.

Take any $X \in \text{Lie } \mathcal{G} = C^\infty(B)$. We need to check that

$$d\mu^X(a) = \omega(X^\#, a), \quad \forall a \in \Omega^1(B). \quad (\star)$$

As for the left-hand side of $(\star)$, the map μ^X ,

$$\begin{aligned}\mu^X : \mathcal{A} &\longrightarrow \mathbb{R} \\ A &\longmapsto \left\langle \underbrace{X}_{\in C^\infty(B)}, \underbrace{dA}_{\in \Omega^2(B)} \right\rangle = \int_B X \cdot dA,\end{aligned}$$

is linear in A . Consequently,

$$\begin{aligned}d\mu^X : \mathfrak{a} &\longrightarrow \mathbb{R} \\ a &\longmapsto \int_B X \cdot da.\end{aligned}$$

As for the right-hand side of $(\star)$, by definition of ω , we have

$$\omega(X^\#, a) = \int_B X^\# \cdot a = - \int_B dX \cdot a.$$

But, by Stokes theorem, the last integral is

$$- \int_B dX \cdot a = \int_B X \cdot da,$$

so we are done in proving that μ is the moment map.

Homework 19: Examples of Moment Maps

1. Suppose that a Lie group G acts in a hamiltonian way on two symplectic manifolds (M_j, ω_j) , $j = 1, 2$, with moment maps $\mu_j : M_j \rightarrow \mathfrak{g}^*$. Prove that the diagonal action of G on $M_1 \times M_2$ is hamiltonian with moment map $\mu : M_1 \times M_2 \rightarrow \mathfrak{g}^*$ given by

$$\mu(p_1, p_2) = \mu_1(p_1) + \mu_2(p_2) , \text{ for } p_j \in M_j .$$

2. Let $\mathbb{T}^n = \{(t_1, \dots, t_n) \in \mathbb{C}^n : |t_j| = 1, \text{ for all } j\}$ be a torus acting on $\mathbb{C}^n$ by

$$(t_1, \dots, t_n) \cdot (z_1, \dots, z_n) = (t_1^{k_1} z_1, \dots, t_n^{k_n} z_n) ,$$

where $k_1, \dots, k_n \in \mathbb{Z}$ are fixed. Check that this action is hamiltonian with moment map $\mu : \mathbb{C}^n \rightarrow (\mathfrak{t}^n)^* \simeq \mathbb{R}^n$ given by

$$\mu(z_1, \dots, z_n) = -\frac{1}{2}(k_1|z_1|^2, \dots, k_n|z_n|^2) (+ \text{constant}) .$$

3. The vector field $X^\#$ generated by $X \in \mathfrak{g}$ for the coadjoint representation of a Lie group G on $\mathfrak{g}^*$ satisfies $\langle X^\#, Y \rangle = \langle \xi, [Y, X] \rangle$, for any $Y \in \mathfrak{g}$. Equip the coadjoint orbits with the canonical symplectic forms. Show that, for each $\xi \in \mathfrak{g}^*$, the coadjoint action on the orbit $G \cdot \xi$ is hamiltonian with moment map the inclusion map:

$$\mu : G \cdot \xi \hookrightarrow \mathfrak{g}^* .$$

4. Consider the natural action of $U(n)$ on $(\mathbb{C}^n, \omega_0)$. Show that this action is hamiltonian with moment map $\mu : \mathbb{C}^n \rightarrow \mathfrak{u}(n)$ given by

$$\mu(z) = \frac{i}{2} z z^* ,$$

where we identify the Lie algebra $\mathfrak{u}(n)$ with its dual via the inner product $(A, B) = \text{trace}(A^* B)$.

Hint: Denote the elements of $U(n)$ in terms of real and imaginary parts $g = h + i k$. Then g acts on $\mathbb{R}^{2n}$ by the linear symplectomorphism $\begin{pmatrix} h & -k \\ k & h \end{pmatrix}$. The Lie algebra $\mathfrak{u}(n)$ is the set of skew-hermitian matrices $X = V + i W$ where $V = -V^t \in \mathbb{R}^{n \times n}$ and $W = W^t \in \mathbb{R}^{n \times n}$. Show that the infinitesimal action is generated by the hamiltonian functions

$$\mu^X(z) = -\frac{1}{2}(x, Wx) + (y, Vx) - \frac{1}{2}(y, Wy)$$

where $z = x + i y$, $x, y \in \mathbb{R}^n$ and $(\cdot, \cdot)$ is the standard inner product. Show that

$$\mu^X(z) = \frac{1}{2} i z^* X z = \frac{1}{2} i \text{trace}(z z^* X) .$$

Check that μ is equivariant.

5. Consider the natural action of $U(k)$ on the space $(\mathbb{C}^{k \times n}, \omega_0)$ of complex $(k \times n)$ -matrices. Identify the Lie algebra $\mathfrak{u}(k)$ with its dual via the inner product $(A, B) = \text{trace}(A^* B)$. Prove that a moment map for this action is given by

$$\mu(A) = \frac{i}{2} AA^* + \frac{\text{Id}}{2i}, \quad \text{for } A \in \mathbb{C}^{k \times n}.$$

(The choice of the constant $\frac{\text{Id}}{2i}$ is for convenience in Homework 20.)

Hint: Exercises 1 and 4.

6. Consider the $U(n)$ -action by conjugation on the space $(\mathbb{C}^{n^2}, \omega_0)$ of complex $(n \times n)$ -matrices. Show that a moment map for this action is given by

$$\mu(A) = \frac{i}{2} [A, A^*].$$

Hint: Previous exercise and its “transpose” version.

26 Existence and Uniqueness of Moment Maps

26.1 Lie Algebras of Vector Fields

Let (M, ω) be a symplectic manifold and $v \in \chi(M)$ a vector field on M .

$$\begin{aligned} v \text{ is symplectic} &\iff \iota_v \omega \text{ is closed ,} \\ v \text{ is hamiltonian} &\iff \iota_v \omega \text{ is exact .} \end{aligned}$$

The spaces

$$\begin{aligned} \chi^{\text{symp}}(M) &= \text{symplectic vector fields on } M , \\ \chi^{\text{ham}}(M) &= \text{hamiltonian vector fields on } M . \end{aligned}$$

are Lie algebras for the Lie bracket of vector fields. $C^\infty(M)$ is a Lie algebra for the Poisson bracket, $\{f, g\} = \omega(v_f, v_g)$. $H^1(M; \mathbb{R})$ and $\mathbb{R}$ are regarded as Lie algebras for the trivial bracket. We have two exact sequences of Lie algebras:

$$\begin{array}{ccccccc} 0 & \longrightarrow & \chi^{\text{ham}}(M) & \hookrightarrow & \chi^{\text{symp}}(M) & \longrightarrow & H^1(M; \mathbb{R}) \longrightarrow 0 \\ & & & & v \longmapsto & [\iota_v \omega] & \\ 0 & \longrightarrow & \mathbb{R} & \hookrightarrow & C^\infty(M) & \longrightarrow & \chi^{\text{ham}}(M) \longrightarrow 0 \\ & & & & f \longmapsto & v_f . & \end{array}$$

In particular, if $H^1(M; \mathbb{R}) = 0$, then $\chi^{\text{ham}}(M) = \chi^{\text{symp}}(M)$.

Let G be a connected Lie group. A symplectic action $\psi : G \rightarrow \text{Symp}(M, \omega)$ induces an infinitesimal action

$$\begin{aligned} d\psi : \mathfrak{g} &\longrightarrow \chi^{\text{symp}}(M) \\ X &\longmapsto X^\# = \text{vector field generated by the} \\ &\quad \text{one-parameter group } \{\exp tX(e) \mid t \in \mathbb{R}\} . \end{aligned}$$

Exercise. Check that the map $d\psi$ is a Lie algebra anti-homomorphism. $\diamond$

The action ψ is **hamiltonian** if and only if there is a Lie algebra homomorphism $\mu^* : \mathfrak{g} \rightarrow C^\infty(M)$ lifting $d\psi$, i.e., making the following diagram commute.

$$\begin{array}{ccc} C^\infty(M) & \xrightarrow{\quad} & \chi^{\text{symp}}(M) \\ & \nwarrow \mu^* \quad \nearrow d\psi & \\ & \mathfrak{g} & \end{array}$$

The map μ^* is then called a **comoment map** (defined in Lecture 22).

$$\begin{array}{ccc} \text{Existence of } \mu^* & \iff & \text{Existence of } \mu \\ \text{comoment map} & & \text{moment map} \end{array}$$

$$\text{Lie algebra homomorphism} \iff \text{equivariance}$$

26.2 Lie Algebra Cohomology

Let $\mathfrak{g}$ be a Lie algebra, and

$$\begin{aligned} C^k &:= \Lambda^k \mathfrak{g}^* = k\text{-cochains on } \mathfrak{g} \\ &= \text{alternating } k\text{-linear maps } \underbrace{\mathfrak{g} \times \dots \times \mathfrak{g}}_k \longrightarrow \mathbb{R} . \end{aligned}$$

Define a linear operator $\delta : C^k \rightarrow C^{k+1}$ by

$$\delta c(X_0, \dots, X_k) = \sum_{i < j} (-1)^{i+j} c([X_i, X_j], X_0, \dots, \widehat{X}_i, \dots, \widehat{X}_j, \dots, X_k) .$$

Exercise. Check that $\delta^2 = 0$. ◇

The **Lie algebra cohomology groups** (or **Chevalley cohomology groups**) of $\mathfrak{g}$ are the cohomology groups of the complex $0 \xrightarrow{\delta} C^0 \xrightarrow{\delta} C^1 \xrightarrow{\delta} \dots$:

$$H^k(\mathfrak{g}; \mathbb{R}) := \frac{\ker \delta : C^k \longrightarrow C^{k+1}}{\operatorname{im} \delta : C^{k-1} \longrightarrow C^k} .$$

Theorem 26.1 *If $\mathfrak{g}$ is the Lie algebra of a compact connected Lie group G , then*

$$H^k(\mathfrak{g}; \mathbb{R}) = H_{\text{deRham}}^k(G) .$$

Proof. Exercise. Hint: by averaging show that the de Rham cohomology can be computed from the subcomplex of G -invariant forms. □

Meaning of $H^1(\mathfrak{g}; \mathbb{R})$ and $H^2(\mathfrak{g}; \mathbb{R})$:

- An element of $C^1 = \mathfrak{g}^*$ is a linear functional on $\mathfrak{g}$. If $c \in \mathfrak{g}^*$, then $\delta c(X_0, X_1) = -c([X_0, X_1])$. The **commutator ideal** of $\mathfrak{g}$ is

$$[\mathfrak{g}, \mathfrak{g}] := \{\text{linear combinations of } [X, Y] \text{ for any } X, Y \in \mathfrak{g}\} .$$

Since $\delta c = 0$ if and only if c vanishes on $[\mathfrak{g}, \mathfrak{g}]$, we conclude that

$$H^1(\mathfrak{g}; \mathbb{R}) = [\mathfrak{g}, \mathfrak{g}]^0$$

where $[\mathfrak{g}, \mathfrak{g}]^0 \subseteq \mathfrak{g}^*$ is the annihilator of $[\mathfrak{g}, \mathfrak{g}]$.

- An element of C^2 is an alternating bilinear map $c : \mathfrak{g} \times \mathfrak{g} \rightarrow \mathbb{R}$.

$$\delta c(X_0, X_1, X_2) = -c([X_0, X_1], X_2) + c([X_0, X_2], X_1) - c([X_1, X_2], X_0) .$$

If $c = \delta b$ for some $b \in C^1$, then

$$c(X_0, X_1) = (\delta b)(X_0, X_1) = -b([X_0, X_1]) .$$

26.3 Existence of Moment Maps

Theorem 26.2 *If $H^1(\mathfrak{g}; \mathbb{R}) = H^2(\mathfrak{g}; \mathbb{R}) = 0$, then any symplectic G -action is hamiltonian.*

Proof. Let $\psi : G \rightarrow \text{Symp}(M, \omega)$ be a symplectic action of G on a symplectic manifold (M, ω) . Since

$$H^1(\mathfrak{g}; \mathbb{R}) = 0 \iff [\mathfrak{g}, \mathfrak{g}] = \mathfrak{g}$$

and since commutators of symplectic vector fields are hamiltonian, we have

$$d\psi : \mathfrak{g} = [\mathfrak{g}, \mathfrak{g}] \longrightarrow \chi^{\text{ham}}(M).$$

The action ψ is hamiltonian if and only if there is a Lie algebra homomorphism $\mu^* : \mathfrak{g} \rightarrow C^\infty(M)$ such that the following diagram commutes.

$$\begin{array}{ccccc} \mathbb{R} & \longrightarrow & C^\infty(M) & \longrightarrow & \chi^{\text{ham}}(M) \\ & & \nwarrow \tau & & \nearrow d\psi \\ & & \mathfrak{g} & & \end{array}$$

We first take an arbitrary vector space lift $\tau : \mathfrak{g} \rightarrow C^\infty(M)$ making the diagram commute, i.e., for each basis vector $X \in \mathfrak{g}$, we choose

$$\tau(X) = \tau^X \in C^\infty(M) \quad \text{such that} \quad v_{(\tau^X)} = d\psi(X).$$

The map $X \mapsto \tau^X$ may not be a Lie algebra homomorphism. By construction, $\tau^{[X,Y]}$ is a hamiltonian function for $[X, Y]^\#$, and (as computed in Lecture 16) $\{\tau^X, \tau^Y\}$ is a hamiltonian function for $-[X^\#, Y^\#]$. Since $[X, Y]^\# = -[X^\#, Y^\#]$, the corresponding hamiltonian functions must differ by a constant:

$$\tau^{[X,Y]} - \{\tau^X, \tau^Y\} = c(X, Y) \in \mathbb{R}.$$

By the Jacobi identity, $\delta c = 0$. Since $H^2(\mathfrak{g}; \mathbb{R}) = 0$, there is $b \in \mathfrak{g}^*$ satisfying $c = \delta b$, $c(X, Y) = -b([X, Y])$. We define

$$\begin{aligned} \mu^* : \mathfrak{g} &\longrightarrow C^\infty(M) \\ X &\longmapsto \mu^*(X) = \tau^X + b(X) = \mu^X. \end{aligned}$$

Now μ^* is a Lie algebra homomorphism:

$$\mu^*([X, Y]) = \tau^{[X,Y]} + b([X, Y]) = \{\tau^X, \tau^Y\} = \{\mu^X, \mu^Y\}.$$

□

So when is $H^1(\mathfrak{g}; \mathbb{R}) = H^2(\mathfrak{g}; \mathbb{R}) = 0$?

A compact Lie group G is **semisimple** if $\mathfrak{g} = [\mathfrak{g}, \mathfrak{g}]$.

Examples. The unitary group $U(n)$ is not semisimple because the multiples of the identity, $S^1 \cdot \text{Id}$, form a nontrivial center; at the level of the Lie algebra, this corresponds to the 1-dimensional subspace $\mathbb{R} \cdot \text{Id}$ of scalar matrices which are not commutators since they are not traceless.

Any direct product of the other compact classical groups $SU(n)$, $SO(n)$ and $Sp(n)$ is semisimple ($n > 1$). Any commutative Lie group is not semisimple. $\diamond$

Theorem 26.3 (Whitehead Lemmas) *Let G be a compact Lie group.*

$$G \text{ is semisimple} \iff H^1(\mathfrak{g}; \mathbb{R}) = H^2(\mathfrak{g}; \mathbb{R}) = 0.$$

A proof can be found in [67, pages 93-95].

Corollary 26.4 *If G is semisimple, then any symplectic G -action is hamiltonian.*

26.4 Uniqueness of Moment Maps

Let G be a compact connected Lie group.

Theorem 26.5 *If $H^1(\mathfrak{g}; \mathbb{R}) = 0$, then moment maps for hamiltonian G -actions are unique.*

Proof. Suppose that μ_1^* and μ_2^* are two comoment maps for an action ψ :

$$\begin{array}{ccc} C^\infty(M) & \xrightarrow{\quad} & \chi^{\text{ham}}(M) \\ & \swarrow \mu_1^* \quad \searrow \mu_2^* & \\ & \mathfrak{g} & \end{array} \quad \begin{array}{c} \nearrow d\psi \\ \nwarrow \end{array}$$

For each $X \in \mathfrak{g}$, μ_1^X and μ_2^X are both hamiltonian functions for $X^\#$, thus $\mu_1^X - \mu_2^X = c(X)$ is locally constant. This defines $c \in \mathfrak{g}^*$, $X \mapsto c(X)$.

Since μ_1^* , μ_2^* are Lie algebra homomorphisms, we have $c([X, Y]) = 0$, $\forall X, Y \in \mathfrak{g}$, i.e., $c \in [\mathfrak{g}, \mathfrak{g}]^0 = \{0\}$. Hence, $\mu_1^* = \mu_2^*$. $\square$

Corollary of this proof. *In general, if $\mu : M \rightarrow \mathfrak{g}^*$ is a moment map, then given any $c \in [\mathfrak{g}, \mathfrak{g}]^0$, $\mu_1 = \mu + c$ is another moment map.*

In other words, moment maps are unique up to elements of the dual of the Lie algebra which annihilate the commutator ideal.

The two extreme cases are:

- G semisimple: any symplectic action is hamiltonian ,
moment maps are unique .
- G commutative: symplectic actions may not be hamiltonian ,
moment maps are unique up to any constant $c \in \mathfrak{g}^*$.

Example. The circle action on $(\mathbb{T}^2, \omega = d\theta_1 \wedge d\theta_2)$ by rotations in the θ_1 direction has vector field $X^\# = \frac{\partial}{\partial \theta_1}$; this is a symplectic action but is not hamiltonian. $\diamond$

Homework 20: Examples of Reduction

1. For the natural action of $U(k)$ on $\mathbb{C}^{k \times n}$ with moment map computed in exercise 5 of Homework 19, we have $\mu^{-1}(0) = \{A \in \mathbb{C}^{k \times n} \mid AA^* = \text{Id}\}$. Show that the quotient

$$\mu^{-1}(0)/U(k) = \mathbb{G}(k, n)$$

is the grassmannian of k -planes in $\mathbb{C}^n$.

2. Consider the S^1 -action on $(\mathbb{R}^{2n+2}, \omega_0)$ which, under the usual identification of $\mathbb{R}^{2n+2}$ with $\mathbb{C}^{n+1}$, corresponds to multiplication by e^{it} . This action is hamiltonian with a moment map $\mu : \mathbb{C}^{n+1} \rightarrow \mathbb{R}$ given by

$$\mu(z) = -\frac{1}{2}|z|^2 + \frac{1}{2}.$$

Prove that the reduction $\mu^{-1}(0)/S^1$ is $\mathbb{CP}^n$ with the Fubini-Study symplectic form $\omega_{\text{red}} = \omega_{\text{FS}}$.

Hint: Let $\text{pr} : \mathbb{C}^{n+1} \setminus \{0\} \rightarrow \mathbb{CP}^n$ denote the standard projection. Check that

$$\text{pr}^* \omega_{\text{FS}} = \frac{i}{2} \partial \bar{\partial} \log(|z|^2).$$

Prove that this form has the same restriction to S^{2n+1} as ω_0 .

3. Show that the natural actions of $\mathbb{T}^{n+1}$ and $U(n+1)$ on $(\mathbb{CP}^n, \omega_{\text{FS}})$ are hamiltonian, and find formulas for their moment maps.

Hint: Previous exercise and exercises 2 and 4 of Homework 19.

27 Convexity

27.1 Convexity Theorem

From now on, we will concentrate on actions of a torus $G = \mathbb{T}^m = \mathbb{R}^m / \mathbb{Z}^m$.

Theorem 27.1 (Atiyah [6], Guillemin-Sternberg [57])

Let (M, ω) be a compact connected symplectic manifold, and let $\mathbb{T}^m$ be an m -torus. Suppose that $\psi : \mathbb{T}^m \rightarrow \text{Symp}(M, \omega)$ is a hamiltonian action with moment map $\mu : M \rightarrow \mathbb{R}^m$. Then:

1. the levels of μ are connected;
2. the image of μ is convex;
3. the image of μ is the convex hull of the images of the fixed points of the action.

The image $\mu(M)$ of the moment map is hence called the **moment polytope**.

Proof. This proof (due to Atiyah) involves induction over $m = \dim \mathbb{T}^m$. Consider the statements:

A_m : “the levels of μ are connected, for any $\mathbb{T}^m$ -action;”

B_m : “the image of μ is convex, for any $\mathbb{T}^m$ -action.”

Then

$$\begin{aligned} (1) \quad & \Longleftrightarrow A_m \text{ holds for all } m, \\ (2) \quad & \Longleftrightarrow B_m \text{ holds for all } m. \end{aligned}$$

- A_1 is a non-trivial result in Morse theory.
- $A_{m-1} \implies A_m$ (induction step) is in Homework 21.
- B_1 is trivial because in $\mathbb{R}$ connectedness is convexity.
- $A_{m-1} \implies B_m$ is proved below.

Choose an injective matrix $A \in \mathbb{Z}^{m \times (m-1)}$. Consider the action of an $(m-1)$ -subtorus

$$\begin{aligned} \psi_A : \mathbb{T}^{m-1} &\longrightarrow \text{Symp}(M, \omega) \\ \theta &\longmapsto \psi_{A\theta}. \end{aligned}$$

Exercise. The action ψ_A is hamiltonian with moment map $\mu_A = A^t \mu : M \rightarrow \mathbb{R}^{m-1}$. $\diamond$

Given any $p_0 \in \mu_A^{-1}(\xi)$,

$$p \in \mu_A^{-1}(\xi) \Longleftrightarrow A^t \mu(p) = \xi = A^t \mu(p_0)$$

so that

$$\mu_A^{-1}(\xi) = \{p \in M \mid \mu(p) - \mu(p_0) \in \ker A^t\}.$$

By the first part (statement A_{m-1}), $\mu_A^{-1}(\xi)$ is connected. Therefore, if we connect p_0 to p_1 by a path p_t in $\mu_A^{-1}(\xi)$, we obtain a path $\mu(p_t) - \mu(p_0)$ in $\ker A^t$. But $\ker A^t$ is 1-dimensional. Hence, $\mu(p_t)$ must go through any convex combination of $\mu(p_0)$ and $\mu(p_1)$, which shows that any point on the line segment from $\mu(p_0)$ to $\mu(p_1)$ must be in $\mu(M)$:

$$(1-t)\mu(p_0) + t\mu(p_1) \in \mu(M), \quad 0 \leq t \leq 1.$$

Any $p_0, p_1 \in M$ can be approximated arbitrarily closely by points p'_0 and p'_1 with $\mu(p'_1) - \mu(p'_0) \in \ker A^t$ for some injective matrix $A \in \mathbb{Z}^{m \times (m-1)}$. Taking limits $p'_0 \rightarrow p_0, p'_1 \rightarrow p_1$, we obtain that $\mu(M)$ is convex.¹⁵

To prove part 3, consider the fixed point set C of ψ . Homework 21 shows that C is a finite union of connected symplectic submanifolds, $C = C_1 \cup \dots \cup C_N$. The moment map is constant on each C_j , $\mu(C_j) = \eta_j \in \mathbb{R}^m$, $j = 1, \dots, N$. By the second part, the convex hull of $\{\eta_1, \dots, \eta_N\}$ is contained in $\mu(M)$.

For the converse, suppose that $\xi \in \mathbb{R}^m$ and $\xi \notin \text{convex hull of } \{\eta_1, \dots, \eta_N\}$. Choose $X \in \mathbb{R}^m$ with rationally independent components and satisfying

$$\langle \xi, X \rangle > \langle \eta_j, X \rangle, \text{ for all } j.$$

By the irrationality of X , the set $\{\exp tX(e) \mid t \in \mathbb{R}\}$ is dense in $\mathbb{T}^m$, hence the zeros of the vector field $X^\#$ on M are the fixed points of the $\mathbb{T}^m$ -action. Since $\mu^X = \langle \mu, X \rangle$ attains its maximum on one of the sets C_j , this implies

$$\langle \xi, X \rangle > \sup_{p \in M} \langle \mu(p), X \rangle,$$

hence $\xi \notin \mu(M)$. Therefore,

$$\mu(M) = \text{convex hull of } \{\eta_1, \dots, \eta_N\}.$$

□

27.2 Effective Actions

An action of a group G on a manifold M is called **effective** if each group element $g \neq e$ moves at least one $p \in M$, that is,

$$\bigcap_{p \in M} G_p = \{e\},$$

where $G_p = \{g \in G \mid g \cdot p = p\}$ is the stabilizer of p .

¹⁵Clearly $\mu(M)$ is closed because it is compact.

Corollary 27.2 *Under the conditions of the convexity theorem, if the $\mathbb{T}^m$ -action is effective, then there must be at least $m + 1$ fixed points.*

Proof. If the $\mathbb{T}^m$ -action is effective, there must be a point p where the moment map is a submersion, i.e., $(d\mu_1)_p, \dots, (d\mu_m)_p$ are linearly independent. Hence, $\mu(p)$ is an interior point of $\mu(M)$, and $\mu(M)$ is a nondegenerate convex polytope. Any nondegenerate convex polytope in $\mathbb{R}^m$ must have at least $m + 1$ vertices. The vertices of $\mu(M)$ are images of fixed points. $\square$

Theorem 27.3 *Let $(M, \omega, \mathbb{T}^m, \mu)$ be a hamiltonian $\mathbb{T}^m$ -space. If the $\mathbb{T}^m$ -action is effective, then $\dim M \geq 2m$.*

Proof. On an orbit $\mathcal{O}$, the moment map $\mu(\mathcal{O}) = \xi$ is constant. For $p \in \mathcal{O}$, the exterior derivative

$$d\mu_p : T_p M \longrightarrow \mathfrak{g}^*$$

maps $T_p \mathcal{O}$ to 0. Thus

$$T_p \mathcal{O} \subseteq \ker d\mu_p = (T_p \mathcal{O})^\omega,$$

which shows that *orbits $\mathcal{O}$ of a hamiltonian torus action are always isotropic submanifolds of M* . In particular, $\dim \mathcal{O} \leq \frac{1}{2} \dim M$.

Fact: If $\psi : \mathbb{T}^m \rightarrow \text{Diff}(M)$ is an effective action, then it has orbits of dimension m ; a proof may be found in [17]. $\square$

Definition 27.4 A **(symplectic) toric manifold**¹⁶ is a compact connected symplectic manifold (M, ω) equipped with an effective hamiltonian action of a torus $\mathbb{T}$ of dimension equal to half the dimension of the manifold:

$$\dim \mathbb{T} = \frac{1}{2} \dim M$$

and with a choice of a corresponding moment map μ .

Exercise. Show that an effective hamiltonian action of a torus $\mathbb{T}^n$ on a $2n$ -dimensional symplectic manifold gives rise to an integrable system.

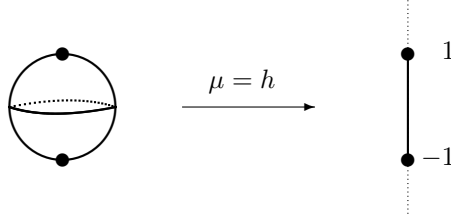
Hint: The coordinates of the moment map are commuting integrals of motion.

$\diamond$

¹⁶In these notes, a toric manifold is always a *symplectic* toric manifold.

27.3 Examples

1. The circle S^1 **acts on the 2-sphere** $(S^2, \omega_{\text{standard}} = d\theta \wedge dh)$ by rotations with moment map $\mu = h$ equal to the height function and moment polytope $[-1, 1]$.



- 1.' The circle S^1 **acts on** $\mathbb{CP}^1 = \mathbb{C}^2 - 0 / \sim$ with the Fubini-Study form $\omega_{\text{FS}} = \frac{1}{4}\omega_{\text{standard}}$, by $e^{i\theta} \cdot [z_0, z_1] = [z_0, e^{i\theta} z_1]$. This is hamiltonian with moment map $\mu[z_0, z_1] = -\frac{1}{2} \cdot \frac{|z_1|^2}{|z_0|^2 + |z_1|^2}$, and moment polytope $[-\frac{1}{2}, 0]$.

2. The $\mathbb{T}^2$ -**action on** $\mathbb{CP}^2$ by

$$(e^{i\theta_1}, e^{i\theta_2}) \cdot [z_0, z_1, z_2] = [z_0, e^{i\theta_1} z_1, e^{i\theta_2} z_2]$$

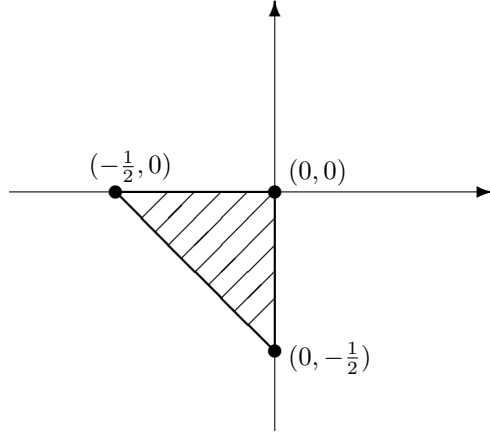
has moment map

$$\mu[z_0, z_1, z_2] = -\frac{1}{2} \left(\frac{|z_1|^2}{|z_0|^2 + |z_1|^2 + |z_2|^2}, \frac{|z_2|^2}{|z_0|^2 + |z_1|^2 + |z_2|^2} \right).$$

The fixed points get mapped as

$$\begin{aligned} [1, 0, 0] &\longmapsto (0, 0) \\ [0, 1, 0] &\longmapsto \left(-\frac{1}{2}, 0\right) \\ [0, 0, 1] &\longmapsto \left(0, -\frac{1}{2}\right). \end{aligned}$$

Notice that the stabilizer of a preimage of the edges is S^1 , while the action is free at preimages of interior points of the moment polytope.



Exercise. What is the moment polytope for the $\mathbb{T}^3$ -action on $\mathbb{CP}^3$ as

$$(e^{i\theta_1}, e^{i\theta_2}, e^{i\theta_3}) \cdot [z_0, z_1, z_2, z_3] = [z_0, e^{i\theta_1} z_1, e^{i\theta_2} z_2, e^{i\theta_3} z_3] ?$$

◇

Exercise. What is the moment polytope for the $\mathbb{T}^2$ -action on $\mathbb{CP}^1 \times \mathbb{CP}^1$ as

$$(e^{i\theta}, e^{i\eta}) \cdot ([z_0, z_1], [w_0, w_1]) = ([z_0, e^{i\theta} z_1], [w_0, e^{i\eta} w_1]) ?$$

◇

Homework 21: Connectedness

Consider a hamiltonian action $\psi : \mathbb{T}^m \rightarrow \text{Symp}(M, \omega)$, $\theta \mapsto \psi_\theta$, of an m -dimensional torus on a $2n$ -dimensional compact connected symplectic manifold (M, ω) . If we identify the Lie algebra of $\mathbb{T}^m$ with $\mathbb{R}^m$ by viewing $\mathbb{T}^m = \mathbb{R}^m / \mathbb{Z}^m$, and we identify the Lie algebra with its dual via the standard inner product, then the moment map for ψ is $\mu : M \rightarrow \mathbb{R}^m$.

1. Show that there exists a compatible almost complex structure J on (M, ω) which is invariant under the $\mathbb{T}^m$ -action, that is, $\psi_\theta^* J = J \psi_\theta^*$, for all $\theta \in \mathbb{T}^m$.

Hint: We cannot average almost complex structures, but we can average riemannian metrics (why?). Given a riemannian metric g_0 on M , its $\mathbb{T}^m$ -average $g = \int_{\mathbb{T}^m} \psi_\theta^* g_0 d\theta$ is $\mathbb{T}^m$ -invariant.

2. Show that, for any subgroup $G \subseteq \mathbb{T}^m$, the fixed-point set for G ,

$$\text{Fix}(G) = \bigcap_{\theta \in G} \text{Fix}(\psi_\theta),$$

is a symplectic submanifold of M .

Hint: For each $p \in \text{Fix}(G)$ and each $\theta \in G$, the differential of ψ_θ at p ,

$$d\psi_\theta(p) : T_p M \longrightarrow T_p M,$$

preserves the complex structure J_p on $T_p M$. Consider the exponential map $\exp_p : T_p M \rightarrow M$ with respect to the invariant riemannian metric $g(\cdot, \cdot) = \omega(\cdot, J\cdot)$. Show that, by uniqueness of geodesics, $\exp_p$ is equivariant, i.e.,

$$\exp_p(d\psi_\theta(p)v) = \psi_\theta(\exp_p v)$$

for any $\theta \in G$, $v \in T_p M$. Conclude that the fixed points of ψ_θ near p correspond to the fixed points of $d\psi_\theta(p)$ on $T_p M$, that is

$$T_p \text{Fix}(G) = \bigcap_{\theta \in G} \ker(\text{Id} - d\psi_\theta(p)).$$

Since $d\psi_\theta(p) \circ J_p = J_p \circ d\psi_\theta(p)$, the eigenspace with eigenvalue 1 is invariant under J_p , and is therefore a symplectic subspace.

3. A smooth function $f : M \rightarrow \mathbb{R}$ on a compact riemannian manifold M is called a **Morse-Bott function** if its critical set $\text{Crit}(f) = \{p \in M \mid df(p) = 0\}$ is a submanifold of M and for every $p \in \text{Crit}(f)$, $T_p \text{Crit}(f) = \ker \nabla^2 f(p)$ where $\nabla^2 f(p) : T_p M \rightarrow T_p M$ denotes the linear operator obtained from the hessian via the riemannian metric. This is the natural generalization of the notion of Morse function to the case where the critical set is not just isolated points. If f is a Morse-Bott function, then $\text{Crit}(f)$ decomposes into finitely many connected critical manifolds C . The tangent space $T_p M$ at $p \in C$ decomposes as a direct sum

$$T_p M = T_p C \oplus E_p^+ \oplus E_p^-$$

where E_p^+ and E_p^- are spanned by the positive and negative eigenspaces of $\nabla^2 f(p)$. The *index* of a connected critical submanifold C is $n_C^- = \dim E_p^-$, for any $p \in C$, whereas the *coindex* of C is $n_C^+ = \dim E_p^+$.

For each $X \in \mathbb{R}^m$, let $\mu^X = \langle \mu, X \rangle : M \rightarrow \mathbb{R}$ be the component of μ along X . Show that μ^X is a Morse-Bott function with even-dimensional critical manifolds of even index. Moreover, show that the critical set

$$\text{Crit}(\mu^X) = \bigcap_{\theta \in \mathbb{T}^X} \text{Fix}(\psi_\theta)$$

is a symplectic manifold, where $\mathbb{T}^X$ is the closure of the subgroup of $\mathbb{T}^m$ generated by X .

Hint: Assume first that X has components independent over $\mathbb{Q}$, so that $\mathbb{T}^X = \mathbb{T}^m$ and $\text{Crit}(\mu^X) = \text{Fix}(\mathbb{T}^m)$. Apply exercise 2. To prove that $T_p \text{Crit}(\mu^X) = \ker \nabla^2 \mu^X(p)$, show that $\ker \nabla^2 \mu^X(p) = \bigcap_{\theta \in \mathbb{T}^m} \ker(\text{Id} - d\psi_\theta(p))$. To see this, notice that the 1-parameter group of matrices $(d\psi_{\exp tX})_p$ coincides with $\exp(tv_p)$, where $v_p = -J_p \nabla^2 \mu^X(p) : T_p M \rightarrow T_p M$ is a vector field on $T_p M$. The kernel of $\nabla^2 \mu^X(p)$ corresponds to the fixed points of $d\psi_{tX}(p)$, and since X has rationally independent components, these are the common fixed points of all $d\psi_\theta(p)$, $\theta \in \mathbb{T}^m$. The eigenspaces of $\nabla^2 \mu^X(p)$ are even-dimensional because they are invariant under J_p .

4. The moment map $\mu = (\mu_1, \dots, \mu_m)$ is called **effective** if the 1-forms $d\mu_1, \dots, d\mu_m$ of its components are linearly independent. Show that, if μ is not effective, then the action reduces to that of an $(m-1)$ -subtorus.

Hint: If μ is not effective, then the function $\mu^X = \langle \mu, X \rangle$ is constant for some nonzero $X \in \mathbb{R}^m$. Show that we can neglect the direction of X .

5. Prove that the level set $\mu^{-1}(\xi)$ is connected for every regular value $\xi \in \mathbb{R}^m$.

Hint: Prove by induction over $m = \dim \mathbb{T}^m$. For the case $m = 1$, use the lemma that all level sets $f^{-1}(c)$ of a Morse-Bott function $f : M \rightarrow \mathbb{R}$ on a compact manifold M are necessarily connected, if the critical manifolds all have index and coindex $\neq 1$ (see [83, p.178-179]). For the induction step, you can assume that ψ is effective. Then, for every $0 \neq X \in \mathbb{R}^m$, the function $\mu^X : M \rightarrow \mathbb{R}$ is not constant. Show that $\mathcal{C} := \bigcup_{X \neq 0} \text{Crit} \mu^X = \bigcup_{0 \neq X \in \mathbb{Z}^m} \text{Crit} \mu^X$ where each $\text{Crit} \mu^X$ is an even-dimensional proper submanifold, so the complement $M \setminus \mathcal{C}$ must be dense in M . Show that $M \setminus \mathcal{C}$ is open. Hence, by continuity, to show that $\mu^{-1}(\xi)$ is connected for every regular value $\xi = (\xi_1, \dots, \xi_m) \in \mathbb{R}^m$, it suffices to show that $\mu^{-1}(\xi)$ is connected whenever $(\xi_1, \dots, \xi_{m-1})$ is a regular value for a reduced moment map $(\mu_1, \dots, \mu_{m-1})$. By the induction hypothesis, the manifold $Q = \bigcap_{j=1}^{m-1} \mu_j^{-1}(\xi_j)$ is connected whenever $(\xi_1, \dots, \xi_{m-1})$ is a regular value for $(\mu_1, \dots, \mu_{m-1})$. It suffices to show that the function $\mu_m : Q \rightarrow \mathbb{R}$ has only critical manifolds of even index and coindex (see [83, p.183]), because then, by the lemma, the level sets $\mu^{-1}(\xi) = Q \cap \mu_m^{-1}(\xi_m)$ are connected for every ξ_m .

Part XI

Symplectic Toric Manifolds

Native to algebraic geometry, toric manifolds have been studied by symplectic geometers as examples of extremely symmetric hamiltonian spaces, and as guinea pigs for new theorems. Delzant showed that symplectic toric manifolds are classified (as hamiltonian spaces) by a set of special polytopes.

28 Classification of Symplectic Toric Manifolds

28.1 Delzant Polytopes

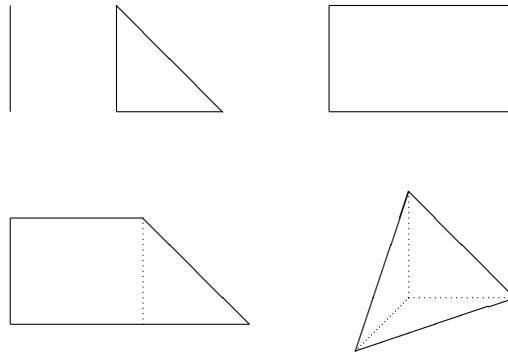
A $2n$ -dimensional **(symplectic) toric manifold** is a compact connected symplectic manifold (M^{2n}, ω) equipped with an effective hamiltonian action of an n -torus $\mathbb{T}^n$ and with a corresponding moment map $\mu : M \rightarrow \mathbb{R}^n$.

Definition 28.1 A **Delzant polytope** Δ in $\mathbb{R}^n$ is a convex polytope satisfying:

- it is **simple**, i.e., there are n edges meeting at each vertex;
- it is **rational**, i.e., the edges meeting at the vertex p are rational in the sense that each edge is of the form $p + tu_i$, $t \geq 0$, where $u_i \in \mathbb{Z}^n$;
- it is **smooth**, i.e., for each vertex, the corresponding $u_1, \dots, u_n$ can be chosen to be a $\mathbb{Z}$ -basis of $\mathbb{Z}^n$.

Remark. The Delzant polytopes are the simple rational smooth polytopes. These are closely related to the **Newton polytopes** (which are the nonsingular n -valent polytopes), except that the vertices of a Newton polytope are required to lie on the integer lattice and for a Delzant polytope they are not. $\diamond$

Examples of Delzant polytopes:



The dotted vertical line in the trapezoidal example means nothing, except that it is a picture of a rectangle plus an isosceles triangle. For “taller” triangles, smoothness would be violated. “Wider” triangles (with integral slope) may still be Delzant. The family of the Delzant trapezoids of this type, starting with the rectangle, correspond, under the Delzant construction, to **Hirzebruch surfaces**; see Homework 22.

Examples of polytopes which are **not Delzant**:



The picture on the left fails the smoothness condition, whereas the picture on the right fails the simplicity condition.

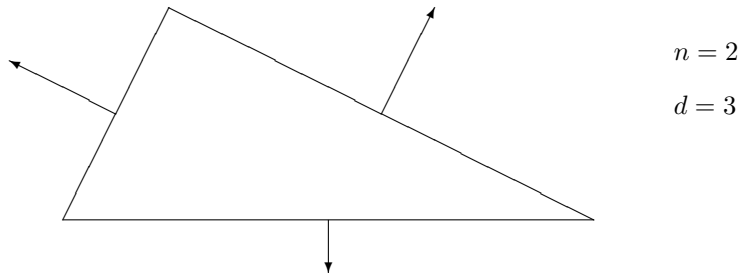
Algebraic description of Delzant polytopes:

A **facet** of a polytope is a $(n - 1)$ -dimensional face.

Let Δ be a Delzant polytope with $n = \dim \Delta$ and $d = \text{number of facets}$.

A lattice vector $v \in \mathbb{Z}^n$ is **primitive** if it cannot be written as $v = ku$ with $u \in \mathbb{Z}^n$, $k \in \mathbb{Z}$ and $|k| > 1$; for instance, $(1, 1)$, $(4, 3)$, $(1, 0)$ are primitive, but $(2, 2)$, $(3, 6)$ are not.

Let $v_i \in \mathbb{Z}^n$, $i = 1, \dots, d$, be the primitive outward-pointing normal vectors to the facets.

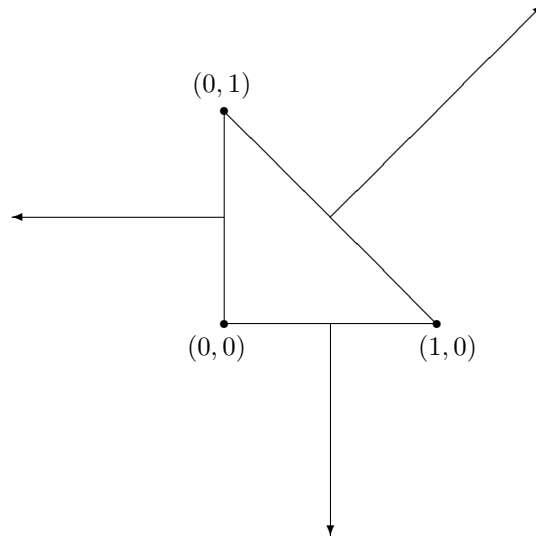


Then we can describe Δ as an intersection of halfspaces

$$\Delta = \{x \in (\mathbb{R}^n)^* \mid \langle x, v_i \rangle \leq \lambda_i, \ i = 1, \dots, d\} \quad \text{for some } \lambda_i \in \mathbb{R}.$$

Example. For the picture below, we have

$$\begin{aligned}\Delta &= \{x \in (\mathbb{R}^2)^* \mid x_1 \geq 0, x_2 \geq 0, x_1 + x_2 \leq 1\} \\ &= \{x \in (\mathbb{R}^2)^* \mid \langle x, (-1, 0) \rangle \leq 0, \langle x, (0, -1) \rangle \leq 0, \langle x, (1, 1) \rangle \leq 1\} .\end{aligned}$$



◇

28.2 Delzant Theorem

We do not have a classification of symplectic manifolds, but we do have a classification of toric manifolds in terms of combinatorial data. This is the content of the Delzant theorem.

Theorem 28.2 (Delzant [23]) *Toric manifolds are classified by Delzant polytopes. More specifically, there is the following one-to-one correspondence*

$$\begin{aligned}\{\text{toric manifolds}\} &\xrightarrow{1-1} \{\text{Delzant polytopes}\} \\ (M^{2n}, \omega, \mathbb{T}^n, \mu) &\longmapsto \mu(M).\end{aligned}$$

We will prove the existence part (or surjectivity) in the Delzant theorem following [54]. Given a Delzant polytope, what is the corresponding toric manifold?

$$(M_\Delta, \omega_\Delta, \mathbb{T}^n, \mu) \xleftarrow{?} \Delta^n$$

28.3 Sketch of Delzant Construction

Let Δ be a Delzant polytope with d facets. Let $v_i \in \mathbb{Z}^n$, $i = 1, \dots, d$, be the primitive outward-pointing normal vectors to the facets. For some $\lambda_i \in \mathbb{R}$,

$$\Delta = \{x \in (\mathbb{R}^n)^* \mid \langle x, v_i \rangle \leq \lambda_i, i = 1, \dots, d\}.$$

Let $e_1 = (1, 0, \dots, 0), \dots, e_d = (0, \dots, 0, 1)$ be the standard basis of $\mathbb{R}^d$. Consider

$$\begin{aligned} \pi : \mathbb{R}^d &\longrightarrow \mathbb{R}^n \\ e_i &\longmapsto v_i. \end{aligned}$$

Claim. The map π is onto and maps $\mathbb{Z}^d$ onto $\mathbb{Z}^n$.

Proof. The set $\{e_1, \dots, e_d\}$ is a basis of $\mathbb{Z}^d$. The set $\{v_1, \dots, v_d\}$ spans $\mathbb{Z}^n$ for the following reason. At a vertex p , the edge vectors $u_1, \dots, u_n \in (\mathbb{R}^n)^*$, form a basis for $(\mathbb{Z}^n)^*$ which, without loss of generality, we may assume is the standard basis. Then the corresponding primitive normal vectors to the facets meeting at p are symmetric (in the sense of multiplication by -1) to the u_i 's, hence form a basis of $\mathbb{Z}^n$. $\square$

Therefore, π induces a surjective map, still called π , between tori:

$$\begin{array}{ccc} \mathbb{R}^d / \mathbb{Z}^d & \xrightarrow{\pi} & \mathbb{R}^n / \mathbb{Z}^n \\ \parallel & & \parallel \\ \mathbb{T}^d & \longrightarrow & \mathbb{T}^n \longrightarrow 0. \end{array}$$

Let

$$\begin{aligned} N &= \text{kernel of } \pi \text{ (} N \text{ is a Lie subgroup of } \mathbb{T}^d \text{)} \\ \mathfrak{n} &= \text{Lie algebra of } N \\ \mathbb{R}^d &= \text{Lie algebra of } \mathbb{T}^d \\ \mathbb{R}^n &= \text{Lie algebra of } \mathbb{T}^n. \end{aligned}$$

The exact sequence of tori

$$0 \longrightarrow N \xrightarrow{i} \mathbb{T}^d \xrightarrow{\pi} \mathbb{T}^n \longrightarrow 0$$

induces an exact sequence of Lie algebras

$$0 \longrightarrow \mathfrak{n} \xrightarrow{i} \mathbb{R}^d \xrightarrow{\pi} \mathbb{R}^n \longrightarrow 0$$

with dual exact sequence

$$0 \longrightarrow (\mathbb{R}^n)^* \xrightarrow{\pi^*} (\mathbb{R}^d)^* \xrightarrow{i^*} \mathfrak{n}^* \longrightarrow 0.$$

Now consider $\mathbb{C}^d$ with symplectic form $\omega_0 = \frac{i}{2} \sum dz_k \wedge d\bar{z}_k$, and standard hamiltonian action of $\mathbb{T}^d$

$$(e^{2\pi i t_1}, \dots, e^{2\pi i t_d}) \cdot (z_1, \dots, z_d) = (e^{2\pi i t_1} z_1, \dots, e^{2\pi i t_d} z_d).$$

The moment map is $\phi : \mathbb{C}^d \longrightarrow (\mathbb{R}^d)^*$

$$\phi(z_1, \dots, z_d) = -\pi(|z_1|^2, \dots, |z_d|^2) + \text{constant} ,$$

where we choose the constant to be $(\lambda_1, \dots, \lambda_d)$. What is the moment map for the action restricted to the subgroup N ?

Exercise. Let G be any compact Lie group and H a closed subgroup of G , with $\mathfrak{g}$ and $\mathfrak{h}$ the respective Lie algebras. The inclusion $i : \mathfrak{h} \hookrightarrow \mathfrak{g}$ is dual to the projection $i^* : \mathfrak{g}^* \rightarrow \mathfrak{h}^*$. Suppose that (M, ω, G, ϕ) is a hamiltonian G -space. Show that the restriction of the G -action to H is hamiltonian with moment map

$$i^* \circ \phi : M \longrightarrow \mathfrak{h}^* .$$

◇

The subtorus N acts on $\mathbb{C}^d$ in a hamiltonian way with moment map

$$i^* \circ \phi : \mathbb{C}^d \longrightarrow \mathfrak{n}^* .$$

Let $Z = (i^* \circ \phi)^{-1}(0)$ be the zero-level set.

Claim. The set Z is compact and N acts freely on Z .

This claim will be proved in the next lecture.

By the first claim, $0 \in \mathfrak{n}^*$ is a regular value of $i^* \circ \phi$. Hence, Z is a compact submanifold of $\mathbb{C}^d$ of dimension

$$\dim_{\mathbb{R}} Z = 2d - \underbrace{(d - n)}_{\dim \mathfrak{n}^*} = d + n .$$

The orbit space $M_{\Delta} = Z/N$ is a compact manifold of dimension

$$\dim_{\mathbb{R}} M_{\Delta} = d + n - \underbrace{(d - n)}_{\dim N} = 2n .$$

The point-orbit map $p : Z \rightarrow M_{\Delta}$ is a principal N -bundle over M_{Δ} . Consider the diagram

$$\begin{array}{ccc} Z & \xrightarrow{j} & \mathbb{C}^d \\ p \downarrow & & \\ M_{\Delta} & & \end{array}$$

where $j : Z \hookrightarrow \mathbb{C}^d$ is inclusion. The Marsden-Weinstein-Meyer theorem guarantees the existence of a symplectic form ω_{Δ} on M_{Δ} satisfying

$$p^* \omega_{\Delta} = j^* \omega_0 .$$

Exercise. Work out all details in the following simple example.

Let $\Delta = [0, a] \subset \mathbb{R}^*$ ($n = 1, d = 2$). Let $v (= 1)$ be the standard basis vector in $\mathbb{R}$. Then

$$\Delta : \begin{array}{ll} \langle x, v_1 \rangle \leq 0 & v_1 = -v \\ \langle x, v_2 \rangle \leq a & v_2 = v . \end{array}$$

The projection

$$\begin{array}{ccc} \mathbb{R}^2 & \xrightarrow{\pi} & \mathbb{R} \\ e_1 & \mapsto & -v \\ e_2 & \mapsto & v \end{array}$$

has kernel equal to the span of $(e_1 + e_2)$, so that N is the diagonal subgroup of $\mathbb{T}^2 = S^1 \times S^1$. The exact sequences become

$$\begin{array}{ccccccc} 0 & \longrightarrow & N & \xrightarrow{i} & \mathbb{T}^2 & \xrightarrow{\pi} & S^1 \longrightarrow 0 \\ 0 & \longrightarrow & \mathbb{R}^* & \xrightarrow{\pi^*} & (\mathbb{R}^2)^* & \xrightarrow{i^*} & \mathfrak{n}^* \longrightarrow 0 \\ & & & & (x_1, x_2) & \longmapsto & x_1 + x_2 . \end{array}$$

The action of the diagonal subgroup $N = \{(e^{2\pi it}, e^{2\pi it}) \in S^1 \times S^1\}$ on $\mathbb{C}^2$,

$$(e^{2\pi it}, e^{2\pi it}) \cdot (z_1, z_2) = (e^{2\pi it} z_1, e^{2\pi it} z_2) ,$$

has moment map

$$(i^* \circ \phi)(z_1, z_2) = -\pi(|z_1|^2 + |z_2|^2) + a ,$$

with zero-level set

$$(i^* \circ \phi)^{-1}(0) = \{(z_1, z_2) \in \mathbb{C}^2 : |z_1|^2 + |z_2|^2 = \frac{a}{\pi}\} .$$

Hence, the reduced space is

$$(i^* \circ \phi)^{-1}(0)/N = \mathbb{CP}^1 \quad \text{projective space!}$$

◇

29 Delzant Construction

29.1 Algebraic Set-Up

Let Δ be a Delzant polytope with d facets. We can write Δ as

$$\Delta = \{x \in (\mathbb{R}^n)^* \mid \langle x, v_i \rangle \leq \lambda_i, \ i = 1, \dots, d\},$$

for some $\lambda_i \in \mathbb{R}$. Recall the exact sequences from the previous lecture

$$\begin{array}{ccccccccc} 0 & \longrightarrow & N & \xrightarrow{i} & \mathbb{T}^d & \xrightarrow{\pi} & \mathbb{T}^n & \longrightarrow & 0 \\ 0 & \longrightarrow & \mathfrak{n} & \xrightarrow{i} & \mathbb{R}^d & \xrightarrow{\pi} & \mathbb{R}^n & \longrightarrow & 0 \\ & & & & e_i & \longmapsto & v_i & & \end{array}$$

and the dual sequence

$$0 \longrightarrow (\mathbb{R}^n)^* \xrightarrow{\pi^*} (\mathbb{R}^d)^* \xrightarrow{i^*} \mathfrak{n}^* \longrightarrow 0.$$

The standard hamiltonian action of $\mathbb{T}^d$ on $\mathbb{C}^d$

$$(e^{2\pi i t_1}, \dots, e^{2\pi i t_d}) \cdot (z_1, \dots, z_d) = (e^{2\pi i t_1} z_1, \dots, e^{2\pi i t_d} z_d)$$

has moment map $\phi : \mathbb{C}^d \rightarrow (\mathbb{R}^d)^*$ given by

$$\phi(z_1, \dots, z_d) = -\pi(|z_1|^2, \dots, |z_d|^2) + (\lambda_1, \dots, \lambda_d).$$

The restriction of this action to N has moment map

$$i^* \circ \phi : \mathbb{C}^d \longrightarrow \mathfrak{n}^*.$$

29.2 The Zero-Level

Let $Z = (i^* \circ \phi)^{-1}(0)$.

Proposition 29.1 *The level Z is compact and N acts freely on Z .*

Proof. Let Δ' be the image of Δ by π^* . We will show that $\phi(Z) = \Delta'$. Since ϕ is a proper map and Δ' is compact, it will follow that Z is compact.

Lemma 29.2 *Let $y \in (\mathbb{R}^d)^*$. Then:*

$$y \in \Delta' \iff y \text{ is in the image of } Z \text{ by } \phi.$$

Proof of the lemma. The value y is in the image of Z by ϕ if and only if both of the following conditions hold:

1. y is in the image of ϕ ;

2. $i^*y = 0$.

Using the expression for ϕ and the third exact sequence, we see that these conditions are equivalent to:

1. $\langle y, e_i \rangle \leq \lambda_i$ for $i = 1, \dots, d$.
2. $y = \pi^*(x)$ for some $x \in (\mathbb{R}^n)^*$.

Suppose that the second condition holds, so that $y = \pi^*(x)$. Then

$$\begin{aligned} \langle y, e_i \rangle \leq \lambda_i, \forall i &\iff \langle \pi^*(x), e_i \rangle \leq \lambda_i, \forall i \\ &\iff \langle x, \pi(e_i) \rangle \leq \lambda_i, \forall i \\ &\iff x \in \Delta. \end{aligned}$$

Thus, $y \in \phi(Z) \iff y \in \pi^*(\Delta) = \Delta'$. □

Hence, we have a surjective proper map $\phi : Z \rightarrow \Delta'$. Since Δ' is compact, we conclude that Z is compact. It remains to show that N acts freely on Z .

We define a stratification of Z with three equivalent descriptions:

- Define a stratification on Δ' whose i th stratum is the closure of the union of the i -dimensional faces of Δ' . Pull this stratification back to Z by ϕ .

We can obtain a more explicit description of the stratification on Z :

- Let F be a face of Δ' with $\dim F = n - r$. Then F is characterized (as a subset of Δ') by r equations

$$\langle y, e_i \rangle = \lambda_i, \quad i = i_1, \dots, i_r.$$

We write $F = F_I$ where $I = (i_1, \dots, i_r)$ has $1 \leq i_1 < i_2 < \dots < i_r \leq d$.

Let $z = (z_1, \dots, z_d) \in Z$.

$$\begin{aligned} z \in \phi^{-1}(F_I) &\iff \phi(z) \in F_I \\ &\iff \langle \phi(z), e_i \rangle = \lambda_i, \quad \forall i \in I \\ &\iff -\pi|z_i|^2 + \lambda_i = \lambda_i, \quad \forall i \in I \\ &\iff z_i = 0, \quad \forall i \in I. \end{aligned}$$

- The $\mathbb{T}^d$ -action on $\mathbb{C}^d$ preserves ϕ , so the $\mathbb{T}^d$ -action takes $Z = \phi^{-1}(\Delta')$ onto itself, so $\mathbb{T}^d$ acts on Z .

Exercise. The stratification of Z is just the stratification of Z into $\mathbb{T}^d$ orbit types. More specifically, if $z \in Z$ and $\phi(z) \in F_I$ then the stabilizer of z in $\mathbb{T}^d$ is $(\mathbb{T}^d)_I$ where

$$\begin{aligned} I &= (i_1, \dots, i_r), \\ F_I &= \{y \in \Delta' \mid \langle y, e_i \rangle = \lambda_i, \forall i \in I\}, \end{aligned}$$

and

$$(\mathbb{T}^d)_I = \{(e^{2\pi i t_1}, \dots, e^{2\pi i t_d}) \mid e^{2\pi i t_s} = 1, \forall s \notin I\}$$

Hint: Suppose that $z = (z_1, \dots, z_d) \in \mathbb{C}^d$. Then

$$(e^{2\pi i t_1} z_1, \dots, e^{2\pi i t_d} z_d) = (z_1, \dots, z_d)$$

if and only if $e^{2\pi i t_s} = 1$ whenever $z_s \neq 0$.

◇

In order to show that N acts freely on Z , consider the worst case scenario of points $z \in Z$ whose stabilizer under the action of $\mathbb{T}^d$ is as large as possible. Now $(\mathbb{T}^d)_I$ is largest when $F_I = \{y\}$ is a vertex of Δ' . Then y satisfies n equations

$$\langle y, e_i \rangle = \lambda_i, \quad i \in I = \{i_1, \dots, i_n\}.$$

Lemma 29.3 *Let $z \in Z$ be such that $\phi(z)$ is a vertex of Δ' . Let $(\mathbb{T}^d)_I$ be the stabilizer of z . Then the map $\pi : \mathbb{T}^d \rightarrow \mathbb{T}^n$ maps $(\mathbb{T}^d)_I$ bijectively onto $\mathbb{T}^n$.*

Since $N = \ker \pi$, this lemma shows that in the worst case, the stabilizer of z intersects N in the trivial group. It will follow that N acts freely at this point and hence on Z .

Proof of the lemma. Suppose that $\phi(z) = y$ is a vertex of Δ' . Renumber the indices so that

$$I = (1, 2, \dots, n).$$

Then

$$(\mathbb{T}^d)_I = \{(e^{2\pi i t_1}, \dots, e^{2\pi i t_n}, 1, \dots, 1) \mid t_i \in \mathbb{R}\}.$$

The hyperplanes meeting at y are

$$\langle y', e_i \rangle = \lambda_i, \quad i = 1, \dots, n.$$

By definition of Delzant polytope, the set $\pi(e_1), \dots, \pi(e_n)$ is a basis of $\mathbb{Z}^n$. Thus, $\pi : (\mathbb{T}^d)_I \rightarrow \mathbb{T}^n$ is bijective. □

This proves the theorem in the worst case scenario, and hence in general. □

29.3 Conclusion of the Delzant Construction

We continue the construction of $(M_\Delta, \omega_\Delta)$ from Δ . We already have that

$$M_\Delta = Z/N$$

is a compact $2n$ -dimensional manifold. Let ω_Δ be the reduced symplectic form.

Claim. The manifold $(M_\Delta, \omega_\Delta)$ is a hamiltonian $\mathbb{T}^n$ -space with a moment map μ having image $\mu(M_\Delta) = \Delta$.

Suppose that $z \in Z$. The stabilizer of z with respect to the $\mathbb{T}^d$ -action is $(\mathbb{T}^d)_I$, and

$$(\mathbb{T}^d)_I \cap N = \{e\}.$$

In the worst case scenario, F_I is a vertex of Δ' and $(\mathbb{T}^d)_I$ is an n -dimensional subgroup of $\mathbb{T}^d$. In any case, there is a right inverse map $\pi^{-1} : \mathbb{T}^n \rightarrow (\mathbb{T}^d)_I$. Thus, the exact sequence

$$0 \longrightarrow N \longrightarrow \mathbb{T}^d \longrightarrow \mathbb{T}^n \longrightarrow 0$$

splits, and $\mathbb{T}^d = N \times \mathbb{T}^n$.

Apply the results on reduction for product groups (Section 24.3) to our situation of $\mathbb{T}^d = N \times \mathbb{T}^n$ acting on $(M_\Delta, \omega_\Delta)$. The moment map is

$$\phi : \mathbb{C}^d \longrightarrow (\mathbb{R}^d)^* = \mathfrak{n}^* \oplus (\mathbb{R}^n)^* .$$

Let $j : Z \hookrightarrow \mathbb{C}^d$ be the inclusion map, and let

$$\text{pr}_1 : (\mathbb{R}^d)^* \longrightarrow \mathfrak{n}^* \quad \text{and} \quad \text{pr}_2 : (\mathbb{R}^d)^* \longrightarrow (\mathbb{R}^n)^*$$

be the projection maps. The map

$$\text{pr}_2 \circ \phi \circ j : Z \longrightarrow (\mathbb{R}^n)^*$$

is constant on N -orbits. Thus there exists a map

$$\mu : M_\Delta \longrightarrow (\mathbb{R}^n)^*$$

such that

$$\mu \circ p = \text{pr}_2 \circ \phi \circ j .$$

The image of μ is equal to the image of $\text{pr}_2 \circ \phi \circ j$. We showed earlier that $\phi(Z) = \Delta'$. Thus

$$\text{Image of } \mu = \text{pr}_2(\Delta') = \underbrace{\text{pr}_2 \circ \pi^*}_{\text{id}}(\Delta) = \Delta .$$

Thus $(M_\Delta, \omega_\Delta)$ is the required toric manifold corresponding to Δ .

29.4 Idea Behind the Delzant Construction

We use the idea that $\mathbb{R}^d$ is “universal” in the sense that any n -dimensional polytope Δ with d facets can be obtained by intersecting the negative orthant $\mathbb{R}_-^d$ with an affine plane A . Given Δ , to construct A first write Δ as:

$$\Delta = \{x \in \mathbb{R}^n \mid \langle x, v_i \rangle \leq \lambda_i, \ i = 1, \dots, d\} .$$

Define

$$\begin{array}{ccc} \pi : \mathbb{R}^d & \longrightarrow & \mathbb{R}^n \\ e_i & \longmapsto & v_i \end{array} \quad \text{with dual map} \quad \pi^* : \mathbb{R}^n \longrightarrow \mathbb{R}^d .$$

Then

$$\pi^* - \lambda : \mathbb{R}^n \longrightarrow \mathbb{R}^d$$

is an affine map, where $\lambda = (\lambda_1, \dots, \lambda_d)$. Let A be the image of $\pi^* - \lambda$. Then A is an n -dimensional affine plane.

Claim. We have the equality $(\pi^* - \lambda)(\Delta) = \mathbb{R}_-^d \cap A$.

Proof. Let $x \in \mathbb{R}^n$. Then

$$\begin{aligned} (\pi^* - \lambda)(x) \in \mathbb{R}_-^d &\iff \langle \pi^*(x) - \lambda, e_i \rangle \leq 0, \forall i \\ &\iff \langle x, \pi(e_i) \rangle - \lambda_i \leq 0, \forall i \\ &\iff \langle x, v_i \rangle \leq \lambda_i, \forall i \\ &\iff x \in \Delta. \end{aligned}$$

□

We conclude that $\Delta \simeq \mathbb{R}_-^d \cap A$. Now $\mathbb{R}_-^d$ is the image of the moment map for the standard hamiltonian action of $\mathbb{T}^d$ on $\mathbb{C}^d$

$$\begin{aligned} \phi : \mathbb{C}^d &\longrightarrow \mathbb{R}^d \\ (z_1, \dots, z_d) &\longmapsto -\pi(|z_1|^2, \dots, |z_d|^2). \end{aligned}$$

Facts.

- The set $\phi^{-1}(A) \subset \mathbb{C}^d$ is a compact submanifold. Let $i : \phi^{-1}(A) \hookrightarrow \mathbb{C}^d$ denote inclusion. Then $i^*\omega_0$ is a closed 2-form which is degenerate. Its kernel is an integrable distribution. The corresponding foliation is called the **null foliation**.
- The null foliation of $i^*\omega_0$ is a principal fibration, so we take the quotient:

$$\begin{array}{ccc} N & \hookrightarrow & \phi^{-1}(A) \\ & & \downarrow \\ & & M_\Delta = \phi^{-1}(A)/N \end{array}$$

Let ω_Δ be the reduced symplectic form.

- The (non-effective) action of $\mathbb{T}^d = N \times \mathbb{T}^n$ on $\phi^{-1}(A)$ has a “moment map” with image $\phi(\phi^{-1}(A)) = \Delta$. (By “moment map” we mean a map satisfying the usual definition even though the closed 2-form is not symplectic.)

Theorem 29.4 For any $x \in \Delta$, we have that $\mu^{-1}(x)$ is a single $\mathbb{T}^n$ -orbit.

Proof. Exercise.

First consider the standard $\mathbb{T}^d$ -action on $\mathbb{C}^d$ with moment map $\phi : \mathbb{C}^d \rightarrow \mathbb{R}^d$. Show that $\phi^{-1}(y)$ is a single $\mathbb{T}^d$ -orbit for any $y \in \phi(\mathbb{C}^d)$. Now observe that

$$y \in \Delta' = \pi^*(\Delta) \iff \phi^{-1}(y) \subseteq Z.$$

Suppose that $y = \pi^*(x)$. Show that $\mu^{-1}(x) = \phi^{-1}(y)/N$. But $\phi^{-1}(y)$ is a single $\mathbb{T}^d$ -orbit where $\mathbb{T}^d = N \times \mathbb{T}^n$, hence $\mu^{-1}(x)$ is a single $\mathbb{T}^n$ -orbit. $\square$

Therefore, for toric manifolds, Δ is the orbit space.

Now Δ is a *manifold with corners*. At every point p in a face F , the tangent space $T_p\Delta$ is the subspace of $\mathbb{R}^n$ tangent to F . We can visualize $(M_\Delta, \omega_\Delta, \mathbb{T}^n, \mu)$ from Δ as follows. First take the product $\mathbb{T}^n \times \Delta$. Let p lie in the interior of $\mathbb{T}^n \times \Delta$. The tangent space at p is $\mathbb{R}^n \times (\mathbb{R}^n)^*$. Define ω_p by:

$$\omega_p(v, \xi) = \xi(v) = -\omega_p(\xi, v) \quad \text{and} \quad \omega_p(v, v') = \omega(\xi, \xi') = 0.$$

for all $v, v' \in \mathbb{R}^n$ and $\xi, \xi' \in (\mathbb{R}^n)^*$. Then ω is a closed nondegenerate 2-form on the interior of $\mathbb{T}^n \times \Delta$. At the corner there are directions missing in $(\mathbb{R}^n)^*$, so ω is a degenerate pairing. Hence, we need to eliminate the corresponding directions in $\mathbb{R}^n$. To do this, we collapse the orbits corresponding to subgroups of $\mathbb{T}^n$ generated by directions orthogonal to the annihilator of that face.

Example. Consider

$$(S^2, \omega = d\theta \wedge dh, S^1, \mu = h),$$

where S^1 acts on S^2 by rotation. The image of μ is the line segment $I = [-1, 1]$. The product $S^1 \times I$ is an open-ended cylinder. By collapsing each end of the cylinder to a point, we recover the 2-sphere. $\diamond$

Exercise. Build $\mathbb{CP}^2$ from $\mathbb{T}^2 \times \Delta$ where Δ is a right-angled isosceles triangle. $\diamond$

Finally, $\mathbb{T}^n$ acts on $\mathbb{T}^n \times \Delta$ by multiplication on the $\mathbb{T}^n$ factor. The moment map for this action is projection onto the Δ factor.

Homework 22: Delzant Theorem

1. (a) Consider the standard $(S^1)^3$ -action on $\mathbb{CP}^3$:

$$(e^{i\theta_1}, e^{i\theta_2}, e^{i\theta_3}) \cdot [z_0, z_1, z_2, z_3] = [z_0, e^{i\theta_1} z_1, e^{i\theta_2} z_2, e^{i\theta_3} z_3] .$$

Exhibit explicitly the subsets of $\mathbb{CP}^3$ for which the stabilizer under this action is $\{1\}$, S^1 , $(S^1)^2$ and $(S^1)^3$. Show that the images of these subsets under the moment map are the interior, the facets, the edges and the vertices, respectively.

- (b) Classify all 2-dimensional Delzant polytopes with 4 vertices, up to translation and the action of $SL(2; \mathbb{Z})$.

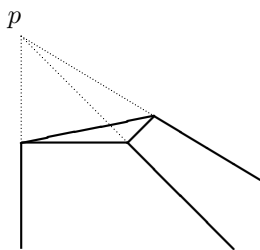
Hint: By a linear transformation in $SL(2; \mathbb{Z})$, you can make one of the angles in the polytope into a square angle. Check that automatically another angle also becomes 90° .

- (c) What are all the 4-dimensional symplectic toric manifolds that have four fixed points?

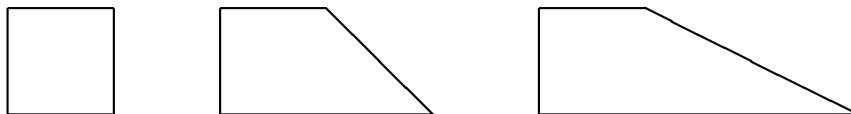
2. Take a Delzant polytope in $\mathbb{R}^n$ with a vertex p and with primitive (inward-pointing) edge vectors $u_1, \dots, u_n$ at p . Chop off the corner to obtain a new polytope with the same vertices except p , and with p replaced by n new vertices:

$$p + \varepsilon u_j, \quad j = 1, \dots, n,$$

where ε is a small positive real number. Show that this new polytope is also Delzant. The corresponding toric manifold is the ε -**symplectic blowup** of the original one.

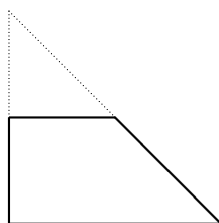


3. The toric 4-manifold $\mathcal{H}_n$ corresponding to the polygon with vertices $(0, 0)$, $(n+1, 0)$, $(0, 1)$ and $(1, 1)$, for n a nonnegative integer, is called a **Hirzebruch surface**.



- (a) What is the manifold $\mathcal{H}_0$? What is the manifold $\mathcal{H}_1$?

Hint:



- (b) Construct the manifold $\mathcal{H}_n$ by symplectic reduction of $\mathbb{C}^4$ with respect to an action of $(S^1)^2$.
- (c) Exhibit $\mathcal{H}_n$ as a $\mathbb{CP}^1$ -bundle over $\mathbb{CP}^1$.
4. Which $2n$ -dimensional toric manifolds have exactly $n + 1$ fixed points?

30 Duistermaat-Heckman Theorems

30.1 Duistermaat-Heckman Polynomial

Let (M^{2n}, ω) be a symplectic manifold. Then $\frac{\omega^n}{n!}$ is the symplectic volume form.

Definition 30.1 *The Liouville measure (or symplectic measure) of a Borel subset¹⁷ $\mathcal{U}$ of M is*

$$m_\omega(\mathcal{U}) = \int_{\mathcal{U}} \frac{\omega^n}{n!}.$$

Let G be a torus. Suppose that (M, ω, G, μ) is a hamiltonian G -space, and that the moment map μ is proper.

Definition 30.2 *The Duistermaat-Heckman measure, m_{DH} , on $\mathfrak{g}^*$ is the push-forward of m_ω by $\mu : M \rightarrow \mathfrak{g}^*$. That is,*

$$m_{DH}(U) = (\mu_* m_\omega)(U) = \int_{\mu^{-1}(U)} \frac{\omega^n}{n!}$$

for any Borel subset U of $\mathfrak{g}^*$.

For a compactly-supported function $h \in C^\infty(\mathfrak{g}^*)$, we define its integral with respect to the Duistermaat-Heckman measure to be

$$\int_{\mathfrak{g}^*} h \, dm_{DH} = \int_M (h \circ \mu) \frac{\omega^n}{n!}.$$

On $\mathfrak{g}^*$ regarded as a vector space, say $\mathbb{R}^n$, there is also the Lebesgue (or euclidean) measure, m_0 . The relation between m_{DH} and m_0 is governed by the Radon-Nikodym derivative, denoted by $\frac{dm_{DH}}{dm_0}$, which is a generalized function satisfying

$$\int_{\mathfrak{g}^*} h \, dm_{DH} = \int_{\mathfrak{g}^*} h \frac{dm_{DH}}{dm_0} \, dm_0.$$

Theorem 30.3 (Duistermaat-Heckman, 1982 [31]) *The Duistermaat-Heckman measure is a piecewise polynomial multiple of Lebesgue (or euclidean) measure m_0 on $\mathfrak{g}^* \simeq \mathbb{R}^n$, that is, the Radon-Nikodym derivative*

$$f = \frac{dm_{DH}}{dm_0}$$

is piecewise polynomial. More specifically, for any Borel subset U of $\mathfrak{g}^*$,

$$m_{DH}(U) = \int_U f(x) \, dx,$$

where $dx = dm_0$ is the Lebesgue volume form on U and $f : \mathfrak{g}^* \simeq \mathbb{R}^n \rightarrow \mathbb{R}$ is polynomial on any region consisting of regular values of μ .

¹⁷The set $\mathcal{B}$ of **Borel subsets** is the σ -ring generated by the set of compact subsets, i.e., if $A, B \in \mathcal{B}$, then $A \setminus B \in \mathcal{B}$, and if $A_i \in \mathcal{B}$, $i = 1, 2, \dots$, then $\cup_{i=1}^\infty A_i \in \mathcal{B}$.

The proof of Theorem 30.3 for the case $G = S^1$ is in Section 30.3. The proof for the general case, which follows along similar lines, can be found in, for instance, [54], besides the original articles.

The Radon-Nikodym derivative f is called the **Duistermaat-Heckman polynomial**. In the case of a toric manifold, the Duistermaat-Heckman polynomial is a universal constant equal to $(2\pi)^n$ when Δ is n -dimensional. Thus the symplectic volume of $(M_\Delta, \omega_\Delta)$ is $(2\pi)^n$ times the euclidean volume of Δ .

Example. Consider $(S^2, \omega = d\theta \wedge dh, S^1, \mu = h)$. The image of μ is the interval $[-1, 1]$. The Lebesgue measure of $[a, b] \subseteq [-1, 1]$ is

$$m_0([a, b]) = b - a .$$

The Duistermaat-Heckman measure of $[a, b]$ is

$$m_{DH}([a, b]) = \int_{\{(\theta, h) \in S^2 \mid a \leq h \leq b\}} d\theta \, dh = 2\pi(b - a) .$$

Consequently, the spherical area between two horizontal circles depends only on the vertical distance between them, a result which was known to Archimedes around 230 BC.

Corollary 30.4 *For the standard hamiltonian action of S^1 on (S^2, ω) , we have*

$$m_{DH} = 2\pi \, m_0 .$$

◇

30.2 Local Form for Reduced Spaces

Let (M, ω, G, μ) be a hamiltonian G -space, where G is an n -torus.¹⁸ Assume that μ is proper. If G acts freely on $\mu^{-1}(0)$, it also acts freely on nearby levels $\mu^{-1}(t)$, $t \in \mathfrak{g}^*$ and $t \approx 0$. Consider the reduced spaces

$$M_{\text{red}} = \mu^{-1}(0)/G \quad \text{and} \quad M_t = \mu^{-1}(t)/G$$

with reduced symplectic forms ω_{red} and ω_t . What is the relation between these reduced spaces as symplectic manifolds?

For simplicity, we will assume G to be the circle S^1 . Let $Z = \mu^{-1}(0)$ and let $i : Z \hookrightarrow M$ be the inclusion map. We fix a connection form $\alpha \in \Omega^1(Z)$ for the principal bundle

$$\begin{array}{ccc} S^1 & \hookrightarrow & Z \\ & & \downarrow \pi \\ & & M_{\text{red}} \end{array}$$

¹⁸The discussion in this section may be extended to hamiltonian actions of other compact Lie groups, not necessarily tori; see [54, Exercises 2.1-2.10].

that is, $\mathcal{L}_{X^\#}\alpha = 0$ and $\iota_{X^\#}\alpha = 1$, where $X^\#$ is the infinitesimal generator for the S^1 -action. From α we construct a 2-form on the product manifold $Z \times (-\varepsilon, \varepsilon)$ by the recipe

$$\sigma = \pi^*\omega_{\text{red}} - d(x\alpha) ,$$

x being a linear coordinate on the interval $(-\varepsilon, \varepsilon) \subset \mathbb{R} \simeq \mathfrak{g}^*$. (By abuse of notation, we shorten the symbols for forms on $Z \times (-\varepsilon, \varepsilon)$ which arise by pullback via projection onto each factor.)

Lemma 30.5 *The 2-form σ is symplectic for ε small enough.*

Proof. The form σ is clearly closed. At points where $x = 0$, we have

$$\sigma|_{x=0} = \pi^*\omega_{\text{red}} + \alpha \wedge dx ,$$

which satisfies

$$\sigma|_{x=0} \left(X^\#, \frac{\partial}{\partial x} \right) = 1 ,$$

so σ is nondegenerate along $Z \times \{0\}$. Since nondegeneracy is an open condition, we conclude that σ is nondegenerate for x in a sufficiently small neighborhood of 0. $\square$

Notice that σ is invariant with respect to the S^1 -action on the first factor of $Z \times (-\varepsilon, \varepsilon)$. In fact, this S^1 -action is hamiltonian with moment map given by projection onto the second factor,

$$x : Z \times (-\varepsilon, \varepsilon) \longrightarrow (-\varepsilon, \varepsilon) ,$$

as is easily verified:

$$\iota_{X^\#}\sigma = -\iota_{X^\#}d(x\alpha) = -\underbrace{\mathcal{L}_{X^\#}(x\alpha)}_0 + d\underbrace{\iota_{X^\#}(x\alpha)}_x = dx .$$

Lemma 30.6 *There exists an equivariant symplectomorphism between a neighborhood of Z in M and a neighborhood of $Z \times \{0\}$ in $Z \times (-\varepsilon, \varepsilon)$, intertwining the two moment maps, for ε small enough.*

Proof. The inclusion $i_0 : Z \hookrightarrow Z \times (-\varepsilon, \varepsilon)$ as $Z \times \{0\}$ and the natural inclusion $i : Z \hookrightarrow M$ are S^1 -equivariant coisotropic embeddings. Moreover, they satisfy $i_0^*\sigma = i^*\omega$ since both sides are equal to $\pi^*\omega_{\text{red}}$, and the moment maps coincide on Z because $i_0^*x = 0 = i^*\mu$. Replacing ε by a smaller positive number if necessary, the result follows from the equivariant version of the coisotropic embedding theorem stated in Section 8.3.¹⁹ $\square$

¹⁹The equivariant version of Theorem 8.6 needed for this purpose may be phrased as follows: Let (M_0, ω_0) , (M_1, ω_1) be symplectic manifolds of dimension $2n$, G a compact Lie group acting on (M_i, ω_i) , $i = 0, 1$, in a hamiltonian way with moment maps μ_0 and μ_1 , respectively, Z a manifold of dimension $k \geq n$ with a G -action, and $\iota_i : Z \hookrightarrow M_i$, $i = 0, 1$, G -equivariant coisotropic embeddings. Suppose that $\iota_0^*\omega_0 = \iota_1^*\omega_1$ and $\iota_0^*\mu_0 = \iota_1^*\mu_1$. Then there exist G -invariant neighborhoods $\mathcal{U}_0$ and $\mathcal{U}_1$ of $\iota_0(Z)$ and $\iota_1(Z)$ in M_0 and M_1 , respectively, and a G -equivariant symplectomorphism $\varphi : \mathcal{U}_0 \rightarrow \mathcal{U}_1$ such that $\varphi \circ \iota_0 = \iota_1$ and $\mu_0 = \varphi^*\mu_1$.

Therefore, in order to compare the reduced spaces

$$M_t = \mu^{-1}(t)/S^1, \quad t \approx 0,$$

we can work in $Z \times (-\varepsilon, \varepsilon)$ and compare instead the reduced spaces

$$x^{-1}(t)/S^1, \quad t \approx 0.$$

Proposition 30.7 *The reduced space (M_t, ω_t) is symplectomorphic to*

$$(M_{\text{red}}, \omega_{\text{red}} - t\beta),$$

where β is the curvature form of the connection α .

Proof. By Lemma 30.6, (M_t, ω_t) is symplectomorphic to the reduced space at level t for the hamiltonian space $(Z \times (-\varepsilon, \varepsilon), \sigma, S^1, x)$. Since $x^{-1}(t) = Z \times \{t\}$, where S^1 acts on the first factor, all the manifolds $x^{-1}(t)/S^1$ are diffeomorphic to $Z/S^1 = M_{\text{red}}$. As for the symplectic forms, let $\iota_t : Z \times \{t\} \hookrightarrow Z \times (-\varepsilon, \varepsilon)$ be the inclusion map. The restriction of σ to $Z \times \{t\}$ is

$$\iota_t^* \sigma = \pi^* \omega_{\text{red}} - t d\alpha.$$

By definition of curvature, $d\alpha = \pi^* \beta$. Hence, the reduced symplectic form on $x^{-1}(t)/S^1$ is

$$\omega_{\text{red}} - t\beta.$$

□

In loose terms, Proposition 30.7 says that the reduced forms ω_t vary linearly in t , for t close enough to 0. However, the identification of M_t with M_{red} as abstract manifolds is not natural. Nonetheless, any two such identifications are isotopic. By the homotopy invariance of de Rham classes, we obtain:

Theorem 30.8 (Duistermaat-Heckman, 1982 [31]) *The cohomology class of the reduced symplectic form $[\omega_t]$ varies linearly in t . More specifically,*

$$[\omega_t] = [\omega_{\text{red}}] + tc,$$

where $c = [-\beta] \in H_{\text{deRham}}^2(M_{\text{red}})$ is the first Chern class of the S^1 -bundle $Z \rightarrow M_{\text{red}}$.

Remark on conventions. Connections on principal bundles are Lie algebra-valued 1-forms; cf. Section 25.2. Often the Lie algebra of S^1 is identified with $2\pi i\mathbb{R}$ under the exponential map $\exp : \mathfrak{g} \simeq 2\pi i\mathbb{R} \rightarrow S^1$, $\xi \mapsto e^\xi$. Given a principal S^1 -bundle, by this identification the infinitesimal action maps the generator $2\pi i$ of $2\pi i\mathbb{R}$ to the generating vector field $X^\#$. A connection form A is then an imaginary-valued 1-form on the total space satisfying $\mathcal{L}_{X^\#} A = 0$ and $\iota_{X^\#} A = 2\pi i$. Its curvature form B is an imaginary-valued 2-form on the base satisfying $\pi^* B = dA$. By the Chern-Weil isomorphism, the **first Chern class** of the principal S^1 -bundle is $c = [\frac{i}{2\pi} B]$.

In this lecture, we identify the Lie algebra of S^1 with $\mathbb{R}$ and implicitly use the exponential map $\exp : \mathfrak{g} \simeq \mathbb{R} \rightarrow S^1$, $t \mapsto e^{2\pi it}$. Hence, given a principal S^1 -bundle, the infinitesimal action maps the generator 1 of $\mathbb{R}$ to $X^\#$, and here a connection form α is an ordinary 1-form on the total space satisfying $\mathcal{L}_{X^\#}\alpha = 0$ and $\iota_{X^\#}\alpha = 1$. The curvature form β is an ordinary 2-form on the base satisfying $\pi^*\beta = d\alpha$. Consequently, we have $A = 2\pi i\alpha$, $B = 2\pi i\beta$ and the first Chern class is given by $c = [-\beta]$. $\diamond$

30.3 Variation of the Symplectic Volume

Let (M, ω, S^1, μ) be a hamiltonian S^1 -space of dimension $2n$ and let (M_x, ω_x) be its reduced space at level x . Proposition 30.7 or Theorem 30.8 imply that, for x in a sufficiently narrow neighborhood of 0, the symplectic volume of M_x ,

$$\text{vol}(M_x) = \int_{M_x} \frac{\omega_x^{n-1}}{(n-1)!} = \int_{M_{\text{red}}} \frac{(\omega_{\text{red}} - x\beta)^{n-1}}{(n-1)!},$$

is a polynomial in x of degree $n-1$. This volume can be also expressed as

$$\text{vol}(M_x) = \int_Z \frac{\pi^*(\omega_{\text{red}} - x\beta)^{n-1}}{(n-1)!} \wedge \alpha.$$

Recall that α is a chosen connection form for the S^1 -bundle $Z \rightarrow M_{\text{red}}$ and β is its curvature form.

Now we go back to the computation of the Duistermaat-Heckman measure. For a Borel subset U of $(-\varepsilon, \varepsilon)$, the Duistermaat-Heckman measure is, by definition,

$$m_{DH}(U) = \int_{\mu^{-1}(U)} \frac{\omega^n}{n!}.$$

Using the fact that $(\mu^{-1}(-\varepsilon, \varepsilon), \omega)$ is symplectomorphic to $(Z \times (-\varepsilon, \varepsilon), \sigma)$ and, moreover, they are isomorphic as hamiltonian S^1 -spaces, we obtain

$$m_{DH}(U) = \int_{Z \times U} \frac{\sigma^n}{n!}.$$

Since $\sigma = \pi^*\omega_{\text{red}} - d(x\alpha)$, its power is

$$\sigma^n = n(\pi^*\omega_{\text{red}} - x d\alpha)^{n-1} \wedge \alpha \wedge dx.$$

By the Fubini theorem, we then have

$$m_{DH}(U) = \int_U \left[\int_Z \frac{\pi^*(\omega_{\text{red}} - x\beta)^{n-1}}{(n-1)!} \wedge \alpha \right] \wedge dx.$$

Therefore, the Radon-Nikodym derivative of m_{DH} with respect to the Lebesgue measure, dx , is

$$f(x) = \int_Z \frac{\pi^*(\omega_{\text{red}} - x\beta)^{n-1}}{(n-1)!} \wedge \alpha = \text{vol}(M_x).$$

The previous discussion proves that, for $x \approx 0$, $f(x)$ is a polynomial in x . The same holds for a neighborhood of any other regular value of μ , because we may change the moment map μ by an arbitrary additive constant.

Homework 23: S^1 -Equivariant Cohomology

1. Let M be a manifold with a circle action and $X^\#$ the vector field on M generated by S^1 . The algebra of S^1 -**equivariant forms** on M is the algebra of S^1 -invariant forms on M tensored with complex polynomials in x ,

$$\Omega_{S^1}^\bullet(M) := (\Omega^\bullet(M))^{S^1} \otimes_{\mathbb{R}} \mathbb{C}[x] .$$

The product $\wedge$ on $\Omega_{S^1}^\bullet(M)$ combines the wedge product on $\Omega^\bullet(M)$ with the product of polynomials on $\mathbb{C}[x]$.

- (a) We grade $\Omega_{S^1}^\bullet(M)$ by adding the usual grading on $\Omega^\bullet(M)$ to a grading on $\mathbb{C}[x]$ where the monomial x has degree 2. Check that $(\Omega_{S^1}^\bullet(M), \wedge)$ is then a supercommutative graded algebra, i.e.,

$$\underline{\alpha} \wedge \underline{\beta} = (-1)^{\deg \underline{\alpha} \cdot \deg \underline{\beta}} \underline{\beta} \wedge \underline{\alpha}$$

for elements of pure degree $\underline{\alpha}, \underline{\beta} \in \Omega_{S^1}^\bullet(M)$.

- (b) On $\Omega_{S^1}^\bullet(M)$ we define an operator

$$d_{S^1} := d \otimes 1 - \iota_{X^\#} \otimes x .$$

In other words, for an elementary form $\underline{\alpha} = \alpha \otimes p(x)$,

$$d_{S^1} \underline{\alpha} = d\alpha \otimes p(x) - \iota_{X^\#} \alpha \otimes xp(x) .$$

The operator d_{S^1} is called the **Cartan differentiation**. Show that d_{S^1} is a superderivation of degree 1, i.e., check that it increases degree by 1 and that it satisfies the *super* Leibniz rule:

$$d_{S^1}(\underline{\alpha} \wedge \underline{\beta}) = (d_{S^1} \underline{\alpha}) \wedge \underline{\beta} + (-1)^{\deg \underline{\alpha}} \underline{\alpha} \wedge d_{S^1} \underline{\beta} .$$

- (c) Show that $d_{S^1}^2 = 0$.

Hint: Cartan magic formula.

2. The previous exercise shows that the sequence

$$0 \longrightarrow \Omega_{S^1}^0(M) \xrightarrow{d_{S^1}} \Omega_{S^1}^1(M) \xrightarrow{d_{S^1}} \Omega_{S^1}^2(M) \xrightarrow{d_{S^1}} \dots$$

forms a graded complex whose cohomology is called the **equivariant cohomology**²⁰ of M for the given action of S^1 . The k th equivariant cohomology group of M is

$$H_{S^1}^k(M) := \frac{\ker d_{S^1} : \Omega_{S^1}^k \longrightarrow \Omega_{S^1}^{k+1}}{\operatorname{im} d_{S^1} : \Omega_{S^1}^{k-1} \longrightarrow \Omega_{S^1}^k} .$$

²⁰The **equivariant cohomology** of a topological space M endowed with a continuous action of a topological group G is, by definition, the cohomology of the diagonal quotient $(M \times EG)/G$, where EG is the *universal bundle* of G , i.e., EG is a contractible space where G acts freely. H. Cartan [21, 59] showed that, for the action of a compact Lie group G on a manifold M , the de Rham model $(\Omega_G^\bullet(M), d_G)$ computes the equivariant cohomology, where $\Omega_G^\bullet(M)$ are the G -equivariant forms on M . [8, 9, 29, 54] explain equivariant cohomology in the symplectic context and [59] discusses equivariant de Rham theory and many applications.

- (a) What is the equivariant cohomology of a point?
- (b) What is the equivariant cohomology of S^1 with its multiplication action on itself?
- (c) Show that the equivariant cohomology of a manifold M with a free S^1 -action is isomorphic to the ordinary cohomology of the quotient space M/S^1 .

Hint: Let $\pi : M \rightarrow M/S^1$ be projection. Show that

$$\begin{aligned} \pi^* : H^\bullet(M/S^1) &\longrightarrow H_{S^1}^\bullet(M) \\ [\alpha] &\longmapsto [\pi^*\alpha \otimes 1] \end{aligned}$$

is a well-defined isomorphism. It helps to choose a connection on the principal S^1 -bundle $M \rightarrow M/S^1$, that is, a 1-form θ on M such that $\mathcal{L}_{X^\#}\theta = 0$ and $\iota_{X^\#}\theta = 1$. Keep in mind that a form β on M is of type $\pi^*\alpha$ for some α if and only if it is *basic*, that is $\mathcal{L}_{X^\#}\beta = 0$ and $\iota_{X^\#}\beta = 0$.

3. Suppose that (M, ω) is a symplectic manifold with an S^1 -action. Let $\mu \in C^\infty(M)$ be a real function. Consider the equivariant form

$$\underline{\omega} := \omega \otimes 1 + \mu \otimes x.$$

Show that $\underline{\omega}$ is **equivariantly closed**, i.e., $d_{S^1}\underline{\omega} = 0$ if and only if μ is a moment map. The equivariant form $\underline{\omega}$ is called the **equivariant symplectic form**.

4. Let M^{2n} be a compact oriented manifold, not necessarily symplectic, acted upon by S^1 . Suppose that the set M^{S^1} of fixed points for this action is finite. Let $\alpha^{(2n)}$ be an S^1 -invariant form which is the top degree part of an equivariantly closed form of even degree, that is, $\alpha^{(2n)} \in \Omega^{2n}(M)^{S^1}$ is such that there exists $\underline{\alpha} \in \Omega_{S^1}^\bullet(M)$ with

$$\underline{\alpha} = \alpha^{(2n)} + \alpha^{(2n-2)} + \dots + \alpha^{(0)}$$

where $\alpha^{(2k)} \in (\Omega^{2k}(M))^{S^1} \otimes \mathbb{C}[x]$ and $d_{S^1}\underline{\alpha} = 0$.

- (a) Show that the restriction of $\alpha^{(2n)}$ to $M \setminus M^{S^1}$ is exact.

Hint: The generator $X^\#$ of the S^1 -action does not vanish on $M \setminus M^{S^1}$. Hence, we can define a *connection* on $M \setminus M^{S^1}$ by $\theta(Y) = \frac{\langle Y, X^\# \rangle}{\langle X^\#, X^\# \rangle}$, where $\langle \cdot, \cdot \rangle$ is some S^1 -invariant metric on M . Use $\theta \in \Omega^1(M \setminus M^{S^1})$ to chase the primitive of $\alpha^{(2n)}$ all the way up from $\alpha^{(0)}$.

- (b) Compute the integral of $\alpha^{(2n)}$ over M .

Hint: Stokes' theorem allows to *localize* the answer near the fixed points.

This exercise is a very special case of the Atiyah-Bott-Berline-Vergne localization theorem for equivariant cohomology [8, 14].

5. What is the integral of the symplectic form ω on a surface with a hamiltonian S^1 -action, knowing that the S^1 -action is free outside a finite set of fixed points?

Hint: Exercises 3 and 4.

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Modular Invariants and their Fusion Rules

David E Evans and Paulo R Pinto

This paper is dedicated to Robert T. Powers on the occasion of his sixtieth birthday

ABSTRACT. The subfactor approach to modular invariants gives insight into the fusion rule structure of the modular invariants.

1. Introduction and Background

We are going to use the inter-relations between quantum and classical dynamical systems and to use tools from non-commutative geometry or non-commutative operator algebras to understand phenomena in classical statistical mechanics.

A good starting point is the two dimensional Ising model and its C*-treatment [1, 23, 25]. The classical model is set in the configuration space $\mathcal{P} = \{\pm\}^{\mathbb{Z}^2}$ of distributions of + and - on the two dimensional lattice $\mathbb{Z}^2$, and the nearest neighbour Hamiltonian

$$H = - \sum_{\alpha, \beta \text{ } nn} J \sigma^\alpha \sigma^\beta$$

where the sum is over nearest neighbours α and β in $\mathbb{Z}^2$, and $\sigma = (\sigma^\alpha)$ in $\mathcal{P}$. The transfer matrix method allows us to study the classical model set in $C(P) = \bigotimes_{\mathbb{Z}^2} \mathbb{C}^2$ and its equilibrium states, characterized by say the Dobrushin-Lanford-Ruelle equations or a variational principle by a quantum system of noncommuting observables $\mathcal{A} = \bigotimes_{\mathbb{Z}} M_2$ in one dimension with dynamics $\alpha_t = T^{it}(-)T^{-it}$ and associated equilibrium states or more precisely its ground states. Associated to an equilibrium state μ at inverse temperature β for the Ising model is a ground state φ_μ on the Pauli algebra $\mathcal{A}$ and to each local observable F in $C(\mathcal{P})$, a quantum observable F_β depending only on the temperature such that we can describe the classical correlation values in terms of quantum ones: $\mu(F) = \varphi_\mu(F_\beta)$. The key

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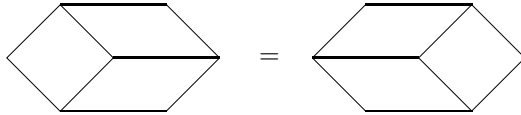


FIGURE 1. Yang-Baxter equation

element in this reduction is to identify the classical partition Z function in the denominator of the classical expectation value

$$(1.1) \quad \mu(F) = \sum_{\sigma} F(\sigma) \exp(-\beta H(\sigma)) / \sum_{\sigma} \exp(-\beta H(\sigma))$$

in quantum terms. We write for horizontal periodic boundary conditions:

$$\begin{aligned} Z &= \sum_{\sigma} \exp(-\beta H(\sigma)) \\ &= \begin{array}{|c|c|c|c|c|c|} \hline & & & & & \\ \hline & & & & & \\ \hline & & & & & \\ \hline & & & & & \\ \hline & & & & & \\ \hline & & & & & \\ \hline \end{array} = \sum_{\sigma} \begin{array}{|c|} \hline \\ \hline \end{array} \begin{array}{|c|} \hline \\ \hline \end{array} \begin{array}{|c|} \hline \\ \hline \end{array} \begin{array}{|c|} \hline \\ \hline \end{array} \begin{array}{|c|} \hline \\ \hline \end{array} \begin{array}{|c|} \hline \\ \hline \end{array} \\ &= \sum_{\sigma} T_{\xi_1 \xi_2} T_{\xi_2 \xi_3} \dots T_{\xi_N \xi_1} = \text{trace } T \dots T = \text{trace } T^N = \text{trace } e^{-N\mathcal{H}}. \end{aligned}$$

The partition function of a square of size M by N decomposes as the exponential of a sum factorising into a product of the partition functions of columns summed over column configurations and hence can be understood as a matrix product, computed in the trace as we have imposed periodic boundary conditions. Taking the scaling limit at criticality, we obtain a field theory, and for a torus model with periodic boundary conditions in addition vertically, we have

$$(1.2) \quad Z(\tau) = \text{tr}(q^{(L_0 - c/24)} \bar{q}^{(\bar{L}_0 - c/24)}).$$

Here L_0 and $\bar{L}_0$ are commuting Hamiltonians, usually part of commuting Virasoro algebras as for the Ising model here with associated central charge c , or multiplier of the projective representation L_m of the vector fields $l_m = -z^{m+1}d/dz$ on the circle. Much of this structure can be understood or is present in the statistical mechanical model itself, and one of our aims is to lift structures or understanding of the conformal field theories at criticality back to the original statistical mechanical models. For example, the central charge c itself can be seen from the asymptotics of the partition function Z : when $1 \ll N \ll M$, then $Z \approx \exp(-NMf + M\pi c/N6)$, where f is the free energy $\log(Z)/MN$ after $N, M \rightarrow \infty$ [15]. To generalise the Ising model we need some integrability of solutions of the Yang-Baxter equation as in Fig. 1. for local Boltzmann weights. Such solutions are naturally provided by Hecke algebras or quantum groups associated to $SU(n)_k$ particularly at roots of unity - so that $SU(2)_2$, the Ising model is the first in a double series of examples.

The Boltzmann weights lie in

$$(\bigotimes M_n)^{SU(n)_q} = \pi(\text{Hecke}).$$

The justification of the term $SU(n)$ models is as follows. By Weyl duality, the representation of the permutation group in $\bigotimes M_n$ is the fixed point algebra of the product action of $SU(n)$. Deforming this, there is a representation of the Hecke algebra in $\bigotimes M_n$ whose commutant is a representation of a deformation of $SU(n)$, the quantum group $SU(n)_q$ [31]. The braid relations are then precisely the YBE at criticality. The representations which appear here are labelled by Young tableaux of at most $n - 1$ rows but a further constraint implying rationality or finiteness of the representation labels occur at roots of unity $q = e^{\pi i/(n+k)}$ when only labels with at most k columns appear. The $SU(2)_k$ models are constructed by distributing edges of the Dynkin diagram A_{k+1} on square lattice. The first non trivial example here is described by A_2 , the two extreme vertices representing our previous symbols $\{\pm\}$ and the internal symbol is a dummy variable. The labels thus have two meanings coming from the representation theory of $SU(n)$ of symmetric group (and their deformations) and which point of view we want to emphasise may depend on whether we are more interested in the statistical mechanics or the conformal field theory picture:

statistical mechanics	conformal field theory
Hecke algebras	$SU(N)$ loop groups
λ symmetric or braided group representation	λ positive energy representation of $SU(N)$
II_1 bimodules Jones-Wenzl subfactors	III_1 sectors Jones-Wassermann subfactors
Ocneanu paragroup and connections	Longo Q-systems and canonical endomorphism

2. Modular Invariants

The link between the two frameworks are deformed Weyl duality, Popa's classification of type III inclusion in terms of those of type II and of course the partition function Z . Decomposing Eq. (1.2) according to our underlying loop group symmetries we have:

$$Z(\tau) = \sum_{\lambda, \mu} Z_{\lambda, \mu} \chi_\lambda(\tau) \chi_\mu(\tau)^*,$$

where $\chi_\lambda(\tau) = \text{tr}_{H_\lambda} \exp(2\pi i \tau L_0)$ is the character in the positive energy representation λ and $q = e^{2\pi i \tau}$. Modular invariance under reparameterisation of the torus then becomes $Z(\tau) = Z((a\tau + b)/(c\tau + d))$ for $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ in $SL(2; \mathbb{Z})$ which is generated

by $\mathcal{S} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ and $\mathcal{T} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. Since $SL(2; \mathbb{Z})$ acts linearly on characters, there is a representation of $SL(2; \mathbb{Z})$ by $\mathcal{S} \mapsto S = [S_{\lambda, \mu}]$, and $\mathcal{T} \mapsto T = [T_{\lambda, \mu}]$, where

$$\chi_\lambda(-1/\tau) = \sum_\mu S_{\lambda, \mu} \chi_\mu(\tau), \quad \chi_\lambda(\tau + 1) = \sum_\mu T_{\lambda, \mu} \chi_\mu(\tau).$$

Then modular invariance is

$$(2.1) \quad ZS = SZ, ZT = TZ.$$

Since the coefficients $Z_{\lambda\mu}$ appeared as multiplicities,

$$(2.2) \quad Z_{\lambda\mu} \in 0, 1, 2, \dots,$$

and usually through uniqueness of the vacuum we have

$$(2.3) \quad Z_{00} = 1.$$

A modular invariant is then a matrix $Z = [Z_{\lambda\mu}]$ satisfying the above constraints Eq. (2.1), Eq. (2.2) and Eq. (2.3), although we will see that much interesting structure is uncovered if we relax the normalization condition $Z_{00} = 1$. Usually as in the $SU(n)_k$ case, the modular data is such that S is symmetric and $S_{0\lambda} > 0$.

In this case [9, 28]

$$(2.4) \quad 0 \leq Z_{\lambda, \mu} \leq d_\lambda d_\mu$$

where $d_\lambda = S_{0, \lambda}/S_{00}$, and thus there are at most finitely many normalized modular invariants.

There always exists the trivial modular invariant $Z = 1$. For $SU(2)$ there is the celebrated A-D-E classification [14], e.g. $SU(2)_{16}$ have three modular invariants labelled by the Dynkin diagrams A_{17}, D_{10}, E_7 with corresponding invariants $Z_{A_{17}} = 1$, $Z_{D_{10}}$ is an orbifold obtained by folding the A_{17} -diagram which has a fixed point, and Z_{E_7} is a twist of the orbifold invariant. These are the normalized modular invariants, and if one relaxes the normalization condition, $Z_{00} = 1$, to consider for example products of normalized modular invariants then an interesting fusion rule algebra appears as in Table 1.

	$Z_{A_{17}}$	$Z_{D_{10}}$	Z_{E_7}
$Z_{A_{17}}$	$Z_{A_{17}}$	$Z_{D_{10}}$	Z_{E_7}
$Z_{D_{10}}$	$Z_{D_{10}}$	$2Z_{D_{10}}$	$2Z_{E_7}$
Z_{E_7}	Z_{E_7}	$2Z_{E_7}$	$Z_{D_{10}} + Z_{E_7}$

TABLE 1. Fusion of $SU(2)_{16}$ modular invariants

It is then natural to ask whether the (unnormalized) modular invariants form a fusion rule algebra generated by the normalized modular invariants and get a better understanding of these fusion rule algebras. The subfactor approach can assist us in this [21, 26].

3. Moore Seiberg dilation

Here we will have a type III factor N , endowed with a system $\mathcal{A}$ of endomorphisms which are taken to be braided and so yield a modular data. As we have seen we always have the trivial modular invariant

$$(3.1) \quad Z = \sum_\lambda |\chi_\lambda|^2,$$

where we now interpret χ_λ as a formal character $\text{tr} q^{L_0 - c/24}$ even when the Hamiltonian L_0 may not exist. More generally we may introduce twists or permutations of the fusion rules preserving S and T and the vacuum 0 so that we should also consider

$$(3.2) \quad Z = \sum_\lambda \chi_\lambda \chi_{\vartheta(\lambda)}^*.$$

In some sense every modular invariant can be dilated or brought to this form. Suppose we can extend the system in the following sense. We have a subfactor $N \subset M$ with a system $\mathcal{B}$ of endomorphisms of M . One can emphasise the extension aspect of moving from endomorphisms of N to those of M (α -induction [34, 6]). One needs the system $\mathcal{A}$ to be braided and the sector of $\bar{u}$, the dual canonical endomorphism to lie in $\Sigma(\mathcal{A})$, the set of finite integral sums of endomorphism of $\mathcal{A}$.

Under these conditions we emphasise the restriction aspect of this extension. The trivial or twisted modular invariants Eq. (3.2) for the $\mathcal{B}$ -system can be restricted to $\mathcal{A}$, written formally in terms of characters as $\chi_\tau = \sum_\lambda b_{\tau,\lambda} \chi_\lambda$, with branching coefficients $b_{\tau,\lambda}$. Then restricting the diagonal modular invariant

$$(3.3) \quad Z^{\text{ext}} = \sum_{\tau \in \mathcal{B}} |\chi_\tau|^2,$$

to the original system we have:

$$(3.4) \quad \sum_{\tau \in \mathcal{B}} |\chi_\tau|^2 = \sum_{\tau \in \mathcal{B}} \left| \sum_{\lambda \in \mathcal{A}} b_{\tau,\lambda} \chi_\lambda \right|^2,$$

with mass matrix

$$(3.5) \quad Z_{\lambda,\mu} = \sum_\tau b_{\tau,\lambda} b_{\tau,\mu}.$$

These invariants are called type I (or more precisely the inclusion $N \subset M$ describing this modular invariant is type I) and are necessarily symmetric $Z_{\lambda\mu} = Z_{\mu\lambda}$. In the presence of non-trivial twist ϑ of the $\mathcal{B}$ system, we have type II invariants

$$(3.6) \quad Z_{\lambda,\mu} = \sum_\tau b_{\tau,\lambda} b_{\vartheta(\tau),\mu}.$$

which have symmetric vacuum coupling $Z_{0\lambda} = Z_{\lambda 0}$.

A type III phenomena occurs, due to some underlying heterotic structure which results in needing different labellings $\mathcal{B}_+$ and $\mathcal{B}_-$ on left and right extended systems $\mathcal{A} \subset \mathcal{B}_\pm$. In the subfactor framework, this is found in two intermediate subfactors $N \subset M_\pm \subset M$, where $M_\pm$ carry systems of endomorphisms $\mathcal{B}_\pm$. Not only can this situation be found with modular invariants with non-symmetric vacuum coupling as in the orthogonal at loop groups low levels [8], but also with quantum doubles of finite groups [26]. Turning to the extension point of view $N \subset M$, when $\mathcal{A} \subset \text{End}(N)$ is non-degenerately braided and the dual canonical endomorphism θ lies in $\Sigma(\mathcal{A})$, we form $\mathcal{C}^\pm$ to be the systems of endomorphisms on M from the irreducible components of the sectors of $\{\alpha_\lambda^\pm : \lambda \in \mathcal{A}\}$. These generate the full system $\mathcal{C} = \mathcal{C}^+ \vee \mathcal{C}^-$, not necessarily even commutative, but $\mathcal{B} = \mathcal{C}^0 = \mathcal{C}^+ \cap \mathcal{C}^-$ is not only commutative but non-degenerately braided [8].

4. Izumi quantum E_6 model

Non-degenerately braided systems and the corresponding modular data can be obtained in the operator algebra setting in at least two ways. One is to take the Jones-Wassermann loop group examples in algebraic quantum field theory analysed by Wasserman [43] and students e.g. Laredo-Toledano [41]. The other is to take the quantum double of a system of endomorphisms - which may not even be commutative let alone braided or may be braided but the braiding has some degeneracy. Here we focus on one such example the quantum double of the even E_6 system. (As explained in [13, Section 5], this is more natural than taking the double of the entire E_6 system as the Longo-Rehren construction then gives the double of the E_6 subfactor). The E_6 system can be realised through a conformal embedding of $SU(2)_{10} \subset SO(5)_1$ [44, 6, 7]. The modular data obtained from the Longo-Rehren inclusion of the even system of E_6 is explicitly written down as follows in [30, Page 648] where $d = 1 + \sqrt{3}$, $\lambda = 2 + d^2$, $i = \sqrt{-1}$. The S and T matrices are

$$S = \frac{1}{\lambda} \begin{pmatrix} 1 & 1 & 1+d & 1+d & 2+d & d & d & d & d & d \\ 1 & 1 & 1+d & 1+d & -2-d & d & d & -d & -d & -d \\ 1+d & 1+d & 1 & 1 & 2+d & -d & -d & -d & -d & -d \\ 1+d & 1+d & 1 & 1 & -2-d & -d & -d & d & d & d \\ 2+d & -2-d & 2+d & -2-d & 0 & 0 & 0 & 0 & 0 & 0 \\ d & d & -d & -d & 0 & d & d & -d & -d & 2d \\ d & d & -d & -d & 0 & d & d & -d & -d & -2d \\ d & -d & -d & d & 0 & -d & d & -2i-di & 2i+di & 0 \\ d & -d & -d & d & 0 & -d & d & 2i+di & -2i-di & 0 \\ d & -d & -d & d & 0 & 2d & -2d & 0 & 0 & 0 \end{pmatrix}$$

$$T = \text{diag}(1, -1, 1, -1, 1, e^{\pi i/3}, e^{4\pi i/3}, e^{5\pi i/6}, e^{5\pi i/6}, -i).$$

4.1. Verlinde matrices of the quantum E_6 model. We computed numerically the Verlinde matrices of the quantum double of E_6 : N_0 is the identity matrix, and the others can be written as the following quadratic expressions

$$\begin{aligned} N_1 &= \chi_0\chi_1^* + \chi_1\chi_0^* + \chi_2\chi_3^* + \chi_3\chi_2^* + |\chi_4|^2 + \chi_5\chi_6^* + \chi_6\chi_5^* + \chi_7\chi_8^* + \chi_8\chi_7^* + |\chi_9|^2, \\ N_2 &= \chi_0\chi_2^* + \chi_2\chi_0^* + \chi_1\chi_3^* + \chi_3\chi_1^* + |\chi_2 + \chi_3|^2 + (\chi_2 + \chi_3)(\chi_5 + \chi_6)^* \\ &\quad + (\chi_5 + \chi_6)(\chi_2 + \chi_3)^* + 2|\chi_4|^2 + \chi_4(\chi_7 + \chi_8 + \chi_9)^* + (\chi_7 + \chi_8 + \chi_9)\chi_4^* \\ &\quad + \chi_5\chi_6^* + \chi_6\chi_5^* + \chi_7(\chi_8 + \chi_9)^* + (\chi_8 + \chi_9)\chi_7^* + \chi_8\chi_9^* + \chi_9\chi_8^*, \\ N_3 &= \chi_0\chi_3^* + \chi_3\chi_0^* + \chi_1\chi_2^* + \chi_2\chi_1^* + |\chi_2 + \chi_3|^2 + (\chi_2 + \chi_3)(\chi_5 + \chi_6)^* \\ &\quad + (\chi_5 + \chi_6)(\chi_2 + \chi_3)^* + 2|\chi_4|^2 + \chi_4(\chi_7 + \chi_8 + \chi_9)^* + (\chi_7 + \chi_8 + \chi_9)\chi_4^* \\ &\quad + |\chi_5|^2 + |\chi_6|^2 + |\chi_7|^2 + |\chi_8|^2 + \chi_9(\chi_7 + \chi_8)^* + (\chi_7 + \chi_8)\chi_9^*, \\ N_4 &= (\chi_0 + \chi_1 + 2\chi_2 + 2\chi_3)\chi_4^* + \chi_4(\chi_0 + \chi_1 + 2\chi_2 + 2\chi_3)^* + \chi_4(\chi_5 + \chi_6)^* \\ &\quad + (\chi_5 + \chi_6)\chi_4^* + (\chi_2 + \chi_3 + \chi_5 + \chi_6)(\chi_7 + \chi_8 + \chi_9)^* \\ &\quad + (\chi_7 + \chi_8 + \chi_9)(\chi_2 + \chi_3 + \chi_5 + \chi_6)^*, \\ N_5 &= (\chi_0 + \chi_3 + \chi_5)\chi_5^* + \chi_5(\chi_0 + \chi_3)^* + |\chi_2 + \chi_3|^2 \\ &\quad + (\chi_4 + \chi_7 + \chi_8 + \chi_9)\chi_4^* + \chi_4(\chi_7 + \chi_8 + \chi_9)^* + (\chi_1 + \chi_2 + \chi_6)\chi_6^* \\ &\quad + \chi_6(\chi_1 + \chi_2 + \chi_6)^* + \chi_7\chi_8^* + \chi_8\chi_7^* + |\chi_9|^2, \end{aligned}$$

$$\begin{aligned}
N_6 &= \chi_0\chi_6^* + \chi_6\chi_0^* + \chi_1\chi_5^* + \chi_5\chi_1^* + |\chi_2 + \chi_3|^2 + \chi_2\chi_5^* + \chi_5\chi_2^* + \chi_3\chi_6^* \\
&\quad + \chi_6\chi_3^* + (\chi_4 + \chi_7 + \chi_8 + \chi_9)\chi_4^* + \chi_4(\chi_7 + \chi_8 + \chi_9)^* + \chi_5\chi_6^* \\
&\quad + \chi_6\chi_5^* + |\chi_7|^2 + |\chi_8|^2 + |\chi_9|^2, \\
N_7 &= \chi_0\chi_7^* + \chi_8\chi_0^* + \chi_1\chi_8^* + \chi_7\chi_1^* + \chi_2(\chi_4 + \chi_8 + \chi_9)^* + (\chi_4 + \chi_7 + \chi_9)\chi_2^* \\
&\quad + \chi_3(\chi_4 + \chi_7 + \chi_9)^* + (\chi_4 + \chi_8 + \chi_9)\chi_3^* + (\chi_5 + \chi_6)\chi_4^* + \chi_4(\chi_5 + \chi_6)^* \\
&\quad + \chi_5\chi_8^* + \chi_7\chi_5^* + \chi_6\chi_7^* + \chi_8\chi_6^*, \\
N_8 &= \chi_0\chi_8^* + \chi_7\chi_0^* + \chi_1\chi_7^* + \chi_8\chi_1^* + \chi_2(\chi_4 + \chi_7 + \chi_9)^* \\
&\quad + (\chi_4 + \chi_8 + \chi_9)\chi_2^* + (\chi_4 + \chi_7 + \chi_9)\chi_3^* + \chi_3(\chi_4 + \chi_8 + \chi_9)^* \\
&\quad + (\chi_5 + \chi_6)\chi_4^* + \chi_4(\chi_5 + \chi_6)^* + \chi_5\chi_7^* + \chi_7\chi_5^* + \chi_6\chi_8^* + \chi_8\chi_6^*, \\
N_9 &= \chi_0\chi_9^* + \chi_9\chi_0^* + (\chi_2 + \chi_3)(\chi_4 + \chi_7 + \chi_8)^* + (\chi_4 + \chi_7 + \chi_8)(\chi_2 + \chi_3)^* \\
&\quad + (\chi_5 + \chi_6)(\chi_4 + \chi_9)^* + (\chi_4 + \chi_9)(\chi_5 + \chi_6)^* + \chi_1\chi_9^* + \chi_9\chi_1^*.
\end{aligned}$$

All the N_i 's are symmetric apart from N_7 and N_8 that are transpose of each other. The first two matrices N_0 and N_1 are permutation matrices as the primary fields 0 and 1 are the simple currents of our present modular data. The quantum E_6 modular invariant Z_2 is the charge conjugation invariant, i.e. $[Z_2]_{\lambda, \bar{\mu}} = \delta_{\lambda, \bar{\mu}}$. The Frobenius-Schur indicator $\text{FS}_\lambda = 1$ for all $\lambda \in \mathcal{A}$ except for $\lambda = 7, 8$ where it is zero.

4.2. Modular invariants of the quantum E_6 subfactor(s). The dimension of the commutant $\{S, T\}'$ is four and spanned by the modular invariants which are exactly four and were computed numerically using [10, Eq. (1.3)]. We obtain $Z_1 = N_0$, and the others are given by the following quadratic expressions:

$$\begin{aligned}
Z_2 &= |\chi_0|^2 + |\chi_1|^2 + |\chi_2|^2 + |\chi_3|^2 + |\chi_4|^2 + |\chi_5|^2 + |\chi_6|^2 + |\chi_9|^2 + \chi_7\chi_8^* + \chi_8\chi_7^*, \\
Z_3 &= |\chi_0 + \chi_2|^2 + |\chi_1 + \chi_3|^2 + 2|\chi_4|^2, \\
Z_4 &= |\chi_0 + \chi_2 + \chi_4|^2.
\end{aligned}$$

4.3. Subfactors for quantum E_6 modular invariants. Here we consider the Longo-Rehren $N \subset M$ inclusion of the even system of the E_6 Dynkin diagram, and let $\mathcal{A}$ be the system of endomorphisms that yield the above modular data. By [26, Lemma 3.8] (see also [40]) we can choose an endomorphism λ_1 from the sector $[\lambda_1]$ such that $\lambda_1^2 = 1$ since $T_{\lambda_1, \lambda_1} = 1$. Therefore $[\theta] = [\lambda_0] \oplus [\lambda_1]$ is a dual sector of the braided subfactor $N \subset N \rtimes \mathbb{Z}_2$. The irreducible decomposition of the N - M sectors is as follows: $[\iota\lambda_0] = [\iota\lambda_1]$, $[\iota\lambda_2] = [\iota\lambda_3]$, $[\iota\lambda_4] = [a_1] \oplus [a_2]$, $[\iota\lambda_5] = [\iota\lambda_6]$, $[\iota\lambda_7] = [\iota\lambda_8]$, and $[\iota\lambda_9] = [a_3] \oplus [a_4]$. Hence gives rise to a trace 8 modular invariant. So $Z_{N \subset N \rtimes \mathbb{Z}_2} = Z_2$ the permutation invariant. Second the $\text{LR}(E_6)$ dual canonical sector is $[\theta] = [\lambda_0] \oplus [\lambda_2] \oplus [\lambda_4]$. The irreducible decomposition of the N - M sectors are as follows: $[\iota\lambda_0], [\iota\lambda_1], [\iota\lambda_2] = [\iota\lambda_0] \oplus [\iota\lambda_5]$, $[\iota\lambda_3] = [\iota\lambda_1] \oplus [\iota\lambda_5]$, $[\iota\lambda_4] = [\iota\lambda_0] \oplus [\iota\lambda_1] \oplus [\iota\lambda_5]$, $[\iota\lambda_5] = [\iota\lambda_6] = [\iota\lambda_7] = [\iota\lambda_8] = [\iota\lambda_9]$. Therefore $\text{LR}(E_6)$ produces a trace 3 modular invariant. Hence $Z_{\text{LR}(E_6)} = Z_4$. By [13, Sect. 5] the notation of Fig. 1 on page 28, $[(0, 0)] \oplus [(6, 0)]$ is the $[\lambda_0] \oplus [\lambda_2]$ in our current notation. In [13, Example 5.1] instead of taking the three even vertices of E_6 one can take the two extreme endpoints (which are even and discard the internal point) following Izumi's Galois correspondence [29, Proposition 2.4]. In this way, we get an intermediate subfactor $N \subset P \subset M$ with inclusion map ι_P such that $[\theta] = [\lambda_0] \oplus [\lambda_2]$ is the sector of the dual canonical endomorphism of $N \subset P$. Then by the Verlinde matrices we find that: $[\iota_P\lambda_0], [\iota_P\lambda_1], [\iota_P\lambda_2] = [\iota_P\lambda_0] \oplus [\iota_P\lambda_5]$,

$[\iota_P \lambda_4] = [\iota_P \lambda_9] \oplus [a] \oplus [b]$, $[\iota_P \lambda_5] = [\iota_P \lambda_6]$, $[\iota_P \lambda_7] = [\iota_P \lambda_8]$, $[\iota_P \lambda_9]$, which gives trace 6 modular invariant. Thus the subfactor $N \subset P$ produces Z_3 . Moreover, chiral locality holds true here. We thus have proven the following.

THEOREM 4.1. *All the modular invariants of the quantum double of the E_6 system are realised by braided subfactors.*

The fusion rules between the quantum E_6 double (sufferable) modular invariants are as in Table 2. It is worth mentioning that as the modular invariants form a basis of the commutant $\{S, T\}'$ the coefficients exist and are uniquely determined, but due to a fusion rule algebra phenomenon those coefficients are non-negative integers [26, Corollary 3.6] (see also [21]).

	Z_1	Z_2	Z_3	Z_4
Z_1	Z_1	Z_2	Z_3	Z_4
Z_2	Z_2	Z_1	Z_3	Z_4
Z_3	Z_3	Z_3	$2Z_3$	$2Z_4$
Z_4	Z_4	Z_4	$2Z_3$	$3Z_4$

TABLE 2. Fusion $Z_a Z_b^t$ of modular invariants

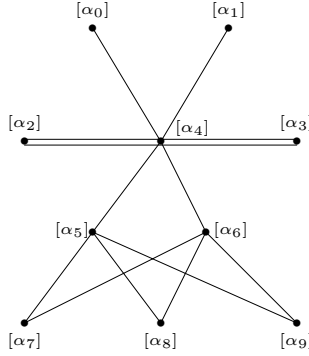


FIGURE 2. Z_1 , fusion graph of $[\alpha_4^+]$

4.4. Full systems of the quantum E_6 modular invariants. The global index of our system is $\omega = 8(1 + d + d^2) \sim 89.5692$.

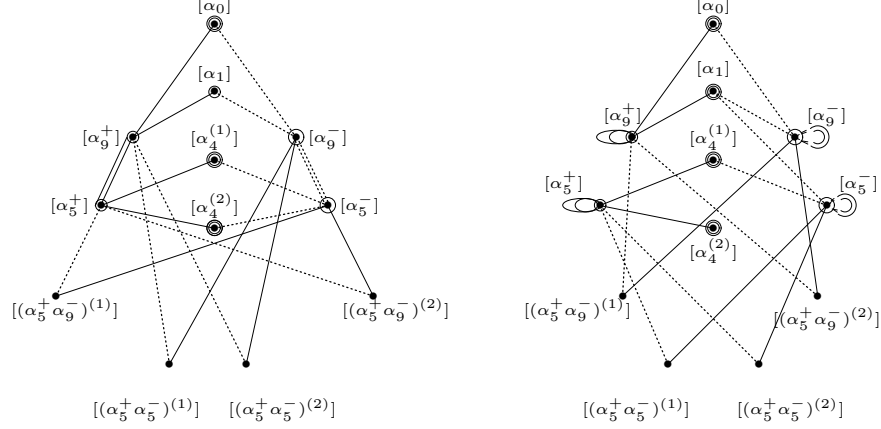
Case Z_1 . For the trivial modular invariant Z_1 we have $\mathcal{A} \simeq \mathcal{C}^0 \simeq \mathcal{C}$. We display the fusion graph of $[\alpha_4]$ in Fig. 2.

Case Z_2 . For the permutation invariant Z_2 , the full system $\mathcal{C}$ is obtained by permuting the sectors of those in the case Z_1 .

Case Z_3 . For the sufferable quantum E_6 modular invariant Z_3 we have by [12]:

$$\omega_{\pm} = 8(1 + d + d^2)/(2 + d) \sim 18.9282, \quad \omega_0 = 4.$$

Computing using the Verlinde matrices we find: $\langle \alpha_0, \alpha_0 \rangle = \langle \alpha_1^{\pm}, \alpha_1^{\pm} \rangle = \langle \alpha_5^{\pm}, \alpha_5^{\pm} \rangle = \langle \alpha_6^{\pm}, \alpha_6^{\pm} \rangle = \langle \alpha_7^{\pm}, \alpha_7^{\pm} \rangle = \langle \alpha_8^{\pm}, \alpha_8^{\pm} \rangle = 1$, $\langle \alpha_2^{\pm}, \alpha_2^{\pm} \rangle = \langle \alpha_3^{\pm}, \alpha_3^{\pm} \rangle = 2$, $\langle \alpha_4^{\pm}, \alpha_4^{\pm} \rangle = 3$, $\langle \alpha_0, \alpha_i^{\pm} \rangle = \delta_{0,i} + \delta_{2,i}$, $\langle \alpha_1^{\pm}, \alpha_i^{\pm} \rangle = \delta_{1,i} + \delta_{3,i}$, $\langle \alpha_2^{\pm}, \alpha_3^{\pm} \rangle = \langle \alpha_2^{\pm}, \alpha_5^{\pm} \rangle = \langle \alpha_2^{\pm}, \alpha_6^{\pm} \rangle = \langle \alpha_3^{\pm}, \alpha_5^{\pm} \rangle = \langle \alpha_3^{\pm}, \alpha_6^{\pm} \rangle = 1$, $\langle \alpha_4^{\pm}, \alpha_7^{\pm} \rangle = \langle \alpha_4^{\pm}, \alpha_8^{\pm} \rangle = \langle \alpha_4^{\pm}, \alpha_9^{\pm} \rangle = 1$, $\langle \alpha_5^{\pm}, \alpha_6^{\pm} \rangle = \langle \alpha_7^{\pm}, \alpha_8^{\pm} \rangle = \langle \alpha_7^{\pm}, \alpha_9^{\pm} \rangle = \langle \alpha_8^{\pm}, \alpha_9^{\pm} \rangle = 1$, and the others vanish. We then conclude

FIGURE 3. Z_3 , fusion graphs of $[\alpha_9^\pm]$ and $[\alpha_5^\pm]$ where $Z_3^2 = 2Z_3$

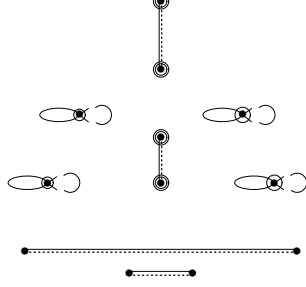
that $[\alpha_0]$ and $[\alpha_1^\pm]$ are irreducible, and $[\alpha_2^\pm] = [\alpha_0] \oplus [\alpha_5^\pm]$, $[\alpha_3^\pm] = [\alpha_1^\pm] \oplus [\alpha_5^\pm]$, $[\alpha_4^\pm] = [\alpha_9^\pm] \oplus [\alpha_4^{\pm(1)}] \oplus [\alpha_4^{\pm(2)}]$, $[\alpha_5^\pm] = [\alpha_6^\pm]$, $[\alpha_7^\pm] = [\alpha_8^\pm] = [\alpha_9^\pm]$. Using also the matrix $Z=Z_3$ and noting that $[\alpha_4^{\pm(i)}] = [\alpha_4^{\mp(i)}]$ for $i = 1, 2$ we conclude that $\mathcal{C}^0 = \{\alpha_0, \alpha_1, \alpha_4^{(1)}, \alpha_4^{(2)}\}$, and $\mathcal{C}^\pm = \mathcal{C}^0 \cup \{\alpha_5^\pm, \alpha_9^\pm\}$. Moreover since

$$\langle \alpha_4^+ \alpha_1^-, \alpha_4^+ \rangle = N_{4,4}^\xi Z_{1,\xi} = 3 = \langle \alpha_4^+ \alpha_1^-, \alpha_4^+ \alpha_1^- \rangle$$

we have the sectors of $\mathcal{C}^0$ equal $\mathbb{Z}_2 \times \mathbb{Z}_2$ with $[\alpha_4^{(i)}][\alpha_4^{(i)}] = [\text{id}]$. We need to find four more irreducible sectors in the full system $\mathcal{C}$ because $\#\mathcal{C} = \text{Tr}(ZZ^t) = 12$. We compute and find that

$$\langle \alpha_5^+ \alpha_5^-, \alpha_5^+ \alpha_5^- \rangle = \sum N_{5,5}^\xi N_{5,5}^\eta Z_{\xi,\eta} = 2$$

and the dimensions of the intertwiner spaces between $\alpha_5^+ \alpha_5^-$ and any sector in both chiral systems $\mathcal{C}^\pm$ vanish. Hence $[\alpha_5^+ \alpha_5^-]$ decomposes into two new irreducible sectors $[(\alpha_5^+ \alpha_5^-)^{(1)}]$ and $[(\alpha_5^+ \alpha_5^-)^{(2)}]$. The other two sectors are similarly obtained from the decomposition of $\alpha_9^+ \alpha_5^-$ further noting that $\langle \alpha_5^+ \alpha_5^-, \alpha_9^+ \alpha_5^- \rangle = 0$. We denote these new irreducible sectors by $[(\alpha_9^+ \alpha_5^-)^{(1)}]$ and $[(\alpha_9^+ \alpha_5^-)^{(2)}]$. We have the fusion graphs of both $[\alpha_9^+]$ and $[\alpha_9^-]$ displayed on the LHS of Fig. 3. We use straight lines for the fusion graph of $[\alpha_9^+]$ and dashed lines for that of $[\alpha_9^-]$. On the RHS of Fig. 3 the fusion graph of $[\alpha_5^+]$ and $[\alpha_5^-]$ is displayed. We display the fusion graph of $[\alpha_1^+]$ and $[\alpha_1^-]$ in Fig. 4. When θ is the local dual canonical endomorphism $\lambda_0 \oplus \lambda_2$, let γ denote the canonical sector of $N \subset M$. Since $\langle \alpha_5^+ \alpha_5^-, \gamma \rangle = \langle \alpha_5^+, \alpha_5^+ \rangle = 1$ we conclude from [7, Corollary 3.19] that $[\gamma] = [\alpha_0] \oplus [(\alpha_5^+ \alpha_5^-)^{(1)}]$. The full system $\mathcal{C}$ decomposes as two sheets. The first is the chiral system $\mathcal{C}^+$ and the second sheet comprises the irreducible components of $\mathcal{C}^+ \alpha_5^-$ in accordance with the decomposition of $Z_3^2 = 2Z_3$ and $[\gamma] = [\alpha_0] \oplus [(\alpha_5^+ \alpha_5^-)^{(1)}]$ and [26, Corollary 3.6] and [21].

FIGURE 4. Z_3 , fusion graph of $[\alpha_1^\pm]$ where $Z_3^2 = 2Z_3$

Case Z_4 . The global indices of the chiral systems $\mathcal{C}^\pm$ and $\mathcal{C}^0$ are by [12]:

$$\omega_\pm = 8(1 + d + d^2)/(4 + 2d) = 2 + d^2 \sim 9.4641, \quad \omega_0 = \omega_\pm^2/\omega = 1$$

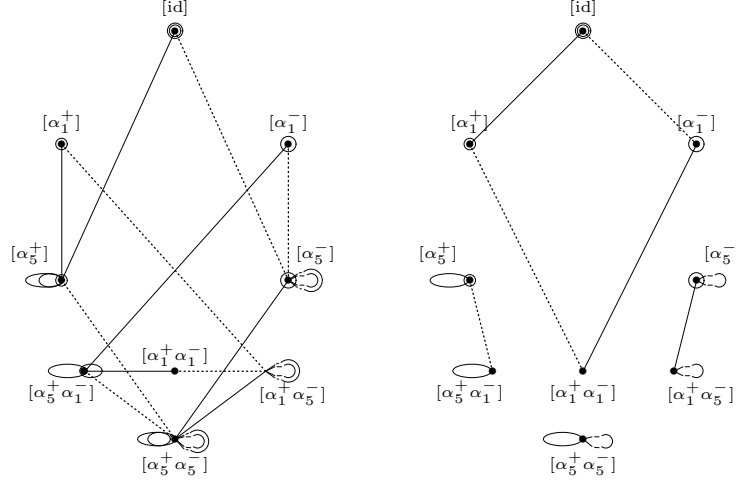
respectively. There are $9 = \text{Tr}(Z_4 Z_4^t)$ M - M irreducible sectors in the full commutative chiral system $\mathcal{C}$. Note that $\mathcal{C}^0 = \{\alpha_0\}$ since $\omega_0 = 1$. We next compute the chiral systems $\mathcal{C}^\pm$. We get (using the streamlined notation $\alpha_{\lambda_i}^\pm =: \alpha_i^\pm$): $\langle \alpha_0^\pm, \alpha_0^\pm \rangle = \langle \alpha_1^\pm, \alpha_1^\pm \rangle = \langle \alpha_4^\pm, \alpha_4^\pm \rangle = \langle \alpha_5^\pm, \alpha_5^\pm \rangle = \langle \alpha_6^\pm, \alpha_6^\pm \rangle = \langle \alpha_7^\pm, \alpha_7^\pm \rangle = \langle \alpha_8^\pm, \alpha_8^\pm \rangle = \langle \alpha_9^\pm, \alpha_9^\pm \rangle = 1$, $\langle \alpha_2^\pm, \alpha_2^\pm \rangle = \langle \alpha_3^\pm, \alpha_3^\pm \rangle = 2$, $\langle \alpha_4^\pm, \alpha_4^\pm \rangle = 3$, $\langle \alpha_0^\pm, \alpha_i^\pm \rangle = \delta_{0,i} + \delta_{2,i} + \delta_{4,i}$, $\langle \alpha_1^\pm, \alpha_i^\pm \rangle = \delta_{1,i} + \delta_{3,i} + \delta_{4,i}$, $\langle \alpha_2^\pm, \alpha_3^\pm \rangle = \langle \alpha_2^\pm, \alpha_5^\pm \rangle = 1$, $\langle \alpha_2^\pm, \alpha_4^\pm \rangle = 2$, $\langle \alpha_3^\pm, \alpha_4^\pm \rangle = 2$, $\langle \alpha_3^\pm, \alpha_5^\pm \rangle = \langle \alpha_3^\pm, \alpha_6^\pm \rangle = 1$, $\langle \alpha_4^\pm, \alpha_5^\pm \rangle = \langle \alpha_4^\pm, \alpha_6^\pm \rangle = \langle \alpha_5^\pm, \alpha_6^\pm \rangle = \langle \alpha_5^\pm, \alpha_7^\pm \rangle = \langle \alpha_5^\pm, \alpha_8^\pm \rangle = \langle \alpha_5^\pm, \alpha_9^\pm \rangle = 1$.

Therefore $\mathcal{C}^\pm = \{\alpha_0, \alpha_1^\pm, \alpha_5^\pm\}$ and as sectors $[\alpha_2^\pm] = [\alpha_0] \oplus [\alpha_5^\pm]$, $[\alpha_3^\pm] = [\alpha_1^\pm] \oplus [\alpha_5^\pm]$, $[\alpha_4^\pm] = [\alpha_0] \oplus [\alpha_1^\pm] \oplus [\alpha_5^\pm]$, $[\alpha_5^\pm] = [\alpha_6^\pm] = [\alpha_7^\pm] = [\alpha_8^\pm] = [\alpha_9^\pm]$.

The fusion rules are as the well known fusion rules of the even vertices of the graph E_6 [13]: $[\alpha_5^\pm \alpha_5^\pm] = [\alpha_0] \oplus [\alpha_1^\pm] \oplus 2[\alpha_5^\pm]$, $[\alpha_1^\pm][\alpha_5^\pm] = [\alpha_5^\pm]$, $[\alpha_1^\pm][\alpha_1^\pm] = [\alpha_0]$. We find that $\mathcal{C} = \mathcal{C}^+ \times \mathcal{C}^-$ and remark that $d_{\alpha_5^\pm} = 1 + \sqrt{3}$. We display the fusion the all M - M system together with the fusion graphs of both $[\alpha_5^-]$ and $[\alpha_5^+]$ in the LHS Fig. 5. In this figure, we use straight lines for the fusion graph of $[\alpha_5^+]$ whereas dashed lines for that of $[\alpha_5^-]$. We also encircled the $\mathcal{C}^+$ -chiral sectors with small circles and with larger circles those for the $\mathcal{C}^-$ -chiral system. The RHS of Fig. 5 we display the fusion graphs of $[\alpha_1^\pm]$. If θ is the local dual canonical endomorphism $\lambda_0 \oplus \lambda_2 \oplus \lambda_4$, let γ be the corresponding canonical endomorphism of $N \subset M$. As an application of [7, Corollary 3.19], we can compute that $[\gamma] = [\alpha_0] \oplus [\alpha_1^+ \alpha_1^-] \oplus [\alpha_5^+ \alpha_5^-]$. In accordance with $Z_4^2 = 3Z_4$ and [26, Corollary 3.6] the full system $\mathcal{C}$ decomposes as three sheets $\mathcal{C}^+$, $\mathcal{C}^+ \alpha_1^-$ and $\mathcal{C}^+ \alpha_5^-$.

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FIGURE 5. Z_4 , fusion graphs of $[\alpha_5^\pm]$ and $[\alpha_1^\pm]$ where $Z_4^2 = 3Z_4$

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SYMPLECTIC GEOMETRY

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Lecture Notes, University of Toronto

These are lecture notes for two courses, taught at the University of Toronto in Spring 1998 and in Fall 2000. Our main sources have been the books “Symplectic Techniques” by Guillemin-Sternberg and “Introduction to Symplectic Topology” by McDuff-Salamon, and the paper “Stratified symplectic spaces and reduction”, Ann. of Math. 134 (1991) by Sjamaar-Lerman.

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CHAPTER 1

Linear symplectic algebra

1. Symplectic vector spaces

Let E be a finite-dimensional, real vector space and E^* its dual. The space $\wedge^2 E^*$ can be identified with the space of skew-symmetric bilinear forms $\omega : E \times E \rightarrow \mathbb{R}$, $\omega(v, w) = -\omega(w, v)$.

DEFINITION 1.1. The pair (E, ω) is called a symplectic vector space if $\omega \in \wedge^2 E^*$ is non-degenerate, that is, if the kernel

$$\ker \omega := \{v \in E \mid \omega(v, w) = 0 \text{ for all } w \in E\}$$

is trivial. Two symplectic vector spaces (E_1, ω_1) and (E_2, ω_2) are called *symplectomorphic* if there is an isomorphism $A : E_1 \rightarrow E_2$ with $A^* \omega_2 = \omega_1$. The group of symplectomorphisms of (E, ω) is denoted $\text{Sp}(E)$.

Since $\text{Sp}(E)$ is a closed subgroup of $\text{Gl}(E)$, it is (by a standard theorem of Lie group theory) a Lie subgroup.

EXAMPLE 1.2. Let $E = \mathbb{R}^{2n}$ with basis $\{e_1, \dots, e_n, f_1, \dots, f_n\}$. Then

$$(1) \quad \omega(e_i, e_j) = 0, \quad \omega(f_i, f_j) = 0, \quad \omega(e_i, f_j) = \delta_{i,j}.$$

defines a symplectic structure on E . Examples of symplectomorphisms are $A(e_j) = f_j$, $A(f_j) = -e_j$ or $A(e_j) = e_j + f_j$, $A(f_j) = f_j$. Also

$$A(e_j) = \sum_k B_{jk} e_k, \quad A(f_j) = \sum_k (B^{-1})_{kj} f_k,$$

for any invertible $n \times n$ -matrix B , is a symplectomorphism.

EXAMPLE 1.3. Let V be a real vector space of dimension n , and V^* its dual space. Then $E = V \oplus V^*$ has a natural symplectic structure: $\omega((v, \alpha), (v', \alpha')) = \alpha'(v) - \alpha(v')$. If $B : V \rightarrow V$ is any isomorphism and $B^* : V^* \rightarrow V^*$ the dual map, $B \oplus (B^*)^{-1} : E \rightarrow E$ is a symplectomorphism.

EXAMPLE 1.4. Let E be a complex vector space of complex dimension n , with complex, positive definite inner product (=Hermitian metric) $h : E \times E \rightarrow \mathbb{C}$. Then E , viewed as a real vector space, with bilinear form the imaginary part $\omega = \text{Im}(h)$ is a symplectic vector space. Every unitary map $E \rightarrow E$ preserves h , hence also ω and is therefore symplectic.

EXERCISE 1.5. Show that these three examples of symplectic vector spaces are in fact symplectomorphic.

2. Subspaces of a symplectic vector space

DEFINITION 2.1. Let (E, ω) be a symplectic vector space. For any subspace $F \subseteq E$, we define the ω -perpendicular space F^ω by

$$F^\omega = \{v \in E, \omega(v, w) = 0 \text{ for all } w \in F\}$$

With our assumption that E is finite dimensional, ω is non-degenerate if and only if the map

$$\omega^\flat : E \rightarrow E^*, \langle \omega^\flat(v), w \rangle = \omega(v, w)$$

is an isomorphism. F^ω is the pre-image of the annihilator $\text{ann}(F) \subset E^*$ under $\omega^\flat$. From this it follows immediately that

$$\dim F^\omega = \dim E - \dim F$$

and

$$(F^\omega)^\omega = F.$$

DEFINITION 2.2. A subspace $F \subseteq E$ of a symplectic vector space is called

- (a) isotropic if $F \subseteq F^\omega$,
- (b) co-isotropic if $F^\omega \subseteq F$
- (c) Lagrangian if $F = F^\omega$,
- (d) symplectic if $F \cap F^\omega = \{0\}$.

The set of Lagrangian subspaces of E is called the *Lagrangian Grassmannian* and denoted $\text{Lag}(E)$.

Notice that F is isotropic if and only if F^ω is co-isotropic. For example, every 1-dimensional subspace is isotropic and every codimension 1 subspace is co-isotropic.

EXAMPLE 2.3. In the above example $E = \mathbb{R}^{2n}$, let $L = \text{span}\{g_1, \dots, g_n\}$ where for all i , $g_i = e_i$ or $g_i = f_i$. Then L is a Lagrangian subspace.

LEMMA 2.4. For any symplectic vector space (E, ω) there exists a Lagrangian subspace $L \in \text{Lag}(E)$.

PROOF. Let L be an isotropic subspace of E , which is maximal in the sense that it is not contained in any isotropic subspace of strictly larger dimension. Then L is Lagrangian: For if $L^\omega \neq L$, then choosing any $v \in L^\omega \setminus L$ would produce a larger isotropic subspace $L \oplus \text{span}(v)$. $\square$

An immediate consequence is that any symplectic vector space E has even dimension: For if L is a Lagrangian subspace, $\dim E = \dim L + \dim L^\omega = 2 \dim L$.

Lemma 2.4 can be strengthened as follows.

LEMMA 2.5. *Given any finite collection of Lagrangian subspaces $M_1, \dots, M_r$, one can find a Lagrangian subspace L with $L \cap M_j = \{0\}$ for all j .*

PROOF. Let L be an isotropic subspace with $L \cap M_j = \{0\}$ and not properly contained in a larger isotropic subspace with this property. We claim that L is Lagrangian. If not, L^ω is a coisotropic subspace properly containing L . Let $\pi : L^\omega \rightarrow L^\omega/L$ be the quotient map. Choose any 1-dimensional subspace $F \subset L^\omega/L$, such that both F is transversal to all $\pi(M_j \cap L^\omega)$. This is possible, since $\pi(M_j \cap L^\omega)$ is isotropic and therefore has positive codimension. Then $L' = \pi^{-1}(F)$ is an isotropic subspace with $L \subset L'$ and $L' \cap M_j = \{0\}$. This contradiction shows $L = L^\omega$. $\square$

3. Symplectic bases

THEOREM 3.1. *Every symplectic vector space (E, ω) of dimension $2n$ is symplectomorphic to $\mathbb{R}^{2n}$ with the standard symplectic form from Example 1.2.*

PROOF. Pick two transversal Lagrangian subspaces $L, M \in \text{Lag}(E)$. The pairing

$$L \times M \rightarrow \mathbb{R}, (v, w) \mapsto \omega(v, w)$$

is non-degenerate. In other words, the composition

$$M \hookrightarrow E \xrightarrow{\omega^\flat} E^* \rightarrow L^*$$

(where the last map is dual to the inclusion $L \hookrightarrow E$) is an isomorphism. Let $e_1, \dots, e_n$ be a basis for L and $f_1, \dots, f_n$ the dual basis for $L^* \cong M$. By definition of the pairing, ω is given in this basis by (1). $\square$

DEFINITION 3.2. A basis $\{e_1, \dots, e_n, f_1, \dots, f_n\}$ of (E, ω) for which ω has the standard form (1) is called a *symplectic basis*.

Our proof of Theorem 3.1 has actually shown a little more:

COROLLARY 3.3. *Let (E_i, ω_i) , $i = 1, 2$ be two symplectic vector spaces of equal dimension, and $L_i, M_i \in \text{Lag}(E_i)$ such that $L_i \cap M_i = \{0\}$. Then there exists a symplectomorphism $A : E_1 \rightarrow E_2$ such that $A(L_1) = L_2$ and $A(M_1) = M_2$.*

(Compare with isometries of inner product spaces, which are much more rigid!) In the following section we give an alternative proof of Theorem 3.1 using complex structures.

4. Compatible complex structures

Recall that a complex structure on a vector space V is an automorphism $J : V \rightarrow V$ such that $J^2 = -\text{Id}$.

DEFINITION 4.1. A complex structure J on a symplectic vector space (E, ω) is called ω -compatible if

$$g(v, w) = \omega(v, Jw)$$

defines a positive definite inner product. This means in particular that J is a symplectomorphism:

$$(J^*\omega)(v, w) = \omega(Jv, Jw) = g(Jv, w) = g(w, Jv) = \omega(w, J^2v) = -\omega(w, v) = \omega(v, w).$$

We denote by $\mathcal{J}(E, \omega)$ the space of compatible complex structures.

We equip $\mathcal{J}(E, \omega)$ with the subset topology induced from $\text{End}(E)$. Later we will see that it is in fact a smooth submanifold.

EXAMPLE 4.2. In Example 1.2, a compatible almost complex structure J is given by $Je_i = f_i$, $Jf_i = -e_i$. This identifies $(\mathbb{R}^{2n}, \omega, J)$ with $\mathbb{C}^n$.

A compatible complex complex structure makes E into a Hermitian vector space (= complex inner product space), with Hermitian metric

$$h(v, w) = g(v, w) + \sqrt{-1}\omega(v, w).$$

That is, h is complex-linear with respect to the second entry and complex-antilinear with respect to the first entry,

$$h(v, Jw) = \sqrt{-1}h(v, w), \quad h(Jv, w) = -\sqrt{-1}h(v, w),$$

and $h(v, v) > 0$ for $v \neq 0$. We will show below that $\mathcal{J}(E, \omega) \neq \emptyset$. Assuming this for a moment, let $J \in \mathcal{J}(E, \omega)$ and pick an orthonormal complex basis $e_1 \dots e_n$. Let $f_i = Je_i$. Then $e_1, \dots, e_n, f_1, \dots, f_n$ is a *symplectic* basis:

$$\omega(e_i, f_j) = \text{Im}(h(e_i, Je_j)) = \text{Im}(\sqrt{-1}h(e_i, e_j)) = \delta_{ij}, \quad \omega(e_i, e_j) = \text{Im}(h(e_i, e_j)) = 0$$

and similarly $\omega(f_i, f_j) = 0$. This is the promised alternative proof of Theorem 3.1.

The next Theorem gives a convenient method for constructing compatible complex structures. For any vector space V let $\text{Riem}(V)$ denote the convex open subset of the space S^2V^* of symmetric bilinear forms, consisting of positive definite inner products.

THEOREM 4.3. *Let (E, ω) be a symplectic vector space. There is a canonical continuous surjective map*

$$F : \text{Riem}(E) \rightarrow \mathcal{J}(E, \omega).$$

The map $G : \mathcal{J}(E, \omega) \rightarrow \text{Riem}(E)$, $J \mapsto g$ associating to each compatible complex structure the corresponding Riemannian structure is a section, i.e., $F \circ G(J) = J$.

PROOF. Given $k \in \text{Riem}(E)$ let $A \in \text{Gl}(E)$ be defined by

$$k(v, w) = \omega(v, Aw)$$

Since ω is skew-symmetric, A is skew-adjoint (with respect to k): $A^T = -A$. It follows that in the polar decomposition $A = J|A|$ with $|A| = (A^T A)^{1/2} = (-A^2)^{1/2}$, J and $|A|$ commute. Therefore $J^2 = -\text{Id}$. The equation

$$\omega(v, Jw) = \omega(v, A|A|^{-1}w) = k(v, |A|^{-1}w) = k(|A|^{-1/2}v, |A|^{-1/2}w)$$

shows that $g(v, w) = \omega(v, Jw)$ defines a positive definite inner product. We thus obtain a continuous map $F : \mathrm{Riem}(E) \rightarrow \mathcal{J}(E, \omega)$. By construction it satisfies $F \circ G = \mathrm{id}$, in particular it is surjective. $\square$

COROLLARY 4.4. *The space $\mathcal{J}(E, \omega)$ is contractible. (In particular, any two compatible complex structures can be deformed into each other.)*

PROOF. Let $X = \mathrm{Riem}(E)$ and $Y = \mathcal{J}(E, \omega)$. The space X is contractible since it is a convex subset of a vector space. Choose a contraction $\phi : I \times X \rightarrow X$, where $\phi_0 = \mathrm{Id}_X$ and ϕ_1 is the map onto some point in X . Then $\psi = F \circ \phi \circ (\mathrm{Id} \times G)$ is the required retraction of Y . $\square$

Given a Lagrangian subspace L of E , any orthonormal basis $e_1, \dots, e_n$ of L is an orthonormal basis for E viewed as a complex Hermitian vector space. The map taking this to an orthonormal basis $e'_1, \dots, e'_n$ of $L' \in \mathrm{Lag}(E)$ is unitary. Hence $\mathrm{U}(E)$ acts transitively on the set of Lagrangian subspaces. The stabilizer in $\mathrm{U}(E)$ of $L \in \mathrm{Lag}(E)$ is canonically identified with the orthogonal group $\mathrm{O}(L)$. This shows:

COROLLARY 4.5. *Any choice of $L \in \mathrm{Lag}(E)$ and $J \in \mathcal{J}(E, \omega)$ identifies the set of Lagrangian subspaces of E with the homogeneous space*

$$\mathrm{Lag}(E) = \mathrm{U}(E) / \mathrm{O}(L).$$

Hence $\mathrm{Lag}(E)$ is a manifold of dimension $n^2 - \frac{n(n-1)}{2} = \frac{n(n+1)}{2}$.

5. The group $\mathrm{Sp}(E)$ of linear symplectomorphisms

Let (E, ω) be a symplectic vector space of dimension $\dim E = 2n$, and $\mathrm{Sp}(E) \subset \mathrm{Gl}(E)$ the Lie group of symplectomorphisms $A : E \rightarrow E$, $A^*\omega = \omega$. Its dimension can be found as follows: Since any two symplectic vector spaces of the same dimension are symplectomorphic, the general linear group $\mathrm{Gl}(E)$ acts transitively on the open subset of $\wedge^2 E^*$ consisting of non-degenerate 2-forms. The stabilizer at ω is $\mathrm{Sp}(E)$. It follows that

$$\dim \mathrm{Sp}(E) = \dim \mathrm{Gl}(E) - \dim \wedge^2 E^* = (2n)^2 - \frac{(2n)(2n-1)}{2} = 2n^2 + n.$$

The Lie algebra $\mathfrak{sp}(E)$ of $\mathrm{Sp}(E)$ consists of all $\xi \in \mathfrak{gl}(E)$ such that $\omega(\xi v, w) + \omega(v, \xi w) = 0$.

The following example (really a repetition of example 1.3) shows in particular that $\mathrm{Sp}(E)$ is not compact.

EXAMPLE 5.1. Let $L, M \in \mathrm{Lag}(E)$ be transversal Lagrangian subspaces, and identify $M = L^*$ so that $E = L \oplus L^*$. Given $B \in \mathrm{Gl}(L)$ let $B^* \in \mathrm{Gl}(L^*)$ the dual map. Then $A = B \oplus (B^{-1})^*$ is a symplectomorphism. Thus for any splitting $E = L \oplus M$ there is a canonical embedding

$$\mathrm{Gl}(L) \rightarrow \mathrm{Sp}(E).$$

as a closed subgroup. Note that any $A \in \mathrm{Sp}(E)$ preserving L, M is of this type.

EXAMPLE 5.2. Another natural subgroup of $\mathrm{Sp}(E)$ is the group $U(E) \subset \mathrm{Sp}(E)$ of automorphisms preserving the Hermitian structure for a given compatible complex structure $J \in \mathcal{J}(E, \omega)$.

Let us now fix a compatible complex structure $J \in \mathcal{J}(E, \omega)$. Let $U(E) \subset \mathrm{Sp}(E)$ denote the unitary group and g the inner product defined by J, ω . If J' is another compatible complex structure, the map $A : E \rightarrow E$ taking an orthonormal basis with respect to (J, g) into one for (J', g') is symplectic and satisfies $A^* J' = J$. This shows:

COROLLARY 5.3. *The action of the symplectic group $\mathrm{Sp}(E)$ on the space $\mathcal{J}(E, \omega)$ of compatible complex structures is transitive, with stabilizer at $J \in \mathcal{J}(E, \omega)$ equal to the unitary group $U(E)$. That is, $\mathcal{J}(E, \omega)$ may be viewed as a homogeneous space*

$$(2) \quad \mathcal{J}(E, \omega) = \mathrm{Sp}(E)/U(E).$$

This shows in particular that $\mathrm{Sp}(E)$ is *connected*. We see that $\mathcal{J}(E, \omega)$ is a non-compact smooth manifold of dimension $(2n^2 + n) - n^2 = n^2 + n$. We will show below that the choice of J actually identifies $\mathcal{J}(E, \omega)$ with a vector space. Let $(\)^T$ denote the transpose of an endomorphism with respect to g .

LEMMA 5.4. *An automorphism $A \in \mathrm{Gl}(E, \omega)$ is in $\mathrm{Sp}(E)$ if and only if*

$$A^T = J A^{-1} J^{-1}$$

where A^T is the transpose of A with respect to g . An endomorphism $\xi \in \mathfrak{gl}(E)$ is in $\mathfrak{sp}(E)$ if and only if

$$\xi^T = J \xi J.$$

PROOF. $A \in \mathrm{Sp}(E)$ if and only if for all $v, w \in E$, $\omega(Av, Aw) = \omega(v, w)$, or equivalently $g(JAv, Aw) = g(Jv, w)$, i.e. $A^T J A = J$. The other identity is checked similarly. $\square$

EXERCISE 5.5. For $E = \mathbb{R}^{2n}$ with the standard symplectic basis and the standard symplectic structure, J is given by a matrix in block form, $J = \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}$. Writing

$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, verify that $A \in \mathrm{Sp}(E)$ is symplectic if and only if $a^T c$, $b^T d$ are symmetric and $a^T d - b^T c = I$. In particular, for $n = 1$ we have $\mathrm{Sp}(\mathbb{R}^2) = \mathrm{Sl}(2, \mathbb{R})$.

THEOREM 5.6 (Symplectic eigenvalue Theorem). *Let $A \in \mathrm{Sp}(E)$. Then $\det(A) = 1$, and all eigenvalues of A other than $1, -1$ come in either pairs*

$$\lambda, \bar{\lambda}, \quad |\lambda| = 1$$

or quadruples

$$\lambda, \bar{\lambda}, \lambda^{-1}, \bar{\lambda}^{-1}, \quad |\lambda| \neq 1.$$

The members of each multiplet all appear with the same multiplicity. The multiplicities of eigenvalues -1 and $+1$ are even.

PROOF. Since $\det(J) = 1$, the Lemma shows that $\det(A)^2 = 1$. Hence $\det(A) = 1$ since $\mathrm{Sp}(E)$ is connected. For any $A \in \mathrm{Gl}(E)$ the eigenvalues appear in complex-conjugate pairs of equal multiplicity. For $A \in \mathrm{Sp}(E)$, the eigenvalues λ, λ^{-1} have equal multiplicity since by the Lemma, the matrices A and A^{-1} are similar: $A^T = J A^{-1} J^{-1}$. The multiplicities of eigenvalues -1 and $+1$ have to be even since the product of all eigenvalues equals $\det A = 1$. $\square$

LEMMA 5.7. *Suppose $A \in \mathrm{Sp}(E)$ is symmetric, $A = A^T$ so that A is diagonalizable and all eigenvalues are real. Let $E_\lambda = \ker(A - \lambda)$ denote the eigenspace. Then*

$$E_\lambda^\omega = \oplus_{\lambda\mu \neq 1} E_\mu.$$

In particular, all E_λ for eigenvalues $\lambda \notin \{1, -1\}$ are isotropic while the eigenspaces for $\lambda \in \{1, -1\}$ are symplectic. Moreover, $E_\lambda \oplus E_{\lambda^{-1}}$ is symplectic.

PROOF. For $v \in E_\lambda, w \in E_\mu$ we have

$$\omega(v, w) = \omega(Av, Aw) = \lambda\mu\omega(v, w).$$

This, together with a check of dimensions proves the Lemma. $\square$

6. Polar decomposition of symplectomorphisms

We will use Lemma 5.7 to obtain the polar decomposition of symplectomorphisms. Recall that for any $A \in \mathrm{Gl}(E)$, the polar decomposition is the unique decomposition $A = CB$ into an orthogonal matrix C and a positive definite symmetric matrix B . Explicitly, $B = |A| := (A^T A)^{1/2}$ and $C = A|A|^{-1}$. Since the exponential map defines a diffeomorphism from the space of symmetric matrices onto positive definite symmetric matrices, this shows that $\mathrm{Gl}(E)$ is diffeomorphic to a product of $\mathrm{O}(E)$ and a vector space. We want to show that if $A \in \mathrm{Sp}(E)$, then both factors in the polar decomposition are in $\mathrm{Sp}(E)$. Thus let

$$\mathfrak{p} = \{\xi \in \mathfrak{sp}(E) \mid \xi = \xi^T\}, \quad P = \{A \in \mathrm{Sp}(E) \mid A = A^T, A > 0\}$$

be the intersection of $\mathrm{Sp}(E)$ with the set of positive definite automorphism and of $\mathfrak{sp}(E)$ with the space of symmetric endomorphisms.

LEMMA 6.1. *The exponential map restricts to a diffeomorphism $\exp : \mathfrak{p} \rightarrow P$.*

PROOF. Since clearly $\exp(\mathfrak{p}) \subset P$, it suffices to show that $\exp : \mathfrak{p} \rightarrow P$ is onto. Given $A \in P$, let $\xi \in \mathfrak{gl}(E)$ the unique symmetric endomorphism with $\exp(\xi) = A$. We have to show $\xi \in \mathfrak{sp}(E)$, or equivalently that $A^s = \exp(s\xi) \in \mathrm{Sp}(E)$ for all $s \in \mathbb{R}$. The power A^s acts on $v \in E_\lambda$ as $\lambda^s \mathrm{Id}$. Let $v \in E_\lambda, w \in E_\mu$. If $\lambda\mu \neq 1$ then $\omega(v, w) = 0$ and also $\omega(A^s v, A^s w) = (\lambda\mu)^s \omega(v, w) = 0$. If $\lambda\mu = 1$ then $\omega(A^s v, A^s w) = (\lambda\mu)^s \omega(v, w) = \omega(v, w)$. This shows $A^s \in \mathrm{Sp}(E)$. $\square$

THEOREM 6.2 (Polar decomposition). *The map $\mathrm{U}(E) \times P \rightarrow \mathrm{Sp}(E)$, $(C, B) \mapsto A = CB$ is a diffeomorphism.*

PROOF. We have to show that the map is onto. Let $A \in \mathrm{Sp}(E)$. By Lemma 5.4, $A^T = JAJ^{-1} \in \mathrm{Sp}(E)$. Therefore, using Lemma 6.1, $|A| = (A^T A)^{1/2} \in \mathrm{Sp}(E)$ and $A|A|^{-1} \in \mathrm{Sp}(E) \cap \mathrm{O}(E) = \mathrm{U}(E)$. $\square$

Since $\mathcal{J}(E, \omega) = \mathrm{Sp}(E)/\mathrm{U}(E)$, we have thus shown:

COROLLARY 6.3. *Any fixed $J \in \mathcal{J}(E, \omega)$ defines a canonical diffeomorphism between $\mathcal{J}(E, \omega)$ and the vector space $\mathfrak{p}$.*

In particular, we see once again that $\mathcal{J}(E, \omega)$ is contractible.

COROLLARY 6.4. *$\mathrm{Sp}(E)$ is homotopically equivalent to $\mathrm{U}(E)$. In particular, it is connected and has fundamental group $\pi_1(\mathrm{Sp}(E)) = \mathbb{Z}$.*

REMARK 6.5. Let $\mathfrak{g} = \mathfrak{sp}(E)$ the Lie algebra of the symplectic group and $\mathfrak{k} = \mathfrak{u}(E) = \mathfrak{sp}(E) \cap \mathfrak{o}(E)$ the Lie algebra of the orthogonal group. Then $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}$ as vector spaces, and

$$[\mathfrak{k}, \mathfrak{k}] \subseteq \mathfrak{k}, \quad [\mathfrak{k}, \mathfrak{p}] \subseteq \mathfrak{p}, \quad [\mathfrak{p}, \mathfrak{p}] \subseteq \mathfrak{k}.$$

The Killing form on $\mathfrak{g}$ is positive definite on $\mathfrak{k}$ and negative definite on $\mathfrak{p}$. From these facts it follows that $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}$ is a Cartan decomposition of the semi-simple Lie algebra $\mathfrak{sp}(E)$. In particular, $K = \mathrm{U}(E)$ is a maximal compact subgroup of $G = \mathrm{Sp}(E)$.

REMARK 6.6. The symplectic group $\mathrm{Sp}(E)$ should not be confused with the symplectic group $\mathrm{Sp}(n)$ from the theory of compact Lie groups. They are however two different real forms of the same complex Lie group.

7. Maslov indices and the Lagrangian Grassmannian

Let E, ω, J, g as before, $\dim E = 2n$. The determinant map $\det : \mathrm{U}(E) \rightarrow S^1$ induces an isomorphism of fundamental groups $\pi_1(\mathrm{U}(E)) \rightarrow \pi_1(S^1) = \mathbb{Z}$. Composing with the identification $\pi_1(\mathrm{U}(E)) \cong \pi_1(\mathrm{Sp}(E))$, we obtain an isomorphism

$$\mu : \pi_1(\mathrm{Sp}(E)) \rightarrow \mathbb{Z}$$

called the *Maslov index* of a loop of symplectomorphisms. It is independent of the choice of J , since any two choices are homotopic. If A, B are two loops and AB there pointwise product, $\mu(AB) = \mu(A) + \mu(B)$. Dually, we can pull-back the generator $\alpha \in H^1(S^1, \mathbb{Z}) \cong \mathbb{Z}$ by $\det$ to find a canonical class $\hat{\mu} \in H^1(\mathrm{Sp}(E), \mathbb{Z})$, called the *Maslov class*.

There are other ‘‘Maslov indices’’ related to the geometry of the Lagrangian Grassmannian

$$\mathrm{Lag}(E) = \mathrm{U}(E)/\mathrm{O}(L).$$

Since $\det : \mathrm{U}(E) \rightarrow S^1$ takes values ± 1 on $\mathrm{O}(L)$, its square descends to a well-defined function $\det^2 : \mathrm{Lag}(E) \rightarrow S^1$.

THEOREM 7.1 (Arnold). *The map $\det^2 : \text{Lag}(E) \rightarrow S^1$ defines an isomorphism of fundamental groups, $\mu : \pi_1(\text{Lag}(E)) \cong \pi_1(S^1) = \mathbb{Z}$ (independent of the choice of L or J). It is called the Maslov index of a loop of Lagrangian subspaces.*

PROOF. Choose an orthonormal basis for L to identify $L = \mathbb{R}^n \subset \mathbb{C}^n$ and $E \cong \mathbb{C}^n$ so that $\text{Lag}(E) = \text{U}(n)/\mathcal{O}(n)$. For $t \in [0, 1]$ let $A_k(t) \in \text{U}(n)$ be the diagonal matrix with entries $(\exp(\sqrt{-1}k\pi t), 1, \dots, 1)$. Since $A_k(1) \in \mathcal{O}(n)$, we obtain a loop $L_k(t) = A_k(t)\mathbb{R}^n$. This loop has Maslov index k , which shows that μ is surjective.

To show that μ is injective, it is enough to show that any loop $L(t)$ with $L(0) = L(1) = \mathbb{R}^n$ can be deformed into one of the loops L_k . To see this lift $L(t)$ to a path $A(t) \in \text{U}(n)$ (not necessarily closed) with endpoints $A(0) = 1$ and $A(1)$ equal to a diagonal matrix with entries $(\pm 1, 1, \dots, 1)$. We have to show that $A(t)$ can be deformed into one of the paths $A_k(t)$ while keeping the endpoints fixed. If the endpoint is the identity matrix $\text{diag}(1, 1, \dots, 1)$ so that $A(t)$ is actually a loop, this is clear because $\pi_1(\text{U}(n)) \rightarrow \pi_1(S^1) = \mathbb{Z}$ is an isomorphism; in this case k must be even. If $A(1) = \text{diag}(-1, 1, \dots, 1)$ the path $B(t) = A(t)A_{-1}(t)$ (pointwise product) is a loop, i.e. can be deformed into A_{2l} for some l . Thus $A(t) = B(t)A_1(t)$ can be deformed into $A_{2l}A_1 = A_{2l+1}$. $\square$

Pulling back the generator $\alpha \in H^1(S^1, \mathbb{Z})$ by $\det^2$ we find an integral cohomology class $\hat{\mu} \in H^1(\text{Lag}(E), \mathbb{Z})$ called the Maslov class.

PROPOSITION 7.2. *Let $A : S^1 \rightarrow \text{Sp}(E)$ and $L : S^1 \rightarrow \text{Lag}(E)$ be loops of symplectomorphisms resp. of Lagrangian subspaces. Then*

$$\mu(A(L)) = \mu(L) + 2\mu(A).$$

PROOF. Using the notation from the previous proof we may assume that A takes values in $\text{U}(n)$ since $\text{Sp}(E) = \text{U}(E) \times \mathfrak{p}$. Any such A is homotopic to a loop A_{2l} where $l = \mu(A)$. The Proposition follows since $A_{2l} \circ A_k = A_{k+2l}$. $\square$

Given $M \in \text{Lag}(E)$ consider the subset

$$\text{Lag}(E; M) = \{L \in \text{Lag}(E) \mid L \cap M = \{0\}\}.$$

LEMMA 7.3. *For any fixed $L \in \text{Lag}(E; M)$ one has a canonical diffeomorphism*

$$\text{Lag}(E; M) \cong S^2(L^*), \quad N \mapsto S_N$$

with the space of symmetric bilinear forms on L . In particular it is contractible. The kernel $\ker(S_N) \subset L$ is the intersection $N \cap L$.

PROOF. Recall that the non-degenerate pairing $L \times M \rightarrow \mathbb{R}$ defined by ω identifies $M = L^*$. Any n -dimensional subspace N transversal to M is of the form $N = \{v + \bar{S}_N(v) \mid v \in L\}$ for some linear map $\bar{S}_N : L \rightarrow M$. One has $N \cap L = \ker(\bar{S}_N)$. Since $L^* = M$ we may view S as a bilinear form $S_N \in L^* \otimes L^*$. The condition that N is Lagrangian is equivalent to

$$0 = \omega(v + \bar{S}_N(v), w + \bar{S}_N(w)) = \omega(v, \bar{S}_N(w)) - \omega(w, \bar{S}_N(v))$$

for all $v, w \in L$. In terms of S_N this says precisely that $S_N \in S^2 L^*$ is symmetric. $\square$

We note in passing that this Lemma defines coordinate charts on the Lagrangian Grassmannian.

REMARK 7.4. The coordinate version of this Lemma is as follows: Let $L \in \text{Lag}(\mathbb{R}^{2n})$ such that L is transversal to the span of $f_1, \dots, f_n$. Then L has a unique basis of the form $g_i = e_i + \sum_j S_{ij} f_j$. The condition $L^\omega = L$ translates into S being a symmetric matrix. If L, L' are two such Lagrangian subspaces and g_j, g'_j the corresponding bases, the pairing $\omega : L \times L' \rightarrow \mathbb{R}$ is given by

$$\omega(g_i, g'_j) = S_{ij} - S'_{ij}.$$

That is, the dimension of the intersection $L \cap L' = L^\omega \cap L'$ equals the nullity of $S - S'$.

The fact that $\text{Lag}(E; M)$ is contractible can be used to generalize the Maslov index to paths $L : [0, 1] \rightarrow \text{Lag}(E)$ which are not loops but satisfy the boundary conditions $L(0), L(1) \in \text{Lag}(E; M)$. Indeed, we can complete L to a loop $\tilde{L} : S^1 = \mathbb{R}/\mathbb{Z} \rightarrow \text{Lag}(E)$ with $\tilde{L}(t) = L(2t)$ for $0 \leq t \leq 1/2$ and $\tilde{L}(t) \in \text{Lag}(E; M)$ for $1/2 \leq t \leq 1$ such that $\tilde{L}(1) = L(0)$. The *Maslov intersection index* is defined as

$$[L : M] := \mu(\tilde{L}) \in \mathbb{Z}$$

which is independent of the choice of $\tilde{L}$.

REMARK 7.5. Maslov's index can be interpreted as a (signed) intersection number of L with the “singular cycle” $\text{Lag}(E) \setminus \text{Lag}(E; M)$. It was in this form that Maslov originally introduced his index. The difficulty of this approach is that the singular cycle is not a smooth submanifold of $\text{Lag}(E)$. Given a path in $\text{Lag}(E)$, one perturbs this path until it intersects only the smooth part of the singular cycle and all intersections are transversal. It is then necessary to prove that the index is independent of the choice of perturbation.

Maslov invented his index in the context of geometrical optics (“high frequency asymptotics”) and quantum mechanics “semi-classical approximation”. It appears physically as a phase shift when a light ray passes through a focal point; a phenomenon discovered in the 19th century. Mathematically Maslov's theory gave rise to Hörmander's theory of Fourier integral operators in PDE.

Maslov's index can be generalized to paths L that are not necessarily transversal to M at the end points. This was first done by Dazord in a 1979 paper and re-discovered several times since then. We will describe one such construction in the following section.

8. The index of a Lagrangian triple

In this section we describe a different approach towards Maslov indices, using the Hörmander-Kashiwara index of a Lagrangian triple. As a motivation, consider the action of $\text{Sp}(E)$ on $\text{Lag}(E)$. We have seen that this action is transitive. Moreover, any two ordered pairs of transversal Lagrangian subspaces can be carried into each other by some

symplectomorphism. An analogous statement is true if one fixes the dimension of the intersection $\dim(L_1 \cap L_2)$.

EXERCISE 8.1. Show that for any $L_1, L_2 \in \text{Lag}(E)$ there exists a symplectic basis in which L_1 is spanned by the $e_1, \dots, e_n$ and L_2 by $e_1, \dots, e_k, f_{k+1}, \dots, f_n$ where $k = \dim(L_1 \cap L_2)$. It follows that the action of $\text{Sp}(E)$ on $\text{Lag}(E) \times \text{Lag}(E)$ has $n + 1$ orbits, labeled by the dimension of intersections.

Is this true also for *triples* of Lagrangian sub-spaces?

EXERCISE 8.2. Let $E = \mathbb{R}^2$ with symplectic basis e, f . Let $L_1 = \text{span}\{e\}$, $L_2 = \text{span}\{f\}$. What is the form of a matrix of the most general symplectomorphism preserving L_1, L_2 ? Let $L_3 = \text{span}\{e + f\}$, and (L'_1, L'_2, L'_3) a second triple of Lagrangian subspaces with $L'_1 = L_1, L'_2 = L_2$. Show by direct computation that there exists a symplectic transformation $A \in \text{Sp}(E)$ with $L'_j = A(L_j)$ for all $j = 1, 2, 3$, if and only if $L'_3 = \text{span}\{e + \lambda f\}$ with $\lambda > 0$.

Thus, specifying the dimensions of intersections is insufficient for describing the orbit of a Lagrangian triple L_1, L_2, L_3 . There is another invariant called the Hörmander-Kashiwara index of a Lagrangian triple.

Before we define the index, let us recall that the signature $\text{Sig}(B) \in \mathbb{Z}$ of a symmetric matrix B is defined to be the number of its positive eigenvalues, minus the number of its negative eigenvalues. More abstractly, letting $\text{sign} : \mathbb{R} \rightarrow \mathbb{R}$ denote the sign function

$$\text{sign}(t) = \begin{cases} -1 & \text{if } t < 0 \\ 0 & \text{if } t = 0 \\ +1 & \text{if } t > 0 \end{cases}$$

and defining $\text{sign}(B)$ by functional calculus, we have $\text{Sig}(B) = \text{tr}(\text{sign}(B))$. The signature has the property $\text{Sig}(ABA^t) = \text{Sig}(B)$ for any invertible matrix A . If $k \in S^2(V^*)$ is a symmetric bi-linear form (equivalently, a quadratic form) on a vector space V , one defines

$$\text{Sig}(k) := \text{Sig}(B)$$

where B is the matrix of k in a given basis of V . The signature and the nullity are the only invariants of a symmetric bilinear form: That is, the action of $\text{Gl}(V)$ on $S^2(V^*)$ has a finite number of orbits labeled by $\dim(\ker(V))$ and $\text{Sig}(k)$.

Given three Lagrangian subspaces (not necessarily transversal) consider the symmetric bilinear form $Q(L_1, L_2, L_3)$ on their direct sum $L_1 \oplus L_2 \oplus L_3$, given by

$$Q(L_1, L_2, L_3)((v_1, v_2, v_3), (v_1, v_2, v_3)) = \omega(v_1, v_2) + \omega(v_2, v_3) + \omega(v_3, v_1).$$

The *index of the the Lagrangian triple* (L_1, L_2, L_3) is the signature,

$$s(L_1, L_2, L_3) := \text{Sig}(Q(L_1, L_2, L_3)) \in \mathbb{Z}.$$

It is due to Hörmander (in his famous 1971 paper on Fourier integral operators) and, in greater generality, Kashiwara (according to the book by Lion-Vergne). Clearly s is invariant under the action of $\text{Sp}(E)$ on $\text{Lag}(E)^3$.

Choosing bases for L_1, L_2, L_3 , the definition gives $Q(L_1, L_2, L_3)$ as a symmetric $3n \times 3n$ -matrix. One can reduce to signatures of $n \times n$ -matrices as follows. Choose a symplectic basis $e_1, \dots, e_n, f_1, \dots, f_n$ of E , such that L_1, L_2, L_3 are transversal to the span of $f_1, \dots, f_n$. Let S_j denote the symmetric bilinear forms on the span of $e_1, \dots, e_n$ corresponding to S_j . In terms of the basis, S_j is just a matrix, and $Q(L_1, L_2, L_3)$ is given by a symmetric matrix,

$$Q(L_1, L_2, L_3) = \frac{1}{2} \begin{pmatrix} 0 & S_1 - S_2 & S_3 - S_1 \\ S_1 - S_2 & 0 & S_2 - S_3 \\ S_3 - S_1 & S_2 - S_3 & 0 \end{pmatrix}.$$

LEMMA 8.3. $s(L_1, L_2, L_3) = \text{Sig}(S_1 - S_2) + \text{Sig}(S_2 - S_3) + \text{Sig}(S_3 - S_1)$.

PROOF. (Brian Feldstein) Let

$$T = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}.$$

An elementary calculation shows that $\det(T) \neq 0$ and that $TQ(L_1, L_2, L_3)T^t$ is the symmetric matrix,

$$\begin{pmatrix} S_3 - S_1 & 0 & 0 \\ 0 & S_2 - S_3 & 0 \\ 0 & 0 & S_1 - S_2 \end{pmatrix}.$$

From this the Lemma is immediate. $\square$

THEOREM 8.4. *The signature $s : \text{Lag}(E)^3 \rightarrow \mathbb{Z}$ of a Lagrangian triple has the following properties:*

- (a) *s is anti-symmetric under permutations of L_1, L_2, L_3 .*
- (b) *(Cocycle Identity) For all quadruples $L_1, L_2, L_3, L_4 \in \text{Lag}(E)$,*

$$s(L_1, L_2, L_3) - s(L_2, L_3, L_4) + s(L_3, L_4, L_1) - s(L_4, L_1, L_2) = 0.$$

- (c) *If $M(t)$ is a continuous path of Lagrangian subspaces such that $M(t)$ is always transversal to $L_1, L_2 \in \text{Lag}(E)$, then $s(L_1, L_2, M(t))$ is constant as a function of t .*
- (d) *Any ordered triple of Lagrangian subspaces is determined up to symplectomorphism by the five numbers*

$$\dim(L_1 \cap L_2), \dim(L_2 \cap L_3), \dim(L_3 \cap L_1), \dim(L_1 \cap L_2 \cap L_3), s(L_1, L_2, L_3).$$

PROOF. The first property is immediate from the definition, while the second and third property follow from the Lemma. The fourth property is left as a non-trivial exercise. (Perhaps try it first for the case that the L_j are pairwise transversal.) $\square$

LEMMA 8.5. *Suppose $L_1(t), L_2(t) \in \text{Lag}(E)$ are two paths of Lagrangian subspaces, $a \leq t \leq b$. Suppose there exists $M \in \text{Lag}(E)$ transversal to $L_1(t)$ and $L_2(t)$ for all $t \in [a, b]$. then the difference*

$$[L_1 : L_2] := \frac{1}{2}(s(L_1(a), L_2(a), M) - s(L_1(b), L_2(b), M))$$

is independent of the choice of such M .

PROOF. Let M, M' be two choices. By the cocycle identity, the first term changes by

$$s(L_1(a), L_2(a), M) - s(L_1(a), L_2(a), M') = s(L_1(a), M, M') - s(L_2(a), M, M').$$

We have to show that this equals the change of the second term,

$$s(L_1(b), L_2(b), M) - s(L_1(b), L_2(b), M') = s(L_1(b), M, M') - s(L_2(b), M, M').$$

But $s(L_1(t), M, M')$ and $s(L_2(t), M, M')$ are independent of t , since L_i stay transversal to M, M' . $\square$

We define the Maslov intersection index for two arbitrary paths $L_1, L_2 : [a, b] \rightarrow \text{Lag}(E)$ as follows: Choose a subdivision $a = t_0 < t_1 < \dots < t_k = b$ such that, for all $j = 1, \dots, k$, there is a Lagrangian subspace M^j transversal to $L_1(t), L_2(t)$ for $t \in [t_{j-1}, t_j]$. Then put

$$[L_1 : L_2] = \frac{1}{2} \sum_{j=1}^k (s(L_1(t_{j-1}), L_2(t_{j-1}), M^j) - s(L_1(t_j), L_2(t_j), M^j)).$$

Clearly, this is independent of the choice of subdivision and of the choice of the M^j . Note that this definition does not require transversality at the endpoints. The intersection is additive under concatenation of paths. It is anti-symmetric $[L_1 : L_2] = -[L_2 : L_1]$.

EXERCISE 8.6. Show that for any path of symplectomorphisms $A : [a, b] \rightarrow \text{Sp}(E)$, $[A(L_1) : A(L_2)] = [L_1 : L_2]$.

EXERCISE 8.7. Let $E = \mathbb{R}^2$, and let $L_1, L_2 : [a, b] \rightarrow \text{Lag}(E)$ be defined by $L_1(t) = \text{span}(f + te)$ and $L_2(t) = \text{span}(f)$. Find $[L_1 : L_2]$. How does it depend on a, b ?

EXERCISE 8.8. Let $L_1, L_2, L_3 : [a, b] \rightarrow \text{Lag}(E)$ be three paths of Lagrangian subspaces. Show that

$$[L_1 : L_2] + [L_2 : L_3] + [L_3 : L_1] = \frac{1}{2}(s(L_1(a), L_2(a), L_3(a)) - s(L_1(b), L_2(b), L_3(b))).$$

The approach can also be used to define Maslov indices of paths (not necessarily loops) of symplectomorphisms. Let E^- denote E with minus the symplectic form, and let $E \oplus E^-$ be equipped with the symplectic form $\text{pr}_1^* \omega - \text{pr}_2^* \omega$ where pr_i are the projections to the first and second factor.

PROPOSITION 8.9. *For any symplectomorphism $A \in \text{Sp}(E)$, the graph*

$$\Gamma_A := \{(Av, v) \mid v \in E\} \subset E \oplus E^-$$

is a Lagrangian subspace.

PROOF. Let pr_1, pr_2 denote the projections from $E \oplus E^-$ to the respective factor. For all $v_1, v_2 \in E$, we have

$$(\text{pr}_1^* \omega - \text{pr}_2^* \omega)((Av_1, v_1), (Av_2, v_2)) = -\omega(v_1, v_2) + \omega(Av_1, Av_2) = 0.$$

□

In this sense Lagrangian subspaces of $E^- \oplus E$ may be viewed as generalized symplectomorphisms. If $A : [a, b] \rightarrow \text{Sp}(E)$ is a path of symplectomorphisms, one can define a Maslov index $[\Gamma_A : \Delta]$ where $\Delta \subset E \oplus E^-$ is the diagonal. For loops based at the identity this reduces (up to a factor of 2) to the index $\mu(A)$ introduced earlier.

9. Linear Reduction

Suppose $F \subseteq E$ is a subspace. Then the kernel of the restriction of ω to F is just $F \cap F^\omega$ (by the very definition of F^ω). It follows that the quotient space $E_F = F/(F \cap F^\omega)$ inherits a natural symplectic form ω_F : Letting $\pi : F \rightarrow F/(F \cap F^\omega)$ be the quotient map we have

$$\omega_F(\pi(v), \pi(w)) = \omega(v, w)$$

for all $v, w \in F$. The space E_F is called the *reduced space* or *symplectic quotient*.

PROPOSITION 9.1. *Suppose $F \subseteq E$ is co-isotropic and $L \in \text{Lag}(E)$ Lagrangian. Let L_F be the image of $L \cap F$ under the reduction map $\pi : F \rightarrow F/F^\omega = E_F$. Then $L_F \in \text{Lag}(E_F)$.*

PROOF. Since $L \cap F$ is isotropic, it is immediate that L_F is isotropic. To verify that L_F is Lagrangian we just have to count dimensions: Using $(F_1 \cap F_2)^\omega = F_1^\omega + F_2^\omega$ for any $F_1, F_2 \subseteq E$ we compute

$$\begin{aligned} \dim(L \cap F^\omega) &= \dim E - \dim(L \cap F^\omega)^\omega \\ &= \dim E - \dim(L + F) \\ &= \dim E - \dim L - \dim F + \dim(L \cap F) \\ &= \dim(L \cap F) - \dim F + \dim L. \end{aligned}$$

This shows that

$$\dim L_F = \dim(L \cap F) - \dim(L \cap F^\omega) = \dim F - \dim L,$$

on the other hand

$$\dim E_F = \dim F - \dim F^\omega = 2 \dim F - \dim E = 2 \dim F - 2 \dim L.$$

□

For any symplectic vector space (E, ω) let E^- denote E with symplectic form $-\omega$. Suppose E_1, E_2, E_3 are symplectic vector spaces. and let

$$E = E_3 \oplus E_2^- \oplus E_2 \oplus E_1^-.$$

Then the diagonal $\Delta \subset E$, consisting of vectors (v_3, v_2, v_2, v_1) is co-isotropic. Given Lagrangian sub-spaces $\Gamma_1 \in \text{Lag}(E_2 \oplus E_1^-)$ and $\Gamma_2 \in \text{Lag}(E_3 \oplus E_2^-)$, the direct product is a Lagrangian subspace of E . The *composition* of Γ_1, Γ_2 is defined as

$$\Gamma_2 \circ \Gamma_1 = (\Gamma_2 \times \Gamma_1)_\Delta \in \text{Lag}(E_3 \oplus E_1^-).$$

This is really the composition of relations:

$$\Gamma_2 \circ \Gamma_1 = \{(v_3, v_1) \mid \exists v_2 \in E_2 \text{ with } (v_3, v_2) \in \Gamma_2, (v_2, v_1) \in \Gamma_1\}.$$

If A_1, A_2 are symplectomorphisms,

$$\Gamma_{A_2 \circ A_1} = \Gamma_{A_2} \circ \Gamma_{A_1}.$$

Similarly, for $L \in \text{Lag}(E)$ we have $\Gamma_A \circ L = A(L)$.

EXERCISE 9.2. Let $F \subseteq E$ be co-isotropic. Show that

$$\Gamma_F = \{(v, w) \in E_F \oplus E^- \mid w \in F, v = \pi(w)\}$$

(where $\pi : F \rightarrow E_F$ is the projection) is Lagrangian. It satisfies $\Gamma_F \circ L = L_F$ for all $L \in \text{Lag}(E)$.

CHAPTER 2

Review of Differential Geometry

1. Vector fields

We assume familiarity with the definition of a manifold (charts, smooth maps etc.). We will always take *paracompactness* as part of the definition – this condition ensures that every open cover U_α of M admits a subordinate partition of unity $f_\alpha \in C^\infty(M)$. That is, f_α is a non-negative function supported in U_α , near every point only a finite number of f_α 's are non-zero, and $\sum_\alpha f_\alpha = 1$.

Let $\mathfrak{X}(M)$ denote the vector space of derivations $X : C^\infty(M) \rightarrow C^\infty(M)$. That is $X \in \mathfrak{X}(M)$ if and only is

$$X(fg) = X(f)g + fX(g)$$

for all $f, g \in C^\infty(M)$. From this definition it follows easily that the value of $X(f)$ at $m \in M$ depends only on the behavior of f in an arbitrarily small neighborhood of m (i.e. on the “germ” of f at m). One can show that in any chart $U \subset M$, with local coordinates $x_1, \dots, x_n$ every $X \in \mathfrak{X}(M)$ has the form

$$X(f) = \sum_{i=1}^n a_i \frac{\partial f}{\partial x_i},$$

where a_i are smooth functions. Elements of $\mathfrak{X}(M)$ are called *vector fields*. The space $\mathfrak{X}(M)$ is a $C^\infty(M)$ -module (that is, for all $f \in C^\infty(M)$, $X \in \mathfrak{X}(M)$ one has $fX \in \mathfrak{X}(M)$), and it is a Lie algebra with bracket

$$[X, Y] = X \circ Y - Y \circ X.$$

In local coordinates, if $X = \sum_i a_i \frac{\partial}{\partial x_i}$ and $Y = \sum_i b_i \frac{\partial}{\partial x_i}$ then

$$[X, Y] = \sum_i \left(\sum_j a_j \frac{\partial b_i}{\partial x_j} - b_j \frac{\partial a_i}{\partial x_j} \right) \frac{\partial}{\partial x_i}.$$

A tangent vector at $m \in M$ is a linear map $v : C^\infty(M) \rightarrow \mathbb{R}$ such that

$$v(fg) = v(f)g(m) + f(m)v(g)$$

for all smooth f, g . The space of tangent vectors at m is denoted $T_m M$, and the union

$$TM := \coprod_{m \in M} T_m M$$

is the tangent bundle. In local coordinates, every $v \in T_m M$ is of the form

$$v(f) = \sum_{i=1}^n v_i \frac{\partial f}{\partial x_i}(m).$$

One can use this to define a manifold structure on TM , with local coordinates $x_1, \dots, x_n, v_1, \dots, v_n$. The natural projection map $\tau : TM \rightarrow M$ is smooth.

Clearly, every vector field X defines a tangent vector $v = X_m$ by $X_m(f) := X(f)(m)$. Conversely, every smooth map $X : M \rightarrow TM$ with $\tau \circ X = \text{id}_M$ defines a vector field. Thus vector fields can be viewed as sections of the tangent bundle TM .

For any smooth function $F : M \rightarrow N$ one has a linear *pull-back* map

$$F^* : C^\infty(N) \rightarrow C^\infty(M), \quad F^*g = g \circ F.$$

This defines a smooth push-forward map

$$F_* : TM \rightarrow TN, \quad v \mapsto F_*(v)$$

where $F_*(v)(g) = v(F^*g)$. This map is fiberwise linear, but does not in general carry vector fields to vector fields. (There are two problems: If F is not surjective we have no candidate for the section $N \rightarrow TN$ away from the image of F . If F is not injective, we may have more than one candidate over some points in the image of F .) Vector fields $X \in \mathfrak{X}(M)$ and $Y \in \mathfrak{X}(N)$ are called F -related (write $X \sim_F Y$) if for all $g \in C^\infty(N)$,

$$X(F^*g) = F^*(Y(g)).$$

This is equivalent to $F_*(X_m) = Y_{F(m)}$ for all $m \in M$. From the definition one sees easily the important fact,

$$X_1 \sim_F Y_1, X_2 \sim_F Y_2 \Rightarrow [X_1, X_2] \sim_F [Y_1, Y_2].$$

If F is a diffeomorphism, we denote by F_*X the unique vector field that is F -related to X .

A (global) *flow* on M is a smooth map

$$\phi : \mathbb{R} \times M \rightarrow M, (t, m) \mapsto \phi_t(m)$$

with

$$\phi_0 = \text{id}_M, \quad \phi_t \circ \phi_s = \phi_{t+s}.$$

Every global flow on M defines a vector field $X \in \mathfrak{X}(M)$ by

$$X(f) = \left. \frac{\partial}{\partial t} \right|_{t=0} \phi_t^* f.$$

For compact manifold M every vector field arises in this way. For non-compact M , one has to allow for “incomplete” flows, that is one has to restrict the definition of ϕ to some open neighborhood of $\{0\} \times M$ in $\mathbb{R} \times M$, and require $\phi_t(\phi_s(m)) = \phi_{t+s}(m)$ whenever

these terms are defined. Suppose ϕ_t is the flow defined by $X \in \mathfrak{X}(M)$. We define the *Lie derivative* of vector fields $Y \in \mathfrak{X}(M)$ by

$$L_X(Y) = \left. \frac{\partial}{\partial t} \right|_{t=0} (\phi_{-t})_*(Y) \in \mathfrak{X}(M).$$

It is a very important fact that for all vector fields X, Y ,

$$L_X(Y) = [X, Y].$$

2. Differential forms

Let E be a vector space of dimension n over $\mathbb{R}$. A k -form on E is a multi-linear map

$$\alpha : \underbrace{E \times \dots \times E}_{k \text{ times}} \rightarrow \mathbb{R},$$

anti-symmetric in all entries. The space of k -forms is denoted $\wedge^k E^*$. One has $\wedge^k E^* = \{0\}$ for $k > n$, $\wedge^0 E^* = \mathbb{R}$ and

$$\dim \wedge^k E^* = \binom{n}{k}$$

for $0 \leq k \leq n$. The space $\wedge E^* = \bigoplus_{k=0}^n \wedge^k E^*$ is an algebra with product given by anti-symmetrization of $\alpha \otimes \beta$:

$$(\alpha \wedge \beta)(X_1, \dots, X_{k+l}) = \sum_{\sigma \in S_{k+l}} \frac{(-1)^\sigma}{k!l!} \alpha(X_{\sigma(1)}, \dots, X_{\sigma(k)}) \beta(X_{\sigma(k+1)}, \dots, X_{\sigma(k+l)})$$

for $\alpha \in \wedge^k E^*$ and $\beta \in \wedge^l E^*$. Given $v \in E$ there is a natural *contraction map* $\iota_v : \wedge^k(E^*) \rightarrow \wedge^{k-1} E^*$ given by

$$\iota_v \alpha = \alpha(v, \cdot, \dots, \cdot).$$

This operator is a graded derivation:

$$\iota_v(\alpha \wedge \beta) = (\iota_v \alpha) \wedge \beta + (-1)^k \alpha \wedge \iota_v \beta.$$

REMARK 2.1. In general, given a $\mathbb{Z}$ -graded algebra $\mathcal{A} = \bigoplus_{k \in \mathbb{Z}} \mathcal{A}^k$ over a commutative ring $\mathcal{R}$, a derivation of degree r of $\mathcal{A}$ is a linear map $D : \mathcal{A} \rightarrow \mathcal{A}$ taking $\mathcal{A}^k$ into $\mathcal{A}^{k+r}$ and satisfying the graded Leibniz rule,

$$D(ab) = D(a)b + (-1)^{kr} a D(b)$$

for $a \in \mathcal{A}^k$, $b \in \mathcal{A}^l$. Let $\text{Der}(\mathcal{A}) = \bigoplus_{r \in \mathbb{Z}} \text{Der}^r(\mathcal{A})$ denote the graded space of graded derivations of $\mathcal{A}$. Then $\text{Der}(\mathcal{A})$ is a graded Lie algebra over $\mathcal{R}$: That is, for $D_j \in \text{Der}^{r_j}(\mathcal{A})$, $j = 1, 2$ one has

$$[D_1, D_2] := D_1 D_2 - (-1)^{r_1 r_2} D_2 D_1 \in \text{Der}^{r_1 + r_2}(\mathcal{A})$$

Suppose now that M is a manifold. Then we can define vector bundles

$$\wedge^k T^*M := \coprod_{m \in M} \wedge^k (T_m^*M).$$

Smooth sections of $\wedge^k T^*M$ are called k -forms, and the space of k -forms is denoted $\Omega^k(M)$. The space $\Omega^*(M) = \bigoplus_k \Omega^k(M)$ is a graded algebra over $C^\infty(M)$. Equivalently, k -forms can be viewed as $C^\infty(M)$ -multilinear anti-symmetric maps

$$\alpha : \underbrace{\mathfrak{X}(M) \times \cdots \times \mathfrak{X}(M)}_{k \text{ times}} \rightarrow C^\infty(M).$$

Note $\Omega^0(M) = C^\infty(M)$. For any vector field $X \in \mathfrak{X}(M)$ there is a contraction operator $\iota_X : \Omega^k(M) \rightarrow \Omega^{k-1}(M)$ which is a graded derivation.

For any function $f \in C^\infty(M)$, denote by $df \in \Omega^1(M)$ the 1-form such that $df(X) = X(f)$.

THEOREM 2.2. *The map $d : \Omega^0(M) \rightarrow \Omega^1(M)$ extends uniquely to a graded derivation $d : \Omega(M) \rightarrow \Omega(M)$ of degree 1, in such a way that $d^2 = 0$.*

The quotient space $H^k(M) = \ker(d|\Omega^k) / \text{im}(d|\Omega^{k-1})$ is called the k th de Rham cohomology group of M . Wedge product gives $H^*(M)$ the structure of a graded algebra. For M compact, the de Rham cohomology groups are always finite-dimensional vector spaces.

Suppose now that $F : M \rightarrow N$ is smooth. Then F defines a unique pull-back map $F^* : \Omega^k(N) \rightarrow \Omega^k(M)$, by

$$F^*\beta(v_1, \dots, v_k) = \beta(F_*(v_1), \dots, F_*(v_k))$$

for all $v_j \in T_m M$. The exterior differential respects F , that is

$$dF^*\beta = F^*d\beta.$$

Suppose ϕ_t is the flow of $X \in \mathfrak{X}$. The Lie derivative of $\alpha \in \Omega^k(M)$ with respect to X is defined as follows:

$$L_X \alpha = \left. \frac{\partial}{\partial t} \right|_{t=0} \phi_t^* \alpha.$$

Since pull-backs commute with the exterior differential, $L_X \circ d = d \circ L_X$. The operator L_X is a graded derivation of degree 0:

$$L_X(\alpha \wedge \beta) = (L_X \alpha) \wedge \beta + \alpha \wedge (L_X \beta).$$

One has the following relations between the derivations d, L_X, ι_X .

$$[d, d] = 2d^2 = 0, \quad [d, L_X] = 0, \quad [L_X, L_Y] = L_{[X, Y]},$$

$$[\iota_X, \iota_Y] = 0, \quad [L_X, \iota_Y] = \iota_{[X, Y]}, \quad [d, \iota_X] = L_X.$$

The last formula

$$d \circ \iota_X + \iota_X \circ d = L_X$$

is known as Cartan's formula. These relations show that the linear subspace of $\text{Der}(\Omega(M))$ spanned by ι_X, L_X, d is in fact a subalgebra of the graded Lie algebra $\text{Der}(\Omega(M))$.

Let us re-express some of the above in local coordinates $x_1, \dots, x_n$ for a chart $U \subset M$. At every point $m \in M$, the 1-forms $dx_1, \dots, dx_n$ are a basis of T_m^*M dual to the basis $\frac{\partial}{\partial x_1}, \dots, \frac{\partial}{\partial x_n}$ of $T_m M$. Every 1-form α is given on U by an expression

$$\alpha = \sum_{i=1}^n \alpha_i dx_i$$

where α_i are smooth functions. For $X = \sum_i X_i \frac{\partial}{\partial x_i}$ we have

$$\alpha(X) = \sum_{i=1}^n \alpha_i X_i.$$

More generally, for every ordered tuple $I = (i_1, \dots, i_k)$ of cardinality $|I| = k$ put

$$dx_I = dx_{i_1} \wedge \dots \wedge dx_{i_k}.$$

Then every $\alpha \in \Omega^k(M)$ has the coordinate expression

$$\alpha = \sum_I \alpha_I dx_I$$

where α_I are smooth functions and the sum is over all *ordered* tuples $i_1 < \dots < i_k$. One has

$$d\alpha = \sum_{j=1}^n \sum_I \frac{\partial \alpha_I}{\partial x_j} dx_j \wedge dx_I,$$

indeed, this formula is forced on us by the derivation property of d and the conditions $d^2=0$.

EXERCISE 2.3. Verify that this formula indeed gives $d^2 = 0$, using the equality of mixed partials.

EXERCISE 2.4. Work out the coordinate expressions for Lie derivatives L_X and contractions ι_X .

A volume form on a manifold M of dimension n is an n -form $\Lambda \in \Omega^n(M)$ with $\Lambda_m \neq 0$ for all m . If M admits a volume form it is called orientable. The choice of an equivalence class of volume forms, where Λ_1, Λ_2 are equivalent if $\Lambda_2 = f\Lambda_1$ with $f > 0$ everywhere, is called an orientation.

EXERCISE 2.5. For any volume form Λ and $X \in \mathfrak{X}(M)$, the divergence of X with respect to Λ is the function $\text{div}_\Lambda(X)$ such that $L_X \Lambda = \text{div}_\Lambda(X) \Lambda$. Find a formula for $L_X \Lambda$ in local coordinates.

CHAPTER 3

Foundations of symplectic geometry

1. Definition of symplectic manifolds

DEFINITION 1.1. A symplectic manifold is a pair (M, ω) consisting of a manifold M together with a closed, non-degenerate 2-form $\omega \in \Omega^2(M)$. Given two symplectic manifolds (M_i, ω_i) , a symplectomorphism is a diffeomorphism $F: M_1 \rightarrow M_2$ such that $F^*\omega_2 = \omega_1$. The group of symplectomorphism of M onto itself is denoted $\text{Symp}(M, \omega)$. The space of vector fields X with $L_X\omega = 0$ is denoted $\mathfrak{X}(M, \omega)$.

Non-degeneracy means that for each $m \in M$, the form ω_m is a symplectic form on $T_m M$, in particular $\dim M = 2n$ is even. Equivalently, the top exterior power ω_m^n is non-zero.

DEFINITION 1.2. The volume form $\Lambda = \exp(\omega)_{[\dim M]} = \frac{1}{n!} \omega^n$ is called the Liouville form.

DEFINITION 1.3. For any $H \in C^\infty(M, \mathbb{R})$, the corresponding Hamiltonian vector field X_H is the unique vector field satisfying

$$\iota_{X_H} \omega = dH$$

The space of vector fields X of the form $X = X_H$ is denoted $\mathfrak{X}_{\text{Ham}}(M, \omega)$.

PROPOSITION 1.4. *Every Hamiltonian vector field is a symplectic vector field. That is,*

$$\mathfrak{X}_{\text{Ham}}(M, \omega) \subseteq \mathfrak{X}(M, \omega).$$

PROOF. Suppose $X = X_H$, that is $\iota_X \omega = dH$. Then

$$L_X \omega = d\iota_X \omega = ddH = 0.$$

□

2. Examples

2.1. Example: open subsets of $\mathbb{R}^{2n}$.

EXAMPLE 2.1. The basic example are open subsets $U \subseteq \mathbb{R}^{2n}$. Let $q_1, \dots, q_n, p_1, \dots, p_n$ be coordinates with respect to a symplectic basis $e_1, \dots, e_n, f_1, \dots, f_n$ for $\mathbb{R}^{2n}$. This

identifies $e_j = \frac{\partial}{\partial q_j}$ and $f_j = \frac{\partial}{\partial p_j}$. In terms of the dual 1-forms $dq_1, \dots, dp_n$, the symplectic form is given by

$$\omega = \sum_{j=1}^n dq_j \wedge dp_j$$

and the Liouville form reads

$$\Lambda = dq_1 \wedge dp_1 \wedge \dots \wedge dq_n \wedge dp_n.$$

Given a smooth function H on U , we have

$$X_H = \sum_{j=1}^n \left(\frac{\partial H}{\partial p_j} \frac{\partial}{\partial q_j} - \frac{\partial H}{\partial q_j} \frac{\partial}{\partial p_j} \right).$$

Hence the ordinary differential equation defined by X_H is

$$\dot{q}_j = \frac{\partial H}{\partial p_j}, \quad \dot{p}_j = -\frac{\partial H}{\partial q_j}.$$

Note that Hamiltonians $H(q, p) = p_j$ generate translation in q_j -direction while $H(q, p) = q_j$ generate translation in minus p_j -direction.

DEFINITION 2.2. Let (M, ω) be a symplectic manifold. We denote by $\mathfrak{X}_{\text{Ham}}(M, \omega)$ the space of Hamiltonian vector fields on M , and by $\mathfrak{X}(M, \omega)$ the space of vector fields X on M preserving ω , i.e. $L_X \omega = 0$.

2.2. Example: Surfaces. Let Σ be an orientable 2-manifold, and $\omega \in \Omega^2(\Sigma)$ a volume form. Then ω is non-degenerate (since $\omega^n = \omega \neq 0$ everywhere) and closed (since every top degree form is obviously closed). A symplectomorphism is just a volume-preserving diffeomorphism in this case. By a result of Moser, any two volume forms on a compact manifold M , defining the same orientation and having the same total volume, are related by some diffeomorphism of M . In particular, every closed symplectic 2-manifold is determined up to symplectomorphism by its genus and total volume.

2.3. Example: Cotangent bundles. A very important example for symplectic manifolds are cotangent bundles. Let Q be a manifold, $M = T^*Q$ its cotangent bundle. There is a *canonical* 1-form $\theta \in \Omega^1(T^*Q)$ given as follows: Let $\pi : M = T^*Q \rightarrow Q$ the bundle projection, $d\pi : TM \rightarrow TQ$ its tangent map. Then for vectors $X_m \in T_m M$,

$$\langle \theta_m, X_m \rangle := \langle m, d_m \pi(X_m) \rangle$$

(since $m \in T_{\pi(m)}^*Q$ is a covector at $\pi(m)$, we can pair it with the projection of X_m to the base!) An alternative characterization of the form θ is as follows.

PROPOSITION 2.3. θ is the unique 1-form $\theta \in \Omega^1(T^*Q)$ with the property that for any 1-form $\alpha \in \Omega^1(Q)$ on the base,

$$\alpha = \alpha^* \theta$$

where on the right hand side, α is viewed as a section $\alpha : Q \rightarrow T^*Q = M$.

PROOF. To check the property let $w \in T_q Q$. Then $d_q \alpha(Y)$ projects to w . We have, therefore,

$$\langle (\alpha^* \theta)_q, w \rangle = \langle \theta_{\alpha_q}, d_q \alpha(w) \rangle = \langle \alpha_q, w \rangle.$$

Uniqueness follows since every tangent vector $v \in T_m M$ (except the vertical ones, i.e. those in the kernel of $d_m \pi$) can be written in the form $d_q \alpha(w)$, where $\alpha \in \Omega^1(Q)$, with $\alpha(q) = m$ and $w \in T_q Q$. (Non-vertical vectors span all of $T_m M$). $\square$

It is useful to work out the form θ in local coordinates.

For any vector bundle $\pi : E \rightarrow Q$ of rank k , one obtains local coordinates over a bundle chart $W \subset Q$ by choosing local coordinates $q_1, \dots, q_n$ on Q and a basis $\epsilon_1, \dots, \epsilon_k : Q \rightarrow E$ for the $C^\infty(W)$ -module of sections of $E|_W$. Any point $m \in E$ is then given by the coordinates q_i of its base point $q = \pi(m)$ and the coordinates $p_1, \dots, p_k$ such that $m = \sum_j p_j \epsilon_j(q)$. Thus if $\sigma = \sum_i \sigma_i \epsilon_i : W \rightarrow E|_W$ is any section over W , the pull-backs of q_i, p_i viewed as functions on $E|_W$ are we have $\sigma^* p_i = \sigma_i$ and $\sigma^* q_i = q_i$.

In our case, $E = T^*Q$, $k = n$ and a natural basis for the space of sections is given by the 1-forms $\epsilon_i = dq_i$. The corresponding coordinates $q_1, \dots, q_n, p_1, \dots, p_n$ on $T^*Q|_W$ are called *cotangent coordinates*.

LEMMA 2.4. *In local cotangent coordinates $q_1, \dots, q_n, p_1, \dots, p_n$ on T^*Q , the canonical 1-form θ is given by*

$$\theta|(T^*W) = \sum_j p_j dq_j.$$

PROOF. Let $\alpha = \sum_j \alpha_j dq_j$ be a 1-form on W . Then $\alpha^* p_j = \alpha_j$, $\alpha^* q_j = q_j$, thus

$$\alpha^* \sum_j p_j dq_j = \sum_j \alpha_j dq_j = \alpha.$$

$\square$

THEOREM 2.5. *Let $M = T^*Q$ be a cotangent bundle and θ its canonical 1-form. Then $\omega = -d\theta$ is a symplectic structure on M .*

PROOF. In local cotangent coordinates, $\omega = \sum_j dq_j \wedge dp_j$. $\square$

We will now describe some natural symplectomorphisms and Hamiltonian vector fields on $M = T^*Q$.

Let $f : Q_1 \rightarrow Q_2$ be a diffeomorphism. Then the tangent map df is a diffeomorphism $TQ_1 \rightarrow TQ_2$, and dually there is a diffeomorphism

$$F = (df^{-1})^* : T^*Q_1 \rightarrow T^*Q_2$$

(called cotangent lift of f) covering f . For all $\alpha \in \Omega^1(Q_1)$ one has a commutative diagram,

$$\begin{array}{ccc} T^*Q_1 & \xrightarrow{F} & T^*Q_2 \\ \uparrow_{\alpha} & & \uparrow_{(f^{-1})^*\alpha} \\ Q_1 & \xrightarrow{f} & Q_2 \end{array}.$$

PROPOSITION 2.6 (Naturality of the canonical 1-form). *Let $F : T^*Q_1 \rightarrow T^*Q_2$ be the cotangent lift of f . Then F preserves the canonical 1-form, $F^*\theta_2 = \theta_1$, hence F is a symplectomorphism: $F^*\omega_2 = \omega_1$,*

PROOF. This is clear since our definition of the canonical 1-form was coordinate-free. For the sceptic, check the property $\alpha^*\theta_1 = \alpha$: We have

$$\alpha^*(F^*\theta_2) = (F \circ \alpha)^*\theta_2 = ((f^{-1})^*\alpha \circ f)^*\theta_2 = f^*((f^{-1})^*\alpha)^*\theta_2 = f^*(f^{-1})^*\alpha = \alpha.$$

□

This gives a natural group homomorphism

$$(3) \quad \text{Diff}(Q) \rightarrow \text{Symp}(T^*Q, \omega), \quad f \mapsto (df^{-1})^*.$$

Another subgroup of $\text{Symp}(T^*Q)$ is obtained from the space of closed 1-forms $Z^1(Q) \subset \Omega^1(Q)$. For any $\alpha \in \Omega^1(Q)$ let $G_\alpha : T^*Q \rightarrow T^*Q$ be the diffeomorphism obtained by adding α .

PROPOSITION 2.7. *For all $\alpha \in \Omega^1(Q)$,*

$$G_\alpha^*\theta - \theta = \pi^*\alpha$$

Thus G_α is a symplectomorphism if and only if $d\alpha = 0$, that is $\alpha \in Z^1(Q)$.

PROOF. Let $\beta \in \Omega^1(Q)$. Then

$$\beta^*G_\alpha^*\theta = (G_\alpha \circ \beta)^*\theta = (\alpha + \beta)^*\theta = \alpha + \beta = \beta^*\pi^*\alpha + \beta.$$

By the characterizing property of θ this proves the Proposition. □

We thus find a group homomorphism

$$(4) \quad Z^1(Q) \rightarrow \text{Symp}(T^*Q)$$

Recall that for any representation of a group G on a vector space V , one defines $G \ltimes V$ to be the group whose underlying set is $G \times V$ and with product structure,

$$(g_1, v_1)(g_2, v_2) = (g_1g_2, v_1 + g_1.v_2).$$

In this case, we can let $\text{Diff}(Q)$ act on $Z^1(Q)$ by $f \cdot \alpha = (f^{-1})^*\alpha$. It is easy to check that the homomorphisms (3) and (4) combine into a group homomorphism

$$\text{Diff}(Q) \ltimes Z^1(Q) \rightarrow \text{Symp}(T^*Q).$$

The semi-direct product $\text{Diff}(Q) \ltimes Z^1(Q)$ may be viewed as an infinite-dimensional generalization of the Euclidean group of motions $O(n) \ltimes \mathbb{R}^n$.

We will now show that the generators of the action of $\text{Diff}(Q)$ are Hamiltonian vector fields. Let Y be a vector field on Q . Then there is a unique vector field $X = \text{Lift}(Q)$ on T^*Q with the property that the flow of X is the cotangent lift of the flow of Y . We call the map

$$\text{Lift} : \mathfrak{X}(Q) \rightarrow \mathfrak{X}(T^*Q)$$

the cotangent lift of a vector field. Note that X projects onto Y , that is $X \sim_\pi Y$. Let $H \in C^\infty(T^*Q, \mathbb{R})$ be defined as the contraction $H = \iota_X \theta$.

LEMMA 2.8. *The cotangent lift X of Y is a Hamiltonian vector field, with Hamiltonian $H = \iota_X \theta$.*

PROOF. Since the flow of X preserves θ , $L_X \theta = 0$. Therefore,

$$dH = d\iota_X \theta = -\iota_X d\theta + L_X \theta = -\iota_X d\theta = \iota_X \omega.$$

□

Suppose Y is given in local coordinates by $Y = \sum_j Y_j \frac{\partial}{\partial q_j}$. What are X and H in these coordinates? Since X projects to Y under π , we know that $X - \sum_j Y_j \frac{\partial}{\partial q_j}$ is a vertical vector field, i.e. a linear combination of $\frac{\partial}{\partial p_j}$. The vertical part does not contribute to $\iota_X \theta$ since $\theta = \sum_j p_j dq_j$ is a *horizontal* 1-form. Hence

$$H(q, p) = \iota_X \theta = \sum_j Y_j(q) \iota_{\frac{\partial}{\partial q_j}} \theta = \sum_j Y_j(q) p_j.$$

From this we recover,

$$X = \sum_{j=1}^n Y_j \frac{\partial}{\partial q_j} - \sum_{j,k=1}^n p_j \frac{\partial Y_j}{\partial q_k} \frac{\partial}{\partial p_k}.$$

From these equations, we see that the Hamiltonians corresponding to cotangent lifts are those which are *linear along the fibers* of T^*Q . Other interesting flows are generated by Hamiltonians that are *constant along the fibers* of T^*Q , i.e. of the form $H = \pi^* f$, with $f \in C^\infty(Q)$. The flow generated by such an H is given, in terms of the notation G_α introduced above, by $\phi_t = G_{-t} df$. The Hamiltonian vector field corresponding to H is, in local cotangent coordinates,

$$X_{\pi^* f} = - \sum_j \frac{\partial f}{\partial q_j} \frac{\partial}{\partial p_j}.$$

EXERCISE 2.9. Verify these claims!

On the total space of any vector bundle $E \rightarrow Q$ there is a canonical vector field $\mathcal{E} \in \mathfrak{X}(E)$, called the Euler vector field; its flow Φ_t is fiberwise multiplication by e^t . In our case $E = T^*Q$, we have in local cotangent coordinates

$$\mathcal{E} = \sum_j p_j \frac{\partial}{\partial p_j}.$$

PROPOSITION 2.10. *The Euler vector field satisfies*

$$L_{\mathcal{E}}\omega = \omega, \quad \iota_{\mathcal{E}}\omega = -\theta.$$

*In particular, $\mathcal{E}$ is the vector field corresponding to $-\theta$ under the isomorphism $\omega^\flat : TQ \rightarrow T^*Q$.*

PROOF. The 1-form θ is homogeneous of degree 1 along the fibers. That is, under fiberwise multiplication by e^t it transforms according to $(e^t)^*\theta = e^t\theta$. Taking the derivative this shows

$$L_{\mathcal{E}}\theta = \theta.$$

Applying d gives the first formula in the Proposition. The second formula is obtained using the Cartan formula for the Lie derivative, together with $\iota_{\mathcal{E}}\theta = 0$. $\square$

REMARK 2.11. A symplectic manifold M , together with a free $\mathbb{R}$ -action whose generating vector field $\mathcal{E}$ satisfies such that $L_{\mathcal{E}}\omega = \omega$ is called a *symplectic cone*. Thus, cotangent bundles minus their zero section are examples of symplectic cones. Another example is $\mathbb{R}^{2n} - \{0\}$.

PROPOSITION 2.12. *For any closed 2-form $\sigma \in \Omega^2(Q)$, the sum $\omega + \pi^*\sigma$ is a symplectic form on T^*Q . The Liouville form of $\omega + \pi^*\sigma$ equals that for ω .*

PROOF. Since $\pi^*\sigma$ vanishes on tangent vectors to fibers of π , its kernel at $m \in T^*Q$ contains a Lagrangian subspace, i.e. its kernel is co-isotropic. The claim now follows from the following Lemma. $\square$

LEMMA 2.13. *Let (E, ω) be a symplectic vector space and $\tau \in \wedge^2 E^*$ a 2-form such that $\ker \tau$ is co-isotropic. Then $(\omega + \tau)^n = \omega^n$. In particular, $\omega + \tau$ is non-degenerate.*

PROOF. Since $\ker \tau$ is co-isotropic it contains a Lagrangian subspace L . Let $e_1, \dots, e_n$ a basis for L . For $k < n$ we have $\iota(e_1) \dots \iota(e_n)\omega^k \tau^{n-k} = 0$. Therefore $\omega^k \tau^{n-k} = 0$, and it follows that $(\omega + \tau)^n = \omega^n$. In particular, $\omega + \tau$ is symplectic since its top power is a volume form. $\square$

This has the following somewhat silly corollary: For any manifold Q with a closed 2-form σ there exists a symplectic manifold (M, ω) and an embedding $\iota : Q \rightarrow M$ such that $\iota^*\omega = \sigma$. (Proof: Take $M = T^*Q$ with symplectic form $\omega = -d\theta + \pi^*\sigma$.)

2.4. Example: Kähler manifolds. An *almost complex manifold* is a manifold Q together with a smoothly varying complex structure on each tangent space; i.e. a smooth section $J : Q \rightarrow \text{Gl}(TQ)$ satisfying $J^2 = -\text{id}$. A *complex manifold* is a manifold, together with an atlas consisting of open subsets of $\mathbb{C}^n$, in such a way that the transition functions are holomorphic maps. Every complex manifold is almost complex, the automorphism J given by multiplication by $\sqrt{-1}$ in complex coordinate charts.

The Newlander-Nirenberg theorem (see e.g. the book by Kobayashi-Nomizu) gives a necessary and sufficient criterion (vanishing of the *Nijenhuis tensor*) for when an almost complex structures is integrable, i.e. comes from a complex manifold.

An almost complex structure J on a symplectic manifold (M, ω) is called ω -compatible if it is ω -compatible on every tangent space $T_m M$. We denote by $\mathcal{J}(M, \omega)$ the space of ω -compatible almost complex structures on M . The constructions from linear symplectic algebra can be carried out fiberwise: Letting $\text{Riem}(M)$ denote the space of Riemannian metrics (the space of sections $g : M \rightarrow S^2(T^*M)$ such that each g_m is an inner product on $T_m M$) we have a canonical surjective map

$$\text{Riem}(M) \rightarrow \mathcal{J}(M, \omega)$$

which is a left inverse to the map $\mathcal{J}(M, \omega) \rightarrow \text{Riem}(M)$ associating to J_m the corresponding inner products on $T_m M$. In particular $\mathcal{J}(M, \omega)$ is non-empty. Similar to the linear case, one finds that any two $J_0, J_1 \in \mathcal{J}(M, \omega)$ can be smoothly deformed within $\mathcal{J}(M, \omega)$. More precisely, there exists a smooth map $J : [0, 1] \times M \rightarrow \text{Gl}(TM)$ such that $J(t, m) \in \mathcal{J}(T_m M, \omega_m)$ and $J(0, \cdot) = J_0$, $J(1, \cdot) = J_1$.

For any $J \in \mathcal{J}(M, \omega)$ the triple (M, ω, J) is called an almost Kähler manifold (sometimes also almost Hermitian manifold). If J comes from an honest complex structure then (M, ω, J) is called a Kähler manifold. An example of a Kähler manifold is $M = \mathbb{C}^n$.

PROPOSITION 2.14. *Let (M, ω, J) be a Kähler manifold. Let (N, J_N) be a complex manifold and $\iota : N \rightarrow M$ a complex immersion: That is, $J \circ d\iota = d\iota \circ J_N$. Then $(N, \iota^* \omega, J_N)$ is a Kähler manifold. Similar assertions hold for the almost Kähler category and almost complex immersions.*

PROOF. Obviously, every complex subspace of a Hermitian vector space is Hermitian. Applying this to each $d\iota_n(T_n N) \subset T_{\iota(n)} M$ we see that the closed 2-form $\iota^* \omega$ is non-degenerate, and $J_N \in \mathcal{J}(N, \iota^* \omega)$. $\square$

This shows in particular that every complex submanifold of $\mathbb{C}^n$ is a symplectic manifold. Notice that if $N \subset \mathbb{C}^n$ is the zero locus of a collection of homogeneous polynomials, such that N is smooth away from $\{0\}$ then $N \setminus \{0\}$ is a symplectic cone.

We next consider complex projective space,

$$\mathbb{C}P(n) = (\mathbb{C}^{n+1} \setminus \{0\}) / (\mathbb{C} \setminus \{0\}) = S^{2n+1} / S^1.$$

Let $\iota : S^{2n+1} \rightarrow \mathbb{C}^{n+1}$ the embedding and $\pi : S^{2n+1} \rightarrow \mathbb{C}P(n)$ the projection. At every point $z \in S^{2n+1}$, we have a canonical splitting of tangent spaces

$$T_z \mathbb{C}^{n+1} = T_{\pi(z)} \mathbb{C}P(n) \oplus \text{span}_{\mathbb{C}}\{z\}$$

as complex vector spaces. Since $T_{\pi(z)} \mathbb{C}P(n)$ is a complex subspace, it is also symplectic. This induces a non-degenerate 2-form ω on $\mathbb{C}P(n)$ which by construction is compatible with the complex structure. Letting $\bar{\omega}$ be the symplectic structure of $\mathbb{C}^n$, we have $\iota^* \bar{\omega} = \pi^* \omega$. Therefore $\pi^* d\omega = \iota^* d\bar{\omega} = 0$, showing that ω is closed. This shows that $\mathbb{C}P(n)$ is a Kähler manifold. The 2-form ω is called Fubini-Study form. (Later we will see this construction of ω more systematically as a symplectic reduction.)

By the above proposition, every nonsingular projective variety is a Kähler manifold.

We have seen that every symplectic vector space (E, ω) admits a compatible almost complex structure. It is natural to ask whether it also admits a compatible complex structure, i.e. whether every symplectic manifold is Kähler. The answer is negative: A first counterexample was found by Kodaira and later rediscovered by Thurston. By now there are large families of counterexamples, see e.g. work of McDuff and Gompf. For a discussion of the Kodaira-Thurston counterexample, see the book McDuff-Salamon, “Introduction to Symplectic Topology”.

3. Basic properties of symplectic manifolds

3.1. Hamiltonian and symplectic vector fields. We will now study the Lie algebras of Hamiltonian and symplectic vector fields in more detail. Let (M, ω) be a symplectic manifold. By definition, a vector field X is Hamiltonian if $\iota_X \omega = dH$ for some smooth function H . This means that the isomorphism between vector fields and 1-forms $\omega^\flat : \mathfrak{X}(M) \rightarrow \Omega^1(M)$ defined by ω restricts to an isomorphism

$$\omega^\flat : \mathfrak{X}_{\text{Ham}}(M, \omega) \rightarrow B^1(M)$$

with the space $B^1(M) = \Omega^1(M) \cap \text{im}(d)$ of exact 1-forms. Similarly a vector field is symplectic if and only if $L_X \omega = 0$, which by Cartan’s identity means $d\iota_X \omega = 0$. Thus we have an isomorphism

$$\omega^\flat : \mathfrak{X}(M, \omega) \rightarrow Z^1(M)$$

with the space $Z^1(M) = \Omega^1(M) \cap \ker(d)$ of closed 1-forms. Thus the quotient $\mathfrak{X}(M, \omega)/\mathfrak{X}_{\text{Ham}}(M, \omega)$ is just the first deRham cohomology group $H^1(M) = Z^1(M)/B^1(M)$, and we have an exact sequence of vector spaces

$$(5) \quad 0 \rightarrow \mathfrak{X}_{\text{Ham}}(M, \omega) \rightarrow \mathfrak{X}(M, \omega) \rightarrow H^1(M) \rightarrow 0.$$

We conclude that if $H^1(M) = \{0\}$ (e.g. for simply connected spaces such as $M = \mathbb{C}^n$ or $M = \mathbb{C}P(n)$) every symplectic vector field is Hamiltonian.

PROPOSITION 3.1. *For all $Y_1, Y_2 \in \mathfrak{X}(M, \omega)$, one has*

$$[Y_1, Y_2] = -X_{\omega(Y_1, Y_2)}.$$

PROOF. Let $Y_1, Y_2 \in \mathfrak{X}(M, \omega)$. Then

$$\begin{aligned} d(\omega(Y_1, Y_2)) &= d\iota_{Y_2}\iota_{Y_1}\omega \\ &= L_{Y_2}\iota_{Y_1}\omega - \iota_{Y_2}d\iota_{Y_1}\omega \\ &= L_{Y_2}\iota_{Y_1}\omega \\ &= \iota(L_{Y_2}Y_1)\omega = -\iota([Y_1, Y_2])\omega. \end{aligned}$$

□

Proposition 3.1 shows that $[\mathfrak{X}(M, \omega), \mathfrak{X}(M, \omega)] \subseteq \mathfrak{X}_{\text{Ham}}(M, \omega)$. In particular, $\mathfrak{X}_{\text{Ham}}(M, \omega)$ is an ideal in the Lie algebra $\mathfrak{X}(M, \omega)$ and the quotient Lie algebra $\mathfrak{X}(M, \omega)/\mathfrak{X}_{\text{Ham}}(M, \omega)$ is abelian. It follows that (5) is an exact sequence of Lie algebras, where $H^1(M)$ carries the trivial Lie algebra structure.

3.2. Poisson brackets. Consider next the surjective map $C^\infty(M) \rightarrow \mathfrak{X}_{Ham}(M, \omega), H \mapsto X_H$. Its kernel is the space $Z^0(M) = H^0(M)$ of locally constant functions. We thus have an exact sequence of vector spaces

$$(6) \quad 0 \longrightarrow Z^0(M) \longrightarrow C^\infty(M) \longrightarrow \mathfrak{X}_{Ham}(M, \omega) \longrightarrow 0.$$

We will now define a Lie algebra structure on $C^\infty(M)$ to make this into an exact sequence of Lie algebras. Proposition 3.1 indicates what the right definition of the Lie bracket should be.

DEFINITION 3.2. Let (M, ω) be a symplectic manifold. The Poisson bracket of two functions $F, G \in C^\infty(M, \mathbb{R})$ is defined as

$$\{F, G\} = -\omega(X_F, X_G).$$

From the definition it is immediate that the Poisson bracket is anti-symmetric. Using that $\iota(X_G)\omega = dG$ by definition together with Cartan's identity, one has the alternative formulas

$$\{F, G\} = L_{X_F}G = -L_{X_G}F.$$

These formulas show for example that if F Poisson commutes with a given Hamiltonian G , then F is an *integral of motion* for X_G : That is, F is constant along solution curves of X_G .

PROPOSITION 3.3. *The Poisson bracket defines a Lie algebra structure on $C^\infty(M, \mathbb{R})$: That is, it is anti-symmetric and satisfies the Jacobi identity*

$$\{F, \{G, H\}\} + \{G, \{H, F\}\} + \{H, \{F, G\}\} = 0$$

for all F, G, H . The map $C^\infty(M) \rightarrow \mathfrak{X}(M), F \mapsto X_F$ is a Lie algebra homomorphism:

$$(7) \quad X_{\{F, G\}} = [X_F, X_G].$$

PROOF. Equation (7) is just a special case of Proposition 3.1. The first statement follows from the calculation,

$$\begin{aligned} \{F, \{G, H\}\} &= L_{X_F}\{G, H\} \\ &= -L_{X_F}(\omega(X_G, X_H)) \\ &= -\omega([X_F, X_G], X_H) - \omega(X_G, [X_F, X_H]) \\ &= -\omega(X_{\{F, G\}}, X_H) - \omega(X_G, X_{\{F, H\}}) \\ &= -\{H, \{F, G\}\} + \{G, \{F, H\}\}. \end{aligned}$$

□

An immediate consequence of (7) is:

COROLLARY 3.4. *If $F, G \in C^\infty(M)$ Poisson-commute, the flows of their Hamiltonian vector fields X_F, X_G commute.*

DEFINITION 3.5. An algebra A together with a Lie structure $[\cdot, \cdot]$ is called a Poisson algebra if

$$[FG, H] = F[G, H] + [F, H]G.$$

For any algebra A , the canonical Lie bracket $[F, G] = FG - GF$ satisfies this property.

PROPOSITION 3.6. *The algebra $(C^\infty(M, \mathbb{R}), \{\cdot, \cdot\})$ is a Poisson algebra.*

PROOF.

$$\{FG, H\} = L_{X_H}(FG) = (L_{X_H}F)G + F(L_{X_H}G) = \{F, H\}G + F\{G, H\}.$$

□

PROPOSITION 3.7. *For any compact connected symplectic manifold, Lie algebra extension (6) has a canonical splitting. That is, there exists a canonical Lie algebra homomorphism $\mathfrak{X}_{Ham}(M, \omega) \rightarrow C^\infty(M, \mathbb{R})$ that is a right inverse to the map $F \mapsto X_F$.*

PROOF. The required map associates to every $X \in \mathfrak{X}_{Ham}(M, \omega)$ the unique H such that $X_H = X$ and $\int_M H\Lambda = 0$ (where Λ is the Liouville form). The equality

$$\int_M \{F, G\} \Lambda = \int_M (L_{X_F}G)\Lambda = \int_M L_{X_F}(G\Lambda) = 0$$

shows that this is indeed a Lie homomorphism. □

Let us give the expression for the Poisson bracket for open subsets $U \subset \mathbb{R}^{2n}$, with symplectic coordinates $q_1, \dots, q_n, p_1, \dots, p_n$. We have

$$X_F = \sum_{j=1}^n \left(\frac{\partial F}{\partial p_j} \frac{\partial}{\partial q_j} - \frac{\partial F}{\partial q_j} \frac{\partial}{\partial p_j} \right),$$

hence $\{F, G\} = X_F(G)$ is given by

$$\{F, G\} = \sum_{j=1}^n \left(\frac{\partial F}{\partial p_j} \frac{\partial G}{\partial q_j} - \frac{\partial F}{\partial q_j} \frac{\partial G}{\partial p_j} \right).$$

EXERCISE 3.8. Verify directly, in local coordinates, that the right hand side of this formula defines a Lie bracket.

3.3. Review of some differential geometry. As a preparation for the following section we briefly review the notions of submersions, fibrations, and foliations. Let Q be an n -dimensional manifold.

3.3.1. *Submersions.* A *submersion* is a smooth map $f : Q \rightarrow B$ to a manifold B such that for all $q \in Q$, the tangent map $d_q f = f_*(q) : T_q Q \rightarrow T_{f(q)} B$ is surjective. For any submersion, the fibers $f^{-1}(a)$ are smooth embedded submanifolds of Q of dimension $k = n - \dim B$. In fact Q is *foliated* by such submanifolds, in the following sense: Every point $q \in Q$ has a coordinate neighborhood U with coordinates $x_1, \dots, x_n$, (with q corresponding to $x = 0$) and $f(q)$ a coordinate neighborhood with coordinates $y_1, \dots, y_{n-k}$ (with $f(q)$ corresponding to $y = 0$) such that the map f is given by projection to the first $n - k$ coordinates.

Let $\pi : Q \rightarrow B$ be a submersion. Let the space of vertical vector fields

$$\mathfrak{X}_{\text{vert}}(Q) = \{X \in \mathfrak{X}(Q) \mid X \sim_\pi 0\}$$

be the space of all vector fields taking values in $\ker \pi_*$, i.e tangent to the fibers. Let

$$\Omega_{\text{hor}}(Q) = \{\alpha \in \Omega(Q) \mid \iota_X \alpha = 0 \quad \forall X \in \mathfrak{X}_{\text{vert}}(Q)\}$$

the space of horizontal forms, and

$$\Omega_{\text{basic}}(Q) = \{\alpha \in \Omega(Q) \mid L_X \alpha = 0, \quad \iota_X \alpha = 0 \quad \forall X \in \mathfrak{X}_{\text{vert}}(Q)\}$$

the space of *basic* forms. Notice that $\Omega_{\text{basic}}(Q)$ preserved by the exterior differential d , that is, it is a subcomplex of $(\Omega(Q), d)$.

3.3.2. *Fibrations.* A submersion is called a *fibration* if it is surjective and has the local triviality condition: There exists a manifold F (called standard fiber) such that every point in M there is a neighborhood U and a diffeomorphism $\phi : U \rightarrow f(U) \times F$, in such a way that f is just projection to the first factor. One can show that every surjective submersion with compact fibers is a fibration. In particular, if Q is compact every submersion from Q is fibrating.

For any fibration, $\pi : Q \rightarrow B$, pull-back defines an isomorphism

$$\pi^* : \Omega^k(B) \rightarrow \Omega_{\text{basic}}^k(Q).$$

(This is easily verified in bundle charts $\pi : U \times F \rightarrow U$.)

3.3.3. *Distributions and foliations.* A vector subbundle $E \subset TQ$ of rank k is called a *distribution*. For example, if $f : Q \rightarrow B$ is a submersion (or more generally if the tangent map f_* has constant rank), then $E = \ker(f_*) \subset TQ$ is a distribution. Also, if $X \in \mathfrak{X}(Q)$ is nowhere vanishing vector field, $\text{span}(X)$ is a distribution of rank 1. A submanifold $S \subset Q$ is called an *integral submanifold* if $TS = E|_S$. For example, the integral submanifolds of $\text{span}(X)$ are just integral curves of X . A distribution E of rank k is called *integrable* if through every point $q \in Q$ there passes a k -dimensional integral submanifold. As one can show, this is equivalent to the condition that every point has a neighborhood U and a submersion $f : U \rightarrow \mathbb{R}^{n-k}$ such that $E|_U = \ker(f_*)$. For this reason integrable distributions are also called *foliations*.

EXAMPLE 3.9. Let $S^3 \subset \mathbb{C}^2$ the unit sphere. For each $z \in S^3$ let $E_z \subset T_z S^3$ denote the orthogonal complement of the complex line through z . The distribution E is not integrable.

A necessary condition for integrability of a distribution is that for every two vector fields X_1, X_2 taking values in E , their Lie bracket $[X_1, X_2]$ takes values in E . This follows because the Lie bracket of two vector fields tangent to a submanifold $S \subset Q$ is also tangent to S . Frobenius' theorem states that this condition is also sufficient:

THEOREM 3.10 (Frobenius' criterion). *A distribution $E \subset TQ$ is integrable if and only if the space of sections of E is closed under the Lie bracket operation.*

EXAMPLE 3.11. Let $Q = \mathbb{R}^3$ with coordinates x, y, z , and let $E \subset TQ$ the vector subbundle spanned by the two vector fields,

$$X_1 = \frac{\partial}{\partial x}, \quad X_2 = x \frac{\partial}{\partial y} + \frac{\partial}{\partial z}.$$

Since $[X_1, X_2] = \frac{\partial}{\partial y}$ is not in E , the distribution E is not integrable.

EXAMPLE 3.12. A smooth map $f : Y_1 \rightarrow Y_2$ is called a constant rank map if the image of the tangent map $f_* : TY_1 \rightarrow TY_2$ has constant dimension. Examples are submersions (f_* surjective) or immersions (f_* injective). The kernel of every constant rank map defines an integrable distribution. Indeed, two vector fields X_1, X_2 are in the kernel if and only if $X_j \sim_f 0$. Hence also $[X_1, X_2] \sim_f 0$.

(Here we have used that for any vector bundle map $F : E_1 \rightarrow E_2$ of constant rank, the kernel and image of F are smooth vector subbundles. In particular $\ker(f_*)$ is a smooth vector subbundle.)

3.4. Lagrangian submanifolds. Let (M, ω) be a symplectic manifold.

DEFINITION 3.13. A submanifold (or, more generally, an immersion) $\iota : N \hookrightarrow M$ is called co-isotropic (resp. isotropic, Lagrangian, symplectic) if at any point $n \in N$, $T_n N$ is a co-isotropic (resp. isotropic, Lagrangian, symplectic) subspace of $T_n M$.

For example $\mathbb{R}P(n) \subset \mathbb{C}P(n)$ is a Lagrangian submanifold. Also, the fibers of a cotangent bundle T^*Q are Lagrangian. Another important example is:

PROPOSITION 3.14. *The graph $\Gamma_\alpha \subset T^*Q$ of a 1-form $\alpha : Q \rightarrow T^*Q$ is Lagrangian if and only if α is closed.*

PROOF.

$$\alpha^* \omega = -\alpha^* d\theta = -d\alpha^* \theta = -d\alpha.$$

□

In local coordinates: If $\alpha = \sum_j \alpha_j dq_j$, the pull-back of $\omega = \sum_j dq_j \wedge dp_j$ to the graph of α is given by

$$\sum_j dq_j \wedge d\alpha_j = \sum_{j,k} \frac{\partial \alpha_j}{\partial q_k} dq_j \wedge dq_k = -d\alpha.$$

Hence α is closed if and only if the pull-back of ω to the graph vanishes.

Let $N \subset Q$ be a submanifold of Q . Dual to the inclusion $\iota_* = d\iota : TN \rightarrow TQ|_N$ there is a surjective vector bundle map,

$$(d\iota)^* : T^*Q|_N \rightarrow T^*N.$$

Its kernel $\ker((d\iota)^*)$ (i.e. the pre-image of the zero section $N \subset T^*Q|_N$) consists of covectors that vanish on all tangent vectors to N . One calls $\text{Ann}(TN) := \ker((d\iota)^*)$ the annihilator bundle of TN , or also the *conormal bundle*.

PROPOSITION 3.15. *The conormal bundle to any submanifold $N \subset Q$ is a Lagrangian submanifold of T^*Q . More generally, let $\alpha \in \Omega^1(N)$ be a 1-form on N . Then the pre-image of the graph $\Gamma_\alpha \subset T^*N$ under the map $(d\iota)^* : T^*Q|_N \rightarrow T^*N$ is a Lagrangian submanifold of T^*Q .*

PROOF. Locally, near any point of N we can choose coordinates $q_1, \dots, q_n$ on Q such that N is given by equations $q_{k+1} = 0, \dots, q_n = 0$. In the corresponding cotangent coordinates q_j, p_j on T^*Q , $\text{Ann}(TN)$ is given by equations $q_{k+1} = 0, \dots, q_n = 0, p_1, \dots, p_k = 0$. Clearly each summand in $\omega = \sum_j dq_j \wedge dp_j$ vanishes on this submanifold. More generally, the pre-image $((d\iota)^*)^{-1}(\Gamma_\alpha)$ is given by equations

$$q_{k+1} = 0, \dots, q_n = 0, p_1 = \alpha_1, \dots, p_k = \alpha_k.$$

Hence the pull-back of ω to this submanifold is given by

$$\sum_{i=1}^k dq_i \wedge d\alpha_j = \sum_{i,j=1}^k \frac{\partial \alpha_i}{\partial q_j} dq_i \wedge dq_j$$

so that ω vanishes on this submanifold if and only if α is closed. $\square$

PROPOSITION 3.16. *Let (M_j, ω_j) be symplectic manifolds, and let M_1^- denote M_1 with symplectic form $-\omega_1$. A diffeomorphism $F : M_1 \rightarrow M_2$ is a symplectomorphism if and only if its graph*

$$\Gamma_F := \{(F(m), m) \mid m \in M\} \subset M_2 \times M_1^-$$

is Lagrangian.

PROOF. Similar to the linear case. $\square$

Here is a nice application of these considerations.

THEOREM 3.17 (Tulczyjew). *Let $E \rightarrow B$ be a vector bundle, $E^* \rightarrow B$ its dual bundle. There is a canonical symplectomorphism $T^*E \cong T^*E^*$.*

OUTLINE OF PROOF. Consider the vector space direct sum $N = E \oplus E^*$ as a smooth submanifold of $Q = E \times E^*$. The natural pairing between E and E^* defines a function $f : N = E \oplus E^* \rightarrow \mathbb{R}$, let $\alpha = df$. By the above, α defines a Lagrangian submanifold L of $T^*Q = T^*E \times T^*E^*$. One checks that the projections from L onto both factors T^* and T^*E^* are diffeomorphisms, hence L is the graph of a symplectomorphism. $\square$

Tulczyjew only considered the case $E = TQ$. It was pointed out by D. Roytenberg in his thesis that the argument works for any vector bundles. (In fact, he uses a generalization to super-vector bundles.)

3.5. Constant rank submanifolds. A submanifold $\iota : N \rightarrow M$ is called a “constant rank submanifold” if the dimension of the kernel of $(\iota^*\omega)_n$ is independent of $n \in N$.

PROPOSITION 3.18. *Let N be a manifold together with a closed 2-form σ of constant rank. Then the subbundle $\ker(\sigma)$ is integrable, i.e. defines a foliation.*

PROOF. We use Frobenius’ criterion. Suppose $X_1, X_2 \in \mathfrak{X}(N)$ with $\iota_{X_j}\sigma = 0$. Since σ is closed, this implies $L_{X_j}\sigma = d\iota_{X_j}\sigma = 0$. Hence

$$\iota_{[X_1, X_2]}\sigma = L_{X_1}\iota_{X_2}\sigma - \iota_{X_2}L_{X_1}\sigma = 0.$$

□

If this so-called null-foliation of N is fibrating, i.e. if the leaves of the foliation are the fibers of a submersion $\pi : N \rightarrow B$, where B is the space of leaves of the foliation. The form σ is basic for this fibration since $\iota_X\sigma = 0$ and $L_X\sigma = 0$ for all vertical vector fields. It follows that B inherits a unique 2-form ω_B such that

$$\pi^*\omega_B = \sigma$$

DEFINITION 3.19. The symplectic manifold (B, ω_B) is called the symplectic reduction of (N, σ) .

REMARK 3.20. Note that the above discussion carries over for any closed differential form of constant rank on N .

In typical applications, N is a constant rank submanifold of a symplectic manifold with $\sigma = \iota^*\omega$. Of course, for a random constant rank submanifold it rarely happens that the null foliation is fibrating, unless additional symmetries are at work.

3.6. Co-isotropic submanifolds. An important special case of constant rank submanifolds of a symplectic manifold (M, ω) are co-isotropic submanifolds, i.e. submanifolds $N \subset M$ with $TN^\omega \subset TN$. For any submanifold $N \subset M$, let

$$C^\infty(M)_N = \{F \in C^\infty(M) \mid F|_N = 0\}$$

denote its vanishing ideal. The tangent bundle $TN \subset TM|_N$ and its annihilator have the following algebraic characterizations:

$$T_n N = \{v \in T_n M \mid v(F) = 0 \text{ for all } F \in C^\infty(M)_N\},$$

$$\text{Ann}(T_n N) = \{\alpha \in T_n^* M \mid \alpha = dF|_n \text{ for some } F \in C^\infty(M)_N\}.$$

The map $\omega^\flat : T_n M \rightarrow T_n^* M$ identifies the annihilator bundle with $TN^\omega \subset TM|_N$, and dF with X_F . Thus

$$(8) \quad T_n N^\omega = \{X_F(n) \mid F \in C^\infty(M)_N\}.$$

THEOREM 3.21. *The following three statements are equivalent:*

- (a) *For all $F \in C^\infty(M)_N$, the Hamiltonian vector field X_F is tangent to N .*
- (b) *The space $C^\infty(M)_N$ is a Poisson subalgebra of $C^\infty(M)$.*
- (c) *N is a coisotropic submanifold of M .*

PROOF. X_F is tangent to N if and only if $X_F(G) = 0$ for all $G \in C^\infty(M)_N$. Since $X_F(G) = \{F, G\}$ this shows that (a) and (b) are equivalent. We have seen above that TN^ω is spanned by restrictions of Hamiltonian vector fields $X_F|_N$ with $F \in C^\infty(M)_N$. Hence N is coisotropic ($TN^\omega \subset TN$) if and only if every such vector field is tangent to N . This shows that (a) and (c) are equivalent. $\square$

LEMMA 3.22. *Suppose $F : M \rightarrow \mathbb{R}^k$ is a submersion. Suppose that the components of F Poisson-commute, $\{F_i, F_j\} = 0$. Then the fibers of F are co-isotropic submanifolds of M of codimension k .*

PROOF. Let $N = F^{-1}(a)$. The vector fields X_{F_j} are tangent to N since $X_{F_j}(F_i) = \{F_j, F_i\} = 0$. Since $dF_1, \dots, dF_k$ span $\text{Ann}(TN)$ at each point of N , the Hamiltonian vector fields X_{F_j} span TN^ω . This shows $TN^\omega \subset TN$. $\square$

REMARK 3.23. (a) In particular, given a submersion $F : M \rightarrow \mathbb{R}^n$ where $\dim M = 2n$, with Poisson-commuting components, the fibers of F define a Lagrangian foliation of M . This is the setting for completely integrable systems. We will discuss this case later in much more detail.

- (b) The assumptions of the Proposition can be made more geometric, by saying that F should be a Poisson map, for the trivial Poisson structure on $\mathbb{R}^k$.

The following Proposition shows that locally, any coisotropic submanifold is obtained in this way:

PROPOSITION 3.24. *Let $\iota : N \hookrightarrow M$ be a codimension k co-isotropic submanifold. For any $m \in N$ there exists a neighborhood $U \subset M$ containing m , and a smooth submersion $F : U \rightarrow \mathbb{R}^k$ such that the components of F Poisson-commute, $\{F_i, F_j\} = 0$ for all i, j , and $N \cap U = F^{-1}(0)$.*

PROOF. The proof is by induction. Suppose that we have constructed Poisson-commuting functions $F_1, \dots, F_l : U \rightarrow \mathbb{R}$, $l < k$ such that $(F_1, \dots, F_l) : U \rightarrow \mathbb{R}^l$ is a submersion and $(F_1, \dots, F_l)^{-1}(0) \supseteq U \cap N$. By the previous Lemma the fibers of $F' := (F_1, \dots, F_l)$ define a foliation of U by co-isotropic submanifolds. Let $X_i = X_{F_i}$. The fact that the F_i vanish on N means that X_i is tangent to N , i.e. N is invariant under the flow, and is contained in a unique fiber N' of F' . Choosing U smaller if necessary, we can pick a codimension l submanifold $S \subset U$ transverse to N' , and a submersion $F_{l+1} : S \rightarrow \mathbb{R}$ such that $N' \cap S \subset F_{l+1}^{-1}(0)$. Choosing U and S even smaller, we can extend F_{l+1} to a function on U , invariant under the flows of the commuting vector fields $X_1, \dots, X_l$. This means that $F_1, \dots, F_{l+1}$ all Poisson commute, and the map $(F_1, \dots, F_{l+1})$ is a submersion near N' . $\square$

Later we will obtain a much better version of this result (local normal form theorem), showing that one can actually take U to be a tubular neighborhood of N .

CHAPTER 4

Normal Form Theorems

1. Moser's trick

Moser's trick was used by Moser in a very short paper (1965) to show that on any compact oriented manifold any two normalized volume forms are diffeomorphism equivalent. (A volume form is a top degree nowhere vanishing differential form.) In order to describe his proof we recall the following fact from differential geometry:

LEMMA 1.1. *Let $X_t \in \text{Vect}(Q)$ ($t \in \mathbb{R}$) a time-dependent vector field on a manifold Q , with flow ϕ_t . For every differential form $\alpha \in \Omega^*(Q)$,*

$$\phi_t^* L_{X_t} \alpha = \frac{\partial}{\partial t} \phi_t^* \alpha$$

(on the region $U \subset Q$ where $\phi_t : U \rightarrow Q$ is defined).

Note that if α is a 0-form, this is just the definition of the flow ϕ_t . The general case follows because both sides are derivations of $\Omega(M)$ commuting with d .

THEOREM 1.2 (Moser). *Let Q be a compact, oriented manifold, and Λ_0, Λ_1 two volume forms such that $\int_M \Lambda_0 = \int_M \Lambda_1$. Then there exists a smooth isotopy $\phi_t \in \text{Diff}(Q)$ such that $\phi_1^* \Lambda_1 = \Lambda_0$.*

PROOF. Moser's argument is as follows. First, note that every $\Lambda_t = (1-t)\Lambda_0 + t\Lambda_1$ is a volume form. Second, since Λ_0 and Λ_1 have the same integral they define the same cohomology class: $\Lambda_1 = \Lambda_0 + d\beta$ for some $n-1$ -form β . Thus

$$\Lambda_t = \Lambda_0 + t d\beta.$$

Since each Λ_t is a volume form, the map $X \mapsto \iota_X \Lambda_t$ from vector fields to $n-1$ -forms ($n = \dim Q$) is an isomorphism. It follows that there is a unique time dependent vector field X_t solving

$$\iota_{X_t} \Lambda_t + \beta = 0.$$

Let ϕ_t denote the flow of X_t . Then

$$\begin{aligned} \frac{\partial}{\partial t} \phi_t^* \Lambda_t &= \phi_t^* (d\beta + L_{X_t} \Lambda_t) \\ &= d(\beta + \iota_{X_t} \Lambda_t) \\ &= 0. \end{aligned}$$

This shows $\phi_t^* \Lambda_t = \Lambda_0$. Now put $\phi = \phi_1$. □

Moser's theorem shows that volume forms on a given compact oriented manifold Q are classified up to diffeomorphism by their integral. The idea from Moser's proof applies to many similar problems. A typical application in symplectic geometry is as follows.

THEOREM 1.3. *Let ω_t be a family of symplectic 2-forms on a compact manifold M , depending smoothly on $t \in [0, 1]$. Suppose that*

$$\omega_t = \omega_0 + d\beta_t$$

for some smooth family of 1-forms $\beta_t \in \Omega^1(M)$. Then there exists a family of diffeomorphisms (i.e. a smooth isotopy) ϕ_t such that

$$\phi_t^* \omega_t = \omega_0$$

for all t .

PROOF. Define a time-dependent vector field X_t by

$$\iota_{X_t} \omega_t + \frac{\partial}{\partial t} \beta_t = 0.$$

Let ϕ_t denote its flow. Then $\phi_t^* \omega_t$ is independent of t :

$$\frac{\partial}{\partial t} \phi_t^* \omega_t = \phi_t^* \left(\frac{\partial}{\partial t} \omega_t + L_{X_t} \omega_t \right) = d \left(\frac{\partial}{\partial t} \beta_t + \iota_{X_t} \omega_t \right) = 0.$$

□

Alan Weinstein used Moser's argument to give a simple proof of Darboux's theorem, saying that symplectic manifolds have no local invariants, and some generalizations. We will present Weinstein's proof below, after some review of homotopy operators in de Rham theory.

2. Homotopy operators

Let Q_1, Q_2 be smooth manifolds and $f_0, f_1 : Q_1 \rightarrow Q_2$ two smooth maps. Suppose f_0, f_1 are homotopic, i.e. that they are the boundary values of a continuous map

$$f : [0, 1] \times Q_1 \rightarrow Q_2.$$

As one can show f can always be taken to be smooth. Define the homotopy operator

$$\mathcal{H} : \Omega^k(Q_2) \rightarrow \Omega^{k-1}(Q_1)$$

as a composition

$$\mathcal{H}(\alpha) = \int_{[0,1]} f^* \alpha.$$

Here $\int_{[0,1]} : \Omega^k([0, 1] \times Q_1) \rightarrow \Omega^{k-1}(Q_1)$ is fiber integration, i.e. integrating out the $s \in [0, 1]$ variable. (The integral of a form not containing ds is defined to be 0.)

EXERCISE 2.1. Verify that for any form β on $[0, 1] \times Q$,

$$\int_{[0,1]} d\beta = -d \int_{[0,1]} \beta + \iota_1^* \beta - \iota_0^* \beta$$

where $\iota_j : Q \rightarrow Q \times \{j\}$ are the two inclusions. (Hint: fundamental theorem of calculus!)

As a consequence the map $\mathcal{H}$ has the property,

$$d \circ \mathcal{H} + \mathcal{H} \circ d = f_1^* - f_0^* : \Omega^k(Q_2) \rightarrow \Omega^k(Q_1)$$

Thus if α is a closed form on Q_2 , then $\beta = \mathcal{H}(\alpha)$ solves

$$f_1^* \alpha - f_0^* \alpha = d\beta.$$

Hence f_0^* and f_1^* induce the same map in cohomology.

EXAMPLE 2.2. (Poincare lemma.) Let $U \subset \mathbb{R}^m$ be an open ball around 0. Let $\iota : \{0\} \rightarrow U$ be the inclusion and $\pi : U \rightarrow \{0\}$ the projection. Then ι^* induces an isomorphism $H^k(U) = H^k(\text{pt})$, with inverse π^* . That is, every closed form $\alpha \in \Omega^k(U)$ with $k > 0$ is a coboundary: $\alpha = d\beta$.

PROOF. Since it is obvious that $\iota^* \circ \pi^* = (\pi \circ \iota)^*$ is the identity map, we only need to show that $\pi^* \circ \iota^* = (\iota \circ \pi)^*$ is the identity map in cohomology. Let $f_t : U \rightarrow U$ be multiplication by $t \in [0, 1]$. The de Rham homotopy operator $\mathcal{H}$ shows that f_1^*, f_0^* induce the same map in cohomology. The claim follows since $f_1 = \text{id}$ and $f_0 = \iota \circ \pi$. $\square$

3. Darboux-Weinstein theorems

THEOREM 3.1 (Darboux). *Let (M, ω) be a symplectic manifold of dimension $\dim M = 2n$ and $m \in M$. Then there exist open neighborhoods U of m and V of $\{0\} \in \mathbb{R}^{2n}$, and a diffeomorphism $\phi : V \rightarrow U$ such that $\phi(0) = m$ and $\phi^* \omega = \sum_j dq_j \wedge dp_j$.*

Coordinate charts of this type are called *Darboux charts*.

PROOF. Using any coordinate chart centered at m , we may assume that M is an open ball U around $m = 0 \in \mathbb{R}^{2n}$, with ω some possibly non-standard symplectic form. Let $\omega_1 = \omega$ and ω_0 the standard symplectic form. Since any two symplectic forms on the vector space $T_0 \mathbb{R}^{2n}$ are related by a linear transformation, we may assume that ω_1 agrees with ω_0 at the origin. Using the homotopy operator from Example 2.2 let

$$\beta := \mathcal{H}(\omega_1 - \omega_0) \in \Omega^1(U).$$

As in the original Moser trick put

$$\omega_t = \omega_0 + t d\beta.$$

For all $t \in [0, 1]$, ω_t agrees with ω_0 at 0. Hence, taking U smaller if necessary, we may assume ω_t is non-degenerate on U for all $t \in [0, 1]$. Define a time-dependent vector field X_t on U by

$$\iota_{X_t} \omega_t = -\beta.$$

The flow of this vector field will not be complete in general. Since $\omega_1 - \omega_0$ vanishes at 0, the 1-form β and therefore the vector field X_t also vanish at 0. Hence we can find a smaller neighborhood U' of 0 such that the flow $\phi_t : U' \rightarrow U$ is defined for all $t \in [0, 1]$. The flow satisfies

$$\frac{\partial}{\partial t} \phi_t^* \omega_t = \phi_t^* (L_{X_t} \omega_t + d\beta) = \phi_t^* (d\iota_{X_t} \omega_t + d\beta) = 0,$$

hence by integration $\phi_t^* \omega_t = \omega_0$. Darboux's theorem follows by setting $\phi = \phi_1$. $\square$

Again, the proof has shown a bit more: In a sufficiently small neighborhood of $0 \in \mathbb{R}^{2n}$ any two symplectic forms are isotopic.

Darboux's theorem says that *symplectic manifolds have no local invariants*, in sharp contrast to Riemannian geometry where there are many local invariants (curvature invariants). Darboux's theorem can be strengthened to the statement that the symplectic form near any submanifold N of a symplectic manifold M is determined by the restriction of ω to $TM|_N$:

THEOREM 3.2. *Let (M_j, ω_j) , $j = 0, 1$ be two symplectic manifolds, and $\iota_j : N_j \rightarrow M_j$ given submanifolds. Suppose there exists a diffeomorphism $\psi : N_0 \rightarrow N_1$ covered by a symplectic vector bundle isomorphism*

$$\hat{\psi} : TM_0|_{N_0} \rightarrow TM_1|_{N_1}.$$

such that $\hat{\psi}$ restricts to the tangent map $\psi_ : TN_0 \rightarrow TN_1$. Then ψ extends to a symplectomorphism ϕ from a neighborhood of N_0 in M_0 to a neighborhood N_1 in M_1 .*

PROOF. Any submanifold N of a manifold M has a “tubular neighborhood” diffeomorphic to the total space of the normal bundle $\nu_N = TM|_N / TN$. Since $\hat{\psi}$ induces an isomorphism $\nu_{N_1} \rightarrow \nu_{N_0}$, we may assume that $M_1 = M_0 =: M$ is the total spaces of a vector bundle $\pi : M \rightarrow N$ over a given manifold $N_1 = N_0 =: N$, with two given symplectic forms ω_0, ω_1 that agree along N . Let $\mathcal{H} : \Omega^k(M) \rightarrow \Omega^{k-1}(M)$ be the standard homotopy operator for the vector bundle $\pi : M \rightarrow N$, and put $\beta = \mathcal{H}(\omega_1 - \omega_0)$ and $\omega_t = \omega_0 + td\beta$. Since ω_t agrees with ω_0 along N , it is in particular symplectic on a neighborhood of N in M . On that neighborhood we can define a time-dependent vector field X_t with $\iota_{X_t} \omega_t + \beta = 0$. Let ϕ_t be its flow (defined on an even small neighborhood for all $t \in [0, 1]$), and put $\phi_1 =: \phi$. By Moser's argument $\phi^* \omega_1 = \omega_0$. $\square$

THEOREM 3.3. *Let (M, ω) be a symplectic manifold, $\iota : L \hookrightarrow M$ a Lagrangian submanifold. There exists a neighborhood U_0 of L in M , a neighborhood U_1 of L in T^*L , and a symplectomorphism from U_0 to U_1 fixing L .*

PROOF. By theorem 3.2 it is enough find a symplectic bundle isomorphism $TM|_L \cong T(T^*L)|_L$. Choose a compatible almost complex structure J on M . Then $J(TL) \subset TM|_L$ is a Lagrangian subbundle complementary to TL , and is therefore isomorphic (by means of the symplectic form) to the dual bundle. It follows that

$$TM|_L \cong TL \oplus T^*L.$$

as a symplectic vector bundle. The same argument applies to M replaced with T^*L . Thus

$$TM|_L \cong TL \oplus T^*L \cong T(T^*L)|_L.$$

□

This type of result generalizes to constant rank submanifolds, as follows. First we need a definition.

DEFINITION 3.4. For any constant rank submanifold $\iota : N \rightarrow M$ of a symplectic manifold (M, ω) , the *symplectic normal bundle* is the symplectic vector bundle

$$TN^\omega / TN \cap TN^\omega$$

Note that for co-isotropic submanifolds the symplectic normal bundle is just 0. For an isotropic submanifold of dimension k it has rank $2(n - k)$ where $2n = \dim M$. The following theorem is due to Marle, extending earlier results of Weinstein (for the cases N Lagrangian or symplectic) and Gotay (for the case N co-isotropic).

THEOREM 3.5 (Constant rank embedding theorem). *Let $\iota_j : N_j \hookrightarrow M_j$ ($j = 1, 2$) two constant rank submanifolds of symplectic manifolds (M_j, ω_j) . Let*

$$F_j = TN_j^{\omega_j} / (TN_j^{\omega_j} \cap TN_j)$$

be their symplectic normal bundles. Suppose there exists a symplectic bundle isomorphism

$$\hat{\psi} : F_0 \rightarrow F_1$$

covering a diffeomorphism $\psi : N_0 \rightarrow N_1$ such that

$$\psi^* \iota_1^* \omega_1 = \iota_0^* \omega_0.$$

Then ψ extends to a symplectomorphism ϕ of neighborhoods of N_j in M_j , such that ϕ induces $\hat{\psi}$.

Thus, a neighborhood of a constant rank submanifold $\iota : N \rightarrow M$ is characterized up to symplectomorphism by $\iota^* \omega$ together with the symplectic normal bundle. In particular, if N is co-isotropic a neighborhood is completely determined by $\iota^* \omega$.

PROOF. Suppose $\iota : N \rightarrow M$ is a compact constant rank submanifold of a symplectic manifold (M, ω) . There are three natural symplectic vector bundles over N :

$$\begin{aligned} E &= TN / (TN \cap TN^\omega), \\ F &= TN^\omega / (TN \cap TN^\omega), \\ G &= (TN \cap TN^\omega) \oplus (TN \cap TN^\omega)^* \end{aligned}$$

Identifying E with a complementary subbundle to $TN \cap TN^\omega$ in TN we have

$$TN \cong E \oplus (TN \cap TN^\omega)$$

(isomorphism of bundles with 2-forms) likewise

$$TN^\omega \cong F \oplus (TN \cap TN^\omega),$$

Therefore $TN + TN^\omega = E \oplus F \oplus (TN \cap TN^\omega)$. Let J be an ω -compatible complex structure on $TM|_N$, preserving the two subbundles E, F . Then the isotropic subbundle $J(TN \cap TN^\omega) \subset TM|_N$ is a complement to $TN + TN^\omega$, which by means of ω is identified with $(TN \cap TN^\omega)^*$. This shows

$$TM|_N \cong E \oplus F \oplus G$$

as symplectic vector bundles. To prove the constant rank embedding theorem, choose isomorphisms of this type for both $TM_i|_{N_i}$. Then E_0, E_1 and G_0, G_1 are symplectomorphic since $\psi : N_0 \rightarrow N_1$ preserves two-forms, and $F_0 \cong F_1$ by assumption of the theorem. Now apply Theorem 3.2 □

CHAPTER 5

Lagrangian fibrations and action-angle variables

1. Lagrangian fibrations

We had seen that for any submersion $F : M \rightarrow \mathbb{R}^k$ from a symplectic manifold M , such that the components F_i Poisson-commute, the fibers of F are co-isotropic submanifolds of codimension k . In particular, if $k = n = \frac{1}{2} \dim M$, the fibers are Lagrangian submanifolds.

DEFINITION 1.1. Let (M, ω) be a symplectic manifold. A Lagrangian fibration is a fibration $\pi : M \rightarrow B$ such that every fiber is a Lagrangian submanifold of M .

This implies in particular $\dim B = n = \frac{1}{2} \dim M$.

- EXAMPLES 1.2.** (a) The fibers of a cotangent bundle $\pi : M = T^*Q \rightarrow Q$ are a Lagrangian fibration.
 (b) If $Q = (\mathbb{R}/\mathbb{Z})^n = T^n$ is an n -torus, we have a natural trivialization $T^*(T^n) = (T^n) \times \mathbb{R}^n$ and the map $\pi : T^*(T^n) \rightarrow \mathbb{R}^n$ defines a Lagrangian fibration of $T^*(T^n)$.

Is it possible to generalize the second example to compact manifolds Q other than a torus? That is, is it possible to find a Lagrangian fibration of T^*Q such that the zero section $Q \subset T^*Q$ is one of the leaves? We will show in this section that the answer is “no”: The leaves of a Lagrangian fibration are *always* diffeomorphic to an open subsets of products of vector spaces with tori.

We will need some terminology group actions on manifold. If G is a Lie group and Q a manifold, a group action is a smooth map

$$\mathcal{A} : G \times Q \rightarrow Q, (g, q) \mapsto \mathcal{A}(g, q) \equiv g \cdot q$$

such that $e \cdot q = q$ and $g_1 \cdot (g_2 \cdot q) = (g_1 g_2) \cdot q$. For $q \in Q$ the set $G \cdot q = \mathcal{A}(G, q)$ is called the orbit, the subgroup $G_q = \{g \in G \mid g \cdot q = q\}$ the stabilizer. Note that $G \cdot q = G/G_q$. If G_q is compact then $G \cdot q$ is a manifold.

LEMMA 1.3. Let G be a connected Lie group and $\mathcal{A} : G \times Q \rightarrow Q$ a group action on a connected manifold Q . Then the following are equivalent:

- (a) The action is transitive: For some (hence all) $q \in Q$, $G \cdot q = Q$.
- (b) The action is locally transitive: That is, for all $q \in Q$ there is a neighborhood U such that $(G \cdot q) \cap U = U$.
- (c) There exists $q \in Q$ such that the tangent map to $G \rightarrow Q, g \mapsto g \cdot q$ is surjective.

PROOF. (a) just means that the orbits are open and closed. Both (b) and (c) imply that condition. The converse is obvious. $\square$

Of particular interest is the case $\dim Q = \dim G$. We will say that Q is an *principal homogeneous G -space* if Q comes equipped with a free, transitive action of G . Any choice of a base point $q \in Q$ identifies $Q = G.q \cong G$. Any two such identifications differ by a translation in G . If G is a torus we call Q an *affine torus*, if G is a vector space we call Q an *affine vector space*.

Suppose V is an n -dimensional vector space acting transitively on an n -dimensional manifold Q . The stabilizer V_q is a discrete subgroup of V , independent of q . Thus Q carries the structure of a principal homogeneous $H = V/V_q$ -space. Note that since H is compact, connected and abelian, it is a product of a vector space and a torus. That is, Q is a product of an *affine torus* and an *affine vector space*.

The above considerations also make sense for fiber bundles: If $\mathcal{G} \rightarrow B$ is a group bundle (i.e. a fiber bundle where the fibers carry group structures, and admitting bundle charts $\pi^{-1}(U) \cong U \times F$ that are fiberwise group isomorphisms with a fixed group F), one can define actions on fiber bundles $E \rightarrow B$ to be smooth maps

$$\mathcal{G} \times_B E \rightarrow E$$

that are fiberwise group actions. For example, if E is a vector bundle and $\mathrm{Gl}(E)$ is the bundle of general linear groups, one has a natural action of $\mathrm{Gl}(E)$ on E . In particular, an affine torus bundle $\pi : M \rightarrow B$ is a fiber bundle with a fiberwise free, transitive action of a torus bundle $\mathcal{T} \rightarrow B$. Thus if $\pi : M \rightarrow B$ has compact fibers, then every fiberwise transitive action of a vector bundle $E \rightarrow B$ with $\dim E = \dim M$ gives M the structure of an affine torus bundle. The torus bundle $\mathcal{T} \rightarrow B$ is the quotient bundle E/Λ , where $\Lambda \rightarrow B$ is the bundle of stabilizer groups. Any section $\sigma : B \rightarrow M$ makes M into a torus bundle, however in general there are obstructions to the existence of such a section. On the other hand, if $\mathcal{T}$ is trivial then $M \rightarrow B$ becomes a T -principal bundle.

Let us now consider a Lagrangian fibration $\pi : M \rightarrow B$. The following discussion is based mainly on the paper, J. J. Duistermaat: “*On global action-angle variables*”, Comm. Pure Appl. Math. **33** (1980), 687–706. For simplicity we will usually assume that the fibers of π are compact and connected.

THEOREM 1.4. *Let (M, ω) be a symplectic manifold and $\pi : M \rightarrow B$ be a Lagrangian fibration with compact, connected fibers. Then there is a canonical, fiberwise transitive vector bundle action $T^*B \times M \rightarrow M$. Thus $\pi : M \rightarrow B$ has canonically the structure of an affine torus bundle.*

PROOF. We denote by $VM \subset TM$ the vertical tangent bundle, $V_m M = \ker(d_m \pi)$. By assumption VM is a Lagrangian subbundle of the symplectic vector bundle TM . For any 1-form $\alpha \in \Omega^1(B)$ let $X_\alpha \in \mathfrak{X}(M)$ be the vector field defined by

$$\iota_{X_\alpha} \omega = -\pi^* \alpha.$$

For all vertical vector fields $Y \in \mathfrak{X}(M)$,

$$\omega(X_\alpha, Y) = \iota_Y \iota_{X_\alpha} \omega = -\iota_Y \pi^* \alpha = 0.$$

Since VM is a Lagrangian subbundle, it follows that X_α is a vertical vector field. Note that the value of X_α at $m \in M$ depends only on $\alpha_{\pi(m)}$. Thus we have constructed a linear map $V_m M \cong T_{\pi(m)}^* B$. Clearly this map is an isomorphism. Taking all of these isomorphisms together we have constructed a canonical bundle isomorphism

$$VM \cong \pi^* T^* B.$$

Let F_α^t denote the flow of the vector field X_α , and $F_\alpha = F_\alpha^1 : M \rightarrow M$ the time one flow. Since the X_α are vertical the flow preserves the fibers, and again $F_\alpha(m)$ depends only on $\alpha_{\pi(m)}$. Thus we obtain a fiber bundle map

$$T^* B \times_B M \rightarrow M, \quad (\alpha_b, m) \mapsto F_\alpha(m).$$

To show that this is a vector bundle action, we need to show that the vector fields X_α all commute. Thus let α_1, α_2 be two 1-forms and $X_j = X_{\alpha_j}$ the vector fields they define. We have

$$\begin{aligned} \iota_{[X_1, X_2]} \omega &= (L_{X_1} \iota_{X_2} - \iota_{X_2} L_{X_1}) \omega \\ &= -L_{X_1} \pi^* \alpha_2 - \iota_{X_2} d\iota_{X_1} \omega \\ &= -L_{X_1} \pi^* \alpha_2 + \iota_{X_2} \pi^* d\alpha_1 \\ &= 0 \end{aligned}$$

since $\pi^* \alpha_j$ and $\pi^* d\alpha_j$ are basic forms on $\pi : M \rightarrow B$. Since ω is non-degenerate this verifies $[X_1, X_2] = 0$. Since each map $T_{\pi(m)}^* B \rightarrow V_m$ is an isomorphism, it follows that the action is fiberwise transitive. $\square$

LEMMA 1.5. *Let $\alpha \in \Omega^1(B)$ be a 1-form, and X_α, F_α^t the corresponding vector field and its flow. Then*

$$(F_\alpha^t)^* \omega = \omega - t \pi^* d\alpha.$$

In particular, F_α is a symplectomorphism if and only if α is closed.

PROOF. The Lemma follows by integrating

$$\frac{\partial}{\partial t} (F_\alpha^t)^* \omega = (F_\alpha^t)^* L_{X_\alpha} \omega = -(F_\alpha^t)^* \pi^* d\alpha = -\pi^* d\alpha$$

from 0 to t . $\square$

Let $\Lambda \subset T^* B$ be bundle of stabilizers for the $T^* B$ -action, and

$$\tau : \mathcal{T} = T^* B / \Lambda \rightarrow B$$

the torus bundle.

PROPOSITION 1.6. *Λ is a Lagrangian submanifold of $T^* B$.*

PROOF. Suppose $U \subseteq B$ is an open subset such that Λ is trivial over U , $\Lambda|_U \cong U \times \mathbb{Z}^n$. Any sheet of $\Lambda|_U \rightarrow U$ is given by the graph of a 1-form $\alpha : B \rightarrow T^*B$ such that $F_\alpha = \text{id}_{\pi^{-1}(U)}$. By the previous Lemma, this means $\pi^*d\alpha = 0$. Thus $d\alpha = 0$ showing that the sheet corresponding to α is a Lagrangian submanifold. $\square$

Consider the Lagrangian fibration $p : T^*B \rightarrow B$, with standard symplectic form $-d\theta$ on T^*B . Any $\alpha \in \Omega^1(B)$ defines a vertical vector field $\hat{X}_\alpha$ on T^*B , and a flow $\hat{F}_\alpha^t$. We had discussed this flow in the section on cotangent bundles, where it was denoted G_α^t . We had proved that $\hat{F}_\alpha^t$ is the diffeomorphism of T^*B given by “adding $t\alpha$ ”. Let us briefly recall the argument: For all 1-forms α the canonical 1-form θ on T^*B transforms according to

$$(\hat{F}_\alpha^t)^*\theta = \theta + tp^*\alpha.$$

This follows by integrating

$$\frac{\partial}{\partial t}(\hat{F}_\alpha^t)^*\theta = (\hat{F}_\alpha^t)^*L_{\hat{X}_\alpha}\theta = (\hat{F}_\alpha^t)^*(\iota(\hat{X}_\alpha)d\theta) = (\hat{F}_\alpha^t)^*p^*\alpha = p^*\alpha.$$

from 0 to t . By the property $\beta^*\theta = \beta$ of the canonical 1-form we find,

$$\hat{F}_\alpha^t \circ \beta = (\hat{F}_\alpha^t \circ \beta)^*\theta = \beta^*(\theta + tp^*\alpha) = \beta + t\alpha$$

showing that $\hat{F}_\alpha^t$ adds on $t\alpha$.

PROPOSITION 1.7. *The symplectic form on T^*B descends to $\mathcal{T} = T^*B/\Lambda$. The projection $\tau : \mathcal{T} \rightarrow B$ is a Lagrangian fibration.*

PROOF. We have seen that local sections of Λ are given by closed 1-forms, but adding a closed 1-form in T^*B is a symplectic transformation. $\square$

To summarize the discussion up to this point: Given any symplectic fibration $\pi : M \rightarrow B$ with compact connected fibers, we constructed a torus bundle $\tau : \mathcal{T} \rightarrow B$ with a transitive bundle action of

$$\mathcal{T} \times_B M \rightarrow M.$$

This shows that M is an *affine* torus bundle. Moreover, $\mathcal{T}$ itself was found to carry a symplectic structure, such that $\tau : \mathcal{T} \rightarrow B$ is a Lagrangian fibration! Nonetheless, in general $\mathcal{T}$ is different from M . The affine torus bundle M becomes a torus bundle only after we choose a global section $\sigma : B \rightarrow M$, but in general there may be obstructions to the existence of such a section.

Let us assume that these obstructions vanish, and let $\sigma : B \rightarrow M$ be a global section. (Any other section differs by the action of a section of $\mathcal{T}$). The choice of σ sets up a unique $\mathcal{T}$ -equivariant fiber bundle isomorphism

$$A : \mathcal{T} = T^*B/\Lambda \cong M,$$

sending the identity-section of $\mathcal{T}$ to σ . Equivariance means that

$$A \circ \hat{F}_\alpha = F_\alpha \circ A$$

for all 1-forms $\alpha \in \Omega^1(B)$.

The cohomology class $[\sigma^*\omega] \in H^2(B)$ is independent of the choice of σ . Indeed, if $\tilde{\sigma} = F_\beta \circ \sigma$ is another choice of σ , then

$$\tilde{\sigma}^*\omega = \sigma^*(F_\beta^*\omega) = \sigma^*(\omega - \pi^*d\beta) = \sigma^*\omega - d\beta.$$

Let us assume that $[\sigma^*\omega] = 0$ (of course, this is automatic if ω is exact, e.g. if M is an open subset of a cotangent bundle T^*Q .) Then $\sigma^*\omega = d\beta$ for some 1-form β , and replacing σ with $F_\beta \circ \sigma$ we can assume $\sigma^*\omega = 0$, i.e. the graph of σ is a Lagrangian submanifold. Note that both the existence of σ and the condition $[\sigma^*\omega] = 0$ are automatic if B is contractible, i.e. always locally.

PROPOSITION 1.8. *The choice of any section $\sigma : B \rightarrow M$ with $\sigma^*\omega = 0$ defines a symplectomorphism $A : \mathcal{T} = T^*B/\Lambda \cong M$.*

PROOF. Let $\tilde{A} : T^*B \rightarrow M$ be the map covering A . It suffices to show that $\tilde{A}$ is a local symplectomorphism, i.e. $\tilde{A}^*\omega = -d\theta$. The map $\tilde{A}$ is uniquely defined by its property

$$\tilde{A} \circ \alpha = F_\alpha \circ \sigma$$

for every 1-form $\alpha : B \rightarrow T^*B$. We have

$$\begin{aligned} \alpha^*\tilde{A}^*\omega &= (\tilde{A} \circ \alpha)^*\omega \\ &= (F_\alpha \circ \sigma)^*\omega \\ &= \sigma^*F_\alpha^*\omega \\ &= \sigma^*(\omega - d\pi^*\alpha) \\ &= -d\sigma^*\pi^*\alpha \\ &= -d\alpha \\ &= -d\alpha^*\theta \\ &= \alpha^*(-d\theta) \end{aligned}$$

Since this is true for any α , we conclude¹ $\tilde{A}^*\omega = -d\theta$. □

2. Action-angle coordinates

We are now ready to define *action-angle coordinates*. Recall that up to this point, we have made two assumptions on the fibration $\pi : M \rightarrow B$: First, we assume that it admits a section $\sigma : B \rightarrow M$, second we assume that the cohomology class $[\sigma^*\omega] \in H^2(M)$ (defined independent of the choice of σ) is zero. Then we choose σ to be a Lagrangian section and this gives a symplectomorphism $\mathcal{T} \rightarrow M$.

Let us now also assume that the base B is simply connected. This implies that the group bundle $\Lambda \rightarrow B$ is trivial: Choose an isomorphism $\Lambda_b \cong \mathbb{Z}^n$ for some b , and identify

¹Actually, with a little bit of cheating: A k -form on a fiber bundle over B is determined by its pull-backs under sections of the fiber bundle, if and only if $\dim B \geq k$. Hence we need $\dim B \geq 2$. However the case $\dim B = 1$ is obvious anyway.

$\Lambda_{b'} \cong \Lambda_b$ by choosing a path from b to b' . (This is independent of the choice of path if B is simply connected.) Thus we have an isomorphism

$$\Lambda = B \times \mathbb{Z}^n,$$

and any two such isomorphisms differ by an action of $\text{Aut}(\mathbb{Z}^n) = \text{Gl}(n, \mathbb{Z})$, the group of invertible matrices A such that both A and A^{-1} have integer coefficients (this implies $\det A = \pm 1$). Let $\beta_1, \dots, \beta_n \in \Omega^1(B)$ be the closed 1-forms corresponding to this trivialization. At any point $b \in B$ they define a basis for Λ_b , hence also for T_b^*B . Thus we have also trivialized

$$T^*B \cong B \times \mathbb{R}^n.$$

and

$$M \cong \mathcal{T} \cong B \times (\mathbb{R}^n / \mathbb{Z}^n) = B \times T^n.$$

The map $s : M \rightarrow (\mathbb{R}/\mathbb{Z})^n = T^n$ given by projection to the second factor are the *angle coordinates*. Since the 1-forms β_i take values in Λ they are closed. They define symplectic vector fields $X_i \in \text{Vect}(\pi^{-1}(U))$ whose flows F_i^t in terms of the angle coordinates are given by

$$\begin{aligned} s_j &\mapsto s_j \text{ if } j \neq i, \\ s_i &\mapsto s_i + t. \end{aligned}$$

Since B is by assumption simply connected the β_i are in fact exact:

$$\beta_i = dI_i.$$

The I_i 's (or their pull-backs to $\pi^{-1}(U)$) are called *action coordinates*. The choice of I_i defines an embedding $B \hookrightarrow \mathbb{R}^n$. That is, B is diffeomorphic to an open subset of $\mathbb{R}^n$!. It is clear that $I_1, s_1, \dots, I_n, s_n$ lift to the standard cotangent coordinates on $T^*B \subset T^*(\mathbb{R}^n)$. Therefore ω takes on the form, in action-angle coordinates,

$$\omega = \sum_i dI_i \wedge ds_i.$$

Notice that the choice of action-angle coordinates is very rigid: The β_i 's are determined up to the action of a matrix $C \in \text{Gl}(n, \mathbb{Z})$, and the choice of I_i is determined up to a constant. That is, any other set of action-angle coordinates is of the form,

$$s'_i = \sum_j C_{ij} s_j + \pi^* c_i, \quad I'_i = \sum_j (C^{-1})_{ji} I_j + d_i$$

where the c_i are functions on B and d_i are constants. The I_j can be viewed as functions on B ; note that this induces, in particular, an affine-linear structure on B .

If we drop the assumption that B is simply connected, there are obstructions to the existence of global action-angle coordinates. (Even if the Lagrangian section $\sigma : B \rightarrow M$ exists.) The first, most serious, obstruction is the *monodromy obstruction*: To introduce angle coordinates we have to assume that the bundle Λ is trivial; this however will only be true if the monodromy map

$$\pi_1(B) \rightarrow \text{Aut}(\Lambda_b) \cong \text{Gl}(n, \mathbb{Z})$$

is trivial. If the monodromy obstruction vanishes, there can still be an obstruction to the existence of action coordinates: The forms β_i defined by the angle coordinates are closed, but they need not be exact in general.

Of course, all obstructions vanish locally, e.g. over contractible open subsets $U \subset B$. On the other hand, Duistermaat shows that for a very standard integrable system, the spherical pendulum, the monodromy obstruction is non-zero.

EXERCISE 2.1. Suppose (M, ω) is a symplectic manifold such that ω is exact: $\omega = d\gamma$ for some 1-form γ . Let $\pi : M \rightarrow B$ be a Lagrangian fibration with compact connected fibers, with B simply connected. Given $b \in B$ let

$$A_1(b), \dots, A_n(b) : \mathbb{R}/\mathbb{Z} \rightarrow \pi^{-1}(b)$$

be smooth loops in $\pi^{-1}(b)$ generating the fundamental group of the fiber. Suppose that the $A_i(b)$ define continuous functions $A_i : B \times \mathbb{R}/\mathbb{Z} \rightarrow M$. Show that the formula

$$I_j(m) := \int_{A_j(\pi(m))} \gamma$$

defines a set of action variables.

For an up-to-date discussion of Lagrangian foliations, including a review of recent developments, see the preprint Nguyen Tien Zung: “*Symplectic Topology of Integrable Hamiltonian Systems, II: Characteristic Classes*”, posted as math.DG/0010181.

3. Integrable systems

After this lengthy general discussion let us finally make the connection with the theory of integrable systems. Let (M, ω) be a compact symplectic manifold, $H \in C^\infty(M, \mathbb{R})$ a Hamiltonian and X_H its vector field. In general the flow of X_H can be very complicated, unless there are many “integrals of motion”. An integral of motion is a function $G \in C^\infty(M, \mathbb{R})$ such that $X_H(G) = 0$, or equivalently $\{H, G\} = 0$. An integral of motion defines itself a Hamiltonian flow X_G , which commutes with the flow of X_H since $[X_H, X_G] = -X_{\{H, G\}} = 0$.

DEFINITION 3.1. The dynamical system (M, ω, H) is called integrable over if there exists n integrals of motion $G_1, \dots, G_n \in C^\infty(M, \mathbb{R})$, $\{G_j, H\} = 0$ such that

- (a) The G_i are “in involution”, i.e. they Poisson-commute: $\{G_i, G_j\} = 0$.
- (b) The map $G : M \rightarrow \mathbb{R}^n$ is a submersion almost everywhere, i.e. $dG_1 \wedge \dots \wedge dG_n \neq 0$ on an open dense subset.

THEOREM 3.2 (Liouville-Arnold). *Let (M, ω, H) be a completely integrable Hamiltonian dynamical system, with integrals of motion G_j . Suppose G is a proper map. Let $M' \subset M$ be the subset on which G is a submersion, let B be the set of connected components of fibers of $G|_{M'}$, and $\pi : M' \rightarrow B$ the induced map. Then $\pi : M' \rightarrow B$ is a Lagrangian fibration with compact connected fibers, hence it is an affine torus bundle.*

The flow F^t of X_H is vertical and preserves the affine structure; in local action-angle coordinates it is given by

$$\begin{aligned} I_j(t) &= I_j(0), \\ s_j(t) &= s_j(0) + t \frac{\partial H}{\partial I_j}. \end{aligned}$$

PROOF. It remains to prove the description of the flow of X_H . Since $\{G_j, H\} = 0$ for all j , $H|_{M'}$ is constant along the fibers of G , i.e. is the pull-back of a function on B . In local action-angle coordinates (I, s) for the fibration, this means that H is a function of the I_i 's, and the Hamiltonian vector field becomes

$$X_H = \sum_j \frac{\partial H}{\partial I_j} \frac{\partial}{\partial s_j}.$$

□

4. The spherical pendulum

As one of the simplest non-trivial examples of an integrable system let us briefly discuss the spherical pendulum. We first give the general description of the motion of a particle on a Riemannian manifold Q in a potential $V : Q \rightarrow \mathbb{R}$.

One defines the kinetic energy $T \in C^\infty(TQ)$ by $T(v) = \frac{1}{2}||v||^2$. Using the identification $g^b : TQ \rightarrow T^*Q$ given by the metric, view T as a function on T^*Q . The Hamiltonian is the total energy $H = T + V \in C^\infty(T^*Q)$.

In local coordinates q_i on Q ,

$$T(v) = \frac{1}{2} \sum_{ij} g(q)_{ij} \dot{q}_i \dot{q}_j,$$

where g_{ij} is the metric tensor. The relation between velocities and momenta in local coordinates is $p_i = \sum_j g_{ij} \dot{q}_j$. Thus

$$T(q, p) = \frac{1}{2} \sum_{ij} h(q)_{ij} p_i p_j$$

where $h(q)_{ij}$ is the inverse matrix to $g(q)_{ij}$, and

$$H(q, p) = \frac{1}{2} \sum_{ij} h(q)_{ij} p_i p_j + V(q).$$

The configuration space Q of the spherical pendulum is the 2-sphere, which by an appropriate normalization we can take to be the unit sphere in $\mathbb{R}^3$. Let $\phi \in [0, 2\pi]$, $\psi \in (0, \pi)$ be polar coordinates on S^2 , that is

$$x_1 = \sin \psi \cos \phi, \quad x_2 = \sin \psi \sin \phi, \quad x_3 = \cos \psi.$$

The potential energy is

$$V = \cos \psi$$

and the kinetic energy is

$$T = \frac{1}{2} \sum_{i=1}^3 \dot{x}_i^2 = \frac{1}{2} (\dot{\psi}^2 + \sin^2 \psi \dot{\phi}^2).$$

Thus

$$H = \frac{1}{2}(p_\psi^2 + \frac{1}{\sin^2 \psi} p_\phi^2) + \cos \psi$$

(The apparent singularity at $\psi = 0, \pi$ comes only from the choice of coordinates.) An integral of motion for this system is given by $G = p_\phi$. Indeed,

$$\{H, G\} = 0$$

because H does not depend on ϕ (i.e. because the problem has rotational symmetry around the x_3 -axis). Since $\dim T^*S^2 = 4$, it follows that the spherical pendulum is a completely integrable system.

The image of the map (G, H) has the form $H \geq f(G)$ where $f(G)$ is a symmetric function shaped roughly like a parabola. The minimum of f is the point $(G, H) = (0, -1)$, corresponding to the stable equilibrium. The set of singular values of (G, H) consists of the boundary of the region, i.e. the set of points $(G, F(G))$, together with the *unstable* equilibrium $(0, 1)$.

Removing these singular points from the image of (G, H) , we obtain a non-simply connected region B , and one can raise the question about existence of global action-angle variables. Duistermaat shows that they do not exist in this system: The lattice bundle $\Lambda \rightarrow B$ is non-trivial, i.e. the monodromy obstruction does not vanish.

CHAPTER 6

Symplectic group actions and moment maps

1. Background on Lie groups

A Lie group is a group G with a manifold structure on G such that group multiplication is a smooth map. (This implies that inversion is a smooth map also.) A Lie subgroup $H \subset G$ is a subgroup which is also a submanifold. By a theorem of Cartan, every closed subgroup of a Lie group is an (embedded) Lie subgroup (i.e, smoothness is automatic). In this case the homogeneous space G/H inherits a unique manifold structure such that the quotient map is smooth.

Let $L_g : G \rightarrow G$, $a \mapsto ga$ denote left translation and $R_g : G \rightarrow G$, $a \mapsto ag$ right translation. A vector field X on G is called left-invariant if it is L_g -related to itself, for all $g \in G$. Any left-invariant vector field is determined by its value $X(e) = \xi$ at the identity element. Thus evaluation at e gives a vector space isomorphism $\mathfrak{X}^L(G) \cong \mathfrak{g} := T_e G$. But $\mathfrak{X}^L(G)$ is closed under the Lie bracket operation of vector fields. The space $\mathfrak{g} = T_e G$ with Lie bracket induced in this way is called the Lie algebra of G . That is, we define the Lie bracket on $\mathfrak{g}$ by the formula

$$[\xi, \eta]^L := [\xi^L, \eta^L],$$

where ξ^L is the left-invariant vector field with $\xi^L(e) = \xi$. For matrix Lie groups (i.e. closed subgroups of $\mathrm{GL}(n, \mathbb{R})$), the Lie bracket coincides with the commutator of matrices. Working with the space of right-invariant vector fields would have produced the *opposite* bracket: We have

$$[\xi, \eta]^R = -[\xi^R, \eta^R].$$

Let $F_\xi^t : G \rightarrow G$ be the flow of ξ^L . One defines the exponential map

$$\exp : \mathfrak{g} \rightarrow G$$

by $\exp(\xi) = F_\xi^1(e)$. In terms of $\exp$, the flow of ξ^L is $g \mapsto g \exp(t\xi)$, while the flow of ξ^R is $g \mapsto \exp(t\xi)g$. For matrix Lie groups, $\exp$ is the usual exponential of matrices.

Let $\mathrm{Ad}_g = L_g \circ R_{g^{-1}}$, i.e. $\mathrm{Ad}_g(a) = gAg^{-1}$. Clearly Ad_g fixes e , so it induces a linear transformation (still denoted Ad_g) of $\mathfrak{g} = T_e G$. One defines

$$\mathrm{ad}_\xi(\eta) = \left. \frac{\partial}{\partial t} \right|_{t=0} \mathrm{Ad}_{\exp(t\xi)}(\eta).$$

It turns out that $\mathrm{ad}_\xi(\eta) = [\xi, \eta]$, which gives an alternative way of defining the Lie bracket on $\mathfrak{g}$.

2. Generating vector fields for group actions

DEFINITION 2.1. Let G be a Lie group. An action of G on a manifold Q is a smooth map

$$\mathcal{A} : G \times Q \rightarrow Q, (g, q) \mapsto \mathcal{A}_g(q) = g \cdot q$$

such that the map $G \rightarrow \text{Diff}(Q)$, $g \mapsto \mathcal{A}_g$ is a group homomorphism.

A manifold Q together with a G -action is called a G -manifold. A map $F : Q_1 \rightarrow Q_2$ between two G -manifolds is called equivariant if it intertwines the G -actions, that is, $g.F(q_1) = F(g.q_1)$.

- EXAMPLE 2.2. (a) There are three natural actions of any Lie group G on itself: The left-action $g.a = ga$, the right action $g.a = ag^{-1}$, and the adjoint action $g.a = gag^{-1}$.
 (b) Any finite dimensional linear representation of G on a vector space V is a G -action on V .

DEFINITION 2.3. Let $\mathfrak{g}$ be a Lie algebra. A *Lie algebra action* of $\mathfrak{g}$ on Q is a smooth vector bundle map

$$Q \times \mathfrak{g} \rightarrow TQ, (q, \xi) \mapsto \xi_Q(q)$$

such that the map $\mathfrak{g} \rightarrow \mathfrak{X}(Q)$, $\xi \mapsto \xi_Q$ is a Lie algebra homomorphism.

Suppose Q is a G -manifold. For $\xi \in \mathfrak{g}$ let the *generating vector field* ξ_Q be the unique vector field with flow

$$q \mapsto \exp(-t\xi).q \quad .$$

If $F : Q_1 \rightarrow Q_2$ is an equivariant map, then $\xi_{Q_1} \sim_F \xi_{Q_2}$.

EXAMPLE 2.4. For any $\xi \in \mathfrak{g}$ let ξ^L denote the unique left-invariant vector field with $\xi^L(e) = \xi$. Similarly let ξ^R be the right-invariant vector field with $\xi^R(e) = \xi$. The generating vector field for the left-action is right-invariant (since the left-action commutes with the right-action), and its value at e is $-\xi$. Hence the left-action is generated by $-\xi^R$. Similarly the right-action is generated by ξ^L , and the adjoint action by $\xi^L - \xi^R$.

PROPOSITION 2.5. Let G be a Lie group with Lie algebra $\mathfrak{g}$. For any action of a Lie group G on a manifold Q the map

$$\mathfrak{g} \rightarrow \mathfrak{X}(Q), \quad \xi \mapsto \xi_Q$$

is a Lie algebra action of $\mathfrak{g}$ on Q . For $g \in G$ one has

$$g_*\xi_Q = (\text{Ad}_g \xi)_Q.$$

PROOF. The idea of proof is to reduce to the case of the action $R_{g^{-1}}$ of G on itself. Let $\tilde{Q} = G \times Q$ with G -action $g.(a, q) = (ag^{-1}, q)$. The generating vector fields for this action are $\xi_{\tilde{Q}} = (\xi^L, 0)$. Since $[\xi^L, \eta^L] = [\xi, \eta]^L$ by definition of the Lie bracket,

$$(9) \quad [\xi_{\tilde{Q}}, \eta_{\tilde{Q}}] = [\xi, \eta]_{\tilde{Q}}.$$

The map $F : G \times Q \rightarrow Q$, $(a, q) \mapsto a^{-1}.q$ is a G -equivariant fibration. In particular, $\xi_{\tilde{Q}} \sim_F \xi_Q$ and (9) implies $[\xi_Q, \eta_Q] = [\xi, \eta]_Q$. For the second equation, we note that for the action of G on itself by left multiplication,

$$(R_{g^{-1}})_*\xi^L = (\text{Ad}_g)_*\xi^L = (\text{Ad}_g \xi)^L.$$

Hence $g_*\xi_{\tilde{Q}} = (\text{Ad}_g \xi)_{\tilde{Q}}$. Again, since F is G -equivariant, this implies $g_*\xi_Q = (\text{Ad}_g \xi)_Q$. $\square$

If G is simply connected and Q is compact, the converse is true: every $\mathfrak{g}$ -action on Q integrates to a G -action.

Any action of G on Q gives rise to an action on TQ and T^*Q . If $q \in Q$ is fixed under the action of G , then these actions induce linear G -actions (i.e. representations) on T_qQ and T_q^*Q . In particular, the conjugation action of G on itself induces an action on the Lie algebra $\mathfrak{g} = T_eG$, called the adjoint action, and on $\mathfrak{g}^*$, called the co-adjoint action. The two actions are related by

$$\langle g.\mu, \xi \rangle = \langle \mu, g^{-1}.\xi \rangle, \quad \mu \in \mathfrak{g}^*, \xi \in \mathfrak{g}.$$

EXERCISE 2.6. Using the identifications $T\mathfrak{g} = \mathfrak{g} \times \mathfrak{g}$ and $T^*\mathfrak{g} = \mathfrak{g}^* \times \mathfrak{g}^*$ determine the generating vector fields for the adjoint and co-adjoint actions.

3. Hamiltonian group actions

A G -action $g \mapsto \mathcal{A}_g$ on a symplectic manifold (M, ω) is called *symplectic* if $\mathcal{A}_g \in \text{Symp}(M, \omega)$ for all g . Similarly a $\mathfrak{g}$ -action $\xi \mapsto \xi_M$ is called symplectic if $\xi_M \in \mathfrak{X}(M, \omega)$ for all ξ . Clearly, the $\mathfrak{g}$ -action defined by a symplectic G -action is symplectic. The G -action or $\mathfrak{g}$ -action is called weakly Hamiltonian if all ξ_M are Hamiltonian vector fields. That is, in the Hamiltonian case there exists a function $\Phi(\xi) \in C^\infty(M)$ for all ξ such that $\xi_M = X_{\Phi(\xi)}$. One can always choose $\Phi(\xi)$ to depend linearly on ξ (define Φ first for a basis of $\mathfrak{g}$ and then extend by linearity). The map $\xi \mapsto \Phi(\xi)$ can then be viewed as a function $\Phi \in C^\infty(M) \otimes \mathfrak{g}^*$.

DEFINITION 3.1. A symplectic G -action on a symplectic manifold (M, ω) is called weakly Hamiltonian if there exists a map (called *moment map*)

$$\Phi \in C^\infty(M) \otimes \mathfrak{g}^*$$

such that for all ξ , the function $\langle \Phi, \xi \rangle \in C^\infty(M)$ is a Hamiltonian for ξ_M :

$$d\langle \Phi, \xi \rangle = \iota(\xi_M)\omega.$$

It is called Hamiltonian if Φ is equivariant with respect to the G -action on M and the co-adjoint action of G on $\mathfrak{g}^*$.

Similarly one defines moment maps for Hamiltonian $\mathfrak{g}$ -actions; one requires Φ to be $\mathfrak{g}$ -equivariant in this case.

It is obvious that if a group G acts in a Hamiltonian way, and if $H \rightarrow G$ is a homomorphism (e.g. inclusion of a subgroup) then the action of H is Hamiltonian; the moment map is the composition of the G -moment map with the dual map $\mathfrak{g}^* \rightarrow \mathfrak{h}^*$.

We sometimes write $\Phi^\xi = \langle \Phi, \xi \rangle$ for the ξ -component of Φ . The equivariance condition means that $g^*\Phi = g.\Phi = (\text{Ad}_{g^{-1}})^*\Phi$, or in detail:

$$\langle g^*\Phi, \eta \rangle = \langle \text{Ad}_{g^{-1}}^* \Phi, \eta \rangle = \langle \Phi, \text{Ad}_{g^{-1}}(\eta) \rangle$$

for all $m \in M$, $\eta \in \mathfrak{g}$. Writing $g = \exp(t\xi)$ and taking the derivative at $t = 0$ we find the infinitesimal version of this condition reads,

$$\xi_M \Phi^\eta = \Phi^{[\xi, \eta]}.$$

For an abelian group, the conjugation action is trivial so that equivariance simply means invariance.

One way of interpreting the equivariance condition is as follows. For a weakly Hamiltonian G -action the fundamental vector fields define a Lie homomorphism $\mathfrak{g} \rightarrow \mathfrak{X}_{\text{Ham}}(M, \omega) \subseteq \mathfrak{X}(M, \omega)$. This can always be lifted to a linear map $\mathfrak{g} \rightarrow C^\infty(M, \mathbb{R})$. The action is Hamiltonian if it can be lifted to an *equivariant* map, which for a connected group is equivalent to the following:

LEMMA 3.2. *For a Hamiltonian G -action, the map*

$$\mathfrak{g} \rightarrow C^\infty(M), \xi \mapsto \langle \Phi, \xi \rangle$$

is a Lie homomorphism.

PROOF.

$$\{\Phi^\xi, \Phi^\eta\} = X_{\Phi^\xi}(\Phi^\eta) = \xi_M \Phi^\eta = \Phi^{[\xi, \eta]}.$$

□

Notice that we defined the the Poisson bracket in such a way that $C^\infty(M) \rightarrow \mathfrak{X}(M)$ is a Lie homomorphism.

From now on, we will always assume that the moment map is equivariant unless stated otherwise.

LEMMA 3.3. *Any weakly Hamiltonian action of a Lie group G on (M, ω) is Hamiltonian if*

- (a) *G is compact, or*
- (b) *M is compact.*

PROOF. Suppose Φ is any moment map, not necessarily equivariant. Define

$$g \cdot \Phi = (g^{-1})^*(\text{Ad}_{g^{-1}}^*)\Phi$$

so that Φ is equivariant if and only if $g \cdot \Phi = \Phi$ for all g . We claim that $g \cdot \Phi$ is also a moment map for the G -action. Indeed,

$$\begin{aligned} d\langle g \cdot \Phi, \xi \rangle &= (g^{-1})^* d\langle \Phi, \text{Ad}_g(\xi) \rangle \\ &= (g^{-1})^* g^* \iota(\xi_M) \omega \\ &= \iota_{\xi_M} \omega. \end{aligned}$$

If G is compact, we obtain a G -equivariant moment map by averaging over the group. If M is compact and connected, normalize Φ by the condition $\int_M \Phi = 0$. Then Φ is invariant since $\int_M g \cdot \Phi = 0$. $\square$

We remark that the above Lemma also holds for actions of connected, simply connected semi-simple Lie groups (i.e. $[\mathfrak{g}, \mathfrak{g}] = \mathfrak{g}$), or more generally for any connected, simply connected group G for which the first and second Lie algebra cohomology vanish. See Theorem 26.1 in Guillemin-Sternberg, *Symplectic techniques in Physics*, Cambridge University Press 1984, for details.

4. Examples of Hamiltonian G -spaces

4.1. Linear momentum and angular momentum. Recall that for any vector field Y on a manifold Q , the cotangent lift $\hat{Y} \in \mathfrak{X}(T^*Q)$ is a Hamiltonian vector field $\hat{Y} = X_H$, with $H = \iota_{\hat{Y}}\theta$. We leave it as an exercise to check that linear map

$$\mathfrak{X}(Q) \rightarrow C^\infty(T^*Q), \quad Y \mapsto \iota_Y \theta$$

is actually a Lie algebra homomorphism (using the Poisson bracket on T^*Q). Hence if Q is a G -manifold, we obtain a moment map for the cotangent lift of the G -action to T^*Q by composing this map with the generating vector fields, $\mathfrak{g} \mapsto \mathfrak{X}(Q)$. Recall that in local cotangent coordinates, if $Y = \sum_j Y_j(q) \frac{\partial}{\partial q_j}$ then the corresponding Hamiltonian is $H(q, p) = \sum_j Y_j(q) p_j$.

For example, let $M = T^*(\mathbb{R}^n) = \mathbb{R}^{2n}$ with standard symplectic coordinates q_j, p_j . Let $G = \mathbb{R}^n$ act on itself by translation. The Lie algebra is $\mathfrak{g} = \mathbb{R}^n$, with exponential map $\exp : \mathfrak{g} \rightarrow G$ the identity map of $\mathbb{R}^n$.

The generating vector fields $b_{\mathbb{R}^n}$ for $b \in \mathbb{R}^n$ are obtained from the calculation,

$$(b_{\mathbb{R}^n} f)(q) = \left. \frac{\partial}{\partial t} \right|_{t=0} \exp(-tb)^* f(q) = \left. \frac{\partial}{\partial t} \right|_{t=0} f(q - tb) = - \sum_j b_j \frac{\partial f}{\partial q_j}.$$

Thus $b_{\mathbb{R}^n} = - \sum_j b_j \frac{\partial}{\partial q_j}$ and the moment map for the cotangent lift is

$$\langle \Phi, b \rangle = - \sum_j b_j p_j.$$

Using the standard inner product $(b, b') = b \cdot b'$ on $\mathbb{R}^n$ to identify $(\mathbb{R}^n) \cong (\mathbb{R}^n)^*$, we find that

$$\Phi(q, p) = -p.$$

That is, the moment map is just linear momentum (up to an irrelevant sign that just comes from the chosen identification $(\mathbb{R}^n) \cong (\mathbb{R}^n)^*$).

Consider next the cotangent lift of the action of $G = \mathrm{Gl}(n, \mathbb{R})$ on $\mathbb{R}^n$. The Lie algebra $\mathfrak{g} = \mathfrak{gl}(n, \mathbb{R})$ is the space of all real matrices, with exponential map the exponential map for matrices. We compute the generating vector field $A_{\mathbb{R}^n}$ for the action of $A \in \mathfrak{gl}(n, \mathbb{R})$ as follows:

$$\begin{aligned} (A_{\mathbb{R}^n} f)(q) &= \left. \frac{\partial}{\partial t} \right|_{t=0} \exp(-tA)^* f(q) \\ &= \left. \frac{\partial}{\partial t} \right|_{t=0} f(\exp(-tA)q) \\ &= - \sum_j (Aq)_j \frac{\partial f}{\partial q_j} \\ &= - \sum_{j,k} A_{jk} q_k \frac{\partial f}{\partial q_j}, \end{aligned}$$

that is, $A_{\mathbb{R}^n} = - \sum_{j,k} A_{jk} q_k \frac{\partial}{\partial q_j}$. A moment map for the cotangent lift of the action is

$$\langle \Phi, A \rangle = - \sum_{j,k} A_{jk} p_j q_k.$$

Hence, using the non-degenerate bilinear form $(A, A') = \mathrm{tr}(A^t A')$ on $\mathfrak{gl}(n, \mathbb{R})$ to identify $\mathfrak{gl}(n, \mathbb{R}) \cong \mathfrak{gl}(n, \mathbb{R})^*$, we have

$$\Phi(q, p)_{ij} = p_i q_j.$$

Note that the pairing on $\mathfrak{gl}(n, \mathbb{R})$ is not invariant under the adjoint action, i.e. it does not identify the adjoint and coadjoint action. Instead, the coadjoint action becomes the contragredient action $g.A = \mathrm{Ad}((g^{-1})^t)A$.

The pairing does, however, restrict to an invariant pairing on the sub-algebra $\mathfrak{h} = \mathfrak{o}(n)$ of skew-symmetric matrices $A = -A^t$. The corresponding connected Lie subgroup of $\mathrm{Gl}(n, \mathbb{R})$ is the special orthogonal group $H = \mathrm{SO}(n)$. The moment map $\Psi : T^*\mathbb{R}^n \rightarrow \mathfrak{o}(n)^*$ for the action of $\mathrm{SO}(n)$ reads,

$$\Psi(q, p)_{ij} = \frac{1}{2}(p_i q_j - p_j q_i).$$

For $n = 3$, we can further identify $\mathfrak{so}(3)^* \cong \mathbb{R}^3$ with the standard rotation action of $\mathrm{SO}(3)$, and Ψ just becomes just *angular momentum* $\Psi(q, p) = \vec{p} \times \vec{q}$ (up to an irrelevant factor, which again just depends on the chosen identification $\mathfrak{so}(3) \cong \mathfrak{so}(3)^*$).

4.2. Exact symplectic manifolds. The previous examples generalize to cotangent bundles T^*Q of G -manifolds Q , or more generally to what is called *exact* symplectic manifolds. A symplectic manifold (M, ω) is called exact if $\omega = -d\theta$ for a 1-form θ (which is sometimes called a symplectic potential). Note that a compact symplectic manifold is never exact (unless it is 0-dimensional): Indeed, if $\omega = -d\theta$ is exact, then

also the Liouville form $\omega^n = -d\theta\omega^{n-1}$ is exact. Hence if M were compact, Stokes' theorem would show that M has zero volume.

PROPOSITION 4.1. *Suppose (M, ω) is an exact symplectic manifold, $\omega = -d\theta$. Then any G -action on M preserving θ is Hamiltonian, with moment map*

$$\langle \Phi, \xi \rangle = \iota(\xi_M)\theta.$$

Note that if G is compact one can construct an invariant θ by averaging. If $H^1(M, \mathbb{R}) = 0$ then Φ is independent of the choice of invariant θ .

PROOF. We calculate $d\iota(\xi_M)\theta = -\iota(\xi_M)d\theta + L(\xi_M)\theta = \iota(\xi_M)\omega$. The resulting moment map is equivariant:

$$g^*\langle \Phi, \xi \rangle = g^*(\iota(\xi_M)\theta) = \iota(g_*\xi_M)g^*\theta = \iota(g_*\xi_M)\theta = \iota((\text{Ad}_g \xi)_M)\theta = \langle \Phi, \text{Ad}_g \xi \rangle.$$

□

The examples considered above were of the form $M = T^*Q$, with G acting by the cotangent lift of a G -action on Q . Another example is provided by the defining action of $U(n)$ on $\mathbb{C}^n = \mathbb{R}^{2n}$:

4.3. Unitary representations. Introduce complex coordinates $z_j = q_j + ip_j$ on $\mathbb{C}^n = \mathbb{R}^{2n}$ and write the symplectic form as

$$\omega = \sum_j dq_j \wedge dp_j = \frac{i}{2} \sum_j dz_j \wedge d\bar{z}_j = \frac{1}{2i} h(dz, dz),$$

where $h(w, w') = \sum_i \bar{w}_i w'_i$ is the Hermitian inner product. Then $\omega = -d\theta$ with

$$\theta = \frac{i}{4} \sum_j (\bar{z}_j dz_j - z_j d\bar{z}_j) = -\frac{1}{2} \text{Im}(h(z, dz)),$$

This choice of θ is preserved under the unitary group. The Lie algebra $\mathfrak{u}(n)$ of $U(n)$ consists of skew-adjoint matrices ξ . The generating vector fields are

$$\xi_{\mathbb{C}^n} = \sum_{j,k} (\xi_{jk} z_k \frac{\partial}{\partial z_j} - \xi_{kj} \bar{z}_k \frac{\partial}{\partial \bar{z}_j})$$

where $\xi \in \mathfrak{u}(n)$ is a skew-Hermitian matrix. Hence $\langle \Phi, \xi \rangle = -\iota(\xi_{\mathbb{C}^n})\theta$ is given by

$$\langle \Phi(z), \xi \rangle = \frac{i}{2} \sum_{j,k} \xi_{jk} \bar{z}_j z_k = \frac{i}{2} h(z, \xi z)$$

(where h is the Hermitian metric) defines a moment map. Using the inner product on $\mathfrak{u}(n)$, $(\xi, \eta) = -\text{Tr}(\xi\eta)$ to identify the Lie algebra and its dual,

$$\Phi(z)_{kj} = -\frac{1}{4} (\bar{z}_j z_k - \bar{z}_k z_j).$$

This example also shows that any finite dimensional unitary representation $G \rightarrow U(n)$ defines a Hamiltonian action of G on $\mathbb{C}^n$; the moment map is the composition of the

$U(n)$ moment map with the projection $\mathfrak{u}(n)^* \rightarrow \mathfrak{g}^*$ dual to $\mathfrak{g} \rightarrow \mathfrak{u}(n)$. For example, the moment map for the scalar S^1 -action is given by $-\frac{1}{2}||z||^2$.

4.4. Projective Representations. The action of $U(n+1)$ on $\mathbb{C}^{n+1}$ induces an action on $\mathbb{C}P(n)$ which also turns out to be Hamiltonian. In homogeneous coordinates $[z_0 : \dots : z_n]$, the moment map is

$$\Phi([z], \xi) = \frac{i}{2} \frac{\sum_{j,k} \xi_{jk} \bar{z}_j z_k}{\sum_j |z_j|^2}.$$

We will verify this fact later in the context of symplectic reduction.

4.5. Symplectic representations. Generalizing the case of unitary representations, consider any *symplectic representation* of G on a symplectic vector space (E, ω) . That is, G acts by a homomorphisms $G \rightarrow \text{Sp}(E)$ into the symplectic group. A moment map for such an action is given by the formula,

$$\langle \Phi(v), \xi \rangle = \frac{1}{2} \omega(v, \xi.v).$$

Indeed, if we identify $T_v E = E$ the generating vector field for $\xi \in \mathfrak{g}$ is just $\xi_E(v) = -\xi.v$. Thus for $w \in E$ we have

$$\omega(\xi_E(v), w) = \omega(w, \xi.v).$$

On the other hand, the map Φ defined above satisfies

$$\begin{aligned} d_v \Phi^\xi(w) &= \left. \frac{1}{2} \frac{\partial}{\partial t} \right|_{t=0} \omega(v + tw, \xi.(v + tw)) \\ &= \frac{1}{2} (\omega(w, \xi.v) + \omega(v, \xi.w)) \\ &= \omega(w, \xi.v), \end{aligned}$$

verifying that Φ is a moment map.

4.6. Coadjoint Orbits. As a preparation, let us note that for any Hamiltonian G -space (M, ω, Φ) , the moment map determines the pull-back of ω to any G -orbit. Indeed, since ξ_M is the Hamiltonian vector field for Φ^ξ ,

$$\omega(\xi_M, \eta_M) = -\{\Phi^\xi, \Phi^\eta\} = -\Phi^{[\xi, \eta]}.$$

In particular if M is a homogeneous Hamiltonian G -manifold (i.e. if M is a single G -orbit), ω is completely determined by Φ !

THEOREM 4.2 (Kirillov-Kostant-Souriau). *Let $\mathcal{O} \subset \mathfrak{g}^*$ be an orbit for the coadjoint action of G on $\mathfrak{g}^*$. There exists a unique symplectic structure on $\mathcal{O}$ for which the action is Hamiltonian and the moment map is the inclusion $\Phi : \mathcal{O} \hookrightarrow \mathfrak{g}^*$.*

PROOF. For any $\mu \in \mathcal{O}$ consider the skew-symmetric bilinear form on $\mathfrak{g}$,

$$B_\mu(\xi, \eta) = \langle \mu, [\xi, \eta] \rangle.$$

Writing $B_\mu(\xi, \eta) = \langle \mu, \text{ad}_\xi \eta \rangle = \langle (\text{ad}_\xi)^* \mu, \eta \rangle$ we see that the kernel of B_μ consists of all ξ with $\text{ad}_\xi^* \mu = 0$, i.e. $\xi_{\mathcal{O}}(\mu) = 0$. It follows that

$$(10) \quad \omega_\mu(\xi_{\mathcal{O}}(\mu), \eta_{\mathcal{O}}(\mu)) = -\langle \mu, [\xi, \eta] \rangle.$$

is a well-defined symplectic 2-form on $T_\mu \mathcal{O}$. The calculation

$$B_{g\mu}(g \cdot \xi, g \cdot \eta) = \langle g \cdot \mu, [g \cdot \xi, g \cdot \eta] \rangle = \langle g \cdot \mu, g \cdot [\xi, \eta] \rangle = \langle \mu, [\xi, \eta] \rangle = B_\mu(\xi, \eta)$$

shows that the resulting 2-form ω on $\mathcal{O}$ is G -invariant (and therefore, smooth!), and equation (10) gives the moment map condition $\iota(\xi_{\mathcal{O}})\omega = d\langle \Phi, \xi \rangle$ for the inclusion map $\Phi : \mathcal{O} \hookrightarrow \mathfrak{g}^*$:

$$\iota(\eta_{\mathcal{O}})d\langle \Phi, \xi \rangle = L(\eta_{\mathcal{O}})\langle \Phi, \xi \rangle = \langle (\text{ad}_\eta)^* \Phi, \xi \rangle = -\langle \Phi, [\xi, \eta] \rangle = \omega(\xi_{\mathcal{O}}, \eta_{\mathcal{O}}).$$

To check $d\omega = 0$, we compute:

$$\iota(\xi_{\mathcal{O}})d\omega = L(\xi_{\mathcal{O}})\omega - d\iota(\xi_{\mathcal{O}})\omega = 0.$$

As remarked above, the moment map uniquely determines the symplectic form. $\square$

EXAMPLE 4.3. Let $G = \text{SO}(3)$. Identify the Lie algebra $\mathfrak{so}(3)$ with $\mathbb{R}^3$, by identifying the standard basis vectors of $\mathfrak{so}(3)$ as follows:

$$e_1 \mapsto \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix} \quad e_2 \mapsto \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix} \quad e_3 \mapsto \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

This identification takes the adjoint action of $\text{SO}(3)$ to the standard rotation action on $\mathbb{R}^3$, and takes the invariant inner product $(A, B) \mapsto -\frac{1}{2} \text{tr}(AB)$ on $\mathfrak{so}(3)$ to the standard inner product on $\mathbb{R}^3$. The inner product also identifies $\mathfrak{so}(3)^* \cong \mathbb{R}^3$. The coadjoint orbits for $\text{SO}(3)$ are the 2-spheres around 0, together with the origin $\{0\}$.

THEOREM 4.4 (Kostant-Souriau). *Let G be a Lie group, and (M, ω, Φ) is a Hamiltonian G -space on which G acts transitively. Then M is a covering space of a coadjoint orbit, with 2-form obtained by pull-back of the KKS form on $\mathcal{O}$.*

PROOF. Let $\mathcal{O} = \Phi(M)$. It is clear that the map $\Phi : M \rightarrow \mathcal{O}$ is a submersion. We had already seen that the 2-form on M is determined by the moment map condition, and the formula shows that the map $\Phi : M \rightarrow \mathcal{O}$ preserves the symplectic form. Hence its tangent map is a bijection everywhere, and so $\Phi : M \rightarrow \mathcal{O}$ is a local diffeomorphism. $\square$

Note that while M is a homogeneous space G/G_m its image under the moment map is a homogeneous space $G/G_{\Phi(m)}$, and the covering map is a fibration with (discrete !) fiber $G_{\Phi(m)}/G_m$. Hence, non-trivial coverings can be obtained only if the stabilizer $G_{\Phi(m)}$ is disconnected (for compact connected Lie group do not have proper subgroups of the same dimension).

If G is a *compact, connected* Lie group then it is known that all stabilizer groups G_μ for the coadjoint action are connected, so one has:

THEOREM 4.5. *If G is compact, connected and (M, ω, Φ) is a homogeneous Hamiltonian G -space, the moment map induces a symplectomorphism of M with the coadjoint orbit $\mathcal{O} = \Phi(M)$.*

The connectedness of G_μ can be shown as follows: Using the fibration $G \rightarrow G/G_\mu$, and the fact that G and G/G_μ are connected it suffices to show that the base G/G_μ is simply connected. (Given $p_0, p_1 \in G_\mu$ choose a path $\gamma : [0, 1] \rightarrow G$ connecting them. γ projects to a closed path in G/G_μ , and can be contracted to a constant path. By the homotopy lifting property this homotopy can be lifted to a homotopy with fixed end points of γ . This will then be a path in G_μ connecting p_0, p_1 .) The simply-connectedness of G/G_μ in turn follows from Morse theory; this will be explained later.

4.7. Poisson manifolds. Moment maps fit very nicely into the more general category of Poisson manifolds.

DEFINITION 4.6. A Poisson manifold is a manifold M together with a bilinear map $\{\cdot, \cdot\} : C^\infty(M) \times C^\infty(M) \rightarrow C^\infty(M)$ such that

- (a) $\{\cdot, \cdot\}$ is a Lie algebra structure on $C^\infty(M)$, and
- (b) for all $H \in C^\infty(M)$, the map $C^\infty(M) \rightarrow C^\infty(M), F \mapsto \{H, F\}$ is a derivation.

A smooth map $\phi : M_1 \rightarrow M_2$ between Poisson manifolds is called a Poisson map if

$$\phi^*\{F_1, F_2\} = \{\phi^*F_1, \phi^*F_2\}.$$

A vector field $X \in \mathfrak{X}(M)$ is called Poisson if

$$X\{F_1, F_2\} = \{X(F_1), F_2\} + \{F_1, X(F_2)\}.$$

Since any derivation of $C^\infty(M)$ is given by a vector field, any H defines a so-called Hamiltonian vector field X_H by $X_H(F) = \{H, F\}$.

EXERCISE 4.7. Show that X_H is a Poisson vector field. Show that the flow of any complete Poisson vector field is Poisson.

Examples of Poisson manifolds are of course symplectic manifolds, with the Poisson bracket associated to the symplectic structure. Another important example, due to Kirillov, is the dual $\mathfrak{g}^*$ of a Lie algebra $\mathfrak{g}$. For any $\mu \in \mathfrak{g}^*$ and any function $F \in C^\infty(\mathfrak{g}^*)$ identify

$$dF_\mu \in T_\mu^* \mathfrak{g}^* \cong (\mathfrak{g}^*)^* = \mathfrak{g}.$$

Then define

$$\{F, G\}(\mu) := \langle \mu, [dF_\mu, dG_\mu] \rangle$$

EXERCISE 4.8. Verify that this is a Poisson structure on $\mathfrak{g}^*$. Show that the inclusion $\mathcal{O} \hookrightarrow \mathfrak{g}^*$ of a coadjoint orbit is a Poisson map.

The definition of a moment map carries over to Poisson manifolds: A G -action is Poisson if it preserves Poisson brackets, and any such action is Hamiltonian if there exists an equivariant smooth map $\Phi : M \rightarrow \mathfrak{g}^*$ such that

$$\xi_M = X_{\langle \Phi, \xi \rangle}$$

for all $\xi \in \mathfrak{k}$.

EXERCISE 4.9. Show that the coadjoint action of G on $\mathfrak{g}^*$ is Hamiltonian, with moment map the identity map.

EXERCISE 4.10. Let (M, ω) be a symplectic manifold and G a connected Lie group acting symplectically on M . Suppose there exists a moment map $\Phi : M \rightarrow \mathfrak{g}^*$ in the weak sense. Show that Φ is equivariant if and only if Φ is a Poisson map for the Kirillov Poisson structure. Conversely, show that for every Poisson map $\Phi : M \rightarrow \mathfrak{g}^*$, the equation

$$\iota(\xi_M)\omega = d\langle \Phi, \xi \rangle$$

defines a symplectic Lie algebra action of $\mathfrak{g}$ on M . More generally, show that for any Poisson manifold M , any Poisson map $\Phi : M \rightarrow \mathfrak{g}^*$ defines a Poisson $\mathfrak{g}$ -action on M .

4.8. 2d Gauge Theory. Let us now spend some time discussing, somewhat informally, an interesting ∞ -dimensional example. We begin with a rapid introduction to what we call (with some exaggeration) gauge theory.

Let Σ be a compact oriented manifold, G a compact Lie group, and $\Omega^k(\Sigma, \mathfrak{g})$ the k -forms on Σ with values in $\mathfrak{g}$. The space $\mathcal{A}(\Sigma) := \Omega^1(\Sigma, \mathfrak{g})$ will be viewed as an infinite dimensional manifold, called the space of *connections*. For any $A \in \mathcal{A}(\Sigma)$ there is a corresponding *covariant derivative*

$$d_A : \Omega^k(\Sigma, \mathfrak{g}) \rightarrow \Omega^{k+1}(\Sigma, \mathfrak{g}),$$

defined

$$d_A \xi = d\xi + [A, \xi]$$

(using the wedge product). The *gauge action* of $\mathcal{G}(\Sigma) := C^\infty(\Sigma, G)$ on $\mathcal{A}(\Sigma)$ is defined by

$$g \cdot A = \text{Ad}_g(A) - dg g^{-1}$$

where the first term is the pointwise adjoint action. The second term is written for matrix-groups so that $dg g^{-1}$ makes sense as a 1-form on Σ with values in $\mathfrak{g}$. More invariantly, it is the pull-back under $g : \Sigma \rightarrow G$ of the right-invariant Maurer-Cartan form $\bar{\theta} \in \Omega^1(G, \mathfrak{g})$; i.e. $\bar{\theta}$ is the unique right-invariant form such that for any right-invariant vector field ξ^R , $\iota(\xi^R)\bar{\theta} = \xi$. The gauge action is defined in such a way that the following is true:

LEMMA 4.11. For all $\xi \in \Omega^k(\Sigma, \mathfrak{g})$,

$$d_{g \cdot A} \text{Ad}_g(\xi) = \text{Ad}_g(d_A \xi).$$

PROOF. Using the definition, and using that for all $\xi \in \Omega^k(\Sigma, \mathfrak{g})$,

$$\begin{aligned} d(\text{Ad}_g \xi) &= d(g\xi g^{-1}) \\ &= dg\xi g^{-1} + \text{Ad}_g d\xi + (-1)^k g\xi dg^{-1} \\ &= dgg^{-1} \text{Ad}_g \xi - (-1)^k \text{Ad}_g \xi dgg^{-1} + \text{Ad}_g(d\xi) \\ &= [dgg^{-1}, \text{Ad}_g \xi] + \text{Ad}_g(d\xi) \end{aligned}$$

we have

$$\begin{aligned} d_{g \cdot A} \text{Ad}_g(\xi) &= d(\text{Ad}_g(\xi)) + [\text{Ad}_g(A) - dgg^{-1}, \text{Ad}_g \xi] \\ &= \text{Ad}_g(d\xi) + [dgg^{-1}, \text{Ad}_g \xi] + [\text{Ad}_g(A) - dgg^{-1}, \text{Ad}_g \xi] \\ &= \text{Ad}_g(d_A \xi). \end{aligned}$$

□

Due to the presence of the gauge term the square of d_A is usually not zero:

LEMMA 4.12. *Let $\text{curv}(A) = dA + \frac{1}{2}[A, A] \in \Omega^2(\Sigma, \mathfrak{g})$ be the curvature of A . Then for all $\xi \in \Omega^k(\Sigma, \mathfrak{g})$,*

$$d_A^2 \xi = [\text{curv}(A), \xi]$$

The curvature transforms equivariantly:

$$\text{curv}(g \cdot A) = \text{Ad}_g \text{curv}(A).$$

PROOF.

$$\begin{aligned} d_A^2 \xi &= d \circ [A, \cdot] + [A, \cdot] \circ d + [A, [A, \cdot]] \\ &= [dA, \cdot] + \frac{1}{2}[[A, A], \cdot] \end{aligned}$$

where the last term is obtained by the Jacobi identity. Equivariance follows by the transformation property for d_A . □

By the lemma, $d_A^2 = 0$ is precisely if the curvature is zero.

We now view $\mathcal{A}(\Sigma)$ as a $\mathcal{G}(\Sigma)$ -space. What are the generating vector fields? The Lie algebra of the gauge group is identified with $\Omega^0(\Sigma, \mathfrak{g})$.

LEMMA 4.13. *The generating vector fields for $\xi \in \Omega^0(\Sigma, \mathfrak{g})$ are*

$$\xi_{\mathcal{A}(\Sigma)}(A) = d_A \xi.$$

PROOF.

$$\begin{aligned} \xi_{\mathcal{A}(\Sigma)}(A) &= \left. \frac{\partial}{\partial t} \right|_{t=0} (\text{Ad}_{\exp(-t\xi)}(A) - d(\exp(-t\xi)) \exp(t\xi)) \\ &= -[\xi, A] + d\xi = d_A \xi. \end{aligned}$$

□

We now specialize to two dimensions: $\dim \Sigma = 2$. Let us fix an invariant inner product on $\mathfrak{g}$ (unique up to scalar if G is simple). Then $\mathcal{A}(\Sigma) = \Omega^1(\Sigma, \mathfrak{g})$ is an ∞ -dimensional symplectic manifold: The 2-form is

$$\omega_A(a, b) = \int_{\Sigma} a \wedge b$$

for all $a, b \in T_A \Omega^1(\Sigma, \mathfrak{g}) \cong \Omega^1(\Sigma, \mathfrak{g})$, using the inner product.

THEOREM 4.14. *The gauge action of $\mathcal{G}(\Sigma)$ on $\mathcal{A}(\Sigma)$ is Hamiltonian, with moment map minus the curvature $A \mapsto \text{curv}(A)$, i.e.*

$$\langle \Phi(A), \xi \rangle = - \int_{\Sigma} \text{curv}(A) \cdot \xi.$$

PROOF. We calculate: For all $a \in \Omega^1(\Sigma, \mathfrak{g})$, viewed as a constant vector field,

$$\begin{aligned} \langle d\langle \Phi, \xi \rangle \Big|_A, a \rangle &= \frac{\partial}{\partial t} \Big|_{t=0} \langle \Phi(A + ta), \xi \rangle \\ &= - \frac{\partial}{\partial t} \Big|_{t=0} \int_{\Sigma} \text{curv}(A + ta) \cdot \xi \\ &= - \frac{\partial}{\partial t} \Big|_{t=0} \int_{\Sigma} (dA + tda + t[A, a] + \frac{1}{2}([A, A] + t^2[a, a])) \cdot \xi \\ &= - \int_{\Sigma} d_A a \cdot \xi \\ &= - \int_{\Sigma} a \cdot d_A \xi \\ &= \omega(\xi_{\mathcal{A}(\Sigma)}(A), a). \end{aligned}$$

□

It is interesting to extend this calculation to 2-manifolds with boundary $\partial \Sigma$. Everything carries over, but the partial integration produces an extra boundary term so that the moment map is

$$\langle \Phi(A), \xi \rangle = - \int_{\Sigma} \text{curv}(A) \cdot \xi - \int_{\partial \Sigma} A \cdot \xi.$$

That is, the (informally) dual space to the Lie algebra $\Omega^0(\Sigma, \mathfrak{g})$ of the gauge group is identified with $\Omega^2(\Sigma, \mathfrak{g}) \oplus \Omega^1(\partial \Sigma, \mathfrak{g})$ with the natural pairing, and the moment map is

$$\Omega^0(\Sigma, \mathfrak{g}) \rightarrow \Omega^2(\Sigma, \mathfrak{g}) \oplus \Omega^1(\partial \Sigma, \mathfrak{g}), \quad A \mapsto (\text{curv}(A), \iota_{\partial \Sigma}^* A).$$

Notice however that this moment map is no longer equivariant in the usual sense, for the action on the second summand is still the gauge action! This leads one to define a central extension of the gauge group. Define

$$c : \mathcal{G}(\Sigma) \times \mathcal{G}(\Sigma) \rightarrow \text{U}(1), \quad c(g_1, g_2) = \exp(-i\pi \int_{\Sigma} g_1^{-1} dg_1 \wedge dg_2 g_2^{-1}),$$

and let $\widehat{\mathcal{G}(\Sigma)} = \mathcal{G}(\Sigma) \times \mathrm{U}(1)$ with product

$$(g_1, z_1)(g_2, z_2) := (g_1 g_2, z_1 z_2 c(g_1, g_2)).$$

One can show that this does indeed define a group structure (i.e. c is a cocycle). The Lie algebra of this new group is $\Omega^0(\Sigma, \mathfrak{g}) \oplus \mathbb{R}$ with defining cocycle $\int_{\Sigma} d\xi_1 d\xi_2 = \int_{\partial\Sigma} \xi_1 d\xi_2$, and its dual is $\Omega^2(\Sigma, \mathfrak{g}) \oplus \Omega^1(\partial\Sigma, \mathfrak{g}) \oplus \mathbb{R}$, with action

$$(g, z) \cdot (\alpha, \beta, \lambda) = (\mathrm{Ad}_g \alpha, \mathrm{Ad}_g \beta - \lambda dg g^{-1}, \lambda).$$

It follows that the moment map for the action of the extended gauge group (where the extra circle acts trivially) is equivariant, the image of the original moment map is identified with the hyperplane $\lambda = 1$.

5. Symplectic Reduction

5.1. The Meyer-Marsden-Weinstein Theorem. Let (M, ω, Φ) be a Hamiltonian G -space. As usual we assume that Φ is an equivariant moment map. One of the basic properties of the moment map is the following:

LEMMA 5.1. *For all $m \in M$, the kernel and image of the tangent map to Φ are given by*

$$\begin{aligned} \ker(d_m \Phi) &= T_m(G \cdot m)^\omega, \\ \mathrm{im}(d_m \Phi) &= \mathrm{ann}(\mathfrak{g}_m). \end{aligned}$$

PROOF. By the defining condition of the moment map,

$$\omega_m(\xi_M(m), X) = \iota(X) d\langle \Phi, \xi \rangle \Big|_m = \langle d_m \Phi(X), \xi \rangle.$$

Therefore $\ker(d_m \Phi) = \{\xi_M(m) | \xi \in \mathfrak{g}\}^\omega = T_m(G \cdot m)^\omega$.

By the same equation, $\xi \in \mathfrak{g}_m$ implies that $\langle d_m \Phi(X), \xi \rangle = 0$ for all X , hence $\mathrm{im}(d_m \Phi) \subseteq \mathrm{ann}(\mathfrak{g}_m)$. Equality follows by dimension count, using non-degeneracy of ω :

$$\begin{aligned} \dim(\mathrm{im}(d_m \Phi)) &= \dim M - \dim(\ker(d_m \Phi)) \\ &= \dim M - \dim(T_m(G \cdot m))^\omega \\ &= \dim(G \cdot m) = \dim G - \dim G_m = \dim \mathrm{ann}(\mathfrak{g}_m). \end{aligned}$$

□

THEOREM 5.2. *A point $\mu \in \mathfrak{g}^*$ is a regular value of Φ if and only if for all $m \in \Phi^{-1}(\mu)$, the stabilizer group G_m is discrete. In this case, $\Phi^{-1}(\mu)$ is a constant rank submanifold. The leaf of the null foliation through $m \in \Phi^{-1}(\mu)$ is the orbit $G_\mu m$.*

PROOF. Since μ is a regular value, $d_m \Phi$ is surjective for all $m \in M$. By the Lemma, this means that $\mathrm{ann}(\mathfrak{g}_m) = \mathfrak{g}^*$ or equivalently $\mathfrak{g}_m = \{0\}$. This shows that $G_m \subseteq G_\mu$ is

discrete. Now let $\iota_\mu : \Phi^{-1}(\mu) \hookrightarrow M$ be the inclusion. Using $T_m\Phi^{-1}(\mu) = \ker(d_m\Phi)$ the kernel of $\iota_\mu^*\omega$ is

$$\begin{aligned} \ker \iota_\mu^*\omega \Big|_m &= T_m\Phi^{-1}(\mu) \cap T_m\Phi^{-1}(\mu)^\omega \\ &= T_m\Phi^{-1}(\mu) \cap T_m(G \cdot m) \\ &= T_m(G_\mu \cdot m). \end{aligned}$$

□

THEOREM 5.3 (Marsden-Weinstein, Meyer). *Let (M, ω, Φ) be a Hamiltonian G -space. Suppose μ is a regular value of Φ and that the foliation of $\Phi^{-1}(\mu)$ by G_μ -orbits is a fibration. (This assumption is satisfied if G_μ is compact and the G_μ -action is free.) Let*

$$\pi_\mu : \Phi^{-1}(\mu) \rightarrow \Phi^{-1}(\mu)/G_\mu =: M_\mu$$

be the quotient map onto the orbit space. There exists a unique symplectic form ω_μ on the reduced space M_μ such that

$$\iota_\mu^*\omega = \pi_\mu^*\omega.$$

PROOF. Since the null-foliation is given by the G_μ -orbits, this is a special case of the theorem on reduction of constant rank submanifolds. □

The reduced space at 0 is often denoted $M_0 = M//G$, this notation is useful if several groups are involved. The symplectic quotients M_μ depend only on the coadjoint orbit $\mathcal{O} = G \cdot \mu$. Let $\mathcal{O}^-$ be the same G -space but with minus the KKS form as a symplectic form and minus the inclusion as a moment map. The moment map for the diagonal action on $M \times \mathcal{O}^-$ is

$$\tilde{\Phi} : M \times \mathcal{O}^- \rightarrow \mathfrak{g}^*, (m, \mu) \mapsto \Phi(m) - \mu.$$

PROPOSITION 5.4 (Shifting-trick). *μ is a regular value of Φ if and only if 0 is a regular value of*

$$\tilde{\Phi} : M \times \mathcal{O}^- \rightarrow \mathfrak{g}^*, (m, \mu) \mapsto \Phi(m) - \mu.$$

Moreover the G_μ -action on $\Phi^{-1}(\mu)$ is free if and only if the G -action on $\tilde{\Phi}^{-1}(0)$ is free. There is a canonical symplectomorphism,

$$M_\mu \cong (M \times \mathcal{O}^-)//G.$$

PROOF. Consider the map $\Phi^{-1}(\mathcal{O}) \rightarrow \tilde{\Phi}^{-1}(0)$ by $m \mapsto (m, \Phi(m))$. Clearly, this map is a G -equivariant bijection. Since $\Phi^{-1}(\mathcal{O}) = G \cdot \Phi^{-1}(\mu)$, the G -action on $\Phi^{-1}(\mathcal{O})$ has discrete (resp. trivial) stabilizers if and only if the G_μ -action on $\Phi^{-1}(\mu)$ has discrete (resp. trivial) stabilizers. The map $M \rightarrow M \times \mathcal{O}^-$, $m \mapsto (m, \mu)$ preserves 2-forms, hence so does the map $\Phi^{-1}(\mu) \rightarrow \tilde{\Phi}^{-1}(0)$, $m \mapsto (m, \mu)$. It follows that the maps

$$M_\mu = \Phi^{-1}(\mu)/G_\mu \rightarrow (\tilde{\Phi}^{-1}(0) \cap M \times \{\mu\})/G_\mu \rightarrow \tilde{\Phi}^{-1}(0)/G$$

are all symplectomorphisms. □

EXAMPLE 5.5. Let $M = \mathbb{C}^{N+1}$, with the standard symplectic form

$$\omega = \frac{i}{2} \sum_{j=0}^N dz_j \wedge d\bar{z}_j,$$

and scalar $S^1 = \mathbb{R}/\mathbb{Z}$ -action (multiplication by $\exp(2\pi it)$). We recall that a moment map for this action is given by $\Phi(z) = \pi||z||^2$. The reduced space at level π is $\mathbb{C}P(N)$. The reduced form is known as the Fubini-Study form. Reducing at a different value $\lambda\pi$ amounts to rescaling the symplectic form by λ .

EXAMPLE 5.6. Let (M, ω) be an exact symplectic manifold, $\omega = -d\theta$, and suppose θ is invariant under some G -action, where G is a compact Lie group. Let Φ be the corresponding moment map $\langle \Phi, \xi \rangle = \iota(\xi_M)\theta$. Then if 0 is a regular value of Φ and the G -action on $\Phi^{-1}(0)$ is free, the pull-back $\iota^*\theta$ is invariant and horizontal, hence descends to a 1-form θ_0 on M_0 such that $\pi^*\theta_0 = \iota^*\theta$, and one has

$$\omega_0 = -d\theta_0.$$

It follows that the symplectic quotient of an exact Hamiltonian G -space at 0 is an exact symplectic manifold.

EXAMPLE 5.7. As a sub-example, consider the case $M = T^*Q$, where G acts by the cotangent lift of some G -action on Q . Let $\rho : T^*Q \rightarrow Q$ denote the projection. The moment map is given by

$$\langle \Phi(m), \xi \rangle = \langle m, \xi_Q(q) \rangle$$

where $q = \rho(m)$. This shows that the zero level set is the union of covectors orthogonal to orbits:

$$\Phi^{-1}(0) = \coprod_{q \in Q} \text{ann}(T_q(G \cdot q)).$$

Since $\Phi^{-1}(0)$ contains the zero section Q , it is clear that the G -action on $\Phi^{-1}(0)$ is locally free if and only if the action on Q is locally free. If the action is free, we have $(T^*Q)//G = T^*(Q/G)$. To see this (at least set-theoretically), note that

$$T(Q/G) = \left(\coprod_q T_q Q / T_q(G \cdot q) \right) / G$$

so that

$$T^*(Q/G) = \left(\coprod_q \text{ann}(T_q(G \cdot q)) \right) / G.$$

To identify the symplectic forms one has to identify the reduced canonical 1-form θ_0 with the canonical 1-form on $T^*(Q/G)$, we leave this as an exercise.

For an arbitrary G -space Q the singular reduced space $(T^*Q)//G$ may be viewed as a cotangent bundle for the singular space Q/G .

EXAMPLE 5.8. Returning to our 2-d gauge theory example, the reduction $\mathcal{A}(\Sigma)//\mathcal{G}(\Sigma)$ is the moduli space of flat connections on Σ .

5.2. Reduced Hamiltonians. Let G be a compact Lie group. Suppose (M, ω, Φ) is a Hamiltonian G -space, that $\mu \in \mathfrak{g}^*$ is a regular value of the moment map, and that the action of G_μ on the level set $\Phi^{-1}(\mu)$ is free. Then every invariant Hamiltonian $H \in C^\infty(M)^G$ descends to a unique function $H_\mu \in C^\infty(M_\mu)$ with $\pi_\mu^* H_\mu = \iota_\mu^* H$. Passing to the reduced Hamiltonian H_μ is often a first step in solving the equations of motion for H . From H -invariance of X_H it follows that the restriction $(X_H)|_{\Phi^{-1}(\mu)} \in \mathfrak{X}(\Phi^{-1}(\mu))$ is π_μ -related to $X_{H_\mu} \in \mathfrak{X}(M_\mu)$, that is its flow projects down to the flow on M_μ . After one has solved the reduced system (i.e. determined its flow $F_\mu(t)$) it is a second step to lift $F_\mu(t)$ up to the level set $\Phi^{-1}(\mu)$.

EXAMPLE 5.9. Consider the motion of a particle on $\mathbb{R}^2$ in a potential $V(q)$. It is described by the Hamiltonian on $T^*\mathbb{R}^2$,

$$H(q, p) = \frac{\|p\|^2}{2} + V(q).$$

Suppose the potential has rotational symmetry, i.e. that it depends only on $r = \|q\|$. Then H is invariant under the cotangent lift of the rotation action of $G = S^1$. We had seen that the moment map for this action is angular momentum, $\Phi(q, p) = p_2 q_1 - q_2 p_1$. In polar coordinates, (r, θ) on $\mathbb{R}^2$ and corresponding cotangent coordinates on $T^*\mathbb{R}^2$,

$$H(r, \theta, p_r, p_\theta) = \frac{1}{2}(p_r^2 + \frac{1}{r^2}p_\theta^2) + V(r)$$

and $\Phi = p_\theta$. The symplectic form on $T^*\mathbb{R}^2$ is $\omega = dr \wedge p_r + d\theta \wedge p_\theta$. Every value $\mu \neq 0$ is a regular value of Φ (since S^1 acts freely on the set where $p_\theta = r^2 \dot{\theta} \neq 0$). On $\iota_\mu : \Phi^{-1}(\mu) \hookrightarrow T^*\mathbb{R}^2$ the second term disappears, i.e. $\iota_\mu^* \omega = dr \wedge p_r$. It follows that $M_\mu \cong T^*\mathbb{R}_{>0}$ symplectically, and the reduced Hamiltonian is

$$H_\mu(r, p_r) = \frac{1}{2}p_r^2 + V_{eff}(r)$$

with the effective potential,

$$V_{eff}(r) = V(r) + \frac{\mu^2}{2r^2}.$$

Using conservation of energy

$$\frac{p_r^2}{2} + V_{eff}(r) = \frac{\dot{r}^2}{2} + V_{eff}(r) = E,$$

i.e. $\dot{r}^2 = 2(E - V_{eff}(r))$, one obtains the solution in implicit form,

$$t - t_0 = \int_{r_0}^r \frac{dr}{\sqrt{2(E - V_{eff}(r))}}.$$

Using $r^2 \dot{\theta} = p_\theta = \mu$, one also obtains a differential equation for the trajectories,

$$\frac{\partial r}{\partial \theta} = \frac{r^2}{\mu} \sqrt{2(E - V_{eff}(r))},$$

with solutions,

$$\theta - \theta_0 = \int_{r_0}^r \frac{\mu dr}{r^2 \sqrt{2(E - V_{eff}(r))}}.$$

In the special case $V(r) = -\frac{1}{r}$ (Kepler problem) this integral can be solved and leads to conic sections – see any textbook on classical mechanics.

5.3. Reduction in stages. As a special case of “reduced Hamiltonian” one sometimes has a reduced moment map. For the simplest situation, suppose G, H are compact Lie groups and (M, ω) is a Hamiltonian $G \times H$ -space, with moment map (Φ, Ψ) . Since the two actions commute, $\langle \Phi^\xi, \Psi^\eta \rangle = 0$ for all $\xi \in \mathfrak{g}$, $\eta \in \mathfrak{h}$. In particular, Φ is H -invariant and Ψ is G -invariant. Let μ be a regular value of Φ , so that the reduced space M_μ is defined. Since Ψ is G -invariant, it descends to a map $\Psi_\mu : M_\mu \rightarrow \mathfrak{h}^*$. It is the moment map for the H -action on M_μ induced from the H -action on $\Phi^{-1}(\mu) \subset M$.

LEMMA 5.10 (Reduction in Stages). *Suppose μ is a regular value of Φ and (μ, ν) a regular value for (Φ, Ψ) . Then ν is a regular value for Ψ_μ . If G_μ acts freely on $\Phi^{-1}(\mu)$ and $G_\mu \times H_\nu$ acts freely on $\Phi^{-1}(\mu) \cap \Psi^{-1}(\nu)$, then H_ν acts freely on $\Psi_\mu^{-1}(\nu)$, and there is a natural symplectomorphism*

$$(M_\mu)_\nu \cong M_{(\mu, \nu)}.$$

PROOF. Clearly, if G_μ acts with finite (resp. trivial) stabilizers on $\Phi^{-1}(\mu)$ and $G_\mu \times H_\nu$ acts with finite (resp. trivial) stabilizers on $\Phi^{-1}(\mu) \cap \Psi^{-1}(\nu)$, the same is true for the H_ν -action on $\Psi_\mu^{-1}(\nu)$. This proves the first part since a level set having finite stabilizers is equivalent to the level being a regular value. The second part follows because the natural identifications

$$(M_\mu)_\nu = \Psi_\mu^{-1}(\nu)/H_\nu = (\Phi^{-1}(\mu) \cap \Psi^{-1}(\nu))/(G_\mu \times H_\nu) = M_{(\mu, \nu)}$$

all preserve 2-forms. □

5.4. The cotangent bundle of a Lie group. Let G be a compact Lie group. For all $\xi \in \mathfrak{g}$ the left- and right invariant vector fields are related by the adjoint action, as follows:

$$\xi^L(g) = (\text{Ad}_g \xi)^R(g).$$

The map

$$G \times \mathfrak{g} \rightarrow TG, \quad (g, \xi) \mapsto \xi^L(g)$$

is a vector bundle isomorphism called left trivialization of TG . In left trivialization, ξ^L becomes the constant vector field ξ and ξ^R becomes the vector field $\text{Ad}_{g^{-1}}(\xi)$. Dual to the left-trivialization of TG there is the trivialization $T^*G \cong G \times \mathfrak{g}^*$ by left-invariant 1-forms.

EXERCISE 5.11. Show that in left trivialization of TG , the tangent maps to inversion $\text{Inv} : G \rightarrow G$, $g \mapsto g^{-1}$, left action $L_h : G \rightarrow G$, $g \mapsto hg$ and right action $R_{h^{-1}} : G \rightarrow G$, $g \mapsto gh^{-1}$ are given by

$$\begin{aligned}\text{Inv}_*(g, \xi) &= (g^{-1}, \text{Ad}_g \xi), \\ (L_h)_*(g, \xi) &= (hg, \xi), \\ (R_{h^{-1}})_*(g, \xi) &= (gh^{-1}, \text{Ad}_h \xi).\end{aligned}$$

Show that the respective cotangent lifts are

$$\begin{aligned}\text{Inv}_*(g, \mu) &= (g^{-1}, (\text{Ad}_{g^{-1}})^* \mu), \\ (L_h)_*(g, \mu) &= (hg, \mu), \\ (R_{h^{-1}})_*(g, \mu) &= (gh^{-1}, (\text{Ad}_{h^{-1}})^* \mu).\end{aligned}$$

Since the generating vector fields for the left- and right action are $-\xi^R, \xi^L$ respectively, we find that the moment maps for these actions are

$$\Phi_L(g, \mu) = -\text{Ad}_g^*(\mu), \quad \Phi_R(g, \mu) = \mu.$$

Note that the cotangent lifts of both the left action and the right action are free. In particular, every $\nu \in \mathfrak{g}^*$ is a regular value for both moment maps.

THEOREM 5.12. *The symplectic reduction $(T^*G)_\nu$ by the right action, with G -action inherited from the left-action, is the coadjoint orbit $G \cdot (-\nu)$.*

PROOF. It suffices to note that the left action of G on the level set $\Phi_R^{-1}(\mu)$ is free and transitive, and the action on the quotient has stabilizer conjugate to G_μ . The moment map induced by Φ_L identifies gives a symplectomorphism onto $G \cdot (-\nu)$. $\square$

Of course, the reduced spaces with respect to the left action are coadjoint orbits $G \cdot (-\nu)$ as well: the cotangent lift of the inversion map $G \rightarrow G$, $g \mapsto g^{-1}$ exchanges the roles of the left- and right action.

THEOREM 5.13. *Let (M, ω, Φ) be a Hamiltonian G -space. Let G act diagonally on $T^*G \times M$, where the action on T^*G is the right action. Consider the reduced space at 0 as a Hamiltonian G -space, with G -action induced from the left- G -action on T^*G . Then there is a canonical isomorphism of Hamiltonian G -spaces,*

$$(T^*G \times M) // G \cong M$$

PROOF. Use left trivialization $T^*G \cong G \times \mathfrak{g}^*$. The map

$$T^*G \times M \rightarrow T^*G \times M, \quad (g, \mu, m) \mapsto (g, \mu, g.m)$$

is symplectic, and takes the diagonal action to the right-action on the first factor, and the left-action becomes a diagonal action. Hence it induces a symplectomorphism

$$(T^*G \times M) // G \cong T^*G // G \times M = M.$$

$\square$

6. Normal forms and the Duistermaat-Heckman theorem

Let (M, ω, Φ) be a Hamiltonian G -space, where G is a compact Lie group. If 0 is a regular value of the moment map then so are nearby values $\mu \in \mathfrak{g}^*$. What is the relation between M_0 and reduced spaces M_μ at nearby values?

To investigate this question we describe the reduction process in terms of a normal form.

Let $Z = \Phi^{-1}(0)$ be the zero level set, $\iota : Z \hookrightarrow M$ the inclusion and $\pi : Z \rightarrow M_0$ the projection. Since the G -action on $\Phi^{-1}(0)$ has finite stabilizers, there exists a connection 1-form

$$\alpha \in \Omega^1(Z, \mathfrak{g}),$$

that is, α satisfies the two conditions

$$g^* \alpha = \text{Ad}_g^* \alpha, \quad \iota(\xi_Z) \alpha = \xi.$$

Such a form can be constructed as follows. Consider the embedding $Z \times \mathfrak{g} \rightarrow TZ$, $(m, \xi) \mapsto \xi_Z(m)$ as a vector subbundle. Choose a G -invariant Riemannian metric on Z , and let $p : TZ \rightarrow Z \times \mathfrak{g}$ be the orthogonal projection with respect to that metric. The 1-form α is defined by $p(v) = (m, \alpha(v))$ for $v \in T_m M$.

Let pr_1, pr_2 be the projections from $Z \times \mathfrak{g}^*$ to the first and second factor. Define a 2-form on the product $X = Z \times \mathfrak{g}^*$ by

$$\sigma = \text{pr}_1^* \pi^* \omega_0 + d\langle \text{pr}_2, \alpha \rangle,$$

let G -action act diagonally (using the coadjoint action on $\mathfrak{g}^*$).

THEOREM 6.1 (Local normal form near the zero level set). *The 2-form σ is non-degenerate (i.e. symplectic) on some neighborhood of Z , and satisfies*

$$\iota(\xi_X) \sigma = d\langle \text{pr}_2, \xi \rangle$$

for all $\xi \in \mathfrak{g}$ (that is, pr_2 is a moment map). There exists an equivariant symplectomorphism between neighborhoods of Z in M and in X , intertwining the two moment maps.

PROOF. Notice that the 2-form

$$\text{pr}_1^* \pi^* \omega_0 + \langle d \text{pr}_2, \alpha \rangle$$

is non-degenerate. Since σ differs from this 2-form by a term $\langle \text{pr}_2, d\alpha \rangle$ which vanishes along Z , it follows that σ is non-degenerate near Z . For the second part, notice that the tangent bundle to the null foliation $\ker(\iota^* \omega) \subset TZ$ is a trivial bundle $Z \times \mathfrak{g} \rightarrow Z$, where G acts on $\mathfrak{g}$ by the adjoint action. By the equivariant version of the co-isotropic embedding theorem, it follows that neighborhoods of Z in M and of Z in X are equivariantly symplectomorphic. Since both moment maps vanish on Z , it is automatic that the symplectomorphism intertwines the moment maps. $\square$

We can view $Z \times \mathfrak{g}^*$ also as a quotient $(Z \times T^*G)/G$, using left trivialization to identify $T^*G \cong G \times \mathfrak{g}^*$; the G -action on $Z \times \mathfrak{g}^*$ is induced from the cotangent lift of the right G -action.

It follows that the reduced space at μ close to 0 is symplectomorphic to $(Z \times G \cdot (-\mu))/G$, that is:

COROLLARY 6.2. *The reduced space M_μ for μ close to 0 fiber over M_0 , with fibers coadjoint orbits:*

$$M_\mu \cong (Z \times G \cdot (-\mu))/G.$$

Letting $\Psi : G \cdot (-\mu) \hookrightarrow \mathfrak{g}^*$ be the embedding, the pull-back of the 2-form ω_μ to $Z \times G \cdot (-\mu)$ is

$$\pi^* \omega_0 + d\langle \Psi, \alpha \rangle.$$

Consider in particular the case that G is a torus $T \cong (\mathbb{R}/\mathbb{Z})^k$. Then the coadjoint action is trivial, and the reduced spaces at 0 and at nearby values are diffeomorphic. Notice that $d\alpha$ descends to a 2-form M_0 which is just the curvature form $F^\alpha \in \Omega^2(M_0, \mathfrak{t})$ of the torus bundle $\Phi^{-1}(0) \rightarrow M_0$. As a consequence we find that the symplectic form changes according to

$$\omega_\mu = \omega_0 + \langle \mu, F^\alpha \rangle$$

In particular this change is *linear in μ !!* This result depends on our identification of $M_\mu \cong M_0$. This identification is not natural, but any two identifications are related by an isotopy of M_0 . Since cohomology classes are stable under isotopies, it makes sense to compare cohomology classes, and the above discussion proves:

THEOREM 6.3 (Duistermaat-Heckman). *The cohomology class of the symplectic form changes according to*

$$[\omega_\mu] = [\omega_0] + \langle \mu, c \rangle$$

where $c \in H^*(M_0) \otimes \mathfrak{t}$ is the first Chern class of the torus bundle $\Phi^{-1}(0) \rightarrow M_0$.

In particular this change is linear in μ !!! As an immediate consequence one has:

COROLLARY 6.4. *Let (M, ω, Φ) be a Hamiltonian T -space, and let U be a connected component of the set of regular values of Φ . Then the volume function $U \rightarrow \mathbb{R}, \mu \mapsto \text{Vol}(M_\mu)$ is given by a polynomial of degree at most $\frac{1}{2} \dim M - \dim T$.*

(Here the maximum degree is obtained as half the dimension of a reduced space.)

7. The symplectic slice theorem

7.1. The slice theorem for G -manifolds. Let G be a compact Lie group, and H a closed Lie subgroup. Then every G -equivariant vector bundle over the homogeneous space G/H is of the form

$$E = G \times_H W \equiv (G \times W)/H$$

where W is an H -representation, the quotient is taken by the H -action $h.(g, w) = (gh^{-1}, h.w)$ and the G -action is given by $g_1.[g, w] = [g_1g, w]$. Indeed, given E one

defines $W = E_m$ to be the fiber over the identity coset $m = eH$, and the map $G \times_H W \mapsto E$, $[g, w] \mapsto g.w$ is easily seen to be a well-defined, equivariant vector bundle isomorphism.

For example, suppose M is a G -manifold, $m \in M$, and $H = G_m$. Let $W = T_m M / T_m(G.m)$ be the so-called *slice representation* of H . Then $E = G \times_H W$ is the normal bundle $\nu_{\mathcal{O}} = TM|_{\mathcal{O}} / T\mathcal{O}$ to the orbit $\mathcal{O} = G.m = G/H$. Recall now that by the tubular neighborhood theorem, if N is any submanifold of M there is a diffeomorphism of neighborhoods of N in M and in ν_N . (The diffeomorphism is constructed using geodesic flow with respect to a Riemannian metric on M .) If N is G -invariant, this diffeomorphism can be chosen G -equivariant. (Just take the Riemannian metric to be G -invariant.) We conclude:

THEOREM 7.1 (Slice theorem). *Let M be a G -manifold, and $m \in M$ a point with stabilizer $H = G_m$ and slice representation $W = T_m M / T_m(G.m)$. There exists a G -equivariant diffeomorphism from an invariant open neighborhood of the orbit $\mathcal{O} = G.m$ to a neighborhood of the zero section of $E = G \times_H W$.*

COROLLARY 7.2. *There exists a neighborhood U of $\mathcal{O} = G.m$ with the property that all stabilizer groups G_x , $x \in U$ are conjugate to subgroups of $H = G_m$. In particular, if M is compact there are only finitely many conjugacy classes of stabilizer groups.*

PROOF. Identify some neighborhood of the orbit with the model $E = G \times_H W$. Let $x = g.y$ with $y \in W$. Then $G_x = \text{Ad}_g(G_y)$. But G_y is a subgroup of H , since it preserves the fiber $W = E_m$. $\square$

COROLLARY 7.3. *If G is compact abelian, and M is compact, the number of stabilizer subgroups $\{H \subset G \mid H = G_m \text{ for some } m \in M\}$ is finite.*

PROOF. For an abelian group, conjugation is trivial. $\square$

DEFINITION 7.4. For any subgroup H of G one denotes its conjugacy class by (H) , and calls the G -invariant subset

$$M_{(H)} = \{m \in M \mid G_m \text{ is } G\text{-conjugate to } H\},$$

the points of orbit type (H) . One also defines

$$M^H = \{m \in M \mid G_m \supset H\}, \quad M_H = \{m \in M \mid G_m = H\}.$$

PROPOSITION 7.5. *The connected components of $M_{(H)}$, M_H and M^H are smooth submanifolds of M .*

PROOF. Any orbit $\mathcal{O} \subset M_{(H)}$ contains a point $m \in M$ with $G_m = H$. In the model $E = G \times_H W$ near $\mathcal{O}$, we have

$$E_{(H)} = G \times_H W^H = G/H \times W^H$$

since W^H is a vector subspace of W , this is clearly a smooth subbundle of E . The connected components of M^H are smooth submanifolds, since for all $m \in M^H$, a neighborhood is H -equivariantly modeled by the H -action on $T_m M$ and $(T_m M)^H$ is a vector

subspace. The closure $\overline{M}^H$ is a union of connected components of M^H . Since M_H is open in its closure, it is in particular a submanifold. $\square$

The decomposition $M = \bigcup_{(H)} M_{(H)}$ is called the orbit type stratification of M . Using the local model near orbits, one can show that it is indeed a stratification in the technical sense. Note that since each $M_{(H)}/G$ is a (union of) smooth manifolds, it induces a decomposition (in fact, stratification) of the orbit space M/G .

7.2. The slice theorem for Hamiltonian G -manifolds. In symplectic geometry one can go one step further and try to equip the total space to the normal bundle with a symplectic structure. Thus let (M, ω, Φ) be a Hamiltonian G -space. Assume that $m \in \Phi^{-1}(0)$ is in the zero level set. This implies that the orbit is an isotropic submanifold: Since Φ vanishes on $\mathcal{O}$ we have $T_m \mathcal{O} \subset \ker(d_m \Phi)$, on the other hand $\ker(d_m \Phi) = T_m \mathcal{O}^\omega$. The symplectic vector space $V = T_m \mathcal{O}^\omega / T_m \mathcal{O}$ with the action of $H = G_m$ is called the symplectic slice representation at m .

DEFINITION 7.6. Let H act on a symplectic vector space (V, ω_V) by linear symplectic transformations, and let $\Phi_V : V \rightarrow \mathfrak{h}^*$ be the unique moment map vanishing at 0 (cf. 4.5),

$$\Phi_V : V \rightarrow \mathfrak{h}^*, \quad \langle \Phi(v), \xi \rangle = \frac{1}{2} \omega(v, \xi.v)$$

One calls the symplectic quotient

$$E = (T^*G \times V) // H$$

the *model* defined by V . Here H acts on $T^*G \times V$ by the diagonal action, where the action on T^*G is given by the cotangent lift of the right-action of H on G . We let $\Phi_E : E \rightarrow \mathfrak{g}^*$ be the moment map for the G -action on E inherited from the cotangent lift of the left- G -action on T^*G .

The orbit $\mathcal{O} = G/H$ is naturally embedded as an isotropic submanifold of E , namely as the zero section of $T^*G // H = T^*(G/H)$. Its symplectic normal bundle in E is an associated bundle, $G \times_H V$.

REMARK 7.7. The model can also be written as an associated bundle: Identify $T^*G = G \times \mathfrak{g}^*$ using left trivialization. The zero level set for the H -action consists of points (g, μ, v) such that $-\text{pr}_{\mathfrak{h}^*} \mu + \Phi_V(v) = 0$. Thus, if we choose an H -equivariant complement to $\text{ann}(\mathfrak{h})$ in $\mathfrak{g}^*$, identifying $\mathfrak{g}^* = \mathfrak{h}^* \oplus \text{ann}(\mathfrak{h})$, we see that the zero level set consists of points $(g, \Phi_V(v) + \nu, v)$ with $\nu \in \text{ann}(\mathfrak{h})$, and is therefore isomorphic to $G \times \text{ann}(\mathfrak{h}) \times V$. Thus

$$E \cong G \times_H (\text{ann}(\mathfrak{h}) \times V).$$

In this description the moment map Φ_E is given by

$$\Phi_E([g, \nu, v]) = g.(\nu + \Phi_V(v)).$$

Note however that this identification depends on the choice of splitting.

THEOREM 7.8. *Let (M, ω, Φ) be a Hamiltonian G -manifold, and $\mathcal{O} = G.m$ an orbit in the zero level set of Φ . There exists a G -equivariant symplectomorphism between neighborhoods of $\mathcal{O}$ in M and in the model E defined by the symplectic slice representation $V = T_m\mathcal{O}^\omega/T_m\mathcal{O}$ of $H = G_m$, intertwining the two moment maps.*

PROOF. This follows from (equivariant version of) the constant rank embedding theorem: The symplectic normal bundles of $\mathcal{O}$ in both spaces are $G \times_H V$. $\square$

The symplectic slice theorem is extremely useful: For example we obtain a model for the singularities of $M//G$ in case 0 is a singular value. Indeed, by reduction in stages we have

$$(T^*G \times V//H)//G = (T^*G \times V//G)//H = V//H$$

which shows that the singularities are modeled by symplectic reductions of unitary representations. Since the moment map for a unitary representation is homogeneous, the zero level set $\Phi_V^{-1}(0)$ is a cone and hence the singularities are *conic singularities*. This discussion can be carried much further, see the paper Sjamaar-Lerman, “Stratified symplectic spaces and reduction”, Ann. of Math. (2) 134 (1991), no. 2, 375–422.

PROPOSITION 7.9. *Let (M, ω, Φ) be a Hamiltonian G -space, $H \subset G$ a closed subgroup. The connected components of M^H and M_H are symplectic submanifolds of M . For every connected open subset $U \subset M_H$, the image $\Phi(U)$ is an open subset of an affine subspace $\mu + \text{ann}(\mathfrak{h})^H \subset \mathfrak{g}^*$ for some $\mu \in (\mathfrak{g}^*)^H$.*

PROOF. For all $m \in M^H$, the tangent space $T_m(M^H)$ is equal to $(T_m M)^H$. But for any symplectic representation V of a compact Lie group H , the subspace V^H is symplectic. (Proof: Choose an H -invariant compatible complex structure. Then V^H is a complex, hence symplectic, subspace.) This shows that M^H and the open subset $M_H \subset M^H$ are symplectic.

The second part follows from the local model, or alternatively follows: Let $Z = Z_G(H)$ be the centralizer and $K = N_G(H)$ the normalizer of H in G , respectively. Thus $Z \subset K \subset G$ and $\mathfrak{z} = \mathfrak{k}^H = \mathfrak{g}^H$. Dually, identify $\mathfrak{z}^* = (\mathfrak{k}^*)^H = (\mathfrak{g}^*)^H$. By equivariance of the moment map, $\Phi(M_H) \subset \Phi(M^H) \subset (\mathfrak{g}^*)^H = \mathfrak{z}^*$. The action of $K \subset G$ preserves M_H . Its moment map $\Psi : M_H \rightarrow \mathfrak{k}^*$ is the restriction of Φ followed by projection $\mathfrak{g}^* \rightarrow \mathfrak{k}^*$, but since it takes values in $(\mathfrak{k}^*)^H = \mathfrak{z}^*$ it is actually just the restriction of Φ . Since $\text{im } d_m \Psi = \text{ann}_{\mathfrak{k}^*}(\mathfrak{h}) = \text{ann}(\mathfrak{h})^H$ is independent of $m \in U$, we conclude that $\Phi(U)$ is an open neighborhood of $\Phi(m)$ in $\Phi(m) + \text{ann}(\mathfrak{h})^H$. $\square$

CHAPTER 7

Hamiltonian torus actions

1. The Atiyah-Guillemin-Sternberg convexity theorem

1.1. Motivation. As a motivating example, which on first sight seems quite unrelated to symplectic geometry, consider the following problem about self-adjoint matrices. Let $\lambda = (\lambda_1, \dots, \lambda_n) \in \mathbb{R}^n$ be an n -tuple of real numbers, and let $\mathcal{O}(\lambda)$ be the set of all self-adjoint complex $n \times n$ -matrices having eigenvalues $\lambda_1, \dots, \lambda_n$. Let $\pi : \mathcal{O}(\lambda) \rightarrow \mathbb{R}^n$ be the projection to the diagonal.

THEOREM 1.1. *The image $\pi(\mathcal{O}(\lambda))$ is the convex hull*

$$\Delta := \text{hull}\{(\lambda_{\sigma(1)}, \dots, \lambda_{\sigma(n)}), \sigma \in \mathfrak{S}_n\}$$

where $\mathfrak{S}_n$ is the permutation group.

This and related results were proved by Schur and Horn, later greatly generalized by Kostant and Heckman.

The relation to symplectic geometry is as follows. First, instead of self-adjoint matrices we can equivalently consider skew-adjoint matrices, i.e. the Lie algebra $\mathfrak{g}$ of $G = \text{U}(n)$. Since all matrices with given eigenvalues are conjugate, $\mathcal{O}(\lambda)$ is an orbit for the action of $\text{U}(n)$. Using the inner product

$$(A, B) = -\text{tr}(AB)$$

on $\mathfrak{g}$ we can also view it as a coadjoint orbit.

The projection π is just orthogonal projection onto the diagonal matrices, which are a maximal commutative subalgebra $\mathfrak{t} \subseteq \mathfrak{g}$. Using the inner product to identify $\mathfrak{t} \cong \mathfrak{t}^*$ it becomes the moment map for the induced $T \subset G$ action.

For this reason the Schur-Horn theorem can be viewed as a convexity theorem for Hamiltonian torus actions on coadjoint orbits of $\text{U}(n)$. Nothing is special about $\text{U}(n)$, analogous results hold for arbitrary compact groups.

1.2. Local convexity. Our discussion of the convexity theorem will follow the exposition in Guillemin-Sternberg, “Symplectic Techniques in Physics”. In order to understand how images of moment maps for Hamiltonian T -spaces look like, we first have to understand how they look like “locally”. We will work with the local model E for the T -action near any orbit $\mathcal{O} = T.m$. (In that section we assumed $T.m \subset \Phi^{-1}(0)$, but this can be arranged by adding a constant to the moment map.) Letting $H = T_m$ be the

stabilizer, we have the model

$$E = (T^*(T) \times V) // H$$

where V is a unitary H -representation. Using the identification $T^*(T) = T \times \mathfrak{t}^*$, the moment map for the H -action on $T^*(T) \times V$ is $(t, \nu, v) \mapsto -\text{pr}_{\mathfrak{h}^*}(\nu) + \Phi_V(v)$ and the moment map for the T -action on E is induced from the map $(t, \nu, v) \mapsto \mu + \nu$. The zero level set condition for the H -action reads $\text{pr}_{\mathfrak{h}^*}(\nu) = \Phi_V(v)$. This shows:

LEMMA 1.2. *The image of the moment map Φ_E is of the form*

$$\Phi_E(E) = \mu + (\text{pr}_{\mathfrak{h}^*})^{-1}(\Phi_V(V)).$$

To understand the shape of this set we need to describe the moment map images of unitary torus representations (here for the identity component of H acting on V).

Let T be a torus, $\Lambda \subset \mathfrak{t}$ the integral lattice. Every 1-dimensional unitary representation

$$T \rightarrow \text{U}(1) \cong S^1 = \mathbb{R}/\mathbb{Z}$$

defines a map $\mathfrak{t} \rightarrow \mathbb{R}$, which restricts to a homomorphism of lattices,

$$\alpha : \Lambda \rightarrow \mathbb{Z}$$

called the *weight* of the representation. One calls

$$\Lambda^* = \text{Hom}(\Lambda, \mathbb{Z}) \subset \mathfrak{t}^*$$

Given $\alpha \in \Lambda^*$ one defines a 1-dimensional representation $\mathbb{C}_\alpha$ by

$$\exp(\xi) \cdot z = e^{2\pi i \langle \alpha, \xi \rangle} z.$$

By Schur's Lemma, any unitary T -representation V splits into a sum of 1-dimensional representations, i.e. is isomorphic to a representation of the form

$$V = \bigoplus_{j=1}^n \mathbb{C}_{\alpha_j}$$

where α_j are called the weights of V . Given a *symplectic* vector space V with a symplectic T -representation, one chooses an invariant compatible complex structure I , which makes V into a unitary T -representation. The weights α_j for this representation are independent of the choice of I , since any two I 's are deformation equivalent. They are called the weights of the symplectic T -representation.

LEMMA 1.3. *Let (V, ω_V, Φ_V) be a symplectic T -representation with moment map $\langle \Phi_V(v), \xi \rangle = \frac{1}{2} \omega(v, \xi v)$. The image of the moment map Φ_V is a convex, rational polyhedral cone $\Phi_V(V) = C_V$ spanned by the weight $\alpha_j \in \Lambda^*$ of the representation:*

$$C_V = \text{cone}\{\alpha_1, \dots, \alpha_n\}$$

For any open subset $U \subset V$, the image $\Phi_V(U)$ is open in C_V ; in particular any open neighborhood of the origin gets mapped onto an open neighborhood of the tip of the cone.

PROOF. It is convenient to write the moment map in a slightly different form. By the above discussion, we can choose an equivariant symplectomorphism $V \cong \mathbb{C}^n$ such that T acts on $\mathbb{C}^n$ by the homomorphism $T \rightarrow (S^1)^n$ determined by the weights. The moment map for the standard $(S^1)^n$ -action on $\mathbb{C}^n$ is

$$\Phi_0(z_1, \dots, z_n) = \pi(|z_1|^2, \dots, |z_n|^2) = \pi \sum_j |z_j|^2 e_j$$

where e_j is the j th standard basis vector for $\mathbb{R}^n \cong (\mathbb{R}^n)^*$. Hence Φ_V is a composition of Φ_0 with the map $(\mathbb{R}^n)^* \rightarrow \mathfrak{t}^*$ dual to the tangent map. The latter map takes e_j to α_j . Thus

$$\Phi_V(z_1, \dots, z_n) = \pi \sum_j |z_j|^2 \alpha_j.$$

The description of the image of Φ_V is now immediate. If $U \subset V$ is open, then $\Phi_0(U)$ is open in the positive orthant $\mathbb{R}_+^n$, i.e. is the intersection of $\mathbb{R}_+^n$ with an open set $U' \subset (\mathbb{R}^n)$. (This amounts to saying that the quotient topology on $\mathbb{R}_+^n = \mathbb{C}^n / (S^1)^n$ coincides with the subset topology.) Since Φ is obtained from Φ_0 by composition with a linear map, it follows that Φ is open onto its image. $\square$

Following Sjamaar, we find it useful to make the following definition:

DEFINITION 1.4. Let (M, ω, Φ) be a Hamiltonian T -space, $m \in M$. Let $H = T_m$ be the stabilizer of m and H_0 its identity component. Let $C \subset \mathfrak{h}^*$ be the cone spanned by the weights for the H_0 -action on $T_m M$. The affine cone

$$C_m = \Phi(m) + (\text{pr}_{\mathfrak{h}^*})^{-1}(C)$$

is called the *local moment cone* at $m \in M$.

Thus C_m is exactly the moment map image for the local model at m . We have shown:

THEOREM 1.5 (Local convexity theorem). *Let (M, ω, Φ) be a Hamiltonian T -space, $m \in M$, $\mathcal{O} = T \cdot m$. Then for any sufficiently small T -invariant neighborhood U of $\mathcal{O}$ there is a neighborhood V of $\Phi(m) \in C_m$ such that*

$$\Phi(U) = C_m \cap V.$$

1.3. Global convexity. We now come to the key observation of Guillemin-Sternberg. Given $\xi \in \mathfrak{t}$ consider the corresponding component $\Phi^\xi = \langle \Phi, \xi \rangle$ of the moment map. A value $s \in \mathbb{R}$ is called a local minimum for Φ^ξ if there exists $m \in M$ with $\Phi^\xi(m) = s$ and $\Phi^\xi \geq s$ on some neighborhood.

LEMMA 1.6 (Guillemin-Sternberg). *Let (M, ω, Φ) be a compact connected Hamiltonian T -space. Then all fibers of Φ are connected. Moreover, the function Φ^ξ has a unique local minimum/maximum.*

We will prove this Lemma in the next section. For any subset $S \subset \mathfrak{t}^*$ and $\mu \in \mathfrak{t}^*$ let

$$\text{cone}_\mu(S) = \{\mu + t(\nu - \mu) \mid \nu \in S\}$$

be the cone over S at μ .

THEOREM 1.7 (Guillemin-Sternberg, Atiyah). *Let (M, ω, Φ) be a compact, connected Hamiltonian T -space on which T acts effectively. The image $\Delta = \Phi(M)$ is a convex rational polytope of dimension $\dim T$. For all $m \in M$, one has*

$$(11) \quad C_m = \text{cone}_{\Phi(m)}(\Delta).$$

PROOF. Since local convexity of a compact set implies global convexity it suffices to prove Equation (11). Since a neighborhood of the tip of the cone C_m gets mapped into Δ , it follows that $\text{cone}_m(\Delta) \supset C_m$. To see the opposite inclusion, we define, for all $\xi \in \mathfrak{t}$, the affine linear functional $f_\xi = \langle \cdot, \xi \rangle - \langle \mu, \xi \rangle$ on $\mathfrak{t}^*$. We have to show that for all ξ ,

$$f_\xi|_{C_m} \geq 0 \Rightarrow f_\xi|_\Delta \geq 0.$$

But $f_\xi \geq 0$ on C_m means, by the model, that $\langle \mu, \xi \rangle$ is a local minimum for Φ^ξ . By the Lemma, this has to be a global minimum, or equivalently $f_\xi \geq 0$ on Δ . This proves (11). $\square$

We note that (11) was first observed by Sjamaar, who generalized it also to the non-abelian case. Since C_m is the moment map image for the local model, it shows that Φ is *open* as a map onto Δ .

We obtain the following description of the faces and the “fine structure” of Δ . Let $H \subset T$ be in the (finite) list of stabilizer groups, and M_H the points with stabilizer H . Recall again that M_H is an open subset of the symplectic submanifold M^H . Each connected component of M^H is a Hamiltonian T -space in its own right, with H acting trivially. Thus its moment map image is a convex polytope of dimension $\dim(T/H)$ inside an affine subspace $\mu + \text{ann}(\mathfrak{h})$, with the corresponding component of M_H mapping to its interior. That is, the (open) faces of Δ correspond to orbit type strata, and in particular the vertices of Δ correspond to fixed points M^T . That is,

$$\Delta = \text{hull}(\Phi(M^T))$$

is the convex hull of the fixed point set. Note however that some of the polytopes $\Phi(M^H)$ get mapped to the *interior* of Δ . Thus Δ gets subdivided into polyhedral subregions, consisting of regular values of Φ .

THEOREM 1.8. *Let (M, ω, Φ) be a Hamiltonian T -space, with T acting effectively, and $\Delta \subset \mathfrak{t}^*$ its moment polytope. For any closed face Δ_i of Δ of codimension d_i , the pre-image $\Phi^{-1}(\Delta_i)$ is symplectic, and is a connected component of the fixed point set for some d_i -dimensional stabilizer group $H_i \subset T$ where $\mathfrak{h}_i$ is the subspace orthogonal to Δ_i . In particular, the vertices of Δ correspond to fixed point manifolds.*

PROOF. We note that each $\Phi^{-1}(\Delta_i) \subset M$ is closed and connected, by connectedness of the fibers of Φ . Hence it is a connected component of some M^{H_i} , where $\text{ann}(\mathfrak{h}_i)$ is parallel to Δ_i . $\square$

In particular, Hamiltonian torus actions on compact symplectic manifolds are *never fixed point free*. (This shows immediately that the standard $2k$ -torus action on itself cannot be Hamiltonian.)

EXERCISE 1.9. Let (M, ω, Φ) be a compact, connected Hamiltonian T -space where T acts effectively. Let $M_* = M_{\{e\}}$ be the subset on which the action is free. Show that M_* is connected, and that its image $\Phi(M_*)$ is precisely the interior of the moment polytope $\Delta = \Phi(M)$.

1.4. Duistermaat-Heckman measure. Let us assume (with no loss of generality) that the T -action on (M, ω, Φ) is effective. The images of the fixed point manifolds for non-trivial stabilizer groups define a subdivision of the polytope Δ into chambers, given as the connected components of the set of regular values of Φ . By the Duistermaat-Heckman theorem, the volume function $\mu \mapsto \text{Vol}(M_\mu)$ is a polynomial on each of these connected components.

The volume function is equivalent to the *Duistermaat-Heckman measure* ϱ , defined as the push-forward

$$\varrho := \Phi_* \left| \frac{\omega^n}{n!} \right|$$

of the Liouville measure under the moment map. Thus ϱ is the measure such that for every continuous function ϕ on $\mathfrak{t}^*$,

$$\int_{\mathfrak{t}^*} \phi \varrho = \int_M \Phi^* \phi \left| \frac{\omega^n}{n!} \right|.$$

Let ϱ_{Leb} be Lebesgue measure on $\mathfrak{t}^*$, normalized in such a way that $\mathfrak{t}^*/\Lambda^*$ (where Λ^* is the weight lattice) has volume 1.

EXERCISE 1.10. Let (M, ω, Φ) be a compact Hamiltonian T -space, with T acting effectively. Show that at regular values of Φ , ϱ is smooth with respect to the normalized Lebesgue measure ϱ_{Leb} , and

$$\varrho(\mu) = \text{Vol}(M_\mu) \varrho_{Leb}$$

The proof of this fact is left as an exercise. Hint: Use the local model for reduction near a regular level set. The Duistermaat-Heckman theorem may thus be re-phrased by saying that ϱ is a *piecewise polynomial measure*.

Duistermaat-Heckman in their paper use this fact to derive a remarkable “exact integration formula”, which we will in Section 3.

2. Some basic Morse-Bott theory

The proof of the fact that every component $f = \Phi^\xi$ of the moment map has a unique local minimum relies on the idea of viewing f as a Morse-Bott function. For any function $f \in C^\infty(M, \mathbb{R})$ on a manifold M , the set of critical points is the closed subset

$$\mathcal{C} = \{m \mid df(m) = 0\}.$$

For all $m \in \mathcal{C}$ there is a well-defined symmetric bilinear form on $T_m M$, called the Hessian

$$d^2 f(m)(X_m, Y_m) = (L_X L_Y f)(m)$$

for all $X, Y \in \text{Vect}(M)$. In local coordinates, the Hessian is simply given by the matrix of second derivatives of f .

The function f is called a Morse function if $\mathcal{C}$ is discrete and for all $m \in \mathcal{C}$ the Hessian is non-degenerate. More generally, f is called Morse-Bott if the connected components $\mathcal{C}^j$ of $\mathcal{C} = \{m \mid df(m) = 0\}$ are smooth manifolds, and for all $m \in \mathcal{C}^j$ we have

$$\ker(d^2 f(m)) = T_m \mathcal{C}^j.$$

Given a Riemannian metric on M , consider the negative gradient flow of f , i.e. the flow F^t of the vector field $-\nabla(f) \in \text{Vect}(M)$. For all connected components $\mathcal{C}^j$ we can consider the sets

$$W_+^j = \{m \in M, \lim_{t \rightarrow \infty} F^t(m) \in \mathcal{C}^j\}$$

and

$$W_-^j = \{m \in M, \lim_{t \rightarrow \infty} F^t(m) \in \mathcal{C}^j\}.$$

If f is Morse-Bott then all $W_\pm^j$ are smooth manifolds, and one has natural finite decompositions

$$M = \cup_j W_-^j = \cup_j W_+^j$$

into unstable/stable manifolds. The dimension of W_-^j (resp. W_+^j) is equal to the dimension of $\mathcal{C}^j$ plus the dimension of the negative eigenspace of $\text{Hess}(f)$, denoted $n_\pm^j$. Thus

$$n_\mp^j = \text{codim}(W_\pm^j).$$

The number n_-^j is called the index of $\mathcal{C}^j$.

PROPOSITION 2.1. *If none of the indices n_-^j is equal to 1, there exists a unique critical manifold of index 0, i.e. a unique local minimum of f . If moreover all $n_+^j \neq 1$ then all level sets $f^{-1}(c)$ are connected.*

PROOF. The condition $n_-^j \neq 1$ means that all W_j^+ of positive index have codimension at least 2, so that their complement is connected. Hence there is a unique stable manifold W_j^+ with $n_j^- = 0$. If in addition $n_+^j \neq 1$, the set M_* obtained from M by removing all M_+^j with $n_-^j > 0$ and all M_-^j with $n_+^j > 0$ is open, dense and connected in M . Notice that M_* consists of all points which flow to the (unique) minimum of f for $t \rightarrow \infty$ and to the (unique) maximum of f for $t \rightarrow -\infty$. If $\min(f) < c < \max(f)$ then every trajectory of the gradient flow of a point in M_* intersects $f^{-1}(c)$ in a unique point. Therefore the map

$$(f^{-1}(c) \cap M_*) \times \mathbb{R} \rightarrow M_*, (m, t) \rightarrow F^t(m)$$

is a diffeomorphism, and in particular $f^{-1}(c) \cap M_*$ is connected. To prove the proposition it suffices to show that $f^{-1}(c) \cap M_*$ is dense in $f^{-1}(c)$. Let $m \in f^{-1}(c)$ and U a connected

open neighborhood of m . Since c is neither maximum or minimum, $U \cap M^*$ meets both the sets where $f < c$ and $f > c$, and since it is connected it meets $f^{-1}(c)$. $\square$

Returning to the symplectic geometry context, we need to show:

THEOREM 2.2. *Let (M, ω, Φ) be a Hamiltonian G -space, $\xi \in \mathfrak{g}$. Then $f = \Phi^\xi$ is a Morse-Bott function. Moreover all critical manifolds $\mathbb{C}^j$ are symplectic submanifolds of M , and the indices n_-^j are all even.*

PROOF. Let $H \subset G$ be the closure of the 1-parameter subgroup generated by ξ . Then H is a torus. The critical set of f is given by the condition

$$0 = d\langle \Phi, \xi \rangle(m) = \iota(\xi_M(m))\omega_m.$$

Since ω is non-degenerate, it is precisely the set of zeroes of the vector field ξ_M , or equivalently the fixed point set for the 1-parameter subgroup $\{\exp(t\xi) | t \in \mathbb{R}\} \subseteq G$. Let

$$H = \overline{\{\exp(t\xi) | t \in \mathbb{R}\}}.$$

then H is abelian and connected, hence is a torus, and $\mathbb{C}$ is just the set of fixed points for this torus action. Let $m \in \mathbb{C}$, and equip $T_m M = V$ with an H -invariant compatible complex structure. As a unitary representation, V is equivalent to $V = \oplus \mathbb{C}_{\alpha_j}$ where α_j are the weights for the action. By the equivariant Darboux-theorem, V serves as a model for the H -action near m . In particular the fixed point manifold $\mathbb{C} = M^H$ gets modeled by the space of fixed vectors V^H , which is a complex, hence also symplectic subspace. This shows that all $\mathbb{C}_j$ are symplectic manifolds. Moreover the moment map in this model is (a constant plus)

$$z \mapsto \pi \sum_j |z_j|^2 \alpha_j = \pi \sum_j (q_j^2 + p_j^2) \alpha_j,$$

in particular

$$f = \pi \sum_j |z_j|^2 \alpha_j = \pi \sum_j (q_j^2 + p_j^2) \langle \alpha_j, \xi \rangle.$$

From this it is evident that f is Morse-Bott and that all indices are even. $\square$

The fact that all indices are even has very strong implications in Morse theory: It implies that the so-called lacunary principle applies, and the Morse-Bott polynomial is equal to the Poincare polynomial. (I.e. the Morse inequalities are equalities – Morse functions for which this is the case are called *perfect*.) This gives a powerful tool to calculate the cohomology of Hamiltonian G -spaces: in particular for isolated fixed points, this gives

$$\dim H^k(M, \mathbb{Q}) = \#\{\text{critical points of index } k\};$$

in particular all cohomology sits in even degree if all indices are even.

COROLLARY 2.3. *Suppose M admits a Morse-Bott function f such that the minimum of f is an isolated point and all $n_-^j \neq 1$. Then M is simply connected.*

PROOF. Given any $m \in X$ and a loop $\gamma \in X$ based at m , one can always perturb γ so that it does not meet the stable manifolds of index > 0 . Applying the gradient flow to γ contracts γ to the minimum. $\square$

Examples are coadjoint orbits of a compact Lie group (the fact that coadjoint orbits are compact submanifolds of a vector space allows one to show that for generic components of the moment map the minimum is isolated.) Thus coadjoint orbits are simply connected. (We remark that this is not true in general for conjugacy classes.) Let G/G_μ be a coadjoint orbit where G is compact, connected. View G_μ as the fiber over the identity coset. Given any two points in G_μ they can be joined by a path in G . The projection to G/G_μ is a closed path, hence can be contracted. Lifting the contraction to G produces a path in G_μ connecting the two points. Thus all stabilizer groups for the (co)-adjoint action are connected.

3. Localization formulas

Let (M, ω, Φ) be a compact Hamiltonian T -space. For simplicity we assume that the set M^T of fixed points is finite. (This is for example the case for the action of a maximal torus $T \subset G$ on a coadjoint orbit $\mathcal{O} = G \cdot \mu$.) Given $p \in M^T$ let $a_1(p), \dots, a_n(p) \in \Lambda^* \subset \mathfrak{t}^*$ be the weights for the action on $T_p M$.

THEOREM 3.1 (Duistermaat-Heckman). *Let $\xi \in \mathfrak{t}^\mathbb{C}$ be such that $\langle a_j(p), \xi \rangle \neq 0$ for all p, j . Then one has the exact integration formula*

$$\int_M e^{\langle \Phi, \xi \rangle} \frac{\omega^n}{n!} = \sum_{p \in M^T} \frac{e^{\langle \Phi(p), \xi \rangle}}{\prod_j \langle a_j(p), \xi \rangle}.$$

One way of looking at this result is to say that the stationary phase approximation for the integral $\int_M e^{it\langle \Phi, \xi \rangle} \frac{\omega^n}{n!}$ is *exact*!

Our proof of the DH-formula will follow an argument of Berline-Vergne. Notice first that the integrand is just the top form degree part of

$$e^{\omega + \langle \Phi, \xi \rangle} = e^{\langle \Phi, \xi \rangle} \sum_{j=0}^n \frac{\omega^j}{j!} \in \Omega^*(M).$$

Consider the derivation

$$d_\xi : \Omega^*(M) \rightarrow \Omega^*(M), d_\xi := d - \iota(\xi_M).$$

The differential form $\omega + \langle \Phi, \xi \rangle$ is d_ξ -closed, i.e. killed by d_ξ :

$$d_\xi(\omega + \langle \Phi, \xi \rangle) = -\iota(\xi_M)\omega + d\langle \Phi, \xi \rangle = 0.$$

Moreover $\alpha := e^{\omega + \langle \Phi, \xi \rangle}$ is d_ξ -closed as well. Berline-Vergne prove the following generalization of the DH-formula:

THEOREM 3.2. *Let M be a compact, oriented T -manifold with isolated fixed point set. Given $p \in M^T$ let $a_j(p)$ be the weights for the action on $T_p M$, defined with respect to some choice of T -invariant complex structure on $T_p M$. Suppose $\xi_M \neq 0$ on $M \setminus M^T$. Then for all forms $\alpha \in \Omega^*(M)$ such that $d_\xi \alpha = 0$, one has the integration formula*

$$\int_M \alpha_{[\dim M]} = \sum_{p \in M^T} \frac{\alpha_{[0]}(p)}{\prod_j \langle a_j(p), \xi \rangle}.$$

In the proof we will use the useful notion of *real blow-ups*. Consider first the case of a real vector space V . Let

$$S(V) = V \setminus \{0\} / \mathbb{R}_{>0}$$

be its sphere, thought of as the space of rays based at 0. Define $\hat{V}$ as the subset of $V \times S(V)$,

$$\hat{V} := \{(v, x) \in V \times S(V) \mid v \text{ lies on the ray parametrized by } x\}.$$

Then $\hat{V}$ is a manifold with boundary. (In fact, if one introduces an inner product on V then $\hat{V} = S(V) \times \mathbb{R}_{\geq 0}$). There is a natural smooth map $\pi : \hat{V} \rightarrow V$ which is a diffeomorphism away from $S(V)$. If M is a manifold and $m \in M$, one can define its blow-up $\pi : \hat{M} \rightarrow M$ by using a coordinate chart based at m . Just as in the complex category, one shows that this is independent of the choice of chart (although this is actually not important for our purposes).

Suppose now that M is a T -space as above. Let $\pi : \hat{M} \rightarrow M$ be the manifold with boundary obtained by real blow-up at all the fixed points M^T . The T -action on M lifts to a T -action on $\hat{M}$ with no fixed points. In particular $\xi_{\hat{M}}$ has no zeroes. Choose an invariant Riemannian metric g on $\hat{M}$, and define

$$\theta := \frac{g(\xi_{\hat{M}}, \cdot)}{g(\xi_{\hat{M}}, \xi_{\hat{M}})} \in \Omega^1(\hat{M}).$$

Then θ satisfies $\iota(\xi_{\hat{M}})\theta = 1$ and $d_\xi^2 \theta = L_{\xi_M} \theta = 0$. Therefore

$$\gamma := \frac{\theta}{d_\xi \theta} = \frac{\theta}{d\theta - 1} = -\theta \wedge \sum_j (d\theta)^j$$

is a well-defined form satisfying $d_\xi \gamma = 1$. The key idea of Berline-Vergne is to use this form for partial integration:

$$\begin{aligned}
\int_M \alpha &= \int_{\hat{M}} \pi^* \alpha \\
&= \int_{\hat{M}} \pi^* \alpha \wedge d_\xi \gamma \\
&= \int_{\hat{M}} d_\xi (\pi^* \alpha \wedge \gamma) \\
&= \int_{\hat{M}} d(\pi^* \alpha \wedge \gamma) \\
&= \sum_{p \in M^T} \int_{S(T_p M)} \pi^* \alpha \wedge \gamma \\
&= \sum_{p \in M^T} \alpha_{[0]}(p) \int_{S(T_p M)} \gamma
\end{aligned}$$

Thus, to complete the proof we have to carry out the remaining integral over the sphere. We will do this by a trick, defining a d_ξ -closed form α where we can actually compute the integral by hand.

Consider the T -action on $T_p M = \sum_{j=1}^n \mathbb{C}_{a_j(p)}$ for a given $p \in M^T$. Introduce coordinates $r_j \geq 0, t_j \in [0, 1]$ by $z_j = r_j e^{2\pi\sqrt{-1}t_j}$. Given $\epsilon > 0$ let $\chi \in C^\infty(\mathbb{R}_{\geq 0})$ be a cut-off function, with $\chi(r) = 1$ for $r \leq \epsilon$ and $\sigma = 0$ for $\epsilon \geq 2$. Define a form

$$\alpha = \prod_{j=1}^n (-d_\xi(\chi(r_j) dt_j)) = \prod_{j=1}^n (\langle a_j(p), \xi \rangle - \chi'(r_j) dr_j \wedge dt_j).$$

Note that this form is well-defined (even though the coordinates are not globally well-defined), compactly supported and d_ξ -closed. Its integral is equal to

$$\int_{T_p M} \alpha = \prod_{j=1}^n (-\chi'(r_j) dr_j) = 1.$$

On the other hand $\alpha_{[0]} = \prod_{j=1}^n (\langle a_j(p), \xi \rangle)$.

Choosing ϵ sufficiently small, we can consider α as a form on M , vanishing at all the other fixed points. Applying the localization formula we find

$$1 = \int_M \alpha = \prod_{j=1}^n (\langle a_j(p), \xi \rangle) \int_{S(T_p M)} \gamma,$$

thus

$$\int_{S(T_p M)} \gamma = \frac{1}{\prod_{j=1}^n \langle a_j(p), \xi \rangle}.$$

Q.E.D.

The above discussion extends to non-isolated fixed points, in this case the product $\prod_{j=1}^n \langle a_j(p), \xi \rangle$ is replaced by the equivariant Euler class of the normal bundle of the fixed point manifold.

One often applies the Duistermaat-Heckman theorem in order to compute Liouville volumes of symplectic manifolds with Hamiltonian group action. Consider for example a Hamiltonian $S^1 = \mathbb{R}/\mathbb{Z}$ -action with isolated fixed points. Identify $\text{Lie}(S^1)$, so that the integral lattice and its dual are just $\Lambda = \mathbb{Z}$, $\Lambda^* = \mathbb{Z}$. Let $H = \langle \Phi, \xi \rangle$ where ξ corresponds to $1 \in \mathbb{R}$. By Duistermaat-Heckman,

$$\int_M e^{tH} \frac{\omega^n}{n!} = \frac{1}{t^n} \sum_{p \in M^{S^1}} \frac{e^{tH(p)}}{\prod_j a_j(p)}.$$

Notice by the way that the individual terms on the right hand side are singular for $t = 0$. This implies very subtle relationships between the weight, for example one must have

$$\sum_{p \in M^{S^1}} \frac{H(p)^k}{\prod_j a_j(p)} = 0$$

for all $k < n$. For the volume one reads off,

$$\text{Vol}(M) = \frac{1}{n!} \sum_{p \in M^{S^1}} \frac{H(p)^n}{\prod_j a_j(p)}.$$

4. Frankel's theorem

As we have seen, Hamiltonian torus actions are very special in many respects: In particular they always have fixed points. It is a classical result of Frankel (long before moment maps were invented) that on Kähler manifolds the converse is true:

THEOREM 4.1. *Let M be a compact Kähler manifold, with Kähler form ω . Consider a symplectic S^1 -action on M with at least one fixed point. Then the action is Hamiltonian.*

PROOF. Let $\dim M = 2n$. We need one non-trivial result from complex geometry, which is a particular case of the hard Lefschetz theorem: Wedge product with ω^{n-1} induces an isomorphism in cohomology,

$$\wedge [\omega]^{n-1} : H^1(M) \cong H^{2n-1}(M).$$

Let $X \in \text{Vect}(M)$ be the vector field corresponding to $1 \in \mathbb{R} = \text{Lie}(S^1)$. We need to show that $\iota_X \omega$ is exact. By hard Lefschetz, this is equivalent to showing that $\iota_X \omega^n$ is exact. Let $m \in M^{S^1}$ be a fixed point. In a neighborhood of m we can identify M as a T -space with $T_m M$. Let $\sigma \in \Omega^{2n}(T_m M)$ be an invariant form supported in an ϵ -ball around $T_m M$, normalized so that $\int_{T_m M} \sigma = \int_M \omega^n$. Choosing ϵ sufficiently small we can view σ as a form on M . Since σ and ω^n have the same integral, it follows that $\omega^n - \sigma = d\beta$ for some invariant form $\beta \in \omega^{2n-1} M$. Then

$$\iota(X)(\sigma - \omega^n) = \iota(X)d\beta = L_X \beta - d\iota(X)\beta = -d\iota(X)\beta,$$

showing that $\iota(X)(\sigma - \omega^n)$ is exact. We thus need to show that $\iota(X)\sigma$ is exact. This, however, follows from the Poincare lemma since it is supported in a ball around m , where one can just apply the homotopy operator. $\square$

5. Delzant spaces

DEFINITION 5.1. A Hamiltonian T -space (M, ω, Φ) with proper moment map Φ is called multiplicity-free if all reduced space M_μ are either empty or 0-dimensional. We call (M, ω, Φ) a *Delzant*-space if in addition M is connected, the moment map is proper, and the number of orbit type strata is finite.¹

Thus, if T acts effectively, (M, ω, Φ) is Delzant if and only if $\dim M = 2 \dim T$.

- EXAMPLES 5.2.**
- (a) $M = \mathbb{C}^n$ with the standard action of $T = (S^1)^n$. The moment map image is the positive orthant $\mathbb{R}_+^n \subset \mathbb{R}^n \cong \mathfrak{t}^*$. More abstractly, if V is a Hermitian vector space, the action of the maximal torus $T \subset U(V)$ on V is Delzant.
 - (b) $M = \mathbb{C}P(n)$ with the action of $T = (S^1)^{n+1}/S^1$ (quotient by diagonal subgroup) coming from the action of $(S^1)^{n+1}$ on $\mathbb{C}^{n+1}$. The moment map image is a simplex, given as the intersection of the positive orthant $\mathbb{R}_+^{n+1}$ with the hyperplane $\sum_{i=0}^n t_i = \pi$. More generally, if V is a Hermitian vector space, the action of the maximal torus $T \subset U(V)$ on the projectivization $P(V)$ is Delzant.
 - (c) $M = T^*(T)$ with the cotangent lift of the left-action of T on itself. The moment map image is all of $\mathfrak{t}^*$. We will call this, from now on, the standard T -action on $T^*(T)$.
 - (d) Suppose (M, ω, Φ) is a Delzant T -space, and $H \subset T$ is a subgroup acting freely on the level set of $\mu \in \mathfrak{h}^*$. Then the H -reduced space $(M_\mu, \omega_\mu, \Phi_\mu)$ is Delzant. The moment map image $\Phi(M_\mu) \in \mathfrak{t}^*$ is the intersection of $\Phi(M)$ with the affine subspace $\text{pr}_{\mathfrak{h}^*}^{-1}(\mu)$. We can view M_μ as a Delzant T/H -space, after choosing a moment map for the T/H -action; such a choice amounts to choosing a point in $\text{pr}_{\mathfrak{h}^*}^{-1}(\mu)$.

The moment map images for Delzant spaces can be characterized as follows. Let $\Lambda \subset \mathfrak{t}$ be the integral lattice, i.e. the kernel of $\exp : \mathfrak{t} \rightarrow T$. Let $\Delta \subset \mathfrak{t}^*$ be a rational convex polyhedral set of dimension $d = \dim T$, with k boundary hyperplanes. That is, Δ is of the form

$$(12) \quad \Delta = \bigcap_{i=1}^k \mathcal{H}_{v_i, \lambda_i}$$

where $v_i \in \Lambda$ are primitive lattice vectors and $\lambda_i \in \mathbb{R}$, and

$$\mathcal{H}_{v_i, \lambda_i} = \{\mu \in \mathfrak{t}^* \mid \langle \mu, v_i \rangle \leq \lambda_i\}.$$

¹The finiteness assumption is not very important, and is of course automatic if M is compact.

For any subset $I \subset \{1, \dots, k\}$ let Δ_I be the set of all μ with $\langle \mu, v_i \rangle = \lambda_i$ for $i \in I$. We set $\Delta_\emptyset = \text{int}(\Delta)$.

DEFINITION 5.3. The polyhedral set $\Delta \subset \mathfrak{t}^*$ is called *Delzant* if for all I with $\Delta_I \neq \emptyset$, the vectors v_i , $i \in I$ are linearly independent, and

$$\text{span}_{\mathbb{Z}}\{v_i \mid i \in I\} = \Lambda \cap \text{span}_{\mathbb{R}}\{v_i \mid i \in I\}.$$

REMARK 5.4. For compact polyhedral sets, (that is, polytopes) it is enough to check the Delzant condition at the vertices. The Delzant condition means in particular that each v_i has to be a *primitive* normal vector, i.e. is not of the form $v_i = a v'_i$ where $v'_i \in \Lambda$ and $a \in \mathbb{Z}_{>0}$.

EXAMPLE 5.5. Let $T = (S^1)^2$ and identify $\mathfrak{t} = \mathfrak{t}^* = \mathbb{R}^2$ and $\Lambda \cong \Lambda^* = \mathbb{Z}^2$. The polytope with vertices at $(0, 0), (0, 1), (1, 0)$ is Delzant. However, the polytope with vertices at $(0, 0), (0, 2), (1, 0)$ is not Delzant. Indeed, for the vertex at $(1, 0)$ the two primitive normal vectors are $v_1 = (0, -1)$ and $v_2 = (2, 1)$, and they do not span the lattice $\mathbb{Z}^2$.

The Delzant condition for $\Delta_I \neq \emptyset$ says that $\sum_{j \in I} s_j v_j \in \Lambda \Leftrightarrow s_j \in \mathbb{Z}$ for all $j \in I$, or equivalently,

$$\exp\left(\sum_{j \in I} s_j v_j\right) = 1 \Leftrightarrow s_j = 0 \bmod \mathbb{Z} \text{ for all } j \in I.$$

Thus if we define a homomorphism

$$\phi_\Delta : (S^1)^k \rightarrow T, [(s_1, \dots, s_k)] \mapsto \exp\left(\sum_{i=1}^k s_i v_i\right)$$

and let

$$(S^1)^I = \{[(s_1, \dots, s_k)] \in (S^1)^k \mid s_j = 0 \bmod \mathbb{Z} \text{ for } j \notin I\}$$

be the product of S^1 -factors corresponding to indices $j \in I$, the Delzant condition is equivalent to saying that ϕ_Δ restricts to an inclusion $\phi_\Delta : (S^1)^I \hookrightarrow T$. The image $H_I = \phi_\Delta((S^1)^I) \subset T$ is obtained by exponentiating $\mathfrak{h}_I = \text{span}_{\mathbb{R}}\{v_j \mid j \in I\}$; by definition it is the subspace perpendicular to $\Delta_I \subset \mathfrak{t}^*$.

THEOREM 5.6. *Let (M, ω, Φ) be a Delzant T -space with effective T -action. Then $\Delta = \Phi(M)$ is a Delzant polyhedron. For all open faces $F \subset \Delta$, the pre-image $\Phi^{-1}(F)$ is a connected component of the orbit type stratum $M_H \subset M$ for $H = \exp(\mathfrak{h}_F)$, where $\mathfrak{h}_F \subset \mathfrak{t}$ is the subspace perpendicular to F . In particular, all stabilizer groups are connected.*

PROOF. Let $\mu \in F$, $\mathcal{O} = T.m \in \Phi^{-1}(\mu)$ an orbit, and $H = T_m$ the stabilizer group. We had seen that the cone $\mu(\Delta)$ is equal to the local moment cone

$$C_m = \mu + (\text{pr}_{\mathfrak{h}}^*)^{-1}(C),$$

where $C \subset \mathfrak{h}^*$ is the cone spanned by the weights $\alpha_1, \dots, \alpha_k \in \mathfrak{h}^*$ for the H -action on the symplectic vector space $V = T_m(\mathcal{O})^\omega / T_m(\mathcal{O})$. By dimension count, $k = \dim_{\mathbb{C}} V =$

$\frac{1}{2} \dim M - \dim(T/H) = \dim H$. It follows that α_i are a basis of $\mathfrak{h}^*$. Since $\text{ann}(\mathfrak{h}) \subset \mathfrak{t}^*$ is the maximal linear subspace inside the cone $(\text{pr}_{\mathfrak{h}^*})^{-1}(C)$, it must coincide with the space parallel to F . That is, $\mathfrak{h} = \mathfrak{h}_F$.

The action of H on V must be effective since the T -action on E is effective. Thus H acts as a compact abelian subgroup of $U(V)$ of dimension $\dim H = \dim_{\mathbb{C}} V$. So its identity component H_0 is a maximal torus. But it is a well-known fact from Lie group theory that maximal tori are maximal abelian, so $H = H_0$. In particular, we have shown that all points in $\Phi^{-1}(F)$ have the same stabilizer group.

It follows that the map $H \rightarrow (S^1)^k$ defined by the roots is an isomorphism. This means that $\alpha_1, \dots, \alpha_k$ are a basis for the weight lattice $(\Lambda \cap \mathfrak{h})^*$ in $\mathfrak{h}^*$. Equivalently, the dual basis $w_1, \dots, w_k \in \mathfrak{h}$ are a basis for $\Lambda \cap \mathfrak{h}$. We have

$$C = \text{cone}\{\alpha_1, \dots, \alpha_n\} = \{\nu \in \mathfrak{h}^* \mid \langle \nu, w_i \rangle \geq 0\},$$

which identifies the $\{w_1, \dots, w_k\}$ with $\{v_i \mid i \in I\}$. $\square$

Delzant gave an explicit recipe for constructing a Delzant space with moment polytope a given Delzant polyhedron. The following version of Delzant's construction is due to Eugene Lerman.

Let $(S^1)^k$ act on the cotangent bundle $T^*(T)$ via the composition of ϕ_{Δ} with the standard T -action on $T^*(T)$. In the left trivialization $T^*(T) = T \times \mathfrak{t}^*$, a moment map for the T -action is projection to $\mathfrak{t}^*$. Hence

$$\Psi_{\Delta}(t, \mu) = \sum_{j=1}^k \langle \mu, v_j \rangle e_j - \sum_j \lambda_j e_j$$

is a moment map for the action of $(S^1)^k$. Let $(S^1)^k$ act on $\mathbb{C}^k$ in the standard way, with moment map $\pi \sum_j |z_j|^2 e_j$.

DEFINITION 5.7. For any polyhedron Δ let D_{Δ} be the symplectic quotient

$$D_{\Delta} = (T^*(T) \times \mathbb{C}^k) // (S^1)^k,$$

by the diagonal action, with T -action induced from the standard T -action on $T^*(T)$.

THEOREM 5.8. *Suppose Δ is a Delzant polyhedron. Then the action of $(S^1)^k$ on the zero level set of $(T^*(T) \times \mathbb{C}^k)$ is free, and the quotient D_{Δ} is a Delzant- T -space. The moment map image of D_{Δ} is exactly Δ .*

PROOF. Let $((t, \mu), z)$ in the zero level set. Thus

$$\langle \mu, v_i \rangle = \lambda_i - \pi |z_i|^2.$$

If $z_i \neq 0$ then the i th factor of $(S^1)^k$ acts freely at $((t, \mu), z)$. Thus we need only worry about the set I of indices i with $z_i = 0$. For these indices $\langle \mu, v_i \rangle = \lambda_i$. Let $(S^1)^I$ be the product of copies of S^1 corresponding to these indices. By the Delzant condition, ϕ_{Δ} restricts to an *embedding* $(S^1)^I \rightarrow T$. Since T acts freely on $T^*(T)$, so does $(S^1)^I$. This shows that the action is free, and D_{Δ} is a smooth symplectic manifold. To identify the

image of the T -moment map note that, given $\mu \in \mathfrak{t}^*$, one can find t, z with $((t, \mu), z)$ is in the zero level set if and only if $\langle \mu, v_i \rangle \leq \lambda_i$. $\square$

DEFINITION 5.9 (Lerman). Let Δ be a Delzant polyhedron, and (M, ω, Φ) a Hamiltonian T -space. The *cut space* defined by Δ is the symplectic quotient

$$M_\Delta = (M \times D_\Delta^-) // T$$

with T -action induced from the action on the first factor.

It is immediate that $T^*(T)_\Delta = D_\Delta$: In particular, $T^*(S^1)_{[0, \infty)} \cong \mathbb{C}$. We will now use these two facts to prove:

THEOREM 5.10 (Delzant). *Every Delzant space (M, ω, Φ) is determined by its moment polyhedron $\Delta = \Phi(M)$, up to equivariant symplectomorphism intertwining the moment maps.*

PROOF. Usually this is proved using a Čech theoretic argument. Below we sketch a more elementary (?) approach. The idea is to present M as a symplectic cut $\tilde{M}_\Delta$ of a connected, multiplicity free Hamiltonian T -space $\tilde{M}$ with *free* T -action. Since the action of T on $\tilde{M}$ is free, the map $\tilde{\Phi}$ is a Lagrangian fibration over its image. Thus we can introduce action-angle variables which identifies $\tilde{M}$ as an open subset of $T^*(T)$. Therefore, $M = \tilde{M}_\Delta = T^*(T)_\Delta = D_\Delta$.

We now indicate how to construct such a space $\tilde{M}$. Let $i_1 \in \{1, \dots, k\}$ be an index such that $\Delta_{i_1} \neq 0$, and $S = \Phi^{-1}(\Delta_{i_1})$ the symplectic submanifold obtained as its preimage. It is a connected component of the fixed point set of H_{i_1} , and has codimension 2. Let $\nu_S = TS^\omega$ be its symplectic normal bundle. After choosing a compatible complex structure it can be viewed as a Hermitian line bundle. Let $Q \subset \nu_S$ be the unit circle bundle inside Q . It is a T -equivariant principal S^1 -bundle, and $\nu_S = Q \times_{S^1} \mathbb{C}$. Let $\pi_Q : Q \rightarrow S$ be the projection map. Let $\alpha \in \Omega^1(Q)^T$ be a T -invariant connection 1-form, and consider the closed 2-form

$$\omega_{Q \times \mathbb{C}} := \pi_Q^* \omega_S + \omega_{\mathbb{C}} + \pi d(|z|^2 \alpha).$$

It is easy to check that this 2-form is basic for the S^1 -action, so it descends to a closed 2-form

$$\omega_{\nu_S} \in \Omega^2(\nu_S).$$

Furthermore, ω_{ν_S} is non-degenerate near $S = Q/S^1$. It follows that there exists an equivariant symplectomorphism between open neighborhoods of S in M and in ν_S . Now $\nu_S = (Q \times \mathbb{R})_{[0, \infty)}$ (cut with respect to the S^1 -action), where $Q \times \mathbb{R}$ is equipped with the 2-form

$$\omega_{Q \times \mathbb{R}} = \pi_Q^* \omega_S + \omega_{\mathbb{C}} + d(s\alpha).$$

We have a natural diffeomorphism between $Q \times \mathbb{R}_{>0}$ and $\nu_S \setminus S$, preserving 2-forms. We can thus glue $M \setminus S$ with a small neighborhood of Q in $Q \times \mathbb{R}_-$, to obtain a new connected

multiplicity free Hamiltonian T -space (M_1, ω_1, Φ_1) with one orbit type stratum less. The original space is obtained from M_1 by cutting,

$$M = (M_1)_{\mathcal{H}_1}$$

where $\mathcal{H}_1$ is the affine half-space $\langle \mu, v_i \rangle \geq \lambda_i$. Continuing in this fashion, construct spaces $M_1, M_2, \dots, M_n = \tilde{M}$ where n is the number of faces of Δ . We have

$$M = (M_1)_{\mathcal{H}_1} = (M_2)_{\mathcal{H}_1 \cap \mathcal{H}_2} = \dots = (M_n)_{\Delta},$$

The final space $M_n = \tilde{M}$ no longer has 1-dimensional stabilizer groups, so the T -action is free as required. $\square$

Batalin-Vilkovisky formalism in topological field theory

Habilitationsschrift

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Abstract

This thesis is comprised of 8 papers where we develop different aspects of perturbative topological field theory revolving around the Batalin-Vilkovisky framework for cohomological treatment of gauge symmetry in the perturbative path integral. In particular, we present interesting examples of the pushforward construction for solutions of the Batalin-Vilkovisky quantum master equation; we develop an extension of the Batalin-Vilkovisky formalism to topological field theories (in fact, to general gauge theories) on spacetime manifolds with boundary. We also develop an algebraic version of a special class of topological field theories – the AKSZ sigma models – based on differential graded Frobenius algebras, with the example of Chern-Simons theory studied in particular detail.

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Preface

Here we sum up some basic background material on topological field theories and on the Batalin-Vilkovisky formalism and give a brief account of the results of the papers included in the present Habilitationsschrift.

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1. INTRODUCTION

1.1. Topological quantum field theory: historical background. Topological quantum field theory was conceived in the works of A. S. Schwarz [43] and E. Witten [49] as a tool for constructing invariants of manifolds and knots. It arose as a spin-off of the quantum field theoretic idea of the integral of the exponential of a local (and diffeomorphism-invariant, in case of a topological theory) action functional over a space of local fields over the spacetime manifold – the functional integral (or “path integral” by analogy with Feynman’s path integrals representing evolution kernels in quantum mechanics). The path integral studied by Schwarz in [43] corresponds to the so-called “abelian BF theory” and produces the Ray-Singer torsion of a manifold, which is known, by comparison to Reidemeister torsion via the Cheeger-Müller theorem, to be invariant under simple homotopy equivalences. In [49], Witten studied the path integral over the space of connections in a principal bundle over a closed oriented 3-manifold with the action given by the integral of the Chern-Simons 3-form associated to the connection. Witten showed that the value of the path integral in this case is a diffeomorphism invariant of framed oriented closed 3-manifolds and presented a way to calculate this invariant through surgery. He also suggested a deformation of this path integral by the product of holonomies of the connection along components of a fixed link and showed that this deformation yields the Jones polynomial of the link evaluated at a root of unity (in the case of connections in principal $SU(2)$ -bundles and with holonomy evaluated in the fundamental representation).

Based on the axiomatic definition of conformal field theory by G. Segal [45], M. Atiyah [5] suggested an axiomatic definition of an $(n + 1)$ -dimensional topological quantum field theory (TQFT) as an association, mapping every closed n -manifold to a vector space (the “space of states”) and every $(n + 1)$ -manifold with boundary to a vector (the “partition function”) in the space of states associated to the boundary n -manifold. This association has to satisfy a set of axioms, in particular the multiplicativity axiom and the gluing axiom, expressing the partition function of an $(n + 1)$ -manifold with two diffeomorphic boundary components glued together in terms of the partition function for the original $(n + 1)$ -manifold. An immediate consequence of the axioms is that the partition function for a closed $(n + 1)$ -manifold is a number, invariant under diffeomorphisms of the manifold; using the gluing axiom this number can be recovered, once the manifold is cut into simpler pieces, from values of the TQFT on the pieces. In algebraic language, Atiyah’s topological quantum field theory is a functor of symmetric monoidal categories from the category of $(n + 1)$ -cobordisms to the category of vector spaces. Atiyah’s definition summarized the known properties of a collection of manifold invariants, e.g.

Floer-Gromov theory ($n = 1$) and Donaldson-Floer invariants ($n = 3$), as well as properties of the Chern-Simons path integral ($n = 2$) studied by Witten.

The idea of constructing topological quantum field theories via path integrals immediately runs into a number of problems:

- (I) Path integrals over spaces of local fields typically cannot be defined as measure theoretic integrals. Instead one can try to define the “perturbative” path integral by postulating the stationary phase approximation formula as the definition of the path integral, giving the latter as a sum over critical points of the action of sums of contributions of Feynman diagrams evaluated at a critical point (cf. e.g. [27], see also [39] for a modern exposition).
- (II) To apply, even formally, the stationary phase formula, one needs the critical points of the action to be isolated. Typically this is not the case with topological Lagrangians due to the presence of a local symmetry – the gauge symmetry. There are several ways out: Faddeev-Popov [25], BRST [12, 48] and Batalin-Vilkovisky [10, 11] constructions of gauge-fixing (in the chronological order, each a generalization of previous one). Generally they amount to extending the space of fields to a superspace and extending the action by additional terms depending on the newly introduced fields in such a way that the critical points of the new action are isolated and thus one can proceed with the stationary phase formula.
- (III) The gauge fixing procedure needed to define Feynman diagrams for the path integral typically destroys the manifest diffeomorphism invariance of the Lagrangian by introducing dependence on an additional geometric structure, e.g. a Riemannian metric on the spacetime manifold (chosen arbitrarily for the gauge fixing). One has to prove then that the resulting partition functions are independent on the chosen geometric structure. (In fact, the situation can be more subtle, e.g. Witten [49] discovered that the phase of the 1-loop Chern-Simons partition function is metric-dependent, though the metric dependence can be cancelled by a local counterterm, at the cost of introducing dependence on the framing.)
- (IV) Values of Feynman diagrams in the perturbative path integral contain integrals over configuration spaces of points on the spacetime manifold, with the integrand a product of propagators, singular on diagonals. One has to prove finiteness (convergence) of the Feynman diagrams, possibly after a suitable renormalization procedure.
- (V) Finally, one has to extend the path integral construction of TQFT partition functions to allow the spacetime manifold to have a boundary, in such a way that Atiyah’s gluing axiom is satisfied.

A purely perturbative treatment of the Chern-Simons path integral (as opposed to Witten’s treatment of [49], where certain properties of the path integral, e.g. extendability to an Atiyah’s TQFT on 3-manifolds with boundary and relation to 2-dimensional conformal field theory on the boundary, are assumed, instead of being derived from some definition) was given by S. Axelrod and I. M. Singer in [6, 7]. They constructed the perturbative Chern-Simons partition function on a *closed* oriented 3-manifold on the background of a fixed *acyclic* flat connection (a chosen critical point of the Chern-Simons action functional) as a formal power series in formal parameter $\hbar$, with finitely many Feynman graphs contributing to each order in $\hbar$. Feynman diagrams depend explicitly, via the propagator, on the

choice of an arbitrary Riemannian metric on the 3-manifold, necessary to do the Faddeev-Popov gauge fixing. The main results of Axelrod and Singer are:

- Finiteness of Feynman diagrams – Axelrod and Singer proved that the products of propagators involved lift to smooth differential forms on the Fulton-Macpherson compactification of the configuration space of points on the 3-manifold.
- Metric independence of the resulting partition function. More precisely, Axelrod and Singer proved (extending Witten’s 1-loop result to higher loop corrections) that there is metric dependence, but it can be cancelled by a local counterterm, proportional to the Chern-Simons action functional evaluated on the Levi-Civita connection, dependent on the choice of an arbitrary framing of the 3-manifold.

Another important example of topological quantum field theory constructed via path integral quantization is the 2-dimensional Poisson sigma model (originally considered in [29, 42]) which, in the case when the spacetime manifold is a disk, with certain boundary condition for fields imposed at the boundary circle, plays a crucial role in Kontsevich’s deformation quantization of Poisson manifolds [32, 17]. Here, in order to make sense of perturbation theory for the path integral, one is forced to employ the Batalin-Vilkovisky (BV) formalism [10], since the gauge symmetry is not given by a group action on the space of fields, but is instead a non-integrable distribution on the space of fields, and thus the less general Faddeev-Popov and BRST constructions do not work.

We should stress that neither quantum Chern-Simons theory, nor quantum Poisson sigma model are constructed by a perturbative path integral on manifolds with boundary, as an Atiyah’s TQFT (in Poisson sigma model on a disk one does indeed have a boundary, but fixing *one* boundary condition is not enough to be able to glue; one should instead allow a family of boundary conditions given by a Lagrangian foliation of the boundary phase space). In fact, the only example of perturbative Atiyah’s TQFT constructed so far seems to be the 1-dimensional Chern-Simons theory [4]. Papers [4, 21, 22, 3] are the first steps in the project of the author together with A. S. Cattaneo and N. Reshetikhin, aimed at constructing Atiyah’s TQFTs by path integral quantization of diffeomorphism-invariant Lagrangians (e.g. those provided by the AKSZ construction [1]) on manifolds with boundary. One outcome of the project would be the extension of Axelrod-Singer’s construction of the Chern-Simons path integral on closed 3-manifolds to a full Atiyah’s TQFT.

We should also mention that there are well known constructions of Atiyah’s TQFTs not coming from quantization (of a classical field theory) and not relying on perturbative path integrals. In particular, there are Turaev-Viro [47] and Reshetikhin-Turaev [40] constructions of 3-dimensional Atiyah’s TQFTs based on the representation theory of quantum groups.

In [1], Aleksandrov, Kontsevich, Schwarz and Zaboronsky (AKSZ) suggested a geometric construction of local diffeomorphism-invariant functionals on mapping spaces of the form $\text{Map}(T[1]\Sigma, \mathcal{M})$ where Σ is a manifold (the spacetime) and the target $\mathcal{M}$ is a graded supermanifold endowed with differential graded symplectic structure of appropriate degree; $T[1]\Sigma$ stands for the tangent bundle to Σ with fiber coordinates prescribed degree 1. The AKSZ construction is from the start consistent with the Batalin-Vilkovisky formalism: the resulting action functionals automatically satisfy the classical master equation. As particular examples of the

AKSZ construction, one gets Batalin-Vilkovisky extensions of the Chern-Simons action, the Poisson sigma model, the BF theory, the Courant sigma model [41], Witten's A and B models and more.

1.2. What is done in the papers. We will give a more detailed account of the results in Section 4, here we just outline the general logic of the development.

- In [20] we construct a finite-dimensional algebraic model for Chern-Simons action on closed 3-manifolds, based on differential graded Frobenius algebras (in [15] this construction is extended to general AKSZ sigma models). We study the BV pushforward to cohomology in this model and the arising invariants of dg Frobenius algebras. Then we apply this construction in the case of standard Chern-Simons theory, obtaining a generalization of Axelrod-Singer's treatment of Chern-Simons path integral to non-acyclic flat background connections.
- In [13] we study the BV pushforward to zero modes for the Poisson sigma model on a closed surface. We prove that in genus 1 or for a regular unimodular target Poisson structure there are no quantum corrections in the effective action on zero modes. In genus 1 case, we perform the non-perturbative integration over zero-modes, yielding (under further assumptions on the target Poisson structure) the Euler characteristic of the target.
- In [4] we construct simplicial 1-dimensional Chern-Simons theory as Atiyah's TQFT on triangulated 1-cobordisms. Two definitions of the partition functions are given (and proven to coincide): a perturbative path integral definition and a non-perturbative one, expressing partition functions in terms of exponentials in Clifford algebra of the gauge Lie algebra. The partition functions are proven to satisfy the Batalin-Vilkovisky quantum master equation extended by boundary terms. The space of states associated to a point here is the spinor module, which arises as the geometric quantization of the parity-shifted gauge Lie algebra in operator formalism and as the space of functions on the parameter space of the chosen family of boundary conditions in the path integral formulation. It is an important observation that the cubic Dirac operator on the spinor module arising as the boundary term of the quantum master equation in this model is the geometric quantization of the target Hamiltonian in the AKSZ formulation of 1-dimensional Chern-Simons theory.
- In [21, 22] we extend the classical Batalin-Vilkovisky formalism to gauge theories on manifolds with boundary. In this formalism, the graded phase space (the space of boundary fields) associated to the boundary of the spacetime manifold carries a Batalin-Fradkin-Vilkovisky structure, i.e. is equipped with a degree 0 symplectic form and a degree 1 function (the BFV action) generating, as its Hamiltonian vector field, the cohomological vector field. This structure is in perfect accord with the quantum structure appearing in [4]. The work [21] is the first step in constructing perturbative Atiyah's TQFTs, e.g. starting from AKSZ Lagrangians.
- In [3] we apply the construction of [21] to Chern-Simons theory on a 3-manifold with boundary. We also incorporate the Wilson lines, possibly ending on the boundary, using the enhancement of AKSZ sigma models by observables suggested in [37]. Then we proceed with geometric quantization of the boundary BFV phase space. The cohomology in degree 0 of the

corresponding quantized BFV action is, at least in the case of a boundary surface of genus 0 or 1, the space of conformal blocks of the Wess-Zumino-Witten model.

Overall, the results group around the following subjects:

- (i) Effective BV actions for topological field theories (examples of pushforward construction for solutions of the master equation), [20, 13, 4, 37].
- (ii) Incorporation of spacetime boundary and Atiyah's gluing in the BV formalism, [4, 21, 22, 3].
- (iii) Algebraic finite-dimensional models for topological actions, [20, 15].
- (iv) Observables in AKSZ theories [37, 3].

1.3. Prospects. The results of the papers presented in this thesis (in particular, [4, 21, 22, 3]) can be regarded as steps towards the construction of a class of examples of functorial topological quantum field theories in the sense of Atiyah (and, more ambitiously, extended TQFTs in the sense of Baez-Dolan-Lurie) from perturbative path integral quantization of gauge- and diffeomorphism-invariant action functionals (in particular, AKSZ sigma models provide natural testing grounds for such a quantization) on manifolds with boundary or corners, in a way consistent with gluing. The natural framework for the treatment of gauge symmetry is provided by the Batalin-Vilkovisky formalism, extended to the setting with boundary/corners as in [21].

As a part of this program, Axelrod-Singer's perturbative treatment of Chern-Simons theory on closed 3-manifolds [6, 7] can be extended to 3-manifolds with boundary, constructing Chern-Simons theory as Atiyah's TQFT by the perturbative path integral and providing a mathematically transparent explanation for Witten's non-perturbative results of [49]. Another anticipated result is the proof of a long-standing conjecture that the $k \rightarrow \infty$ asymptotics of the Reshetikhin-Turaev invariant [40] coincides with the perturbative partition function in Chern-Simons theory (with Atiyah's canonical framing on the 3-manifold).

The output of the quantization program outlined above in algebraic topology would be in constructing the gluing-cutting formulae for invariants of manifolds and knots.

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2. PAPERS INCLUDED

Here we list the abstracts of the papers comprising the thesis.

1. [20] *Remarks on Chern-Simons invariants*, with A. S. Cattaneo.

The perturbative Chern-Simons theory is studied in a finite-dimensional version or assuming that the propagator satisfies certain properties (as is the case, e.g., with the propagator defined by Axelrod and Singer). It turns out that the effective BV action is a function on cohomology (with shifted degrees) that solves the quantum master equation and is defined modulo certain canonical transformations that can be characterized completely. Out of it one obtains invariants.

2. [15] *Finite-dimensional AKSZ-BV theories*, with F. Bonechi and M. Zabzine.

We describe a canonical reduction of AKSZ-BV theories to the cohomology of the source manifold. We get a finite-dimensional BV theory that describes the contribution of the zero modes to the full QFT. Integration can be defined and correlators can be computed. As an illustration of the general construction, we consider two-dimensional Poisson sigma model and three-dimensional Courant sigma model. When the source manifold is compact, the reduced theory is a generalization of the AKSZ construction where we take as source the cohomology ring. We present the possible generalizations of the AKSZ theory.

3. [4] *One-dimensional Chern-Simons theory*, with A. Alekseev.

We study a one-dimensional toy version of the Chern-Simons theory. We construct its simplicial version which comprises features of a low-energy effective gauge theory and of a topological quantum field theory in the sense of Atiyah.

4. [13] *The Poisson sigma model on closed surfaces*, with F. Bonechi and A. S. Cattaneo.

Using methods of formal geometry, the Poisson sigma model on a closed surface is studied in perturbation theory. The effective action, as a function on vacua, is shown to have no quantum corrections if the surface is a torus or if the Poisson structure is regular and unimodular (e.g., symplectic). In the case of a Kähler structure or of a trivial Poisson structure, the partition function on the torus is shown to be the Euler characteristic of the target; some evidence is given for this to happen more generally. The methods of formal geometry introduced in this paper might be applicable to other sigma models, at least of the AKSZ type.

5. [21] *Classical BV theories on manifolds with boundary*, with A. S. Cattaneo and N. Reshetikhin.

In this paper we extend the classical BV framework to gauge theories on spacetime manifolds with boundary. In particular, we connect the BV construction in the bulk with the BFV construction on the boundary and we develop its extension to strata of higher codimension in the case of manifolds with corners. We present several examples including electrodynamics, Yang-Mills theory and topological field theories coming from the AKSZ construction, in particular, the Chern-Simons theory, the BF theory, and the Poisson sigma model. This paper is the first step towards developing the perturbative quantization of such theories on manifolds with boundary in a way consistent with gluing.

6. [22] *Classical and quantum Lagrangian field theories with boundary*, with A. S. Cattaneo and N. Reshetikhin.

This note gives an introduction to Lagrangian field theories in the presence of boundaries. After an overview of the classical aspects, the cohomological formalisms to resolve singularities in the bulk and in the boundary theories (the BV and the BFV formalisms, respectively) are recalled. One of the goals here (and in [21]) is to show how the latter two formalisms can be put together in a consistent way, also in view of perturbative quantization.

7. [37] *A construction of observables for AKSZ sigma models*.

A construction of gauge-invariant observables is suggested for a class of topological field theories, the AKSZ sigma-models. The observables are associated to extensions of the target Q-manifold of the sigma model to a Q-bundle over it with additional Hamiltonian structure in fibers.

8. [3] *Chern-Simons theory with Wilson lines and boundary in the BV-BFV formalism*, with A. Alekseev and Y. Barmaz.

We consider the Chern-Simons theory with Wilson lines in 3D and in 1D in the BV-BFV formalism of Cattaneo-Mnev-Reshetikhin. In particular, we allow for Wilson lines to end on the boundary of the spacetime manifold. In the toy model of 1D Chern-Simons theory, the quantized BFV boundary action coincides with the Kostant cubic Dirac operator which plays an important role in representation theory. In the case of 3D Chern-Simons theory, the boundary action turns out to be the odd (degree 1) version of the BF model with source terms for the B field at the points where the Wilson lines meet the boundary. The boundary space of states arising as the cohomology of the quantized BFV action coincides with the space of conformal blocks of the corresponding WZW model.

3. BACKGROUND

3.1. Batalin-Vilkovisky formalism. The Batalin-Vilkovisky formalism [10, 11], sometimes also referred to as “symplectic BRST formalism”, was suggested as an enhancement of the BRST formalism [12, 48] for the gauge fixing of Lagrangian field theories with local symmetry, thus curing the problem of non-isolated critical points of the action obstructing the construction of the perturbative path integral. The Batalin-Vilkovisky formalism is applicable to a larger class of field theories than the BRST formalism and is the most general known method of gauge-fixing. Apart from being unavoidable in some examples (e.g. in case of the Poisson sigma model [32, 17] and in non-abelian BF theory in dimension ≥ 4 , cf. [23]), the BV formalism turned out to be a particularly elegant language in case of topological field theories, emphasizing the supergeometric origins (and permitting the superfield formulation) of diffeomorphism-invariant Lagrangians [1]. The geometric structure underlying the Batalin-Vilkovisky formalism was elucidated in the works of E. Witten [50], A. Schwarz [44], H. Khudaverdian [30], P. Ševera [46].

3.1.1. BV algebra. One can summarize the structure of the BV formalism as follows. First, one defines a *BV algebra* as a quadruple $\{\mathcal{A}^\bullet, \cdot, \{, \}, \Delta\}$ consisting of:

- A $\mathbb{Z}$ -graded vector space $\mathcal{A}^\bullet$.¹
- A degree 0 multiplication $\cdot : \mathcal{A}^p \otimes \mathcal{A}^q \rightarrow \mathcal{A}^{p+q}$ satisfying supercommutativity $yx = (-1)^{pq}xy$ and associativity $(xy)z = x(yz)$ properties; one also assumes that the multiplication possesses a unit.
- A degree 1 bi-derivation, the “odd Poisson bracket”² $\{\bullet, \bullet\} : \mathcal{A}^p \otimes \mathcal{A}^q \rightarrow \mathcal{A}^{p+q+1}$ satisfying

$$\begin{aligned} \{y, x\} &= -(-1)^{(p+1)(q+1)}\{x, y\}, \\ \{xy, z\} &= x\{y, z\} + (-1)^{q(r+1)}\{x, z\}y, \\ \{x, \{y, z\}\} &= \{\{x, y\}, z\} + (-1)^{(p+1)(q+1)}\{y, \{x, z\}\} \end{aligned}$$

– graded skew-symmetry, derivation property and Jacobi identity. Here we assume that $x \in \mathcal{A}^p, y \in \mathcal{A}^q, z \in \mathcal{A}^r$ are homogeneous elements.

¹I.e. a collection of vector spaces $\mathcal{A}^p$ indexed by $p \in \mathbb{Z}$.

²Also known as “anti-bracket”, “BV bracket” or “Gerstenhaber bracket”.

- A degree 1 linear operator $\Delta : \mathcal{A}^p \rightarrow \mathcal{A}^{p+1}$ (the “BV Laplacian”) squaring to zero

$$\Delta^2 = 0$$

satisfying the second order derivation property

$$\Delta(xyz) = \Delta(xy)z + (-1)^{qr} \Delta(xz)y + (-1)^p x \Delta(yz) - (\Delta x)yz - (-1)^p x(\Delta y)z - (-1)^{p+q} xy(\Delta z)$$

and the normalization property $\Delta(1) = 0$, and generating the odd Poisson bracket as the defect of the first order Leibnitz identity:

$$\Delta(xy) = (\Delta x)y + (-1)^p x(\Delta y) + (-1)^p \{x, y\}$$

(The set of axioms above is redundant: properties of $\{, \}$ follow from axioms for Δ ; excessive axioms are provided for exposition purposes.) The set of data $(\mathcal{A}^\bullet, \cdot, \{, \})$ comprises the so-called Gerstenhaber (or odd Poisson) algebra. Thus a BV algebra is in particular a Gerstenhaber algebra enriched by an additional structure – the BV Laplacian. Another remark is that, by forgetting the multiplication, a BV algebra $(\mathcal{A}^\bullet, \cdot, \{, \}, \Delta)$ induces a differential graded Lie algebra (dgLa) $(\mathcal{A}^{\bullet-1}, \{, \}, \Delta)$.

3.1.2. *Quantum and classical master equation.* Given a BV algebra $(\mathcal{A}^\bullet, \cdot, \{, \}, \Delta)$ one calls the equation

$$\{S, S\} = 0$$

for an element $S \in \mathcal{A}^0$ the *classical master equation* (CME).

The equation

$$\Delta e^{\frac{i}{\hbar} S} = 0$$

for an element $S \in \mathcal{A}^0[[\hbar]]$ in the ring of formal power series in a formal variable $\hbar$ with coefficients in $\mathcal{A}^0$ is called the *quantum master equation* (QME). Equivalently, it can be written in the Maurer-Cartan form:

$$(1) \quad \frac{1}{2} \{S, S\} - i\hbar \Delta S = 0$$

Given a solution $S = S^{(0)} + \hbar S^{(1)} + \hbar^2 S^{(2)} + \dots$ of the quantum master equation with $S^{(k)} \in \mathcal{A}^0$, the zeroth term $S^{(0)}$ automatically satisfies the classical master equation and the higher order corrections satisfy the sequence of equations

$$\{S^{(0)}, S^{(1)}\} - i\Delta S^{(0)} = 0, \quad \{S^{(0)}, S^{(2)}\} + \frac{1}{2} \{S^{(1)}, S^{(1)}\} - i\Delta S^{(1)} = 0, \quad \dots$$

The obstructions to order-by-order extension of a solution $S^{(0)}$ of CME to a solution of $S = S^{(0)} + O(\hbar)$ of QME lie in cohomology of the coboundary operator $\{S^{(0)}, \bullet\} : \mathcal{A}^p \rightarrow \mathcal{A}^{p+1}$ in degree 1. (Hence if this obstruction space vanishes, all solutions of CME extend to solutions of QME.) In the special case $S = S^{(0)}$ – a solution of QME independent of $\hbar$ – QME is equivalent to the pair of equations $\{S^{(0)}, S^{(0)}\} = 0$, $\Delta S^{(0)} = 0$ holding simultaneously.

For a solution $S^{(0)}$ of the classical master equation, the corresponding degree 1 derivation $Q = \{S^{(0)}, \bullet\} \in \text{Der}(\mathcal{A}^\bullet)_1$ (sometimes called the BRST operator) automatically satisfies the property $Q^2 = 0$.

If S, S' are two solutions of QME related by

$$e^{\frac{i}{\hbar} S'} = e^{\frac{i}{\hbar} S} + \Delta \left(e^{\frac{i}{\hbar} S} R \right)$$

one says that S and S' are *equivalent*, or related by a (*quantum*) *canonical BV transformation* with generator $R \in \mathcal{A}^{-1}[[\hbar]]$. An infinitesimal canonical transformation (i.e. with an infinitesimal generator) can be expressed as³

$$S' = S + \{S, R\} - i\hbar \Delta R + O(R^2)$$

3.1.3. Examples of BV algebras. The main source of examples of BV algebras is the following construction [44]: assume that $\mathcal{F}$ is a finite-dimensional⁴ graded supermanifold endowed with a degree -1 symplectic structure $\omega \in \Omega^2(\mathcal{F})$ (the *BV 2-form*) and a volume element (Berezinian) μ . Then one constructs the BV algebra structure $\mathcal{A}^\bullet = C^\infty(\mathcal{F})$ with grading coming from the grading of $\mathcal{F}$; the product is the pointwise product of functions, the odd Poisson bracket $\{, \}$ is induced from ω by $\{f, g\} = \check{f}g$ where the Hamiltonian vector field $\check{f} \in \mathfrak{X}(\mathcal{F})$ is defined by $\iota_{\check{f}}\omega = \delta f$.⁵ The BV Laplacian is constructed as

$$\Delta f = \frac{1}{2} \operatorname{div}_\mu \check{f}$$

where $\operatorname{div}_\mu : \mathfrak{X}(\mathcal{F}) \rightarrow C^\infty(\mathcal{F})$ is the divergence of a vector field associated to the chosen volume element μ by $\int_{\mathcal{F}} \mu v(g) = - \int_{\mathcal{F}} \mu (\operatorname{div}_\mu v) g$ for a fixed vector field v and any test function g . The requirement that $\Delta^2 = 0$ imposes a restriction on ω and μ . Following [44], we will call the set of data $(\mathcal{F}, \omega, \mu)$ (assuming that the induced BV Laplacian does square to zero) an *SP manifold*.

In this example, if $S^{(0)}$ is a solution of CME, the degree 1 Hamiltonian vector field $Q = \{S^{(0)}, \bullet\} \in \mathfrak{X}(\mathcal{F})_1$ satisfies $Q^2 = \frac{1}{2}[Q, Q] = 0$, i.e. it is (by definition) a cohomological vector field on $\mathcal{F}$.

A special example of the construction above arises when one takes M an (ordinary) manifold with a volume form $\nu \in \Omega^{\text{top}}(M)$ and constructs the BV algebra $\mathcal{V}^{-\bullet}(M)$ – polyvector fields on M with reversed grading. The product is the wedge product of polyvectors, the odd Poisson bracket $\{, \} = [\cdot, \cdot]_{NS}$ is the Nijenhuis-Schouten bracket of polyvectors and the BV Laplacian is $\Delta = \operatorname{div}_\nu : \mathcal{V}^{-\bullet}(M) \rightarrow \mathcal{V}^{-\bullet-1}(M)$ – the divergence of a polyvector field with respect to ν , lowering the degree of a polyvector by 1. This example embeds into the construction above by setting $\mathcal{F} = T^*[-1]M$ – the cotangent bundle with shifted homological degree of the fiber coordinates, carrying a canonical degree -1 symplectic form ω (as a cotangent bundle) and endowed with the Berezinian $\mu = \nu^{\otimes 2}$ (we are using here the canonical isomorphism of line bundles over M , $\operatorname{Ber}(T^*[-1]M) \cong \operatorname{Det}(M)^{\otimes 2}$).

Another important special case is when $\mathcal{F}$ is a graded vector space $\mathcal{F}^\bullet$ with a translation-invariant symplectic form $\omega \in \Omega^2(\mathcal{F}^\bullet)$. Then there is a canonical choice of translation-invariant volume element μ , defined up to normalization. This induces a canonical BV algebra structure on $C^\infty(\mathcal{F}^\bullet)$.

³Note that modulo $\hbar$, this is a Hamiltonian transformation of S with generator R (hence the name “canonical transformation”).

⁴Finite-dimensionality is an important simplifying assumption that we will make to clarify the structure, despite the fact that it excludes spaces of fields of local field theories (which are typically Fréchet manifolds). Local field theories can be cast in BV formalism with an appropriate renormalization procedure, cf. e.g. the approaches of [24, 35, 36].

⁵We denote the de Rham differential on (forms on) $\mathcal{F}$ by δ , since we want to reserve d for the de Rham differential on spacetime manifolds.

3.1.4. *BV integrals.* Let $(\mathcal{F}, \omega, \mu)$ be an *SP* manifold and $S \in C^\infty(\mathcal{F})_0$ (subscript denotes the degree of the function) a solution of the quantum master equation in the associated BV algebra. A *BV integral* is an expression of the form

$$Z_{S, \mathcal{L}} = \int_{\mathcal{L} \subset \mathcal{F}} e^{\frac{i}{\hbar} S} \sqrt{\mu}|_{\mathcal{L}}$$

where $\mathcal{L}$ is a Lagrangian submanifold of $\mathcal{F}$ with respect to the odd symplectic structure ω and $\sqrt{\mu}|_{\mathcal{L}}$ is the volume element on $\mathcal{L}$ canonically induced from μ , cf. [44]. We assume that S and $\mathcal{L}$ are such that the integral converges. The crucial properties of the BV integral (known as the BV-Stokes theorem, [10, 44]) are:

- (i) For two solutions S, S' of QME related by a canonical transformation and $\mathcal{L} \subset \mathcal{F}$ a fixed Lagrangian, one has $Z_{S, \mathcal{L}} = Z_{S', \mathcal{L}}$.
- (ii) For S a fixed solution of QME and $\mathcal{L}, \mathcal{L}' \subset \mathcal{F}$ two Lagrangians which can be connected by a smooth family of Lagrangians in $\mathcal{F}$ (we will call, somewhat abusively, the corresponding equivalence relation on Lagrangians the *Lagrangian homotopy*), one has $Z_{S, \mathcal{L}} = Z_{S, \mathcal{L}'}$.

3.1.5. *Fiber BV integrals (pushforward of solutions of the quantum master equation).* Assume $\mathcal{F} = \mathcal{F}' \times \mathcal{F}''$ is a direct product of two *SP* manifolds, with $\omega = \omega' + \omega''$ and $\mu = \mu' \cdot \mu''$. The corresponding BV algebra is the (completed) tensor product $\mathcal{A} = \mathcal{A}' \hat{\otimes} \mathcal{A}''$ of the BV algebras for $\mathcal{F}', \mathcal{F}''$, with BV Laplacian on $\mathcal{F}$ splitting as $\Delta = \Delta' + \Delta''$. Then, given a solution S of QME on $\mathcal{F}$ can construct a solution of QME on $\mathcal{F}'$, $S' \in C^\infty(\mathcal{F}')_0[[\hbar]]$ by means of the *fiber BV integral* [33, 36, 37]:

$$(2) \quad e^{\frac{i}{\hbar} S'} = \int_{\mathcal{L}'' \subset \mathcal{F}''} e^{\frac{i}{\hbar} S} \sqrt{\mu''}|_{\mathcal{L}''}$$

Here $\mathcal{L}''$ is a Lagrangian submanifold of $\mathcal{F}''$ and the r.h.s. is to be understood as an integral over a Lagrangian submanifold of the fiber of the bundle $\mathcal{F}' \times \mathcal{F}'' \rightarrow \mathcal{F}'$ parameterized by a point of $\mathcal{F}'$. One calls S' an *effective BV action* induced from S on $\mathcal{F}'$. The main properties of this construction (cf. [36, 37] for details), generalizing the properties of the BV integrals (the BV-Stokes theorem) are:

- (i) The effective action S' as defined by (2) is indeed a solution of QME on $\mathcal{F}'$, provided that S is a solution of QME on $\mathcal{F}$.
- (ii) If $\mathcal{L}''$ and $\tilde{\mathcal{L}}''$ are two Lagrangian submanifolds of $\mathcal{F}''$ connected by a Lagrangian homotopy and S is a solution of QME on $\mathcal{F}$, then the two corresponding effective actions $S', \tilde{S}'$ are related by a quantum BV canonical transformation. Moreover, if $\tilde{\mathcal{L}}''$ is a graph of the 1-form $\delta\Psi$ for $\Psi \in C^\infty(\mathcal{L}'')_{-1}$, then one can write the generator of the canonical transformation $R' \in C^\infty(\mathcal{F}')_{-1}[[\hbar]]$ explicitly in terms of Ψ :

$$(3) \quad R' = e^{-\frac{i}{\hbar} S'} \int_{\mathcal{L}''} \Psi e^{\frac{i}{\hbar} S} \sqrt{\mu''}|_{\mathcal{L}''}$$

- (iii) If S and $\tilde{S}$ are two solutions of QME on $\mathcal{F}$ related by a canonical transformation with generator R and the Lagrangian $\mathcal{L}''$ is fixed, then the two corresponding effective actions $S', \tilde{S}'$ are related by a canonical transformation with generator given again by formula (3) where one substitutes $\Psi \mapsto R$.

Thus the fiber BV integral construction is a *pushforward*, mapping solutions of QME on $\mathcal{F}$ modulo canonical transformations to solutions of QME on $\mathcal{F}'$ modulo

canonical transformation. The pushforward depends on a choice of Lagrangian $\mathcal{L}'' \subset \mathcal{F}''$ modulo Lagrangian homotopy.

3.1.6. The BV formalism and gauge-fixing. The general abstract setup for a gauge system is a manifold F_{cl} – the “space of classical fields” (for a bosonic theory, this is an ordinary, non-super, manifold; we are assuming finite-dimensionality here to elucidate the geometric content of the formalism) endowed with a volume form μ_{cl} , a function S_{cl} and a distribution E – a subbundle

$$\begin{array}{ccc} E & \xrightarrow[\rho]{\hookrightarrow} & TF_{\text{cl}} \\ \downarrow & & \downarrow \\ F_{\text{cl}} & \xlongequal{\quad} & F_{\text{cl}} \end{array}$$

– the gauge symmetry. One requires that the following conditions hold:

- S_{cl} and μ_{cl} are E -invariant,
- distribution E is integrable on the critical locus of S_{cl} .

One is interested in the integral

$$(4) \quad Z = \int_{F_{\text{cl}}} e^{\frac{i}{\hbar} S_{\text{cl}}} \mu_{\text{cl}}$$

Due to E -invariance of the integrand, critical points are not isolated and the stationary phase formula is not applicable.

One solution of the problem is to extend the space of fields F_{cl} to the “space of BV fields” – an SP manifold

$$(5) \quad \mathcal{F} = T^*[-1](E[1])$$

where $E[1]$ is the total space of the vector bundle $E \rightarrow F_{\text{cl}}$ with fiber coordinates prescribed degree (“ghost number”) 1. The BV 2-form is the canonical symplectic structure on the cotangent bundle, the Berezinian is constructed as $(\mu_{\text{cl}} \cdot \mu_{\text{ghost}})^{\otimes 2}$ where μ_{ghost} is a fiber Berezinian on $E[1]$. If x^i are local coordinates on F_{cl} (“classical fields”) and c^a are fiber coordinates on $E[1]$ (“ghosts”), associated to a choice of basis $\{e_a\}$ in the fibers of $E \rightarrow F_{\text{cl}}$, then on $\mathcal{F}$ one has coordinates c^a, x^i, x_i^+, c_a^+ (plus denotes the canonically conjugate coordinate on the cotangent fiber; the coordinates x_i^+, c_a^+ are called “anti-fields” and “anti-ghosts”, respectively) of degrees 1, 0, $-1, -2$ respectively. The BV 2-form is locally written as

$$\omega = \delta x_i^+ \wedge \delta x^i + \delta c_a^+ \wedge \delta c^a$$

Next, one constructs a solution of the classical master equation $S^{(0)} \in C^\infty(\mathcal{F})_0$ satisfying the properties

- The restriction of $S^{(0)}$ to $F_{\text{cl}} \subset \mathcal{F}$ is the original classical action S_{cl} .
- The cohomological vector field $Q = \{S^{(0)}, \bullet\} \in \mathfrak{X}(\mathcal{F})_1$ induces⁶ the distribution E on $F_{\text{cl}} \subset \mathcal{F}$.

⁶In the sense that, locally, $Q = \sum_a c^a \rho(e_a) + \text{higher order monomials in fields of degree } \neq 0$. Here ρ is the inclusion of E as a subbundle of TF_{cl} (the “anchor map” in the terminology of Lie algebroids).

The construction of $S^{(0)}$ proceeds by adding corrections order by order in anti-fields, cf. [10] (see also [26]⁷). The next step is to extend $S^{(0)}$ to a solution of the quantum master equation $S = S^{(0)} + \hbar S^{(1)} + \hbar^2 S^{(2)} + \dots$ order by order in $\hbar$.

Then one regularizes the expression (4) to the BV integral

$$(6) \quad Z_{\text{BV}} = \int_{\mathcal{L} \subset \mathcal{F}} e^{\frac{i}{\hbar} S} \sqrt{\mu}|_{\mathcal{L}}$$

over a Lagrangian submanifold $\mathcal{L} \subset \mathcal{F}$, which now plays the role of the gauge-fixing condition. If $\mathcal{L}$ is such that the critical points of the integrand of (6) are isolated, one can proceed with the stationary phase formula and construct Feynman diagrams for the BV integral.

An important special case is when the gauge symmetry is given by a faithful action of a Lie group G on F_{cl} . Here E is a trivial bundle over F_{cl} with fiber the Lie algebra $\mathfrak{g} = \text{Lie}(G)$ and the anchor map ρ is the differential of the group action. In this case, the BV action is, locally:

$$(7) \quad S = S_{\text{cl}}(x) + x_i^+ c^a v_a^i(x) + \frac{1}{2} f_{bc}^a c_a^+ c^b c^c$$

where $v_a = \rho(e_a)$ are the vector fields on F_{cl} representing the infinitesimal action of the chosen generators of $\mathfrak{g}$; f_{bc}^a are the structure constants of $\mathfrak{g}$. The quantum master equation for S follows from: G -invariance of S_{cl} , the fact that $\rho : \mathfrak{g} \rightarrow \mathfrak{X}(F_{\text{cl}})$ is a Lie algebra homomorphism, the Jacobi identity for structure constants $\mathfrak{g}$, G -invariance of the volume form μ_{cl} (or equivalently, divergence-free property of vector fields v_a), unimodularity of $\mathfrak{g}$ (following from existence of a Haar measure on G). Choosing $\mathcal{L} \subset \mathcal{F} = T^*[-1](F_{\text{cl}} \oplus \mathfrak{g}[1])$ to be the conormal bundle to $\phi^{-1}(0) \oplus \mathfrak{g}[1] \subset F_{\text{cl}} \oplus \mathfrak{g}[1]$ with $\phi : F_{\text{cl}} \rightarrow \mathfrak{g}$ the gauge-fixing function (in the Faddeev-Popov sense), reduces the BV integral (6) to the Faddeev-Popov gauge-fixing of the integral (4). In this case the BV integral (6) coincides, up to a factor of the volume of G , with the original integral (4).

In the more general case, when the gauge symmetry does not come from a group action, one cannot directly compare (6) with (4), rather one should view the BV integral as a redefinition/regularization of (4).

An important generalization of the construction, see [11], is to allow the anchor map $\rho : E \rightarrow TF_{\text{cl}}$ to be non-injective and to introduce a resolution of the gauge symmetry – an exact⁸ sequence of vector bundles $0 \rightarrow E_k \rightarrow \dots \rightarrow E_1 = E \rightarrow TF_{\text{cl}}$. Then one constructs the space of BV fields as

$$(8) \quad \mathcal{F} = T^*[-1](E_1[1] \oplus \dots \oplus E_k[k])$$

The fiber coordinates of $E_i[i]$ are called “ i -th ghosts”. Examples of gauge systems where higher ghosts appear include the BF theory in dimension ≥ 4 [23] and the Courant sigma model [41].

⁷In the setting we outlined here, there may be obstructions for the construction of $S^{(0)}$ order by order in anti-fields as a solution of the classical master equation. Felder and Kazhdan [26] consider a different setup where the distribution E is not a part of initial data, but is recovered as the “maximal” symmetry of S . In this setting they prove existence and uniqueness (up to stable equivalence) of the solution of CME compatible with given initial data.

⁸In fact, in the context of local field theory, when each E_i is required to be the space of sections of some locally free sheaf over the spacetime manifold, one has to allow the resolution to have finite-dimensional cohomology.

In case when the gauge system has zero-modes, i.e. (set of critical points of $S_{\text{cl}})/E$ is not a discrete set of points, the procedure above does not lead directly to a perturbatively well-defined integral. Instead, one should use the construction of fiber BV integral (2) to pushforward the action S to zero-modes and then integrate over the zero-modes non-perturbatively (thus the problem of defining the path integral in this case becomes split into a perturbatively well-defined fiber BV integral, parameterized by zero-modes, and a finite-dimensional non-perturbative integral over zero-modes).

3.1.7. The classical part of the BV formalism. The classical part of the BV formalism, i.e. the structure modulo $\hbar$ -corrections is as follows:

- The space of BV fields $\mathcal{F}$ is a graded supermanifold with a degree -1 symplectic form ω . Functions on $\mathcal{F}$ form a Gerstenhaber (odd Poisson) algebra $(C^\infty(\mathcal{F}), \cdot, \{, \})$.
- The BV action $S \in C^\infty(\mathcal{F})$ is a solution of CME: $\{S, S\} = 0$. The Hamiltonian vector field $Q = \{S, \bullet\}$ on $\mathcal{F}$ is a cohomological vector field, $Q^2 = 0$.

3.1.8. Remarks.

- Above we defined, following [44], the BV Laplacian Δ on an SP manifold $\mathcal{F}$ as an operator acting on functions and depending on a choice of a Berezinian μ . An alternative viewpoint developed in [30, 46] is to define a *canonical* (independent on a choice of a Berezinian) BV Laplacian Δ_{can} acting on *semi-densities* on $\mathcal{F}$ (which is just required to be an odd-symplectic supermanifold in this setting), i.e. on sections of $\text{Ber}(\mathcal{F})^{\otimes \frac{1}{2}}$. The drawback of this approach is that there is no natural BV algebra associated to Δ_{can} .
- The fiber BV integral can be viewed as a quantum version of the homological perturbation theory ([28]): the classical part of the induced BRST operator Q' on $\mathcal{F}'$ can be recovered from the classical part of the original BRST operator Q on $\mathcal{F}$ by means of homological perturbation theory, which corresponds to the tree-level approximation to the fiber BV integral (see [36, 20] for details).
- For $\mathcal{F}$ a graded vector space with constant BV 2-form, a solution of the classical master equation is a generating function for a cyclic L_∞ algebra structure on $\mathcal{F}[-1]$, see [20].
- In much of the discussion in this section we were working on the premise that $\mathcal{F}$ is a finite-dimensional supermanifold and BV integrals make sense as measure-theoretic integrals. In the context of local field theory this is no longer true and certain corrections have to be made to the discussion. In particular, the BV Laplacian is not defined on local functionals, thus the classical master equation makes sense right away, but QME has to be understood in a regularized/renormalized sense, see [24]. BV integrals over Lagrangian submanifolds $\mathcal{L} \subset \mathcal{F}$ now become perturbative path integrals and the Stokes' theorem (independence on deformations of $\mathcal{L}$) for them has to be verified via the definition through Feynman diagrams. In the cases of Chern-Simons theory on closed 3-manifolds and Poisson sigma model on a disk, this analysis was performed in [7, 32, 17]. The analysis showed that arguments proving the Stokes' theorem for measure-theoretic BV integrals translate, in the setup of path integrals, to cancellation of the contribution of part of the boundary (so-called principal boundary strata) of configuration spaces of points on the spacetime manifold to the variation of partition function. Cancellation

of the contribution of rest of the boundary (the hidden boundary strata) has to be checked separately and does not hold automatically. In case of Chern-Simons theory with Hodge-theoretic propagator [7], certain hidden strata do contribute and lead to the “gravitational anomaly” in Chern-Simons theory – dependence of the partition function on the choice of a Riemannian metric, which can be cancelled at the cost of introducing framing on the spacetime manifold [49, 7]. Different constructions of propagator in Chern-Simons theory were suggested by Kontsevich [31] and by Bott and Cattaneo [14]. In the first case, the propagator depends on a choice of framing from the start, but leads to hidden strata not contributing to the variation of the partition function. In the construction of [14], one constructs the propagator using a choice of metric connection on the 3-manifold and corrects for the contributions of the hidden strata using a framing (see e.g. [20] for an in-depth discussion).

- (v) The classical BV formalism of Section 3.1.7 makes sense in the context of local field theory on closed spacetime manifolds without any regularization, if one restricts to variation calculus type functionals on the space of fields.

3.2. Chern-Simons theory.

3.2.1. *Classical Chern-Simons theory on closed 3-manifolds.* For M a closed 3-manifold and G a compact simply-connected Lie group with Lie algebra $\mathfrak{g}$, one defines the space of classical fields of Chern-Simons theory to be the space of connections in the trivial G -bundle over M , $\text{Conn}(M) \simeq \mathfrak{g} \otimes \Omega^1(M)$. The action functional is

$$(9) \quad S_{\text{CS}}(A) = \int_M \text{tr}_{\mathfrak{g}} \left(\frac{1}{2} A \wedge dA + \frac{1}{3} A \wedge A \wedge A \right)$$

where $A \in \text{Conn}(M)$ is a connection viewed as a 1-form on M with values in $\mathfrak{g}$, and we view $\mathfrak{g}$ as a matrix Lie algebra using the fundamental representation of G .

The critical point equation (Euler-Lagrange equation) for the action (9) is the zero curvature equation for the connection:

$$F_A = dA + A \wedge A = 0$$

Gauge transformations are given by the action of $\text{Map}(M, G)$ on the connection. For $g : M \rightarrow G$, the transformation is:

$$(10) \quad A \mapsto A^g = g^{-1} A g + g^{-1} dg$$

Infinitesimal gauge transformations are given by

$$(11) \quad A \mapsto A + \epsilon(d\alpha + [A, \alpha]) + O(\epsilon^2)$$

for the generator $\alpha : M \rightarrow \mathfrak{g}$.

Gauge transformations preserve the zero curvature condition and, moreover, preserve the action S_{CS} , provided that the generator $g \in \text{Map}(M, G)$ is in the connected component of identity in the mapping space. For a general gauge transformation, one has (with conventional normalization of trace in (9)):

$$S_{\text{CS}}(A^g) = S_{\text{CS}}(A) + 4\pi^2 n$$

with n an integer, $n = \langle g^*[\theta], [M] \rangle$ where $[M]$ is the fundamental class of M and $\theta = \frac{1}{24\pi^2} \text{tr}(g^{-1} dg)^{\wedge 3} \in \Omega^3(G)$ is the Cartan integral 3-form on the Lie group G with $[\theta]$ its cohomology class.

The space of critical points of the action modulo gauge transformations is the moduli space of flat G -connections on M and is, generally, a finite-dimensional singular variety $\mathcal{M}(M, G) \simeq \text{Hom}(\pi_1(M), G)/G$.

3.2.2. *Chern-Simons theory in the Batalin-Vilkovisky formalism.* The (minimal) space of BV fields is, by construction (5):

$$\begin{aligned}\mathcal{F} &= T^*[-1](\mathfrak{g} \otimes \Omega^1(M) \oplus \mathfrak{g} \otimes \Omega^0(M)[1]) \simeq \\ &\simeq \mathfrak{g} \otimes \Omega^0(M)[1] \oplus \mathfrak{g} \otimes \Omega^1(M) \oplus \mathfrak{g} \otimes \Omega^2(M)[-1] \oplus \mathfrak{g} \otimes \Omega^3(M)[-2]\end{aligned}$$

where we identify g^* with $\mathfrak{g}$ using the non-degenerate pairing $\text{tr}_{\mathfrak{g}} XY$. A point in $\mathcal{F}$ is a quadruple (c, A, A^+, c^+) of a ghost, a connection, an anti-field and an anti-ghost, i.e., $\mathfrak{g}$ -valued differential forms on M of degrees 0, 1, 2, 3 with ghost number 1, 0, -1, -2 respectively. The BV 2-form is:

$$\omega = \int_M \text{tr} (\delta A \wedge \delta A^+ + \delta c \wedge \delta c^+)$$

and the BV action (7) is:

$$(12) \quad S_{\text{BV CS}} = \int_M \text{tr} \left(\frac{1}{2} A \wedge dA + \frac{1}{6} A \wedge [A, A] + A^+ \wedge d_A c + \frac{1}{2} c^+ \wedge [c, c] \right)$$

where $d_A c = dc + [A, c]$ is the de Rham differential twisted by connection A . Lorentz gauge-fixing employed in [49, 6, 7] consists in choosing a Riemannian metric g on M and defining the gauge-fixing Lagrangian subspace $\mathcal{L} \subset \mathcal{F}$ as

$$(13) \quad \mathcal{L} = \{(c, A, A^+, c^+) \in \mathcal{F} \mid d^* A = 0, A^+ = d^* \bar{c}, c^+ = 0\}$$

Here $d^* = - * d *$ is the Hodge adjoint to the de Rham operator, and $\bar{c} \in \mathfrak{g} \otimes \Omega^3(M)[-1]$ is a new field introduced to parameterize the allowed values of the antifield A^+ on $\mathcal{L}$. Restriction of the BV action (12) to the Lagrangian $\mathcal{L}$ yields

$$S_{\text{BV CS}}|_{\mathcal{L}} = \int_M \text{tr} \left(\frac{1}{2} A \wedge dA + \frac{1}{6} A \wedge [A, A] + \bar{c} \wedge d^* d_A c \right)$$

the Faddeev-Popov action for Chern-Simons theory in Lorentz gauge.

3.2.3. *The perturbative Chern-Simons path integral on closed 3-manifolds.* Witten's idea in [49] was to use the action (9) to define diffeomorphism invariants of oriented 3-manifolds as

$$(14) \quad Z(M) = \int_{\text{Conn}(M)} \mathcal{D}A e^{\frac{i}{\hbar} S_{\text{CS}}(A)} \in \mathbb{C}$$

where $\int \mathcal{D}A$ should be understood as a single symbol for the path integral, which is to be made sense of. The integrand in (14) is manifestly independent on any geometric structure on M other than the orientation, so it is reasonable to expect $Z(M)$ to be a diffeomorphism invariant of M .

Under the simplifying assumption that the moduli space of flat connections $\mathcal{M}(M, G)$ is a discrete set of points (e.g. for M a rational homology sphere, for instance a lens space $L(p, q)$), the perturbative Chern-Simons path integral (14) splits into contributions of the points α of $\mathcal{M}(M, G)$:

$$Z(M) = \sum_{\alpha \in \mathcal{M}(M, G)} Z(M, A^{(\alpha)})$$

where $A^{(\alpha)}$ are some flat connections representing points $\alpha \in \mathcal{M}(M, G)$.

The contribution of a flat connection $A^{(\alpha)}$ to the perturbative partition function can be evaluated through the BV formalism as

$$(15) \quad Z(M, A^{(\alpha)}) = \int_{\mathcal{L}_{A^{(\alpha)}}} e^{\frac{i}{\hbar} S_{\text{BV CS}}}$$

where

$$\mathcal{L}_{A^{(\alpha)}} = \{(c, A^{(\alpha)} + a, A^+, c^+) \in \mathcal{F} \mid d_{A^{(\alpha)}}^* a = 0, A^+ = d_{A^{(\alpha)}}^* \bar{c}, c^+ = 0\}$$

– the construction (13) twisted by the flat connection $A^{(\alpha)}$. In (15), the right hand side is understood as the perturbative contribution of a single critical point, $c = a = A^+ = c^+ = 0$.

The analysis of the semiclassical approximation of the r.h.s. of (15) was done in [49] and extended to higher loop corrections in [6, 7] (without using the BV language explicitly), in case of an *acyclic* flat connection $A^{(\alpha)}$. The result is:

$$(16) \quad Z(M, A^{(\alpha)}, s) = e^{\frac{i}{\hbar} S_{\text{CS}}(A^{(\alpha)})} \tau(M, A^{(\alpha)}) e^{\frac{i\pi}{4} \eta(M, A^{(\alpha)}, g)} e^{ic(\hbar) S_{\text{grav}}(g, s)} \cdot \exp \left(\frac{i}{\hbar} \sum_{\text{connected 3-valent graphs } \Gamma} \frac{(i\hbar)^{l(\Gamma)}}{|\text{Aut}(\Gamma)|} \int_{\text{Conf}_{V(\Gamma)}(M)} \prod_{\text{edges } e \text{ of } \Gamma} \pi_{e_1 e_2}^* \eta \right)$$

Here the notations are as follows:

- $\tau(M, A^{(\alpha)})$ is the Ray-Singer torsion of M with respect to the local system defined by $A^{(\alpha)}$.
- $\eta(M, A^{(\alpha)}, g)$ is the Atiyah η -invariant of the elliptic operator $L_- = *d_{A^{(\alpha)}} + d_{A^{(\alpha)}}^* : \mathfrak{g} \otimes \Omega^{\text{odd}}(M) \rightarrow \mathfrak{g} \otimes \Omega^{\text{odd}}(M)$. It depends explicitly on the chosen Riemannian metric on M . By the Atiyah-Patodi-Singer theorem [49], the dependence of $\eta(M, A^{(\alpha)}, g)$ on the flat connection $A^{(\alpha)}$ can be spelled out explicitly: $\frac{\pi}{4}(\eta(M, A^{(\alpha)}, g) - \eta(M, 0, g)) = \frac{\hbar}{2\pi} S_{\text{CS}}(A^{(\alpha)})$ where h is the dual Coxeter number of the Lie algebra $\mathfrak{g}$.
- $V(\Gamma)$ and $l(\Gamma)$ are the number of vertices and the number of loops in the graph Γ ; $|\text{Aut}(\Gamma)|$ is the order of the automorphism group of Γ .
- $\text{Conf}_n(M)$ is the Fulton-Macpherson-Axelrod-Singer compactification of the configuration space of n -tuples of distinct points on M .
- $\eta \in \Omega^2(\text{Conf}_2(M))$ is the parametrix for the Hodge-theoretic inverse for the de Rham operator twisted by the flat connection, $\frac{d_{A^{(\alpha)}}^*}{[d_{A^{(\alpha)}}^*, d_{A^{(\alpha)}}^*]}$.
- $\pi_{ij} : \text{Conf}_n(M) \rightarrow \text{Conf}_2(M)$ is the projection by forgetting all points except the i -th and j -th.
- $S_{\text{grav}}(g, s)$ is the Chern-Simons invariant of the Levi-Civita connection on the spin bundle over M ; it depends on the trivialization of TM up to homotopy, i.e. on a choice of *framing* s of the 3-manifold M . $c(\hbar) \in \mathbb{C}[[\hbar]]$ is some universal power series.

The term $e^{ic(\hbar) S_{\text{grav}}(g, s)}$ in (16) is a local counterterm introduced by hand to cancel the dependence of the result on the choice of metric g on M , which comes in in 1-loop approximation because of Atiyah's η -invariant and in higher loops due to the contribution of certain hidden boundary strata of $\text{Conf}_n(M)$ (namely, those corresponding to collapse of all n points together) to the variation of $Z(M, A^{(\alpha)})$ with the variation of the metric. A theorem of Axelrod and Singer [7] asserts the existence of such a power series $c(\hbar)$ that the r.h.s. of (16) becomes metric

independent. However, this metric independence comes at the cost of introducing an explicit dependence on the choice of a framing s of M .

An important result of Axelrod and Singer implicit in (16) is the *finiteness* of Chern-Simons perturbation theory: in each order in $\hbar$ in (16) only finitely many Feynman graphs Γ contribute, and their contributions are given by integrals of smooth differential forms (products of propagators) over compact manifolds $\text{Conf}_n(M)$.

3.2.4. The boundary structure of classical Chern-Simons theory. The phase space associated to a closed 2-dimensional surface Σ in Chern-Simons theory is the space $\Phi_\Sigma = \text{Conn}(\Sigma)$ of G -connections on Σ and the constraint subspace is the subspace of *flat* G -connections $C_\Sigma = \text{FlatConn}(\Sigma) \subset \text{Conn}(\Sigma)$. The symplectic structure on Φ_Σ is given by

$$(17) \quad \omega_\Sigma = \frac{1}{2} \int_\Sigma \text{tr } \delta A \wedge \delta A$$

The subspace of flat connections is coisotropic with respect to ω_Σ and the characteristic foliation is given by gauge transformations (11) on Σ .

One construction leading to this phase space is through the Hamiltonian formalism for Chern-Simons theory on a cylinder $\Sigma \times [0, 1]$ (as e.g. in [49]). The coisotropic C_Σ arises in this approach as the image of the Legendre transform of the axial density of Chern-Simons action.

The other construction (cf. e.g. [22]) is by regarding Σ as a component of the boundary of some 3-manifold M (not necessarily a cylinder) and then constructing Φ_Σ as the space of pullbacks of bulk fields to the boundary; in this approach the symplectic form ω_Σ is constructed as the differential of

$$(18) \quad \alpha_\Sigma = \frac{1}{2} \int_\Sigma \text{tr } A \wedge \delta A \in \Omega^1(\Phi_\Sigma)$$

– the 1-form arising from the boundary term of the variation of the action S_{CS} . The coisotropic C_Σ arises as the subspace of boundary fields which can be extended to a solution of the Euler-Lagrange equation in a neighborhood of Σ in M .

The space of leaves of the characteristic foliation of C_Σ (i.e. the symplectic reduction of C_Σ) constitutes the *reduced* phase space of Chern-Simons theory, $\Phi_\Sigma^{\text{red}} = \underline{C}_\Sigma = \mathcal{M}(\Sigma, G)$ – the moduli space of flat connections on Σ . The symplectic structure (17) induces the Atiyah-Bott symplectic structure ω_Σ on the moduli space $\mathcal{M}(\Sigma, G)$ via symplectic reduction.

The trivial $U(1)$ -bundle L over Φ_Σ with connection ∇ defined by the 1-form $\frac{1}{\hbar} \alpha_\Sigma$ pushes forward to the reduction in case $\hbar = 2\pi/k$ with k a nonzero integer (the “level” of Chern-Simons theory) by identifying the $U(1)$ -fibers of L over gauge orbits in C_Σ (integrality condition for the level exactly ensures that the monodromy of ∇ in gauge orbits in C_Σ is trivial, thus the identification of fibers can be performed globally). The pushforward of L to the reduction is the Quillen $U(1)$ -bundle $\underline{L}_k = (\underline{L}_1)^{\otimes k}$ over the moduli space $\mathcal{M}(\Sigma, G)$. By construction, it is endowed with a connection $\underline{\nabla}_k$ of curvature $\frac{k}{2\pi} \omega_\Sigma$.

If Σ is the boundary of a 3-manifold M , one has a restriction map between the moduli space of flat connections $\pi : \mathcal{M}(M, G) \rightarrow \mathcal{M}(\Sigma, G)$ with Lagrangian image – the “evolution relation” (cf. e.g. [21]), and the exponential of the Chern-Simons action restricted to flat connections on M defines a horizontal section of the pullback $U(1)$ -bundle $\pi^* \underline{L}_k$ over $\mathcal{M}(M, G)$ – the “Hamilton-Jacobi action”.

Another important remark on classical Chern-Simons theory on manifolds with boundary is that if $\partial M = \Sigma \neq \emptyset$, then the action (9) is not gauge-invariant. Instead one has

$$S_{\text{CS}}(A^g) = S_{\text{CS}}(A) + \frac{1}{2} \int_{\Sigma} \text{tr } A \wedge dg g^{-1} - \frac{1}{6} \int_M \text{tr } (g^{-1} dg)^{\wedge 3}$$

where a Wess-Zumino term makes an appearance. Under infinitesimal gauge transformations (11), the action transforms as

$$S_{\text{CS}} \rightarrow S_{\text{CS}} + \epsilon \frac{1}{2} \int_{\Sigma} \text{tr } A \wedge d\alpha + O(\epsilon^2)$$

3.2.5. The space of states. The space of states $\mathcal{H}_{\Sigma}$ that Chern-Simons *quantum* theory associates to a surface Σ is the geometric quantization of the moduli space $\mathcal{M}(\Sigma, G)$ of flat connections on the surface, endowed with the Atiyah-Bott symplectic form ω_{Σ} and a prequantum Hermitian line bundle $\underline{L}_k$ with unitary connection ∇_k with $k \in \mathbb{Z} - \{0\}$ the fixed *level* of Chern-Simons theory. A complex polarization on $\mathcal{M}(\Sigma, G)$ required for the geometric quantization is inferred from an arbitrary choice of a complex structure J on Σ , which induces a splitting of 1-forms on Σ into $(1, 0)$ - and $(0, 1)$ -forms:

$$\mathfrak{g}_{\mathbb{C}} \otimes \Omega^1(\Sigma) = \mathfrak{g} \otimes \Omega^{1,0}(\Sigma) \oplus \mathfrak{g} \otimes \Omega^{0,1}(\Sigma)$$

where $\mathfrak{g}_{\mathbb{C}} = \mathbb{C} \otimes \mathfrak{g}$ the complexified Lie algebra. Performing the Marsden-Weinstein reduction with the moment map $\mathfrak{g} \otimes \Omega^1(\Sigma) \rightarrow \mathfrak{g} \otimes \Omega^2(\Sigma)$, sending a connection to its curvature 2-form, one obtains as the symplectic quotient the moduli space $\mathcal{M}(\Sigma, G)$ endowed with a Kähler structure (dependent on the choice of complex structure J on Σ). The output of geometric quantization of $\mathcal{M}(\Sigma, G)$ with this structure (cf. [8] for details) is the space of holomorphic sections of $\underline{L}_k$.

Let us summarize some properties of $\mathcal{H}_{\Sigma}$.

- $\mathcal{H}_{\Sigma}$ is a finite-dimensional Hilbert space (we are assuming that G is compact), with dimension depending on the level k . In particular, for $G = SU(2)$ the dimension is given by the Verlinde formula:

$$\dim \mathcal{H}_{\Sigma} = \left(\frac{k+2}{2} \right)^{g-1} \sum_{j=0}^k \frac{1}{\left(\sin \frac{\pi(j+1)}{k+2} \right)^{2g-2}}$$

where g is the genus of Σ .

- On the other hand, $\dim \mathcal{H}_{\Sigma}$ coincides with the Euler characteristic of the complex $\Omega^{0,\bullet}(\mathcal{M}(\Sigma, G), \underline{L}_k)$ with Dolbeault differential twisted by the connection ∇_k and is given by the Riemann-Roch-Hirzebruch formula as

$$\dim \mathcal{H}_{\Sigma} = \int_{\mathcal{M}(\Sigma, G)} \text{Td}(\mathcal{M}) e^{k\omega}$$

with $\text{Td}(\mathcal{M})$ the Todd class of the tangent bundle of $\mathcal{M}(\Sigma, G)$. Thus $\dim \mathcal{H}_{\Sigma}$ is a polynomial in k of degree $\frac{1}{2} \dim \mathcal{M} = (g-1) \dim(G)$ for $g \geq 2$, with the leading coefficient given by the symplectic volume of the moduli space of flat connections. This volume was separately calculated in [51] as

$$\text{Vol}(\mathcal{M}(\Sigma, G)) = \frac{\#Z(G) \cdot \text{Vol}(G)^{2g-2}}{(2\pi)^{\dim \mathcal{M}}} \cdot \sum_R \frac{1}{(\dim R)^{2g-2}}$$

where the sum is over irreducible unitary representations of G and $Z(G)$ stands for the center of G .

- $\mathcal{H}_\Sigma$ coincides with the space of conformal blocks of the Wess-Zumino-Witten model of conformal field theory on the Riemann surface (Σ, J) , with affine Lie algebra $\hat{\mathfrak{g}}$ acting on the (chiral) space of states for a circle $\mathcal{H}_{S^1}^{WZW}$.
- $\mathcal{H}_\Sigma$ is endowed with a projective representation of the mapping class group of Σ . In the non-perturbative approach to Chern-Simons theory [49] this representation is used to relate Chern-Simons partition functions for 3-manifolds related by surgery.
- One can consider $\mathcal{H}_\Sigma$ for a surface Σ with boundary (or punctures), which corresponds to the inclusion of Wilson line observables supported on a tangle in the 3-manifold in Chern-Simons theory. Then in addition, $\mathcal{H}_\Sigma$ carries an action of the affine Lie algebra $\hat{\mathfrak{g}}$ for each boundary component of Σ .
- Varying the complex structure J on Σ , one obtains a Hermitian vector bundle over the moduli space of complex structures on Σ with fiber $\mathcal{H}_\Sigma$. This vector bundle is endowed with a canonical projective connection – the Hitchin’s connection (corresponding to the stress-energy tensor of the WZW model).

3.3. AKSZ construction. In this Section we give a brief review of the Aleksandrov-Kontsevich-Schwarz-Zaboronsky (AKSZ) construction of topological sigma models in the Batalin-Vilkovisky formalism. This account is taken from [37]; we refer the reader to the original paper [1] and the later expositions in [18], [41] for details.

3.3.1. Target data. Let $\mathcal{M}$ be a degree n symplectic Q -manifold, i.e. a $\mathbb{Z}$ -graded manifold endowed with a degree 1 vector field Q satisfying $Q^2 = 0$ (the cohomological vector field) and with a degree n symplectic form $\omega \in \Omega^2(\mathcal{M})_n$ which is compatible with Q , i.e. $L_Q\omega = 0$.

Assume⁹ that Q has a Hamiltonian function $\Theta \in C^\infty(\mathcal{M})_{n+1}$ with $\{\Theta, \bullet\}_\omega = Q$ satisfying

$$(19) \quad \{\Theta, \Theta\}_\omega = 0$$

Also assume that ω is exact, with $\alpha \in \Omega^1(\mathcal{M})_n$ a primitive.

We call the set of data $(\mathcal{M}, Q, \omega = \delta\alpha, \Theta)$ a *Hamiltonian Q -manifold of degree n* .

3.3.2. AKSZ sigma model. Fix a Hamiltonian Q -manifold $\mathcal{M}$ of degree $n \geq -1$ and let Σ be an oriented closed manifold, $\dim \Sigma = n + 1$. Then one constructs the space of fields as the space of graded maps between graded manifolds from the degree-shifted tangent bundle $T[1]\Sigma$ to $\mathcal{M}$:

$$(20) \quad \mathcal{F}_\Sigma = \text{Map}(T[1]\Sigma, \mathcal{M})$$

⁹In fact (cf. [41]), for $n \neq -1$ the symplectic property of the cohomological vector field ($L_Q\omega = 0$) implies existence and uniqueness of the Hamiltonian $\Theta = \frac{1}{n+1} \iota_E \iota_Q \omega$ where E is the Euler vector field. The Maurer-Cartan equation (19) follows from $Q^2 = 0$ for $n \neq -2$. Also, for $n \neq 0$, a closed form ω is automatically exact.

It is a Q -manifold with the cohomological vector field coming from the lifting of Q on the target and of the de Rham operator d_Σ on Σ (viewed as a cohomological vector field on $T[1]\Sigma$) to the mapping space:

$$(21) \quad Q_\Sigma = (d_\Sigma)^{\text{lifted}} + (Q)^{\text{lifted}} \in \mathfrak{X}(\mathcal{F}_\Sigma)_1$$

The transgression map. The following natural maps

$$(22) \quad \begin{array}{ccc} \mathcal{F}_\Sigma \times T[1]\Sigma & \xrightarrow{\text{ev}} & \mathcal{M} \\ p \downarrow & & \\ & & \mathcal{F}_\Sigma \end{array}$$

(where p is the projection to the first factor) allow us to define the transgression map

$$(23) \quad \tau_\Sigma = p_* \text{ev}^* : \Omega^\bullet(\mathcal{M}) \rightarrow \Omega^\bullet(\mathcal{F}_\Sigma)$$

Here p_* is the fiber integration over $T[1]\Sigma$ (with the canonical integration measure). The map τ_Σ preserves the de Rham degree of a form, but changes the internal grading (the ghost number) by $-\dim \Sigma$.

If $u \in \mathfrak{X}(T[1]\Sigma)$, $v \in \mathfrak{X}(\mathcal{M})$ are any vector fields on the source and the target and $\psi \in \Omega^\bullet(\mathcal{M})$ is a form on the target, then for the Lie derivatives of the transgressed form along the lifted vector fields we have

$$(24) \quad L_{u^{\text{lifted}}} \tau_\Sigma(\psi) = 0$$

$$(25) \quad L_{v^{\text{lifted}}} \tau_\Sigma(\psi) = (-1)^{|v| \dim \Sigma} \tau_\Sigma(L_v \psi)$$

The integrated version of (24) is that for $\Phi : T[1]\Sigma \rightarrow T[1]\Sigma$ a diffeomorphism of the source, we have

$$(26) \quad (\Phi^*)^* \tau_\Sigma(\psi) = \tau_\Sigma(\psi)$$

where $\Phi^* : \mathcal{F}_\Sigma \rightarrow \mathcal{F}_\Sigma$ is the lifting of Φ to the mapping space and $(\Phi^*)^* : \Omega^\bullet(\mathcal{F}_\Sigma) \rightarrow \Omega^\bullet(\mathcal{F}_\Sigma)$ is the pull-back by Φ^* .

The BV 2-form and the master action. One obtains the degree -1 symplectic form (the “BV 2-form”) on $\mathcal{F}_\Sigma$ from the target by transgression¹⁰:

$$(27) \quad \Omega_\Sigma = (-1)^{\dim \Sigma} \tau_\Sigma(\omega) \in \Omega^2(\mathcal{F}_\Sigma)_{-1}$$

The Hamiltonian function for Q_Σ (the master action) is constructed as

$$(28) \quad S_\Sigma = \underbrace{\iota_{d_\Sigma^{\text{lifted}}} \tau_\Sigma(\alpha)}_{S_\Sigma^{\text{kin}}} + \underbrace{\tau_\Sigma(\Theta)}_{S_\Sigma^{\text{target}}} \in C^\infty(\mathcal{F}_\Sigma)_0$$

It automatically satisfies the classical master equation

$$\{S_\Sigma, S_\Sigma\}_{\Omega_\Sigma} = 0$$

One can summarize the construction above by saying that we have a degree -1 Hamiltonian Q -manifold structure on the mapping space (20): $(\mathcal{F}_\Sigma, Q_\Sigma, \Omega_\Sigma, S_\Sigma)$. The primitive 1-form for the BV 2-form can also be constructed by transgression as $\tau_\Sigma(\alpha)$.

¹⁰We introduce the sign $(-1)^{\dim \Sigma}$ in this definition to avoid signs in the formula for the action below. The reader may encounter different sign conventions for the AKSZ construction in the literature.

Note that exact shifts of the target 1-form, $\alpha \mapsto \alpha + \delta f$, with $f \in C^\infty(M)$ leave S_Σ unchanged.

Why AKSZ theory is topological. For $\phi \in \text{Diff}(\Sigma)$ a diffeomorphism of Σ , denote $\tilde{\phi} \in \text{Diff}(T[1]\Sigma)$ the tangent lift of ϕ to $T[1]\Sigma$. Then

$$(29) \quad (\tilde{\phi}^*)^* S_\Sigma = S_\Sigma, \quad (\tilde{\phi}^*)^* \Omega_\Sigma = \Omega_\Sigma$$

because of (26) and because $d_\Sigma \in \mathfrak{X}(T[1]\Sigma)$ commutes with $\tilde{\phi}$, since the latter is a tangent lift.

In coordinates. Let x^a be local homogeneous coordinates on the target $\mathcal{M}$, let u^μ be local coordinates on Σ and $\theta^\mu = du^\mu$ be the associated degree 1 fiber coordinates on $T[1]\Sigma$. Then locally an element of $\mathcal{F}_\Sigma$ is parameterized by:

$$(30) \quad X^a(u, \theta) = \sum_{k=0}^{\dim \Sigma} \underbrace{\sum_{1 \leq \mu_1 < \dots < \mu_k \leq \dim \Sigma} X_{\mu_1 \dots \mu_k}^a(u) \theta^{\mu_1} \dots \theta^{\mu_k}}_{X_{(k)}^a(u, \theta)}$$

The coefficient functions $X_{\mu_1 \dots \mu_k}^a(u)$ are local coordinates of degree $|x^a| - k$ on the mapping space $\mathcal{F}_\Sigma$. The expression (30) is known as (the component of) the *superfield*, and it can be regarded as a generating function for the coordinates on the mapping space $\mathcal{F}_\Sigma$.

For any function $f \in C^\infty(\mathcal{M})$, we have

$$(31) \quad \begin{aligned} \text{ev}^* f &= f(X) = \\ &= f(X_{(0)}) + X_{\geq 1}^a \cdot (\partial_a f)(X_{(0)}) + \frac{1}{2} X_{\geq 1}^a X_{\geq 1}^b \cdot (\partial_b \partial_a f)(X_{(0)}) + \dots \\ &\in C^\infty(\mathcal{F}_\Sigma \times T[1]\Sigma) \end{aligned}$$

where we denote by $X_{\geq 1}^a = \sum_{k \geq 1} X_{(k)}^a$ the part of the superfield of positive de Rham degree with respect to Σ ; ev is the horizontal arrow in (22).

Let α and ω be locally given as $\alpha = \alpha_a(x) \delta x^a$ and $\omega = \frac{1}{2} \omega_{ab}(x) \delta x^a \wedge \delta x^b$. Then the BV 2-form and its primitive are:

$$\Omega_\Sigma = (-1)^{\dim \Sigma} \int_\Sigma \frac{1}{2} \omega_{ab}(X) \delta X^a \wedge \delta X^b, \quad \alpha_\Sigma = \int_\Sigma \alpha_a(X) \delta X^a$$

(Note that we use δ to denote the de Rham differential on the target $\mathcal{M}$ and on the mapping space $\mathcal{F}_\Sigma$. We reserve symbol d for the de Rham differential on the source Σ .)

The master action is:

$$S_\Sigma(X) = \underbrace{\int_\Sigma \alpha_a(X) dX^a}_{S_\Sigma^{\text{kin}}} + \underbrace{\int_\Sigma \Theta(X)}_{S_\Sigma^{\text{target}}}$$

If the cohomological vector field on the target is locally written as $Q = Q^a(x) \frac{\partial}{\partial x^a}$ then the cohomological vector field (21) is determined by its action on the components of the superfield:

$$Q_\Sigma X^a = dX^a + Q^a(X)$$

The critical points of S_Σ are (with our sign conventions) Q -anti-morphisms between $T[1]\Sigma$ and $\mathcal{M}$, i.e. Q -morphisms between $(T[1]\Sigma, d)$ and $(\mathcal{M}, -Q)$.

3.3.3. *Examples.* Here we recall some of the standard examples of the AKSZ construction.

Chern-Simons theory [1]. Let $\mathfrak{g}$ be a quadratic Lie algebra, i.e. a Lie algebra with a non-degenerate invariant pairing $(\cdot, \cdot)$. Denote by $\psi : \mathfrak{g}[1] \rightarrow \mathfrak{g}$ the degree 1 $\mathfrak{g}$ -valued coordinate on $\mathfrak{g}[1]$. We choose the target Hamiltonian Q -manifold of degree 2 as

$$(32) \quad \mathcal{M} = \mathfrak{g}[1], \quad Q = \left\langle \frac{1}{2}[\psi, \psi], \frac{\partial}{\partial \psi} \right\rangle, \\ \omega = \frac{1}{2}(\delta\psi, \delta\psi), \quad \alpha = \frac{1}{2}(\psi, \delta\psi), \quad \Theta = \frac{1}{6}(\psi, [\psi, \psi])$$

where $\langle \cdot, \cdot \rangle$ is the canonical pairing between $\mathfrak{g}$ and $\mathfrak{g}^*$. The associated AKSZ sigma model on a closed oriented 3-manifold Σ has the space of fields

$$\mathcal{F}_\Sigma = \text{Map}(T[1]\Sigma, \mathfrak{g}[1]) \cong \mathfrak{g}[1] \otimes \Omega^\bullet(\Sigma)$$

The superfield is

$$(33) \quad A = A_{(0)} + A_{(1)} + A_{(2)} + A_{(3)}$$

with $A_{(k)}$ a coordinate on $\mathcal{F}_\Sigma$ with values in $\mathfrak{g}$ -valued k -forms on Σ , with internal degree (ghost number) $1 - k$, for $k = 0, 1, 2, 3$. The BV 2-form is

$$\Omega_\Sigma = -\frac{1}{2} \int_\Sigma (\delta A, \delta A)$$

and the action is

$$S_\Sigma = \int_\Sigma \frac{1}{2}(A, dA) + \frac{1}{6}(A, [A, A])$$

This is the action of Chern-Simons theory in the Batalin-Vilkovisky formalism (12) written in terms of the superfield (33).

In case $\mathfrak{g} = \mathbb{R}$ with abelian Lie algebra structure, we have $Q = \Theta = 0$ on the target and

$$(34) \quad \mathcal{F}_\Sigma = \Omega^\bullet(\Sigma)[1], \quad \Omega_\Sigma = -\frac{1}{2} \int_\Sigma \delta A \wedge \delta A, \quad S_\Sigma = \frac{1}{2} \int_\Sigma A \wedge dA$$

This is the *abelian* Chern-Simons theory in the BV formalism.

BF theory. For $\mathfrak{g}$ a Lie algebra (not necessarily quadratic¹¹) and D a non-negative integer, we define the target Hamiltonian Q -manifold of degree $D - 1$ as

$$(35) \quad \mathcal{M} = \mathfrak{g}[1] \oplus \mathfrak{g}^*[D - 2], \quad Q = \left\langle \frac{1}{2}[\psi, \psi], \frac{\partial}{\partial \psi} \right\rangle + \left\langle \text{ad}_\psi^* \xi, \frac{\partial}{\partial \xi} \right\rangle, \\ \omega = \langle \delta \xi, \delta \psi \rangle, \quad \alpha = \langle \xi, \delta \psi \rangle, \quad \Theta = \frac{1}{2} \langle \xi, [\psi, \psi] \rangle$$

Here $\psi : \mathfrak{g}[1] \rightarrow \mathfrak{g}$ is as before and $\xi : \mathfrak{g}^*[D - 2] \rightarrow \mathfrak{g}^*$ is the $\mathfrak{g}^*$ -valued coordinate on $\mathfrak{g}^*[D - 2]$ of degree $D - 2$; ad^* is the coadjoint action of $\mathfrak{g}$ on $\mathfrak{g}^*$.

The associated AKSZ sigma model on a closed oriented D -manifold Σ has the space of fields:

$$(36) \quad \mathcal{F}_\Sigma = \mathfrak{g}[1] \otimes \Omega^\bullet(\Sigma) \oplus \mathfrak{g}^*[D - 2] \otimes \Omega^\bullet(\Sigma)$$

¹¹However, for the consistency of quantization, in particular for the quantum master equation (1), one should require that $\mathfrak{g}$ is unimodular.

The superfields associated to ψ and ξ are respectively

$$A = \sum_{k=0}^D A_{(k)}, \quad B = \sum_{k=0}^D B_{(k)}$$

with $A_{(k)}$ a $\mathfrak{g}$ -valued k -form on Σ of internal degree $1-k$; $B_{(k)}$ is a $\mathfrak{g}^*$ -valued k -form of internal degree $D-2-k$. The BV 2-form and the action are:

$$(37) \quad \Omega_\Sigma = (-1)^D \int_\Sigma \langle \delta B, \delta A \rangle, \quad S_\Sigma = \int_\Sigma \left\langle B, dA + \frac{1}{2}[A, A] \right\rangle$$

This is the BF theory in BV formalism.

In the abelian case, $\mathfrak{g} = \mathbb{R}$, we have $Q = \Theta = 0$ on the target and

$$(38) \quad \mathcal{F}_\Sigma = \Omega^\bullet(\Sigma)[1] \oplus \Omega^\bullet(\Sigma)[D-2],$$

$$\Omega_\Sigma = (-1)^D \int_\Sigma \delta B \wedge \delta A, \quad S_\Sigma = \int_\Sigma B \wedge dA$$

The Poisson sigma model [18]. Let M be a manifold endowed with a Poisson bivector $\pi \in \Gamma(M, \wedge^2 TM)$. We construct the target Hamiltonian Q -manifold of degree 1 as

$$(39) \quad \mathcal{M} = T^*[1]M,$$

$$Q = \left\langle \pi(x), p \wedge \frac{\partial}{\partial x} \right\rangle_{\wedge^2 T_x M} + \frac{1}{2} \left\langle \frac{\partial}{\partial x} \pi(x), (p \wedge p) \otimes \frac{\partial}{\partial p} \right\rangle_{(\wedge^2 T \otimes T^*)_x M},$$

$$\omega = \langle \delta p, \delta x \rangle, \quad \alpha = \langle p, \delta x \rangle, \quad \Theta = \frac{1}{2} \langle \pi(x), p \wedge p \rangle_{\wedge^2 T_x M}$$

Here x and p stand for the local base and fiber coordinates on $T^*[1]M$ respectively. Note that all objects in (39) are globally well defined.

The corresponding AKSZ sigma model on an oriented closed surface Σ has the space of fields

$$\mathcal{F}_\Sigma = \text{Map}(T[1]\Sigma, T^*[1]M)$$

with superfields

$$X = X_{(0)} + X_{(1)} + X_{(2)}, \quad \eta = \eta_{(0)} + \eta_{(1)} + \eta_{(2)}$$

associated to local coordinates x and p on the target, respectively. Here $X_{(k)}$ and $\eta_{(k)}$ are k -forms on Σ with internal degrees $-k$ and $1-k$ respectively. The BV 2-form and the action are:

$$\Omega_\Sigma = \int_\Sigma \langle \delta \eta, \delta X \rangle, \quad S_\Sigma = \int_\Sigma \langle \eta, dX \rangle + \frac{1}{2} \langle \pi(X), \eta \wedge \eta \rangle$$

3.4. Functorial topological field theory. In Atiyah's axiomatic approach [5], an $(n+1)$ -dimensional topological quantum field theory associates:

- To a closed oriented n -manifold Σ – a Hilbert space $\mathcal{H}_\Sigma$ (the *space of states*).
- To an oriented $(n+1)$ -cobordism M from Σ_{in} to Σ_{out} (i.e. with a splitting of the boundary as $\partial M = \bar{\Sigma}_{\text{in}} \sqcup \Sigma_{\text{out}}$) – a linear map of vector spaces $Z_M : \mathcal{H}_{\Sigma_{\text{in}}} \rightarrow \mathcal{H}_{\Sigma_{\text{out}}}$ (the *partition function*).
- Orientation-preserving diffeomorphisms $\phi : \Sigma \rightarrow \Sigma'$ act on the spaces of states by unitary maps $\rho(\phi) : \mathcal{H}_\Sigma \rightarrow \mathcal{H}_{\Sigma'}$. The orientation-reversing identity diffeomorphism $s_\Sigma : \Sigma \rightarrow \bar{\Sigma}$ (the bar denotes the opposite orientation) acts by a $\mathbb{C}$ -anti-linear map $\sigma_\Sigma = \rho(s_\Sigma) : \mathcal{H}_\Sigma \rightarrow \mathcal{H}_{\bar{\Sigma}}$.

One requires this association to satisfy the following axioms:

- (i) (Multiplicativity) $\mathcal{H}_{\Sigma \sqcup \Sigma'} = \mathcal{H}_{\Sigma} \otimes \mathcal{H}_{\Sigma'}$ for the spaces of states and

$$Z_{M \sqcup M'} = Z_M \otimes Z_{M'} : \mathcal{H}_{\Sigma_{\text{in}}} \otimes \mathcal{H}_{\Sigma'_{\text{in}}} \rightarrow \mathcal{H}_{\Sigma_{\text{out}}} \otimes \mathcal{H}_{\Sigma'_{\text{out}}}$$

for the partition functions.

- (ii) (Gluing) For two cobordisms $M_1 : \Sigma_1 \rightarrow \Sigma_2$ and $M_2 : \Sigma_2 \rightarrow \Sigma_3$, one can construct the glued cobordism $M_1 \cup_{\Sigma_2} M_2 : \Sigma_1 \rightarrow \Sigma_3$ and the corresponding partition function is the composition of partition functions for M_1 and M_2 as linear maps:

$$Z_{M_1 \cup_{\Sigma_2} M_2} = Z_{M_2} \circ Z_{M_1} : \mathcal{H}_{\Sigma_1} \rightarrow \mathcal{H}_{\Sigma_3}$$

- (iii) (Normalization) $\mathcal{H}_{\emptyset} = \mathbb{C}$, $Z_{\Sigma \times [0,1]} = \text{id} : \mathcal{H}_{\Sigma} \rightarrow \mathcal{H}_{\Sigma}$.

- (iv) For $\phi : M \rightarrow M'$ a diffeomorphism, the following diagram commutes:

$$\begin{array}{ccc} \mathcal{H}_{\Sigma_{\text{in}}} & \xrightarrow{Z_M} & \mathcal{H}_{\Sigma_{\text{out}}} \\ \rho(\phi|_{\Sigma_{\text{in}}}) \downarrow & & \downarrow \rho(\phi|_{\Sigma_{\text{out}}}) \\ \mathcal{H}_{\Sigma'_{\text{in}}} & \xrightarrow{Z_{M'}} & \mathcal{H}_{\Sigma'_{\text{out}}} \end{array}$$

(In particular, Z_M is invariant under diffeomorphisms of M relative to the boundary of M .)

- (v) (Symmetry) The natural diffeomorphism $\Sigma \sqcup \Sigma' \rightarrow \Sigma' \sqcup \Sigma$ is sent by ρ to the natural isomorphism $\mathcal{H}_{\Sigma} \otimes \mathcal{H}_{\Sigma'} \rightarrow \mathcal{H}_{\Sigma'} \otimes \mathcal{H}_{\Sigma}$.
- (vi) The partition function for the cylinder $\Sigma \times [0,1]$, regarded as a cobordism $\Sigma \times \bar{\Sigma} \rightarrow \emptyset$, together with the anti-linear map $(\sigma_{\Sigma})^{-1} : \mathcal{H}_{\bar{\Sigma}} \rightarrow \mathcal{H}_{\Sigma}$, yields the Hermitian inner product $\langle, \rangle : \mathcal{H}_{\Sigma} \times \mathcal{H}_{\Sigma} \rightarrow \mathbb{C}$.

The following are immediate consequences of the axioms:

- For M a closed $(n+1)$ -manifold, regarded as a cobordism $\emptyset \rightarrow \emptyset$, the partition function $Z_M \in \mathbb{C}$ is a number invariant under orientation-preserving diffeomorphisms of M .
- $Z_{\Sigma \times S^1} = \dim \mathcal{H}_{\Sigma} \in \mathbb{Z}$. In particular, this implies that spaces of states have to be finite-dimensional. More generally, for a *mapping torus* $\frac{\Sigma \times [0,1]}{\Sigma \times \{0\} \xrightarrow{\phi} \Sigma \times \{1\}}$ with $\phi : \Sigma \rightarrow \Sigma$ a gluing diffeomorphism, the partition function is $Z = \text{Tr}_{\mathcal{H}_{\Sigma}} \rho(\phi)$.

A short way to formulate Atiyah's axioms is to say that a TQFT is a functor of symmetric monoidal categories from the category of oriented cobordisms (with composition given by gluing and monoidal structure given by disjoint unions) to the category of Hilbert spaces, and diffeomorphisms act by natural transformations of the functor (alternatively, one can view diffeomorphisms as 2-morphisms; then $(\mathcal{H}, Z, \rho)$ becomes a functor of 2-categories).

In dimensions 1 and 2, Atiyah's TQFTs admit a straightforward complete classification. For $n+1=1$, a TQFT, up to equivalence, is specified by a non-negative integer – the dimension of the space of states for a point. For $n+1=2$, the classification of TQFTs is given by a result of Dijkgraaf by isomorphism classes of unital Frobenius algebras (in particular, the underlying vector space of the algebra is $\mathcal{H}_{S^1}$, multiplication is the partition function for a pair of pants regarded as a cobordism $S^1 \sqcup S^1 \rightarrow S^1$, the unit is the partition function for the disk viewed as

a cobordism $\emptyset \rightarrow S^1$ etc.) In dimension 3 and above the complete classification of Atiyah's TQFTs is not known.

Baez and Dolan [9] suggested an enhancement of Atiyah's notion of TQFT as a functor from the (∞, n) -extension of the cobordism category (which essentially allows gluing-cutting with higher codimension strata) – the “extended” TQFT. Lurie proved in [34] that a fully extended TQFT (i.e. extended down to strata of maximal codimension – points) is determined by its values on points, thus providing a complete classification result for fully extended TQFTs. It is not known however whether e.g. Chern-Simons theory can be embedded in a fully extended TQFT.

4. MAIN RESULTS

4.1. Paper “Remarks on Chern-Simons invariants” [20], with A. S. Cattaneo. Given a differential graded Frobenius algebra $C^\bullet$ with invariant pairing of degree -3 , and a quadratic Lie algebra $\mathfrak{g}$, we construct the algebraic model for Chern-Simons theory in the framework of the Batalin-Vilkovisky formalism, with the space of BV fields $\mathcal{F} = \mathfrak{g} \otimes C^\bullet[1]$, with BV 2-form coming from the pairing on $C^\bullet$ tensored with the ad-invariant pairing on $\mathfrak{g}$. Out of the structure constants of algebraic operations on $C^\bullet$ and $\mathfrak{g}$, we construct the BV action $S \in C^\infty(\mathcal{F})_0$ – a cubic polynomial in fields solving the quantum master equation. In the case $C^\bullet = \Omega^\bullet(M)$, the de Rham algebra of M , the action S becomes the master action for Chern-Simons theory (12).

In case of a finite-dimensional dg Frobenius algebra $C^\bullet$, we analyze in detail the effective BV action induced on the cohomology $\mathfrak{g} \otimes H^\bullet(C^\bullet)[1]$ via the pushforward construction (2), and the canonical transformations induced by the deformations of the induction data (deformations of the gauge-fixing Lagrangian and of the choice of representatives of cohomology classes of $H^\bullet(C^\bullet)$ in $C^\bullet$). Then we analyze the invariants coming from the effective action on cohomology by considering it modulo canonical transformations coming from deformations of the induction data (taking a quotient over *all* canonical transformation, one would kill interesting invariants).

In the special case of $H^1(C^\bullet) = 0$, we prove that the effective action is of the form $W_{\text{prod}} + F(\hbar)$ where W_{prod} is a cubic polynomial on $\mathfrak{g} \otimes H^\bullet(C^\bullet)[1]$ corresponding to the Lie bracket on $\mathfrak{g}$ -valued cohomology of $C^\bullet$, while $F(\hbar) \in \hbar^2\mathbb{C}[[\hbar]]$ is a formal power series, comprising the complete invariant of the effective action under admissible (in the sense above) canonical transformations.

In the case when $\dim H^1(C^\bullet) = 1$ and $C^\bullet$ is a formal dg algebra, we prove that the 1-loop part of the effective action restricted to the Maurer-Cartan set of the graded Lie algebra $\mathfrak{g} \otimes H^\bullet(C^\bullet)$ is an invariant.

Then we extend the results developed for $C^\bullet$ finite-dimensional, to the differential-geometric Chern-Simons theory with $C^\bullet = \Omega^\bullet(M)$, taking care of the translation of proofs valid for finite-dimensional BV integrals in the language of perturbative path integrals, with Feynman diagrams given by integrals over configuration spaces. We prove that one gets invariants of closed, oriented, framed 3-manifolds from the effective BV action for Chern-Simons theory induced on de Rham cohomology of the 3-manifold.

This treatment generalizes Axelrod-Singer's treatment of perturbative Chern-Simons path integral to the case of non-acyclic flat background connections.

4.2. Paper “Finite-dimensional AKSZ-BV theories” [15], with F. Bonechi and M. Zabzine. In the framework of AKSZ sigma models, we consider the symplectic reduction of the coisotropic subspace of the mapping space $\text{Map}(T[1]\Sigma, \mathcal{M})$ given by the zero locus of Q_0 – the lifting of the de Rham differential on Σ to a cohomological vector field on the mapping space (the source term of the full AKSZ cohomological vector field (21)). The action of the sigma model is a reducible function with respect to this reduction. The reduced theory provides the leading term in the perturbation expansion for the zero-mode effective action for the AKSZ sigma model. The reduced mapping space is finite-dimensional and can be viewed as a mapping space of sheaves from the de Rham cohomology of Σ as a sheaf over a point to the structure sheaf of $\mathcal{M}$. We develop the corresponding generalization of the AKSZ construction, based on replacing the source by a differential graded Frobenius algebra. We also consider in detail two examples of the symplectic reduction to source zero-modes: the Poisson sigma model (on closed surfaces and on surfaces with boundary, with coisotropic boundary conditions introduced in [19]) and the Courant sigma-model.

4.3. Paper “One-dimensional Chern-Simons theory” [4], with A. Alekseev. We start by considering the “1-dimensional Chern-Simons theory” on a circle, a model in BV formalism with $\mathbb{Z}_2$ -graded space of BV fields $\mathcal{F} = \Pi\mathfrak{g} \otimes \Omega^0(S^1) \oplus \mathfrak{g} \otimes \Omega^1(S^1)$ (where Π is the parity-reversal operation and $\mathfrak{g}$ a fixed quadratic Lie algebra) and the action $S = \int_{S^1} \langle \psi, d\psi + [A, \psi] \rangle$ with ψ and A the odd 0-form field and the even 1-form field, respectively. This theory arises from the AKSZ construction (Section 3.3) with the target supermanifold $\Pi\mathfrak{g}$ (same as in 3-dimensional Chern-Simons theory), but 1-dimensional source instead of a 3-dimensional one.

We calculate the BV pushforward to simplicial cochains of a polygon (triangulation of S^1), in the spirit of the work [35, 36], obtaining an explicit result. The resulting effective action on cochains depends on the simplicial fields non-locally.

The problem of splitting the result for the polygon into contributions of individual 1-simplices is solved by going to operator formalism and constructing 1-dimensional simplicial Chern-Simons theory as Atiyah’s topological quantum field theory on the monoidal category of triangulated 1-cobordisms with additional operations given by merging of neighboring 1-simplices (simplicial aggregations). The value of 1-dimensional Chern-Simons theory on a point (the space of states) is the spinor module and the value on a triangulated 1-cobordism (the partition function) is given in terms of Clifford exponentials. Concatenation of triangulated 1-cobordisms corresponds to the multiplication in the Clifford algebra $\text{Cl}(\mathfrak{g})$ and simplicial aggregations corresponds to certain finite-dimensional fiber BV integrals. Moreover, the resulting partition functions satisfy a version of the quantum master equation extended by boundary terms.

We prove that the value of 1-dimensional Chern-Simons theory, as constructed via Clifford exponentials, on a polygon (triangulated circle) is the same as calculated originally via perturbative path integral. Also, we prove that the partition function for a triangulated interval can be written as a path integral with boundary conditions given by a complex Lagrangian foliation of $\Pi\mathfrak{g}$ imposed on the restrictions of field ψ to the boundary points of the interval. In the usual way for the BV formalism, one also imposes the gauge-fixing for fields in the bulk by specifying a Lagrangian subspace in the bulk fields (with fixed boundary conditions).

4.4. Paper “The Poisson sigma model on closed surfaces” [13], with F. Bonechi and A. S. Cattaneo. We study the effective action of the Poisson sigma model on a closed 2-dimensional surface, regarding the Poisson structure as a perturbation, induced on the zero-modes of the unperturbed theory. The perturbation theory is developed around constant maps and we employ the tools of formal geometry of the target to glue the perturbative results into the global effective action.

In two important cases, the case of the surface being a torus and the case of a regular unimodular Poisson structure, we manage to prove vanishing of the quantum corrections in the effective action. In the case of a torus we employ a version of the axial gauge, while in the case of a regular unimodular Poisson structure we make use of the appropriate local normal form theorem.

In the case of a torus, with additional assumption that the target Poisson structure is Kähler, we are able to prove that the partition function (obtained from the effective action by evaluating the BV integral over the zero-mode space) is the Euler characteristic of the target. We also present an argument, using a regularization of the effective action based on the source supersymmetry, that the same result for the torus partition function holds for a broader class of Poisson structures.

Finally, in the case of a general surface and a general Poisson structure, we prove that perturbative effective actions constructed in formal neighborhoods of points of the target can indeed be glued, using quantum canonical transformations, into a global object.

4.5. Papers “Classical BV theories on manifolds with boundary” [21] and “Classical and quantum Lagrangian field theories with boundary” [22], with A. S. Cattaneo and N. Reshetikhin. We extend the Batalin-Vilkovisky formalism for gauge theories to spacetime manifolds with boundary (and corners) in a way consistent with gluing.

In this setup to a boundary component Σ of the spacetime one associates a phase space $\mathcal{F}_\Sigma$ – a Hamiltonian Q -manifold with an exact degree 0 symplectic form $\omega_\Sigma = \delta\alpha_\Sigma$ (this set of data is known as a BFV manifold, for the Batalin-Fradkin-Vilkovisky cohomological resolution of coisotropic reductions in symplectic geometry). To the spacetime manifold M one associates a Q -manifold $\mathcal{F}_M$ with a degree -1 symplectic form ω and an action function S which, instead of being a Hamiltonian for the cohomological vector field, satisfies the equation $\iota_Q\omega = dS + \pi^*\alpha_\partial$ where $\pi : \mathcal{F}_M \rightarrow \mathcal{F}_{\partial M}$ is a surjective submersion – the pullback of bulk fields to the boundary (in first-order theories; more generally $\mathcal{F}_{\partial M}$ is constructed as a symplectic reduction of the space of normal jets of bulk fields at the boundary and thus by construction comes with a surjective submersion π). Instead of the classical master equation in the bulk, one has the equation $\iota_Q\iota_Q\omega = \pi^*S_\partial$ where S_∂ is the BFV action on the boundary – the degree 1 Hamiltonian for Q_∂ .

A classical gauge theory in this setup is therefore a functor of symmetric monoidal categories from the category of spacetime cobordisms (endowed with the local geometric data appropriate to the model in question) to the *BFV category*. On the target side, the objects are the BFV phase spaces and the morphisms are the BV spaces of bulk fields. Composition is given by fiber products and monoidal structure is given by Cartesian products of BFV manifolds. This functorial picture admits

a higher category extension (where on the source side one essentially allows space-time manifolds with corners); we construct the appropriate extension of the target category.

All AKSZ sigma models fit naturally in the *BV-BFV formalism* which we develop. In particular, the boundary BFV phase space is also a mapping space with the same AKSZ target, $\mathcal{F}_\Sigma = \text{Map}(T[1]\Sigma, \mathcal{M})$, and the BFV action is again given by the formula (28), where one replaces the bulk by the boundary. We also analyze several other examples of non-topological gauge theories, in particular the electrodynamics and the pure Yang-Mills theory on Riemannian manifolds with boundary.

We also analyze the reduced picture where to a bulk manifold and to its boundary components one associates the “Euler-Lagrange moduli spaces” arising as a reduction of the zero locus of the cohomological vector field Q on the space of fields (or the boundary phase space) by the distribution induced by Q . Under a certain Hodge-theoretic assumption, which holds for a broad class of gauge theories, we prove a version of Lefschetz duality for the moduli spaces. In particular, for a bulk manifold M , denoting the bulk moduli space $\mathcal{M}_M$ and the boundary moduli space $\mathcal{M}_{\partial M}$, we have a degree 0 symplectic structure on $\mathcal{M}_{\partial M}$, a degree 1 Poisson structure on $\mathcal{M}_M$ and a map $\pi_* : \mathcal{M}_M \rightarrow \mathcal{M}_{\partial M}$ whose image is Lagrangian in $\mathcal{M}_{\partial M}$ and whose fibers are the symplectic leaves of the Poisson structure on $\mathcal{M}_M$. In the case of Chern-Simons theory, the Euler-Lagrange moduli spaces are certain natural graded enhancements of the moduli spaces of flat connections on the bulk 3-manifold and the boundary surface. In the simplest example, the abelian Chern-Simons theory with gauge group $\mathbb{R}$, the moduli spaces are given by the de Rham cohomology (in all degrees) of the bulk and the boundary.

Our results in the case of AKSZ sigma models were subsequently generalized in the context of derived symplectic geometry of mapping stacks in [16, 38].

The formalism developed in [21, 22] is a step towards perturbative quantization of gauge theories on manifolds with boundary consistent with the Atiyah-Segal axiomatic. The motivating quantum example, where the interaction of bulk BV and boundary BFV structures was first noticed, was constructed by the author and A. Alekseev in [4].

4.6. Paper “A construction of observables for AKSZ sigma models” [37].

We present a general construction of observables for BV theories based on the construction of BV pushforward. Namely, one extends a BV theory by auxiliary fields and extends the original BV action by a term depending on the original and auxiliary fields, so that the extended action still satisfies the master equation (we call this auxiliary data a *pre-observable*). One then produces an observable out of the pre-observable by performing the BV integral over auxiliary fields.

The construction is then specialized to the setting of AKSZ theories where it is shown that pre-observables can be constructed out of Hamiltonian Q -bundles over the AKSZ target. The corresponding observables are associated to embedded submanifolds of the source manifolds and their expectation values are expected to produce higher-dimensional generalized knot invariants (or more generally, cocycles on the spaces of embeddings).

As a special case, the construction produces the Alekseev-Faddeev-Shatashvili path integral representation [2] for the Wilson loop observable in Chern-Simons theory, associated to a knot in a 3-manifold. Another special case is the observable

in BF theory associated to codimension 2 “long knots” in $\mathbb{R}^n$, constructed by Cattaneo and Rossi [23].

4.7. Paper “Chern-Simons theory with Wilson lines and boundary in the BV-BFV formalism” [3], with A. Alekseev and Y. Barmaz. We analyze the Chern-Simons theory on 3-manifolds with boundary enriched by Wilson lines which are allowed to end on the boundary in the BV-BFV formalism developed in [21]. Wilson lines are represented by BV integrals over auxiliary fields supported on a tangle in the 3-manifold by specialization of the construction of [37]. We study the toy example of a 1-dimensional Chern-Simons theory of [4] with a Wilson line where the quantized BFV operator on the space of states turns out to be the Kostant’s cubic Dirac operator. In the case of 3-dimensional Chern-Simons theory with Wilson lines ending on the boundary, we construct the quantum BFV operator on the space of states. Its cohomology in degree zero yields the space of conformal blocks of the Wess-Zumino-Witten model on the boundary (or equivalently the geometric quantization of the moduli space of flat connections on the boundary [8]), at least in genera 0 and 1.

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REMARKS ON CHERN–SIMONS INVARIANTS

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ABSTRACT. The perturbative Chern–Simons theory is studied in a finite-dimensional version or assuming that the propagator satisfies certain properties (as is the case, e.g., with the propagator defined by Axelrod and Singer). It turns out that the effective BV action is a function on cohomology (with shifted degrees) that solves the quantum master equation and is defined modulo certain canonical transformations that can be characterized completely. Out of it one obtains invariants.

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1. INTRODUCTION

Since its proposal in Witten's paper [18], Chern–Simons theory has been a source of fruitful constructions for 3-manifold invariants. In the perturbative framework one would like to get the invariants from the Feynman diagrams of the theory. These may be shown to be finite, see [2]. However, as in every gauge theory, one has to fix a gauge and then one has to show that the result, the invariant, is independent of the gauge fixing. In the case when one works around an acyclic connection, this was proved in [2], but this assumption rules out the trivial connection. Gauge-fixing independence for perturbation theory around the trivial connection of a rational homology sphere was proved in [1] and, for more general definitions for the propagator, in [11] and in [3]. The flexibility in the choice of propagator allows one to show that the invariant is of finite type [13].

The case of general 3-manifolds was not treated in detail, even though the propagators described in [2, 11, 3] are defined in general. The main point is the presence of zero modes, namely—working around the trivial connection—elements of de Rham cohomology (with shifted degree) of the manifold tensor the given Lie algebra. Out of formal properties of the BV formalism, it is however clear [14, 15, 7, 6] what the invariant in the general case (of a compact manifold) should be: a solution of the quantum master equation on the space of zero modes modulo certain BV canonical transformations. This effective action has already been studied—but only modulo constants—in [7], which has been a source of inspiration for us.

In the first part of this note, we make this precise and mathematically rigorous working with a finite-dimensional version [17] of Chern–Simons theory, where the algebra of smooth functions on the manifold is replaced by an arbitrary finite-dimensional dg Frobenius algebra (of appropriate degrees). We are able to produce the solution to the quantum master equation on cohomology and to describe the BV canonical transformations that occur. Out of this we are able to describe the invariant in the case of cohomology concentrated in degree zero and three (the algebraic version of a rational homology sphere) and to extract invariants in case the first Betti number is one or, more generally, when the Frobenius algebra is formal.

In the second part we revert to the infinite-dimensional case and show that whatever we did in the finite-dimensional case actually goes through if the propagator satisfies certain properties. It is good news that the propagator introduced by Axelrod and Singer in [2] does indeed satisfy them. In particular, we get an invariant for framed 3-manifolds as described in Theorem 1 on page 29.

The problems with this scheme are that there is little flexibility in the choice of propagator and that the invariants are defined up to a universal constant that is very difficult to compute. (Notice that this constant is the same that appears anyway in the case of rational homology spheres in [2, 3]). For the general case, that is of a propagator as in [11] or [3], we are able to show that all properties but one can easily be achieved. We reduce the last property to Conjecture 1 on page 27 which we hope to be able to prove in a forthcoming paper.

During the preparation of this note, we have become aware of independent work by Iacovino [10] on the same topic.

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2. EFFECTIVE BV ACTION

Let¹ $(\mathcal{F}, \sigma)$ be a finite-dimensional graded vector space endowed with an odd symplectic form $\sigma \in \Lambda^2 \mathcal{F}^*$ of degree -1, which means $\sigma(u, v) \neq 0 \Rightarrow |u| + |v| = 1$ for $u, v \in \mathcal{F}$. The space of polynomial functions $\text{Fun}(\mathcal{F}) := S^\bullet \mathcal{F}^*$ is a BV algebra with anti-bracket $\{\bullet, \bullet\}$ and BV Laplacian² Δ generated by the odd symplectic form σ .

In coordinates: let $\{u_i\}$ be a basis in $\mathcal{F}$ and $\{x^i\}$ be the dual basis in $\mathcal{F}^*$. Let us denote $\sigma_{ij} = \sigma(u_i, u_j)$. Then

$$\begin{aligned}\sigma &= \sum_{i,j} (-1)^{\text{gh}(x^i)} \sigma_{ij} \delta x^i \wedge \delta x^j \\ \Delta f &= \frac{1}{2} \sum_{i,j} (-1)^{\text{gh}(x^i)} (\sigma^{-1})^{ij} \frac{\partial}{\partial x^i} \frac{\partial}{\partial x^j} f \\ \{f, g\} &= f \left(\sum_{i,j} (\sigma^{-1})^{ij} \frac{\overleftarrow{\partial}}{\partial x^i} \frac{\overrightarrow{\partial}}{\partial x^j} \right) g\end{aligned}$$

In our convention $\sigma_{ij} = -\sigma_{ji}$ and we call grading on $\text{Fun}(\mathcal{F})$ the “ghost number”: $\text{gh}(x^i) = -|u_i|$.

Suppose $(\mathcal{F}', \sigma')$ is another odd symplectic vector space and $\iota : \mathcal{F}' \hookrightarrow \mathcal{F}$ is an embedding (injective linear map of degree 0) that agrees with the odd symplectic structure:

$$\sigma'(u', v') = \sigma(\iota(u'), \iota(v'))$$

for $u', v' \in \mathcal{F}'$. Then $\mathcal{F}$ can be represented as

$$(1) \quad \mathcal{F} = \iota(\mathcal{F}') \oplus \mathcal{F}''$$

where $\mathcal{F}'' := \iota(\mathcal{F}')^\perp$ is the symplectic complement of the image of ι in $\mathcal{F}$ with respect to σ . Hence the algebra of functions on $\mathcal{F}$ factors

$$\text{Fun}(\mathcal{F}) \cong \text{Fun}(\mathcal{F}') \otimes \text{Fun}(\mathcal{F}'')$$

(the isomorphism depends on the embedding ι). Since (1) is an (orthogonal) decomposition of odd symplectic vector spaces, the BV Laplacian also splits:

$$(2) \quad \Delta = \Delta' + \Delta''$$

Here Δ' is the BV Laplacian on $\text{Fun}(\mathcal{F}')$, associated to the odd symplectic form σ' , and Δ'' is the BV Laplacian on $\mathcal{F}''$ associated to the restricted odd symplectic form $\sigma|_{\mathcal{F}''}$.

Let $S \in \text{Fun}(\mathcal{F})[[\hbar]]$ be a solution to the quantum master equation (QME):

$$\Delta e^{S/\hbar} = 0 \quad \Leftrightarrow \quad \frac{1}{2} \{S, S\} + \hbar \Delta S = 0$$

(the BV action on $\mathcal{F}$). Let also $\mathcal{L} \subset \mathcal{F}''$ be a Lagrangian subspace in $\mathcal{F}''$. We define the effective (or “induced”) BV action $S' \in \text{Fun}(\mathcal{F}')[[\hbar]]$ by the fiber BV integral

$$(3) \quad e^{S'(x')/\hbar} = \int_{\mathcal{L}} e^{S(\iota(x') + x'')/\hbar} \mu_{\mathcal{L}}$$

¹This Section is an adaptation of Section 4.2 from [16].

²In the general case, on an odd-symplectic graded manifold, the BV Laplacian Δ is constructed from the symplectic form and a consistent measure (the “*SP*-structure”). But here we treat only the linear case, and an odd-symplectic graded vector space $(\mathcal{F}, \sigma)$ is automatically an *SP*-manifold with Lebesgue measure (the constant Berezinian) $\mu_{\mathcal{F}}$.

where $\mu_{\mathcal{L}}$ is the Lebesgue measure on $\mathcal{L}$ and we use the notation $x = \sum_i u_i x^i$ for the canonical element of $\mathcal{F} \otimes \text{Fun}(\mathcal{F})$, corresponding to the identity map $\mathcal{F} \rightarrow \mathcal{F}$, and analogously for $x' \in \mathcal{F}' \otimes \text{Fun}(\mathcal{F}')$, $x'' \in \mathcal{F}'' \otimes \text{Fun}(\mathcal{F}'')$

Proposition 1. *The Effective action S' defined by (3) satisfies the QME on $\mathcal{F}'$, i.e.*

$$\Delta' e^{S'/\hbar} = 0$$

Proof. This is a direct consequence of (2) and the BV-Stokes theorem (integrals over Lagrangian submanifolds of BV coboundaries vanish):

$$\begin{aligned} \Delta' e^{S'(x')/\hbar} &= \Delta' \int_{\mathcal{L}} e^{S(\iota(x') + x'')/\hbar} \mu_{\mathcal{L}} = \int_{\mathcal{L}} (\Delta - \Delta'') e^{S(\iota(x') + x'')/\hbar} \mu_{\mathcal{L}} = \\ &= \int_{\mathcal{L}} \Delta e^{S(\iota(x') + x'')/\hbar} \mu_{\mathcal{L}} - \int_{\mathcal{L}} \Delta'' e^{S(\iota(x') + x'')/\hbar} \mu_{\mathcal{L}} = 0 \end{aligned}$$

In the last line the first term vanishes due to QME for S , while the second term is zero due to the BV-Stokes theorem. $\square$

The BV integral (3) depends on a choice of embedding $\iota : \mathcal{F}' \hookrightarrow \mathcal{F}$ and Lagrangian subspace $\mathcal{L} \subset \mathcal{F}''$ (notice that $\mathcal{F}'' = \iota(\mathcal{F}')^\perp$ itself depends on ι). We call the pair $(\iota, \mathcal{L})$ the “induction data” in this setting. We are interested in the dependence of the effective BV action S' on deformations of induction data.

Recall that a generic small Lagrangian deformation $\mathcal{L}_\Psi$ of a Lagrangian submanifold $\mathcal{L} \subset \mathcal{F}''$ is given by a gauge fixing fermion $\Psi \in \text{Fun}(\mathcal{L})$ of ghost number -1. If (q^a, p_a) are Darboux coordinates on $\mathcal{F}''$ such that $\mathcal{L}$ is given by $p = 0$, then $\mathcal{L}_\Psi$ is

$$\mathcal{L}_\Psi = \left\{ (q, p) : p_a = -\frac{\partial}{\partial q^a} \Psi \right\} \subset \mathcal{F}''$$

In our case we are interested in linear Lagrangian subspaces and thus only allow quadratic gauge fixing fermions.

A general small deformation of induction data $(\iota, \mathcal{L})$ from $(\mathcal{F}, \sigma)$ to $(\mathcal{F}', \sigma')$ can be written as

$$(4) \quad \iota \mapsto \iota + \delta\iota_\perp + \delta\iota_\parallel, \quad \mathcal{L} \mapsto (\text{id}_{\mathcal{F}} - \iota \circ (\delta\iota_\perp)^T) \mathcal{L}_\Psi$$

where the “perpendicular part” of the deformation of the embedding $\delta\iota_\perp : \mathcal{F}' \rightarrow \mathcal{F}$ is a linear map of degree 0 satisfying $\delta\iota_\perp(\mathcal{F}') \subset \mathcal{F}''$, while the “parallel part” is of the form $\delta\iota_\parallel = \iota \circ \delta\phi_\parallel$ with $\delta\phi_\parallel : \mathcal{F}' \rightarrow \mathcal{F}'$ a linear map of degree 0 satisfying $(\delta\phi_\parallel)^T = -\delta\phi_\parallel$ (i.e. $\delta\phi_\parallel$ lies in the Lie algebra of the group of linear symplectomorphisms $\delta\phi_\parallel \in \text{sp}(\mathcal{F}', \sigma')$). We use superscript T to denote transposition w.r.t. symplectic structure. Thus it is reasonable to classify small deformations into the three following types:

- Type I: small Lagrangian deformations of Lagrangian subspace $\mathcal{L} \mapsto \mathcal{L}_\Psi$ leaving the embedding ι intact.
- Type II: small perpendicular deformations of the embedding $\iota \mapsto \iota + \delta\iota_\perp$ accompanied by an associated deformation³ of the Lagrangian subspace

$$(5) \quad \mathcal{L} \mapsto (\text{id}_{\mathcal{F}} - \iota \circ (\delta\iota_\perp)^T) \mathcal{L}$$

³Formula (5) is explained as follows: one can express the deformation of the embedding as $\iota + \delta\iota_\perp = (\text{id}_{\mathcal{F}} + \delta\Phi) \circ \iota$, where $\delta\Phi = \delta\iota_\perp \circ \iota^T - \iota \circ \delta\iota_\perp^T \in \text{sp}(\mathcal{F}, \sigma)$, so that $\text{id}_{\mathcal{F}} + \delta\Phi$ is an infinitesimal symplectomorphism of the space $\mathcal{F}$, accounting for the deformation of ι . Then (5) just means $\mathcal{L} \mapsto (\text{id}_{\mathcal{F}} + \delta\Phi) \circ \mathcal{L}$.

(this is necessary since here we deform the splitting (1) and $\mathcal{L}$ is supposed to be a subspace of the deformed $\mathcal{F}''$).

- Type III: small parallel deformations of the embedding

$$\iota \mapsto \iota + \delta\iota_{||} = \iota \circ (\text{id}_{\mathcal{F}'} + \delta\phi_{||})$$

leaving $\mathcal{L}$ intact.

A general small deformation (4) is the sum of deformations of types I, II, III.

We call the following transformation of the action

$$(6) \quad S \mapsto \tilde{S} = S + \{S, R\} + \hbar\Delta R$$

(regarded in first order in R) the infinitesimal canonical transformation of the action with (infinitesimal) generator $R \in \text{Fun}(\mathcal{F})[[\hbar]]$ of ghost number -1. Equivalently

$$(7) \quad e^{S/\hbar} \mapsto e^{\tilde{S}/\hbar} = e^{S/\hbar} + \Delta(e^{S/\hbar} R)$$

The transformed action also solves the QME (in first order in R). Infinitesimal canonical transformations generate the equivalence relation on solutions of QME.

Lemma 1. *If the action S is changed by an infinitesimal canonical transformation $S \mapsto S + \{S, R\} + \hbar\Delta R$, then the effective BV action is also changed by an infinitesimal canonical transformation*

$$(8) \quad S' \mapsto S' + \{S', R'\}' + \hbar\Delta' R'$$

with generator $R' \in \text{Fun}(\mathcal{F}')[[\hbar]]$ given by fiber BV integral

$$(9) \quad R' = e^{-S'/\hbar} \int_{\mathcal{L}} e^{S/\hbar} R \mu_{\mathcal{L}}$$

Proof. This follows straightforwardly from (2), the BV-Stokes theorem and the exponential form of canonical transformations (7):

$$\begin{aligned} e^{S'/\hbar} \mapsto e^{\tilde{S}'/\hbar} &= \int_{\mathcal{L}} (e^{S'/\hbar} + \Delta(e^{S'/\hbar} R')) \mu_{\mathcal{L}} = \\ &= e^{S'/\hbar} + \Delta' \int_{\mathcal{L}} e^{S'/\hbar} R \mu_{\mathcal{L}} + \underbrace{\int_{\mathcal{L}} \Delta''(e^{S'/\hbar} R) \mu_{\mathcal{L}}}_0 = e^{S'/\hbar} + \Delta'(e^{S'/\hbar} R') \end{aligned}$$

with R' given by (9). □

Proposition 2. *Under general infinitesimal deformation of the induction data $(\iota, \mathcal{L})$ as in (4) the effective BV action S' is transformed canonically (up to constant shift):*

$$(10) \quad S' \mapsto S' + \{S', R'\}' + \hbar\Delta'(R' - R'_{III})$$

with generator

$$(11) \quad R' = \underbrace{\sigma'(x', \delta\phi_{||} x')}_{R'_{III}} + e^{-S'/\hbar} \int_{\mathcal{L}} e^{S/\hbar} (\Psi + \sigma(x'', \delta\iota_{\perp} x')) \mu_{\mathcal{L}}$$

Proof. The deformation (4) can be represented in the form

$$\iota \mapsto (\text{id}_{\mathcal{F}} + \delta\Phi) \circ \iota \quad , \quad \mathcal{L} \mapsto (\text{id}_{\mathcal{F}} + \delta\Phi)\mathcal{L}$$

where $\delta\Phi \in \text{sp}(\mathcal{F}, \sigma)$ is the infinitesimal symplectomorphism given by

$$\delta\Phi = \{\bullet, \Psi\} + (\delta\iota_{\perp} \circ \iota^T - \iota \circ \delta\iota_{\perp}^T) + \delta\iota_{||} \circ \iota^T$$

Here $\{\bullet, \Psi\} = \exp(\{\bullet, \Psi\}) - \text{id}_{\mathcal{F}}$ is understood as the (infinitesimal) flow generated by Hamiltonian vector field $\{\bullet, \Psi\}$ in unit time. The pull-back $(\text{id}_{\mathcal{F}} + \delta\Phi)^* : \text{Fun}(\mathcal{F}) \rightarrow \text{Fun}(\mathcal{F})$ acts on functions as canonical transformation

$$(12) \quad f \mapsto f + \{f, R\}$$

with generator given by

$$R = \frac{1}{2}\sigma(x, \delta\Phi x) = \underbrace{\Psi}_{R_I} + \underbrace{\sigma(x, (\delta\iota_{\perp} \circ \iota^T)x)}_{R_{II}} + \underbrace{\frac{1}{2}\sigma(x, (\delta\iota_{||} \circ \iota^T)x)}_{R_{III}}$$

It is important to note that only the third term (the effect of type III deformation) contributes to ΔR :

$$\Delta R = \Delta' R = \Delta R_{III} = \frac{1}{2} \text{Str}_{\mathcal{F}'} \delta\phi_{||}$$

($\text{Str}_{\mathcal{F}'}$ denotes supertrace over $\mathcal{F}'$) and $\Delta'' R = 0$. The latter implies that $(\text{id}_{\mathcal{F}} + \delta\Phi)^* \mu_{\mathcal{L}} = \mu_{\mathcal{L}}$. Now we can compute the transformation of the effective BV action S' due to infinitesimal change of induction data:

$$(13) \quad e^{S'/\hbar} \mapsto e^{\tilde{S}'/\hbar} = \int_{\mathcal{L}} (\text{id}_{\mathcal{F}} + \delta\Phi)^* e^{S/\hbar} \mu_{\mathcal{L}} = \\ = e^{-\text{Str}_{\mathcal{F}'} \delta\phi_{||}} \int_{\mathcal{L}} e^{(S + \{S, R\} + \hbar \Delta R)/\hbar} \mu_{\mathcal{L}} = e^{-\text{Str}_{\mathcal{F}'} \delta\phi_{||}} e^{(S' + \{S', R'\}' + \hbar \Delta' R')/\hbar}$$

Here we used Lemma 1 and the generator is given by (9):

$$R' = e^{-S'/\hbar} \int_{\mathcal{L}} e^{S/\hbar} R \mu_{\mathcal{L}} = e^{-S'/\hbar} \int_{\mathcal{L}} e^{S/\hbar} (\Psi + \sigma(x, (\delta\iota_{\perp} \circ \iota^T)x) + \frac{1}{2}\sigma(x, (\delta\iota_{||} \circ \iota^T)x)) \mu_{\mathcal{L}}$$

which yields (11). At last note that $\frac{1}{2} \text{Str}_{\mathcal{F}'} \delta\phi_{||} = \Delta' \frac{1}{2} \sigma'(x', \delta\phi_{||} x') = \Delta' R'_{III}$ which explains the constant shift in (10). $\square$

Remark 1. If we were treating BV actions as log-half-densities (meaning that $e^{S/\hbar}$ is a half-density), we would write an honest canonical transformation (8) instead of (10), with no $-\hbar \Delta' R'_{III}$ shift. This is because the pull-back $(\text{id}_{\mathcal{F}} + \delta\Phi)^*$ would then be acting on S by transformation (6) instead of (12). But in practice one works with effective BV actions defined by a normalized BV integral: if the initial BV action is of the form $S = S_0 + S_{\text{int}}$ with S_0 quadratic in fields (the “free part” of BV action) and S_{int} the “interaction part”, one usually defines

$$e^{S'(x')/\hbar} = \frac{1}{N} \int_{\mathcal{L}} e^{S(\iota(x') + x'')/\hbar} \mu_{\mathcal{L}}$$

with the normalization factor $N = \int_{\mathcal{L}} e^{S_0(x'')/\hbar} \mu_{\mathcal{L}}$. The effective action defined via a normalized BV integral is indeed a function rather than log-half-density, and transforms according to (10) under change of induction data.

3. TOY MODEL FOR EFFECTIVE CHERN–SIMONS THEORY ON ZERO-MODES: EFFECTIVE BV ACTION ON COHOMOLOGY OF DG FROBENIUS ALGEBRAS

By a dg Frobenius algebra $(\mathcal{C}, d, m, \pi)$ we mean a unital differential graded commutative algebra $\mathcal{C}$ with differential $d : \mathcal{C}^{\bullet} \rightarrow \mathcal{C}^{\bullet+1}$ and (super-commutative, associative) product $m : S^2 \mathcal{C} \rightarrow \mathcal{C}$, endowed in addition with a non-degenerate pairing

$\pi : S^2\mathcal{C} \rightarrow \mathbb{R}$ of degree $-k$ (which means $\pi(a, b) \neq 0 \Rightarrow |a| + |b| = k$; and k is some fixed integer), satisfying the following consistency conditions:

$$(14) \quad \pi(da, b) + (-1)^{|a|}\pi(a, db) = 0$$

$$(15) \quad \pi(a, m(b, c)) = \pi(m(a, b), c)$$

for $a, b, c \in \mathcal{C}$.

By a dg Frobenius–Lie⁴ algebra $(\mathcal{A}, d, l, \pi)$ we mean a differential graded Lie algebra $\mathcal{A}$ with differential $d : \mathcal{A}^\bullet \rightarrow \mathcal{A}^{\bullet+1}$ and Lie bracket $l : \wedge^2 \mathcal{A} \rightarrow \mathcal{A}$, endowed with a non-degenerate pairing $\pi : S^2 \mathcal{A} \rightarrow \mathbb{R}$ of degree $-k$ satisfying the conditions

$$(16) \quad \pi(A, l(B, C)) = \pi(l(A, B), C)$$

$$(17) \quad \pi(dA, B) + (-1)^{|A|}\pi(A, dB) = 0$$

for $A, B, C \in \mathcal{A}$.

If $\mathfrak{g}$ is an (ordinary) Lie algebra with non-degenerate ad-invariant inner product $\pi_{\mathfrak{g}} : S^2 \mathfrak{g} \rightarrow \mathbb{R}$ (one calls such Lie algebras “quadratic”) and $(\mathcal{C}, d, m, \pi)$ is a dg Frobenius algebra, then $(\mathfrak{g} \otimes \mathcal{C}, d, l, \pi_{\mathfrak{g} \otimes \mathcal{C}})$ is a dg Frobenius–Lie algebra. Here one defines $d(X \otimes a) := X \otimes da$, $l(X \otimes a, Y \otimes b) := [X, Y] \otimes m(a, b)$, $\pi_{\mathfrak{g} \otimes \mathcal{C}}(X \otimes a, Y \otimes b) := \pi_{\mathfrak{g}}(X, Y) \pi(a, b)$. We will usually write π instead of $\pi_{\mathfrak{g} \otimes \mathcal{C}}$.

Example 1. If M is a closed (compact, without boundary) orientable smooth manifold of dimension D , then de Rham algebra $\Omega^\bullet(M)$ is a dg Frobenius algebra with de Rham differential, wedge product and Poincare pairing $\int_M \bullet \wedge \bullet$ of degree $-D$. If $\mathfrak{g}$ is a finite-dimensional Lie algebra with invariant non-degenerate trace tr , then the algebra $\Omega^\bullet(M, \mathfrak{g}) = \mathfrak{g} \otimes \Omega^\bullet(M)$ of $\mathfrak{g}$ -valued differential forms on M is a dg Frobenius–Lie algebra with pairing $\text{tr} \int_M \bullet \wedge \bullet$ of degree $-D$.

3.1. Abstract Chern–Simons action from a dg Frobenius algebra. Let $(\mathcal{C}, d, m, \pi)$ be a finite dimensional non-negatively graded dg Frobenius algebra with pairing π of degree -3

$$\mathcal{C} = \mathcal{C}^0 \oplus \mathcal{C}^1[-1] \oplus \mathcal{C}^2[-2] \oplus \mathcal{C}^3[-3]$$

We denote by $B_i = \dim H^i(\mathcal{C})$ the Betti numbers. Due to non-degeneracy of π , there is an isomorphism $\pi_b : \mathcal{C}^\bullet \cong (\mathcal{C}^{3-\bullet})^*$ (the Poincare duality). Induced pairing on cohomology is also automatically non-degenerate, and so Poincare duality descends to cohomology: $\pi_b : H^\bullet(\mathcal{C}) \cong (H^{3-\bullet}(\mathcal{C}))^*$. Hence $B_0 = B_3$, $B_1 = B_2$. We will suppose in addition that $B_0 = B_3 = 1$ (so that $\mathcal{C}$ models the de Rham algebra of a connected manifold).

Let $\mathfrak{g}$ be a finite dimensional quadratic Lie algebra of coefficients and let $(\mathfrak{g} \otimes \mathcal{C}, d, l, \pi)$ be the corresponding dg Frobenius–Lie structure on $\mathfrak{g} \otimes \mathcal{C}$. Then we can construct an odd symplectic space of BV fields

$$\mathcal{F} = \mathfrak{g} \otimes \mathcal{C}[1]$$

with odd symplectic structure of degree -1 given by

$$(18) \quad \sigma(sA, sB) = (-1)^{|A|}\pi(A, B)$$

Here $s : \mathfrak{g} \otimes \mathcal{C} \rightarrow \mathfrak{g} \otimes \mathcal{C}[1]$ is the suspension map. Let us also introduce the notation ω for the canonical element of $(\mathfrak{g} \otimes \mathcal{C}) \otimes \text{Fun}(\mathcal{F})$ corresponding to the desuspension

⁴Alternatively one could use the term “cyclic dg Lie algebra”.

map $s^{-1} : \mathcal{F} \rightarrow \mathfrak{g} \otimes \mathcal{C}$. If $\{e_I\}$ is a basis in $\mathcal{C}$ and $\{T_a\}$ is an orthonormal basis in $\mathfrak{g}$, then we can write

$$\omega = \sum_{I,a} T_a e_I \omega^{Ia}$$

where $\{\omega^{Ia}\}$ are the corresponding coordinates on $\mathcal{F}$. By abuse of terminology we call ω the “BV field”. Let us introduce notations for the structure constants: $\pi_{IJ} = \pi(e_I, e_J)$, $m_{IJK} = \pi(e_I, m(e_J, e_K))$, $f_{abc} = \pi_{\mathfrak{g}}(T_a, [T_b, T_c])$, $d_{IJ} = \pi(e_I, de_J)$. We will also use the shorthand notation for degrees $|I| = |e_I|$. In terms of π the BV Laplacian and the anti-bracket are

$$\begin{aligned} \Delta f &= \frac{1}{2} \sum_{I,J,a} (\pi^{-1})^{IJ} \frac{\partial}{\partial \omega^{Ia}} \frac{\partial}{\partial \omega^{Ja}} f \\ \{f, g\} &= f \left(\sum_{I,J,a} (-1)^{|I|+1} (\pi^{-1})^{IJ} \frac{\overleftarrow{\partial}}{\partial \omega^{Ia}} \frac{\overrightarrow{\partial}}{\partial \omega^{Ja}} \right) g \end{aligned}$$

Proposition 3. *The action $S \in \text{Fun}(\mathcal{F})$ defined as*

$$\begin{aligned} (19) \quad S &:= \frac{1}{2} \pi(\omega, d\omega) + \frac{1}{6} \pi(\omega, l(\omega, \omega)) = \\ &= \frac{1}{2} \sum_{I,J,a} (-1)^{|I|+1} d_{IJ} \omega^{Ia} \omega^{Ja} + \frac{1}{6} \sum_{I,J,K,a,b,c} (-1)^{|J|(|K|+1)} f_{abc} m_{IJK} \omega^{Ia} \omega^{Jb} \omega^{Kc} \end{aligned}$$

satisfies the QME with BV Laplacian defined by the odd symplectic structure (18) on $\mathcal{F}$.

Proof. Indeed, let us check the CME:

$$\begin{aligned} \frac{1}{2} \{S, S\} &= \\ \frac{1}{8} \{ \pi(\omega, d\omega), \pi(\omega, d\omega) \} &+ \frac{1}{12} \{ \pi(\omega, d\omega), \pi(\omega, l(\omega, \omega)) \} + \frac{1}{72} \{ \pi(\omega, l(\omega, \omega)), \pi(\omega, l(\omega, \omega)) \} \\ &= -\frac{1}{2} \pi(\omega, d^2 \omega) - \frac{1}{2} \pi(\omega, dl(\omega, \omega)) - \frac{1}{8} \pi(\omega, l(\omega, l(\omega, \omega))) = 0 \end{aligned}$$

The first term vanishes due to $d^2 = 0$, the second — due to the Leibniz identity for $\mathfrak{g} \otimes \mathcal{C}$, since property (16) implies

$$\frac{1}{2} \pi(\omega, dl(\omega, \omega)) = \frac{1}{6} \pi(\omega, dl(\omega, \omega) - l(d\omega, \omega) + l(\omega, d\omega)),$$

and the third term is zero due to the Jacobi identity for $\mathfrak{g} \otimes \mathcal{C}$. Next, check the quantum part of the QME:

$$\hbar \Delta S = -\hbar \frac{1}{2} \text{Str}_{\mathfrak{g} \otimes \mathcal{C}} d - \hbar \frac{1}{2} \text{Str}_{\mathfrak{g} \otimes \mathcal{C}} l(\omega, \bullet) = 0$$

Here the first term vanishes since d raises degree and the second term vanishes due to unimodularity of Lie algebra $\mathfrak{g}$. $\square$

The BV action (19) can be viewed as an abstract model (or toy model, since $\mathcal{C}$ is finite dimensional) for Chern–Simons theory on a connected closed orientable 3-manifold. We associate such a model to any finite dimensional non-negatively graded dg Frobenius algebra $\mathcal{C}$ with pairing of degree -3 and $B_0 = B_3 = 1$, and arbitrary finite dimensional quadratic Lie algebra of coefficients $\mathfrak{g}$. We are interested

in the effective BV action for (19) induced on cohomology $\mathcal{F}' = H^\bullet(\mathcal{C}, \mathfrak{g})[1]$. We will now specialize the general induction procedure sketched in Section 2 to this case.

3.2. Effective action on cohomology. Let $\iota : H^\bullet(\mathcal{C}) \hookrightarrow \mathcal{C}^\bullet$ be an embedding of cohomology into $\mathcal{C}$. Note that ι is not just an arbitrary chain map between two fixed complexes, but is also subject to condition $\iota([a]) = a + d(\dots)$ for any cocycle $a \in \mathcal{C}$. This implies in particular that the only allowed deformations of ι are of the form $\iota \mapsto \iota + d \delta I$ where $\delta I : H^\bullet(\mathcal{C}) \rightarrow \mathcal{C}^{\bullet-1}$ is an arbitrary degree -1 linear map. This is indeed a type II deformation (in the terminology of Section 2), while type III deformations are prohibited in this setting.

Let also $K : \mathcal{C}^\bullet \rightarrow \mathcal{C}^{\bullet-1}$ be a symmetric chain homotopy retracting $\mathcal{C}^\bullet$ to $H^\bullet(\mathcal{C})$, that is a degree -1 linear map satisfying

$$(20) \quad dK + Kd = \text{id}_{\mathcal{C}} - \mathcal{P}'$$

$$(21) \quad \pi(Ka, b) + (-1)^{|a|} \pi(a, Kb) = 0$$

$$(22) \quad K \circ \iota = 0$$

where $\mathcal{P}' = \iota \circ \iota^T : \mathcal{C} \rightarrow \mathcal{C}$ is the orthogonal (w.r.t. π) projection to the representatives of cohomology in $\mathcal{C}$. We require the additional property

$$(23) \quad K^2 = 0$$

Remark 2 (cf. [9]). An arbitrary linear map $K_0 : \mathcal{C}^\bullet \rightarrow \mathcal{C}^{\bullet-1}$ satisfying just (20) can be transformed into a chain homotopy K with all the properties (20,21,22,23) via a chain of transformations $K_0 \mapsto K_1 \mapsto K_2 \mapsto K_3 = K$ where

$$(24) \quad \begin{aligned} K_1 &= \frac{1}{2}(K_0 - K_0^T) \\ K_2 &= (\text{id}_{\mathcal{C}} - \mathcal{P}') K_1 (\text{id}_{\mathcal{C}} - \mathcal{P}') \end{aligned}$$

$$(25) \quad K_3 = K_2 d K_2$$

Having ι and K we can define a Hodge decomposition for $\mathcal{C}$ into representatives of cohomology, d -exact part and K -exact part:

$$(26) \quad \mathcal{C} = \text{im}(\iota) \oplus \underbrace{\mathcal{C}_{d-ex}}_{\text{im}(d)} \oplus \underbrace{\mathcal{C}_{K-ex}}_{\text{im}(K)}$$

Properties (21), (22), (23) and skew-symmetry of differential (14) imply the orthogonality properties for Hodge decomposition (26):

$$\text{im}(\iota)^\perp = \mathcal{C}_{d-ex} \oplus \mathcal{C}_{K-ex}, \quad (\mathcal{C}_{d-ex})^\perp = \text{im}(\iota) \oplus \mathcal{C}_{K-ex}, \quad (\mathcal{C}_{K-ex})^\perp = \text{im}(\iota) \oplus \mathcal{C}_{d-ex}$$

In terms of Hodge decomposition (26) the splitting (1) of the space of BV fields $\mathcal{F} = \mathfrak{g} \otimes \mathcal{C}[1]$ is given by

$$\underbrace{\mathfrak{g} \otimes \mathcal{C}[1]}_{\mathcal{F}} = \underbrace{\iota(H^\bullet(\mathcal{C}, \mathfrak{g})[1])}_{\iota(\mathcal{F}')} \oplus \underbrace{\mathfrak{g} \otimes \mathcal{C}_{d-ex}[1] \oplus \mathfrak{g} \otimes \mathcal{C}_{K-ex}[1]}_{\mathcal{F}''}$$

and we choose the Lagrangian subspace

$$(27) \quad \mathcal{L}_K = \mathfrak{g} \otimes \mathcal{C}_{K-ex}[1] \subset \mathcal{F}''$$

We define the “effective BV action on cohomology” (or “on zero-modes”) $W \in \text{Fun}(\mathcal{F}')[[\hbar]]$ for an abstract Chern–Simons action (19) by a normalized fiber BV

integral

$$(28) \quad e^{W(\alpha)/\hbar} = \frac{1}{N} \int_{\mathcal{L}_K} e^{S(\iota(\alpha)+\omega'')/\hbar} \mu_{\mathcal{L}_K}$$

where

$$(29) \quad N = \int_{\mathcal{L}_K} e^{S_0(\omega'')/\hbar} \mu_{\mathcal{L}_K}$$

is the normalization factor and

$$S_0(\omega'') = \frac{1}{2} \pi(\omega'', d\omega'')$$

is the free part of the action S . To lighten somewhat the notation, we denoted the effective action by W instead of S' and the BV field associated to $\mathcal{F}' = H^\bullet(\mathcal{C}, \mathfrak{g})[1]$ by α instead of ω' . Let $\{e_p\}$ be a basis of the cohomology $H^\bullet(\mathcal{C})$. Then $\alpha = \sum_{a,p} T_a e_p \alpha^{pa}$ where α^{pa} are coordinates on $\mathcal{F}'$ with ghost numbers $\text{gh}(\alpha^{pa}) = 1 - |e_p|$. We have the following decomposition of $S(\iota(\alpha) + \omega'')$:

$$(30) \quad S(\iota(\alpha) + \omega'') = \underbrace{\frac{1}{6} \pi(\iota(\alpha), l(\iota(\alpha), \iota(\alpha)))}_{W_{\text{prod}}(\alpha)} + \underbrace{\frac{1}{2} \pi(\omega'', d\omega'')}_{S_0(\omega'')} + \underbrace{\frac{1}{2} \pi(\omega'', l(\iota(\alpha), \iota(\alpha))) + \frac{1}{2} \pi(\iota(\alpha), l(\omega'', \omega'')) + \frac{1}{6} \pi(\omega'', l(\omega'', \omega''))}_{S_{\text{int}}(\alpha, \omega'')}$$

The perturbation expansion for (28) is obtained in a standard way and can be written as

$$(31) \quad W(\alpha) = W_{\text{prod}}(\alpha) + \hbar \log \left(e^{-\hbar \frac{1}{2} \pi^{-1} \left(\frac{\partial}{\partial \omega''}, K \frac{\partial}{\partial \omega''} \right) \Big|_{\omega''=0}} \circ e^{S_{\text{int}}(\alpha, \omega'')/\hbar} \right) = \sum_{l=0}^{\infty} \hbar^l \sum_{n=0}^{\infty} \sum_{\Gamma \in G_{l,n}} \frac{1}{|\text{Aut}(\Gamma)|} W_{\Gamma}(\alpha)$$

Where $G_{l,n}$ denotes the set of connected non-oriented Feynman graphs with vertices of valence 1 and 3 (we would like to understand them as trivalent graphs with “leaves” allowed, i.e. external edges), with l loops and n leaves. The contribution $W_{\Gamma}(\alpha)$ of each Feynman graph $\Gamma \in G_{l,n}$ is a homogeneous polynomial of degree n in $\{\alpha^{pa}\}$ and of ghost number 0 obtained by decorating each leaf of Γ by $\iota_p^I \alpha^{pa}$, each trivalent vertex by $f_{abc} m_{IJK}$ and each (internal) edge by $\delta^{ab} K^{IJ}$, and taking contraction of all indices, corresponding to incidence of vertices and edges in Γ . One should also take into account signs for contributions, which can be obtained from the exponential formula for perturbation series (31). The cubic term

$$(32) \quad W_{\text{prod}}(\alpha) = \frac{1}{6} \pi(\iota(\alpha), l(\iota(\alpha), \iota(\alpha))) = \frac{1}{6} \sum_{a,b,c} \sum_{p,q,r} (-1)^{|e_q|(|e_r|+1)} f_{abc} \mu_{pqr} \alpha^{pa} \alpha^{qb} \alpha^{rc}$$

is the contribution of the simplest Feynman diagram $\Gamma_{0,3}$, the only element of $G_{0,3}$. Here $\mu_{pqr} = \pi(\iota(e_p), m(\iota(e_q), \iota(e_r)))$ are structure constants of the induced associative product on $H^\bullet(\mathcal{C})$ (hence the notation W_{prod}).

Remark 3. The perturbative expansion (31) is related to homological perturbation theory (HPT) in the following way. Denote by $c_{l,n}(\alpha) = n! \sum_{\Gamma \in G_{l,n}} \frac{1}{|\text{Aut}(\Gamma)|} W_\Gamma(\alpha)$ the total contribution of Feynman graphs with l loops and n leaves to the effective action (31), with additional factor $n!$. Then each

$$c_{l,n} \in S^n(\mathcal{F}'^*) \cong \text{Hom}(\wedge^n(H^\bullet(\mathcal{C}, \mathfrak{g})), \mathbb{R})$$

can be understood as a (super-)anti-symmetric n -ary operation on cohomology $H^\bullet(\mathcal{C}, \mathfrak{g})$, taking values in numbers. Now suppose $l_n \in \text{Hom}(\wedge^n(H^\bullet(\mathcal{C}, \mathfrak{g})), H^\bullet(\mathcal{C}, \mathfrak{g}))$ are the L_∞ operations on cohomology, induced from dg Lie algebra $\mathfrak{g} \otimes \mathcal{C}$ (by means of HPT). Then it is easy to see that $c_{0,n+1}(\alpha_0, \dots, \alpha_n) = \pi'(\alpha_0, l_n(\alpha_1, \dots, \alpha_n))$ and trees for HPT (the Lie version of trees from [12], cf. also [9]) are obtained from Feynman trees for $W(\alpha)$ by assigning one leaf as a root and inserting the inverse of pairing⁵ $(\pi')^{-1}$ there, or vice versa: Feynman trees are obtained from trees of HPT by reverting the root⁶ with π' and forgetting the orientation of edges (cf. Section 7.2.1 of [16]). Thus we can loosely say that the BV integral (28) defines a sort of “loop enhancement” of HPT for a cyclic dg Lie algebra $\mathfrak{g} \otimes \mathcal{C}$. Also, in this language (due to A. Losev), using the BV-Stokes theorem to prove that the effective action W satisfies the quantum master equation can be viewed as the loop-enhanced version of using the HPT machinery to prove the system of quadratic relations (homotopy Jacobi identities) on induced L_∞ operations l_n .

Let us introduce a Darboux basis in $H^\bullet(\mathcal{C})$. Namely, let $e_{(0)} = [1]$ be the basis vector in $H^0(\mathcal{C})$, the cohomology class of unit $1 \in \mathcal{C}^0$ (recall that we assume $B_0 = \dim H^0(\mathcal{C}) = 1$) and let $e_{(3)}$ be the basis vector in $H^3(\mathcal{C})$, satisfying $\pi'(e_{(0)}, e_{(3)}) = 1$ (i.e. $e_{(3)}$ is represented by some top-degree element $v = \iota(e_{(3)}) \in \mathcal{C}^3$, normalized by the condition $\pi(1, v) = 1$). Let also $\{e_{(1)i}\}$ be some basis in $H^1(\mathcal{C})$ and $\{e_{(2)}^i\}$ the dual basis in $H^2(\mathcal{C})$, so that $\pi'(e_{(1)i}, e_{(2)}^j) = \delta_i^j$. The BV field α is then represented as

$$\alpha = \sum_a e_{(0)} T_a \alpha_{(0)}^a + \sum_{a,i} e_{(1)i} T_a \alpha_{(1)}^{ia} + \sum_{a,i} e_{(2)}^i T_a \alpha_{(2)i}^a + \sum_a e_{(3)} T_a \alpha_{(3)}^a$$

In Darboux coordinates $\{\alpha_0^a, \alpha_{(1)}^{ia}, \alpha_{(2)i}^a, \alpha_{(3)}^a\}$ the BV Laplacian on $\mathcal{F}'$ is

$$\Delta' = \sum_a \frac{\partial}{\partial \alpha_{(0)}^a} \frac{\partial}{\partial \alpha_{(3)}^a} + \sum_{a,i} \frac{\partial}{\partial \alpha_{(1)}^{ia}} \frac{\partial}{\partial \alpha_{(2)i}^a}$$

It is also convenient to introduce $\mathfrak{g}$ -valued coordinates on $\mathcal{F}'$:

$$\alpha_{(0)} = \sum_a T_a \alpha_{(0)}^a, \quad \alpha_{(1)}^i = \sum_a T_a \alpha_{(1)}^{ia}, \quad \alpha_{(2)i} = \sum_a T_a \alpha_{(2)i}^a, \quad \alpha_{(3)} = \sum_a T_a \alpha_{(3)}^a$$

⁵We use notation $\pi' = \pi(\iota(\bullet), \iota(\bullet)) : \mathcal{C} \otimes \mathcal{C} \rightarrow \mathbb{R}$ for the induced pairing on cohomology $H^\bullet(\mathcal{C})$. By a slight abuse of notation, we also use π' to denote the pairing on $H^\bullet(\mathcal{C}, \mathfrak{g})$ induced from $\pi_{\mathfrak{g} \otimes \mathcal{C}}$.

⁶This means the following: let T be a binary rooted tree with n leaves, oriented towards the root; let $\bar{T}$ be the non-oriented (and non-rooted) tree with $n+1$ leaves, obtained from T by forgetting the orientation (and treating the root as additional leaf). Then the weight $W_{\bar{T}}(\alpha)$ of $\bar{T}$ as a Feynman graph and the contribution l_T of tree T (without the symmetry factor) to the induced L_∞ operation l_n are related by $W_{\bar{T}}(\alpha) = \pi'(\alpha, l_T(\underbrace{\alpha, \dots, \alpha}_n))$. Pictorially this is

represented by inserting a bivalent vertex (associated to the operation $\pi'(\bullet, \bullet)$) at the root of T . Both edges incident to this vertex are incoming, thus we say that the root becomes reverted.

The ghost numbers of $\alpha_{(0)}, \alpha_{(1)}^i, \alpha_{(2)i}, \alpha_{(3)}$ are 1, 0, -1, -2 respectively. In terms of this Darboux basis, the trivial part (32) of the effective action is

$$\begin{aligned} W_{\text{prod}}(\alpha) &= \\ &= \frac{1}{2} \sum_{a,b,c} f_{abc} \alpha_{(0)}^a \alpha_{(0)}^b \alpha_{(3)}^c - \sum_{a,b,c} \sum_i f_{abc} \alpha_{(0)}^a \alpha_{(1)}^{ib} \alpha_{(2)i}^c + \frac{1}{6} \sum_{a,b,c} \sum_{i,j,k} f_{abc} \mu_{ijk} \alpha_{(1)}^{ia} \alpha_{(1)}^{jb} \alpha_{(1)}^{kc} = \\ &= \underbrace{\frac{1}{2} \pi_{\mathfrak{g}}(\alpha_{(3)}, [\alpha_{(0)}, \alpha_{(0)}])}_{W_{\text{prod}}^{003}} - \underbrace{\sum_i \pi_{\mathfrak{g}}(\alpha_{(0)}, [\alpha_{(1)}^i, \alpha_{(2)i}])}_{W_{\text{prod}}^{012}} + \underbrace{\frac{1}{6} \sum_{i,j,k} \mu_{ijk} \pi_{\mathfrak{g}}(\alpha_{(1)}^i, [\alpha_{(1)}^j, \alpha_{(1)}^k])}_{W_{\text{prod}}^{111}} \end{aligned}$$

Here μ_{ijk} is the totally antisymmetric tensor of structure constants of multiplication of 1-cohomologies: $\mu_{ijk} = \pi(\iota(e_{(1)i}), m(\iota(e_{(1)j}), \iota(e_{(1)k})))$.

Proposition 4. *The effective BV action W , induced from the abstract Chern–Simons action (19) on $\mathcal{F}' = H^\bullet(\mathcal{C}, \mathfrak{g})[1]$ has the form*

$$(33) \quad W(\alpha) = W_{\text{prod}}(\alpha) + F(\alpha_{(1)}^1, \dots, \alpha_{(1)}^{B_1}; \hbar)$$

where $F \in (\text{Fun}(\mathfrak{g}^{B_1})[[\hbar]])^{\mathfrak{g}}$ is some $\hbar$ -dependent function on $H^1(\mathcal{C}, \mathfrak{g})[1] \cong \mathfrak{g}^{B_1}$, invariant under the diagonal adjoint action of $\mathfrak{g}$, i.e.

$$(34) \quad F(\alpha_{(1)}^1 + [X, \alpha_{(1)}^1], \dots, \alpha_{(1)}^{B_1} + [X, \alpha_{(1)}^{B_1}]; \hbar) = F(\alpha_{(1)}^1, \dots, \alpha_{(1)}^{B_1}; \hbar) \mod X^2$$

at the first order in $X \in \mathfrak{g}$.

Proof. Ansatz (33) follows from the observation that the values of individual Feynman graphs $\Gamma \neq \Gamma_{0,3}$ in (31) depend only on the 1-cohomology: $W_\Gamma = W_\Gamma(\{\alpha_{(1)}^i\})$. The argument is as follows: suppose not all leaves of Γ are decorated with insertions of 1-cohomology. Then, since $\text{gh}(W_\Gamma) = 0$, there is at least one leaf decorated with insertion of 0-cohomology. Since $\iota(H^0(\mathcal{C})) = \mathbb{R} \cdot 1$, the value of Γ will contain one of the expressions $K^2(\dots)$, $K(\iota(\dots))$. Hence Feynman diagrams with insertion of cohomology of degree $\neq 1$ vanish due to (22), (23). This proves (33).

By construction, W has to satisfy the QME (Proposition 1). The QME for an action satisfying ansatz (33) is equivalent to ad-invariance of F (34):

$$\begin{aligned} \frac{1}{2} \{W, W\}' + \hbar \Delta' W &= \{W_{\text{prod}}^{012}, F\}' = \sum_i \sum_{a,b,c} f_{abc} \alpha_{(0)}^a \alpha_{(1)}^{ib} \frac{\partial}{\partial \alpha_{(1)}^{ic}} F = \\ &= \sum_i \left\langle [\alpha_{(0)}, \alpha_{(1)}^i], \frac{\partial}{\partial \alpha_{(1)}^i} \right\rangle_{\mathfrak{g}} F \end{aligned}$$

where $\langle \bullet, \bullet \rangle_{\mathfrak{g}}$ denotes the canonical pairing between $\mathfrak{g}$ and $\mathfrak{g}^*$.

Another explanation of (34) is the following: if $\mathfrak{g}$ is the Lie algebra of the Lie group $\mathcal{G}$, then the original abstract Chern–Simons action (19) is invariant under the adjoint action of $\mathcal{G}$, i.e. $\omega \mapsto g\omega g^{-1}$ for $g \in \mathcal{G}$. The embedding ι and the choice of the Lagrangian subspace $\mathcal{L} \subset \mathcal{F}''$ are also compatible with this symmetry. Hence $W(\alpha)$ is also invariant under the adjoint action of $\mathcal{G}$: $\alpha \mapsto g\alpha g^{-1}$, and (34) is the infinitesimal form of this symmetry. $\square$

Remark 4. There is another argument for ansatz (33) that can be formulated on the level of BV integral (28) itself, rather than on the level of Feynman diagrams.

Namely, the restriction of $S_{\text{int}}(\alpha, \omega'')$ (we refer to decomposition (30)) to the Lagrangian subspace $\mathcal{L}_K$ does not depend on $\alpha_{(0)}$. This means that the only term depending on $\alpha_{(0)}$ in (30) is the trivial one $W_{\text{prod}}(\alpha)$, constant on $\mathcal{L}_K$. Hence the non-trivial part of the effective action $W(\alpha) - W_{\text{prod}}(\alpha)$ does not depend on $\alpha_{(0)}$. Since it also has to be of ghost number zero, it can only depend on $\alpha_{(1)}$.

3.3. Dependence of the effective action on cohomology on induction data.

The effective action $W(\alpha)$ depends on induction data (ι, K) , and we are interested in describing how $W(\alpha)$ changes due to the deformation of (ι, K) .

In the terminology of Section 2, the type I deformations of induction data are of the form $\iota \mapsto \iota$, $K \mapsto K + [d, \delta\kappa]$ where $\delta\kappa : \mathcal{C}_{d-ex}^\bullet \rightarrow \mathcal{C}_{K-ex}^{\bullet-2}$ is a skew-symmetric linear map of degree -2. The corresponding deformation of the Lagrangian subspace $\mathcal{L}_K$ is described by the gauge fixing fermion $\Psi = \frac{1}{2}\pi(\omega, d\delta\kappa d\omega)$. Type II deformations change embedding as $\iota \mapsto \iota + d\delta I$ where $\delta I : H^\bullet(\mathcal{C}) \rightarrow \mathcal{C}_{K-ex}^{\bullet-1}$, and change chain homotopy in a minimal way (so as not to spoil properties (22), (20)): $K \mapsto K + \iota\delta I^T - \delta I\iota^T$. Type III transformations are forbidden in this setting, as discussed above (we have a canonical surjective map from $\ker d \subset \mathcal{C}$ to $H^\bullet(\mathcal{C})$ that sends the cocycle α to its cohomology class $[\alpha]$).

Due to Proposition 2, an infinitesimal deformation of (ι, K) induces an infinitesimal canonical transformation of $W(\alpha)$

$$W \mapsto W + \{W, R'\} + \hbar\Delta'R'$$

with generator given by the fiber BV integral

(35)

$$R'(\alpha) = e^{-W(\alpha)/\hbar} \int_{\mathcal{L}_K} e^{S(\iota(\alpha) + \omega'')/\hbar} \left(\frac{1}{2}\pi(\omega'', d\delta\kappa d\omega'') + \pi(\omega'', d\delta I\alpha) \right) \mu_{\mathcal{L}_K}$$

This integral is evaluated perturbatively in analogy with (31):

(36) $R'(\alpha) =$

$$\begin{aligned} & e^{-W(\alpha)/\hbar} \cdot e^{-\hbar \frac{1}{2}\pi^{-1}(\frac{\partial}{\partial\omega''}, K \frac{\partial}{\partial\omega''})} \Big|_{\omega''=0} \circ \\ & \circ \left(e^{S_{\text{int}}(\alpha, \omega'')/\hbar} \cdot \left(\frac{1}{2}\pi(\omega'', d\delta\kappa d\omega'') + \pi(\omega'', d\delta I\alpha) \right) \right) = \\ & \sum_{l=0}^{\infty} \hbar^l \sum_{n=0}^{\infty} \sum_{\Gamma^M \in G_{l,n}^M} \frac{1}{|\text{Aut}(\Gamma^M)|} R'_{\Gamma^M}(\alpha) \end{aligned}$$

Here the superscript M stands for “marked edge”, $G_{l,n}^M$ is the set of connected non-oriented trivalent graphs with l loops and n leaves and either one leaf or one internal edge marked. Values of Feynman graphs $R'_{\Gamma^M}(\alpha)$ are now homogeneous polynomials of degree n and ghost number -1 on $\mathcal{F}'$, obtained by the same Feynman rules as for $W_\Gamma(\alpha)$, supplemented with a Feynman rule for marked edge: we decorate the marked leaf with $\delta I_p^I \alpha^{pa}$ and marked internal edge with $\delta^{ab} \delta \kappa^{IJ}$.

Proposition 5. *The generator of the the infinitesimal canonical transformation induced on the effective BV action $W(\alpha)$ by the infinitesimal change of induction data (ι, K) has the following form:*

$$(37) \quad R'(\alpha) = \sum_{a,i} \alpha_{(2)i}^a G^{ia}(\alpha_{(1)}^1, \dots, \alpha_{(1)}^{B_1}; \hbar)$$

where $G = \sum_{a,i} G^{ia} \frac{\partial}{\partial \alpha_{(1)}^a} \in (\text{Vect}(\mathfrak{g}^{B_1})[[\hbar]])^{\mathfrak{g}}$ is some $\hbar$ -dependent vector field on $H^1(\mathcal{C}, \mathfrak{g})[1] \cong \mathfrak{g}^{B_1}$, equivariant under the diagonal adjoint action of $\mathfrak{g}$, i.e.

$$(38) \quad G^i(\alpha_{(1)}^1 + [X, \alpha_{(1)}^1], \dots, \alpha_{(1)}^{B_1} + [X, \alpha_{(1)}^{B_1}]; \hbar) = [X, G^i(\alpha_{(1)}^1, \dots, \alpha_{(1)}^{B_1}; \hbar)] \mod X^2$$

at first order in $X \in \mathfrak{g}$, for all $i = 1, \dots, B_1$ (and we set $G^i := \sum_a T_a G^{ia}$). The canonical transformation with generator (37) in terms of ansatz (33) is

$$(39) \quad F \mapsto F + G \circ (W_{\text{prod}}^{111} + F) + \hbar \text{div}(G)$$

Proof. The argument for ansatz (37) is pretty much the same as for (33): the value $R'_{\Gamma^M}(\alpha)$ of each Feynman graph $\Gamma^M \in G_{l,n}^M$ is linear in $\alpha_{(2)}$ and does not depend on $\alpha_{(0)}, \alpha_{(3)}$ for the following reason: Unless we decorate one leaf of Γ^M by $\alpha_{(2)}$ and all other leaves by $\alpha_{(1)}$, some leaf has to be decorated by $\alpha_{(0)}$ (since the total ghost number of $R'_{\Gamma^M}(\alpha)$ has to be -1). Then the contribution of this decoration of Γ^M vanishes due to $\iota(H^0(\mathcal{C})) = \mathbb{R} \cdot 1$ and the vanishing of the expressions $\delta I(e_{(0)})$, K^2 , $K \delta \kappa$, $\delta \kappa K$, $K \iota$, $\delta \kappa \iota$, one of which necessarily appears as contribution of a neighborhood of the place of insertion of 0-cohomology on the Feynman graph. This proves ansatz (37).

The equivariance of G (38) is equivalent to the fact that a canonical transformation with generator (37) preserves ansatz (33) for $W(\alpha)$. Indeed, if G were not equivariant, the term $\{W_{\text{prod}}^{012}, R'\}'$ would produce $\alpha_{(0)}$ -dependence for the canonically transformed effective action. The other explanation is that the equivariance of G is due to the invariance of R' under the adjoint action of the group $\mathcal{G}$, which is due to the fact that a deformation of (ι, K) is trivial in $\mathfrak{g}$ -coefficients and hence consistent with the adjoint $\mathcal{G}$ -action.

Rewriting the canonical transformation of the effective action (33) with generator (37) as (39) is straightforward. $\square$

Remark 5. Analogously to Proposition 4, one can prove ansatz (37) on the level of BV integral (35) instead of using Feynman diagrams. Namely, expressions $S(\iota(\alpha) + \omega'') - W_{\text{prod}}(\alpha)$, $W(\alpha) - W_{\text{prod}}(\alpha)$ and the expression in parentheses in (35) all do not depend on $\alpha_{(0)}$. Hence R' does not depend on $\alpha_{(0)}$. But it also has to be of ghost number -1, which can only be achieved if it is of form (37).

3.4. Invariants. We are interested in describing the effective action $W(\alpha)$ on cohomology modulo changes of induction data (ι, K) . Due to Propositions 4, 5, we can give a complete solution (i.e. describe the complete invariant) in case $B_1 = 0$, and find some partial solution (i.e. describe some, probably incomplete, invariant) for the case of a formal Frobenius algebra $\mathcal{C}$, meaning that we can find representatives for cohomology $\iota : H^\bullet(\mathcal{C}) \hookrightarrow \mathcal{C}$ closed under multiplication. In particular, in case $B_1 = 1$ the algebra $\mathcal{C}$ is necessarily formal.

Proposition 6. *If $B_1 = 0$, the effective action on cohomology is*

$$(40) \quad W(\alpha) = W_{\text{prod}}^{003}(\alpha) + F(\hbar)$$

where $W_{\text{prod}}^{003}(\alpha) = \frac{1}{2} \pi_{\mathfrak{g}}(\alpha_{(3)}, [\alpha_{(0)}, \alpha_{(0)}])$ and $F(\hbar)$ is an $\hbar$ -dependent constant, invariant under deformations of induction data (ι, K) .

Proof. Ansatz (40) is a restriction of (33) to the case $B_1 = 0$. Due to (37) and $B_1 = 0$, the generator of induced canonical transformation necessarily vanishes $R' = 0$. Hence $F(\hbar)$ is invariant under deformation of (ι, K) . $\square$

So $F(\hbar)$ is the complete invariant of $W(\alpha)$ for the $B_1 = 0$ case (which is an abstract model for Chern–Simons theory on a rational homology sphere) and is given by

$$(41) \quad F(\hbar) = \hbar \log \left(e^{-\hbar \frac{1}{2} \pi^{-1} \left(\frac{\partial}{\partial \omega^I}, K \frac{\partial}{\partial \omega^I} \right) \Big|_{\omega''=0}} \circ e^{\frac{1}{\hbar} \frac{1}{6} \pi(\omega'', l(\omega'', \omega''))} \right) = \\ = \sum_{l=2}^{\infty} \hbar^l \sum_{\Gamma^{\text{vac}} \in G_{l,0}} \frac{1}{|\text{Aut}(\Gamma^{\text{vac}})|} F_{\Gamma^{\text{vac}}}$$

Where we sum over trivalent connected non-oriented graphs without leaves Γ^{vac} (the “vacuum loops”). The contribution of a Feynman graph $F_{\Gamma^{\text{vac}}} \in \mathbb{R}$ is a number, computed by the same Feynman rules as for (31), just without the insertions of α .

Example 2 (Chevalley–Eilenberg complex of $\mathfrak{su}(2)$). We obtain an interesting example of abstract Chern–Simons theory with $B_1 = 0$ if we choose

$$\mathcal{C} = S^\bullet(\mathfrak{su}(2)^*[-1]) = \mathbb{R} \oplus \mathfrak{su}(2)^*[-1] \oplus (\wedge^2 \mathfrak{su}(2)^*)[-2] \oplus (\wedge^3 \mathfrak{su}(2)^*)[-3]$$

— the Chevalley–Eilenberg complex of the Lie algebra $\mathfrak{su}(2)$. This $\mathcal{C}$ is naturally a dg Frobenius algebra with super-commutative wedge product, Chevalley–Eilenberg differential

$$d: \quad e_1 \mapsto e_2 e_3, \quad e_2 \mapsto e_3 e_1, \quad e_3 \mapsto e_1 e_2$$

and pairing

$$\pi(\mathbf{1}, e_1 e_2 e_3) = \pi(e_1, e_2 e_3) = \pi(e_2, e_3 e_1) = \pi(e_3, e_1 e_2) = 1$$

Here $\{e_1, e_2, e_3\}$ is the basis in $\mathfrak{su}(2)^*$, dual to the basis $\{-\frac{i}{2}\sigma_1, -\frac{i}{2}\sigma_2, -\frac{i}{2}\sigma_3\}$ in $\mathfrak{su}(2)$, where $\{\sigma_i\}$ are the Pauli matrices; $\mathbf{1}$ is the unit in $\mathcal{C}^0 = \mathbb{R}$. This $\mathcal{C}$ can be understood as the algebra of left-invariant differential forms on the Lie group $SU(2) \sim S^3$ which is indeed quasi-isomorphic (as a dg algebra) to the whole de Rham algebra of the sphere S^3 ; thus the abstract Chern–Simons theory associated to this $\mathcal{C}$ is in a sense a toy model for true Chern–Simons theory on S^3 . The Hodge decomposition (26) for $\mathcal{C}$ is unique:

$$\mathcal{C} = \underbrace{\mathbb{R} \oplus (\wedge^3 \mathfrak{su}(2)^*)[-3]}_{H^\bullet(\mathcal{C})} \oplus \underbrace{\mathfrak{su}(2)^*[-1]}_{\mathcal{C}_{K-ex}} \oplus \underbrace{(\wedge^2 \mathfrak{su}(2)^*)[-2]}_{\mathcal{C}_{d-ex}}$$

and the BV field ω is

$$\omega = \underbrace{\alpha_{(0)} \mathbf{1} + \alpha_{(3)} e_1 e_2 e_3}_{\alpha} + \underbrace{\omega^1 e_1 + \omega^2 e_2 + \omega^3 e_3}_{\omega''_{K-ex}} + \underbrace{\omega^{23} e_2 e_3 + \omega^{31} e_3 e_1 + \omega^{12} e_1 e_2}_{\omega''_{d-ex}}$$

Here $\alpha_{(0)}, \alpha_{(3)}$ are $\mathfrak{g}$ -valued coordinates on $\mathcal{F}' = \mathfrak{g} \otimes H^\bullet(\mathcal{C})[1]$ and $\{\omega^I\}, \{\omega^{IJ}\}$ are $\mathfrak{g}$ -valued coordinates on $\mathcal{F}'' \subset \mathfrak{g} \otimes \mathcal{C}[1]$. The Lagrangian subspace (27) is $\mathcal{L} = \mathfrak{g} \otimes \mathfrak{su}(2)^*$. The effective action on cohomology, as defined by the integral (28), satisfies the ansatz (40) with $F(\hbar)$ given by

$$(42) \quad F(\hbar) = \hbar \log \frac{\int_{\mathfrak{g} \otimes \mathfrak{su}(2)^*} e^{\frac{1}{\hbar} \left(\frac{1}{2} \sum_{I=1}^3 \pi_{\mathfrak{g}}(\omega^I, \omega^I) + \pi_{\mathfrak{g}}(\omega^1, [\omega^2, \omega^3]) \right)} d\omega^1 d\omega^2 d\omega^3}{\int_{\mathfrak{g} \otimes \mathfrak{su}(2)^*} e^{\frac{1}{\hbar} \cdot \frac{1}{2} \sum_{I=1}^3 \pi_{\mathfrak{g}}(\omega^I, \omega^I)} d\omega^1 d\omega^2 d\omega^3}$$

The perturbative expansion (41) now reads

$$(43) \quad F(\hbar) = \hbar \log \left(e^{-\hbar \frac{1}{2} \sum_{I=1}^3 \pi_{\mathfrak{g}}^{-1} \left(\frac{\partial}{\partial \omega^I}, \frac{\partial}{\partial \omega^I} \right) \Big|_{\omega^I=0}} \circ e^{\frac{1}{\hbar} \pi_{\mathfrak{g}}(\omega^1, [\omega^2, \omega^3])} \right) =$$

$$= \sum_{l=2}^{\infty} (-1)^{l+1} \hbar^l \sum_{\Gamma^{\text{vac}} \in G_{l,0}} \frac{1}{|\text{Aut}(\Gamma^{\text{vac}})|} L_{\Gamma^{\text{vac}}}^{\mathfrak{su}(2)} \cdot L_{\Gamma^{\text{vac}}}^{\mathfrak{g}}$$

Here $L_{\Gamma^{\text{vac}}}^{\mathfrak{g}}$ denotes the “Lie algebra graph” (or the “Jacobi graph”), i.e. the number⁷ obtained by decorating vertices of Γ^{vac} with the structure constants f_{abc} of the Lie algebra $\mathfrak{g}$ and taking contraction of indices over edges of Γ^{vac} . In particular, for $L_{\Gamma^{\text{vac}}}^{\mathfrak{su}(2)}$ vertices are decorated with structure constants of $\mathfrak{su}(2)$ — the Levi-Civita symbol⁸ $\epsilon_{IJK} \in \{\pm 1, 0\}$. First terms in the series (43) are:

$$F(\hbar) = -\hbar^2 \frac{1}{12} \cdot 6 \cdot f_{abc} f_{abc} + \hbar^3 \left(\frac{1}{16} \cdot 12 \cdot f_{acd} f_{bcd} f_{aef} f_{bef} + \frac{1}{24} \cdot 6 \cdot f_{abc} f_{ade} f_{bef} f_{cfd} \right) + \dots$$

In particular, for $\mathfrak{g} = \mathfrak{su}(N)$ we have

$$F(\hbar) = -\hbar^2 \frac{1}{2} (N^3 - N) + \hbar^3 \frac{7}{8} (N^4 - N^2) - \hbar^4 \frac{23}{8} (N^5 - N^3) + \dots$$

As a side note, $L_{\Gamma^{\text{vac}}}^{\mathfrak{su}(2)}$ can be interpreted combinatorially as the number of ways to decorate edges of Γ^{vac} with 3 colors, such that in each vertex edges of all 3 colors meet (these decorations should be counted with signes, determined by the cyclic order of colors in each vertex). Thus for $\mathfrak{g} = \mathfrak{su}(2)$ the invariant $F(\hbar)$ is given by

$$F(\hbar) = \sum_{l=2}^{\infty} (-1)^{l+1} \hbar^l \sum_{\Gamma^{\text{vac}} \in G_{l,0}} \frac{1}{|\text{Aut}(\Gamma^{\text{vac}})|} (L_{\Gamma^{\text{vac}}}^{\mathfrak{su}(2)})^2$$

and it can be viewed as a generating function for certain interesting combinatorial quantities (and on the other hand, there is an “explicit” integral formula (42) for $F(\hbar)$ in terms of a 9-dimensional Airy-type integral).

Proposition 7. *Suppose $\mathcal{C}$ is formal and the embedding $\iota : H^\bullet(\mathcal{C}) \hookrightarrow \mathcal{C}$ is an algebra homomorphism, then:*

- *The tree part of the effective action on cohomology contains only the trivial W_{prod} term, i.e.*

$$(44) \quad W(\alpha) = W_{\text{prod}}(\alpha) + \hbar F^{(1)}(\alpha_{(1)}^1, \dots, \alpha_{(1)}^{B_1}) + \sum_{l=2}^{\infty} \hbar^l F^{(l)}(\alpha_{(1)}^1, \dots, \alpha_{(1)}^{B_1})$$

where $F^{(l)} \in (\text{Fun}(\mathfrak{g}^{B_1}))^{\mathfrak{g}}$ for $l = 1, 2, \dots$

- *The 1-loop part of effective action can be written as*

$$(45) \quad F^{(1)}(\alpha_{(1)}) = \frac{1}{2} \text{Str}_{\mathfrak{g} \otimes \mathcal{C}} \log (1 + K \circ l(\iota(\alpha_{(1)}), \bullet))$$

⁷Strictly speaking, one also has to choose a cyclic ordering of half-edges for each vertex of Γ^{vac} , and this choice affects the total sign of $L_{\Gamma^{\text{vac}}}^{\mathfrak{g}}$. But this ambiguity is cancelled in (43) since the factors $L_{\Gamma^{\text{vac}}}^{\mathfrak{g}}$ and $L_{\Gamma^{\text{vac}}}^{\mathfrak{su}(2)}$ change their signs simultaneously if we change the cyclic ordering of half-edges in any vertex of Γ^{vac} .

⁸We assume that the inner product on $\mathfrak{su}(2)$ is normalized as $\pi_{\mathfrak{su}(2)}(x, y) = -2 \text{tr}(xy)$ (in the fundamental representation of $\mathfrak{su}(2)$), so that the structure constants for the orthonormal basis are really ϵ_{IJK} .

- *Restriction of the 1-loop part of the effective action $F^{(1)}|_{\text{MC}}$ to the Maurer-Cartan set is invariant under deformations of (ι, K) (preserving the homomorphism condition for ι). Here $\text{MC} \subset H^1(\mathcal{C}, \mathfrak{g})[1]$ is given by*

$$\sum_{j,k} \mu_{ijk}[\alpha_{(1)}^j, \alpha_{(1)}^k] = 0.$$

Proof. The fact that ι is a homomorphism implies

$$(46) \quad K \circ l(\iota(\alpha), \iota(\alpha)) = 0$$

This means that all Feynman diagrams for $W(\alpha)$ that contain a vertex adjacent to two leaves and one internal edge, vanish. In particular, all tree diagrams except $\Gamma_{0,3}$ vanish. Together with Proposition 4 this implies (44). Also (46) implies that among 1-loop diagrams only “wheels” survive, and they are summed up to form (45) in standard way.

The next observation is that the tree part R^0 of the generator of the canonical transformation (36) induced by changing (ι, K) vanishes. This is a consequence of properties (46), $\delta\kappa \circ l(\iota(\alpha), \iota(\alpha)) = 0$ and $\pi(\delta I(\alpha), l(\iota(\alpha), \iota(\alpha))) = 0$ (which all follow from the fact that ι is a homomorphism). In terms of the vector field G (37), we have $G = \sum_{l=1}^{\infty} \hbar^l G^{(l)}$ and the l -loop part of effective action is transformed according to (39) as

$$F^{(l)} \mapsto F^{(l)} + G^{(l)} \circ W_{\text{prod}}^{111} + \sum_{i=1}^{l-1} G^{(l-i)} \circ F^{(i)} + \text{div } G^{(l-1)}$$

In particular, the 1-loop part is transformed as

$$F^{(1)} \mapsto F^{(1)} + G^{(1)} \circ W_{\text{prod}}^{111}$$

Since the Maurer-Cartan set is precisely the locus of stationary points of W_{prod}^{111} , the restriction $F^{(1)}|_{\text{MC}}$ is invariant. This finishes the proof. $\square$

Special case of formal $\mathcal{C}$ is the case $B_1 = 1$. Here the Maurer-Cartan set is $\text{MC} = H^1(\mathcal{C}, \mathfrak{g})[1] \cong \mathfrak{g}$, so the 1-loop part of effective action $F^{(1)} \in \text{Fun}(\mathfrak{g})^{\mathfrak{g}}$ is invariant (without any restriction).

3.5. Comments on relaxing the condition $K^2 = 0$ for chain homotopy. Let us introduce the notation

$$(47) \quad \text{Ind}_{\iota, K}(\bar{S})(\alpha) := \hbar \log \left(e^{-\hbar \frac{1}{2} \pi^{-1}(\frac{\partial}{\partial \omega''}, K \frac{\partial}{\partial \omega''})}|_{\omega''=0} \circ e^{(\bar{S}(\iota(\alpha) + \omega'') - S_0(\omega''))/\hbar} \right) \in \text{Fun}(\mathcal{F}')[[\hbar]]$$

for the “effective action” for some (not necessarily abstract Chern–Simons) action $\bar{S} \in \text{Fun}(\mathcal{F})[[\hbar]]$, defined by perturbation series, rather than by BV integral itself. Expression (47) is indeed the perturbation series generated by the BV integral

$$(48) \quad e^{\text{Ind}_{\iota, K}(\bar{S})(\alpha)/\hbar} = \frac{1}{N} \int_{\mathcal{L}_K} e^{\bar{S}(\iota(\alpha) + \omega'')/\hbar} \mu_{\mathcal{L}_K}$$

with normalization N as before (29). In particular for abstract Chern–Simons action $\bar{S} = S$ we recover the definition of W : $\text{Ind}_{\iota, K}(S)(\alpha) = W(\alpha)$.

The important observation with which we are concerned here is that definition (47) makes sense for a chain homotopy K not necessarily satisfying property $K^2 = 0$ (we assume that the other properties we demanded (20,21,22) hold), while the

definition via BV integral (48) is less general and uses essentially $K^2 = 0$ property for construction of Lagrangian subspace $\mathcal{L}_K$. To avoid confusion we will denote by $\hat{K}$ the chain homotopy without property (23) and reserve notation K for the “honest” chain homotopy with property (23). We will call the effective action defined via (47) with $\hat{K}$ as chain homotopy the “relaxed” effective action, while for an honest chain homotopy K we call the effective action (defined equivalently by (47) or by BV integral (48)) “strict”.

Proposition 8. *Let $\hat{K} : \mathcal{C}^\bullet \rightarrow \mathcal{C}^{\bullet-1}$ be a chain homotopy satisfying properties (20,21,22), but with $\hat{K}^2 \neq 0$, and let $K = \hat{K}d\hat{K}$ be the construction (25) applied to $\hat{K}$ (i.e. K satisfies all the properties (20,21,22,23)). Then the relaxed effective action $\text{Ind}_{\iota, \hat{K}}(S)$ is equivalent (i.e. connected by a canonical transformation) to the strict effective action $\text{Ind}_{\iota, K}(S)$.*

Proof. The first observation is that since $K = \hat{K}d\hat{K}$, we can write the relaxed chain homotopy as

$$(49) \quad \hat{K} = K + d\Lambda d$$

where $\Lambda = \hat{K}^3 : \mathcal{C}^3 \rightarrow \mathcal{C}^0$. In fact, formula (49), with arbitrary skew-symmetric, degree -3 linear map $\Lambda : \mathcal{C}^3 \rightarrow \mathcal{C}^0$, gives a general (finite) deformation of honest chain homotopy K , preserving properties (20,21,22), but violating (23) and satisfying in addition $K = \hat{K}d\hat{K}$. In other words, deformation (49) is the inverse of projection (25).

Second, we interpret the Feynman diagram decomposition for $\text{Ind}_{\iota, \hat{K}}$ with propagator $\hat{K}$ given by (49) as sum over graphs with edges decorated either by K or by $d\Lambda d$, and then raise the Feynman subgraphs with edges decorated only by $d\Lambda d$ into action. I.e. we obtain

$$(50) \quad \text{Ind}_{\iota, \hat{K}}(S) = \text{Ind}_{\iota, K}(S + \Phi_\Lambda)$$

where

$$(51) \quad \begin{aligned} \Phi_\Lambda = & \frac{1}{8}\pi(l(\omega, \omega), d\Lambda d l(\omega, \omega)) + \frac{1}{8}\pi(l(\omega, \omega), d\Lambda d l(\omega, d\Lambda d l(\omega, \omega))) + \\ & + \frac{1}{8}\pi(l(\omega, \omega), d\Lambda d l(\omega, d\Lambda d l(\omega, d\Lambda d l(\omega, \omega)))) + \dots + \\ & + \hbar \frac{1}{2} \cdot \frac{1}{2} \text{Str}_{\mathcal{F}} d\Lambda d l(\omega, d\Lambda d l(\omega, \bullet)) + \hbar \frac{1}{2} \cdot \frac{1}{3} \text{Str}_{\mathcal{F}} d\Lambda d l(\omega, d\Lambda d l(\omega, d\Lambda d l(\omega, \bullet))) + \dots \end{aligned}$$

Only the simplest trees (“branches”) and only the simplest one-loop diagrams (“wheels”) contribute to Φ_Λ , because any diagram with 3 incident internal edges decorated by $d\Lambda d$ automatically vanishes due to

$$\pi(d\Lambda d(\dots), l(d\Lambda d(\dots), d\Lambda d(\dots))) = 0$$

(which is implied by Leibniz identity in $\mathfrak{g} \otimes \mathcal{C}$, by $d^2 = 0$ and skew-symmetry of d).

Third, we notice that there is a canonical transformation from S to $S + \Phi_\Lambda$. A convenient way to describe a finite canonical transformation is to present a “homotopy”

$$S_\Lambda(t, dt) = S_\Lambda(t) + dt \cdot R_\Lambda(t) \in \text{Fun}(\mathcal{F})[[\hbar]] \otimes \Omega^\bullet([0, 1])$$

— a differential form on interval $[0, 1]$ with values in functions on $\mathcal{F}$ ($t \in [0, 1]$ is a coordinate on the interval), satisfying the QME on the extended space $\mathcal{F} \oplus$

$T[1]([0, 1])$:

$$(52) \quad (d_t + \hbar \Delta) S_\Lambda(t, dt) + \frac{1}{2} \{S_\Lambda(t, dt), S_\Lambda(t, dt)\} = 0$$

(where $d_t = dt \frac{\partial}{\partial t}$ is the de Rham differential on the interval), and satisfying boundary conditions

$$S_\Lambda(t)|_{t=0} = S, \quad S_\Lambda(t)|_{t=1} = S + \Phi_\Lambda$$

The extended QME is equivalent to the fact that $S_\Lambda(t)$ is a solution to the QME on $\mathcal{F}$ for any given $t \in [0, 1]$ plus the fact that $S_\Lambda(t + \delta t)$ is obtained from $S_\Lambda(t)$ by the infinitesimal canonical transformation with generator $\delta t \cdot R_\Lambda(t)$. In our case it is a straightforward exercise to present the desired homotopy between S and $S + \Phi_\Lambda$:

$$S_\Lambda(t, dt) = S + \Phi_{t\Lambda} + dt \frac{1}{t} R_{t\Lambda}$$

where $\Phi_{t\Lambda}$ is defined by (51) with Λ rescaled by t , and R_Λ is given by

$$R_\Lambda = \frac{1}{4} \pi(d\Lambda d\omega, l(\omega, \omega)) + \frac{1}{4} \pi(d\Lambda d\omega, l(\omega, d\Lambda d l(\omega, \omega))) + \\ + \frac{1}{4} \pi(d\Lambda d\omega, l(\omega, d\Lambda d l(\omega, d\Lambda d l(\omega, \omega)))) + \dots$$

We showed that S and $S + \Phi_\Lambda$ are connected by a homotopy. Due to Proposition 1 and Lemma 1 the effective actions $\text{Ind}_{\iota, K}(S)$ and $\text{Ind}_{\iota, K}(S + \Phi_\Lambda)$ are also connected by a homotopy defined by

$$(53) \quad e^{W_\Lambda(t, dt)(\alpha)/\hbar} = \frac{1}{N} \int_{\mathcal{L}_K} e^{S_\Lambda(t, dt)(\iota(\alpha) + \omega'')/\hbar} \mu_{\mathcal{L}_K}$$

Together with (50) this finishes the proof. $\square$

Remark 6. An immediate consequence of Proposition 8 is that the relaxed effective action $\text{Ind}_{\iota, \hat{K}}(S)$ satisfies the QME on $\mathcal{F}'$.

Remark 7. Expression (51) suggests that it is itself the value of a certain Gaussian integral. Namely, the restriction of $S + \Phi_\Lambda$ to the subspace $\iota(\mathcal{F}') \oplus \mathcal{L}_K \subset \mathcal{F}$ can be written as a fiber integral over fibers $\mathfrak{g} \otimes \mathcal{C}_{d-ex}^1[1]$ of vector bundle $\iota(\mathcal{F}') \oplus \mathcal{L}_K \oplus \mathfrak{g} \otimes \mathcal{C}_{d-ex}^1[1] \rightarrow \iota(\mathcal{F}') \oplus \mathcal{L}_K$ (which is a sub-bundle of $\mathcal{F} \rightarrow \iota(\mathcal{F}') \oplus \mathcal{L}_K$):

$$e^{(S + \Phi_\Lambda)/\hbar} \Big|_{\iota(\mathcal{F}') \oplus \mathcal{L}_K} = \int_{\mathfrak{g} \otimes \mathcal{C}_{d-ex}^1[1]} e^{(S + \frac{1}{2} \pi(\omega, K \Lambda^{-1} K \omega))/\hbar} \mu_{\mathfrak{g} \otimes \mathcal{C}_{d-ex}^1[1]}$$

(by $\mu(\dots)$ we always mean the Lebesgue measure on the vector space). Here we assume for simplicity that $\Lambda : \mathcal{C}^3 \rightarrow \mathcal{C}^0$ is an isomorphism, and we denote $\Lambda^{-1} : \mathcal{C}^0 \rightarrow \mathcal{C}^3$ its inverse. Now we can write the relaxed effective action as

$$(54) \quad e^{\text{Ind}_{\iota, \hat{K}}(S)(\alpha)/\hbar} = \int_{\mathcal{L}_K \oplus \mathfrak{g} \otimes \mathcal{C}_{d-ex}^1[1]} e^{S(\iota(\alpha) + \omega'')/\hbar} e^{\frac{1}{2\hbar} \pi(\omega'', K \Lambda^{-1} K \omega'')} \mu_{\mathcal{L}_K \oplus \mathfrak{g} \otimes \mathcal{C}_{d-ex}^1[1]}$$

where we integrate over the coisotropic subspace $\mathcal{L}_K \oplus \mathfrak{g} \otimes \mathcal{C}_{d-ex}^1[1] \subset \mathcal{F}''$ instead of just the Lagrangian subspace $\mathcal{L}_K \subset \mathcal{F}''$, with measure $e^{\frac{1}{2\hbar} \pi(\omega, K \Lambda^{-1} K \omega)} \mu_{\mathcal{L}_K \oplus \mathfrak{g} \otimes \mathcal{C}_{d-ex}^1[1]}$ that is constant in direction of $\mathcal{L}_K$ and is Gaussian in direction of $\mathfrak{g} \otimes \mathcal{C}_{d-ex}^1[1]$. So expression (54) gives an elegant interpretation of the relaxed effective action via a “thick” fiber BV integral (over a Gaussian-smeared Lagrangian subspace). In the case of Λ of general rank (not necessarily an isomorphism), we should replace $\mathcal{C}_{d-ex}^1$

by $\text{im}(d\Lambda) \subset C_{d-ex}^1$ in this discussion, which leads to a thick fiber BV integral over a smaller coisotropic $\mathcal{L}_K \oplus \mathfrak{g} \otimes \text{im}(d\Lambda)[1] \subset \mathcal{F}''$.

3.5.1. Invariants from the relaxed effective action. We are interested in describing the invariants of the relaxed effective action $\text{Ind}_{\iota, \hat{K}}(S)$ modulo deformations of the “relaxed induction data” $(\iota, \hat{K})$. Since we have a general description (49) for a non-strict chain homotopy, a general (infinitesimal) deformation of the relaxed induction data $(\iota, \hat{K})$ can be described as a deformation of the underlying strict induction data (ι, K) studied in the beginning of Section 3.3, plus a deformation of Λ . Now we can restate some weak version of Proposition 6 for the case of relaxed induction, where we make a special choice for the Lie algebra of coefficients: $\mathfrak{g} = \mathfrak{su}(2)$ (we generalize this in the Remark afterwards). We are able to recover only the two-loop part of the complete invariant $F(\hbar)$ of the strict effective action in this discussion.

Proposition 9. *If $B_1 = 0$ and $\mathfrak{g} = \mathfrak{su}(2)$, the relaxed effective action on cohomology has the form*

$$(55) \quad \text{Ind}_{\iota, \hat{K}}(S)(\alpha) = A(\hbar) \cdot W_{\text{prod}}^{003}(\alpha) + B(\hbar)$$

where $W_{\text{prod}}^{003} = \frac{1}{2} \sum_{a,b,c} \epsilon_{abc} \alpha_{(0)}^a \alpha_{(0)}^b \alpha_{(3)}^c$, and $A(\hbar) = 1 + \hbar A^{(1)} + \hbar^2 A^{(2)} + \dots$, $B(\hbar) = \hbar^2 B^{(2)} + \hbar^3 B^{(3)} + \dots$ are some $\hbar$ -dependent constants. The number

$$(56) \quad F^{(2)} = 3A^{(1)} + B^{(2)}$$

is invariant under deformations of the relaxed induction data $(\iota, \hat{K})$.

Proof. Ansatz (55) follows from the following argument. Since the relaxed effective action $\hat{W} = \text{Ind}_{\iota, \hat{K}}(S) \in \text{Fun}(\mathfrak{g}[1] \oplus \mathfrak{g}[-2])[[\hbar]]$ is a function of ghost number zero and $\mathfrak{g} = \mathfrak{su}(2)$ is 3-dimensional (and hence an at most cubic dependence on $\alpha_{(0)}$ is possible), it has to be of the form

$$\hat{W}(\alpha) = \frac{1}{2} \sum_{a,b,c} A_{abc}(\hbar) \alpha_{(0)}^a \alpha_{(0)}^b \alpha_{(3)}^c + B(\hbar)$$

Since $\hat{W}$ also inherits the invariance under adjoint action of $\mathcal{G} = SU(2)$: $\alpha \mapsto g\alpha g^{-1}$ for $g \in \mathcal{G}$ from the ad-invariance of the original abstract Chern–Simons action S , the tensor $A_{abc}(\hbar)$ has to be of the form $A_{abc}(\hbar) = A(\hbar)\epsilon_{abc}$. This proves (55). The fact that the series for $A(\hbar)$ starts as $A(\hbar) = 1 + O(\hbar)$ is due to the vanishing of all tree diagrams $\Gamma \neq \Gamma_{0,3}$ which follows from formality of $\mathcal{C}$ (it is automatic for the $B_1 = 0$ case) and $\hat{K} \circ \iota = 0$. The series for $B(\hbar)$ starts with an $O(\hbar^2)$ -term just because $G_{l,0}$ is empty for $n = 0, 1$.

Next, we know from Proposition 8 that there is a homotopy $W_\Lambda(t, dt)(\alpha)$ connecting the strict effective action $W = \text{Ind}_{\iota, K}(S) = W_{\text{prod}}^{003}(\alpha) + F(\hbar)$ (ansatz (40)) and the relaxed one $\hat{W} = \text{Ind}_{\iota, \hat{K}}(S)$. The general ansatz for $W_\Lambda(t, dt)$, taking into account that $\mathfrak{g} = \mathfrak{su}(2)$ and that construction (53) is compatible with ad-invariance $\alpha \mapsto g\alpha g^{-1}$, is the following:

$$(57) \quad W_\Lambda(t, dt)(\alpha) = A(\hbar; t) W_{\text{prod}}^{003}(\alpha) + B(\hbar; t) + \\ + dt \cdot \left(C(\hbar; t) \frac{1}{12} \sum_{a,b,c,d,e} \epsilon_{abc} \delta_{de} \alpha_{(0)}^a \alpha_{(0)}^b \alpha_{(0)}^c \alpha_{(3)}^d \alpha_{(3)}^e + D(\hbar; t) \sum_{a,b} \delta_{ab} \alpha_{(0)}^a \alpha_{(3)}^b \right)$$

where A, B, C, D are some functions of $\hbar$ and the homotopy parameter $t \in [0, 1]$. The boundary conditions are:

$$A(\hbar; 0) = 1, \quad A(\hbar; 1) = A(\hbar), \quad B(\hbar; 0) = F(\hbar), \quad B(\hbar; 1) = B(\hbar)$$

The extended QME (52) for homotopy $W_\Lambda(t, dt)$ is equivalent to the system

$$(58) \quad \frac{\partial}{\partial t} A(\hbar; t) = \hbar C(\hbar; t) - A(\hbar; t) \cdot D(\hbar; t)$$

$$(59) \quad \frac{\partial}{\partial t} B(\hbar; t) = 3\hbar D(\hbar; t)$$

Next, the argument of vanishing of non-trivial trees (due to formality of $\mathcal{C}$, $K \circ \iota = 0$ and $d \circ \iota = 0$) applies again to construction (53). Hence, we have

$$A(\hbar; t) = 1 + \hbar A^{(1)}(t) + O(\hbar^2), \quad C(\hbar; t) = \hbar C^{(1)}(t) + O(\hbar^2), \quad D(\hbar; t) = \hbar D^{(1)}(t) + O(\hbar^2)$$

And due to $G_{0,0} = G_{1,0} = \emptyset$, we again have $B(\hbar; t) = \hbar^2 B^{(2)}(t) + O(\hbar^3)$. Equation (58) in order $O(\hbar)$ together with (59) in order $O(\hbar^2)$ yield

$$\frac{\partial}{\partial t} A^{(1)}(t) = -D^{(1)}(t), \quad \frac{\partial}{\partial t} B^{(2)}(t) = 3D^{(1)}(t)$$

Hence the expression $3A^{(1)}(t) + B^{(2)}(t)$ does not depend on t . For $t = 0$ it is the two-loop part of the invariant $F(\hbar)$ from (40), while for $t = 1$ it is the right hand side of (56). This concludes the proof. $\square$

Remark 8 (Generalization). The generalization of Proposition 9 to an arbitrary (quadratic) $\mathfrak{g}$ is straightforward. We no longer have ansatz (55) for $\hat{W}$, since there might be more ad-invariant functions on $\mathfrak{g}[1] \oplus \mathfrak{g}[-2]$, but we still can write

$$\hat{W}(\alpha) = (1 + \hbar A^{(1)}) W_{\text{prod}}^{003}(\alpha) + \hbar^2 B^{(2)} + O(\hbar^3 + \hbar^2(\alpha_{(0)})^2 \alpha_{(3)} + \hbar(\alpha_{(0)})^4 (\alpha_{(3)})^2)$$

(we could prescribe weight 1 to $\hbar$ and weight $1/3$ to $\alpha_{(0)}$ and $\alpha_{(3)}$, then we write explicitly terms with weight ≤ 2). The fact that $O(\hbar \alpha_{(0)} \alpha_{(0)} \alpha_{(3)})$ -contribution is proportional to W_{prod}^{003} (in principle there could be some other invariant tensor of rank 3) is explained by the fact that it is given by a single Feynman diagram $\Gamma \in G_{1,3}$ — the wheel with 3 leaves — and the Lie algebra part of this diagram is described by contraction $\sum_{d,e,f} f_{ade} f_{bef} f_{cfd} \propto f_{abc}$. Following the proof of Proposition 9, for the homotopy we have

$$\begin{aligned} W_\Lambda(t, dt)(\alpha) = & (1 + \hbar A^{(1)}(t)) W_{\text{prod}}^{003}(\alpha) + \hbar^2 B^{(2)}(t) + O(\hbar^3 + \hbar^2(\alpha_{(0)})^2 \alpha_{(3)} + \hbar(\alpha_{(0)})^4 (\alpha_{(3)})^2) + \\ & + dt \cdot \left(\hbar D^{(1)}(t) \sum_a \alpha_{(0)}^a \alpha_{(3)}^a + O(\hbar^2 \alpha_{(0)} \alpha_{(3)} + \hbar(\alpha_{(0)})^3 (\alpha_{(3)})^2) \right) \end{aligned}$$

The extended QME at order $O(dt \hbar (\alpha_{(0)})^2 \alpha_{(3)} + dt \hbar^2)$ yields

$$\frac{\partial}{\partial t} A^{(1)}(t) = -D^{(1)}(t), \quad \frac{\partial}{\partial t} B^{(2)}(t) = \dim \mathfrak{g} \cdot D^{(1)}(t)$$

Hence the two-loop part of the invariant $F(\hbar)$ is expressed in terms of coefficients of the relaxed effective action $\hat{W}$ as

$$(60) \quad F^{(2)} = \dim \mathfrak{g} \cdot A^{(1)} + B^{(2)}$$

The other case discussed in Section 3.4, the case of formal $\mathcal{C}$ with general B_1 , is translated straightforwardly into the setting of relaxed effective actions.

Proposition 10. *Suppose $\mathcal{C}$ is formal and $\iota : H^\bullet(\mathcal{C}) \hookrightarrow \mathcal{C}$ is an algebra homomorphism. Then the relaxed effective action has the form*

$$(61) \quad \text{Ind}_{\iota, \hat{K}}(S)(\alpha) = W_{\text{prod}}(\alpha) + \hbar \hat{F}^{(1)}(\alpha) + O(\hbar^2)$$

where the one-loop part can be expressed as a super-trace:

$$(62) \quad \hat{F}^{(1)}(\alpha) = \frac{1}{2} \text{Str}_{\mathfrak{g} \otimes \mathcal{C}} \log \left(1 + \hat{K} \circ l(\iota(\alpha), \bullet) \right)$$

The restriction of the one-loop relaxed effective action to the Maurer-Cartan set $\hat{F}^{(1)}|_{\text{MC}}$ is invariant under deformations of $(\iota, \hat{K})$, preserving the homomorphism property of ι .

Proof. Analogously to the case of a strict chain homotopy K (Proposition 7), formality of $\mathcal{C}$ together with $\hat{K} \circ \iota = 0$ imply vanishing of non-trivial trees (hence the ansatz (61)) and that the only possibly non-vanishing one-loop diagrams are wheels (hence the super-trace formula (62)).

Second, we know that there is a homotopy $W_\Lambda(t, dt)(\alpha)$ connecting the strict effective action $W(\alpha) = W_{\text{prod}}(\alpha) + \hbar F^{(1)}(\alpha_{(1)}) + \dots$ to the relaxed one $\hat{W}(\alpha) = W_{\text{prod}}(\alpha) + \hbar \hat{F}^{(1)}(\alpha) + \dots$:

$$W_\Lambda(t, dt)(\alpha) = \left(W_{\text{prod}}(\alpha) + \hbar F^{(1)}(t)(\alpha) + O(\hbar^2) \right) + dt \cdot \left(\hbar R'_\Lambda(t)(\alpha) + O(\hbar^2) \right)$$

(here we again exploit the vanishing of trees implied by construction (53), formality of $\mathcal{C}$ and $K \circ \iota = d \circ \iota = 0$). The extended QME for the homotopy at order $O(dt \cdot \hbar)$ is

$$\frac{\partial}{\partial t} F^{(1)}(t)(\alpha) = \{W_{\text{prod}}(\alpha), R'_\Lambda(t)(\alpha)\}'$$

Hence the restriction $F^{(1)}(t)|_{\overline{\text{MC}}}$ does not depend on t , where

$$\overline{\text{MC}} = \{\alpha \mid l(\iota(\alpha), \iota(\alpha)) = 0\} \subset \mathcal{F}'$$

is the set of critical points of W_{prod} (the “non-homogeneous Maurer-Cartan set”). As $\text{MC} \subset \overline{\text{MC}}$, the restriction $F^{(1)}(t)|_{\text{MC}}$ also does not depend on t , hence

$$\hat{F}^{(1)}|_{\text{MC}} = F^{(1)}|_{\text{MC}}$$

As the right hand side is invariant (Proposition 7), so is the left hand side. This concludes the proof. $\square$

Obviously, going from the restriction to $\overline{\text{MC}}$ to the restriction to MC , we do not lose any invariant information, since $F^{(1)}$ depends only on $\alpha_{(1)}$ and not on other components of α , which implies $\hat{F}^{(1)}|_{\overline{\text{MC}}} = \hat{F}^{(1)}|_{\text{MC}}$. Note also that in Proposition 9 we managed to reconstruct only the two-loop part of the invariant $F(\hbar)$ (Proposition 6) in the relaxed setting. On the other hand, we can completely reconstruct the invariant $F^{(1)}|_{\text{MC}}$ of Proposition 7 in the relaxed setting.

3.6. Examples of dg Frobenius algebras. In this Section we will provide some examples of non-negatively graded dg Frobenius algebras with pairing of degree -3 and zeroth Betti number $B_0 = 1$, i.e. algebras suitable for constructing abstract Chern–Simons actions.

Example 1: minimal dg Frobenius algebra. Let V be a vector space and $\mu \in \wedge^3 V^*$ an arbitrary exterior 3-form on V . Then we construct the dg Frobenius algebra $\mathcal{C}$ from this data as

$$\mathcal{C} := \mathbb{R} \cdot 1 \oplus V[-1] \oplus V^*[-2] \oplus \mathbb{R} \cdot v$$

where v is a degree 3 element. The pairing π is defined to be the canonical pairing between $V[-1]$ and $V^*[-2]$, and also we set $\pi(1, v) := 1$. We define the product m as

$$m(1, x) = x, \quad m(x_{(1)}, y_{(1)}) = \langle \mu, \bar{x}_{(1)} \wedge \bar{y}_{(1)} \rangle, \quad m(x_{(1)}, z_{(2)}) = \langle \bar{x}_{(1)}, \bar{z}_{(2)} \rangle$$

where $x \in \mathcal{C}$ (element of arbitrary degree), $x_{(1)}, y_{(1)} \in V[-1]$, $z_{(2)} \in V^*[-2]$ and bar means shifting an element to degree zero, e.g. $\bar{x}_{(1)} = sx_{(1)} \in V$, $\bar{z}_{(2)} = s^2 z_{(2)} \in V^*$; $\langle \bullet, \bullet \rangle$ denotes the canonical pairing. The differential d is set to zero. This construction gives the most general minimal (i.e. with zero differential) dg Frobenius algebra (non-negatively graded, with pairing of degree -3 and $B_0 = 1$). Abstract Chern-Simons action associated to a minimal algebra is a purely cubic polynomial in the fields, with coefficients being the components of μ . Inducing the effective action on cohomology is the identity operation, since here $\mathcal{C} = H^\bullet(\mathcal{C})$.

Example 2: differential concentrated in degree $\mathcal{C}^1 \rightarrow \mathcal{C}^2$. Let V be a vector space, $\mu \in \wedge^3 V^*$ be some 3-form on V and $\delta : S^2 V \rightarrow \mathbb{R}$ some symmetric pairing on V (not necessarily non-degenerate). We define $\mathcal{C}$, the pairing π and product m as in Example 1, but now we construct the differential $d : V[-1] \rightarrow V^*[-2]$ from δ as $x_{(1)} \mapsto s^{-2} \delta(\bar{x}_{(1)}, \bullet)$. The two other components of the differential $\mathbb{R} \cdot 1 \rightarrow V[-1]$ and $V^*[-2] \rightarrow \mathbb{R} \cdot v$ are set to zero as before. This construction gives the general dg Frobenius algebra with differential concentrated in degree $\mathcal{C}^1[-1] \rightarrow \mathcal{C}^2[-2]$ only.

Here the Hodge decomposition $\mathcal{C} = \iota(H^\bullet(\mathcal{C})) \oplus \mathcal{C}_{d-ex} \oplus \mathcal{C}_{K-ex}$ is defined by a choice of projection $p : V \rightarrow \ker \delta$ or equivalently by choosing a complement V'' of $\ker \delta \subset V$ in V (here we understand δ as the self-dual map $V \rightarrow V^*$). We set

$$H^\bullet(\mathcal{C}) = \mathbb{R} \cdot 1 \oplus (\ker \delta)[-1] \oplus (\ker \delta)^*[-2] \oplus \mathbb{R} \cdot v,$$

$$\mathcal{C}_{d-ex} = (\text{im } \delta)[-2], \quad \mathcal{C}_{K-ex} = V''[-1]$$

The embedding $\iota : H^\bullet(\mathcal{C}) \hookrightarrow \mathcal{C}$ is canonical in degrees 0, 1, 3 and given by $p^* : (\ker \delta)^* \hookrightarrow V^*$ in degree 2. The (non-vanishing part of the) chain homotopy K is the inverse map for the isomorphism $d : V''[-1] \rightarrow (\text{im } \delta)[-2]$. So the induction data (ι, K) is completely determined by the choice of p . This means in particular that only the deformations of induction data of type II are possible here. Also there are no relaxed chain homotopies $\hat{K}$ (other than the strict one described above), which is obvious from (49). Despite these simplifications, the effective action on cohomology for such $\mathcal{C}$ is in general non-trivial. A particular example here is the case $\mathcal{C} = S^\bullet(\mathfrak{su}(2)^*[-1])$ discussed in section 3.4.

Example 3: “doubled” commutative dga. Let $\mathcal{V} = V^0 \oplus V^1[-1]$ be a unital commutative associative dg algebra, concentrated in degrees 0 and 1, with differential d_V and multiplication m_V , and satisfying $\dim H^0(\mathcal{V}) = 1$. Then we set

$$\mathcal{C} := \mathcal{V} \oplus \mathcal{V}^*[-3] = V^0 \oplus V^1[-1] \oplus (V^1)^*[-2] \oplus (V^0)^*[-3]$$

with pairing π generated by the canonical pairing between $\mathcal{V}$ and $\mathcal{V}^*$. The component of differential $V^0 \rightarrow V^1[-1]$ is given by d_V , the component $(V^1)^*[-2] \rightarrow (V^0)^*[-3]$ — by the dual map $(d_V)^*$, and the component $V^1[-1] \rightarrow (V^1)^*[-2]$ is

set to zero. Multiplication for elements of $\mathcal{C}$ of degrees 0 and 1 is given by $m_{\mathcal{V}}$ and is extended to other degrees by the cyclicity property (15):

$$m(x_{(0,1)}, y_{(0,1)}) := m_{\mathcal{V}}(x_{(0,1)}, y_{(0,1)}), \quad m(x_{(0,1)}, z_{(2,3)}) := \langle m_{\mathcal{V}}(\bullet, x_{(0,1)}), z_{(2,3)} \rangle$$

for $x_{(0,1)}, y_{(0,1)} \in \mathcal{V}$ and $z_{(2,3)} \in \mathcal{V}^*[-3]$. In particular, the product of elements of degree 1 in $\mathcal{C}$ is zero.

Hodge decomposition for $\mathcal{C}$ is fixed by choosing an embedding $\iota_{\mathcal{V}} : H^{\bullet}(\mathcal{V}) \hookrightarrow \mathcal{V}$ (it is canonical in degree zero since $H^0(\mathcal{V}) = \mathbb{R} \cdot 1$, but non-canonical in degree one) and a retraction $r_{\mathcal{V}} : \mathcal{V} \rightarrow H^{\bullet}(\mathcal{V})$. Equivalently, we choose a splitting of $\mathcal{V}$ into representatives of cohomology and the complement $\mathcal{V} = \iota_{\mathcal{V}}(H^{\bullet}(\mathcal{V})) \oplus \mathcal{V}''$, i.e. $V^0 = \mathbb{R} \cdot 1 \oplus V^{0''}$, $V^1 = \iota_{\mathcal{V}}(H^1(\mathcal{V})) \oplus V^{1''}$. Then the Hodge decomposition for $\mathcal{C}$ is $\mathcal{C} = \iota(H^{\bullet}(\mathcal{C})) \oplus \mathcal{C}_{d-ex} \oplus \mathcal{C}_{K-ex}$ with

$$\begin{aligned} H^{\bullet}(\mathcal{C}) &= \mathbb{R} \oplus H^1(\mathcal{V})[-1] \oplus (H^1(\mathcal{V}))^*[-2] \oplus \mathbb{R} \cdot v, \\ \mathcal{C}_{d-ex} &= V^{1''}[-1] \oplus (V^{0''})^*[-3], \quad \mathcal{C}_{K-ex} = V^{0''} \oplus (V^{1''})^*[-2] \end{aligned}$$

Here $v \in (V^0)^*[-3]$ is the element defined by the component of retraction $r_{\mathcal{V}} : V^0 \rightarrow \mathbb{R} \cdot 1$. The embedding $\iota : H^{\bullet}(\mathcal{C}) \rightarrow \mathcal{C}$ is given by $\iota_{\mathcal{V}}$ in degrees 0,1 and by $(r_{\mathcal{V}})^*$ in degrees 2,3. As in the Example 2, only type II deformations of the induction data (ι, K) are possible here. However, one can introduce the relaxed chain homotopy (49) here with $\Lambda : (V^{0''})^*[-3] \rightarrow V^{0''}$.

The BV integral (28) is Gaussian in this example since the cyclic product $\pi(\bullet, m(\bullet, \bullet))$ vanishes on $(\mathcal{C}_{K-ex})^{\otimes 3}$ (for trivial degree reasons). Hence, the only Feynman graphs contributing to $W(\alpha)$ are wheels and the trivial tree $\Gamma_{0,3}$ (“branches” could also contribute, but they vanish since the component of the multiplication $\iota(H^1(\mathcal{C})) \otimes \iota(H^1(\mathcal{C})) \rightarrow \mathcal{C}^2$ vanishes and the non-trivial part of $W(\alpha)$ depends only on $\alpha_{(1)}$ due to Proposition 4). The effective action can be written as

$$W(\alpha) = W_{\text{prod}}^{003}(\alpha) + W_{\text{prod}}^{012}(\alpha) + 2 \cdot \frac{1}{2} \hbar \operatorname{tr}_{\mathfrak{g} \otimes V^0} \log(1 + K \circ l(\iota(\alpha_{(1)}), \bullet))$$

(the factor 2 accounts for the contribution of the trace over $\mathfrak{g} \otimes (V^{1''})^*$ — the other half of the Lagrangian subspace $\mathcal{L}_K$).

4. THREE-MANIFOLD INVARIANTS

We now wish to extend the results of the previous Sections to the Frobenius algebra $\Omega^{\bullet}(M)$ of differential forms on a smooth compact 3-manifold M (with de Rham differential, wedge product and integration pairing). This will provide us with the effective action (around the trivial connection) of Chern–Simons theory [18]. The invariant of this Frobenius algebra will also constitute an invariant of 3-manifolds modulo diffeomorphisms.

The discussion of the previous Sections, however, does not go through automatically since $\Omega^{\bullet}(M)$ is infinite dimensional. As in previous works [2, 11, 3] the way out is to restrict oneself to a special class of chain homotopies K for which the finite dimensional arguments are simply replaced by the application of Stokes’ theorem. However, at some point of the construction we have to choose a framing and hence we get invariants of framed 3-manifolds.

4.1. Induction data and the propagator. As in the finite-dimensional case we fix an embedding $\iota: H^\bullet(M) \hookrightarrow \Omega^\bullet(M)$. By $\chi \in \Omega^3(M \times M)$ we will denote the representative of the Poincaré dual of the diagonal Δ determined by this embedding. Namely, if $\{1, \{\alpha_i, \beta^i\}_{i=1, \dots, B_1}, v\}$ is a basis of $\iota(H^\bullet(M))$ (with 1 the constant function, $\alpha_i \in \Omega^1$, $\beta^i \in \Omega^2$, $v \in \Omega^3$ and $\int_M \alpha_i \beta^j = \delta_i^j$, $\int_M v = 1$), then

$$\chi = v_2 - \alpha_{i,1} \beta_2^i + \beta_1^i \alpha_{i,2} - v_1,$$

where we have used Einstein's convention over repeated indices and for any form $\gamma \in \Omega^\bullet(M)$ we write γ_i for $\pi_i^* \gamma$ with π_1 and π_2 the two projections $M \times M \rightarrow M$.

The chain homotopy K is assumed to be determined by a smooth integral kernel $\hat{\eta}$. Namely, let $C_2^0(M) := M \times M \setminus \Delta = \{(x_1, x_2) \in M \times M : x_1 \neq x_2\}$ be the open configuration space of two points in M and let $C_2(M)$ be its Fulton–MacPherson–Axelrod–Singer (FMAS) compactification [8, 2] obtained by replacing the diagonal Δ with its unit normal bundle. Let $\pi_{1,2}$ be the extensions to the compactification of the projection maps $\pi_i(x_1, x_2) = x_i$. Then $\hat{\eta}$ is a smooth 2-form on $C_2(M)$ and the chain homotopy K is defined by

$$K\alpha = -\pi_{2*}(\hat{\eta} \pi_1^* \alpha), \quad \alpha \in \Omega^\bullet(M),$$

where a lower $*$ denotes integration along the fiber.

For K to be a symmetric chain homotopy satisfying (20), $\hat{\eta}$ must satisfy the following properties (see [3, 5] for more details):

- P1:** $d\hat{\eta} = \pi^* \chi$, with π the extension to $C_2(M)$ of the inclusion of $C_2^0(M)$ into $M \times M$.
- P2:** $\int_{\partial C_2(M)} \hat{\eta} = -1$.
- P3:** $T^* \hat{\eta} = -\hat{\eta}$, where T is the extension to $C_2(M)$ of the involution $(x_1, x_2) \mapsto (x_2, x_1)$ of $C_2^0(M)$.

Observe that $\partial C_2(M)$ is canonically diffeomorphic to the sphere tangent bundle STM of M (STM is the quotient of TM by the action $(x, v) \mapsto (x, \lambda v)$, $\lambda \in \mathbb{R}_*^+$). Condition P2 may then be refined to

- P2':** $\iota_\partial^* \hat{\eta} = -\eta$, where ι_∂ is the inclusion map $STM = \partial C_2(M) \hookrightarrow C_2(M)$ and η is a given odd global angular form on STM .

Recall that a global angular form on a sphere bundle $S \rightarrow M$ is a differential form on the total space S whose restriction to each fiber generates its cohomology and whose differential is minus the pullback of a representative of the Euler class (in our case zero). These two properties are consistent with the restriction to the boundary of P1 and P2. Odd here means $T^* \eta = -\eta$ where T is the antipodal map on each fiber, and this is compatible with the restrictions to the boundary of P3. Global angular forms always exist.

Remark 9 (Kontsevich [11]). Since STM is trivial for every 3-manifold, there is a simple choice for η only depending on a choice of framing or, equivalently, a choice of trivialization $f: STM \xrightarrow{\sim} M \times S^2$. One simply sets $\eta = f^* \omega$ where ω is the normalized $SO(3)$ -invariant volume form on S^2 (tensor 1 $\in \Omega^0(M)$).

Remark 10 ([3]). It is also possible to construct an odd global angular form depending on the choice of a connection but not on a framing. Namely, realize STM as the S^2 -bundle associated to the frame bundle $F(M)$:

$$STM = F(M) \times_{SO(3)} S^2$$

(we reduce the structure group of the frame bundle to $SO(3)$ by picking a Riemannian metric). By choosing a metric connection θ (e.g., the Levi-Civita connection), one defines $\bar{\eta} := \omega + d(\theta_i x^i)/(2\pi)$; here the x^i s, $i = 1, 2, 3$, are the homogeneous coordinates on S^2 , while the θ_i s are the coefficients of the connection in the basis $\{\xi_i\}$ of $\mathfrak{so}(3)$ given by $(\xi_i)_{jk} = \epsilon_{ijk}$. It is then easy to show that $\bar{\eta}$ is a global angular form on $F(M) \times S^2$ and that it is basic. Hence it defines a global angular form η on STM .

Lemma 2 ([3]). *For any given odd global angular form η on STM , there exists a propagator $\hat{\eta} \in \Omega^2(C_2(M))$ satisfying $P1, P2', P3$.*

Proof. The complete proof in the case of a rational homology sphere is contained in [3]. (The general case, whose proof is a straightforward generalization of the previous case, is spelled out in [5].) For completeness, we briefly recall the idea of the proof. One chooses a tubular neighborhood U of $\partial C_2(M)$ and a subneighborhood V . One picks a compactly supported function ρ which is constant and equal to -1 on V and is even under the action of T . Let p be the projection $U \rightarrow \partial C_2(M)$. The differential of the form $\rho p^* \eta$ is a representative of the Thom class of the normal bundle of Δ . The form $\rho p^* \eta$ may be extended by zero to the whole of $C_2(M)$ and its differential is then a representative of the Poincaré dual of Δ . Thus, its difference from the given representative χ will be exact and actually **regular** on the diagonal (see Appendix B for the definition). Namely, $d\rho p^* \eta = \pi^*(\chi + d\alpha)$, $\alpha \in \Omega^3(M \times M)$. Since both $\rho p^* \eta$ and χ are odd under the action of T , one may also choose α to be odd. Finally, one sets $\hat{\eta} := \rho p^* \eta - \pi^* \alpha$. $\square$

4.2. The improved propagator. In the previous Sections it was also important to assume the conditions $K \circ \iota = 0$ and $K^2 = 0$. In terms of the propagator $\hat{\eta}$ they correspond to additional conditions that are easily expressed by using the

Definition 1 (Compactification of configuration spaces). The FMAS compactification $C_n(M)$ of the configuration space $C_n^0(M) := \{(x_1, \dots, x_n) \in M^n : x_i \neq x_j \forall i \neq j\}$ is obtained by taking the closure

$$C_n(M) := \overline{C_n^0(M)} \subset M^n \times \prod_{S \subset \{1, \dots, n\} : |S| \geq 2} \text{Bl}(M^S, \Delta_S),$$

where $\text{Bl}(M^S, \Delta_S)$ denotes the differential-geometric blowup obtained by replacing the principal diagonal Δ_S in M^S with its unit normal bundle $N(\Delta_S)/\mathbb{R}_*^+$. See [2, 4].

We recall that the $C_n(M)$ are smooth manifolds with corners.

Notation 1. Given a differential form γ on M , we write $\gamma_i := \pi_i^* \gamma$ where π_i is the extension to $C_n(M)$ of the projection $C_n^0(M) \rightarrow M$, $(x_1, \dots, x_n) \mapsto x_i$. We also set $\hat{\eta}_{ij} := \pi_{ij}^* \hat{\eta}$, where π_{ij} is the extension to the compactifications of the projection $C_n^0(M) \rightarrow C_2^0(M)$, $(x_1, \dots, x_n) \mapsto (x_i, x_j)$. Given a product of such forms on a compactified configuration space and a set of indices $i, j, \dots$, we denote by $\int_{i, j, \dots}$ the fiber integral over the points $x_i, x_j, \dots$. This notation also takes care of orientation. Namely, if $i, j, \dots$ are not ordered, the integral carries the sign of the permutation to order them.

We are finally ready to list the two simple properties corresponding to $K \circ \iota = 0$ and $K^2 = 0$:

P4: $\int_2 \hat{\eta}_{12} \gamma_2 = 0$ for all $\gamma \in \iota(H^\bullet(M))$.

P5: $\int_2 \hat{\eta}_{12} \hat{\eta}_{23} = 0$.

Lemma 3. *For any given odd global angular form η on STM , there exists a propagator $\hat{\eta} \in \Omega^2(C_2(M))$ satisfying P1, P2', P3, P4.*

Proof. The idea is to apply transformation (24) to a propagator as the one constructed in Lemma 2. This must however be reformulated in terms of integral kernels. Namely, let $\hat{\eta}$ satisfy P1, P2', P3. Set

$$\begin{aligned} \lambda_{12} := & \int_3 \eta_{13} v_3 - \int_3 \eta_{23} v_3 + \int_3 \eta_{13} \alpha_{3,i} \beta_2^i - \beta_1^i \int_3 \eta_{23} \alpha_{3,i} + \int_3 \eta_{13} \beta_3^i \alpha_{2,i} + \\ & + \alpha_{1,i} \int_3 \eta_{23} \beta_3^i - \alpha_{1,i} \int_{3,4} \beta_3^i \eta_{34} \beta_4^j \alpha_{2,j} - \beta_1^i \int_{3,4} \alpha_{3,i} \eta_{34} v_4 - \int_{3,4} v_3 \eta_{34} \alpha_{4,i} \beta_2^i. \end{aligned}$$

By construction the new propagator $\hat{\eta} - \lambda$ satisfies P3 and P4. It is not difficult to check that λ is closed, so P1 is still satisfied. Finally observe that the integrals on one argument simply produce a form on M , while the integrals on two arguments simply produce a number. As a consequence, λ is (the pullback by π of) a form on $M \times M$. As it is also T -odd, its restriction to the boundary vanishes. Thus, P2' still holds. $\square$

4.3. On property P5.

Lemma 4. *For any given odd global angular form η on STM , there exists a propagator $\hat{\eta} \in \Omega^2(C_2(M))$ satisfying P1, P2, P3, P4, P5.*

Proof. We apply transformation (25) in the equivalent form

$$K_3 = K_2 + [d, dK_2^3]$$

to the propagator $\hat{\eta}$ constructed in Lemma 3. In terms of integral kernels, the new propagator is $\hat{\eta} + \gamma$ with $\gamma_{12} := dd_1 f_{12}$ and

$$f_{12} := \int_{3,4} \hat{\eta}_{13} \hat{\eta}_{34} \hat{\eta}_{42}.$$

Properties P1 and P2 are obviously satisfied as we have changed the propagator by an exact term. As for property P3, observe that equivalently γ may be written as $d_2 d_1 f_{12}$ and that f is even under the action of T . Finally property P4 is easily checked by integration. $\square$

It would be very useful to prove the following

Conjecture 1. *The propagator constructed in Lemma 4 also satisfies property P2'.*

Observe that since γ is T -odd, it would suffice to show that it is regular on the diagonal (i.e., a pullback from $M \times M$). For this it would suffice to show that f or at least df has this property. By its definition f looks rather regular, but at the moment we have no complete proof of this fact.

Remark 11 (The Riemannian propagator). The physicists' treatment of Chern–Simons theory would simply be to choose a Riemannian metric g and use it to impose the Lorentz gauge-fixing. Out of this one gets the propagator $d^* \circ G$, where G is the Green function for $\Delta + \mathcal{P}'$, where Δ is the Laplace operator and $\mathcal{P}'$ is the projection to harmonic forms in the Hodge decomposition. The integral kernel of this propagator is a smooth two-form $\hat{\eta}$ on $C_2(M)$ satisfying properties P1, P2', P3,

P4, P5 with odd angular form on the boundary of the type described in Remark 10. In this case, the metric connection is actually the Levi-Civita connection for the chosen Riemannian metric. See [2] and [3, Remark 3.6].

4.4. The construction of the invariant by a framed propagator. We now fix the boundary value of the propagator as in Remark 9 for a given choice f of framing. We also choose an embedding ι of cohomology and denote by $P_{\iota,f}$ the space of propagators satisfying properties P1,P2',P3. Observe that, by Lemma 3, $P_{\iota,f}$ is not empty. For $\hat{\eta} \in P_{\iota,f}$ we define $\text{Ind}_{\iota,\hat{\eta}} \in \text{Fun}(\mathcal{F}')[[\hbar]]$, $\mathcal{F}' = H^\bullet(M, \mathfrak{g})[1] = H^\bullet(M)[1] \otimes \mathfrak{g}$, analogously to $\text{Ind}_{\iota,K}$ as at the beginning of subsection 3.5 by the following obvious changes of notations:

- (1) Every chain homotopy K is replaced by a propagator $\hat{\eta}$;
- (2) every vertex is replaced by a point in the compactified configuration space over which we eventually integrate.

Signs may be taken care of by choosing an ordering of vertices and of half edges at each vertex (see, e.g., [3]).

All computations in Section 3 go through as they are simply replaced by Stokes theorem:

$$d \int_{C_n(M)} = (-1)^n \int_{C_n(M)} d + \int_{\partial C_n(M)}.$$

Among the codimension-one boundary components of $C_n(M)$ we distinguish between principal and hidden faces: the former correspond to the collapse of exactly two points, the latter to the collapse of more than two points. Principal faces contribute by the same combinatorics as in Section 3, whereas hidden faces do not contribute by our choice of propagators because of Kontsevich's vanishing Lemmata [11]. As a result we conclude that $\text{Ind}_{\iota,\hat{\eta}}$ satisfies the quantum master equation for every choice of induction data as above. For more details, we refer to Appendix A.

By the same reasoning and by the arguments of subsection 3.5, we may prove that $\text{Ind}_{\iota,\hat{\eta}}$ is canonically equivalent to a strict effective action. This allows us to recover at least the two-loop part of the complete invariant of a rational homology sphere as discussed in subsection 3.5.1.

Remark 12. If Conjecture 1 were true, the space $P'_{\iota,f}$ of propagators satisfying properties P1,P2',P3,P4,P5 would not be empty. We could then repeat the above construction using $P'_{\iota,f}$ instead of $P_{\iota,f}$ and get a strict effective action directly. The discussions of subsections 3.2, 3.3 and 3.4 would then go through and, in particular, Propositions 4, 5, 6, 7 would hold.

Remark 13. In [3] an invariant for framed rational homology spheres was introduced. The boundary condition for the propagator was different (see next subsection), but this is immaterial for the present discussion. Namely, choose a propagator in $P_{\iota,f}$ and define $\tilde{\eta}_{123} := \hat{\eta}_{12} + \hat{\eta}_{23} + \hat{\eta}_{31}$. If M is a rational homology sphere, $\tilde{\eta}_{123}$ is closed. Now take the graphs appearing in the constant part of the strict effective action and reinterpret them as follows:

- (1) Each vertex is replaced by a point in the compactified configuration space;
- (2) an extra point x_0 is added on which one puts the representative $v \in \iota(H^3(M))$ with $\int v = 1$;
- (3) each chain homotopy is replaced by $\tilde{\eta}$ (more precisely, the chain homotopy between vertices i and j is replaced by $\tilde{\eta}_{ij0}$).

It is now possible to show that this produces an invariant of (M, f) . This is a different way of getting the invariant corresponding to the constant part of the strict effective action for a choice of propagator not necessarily satisfying property P5. We do not have a direct proof that this invariant is the same. The indirect proof consists of showing that both invariants are finite type with the same normalizations along the lines of [13].

If Conjecture 1 were true, then it would immediately follow that this invariant is exactly the constant part of the strict effective action. In fact, for a propagator as in the Conjecture, it is not difficult to show that only the term $\hat{\eta}_{ij}$ in each $\tilde{\eta}_{ij0}$ would contribute (and the integration over x_0 would then decouple). Since the induced action is constant on $P_{\iota, f}$, by restriction to $P'_{\iota, f} \subset P_{\iota, f}$ we would prove the claim.

4.5. The unframed propagator. Instead of using Kontsevich’s propagator, one may proceed as in [3] and define the propagator by choosing the global angular form on $\partial C_2(M)$ as in Remark 10. Recall that in this case no choice of framing is required. On the other hand, one needs to specify a Riemannian metric g and a metric connection θ . We denote by $P_{\iota, g, \theta}$ the space of propagators corresponding to these choices.

We proceed exactly as in the previous subsection to define the effective action (see Appendix A for more details). In particular we want to check independence on the induction data; so we choose a path in $P_{\iota, g, \theta}$ and consider the effective action W as a function on the shifted cohomology tensor the differential forms on $[0, 1]$ and check whether the extended QME $(d + \hbar\Delta)W + \{W, W\}/2 = 0$ holds. The only difference with respect to the previous subsection is that there is an extra set of boundary components of the configuration spaces that may appear: namely, the most degenerate faces corresponding to the collapse of all points. These faces may be treated exactly as in [2, 3] and one shows that their contribution is a multiple of the first Pontryagin form $-\text{tr } F_\theta^2/(8\pi^2)$, where F_θ is the curvature of θ . The important point is that the coefficient depends only on the graph involved but not on the 3-manifold M . As a result the effective action might not satisfy the extended quantum master equation. However, one may easily compensate for this by adding to it the integral over M of the Chern–Simons 3-form of the connection θ pulled back from $F(M)$ to M by choosing a section f (i.e., a framing). The framing now appears because of this correction but is not present in the propagator.

The main disadvantage of this approach is that one does not know how to compute the universal coefficients. (It is known that the coefficients vanish for graphs with an odd number of loops, while for the graph with two loops one may compute the coefficient explicitly and see that it is not zero.) The advantage is that $P_{\iota, g, \theta}$ contains a subspace of propagators satisfying also property P5: These are the integral kernels constructed in [2], see Remark 11. With these choices, and the addition of the frame-dependent constant as in the previous paragraph, one gets an induced effective action satisfying all properties stated in subsections 3.2, 3.3 and 3.4. More precisely, let

$$\text{CS}(M, \theta, f) := -\frac{1}{8\pi^2} \int_M f^* \text{Tr} \left(\theta d\theta + \frac{2}{3} \theta^3 \right)$$

be the Chern–Simons integral for a connection θ on the frame bundle $F(M)$ of M and a framing f (regarded here as a section of $F(M)$).

Theorem 1. *Let M be a compact 3-manifold and $\mathfrak{g}$ a quadratic Lie algebra. Then*

- (1) For every choice of Riemannian metric g on M , the effective action W constructed using the Riemannian propagator of Remark 11 is a function on $H^\bullet(M, \mathfrak{g})[1]$, solves the quantum master equation and has the properties described in Proposition 4. In addition it has the form given in (40) in case $B_1(M) = 0$ and in (44) in case M is formal.
- (2) There is a universal element $\phi \in \hbar^2 \mathbb{R}[[\hbar^2]]$, depending only on the choice of Lie algebra $\mathfrak{g}$, such that the modified effective action

$$\widetilde{W}(M, g, f) := W(M, g) + \phi \text{CS}(M, \theta_g, f),$$

where θ_g is the Levi-Civita connection for g , solves the QME and is independent of g modulo canonical transformations as in Proposition 5. In particular we get invariants for the framed 3-manifold (M, f) as in Propositions 6 and 7.

Remark 14. The leading contribution to ϕ may be explicitly computed and yields

$$\phi = C_2(\mathfrak{g})\hbar^2/48 + O(\hbar^4)$$

with $C_2(\mathfrak{g}) = f_{abc}f_{abc}$, where the f_{abc} are the structure constants of $\mathfrak{g}$ in an orthonormal basis. It is not known whether there are nonvanishing higher order corrections.

APPENDIX A. THE CHERN–SIMONS MANIFOLD INVARIANT

In this Appendix we give more details on the construction outlined in subsection 4.5. To a graph Γ with $|\Gamma|$ vertices we associate an element ω_Γ of $\Omega^\bullet(C_{|\Gamma|}(M)) \otimes \text{Fun}(\mathcal{F}')$ as follows:⁹

- to each edge we associate the pullback of a propagator by the corresponding projection from $C_{|\Gamma|}(M)$ to $C_2(M)$;
- to each leaf we associate the pullback (by the corresponding projection from $C_{|\Gamma|}(M)$ to M) of $\sum z_\mu^a \gamma^\mu e_a$, where $\{\gamma^\mu\}$ is the chosen basis of $H^\bullet(M)$, $\{e_a\}$ is an orthonormal basis of $\mathfrak{g}$, and the $\{z_\mu^a\}$ s are the corresponding coordinate functions on $\mathcal{F}'$;
- on each vertex we put a structure constant in the orthonormal basis chosen above.

Then we take the wedge product of the differential forms and sum over Lie algebra indices for each edge. The result does not depend on the choice of orthonormal basis for $\mathfrak{g}$ but depends on a choice of numbering of the vertices and of orientation of the edges. If we however make the same choice also to orient $C_{|\Gamma|}(M)$, then

$$\int_{C_{|\Gamma|}(M)} \omega_\Gamma \in \text{Fun}(\mathcal{F}')$$

is well defined. We define Z (the exponential of W) to be the sum of $\omega_{|\Gamma|}/|\text{aut } \Gamma|$ over all trivalent graphs, where $|\text{aut } \Gamma|$ is the order or the group of automorphisms of Γ . Proving the QME for W is equivalent to proving $\Delta Z = 0$.

The main observation is that Property P1 of the propagator and the same combinatorics as in the toy model imply that ΔZ is obtained by replacing the ω_Γ s by $d\omega_\Gamma$ s one by one. We then use Stokes theorem. The contributions of principal faces (i.e., boundary faces of configuration spaces corresponding to the collapse

⁹Here and in the rest of the Appendix, the symbol $\otimes$ is understood as the completed tensor product: i.e., the space of functions on the Cartesian product of the corresponding supermanifolds.

of two vertices) sum up to zero thanks to the Lie algebra contributions (this is also the same combinatorics as in the toy model). Hidden faces (i.e., the other boundary faces) may in principle contribute. Let γ be the subgraph corresponding to a hidden face (i.e., the vertices of γ are those that collapse and its edges are the edges between such vertices). By simple dimensional reasons, the hidden faces corresponding to γ vanishes if γ has a univalent vertex; if γ has a bivalent vertex, its contribution also vanishes by Kontsevich’s lemma thanks to Property P3. Since we only consider trivalent graphs, we are left with contributions coming from the collapse of *all* vertices of a connected component of Γ s with *no leaves*. These are the graphs that contribute to the constant part of Z and thus of W . The latter contributions also vanish by a simple dimensional argument.

More generally, to keep track of the choices involved in the propagator, we consider a one parameter family of choices with parameter $t \in I := [0, 1]$ and show $\Delta' \tilde{Z} = 0$ with $\Delta' := \Delta + dt \frac{d}{dt}$ and $\tilde{Z} \in Fun(\mathcal{F}') \otimes \Omega^\bullet(I)$ constructed as follows. Let $\hat{\eta}$ be a one-parameter family of propagators regarded as an element of $\Omega^2(C_2(M) \times I)$ related, at every $t \in I$, by Property P1 to the one parameter family $\{\gamma^\mu\}$ of bases of $\Omega^\bullet(M)$. Let $\dot{\eta} \in \Omega^2(C_2(M) \times I)$ and $\dot{\gamma}^\mu \in \Omega^\bullet(M \times I)$ be their t -derivatives. We may assume $\int_M \gamma^\mu \dot{\gamma}^\nu = 0, \forall \mu, \nu$, for more general choices may be compensated by a linear transformation of $H^\bullet(M)$. Observe that by Property P2', the restriction of $\hat{\eta}$ to the boundary is fixed, so the restriction of $\dot{\eta}$ vanishes. Actually, by construction $\dot{\eta}$ vanishes in a whole neighborhood of the boundary, so it is a regular form. Let

$$\lambda_{13} := \frac{\int_2 \hat{\eta}_{12} \dot{\eta}_{23} - \dot{\eta}_{12} \hat{\eta}_{23}}{2},$$

which is regular by Lemma 5 in Appendix B. Also set $\xi_1^\mu := (\int \hat{\eta}_{12} \dot{\gamma}_2^\mu)$. We define

$$\begin{aligned} \tilde{\eta} &:= \hat{\eta} - \lambda dt, \\ \tilde{\gamma}^\mu &:= \gamma^\mu + \xi^\mu dt, \\ \tilde{\chi} &:= g_{\mu\nu} \tilde{\gamma}^\mu \tilde{\gamma}^\nu = \chi + O(dt), \end{aligned}$$

where $g_{\mu\nu}$ is the metric on $H^\bullet(M)$ in the given basis. A simple computation then shows $D \tilde{\eta} = \tilde{\chi}$ and $D \tilde{\gamma}^\mu = 0$ with $D := d + dt \frac{d}{dt}$. Finally, we define $\tilde{Z}$ as above by using $\tilde{\eta}$ and $\tilde{\gamma}$ instead of $\hat{\eta}$ and γ , respectively. We now observe that applying Δ' to $\tilde{Z}$ is the same as applying D to the propagators. Reasoning as above by Stokes theorem, we see that the only possible nonvanishing contributions come from hidden faces corresponding to the collapse of all vertices of a connected component with *no leaves*.¹⁰ These contributions also vanish if one uses a framed propagator, whereas the choice of an unframed propagator yields some constant times the integral of the Pontryagin form on M as a one-form on I , see [3].

By writing $\tilde{Z} = Z + \zeta dt$, we may see that the equivalences are produced by ζ . These are very particular kinds of BV equivalences as ζ consists of graphs decorated by propagators and generators of cohomology classes with the exception of one edge that is decorated by λ or one leaf that is decorated by ξ .

If Property P5 holds, the computation of $\tilde{Z}$ simplifies drastically. By construction we obtain $\int_2 \hat{\eta}_{12} \lambda_{23} = \int_2 \lambda_{12} \hat{\eta}_{23} = 0$ and $\int_2 \eta_{12} \xi_2^\mu = 0$. As a result, whenever a vertex is decorated by $1 \in \Omega^0(M)$ the corresponding integral vanishes. Thus, $\tilde{Z}$, as a function on $H^\bullet(M, \mathfrak{g})[1]$ is independent of the coordinates in degree 1 apart

¹⁰Observe that, since λ is regular, only η will appear in the boundary computations.

from the trivial classical term. Since Z is of degree zero, it will only depend on the coordinates of degree zero. Since ζ is of degree -1 , it will be linear in the coordinates of degree -1 with coefficients depending on the coordinates of degree zero.

APPENDIX B. REGULAR FORMS

Let $\varpi: C_n(M) \rightarrow M^n$ be the extension to the compactification of the inclusion $C_n(M)^0 \rightarrow M^n$. We call a form on $C_n(M)$ *regular* if it is a pullback by ϖ . Recall the maps π_i s and π_{ij} s defined in Notation 1 on page 26.

Lemma 5. *Let α and β be differential forms on $C_2(M)$ and let at least one of them be regular. Then their convolution $\alpha * \beta := \int_2 \alpha_{12} \beta_{23} := \pi_{13,*}(\pi_{12}^* \alpha \pi_{23}^* \beta)$ is regular.*

Proof. Suppose that, e.g., α is regular; i.e., $\alpha = \varpi^* \alpha'$, $\alpha' \in \Omega^\bullet(M \times M)$. Define

$$\gamma = (\text{pr}_1 \times \pi_2)_* ((\text{pr}_1 \times \pi_1)^* \alpha' \text{pr}_2^* \beta) \in \Omega^\bullet(M \times M),$$

where pr_1 and pr_2 are the projections from $M \times C_2(M)$ to the two factors. It then follows that $\alpha * \beta = \varpi^* \gamma$. $\square$

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Finite dimensional AKSZ-BV theories

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Abstract

We describe a canonical reduction of AKSZ-BV theories to the cohomology of the source manifold. We get a finite dimensional BV theory that describes the contribution of the zero modes to the full QFT. Integration can be defined and correlators can be computed. As an illustration of the general construction we consider two dimensional Poisson sigma model and three dimensional Courant sigma model. When the source manifold is compact, the reduced theory is a generalization of the AKSZ construction where we take as source the cohomology ring. We present the possible generalizations of the AKSZ theory.

1 Introduction

The Batalin-Vilkovisky (BV) formalism [2] is widely regarded as the most powerful and general approach to the quantization of gauge theories. The idea is to extend the space of fields of the original gauge theory by auxiliary fields (ghosts, antighosts, Lagrangian multipliers etc.) and their conjugate antifields in such way that the total field-antifield space is equipped with two canonical structures: an odd Poisson bracket (antibracket) and an odd second order differential operator Δ (BV Laplacian). The original gauge action should be extended to master action defined on the total field-antifield space in such way that the master action satisfies the master equation (the equation involving antibracket and BV Laplacian). The path integral is defined as an appropriate integral over half of the field-antifield space. The master action can be computed within the homological perturbation theory which can be very difficult to carry out in general. In [1] Alexandrov, Kontsevich, Schwarz and Zaboronsky (AKSZ) proposed a way to construct solutions of the BV classical master equation directly, without any reference to a classical action with a set of gauge symmetries. Their approach (AKSZ method) uses mapping spaces of supermanifolds equipped with the additional structures. The AKSZ method was applied in [1] to the Chern-Simons theory, the Witten A- and B-models. Furthermore the AKSZ approach was applied to two-dimensional Poisson sigma model in [7] and to three-dimensional Courant sigma model in [17], [24]. Moreover the higher dimensional case of open p-branes were discussed in [22, 15, 16].

In the present work we propose the reduction within the AKSZ framework to a finite dimensional BV theory which governs the zero mode contribution and is responsible for the semiclassical approximation in the full theory. These finite dimensional BV theories offer interesting perspective on some of the standard geometry. The proposed reduction naturally suggests a generalization of AKSZ framework.

The paper is organized as follows. In Section 2 we sketch the essentials of the BV formalism. Section 3 reviews the AKSZ formalism and discusses its reduction in general terms. The rest of the paper is devoted to the consideration of the examples of this reduction and to the discussion of finite dimensional BV theories arising as a result of the reduction. Section 4 treats the Poisson sigma model defined over a closed Riemann surface of genus g . Section 5 considers the Courant sigma model, a three dimensional topological sigma model. Section 6 considers the more involved setup of the Poisson sigma model on the Riemann surface with a boundary. In Section 7 we discuss the generalization of the AKSZ construction. In Section 8 we give a summary and discuss the possible developments. In Appendix A we calculate the Berezinian measure for the reduced BV manifold of the Courant sigma model. In Appendix B we collect some facts on the relative

cohomology of the manifolds with boundaries.

2 Summary of BV formalism

In this Section we recall the basic notions of BV formalism and fix the notation. For further details the reader may consult the following reviews [9, 12].

The BV algebra can be defined in many different but equivalent ways. In particular, a Gerstenhaber algebra (odd Poisson algebra) $(\mathcal{A}, \{ , \})$ together with an odd $\mathbb{R}$ -linear map

$$\Delta : \mathcal{A} \longrightarrow \mathcal{A} ,$$

which squares to zero $\Delta^2 = 0$ and generates the bracket $\{ , \}$ as

$$\{f, g\} = (-1)^{|f|} \Delta(fg) + (-1)^{|f|+1} (\Delta f)g - f(\Delta g) ,$$

is called a BV-algebra. Δ is called odd Laplace operator (odd Laplacian). Quite often such odd Poisson bracket is called antibracket.

The canonical example of BV algebra is given by the space of functions on $W \oplus \Pi W^*$, where W is a superspace, W^* is its dual and Π stands for the reversed parity functor. $W \oplus \Pi W^*$ is equipped with an odd non-degenerate pairing. Let y^a be the coordinates on W (the fields) and y_a^+ be the corresponding coordinates on ΠW^* (the antifields). We denote the parity of y^a as $(-1)^{|y^a|}$ and that of y_a^+ as $(-1)^{|y_a^+|} = (-1)^{|y^a|+1}$. Then the odd Laplacian is defined as follows

$$\Delta = (-1)^{|y_a|} \frac{\partial}{\partial y_a^+} \frac{\partial}{\partial y^a} . \quad (2.1)$$

It generates the canonical antibracket on $C^\infty(W \oplus \Pi W^*)$

$$\{f, g\} = (-1)^{|y^a|} \frac{\overleftarrow{\partial} f}{\partial y_a^+} \frac{\overrightarrow{\partial} g}{\partial y^a} + (-1)^{|y^a|} \frac{\overleftarrow{\partial} f}{\partial y^a} \frac{\overrightarrow{\partial} g}{\partial y_a^+} , \quad (2.2)$$

where we use the notation $\overrightarrow{\partial}_v f = \partial_v f$ and $\overleftarrow{\partial}_v f = (-1)^{|v||f|} \partial_v f$. Indeed the bracket (2.2) is non degenerate and defines the canonical odd symplectic structure on $W \oplus \Pi W^*$.

A Lagrangian submanifold $\mathcal{L} \subset W \oplus \Pi W^*$ is an isotropic supermanifold of maximal dimension. The volume form $dy^1 \dots dy^n dy_1^+ \dots dy_n^+$ induces a well defined volume form on $\mathcal{L}$. Thus the integral

$$\int_{\mathcal{L}} f, \quad f \in C^\infty(W \oplus \Pi W^*) \quad (2.3)$$

is defined for any $\mathcal{L}$. If $\Delta f = 0$, then then the integral (2.3) depends only on the homology class of $\mathcal{L}$. Moreover the integral (2.3) is zero for any Lagrangian $\mathcal{L}$ if $f = \Delta g$.

The canonical example $W \oplus \Pi W^*$ can be generalized to the cotangent bundle $T^*[-1]\mathcal{M}$ of any graded manifold $\mathcal{M}$ [25]. By a graded manifold we mean a sheaf of freely generated $\mathbb{Z}$ -graded commutative algebras over a smooth manifold [26]. As a cotangent bundle, $T^*[-1]\mathcal{M}$ is naturally equipped with an odd Poisson bracket that makes $C^\infty(T^*[-1]\mathcal{M})$ to a Gerstenhaber algebra. The idea is that locally one can map $T^*[-1]\mathcal{M}$ to $W \oplus \Pi W^*$, define the bracket on coordinates with (2.2) and then glue the patches in a consistent manner.

Now in order to define the odd Laplacian Δ we need an integration over $T^*[-1]\mathcal{M}$. Namely, the choice of a volume form v on $\mathcal{M}$ produces the corresponding volume¹ form μ_v on $T^*[-1]\mathcal{M}$. The divergence operator is defined as a map from the vector fields on $T^*[-1]\mathcal{M}$ to $C^\infty(T^*[-1]\mathcal{M})$ through the following integral relation

$$\int_{T^*[-1]\mathcal{M}} X(f) \mu_v = - \int_{T^*[-1]\mathcal{M}} \operatorname{div}_{\mu_v} X f \mu_v, \quad \forall f \in C^\infty(T^*[-1]\mathcal{M}), \quad (2.4)$$

with X being a vector field. As one can easily check, for any function f and vector field X the divergence satisfies

$$\operatorname{div}_{\mu_v}(fX) = f \operatorname{div}_{\mu_v}(X) + (-1)^{|f||X|} X(f). \quad (2.5)$$

Now the odd Laplacian of $f \in C^\infty(T^*[-1]\mathcal{M})$ is defined through the divergence of the corresponding Hamiltonian vector field as

$$\Delta_v f = \frac{(-1)^{|f|}}{2} \operatorname{div}_{\mu_v} X_f, \quad \{f, g\} = X_f(g). \quad (2.6)$$

Indeed one can check that thanks to (2.5) Δ_v generates the bracket and $\Delta_v^2 = 0$. Thus $C^\infty(T^*[-1]\mathcal{M})$ is a BV-algebra as defined above, see [18] for the explicit calculations. If the volume form is written in terms of an even density ρ_v as

$$\mu_v = \rho_v dy^1 \cdots dy^n dy_1^+ \cdots dy_n^+,$$

then the Laplacian can be written as

$$\Delta_v = (-1)^{|y_a|} \frac{\partial}{\partial y_a^+} \frac{\partial}{\partial y^a} + \frac{1}{2} \{\log \rho_v, -\}. \quad (2.7)$$

This Laplacian squares to zero since we take a specific Berezinian measure ρ_v which originates from the measure on $\mathcal{M}$. Thus $T^*[-1]\mathcal{M}$ is equipped with the SP-structure in the

¹We require that the measure is well-defined functional on the space of compactly supported functions and locally it is Berezinian measure.

Schwarz's terminology [25] (also see [19] for a nice review of the related issues) and in what follows we refer to them simply as BV-manifolds.

There exists a canonical way (up to a sign) of restricting a volume form μ_v on $T^*[-1]\mathcal{M}$ to a volume form on a Lagrangian submanifold $\mathcal{L}$. We denote such restriction as $\sqrt{\mu_v}$ and consider the integrals of the form

$$\int_{\mathcal{L}} \sqrt{\mu_v} f, \quad f \in C^\infty(T^*[-1]\mathcal{M}). \quad (2.8)$$

Again if f is Δ -closed then the integral depends only on homology class of $\mathcal{L}$ and if f is Δ -exact then the integral is zero. In particular we are interested in the situation when the integrands in (2.8) are of the form

$$\int_{\mathcal{L}} \sqrt{\mu_v} \Psi e^S \equiv \langle \Psi \rangle, \quad (2.9)$$

where we assume naturally that $\Delta_v(\Psi e^S) = 0$. If $\Psi = 1$ then we get the following relation

$$\Delta_v(e^S) = 0 \iff \Delta_v S + \frac{1}{2}\{S, S\} = 0, \quad (2.10)$$

which is known as the *quantum master equation*. In the general case we have

$$\Delta_v(\Psi e^S) = 0 \iff \Delta_{(v,S)}\Psi = \Delta_v\Psi + \{S, \Psi\} = 0, \quad (2.11)$$

where we refer to $\Delta_{(v,S)}$ as the quantum Laplacian. In the derivation of (2.11) we have used the quantum master equation (2.10). A function S that satisfies the quantum master equation is called a quantum BV action and Ψ satisfying (2.11) is a quantum observable. Indeed the quantum observables are elements of the cohomology $H(\Delta_{(v,S)})$; by the above construction it is clear that S defines the isomorphism

$$H^\bullet(\Delta_v) \approx H^\bullet(\Delta_{(v,S)}). \quad (2.12)$$

The integral (2.9) defines a trace ($H^\bullet(\Delta_{(v,S)}) \rightarrow \mathbb{C}$) on this cohomology.

If we change S to $S/\hbar$, we see that in the classical limit ($\hbar \rightarrow 0$) S must satisfy the classical master equation $\{S, S\} = 0$ and the classical observables Ψ are such that $\delta_{BV}\Psi \equiv \{S, \Psi\} = 0$. Due to the classical master equation the vector field δ_{BV} squares to zero and defines the cohomology $H(\delta_{BV})$ of classical observables.

If $\mathcal{M}$ is a finite dimensional manifold then everything is well-defined. In the following Sections we provide several finite dimensional BV manifolds equipped with a solution of the classical (quantum) master equation. However in field theory one deals with $\mathcal{M}$ being infinite dimensional. In fact, $\mathcal{M}$ is usually the space of the physical fields, ghosts and

Lagrange multipliers, that is infinite dimensional. This set of fields is then extended by adding antifields such that together they form $T^*[-1]\mathcal{M}$, where an odd Poisson bracket is well-defined on large enough class of functions, as described above. However there is no well-defined measure on $\mathcal{M}$ and thus there is no well-defined odd Laplace operators. In physics literature, the naive Laplacian of the form (2.2) is used. Moreover the field theory suffers from the problems with renormalization which can be resolved within the perturbative setup.

3 AKSZ formalism and its reduction

In this Section we review the AKSZ construction [1] of the solutions of the classical master equation. Here we will follow the presentation given in [24] and we use the language of graded manifolds. Relevant definitions are postponed to Section 7. For further details the reader may consult [26].

The AKSZ solution of the classical master equation is defined starting from the following data:

The source: A graded manifold $\mathcal{N}$ endowed with a cohomological vector field D and a measure $\int_{\mathcal{N}} \mu$ of degree $-n-1$ for some positive integer n . In what follows the source will be $\mathcal{N} = T[1]\mathcal{N}_0$, for any smooth manifold $\mathcal{N}_0$ of dimension $n+1$, with $D = d$ the de Rham differential over $\mathcal{N}_0$ and the canonical coordinate measure.

The target: A graded symplectic manifold $(\mathcal{M}, \omega)$ with $\deg(\omega) = n$ and an homological vector field Q preserving ω . We require that Q is Hamiltonian, *i.e.* it exists $\Theta \in C_{n+1}(\mathcal{M})$ (functions of degree $n+1$) such that $Q = \{\Theta, -\}$. Therefore Θ satisfies the following Maurer-Cartan equation

$$\{\Theta, \Theta\} = 0 .$$

Since graded manifolds are ringed spaces, we consider the space $\text{Maps}(\mathcal{N}, \mathcal{M})$ to be the space of morphisms of ringed spaces (see Definition 9). We choose a set of coordinates $X^A = \{x^\mu; \xi^m\}$ on the target $\mathcal{M}$, where $\{x^\mu\}$ are the coordinates for an open $U \subset \mathcal{M}_0$ and $\{\xi^m\}$ are the coordinates in the formal directions. We also choose the coordinates $\{u^\alpha; \theta^a\}$ on the source $\mathcal{N}$ correspondently. Let $(\phi, \Phi) \in \text{Maps}(\mathcal{N}, \mathcal{M})$, where $\phi : \mathcal{N}_0 \rightarrow \mathcal{M}_0$. The superfield Φ is defined as an expansion over the formal coordinates of $\mathcal{N}$ for $\phi^{-1}(U)$

$$\Phi^A = \Phi_0^A(u) + \Phi_a^A(u)\theta^a + \Phi_{a_1 a_2}^A(u)\theta^{a_1}\theta^{a_2} + \dots , \quad (3.13)$$

such that $\Phi_0^\mu = x^\mu \circ \phi$. The symplectic form ω of degree n on $\mathcal{M}$ can be written in the Darboux coordinates $\omega = dX^A \omega_{AB} dX^B$. Using this form we define the symplectic form of

degree -1 on $\text{Maps}(\mathcal{N}, \mathcal{M})$ as

$$\Omega = \int_{\mathcal{N}} \mu \delta\Phi^A \omega_{AB} \delta\Phi^B . \quad (3.14)$$

Thus the space of maps $\text{Maps}(\mathcal{N}, \mathcal{M})$ is naturally equipped with the odd Poisson bracket $\{ , \}$. Since the space $\text{Maps}(\mathcal{N}, \mathcal{M})$ is infinite dimensional we cannot define the BV Laplacian properly. We can only talk about the naive Laplacian adapted to the local field-antifield splitting, given by the formula (2.1). However on $\text{Maps}(\mathcal{N}, \mathcal{M})$ we can discuss the solutions of the classical master equation. The AKSZ action then reads

$$S_{BV}[\Phi] = S_{kin}[\Phi] + S_{int}[\Phi] = \int_{\mathcal{N}} \mu \left(\frac{1}{2} \Phi^A \omega_{AB} D\Phi^B + (-1)^{n+1} \Phi^*(\Theta) \right) . \quad (3.15)$$

and solves the classical master equation $\{S_{BV}, S_{BV}\} = 0$ with respect to the bracket defined by the symplectic structure (3.14). We denote with δ_{BV} the Hamiltonian vector field for S_{BV} , which is homological as a consequence of classical master equation. The action (3.15) is invariant under all orientation preserving diffeomorphisms of $\mathcal{N}_0$ and thus defines a topological field theory. The solutions of the classical field equations of (3.15) are graded differentiable maps $(\mathcal{N}, D) \rightarrow (\mathcal{M}, Q)$, i.e. maps which commute with the homological vector fields.

The homological vector field Q on $\mathcal{M}$ defines a complex on $C^\infty(\mathcal{M})$ whose cohomology we denote with $H_Q(\mathcal{M})$. Take $f \in C^\infty(\mathcal{M})$ and expand $\Phi^* f$ in the formal variables on $\mathcal{N}$, i.e.

$$\Phi^* f = O^{(0)}(f) + O_a^{(1)}(f) \theta^a + O_{a_1 a_2}^{(2)}(f) \theta^{a_1} \theta^{a_2} + \dots .$$

We compute

$$\delta_{BV}(\Phi^* f) = \{S_{BV}, \Phi^* f\} = D\Phi^* f + \Phi^* Qf ,$$

so that, if $Qf = 0$ and μ_k is a D -invariant linear functional on $C_k(\mathcal{N})$, i.e. a representative of a homology class of $\mathcal{N}_0$, then $\mu_k(O^{(k)}(f))$ is δ_{BV} -closed, i.e. it is a classical observable. Therefore $H_Q(\mathcal{M})$ naturally defines a set of classical observables in the theory.

Like in the smooth case, the symplectic reduction of the odd symplectic manifold $\text{Maps}(\mathcal{N}, \mathcal{M})$ is specified by the choice of a coisotropic submanifold. The reduced space naturally inherits the odd symplectic structure and any function on $\text{Maps}(\mathcal{N}, \mathcal{M})$ descends to the quotient provided it is invariant once restricted to the coisotropic submanifold. In this paper we consider the symplectic reduction of $\text{Maps}(\mathcal{N}, \mathcal{M})$ to the cohomology $H_D(\mathcal{N}) = H_{dR}(\mathcal{N}_0)$ of the source. We consider the constraint

$$\Lambda = \int_{\mathcal{N}} \mu \Lambda_A D\Phi^A ,$$

which defines the coisotropic submanifold $D\Phi = 0$ of $\text{Maps}(\mathcal{N}, \mathcal{M})$ (the constrained surface). The corresponding infinitesimal gauge transformations

$$\delta_\Lambda \Phi^A = D\Lambda^A \quad (3.16)$$

identify two configurations which differ by a D -exact term. If $\mathcal{N}_0$ has a boundary then boundary conditions must be discussed. We leave the case with boundary to Section 6 and for now we assume that $\mathcal{N}_0$ does not have boundary. The reduced odd symplectic manifold is finite dimensional and can be globally described as the space of maps from the cohomology of $\mathcal{N}_0$ to $\mathcal{M}$. More precisely, we can consider a generalized AKSZ construction where the source is the cohomology of $\mathcal{N}_0$ seen as a sheaf $X_{\mathcal{N}_0}$ of graded commutative algebras over a point, equipped with the zero homological vector field and the integration naturally induced on cohomology. The source is not anymore a graded manifold since the cohomology ring is not freely generated in general. However the space of maps $\text{Maps}(X_{\mathcal{N}_0}, \mathcal{M})$ is still defined; this simply corresponds to interpret $\{\theta^a\}$ in the superfield (3.13) as the generators of the cohomology ring $H_{dR}(\mathcal{N}_0)$. The simplest case is when $\mathcal{N}_0$ has cohomology ring concentrated in degree zero and top degree. In this case the reduced theory will be simply $T^*[-1]\mathcal{M}$.

For any $f \in C^\infty(\mathcal{M})$, $\Phi^*f - (\Phi + D\Lambda)^*f$ is D -exact. Thus any observable $\mu_k(O^{(k)}(f))$ in the full theory is invariant under the gauge transformations (3.16) and defines a function on the reduced odd symplectic manifold. In particular, the BV-action (3.15) $S_{BV} = \int_{\mathcal{N}} \mu \Phi^*(\Theta)$ defines a solution of the classical master equation that coincides with the AKSZ action on $\text{Maps}(X_{\mathcal{N}_0}, \mathcal{M})$.

It is crucial that the reduced odd symplectic manifold is finite dimensional so that integration and BV Laplacian can be well defined. Thus we can define properly the quantum master equation and discuss the possible obstructions to satisfy it. The rest of the paper consists of a detailed account of these reduced BV-theories in several examples.

4 Two dimensional case: Poisson sigma model

Let us consider the AKSZ construction described in the previous section with source $\mathcal{N} = T[1]\Sigma_g$, where Σ_g is the two dimensional compact surface of genus g . Then the target $\mathcal{M}$ is a symplectic graded manifold of degree 1 with a homological vector field Q preserving the symplectic structure. It must be necessarily of the form $\mathcal{M} = T^*[1]M$ where M is a Poisson manifold with Poisson tensor α . By choosing coordinates $\{x^\mu, \beta_\mu\}$, the Hamiltonian for Q is $\Theta = \alpha^{\mu\nu} \beta_\mu \beta_\nu \in C_2(\mathcal{M})$. We have that $C^\infty(\mathcal{M}) = C^\infty(\Lambda TM)$ and the cohomology $H_Q(M)$ of the complex $(C^\infty(\mathcal{M}), Q)$ coincides with the usual Lichnerowicz-Poisson cohomology $H_{LP}(M; \alpha)$. This is the case studied in [6] as a BV quantization of

the Poisson Sigma Model. We will first review the relevant results from [6] and [7] and then study the reduction procedure to the finite dimensional BV -manifold outlined in the previous section. In the case of $\Sigma_0 = \mathbb{S}^2$ the reduction to the finite dimensional theory has been studied in [3].

The Poisson Sigma Model covers many interesting geometrical structures. For example, if A is a Lie algebroid then the dual vector bundle A^* regarded as a manifold is naturally equipped with a Poisson structure. This is an example of a non-compact Poisson manifold where some issues related to the integration require extra care. We assume that for non-compact case the integration is defined as functional over the space of functions with compact support (or with exponential decay along the non-compact directions) and thus formally all our considerations are equally applicable both for compact and non-compact cases.

4.1 BV action

Let $\{u^\alpha\}$ be coordinates on Σ_g and $\{x^\mu\}$ on M . The superfields read as

$$\mathbf{X}^\mu = X^\mu + \theta^\alpha \eta_\alpha^{+\mu} - \frac{1}{2} \theta^\alpha \theta^\beta \beta_{\alpha\beta}^{+\mu} ,$$

$$\boldsymbol{\eta}_\mu = \beta_\mu + \theta^\alpha \eta_{\alpha\mu} + \frac{1}{2} \theta^\alpha \theta^\beta X_{\alpha\beta\mu}^+ ,$$

with $\{\theta^\alpha\}$ being the odd coordinates on $T[1]\Sigma_g$. In the BV language, the components of ghost number 0, X and η are the classical fields, β is a ghost with the ghost number 1, while η^+ , β^+ and X^+ are antifields of ghost number -1 , -2 and -1 respectively. The space of maps $\text{Map}(T[1]\Sigma_g, T^*[1]M)$ can be seen as $T^*[-1]\mathcal{M}$, where $\mathcal{M}$ is the infinite dimensional manifold corresponding to the fields (X, η, β) .

If we change coordinates $y^i = y^i(x)$, the superfields transform as

$$\mathbf{Y}^i = y^i(\mathbf{X}) , \quad \boldsymbol{\eta}_i = \frac{\partial x^\mu}{\partial y^i}(\mathbf{X}) \boldsymbol{\eta}_\mu . \quad (4.17)$$

For later utility we add the explicit component content of (4.17)

$$\begin{aligned} \eta_\alpha^{+i} &= \frac{\partial y^i}{\partial x^\mu} \eta_\alpha^{+\mu} , \quad \beta_{\alpha\beta}^{+i} = \frac{\partial y^i}{\partial x^\mu} \beta_{\alpha\beta}^{+\mu} + \frac{\partial^2 y^i}{\partial x^\mu \partial x^\nu} \eta_\alpha^{+\mu} \eta_\beta^{+\nu} , \quad \beta_i = \frac{\partial x^\mu}{\partial y^i} \beta_\mu , \\ \eta_i &= \frac{\partial x^\mu}{\partial y^i} \eta_\mu + \frac{\partial}{\partial x^\nu} \frac{\partial x^\mu}{\partial y^i} \eta^{+\nu} \beta_\mu , \\ X_{i\alpha\beta}^+ &= \frac{\partial x^\mu}{\partial y^i} X_{\mu\alpha\beta}^+ - 2 \frac{\partial}{\partial x^\nu} \frac{\partial x^\mu}{\partial y^i} \eta_\alpha^{+\nu} \eta_{\mu\beta} - \frac{\partial}{\partial x^\nu} \frac{\partial x^\mu}{\partial y^i} \beta_{\alpha\beta}^{+\nu} \beta_\mu - \frac{\partial^2}{\partial x^\nu \partial x^\rho} \frac{\partial x^\mu}{\partial y^i} \eta_\alpha^{+\nu} \eta_\beta^{+\rho} \beta_\mu . \end{aligned} \quad (4.18)$$

The AKSZ action defined in (3.15) reads

$$S_{BV} = \int d^2\theta d^2u \left(\boldsymbol{\eta}_\mu D\mathbf{X}^\mu + \frac{1}{2} \alpha^{\mu\nu}(\mathbf{X}) \boldsymbol{\eta}_\mu \boldsymbol{\eta}_\nu \right) , \quad (4.19)$$

where $D = \theta^\alpha \partial_\alpha$. The odd symplectic structure is

$$\omega = \int_{\Sigma_g} \left(\delta X \wedge \delta X^+ + \delta \boldsymbol{\eta} \wedge \delta \boldsymbol{\eta}^+ + \delta \beta \wedge \delta \beta^+ \right) . \quad (4.20)$$

The action (4.19) satisfies both classical and naive quantum master equations [6]. The corresponding BV operator δ_{BV} acts on the superfields as follows

$$\delta_{BV} \mathbf{X}^\mu = D\mathbf{X}^\mu + \alpha^{\mu\nu}(\mathbf{X}) \boldsymbol{\eta}_\nu , \quad (4.21)$$

$$\delta_{BV} \boldsymbol{\eta}_\mu = D\boldsymbol{\eta}_\mu + \frac{1}{2} \partial_\mu \alpha^{\nu\rho}(\mathbf{X}) \boldsymbol{\eta}_\nu \boldsymbol{\eta}_\rho . \quad (4.22)$$

The local and non-local classical observables are labelled by the Lichnerowicz- Poisson cohomology [3]. In components the AKSZ-BV action (4.19) has the form

$$\begin{aligned} S_{BV} = \int_{\Sigma_g} & \eta_\mu \wedge dX^\mu + \frac{1}{2} \alpha^{\mu\nu}(X) \eta_\mu \wedge \eta_\nu + X_\mu^+ \alpha^{\mu\nu}(X) \beta_\nu - \eta^{+\mu} \wedge (d\beta_\mu + \partial_\mu \alpha^{\rho\nu}(X) \eta_\rho \beta_\nu) - \\ & - \frac{1}{2} \beta^{+\mu} \partial_\mu \alpha^{\rho\nu}(X) \beta_\rho \beta_\nu - \frac{1}{4} \eta^{+\mu} \wedge \eta^{+\nu} \partial_\mu \partial_\nu \alpha^{\rho\sigma}(X) \beta_\rho \beta_\sigma . \end{aligned} \quad (4.23)$$

4.2 The reduced BV-AKSZ theory

In this subsection we describe the reduction of the BV manifold $\text{Maps}(T[1]\Sigma_g, T^*[1]M)$ down to "the constant configurations" as described in Section 3. We obtain a solution of the classical master equation on the finite dimensional BV manifold. Namely we define the reduction with respect to the following constraints

$$\Lambda = \int_{T[1]\Sigma} \Lambda_\mu D\mathbf{X}^\mu , \quad T = \int_{T[1]\Sigma} T^\mu D\boldsymbol{\eta}_\mu , \quad (4.24)$$

with $\Lambda_\mu(u, \theta) = \Lambda_\mu^{(0)} + \Lambda_{\mu\alpha}^{(1)} \theta^\alpha + \Lambda_{\mu\alpha\beta}^{(2)} \theta^\alpha \theta^\beta$ and $T^\mu(u, \theta) = T_{(0)}^\mu + T_{(1)\alpha}^\mu \theta^\alpha + T_{(2)\alpha\beta}^\mu \theta^\alpha \theta^\beta$. We assign $|\Lambda_\mu| = 0$ and $|T^\mu| = -1$. After a short computation one gets

$$\delta_{\Lambda, T} \mathbf{X}^\mu = \{T, \mathbf{X}^\mu\} = DT^\mu , \quad \delta_{\Lambda, T} \boldsymbol{\eta}_\nu = \{\Lambda, \boldsymbol{\eta}_\nu\} = D\Lambda_\nu , \quad (4.25)$$

or in components

$$\delta_{\Lambda, T} X^\mu = 0 , \quad \delta_{\Lambda, T} \eta^{+\mu} = dT_{(0)}^\mu , \quad \delta_{\Lambda, T} \beta^{+\mu} = dT_{(1)}^\mu$$

$$\delta_{\Lambda,T}\beta_\mu = 0, \quad \delta_{\Lambda,T}\eta_\mu = d\Lambda_\mu^{(0)}, \quad \delta_{\Lambda,T}X_\mu^+ = d\Lambda_\mu^{(1)}. \quad (4.26)$$

We see that the reduction with respect to the constraints $D\mathbf{X} = D\boldsymbol{\eta} = 0$ consists in identifying $(\mathbf{X}, \boldsymbol{\eta})$ with $(\mathbf{X} + D\mathbf{T}, \boldsymbol{\eta} + D\Lambda)$ and then in going to the cohomology of Σ_g .

In order to define the reduced variables, let us choose a basis for the homology $H_\bullet(\Sigma_g)$. Let $u_0 \in \Sigma_g$ be a generator in degree zero and the whole surface Σ_g be the generator in degree two and consider the canonical basis $\{c_I, c^J\}$ in $H_1(\Sigma_g)$, in degree one, with the following intersection numbers

$$\#(c_I, c_J) = \#(c^I, c^J) = 0, \quad \#(c_I, c^J) = -\#(c^J, c_I) = \delta_I^J.$$

The Poincaré dual basis $\{e^J, e_I\}$ in $H^1(\Sigma_g)$ is defined as

$$\int_{c_I} e^J = \int_{c^J} e_I = \delta_I^J, \quad \int_{c_I} e_J = \int_{c^I} e^J = 0,$$

and satisfies

$$\int_{\Sigma_g} e^J \wedge e_I = \delta_I^J.$$

Let us introduce the reduced coordinates obtained from the zero form fields

$$x^\mu = X^\mu(u_0), \quad b_\mu = \beta_\mu(u_0),$$

from the one form fields

$$\eta_I = \int_{c_I} \eta, \quad \eta^J = \int_{c^J} \eta, \quad \eta_I^+ = \int_{c_I} \eta^+, \quad \eta^{+J} = \int_{c^J} \eta^+.$$

and finally from the two forms fields

$$x_\mu^+ = \int_{\Sigma_g} X_\mu^+, \quad b^{+\mu} = \int_{\Sigma_g} \beta^{+\mu}.$$

The global structure, *i.e.* the change of coordinates under the change $y^i = y^i(x)$ can be obtained from (4.18). One explicitly gets

$$\begin{aligned} \eta_I^{+i} &= \frac{\partial y^i}{\partial x^\mu} \eta_I^{+\mu}, \quad \eta^{+iI} = \frac{\partial y^i}{\partial x^\mu} \eta^{+\mu I}, \quad b_i = \frac{\partial x^\mu}{\partial y^i} b_\mu, \quad b^{+i} = \frac{\partial y^i}{\partial x^\mu} b^{+\mu} + \frac{\partial^2 y^i}{\partial x^\mu \partial x^\nu} \eta_I^{+\mu} \eta^{+\nu I}, \\ \eta_{Ii} &= \frac{\partial x^\mu}{\partial y^i} \eta_{I\mu} + \frac{\partial}{\partial x^\nu} \frac{\partial x^\mu}{\partial y^i} \eta_I^{+\nu} b_\mu, \quad \eta_i^I = \frac{\partial x^\mu}{\partial y^i} \eta_\mu^I + \frac{\partial}{\partial x^\nu} \frac{\partial x^\mu}{\partial y^i} \eta^{+I\nu} b_\mu, \\ x_i^+ &= \frac{\partial x^\mu}{\partial y^i} x_\mu^+ - \frac{\partial}{\partial x^\nu} \frac{\partial x^\mu}{\partial y^i} (\eta_I^{+\nu} \eta_\mu^I - \eta^{+I\nu} \eta_{I\mu}) - \frac{\partial}{\partial x^\nu} \frac{\partial x^\mu}{\partial y^i} b^{+\nu} b_\mu - \frac{\partial^2}{\partial x^\nu \partial x^\rho} \frac{\partial x^\mu}{\partial y^i} \eta_I^{+\nu} \eta^{+\rho I} b_\mu. \end{aligned} \quad (4.27)$$

All the BV structure goes to the quotient which is a finite dimensional BV manifold. The odd symplectic structure (4.20) reads

$$\omega = dx^\mu dx_\mu^+ + d\eta_{I\mu} d\eta^{+I\mu} - d\eta_\mu^I d\eta_I^{+\mu} + db_\mu db^{+\mu} . \quad (4.28)$$

Moreover, the BV action S_{BV} defined in (4.23) when restricted to the constraint surface depends only on the reduced variables, *i.e.* it is a pull-back of a function on the reduced manifold. We use the same notation S_{BV} for the reduced action. The reduced BV action is computed from (4.23) as

$$\begin{aligned} S_{BV} = & \alpha^{\mu\nu}(x) \eta_{I\mu} \eta_\nu^I + x_\mu^+ \alpha^{\mu\nu}(x) b_\nu - \eta_I^{+\mu} \partial_\mu \alpha^{\rho\nu}(x) \eta_\rho^I b_\nu + \\ & + \eta^{+I\mu} \partial_\mu \alpha^{\rho\nu}(x) \eta_{I\rho} b_\nu - \frac{1}{2} b^{+\mu} \partial_\mu \alpha^{\rho\nu}(x) b_\rho b_\nu - \frac{1}{2} \eta_I^{+\mu} \eta^{+I\nu} \partial_\mu \partial_\nu \alpha^{\rho\sigma}(x) b_\rho b_\sigma , \end{aligned} \quad (4.29)$$

which obviously satisfies the classical master equation.

The reduced variables can be assembled in the superfields $\underline{\Phi} = (\mathbf{x}^\mu, \mathbf{e}_\mu)$

$$\mathbf{x}^\mu = x^\mu + e^J \eta_J^{+\mu} + e_J \eta^{+J\mu} - s b^{+\mu} , \quad \mathbf{e}_\nu = b_\nu + e_I \eta_\nu^I + e^I \eta_{I\nu} + s x_\nu^+ , \quad (4.30)$$

where s is the generator of $H_{dR}^2(\Sigma_g)$ normalized to $\int_\Sigma s = 1$. The ring structure for $g > 0$ is defined by $e^J \wedge e_I = \delta_I^J s$ and by $s^2 = 0$ for $g = 0$. One can check that the transformations of coordinates (4.18) can be deduced from the transformations of superfields

$$\mathbf{y}^i = y^i(\mathbf{x}) , \quad \mathbf{e}_i = \frac{\partial x^\mu}{\partial y^i}(\mathbf{x}) \mathbf{e}_\mu .$$

The action (4.29) can be written as

$$S_{BV} = \int ds \, \alpha^{\mu\nu}(\mathbf{x}) \mathbf{e}_\mu \mathbf{e}_\nu , \quad (4.31)$$

where $\int ds$ is the induced integral on $H_{dR}^\bullet(\Sigma_g)$. We easily see that this is an AKSZ formulation of the reduced theory, where we take as “coordinates” on the source the generators of the ring $H_{dR}(\Sigma_g)$. Compared with the discussion in Section 3, here coordinates $\{e_J, e^J\}$ on the source for $g \neq 1$ are not free but simply generate the commutative graded algebra $H_{dR}(\Sigma_g)$. We may regard it as the sheaf X_{Σ_g} obtained by putting over a point the commutative graded algebra $H_{dR}(\Sigma_g)$ such that that the reduced manifold is the space $\text{Maps}(X_{\Sigma_g}, T^*[1]M)$ of morphisms between X_{Σ_g} and $T^*[1]M$. The case $g = 1$ is special since the cohomology ring is freely generated and we have a true graded manifold $X_{\Sigma_1} = \mathbb{R}^2[1]$. We will comment more on this in Section 8.

Remark 1 For $g = 0$, $\text{Maps}(X_{\mathbb{S}^2}, T^*[1]M)$ can be equivalently described as $T^*[-1]T^*[1]M$, see [3]. For $g > 0$ the BV manifold can be described as

$$\text{Maps}(X_{\Sigma_g}, T^*[1]M) = T^*[-1](\text{Maps}(X_L, T^*[1]M)) ,$$

where X_L is the subsheaf over a point generated by $(1, e_I)$. The total dimension of the BV manifold is given by the following formula

$$\dim \text{Maps}(X_{\Sigma_g}, T^*[1]M) = 4(g + 1) \dim M .$$

Since the reduced BV-manifold is finite dimensional we can introduce an integral. Let $\Omega = \rho_\Omega dx^1 \dots dx^n$ be a volume form on M . It defines the generator of the Schouten bracket D_Ω on $\Lambda TM = C^\infty(T^*[1]M)$. Moreover it defines a Berezinian integration on the reduced BV manifold $\text{Maps}(X_{\Sigma_g}, T^*[1]M)$ as

$$\mu_\Omega = \rho_\Omega^{2(2-2g)} Dz ,$$

where $Dz = dx \dots dx^+ \dots db \dots db^+ \dots d\eta \dots d\eta^+$ is the coordinate volume form. Since under the change of coordinates $\tilde{z} = \tilde{z}(z)$ the coordinate volume form transforms as

$$D\tilde{z} = \text{Ber} \frac{\partial \tilde{z}}{\partial z} Dz , \quad \text{Ber} \begin{pmatrix} I_{00} & I_{01} \\ I_{10} & I_{11} \end{pmatrix} = \frac{\det(I_{00} - I_{01} I_{11}^{-1} I_{10})}{\det I_{11}} \quad (4.32)$$

it is a tedious but straightforward computation to put (4.27) in (4.32) and check that μ_Ω is well defined.

The corresponding generator of the BV-bracket is

$$\Delta_\Omega = \frac{\partial}{\partial x_\mu^+} \frac{\partial}{\partial x^\mu} - \frac{\partial}{\partial b^{+\mu}} \frac{\partial}{\partial b_\mu} - \frac{\partial}{\partial \eta_I^{+\mu}} \frac{\partial}{\partial \eta_\mu^I} + \frac{\partial}{\partial \eta^{+I\mu}} \frac{\partial}{\partial \eta_{I\mu}} + (2 - 2g) \{ \log \rho_\Omega, - \} . \quad (4.33)$$

We then compute

$$\Delta_\Omega S_{BV} = (g - 1)(D_\Omega \alpha)^\mu b_\mu = (g - 1) \chi_\Omega^\mu b_\mu ,$$

where $\chi_\Omega = D_\Omega \alpha$ is the modular vector field that satisfies $d_{LP} \chi_\Omega = 0$. The class $[\chi_\Omega] \in H_{LP}(M; \alpha)$ is called *modular class* and it is independent from the choice of the volume form Ω . The modular class represents an obstruction for S_{BV} to satisfy the quantum master equation for any genus $g \neq 1$. This extends to any genus the construction in [3]; the result has been already observed in [10]. In order to understand from the point of view of QFT this dependence on the genus, we recall the renormalization of the PSM on the disk in [6]; in fact in the expansion on Feynman graphs, the modular vector field appeared as a factor of the divergent graphs with self insertions. Their renormalization

was obtained by introducing a non vanishing vector field, which in the compact case is possible only for $g = 1$.

Any $w = w^{\mu_1 \dots \mu_k} \beta_{\mu_1} \dots \beta_{\mu_k}$ representing a class in Lichnerowicz-Poisson cohomology defines the chain of observables $\Phi^* w = w^{\mu_1 \dots \mu_k}(\mathbf{X}) \boldsymbol{\eta}_{\mu_1} \dots \boldsymbol{\eta}_{\mu_k} = O^{(0)}(w) + O_\alpha^{(1)}(w) \theta^\alpha + \dots$ of the full theory. They are invariant with respect to gauge transformations (4.25) and correspond to the observables $\underline{\Phi}^* w = w^{\mu_1 \dots \mu_k}(\mathbf{x}) \mathbf{e}_{\mu_1} \dots \mathbf{e}_{\mu_k} = O^{(0)}(w) + O_I^{(1)}(w) e^I + \dots$ of the reduced AKSZ theory. By using (4.33) we compute the following relation

$$\Delta_\Omega \underline{\Phi}^* w = 2s(1 - g) \underline{\Phi}^* D_\Omega w .$$

We conclude that, given a unimodular Poisson structure α , for any genus $g \neq 1$ the finite dimensional BV theory maps any class $[w] \in H_{LP}(M; \alpha)$ represented by a divergenceless vector field w to the class represented by $e^{S_{BV}} \underline{\Phi}^* w$ in the cohomology of the complex $(C^\infty(\text{Maps}(X_{\Sigma_g}, T^*[1]M)), \Delta_\Omega)$. For $g = 1$ the hypothesis of unimodularity is not needed and such invariants are defined for any class in Lichnerowicz Poisson cohomology. The main point is now to characterize these invariants in terms of more canonical cohomologies and mainly understand when they are not trivial. Here we simply sketch some obvious considerations and leave a more systematic analysis for further investigations.

One has to compute the correlators of such observables by choosing a gauge fixing, *i.e.* a lagrangian submanifold $\mathcal{L}$. There is always one canonical choice which consists simply in putting all antifields equal to zero, *i.e.* $x^+ = \eta^+ = b^+ = 0$. The case $g = 0$ has been studied in [3] and we refer to it for the result. The case $g > 0$ is badly defined due to fibrewise integration along η in the degenerate directions of the Poisson tensor.

If one allows on the target M a complex structure and introduces complex coordinates $\{z^a, z^{\bar{a}}\}$, then one can define a gauge fixing by putting $x^+ = b^+ = 0$ and $\eta_{Ia} = \eta_a^I = \eta^{+I\bar{a}} = \eta_I^{+a} = 0$. By looking at (4.18) one can easily check that this is invariant under holomorphic transformation of variables. In the symplectic case the computation of the partition function in this gauge fixing gives the Euler number of M ; due to degeneracy of the Poisson tensor it is not clear how to give a meaning to this integral in the general Poisson case.

5 Three dimensional case: Courant sigma model

In this Section we consider the reduction of three dimensional topological theory associated to any Courant algebroid, the Courant sigma model. We follow closely the presentation given in [24] (see also for the related discussion [17, 15, 16]).

We follow notations of Section 4 of [24]. Let $E \rightarrow M$ be a vector bundle equipped with a fiberwise nondegenerate symmetric inner product $\langle, \rangle$, of arbitrary signature. A *Courant*

algebroid structure on $(E, \langle, \rangle)$ is a bilinear operation $\circ$ on sections of E and a bundle map (the *anchor*) $a : E \rightarrow TM$ satisfying the following properties

- i) $s \circ (s_1 \circ s_2) = (s \circ s_1) \circ s_2 + s_1 \circ (s \circ s_2)$;
- ii) $a(s_1 \circ s_2) = [a(s_1), a(s_2)]$;
- iii) $s_1 \circ (f s_2) = f(s_1 \circ s_2) + (a(s_1)(f))s_2$;
- iv) $\langle s, s_1 \circ s_2 + s_2 \circ s_1 \rangle = a(s)(\langle s_1, s_2 \rangle)$;
- v) $a(e)(\langle s_1, s_2 \rangle) = \langle s \circ s_1, s_2 \rangle + \langle s_1, s \circ s_2 \rangle$,

for $s, s_1, s_2 \in \Gamma(E)$ and $f \in C^\infty(M)$.

We can associate to these data a symplectic graded manifold $(\mathcal{M}, \omega)$ with $\deg \omega = 2$. Locally we introduce a set of coordinates $\{x^\mu\}$ on M and the local basis $\{e_a\}$ of sections of E such that $\langle e_a, e_b \rangle = g_{ab}$, with constant g_{ab} . Then the symplectic non negatively graded manifold $(\mathcal{M}, \omega)$ is described in terms of the local coordinates $\{x^\mu, p_\mu, \xi^a\}$, with $\deg(x) = 0$, $\deg(p) = 2$ and $\deg(\xi) = 1$ with the symplectic form of degree 2

$$\omega = dp_\mu dx^\mu + \frac{1}{2} d\xi^a g_{ab} d\xi^b .$$

Globally $\mathcal{M}$ can be interpreted as the symplectic submanifold of $T^*[2]E[1]$ defined by $\theta_a = \frac{1}{2} g_{ab} \xi^b$ with θ being a momentum for ξ . As shown in [23] the Hamiltonian function of degree 3

$$\Theta = \xi^a P_a^\mu p_\mu - \frac{1}{6} T_{abc} \xi^a \xi^b \xi^c , \quad (5.34)$$

where the structure constants $T_{abc} = \langle e_a \circ e_b, e_c \rangle$ and the anchor $a(e_a) = P_a^\mu \partial_\mu$ defines on $(E, \langle, \rangle)$ a *Courant algebroid* structure if and only if $\{\Theta, \Theta\} = 0$. Due to the AKSZ logic reviewed in Section 3 one can define a three dimensional theory whose space of fields is $\text{Maps}(T[1]\Sigma, \mathcal{M})$ with Σ being any three dimensional manifold. Introducing the superfields $(X^\mu, \mathbf{P}_\mu, \boldsymbol{\xi}^a)$ corresponding to the local description of $\mathcal{M}$ the Courant sigma model is defined by the following BV action

$$S_{BV} = \int d^3\theta d^3u \left(\mathbf{P}_\mu D X^\mu + \frac{1}{2} \boldsymbol{\xi}^a g_{ab} D \boldsymbol{\xi}^b - \boldsymbol{\xi}^a P_a^\mu \mathbf{P}_\mu + \frac{1}{6} T_{abc} \boldsymbol{\xi}^a \boldsymbol{\xi}^b \boldsymbol{\xi}^c \right) . \quad (5.35)$$

The AKSZ construction guarantees that (5.35) is a solution of the classical master equation associated to any three dimensional manifold Σ and the Courant algebroid E . The reader may consult [24] for the explicit expression of this *BV* action in components of the superfield.

Here we are interested in the finite dimensional BV theory obtained through the symplectic reduction of the AKSZ theory described above. The whole construction is done in complete analogy with the two dimensional case, see subsection 4.2. Thus we skip the details of the actual reduction and present the final answer. Introduce the basis in $H_{dR}(\Sigma)$ such that $\{e_I\}$ is a basis in $H_{dR}^1(\Sigma)$, e^I is the dual basis in $H_{dR}^2(\Sigma)$, 1 basis element in $H_{dR}^0(\Sigma)$ and s is a basis element in $H_{dR}^3(\Sigma)$. With this choice the natural ring structure on $H_{dR}(\Sigma)$ is given

$$e_I \wedge e^J = \delta_J^I s, \quad e_I \wedge e_J = f_{IJK} e^K$$

and also it follows that

$$e_I \wedge e_J \wedge e_K = f_{IJK} s.$$

The constants f_{IJK} have the interpretation as intersection numbers of one cycles. Define the sheaf X_Σ by putting over a point the commutative graded algebra $H_{dR}(\Sigma)$. The space of maps (*i.e.* morphisms of ringed spaces) $\text{Maps}(X_\Sigma, \mathcal{M})$ is equipped with an odd symplectic structure. We can introduce the superfields expanded in $(1, e_I, e^I, s)$

$$\begin{aligned} X^\mu &= x^\mu + F^{+\mu I} e_I + \alpha_I^{+\mu} e^I + \gamma^{+\mu} s, \\ \xi^a &= \beta^a + A^{aI} e_I + g^{ab} A_{bI}^+ e^I + g^{ab} \beta_b^+ s, \\ P_\mu &= \gamma_\mu + \alpha_\mu^I e_I + F_{\mu I} e^I + X_\mu^+ s, \end{aligned} \tag{5.36}$$

which correspond to the local description of element from $\text{Maps}(X_\Sigma, \mathcal{M})$. The integration is naturally defined by the relation $\int ds s = 1$ and all other integrals are zero (modulo ring relations). The odd symplectic structure is

$$\omega = \int ds \left(\delta X^\mu \delta P_\mu + \frac{1}{2} \delta \xi^a g_{ab} \delta \xi^b \right). \tag{5.37}$$

Upon the reduction the action (5.35) gives rise to the following BV action

$$\begin{aligned} S_{BV} &= -A^{aI} P_a^\nu(x) F_{\mu I} + \frac{1}{6} T_{abc}(x) f_{IJK} A^{aI} A^{bJ} A^{cK} - \beta^a P_a^\mu(x) X_\mu^+ \\ &+ (-g^{ac} P_a^\mu(x) \alpha_\mu^I - \beta^a A^{bI} T_{abr} g^{rc}) A_{cI}^+ \\ &+ (-\beta^a \partial_\mu P_a^\nu F_{\nu I} + f_{JKI} A^{aJ} \partial_\mu P_a^\nu \alpha_\nu^K + \frac{1}{2} f_{JKI} \partial_\mu T_{abc} \beta^a A^{bJ} A^{cK}) F^{+\mu I} \\ &+ (-\beta^a \partial_\mu P_a^\nu \alpha_\nu^I - A^{aI} \partial_\mu P_a^\nu \gamma_\nu + \frac{1}{2} \partial_\mu T_{abc} \beta^a \beta^b A^{cI}) \alpha_I^{+\mu} \\ &+ (\frac{1}{2} T_{abr} g^{rc} \beta^a \beta^b - g^{ac} P_a^\mu \gamma_\mu) \beta_c^+ + (\frac{1}{6} \partial_\mu T_{abc} \beta^a \beta^b \beta^c - \beta^a \partial_\mu P_a^\nu \gamma_\nu) \gamma^{+\mu} \\ &- (\frac{1}{2} \partial_\mu T_{abr} g^{rc} \beta^a \beta^b - g^{ac} \partial_\mu P_a^\nu \gamma_\nu) F^{+\mu I} A_{cI}^+ + \frac{1}{2} (\beta^a \partial_\mu \partial_\nu P_a^\rho \alpha_\rho^K + A^{aK} \partial_\mu \partial_\nu P_a^\rho \gamma_\rho \\ &- \frac{1}{2} \partial_\mu \partial_\nu T_{abc} \beta^a \beta^b A^{cK}) F^{+\mu I} F^{+\nu J} f_{KIJ} - (\beta^a \partial_\mu \partial_\nu P_a^\rho \gamma_\rho - \frac{1}{6} \partial_\mu \partial_\nu T_{abc} \beta^a \beta^b \beta^c) F^{+\mu I} \alpha_I^{+\nu} \\ &- \frac{1}{6} (-\beta^a \partial_\mu \partial_\nu \partial_\rho P_a^\sigma \gamma_\sigma + \frac{1}{6} \partial_\mu \partial_\nu \partial_\rho T_{abc} \beta^a \beta^b \beta^c) F^{+\mu I} F^{+\nu J} F^{+\rho K} f_{IJK}, \end{aligned} \tag{5.38}$$

hich automatically satisfies the classical master equation if E is equipped with the structure of a Courant algebroid.

On the given BV-manifold $\text{Maps}(X_\Sigma, \mathcal{M})$ the Berezinian integration can be defined as

$$\mu = g^{1-b_1} Dz , \quad (5.39)$$

where $b_1 = \dim H_{dR}^1(\Sigma)$, Dz is the canonical volume form with respect to the coordinates introduced in (5.36) and $g = \det(g^{ab})$. The Berezinian integration is canonically defined (see Appendix A for the details). Thus the corresponding generator (odd Laplacian) of the BV bracket coincides with the naive one

$$\Delta_\mu = \frac{\partial}{\partial x_\mu^+} \frac{\partial}{\partial x^\mu} - \frac{\partial}{\partial \beta_a^+} \frac{\partial}{\partial \beta^a} + \frac{\partial}{\partial A^{aI}} \frac{\partial}{\partial A_{aI}^+} + \frac{\partial}{\partial \gamma^{+\mu}} \frac{\partial}{\partial \gamma_\mu} - \frac{\partial}{\partial \alpha_I^{+\mu}} \frac{\partial}{\partial \alpha_\mu^I} - \frac{\partial}{\partial F^{+\mu I}} \frac{\partial}{\partial F_{\mu I}} \quad (5.40)$$

Using the explicit expression (5.38) and the axioms of Courant algebroid we easily check that

$$\Delta_\mu S_{BV} = 0 ,$$

i.e. S_{BV} satisfies the quantum master equation.

As an example we can consider the case when the source manifold Σ is a three sphere $\mathbb{S}^3$. In this case the reduced BV manifold $\text{Maps}(X_\Sigma, \mathcal{M})$ corresponds to $T^*[-1]\mathcal{M}$ since $H_{dR}(\mathbb{S}^3)$ has only elements of degree 0 and 3. To describe the reduced BV theory for $\mathbb{S}^3$ we have to set fields A^{aI} , α_μ^a , $F_{\mu I}$ together with their antifields A_{bI}^+ , $\alpha_I^{+\mu}$, $F^{+\mu I}$ to zero in (5.36), (5.37), (5.38) and remove them from the measure (5.39) and Laplacian (5.40). It is easy to describe the observables and the correlators in this theory. The Hamiltonian (5.34) defines an homological vector field $Q = \{\Theta, -\}$. The complex $(C^\infty(\mathcal{M})_{pol}, Q)$ of functions polynomial in p is called the *standard complex* and we denote its cohomology with $H_Q(\mathcal{M})_{pol}$. However in what follows we need to consider the complex $(C^\infty(\mathcal{M})_{exp}, Q)$ of functions with exponential decay in p directions. We have to use this complex in order to make sense of the integrals. For $f \in C^\infty(\mathcal{M})_{exp}$, $Q(f) = 0$ the corresponding observable is

$$\Phi^*(f) = O^{(0)}(f) + sO^{(3)}(f) ,$$

which satisfies by construction

$$\delta_{BV}(\Phi^*(f)) = \{S_{BV}, \Phi^*(f)\} = 0 ,$$

where $\Phi \in \text{Maps}(X_{\mathbb{S}^3}, \mathcal{M})$. For $T^*[-1]\mathcal{M}$ we choose $\mathcal{M}$ as a Lagrangian submanifold defined by setting all antifields to zero, *i.e.* $X^+ = \gamma^+ = \beta^+ = 0$. Thus on $\mathcal{M}$ we have that $\Phi^*(f) = O^{(0)}(f) = f$ and the berezinian measure on $\mathcal{M}$ reads

$$\sqrt{\mu} = \sqrt{g} \, dx \, d\gamma \, d\beta .$$

The correlator is defined as integral over the Lagrangian submanifold

$$\langle \Phi^*(f) \rangle = \int_{\mathcal{M}} \sqrt{\mu} f . \quad (5.41)$$

To make this integral well-defined we have to assume that f decays fast enough along non-compact directions. By construction the correlator (5.41) satisfies the version of the Stokes theorem with respect to BV-differential and odd Laplacian.

Example 2 Consider E to be a Lie algebra with invariant metric interpreted as a vector bundle over a point. In this case the Courant sigma model is just standard Chern-Simons theory. The reduced finite dimensional theory is BV version of the matrix theory. The standard complex is just ΛE^* equipped with the Chevalley-Eilenberg differential. The integral defined in (5.41) is different from zero only on top forms.

Example 3 Consider $E = TM + T^*M$ with the canonical inner product and the Dorfman bracket

$$[X_1 \oplus \omega_1, X_2 \oplus \omega_2] = [X_1, X_2] \oplus L_{X_1} \omega_2 - \iota_{X_2} d\omega_1 .$$

The coordinates on the fiber now split $\xi \rightarrow \{\psi^\mu, \theta_\mu\}$ and the corresponding graded manifold is described as the even symplectic manifold $\mathcal{M} = T^*[2]T[1]M$. The Hamiltonian (5.34) reads $\Theta = \psi^\mu p_\mu$ giving rise to the homological vector field

$$Q = \psi^\mu \frac{\partial}{\partial x^\mu} + p_\mu \frac{\partial}{\partial \theta_\mu} .$$

As shown in [23], Q commutes with

$$\epsilon = p_\mu \frac{\partial}{\partial p_\mu} + \theta_\mu \frac{\partial}{\partial \theta_\mu} .$$

The complex of polynomial functions in p ($C^\infty(\mathcal{M})_{pol}, Q$) then decomposes according to the ϵ -degree, *i.e.* $C^\infty(\mathcal{M})_{pol} = \oplus_{k \geq 0} C^\infty(\mathcal{M})_{(k)}$, with the subcomplex of degree zero is $(C^\infty(\mathcal{M})_{(0)}, Q) = (\Omega(M), d)$ being the de Rham complex for M . Moreover, since $\epsilon = Q\iota + \iota Q$ with $\iota = \theta_\mu \frac{\partial}{\partial p_\mu}$ the cohomology $H_Q(\mathcal{M})_{pol}$ is concentrated in degree zero and isomorphic to the de Rham cohomology $H_{dR}(M)$. However for the BV theory we need the complex $(C^\infty(\mathcal{M})_{exp}, Q)$ of functions with the exponential decay in p . It is not clear how the corresponding cohomology is related to $H_{dR}(M)$. Otherwise the correlators can be defined in the way we described above.

6 Two dimensional case with boundary

In this section we discuss the reduction of PSM on the surface with a boundary. Take a surface $\Sigma_{g,n}$, of genus g and n boundary components. We consider the boundary conditions for PSM that have been introduced in [8]. Let $\partial_i \Sigma_{g,n}$ the i -th boundary component of $\Sigma_{g,n}$ and let C_i a coisotropic submanifold of the Poisson manifold (M, α) . We recall that a submanifold C of M is coisotropic if $\alpha(N^*C) \subset TC$, where N^*C is the conormal bundle of C . We assume that the superfields (X, η) restrict to $\text{Maps}(T[1]\partial_i \Sigma, N^*[1]C_i)$ for any $i = 1, \dots, n$. More general boundary conditions have been introduced in [4], where it is required that the rank of $\alpha(N_x^*C) + T_x C \subset T_x M$ is constant as x varies over C .

We apply the same construction as in the closed case, *i.e.* we perform the reduction with respect to the constraints defined in (4.24). Since the bracket between the constraints has now a boundary contribution

$$\{\Lambda, T\} = \int_{T[1]\partial \Sigma} \Lambda_\mu D T^\mu ,$$

we require that

$$\Lambda|_{\partial_i \Sigma_{g,n}} \in \Gamma(\Lambda T^* \partial_i \Sigma_{g,n} \otimes X^* N^* C_i) , \quad T|_{\partial_i \Sigma_{g,n}} \in \Gamma(\Lambda T^* \partial_i \Sigma_{g,n} \otimes X^* T C_i) . \quad (6.42)$$

in order to have a consistent reduction. Due to the presence of the boundary conditions for fields and constraints, cohomologies of $\Sigma_{g,n}$ relative to boundary components will appear in the description of the reduced BV-manifolds. In Appendix B we collect the relevant facts about relative (co)homology.

6.1 The case with one boundary component

Consider the case of a surface $\Sigma_{g,1}$ with one boundary component with the boundary condition corresponding to a coisotropic submanifold C . Let us introduce a set of coordinates of M adapted to C with $\{x^a\}$ tangent to C and $\{x^n\}$ normal. Coisotropy of C is then simply expressed by the condition $\alpha^{mn} = 0$.

The gauge transformations (4.25) defined by the constraints together with boundary conditions (6.42) imply that the reduced BV manifold is described by the following variables

$$X^a \in H_{dR}(\Sigma_{g,1}) , \quad X^n \in H_{dR}(\Sigma_{g,1}, \partial \Sigma_{g,1}) , \quad \eta_a \in H_{dR}(\Sigma_{g,1}, \partial \Sigma_{g,1}) , \quad \eta_n \in H_{dR}(\Sigma_{g,1}) .$$

The covariant meaning of the above statements and the gluing data of the reduced BV-manifold are better understood once that we introduce the reduced variables.

We recall from Appendix B that $H^2(\Sigma_{g,1}) = H^0(\Sigma_{g,1}, \partial\Sigma_{g,1}) = 0$. In order to define the reduced coordinates, let us choose a set of representatives defining a basis of the relevant homologies. Let $u_0 \in \Sigma_{g,1}$ be a representative generating $H_0(\Sigma_{g,1})$, $\{c^I\}$ for $H_1(\Sigma_{g,1})$, $\{g^{I'}\}$ for $H_1(\Sigma_{g,1}, \partial\Sigma_{g,1})$ and the whole surface $\Sigma_{g,1}$ for $H_2(\Sigma_{g,1}, \partial\Sigma_{g,1})$. Let $\{c_I\}$ and $\{g_{I'}\}$ be the dual basis for $H_{dR}^1(\Sigma_{g,1})$ and $H_{dR}^1(\Sigma_{g,1}, \partial\Sigma_{g,1})$ respectively. Remark that beyond the natural pairings between homology and cohomology one can pair $H_1(\Sigma_{g,1})$ with $H_{dR}^1(\Sigma_{g,1}, \partial\Sigma_{g,1})$. The matrix

$$\lambda_{I'}^I = \int_{c^I} g_{I'}$$

describes the natural homomorphism $H_{dR}^1(\Sigma_{g,1}, \partial\Sigma_{g,1}) \rightarrow H_{dR}^1(\Sigma_{g,1})$ as $g_{I'} \rightarrow \lambda_{I'}^I c_I$. Finally we define the following matrices

$$B_{I'J'}^{(g,1)} = \int_{\Sigma_{g,1}} g_{I'} \wedge g_{J'} , \quad A_{II'}^{(g,1)} = \int_{\Sigma_{g,1}} c_I \wedge g_{I'} ,$$

where $A^{(g,1)}$ is non degenerate matrix.

Then we define

$$\begin{aligned} x^a &= X^a(u_0) , \quad b_n = \beta_n(u_0) , \\ \eta_a^{I'} &= \int_{g^{I'}} \eta_a , \quad \eta_n^I = \int_{c^I} \eta_n , \quad \eta^{+aI} = \int_{c^I} \eta^{+a} , \quad \eta^{+nI'} = \int_{g^{I'}} \eta^{+n} \\ x_a^+ &= \int_{\Sigma} X_a^+ , \quad b^{+n} = \int_{\Sigma} \beta^{+n} . \end{aligned}$$

The reduced odd symplectic form reads

$$\omega = dx^a dx_a^+ + d\eta_a^{I'} d\eta^{+aI} A_{II'}^{(g,1)} + d\eta_n^I d\eta^{+nI'} A_{II'}^{(g,1)} + db_n db^{+n} . \quad (6.43)$$

If we take another adapted system of coordinates $\{y^\alpha = y^\alpha(x^a, x^n), y^\nu = y^\nu(x^a, x^n)\}$ with y^α being the coordinates along C and y^ν transverse to C , then the law for the corresponding transformation of the reduced coordinates can be derived from (4.18). We will not explicitly write it here; we simply remark that beyond the matrix $(\partial y / \partial x)$, it involves all matrices $A_{II'}^{(g,1)}$, $B_{I'J'}^{(g,1)}$ and $\lambda_{J'}^{I'}$ introduced above.

We give here a non canonical description of the reduced BV manifold that depends on the choice of a tubular neighborhood for C . We recall that a *tubular neighborhood* for C is an embedding $j : NC \rightarrow M$, where NC is the normal bundle of C inside M , such that $j(C) = C$. Let $X_{\Sigma_{g,1}}$ be the sheaf obtained by putting over a point the commutative graded algebra $H_{dR}(\Sigma_{g,1})$.

Proposition 4 *For any choice of a tubular neighborhood of C inside M , there exists an isomorphism between the reduced BV-manifold and $T^*[-1](\text{Maps}(X_{\Sigma_{g,1}}, N^*[1]C))$. Such isomorphism is canonical in the case $g = 0$, i.e. $\Sigma = D^1$, when the reduced BV manifold is $T^*[-1]N^*[1]C$.*

Proof. The superfields for the graded manifold $\text{Maps}(X_{\Sigma_{g,1}}, N^*[1]C)$ are

$$\mathbf{x}^a = x^a + c_I \eta^{+Ia}, \quad \mathbf{e}_n = b_n + c_I \eta_n^I.$$

One can easily check that the degree of the momenta are correct. Let us choose the tubular neighborhood of C . A trivialization of NC with (t_n^ν) as transition functions of NC , defines the atlas of adapted coordinates $\{x^a, x^n\}$, where $\{x^a\}$ are coordinates of C and $\{x^n\}$ are coordinates on the fibre. The change of coordinates is given by

$$y^\alpha = y^\alpha(x^a), \quad y^\nu = t_n^\nu(x^a) x^n.$$

It is tedious but straightforward to check that the rules of change of variables defined in $\text{Maps}(X_{\Sigma_{g,1}}, N^*[1]C)$ together with those of momenta are the same that we get from (4.18). $\square$

The reduced BV action reads as

$$\begin{aligned} S_{BV} = & x_a^+ \alpha^{an} b_n - \frac{1}{2} b^{+m} \partial_m \alpha^{np} b_n b_p + \frac{1}{2} \alpha^{ab} \eta_a^{I'} \eta_b^{J'} B_{I'J'}^{(g,1)} + \alpha^{am} \eta_a^{I'} \eta_m^J A_{I'J}^{(g,1)} + \\ & \partial_b \alpha^{na} \eta^{+bI} \eta_a^{J'} b_n A_{IJ'}^{(g,1)} + \partial_m \alpha^{na} \eta^{+mI'} \eta_a^{J'} b_n B_{I'J'}^{(g,1)} - \partial_m \alpha^{np} \eta^{+mI'} \eta_p^J b_n A_{JI'}^{(g,1)} + \\ & - \frac{1}{4} \partial_p \partial_q \alpha^{mn} b_m b_n \eta^{+pI'} \eta^{+qJ'} B_{I'J'}^{(g,1)} - \frac{1}{2} \partial_a \partial_p \alpha^{mn} b_m b_n \eta^{+aI} \eta^{pI'} A_{II'}^{(g,1)}, \end{aligned}$$

which automatically satisfies the classical master equation.

Remark 5 Apparently, there is no natural description of these solutions of BV equation as AKSZ-actions.

Recall that $\dim H_{dR}^1(\Sigma_{g,n}) = \dim H_{dR}^1(\Sigma_{g,n}, \partial\Sigma_{g,n}) = 2g + n - 1$. Any choice of volume form $\Omega = \rho_\Omega dx^a dx^n$ on M defines the berezinian

$$\mu_\Omega = \rho_\Omega^{2(1-2g)} dx^a dx_a^+ db_n db^{+n} d\eta d\eta^+$$

and the BV-generator

$$\Delta_\Omega = \frac{\partial}{\partial x_a^+} \frac{\partial}{\partial x^a} - \frac{\partial}{\partial b^{+n}} \frac{\partial}{\partial b_n} + A^{(g,1)I'J} \left(\frac{\partial}{\partial \eta_a^{I'}} \frac{\partial}{\partial \eta^{+aJ}} - \frac{\partial}{\partial \eta_m^J} \frac{\partial}{\partial \eta^{+mI'}} \right) + (1-2g) \{ \log \rho_\Omega, - \}.$$

We easily compute

$$\Delta_\Omega S_{BV} = (1-2g)(\partial_a \alpha^{an} + \partial_m \alpha^{mn} + \partial_a \log \rho_\Omega \alpha^{an}) b_n = (1-2g) \chi_{\Omega, N^*C}^n b_n,$$

where $\chi_{\Omega, N^*C} \in NC = T_C M / TC$ is a representative of the modular class of the Lie algebroid N^*C . We conclude that the solution of the classical master equation S_{BV} satisfies the quantum master equation if and only if N^*C is unimodular.

7 Abstract AKSZ models from dg Frobenius algebras

Inspired by the previous consideration we suggest the extension of the AKSZ idea. The AKSZ construction admits the following straightforward generalization: loosely, one can use any differential graded (“dg”) Frobenius algebra (or a sheaf of dg algebras such that the global sections carry the structure of the Frobenius algebra) instead of differential forms on the worldsheet. In this section we sketch two versions of the construction — vector space version and sheaf version.

7.1 Vector space version

The source: A unital dg Frobenius algebra $\mathcal{C}$, i.e. a $\mathbb{Z}$ -graded vector space² $\mathcal{C} = \mathcal{C}^0 \oplus \dots \oplus \mathcal{C}^{n+1}$ endowed with (super-)commutative associative multiplication $m : S^2\mathcal{C} \rightarrow \mathcal{C}$ of degree 0, differential $D : \mathcal{C}^\bullet \rightarrow \mathcal{C}^{\bullet+1}$ of degree 1 and non-degenerate symmetric pairing $\Pi : S^2\mathcal{C} \rightarrow \mathbb{R}$ of degree $-n-1$, satisfying the following axioms:

- Degree properties:

$$\begin{aligned} |m(u, v)| &= |u| + |v| \\ |Du| &= |u| + 1 \\ \Pi(u, v) \neq 0 &\text{ implies } |u| + |v| = n + 1 \end{aligned}$$

- Symmetry properties:

$$\begin{aligned} m(u, v) &= (-1)^{|u|\cdot|v|} m(v, u) \\ \Pi(u, v) &= (-1)^{|u|\cdot|v|} \Pi(v, u) \end{aligned}$$

- Poincaré, Leibniz and associativity identities for differential and multiplication:

$$\begin{aligned} D^2 &= 0 \\ Dm(u, v) &= m(Du, v) + (-1)^{|u|} m(u, Dv) \\ m(m(u, v), w) &= m(u, m(v, w)) \end{aligned}$$

- Multiplication is cyclic w.r.t. the pairing:

$$\Pi(u, m(v, w)) = \Pi(m(u, v), w)$$

²Our convention is that *elements* of $\mathcal{C}^i$ have degree i (not coordinates on $\mathcal{C}^i$).

- Differential is skew-symmetric w.r.t. the pairing:

$$\Pi(Du, v) + (-1)^{|u|} \Pi(u, Dv) = 0$$

- Pairing Π is non-degenerate, i.e. induces an isomorphism

$$\mathcal{C}^\bullet \xrightarrow{\sim} (\mathcal{C}^{n+1-\bullet})^*$$

Here we assume that $u, v, w \in \mathcal{C}$ are homogeneous elements and $|\cdots|$ denotes the degree of an element. Also, we denote the unit of $\mathcal{C}$ by $\mathbf{1} \in \mathcal{C}^0$.

Equivalently, one can describe $\mathcal{C}$ as a unital (super-)commutative dg algebra with trace $\text{Tr} : \mathcal{C}^{n+1} \rightarrow \mathbb{R}$ (and extended by zero on lower degree components of $\mathcal{C}$) satisfying

$$\text{Tr}(Du) = 0$$

and such that the pairing $\text{Tr}(m(\bullet, \bullet))$ is non-degenerate. The trace is constructed from the pairing as $\text{Tr}(u) = \Pi(\mathbf{1}, u)$ (and vice versa, pairing can be constructed from the trace as $\Pi(u, v) = \text{Tr}(m(u, v))$).

The target: A $\mathbb{Z}$ -graded vector space $\mathcal{W}$ endowed with a (constant) symplectic form $\omega \in S^2(\mathcal{W}[1])^*$ of degree n and a function $\Theta \in S^\bullet(\mathcal{W}^*)$ of degree $n+1$, satisfying $\{\Theta, \Theta\} = 0$.

The space of BV fields of abstract AKSZ model is defined in this setting to be the $\mathbb{Z}$ -graded vector space

$$\mathcal{F} = \mathcal{C} \otimes \mathcal{W} . \quad (7.44)$$

Let $\{e_\zeta\}$ be a basis in $\mathcal{C}$ and $\{\tau_A\}$ be a basis in $\mathcal{W}$ (we denote the corresponding coordinates on $\mathcal{W}$ by $\{X^A\}$). Then $\{e_\zeta \otimes \tau_A\}$ is the basis in $\mathcal{F}$ and we denote the corresponding coordinates on $\mathcal{F}$ by $\{\Phi^{A\zeta}\}$. The degree (ghost number) of $\Phi^{A\zeta}$ is $-|e_\zeta| + |X^A|$. If the symplectic form ω on $\mathcal{W}$ is $\omega = dX^A \omega_{AB} dX^B$, then the degree -1 symplectic form Ω on $\mathcal{F}$ is defined as

$$\Omega = \text{Tr}(\delta\Phi^A \omega_{AB} \delta\Phi^B) = (-1)^{(|A|+1) \cdot |\zeta| + n+1} \delta\Phi^{A\zeta} \Pi(e_\zeta, e_\eta) \omega_{AB} \delta\Phi^{B\eta} ,$$

where $\Phi^A := \Phi^{A\zeta} e_\zeta$ (and we use multiplication m under trace implicitly) and we use the obvious shorthand notation $|A| = |X^A|$, $|\zeta| = |e_\zeta|$. The abstract AKSZ action is

$$S = S_{kin} + S_{int} = \underbrace{\frac{1}{2} \text{Tr}(\Phi^A \omega_{AB} D\Phi^B)}_{=(-1)^{|A| \cdot (|\zeta|+1)} \frac{1}{2} \Phi^{A\zeta} \Pi(e_\zeta, D e_\eta) \omega_{AB} \Phi^{B\eta}} + (-1)^{n+1} \text{Tr}(\Phi^*(\Theta)) , \quad (7.45)$$

where $\Phi^* : S^\bullet(\mathcal{W}^*) \rightarrow \mathcal{C}$ is the ring homomorphism induced by the field $\Phi \in \mathcal{C} \otimes \mathcal{W} \cong \text{Hom}(\mathcal{W}^*, \mathcal{C})$ (i.e. we first interpret Φ as a map of graded vector spaces from $\mathcal{W}^*$ to $\mathcal{C}$ and

then extend it as a ring homomorphism, according to free multiplication in $S^\bullet(\mathcal{W}^*)$ and multiplication m in $\mathcal{C}$). In coordinates: if

$$\Theta = \Theta_A X^A + \frac{1}{2} \Theta_{AB} X^A X^B + \dots \quad (7.46)$$

then

$$S_{int} = (-1)^{n+1} \Theta_A \text{Tr}(e_\zeta) \Phi^{A\zeta} + (-1)^{(|A|+1) \cdot |\zeta| + n+1} \frac{1}{2} \Theta_{AB} \text{Tr}(m(e_\zeta, e_\eta)) \Phi^{A\zeta} \Phi^{B\eta} + \dots \quad (7.47)$$

We claim that the action (7.45) satisfies classical master equation $\{S, S\} = 0$ w.r.t. the anti-bracket on $\mathcal{F}$ associated to the odd symplectic form Ω .

Remark 6 The coordinate-free version of the construction above is as follows. We have a ring homomorphism $\text{ev}^* : S^\bullet(\mathcal{W}^*) \rightarrow S^\bullet(\mathcal{F}^*) \otimes \mathcal{C}$ defined on generators as the canonical map $\mathcal{W}^* \rightarrow \mathcal{F}^* \otimes \mathcal{C}$ associated to the identity $\mathcal{F} \rightarrow \mathcal{C} \otimes \mathcal{W} = \mathcal{F}$. (Notation ev^* should remind of the pull-back by evaluation map $\text{ev} : \text{Maps}(\mathcal{N}, \mathcal{M}) \times \mathcal{N} \rightarrow \mathcal{M}$ in usual AKSZ construction which goes as $\text{ev}^* : C^\infty(\mathcal{M}) \rightarrow C^\infty(\text{Maps}(\mathcal{N}, \mathcal{M})) \otimes C^\infty(\mathcal{N})$.) The interaction part of action (7.45) is then

$$S_{int} = (\text{id} \otimes \text{Tr}) \circ \text{ev}^*(\Theta) .$$

We can also formally extend ev^* to differential forms on $\mathcal{W}$ as a homomorphism of dg algebras $\text{ev}^* : \Omega^\bullet(\mathcal{W}) \rightarrow \Omega^\bullet(\mathcal{F}) \otimes \mathcal{C}$ (here $\mathcal{C}$ is treated as an algebra with zero de Rham differential). Then the odd symplectic form on $\mathcal{F}$ is given by

$$\Omega = (\text{id} \otimes \text{Tr}) \circ \text{ev}^*(\omega) .$$

The kinetic part of action (7.45) is defined as the Hamiltonian function for the cohomological vector field on $\mathcal{F}$, induced by the differential $D : \mathcal{C}^\bullet \rightarrow \mathcal{C}^{\bullet+1}$.

Example 7 (*AKSZ with target a vector space.*) Usual AKSZ models on the space $\text{Maps}(\mathcal{N}, \mathcal{M})$, in the case when $\mathcal{M}$ is a graded vector space with constant symplectic form, can be interpreted as abstract AKSZ with $\mathcal{W} = \mathcal{M}$ and $\mathcal{C} = C^\infty(\mathcal{N})$.

Example 8 (*abstract Chern-Simons.*) Taking arbitrary $\mathcal{C}$ with $n = 2$ (i.e. concentrated in degrees 0,1,2,3 and with pairing of degree -3), taking $\mathcal{W} = \mathfrak{g}[1]$ for a quadratic Lie algebra $\mathfrak{g}$ with invariant pairing $\pi_{\mathfrak{g}}$ and setting

$$\omega = dX^A \pi_{\mathfrak{g}}(\tau_A, \tau_B) dX^B \quad , \quad \Theta = \frac{1}{6} \pi_{\mathfrak{g}}(\tau_A, [\tau_B, \tau_C]) X^A X^B X^C$$

we obtain abstract Chern-Simons in the sense of [11].

In general, in this way (i.e. by allowing $\mathcal{C}$ to be an arbitrary dg Frobenius algebra with pairing of appropriate degree, instead of demanding that it is of form $C^\infty(\mathcal{N})$) we can construct abstract versions of AKSZ models with vector space targets.

7.2 Sheaf version

We recall first some basic definition that can be found in [13]. A ringed space is a couple $(X, \mathcal{O}_X)$ where X is a topological space and $\mathcal{O}_X$ is sheaf of rings on X , see [13]. We denote with $\mathcal{O}_X(U)$ the local sections on the open $U \subset X$. A graded manifold $\mathcal{M}$ is a ringed space $(\mathcal{M}_0, \mathcal{M})$ such that $\mathcal{M}(U)$ is locally isomorphic to $C^\infty(U) \otimes S(V^*)$, for some open $U \subset \mathcal{M}_0$ and graded vector space V .

Definition 9 *A morphism of ringed spaces from $(X, \mathcal{O}_X)$ to $(Y, \mathcal{O}_Y)$ is a couple (ϕ, Φ) , where $\phi : X \rightarrow Y$ and $\Phi : \mathcal{O}_Y \rightarrow \phi_* \mathcal{O}_X$ is a morphism of sheaves on Y .*

The pushforward sheaf is defined as the sheaf on Y with local sections $\phi_* \mathcal{O}_X(U) = \mathcal{O}_X(\phi^{-1}U)$, for any open $U \subset Y$. The morphism (ϕ, Φ) assigns to any open $U \subset Y$ a ring morphism $\Phi^*(U) : \mathcal{O}_Y(U) \rightarrow \mathcal{O}_X(\phi^{-1}U)$.

The abstract AKSZ construction depends on the following data.

The source: A sheaf $\mathcal{J}$ of dg supercommutative algebras over some closed manifold $\mathcal{N}_0$, such that locally, for some open $U \subset \mathcal{N}_0$, we have

$$\mathcal{J}(U) \cong \Omega^\bullet(U) \otimes \mathcal{C} , \quad (7.48)$$

where $\Omega^\bullet(U)$ is the algebra of differential forms on U , $\mathcal{C}$ is some fixed finite dimensional unital dg Frobenius algebra with differential D , product m and pairing Π of degree $-n_{\mathcal{C}}$ (with Tr the corresponding trace). We denote the unit of $\mathcal{C}$ by $\mathbf{1}$ and impose that $\mathcal{C}$ is equipped with the splitting

$$\mathcal{C} = \mathbb{R} \cdot \mathbf{1} \oplus \bar{\mathcal{C}} , \quad (7.49)$$

where $\bar{\mathcal{C}}$ is an ideal. The ring $\Omega^\bullet(U) \otimes \mathcal{C} = C^\infty(U) \oplus \dots$ inherits the splitting and the restriction homomorphisms must respect the splitting. Moreover we require the existence of a morphism of sheaves of complexes

$$\text{Tr}_J : \mathcal{J} \rightarrow \Omega^\bullet ,$$

where $\Omega^\bullet$ is sheaf of differential forms. This morphism has a degree $-n_{\mathcal{C}}$ and locally has the form $\text{id}_U \otimes \text{Tr}$. Therefore the set of the global sections of $\mathcal{J}$ is equipped with the structure of dg Frobenius algebra with the trace given by the composition of Tr_J and integration of differential forms over $\mathcal{N}_0$ and with total differential $d_{\mathcal{N}_0} + D$.

The target: A $\mathbb{Z}$ -graded manifold $\mathcal{M}$ with body $\mathcal{M}_0$, equipped with symplectic form $\omega \in \Omega^2(\mathcal{M})$ of degree n and a function $\Theta \in C^\infty(\mathcal{M})$ of degree $n+1$, satisfying $\{\Theta, \Theta\} = 0$. Here $n := \dim \mathcal{N}_0 + n_{\mathcal{C}} - 1$.

We define the space of BV fields $\mathcal{F} = \text{Maps}(\mathcal{J}, \mathcal{M})$ as the space of morphisms of ringed spaces from $(\mathcal{N}_0, \mathcal{J})$ to $(\mathcal{M}_0, \mathcal{M})$. Let $(\phi, \Phi) \in \mathcal{F}$ and let $U \subset \mathcal{M}_0$ be a coordinate neighborhood such that $\mathcal{M}(U) \sim C^\infty(U) \otimes S(V^*)$. Let $\{e_\zeta\}$ be some basis in $\mathcal{C}$ with $e_0 = \mathbf{1}$ and $e_\zeta \in \bar{\mathcal{C}}$ for $\zeta \neq 0$; let $\{X^A\} = \{x^\mu; \xi^m\}$ be local coordinates on $\mathcal{M}$ (i.e. $\{x^\mu\}$ are coordinates on $U \subset \mathcal{M}_0$ and $\{\xi^m\}$ are coordinates on V), and let $\{u^a\}$ be local coordinates on $\mathcal{N}_0$. Then the BV field $(\phi, \Phi) \in \mathcal{F}$ is locally described by the *superfields* that are the values of $\Phi^*(U)$ on the generators $\{X^A\}$ of $\mathcal{M}(U)$:

$$\Phi^*(U) : X^A \mapsto \Phi_0^{A\zeta}(u)e_\zeta + \Phi_a^{A\zeta}(u)\theta^a e_\zeta + \Phi_{a_1 a_2}^{A\zeta}(u)\theta^{a_1}\theta^{a_2} e_\zeta + \dots = \Phi^{A\zeta}(u, \theta)e_\zeta . \quad (7.50)$$

Here $\theta^a := d_{\mathcal{N}_0} u^a$ are the odd generators of $\Omega^\bullet(\phi^{-1}(U))$. The splitting condition (7.49) allows to define the coefficients of 1 that define a local map $\phi_U : \phi^{-1}(U) \rightarrow U$ as $x^\mu \circ \phi_U(u) = \Phi_0^{\mu 0}(u)$. By using a standard argument we see that $\phi_U(u) = \phi(u)$, i.e. we recover the global map ϕ .

Let us denote the symplectic form as $\omega = dX^A \omega_{AB} dX^B$; we construct the degree -1 symplectic form Ω on $\mathcal{F}$ as

$$\begin{aligned} \Omega &= \int_{\mathcal{N}_0} \text{Tr} (\delta \Phi^A \omega_{AB} \delta \Phi^B) = \\ &= (-1)^{(|A| + \dim \mathcal{N}_0 + 1) \cdot |\zeta| + (n+1)n_{\mathcal{C}}} \int_{\mathcal{N}_0} \delta \Phi^{A\zeta} \Pi(e_\zeta, e_\eta) \omega_{AB} \delta \Phi^{B\eta} , \end{aligned} \quad (7.51)$$

where $\Phi^A = \Phi^*(X^A)$ is the right hand side of (7.50) and the expression under trace in the first line implicitly uses the multiplication m in $\mathcal{C}$. The abstract AKSZ action is

$$S[\Phi] = \underbrace{\frac{1}{2} \int_{\mathcal{N}_0} \text{Tr} (\Phi^A \omega_{AB} (d_{\mathcal{N}_0} + D) \Phi^B)}_{S_{kin}} + (-1)^{n+1} \underbrace{\int_{\mathcal{N}_0} \text{Tr} (\Phi^*(\Theta))}_{S_{int}} . \quad (7.52)$$

The kinetic term may be also written as

$$\begin{aligned} S_{kin} &= (-1)^{(|A| + \dim \mathcal{N}_0) \cdot |\zeta| + n_{\mathcal{C}} \dim \mathcal{N}_0} \cdot \int_{\mathcal{N}_0} \left((-1)^{|A| + \dim \mathcal{N}_0} \frac{1}{2} \Phi^{A\zeta} \Pi(e_\zeta, D e_\eta) \omega_{AB} \Phi^{B\eta} + \right. \\ &\quad \left. + (-1)^{|\zeta|} \frac{1}{2} \Phi^{A\zeta} \Pi(e_\zeta, e_\eta) \omega_{AB} d_{\mathcal{N}_0} \Phi^{B\eta} \right) . \end{aligned} \quad (7.53)$$

If Θ is expanded in local coordinates on $\mathcal{M}$ as (7.46), then the expansion analogous to (7.47) in this setting is

$$S_{int} = (-1)^{n+1+n_{\mathcal{C}}} \int_{\mathcal{N}_0} \Theta_A \text{Tr}(e_\zeta) \Phi^{A\zeta} + (-1)^{(|A| + \dim \mathcal{N}_0 + 1) \cdot |\zeta|} \frac{1}{2} \Theta_{AB} \text{Tr}(m(e_\zeta, e_\eta)) \Phi^{A\zeta} \Phi^{B\eta} + \dots$$

We again claim that $\{S, S\} = 0$.

Example 10 Taking the trivial dg Frobenius algebra $\mathcal{C} = \mathbb{R}$, we obtain standard AKSZ models on $\text{Maps}(T[1]\mathcal{N}_0, \mathcal{M})$.

Example 11 Taking $\mathcal{N}_0 = pt$, specifying some finite dimensional dg Frobenius algebra $\mathcal{C}$ with splitting (7.49) and a finite-dimensional target $(\mathcal{M}, \omega, \Theta)$, we obtain a finite-dimensional abstract AKSZ model. A particular class of such sources is provided by the de Rham cohomology of connected closed orientable manifolds $\mathcal{C} = H_{dR}^\bullet(\Sigma)$, viewed as dg Frobenius algebras with $D = 0$ and Π associated to Poincaré duality. Since Σ is connected, splitting (7.49) is automatic: $\mathcal{C} = \mathbb{R} \cdot \mathbf{1} \oplus H_{dR}^{\geq 1}(\Sigma)$. For these models $S_{kin} = 0$, and the AKSZ action is just the pull-back of the function Θ on the target. These are the examples that we studied in (4.31) when Σ is two-dimensional and in (5.38) when Σ is three-dimensional.

Example 12 (*Source given by fiber cohomology of a fiber bundle.*) Suppose E is a fiber bundle over $\mathcal{N}_0$ with typical fiber F (closed connected orientable manifold), endowed with a flat connection ∇_E . Then we define fiber differential $d_{\text{fib}} : \Omega^\bullet(E) \rightarrow \Omega^{\bullet+1}(E)$ as follows: for 0-forms $f \in C^\infty(E)$ we define d_{fib} by the property $i_v(d_{\text{fib}}f) := v_\perp(f)$ (where i_v is the convolution with arbitrary vector field $v \in \text{Vect}(E)$ and the projection to fiber $v \mapsto v_\perp$ is the projection to second term in the splitting of tangent bundle $TE = T_{||}E \oplus T_\perp E$ defined by the connection ∇_E); then we extend d_{fib} to all forms on E by Leibniz rule and property $d_E d_{\text{fib}} + d_{\text{fib}} d_E = 0$ (where d_E is the de Rham differential on E). Flatness of ∇_E implies that $d_{\text{fib}}^2 = 0$. Then we construct sheaf $\mathcal{J}$ over $\mathcal{N}_0$ as the cohomology of d_{fib} :

$$\Gamma(\mathcal{N}_0, \mathcal{J}) = H_{d_{\text{fib}}}^\bullet(\Omega^\bullet(E)) .$$

Locally $\mathcal{J}$ splits as (7.48) with $\mathcal{C} = H^\bullet(F)$ (de Rham cohomology of the fiber).

In particular we can take a trivial fiber bundle $E = \mathcal{N}_0 \times F$ with canonical flat connection ∇_E . Then d_{fib} is just the de Rham differential along fiber $d_{\text{fib}} = \text{id}_{\mathcal{N}_0} \otimes d_F$ and the fiber cohomology sheaf $\mathcal{J}$ splits globally:

$$\Gamma(\mathcal{N}_0, \mathcal{J}) = \Omega^\bullet(\mathcal{N}_0) \otimes H^\bullet(F) .$$

Abstract AKSZ models with source $\mathcal{J}$ and some target $(\mathcal{M}, \omega, \Theta)$ may in some cases arise as a partial reduction of usual AKSZ models on $\text{Maps}(T[1]E, \mathcal{M})$. The simplest example here is: E is the 2-torus, viewed as a trivial bundle over circle $\mathcal{N}_0 = \mathbb{S}^1$ with fiber a circle $F = \mathbb{S}^1$, and $\mathcal{M}$ is a Poisson manifold. The corresponding abstract AKSZ model is a partial reduction of Poisson sigma model on torus.

We hope to explore this set of examples in more detail in a future publication.

Remark 13 Global sections $\Gamma(\mathcal{N}_0, \mathcal{J})$ of the sheaf $\mathcal{J}$ themselves form a dg Frobenius algebra with differential $D_{\mathcal{J}} = d_{\mathcal{N}_0} + D$ (where $d_{\mathcal{N}_0}$ is the de Rham differential on $\mathcal{N}_0$), multiplication $m_{\mathcal{J}}$ coming from wedge product of forms and multiplication m on $\mathcal{C}$, and with the pairing $\Pi_{\mathcal{J}}(\chi, \psi) = \int_{\mathcal{N}_0} \text{Tr}(m(\chi, \psi))$. When the target is a graded vector space, we can apply the vector space construction of Subsection 7.1 with $\Gamma(\mathcal{N}_0, \mathcal{J})$ as source. We easily verify that the two constructions coincide. Remark that requirement (7.49) is not needed in the vector space version, so that more general Frobenius algebras are allowed in the vector space version.

8 Conclusions

In this paper we studied a canonical reduction of the AKSZ-BV field theory to a finite dimensional BV theory which governs the semi-classical approximation. As illustration of the general construction, we discussed the two dimensional Poisson sigma model and the three dimensional Courant sigma model.

Our main perspective has been the odd symplectic reduction of the infinite dimensional manifold of fields. It is important to remark that one can look at the reduced action S_{BV} as the leading contribution in the effective BV action which controls the low energy fields (we can call them either "constant" maps or zero modes). It is convenient to consider the idea of effective BV theories suggested by Losev [20] (see [21], [5] and [10] for further developments). In fact, given any embedding of the cohomology in the space of forms of the source, one can look at the reduced variables as "infrared" degrees of freedom of the full theory. The effective action is then defined by integrating over the "ultraviolet" degrees of freedom and, in the perturbative approach, is a series in $\hbar$ and in the hamiltonian function Θ of the target. From this point of view, the reduced BV manifolds that we studied in this paper are the spaces of the infrared degrees of freedom and the action S_{BV} is the lowest order in the expansion of the effective action. In principle one can calculate the corrections by applying Feynman diagrams techniques.

From the examples considered in this paper, it is natural to consider the generalization of the AKSZ construction [1]. In Section 4.2 we observed that the reduced BV theory can be described in terms of "supermaps" to the target graded manifold. The novelty is that the formal variables of the source manifold have to satisfy some constraints and thus they cannot be anymore considered as the coordinates of a graded manifold. This is the generalization of the AKSZ construction that we introduced in Section 7. The examples

that appeared in this paper have sources that are commutative graded algebras seen as sheaves over a point and so are zero dimensional TFT's. One can consider generalized AKSZ theories in any dimension, as we described for instance in the Example (12), that will be the object of future study.

Finally, one can consider more general type of BV reductions, not necessary to the constant map configurations. Moreover many ideas presented here can be applied to a wider setup than simply AKSZ-BV theories. For example, it could be interesting to study the reduction of the two dimensional BV theories described in [27, 28].

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A Computation of Berezinian (5.39)

We show here that the volume form introduced in (5.39) is globally defined on $\text{Map}(X_\Sigma, \mathcal{M})$. The coordinates $z = (\mathbf{X}^\mu, \boldsymbol{\xi}_a, \mathbf{P}_\mu)$ defined as coefficients of the superfields (5.36) depend on the choice of coordinates $\{x^\mu\}$ on M and of a trivialization $\{e_a\}$ of E . If we change to coordinates $\{y^i = y^i(x)\}$ on M and to trivialization $\{e_\alpha = t_\alpha^a(x)e_a\}$, the coordinates on $\text{Map}(X_\Sigma, \mathcal{M})$ change to $\tilde{z} = (\mathbf{X}^i, \boldsymbol{\xi}_\alpha, \mathbf{P}_i)$ accordingly as

$$\mathbf{X}^i = y^i(\mathbf{X}) , \quad \boldsymbol{\xi}^\alpha = t_\alpha^a(\mathbf{X})\boldsymbol{\xi}^a , \quad \mathbf{P}_i = \frac{\partial x^\mu}{\partial y^i}(\mathbf{X})\mathbf{P}_\mu + \frac{1}{2}\boldsymbol{\xi}^a\boldsymbol{\xi}^b \frac{\partial t_a^\alpha}{\partial y^i}(\mathbf{X})g_{\alpha\beta}t_b^\beta(\mathbf{X}) . \quad (\text{A.54})$$

The quadratic term in the transformation of $\mathbf{P}_i$ can be removed by introducing a connection on the vector bundle E . In fact, the coordinate $\mathbf{P}_\mu^\Gamma = \mathbf{P}_\mu + \frac{1}{2}\Gamma_{\mu a}^b\boldsymbol{\xi}^a g_{bc}\boldsymbol{\xi}^c$ transforms as a tensor

$$\mathbf{P}_i^\Gamma = \frac{\partial x^\mu}{\partial y^i}(\mathbf{X})\mathbf{P}_\mu^\Gamma .$$

It can be easily checked that the Berezinian of the transformation from $\{\mathbf{X}^\mu, \boldsymbol{\xi}^a, \mathbf{P}_\mu\}$ to $\{\mathbf{X}^\mu, \boldsymbol{\xi}^a, \mathbf{P}_\mu^\Gamma\}$ is one so that the coordinate volume forms are the same. Equivalently, in

order to compute the Berezinian of the transformation $\tilde{z}(z)$ we are allowed to ignore the quadratic term in the transformation of $\mathbf{P}$ in (A.54).

The final result is that the berezinian of the transformation matrix $I = (\frac{\partial \tilde{z}}{\partial z})$ is

$$\text{Ber} I = \text{Ber} \begin{pmatrix} I_{00} & I_{01} \\ I_{10} & I_{11} \end{pmatrix} = (\det t_a^\alpha)^{2(1-b_1)}.$$

We first compute it with respect to the transformation $y^i = y^i(x)$ with fixed trivialization. Let us order the relevant coordinates $z = \{z_0, z_1\}$, where z_0 are the even ones and z_1 are the odd ones, as $z_0 = (x^\mu, \gamma_\mu, F_{\mu I}, \alpha_I^{+\mu})$ and $z_1 = (\alpha_\mu^I, F^{+\mu I}, \gamma^{+\mu}, x_\mu^+)$. It is important to see where the zeros are located in I , so that at the end only few matrix elements (the diagonal ones) enter the result. By inspection of the degree in (A.54) we can easily write

$$I_{00} = \frac{\partial \tilde{z}_0}{\partial z_0} = \begin{pmatrix} \frac{\partial y^i}{\partial x^\mu} & 0 & 0 & 0 \\ * & \frac{\partial \gamma_i}{\partial \gamma_\mu} & 0 & 0 \\ & & \frac{\partial F_{iI}}{\partial F_{\mu J}} & * \\ & * & 0 & \frac{\partial \alpha_I^{+i}}{\partial \alpha_J^{+\mu}} \end{pmatrix} \quad I_{01} = \frac{\partial \tilde{z}_0}{\partial z_1} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ & & 0 & 0 \\ * & & 0 & 0 \end{pmatrix}$$

$$I_{11} = \frac{\partial \tilde{z}_1}{\partial z_1} = \begin{pmatrix} \frac{\partial \alpha_i^I}{\partial \alpha_\mu^J} & * & 0 & 0 \\ 0 & \frac{\partial F^{+iI}}{\partial F^{+\mu J}} & 0 & 0 \\ & & \frac{\partial \gamma^{+i}}{\partial \gamma^{+\mu}} & 0 \\ * & & * & \frac{\partial x_{+i}}{\partial x_{+\mu}} \end{pmatrix} \quad I_{10} = \frac{\partial \tilde{z}_1}{\partial z_0} = \begin{pmatrix} & 0 & 0 \\ * & 0 & 0 \\ & * & \end{pmatrix},$$

where the block structure is easily understood. It is then easy to compute that

$$I_{01} I_{11}^{-1} I_{10} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ & & 0 & 0 \\ * & & 0 & 0 \end{pmatrix}$$

so that

$$\text{Ber} I = \det(I_{00} - I_{01} I_{11}^{-1} I_{10}) / \det I_{11} = \det I_{00} / \det I_{11} = 1.$$

Consider now the change of trivialization $e_\alpha = t_\alpha^a(x) e_a$, without changing coordinates $\{x^\mu\}$. Let us order the relevant coordinates as follows: the even ones are $z_0 = (x^\mu, \alpha_I^{+\mu}, A^{aI}, \beta_a^+)$ and the odd ones are $z_1 = (F^{+\mu I}, \gamma^{+\mu}, \beta^a, A_{aI}^+)$. Since we are ignoring the quadratic terms in (A.54) the coefficients of $\mathbf{P}_\mu$ do not appear. Then we compute

$$I_{00} = \begin{pmatrix} \delta_\nu^\mu & 0 & 0 & 0 \\ 0 & \delta_J^I \delta_\nu^\mu & 0 & 0 \\ & & \frac{\partial A^{aI}}{\partial A^{aJ}} & 0 \\ * & & * & \frac{\partial \beta_a^+}{\partial \beta_a^+} \end{pmatrix} \quad I_{01} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ & & * & 0 \\ * & & * & * \end{pmatrix}$$

$$I_{11} = \begin{pmatrix} \delta^\mu_\nu \delta^I_J & 0 & 0 & 0 \\ 0 & \delta^\nu_\mu & 0 & 0 \\ & & \frac{\partial \beta^\alpha}{\partial \beta^a} & 0 \\ * & & * & \frac{\partial A^+_{\alpha I}}{\partial A^+_{a I}} \end{pmatrix} \quad I_{10} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ & & 0 & 0 \\ * & & * & 0 \end{pmatrix}.$$

We then compute that

$$I_{01} I_{11}^{-1} I_{10} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ & & 0 & 0 \\ * & & * & 0 \end{pmatrix}$$

and finally we get

$$\text{Ber} I = \det(I_{00} - I_{01} I_{11}^{-1} I_{10}) / \det I_{11} = \det I_{00} / \det I_{11} = (\det t_a^\alpha)^{2(b_1-1)}.$$

B Relative (co)homology

We recall in this appendix basic facts about relative (co)homology and Lefschetz duality for manifolds with boundary [14].

Let Σ be a smooth manifold of dimension d with boundary $\partial\Sigma$. The relative k -chains with real coefficients are defined as $C_k(\Sigma, \partial\Sigma) = C_k(\Sigma) / C_k(\partial\Sigma)$, *i.e.* the chains in Σ modulo the chains in $\partial\Sigma$. We will always work with real coefficient and we will omit it in the notation. The usual boundary ∂ goes to the quotient and defines the relative homology $H_k(\Sigma, \partial\Sigma)$. An alternative description of chains is obtained by defining $C'_k(\Sigma, \partial\Sigma) = \{(c_k, \sigma_{k-1}) \mid c_k \in C_k(\Sigma), \sigma_{k-1} \in C_{k-1}(\partial\Sigma)\}$ with boundary $\partial(c_k, \sigma_{k-1}) = (\partial c_k + (-)^k \sigma_{k-1}, \partial \sigma_{k-1})$. It is easy to check that the map $(c_k, \sigma_{k-1}) \in C'_k(\Sigma, \partial\Sigma) \rightarrow \underline{c_k} \in C_k(\Sigma, \partial\Sigma)$ is a quasisomorphism.

The exact sequence $0 \rightarrow C_k(\partial\Sigma) \rightarrow C_k(\Sigma) \rightarrow C_k(\Sigma, \partial\Sigma) \rightarrow 0$ gives rise to the long exact sequence in homology

$$\dots \rightarrow H_k(\partial\Sigma) \rightarrow H_k(\Sigma) \rightarrow H_k(\Sigma, \partial\Sigma) \rightarrow H_{k-1}(\partial\Sigma) \rightarrow \dots, \quad (\text{B.1})$$

where the last map sends $[\underline{c}] \in H_k(\Sigma, \partial\Sigma) \rightarrow [\partial c] \in H_{k-1}(\partial\Sigma)$, for some $c \in C_k(\Sigma)$.

The complex of relative cochains $C^\bullet(\Sigma, \partial\Sigma)$ can be described as the restriction of de Rham complex to those forms ω whose restriction $\omega|_{\partial\Sigma}$ to the boundary is zero. We denote the relative cohomology as $H_{dR}(\Sigma, \partial\Sigma)$. By the universal coefficient theorem we have that $H_{dR}^\bullet(\Sigma) = H(\Sigma)_\bullet^*$. The alternative description for k -relative cochains is $C'^k(\Sigma, \partial\Sigma) = \Omega^k \Sigma \oplus \Omega^{k-1} \partial\Sigma$, with differential $d(\omega_k, \nu_{k-1}) = (d\omega_k, d\nu_{k-1} - (-)^k \omega_k|_{\partial\Sigma})$. The map $\omega_k \in$

$C^k(\Sigma, \partial\Sigma) \rightarrow (\omega_k, 0) \in C''^k(\Sigma, \partial\Sigma)$ is a quasisomorphism. The pairing is then defined as

$$\langle \omega_k, \underline{c_k} \rangle = \int_{c_k} \omega_k .$$

or alternatively as

$$\langle (\omega_k, \nu_{k-1}), (c_k, \sigma_{k-1}) \rangle = \int_{c_k} \omega_k + \int_{\sigma_{k-1}} \nu_{k-1} .$$

The above notion of relative (co)homology makes sense for any subspace of Σ , in particular we can consider (union of) components of $\partial\Sigma$. If $\partial\Sigma = \cup_{i \in I} \partial_i \Sigma$ and $\partial_J \Sigma = \cup_{i \in J} \partial_i \Sigma$, for $J \subset I$, we will consider the relative homology $H(\Sigma, \partial_J \Sigma)$ and cohomology $H_{dR}(\Sigma, \partial_J \Sigma)$. Let $I = I_1 \cup I_2$, with $I_1 \cap I_2 = \emptyset$; the choice of the fundamental class $[\Sigma] \in H_d(\Sigma, \partial\Sigma)$ determines the following isomorphism (for a proof see Theorem 3.43 of [14])

$$H_k(\Sigma, \partial_{I_1} \Sigma) = H_{d-k}^{dR}(\Sigma, \partial_{I_2} \Sigma) .$$

In particular, the case $I_1 = \emptyset$ or $I_2 = \emptyset$ is known as *Lefschetz duality*,

$$H_{dR}^k(\Sigma, \partial\Sigma) \sim H_{d-k}(\Sigma) , \quad H_{dR}^k(\Sigma) \sim H_{d-k}(\Sigma, \partial\Sigma) .$$

Let us describe more explicitly the case $d = 2$ and consider a compact surface $\Sigma_{g,n}$ of genus g and n boundary components.

Since $H_0(\Sigma_{g,n}, \partial\Sigma_{g,n}) = H_2(\Sigma_{g,n}) = 0$, by Lefschetz duality we get that $H_{dR}^2(\Sigma_{g,n}) = H_{dR}^0(\Sigma_{g,n}, \partial\Sigma_{g,n}) = 0$. The Lefschetz duality in degree one can be seen as the non degeneracy of the pairing $H_{dR}^1(\Sigma_{g,n}) \otimes H_{dR}^1(\Sigma_{g,n}, \partial\Sigma_{g,n}) \rightarrow \mathbb{R}$, $(a, b) \rightarrow \int_{\Sigma_{g,n}} a \wedge b$. Equivalently, for any basis $\{c_I\}$ for $H_{dR}^1(\Sigma_{g,n})$ and $\{g_{I'}\}$ for $H_{dR}^1(\Sigma_{g,n}, \partial\Sigma_{g,n})$ the matrix $A_{II'}^{(g,n)} = \int_{\Sigma_{g,n}} c_I \wedge g_{I'}$ is non degenerate. Let us denote with $A^{(g,n)I'J}$ the inverse matrix. It will be useful even the (possibly degenerate) matrix $B_{I'J'}^{(g,n)} = \int_{\Sigma_{g,n}} g_{I'} \wedge g_{J'}$.

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ONE-DIMENSIONAL CHERN-SIMONS THEORY

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ABSTRACT. We study a one-dimensional toy version of the Chern-Simons theory. We construct its simplicial version which comprises features of a low-energy effective gauge theory and of a topological quantum field theory in the sense of Atiyah.

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1. INTRODUCTION

We begin (see section 2) by considering a one-dimensional version of the Chern-Simons theory on a circle. This is a gauge theory in the Batalin-Vilkovisky formalism defined by the action

$$(1) \quad \frac{1}{2} \int (\psi, d\psi) + (\psi, [A, \psi]),$$

where the field ψ is an odd function on the circle with values in a quadratic Lie algebra $\mathfrak{g}$, and the field A is an even 1-form with values in $\mathfrak{g}$. We address the problem of constructing an effective BV action induced on a triangulation of the circle.

This problem is interesting by itself since it is related to discretization of differential geometry. Indeed, the action (1) can be viewed as a generating function for natural operations on differential forms on the circle: the de Rham differential, the wedge product and the integral over the circle (more precisely, it is a generating function for a unimodular cyclic DGLA structure on $\mathfrak{g}$ -valued forms, cf. [7]). In this language, the effective action on a triangulation is a generating function for some discretized (homotopy) version of this structure induced on cochains of the triangulations (viewed as discrete differential forms).

Another motivation for studying the effective action for (1) is that it might give a new insight for constructing a discrete version of the 3-dimensional Chern-Simons theory. Such a discrete Chern-Simons theory would allow to compute invariants of 3-manifolds as finite-dimensional integrals, and it would be compatible with the gauge symmetry (*i.e.* it would satisfy the Batalin-Vilkovisky quantum master equation).

The effective action for the one-dimensional Chern-Simons theory on a triangulated circle turns out to be given by an explicit but somewhat bizarre formula (42). It immediately raises a number of questions. For instance, the result is expected to satisfy the quantum master equation (QME) and to be compatible with simplicial aggregations (merging several 1-simplices of the triangulation). How can we check this directly? Another desire is to represent the result (42) in a “simplicially-local” form.

It turns out that answers to these questions come from the following construction. We give a new definition of the one-dimensional simplicial Chern-Simons theory in the “operator formalism”, *i.e.* in the language of Clifford algebras $Cl(\mathfrak{g})$ (section 3.2.1). The partition function for a simplicial complex is an element of $Cl(\mathfrak{g})^{\otimes i}$ (where i is half the number of boundary points of the simplicial complex), and it given by a product of local $Cl(\mathfrak{g})$ -valued expressions (61) for 1-simplices. In particular, for a triangulated circle the partition function takes values in numbers (and it also depends on simplicial “bulk fields”). In section 4.4.2, we establish the equivalence between the operator formalism and the path integral formalism of section 2. Consistency with simplicial aggregations is checked straightforwardly in the operator language (section 3.2.2). The partition function for an interval can be

shown to satisfy equation (68):

$$\hbar \frac{\partial}{\partial \tilde{\psi}^a} \frac{\partial}{\partial A^a} Z_{\mathcal{I}} + \frac{1}{\hbar} \left[\frac{1}{6} f^{abc} \hat{\psi}^a \hat{\psi}^b \hat{\psi}^c, Z_{\mathcal{I}} \right]_{Cl(\mathfrak{g})} = 0$$

which is interpreted as a version of the quantum master equation adjusted for the presence of the boundary. This immediately implies the QME with boundary contributions for arbitrary one-dimensional simplicial complex (73) and the usual QME for the triangulated circle (44).

To formulate one-dimensional simplicial Chern-Simons theory in the spirit of Atiyah's axioms of TQFT (section 4.2), we choose a complex polarization of $\mathfrak{g}$:

$$\mathfrak{g}_{\mathbb{C}} = \mathfrak{h} \oplus \bar{\mathfrak{h}}$$

(which can always be introduced if $\mathfrak{g}$ is even-dimensional; however, by introducing a complex polarization we break the $O(\mathfrak{g})$ -symmetry of the original problem). The Clifford algebra $Cl(\mathfrak{g})$ is isomorphic to the matrix (super-)algebra $\text{End}(\wedge^{\bullet} \mathfrak{h}) = \text{End}(\text{Fun}(\Pi \mathfrak{h}))$. Therefore, the space of states associated to a point in the one-dimensional Chern-Simons theory is $\mathcal{H}_{pt} = \text{Fun}(\Pi \mathfrak{h})$ — the super vector space of polynomials in $(\dim \mathfrak{g})/2$ odd variables. The super-space $\mathcal{H}_{pt}$ is endowed with an odd third-order differential operator δ . One-dimensional “cobordisms” are now equipped with triangulations. To a triangulated cobordism Θ we associate the “space of bulk fields” $\mathcal{F}_{\Theta}^{\text{bulk}}$, equipped with the BV Laplacian $\Delta_{\Theta}^{\text{bulk}}$. The partition function for a triangulated cobordism satisfies the quantum master equation (80). In addition to the operations of gluing and disjoint union (which are standard in Atiyah's picture), simplicial aggregations are allowed for triangulated cobordisms. The original continuum theory can be thought of as the simplicial theory in the limit of dense triangulation.

Matrix elements of the partition function for a triangulated cobordism can be written as path integrals for the one-dimensional Chern-Simons with BV gauge fixing in the bulk and holomorphic-antiholomorphic boundary conditions (section 4.4.2). This brings us back to the formalism of effective BV actions. The action for an interval is given by a Gaussian integral and it is easy to compute it explicitly (115).

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1.2. Authorship. The idea of looking at the one-dimensional Chern-Simons theory and some parts of section 3 (operator formalism approach) are joint work of both authors (the idea to use operator formalism for the one-dimensional Chern-Simons theory was suggested by A. A.). Other parts of the paper are due to P. M.

2. SIMPLICIAL CHERN-SIMONS THEORY ON THE CIRCLE

In this section we study the Chern-Simons theory on the circle in Batalin-Vilkovisky (BV) formalism and construct an effective BV action induced on cochains of a triangulation. Much of this discussion is inspired by [7] and [14]. In particular,

the reader is referred to sections 2 and 3.2 of [7] for details of the effective BV action construction.

2.1. Continuum theory on the circle: fields, BV structure, action. Let $\mathfrak{g}$ be a quadratic Lie algebra with Lie bracket $[\cdot, \cdot]$ and non-degenerate ad-invariant pairing $(\cdot, \cdot)$. We will denote by $\{T^a\}$ an orthonormal basis in $\mathfrak{g}$ and by $f^{abc} = (T^a, [T^b, T^c])$ the structure constants in this basis. We will also use the Einstein summation convention for the Lie algebra indices.

The Chern-Simons theory on a 3-manifold M can be constructed as an AKSZ sigma model [1] with the space of fields

$$\mathcal{F} = \text{Maps}(\Pi T M, \Pi \mathfrak{g}) = \Pi \mathfrak{g} \otimes \Omega^\bullet(M).$$

That is, $\mathcal{F}$ is the space of maps of super-manifolds from the parity-shifted tangent bundle of M to the parity-shifted Lie algebra. Equivalently, this is the space of differential forms on M with values in $\Pi \mathfrak{g}$. From the canonical integration measure on $\Pi T M$ and the even symplectic structure $\omega_{\Pi \mathfrak{g}} = \frac{1}{2} \delta X^a \wedge \delta X^a$ on $\Pi \mathfrak{g}$ (we denote by $\{X^a\}$ the set of odd coordinates on $\Pi \mathfrak{g}$ associated to the orthonormal basis $\{T^a\}$ on $\mathfrak{g}$) one constructs an odd symplectic form (the “BV 2-form”) on $\mathcal{F}$:

$$(2) \quad \omega = \frac{1}{2} \int_M (\delta \alpha, \delta \alpha).$$

Here the superfield α is the canonical odd map (the parity-shifted identity operator)

$$(3) \quad \alpha : \mathcal{F} \rightarrow \mathfrak{g} \otimes \Omega^\bullet(M)$$

which can be viewed as the generating function for coordinates on $\mathcal{F}$ with values in $\mathfrak{g}$ -valued differential forms on M . By splitting α into components according to the degrees of differential forms we obtain

$$\alpha = A^{(0)} + A^{(1)} + A^{(2)} + A^{(3)}$$

where $A^{(p)}$ takes values in $\mathfrak{g}$ -valued p -forms.¹ Since α is totally odd, $A^{(0)}$ and $A^{(2)}$ are intrinsically odd, $A^{(1)}$ and $A^{(3)}$ are intrinsically even (the intrinsic parity is the total parity minus the de Rham degree modulo 2). The Chern-Simons action is built of the 1-form $\frac{1}{2} X^a \wedge \delta X^a$ on $\Pi \mathfrak{g}$ (which is a primitive for $\omega_{\Pi \mathfrak{g}}$) and the odd function on $\Pi \mathfrak{g}$

$$(4) \quad \theta = \frac{1}{6} f^{abc} X^a X^b X^c$$

which satisfies $\{\theta, \theta\}_{\Pi \mathfrak{g}} = 0$. The action is given by formula,

$$(5) \quad S = \int_M \frac{1}{2} (\alpha, d\alpha) + \frac{1}{6} (\alpha, [\alpha, \alpha]).$$

By the general construction [1], S satisfies the classical master equation

$$\{S, S\} = 0$$

where $\{\cdot, \cdot\}$ is the BV anti-bracket on functions on $\mathcal{F}$ defined by the odd symplectic form ω .

¹In the BV formalism, $A^{(1)}$ is the “classical field”, $A^{(0)}$ is the “ghost”; $A^{(2)}$ and $A^{(3)}$ are the “anti-fields” for $A^{(1)}$ and $A^{(0)}$, respectively.

We would like to define the one-dimensional Chern-Simons theory on the circle $\mathcal{S}^1$ by substituting $M = \mathcal{S}^1$ into the construction described above. Then, the space of fields becomes

$$\mathcal{F} = \text{Maps}(\Pi T\mathcal{S}^1, \Pi \mathfrak{g}) = \Pi \mathfrak{g} \otimes \Omega^0(\mathcal{S}^1) \oplus \Pi \mathfrak{g} \otimes \Omega^1(\mathcal{S}^1)$$

The superfield α can now be written as

$$(6) \quad \alpha = \psi + A,$$

where the component $\psi = T^a \psi^a(\tau)$ takes values in $\mathfrak{g}$ -valued functions on the circle and is intrinsically odd (τ is the coordinate on $\mathcal{S}^1$); and the component $A = d\tau T^a A^a(\tau)$ takes values in $\mathfrak{g}$ -valued 1-forms on the circle and is intrinsically even. Thus, $\{\psi^a(\tau), A^a(\tau)\}$ are odd and even coordinates on $\mathcal{F}$, respectively. The space $\mathcal{F}$ is equipped with an odd symplectic structure (2):

$$(7) \quad \omega = \int_{\mathcal{S}^1} (\delta\psi, \delta A),$$

defining the anti-bracket $\{\bullet, \bullet\} : \text{Fun}(\mathcal{F}) \times \text{Fun}(\mathcal{F}) \rightarrow \text{Fun}(\mathcal{F})$,

$$\{f, g\} = \int_{\mathcal{S}^1} d\tau f \left(\frac{\overleftarrow{\delta}}{\delta\psi^a(\tau)} \frac{\overrightarrow{\delta}}{\delta A^a(\tau)} - \frac{\overleftarrow{\delta}}{\delta A^a(\tau)} \frac{\overrightarrow{\delta}}{\delta\psi^a(\tau)} \right) g$$

and the BV Laplacian $\Delta : \text{Fun}(\mathcal{F}) \rightarrow \text{Fun}(\mathcal{F})$

$$\Delta f = \int_{\mathcal{S}^1} d\tau \frac{\delta}{\delta\psi^a(\tau)} \frac{\delta}{\delta A^a(\tau)} f.$$

Note that the operator Δ is ill-defined on local functionals.

The action (5) can be written in terms of components (6) of the superfield as

$$(8) \quad S = \frac{1}{2} \int_{\mathcal{S}^1} ((\psi, d\psi) + (\psi, [A, \psi])).$$

Here d is the de Rham differential on $\mathcal{S}^1$. By the general AKSZ construction², the action S satisfies the classical master equation

$$\{S, S\} = 0.$$

Naïvely, one could also say that the unimodularity of $\mathfrak{g}$ implies unimodularity of $\mathfrak{g} \otimes \Omega^\bullet(\mathcal{S}^1)$, and therefore the quantum master equation is fulfilled

$$\frac{1}{2} \{S, S\} + \hbar \Delta S = 0.$$

However, ΔS is ill-defined in continuum theory.

Remark 1. The $\mathbb{Z}_2$ -grading on the space of fields of the Chern-Simons theory on a 3-manifold can be promoted to a $\mathbb{Z}$ -grading (by setting $\mathcal{F} = \text{Maps}(T[1]M, \mathfrak{g}[1])$) in such a way that the odd symplectic form Ω attains grade³ -1 (so that the anti-bracket has degree $+1$) and the action S is in degree zero. However, this does not apply to the one-dimensional Chern-Simons theory which is essentially $\mathbb{Z}_2$ -graded: there is no consistent $\mathbb{Z}$ -grading on the space of fields.

2.2. Effective action on the cohomology of the circle.

²Or, in the algebraic language, due to relations (Leibniz identity, Jacobi identity, cyclicity of differential, cyclicity of Lie bracket) in the cyclic dg Lie algebra $\mathfrak{g} \otimes \Omega^\bullet(\mathcal{S}^1)$, cf. [7].

³Grade is defined as the total degree minus the de Rham degree of a differential form.

2.2.1. *Harmonic gauge.* Let's split the space of differential forms on the circle into constant 0- and 1-forms and those with vanishing integral⁴: $\Omega^\bullet(\mathcal{S}^1) = \Omega'^\bullet(\mathcal{S}^1) \oplus \Omega''^\bullet(\mathcal{S}^1)$ where

$$(9) \quad \Omega'^\bullet(\mathcal{S}^1) = \{f + d\tau g \mid f, g \in \mathbb{R}\},$$

$$(10) \quad \Omega''^\bullet(\mathcal{S}^1) = \{f''(\tau) + d\tau g''(\tau) \mid \int_{\mathcal{S}^1} d\tau f''(\tau) = 0, \int_{\mathcal{S}^1} d\tau g''(\tau) = 0\}.$$

It induces the splitting for fields into infrared and ultraviolet parts $\mathcal{F} = \mathcal{F}' \oplus \mathcal{F}''$, where

$$\begin{aligned} \mathcal{F}' &= \{\psi_0 + d\tau A_0 \mid \psi_0 \in \Pi\mathfrak{g}, A_0 \in \mathfrak{g}\}, \\ \mathcal{F}'' &= \{\psi'' + A'' \mid \int_{\mathcal{S}^1} d\tau \psi''(\tau) = 0, \int_{\mathcal{S}^1} A'' = 0\}. \end{aligned}$$

This splitting respects both the BV 2-form and the de Rham differential. We define the Lagrangian subspace $\mathcal{L} \subset \mathcal{F}''$ as

$$(11) \quad \mathcal{L} = \{\psi'' + A'' \mid A'' = 0\}.$$

2.2.2. *Effective action on cohomology.*⁵ We define the effective action W on $\mathcal{F}'$ by the fiber BV integral

$$\begin{aligned} (12) \quad e^{\frac{1}{\hbar}W(\psi_0, A_0, \hbar)} &= \int_{\mathcal{L}} e^{\frac{1}{\hbar}S(\psi_0 + \psi'', d\tau A_0 + A'')} = \\ &= \int \mathcal{D}\psi'' e^{\frac{1}{2\hbar} \int_{\mathcal{S}^1} (\psi_0 + \psi'', (d + d\tau \text{ad}_{A_0})(\psi_0 + \psi''))}. \end{aligned}$$

The Lagrangian subspace $\mathcal{L} \subset \mathcal{F}''$ is uniquely⁶ fixed by the requirement that the “free” part of the action $\frac{1}{2} \int_{\mathcal{S}^1} (\psi, d\psi)$ be non-degenerate when restricted to $\mathcal{L}$.

Integral (12) is Gaussian, and it yields the following result.

Proposition 1. *The effective BV action W of the one-dimensional Chern-Simons theory is given by*

$$(13) \quad e^{\frac{1}{\hbar}W(\psi_0, A_0, \hbar)} = \det_{\mathfrak{g}}^{1/2} \left(\frac{\sinh \frac{\text{ad}_{A_0}}{2}}{\frac{\text{ad}_{A_0}}{2}} \right) \cdot e^{-\frac{1}{2\hbar}(\psi_0, \text{ad}_{A_0} \psi_0)}.$$

A simple form of the 0-loop part is due to the fact that multiplication by constant 1-forms respects the splitting of forms into infrared and ultraviolet parts (9,10).⁷ The functional determinant is easily computed *e.g.* by using the exponential basis $\{e^{2\pi i k \tau}\}$ for 0-forms on the circle.

⁴Properties of being constant for a 1-form and being of integral zero for a 0-form are non-covariant. This is not a problem as choosing a gauge always relies on introducing some additional structure. In the case of harmonic gauge, this extra structure is the parametrization of the circle.

⁵More precisely, we consider the effective action as a function on $\Pi\mathfrak{g} \otimes H^\bullet(\mathcal{S}^1)$, *i.e.* on the parity-shifted de Rham cohomology of the circle (which we represented by harmonic forms) with coefficients in $\mathfrak{g}$.

⁶This is a special property of the one-dimensional theory related to the fact that the de Rham operator $d : \Omega''^0(\mathcal{S}^1) \rightarrow \Omega''^1(\mathcal{S}^1)$ is an isomorphism, and there is unique chain homotopy $K = d^{-1} : \Omega''^1(\mathcal{S}^1) \rightarrow \Omega''^0(\mathcal{S}^1)$.

⁷Higher order terms in the 0-loop effective action would correspond to Massey operations on the de Rham cohomology (cf. [7], [14]). In the case of the circle, Massey operations vanish.

The effective action (13) satisfies the quantum master equation

$$\frac{\partial}{\partial \psi_0^a} \frac{\partial}{\partial A_0^a} e^{\frac{1}{\hbar} W(\psi_0, A_0, \hbar)} = 0.$$

The classical master equation is implied by the Jacobi identity and by cyclicity property of the Lie bracket on $\mathfrak{g}$; the quantum part of master equation follows from the fact that the one-loop part of W is manifestly ad-invariant.

2.3. Simplicial Chern-Simons action on circle. Let's assume that the circle S^1 is glued of n intervals $\mathcal{I}_1 = [\mathbf{p}_1, \mathbf{p}_2], \dots, \mathcal{I}_n = [\mathbf{p}_n, \mathbf{p}_1]$ (where $\mathbf{p}_1, \dots, \mathbf{p}_n$ is a cyclically ordered collection of points on the circle), and that each interval $\mathcal{I}_k$ is equipped with a coordinate function $\tau : \mathcal{I}_k \rightarrow [0, 1]$. We will denote a point of $\mathcal{I}_k$ with coordinate τ by (k, τ) . We will assume n to be odd (otherwise, our gauge will be inconsistent, see remark 2). We denote this “triangulation” of the circle by Ξ_n .

2.3.1. Cyclic Whitney gauge. Splitting of fields into infrared and ultraviolet parts is defined by splitting for differential forms on the circle $\Omega^\bullet(S^1) = \Omega_{\Xi_n}^\bullet(S^1) \oplus \Omega_{\Xi_n}^{\prime\prime\bullet}(S^1)$, where we split 0-forms into continuous piecewise-linear ones and those with vanishing integrals over each $\mathcal{I}_k$, and we split 1-forms into piecewise-constant ones and the orthogonal complement of piecewise-linear 0-forms:

$$(14) \quad \Omega_{\Xi_n}^\bullet(S^1) = \{f' + d\tau g' \mid f'|_{\mathcal{I}_k} = (1 - \tau)f_k + \tau f_{k+1}, g'|_{\mathcal{I}_k} = g_k \forall k\}$$

$$\Omega_{\Xi_n}^{\prime\prime\bullet}(S^1) = \{f'' + d\tau g'' \mid \int_{\mathcal{I}_k} d\tau f'' = 0, \int_{\mathcal{I}_k} \tau d\tau g'' + \int_{\mathcal{I}_{k+1}} (1 - \tau) d\tau g'' = 0 \forall k\}$$

where $f_k, g_k \in \mathbb{R}$ are numbers. Thus, $\Omega_{\Xi_n}^\bullet(S^1) \cong \mathbb{R}^{n|n}$ (as a super-space). As a cochain complex, $\Omega_{\Xi_n}^\bullet(S^1)$ is isomorphic to $C^\bullet(\Xi_n)$, the cochain complex of the simplicial complex Ξ_n . As in section 2.2.1, this splitting agrees with the de Rham differential, and the associated splitting for fields $\mathcal{F} = \mathcal{F}' \oplus \mathcal{F}''$ agrees with the BV 2-form.

We call splitting (14) the “cyclic Whitney gauge”, because our representatives $\Omega^\bullet$ for cell cochains of triangulation are exactly the Whitney forms [16] for Ξ_n . The word “cyclic” indicates that $\Omega^{\prime\prime\bullet}$ is constructed as an orthogonal complement of $\Omega^\bullet$ with respect to the Poincaré pairing $\int_{S^1} \bullet \wedge \bullet$.

Coordinates on $\mathcal{F}'$ are given by values of ψ' at the vertices of triangulation:

$$(15) \quad \psi_k = \psi'(\mathbf{p}_k) \in \Pi \mathfrak{g},$$

and by integrals of A over intervals:

$$(16) \quad A_k = \int_{\mathcal{I}_k} A' \in \mathfrak{g}.$$

The BV 2-form on $\mathcal{F}'$ is given by

$$(17) \quad \omega' = \sum_{k=1}^n \frac{\delta \psi_k^a + \delta \psi_{k+1}^a}{2} \wedge \delta A_k^a.$$

We will denote $\mathcal{F}' = \mathcal{F}_{\Xi_n}$ to emphasize its dependence on n . We have $\mathcal{F}_{\Xi_n} \cong \Pi \mathfrak{g} \otimes C^\bullet(\Xi_n)$.

Remark 2. The requirement that n be odd is needed since for n even the piecewise-linear 0-form

$$f(k, \tau) = (-1)^k (\tau - 1/2)$$

belongs to both the infrared and ultraviolet subspaces.

As in section 2.2.1, we define the Lagrangian subspace $\mathcal{L} \subset \mathcal{F}''$ by setting the 1-form part of the ultraviolet field to zero (11).

2.3.2. Chain homotopy, dressed chain homotopy. Let us define the infrared projector $\mathcal{P}' : \Omega^\bullet(\mathcal{S}^1) \rightarrow \Omega_{\Xi_n}^\bullet(\mathcal{S}^1)$ by formula

$$(18) \quad f + d\tau \cdot g \mapsto \sum_{k=1}^n \left(\int_{\mathcal{I}_k} d\tau' f - (1-2\tau) \cdot \left(\int_{\mathcal{I}_{k+1}} d\tau' f - \int_{\mathcal{I}_{k+2}} d\tau' f + \cdots + \int_{\mathcal{I}_{k-2}} d\tau' f - \int_{\mathcal{I}_{k-1}} d\tau' f \right) \right) \theta_{\mathcal{I}_k} + d\tau \sum_{k=1}^n \left(\int_{\mathcal{I}_k} d\tau' g + \int_{\mathcal{I}_{k+1}} (1-2\tau') d\tau' g - \int_{\mathcal{I}_{k+2}} (1-2\tau') d\tau' g + \cdots - \int_{\mathcal{I}_{k-1}} (1-2\tau') d\tau' g \right) \theta_{\mathcal{I}_k},$$

where $\theta_{\mathcal{I}_k}$ is the function on the circle with value 1 on $\mathcal{I}_k$ and zero elsewhere.

The chain homotopy $\kappa : \Omega^1(\mathcal{S}^1) \rightarrow \Omega^0(\mathcal{S}^1)$ is uniquely defined by the properties

$$(19) \quad d\kappa + \kappa d = \text{id} - \mathcal{P}',$$

$$(20) \quad \mathcal{P}'\kappa = 0,$$

$$(21) \quad \kappa\mathcal{P}' = 0.$$

Lemma 1. *The operator κ defined by relations (19), (20), (21) acts on the 1-form $d\tau \cdot g \in \Omega^1(\mathcal{S}^1)$ by⁸*

$$(22) \quad \kappa(d\tau \cdot g)(k, \tau) = \sum_{k'=1}^n \int_{\mathcal{I}_{k'}} d\tau' \kappa((k, \tau), (k', \tau')) g(k', \tau'),$$

where the integral kernel is given by

$$(23) \quad \kappa((k, \tau), (k', \tau')) = \begin{cases} \theta(\tau - \tau') - \frac{1}{2} - \tau + \tau' & \text{if } k = k' \\ (-1)^{k-k'} 2(\frac{1}{2} - \tau)(\frac{1}{2} - \tau') & \text{if } k' < k < k' + n \end{cases}$$

and θ is the unit step function. In addition, the kernel has the anti-symmetry property:

$$\kappa((k', \tau'), (k, \tau)) = -\kappa((k, \tau), (k', \tau'))$$

To obtain formula (23), one observes that relations (19) and (20) imply the differential equation

$$(24) \quad \frac{\partial}{\partial \tau} \kappa((k, \tau), (k', \tau')) = \delta_{k,k'} \delta(\tau - \tau') + C_k(k', \tau')$$

subject to conditions

$$(25) \quad \int_0^1 d\tau \kappa((k, \tau), (k', \tau')) = 0, \quad \kappa((k, 1), (k', \tau')) = \kappa((k+1, 0), (k', \tau')) \quad \forall k.$$

Here $C_k(k', \tau')$ are some functions independent of τ . Solving (24) together with (25) immediately yields (23). This proves the uniqueness property. In order to prove existence, one checks that (23) satisfies (19), (20), (21).

⁸Recall that (k, τ) denotes a point on the circle which belongs to the interval $\mathcal{I}_k$ and which has a local coordinate τ . Hence, the integral kernel here is actually a function on $\mathcal{S}^1 \times \mathcal{S}^1$.

We will also need the “dressed” chain homotopy (*i.e.* dressed by the connection) $\kappa_{A'} : \mathfrak{g} \otimes \Omega^1(\mathcal{S}^1) \rightarrow \mathfrak{g} \otimes \Omega^0(\mathcal{S}^1)$, where $A' = \sum_{k=1}^n A_k \theta_{T_k}$ is a piecewise-constant $\mathfrak{g}$ -valued 1-form on the circle. The operator $\kappa_{A'}$ is uniquely defined by the properties

$$(26) \quad (\text{id} - \mathcal{P}') d_{A'} \kappa_{A'} + \kappa_{A'} d_{A'} (\text{id} - \mathcal{P}') = \text{id} - \mathcal{P}',$$

$$(27) \quad \mathcal{P}' \kappa_{A'} = 0,$$

$$(28) \quad \kappa_{A'} \mathcal{P}' = 0,$$

where $d_{A'} = d + \text{ad}_{A'}$. One can summarize these properties by saying that $\kappa_{A'} : \Omega_{\Xi_n}^{\prime 1}(\mathcal{S}^1) \rightarrow \Omega_{\Xi_n}^{\prime 0}(\mathcal{S}^1)$ is the inverse of the operator $(\text{id} - \mathcal{P}') d_{A'} (\text{id} - \mathcal{P}') : \Omega_{\Xi_n}^{\prime 0}(\mathcal{S}^1) \rightarrow \Omega_{\Xi_n}^{\prime 1}(\mathcal{S}^1)$. The dressed chain homotopy can either be expressed perturbatively as a series

$$\kappa_{A'} = \kappa - \kappa \text{ad}_{A'} \kappa + \kappa \text{ad}_{A'} \kappa \text{ad}_{A'} \kappa - \dots,$$

or it can be computed explicitly by solving the differential equation (26).

To present an explicit formula for the integral kernel of $\kappa_{A'}$ (defined as in (22)), we have to first introduce some notation. Let $F(\mathcal{A}, \tau)$ be an $\text{End}(\mathfrak{g})$ -valued function on the interval $[0, 1]$ depending on an anti-symmetric matrix $\mathcal{A} \in \mathfrak{so}(\mathfrak{g}) \subset \text{End}(\mathfrak{g})$ and defined by the following properties

$$(29) \quad F(\mathcal{A}, \bullet) \in \text{Span}_{\text{End}(\mathfrak{g})}(1, e^{-\mathcal{A}\tau}),$$

$$\int_0^1 d\tau F(\mathcal{A}, \tau) = 0,$$

$$F(\mathcal{A}, 0) = 1.$$

Property (29) is equivalently stated as $(d + \mathcal{A})F(\mathcal{A}, \bullet) = \text{const}$ (this constant depends on $\mathcal{A}$). It is convenient to introduce notation

$$R(\mathcal{A}) = -F(\mathcal{A}, 1) \in \text{End}(\mathfrak{g}).$$

More explicitly, we have

$$(30) \quad F(\mathcal{A}, \tau) = \frac{\frac{1}{2} (\coth \frac{\mathcal{A}}{2} + 1) e^{-\mathcal{A}\tau} - \mathcal{A}^{-1}}{\frac{1}{2} (\coth \frac{\mathcal{A}}{2} + 1) - \mathcal{A}^{-1}},$$

$$(31) \quad R(\mathcal{A}) = -\frac{\mathcal{A}^{-1} + \frac{1}{2} - \frac{1}{2} \coth \frac{\mathcal{A}}{2}}{\mathcal{A}^{-1} - \frac{1}{2} - \frac{1}{2} \coth \frac{\mathcal{A}}{2}}.$$

We will use notation

$$(32) \quad \mu_k(A') = R(\text{ad}_{A_{k-1}}) R(\text{ad}_{A_{k-2}}) \cdots R(\text{ad}_{A_{k+1}}) R(\text{ad}_{A_k}) \in \text{End}(\mathfrak{g}).$$

The following reflection properties

$$F(-\mathcal{A}, \tau) = -\frac{F(\mathcal{A}, 1 - \tau)}{R(\mathcal{A})}, \quad R(-\mathcal{A}) = R(\mathcal{A})^{-1}, \quad \mu_k(A')^T = \mu_k(A')^{-1}$$

mean that $R(\mathcal{A})$ and $\mu_k(A')$ take values in orthogonal matrices $O(\mathfrak{g}) \subset \text{End}(\mathfrak{g})$.

We can now present the result for the integral kernel of the dressed chain homotopy $\kappa_{A'}$:

Lemma 2. *The operator $\kappa_{A'}$ defined by relations (26), (27), (28) acts on a $\mathfrak{g}$ -valued 1-form $d\tau \cdot \beta \in \mathfrak{g} \otimes \Omega^1(\mathcal{S}^1)$ by formula*

$$(33) \quad \kappa_{A'}(d\tau \cdot \beta)(k, \tau) = \sum_{k'=1}^n \int_{T_{k'}} d\tau' \cdot \kappa_{A'}((k, \tau), (k', \tau')) \circ \beta(k', \tau'),$$

where the integral kernel is given by

$$(34) \quad \kappa_{A'}((k, \tau), (k', \tau')) = \begin{cases} \left(\theta(\tau - \tau') - \frac{1}{2} + \frac{1}{2} \coth \frac{\text{ad}_{A_k}}{2} \right) e^{(\tau' - \tau) \text{ad}_{A_k}} - (\text{ad}_{A_k})^{-1} + \\ \quad + F(\text{ad}_{A_k}, \tau) \left(\frac{1}{1 + \mu_k(A')} - \frac{1}{1 + R(\text{ad}_{A_k})} \right) F(-\text{ad}_{A_k}, \tau') & \text{if } k = k', \\ (-1)^{k-k'} F(\text{ad}_{A_k}, \tau) R(\text{ad}_{A_{k-1}}) \cdots R(\text{ad}_{A_{k'}}) \frac{1}{1 + \mu_{k'}(A')} F(-\text{ad}_{A_{k'}}, \tau') & \text{if } k' < k < k' + n \end{cases}$$

This integral kernel is anti-symmetric:

$$\kappa_{A'}((k', \tau'), (k, \tau))^T = -\kappa_{A'}((k, \tau), (k', \tau'))$$

To obtain (34) we proceed as in the derivation of (23). The differential equation implied by (26) is as follows:

$$(35) \quad \left(\frac{\partial}{\partial \tau} + \text{ad}_{A_k} \right) \kappa_{A'}((k, \tau), (k', \tau')) = \delta_{k,k'} \delta(\tau - \tau') + C_k(k', \tau').$$

Conditions (25) are still fulfilled. Solving (35) with these conditions imposed yields formula (34).

2.3.3. Simplicial action. Simplicial Chern-Simons action S_{Ξ_n} for the triangulation Ξ_n of circle is defined by the fiber BV integral

$$(36) \quad e^{\frac{1}{\hbar} S_{\Xi_n}(\psi', A', \hbar)} = \int_{\mathcal{L}} e^{\frac{1}{\hbar} S(\psi' + \psi'', A' + A'')}.$$

As in the case of induction to cohomology, one puts $A''|_{\mathcal{L}} = 0$, and the integral becomes Gaussian:

$$(37) \quad e^{\frac{1}{\hbar} S_{\Xi_n}(\psi', A', \hbar)} = \int \mathcal{D}\psi'' e^{\frac{1}{2\hbar} \int_{S^1} (\psi' + \psi'', d_{A'}(\psi' + \psi''))}.$$

Expanding the integrand as

$$(\psi', d_{A'} \psi') + (\psi'', d\psi'') + (\psi'', \text{ad}_{A'} \psi'') + 2(\psi'', \text{ad}_{A'} \psi')$$

one can consider the second term as the “free” part of the action and the third and fourth terms as a perturbation. In this way, we arrive at the following Feynman diagram expansion for S_{Ξ_n} :

$$(38) \quad \begin{aligned} S_{\Xi_n} = & \underbrace{\frac{1}{2}(\psi', d\psi') + \frac{1}{2}(\psi', \text{ad}_{A'} \psi') - \frac{1}{2}(\psi', \text{ad}_{A'} \kappa \text{ad}_{A'} \psi') + \frac{1}{2}(\psi', \text{ad}_{A'} \kappa \text{ad}_{A'} \kappa \text{ad}_{A'} \psi') - \cdots}_{S_{\Xi_n}^0} + \\ & \underbrace{\hbar \frac{1}{2 \cdot 2} \text{tr}_{\mathfrak{g} \otimes \Omega^0(S^1)} (\kappa \text{ad}_{A'} \kappa \text{ad}_{A'}) - \hbar \frac{1}{2 \cdot 3} \text{tr}_{\mathfrak{g} \otimes \Omega^0(S^1)} (\kappa \text{ad}_{A'} \kappa \text{ad}_{A'} \kappa \text{ad}_{A'}) + \cdots}_{\hbar S_{\Xi_n}^1} \end{aligned}$$

Here the first line is the sum of “tree” diagrams and the second line is the sum of “wheel diagrams”.

The tree part of S_{Ξ_n} can be expressed in terms of the dressed chain homotopy,

$$(39) \quad S_{\Xi_n}^0 = \frac{1}{2} \int_{S^1} (\psi', d\psi') + \frac{1}{2} \int_{S^1} (\psi', \text{ad}_{A'} \psi') - \frac{1}{2} \int_{S^1} (\psi', \text{ad}_{A'} \kappa_{A'} \text{ad}_{A'} \psi').$$

For the 1-loop part, we first write

$$S_{\Xi_n}^1 = \frac{1}{2} \operatorname{tr}_{\mathfrak{g} \otimes \Omega^0(\mathcal{S}^1)} \log(1 + \kappa \operatorname{ad}_{A'}).$$

Then, by using the general formula

$$\frac{\partial}{\partial s} \operatorname{tr} \log M_s = \operatorname{tr} \left(M_s^{-1} \frac{\partial M_s}{\partial s} \right)$$

one obtains

$$S_{\Xi_n}^1 = \int_0^1 ds \frac{\partial}{\partial s} \left(\frac{1}{2} \operatorname{tr}_{\mathfrak{g} \otimes \Omega^0(\mathcal{S}^1)} \log(1 + \kappa \operatorname{ad}_{sA'}) \right) = \frac{1}{2} \int_0^1 ds \operatorname{tr}_{\mathfrak{g} \otimes \Omega^0(\mathcal{S}^1)} \underbrace{(1 + \kappa \operatorname{ad}_{sA'})^{-1} \kappa \operatorname{ad}_{A'}}_{\kappa_{sA'}}.$$

One can evaluate the functional trace in the integrand in the coordinate representation (*i.e.* in the basis of delta-functions) by expressing it in terms of the integral kernel (34) restricted to the diagonal $\mathcal{S}^1 \subset \mathcal{S}^1 \times \mathcal{S}^1$:

$$(40) \quad S_{\Xi_n}^1 = \int_0^1 ds \frac{1}{2} \sum_{k=1}^n \int_0^1 d\tau \operatorname{tr}_{\mathfrak{g}} (\kappa_{sA'}((k, \tau), (k, \tau)) \operatorname{ad}_{A_k}).$$

There is an ambiguity in this expression since the integral kernel (34) is discontinuous on the diagonal. We regularize this ambiguity by using the convention

$$(41) \quad \theta(0) = \frac{1}{2}$$

Observe that the value assigned to $\theta(0)$ does not really matter: changing the convention for $\theta(0)$ changes $S_{\Xi_n}^1$ by $\propto \operatorname{tr}_{\mathfrak{g}} \operatorname{ad}_{A'} = 0$. It is interesting that the integral over the auxiliary parameter s in (40) can be computed explicitly. By putting together (39) and (40) and substituting (34) we obtain the following explicit result for the simplicial Chern-Simons action on the triangulated circle.

Theorem 1. *Simplicial Chern-Simons action on the triangulated circle is given by*

$$(42) \quad \begin{aligned} S_{\Xi_n} = & -\frac{1}{2} \sum_{k=1}^n \left((\psi_k, \psi_{k+1}) + \frac{1}{3} (\psi_k, \operatorname{ad}_{A_k} \psi_k) + \frac{1}{3} (\psi_{k+1}, \operatorname{ad}_{A_k} \psi_{k+1}) + \frac{1}{3} (\psi_k, \operatorname{ad}_{A_k} \psi_{k+1}) \right) + \\ & + \frac{1}{2} \sum_{k=1}^n (\psi_{k+1} - \psi_k, \left(\frac{1 - R(\operatorname{ad}_{A_k})}{2} \left(\frac{1}{1 + \mu_k(A')} - \frac{1}{1 + R(\operatorname{ad}_{A_k})} \right) \frac{1 - R(\operatorname{ad}_{A_k})}{2R(\operatorname{ad}_{A_k})} + \right. \\ & \quad \left. + (\operatorname{ad}_{A_k})^{-1} + \frac{1}{12} \operatorname{ad}_{A_k} - \frac{1}{2} \coth \frac{\operatorname{ad}_{A_k}}{2} \right) \circ (\psi_{k+1} - \psi_k)) + \\ & + \frac{1}{2} \sum_{k'=1}^n \sum_{k=k'+1}^{k'+n-1} (-1)^{k-k'} (\psi_{k+1} - \psi_k, \frac{1 - R(\operatorname{ad}_{A_k})}{2} R(\operatorname{ad}_{A_{k-1}}) \cdots R(\operatorname{ad}_{A_{k'}}) \cdot \\ & \quad \cdot \frac{1}{1 + \mu_{k'}(A')} \cdot \frac{1 - R(\operatorname{ad}_{A_{k'}})}{2R(\operatorname{ad}_{A_{k'}})} \circ (\psi_{k'+1} - \psi_{k'})) + \\ & + \hbar \frac{1}{2} \operatorname{tr}_{\mathfrak{g}} \log \left((1 + \mu_{\bullet}(A')) \prod_{k=1}^n \left(\frac{1}{1 + R(\operatorname{ad}_{A_k})} \cdot \frac{\sinh \frac{\operatorname{ad}_{A_k}}{2}}{\frac{\operatorname{ad}_{A_k}}{2}} \right) \right). \end{aligned}$$

In the last term (the 1-loop contribution) $\mu_\bullet(A')$ stands for $\mu_l(A')$ for arbitrary l . For different l , these matrices differ by conjugation. Hence, the expression $\det_{\mathfrak{g}}(1 + \mu_\bullet(A'))$ is well defined.

2.3.4. Remarks.

Remark 3. In deriving formula (42) we were sloppy about additive constants (we did not pay attention to normalization of the measure in the functional integral (37)). We chose an ad hoc normalization

$$S_{\Xi_n}(0, 0, \hbar) = -\hbar \frac{n-1}{2} \dim \mathfrak{g} \log 2$$

which turns out to be consistent with the operator formalism (see section 3.1).

Remark 4. Setting $n = 1$ in (42) yields the effective action on cohomology (13).

Remark 5. By expanding (42) in a power series with respect to A' we get back the perturbative expansion (38)),

$$\begin{aligned} (43) \quad S_{\Xi_n} = & -\frac{1}{2} \sum_{k=1}^n (\psi_k, \psi_{k+1}) - \frac{1}{2} \sum_{k=1}^n \left(\frac{1}{3} (\psi_k, \text{ad}_{A_k} \psi_k) + \frac{1}{3} (\psi_{k+1}, \text{ad}_{A_k} \psi_{k+1}) + \frac{1}{3} (\psi_k, \text{ad}_{A_k} \psi_{k+1}) \right) + \\ & + \frac{1}{2} \sum_{k'=1}^n \sum_{k=k'+1}^{k'+n-1} \frac{1}{72} (-1)^{k-k'} (\psi_{k+1} - \psi_k, \text{ad}_{A_k} \text{ad}_{A_{k'}} (\psi_{k'+1} - \psi_{k'})) - \\ & - \hbar \frac{n-1}{2} \dim \mathfrak{g} \log 2 + \hbar \frac{1}{2 \cdot 2} \left(\sum_{k=1}^n \frac{1}{12} \text{tr}_{\mathfrak{g}} (\text{ad}_{A_k})^2 + \sum_{k'=1}^n \sum_{k=k'+1}^{k'+n-1} \frac{1}{36} \text{tr}_{\mathfrak{g}} (\text{ad}_{A_k} \text{ad}_{A_{k'}}) \right) + \mathcal{O}((A')^3). \end{aligned}$$

Remark 6. Naïvely, at large n the simplicial action S_{Ξ_n} can be viewed as a lattice approximation to the continuum action (8). For ψ and A fixed, we have

$$S_{\Xi_n} \left(\{ \psi_k = \psi(k/n) \}, \{ A_k = \int_{k/n}^{(k+1)/n} A \} \right) \longrightarrow S(\psi, A) + \mathcal{O}(1/n)$$

when n tends to infinity. The point is that rather than being just an approximation the simplicial theory $(\mathcal{F}_{\Xi_n}, S_{\Xi_n})$ is exactly equivalent to the continuum theory $(\mathcal{F}, S)$ for any finite n .

Remark 7. The simplicial theory is constructed by the fiber BV integral from the continuum theory. Hence, we expect the simplicial action to satisfy the quantum master equation

$$(44) \quad \Delta_{\Xi_n} e^{\frac{1}{\hbar} S_{\Xi_n}} = 0$$

with BV Laplacian associated to the BV 2-form (17),

$$(45) \quad \Delta_{\Xi_n} = \sum_{k=1}^n \frac{\partial}{\partial \tilde{\psi}_k^a} \frac{\partial}{\partial A_k^a},$$

where

$$(46) \quad \tilde{\psi}_k = \frac{\psi_k + \psi_{k+1}}{2}.$$

However, the BV Laplacian in the continuum theory is ill-defined. Therefore, the quantum master equation (44) is not automatic, and it must be checked independently. It is easy to do it in low degrees in A' by using the expansion (43). We will prove the quantum master equation via the operator approach in section 3.2.3.

Remark 8. Another property we expect from the simplicial theory is its compatibility with simplicial aggregations $\Xi_{n+2} \rightarrow \Xi_n$. We will prove this property in section 3.2.2.

Remark 9. The gauge choice (14) is actually rigid up to diffeomorphisms of intervals $\mathcal{I}_k$. More exactly, if we require compatibility of the splitting $\Omega = \Omega' \oplus \Omega''$ with the de Rham differential and with the pairing $\int_{S^1} \bullet \wedge \bullet$ (i.e. so that Ω' and Ω'' be subcomplexes of Ω , and Ω'' be orthogonal to Ω'), then the splitting is completely determined by the images of basis 1-cochains of Ξ_n in $\Omega^1(\mathcal{S}^1)$. If in addition, we require that the image of each basis 1-cochain $e_k^{(1)}$ be supported exactly on the respective interval $\mathcal{I}_k \subset \mathcal{S}^1$ (a kind of simplicial locality property), we obtain the splitting (14) up to diffeomorphisms of intervals $\mathcal{I}_k$ taking the representatives of basis 1-cochains $e_k^{(1)}$ to constant forms $\theta_{\mathcal{I}_k} d\tau$. It is important to note that we are implicitly assuming that the splitting for fields is induced by the splitting for real-valued differential forms. If we drop this assumption and allow splittings of $\mathfrak{g} \otimes \Omega$ which are not obtained from a splitting of Ω by tensoring with $\mathfrak{g}$, we can introduce other gauges.

Remark 10. The embedding of cell cochains of Ξ_n to the space of differential forms in (14) is the same as in the 1-dimensional simplicial BF theory [14]: 0-cochains are represented by continuous piecewise-linear functions, and 1-cochains are represented by piecewise-constant 1-forms. However, the projection (18) is very different. In fact, it is non-local: for 0-forms instead of evaluation at vertices (as in simplicial BF) we have a sum of integrals over all intervals $\mathcal{I}_k$ with certain signs. And for 1-forms, instead of integration over one interval we have an integral over the whole circle with a certain piecewise-linear integral kernel. Thus, in the splitting $\Omega = \Omega' \oplus \Omega''$ the infrared part is like in the simplicial BF theory, but the ultraviolet part is different.

Remark 11. The action (42) can be viewed as a generating function of a certain infinity-structure on $\mathfrak{g} \otimes C^\bullet(\Xi_n)$. In more detail, this is a structure of a loop-enhanced (or “quantum”, or “unimodular”) cyclic L_∞ algebra with the structure maps (operations) $c_k^{(l)} : \wedge^k(\mathfrak{g} \otimes C^\bullet(\Xi_n)) \rightarrow \mathbb{R}$ related to the action (42) by

$$S_{\Xi_n} = \sum_{l=0}^1 \sum_{k=2}^{\infty} \frac{\hbar^l}{k!} c_k^{(l)}(\underbrace{\psi' + A', \dots, \psi' + A'}_k)$$

where the superfield $\psi' + A'$ is understood as the parity-shifted identity map $\mathcal{F}_{\Xi_n} \rightarrow \mathfrak{g} \otimes C^\bullet(\Xi_n)$ (as in (3)). The quantum master equation (44) generates a family of structure equations on operations $c_k^{(l)}$. In particular, $c_k^{(0)}$ satisfy the structure equations of the usual (nonunimodular) cyclic L_∞ algebra. This algebraic structure on $\mathfrak{g}$ -valued cochains of Ξ_n can be viewed as a homotopy transfer of the unimodular cyclic DGLA structure on $\mathfrak{g} \otimes \Omega^\bullet(\mathcal{S}^1)$. Only the first two cyclic operations, $c_2^{(0)} : \wedge^2(\mathfrak{g} \otimes C^\bullet(\Xi_n)) \rightarrow \mathbb{R}$ and $c_3^{(0)} : \wedge^3(\mathfrak{g} \otimes C^\bullet(\Xi_n)) \rightarrow \mathbb{R}$ are simplicially-local. All the other operations are non-local. We can also use the pairing on $\mathfrak{g} \otimes C^\bullet(\Xi_n)$ (induced by the pairing $\int_{S^1}(\bullet, \bullet)$ on $\mathfrak{g}$ -valued differential forms)

to invert one input in cyclic operations. This gives an oriented (non-cyclic) version of the unimodular L_∞ structure (see e.g. [10]) on $\mathfrak{g} \otimes C^\bullet(\Xi_n)$ with structure maps $l_k^{(0)} : \wedge^k(\mathfrak{g} \otimes C^\bullet(\Xi_n)) \rightarrow \mathfrak{g} \otimes C^\bullet(\Xi_n)$ and $l_k^{(1)} : \wedge^k(\mathfrak{g} \otimes C^\bullet(\Xi_n)) \rightarrow \mathbb{R}$ related to the cyclic operations $c_k^{(l)}$ by the following formula

$$\begin{aligned} c_{k+1}^{(0)}(\underbrace{\psi' + A', \dots, \psi' + A'}_{k+1}) &= \left(\psi' + A', l_k^{(0)}(\underbrace{\psi' + A', \dots, \psi' + A'}_k) \right), \quad k \geq 1 \\ c_k^{(1)} &= l_k^{(1)}, \quad k \geq 2 \end{aligned}$$

In this unimodular L_∞ algebra, only the differential $l_1^{(0)}$ is local while all other operations (including the binary bracket $l_2^{(0)} : \wedge^2(\mathfrak{g} \otimes C^\bullet(\Xi_n)) \rightarrow \mathfrak{g} \otimes C^\bullet(\Xi_n)$) are non-local — unlike in the simplicial BF theory [14] where all higher operations are simplicially-local.

Remark 12. The dressed chain homotopy $\kappa_{A'}$ can be used to construct the non-linear map $U : \mathcal{F}_{\Xi_n} \rightarrow \mathcal{F}$,

$$U : \psi' + A' \mapsto \psi' + A' - \kappa_{A'} \text{ad}_{A'} \psi'.$$

It sends the infrared field $\psi' + A'$ to the conditional extremum of the continuum action S restricted to $\{\psi' + A'\} \oplus \mathcal{L}$. The tree part of the simplicial action can be expressed in terms of U ,

$$\begin{aligned} S_{\Xi_n}^0 &= S(U(\psi' + A')) = \\ &= \int_{\mathcal{S}^1} \frac{1}{2} (U(\psi' + A'), dU(\psi' + A')) + \frac{1}{6} (U(\psi' + A'), [U(\psi' + A'), U(\psi' + A')]) \end{aligned}$$

In the language of infinity algebras, U is an L_∞ morphism intertwining the DGLA structure on $\mathfrak{g} \otimes \Omega^\bullet(\mathcal{S}^1)$ and the L_∞ structure on $\mathfrak{g} \otimes C^\bullet(\Xi_n)$.

Remark 13. There are two natural systems of $\mathfrak{g}$ -valued coordinates on the space of simplicial BV fields $\mathcal{F}_{\Xi_n} : \{\psi_k, A_k\}$ and $\{\tilde{\psi}_k, A_k\}$ (where variables $\tilde{\psi}_k$ are defined by (46)). The first coordinate system is associated to the realization of the space of fields through the cochain complex of a triangulation: $\mathcal{F}_{\Xi_n} = \Pi \mathfrak{g} \otimes C^\bullet(\Xi_n)$. The second coordinate system is associated to the realization through 1-chains and 1-cochains: $\mathcal{F}_{\Xi_n} \cong \mathfrak{g} \otimes C_1(\Xi_n) \oplus \Pi \mathfrak{g} \otimes C^1(\Xi_n)$ (or instead of 1-chains of Ξ_n one can talk of 0-cochains of the dual cell decomposition $\Xi_n^\vee$: $\mathcal{F}_{\Xi_n} \cong \Pi \mathfrak{g} \otimes C^0(\Xi_n^\vee) \oplus \Pi \mathfrak{g} \otimes C^1(\Xi_n)$). The convenience of the first coordinate system is that the abelian part of the simplicial action (42) is local in variables (ψ_k, A_k) . The convenience of the second coordinate system $\{\tilde{\psi}_k, A_k\}$ is that the BV Laplacian becomes diagonal (45).

3. APPROACH THROUGH OPERATOR FORMALISM

In this section, our strategy is to give a new definition of the one-dimensional Chern-Simons theory. It will be inspired by the definition of section 2.1, but we will be able to consider our theory on an interval and to define a concatenation (gluing) procedure. We will check that the results obtained by the new approach are consistent with those of section 2.

For the rest of the paper, we will assume that $\dim \mathfrak{g} = 2m$ is even. This is important for theorem 2 (the correspondence between the operator and path integral

formalism): its proof relies on recovering the path integral by using the fundamental representation of Clifford algebra $Cl(\mathfrak{g})$ (section 4.1) which is simpler for $\dim \mathfrak{g} = 2m$.

3.1. One-dimensional Chern-Simons theory in operator formalism.

3.1.1. *First approximation.* We would like to define the one-dimensional Chern-Simons theory (8) as a quantum mechanics where components of the quantized odd field $\{\hat{\psi}^a\}$ are subject to the anti-commutation relations

$$(47) \quad \hat{\psi}^a \hat{\psi}^b + \hat{\psi}^b \hat{\psi}^a = \hbar \delta^{ab},$$

i.e. $\{\hat{\psi}^a\}$ are generators of the Clifford algebra $Cl(\mathfrak{g})$. The even 1-form (connection) field $A = d\tau T^a A^a(\tau)$ is non-dynamical, and it is treated as a classical background. The evolution operator for the theory on the interval is defined as a path-ordered exponential of A in the spin representation:

$$(48) \quad U_{\mathcal{I}}(A) = \overleftarrow{P} \exp \left(-\frac{1}{2\hbar} \int_{\mathcal{I}} d\tau f^{abc} \hat{\psi}^a A^b(\tau) \hat{\psi}^c \right) \in Cl(\mathfrak{g})$$

Here, the intuition is as follows: the term $\frac{1}{2} \int (\psi, d\psi)$ in the action (8) generates the canonical anti-commutation relations (47) while the term $\frac{1}{2} \int (\psi, [A, \psi])$ generates the time-dependent quantum Hamiltonian $\hat{H}(\tau) = -\frac{1}{2} f^{abc} \hat{\psi}^a A^b(\tau) \hat{\psi}^c$ which appears in (48). The evolution operator for the concatenation of intervals $\mathcal{I}_1 = [\mathfrak{p}_1, \mathfrak{p}_2]$, $\mathcal{I}_2 = [\mathfrak{p}_2, \mathfrak{p}_3]$ with connections A_1, A_2 is naturally given by the product of the corresponding evolution operators in the Clifford algebra:

$$U_{[\mathfrak{p}_1, \mathfrak{p}_3]}(A_1 \theta_{[\mathfrak{p}_1, \mathfrak{p}_2]} + A_2 \theta_{[\mathfrak{p}_2, \mathfrak{p}_3]}) = U_{[\mathfrak{p}_2, \mathfrak{p}_3]}(A_2) \cdot U_{[\mathfrak{p}_1, \mathfrak{p}_2]}(A_1).$$

As in section 2.3.2, $\theta_{[\mathfrak{p}_k, \mathfrak{p}_{k+1}]} = \theta_{\mathcal{I}_k}$ denotes the function taking value 1 on the interval $\mathcal{I}_k$ and zero everywhere else. The partition function for a circle is given by

$$Z_{S^1}(A) = \text{Str}_{Cl(\mathfrak{g})} \overleftarrow{P} \exp \left(-\frac{1}{2\hbar} \int_{\mathcal{I}} d\tau f^{abc} \hat{\psi}^a A^b(\tau) \hat{\psi}^c \right) \in \mathbb{R},$$

where $\text{Str}_{Cl(\mathfrak{g})}$ is the super-trace on $Cl(\mathfrak{g})$ defined as

$$(49) \quad \text{Str}_{Cl(\mathfrak{g})} : \hat{a} \mapsto (i\hbar)^m \cdot \left(\text{Coefficient of } \hat{\psi}^1 \dots \hat{\psi}^{2m} \text{ in } \hat{a} \right).$$

3.1.2. *Imposing the cyclic Whitney gauge.* Our next task is to model in operator formalism the fiber BV integral (36) for the theory on circle. We will look for the analogue of the cyclic Whitney gauge introduced in section 2.3.1. For the connection A , imposing the gauge just amounts to saying that A is now infrared, *i.e.* a piecewise-constant connection $A' = \sum_{k=1}^n d\tau A_k \theta_{\mathcal{I}_k}$ with $A_k \in \mathfrak{g}$. For ψ , we would like to restrict the integration to ψ 's with given integrals (average values) $\tilde{\psi}_k = \frac{\psi_k + \psi_{k+1}}{2}$ over intervals $\mathcal{I}_k$. So, we are interested in the integral

$$(50) \quad e^{\frac{1}{\hbar} S_{\Xi_n}} = \int \mathcal{D}\psi \, e^{\frac{1}{2\hbar} \int_{S^1} ((\psi, d\psi) + (\psi, \text{ad}_{A'} \psi))} \prod_{k=1}^n \delta \left(\int_{\mathcal{I}_k} d\tau \, \psi - \tilde{\psi}_k \right) = \\ = \int \prod_{k=1}^n D\lambda_k \, e^{-\sum_{k=1}^n (\lambda_k, \tilde{\psi}_k)} \underbrace{\int \mathcal{D}\psi \, e^{\frac{1}{2\hbar} \int_{S^1} ((\psi, d\psi) + (\psi, \text{ad}_{A'} \psi)) + \sum_{k=1}^n (\lambda_k, \psi)}}_{Z_{\Xi_n}(\lambda', A')}.$$

Here we got rid of δ -functions at the cost of introducing odd auxiliary variables $\lambda_k = T^a \lambda_k^a \in \Pi \mathfrak{g}$. One can organize them into an odd piecewise-constant function on the circle $\lambda' = \sum_{k=1}^n \lambda_k \theta_{I_k}$ which plays the role of a source for the field ψ . The entity $Z_{\Xi_n}(\lambda', A')$ that appeared in the integrand can be written in the operator formalism as

$$(51) \quad Z_{\Xi_n}(\lambda', A') = \text{Str}_{Cl(\mathfrak{g})} \overleftarrow{\prod}_{k=1}^n \exp \left(-\frac{1}{2\hbar} f^{abc} \hat{\psi}^a A_k^b \hat{\psi}^c + \lambda_k^a \hat{\psi}^a \right).$$

Then, the partition function of the one-dimensional Chern-Simons theory on the circle (in the Whitney gauge) is given by the odd Fourier transform of (51):

$$(52) \quad Z_{\Xi_n}(\psi', A') = (i\hbar)^{-nm} \int \prod_{k=1}^n D\lambda_k e^{-\sum_{k=1}^n \lambda_k^a \tilde{\psi}_k^a} Z_{\Xi_n}(\lambda', A').$$

Remark 14. To be precise with signs, we should introduce an ordering convention for the Berezin measure in (52). We set

$$\prod_{k=1}^n D\lambda_k = \overrightarrow{\prod}_{k=1}^n \overrightarrow{\prod}_{a=1}^{2m} D\lambda_k^a.$$

Theorem 2. For n odd and $\dim \mathfrak{g} = 2m$ even, one has

$$(53) \quad Z_{\Xi_n}(\psi', A') = e^{\frac{1}{\hbar} S_{\Xi_n}(\psi', A')},$$

where the right hand side is given by (42) and the left hand side is defined by (51, 52).

We will prove this theorem in section 4.4.2 by constructing a path integral representation for Z_{Ξ_n} . But first (see sections 3.1.3, 3.1.4) we will perform some direct tests of formula (53).

Remark 15. Note that we can define the right hand side of (53) only for n odd while the definition of left hand side makes sense for both even and odd n . A simple computation shows that for n even the partition function $Z_{\Xi_n}(\psi', A')$ vanishes at $\psi' = 0, A' = 0$. This agrees with the observation that for n even the Whitney gauge does not apply (see section 2).

Remark 16. We chose the normalization for the super-trace in Clifford algebra (49) and for Z_{Ξ_n} (52) in a way consistent with the path integral formalism (see (76), (77)).

3.1.3. Consistency check: the effective action on cohomology.⁹ The first test of the correspondence (53) is the case of $n = 1$. Let us first compute the following expression in the Clifford algebra with two generators Cl_2 :

$$(54) \quad \varphi(\tilde{\psi}^1, \tilde{\psi}^2, a) = (i\hbar)^{-1} \int D\lambda^1 D\lambda^2 e^{-\lambda^1 \tilde{\psi}^1 - \lambda^2 \tilde{\psi}^2} \text{Str}_{Cl_2} \exp \left(-\frac{1}{\hbar} \hat{\psi}^1 a \hat{\psi}^2 + \lambda^1 \hat{\psi}^1 + \lambda^2 \hat{\psi}^2 \right),$$

where $a \in \mathbb{R}$ is a number. For the exponential under the super-trace we have

$$(55) \quad \exp \left(-\frac{1}{\hbar} \hat{\psi}^1 a \hat{\psi}^2 + \lambda^1 \hat{\psi}^1 + \lambda^2 \hat{\psi}^2 \right) = \left(-\frac{2}{\hbar} \sin(a/2) - \frac{\sin(a/2)}{a/2} \lambda^1 \lambda^2 \right) \hat{\psi}^1 \hat{\psi}^2 +$$

⁹For reader's convenience, we present explicit calculations in the Clifford algebra here and in subsequent sections; the general reference for the Clifford calculus is [4].

$$+ \frac{\sin(a/2)}{a/2} (\lambda^1 \hat{\psi}^1 + \lambda^2 \hat{\psi}^2) + \left(\cos(a/2) + \hbar \frac{1}{a} \left(\cos(a/2) - \frac{\sin(a/2)}{a/2} \right) \lambda^1 \lambda^2 \right).$$

In this expression, only the first term contributes to the super-trace in (54), and we obtain

$$\begin{aligned} \varphi(\tilde{\psi}^1, \tilde{\psi}^2, a) &= \int D\lambda^1 D\lambda^2 e^{-\lambda^1 \tilde{\psi}^1 - \lambda^2 \tilde{\psi}^2} \left(-\frac{2}{\hbar} \sin(a/2) - \frac{\sin(a/2)}{a/2} \lambda^1 \lambda^2 \right) = \\ &= \frac{\sin(a/2)}{a/2} - \frac{2}{\hbar} \sin(a/2) \tilde{\psi}^1 \tilde{\psi}^2 = \frac{\sin(a/2)}{a/2} e^{-\frac{1}{\hbar} \tilde{\psi}^1 a \tilde{\psi}^2}. \end{aligned}$$

Next let us consider an anti-symmetric block-diagonal matrix with blocks 2×2

$$(56) \quad \mathcal{A} = \begin{pmatrix} 0 & a^1 & & & & \\ -a^1 & 0 & & & & \\ & & 0 & a^2 & & \\ & & -a^2 & 0 & & \\ & & & & \ddots & \\ & & & & & 0 & a^m \\ & & & & & -a^m & 0 \end{pmatrix} \in \mathfrak{so}(\mathfrak{g}) \subset \text{End}(\mathfrak{g}).$$

Then, we have

$$\begin{aligned} (57) \quad (i\hbar)^{-m} \int \prod_a D\lambda^a e^{-\lambda^a \tilde{\psi}^a} \text{Str}_{Cl(\mathfrak{g})} \exp \left(-\frac{1}{2\hbar} \hat{\psi}^a \mathcal{A}^{ab} \hat{\psi}^b + \lambda^a \hat{\psi}^a \right) = \\ = \prod_{p=1}^m \varphi(\tilde{\psi}^{2p-1}, \tilde{\psi}^{2p}, a^p) = \det_g^{1/2} \left(\frac{\sinh \frac{\mathcal{A}}{2}}{\frac{\mathcal{A}}{2}} \right) \cdot e^{-\frac{1}{2\hbar} \tilde{\psi}^a \mathcal{A}^{ab} \tilde{\psi}^b}. \end{aligned}$$

Since both the left hand side and the right hand side of (57) are $SO(\mathfrak{g})$ -invariant, the equality actually holds for all anti-symmetric matrices $\mathcal{A}$, and in particular for $\mathcal{A} = \text{ad}_{A_0}$, where $A_0 \in \mathfrak{g}$. Thus, we have shown that

$$Z_{\Xi_1}(\tilde{\psi}, A_0) = \det_g^{1/2} \left(\frac{\sinh \frac{\text{ad}_{A_0}}{2}}{\frac{\text{ad}_{A_0}}{2}} \right) \cdot e^{-\frac{1}{2\hbar} (\tilde{\psi}, \text{ad}_{A_0} \tilde{\psi})}$$

The right hand side coincides with (13), so we have checked the correspondence (53) in the case of $n = 1$.

3.1.4. Consistency check: the case of mutually commuting $\{A_k\}$ and $\psi' = 0$. Now we would like to perform a direct check of the correspondence (53) at the point $\psi' = 0$ (i.e. neglecting the tree part of the simplicial action) and assuming that $[A_k, A_{k'}] = 0$ for all $k, k' = 1, \dots, n$. That is, we will check that

$$\begin{aligned} (58) \quad (i\hbar)^{-nm} \int \prod_{k=1}^n D\lambda_k \text{Str}_{Cl(\mathfrak{g})} \prod_{k=1}^{\overleftarrow{n}} \exp \left(-\frac{1}{2\hbar} f^{abc} \hat{\psi}^a A_k^b \hat{\psi}^c + \lambda_k^a \hat{\psi}^a \right) = \\ = \det_g^{1/2} \left(\frac{1 + \prod_{k=1}^n R(\text{ad}_{A_k})}{\prod_{k=1}^n (1 + R(\text{ad}_{A_k}))} \cdot \prod_{k=1}^n \frac{\sinh \frac{\text{ad}_{A_k}}{2}}{\frac{\text{ad}_{A_k}}{2}} \right). \end{aligned}$$

We use the idea of section 3.1.3 to reduce (58) to a computation in Cl_2 . Since all A_k mutually commute, we can choose an orthonormal basis in $\mathfrak{g}$, such that the

matrices ad_{A_k} simultaneously assume the standard form (56). Then, both sides of (58) factorize into contributions of 2×2 blocks, and it suffices to check the identity

$$(59) \quad (i\hbar)^{-n} \int \prod_{k=1}^n (D\lambda_k^1 D\lambda_k^2) \text{Str}_{Cl_2} \prod_{k=1}^{\overleftarrow{n}} \exp \left(-\frac{1}{\hbar} \hat{\psi}^1 a_k \hat{\psi}^2 + \lambda_k^1 \hat{\psi}^1 + \lambda_k^2 \hat{\psi}^2 \right) = \\ = \det^{1/2} \left(\frac{1 + \prod_{k=1}^n R(i\sigma_2 a_k)}{\prod_{k=1}^n (1 + R(i\sigma_2 a_k))} \cdot \prod_{k=1}^n \frac{\sinh \frac{i\sigma_2 a_k}{2}}{\frac{i\sigma_2 a_k}{2}} \right).$$

Here $\sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$ is the second Pauli matrix, and $i\sigma_2 a_k = \begin{pmatrix} 0 & a_k \\ -a_k & 0 \end{pmatrix}$. To evaluate the left hand side of (59), we use the result (55) for the Clifford exponential:

$$\text{l.h.s of (59)} = (i\hbar)^{-n} \text{Str}_{Cl_2} \prod_{k=1}^{\overleftarrow{n}} \left(\frac{\sin(a_k/2)}{a_k/2} \hat{\psi}^1 \hat{\psi}^2 + \hbar \frac{1}{a_k} \left(\frac{\sin(a_k/2)}{a_k/2} - \cos(a_k/2) \right) \right)$$

An easy way to evaluate this expression is to use the matrix representation $Cl_2 \rightarrow \text{End}(\mathbb{C}^{1|1})$ which maps

$$\hat{\psi}^1 \mapsto \left(\frac{\hbar}{2} \right)^{1/2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \hat{\psi}^2 \mapsto \left(\frac{\hbar}{2} \right)^{1/2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \text{Str}_{Cl_2} \mapsto \text{Str}_{\text{End}(\mathbb{C}^{1|1})},$$

and

$$\text{Str}_{\text{End}(\mathbb{C}^{1|1})} : \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \mapsto \alpha - \delta$$

is the standard super-trace on matrices of the size $(1|1) \times (1|1)$. Using this representation, we obtain

$$\begin{aligned} \text{l.h.s of (59)} &= i^{-n} \text{Str}_{\text{End}(\mathbb{C}^{1|1})} \prod_{k=1}^n \left(\frac{\frac{1}{a_k} \left(\frac{\sin(a_k/2)}{a_k/2} - \cos(a_k/2) \right) + \frac{i}{2} \frac{\sin(a_k/2)}{a_k/2}}{0} \middle| \frac{0}{\frac{1}{a_k} \left(\frac{\sin(a_k/2)}{a_k/2} - \cos(a_k/2) \right) - \frac{i}{2} \frac{\sin(a_k/2)}{a_k/2}} \right) \\ &= i^{-n} \prod_{k=1}^n \frac{\sin(a_k/2)}{a_k/2} \cdot 2i \text{Im} \prod_{k=1}^n \left(\frac{i}{2} + \frac{1}{a_k} - \frac{1}{2} \cot(a_k/2) \right) = \\ &= 2 \prod_{k=1}^n \frac{\sin(a_k/2)}{a_k/2} \cdot \text{Re} \prod_{k=1}^n \frac{1}{1 + R(ia_k)}. \end{aligned}$$

In the last line we used the assumption that n is odd.

To evaluate the right hand side of (59), first observe that the matrix

$$\frac{1 + \prod_{k=1}^n R(i\sigma_2 a_k)}{\prod_{k=1}^n (1 + R(i\sigma_2 a_k))} = \frac{1}{\prod_{k=1}^n (1 + R(i\sigma_2 a_k))} + \frac{1}{\prod_{k=1}^n (1 + R(-i\sigma_2 a_k))}$$

is constructed from matrices $i\sigma_2 a_k = \begin{pmatrix} 0 & a_k \\ -a_k & 0 \end{pmatrix}$ and is therefore of form $\begin{pmatrix} \alpha & \beta \\ -\beta & \alpha \end{pmatrix}$ (*i.e.* belongs to $\text{Span}_{\mathbb{R}}(1, i\sigma_2)$), and that it is symmetric. Hence, it is actually a multiple of the identity matrix $\begin{pmatrix} \alpha & 0 \\ 0 & \alpha \end{pmatrix}$, and the determinant may be expressed as

$$\det^{1/2} \left(\frac{1 + \prod_{k=1}^n R(i\sigma_2 a_k)}{\prod_{k=1}^n (1 + R(i\sigma_2 a_k))} \right) = \frac{1}{2} \text{tr} \left(\frac{1 + \prod_{k=1}^n R(i\sigma_2 a_k)}{\prod_{k=1}^n (1 + R(i\sigma_2 a_k))} \right) =$$

$$= \prod_{k=1}^n \frac{1}{1 + R(ia_k)} + \prod_{k=1}^n \frac{1}{1 + R(-ia_k)} = 2 \operatorname{Re} \prod_{k=1}^n \frac{1}{1 + R(ia_k)}$$

Now it is obvious that (59) is verified, and this implies (58).

3.2. One-dimensional Chern-Simons with boundary.

3.2.1. One-dimensional simplicial Chern-Simons in the operator formalism. Concatenations. There is a natural construction of the one-dimensional simplicial Chern-Simons theory in the operator formalism: to a one-dimensional oriented simplicial complex Θ (a collection of $i(\Theta)$ triangulated intervals and $c(\Theta)$ circles) it associates a partition function

$$(60) \quad Z_\Theta \in \operatorname{Fun}(\underbrace{\mathfrak{g} \otimes C_1(\Theta) \oplus \Pi \mathfrak{g} \otimes C^1(\Theta)}_{\mathcal{F}_\Theta^{\text{bulk}}}) \otimes Cl(\mathfrak{g})^{\otimes i(\Theta)},$$

where we think of variables $A_k \in \mathfrak{g}$ as coordinates on $\mathfrak{g}$ -valued simplicial 1-cochains of Θ with parity shifted, and we think of variables $\tilde{\psi}_k \in \Pi \mathfrak{g}$ as coordinates on $\mathfrak{g}$ -valued simplicial 1-chains of Θ . The partition function Z_Θ is defined by the following properties:

- For a single interval with a standard triangulation, we have

$$(61) \quad Z_I = (i\hbar)^{-m} \int \prod_{a=1}^{2m} D\lambda^a e^{-\lambda^a \tilde{\psi}^a} \exp\left(-\frac{1}{2\hbar} f^{abc} \hat{\psi}^a A^b \hat{\psi}^c + \lambda^a \hat{\psi}^a\right).$$

- For a disjoint union of Θ_1 and Θ_2 ,

$$Z_{\Theta_1 \sqcup \Theta_2} = Z_{\Theta_1} \otimes Z_{\Theta_2}.$$

- For a concatenation $\Theta_1 \cup \Theta_2$ with intersection $\Theta_1 \cap \Theta_2 = \mathfrak{p}$ being a single point embedded as a boundary point of positive orientation $\mathfrak{p}_1 \in \Theta_1$ belonging to the i -th triangulated interval of Θ_1 , and embedded as a boundary point of negative orientation $\mathfrak{p}_2 \in \Theta_2$ belonging to the j -th triangulated interval of Θ_2 , we have:

$$(62) \quad Z_{\Theta_1 \cup \Theta_2} = \mathfrak{m}(Z_{\Theta_2} \otimes Z_{\Theta_1}).$$

Here $\mathfrak{m} : Cl(\mathfrak{g}) \otimes Cl(\mathfrak{g}) \rightarrow Cl(\mathfrak{g})$ is the Clifford algebra multiplication, and $\mathfrak{m}$ in (62) acts on Clifford algebras associated to the j -th triangulated interval of Θ_2 and the i -th triangulated interval of Θ_1 .

- If the simplicial complex Θ' is obtained from Θ by closing the i -th triangulated interval into a circle, we have:

$$(63) \quad Z_{\Theta'} = \operatorname{Str}_{Cl(\mathfrak{g})} Z_\Theta,$$

where the super-trace is taken over the Clifford algebra associated to the i -th interval of Θ .

Obviously, this construction gives (52) for a triangulated circle $\Theta = \Xi_n$, and due to the correspondence (53) it is consistent with the results of section 2.

Remark 17. We may regard Z_I as a contribution of an open interval. Then, the concatenation (62) is understood as taking a disjoint union of open triangulated intervals and then gluing them together at the point $\mathfrak{p}$. Likewise, (63) is understood as gluing together of two end-points of the same interval. Thus, a contribution

of point is either the (1,2)-tensor¹⁰ $\mathbf{m}$ on the Clifford algebra or the (0,1)-tensor $\text{Str}_{Cl(\mathfrak{g})}$, depending on whether we are gluing together two different or the same connected component.

Remark 18. Once we come to a description of the one-dimensional Chern-Simons theory in terms of Atiyah-Segal's axioms and specify the vector space associated to a point (the space of states), properties (62) and (63) become a single sewing axiom (see section 4.2 below).

3.2.2. Simplicial aggregations. Let Θ' be the interval $[\mathbf{p}_1, \mathbf{p}_3]$ with the standard triangulation, and let Θ be the subdivision of Θ' with three 0-simplices $\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_3$ and two 1-simplices $[\mathbf{p}_1, \mathbf{p}_2], [\mathbf{p}_2, \mathbf{p}_3]$. The aggregation morphism r^ξ acts on simplicial chains as

$$\begin{aligned} r_{C_\bullet}^\xi : C_\bullet(\Theta) &\rightarrow C_\bullet(\Theta') \\ \alpha_{\mathbf{p}_1} e^{\mathbf{p}_1} + \alpha_{\mathbf{p}_2} e^{\mathbf{p}_2} + \alpha_{\mathbf{p}_3} e^{\mathbf{p}_3} + \alpha_{\mathbf{p}_1 \mathbf{p}_2} e^{\mathbf{p}_1 \mathbf{p}_2} + \alpha_{\mathbf{p}_2 \mathbf{p}_3} e^{\mathbf{p}_2 \mathbf{p}_3} &\mapsto \\ \mapsto (\alpha_{\mathbf{p}_1} + (1 - \xi)\alpha_{\mathbf{p}_2})e'^{\mathbf{p}_1} + (\alpha_{\mathbf{p}_3} + \xi\alpha_{\mathbf{p}_2})e'^{\mathbf{p}_3} + (\xi\alpha_{\mathbf{p}_1 \mathbf{p}_2} + (1 - \xi)\alpha_{\mathbf{p}_2 \mathbf{p}_3})e'^{\mathbf{p}_1 \mathbf{p}_3}, \end{aligned}$$

and on simplicial cochains as

$$\begin{aligned} r_{C^\bullet}^\xi : C^\bullet(\Theta) &\rightarrow C^\bullet(\Theta') \\ \alpha^{\mathbf{p}_1} e_{\mathbf{p}_1} + \alpha^{\mathbf{p}_2} e_{\mathbf{p}_2} + \alpha^{\mathbf{p}_3} e_{\mathbf{p}_3} + \alpha^{\mathbf{p}_1 \mathbf{p}_2} e_{\mathbf{p}_1 \mathbf{p}_2} + \alpha^{\mathbf{p}_2 \mathbf{p}_3} e_{\mathbf{p}_2 \mathbf{p}_3} &\mapsto \alpha^{\mathbf{p}_1} e'_{\mathbf{p}_1} + \alpha^{\mathbf{p}_3} e'_{\mathbf{p}_3} + (\alpha^{\mathbf{p}_1 \mathbf{p}_2} + \alpha^{\mathbf{p}_2 \mathbf{p}_3})e'_{\mathbf{p}_1 \mathbf{p}_3}. \end{aligned}$$

Here $0 < \xi < 1$ is a parameter determining the relative weight of intervals $[\mathbf{p}_1, \mathbf{p}_2]$ and $[\mathbf{p}_2, \mathbf{p}_3]$ inside $[\mathbf{p}_1, \mathbf{p}_3]$ (the symmetric choice corresponds to $\xi = 1/2$). Basis chains and cochains corresponding a simplex σ are denoted by e^σ and e_σ , respectively. We use primes to distinguish the basis of $C_\bullet(\Theta'), C^\bullet(\Theta')$; $\alpha_{\dots}, \alpha^{\dots}$ are numerical coefficients.

One can also introduce the subdivision morphism which is dual to the aggregation morphism:

$$i_{C_\bullet}^\xi = (r_{C^\bullet}^\xi)^* : C_\bullet(\Theta') \rightarrow C_\bullet(\Theta), \quad i_{C^\bullet}^\xi = (r_{C_\bullet}^\xi)^* : C^\bullet(\Theta') \rightarrow C^\bullet(\Theta).$$

More explicitly,

$$\begin{aligned} i_{C_\bullet}^\xi : \alpha'_{\mathbf{p}_1} e'^{\mathbf{p}_1} + \alpha'_{\mathbf{p}_3} e'^{\mathbf{p}_3} + \alpha'_{\mathbf{p}_1 \mathbf{p}_3} e'^{\mathbf{p}_1 \mathbf{p}_3} &\mapsto \alpha'_{\mathbf{p}_1} e^{\mathbf{p}_1} + \alpha'_{\mathbf{p}_3} e^{\mathbf{p}_3} + \alpha'_{\mathbf{p}_1 \mathbf{p}_3} (e^{\mathbf{p}_1 \mathbf{p}_2} + e^{\mathbf{p}_2 \mathbf{p}_3}), \\ i_{C^\bullet}^\xi : \alpha'^{\mathbf{p}_1} e'_{\mathbf{p}_1} + \alpha'^{\mathbf{p}_3} e'_{\mathbf{p}_3} + \alpha'^{\mathbf{p}_1 \mathbf{p}_3} e'_{\mathbf{p}_1 \mathbf{p}_3} &\mapsto \\ \mapsto \alpha'^{\mathbf{p}_1} e_{\mathbf{p}_1} + ((1 - \xi)\alpha'^{\mathbf{p}_1} + \xi\alpha'^{\mathbf{p}_3})e_{\mathbf{p}_2} + \alpha'^{\mathbf{p}_3} e_{\mathbf{p}_3} + \alpha'^{\mathbf{p}_1 \mathbf{p}_3} (\xi e_{\mathbf{p}_1 \mathbf{p}_2} + (1 - \xi)e_{\mathbf{p}_2 \mathbf{p}_3}). \end{aligned}$$

Aggregation and subdivision morphisms are chain maps. Moreover, they are quasi-isomorphisms.

These maps induce an embedding $i_{\Theta, \Theta'}^\xi : \mathcal{F}_{\Theta'}^{\text{bulk}} \hookrightarrow \mathcal{F}_\Theta^{\text{bulk}}$ and projection $r_{\Theta, \Theta'}^\xi : \mathcal{F}_\Theta^{\text{bulk}} \rightarrow \mathcal{F}_{\Theta'}^{\text{bulk}}$ for the spaces of “bulk fields”

$$\mathcal{F}_\Theta^{\text{bulk}} = \mathfrak{g} \otimes C_1(\Theta) \oplus \Pi \mathfrak{g} \otimes C^1(\Theta), \quad \mathcal{F}_{\Theta'}^{\text{bulk}} = \mathfrak{g} \otimes C_1(\Theta') \oplus \Pi \mathfrak{g} \otimes C^1(\Theta').$$

That is, we have a splitting

$$(64) \quad \mathcal{F}_\Theta^{\text{bulk}} = i_{\Theta, \Theta'}^\xi(\mathcal{F}_{\Theta'}^{\text{bulk}}) \oplus \mathcal{F}_{\Theta, \Theta'}^{\prime \text{bulk}, \xi}.$$

In more detail, we split the bulk fields as

$$\tilde{\psi}_1 = \tilde{\psi}' + (1 - \xi)\tilde{\psi}'', \quad \tilde{\psi}_2 = \tilde{\psi}' - \xi\tilde{\psi}'', \quad A_1 = \xi A' - A'', \quad A_2 = (1 - \xi)A' + A''.$$

¹⁰We mean the rank of the tensor: one time covariant, two times contravariant.

The maps $i_{\Theta, \Theta'}^\xi$ and $r_{\Theta, \Theta'}^\xi$ are dual to each other with respect to the odd pairing on $\mathcal{F}_\Theta^{\text{bulk}}$ and $\mathcal{F}_{\Theta'}^{\text{bulk}}$ (thus, (64) is an orthogonal decomposition). We denote by $\mathcal{L}_{\Theta, \Theta'}^\xi \subset \mathcal{F}_{\Theta, \Theta'}^{\text{bulk}, \xi}$ the Lagrangian subspace defined by setting to zero the 1-cochain part of the ultraviolet field A'' .

We define the action of the aggregation map $\Theta \rightarrow \Theta'$ on the partition function of the one-dimensional simplicial Chern-Simons theory by the fiber BV integral associated to the splitting (64):

$$(65) \quad (r_{\Theta, \Theta'}^\xi)_*(Z_\Theta) = \int_{\mathcal{L}_{\Theta, \Theta'}^\xi} Z_\Theta \in \text{Fun}(\mathcal{F}_{\Theta'}^{\text{bulk}}) \otimes Cl(\mathfrak{g}).$$

It is easy to give a direct check of the following statement.

Lemma 3.

$$(66) \quad (r_{\Theta, \Theta'}^\xi)_*(Z_\Theta) = Z_{\Theta'}$$

Proof. Indeed, by definition (65) we have

$$\begin{aligned} & (r_{\Theta, \Theta'}^\xi)_*(Z_\Theta) = \\ &= (i\hbar)^m \int \prod_{a=1}^{\overleftarrow{2m}} D\tilde{\psi}''^a Z_{[\mathfrak{p}_2, \mathfrak{p}_3]}(\tilde{\psi}' - \xi\tilde{\psi}'', (1-\xi)A') \cdot Z_{[\mathfrak{p}_1, \mathfrak{p}_2]}(\tilde{\psi}' + (1-\xi)\tilde{\psi}'', \xi A') = \\ &= (i\hbar)^{-m} \int \prod_{a=1}^{\overleftarrow{2m}} D\tilde{\psi}''^a \prod_{a=1}^{\overrightarrow{2m}} D\lambda_2^a \prod_{a=1}^{\overrightarrow{2m}} D\lambda_1^a e^{-\lambda_1^a(\tilde{\psi}'^a + (1-\xi)\tilde{\psi}''^a) - \lambda_2^a(\tilde{\psi}'^a - \xi\tilde{\psi}''^a)} \cdot \\ & \quad \cdot \exp\left(-\frac{1-\xi}{2\hbar} f^{abc} \hat{\psi}^a A'^b \hat{\psi}^c + \lambda_2^a \hat{\psi}^a\right) \cdot \exp\left(-\frac{\xi}{2\hbar} f^{abc} \hat{\psi}^a A'^b \hat{\psi}^c + \lambda_1^a \hat{\psi}^a\right). \end{aligned}$$

Here we made a change of coordinates $(\lambda_1, \lambda_2) \rightarrow (\lambda = \lambda_1 + \lambda_2, \nu = (1-\xi)\lambda_1 - \xi\lambda_2)$. The integral over $\tilde{\psi}''$ produces the delta-function $\delta(\nu)$; by integrating over ν we obtain

$$\begin{aligned} & (r_{\Theta, \Theta'}^\xi)_*(Z_\Theta) = (i\hbar)^{-m} \int \prod_{a=1}^{\overrightarrow{2m}} D\lambda^a e^{-\lambda^a \tilde{\psi}'^a} \cdot \\ & \cdot \exp\left(-\frac{1-\xi}{2\hbar} f^{abc} \hat{\psi}^a A'^b \hat{\psi}^c + (1-\xi)\lambda^a \hat{\psi}^a\right) \cdot \exp\left(-\frac{\xi}{2\hbar} f^{abc} \hat{\psi}^a A'^b \hat{\psi}^c + \xi\lambda^a \hat{\psi}^a\right) = \\ &= (i\hbar)^{-m} \int \prod_{a=1}^{\overrightarrow{2m}} D\lambda^a e^{-\lambda^a \tilde{\psi}'^a} \cdot \exp\left(-\frac{1}{2\hbar} f^{abc} \hat{\psi}^a A'^b \hat{\psi}^c + \lambda^a \hat{\psi}^a\right) = Z_{\Theta'}. \end{aligned}$$

□

Remark 19. We normalize the measure on $\mathcal{L}_{\Theta, \Theta'}^\xi$ in such a way that relation (66) holds with no additional factors.

Up to now, we only discussed the elementary aggregation which takes an interval subdivided into two smaller intervals and into an interval with the standard triangulation (that is, one removes the middle point $\mathfrak{p}_2$ and merges intervals $[\mathfrak{p}_1, \mathfrak{p}_2]$ and $[\mathfrak{p}_2, \mathfrak{p}_3]$). A general simplicial aggregation for a one-dimensional simplicial complex is a sequence of elementary aggregations made at each step on an incident pair of

intervals. In particular, there are many simplicial aggregations for triangulated circles: $\Xi_n \rightarrow \Xi_{n'}$ with $n > n'$.

The following is an immediate consequence of (66):

Proposition 2. *For a general simplicial aggregation*

$$r = r_{\Theta_{l-1}, \Theta'}^{\xi_l} \circ \cdots \circ r_{\Theta_1, \Theta_2}^{\xi_2} \circ r_{\Theta, \Theta_1}^{\xi_1} : \Theta \rightarrow \Theta'$$

(where Θ is an arbitrary one-dimensional simplicial complex and Θ' is some aggregation of Θ), one has

$$(67) \quad r_*(Z_\Theta) = Z_{\Theta'}.$$

The compatibility with aggregations is an important property expected from a simplicial theory. In particular, (53) implies that the simplicial action for a circle S_{Ξ_n} given by (42) is compatible with simplicial aggregations $\Xi_n \rightarrow \Xi_{n'}$ for n, n' odd.

3.2.3. Quantum master equation.

Lemma 4.¹¹ *The partition function for an interval (61) satisfies the following differential equation:*

$$(68) \quad \hbar \frac{\partial}{\partial \tilde{\psi}^a} \frac{\partial}{\partial A^a} Z_{\mathcal{I}} + \frac{1}{\hbar} \left[\frac{1}{6} f^{abc} \hat{\psi}^a \hat{\psi}^b \hat{\psi}^c, Z_{\mathcal{I}} \right]_{Cl(\mathfrak{g})} = 0,$$

where $[\cdot, \cdot]_{Cl(\mathfrak{g})}$ denotes the super-commutator on $Cl(\mathfrak{g})$.

Proof. We will check (68) using variables (λ, A) . We have,

$$(69) \quad \left(\hbar \lambda^a \frac{\partial}{\partial A^a} + \frac{1}{\hbar} \left[\frac{1}{6} f^{abc} \hat{\psi}^a \hat{\psi}^b \hat{\psi}^c, \bullet \right]_{Cl(\mathfrak{g})} \right) \exp \left(-\frac{1}{2\hbar} f^{abc} \hat{\psi}^a A^b \hat{\psi}^c + \lambda^a \hat{\psi}^a \right) = 0.$$

After the Fourier transform from the variable λ to the variable $\tilde{\psi}$, this expression becomes (68)). Observe that

$$(70) \quad \hbar \lambda^a \frac{\partial}{\partial A^a} \exp \left(-\frac{1}{2\hbar} f^{abc} \hat{\psi}^a A^b \hat{\psi}^c + \lambda^a \hat{\psi}^a \right) = \\ = \int_0^1 d\tau e^{\tau \left(-\frac{1}{2\hbar} f^{abc} \hat{\psi}^a A^b \hat{\psi}^c + \lambda^a \hat{\psi}^a \right)} \cdot \frac{1}{2} f^{abc} \hat{\psi}^a \lambda^b \hat{\psi}^c \cdot e^{(1-\tau) \left(-\frac{1}{2\hbar} f^{abc} \hat{\psi}^a A^b \hat{\psi}^c + \lambda^a \hat{\psi}^a \right)}.$$

Next, compute commutators in the Clifford algebra,

$$(71) \quad \left[\frac{1}{6} f^{abc} \hat{\psi}^a \hat{\psi}^b \hat{\psi}^c, \lambda^{a'} \hat{\psi}^{a'} \right]_{Cl(\mathfrak{g})} = -\frac{\hbar}{2} f^{abc} \hat{\psi}^a \lambda^b \hat{\psi}^c,$$

and

$$(72) \quad \left[\frac{1}{6} f^{abc} \hat{\psi}^a \hat{\psi}^b \hat{\psi}^c, \frac{1}{2} f^{a'b'c'} \hat{\psi}^{a'} A^{b'} \hat{\psi}^{c'} \right]_{Cl(\mathfrak{g})} = \\ = \frac{1}{12} f^{abc} f^{a'b'c'} A^{b'} ([\hat{\psi}^a \hat{\psi}^b \hat{\psi}^c, \hat{\psi}^{a'}] \hat{\psi}^{c'} - \hat{\psi}^{a'} [\hat{\psi}^a \hat{\psi}^b \hat{\psi}^c, \hat{\psi}^{c'}]) = \\ = \frac{\hbar}{4} f^{abc} f^{a'b'c'} A^{b'} (\delta^{ca'} \hat{\psi}^a \hat{\psi}^b \hat{\psi}^{c'} - \delta^{cc'} \hat{\psi}^{a'} \hat{\psi}^a \hat{\psi}^b) = \frac{\hbar}{4} f^{abc} f^{cb'c'} A^{b'} (\hat{\psi}^a \hat{\psi}^b \hat{\psi}^{c'} + \hat{\psi}^{c'} \hat{\psi}^a \hat{\psi}^b) =$$

¹¹In a different setting equation (68) appeared in [2].

$$\begin{aligned}
&= \frac{\hbar}{6} \underbrace{f^{abc} f^{cb'c'} A^{b'} (\hat{\psi}^a \hat{\psi}^b \hat{\psi}^{c'} + \hat{\psi}^{c'} \hat{\psi}^a \hat{\psi}^b + \hat{\psi}^b \hat{\psi}^{c'} \hat{\psi}^a)}_{=0 \text{ by Jacobi identity}} + \\
&\quad + \frac{\hbar}{12} f^{abc} f^{cb'c'} A^{b'} (\hat{\psi}^a \hat{\psi}^b \hat{\psi}^{c'} + \hat{\psi}^{c'} \hat{\psi}^a \hat{\psi}^b - 2\hat{\psi}^b \hat{\psi}^{c'} \hat{\psi}^a) = \\
&= \frac{\hbar}{24} f^{abc} f^{cb'c'} A^{b'} ((\hat{\psi}^a \hat{\psi}^b \hat{\psi}^{c'} + \hat{\psi}^{c'} \hat{\psi}^a \hat{\psi}^b - 2\hat{\psi}^b \hat{\psi}^{c'} \hat{\psi}^a) - (\hat{\psi}^b \hat{\psi}^a \hat{\psi}^{c'} + \hat{\psi}^{c'} \hat{\psi}^b \hat{\psi}^a - 2\hat{\psi}^a \hat{\psi}^{c'} \hat{\psi}^b)) = \\
&= \frac{\hbar}{24} f^{abc} f^{cb'c'} A^{b'} (\hat{\psi}^a [\hat{\psi}^b, \hat{\psi}^{c'}] + [\hat{\psi}^{c'}, \hat{\psi}^a] \hat{\psi}^b - \hat{\psi}^b [\hat{\psi}^{c'}, \hat{\psi}^a] - [\hat{\psi}^b, \hat{\psi}^{c'}] \hat{\psi}^a) = 0.
\end{aligned}$$

For brevity, we are omitting the subscript in $[\cdot, \cdot]_{Cl(\mathfrak{g})}$ in computations. Identities (71,72) imply

$$\begin{aligned}
&\frac{1}{\hbar} \left[\frac{1}{6} f^{abc} \hat{\psi}^a \hat{\psi}^b \hat{\psi}^c, \exp \left(-\frac{1}{2\hbar} f^{abc} \hat{\psi}^a A^b \hat{\psi}^c + \lambda^a \hat{\psi}^a \right) \right]_{Cl(\mathfrak{g})} = \\
&= - \int_0^1 d\tau e^{\tau \left(-\frac{1}{2\hbar} f^{abc} \hat{\psi}^a A^b \hat{\psi}^c + \lambda^a \hat{\psi}^a \right)} \cdot \frac{1}{2} f^{abc} \hat{\psi}^a \lambda^b \hat{\psi}^c \cdot e^{(1-\tau) \left(-\frac{1}{2\hbar} f^{abc} \hat{\psi}^a A^b \hat{\psi}^c + \lambda^a \hat{\psi}^a \right)}.
\end{aligned}$$

Together with (70), this implies (69) which finishes the proof of (68). $\square$

Let us denote¹² by

$$\hat{\theta} := \frac{1}{6} f^{abc} \hat{\psi}^a \hat{\psi}^b \hat{\psi}^c$$

the Clifford element in (68).

Proposition 3. *The partition function for any one-dimensional simplicial complex Θ satisfies the differential equation*

$$(73) \quad (\hbar \Delta_{\Theta}^{\text{bulk}} + \frac{1}{\hbar} \delta_{\Theta}) Z_{\Theta} = 0,$$

where

$$\Delta_{\Theta}^{\text{bulk}} = \sum_k \frac{\partial}{\partial \tilde{\psi}_k^a} \frac{\partial}{\partial A_k^a}$$

(the sum goes over all 1-simplices of Θ), and

$$\delta_{\Theta} = \sum_{j=1}^{i(\Theta)} \left[\hat{\theta}^{(j)}, \bullet \right]_{Cl(\mathfrak{g})}.$$

Here the sum goes over connected components of Θ that are triangulated intervals; $\hat{\theta}^{(j)}$ denotes $\hat{\theta}$ as an element of the j -th copy of $Cl(\mathfrak{g})$.

Proof. Equation (73) follows from (68). The compatibility with disjoint unions is obvious as $\Delta_{\Theta_1 \sqcup \Theta_2}^{\text{bulk}} = \Delta_{\Theta_1}^{\text{bulk}} + \Delta_{\Theta_2}^{\text{bulk}}$, and $\delta_{\Theta_1 \sqcup \Theta_2} = \delta_{\Theta_1} + \delta_{\Theta_2}$. For concatenations, it suffices to check the case of two triangulated intervals Θ_1 and Θ_2 :

$$\begin{aligned}
&(\hbar \Delta_{\Theta_1 \cup \Theta_2}^{\text{bulk}} + \hbar^{-1} \delta_{\Theta_1 \cup \Theta_2}) Z_{\Theta_1 \cup \Theta_2} = \left(\hbar \Delta_{\Theta_1}^{\text{bulk}} + \hbar \Delta_{\Theta_2}^{\text{bulk}} + \hbar^{-1} [\hat{\theta}, \bullet]_{Cl(\mathfrak{g})} \right) \circ (Z_{\Theta_2} \cdot Z_{\Theta_1}) = \\
&= Z_{\Theta_2} \cdot \left(\left(\hbar \Delta_{\Theta_1}^{\text{bulk}} + \hbar^{-1} [\hat{\theta}, \bullet]_{Cl(\mathfrak{g})} \right) \circ Z_{\Theta_1} \right) + \left(\left(\hbar \Delta_{\Theta_2}^{\text{bulk}} + \hbar^{-1} [\hat{\theta}, \bullet]_{Cl(\mathfrak{g})} \right) \circ Z_{\Theta_2} \right) \cdot Z_{\Theta_1} = 0.
\end{aligned}$$

¹²The notation stems from the fact that this is a quantization of the Maurer-Cartan element $\theta \in \text{Fun}(\Pi \mathfrak{g})$ (4).

The compatibility with closure of a triangulated interval into a triangulated circle follows from $\text{Str}_{Cl(\mathfrak{g})}[\hat{\theta}, Z_\Theta]_{Cl(\mathfrak{g})} = 0$. $\square$

Remark 20. We understand (73) as a kind of quantum master equation with the boundary term $\delta_\Theta Z_\Theta$. It is tempting to think of the operator $\Delta_\Theta^{\text{bulk}} + \delta_\Theta$ appearing in (73) as a new BV Laplacian adjusted for the presence of the boundary.

Corollary 1. *In the case of a triangulated circle $\Theta = \Xi_n$, the partition function satisfies the usual (non modified) quantum master equation*

$$\Delta_{\Xi_n} Z_{\Xi_n} = 0.$$

4. BACK TO PATH INTEGRAL

Sections 4.1 and 4.4 mostly go along the lines of the standard derivation of the path integral representation for quantum mechanics, see [8].

4.1. Representation of $Cl(\mathfrak{g})$, complex polarization of $\mathfrak{g}$. The Clifford algebra $Cl(\mathfrak{g})$ admits a representation ρ on the space of polynomials of m odd variables $\text{Fun}(\mathbb{C}^{0|m}) \cong \mathbb{C}[\eta^1, \dots, \eta^m]$. This representation is defined on generators of $Cl(\mathfrak{g})$ as

$$(74) \quad \rho : \hat{\psi}^a \mapsto \begin{cases} \frac{1}{\sqrt{2}} \left(\eta^p + \hbar \frac{\partial}{\partial \eta^p} \right) & \text{if } a = 2p - 1, \\ \frac{i}{\sqrt{2}} \left(\eta^p - \hbar \frac{\partial}{\partial \eta^p} \right) & \text{if } a = 2p. \end{cases}$$

In fact, $\rho : Cl(\mathfrak{g}) \rightarrow \text{End}(\text{Fun}(\mathbb{C}^{0|m})) \cong \text{End}(\mathbb{C}^{2^{m-1}|2^{m-1}})$ is an isomorphism of super-algebras. There is a natural identification

$$\phi : \text{End}(\text{Fun}(\mathbb{C}^{0|m})) \xrightarrow{\sim} \text{Fun}(\mathbb{C}^{0|m} \oplus \mathbb{C}^{0|m}) \cong \mathbb{C}[\eta^1, \dots, \eta^m, \bar{\eta}^1, \dots, \bar{\eta}^m].$$

This is not an algebra morphism with respect to the standard algebra structure on polynomials. Instead, it takes the product of endomorphisms into the convolution; see [3], [8]). We are interested in the composition $\Phi = \phi \circ \rho : Cl(\mathfrak{g}) \rightarrow \text{Fun}(\mathbb{C}^{0|m} \oplus \mathbb{C}^{0|m})$ which maps

$$(75) \quad \Phi : \begin{cases} \hat{1} & \mapsto e^{\frac{1}{\hbar} \sum_q \eta^q \bar{\eta}^q} \\ \hat{\psi}^{2p-1} & \mapsto \frac{1}{\sqrt{2}} (\eta^p + \bar{\eta}^p) e^{\frac{1}{\hbar} \sum_q \eta^q \bar{\eta}^q} \\ \hat{\psi}^{2p} & \mapsto \frac{i}{\sqrt{2}} (\eta^p - \bar{\eta}^p) e^{\frac{1}{\hbar} \sum_q \eta^q \bar{\eta}^q} \end{cases}$$

and sends the product in $Cl(\mathfrak{g})$ into the convolution

$$(76) \quad \begin{aligned} \Phi(\hat{\alpha} \cdot \hat{\beta})(\eta_2, \bar{\eta}_1) &= \\ &= \hbar^m \int \prod_p (D\eta_1^p D\bar{\eta}_2^p) \Phi(\hat{\alpha})(\eta_2, \bar{\eta}_2) \cdot e^{\frac{1}{\hbar} \sum_q \bar{\eta}_2^q \eta_1^q} \cdot \Phi(\hat{\beta})(\eta_1, \bar{\eta}_1). \end{aligned}$$

Formula (76) is a key point in reconstructing the path integral from the operator formalism. Another useful identity is as follows,

$$(77) \quad \text{Str}_{Cl(\mathfrak{g})}(\hat{\alpha}) = \hbar^m \int \prod_p (D\eta^p D\bar{\eta}^p) e^{\frac{1}{\hbar} \sum_q \bar{\eta}^q \eta^q} \cdot \Phi(\hat{\alpha})(\eta, \bar{\eta}).$$

More generally, a representation of type (74) is associated to a choice of a linear complex structure J on $\mathfrak{g}$ compatible with the pairing:

$$J : \mathfrak{g} \rightarrow \mathfrak{g}, \quad J^2 = -\text{id}_{\mathfrak{g}}, \quad (Ja, b) = -(a, Jb) \text{ for } a, b \in \mathfrak{g}.$$

It induces the splitting of the complexified Lie algebra $\mathfrak{g}_{\mathbb{C}} = \mathbb{C} \otimes \mathfrak{g}$ into “holomorphic” and “anti-holomorphic” subspaces:

$$(78) \quad \mathfrak{g}_{\mathbb{C}} = \mathfrak{h} \oplus \bar{\mathfrak{h}},$$

where J acts on $\mathfrak{h}, \bar{\mathfrak{h}}$ by multiplication by $+i$ and $-i$, respectively. (Note that $\mathfrak{h}$ and $\bar{\mathfrak{h}}$ are complex subspaces of $\mathfrak{g}_{\mathbb{C}}$ with respect to the standard complex structure; bar in $\bar{\mathfrak{h}}$ does not mean conjugation.) The subspaces $\mathfrak{h}$ and $\bar{\mathfrak{h}}$ are Lagrangian with respect to the pairing $(\cdot, \cdot)$. The complex Lagrangian polarization (78) induces a polarization for the parity-reversed Lie algebra

$$(79) \quad \Pi \mathfrak{g} = \Pi \mathfrak{h} \oplus \Pi \bar{\mathfrak{h}}.$$

We denote coordinates on $\Pi \mathfrak{h}$ by $\eta^1, \dots, \eta^m$ and coordinates on $\Pi \bar{\mathfrak{h}}$ by $\bar{\eta}^1, \dots, \bar{\eta}^m$. The representation $\rho : Cl(\mathfrak{g}) = \widehat{\text{Fun}(\Pi \mathfrak{g})} \rightarrow \text{End}(\text{Fun}(\Pi \mathfrak{h}))$ sends quantized holomorphic coordinates to multiplication operators and quantized anti-holomorphic coordinates to partial derivatives:

$$\rho : \begin{cases} \hat{\eta}^p \mapsto \eta^p \cdot, \\ \hat{\bar{\eta}}^p \mapsto \hbar \frac{\partial}{\partial \eta^p}. \end{cases}$$

Morphism (75) from $Cl(\mathfrak{g})$ to the convolution algebra $\text{Fun}(\Pi \mathfrak{h} \oplus \Pi \bar{\mathfrak{h}})$ is given on generators by

$$\Phi : \begin{cases} \hat{\eta}^p \mapsto \eta^p e^{\frac{1}{\hbar} \sum_q \eta^q \bar{\eta}^q}, \\ \hat{\bar{\eta}}^p \mapsto \bar{\eta}^p e^{\frac{1}{\hbar} \sum_q \eta^q \bar{\eta}^q}, \end{cases}$$

and it extends to the other elements of $Cl(\mathfrak{g})$ by the convolution formula (76).

We will use notation $\pi, \bar{\pi}$ for projections from $\Pi \mathfrak{g}$ to $\Pi \mathfrak{h}$ and $\Pi \bar{\mathfrak{h}}$, respectively. We denote by $\iota, \bar{\iota}$ embeddings of $\Pi \mathfrak{h}$ and $\Pi \bar{\mathfrak{h}}$ into $\Pi \mathfrak{g}$.

4.2. One-dimensional Chern-Simons in terms of Atiyah-Segal’s axioms.

To a point with positive orientation we associate the vector super-space (the space of states)

$$\mathcal{H}_{pt+} = \text{Fun}(\Pi \mathfrak{h}) \cong \mathbb{C}[\eta^1, \dots, \eta^m],$$

and to a point with negative orientation – the dual space

$$\mathcal{H}_{pt-} = (\mathcal{H}_{pt+})^* = \text{Fun}(\Pi \bar{\mathfrak{h}}) \cong \mathbb{C}[\bar{\eta}^1, \dots, \bar{\eta}^m].$$

To an interval $\mathcal{I} = [\mathfrak{p}_1, \mathfrak{p}_2]$ we associate the partition function

$$Z_{\mathcal{I}}^{\rho} := \rho(Z_{\mathcal{I}}) \in \text{Fun}(\Pi \mathfrak{g} \oplus \mathfrak{g}) \otimes \underbrace{\mathcal{H}_{\mathfrak{p}_2+} \otimes \mathcal{H}_{\mathfrak{p}_1-}}_{\cong \text{End}(\mathcal{H}_{pt+})}$$

given by formula (61) in representation ρ (see equation (74)). In general, to a one-dimensional simplicial complex Θ we associate the partition function (60) of section 3.2.1 taken in representation ρ :

$$Z_{\Theta}^{\rho} := \rho^{\otimes i(\Theta)} \circ Z_{\Theta} \in \text{Fun}(\mathcal{F}_{\Theta}^{\text{bulk}}) \otimes (\mathcal{H}_{pt+} \otimes \mathcal{H}_{pt-})^{\otimes i(\Theta)}.$$

The one-dimensional Chern-Simons theory features three types of operations:

- To a disjoint union $\Theta_1 \sqcup \Theta_2$ corresponds the tensor product for partition functions.
- To a sewing of boundary points $\mathfrak{p}_1^-$ and $\mathfrak{p}_2^+$ in a simplicial complex Θ corresponds the convolution of spaces of states $\mathcal{H}_{\mathfrak{p}_2+}$ and $\mathcal{H}_{\mathfrak{p}_1-}$.

- To a simplicial aggregation $r : \Theta \rightarrow \Theta'$ corresponds a fiber BV integral r_* which reduces the space of bulk fields from $\mathcal{F}_{\Theta}^{\text{bulk}}$ to $\mathcal{F}_{\Theta'}^{\text{bulk}}$.

In addition, $\mathcal{H}_{pt+}$ is equipped with an odd third-order differential operator

$$\delta^\rho = \rho(\hat{\theta}) : \mathcal{H}_{pt+} \rightarrow \mathcal{H}_{pt+},$$

and $\mathcal{H}_{pt-}$ is equipped with minus its dual $-(\delta^\rho)^* : \mathcal{H}_{pt-} \rightarrow \mathcal{H}_{pt-}$. The partition function Z_Θ^ρ satisfies the quantum master equation

$$(80) \quad (\hbar \Delta_\Theta^{\text{bulk}} + \hbar^{-1} \delta_\Theta^\rho) Z_\Theta^\rho = 0,$$

where the “boundary BV operator” δ_Θ^ρ is the sum over boundary points of Θ of operators δ^ρ or $-(\delta^\rho)^*$ acting on the corresponding $\mathcal{H}_{pt}$ (depending on whether the orientation of pt is positive or negative).

Remark 21. δ^ρ is “almost” a coboundary operator: its square is proportional to identity:

$$(81) \quad (\delta^\rho)^2 = -\frac{\hbar^3}{48} f^{abc} f^{abc} \cdot \text{id}_{\mathcal{H}_{pt+}}$$

(see [11]). This implies that the boundary BV operator for an interval

$$\delta_I^\rho = \delta^\rho \otimes \text{id}_{\mathcal{H}_{pt-}} - \text{id}_{\mathcal{H}_{pt+}} \otimes (\delta^\rho)^* : \underbrace{\mathcal{H}_{pt+} \otimes \mathcal{H}_{pt-}}_{\cong \text{End}(\mathcal{H}_{pt+})} \rightarrow \underbrace{\mathcal{H}_{pt+} \otimes \mathcal{H}_{pt-}}_{\cong \text{End}(\mathcal{H}_{pt+})}$$

squares to zero

$$(\delta_I^\rho)^2 = 0$$

Cases when δ^ρ squares to zero (*i.e.* when $f^{abc} f^{abc} = 0$) are quite interesting as then the reduced space of states for a point $\mathcal{H}_{pt+}^{\text{red}}$ emerges (see remarks 25, 34, 35 below).

Remark 22. The space of states $\mathcal{H}_{pt+}$ can be viewed as a geometric quantization of the classical phase super-space $\Pi \mathfrak{g}$ (viewed as an odd Kähler manifold). The operator δ^ρ is the quantization of the Maurer-Cartan element θ (4); the operator $\hbar^{-1} \delta_I^\rho$ is the quantization of the Hamiltonian vector field $\{\theta, \bullet\}$ on $\Pi \mathfrak{g}$.

Remark 23. Topological quantum mechanics (TQM) in the sense of A. Losev [12] assigns to an interval a manifold $Geom$ (the “space of geometric data”) and to a point — a vector superspace $\mathcal{H}$ endowed with an odd coboundary operator Q . The evolution operator U for an interval is a differential form on $Geom$ with values in $\text{End}(\mathcal{H})$ and has to satisfy the “homotopy topologicity” equation

$$(82) \quad (d + \text{ad}_Q) U = 0,$$

where d is the de Rham operator on $Geom$. A standard class of examples of TQMs comes from choosing $Geom = \mathbb{R}_{>0}$ (with coordinate $t > 0$) and setting

$$(83) \quad U(t, dt) = e^{[Q, G] t + dt G} = e^{(d + \text{ad}_Q) \circ (tG)}$$

where G is an odd operator on $\mathcal{H}$. For instance, for the Hodge TQM [12] on a Riemannian manifold M , one sets $\mathcal{H} = \Omega^\bullet(M)$, $Q = d_M$ and $G = d_M^*$ — the Hodge operator on forms on M . For the Morse TQM [17], [9], one takes the same $\mathcal{H}$ and Q , but now $G = \iota_v$ is the substitution of the gradient vector field. The one-dimensional Chern-Simons theory on an interval can be viewed as a TQM: here $Geom = \mathfrak{g}$ (with

coordinates A^a), $\mathcal{H} = \text{Fun}(\Pi\mathfrak{h})$, $Q = \hbar^{-1}\delta^\rho$. The odd Fourier transform in variable $\tilde{\psi}$ of the partition function for an interval (61) is

$$(84) \quad U(A, \lambda) = \rho \left(e^{-\frac{1}{2\hbar} f^{abc} \hat{\psi}^a A^b \hat{\psi}^c + \lambda^a \hat{\psi}^a} \right) = e^{(d + \hbar^{-1} \text{ad}_{\delta^\rho}) \circ \hbar^{-1} \rho(A^a \hat{\psi}^a)}$$

where $d = \hbar \lambda^a \frac{\partial}{\partial A^a}$ is the de Rham operator on Geom . Note that the expression (84) is similar to (83) where we make a substitution $tG \mapsto \hbar^{-1} \rho(A^a \hat{\psi}^a)$. The quantum master equation (68) is equivalent to

$$(85) \quad (d + \hbar^{-1} \text{ad}_{\delta^\rho}) U(A, \lambda) = 0$$

which is exactly the “homotopy topologicity” equation (82). The peculiarity of the one-dimensional Chern-Simons theory viewed as a TQM is that δ^ρ is not necessarily a coboundary operator on $\mathcal{H}$.

4.3. Integrating out the bulk fields. In section 3.2.2, we discussed simplicial aggregations which reduce the space of bulk fields of the 1-dimensional Chern-Simons theory $\mathcal{F}_\Theta^{\text{bulk}} \rightarrow \mathcal{F}_{\Theta'}^{\text{bulk}}$ according to combinatorial moves applied to the triangulation $\Theta \rightarrow \Theta'$. It is interesting to consider the “ultimate aggregation” — integrating out the bulk fields completely. This procedure should yield the partition function in the sense of Atiyah-Segal (*i.e.* without bulk fields). We will denote it by Z° .

For an interval, we have

$$(86) \quad (i\hbar)^m \int D\tilde{\psi} Z_{\mathcal{I}}(\tilde{\psi}, A) = e^{-\frac{1}{2\hbar} f^{abc} \hat{\psi}^a A^b \hat{\psi}^c}.$$

We view (86) as a BV integral over the Lagrangian subspace

$$(87) \quad \mathcal{L}_A = \{\tilde{\psi} + A \mid \tilde{\psi} \text{ is free, } A \text{ fixed}\} \subset \mathcal{F}_{\mathcal{I}}^{\text{bulk}}.$$

This subspace depends on the value of A , and integral (86) also depends on A . However, this dependence is $\text{ad}_{\hat{\theta}}$ -exact:

$$(88) \quad e^{-\frac{1}{2\hbar} f^{abc} \hat{\psi}^a (A + \delta A)^b \hat{\psi}^c} - e^{-\frac{1}{2\hbar} f^{abc} \hat{\psi}^a A^b \hat{\psi}^c} = \frac{1}{\hbar} [\hat{\theta}, e^{-\frac{1}{2\hbar} f^{abc} \hat{\psi}^a A^b \hat{\psi}^c + \frac{1}{\hbar} \delta A^a \hat{\psi}^a}]_{Cl(\mathfrak{g})} + \mathcal{O}((\delta A)^2)$$

(this can be checked analogously to the proof of lemma 4). Therefore, we should understand the partition function $Z_{\mathcal{I}}^\circ$ as an element of cohomology of the operator $\text{ad}_{\hat{\theta}}$ (the fact that (86) is $\text{ad}_{\hat{\theta}}$ -closed is an immediate consequence of the quantum master equation (68)). More exactly, $Z_{\mathcal{I}}^\circ$ is the class of Clifford unit $\hat{1}$ in $\text{ad}_{\hat{\theta}}$ -cohomology:

$$(89) \quad Z_{\mathcal{I}}^\circ = [\hat{1}] \in H_{\text{ad}_{\hat{\theta}}}(Cl(\mathfrak{g}))$$

Equivalently, in terms of representation ρ , we have

$$(90) \quad \rho(Z_{\mathcal{I}}^\circ) = [\text{id}_{\mathcal{H}_{pt+}}] \in H_{\delta_{\mathcal{I}}^\rho}(\text{End}(\mathcal{H}_{pt+})).$$

Remark 24. If the contraction of structure constants $f^{abc} f^{abc}$ for $\mathfrak{g}$ is nonzero, the cohomology class (89), (90) vanishes since

$$\hat{1} = -\frac{24\hbar^{-3}}{f^{abc} f^{abc}} \text{ad}_{\hat{\theta}} \hat{\theta}$$

In fact, whole cohomology group $H_{\text{ad}_{\hat{\theta}}} \cong H_{\delta_{\mathcal{I}}^\rho}$ vanishes since very $\text{ad}_{\hat{\theta}}$ -cocycle $\hat{\alpha} \in Cl(\mathfrak{g})$ is automatically exact:

$$\hat{\alpha} = -\frac{24\hbar^{-3}}{f_{abc}f_{abc}} \text{ad}_{\hat{\theta}}(\hat{\theta} \cdot \hat{\alpha}).$$

Remark 25. If $f^{abc}f^{abc} = 0$, we can define the reduced space of states for a point as the δ^ρ -cohomology:

$$\mathcal{H}_{pt^+}^{\text{red}} := H_{\delta^\rho}(\mathcal{H}_{pt^+}), \quad \mathcal{H}_{pt^-}^{\text{red}} := H_{-(\delta^\rho)^*}(\mathcal{H}_{pt^-}) = (\mathcal{H}_{pt^+}^{\text{red}})^*$$

By Künneth formula, we have

$$(91) \quad H_{\delta_{\mathcal{I}}^\rho}(\text{End}(\mathcal{H}_{pt^+})) \cong \mathcal{H}_{pt^+}^{\text{red}} \otimes \mathcal{H}_{pt^+}^{\text{red}}.$$

The partition function (90) is then represented by the identity operator

$$\rho(Z_{\mathcal{I}}^\circ) : \mathcal{H}_{pt^+}^{\text{red}} \xrightarrow{\text{id}} \mathcal{H}_{pt^+}^{\text{red}}.$$

For a circle, we can obtain the partition function $Z_{S^1}^\circ$ (which is just a number) either as a Clifford super-trace of (86) or as a BV integral of the effective action (13) over the Lagrangian subspace (87). Either way, we have

$$(92) \quad Z_{S^1}^\circ = 0$$

due to non-saturation of fermionic modes either in Clifford super-trace or in the Berezin integral over $\tilde{\psi}$.

4.4. From operator formalism to path integral.

4.4.1. *Abelian one-dimensional Chern-Simons theory.* The one-dimensional abelian Chern-Simons theory associates to an interval $\mathcal{I}$ the unit of $Cl(\mathfrak{g})$ (here $\mathfrak{g}$ can be viewed as a Euclidean vector space; the Lie algebra structure is irrelevant). The path integral arises upon applying the map (75) to this trivial partition function:

$$(93) \quad \Phi(\hat{1})(\eta_{out}, \bar{\eta}_{in}) = \Phi(\underbrace{\hat{1} \cdot \hat{1} \cdots \hat{1}}_N)(\eta_{out}, \bar{\eta}_{in}) = \\ = \int \left(\prod_{k=1}^{N-1} \hbar^m D\eta_k D\bar{\eta}_{k+1} \right) \cdot \\ \cdot \exp \frac{1}{\hbar} (\langle \eta_{out}, \bar{\eta}_N \rangle + \langle \bar{\eta}_N, \eta_{N-1} \rangle + \langle \eta_{N-1}, \bar{\eta}_{N-1} \rangle + \cdots + \langle \eta_2, \bar{\eta}_2 \rangle + \langle \bar{\eta}_2, \eta_1 \rangle + \langle \eta_1, \bar{\eta}_{in} \rangle) = \\ = \int \left(\prod_{k=1}^{N-1} \hbar^m D\eta_k D\bar{\eta}_{k+1} \right) \cdot \exp \frac{1}{\hbar} \left(\langle \eta_1, \bar{\eta}_{in} \rangle + \sum_{k=2}^N \langle \eta_k - \eta_{k-1}, \bar{\eta}_k \rangle \right).$$

For convenience, we set $\bar{\eta}_1 := \bar{\eta}_{in}$, $\eta_N := \eta_{out}$ and introduced a notation

$$\langle \eta, \bar{\eta} \rangle := \sum_q \eta^q \bar{\eta}^q.$$

In the exponential of (93), N terms of type $\langle \eta_k, \bar{\eta}_k \rangle$ correspond to Clifford units, and $N-1$ terms of type $\langle \bar{\eta}_{k+1}, \eta_k \rangle$ correspond to convolutions kernels as in (76); the symbol $D\eta_k D\bar{\eta}_{k+1}$ is defined as $\prod_p (D\eta_k^p D\bar{\eta}_{k+1}^p)$. Expression (93) corresponds to triangulating an interval by N smaller intervals; the terms in the exponential correspond to 0- and 1-simplices of this triangulation. In the limit $N \rightarrow \infty$, one formally writes (93) as a path integral over paths $\eta(\tau)$, $\bar{\eta}(\tau)$ with η at the right end-point and $\bar{\eta}$ at left end-point of $\mathcal{I}$ fixed by the boundary conditions.

Lemma 5.

$$(94) \quad \Phi(\hat{1})(\eta_{out}, \bar{\eta}_{in}) = \int_{\bar{\eta}(0)=\bar{\eta}_{in}, \eta(1)=\eta_{out}} \mathcal{D}\eta \mathcal{D}\bar{\eta} \cdot \exp \frac{1}{\hbar} \left(\langle \eta(0), \bar{\eta}(0) \rangle + \int_{\mathcal{I}} \langle d\eta, \bar{\eta} \rangle \right)$$

A perturbative computation of the path integral (94) is trivial: the integral is given by the contribution of the critical point

$$\eta(\tau) = \eta_{out}, \quad \bar{\eta}(\tau) = \bar{\eta}_{in} \quad \text{for all } \tau \in [0, 1]$$

and yields

$$\Phi(\hat{1})(\eta_{out}, \bar{\eta}_{in}) = e^{\frac{1}{\hbar} \langle \eta_{out}, \bar{\eta}_{in} \rangle}.$$

It is instructive to write the integral (94) in terms of the field ψ instead of fields $\eta, \bar{\eta}$. For simplicity, we first choose a complex polarization (as in (74))

$$(95) \quad \begin{cases} \psi^{2p-1} &= \frac{1}{\sqrt{2}}(\eta^p + \bar{\eta}^p) \\ \psi^{2p} &= \frac{i}{\sqrt{2}}(\eta^p - \bar{\eta}^p) \end{cases} \quad \Leftrightarrow \quad \begin{cases} \eta^p &= \frac{1}{\sqrt{2}}(\psi^{2p-1} - i\psi^{2p}) \\ \bar{\eta}^p &= \frac{1}{\sqrt{2}}(\psi^{2p-1} + i\psi^{2p}) \end{cases}$$

(this corresponds to the complex structure J on $\mathfrak{g}$ which assigns ψ^{2p-1} as “real” coordinates and ψ^{2p} as “imaginary” coordinates on $\Pi\mathfrak{g}$). We have,

$$\begin{aligned} \sum_p \eta_k^p \bar{\eta}_k^p &= \sum_p i \psi_k^{2p-1} \psi_k^{2p}, \\ \sum_p \bar{\eta}_{k+1}^p \eta_k^p &= \sum_p \frac{\psi_{k+1}^{2p-1} \psi_k^{2p-1} + \psi_{k+1}^{2p} \psi_k^{2p}}{2} - \sum_p i \frac{\psi_{k+1}^{2p-1} \psi_k^{2p} - \psi_{k+1}^{2p} \psi_k^{2p-1}}{2}. \end{aligned}$$

Substituting these expressions into the integral representation (93) for $\Phi(1)$, we obtain

$$(96) \quad \begin{aligned} \Phi(\hat{1})(\eta_{out}, \bar{\eta}_{in}) &= \int \left(\hbar^{m/2} D\eta_1 \right) \left(\prod_{k=2}^{N-1} (i\hbar)^m D\psi_k \right) \left(\hbar^{m/2} D\bar{\eta}_N \right) \cdot \\ &\cdot \exp \frac{1}{\hbar} \left(\sum_{k=1}^{N-1} \frac{1}{2} (\psi_{k+1}, \psi_k) + \sum_p \sum_{k=1}^{N-1} \frac{i}{2} (\psi_{k+1}^{2p-1} - \psi_k^{2p-1})(\psi_{k+1}^{2p} - \psi_k^{2p}) + \right. \\ &\quad \left. + \sum_p \frac{i}{2} \psi_1^{2p-1} \psi_1^{2p} + \sum_p \frac{i}{2} \psi_N^{2p-1} \psi_N^{2p} \right). \end{aligned}$$

Here $D\psi_k := \overleftarrow{\prod}_a D\psi_k^a$ is the Berezin measure on $\Pi\mathfrak{g}$; the variable ψ_1 is constructed by formulae (95) from the integration variable η_1 and the boundary value $\bar{\eta}_1 := \bar{\eta}_{in}$, and ψ_N is constructed from the integration variable $\bar{\eta}_N$ and the boundary value $\eta_N := \eta_{out}$.

Remark 26. We think of integral (93) as corresponding to cutting the interval $\mathcal{I} = [\mathbf{p}_{in}, \mathbf{p}_{out}]$ into N intervals $[\mathbf{p}_{in}, \mathbf{p}_2] \cup [\mathbf{p}_2, \mathbf{p}_3] \cup \dots \cup [\mathbf{p}_N, \mathbf{p}_{out}]$. Integration variables $\eta_k, \bar{\eta}_{k+1}$ are associated to the point $\mathbf{p}_{k+1}$ (more specifically, to the right end of interval $[\mathbf{p}_k, \mathbf{p}_{k+1}]$ and to the left end of the interval $[\mathbf{p}_{k+1}, \mathbf{p}_{k+2}]$, respectively); the boundary value $\bar{\eta}_{in}$ corresponds to the point $\mathbf{p}_{in}$, the boundary value η_{out} — to the point $\mathbf{p}_{out}$. However, variables ψ_k are linear combinations of η_k and $\bar{\eta}_k$. Thus, they are not associated to any single point, but rather to a pair of neighboring points $(\mathbf{p}_k, \mathbf{p}_{k+1})$.

Again, we formally write the limit $N \rightarrow \infty$ of the integral (96) as a path integral over paths $\psi : \mathcal{I} \rightarrow \Pi \mathfrak{g}$ with a fixed anti-holomorphic projection of ψ at the right end-point of the interval and a fixed holomorphic projection at the left end-point:

Lemma 6. *The path integral expression for the partition function of the abelian Chern-Simons theory on an interval is given by*

$$(97) \quad \Phi(\hat{1})(\eta_{out}, \bar{\eta}_{in}) = \int_{\bar{\pi}(\psi(0))=\bar{\eta}_{in}, \pi(\psi(1))=\eta_{out}} \mathcal{D}\psi \cdot \exp \frac{1}{\hbar} \left(\sum_p \frac{i}{2} \psi^{2p-1}(0) \psi^{2p}(0) + \int_{\mathcal{I}} \frac{1}{2} (\psi, d\psi) + \sum_p \frac{i}{2} \psi^{2p-1}(1) \psi^{2p}(1) \right).$$

The exact meaning of the conditional measure on paths in (97) is the formal $N \rightarrow \infty$ limit of the measure in (96). The second term in the exponential in (96) does not contribute to the limit $N \rightarrow \infty$: once we assume that ψ_k are values of a differentiable path $\psi(\tau)$ at times $\tau = k/N$, the contribution of this term becomes of order $\mathcal{O}(1/N)$.

For a general complex structure J on $\mathfrak{g}$, path integral (97) becomes

$$\begin{aligned} & \Phi(\hat{1})(\eta_{out}, \bar{\eta}_{in}) = \\ & = \int_{\bar{\pi}(\psi(0))=\bar{\eta}_{in}, \pi(\psi(1))=\eta_{out}} \mathcal{D}\psi \cdot \exp \frac{1}{\hbar} \left(\frac{i}{4} (\psi(0), J\psi(0)) + \int_{\mathcal{I}} \frac{1}{2} (\psi, d\psi) + \frac{i}{4} (\psi(1), J\psi(1)) \right). \end{aligned}$$

4.4.2. *Path integral for the non-abelian one-dimensional Chern-Simons theory in the cyclic Whitney gauge. End of proof of theorem 2.* To obtain a path integral representation for the partition function of the one-dimensional Chern-Simons theory on an interval (61) we use the same strategy as in section 4.4.1: we cut the interval into N smaller intervals and then apply the map Φ (75). The new point here is that for small intervals we have to use the “heat kernel” approximation which gives an exact result only in the limit $N \rightarrow \infty$.

Applying Φ to $Z_{\mathcal{I}}$ (61), we have

$$\begin{aligned} (98) \quad & \Phi(Z_{\mathcal{I}})(\eta_{out}, \bar{\eta}_{in}) = \int (i\hbar)^{-m} D\lambda \cdot e^{-(\lambda, \tilde{\psi})} \cdot \Phi \left(\exp \left(-\frac{1}{2\hbar} (\hat{\psi}, [A, \hat{\psi}]) + (\lambda, \hat{\psi}) \right) \right) \eta_{out}, \bar{\eta}_{in} = \\ & = \int (i\hbar)^{-m} D\lambda \cdot e^{-(\lambda, \tilde{\psi})} \cdot \Phi \left(\left(\exp \left(-\frac{1}{2\hbar N} (\hat{\psi}, [A, \hat{\psi}]) + \frac{1}{N} (\lambda, \hat{\psi}) \right) \right)^N \right) (\eta_{out}, \bar{\eta}_{in}) = \\ & = \int (i\hbar)^{-m} D\lambda \cdot e^{-(\lambda, \tilde{\psi})} \int \left(\prod_{k=1}^{N-1} \hbar^m D\eta_k D\bar{\eta}_{k+1} \right) \cdot \\ & \cdot \Phi \left(\exp \left(-\frac{1}{2\hbar N} (\hat{\psi}, [A, \hat{\psi}]) + \frac{1}{N} (\lambda, \hat{\psi}) \right) \right) (\eta_{out}, \bar{\eta}_N) \cdot e^{\frac{1}{\hbar} \langle \bar{\eta}_N, \eta_{N-1} \rangle} \dots \\ & \dots e^{\frac{1}{\hbar} \langle \bar{\eta}_2, \eta_1 \rangle} \cdot \Phi \left(\exp \left(-\frac{1}{2\hbar N} (\hat{\psi}, [A, \hat{\psi}]) + \frac{1}{N} (\lambda, \hat{\psi}) \right) \right) (\eta_1, \bar{\eta}_{in}). \end{aligned}$$

Next, we need to evaluate the partition function for a small interval in the limit $N \rightarrow \infty$:

$$(99) \quad \Phi \left(\exp \left(-\frac{1}{2\hbar N} (\hat{\psi}, [A, \hat{\psi}]) + \frac{1}{N} (\lambda, \hat{\psi}) \right) \right) (\eta, \bar{\eta}) =$$

$$\begin{aligned}
&= \Phi \left(\hat{1} - \frac{1}{2\hbar N} (\hat{\psi}, [A, \hat{\psi}]) + \frac{1}{N} (\lambda, \hat{\psi}) + \mathcal{O} \left(\frac{1}{N^2} \right) \right) (\eta, \bar{\eta}) = \\
&= e^{\frac{1}{\hbar} \langle \eta, \bar{\eta} \rangle} \left(1 - \frac{1}{2\hbar N} (\psi, [A, \psi]) + \frac{1}{N} (\lambda, \psi) + \frac{i}{4N} \text{tr} (J \cdot \text{ad}_A) + \mathcal{O} \left(\frac{1}{N^2} \right) \right).
\end{aligned}$$

Here ψ is a linear combination of $\eta, \bar{\eta}$, prescribed by the choice of a complex structure J (e.g. (95)); the term with a trace appeared due to the following identity:

$$\Phi(\hat{\psi}^a \hat{\psi}^b)(\eta, \bar{\eta}) = e^{\frac{1}{\hbar} \langle \eta, \bar{\eta} \rangle} \left(\psi^a \psi^b + \frac{\hbar}{2} \delta^{ab} + \frac{i\hbar}{2} J^{ab} \right).$$

Here the third term generates the trace term in (99). Substituting the “heat kernel” asymptotics (99) into (98), we get

$$\begin{aligned}
(100) \quad \Phi(Z_{\mathcal{I}})(\eta_{out}, \bar{\eta}_{in}) &= \\
&= \int \left(\prod_{k=1}^{N-1} \hbar^m D\eta_k D\bar{\eta}_{k+1} \right) \int (i\hbar)^{-m} D\lambda \cdot e^{(\lambda, \frac{1}{N} \sum_{k=1}^N \psi_k - \tilde{\psi})} \cdot \\
&\cdot e^{\frac{i}{4} \text{tr}(J \cdot \text{ad}_A)} \cdot e^{\frac{1}{\hbar} (\langle \eta_{out}, \bar{\eta}_N \rangle + \langle \bar{\eta}_N, \eta_{N-1} \rangle + \dots + \langle \bar{\eta}_2, \eta_1 \rangle + \langle \eta_1, \bar{\eta}_{in} \rangle)} \cdot e^{-\frac{1}{2\hbar N} \sum_{k=1}^N (\psi_k, [A, \psi_k])} + \mathcal{O} \left(\frac{1}{N} \right).
\end{aligned}$$

Taking the limit $N \rightarrow \infty$, we obtain the following.

Proposition 4. *The Chern-Simons partition function for an interval is given by the path integral:*

$$\begin{aligned}
(101) \quad \Phi(Z_{\mathcal{I}})(\eta_{out}, \bar{\eta}_{in}) &= e^{\frac{i}{4} \text{tr}(J \cdot \text{ad}_A)} \int_{\pi(\psi(0))=\bar{\eta}_{in}, \pi(\psi(1))=\eta_{out}, \int_{\mathcal{I}} d\tau \psi(\tau)=\tilde{\psi}} \mathcal{D}\psi \cdot \\
&\cdot \exp \frac{1}{\hbar} \left(\int_{\mathcal{I}} \frac{1}{2} (\psi, (d + d\tau \cdot \text{ad}_A) \psi) + \frac{i}{4} (\psi(0), J\psi(0)) + \frac{i}{4} (\psi(1), J\psi(1)) \right)
\end{aligned}$$

The conditional measure on paths $\psi(\tau)$ with a fixed holomorphic projection at $\tau = 1$, a fixed anti-holomorphic projection at $\tau = 0$ and with a fixed integral over τ in (101) is the $N \rightarrow \infty$ limit of the measure in (100).

Applying concatenation formulae (76), (77) to (101), we obtain a path integral representation of the Chern-Simons partition function for one-dimensional simplicial complexes:

Corollary 2. *For a triangulated interval $\Theta = [\mathbf{p}_{in}, \mathbf{p}_2] \cup [\mathbf{p}_2, \mathbf{p}_3] \cup \dots \cup [\mathbf{p}_n, \mathbf{p}_{out}]$ we have*

$$\begin{aligned}
(102) \quad \Phi(Z_{\Theta})(\eta_{out}, \bar{\eta}_{in}) &= e^{\sum_{k=1}^n \frac{i}{4} \text{tr}(J \cdot \text{ad}_{A_k})} \int_{\pi(\psi(\mathbf{p}_{in}))=\bar{\eta}_{in}, \pi(\psi(\mathbf{p}_{out}))=\eta_{out}, \int_{\mathbf{p}_k}^{\mathbf{p}_{k+1}} d\tau \psi(\tau)=\tilde{\psi}_k} \mathcal{D}\psi \cdot \\
&\cdot \exp \frac{1}{\hbar} \left(\int_{\mathcal{I}} \frac{1}{2} (\psi, (d + d\tau \cdot \text{ad}_A) \psi) + \frac{i}{4} (\psi(\mathbf{p}_{in}), J\psi(\mathbf{p}_{in})) + \frac{i}{4} (\psi(\mathbf{p}_{out}), J\psi(\mathbf{p}_{out})) \right).
\end{aligned}$$

For a triangulated circle $\Xi_n = [\mathbf{p}_1, \mathbf{p}_2] \cup \dots \cup [\mathbf{p}_{n-1}, \mathbf{p}_n] \cup [\mathbf{p}_n, \mathbf{p}_1]$, we obtain:

$$(103) \quad Z_{\Xi_n} = e^{\sum_{k=1}^n \frac{i}{4} \text{tr}(J \cdot \text{ad}_{A_k})} \int_{\int_{\mathbf{p}_k}^{\mathbf{p}_{k+1}} d\tau \psi(\tau)=\tilde{\psi}_k} \mathcal{D}\psi \cdot e^{\frac{1}{2\hbar} \int_{\mathcal{I}} (\psi, (d + d\tau \cdot \text{ad}_A) \psi)}.$$

Remark 27. Expression (103) returns us to the “naïve” path-integral (37) for the simplicial Chern-Simons on a circle, up to a somewhat puzzling factor $e^{\sum_k \frac{i}{4} \text{tr}(J \cdot \text{ad}_{A_k})}$. The explanation is as follows: the path integral in (103) was obtained from the path integral with boundaries (102) by concatenation formula (77). Hence, it is secretly using the normal ordering prescribed by the choice of a complex structure J on $\mathfrak{g}$ (which dictates the regularization for the one-loop determinant in (103)). This implicit dependence on J is exactly cancelled by the factor $e^{\sum_k \frac{i}{4} \text{tr}(J \cdot \text{ad}_{A_k})}$ (indeed, we know that the left hand side of (103) is defined in terms of the Clifford algebra $Cl(\mathfrak{g})$ and therefore cannot possibly depend on J). For the naïve path integral (37), we implicitly assumed the symmetric normal ordering by making a regularization (41) in our computation of the one-loop determinant (the important point is that $\theta(0)$ is a number, and not a matrix).

Path integral representation (103) returns us to perturbative computation of section 2.3 and thus finishes the proof of theorem 2.

Remark 28. It is easy to compute (101) in the case of $A = 0$:

$$(104) \quad \Phi(Z_{\mathcal{I}}|_{A=0})(\eta_{out}, \bar{\eta}_{in}) = 2^{-m} e^{\frac{1}{\hbar}(\langle \eta_{out}, \bar{\eta}_{in} \rangle - 2\langle \eta_{out} - \bar{\eta}, \bar{\eta}_{in} - \bar{\eta} \rangle)},$$

where $\bar{\eta}, \bar{\bar{\eta}}$ are holomorphic and anti-holomorphic components of the bulk field $\tilde{\psi}$. We can also write (104) as

$$\Phi(Z_{\mathcal{I}}|_{A=0})(\eta_{out}, \bar{\eta}_{in}) = 2^{-m} e^{\frac{1}{\hbar}(\frac{i}{2}(\psi_{bd}, J\psi_{bd}) - i(\psi_{bd} - \tilde{\psi}, J(\psi_{bd} - \tilde{\psi})))},$$

where $\psi_{bd} = \iota(\eta_{out}) + \bar{\iota}(\bar{\eta}_{in})$ is the linear combination of boundary fields $\eta_{out}, \bar{\eta}_{in}$.

4.5. Simplicial action on an interval.

Proposition 5. *The path integral for the Chern-Simons partition function on an interval (101) is given by*

$$(105) \quad \Phi(Z_{\mathcal{I}}(\tilde{\psi}, A))(\eta_{out}, \bar{\eta}_{in}) = \det_{\mathfrak{g}}^{1/2} \left(\frac{\sinh \frac{\text{ad}_A}{2}}{\frac{\text{ad}_A}{2}} \right) \cdot \det_{\mathfrak{g}}^{-1/2} M(\text{ad}_A) \cdot \exp \frac{1}{\hbar} \left(\langle \eta_{out}, \bar{\eta}_{in} \rangle - \frac{1}{2}(\tilde{\psi}, \text{ad}_A \tilde{\psi}) + \frac{1}{2} \begin{pmatrix} \bar{\eta} - \eta_{out} & \bar{\eta} - \bar{\eta}_{in} \end{pmatrix} \cdot M(\text{ad}_A) \cdot \begin{pmatrix} \tilde{\eta} - \eta_{out} \\ \tilde{\eta} - \bar{\eta}_{in} \end{pmatrix} \right),$$

where the bilinear form $M(\text{ad}_A)$ in basis $(\eta, \bar{\eta})$ is represented by the block matrix

$$(106) \quad M(\text{ad}_A) = \begin{pmatrix} R_{-+} R_{++}^{-1} & -1 - R_{--} + R_{-+} R_{++}^{-1} R_{+-} \\ 1 + R_{++}^{-1} & R_{++}^{-1} R_{+-} \end{pmatrix}.$$

Here symbols $R_{\pm\pm}$ stand for blocks of $R(\text{ad}_A)$ (defined by formula (31)) in the basis $(\eta, \bar{\eta})$:

$$(107) \quad R(\text{ad}_A) = \begin{pmatrix} R_{++} & R_{+-} \\ R_{-+} & R_{--} \end{pmatrix}.$$

Proof. The path integral (101) is Gaussian with a critical point $\psi^{cr}(\tau)$ being the solution of

$$(108) \quad (d + d\tau \text{ad}_A)\psi^{cr} = \text{const}$$

subject to conditions

$$(109) \quad \int d\tau \psi^{cr}(\tau) = \tilde{\psi},$$

$$(110) \quad \pi(\psi^{cr}(1)) = \eta_{out},$$

$$(111) \quad \bar{\pi}(\psi^{cr}(0)) = \bar{\eta}_{in}.$$

Equation (108) together with (109) gives

$$(112) \quad \psi^{cr}(\tau) = \tilde{\psi} + F(\text{ad}_A, \tau)(\psi^{cr}(0) - \tilde{\psi}),$$

where $F(\text{ad}_A, \tau)$ is defined by (30). The value of the action (together with boundary terms) for this path is

$$(113) \quad S(\psi^{cr}) + \text{boundary terms} = \\ = -\frac{1}{2}(\tilde{\psi}, \text{ad}_A \tilde{\psi}) - \underbrace{\frac{1}{2}(\tilde{\psi}, \psi^{cr}(1) - \psi^{cr}(0))}_{\frac{1}{2} \int (\psi^{cr}, d_A \psi^{cr})} + \frac{1}{2}\langle \eta^{cr}(0), \bar{\eta}_{in} \rangle + \frac{1}{2}\langle \eta_{out}, \bar{\eta}^{cr}(1) \rangle.$$

Boundary values of ψ^{cr} can be found as follows: (112) implies that

$$\psi^{cr}(1) = (1 + R(\text{ad}_A))\tilde{\psi} - R(\text{ad}_A)\psi^{cr}(0).$$

Solving equation this together with (110) and (111) in coordinates $(\eta, \bar{\eta})$, we obtain

$$(114) \quad \begin{pmatrix} \eta_{out} \\ \bar{\eta}^{cr}(1) \end{pmatrix} = \begin{pmatrix} 1 + R_{++} & R_{+-} \\ R_{-+} & 1 + R_{--} \end{pmatrix} \begin{pmatrix} \tilde{\eta} \\ \bar{\tilde{\eta}} \end{pmatrix} - \begin{pmatrix} R_{++} & R_{+-} \\ R_{-+} & R_{--} \end{pmatrix} \begin{pmatrix} \eta^{cr}(0) \\ \bar{\eta}_{in} \end{pmatrix}.$$

Here $\eta^{cr}(0)$ and $\bar{\eta}^{cr}(1)$ are the unknowns. Solving (114), we get

$$\begin{aligned} \eta^{cr}(0) &= \eta_{out} + (1 + R_{++}^{-1})(\tilde{\eta} - \eta_{out}) + R_{++}^{-1}R_{+-}(\bar{\tilde{\eta}} - \bar{\eta}_{in}), \\ \bar{\eta}^{cr}(1) &= \bar{\eta}_{in} - R_{-+}R_{++}^{-1}(\tilde{\eta} - \eta_{out}) - (-1 - R_{--} + R_{-+}R_{++}^{-1}R_{+-})(\bar{\tilde{\eta}} - \bar{\eta}_{in}). \end{aligned}$$

By substituting into (113), we obtain the exponential in (105).

The pre-exponential in (105) can be derived as follows. We know that it is a function of A only (since it is a square root of a functional determinant of the operator¹³ d_A acting on functions with vanishing integral, vanishing holomorphic part at $\tau = 1$ and vanishing anti-holomorphic part at $\tau = 0$); let us denote it by $G(A)$. Closing the interval into a circle (by using (77)), we find

$$\int \hbar^m D\eta_{out} D\bar{\eta}_{in} e^{\frac{1}{\hbar}\langle \bar{\eta}_{in}, \eta_{out} \rangle} \cdot \Phi(Z_{\mathcal{I}}(\tilde{\psi}, A))(\eta_{out}, \bar{\eta}_{in}) = \det_{\mathfrak{g}}^{1/2} M(\text{ad}_A) \cdot G(A) \cdot e^{-\frac{1}{2\hbar}(\tilde{\psi}, \text{ad}_A \tilde{\psi})}.$$

Comparing with the known result for a circle (13), we obtain the pre-exponential in (105). $\square$

Remark 29. In this computation, we neglected the factor of $e^{\frac{i}{4}\text{tr}(J \cdot \text{ad}_A)}$. If we did not, it would anyway be cancelled by the pre-exponential obtained by comparison with (13) (and this result is obtained by an explicit computation in operator formalism in section 3.1.3).

We can define the simplicial Chern-Simons action for the interval $S_{\mathcal{I}}(\tilde{\psi}, A; \eta_{out}, \bar{\eta}_{in})$ by

$$e^{\frac{1}{\hbar} S_{\mathcal{I}}(\tilde{\psi}, A; \eta_{out}, \bar{\eta}_{in})} = \Phi(Z_{\mathcal{I}}(\tilde{\psi}, A))(\eta_{out}, \bar{\eta}_{in}),$$

or explicitly

$$(115) \quad S_{\mathcal{I}}(\tilde{\psi}, A; \eta_{out}, \bar{\eta}_{in}) =$$

¹³More exactly, the determinant of a matrix of the bilinear form $\int(\bullet, d_A \bullet)$.

$$\begin{aligned}
&= \langle \eta_{out}, \bar{\eta}_{in} \rangle - \frac{1}{2}(\tilde{\psi}, \text{ad}_A \tilde{\psi}) + \frac{1}{2} \begin{pmatrix} \tilde{\eta} - \eta_{out} & \tilde{\eta} - \bar{\eta}_{in} \end{pmatrix} \cdot M(\text{ad}_A) \cdot \begin{pmatrix} \tilde{\eta} - \eta_{out} \\ \tilde{\eta} - \bar{\eta}_{in} \end{pmatrix} + \\
&\quad + \frac{\hbar}{2} \text{tr}_{\mathfrak{g}} \log \left(\frac{\sinh \frac{\text{ad}_A}{2}}{\frac{\text{ad}_A}{2}} \right) - \frac{\hbar}{2} \text{tr}_{\mathfrak{g}} \log M(\text{ad}_A).
\end{aligned}$$

Remark 30. Expansion of (115) as a power series in A starts as

$$\begin{aligned}
(116) \quad S_{\mathcal{I}}(\tilde{\psi}, A; \eta_{out}, \bar{\eta}_{in}) &= \\
&= \frac{i}{2}(\psi_{bd}, J\psi_{bd}) - i(\psi_{bd} - \tilde{\psi}, J(\psi_{bd} - \tilde{\psi})) - \frac{1}{2}(\tilde{\psi}, \text{ad}_A \tilde{\psi}) - \frac{1}{6}(\psi_{bd} - \tilde{\psi}, J \text{ad}_A J(\psi_{bd} - \tilde{\psi})) + \mathcal{O}(A^2) - \\
&\quad - \hbar m \log 2 + \hbar \frac{i}{12} \text{tr}(J \text{ad}_A) + \mathcal{O}(\hbar A^2),
\end{aligned}$$

where $\psi_{bd} = \iota(\eta_{out}) + \bar{\iota}(\bar{\eta}_{in})$.

We can obtain the action for a simplicial complex Θ by gluing actions (115) for individual intervals using concatenation formulae (76), (77). E.g. for a triangulated interval $\Theta = [\mathfrak{p}_{in}, \mathfrak{p}_2] \cup [\mathfrak{p}_2, \mathfrak{p}_3] \cup \dots \cup [\mathfrak{p}_n, \mathfrak{p}_{out}]$ we have

$$\begin{aligned}
(117) \quad e^{\frac{1}{\hbar} S_{\Theta}(\tilde{\psi}_1, A_1, \dots, \tilde{\psi}_n, A_n; \eta_{out}, \bar{\eta}_{in})} &= \int \prod_{k=1}^{n-1} (\hbar^m D\eta_k D\bar{\eta}_{k+1}) \cdot \\
&\cdot \exp \frac{1}{\hbar} \left(S_{\mathcal{I}}(\tilde{\psi}_n, A_n; \eta_{out}, \bar{\eta}_n) + \langle \bar{\eta}_n, \eta_{n-1} \rangle + \dots + \langle \bar{\eta}_2, \eta_1 \rangle + S_{\mathcal{I}}(\tilde{\psi}_1, A_1; \eta_1, \bar{\eta}_{in}) \right).
\end{aligned}$$

For a triangulated circle $\Xi_n = [\mathfrak{p}_1, \mathfrak{p}_2] \cup \dots \cup [\mathfrak{p}_{n-1}, \mathfrak{p}_n] \cup [\mathfrak{p}_n, \mathfrak{p}_1]$, we have

$$(118) \quad e^{\frac{1}{\hbar} S_{\Xi_n}(\tilde{\psi}_1, A_1, \dots, \tilde{\psi}_n, A_n)} = \int \prod_{k=1}^n (\hbar^m D\eta_k D\bar{\eta}_{k+1}) \cdot \exp \frac{1}{\hbar} \sum_{k=1}^n \left(S_{\mathcal{I}}(\tilde{\psi}_k, A_k; \eta_k, \bar{\eta}_k) + \langle \bar{\eta}_{k+1}, \eta_k \rangle \right).$$

Remark 31. Looking at formulae (117), (118), it is tempting to identify $\langle \bar{\eta}_{k+1}, \eta_k \rangle$ as a simplicial action for the point $\mathfrak{p}_{k+1}$.

Remark 32. Formula (118) explains how the simplicially non-local expression (42) is produced from a simplicially local expression (the sum of contributions of individual intervals — the integrand in (118)). The key is integration over boundary fields $\{\eta_k, \bar{\eta}_k\}$.

Let us introduce the notation $f_{\pm\pm\pm}^{pqr}$ for structure constants¹⁴ of $\mathfrak{g}$ in the basis $(\eta, \bar{\eta})$:

$$(119) \quad \theta = \frac{1}{6} f^{abc} \psi^a \psi^b \psi^c = \frac{1}{6} f_{+++}^{pqr} \eta^p \eta^q \eta^r + \frac{1}{2} f_{++-}^{pqr} \eta^p \eta^q \bar{\eta}^r + \frac{1}{2} f_{+-}^{pqr} \eta^p \bar{\eta}^q \bar{\eta}^r + \frac{1}{6} f_{---}^{pqr} \bar{\eta}^p \bar{\eta}^q \bar{\eta}^r.$$

(this is the same θ as in (4) rewritten in holomorphic-antiholomorphic coordinates on $\Pi\mathfrak{g}$). Then, the quantum master equation (68) for the action (115) has the following form:

$$(120) \quad \hbar \frac{\partial}{\partial \tilde{\psi}^a} \frac{\partial}{\partial A^a} e^{\frac{1}{\hbar} S_{\mathcal{I}}(\tilde{\psi}, A; \eta, \bar{\eta})} + \frac{1}{\hbar} \left(\frac{1}{6} f_{+++}^{pqr} \eta^p \eta^q \eta^r + \frac{\hbar}{2} f_{++-}^{pqr} \eta^p \eta^q \frac{\partial}{\partial \eta^r} - \frac{\hbar}{2} f_{+-}^{pqr} \eta^p \frac{\partial}{\partial \eta^r} + \right.$$

¹⁴Here we mean the structure constants of the cyclic operation $(\bullet, [\bullet, \bullet]) : \wedge^3 \mathfrak{g} \rightarrow \mathbb{R}$.

$$\begin{aligned}
& + \frac{\hbar^2}{2} f_{+--}^{pqr} \eta^p \frac{\partial}{\partial \eta^q} \frac{\partial}{\partial \eta^r} - \frac{\hbar^2}{2} f_{+--}^{ppr} \frac{\partial}{\partial \eta^r} + \frac{\hbar^3}{6} f_{+--}^{pqr} \frac{\partial}{\partial \eta^p} \frac{\partial}{\partial \eta^q} \frac{\partial}{\partial \eta^r} \Big) e^{\frac{1}{\hbar} S_{\mathcal{I}}(\tilde{\psi}, A; \eta, \bar{\eta})} - \\
& - e^{\frac{1}{\hbar} S_{\mathcal{I}}(\tilde{\psi}, A; \eta, \bar{\eta})} \frac{1}{\hbar} \left(\frac{\hbar^3}{6} f_{+++}^{pqr} \frac{\overleftarrow{\partial}}{\partial \bar{\eta}^p} \frac{\overleftarrow{\partial}}{\partial \bar{\eta}^q} \frac{\overleftarrow{\partial}}{\partial \bar{\eta}^r} + \frac{\hbar^2}{2} f_{+++}^{pqr} \frac{\overleftarrow{\partial}}{\partial \bar{\eta}^p} \frac{\overleftarrow{\partial}}{\partial \bar{\eta}^q} \bar{\eta}^r - \frac{\hbar^2}{2} f_{+++}^{pqq} \frac{\overleftarrow{\partial}}{\partial \bar{\eta}^p} + \right. \\
& \quad \left. + \frac{\hbar}{2} f_{+--}^{pqr} \frac{\overleftarrow{\partial}}{\partial \bar{\eta}^p} \bar{\eta}^q \bar{\eta}^r - \frac{\hbar}{2} f_{+--}^{ppr} \bar{\eta}^r + \frac{\hbar^3}{6} f_{+--}^{pqr} \bar{\eta}^p \bar{\eta}^q \bar{\eta}^r \right) = 0.
\end{aligned}$$

Remark 33. The one-dimensional BF theory is a special case of the one-dimensional Chern-Simons theory where the complex polarization (78) is compatible with the Lie algebra structure on $\mathfrak{g}$. In more detail, let $\mathfrak{h}$ in (78) be a Lie subalgebra, and let the Lie algebra structure on $\mathfrak{g}$ be given by a semidirect product of $\mathfrak{h}$ with its coadjoint module $\bar{\mathfrak{h}}$:

$$\mathfrak{g} = \mathfrak{h} \ltimes \bar{\mathfrak{h}}.$$

In this case, formula (115) for $S_{\mathcal{I}}$ simplifies: the block R_{+-} in (107) vanishes, and the matrix $M(\text{ad}_A)$ (106) becomes

$$M(\text{ad}_A) = \begin{pmatrix} R_{-+} R_{++}^{-1} & -1 - R_{--} \\ 1 + R_{++}^{-1} & 0 \end{pmatrix}, \quad \det_{\mathfrak{g}}^{1/2} M(\text{ad}_A) = \det_{\mathfrak{h}}(1 + R_{++}^{-1}).$$

In (119), only the second term on the right hand side survives:

$$\theta = \frac{1}{2} F_{pq}^r \eta^p \eta^q \bar{\eta}^r.$$

(Here F_{pq}^r are the structure constants of $\mathfrak{h}$; we distinguish between upper and lower indices to emphasize that we do not assume that $\mathfrak{h}$ comes with a pairing). So, the quantum master equation (120) is simplified:

$$\begin{aligned}
(121) \quad & \hbar \frac{\partial}{\partial \tilde{\psi}^a} \frac{\partial}{\partial A^a} e^{\frac{1}{\hbar} S_{\mathcal{I}}(\tilde{\psi}, A; \eta, \bar{\eta})} + \frac{1}{\hbar} \left(\frac{\hbar}{2} F_{pq}^r \eta^p \eta^q \frac{\partial}{\partial \eta^r} - \frac{\hbar}{2} F_{pq}^q \eta^p \right) e^{\frac{1}{\hbar} S_{\mathcal{I}}(\tilde{\psi}, A; \eta, \bar{\eta})} - \\
& - e^{\frac{1}{\hbar} S_{\mathcal{I}}(\tilde{\psi}, A; \eta, \bar{\eta})} \frac{1}{\hbar} \left(\frac{\hbar^2}{2} F_{pq}^r \frac{\overleftarrow{\partial}}{\partial \bar{\eta}^p} \frac{\overleftarrow{\partial}}{\partial \bar{\eta}^q} \bar{\eta}^r - \frac{\hbar^2}{2} F_{pq}^q \frac{\overleftarrow{\partial}}{\partial \bar{\eta}^p} \right) = 0.
\end{aligned}$$

(If in addition $\mathfrak{h}$ is unimodular, the last terms in brackets vanish.) Note that the result for BF theory that we obtain from (115) cannot be directly compared to the result in [14] as the choice of gauge fixing is very different¹⁵.

Remark 34. Another interesting point about the BF case is that $f^{abc} f^{abc} = 0$. Hence, the operator $\delta^\rho : \mathcal{H}_{pt+} \rightarrow \mathcal{H}_{pt+}$ becomes a coboundary operator. If we assume in addition that $\mathfrak{h}$ is unimodular, then $(\mathcal{H}_{pt+}, \delta^\rho)$ can be identified with the Chevalley-Eilenberg complex of Lie algebra $\mathfrak{h}$. Thus, the reduced space of states associated to a point (see remark 25) is the Chevalley-Eilenberg cohomology of $\mathfrak{h}$:

$$\mathcal{H}_{pt+}^{\text{red}} \cong H_{CE}(\mathfrak{h}).$$

¹⁵Indeed, here we fix the field A to be constant on the interval, and we fix the integral $\tilde{\psi}$ of field ψ over the interval, and the holomorphic and anti-holomorphic projections of ψ at the right and left end-points of the interval. The gauge used in [14] fixes $\pi(A)$ to be constant, $\bar{\pi}(A)$ to be a sum of delta-functions at the ends of the interval; and it fixes the values $\pi(\psi)$ at the ends of the interval and the integral for $\bar{\pi}(\psi)$. The latter gauge choice features better simplicial locality properties, but is only $\mathfrak{h}$ -equivariant.

Therefore, the cohomology space

$$(122) \quad H_{\text{ad}_{\mathfrak{g}}}(Cl(\mathfrak{g})) \cong H_{\delta_{\mathcal{I}}}(\text{End}(\mathcal{H}_{pt+})) \cong H_{CE}(\mathfrak{h}) \otimes (H_{CE}(\mathfrak{h}))^*$$

becomes non-trivial. In this case, the partition function $Z_{\mathcal{I}}^{\circ}$ can be understood as an identity operator acting on the Chevalley-Eilenberg cohomology $H_{CE}(\mathfrak{h})$.

Remark 35. One can also view the one-dimensional version of the BF theory with cosmological term [6] as a special case of the one-dimensional Chern-Simons theory for $\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{h}^*$, where $\mathfrak{h}$ is itself a quadratic Lie algebra, and the Lie algebra structure on $\mathfrak{g}$ is given by

$$(123) \quad \theta = \frac{1}{2} F^{pqr} \eta^p \eta^q \bar{\eta}^r + \kappa \frac{1}{6} F^{pqr} \bar{\eta}^p \bar{\eta}^q \bar{\eta}^r.$$

Here F^{pqr} are the structure constants of $\mathfrak{h}$ (in an orthonormal basis) and the parameter κ is the “cosmological constant”. For Lie algebra $\mathfrak{g}$, we automatically have $f^{abc} f^{abc} = 0$, and remark 25 applies in this case.

Let us denote $\mathfrak{g}$ with Lie algebra structure defined by (123) by $\mathfrak{g}_{BF,\kappa}$. Then, one-dimensional Chern-Simons theories with Lie algebras $\mathfrak{g}_{BF,\kappa}$ and $\mathfrak{h}$ are related, similarly to the 3-dimensional case [6]. In particular, for continuum action on the circle we have

$$(124) \quad S_{\mathfrak{g}_{BF,\kappa}}(\underbrace{\iota(\eta) + \bar{\iota}(\bar{\eta})}_{\psi}, \underbrace{\iota(\mathbf{A}) + \bar{\iota}(\bar{\mathbf{A}})}_A) = \\ = \frac{1}{2\kappa} (S_{\mathfrak{h}}(\eta + \kappa \bar{\eta}, \mathbf{A} + \kappa \bar{\mathbf{A}}) - S_{\mathfrak{h}}(\eta - \kappa \bar{\eta}, \mathbf{A} - \kappa \bar{\mathbf{A}}))$$

where ι and $\bar{\iota}$ denote the embeddings of $\mathfrak{h}, \mathfrak{h}^*$ into $\mathfrak{g}_{BF,\kappa}$ and on the right hand side we implicitly use the isomorphism $\mathfrak{h} \cong \mathfrak{h}^*$ given by the pairing on $\mathfrak{h}$. Relation (124) implies the following relation for partition functions for triangulated circle Ξ_n for Lie algebras $\mathfrak{g}_{BF,\kappa}$ and $\mathfrak{h}$:

$$(125) \quad Z_{\mathfrak{g}_{BF,\kappa}, \Xi_n}(\{\iota(\eta_k) + \bar{\iota}(\bar{\eta}_k)\}, \{\iota(\mathbf{A}_k) + \bar{\iota}(\bar{\mathbf{A}}_k)\}; \hbar) = \\ = Z_{\mathfrak{h}, \Xi_n}(\{\eta_k + \kappa \bar{\eta}_k\}, \{\mathbf{A}_k + \kappa \bar{\mathbf{A}}_k\}; 2\kappa \hbar) \cdot Z_{\mathfrak{h}, \Xi_n}(\{\eta_k - \kappa \bar{\eta}_k\}, \{\mathbf{A}_k - \kappa \bar{\mathbf{A}}_k\}; -2\kappa \hbar).$$

Remark 36. Another special case of a one-dimensional Chern-Simons theory can be constructed from a Lie bialgebra $\mathfrak{h}$. Here we set $\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{h}^*$ with the canonical pairing and with Lie algebra structure on $\mathfrak{g}$ defined by

$$\theta = \frac{1}{2} F_{pq}^r \eta^p \eta^q \bar{\eta}_r + \frac{1}{2} G_p^{qr} \eta^p \bar{\eta}_q \bar{\eta}_r.$$

Here F_{pq}^r and G_p^{qr} are structure constants of the Lie bracket and co-bracket on $\mathfrak{h}$. This is a one-dimensional version of the Lie bialgebra BF theory, cf. [13] (the underlying unimodular Lie bialgebra for continuum theory on circle is $\mathfrak{h} \otimes \Omega^{\bullet}(S^1)$). It does not seem to enjoy any particular simplifications with respect to the general case other than having a canonical complex polarization on $\mathfrak{g}$.

Remark 37. The odd third-order differential operator in variables $\tilde{\psi}, A, \eta_{out}, \bar{\eta}_{in}$ that appears in (120) endows the algebra of functions $\text{Fun}(\Pi \mathfrak{g} \oplus \mathfrak{g} \oplus \Pi \mathfrak{h} \oplus \Pi \bar{\mathfrak{h}})$ with a structure of homotopy BV algebra in the sense of Tamarkin-Tsygan [15]. In general, the same applies to $\text{Fun}(\mathcal{F}_{\Theta}^{\text{bulk}} \oplus (\Pi \mathfrak{h})^{\times i(\Theta)} \oplus (\Pi \bar{\mathfrak{h}})^{\times i(\Theta)})$ for any 1-dimensional simplicial complex Θ . If Θ has no boundary, this homotopy BV structure is strict.

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THE POISSON SIGMA MODEL ON CLOSED SURFACES

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ABSTRACT. Using methods of formal geometry, the Poisson sigma model on a closed surface is studied in perturbation theory. The effective action, as a function on vacua, is shown to have no quantum corrections if the surface is a torus or if the Poisson structure is regular and unimodular (e.g., symplectic). In the case of a Kähler structure or of a trivial Poisson structure, the partition function on the torus is shown to be the Euler characteristic of the target; some evidence is given for this to happen more generally. The methods of formal geometry introduced in this paper might be applicable to other sigma models, at least of the AKSZ type.

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1. INTRODUCTION

In this note we study the Poisson sigma model [17, 20] with worldsheet a connected, closed surface Σ . To do so we treat the Poisson structure on the target manifold M as a perturbation and expand around the vacua (a.k.a. zero modes) of the unperturbed action.¹ As a critical point of the latter in particular contains a constant map, we have first to localize around its image $x \in M$. To glue perturbations around different points x , we use formal geometry [16]. Our first result is that the perturbative effective action (as a function on the moduli space of vacua for the unperturbed theory) has no quantum corrections if Σ is the torus or if the Poisson structure is regular and unimodular (e.g., symplectic). In the former case, under the further assumption that the Poisson structure is Kähler, we can also perform the integration over vacua and show that the partition function is the Euler characteristic of M . For a general Poisson structure we can use worldsheet supersymmetry to regularize the effective action² and study it like in [22]; this argument is however a bit formal unless some extra conditions on the Poisson structure are assumed.

Notice that on the torus we need not assume unimodularity. For other genera, the requirement of unimodularity was first remarked in [6] where the leading term of the effective action on the sphere was also computed.

The techniques presented in this note, in particular the way of using formal geometry to get a global effective action, should be applicable to other field theories, in particular of the AKSZ type [1]. The techniques of subsection 4.3 and of subsections 5.2 and 5.3 should also extend to higher dimensional AKSZ theories in which the source manifold is a Cartesian product with a torus.

The torus case may also be understood as follows. Recall that the BV (Batalin–Vilkovisky) action for the Poisson sigma model can be given in terms of the AKSZ construction [10]. It is a function on the infinite dimensional graded manifold $\text{Map}(T[1]\Sigma, T^*[1]M)$. On a cylinder $\Sigma = S^1 \times I$, the partition function should be interpreted as an operator on the Hilbert space associated to the boundary S^1 . As the theory is topological, this operator is the identity and the partition

¹What we compute is then $\langle e^{\frac{i}{\hbar} S_\pi} \rangle_0$ where S_π is the interaction part depending on the target Poisson structure π and $\langle \rangle_0$ denotes the expectation value for the Poisson sigma model with zero Poisson structure.

²We prove that the regularized effective action does not depend on the regularization as long as one is present. However, in principle this is not the same theory as the non regularized one.

function on the torus is just its supertrace. Now, in the case of trivial Poisson structure, the BFV (Batalin–Fradkin–Vilkovisky) reduced phase space associated to the boundary is the graded symplectic manifold $T^*T^*[1]M = T^*T[-1]M$. If we choose the vertical polarization in the second presentation, the Hilbert space will be $C^\infty(T[-1]M)$, i.e., the de Rham complex with opposite grading. It is then to be expected that the partition function on the torus should be the Euler characteristic of M . In the perturbative computation, however, the final result is usually of the form $0 \cdot \infty$, but in the Kähler case we get an unambiguous answer. We might then think of a Kähler structure on M , if it exists, as a regularization of the Poisson sigma model with trivial Poisson structure.³ Notice that, if such structures exist, they are open dense in the space of all Poisson structures on M . Another regularization, which produces the same result, consists in adding the Hamiltonian functions of the supersymmetry generators for the effective action. Formally, this can even be done before integrating over fluctuations around vacua.

Finally, notice that apart from the cases mentioned above we do expect the effective action to have quantum corrections. Moreover, the naively computed effective action in formal coordinates might happen not to be global. We show however that it is always possible to find a quantum canonical transformation which makes it into the Taylor expansion of a global effective action. Its class modulo quantum canonical transformations is then the well-defined object associated to the theory.

Section 2 is a crash course in formal geometry (essentially following [11, §2]). In Section 3, we develop the construction of [11, §6] to define the effective action in formal coordinates. Next using the results of Section 4, we show that, in the two special cases mentioned above, the effective action has no quantum correction and is the expression in formal coordinates of a global effective action. In Section 5, we study the effective action for the case of the torus and perform the computation of the partition function. Finally, in Section 6 we study the globalization of the effective action in general.

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³That is, we regularize $\text{STr}_{C^\infty(T[-1]M)} \text{id} = \langle 1 \rangle_0$ as

$$\langle 1 \rangle_0 := \lim_{\epsilon \rightarrow 0} \langle e^{\frac{i\epsilon}{\hbar} S_\pi} \rangle_0.$$

We then show that, in the Kähler case, $\langle e^{\frac{i\epsilon}{\hbar} S_\pi} \rangle_0$ is independent of ϵ and equal to the Euler characteristic of M .

2. FORMAL LOCAL COORDINATES

We shortly review the notion of formal local coordinates following the simple introduction of [11, §2] (for more on formal geometry see [2, 7, 16]).

A generalized exponential map for a manifold M is just a smooth map $\phi: U \rightarrow M$, where U is some open neighborhood of the zero section of M in TM , $(x, y \in U_x) \mapsto \phi_x(y)$, satisfying $\phi_x(0) = x$ and $d_y\phi_x(0) = \text{id} \forall x \in M$. As an example, one may take the exponential map of a connection.

If f is a smooth function on M , then the function $\phi^*f \in C^\infty(U)$ satisfies $d(\phi^*f) = df \circ d\phi$. Denoting by d_x (d_y) the horizontal (vertical) part of the differential, we then get $d_x(\phi^*f) = df \circ d_x\phi$ and $d_y(\phi^*f) = df \circ d_y\phi$. Because of the assumptions on ϕ , there is an open neighborhood $U' \subset U$ of the zero section of M in TM on which $d_y\phi$ is invertible. As a consequence, on U' we have the formula

$$(2.1) \quad d_x(\phi^*f) = d_y(\phi^*f) \circ (d_y\phi)^{-1} \circ d_x\phi.$$

Notice that, for each x , ϕ_x^*f is a smooth function on U_x . By $T\phi_x^*f \in \widehat{ST}_x^*M$ we then denote its Taylor expansion in the $y \in U_x$ -variables around $y = 0$.⁴ In doing this, we associate to $f \in C^\infty(M)$ a section $T\phi^*f$ of $\widehat{ST}^*M$ over M . We may now reinterpret (2.1) as a condition on the section $T\phi^*f$ simply taking Taylor expansions w.r.t. y on both sides. Notice that in the definition of $T\phi^*f$ and in the resulting condition only the Taylor coefficients of ϕ appears. We are thus let to considering two generalized exponential maps as equivalent if all their partial derivatives in the vertical directions, for each point of the base M , coincide at the zero section. We call formal exponential map an equivalence class of generalized exponential maps.

If ϕ is a formal exponential map, then $T\phi^*f \in \Gamma(\widehat{ST}^*M)$ is constructed as above just by picking any generalized exponential map in the given equivalence class. Choosing local coordinates $\{x^i\}$ on the base and $\{y^i\}$ on the fiber, we have explicit expressions

$$(2.2) \quad \phi_x^i(y) = x^i + y^i + \frac{1}{2}\phi_{x,jk}^i y^j y^k + \frac{1}{3!}\phi_{x,jkl}^i y^j y^k y^l + \cdots,$$

and the class of ϕ is simply given by the collection of coefficients $\phi_{x,\bullet}$. One can easily see that the coefficients $\phi_{x,jk}^i$ of the quadratic term transform as the components of a connection. We will refer to this as

⁴Here $\widehat{S}$ denotes the formal completion of the symmetric algebra.

the connection in ϕ . Also explicitly we may compute

$$(2.3) \quad T\phi^*f = f(x) + y^i \partial_i f(x) + \frac{1}{2} y^j y^k (\partial_j \partial_k f(x) + \phi_{x,jk}^i \partial_i f(x)) + \cdots.$$

Above we have proved that sections of $\widehat{ST}^*M$ of the form $T\phi^*f$ satisfy (the Taylor expansion of) equation (2.1). One can easily prove that the converse is also true. In fact, one has even more. We may think of the Taylor expansion of the r.h.s. as an operator acting on the section $T\phi^*f$. Actually, for every section σ of $\widehat{ST}^*M$ one can define a section $R(\sigma)$ of $T^*M \otimes \widehat{ST}^*M$ by taking the Taylor expansion of $-\mathrm{d}_y \sigma \circ (\mathrm{d}_y \phi)^{-1} \circ \mathrm{d}_x \phi$. Notice that R is $C^\infty(M)$ -linear. As a consequence we have a connection⁵

$$(X, \sigma) \in \Gamma(TM) \otimes \Gamma(\widehat{ST}^*M) \mapsto i_X R(\sigma) \in \Gamma(\widehat{ST}^*M)$$

on $\widehat{ST}^*M$. One can check that this connection is flat. We can also regard R as a one-form on M taking values in the bundle $\mathrm{End}(\widehat{ST}^*M)$. Also notice that $\widehat{ST}^*M$ is a bundle of algebras and that R acts as a derivation; so we can regard R as a one-form on M taking values in the bundle $\mathrm{Der}(\widehat{ST}^*M)$, which is tantamount to saying the bundle of formal vertical vector fields $\widehat{\mathfrak{X}}(TM) := TM \otimes \widehat{ST}^*M$. Notice that the flatness of the connection may be expressed as the MC (Maurer–Cartan) equation

$$(2.4) \quad \mathrm{d}_x R + \frac{1}{2} [R, R] = 0,$$

where $[\ , \]$ is the Lie bracket of vector fields. Finally, equation (2.1) may now be expressed by saying that $\mathrm{d}\sigma + R(\sigma) = 0$ if σ is of the form $T\phi^*f$ for some f . Below we will see that also the converse is true.

We first extend this connection to a differential D on the complex of $\widehat{ST}^*M$ -valued differential forms $\Gamma(\Lambda^\bullet T^*M \otimes \widehat{ST}^*M)$.⁶ The main result is that the cohomology of D is concentrated in degree zero and $H_D^0 = T\phi^*C^\infty(M)$. This can be easily seen working in local coordinates again:

$$R(\sigma)_i = \frac{\partial \sigma}{\partial y^k} \left(\left(\frac{\partial \phi}{\partial y} \right)^{-1} \right)_j^k \frac{\partial \phi^j}{\partial x^i}.$$

⁵This is the Grothendieck connection in the presentation given by the choice of the formal exponential map ϕ .

⁶Since $\Gamma(\Lambda^\bullet T^*M \otimes \widehat{ST}^*M)$ is the algebra of functions on the formal graded manifold $\mathcal{M} := T[1]M \oplus T[0]M$, the differential D gives $\mathcal{M}$ the structure of a differential graded manifold. In particular since D vanishes on the body, we may linearize at each $x \in M$ and get an L_∞ -algebra structure on $T_x M[1] \oplus T_x M \oplus T_x M$.

Using (2.2) we get $R = \delta + R'$ with $\delta = -dx^i \frac{\partial}{\partial y^i}$ and R' a one-form on the base taking value in the vector fields vanishing at $y = 0$. Hence we have $D = \delta + D'$ with

$$D' = dx^i \frac{\partial}{\partial x^i} + R'.$$

Notice that δ is itself a differential and that it decreases the polynomial degree in y , whereas the operator D' does not decrease this degree. The fundamental remark is that the cohomology of δ consists of zero forms constant in y . This is easily shown by introducing $\delta^* := y^i \iota_{\frac{\partial}{\partial x^i}}$ and observing that $(\delta\delta^* + \delta^*\delta)\sigma = k\sigma$ if σ is an r -form of degree s in y and $r + s = k$. By cohomological perturbation theory the cohomology of D is isomorphic to the cohomology of δ , which is what we wanted to prove.

Finally, observe that, if σ is a D -closed section, we can immediately recover the function f for which $\sigma = T\phi^*f$ simply by setting $y = 0$, $f(x) = \sigma_x(0)$, as follows from (2.3).

We can now extend the whole story to other natural objects. Let $\mathcal{V}(M)$ denote the multivector fields on M (i.e., sections of ΛTM), $\Omega(M)$ the differential forms, $\mathcal{W}^j(M) := \Gamma(S^j TM)$ and $\mathcal{O}^j(M) := \Gamma(S^j T^*M)$. We use similar symbols for formal vertical vector fields $\widehat{\mathcal{V}}(TM) := \Gamma(\Lambda TM \otimes \widehat{S}T^*M)$ and formal vertical differential forms $\widehat{\Omega}(TM) := \Gamma(\Lambda T^*M \otimes \widehat{S}T^*M)$. We have injective maps

$$T\phi^* := T(\phi_*)^{-1}: \mathcal{V}(M) \rightarrow \widehat{\mathcal{V}}(TM), \quad T\phi^*: \Omega(M) \rightarrow \widehat{\Omega}(TM).$$

Similarly, we set $\widehat{\mathcal{W}}^j(TM) := \Gamma(S^j TM \otimes \widehat{S}T^*M)$ and $\widehat{\mathcal{O}}^j(TM) := \Gamma(S^j T^*M \otimes \widehat{S}T^*M)$ and get

$$T\phi^* := T(\phi_*)^{-1}: \mathcal{W}^j(M) \rightarrow \widehat{\mathcal{W}}^j(TM), \quad T\phi^*: \mathcal{O}^j(M) \rightarrow \widehat{\mathcal{O}}^j(TM).$$

We can now let R naturally act on $\widehat{\mathcal{V}}(TM)$, $\widehat{\Omega}(TM)$, $\widehat{\mathcal{W}}^j(TM)$ and $\widehat{\mathcal{O}}^j(TM)$ by Lie derivative and hence get a differential D on the corresponding complexes of differential forms. Notice that D respects the Gerstenhaber algebra structure (by the vertical Schouten–Nijenhuis bracket) of $\widehat{\mathcal{V}}(TM)$ and the differential complex structure (by the vertical differential) of $\widehat{\Omega}(TM)$, so that these structures are induced in cohomology. By the same argument as above, we get that all these cohomologies are concentrated in degree zero with $H_D^0(\widehat{\mathcal{V}}(TM)) = T\phi^*\mathcal{V}(M)$, $H_D^0(\widehat{\Omega}(TM)) = T\phi^*\Omega(M)$, $H_D^0(\widehat{\mathcal{W}}^j(TM)) = T\phi^*\mathcal{W}^j(M)$, and $H_D^0(\widehat{\mathcal{O}}^j(TM)) = T\phi^*\mathcal{O}^j(M)$. Notice in particular that a section is

in the image of $T\phi^*$ if and only if

$$d_x\sigma + L_R\sigma = 0.$$

In order to recover, in local coordinates, the global object corresponding to a solution to the above equation, we should only observe that by assumption $d_y\phi_x(0) = \text{id}$, so that it is enough to evaluate the components of σ at $y = 0$ and to replace formally each dy^i by dx^i and each $\frac{\partial}{\partial y^i}$ by $\frac{\partial}{\partial x^i}$. More explicitly, if $\sigma_x(y) = \sigma_{x;i_1,\dots,i_n}(y) dy^{i_1} \cdots dy^{i_n}$ is equal to $T\phi^*\omega$, then

$$\omega(x) = \sigma_{x;i_1,\dots,i_n}(0) dx^{i_1} \cdots dx^{i_n}.$$

If on the other hand, $\sigma_x(y) = \sigma_x^{i_1,\dots,i_n}(y) \frac{\partial}{\partial y^{i_1}} \cdots \frac{\partial}{\partial y^{i_n}}$ is equal to $T\phi^*(Y)$, then

$$Y(x) = \sigma_x^{i_1,\dots,i_n}(0) \frac{\partial}{\partial x^{i_1}} \cdots \frac{\partial}{\partial x^{i_n}}.$$

One can immediately extend these results to direct sums of the vector bundles above. Notice that cohomology also commutes with direct limits. This implies that the cohomology of $\prod_j \widehat{\mathcal{W}}^j(TM)$ is also concentrated in degree zero and coincides with $T\phi^* \prod_j \mathcal{W}^j(M)$. Now we have that $\prod_j \mathcal{W}^j(M) = \Gamma(\widehat{STM})$ whereas $\prod_j \widehat{\mathcal{W}}^j(TM) = \Gamma(\widehat{S}(TM \otimes T^*M))$. Similarly, we see that the cohomology with values in $\widehat{S}(T^*M \otimes T^*M)$ is concentrated in degree zero and coincides with $T\phi^*\Gamma(\widehat{ST^*M})$. To summarize:

$$H_D^\bullet(\Gamma(\widehat{S}(TM \otimes T^*M))) = H_D^0(\Gamma(\widehat{S}(TM \otimes T^*M))) = T\phi^*\Gamma(\widehat{STM})$$

and

$$H_D^\bullet(\Gamma(\widehat{S}(T^*M \otimes T^*M))) = H_D^0(\Gamma(\widehat{S}(T^*M \otimes T^*M))) = T\phi^*\Gamma(\widehat{ST^*M}).$$

2.1. Gauge transformations. We now wish to consider the effects of changing the choice of formal exponential map. Namely, let ϕ be a family of formal exponential maps depending on a parameter t belonging to an open interval I . We may associate to this family a formal exponential map ψ for the manifold $M \times I$ by $\psi(x, t, y, \tau) := ((\phi)_{x,t}(y), t + \tau)$, where τ denotes the tangent variable to t . We want to define the associated connection $\widetilde{R}$: on a section $\widetilde{\sigma}$ of $\widehat{ST^*}(M \times I)$ we have, by definition,

$$\widetilde{R}(\widetilde{\sigma}) = -(d_y\widetilde{\sigma}, d_\tau\widetilde{\sigma}) \circ \begin{pmatrix} (d_y\phi)^{-1} & 0 \\ 0 & 1 \end{pmatrix} \circ \begin{pmatrix} d_x\phi & \dot{\phi} \\ 0 & 1 \end{pmatrix}.$$

So we can write $\widetilde{R} = R + C dt + T$ with R defined as before (but now t -dependent),

$$C(\widetilde{\sigma}) = -d_y\widetilde{\sigma} \circ (d_y\phi)^{-1} \circ \dot{\phi}$$

and $T = -dt \frac{\partial}{\partial \tau}$. We now spell out the MC equation for $\tilde{R}$ observing that $d_x T = d_t T = 0$ and that T commutes with both R and C . The $(2, 0)$ -form component over $M \times I$ yields again the MC equation for R , whereas the $(1, 1)$ -component reads

$$\dot{R} = d_x C + [R, C].$$

Hence, under a change of formal exponential map, R changes by a gauge transformation with generator the section C of $\hat{\mathfrak{X}}(TM)$.

Finally, if σ is a section in the image of $T\phi^*$, then by a simple computation one gets

$$\dot{\sigma} = -L_C \sigma,$$

which can be interpreted as the associated gauge transformation for sections.

3. PSM IN FORMAL COORDINATES

The Poisson sigma model (PSM) [17, 20] is a topological field theory with source a two-manifold Σ and target a Poisson manifold M . Before getting to the BV action for the PSM, we fix the notations and introduce the AKSZ formalism [1] (for a gentle introduction to it, especially suited to the PSM, see [10]). Let $\text{Map}(T[1]\Sigma, T^*[1]M)$ be the infinite dimensional graded manifold of maps from $T[1]\Sigma$ to $T^*[1]M$. It fibers over $\text{Map}(T[1]\Sigma, M)$. We denote by $\mathbf{X}$ a “point” of $\text{Map}(T[1]\Sigma, M)$ and by $\boldsymbol{\eta}$ a “point” of the fiber. In local target coordinates, the super fields $\mathbf{X}$ and $\boldsymbol{\eta}$ have simple expressions:

$$\begin{aligned} \mathbf{X}^i &= X^i + \eta_+^i + \beta_+^i, \\ \boldsymbol{\eta}_i &= \beta_i + \eta_i + X_i^+, \end{aligned}$$

where we have ordered the terms in increasing order of form degree on Σ . The ghost number is 0 for X and η , 1 for β , -1 for η_+ and X^+ , and -2 for β_+ . As unperturbed BV action one considers

$$S_0 := \int_{\Sigma} \boldsymbol{\eta}_i d\mathbf{X}^i.$$

Notice that it satisfies the classical master equation (CME) $(S_0, S_0) = 0$ if Σ has no boundary or if appropriate boundary conditions are taken (which we assume throughout). Here $(\ , \)$ is the BV bracket corresponding to the odd symplectic structure on the space of fields for which the superfield $\boldsymbol{\eta}$ is the momentum conjugate to the superfield $\mathbf{X}$. Formally one may also assume $\Delta S_0 = 0$ where Δ is the BV operator, so S_0 satisfies also the quantum master equation (QME).

To perturb this action, we pick a multivector field Y on M . We may regard it as a function on $T^*[1]M$. We then define S_Y as the integral over $T[1]\Sigma$ of the pullback of Y by the evaluation map

$$\text{ev}: T[1]\Sigma \times \text{Map}(T[1]\Sigma, T^*[1]M) \rightarrow T^*[1]M.$$

Explicitly, for a k -vector field Y , we have

$$S_Y = \frac{1}{k!} \int_{\Sigma} Y^{i_1, \dots, i_k}(\mathbf{X}) \boldsymbol{\eta}_{i_1} \dots \boldsymbol{\eta}_{i_k}.$$

This construction has several interesting properties. First, $(S_0, S_Y) = 0$ (for $\partial\Sigma = \emptyset$ or with appropriate boundary conditions). Second, for any two multivector fields Y and Y' , we have $(S_Y, S_{Y'}) = S_{[Y, Y']}$. The BV action for the PSM with target Poisson structure π is recovered as $S = S_0 + S_{\pi}$. Notice that by the above mentioned properties it satisfies the CME. As for the quantum master equation, we refer to [12], where it is shown that one can assume $\Delta S_{\pi} = 0$ if the Euler characteristic of Σ is zero or if π is unimodular. In the latter case, one picks a volume form v on M such that $\text{div}_v \pi = 0$ and defines Δ according to it.

We consider S_{π} as a perturbation, so we expand the functional integral around the critical points of S_0 . They consist of closed superfields. In particular, the component X of $\mathbf{X}$ will be a constant map, say with image $x \in M$. Fluctuations will explore only a formal neighborhood of x in M , so as in [11, §6], it makes sense to make the change of variables

$$\mathbf{X} = \phi_x(\mathbf{A}), \quad \boldsymbol{\eta} = d\phi_x(\mathbf{A})^{*, -1} \mathbf{B},$$

where ϕ is a formal exponential map and the new superfields $(\mathbf{A}, \mathbf{B})$ are in $\text{Map}(T[1]\Sigma, T^*[1]T_x M)$. Notice that this change of variables

$$\phi_x: \text{Map}(T[1]\Sigma, T^*[1]T_x M) \rightarrow \text{Map}(T[1]\Sigma, T^*[1]M)$$

is a local symplectomorphism and that

$$T\phi_x^* S_0 = \int_{\Sigma} \mathbf{B}_i d\mathbf{A}^i.$$

The moduli space of vacua (i.e., the space of critical points modulo gauge transformations) is now $\mathcal{H}_x := H^{\bullet}(\Sigma) \otimes T_x M \oplus H^{\bullet}(\Sigma) \otimes T_x^* M[1]$. (Here $H^{\bullet}(\Sigma)$ is regarded as a graded vector space with its natural grading.) We should regard $\mathcal{H} = \bigcup_x \mathcal{H}_x$ as a vector bundle over M , but for the moment we concentrate on a single x . Later on we will also consider the remaining integration over vacua and, in particular, over M (which actually shows up as the space of constant maps $\Sigma \rightarrow M$); see Section 5.

We may repeat the AKSZ construction on $\text{Map}(T[1]\Sigma, T^*[1]T_x M)$. In particular, if Y is a function of degree k on $T^*[1]T_x M$ (i.e., a formal vertical k -vector field), we may construct a functional

$$S_Y = \frac{1}{k!} \int_{\Sigma} Y^{i_1, \dots, i_k}(\mathbf{A}) \mathbf{B}_{i_1} \dots \mathbf{B}_{i_k}.$$

In particular, we have

$$T\phi_x^* S_{\pi} = S_{T\phi_x^* \pi}.$$

As a result, we have a solution $S_x := T\phi_x^* S$ of the QME and may compute its partition function Z_x (as a function on $\mathcal{H}_x$) upon integrating over a Lagrangian submanifold $\mathcal{L}$ of a complement of $\mathcal{H}_x$ in $\text{Map}(T[1]\Sigma, T^*[1]T_x M)$:

$$Z_x := \int_{\mathcal{L}} e^{\frac{i}{\hbar} S_x}.$$

Notice [19, 12] that there is an induced BV operator Δ on $\mathcal{H}_x$ and that Z_x satisfies $\Delta Z_x = 0$. Moreover, upon changing the gauge fixing $\mathcal{L}$, Z_x changes by a Δ -exact term. We wish to compare the class of Z_x with the globally defined partition function morally obtained by integrating in $\text{Map}(T[1]\Sigma, T^*[1]M)$. For this we have to understand the collection $\{Z_x\}_{x \in M}$ as a section $\widehat{Z}: x \mapsto Z_x$ of $\widehat{S}\mathcal{H}^*$ (we hide the dependency on $\hbar$ here) and compute how it changes over M . Using all properties above and setting $\widehat{S}: x \mapsto S_x$, we get⁷

$$d_x \widehat{Z} = \frac{i}{\hbar} \int_{\mathcal{L}} e^{\frac{i}{\hbar} \widehat{S}} S_{d_x T\phi^* \pi} = -\frac{i}{\hbar} \int_{\mathcal{L}} e^{\frac{i}{\hbar} \widehat{S}} (S_R, \widehat{S}).$$

Notice that this may also be rewritten as

$$d_x \widehat{Z} = -\Delta \int_{\mathcal{L}} e^{\frac{i}{\hbar} \widehat{S}} S_R$$

if we assume $\Delta S_R = 0$. This is correct if Σ has zero Euler characteristic or if $\text{div}_{T\phi^* v} R = 0$. From the equation $d_x T\phi^* v + L_R T\phi^* v = 0$, we see that the latter condition is satisfied if and only if $d_x T\phi^* v = 0$. Given a volume form v , it is always possible to find a formal exponential map ϕ satisfying this condition; actually, one can even get $T\phi_x^* v = dy^1 \dots dy^d \forall x$.

We can collect the above identities nicely if we define⁸

$$\widetilde{S} := \widehat{S} + S_R.$$

⁷The choice of $\mathcal{L}$ might be different for different x s, but for simplicity we assume it not to be the case.

⁸For $\pi = 0$, $\widetilde{S}$ is also the BV action for the BF_{∞} -theory [19] with target the L_{∞} algebra of footnote 6. See also [14].

Notice that $\tilde{S}$ is of total degree zero (the term S_R has ghost number minus one but is a one-form on M) and satisfies the modified CME

$$d_x \tilde{S} + \frac{1}{2}(\tilde{S}, \tilde{S}) = 0$$

and by assumption also $\Delta \tilde{S} = 0$ (so it satisfies a modified QME as well). We then define

$$(3.1) \quad \tilde{Z} := \int_{\mathcal{C}} e^{\frac{i}{\hbar} \tilde{S}}$$

as a nonhomogeneous differential form on M taking values in $\mathcal{H}$. It satisfies

$$d_x \tilde{Z} - i\hbar \Delta \tilde{Z} = 0.$$

Remark 3.1. We are now also in a position to understand the change of $\tilde{Z}$ under a change of the formal exponential map. Using the results of subsection 2.1, we immediately see that

$$\dot{\tilde{Z}} = (d_x - i\hbar \Delta) \int_{\mathcal{C}} e^{\frac{i}{\hbar} \tilde{S}} \frac{i}{\hbar} S_C,$$

assuming $\Delta S_C = 0$. The assumption is verified if Σ has zero Euler characteristic or if we let ϕ vary only in the class of formal exponential maps that make $T\phi^*v$ constant. Notice that the space of such formal exponential maps is connected. Therefore, the class of $\tilde{Z}$ under these transformations is independent of all choices needed to compute it.

Finally, we consider the effective action $\tilde{S}_{\text{eff}}$ defined by the identity $\tilde{Z} = e^{\frac{i}{\hbar} \tilde{S}_{\text{eff}}}$. It is a differential form taking values in $\hat{S}\mathcal{H}^*[[\hbar]]$ and satisfies the modified QME

$$(3.2) \quad d_x \tilde{S}_{\text{eff}} + \frac{1}{2}(\tilde{S}_{\text{eff}}, \tilde{S}_{\text{eff}}) - i\hbar \Delta \tilde{S}_{\text{eff}} = 0.$$

This equation formally follows from the properties of BV integrals. In the case when Σ has no boundary, it may be proved directly by considering the expansion of $\tilde{S}_{\text{eff}}$ in Feynman diagrams and applying the usual Stokes theorem techniques on integrals over configuration spaces, see [18, 13]. If Σ has a boundary, additional terms corresponding to several points collapsing to the boundary together may appear and spoil (3.2). From now on we therefore assume that Σ is closed.

Equation (3.2) contains both information on the QME satisfied by the zero-form component $\tilde{S}_{\text{eff}}^{(0)}$ and on its global properties. The equations are in general mixed: we do not simply get a flat connection with respect to which $\tilde{S}_{\text{eff}}^{(0)}$ is covariantly constant. However, it is possible to find a modified quantum BV canonical transformation that produces

a flat connection with respect to which the zero form part of the effective action is horizontal and hence global; we postpone this discussion to Section 6. In the remaining of this Section, we concentrate on two special cases where the general theory is not needed.

The first special case is when Σ is a torus. In Section 4, see Lemma 4.4, we will show that, in an appropriate gauge, there are no quantum corrections, so

$$\tilde{S}_{\text{eff}} = \tilde{S}_{\text{eff}}^{(0)} + \tilde{S}_{\text{eff}}^{(1)},$$

where $\tilde{S}_{\text{eff}}^{(0)}$ is the zero-form obtained by restricting $S_{T\phi^*\pi}$ to vacua and $\tilde{S}_{\text{eff}}^{(1)}$ is the one-form obtained by restricting S_R to vacua. One can explicitly check, see Section 5, that $\Delta\tilde{S}_{\text{eff}}^{(0)} = \Delta\tilde{S}_{\text{eff}}^{(1)} = 0$. Hence the modified QME now simply yields the CME for $\tilde{S}_{\text{eff}}^{(0)}$,

$$(\tilde{S}_{\text{eff}}^{(0)}, \tilde{S}_{\text{eff}}^{(0)}) = 0,$$

the flatness condition for $\tilde{S}_{\text{eff}}^{(1)}$,

$$d_x \tilde{S}_{\text{eff}}^{(1)} + \frac{1}{2}(\tilde{S}_{\text{eff}}^{(1)}, \tilde{S}_{\text{eff}}^{(1)}) = 0,$$

and the fact that $\tilde{S}_{\text{eff}}^{(0)}$ is covariantly constant,

$$d_x \tilde{S}_{\text{eff}}^{(0)} + (\tilde{S}_{\text{eff}}^{(1)}, \tilde{S}_{\text{eff}}^{(0)}) = 0.$$

Now notice that $(\tilde{S}_{\text{eff}}^{(1)}, \cdot)$ is just the natural action of R on the sections of $\mathcal{H}$. Hence we can conclude that $\tilde{S}_{\text{eff}}^{(0)}$ is just $T\phi^*(S|_{\text{vacua}})$.

The second special case is when π is regular and unimodular (and Σ is any two-manifold). Also in Section 4 we will show that, upon choosing an appropriate formal exponential map, $S_{\text{eff}}^{(0)}$ and $S_{\text{eff}}^{(1)}$ have no quantum corrections. Therefore, we may use the same reasoning as above and conclude that $\tilde{S}_{\text{eff}}^{(0)}$ is just $T\phi^*(S|_{\text{vacua}})$.

A final important remark is that in the two cases above the effective action depends polynomially on all vacua but, possibly, for those related to the X field; therefore, $S_{\text{eff}}^{(0)}$ is a section of $S\tilde{\mathcal{H}}^* \otimes \widehat{S}T^*M$, where $\tilde{\mathcal{H}}_x = H^{>0}(\Sigma) \otimes T_x M \oplus H^\bullet(\Sigma) \otimes T_x^* M[1]$ is the moduli space of vacua excluding those for X . The corresponding global effective action $S|_{\text{vacua}}$ will then be a section of $S\tilde{\mathcal{H}}^*$, i.e., a function on the vector bundle $\tilde{\mathcal{H}}$ (polynomial in the fibers). Notice that this vector bundle is diffeomorphic, by choosing a connection (e.g., the one contained in the choice of ϕ), to the natural global definition of the moduli space of vacua as presented, e.g., in [5].

4. SOME COMPUTATIONS OF THE EFFECTIVE ACTION

In this Section we discuss the perturbative computation for the effective action and show that it has no quantum corrections in two important cases.

4.1. Factorization of Feynman graphs. Consider the effective action $\tilde{S}_{\text{eff}}$ defined in (3.1) by the identity $\tilde{Z} = e^{\frac{i}{\hbar} \tilde{S}_{\text{eff}}}$. Here the Lagrangian subspace $\mathcal{L}$ in the complement of $\mathcal{H}_x$ inside the space of fields

$$(4.1) \quad \mathcal{F}_x = \text{Map}(T[1]\Sigma, T^*[1]T_x M) \cong \Omega^\bullet(\Sigma) \otimes (T_x M \oplus T_x^*[1]M)$$

accounts for the gauge fixing. Let $\mathcal{L}$ have the factorized form

$$(4.2) \quad \mathcal{L} = \mathcal{L}_K \otimes (T_x M \oplus T_x^*[1]M)$$

where $\mathcal{L}_K \subset \Omega^\bullet(\Sigma)$ is defined as

$$\mathcal{L}_K = \ker \mathcal{P} \cap \ker K$$

with $\mathcal{P}$ the projector from differential forms on Σ to (the chosen representatives of) de Rham cohomology of Σ and

$$K: \Omega^\bullet(\Sigma) \rightarrow \Omega^{\bullet-1}(\Sigma)$$

a linear operator satisfying

$$(4.3) \quad dK + Kd = \text{id} - \mathcal{P}, \quad \mathcal{P}K = K\mathcal{P} = 0, \quad K^T = K, \quad K^2 = 0$$

(i.e., K is the chain homotopy between identity and the projection to cohomology, also known as a parametrix). The transpose is w.r.t. the Poincaré pairing on forms $\int_\Sigma \bullet \wedge \bullet$. We assume the operator K (which now determines the gauge fixing) to be an integral operator with a distributional integral kernel $\omega \in \Omega^1(\Sigma \times \Sigma)$ – the propagator. An explicit construction may be done along the same lines as in [8, 9]. Let us introduce a basis $\{\chi_\alpha\}$ in the cohomology space $H^\bullet(\Sigma)$; denote the matrix of the Poincaré pairing by $\Pi_{\alpha\beta} = \int_\Sigma \chi_\alpha \wedge \chi_\beta$. In terms of ω , properties (4.3) read:

- (1) $d\omega = \delta_{\text{diag}} - \sum_{\alpha,\beta} (\Pi^{-1})^{\alpha\beta} \chi_\alpha \otimes \chi_\beta$, where δ_{diag} is the delta-form supported on the diagonal of $\Sigma \times \Sigma$;
- (2) $\int_{\Sigma_{(1)}} \omega \pi_1^* \chi_\alpha = \int_{\Sigma_{(2)}} \omega \pi_2^* \chi_\alpha = 0$, $\forall \alpha$, where $\Sigma_{(1)}$ and $\Sigma_{(2)}$ denote the two factors of $\Sigma \times \Sigma$, and π_1 and π_2 are the two projections from $\Sigma \times \Sigma$ to its factors;
- (3) $t^* \omega = \omega$, where $t: \Sigma \times \Sigma \rightarrow \Sigma \times \Sigma$ is the map swapping the two copies of Σ ;
- (4) $\int_{\Sigma_{(2)}} \pi_{12}^* \omega \pi_{23}^* \omega = 0$, where $\Sigma_{(2)}$ denotes the middle factor in $\Sigma \times \Sigma \times \Sigma$, whereas π_{12} and π_{23} are the projections from $\Sigma \times \Sigma \times \Sigma$ to the first two and last two factors, respectively.

Notice that the restriction of ω to the configuration space $C_2^0(\Sigma) := \{(u, v) \in \Sigma : u \neq v\}$ is smooth and it extends to the Fulton–MacPherson–Axelrod–Singer compactification as a smooth form. In [13] it is shown how to implement the property $\mathcal{P}K = K\mathcal{P} = 0$ on the propagator. Once this is done, the propagator will also satisfy the property $K^2 = 0$. This is proved exactly as in [12, Lemma 10].

The perturbation expansion for the effective action (3.1) has the form

$$(4.4) \quad S^{\text{eff}}(\mathbf{A}_{\text{z.m.}}, \mathbf{B}_{\text{z.m.}}; \hbar) = \sum_{\Gamma} \frac{(i\hbar)^{l(\Gamma)}}{|\text{Aut}(\Gamma)|} W_{\Gamma}^{\text{target}}(\mathbf{A}_{\text{z.m.}}, \mathbf{B}_{\text{z.m.}}) \cdot W_{\Gamma}^{\text{source}}$$

where the sum is over connected oriented graphs Γ with leaves⁹ decorated by basis cohomology classes $\{\chi_{\alpha}\}$; $l(\Gamma)$ stands for the number of loops, $|\text{Aut}(\Gamma)|$ is the number of graph automorphisms; $\{\mathbf{A}_{\text{z.m.}}, \mathbf{B}_{\text{z.m.}}\} = \{\mathbf{A}^{\alpha i}, \mathbf{B}_i^{\alpha}\}$ are the coordinates on the moduli space of vacua $\mathcal{H}_x$.

The “target part” $W_{\Gamma}^{\text{target}}$ of the contribution of a graph to $\tilde{S}_{\text{eff}}$ is a homogeneous polynomial function on $\mathcal{H}_x$ of degree equal to the number of leaves, computed using the following set of rules:

- (1) to an incoming leaf of Γ decorated by χ_{α} one associates $\mathbf{A}^{\alpha i}$
- (2) to an outgoing leaf decorated by χ_{α} one associates $\mathbf{B}_i^{\alpha}$
- (3) to a vertex with m inputs and n outputs one associates the expression

$$\partial_{i_1} \cdots \partial_{i_m} Y^{j_1 \cdots j_n}$$

– the m -th derivative of n -vector¹⁰ contribution to the action S .

- (4) for every edge contract the dummy Latin indices for the two constituent half-edges.

The result of contraction is a polynomial function on $\mathcal{H}_x$.

The “source” (or “de Rham”) part $W_{\Gamma}^{\text{source}}$ is a number defined as

$$(4.5) \quad W_{\Gamma}^{\text{source}} = \int_{\Sigma \times V(\Gamma)} \left(\prod_{\text{edges } (h_{\text{in}}, h_{\text{out}})} \pi_{v(h_{\text{in}}), v(h_{\text{out}})}^* \omega \right) \cdot \left(\prod_{\text{leaves } l} \pi_{v(l)}^* \chi_{\alpha_l} \right)$$

⁹A leaf for us is a loose half-edge, i.e., one not connected to another half-edge to form an edge.

¹⁰In the standard setup for the Poisson sigma model, only the perturbation by Poisson bivector field $Y = \pi^{ij} \partial_i \wedge \partial_j$ is present in the action, hence all vertices have to have exactly two outputs, otherwise the graph does not contribute. In the present case, we also have the vector field R .

where $V(\Gamma)$ is the number of vertices, $\pi_v : \Sigma^{\times V(\Gamma)} \rightarrow \Sigma$ is the projection to v -th copy of Σ , $\pi_{u,v} : \Sigma^{\times V(\Gamma)} \rightarrow \Sigma \times \Sigma$ is the projection to u -th and v -th copies of Σ ; $v(h)$ is the vertex incident to the half-edge h . Notice that all these integrals converge. The usual way to show this is to observe that the integrals are actually defined on configuration spaces (i.e., the complements of all diagonals in the Cartesian products of copies of Σ) and that the propagators ω extend to their compactifications.

Remark 4.1. The factorization into source- and target contributions for Feynman diagrams in the expansion (4.4) is due to the factorization of the space of fields (4.1) and to the fact that our ansatz for the gauge fixing (4.2) is compatible with this factorization.

Remark 4.2. The orientation of Γ is irrelevant for the source parts W_Γ^{source} .

4.2. Regular Poisson structures. If π is nondegenerate, it is always possible to find a formal exponential map ϕ such that $T\phi^*\pi$ is constant (in the y variables). One simply has to go to formal Darboux coordinates. Notice, moreover, that $\text{div}_v \pi = 0$ if for v one chooses v to be the Liouville volume form $\omega^k/k!$, $k = \dim M/2$. It then follows that $T\phi^*v$ is also constant and that $\text{div}_{T\phi^*v} R = 0$. A slight generalization occurs when π is regular (i.e., its kernel has constant rank) and unimodular (notice that this is not guaranteed if π is degenerate [21]). After choosing v such that $\text{div}_v \pi = 0$, it is again possible to find a formal exponential map ϕ such that $T\phi^*\pi$ and $T\phi^*v$ are both constant and hence $\text{div}_{T\phi^*v} R = 0$.

In the perturbative expansion, we may thus assume that we have a bivalent vertex, corresponding to $T\phi^*\pi$, with no incoming arrows. If one of the outgoing arrows is replaced by a vacuum mode (i.e., a cohomology class), the result is zero by the property $\mathcal{P}K = K\mathcal{P} = 0$, otherwise it is zero by the property $K^2 = 0$. As a result, every graph containing a $T\phi^*\pi$ -vertex will vanish, apart from the one with both outgoing arrows evaluated on vacua. As a consequence $S_{\text{eff}}^{(0)}$ and $S_{\text{eff}}^{(1)}$ have no quantum corrections.

4.3. Axial gauge on the torus $\Sigma = \mathbb{T}^2 := S^1 \times S^1$. In the case of a torus, differential forms have a bigrading with respect to the two circles. One may choose the axial gauge¹¹ by setting the superfields to vanish if they have nonzero degree with respect to the first circle. Naïvely this implies the propagator to be the product of a propagator for the de

¹¹The axial gauge for topological field theories was originally proposed in the context of Chern–Simons theory in [15].

Rham differential on the first circle and the identity operator on the second circle (just plug in the gauge fixed fields into in the unperturbed action to realize this). This argument however does not take vacua into account nor the fact that the axial gauge fixing does not fix all the gauge freedom. In fact, one can prove that the propagator in the axial gauge has one additional term, see (4.7) below.

To start with a rigorous construction of the propagator, observe that differential forms on a circle admit the Hodge decomposition

$$\Omega^\bullet(S^1) = \underbrace{\Omega_{\text{Harm}}^\bullet(S^1)}_{\text{Span}(1, d\tau)} \oplus \underbrace{\tilde{\Omega}^0(S^1)}_{\{f(\tau) \mid \int_{S^1} f(\tau) d\tau = 0\}} \oplus \underbrace{\tilde{\Omega}^1(S^1)}_{\{g(\tau) d\tau \mid \int_{S^1} g(\tau) d\tau = 0\}}$$

(In our convention the coordinate τ on the circle runs from 0 to 1). The associated chain homotopy operator is

$$K_{S^1} : g(\tau) d\tau \mapsto \int_{S^1} \omega_{S^1}(\tau, \tau') g(\tau') d\tau'$$

with the integral kernel

$$\omega_{S^1}(\tau, \tau') = \theta(\tau - \tau') - \tau + \tau' - \frac{1}{2}$$

Projection to harmonic forms on the circle (representatives of cohomology) is

$$\mathcal{P}_{S^1} : f(\tau) + g(\tau) d\tau \mapsto \int_{S^1} (d\tau' - d\tau) \wedge (f(\tau') + g(\tau') d\tau')$$

For the torus we may decompose the de Rham complex in the following way:

$$\begin{aligned} (4.6) \quad \Omega^\bullet(S^1 \times S^1) &= \Omega^\bullet(S^1) \hat{\otimes} \Omega^\bullet(S^1) = \\ &= \underbrace{\Omega_{\text{Harm}}^\bullet(S^1) \otimes \Omega_{\text{Harm}}^\bullet(S^1)}_{\cong H^\bullet(S^1 \times S^1)} \oplus \underbrace{\tilde{\Omega}^0(S^1) \hat{\otimes} \Omega^\bullet + \Omega_{\text{Harm}}^\bullet(S^1) \otimes \tilde{\Omega}^0(S^1)}_{\mathcal{L}_K} \oplus \\ &\quad \oplus \tilde{\Omega}^1(S^1) \hat{\otimes} \Omega^\bullet + \Omega_{\text{Harm}}^\bullet(S^1) \otimes \tilde{\Omega}^1(S^1) \end{aligned}$$

The associated chain homotopy operator is

$$(4.7) \quad K = \underbrace{K_{S^1} \otimes \text{id}_{S^1}}_{K_I} + \underbrace{\mathcal{P}_{S^1} \otimes K_{S^1}}_{K_{II}} : \Omega^\bullet(S^1 \times S^1) \rightarrow \Omega^{\bullet-1}(S^1 \times S^1)$$

Its integral kernel (the propagator) is

$$(4.8) \quad \omega = \underbrace{\left(\theta(\sigma - \sigma') - \sigma + \sigma' - \frac{1}{2} \right) \cdot \delta(\tau - \tau') \cdot (d\tau' - d\tau)}_{\omega_I} + \underbrace{(d\sigma' - d\sigma) \cdot \left(\theta(\tau - \tau') - \tau + \tau' - \frac{1}{2} \right)}_{\omega_{II}}$$

where we denote by $\sigma, \tau \in \mathbb{R}/\mathbb{Z}$ the coordinates on the first and the second circles, respectively.

Remark 4.3. The chain homotopy (4.7) arises from the composition of two quasi-isomorphisms:

$$\begin{aligned} \Omega^\bullet(S^1) \hat{\otimes} \Omega^\bullet(S^1) & \xlongequal{\quad} \Omega_{\text{Harm}}^\bullet(S^1) \otimes \Omega^\bullet(S^1) \oplus \tilde{\Omega}^0(S^1) \hat{\otimes} \Omega^\bullet(S^1) \oplus \tilde{\Omega}^1(S^1) \hat{\otimes} \Omega^\bullet(S^1) \\ & \downarrow \\ \Omega_{\text{Harm}}^\bullet(S^1) \otimes \Omega^\bullet(S^1) & \xlongequal{\quad} \Omega_{\text{Harm}}^\bullet \otimes \Omega_{\text{Harm}}^\bullet \oplus \Omega_{\text{Harm}}^\bullet(S^1) \otimes \tilde{\Omega}^0(S^1) \oplus \Omega_{\text{Harm}}^\bullet(S^1) \otimes \tilde{\Omega}^1(S^1) \\ & \downarrow \\ \Omega_{\text{Harm}}^\bullet(S^1) \otimes \Omega_{\text{Harm}}^\bullet(S^1) & \end{aligned}$$

i.e., we first contract the first circle to cohomology, then the second one.

4.4. Vanishing of quantum corrections.

Lemma 4.4. *For the Poisson sigma model in the axial gauge on torus, the source parts W_Γ^{source} vanish for all connected graphs Γ except for trees with one vertex (“corollas”).*

Proof. Let us introduce the basis in cohomology of the torus:

$$\chi_{(0,0)} = 1, \quad \chi_{1,0} = d\sigma, \quad \chi_{(0,1)} = d\tau, \quad \chi_{(1,1)} = d\sigma \wedge d\tau$$

Define a decoration c of Γ as an assignment of bidegree

$$c(h) \in \{(0,0), (1,0), (0,1), (1,1)\}$$

to each half-edge h of Γ (so that on leaves the bidegree coincides with the prescribed leaf decoration α) together with an assignment of an index $c(e) \in \{I, II\}$ to each edge e . Define the source part for a decorated graph Γ as

$$(4.9) \quad W_{\Gamma,c}^{\text{source}} = \int_{\Sigma \times V(\Gamma)} \left(\prod_{\text{edges } e=(h_{\text{in}}, h_{\text{out}})} \pi_{v(h_{\text{in}}), v(h_{\text{out}})}^* \omega_{c(e)}|_c \right) \cdot \left(\prod_{\text{leaves } l} \pi_{v(l)}^* \chi_{\alpha_l} \right)$$

where the $\omega|_c$ symbol means the component of the propagator (as an element of $\Omega^\bullet(S^1 \times S^1) \hat{\otimes} \Omega^\bullet(S^1 \times S^1)$) of de Rham bidegrees $c(h_{in}), c(h_{out})$ where h_{in}, h_{out} are the constituent half-edges of the edge; $\omega_{c(e)}$ is one of the two pieces of propagator, ω_I or ω_{II} , as defined in (4.8). Then we have

$$W_\Gamma^{\text{source}} = \sum_{\text{decorations } c} W_{\Gamma, c}^{\text{source}}$$

The source part $W_{\Gamma, c}^{\text{source}}$ vanishes automatically unless the following conditions are satisfied simultaneously:

- (i) At every vertex there is exactly one incident half-edge decorated by $(1, \bullet)$, all others are $(0, \bullet)$.
- (ii) At every vertex there is exactly one incident half-edge decorated by $(\bullet, 1)$, all others are $(\bullet, 0)$. (This half-edge may be the same as in (i)).
- (iii) Compatibility between edge decorations and half-edge decorations: for any edge $e = (h_1, h_2)$ we have

$$\begin{aligned} (c(e) = I) &\implies \begin{cases} c(h_1) = (0, 0), c(h_2) = (0, 1) \\ c(h_1) = (0, 1), c(h_2) = (0, 0) \end{cases} \quad \text{or} \\ (c(e) = II) &\implies \begin{cases} c(h_1) = (0, 0), c(h_2) = (1, 0) \\ c(h_1) = (1, 0), c(h_2) = (0, 0) \end{cases} \quad \text{or} \end{aligned}$$

- (iv) Number of edges decorated as I adjacent to any given vertex should be different from one.
- (v) If a vertex has no adjacent I -edges, then the number of adjacent II -edges should be different from one.

Requirements (i,ii) follow directly from degree counting in (4.9); (iii) follows from the formula for propagator (4.8); (iv,v) follow from the property $K_{S^1} \mathcal{P}_{S^1} = 0$ and from the fact that harmonic forms on a circle are closed under wedge multiplication.

Fix some decoration c of Γ satisfying (i–v). Consider the subgraph Γ_I of Γ obtained by deleting all II -edges in Γ ; Γ_I may be disconnected. Let $\Gamma_I = \sqcup_a \Gamma_I^a$ where Γ_I^a are the connected components of Γ_I . Due to (ii), the number of vertices V_I^a of Γ_I^a is equal to the number of $(0, 1)$ -half-edges in Γ_I^a which is in turn greater or equal to the number of edges E_I^a due to (iii). Hence the Euler characteristic of Γ_I^a non-negative: $V_I^a - E_I^a \geq 0$. Therefore Γ_I^a is either a tree or a 1-loop graph. Next, property (iv) shows that Γ_I^a has to be a wheel graph, with arbitrary number of leaves attached at vertices, or a corolla. On the other hand,

if Γ_I contains a wheel then the corresponding source part vanishes:

$$(4.10) \quad \begin{aligned} W_{\Gamma,c}^{\text{source}} &= \int_{(S^1 \times S^1)^{\times V}} (d\tau_1 - d\tau_2) \delta(\tau_2 - \tau_1) \wedge \cdots \wedge (d\tau_n - d\tau_1) \delta(\tau_1 - \tau_n) \wedge F = \\ &= \int_{(S^1 \times S^1)^{\times V}} \underbrace{(d\tau_1 \wedge d\tau_2 \wedge \cdots \wedge d\tau_n + (-1)^n d\tau_2 \wedge \cdots \wedge d\tau_n \wedge d\tau_1)}_{=0} \wedge \\ &\quad \wedge \delta(\tau_2 - \tau_1) \cdots \delta(\tau_1 - \tau_n) \wedge F = 0 \end{aligned}$$

where n is the length of the wheel and $F \in \Omega^\bullet((S^1 \times S^1)^{\times V})$ is some differential form.

Remark 4.5. Argument (4.10) has the fault that the integrand is singular and the result is $0 \cdot \delta(0)$. This can be remedied by regularizing the propagator ω , e.g., by changing $\delta(\tau - \tau')$ in (4.8) to a smeared delta-function. Notice that the source parts of all diagrams except corollas still vanish exactly: in this vanishing argument the chain homotopy equation is never used; we only use the de Rham bigrading properties, $PK = 0$ and the fact that harmonic forms on a circle are closed under multiplication.

Thus we have shown that $W_{\Gamma,c}^{\text{source}}$ vanishes unless Γ_I is a collection of corollas (i.e. there are no I -edges).

Now fix a decoration c satisfying (i-v) with $c(e) = II$ for all edges. Repeating the Euler characteristic argument as above (using properties (i,iii)), we show that Γ has to be either a tree or a 1-loop graph and using property (v) we show that it has to be either a wheel or a corolla. If it is a wheel then

$$(4.11) \quad \begin{aligned} W_{\Gamma,c}^{\text{source}} &= \int_{(S^1 \times S^1)^{\times V}} (d\sigma_1 - d\sigma_2) \wedge \cdots \wedge (d\sigma_V - d\sigma_1) \wedge F = \\ &= \int_{(S^1 \times S^1)^{\times V}} \underbrace{(d\sigma_1 \wedge \cdots \wedge d\sigma_V + (-1)^V d\sigma_2 \wedge \cdots \wedge d\sigma_V \wedge d\sigma_1)}_{=0} \wedge F = \\ &= 0 \end{aligned}$$

Therefore $W_{\Gamma,c}^{\text{source}}$ vanishes for any decoration c unless Γ is a corolla. This concludes the proof of the Lemma. $\square$

An immediate consequence of the Lemma is that the effective action S_x^{eff} is just the restriction of the action S_x to vacua: there are no quantum corrections.

5. THE PARTITION FUNCTION ON THE TORUS

Let S_{eff} be the global effective action on the moduli space of vacua for the torus $\mathbb{T}^2$. In Lemma 4.4, we have shown that it has no quantum corrections. The moduli space of vacua can be viewed as $\text{Map}(\mathbb{R}^2[1], T^*[1]M)$ and in [5] it has been remarked that the action restricted to vacua is the AKSZ action for this mapping space. In local coordinates the superfields are

$$(5.1) \quad x^\mu = x^\mu + e^1 \eta_1^{+\mu} + e^2 \eta_2^{+\mu} - s b^{+\mu}, \quad \mathbf{e}_\nu = b_\nu + e^1 \eta_{\nu 1} + e^2 \eta_{\nu 2} + s x_\nu^+,$$

where $s = e^1 e^2$ is the generator of $H_{dR}^2(\mathbb{T}^2)$ normalized to $\int_{\mathbb{R}^2[1]} ds s = 1$.

If π is the Poisson bivector field on M , then

$$(5.2) \quad S_{\text{eff}} = \frac{1}{2} \int_{\mathbb{R}^2[1]} ds \pi^{\mu\nu}(\mathbf{x}) \mathbf{e}_\mu \mathbf{e}_\nu.$$

There exists a canonical Berezinian given by the coordinate volume form

$$(5.3) \quad \nu = dx \cdots dx^+ \cdots db \cdots db^+ \cdots d\eta_i \cdots d\eta_i^+ \cdots.$$

If we denote with Δ the corresponding Laplacian, the AKSZ action satisfies

$$\Delta e^{\frac{i}{\hbar} S_{\text{eff}}} = 0$$

and defines a class in Δ -cohomology.

5.1. Kähler gauge fixing and Euler class. Now let π be symplectic such that π^{-1} is the Kähler form of the hermitian structure (J, g) . In the complex coordinates $\{x^i\}$ of M we have $\pi^{i\bar{j}} = i g^{i\bar{j}}$. Let us fix a complex structure on $\mathbb{T}^2$ defined by $z = \theta^1 + \tau \theta^2$, for $\tau = \tau_1 + i\tau_2$ and $\tau_2 > 0$. Let

$$\eta_z^{+\mu} = (\eta_2^{+\mu} - \bar{\tau} \eta_1^{+\mu}) / 2i\tau_2, \quad \eta_{z\mu} = (\eta_{2\mu} - \bar{\tau} \eta_{1\mu}) / 2i\tau_2.$$

Let $\mathcal{L}_{\varepsilon, \tau}$ be the following Lagrangian submanifold of $\text{Map}(\mathbb{R}^2[1], T^*[1]M)$:

$$(5.4) \quad \eta_z^{+i} = \eta_{\bar{z}}^{+\bar{i}} = \eta_{zi} = \eta_{\bar{z}\bar{i}} = x^+ = b^+ = 0.$$

Let us define

$$p_k = \eta_{\bar{z}k} + \Gamma_{ki}^j \eta_{\bar{z}}^{+i} b_j,$$

where Γ are the Christoffel symbols of the Levi-Civita connection. All fiber coordinates b, p, η^+ transform tensorially with respect to a transformation of coordinates on M so that $\mathcal{L}_{\varepsilon, \tau} = (T^*[1] + T^*M + T[-1])M$. After a straightforward computation we get

$$S_{\text{eff}} = \tau_2 \left(R_{s\bar{i}k}^j g^{s\bar{r}} \eta_{\bar{z}}^{+k} \eta_z^{+\bar{l}} b_j b_{\bar{r}} + g^{i\bar{j}} p_i p_{\bar{j}} \right)$$

$$= \frac{1}{2} \tau_2 (R^{\mu\lambda} b_\mu b_\lambda + g^{\mu\nu} p_\mu p_\nu) .$$

The induced Berezinian on $\mathcal{L}_{\varepsilon, \tau}$ reads

$$\sqrt{\nu} = \frac{dx^\mu \cdots db_\mu \cdots dp_\mu \cdots d\eta^{+\mu} \cdots}{(2\pi)^m},$$

with $m = \dim M$.¹² If we perform the fiberwise integration with respect to the fibration $\mathcal{L}_{\varepsilon, \tau} \rightarrow T[-1]M$, we get

$$\begin{aligned} \frac{1}{(2\pi)^m} \int dp_\mu \cdots db_\mu \cdots e^{\frac{i}{\hbar} S_{\text{eff}}} &= \\ &= \frac{1}{(2\pi)^m} \frac{1}{(i\hbar)^{\frac{m}{2}}} \int db_\mu \cdots \frac{1}{\det(\tau_2 g^{\mu\nu})^{1/2}} e^{\frac{i}{2\hbar} \tau_2 R^{\mu\nu} b_\mu b_\nu} = \\ &= \frac{1}{(2\pi)^{\frac{m}{2}}} \int db'_\mu \cdots \frac{1}{\det(g^{\mu\nu})^{1/2}} e^{-\frac{1}{2} R^{\mu\nu} b'_\mu b'_\nu} = \\ &= \frac{1}{(2\pi)^{\frac{m}{2}}} \sqrt{g} \text{Pf}(R) \in C^\infty(T[-1]M) = \Omega M, \end{aligned}$$

which, by the Chern–Gauss–Bonnet theorem, is a representative of the Euler class. Notice that $\hbar$ and τ_2 disappear in the final formula. (The main reason for this is that scaling the b and p variables by the same factor preserves the Berezinian since the former are odd and the latter are even variables.) Finally, we can integrate over M getting

$$Z = \chi(M),$$

the Euler characteristic of M . (Actually, by the argument in the Introduction that the partition function should be the Euler characteristic, we might in reverse think of this result as one more physical proof of the Chern–Gauss–Bonnet theorem, in the case of Kähler manifolds.)

Remark 5.1. In this Section we have assumed the existence of a Kähler structure on M . We expect the above results to hold if we just use an almost Kähler structure, but computations become much more involved.

¹²We choose here the standard convention that the measure for a pair of even conjugate coordinates p, q is $dp dq / (2\pi i \hbar)$, whereas the measure for a pair of odd conjugate coordinates b, η^+ is $i\hbar db d\eta^+$. This is consistent with the standard normalization

$$\int e^{\frac{i}{\hbar} pq} \frac{dp dq}{2\pi i \hbar} = \int e^{\frac{i}{\hbar} b\eta^+} i\hbar db d\eta^+ = 1.$$

Remark 5.2. Another possible gauge fixing consists in setting all $+$ variables to zero. The effective action then reduces to $\pi^{\mu\nu}(x)\eta_{\mu 1}\eta_{\nu 2}$ and is independent of the b variables. If M is compact, the integrals over the η and x variables is finite (and proportional to the symplectic volume of M); because of the b -integration, the partition function then vanishes. If M is not compact, the partition function is ambiguous and of the form $0 \cdot \infty$. This gauge fixing is then in general not equivalent to the Kähler one used above. From the considerations in the Introduction, the Kähler gauge fixing is the one compatible with the Hamiltonian interpretation of the theory.

5.2. Regularized effective action. We now show that the symmetries of S_{eff} induce a regularization which allows one to compute the partition function for every Poisson structure and to show that, independently of the Poisson structure, one gets the Euler characteristic of the target.¹³

The main remark is that the effective action and the symplectic form are invariant under the action of the Lie algebra of divergenceless vector fields of $\mathbb{R}^2[1]$ on the moduli space of vacua. This Lie algebra is spanned by the vector fields $\frac{\partial}{\partial e^1}$, $\frac{\partial}{\partial e^2}$, $e^2 \frac{\partial}{\partial e^1}$, $e^1 \frac{\partial}{\partial e^2}$ and $e^1 \frac{\partial}{\partial e^1} - e^2 \frac{\partial}{\partial e^2}$. The fifth vector field is generated by the previous ones and we are not going to need it in the following. We lift the first four vector fields first to $\text{Map}(\mathbb{R}^2[1], M)$ and next to its cotangent bundle shifted by one. We will denote the resulting vector fields by δ_1 , δ_2 , K_1 and K_2 , respectively. Since they have degree -1 or 0 and the symplectic form has degree -1 , they are also automatically Hamiltonian with uniquely defined Hamiltonian functions τ_1 and τ_2 (of degree -2), and ρ_1 and ρ_2 (of degree -1). The Lie algebra relations translate into the Poisson bracket relations

$$(5.5) \quad (\tau_2, \rho_1) = \tau_1, \quad (\tau_1, \rho_2) = \tau_2, \quad (\tau_1, \rho_1) = 0, \quad (\tau_2, \rho_2) = 0.$$

Also notice that we have

$$(5.6) \quad K_1 \circ \delta_1 = K_2 \circ \delta_2 = 0,$$

which implies that ρ_i Poisson commutes with every δ_i -exact function. Finally, since we started with divergenceless vector fields, we get

$$(5.7) \quad \Delta\tau_1 = \Delta\tau_2 = \Delta\rho_1 = \Delta\rho_2 = 0.$$

Remark 5.3. Even though we do not need the explicit form of these vector fields and their Hamiltonian functions, we give them for completeness of our presentation. From the defining formulae $\delta_i \mathbf{x} = \frac{\partial \mathbf{x}}{\partial e^i}$,

¹³We thank T. Johnson-Freyd for pointing out this approach.

$\delta_i \mathbf{e} = \frac{\partial \mathbf{e}}{\partial e^i} K_1 \mathbf{x} = e^2 \frac{\partial \mathbf{x}}{\partial e^1}$, $K_1 \mathbf{e} = e^2 \frac{\partial \mathbf{e}}{\partial e^1}$, $K_2 \mathbf{x} = e^1 \frac{\partial \mathbf{x}}{\partial e^2}$ and $K_2 \mathbf{e} = e^1 \frac{\partial \mathbf{e}}{\partial e^2}$, we get

$$\begin{aligned}\delta_1 &= \eta_1^{+\mu} \frac{\partial}{\partial x^\mu} + b^{+\mu} \frac{\partial}{\partial \eta_2^{+\mu}} + \eta_{1\mu} \frac{\partial}{\partial b_\mu} - x_\mu^+ \frac{\partial}{\partial \eta_{2\mu}}, \\ \delta_2 &= \eta_2^{+\mu} \frac{\partial}{\partial x^\mu} - b^{+\mu} \frac{\partial}{\partial \eta_1^{+\mu}} + \eta_{2\mu} \frac{\partial}{\partial b_\mu} + x_\mu^+ \frac{\partial}{\partial \eta_{1\mu}}, \\ K_1 &= \eta_{1\mu} \frac{\partial}{\partial \eta_{2\mu}} + \eta_1^{+\mu} \frac{\partial}{\partial \eta_2^{+\mu}}, \\ K_2 &= \eta_{2\mu} \frac{\partial}{\partial \eta_{1\mu}} + \eta_2^{+\mu} \frac{\partial}{\partial \eta_1^{+\mu}}.\end{aligned}$$

The corresponding Hamiltonian functions, with respect to the symplectic structure

$$\omega = \int de^2 de^1 \delta \mathbf{x}^\mu \delta \mathbf{e}_\mu = \delta x^\mu \delta x_\mu^+ + \delta \eta_1^{+\mu} \delta \eta_{2\mu} - \delta \eta_2^{+\mu} \delta \eta_{1\mu} - \delta b^{+\mu} \delta b_\mu,$$

are given by

$$\begin{aligned}\tau_1 &= x_\mu^+ \eta_1^{+\mu} - \eta_{1\mu} b^{+\mu}, \\ \tau_2 &= x_\mu^+ \eta_2^{+\mu} - \eta_{2\mu} b^{+\mu}, \\ \rho_1 &= -\eta_{1\mu} \eta_1^{+\mu}, \\ \rho_2 &= \eta_{2\mu} \eta_2^{+\mu}.\end{aligned}$$

We now turn back to the effective action. It turns out that it is not only δ_1 - and δ_2 -closed, but actually exact:

$$S_{\text{eff}} = \delta_2 \delta_1 \sigma, \quad \sigma := \frac{1}{2} \pi^{\mu\nu}(x) b_\mu b_\nu.$$

From all the above it follows that S_{eff} Poisson commutes not only with τ_1 and τ_2 , but also with ρ_1 and ρ_2 . Notice that the Jacobi identity for π implies $(S_{\text{eff}}, \sigma) = 0$.

Now consider the regularized effective action

$$S_{\text{eff}}^{\epsilon, t_1, t_2} := \epsilon S_{\text{eff}} - i\hbar(t_1 \tau_1 + t_2 \tau_2),$$

which satisfies the QME for all ϵ, t_1, t_2 . By all the above it follows that

$$\begin{aligned}\frac{\partial}{\partial t_1} e^{\frac{i}{\hbar} S_{\text{eff}}^{\epsilon, t_1, t_2}} &= \Delta \left(\frac{1}{t_2} e^{\frac{i}{\hbar} S_{\text{eff}}^{\epsilon, t_1, t_2}} \rho_1 \right), \\ \frac{\partial}{\partial t_2} e^{\frac{i}{\hbar} S_{\text{eff}}^{\epsilon, t_1, t_2}} &= \Delta \left(\frac{1}{t_1} e^{\frac{i}{\hbar} S_{\text{eff}}^{\epsilon, t_1, t_2}} \rho_2 \right),\end{aligned}$$

which means that, as long as the parameters t_1 and t_2 are different from zero, the regularized effective action is independent of them up to quantum canonical transformations. We also have

$$\frac{\partial}{\partial \epsilon} e^{\frac{i}{\hbar} S_{\text{eff}}^{\epsilon, t_1, t_2}} = \Delta \left(\frac{i}{\hbar t_2} e^{\frac{i}{\hbar} S_{\text{eff}}^{\epsilon, t_1, t_2}} \delta_1 \sigma \right) = -\Delta \left(\frac{i}{\hbar t_1} e^{\frac{i}{\hbar} S_{\text{eff}}^{\epsilon, t_1, t_2}} \delta_2 \sigma \right),$$

which implies that, as long as one of the parameters t_1 and t_2 is different from zero, the regularized effective action is independent of ϵ up to quantum canonical transformations. This in particular means that the partition function is independent of ϵ and that, in order to compute it, we may simply set ϵ to zero.

To perform the final computation we further deform the regularized effective action by adding one more irrelevant term. Namely, let G be a function on $\text{Map}(\mathbb{R}^2[1], M)$. Then

$$S_{\text{eff}}^{0, t_1, t_2, G} := -i\hbar(t_1 \tau_1 + t_2 \tau_2 + \delta_2 \delta_1 G)$$

satisfies the QME for all t_1, t_2, G . Moreover, if we take a path $G(t)$ of such functions, we get

$$\frac{\partial}{\partial t} e^{\frac{i}{\hbar} S_{\text{eff}}^{0, t_1, t_2, G(t)}} = \Delta \left(\frac{e^{\frac{i}{\hbar} S_{\text{eff}}^{0, t_1, t_2, G(t)}}}{t_2} \delta_1 \dot{G}(t) \right) = -\Delta \left(\frac{e^{\frac{i}{\hbar} S_{\text{eff}}^{0, t_1, t_2, G(t)}}}{t_1} \delta_2 \dot{G}(t) \right),$$

which means that, as long as one of the two parameters t_1 and t_2 is different from zero, adding the new term is irrelevant up to quantum canonical transformations. We are now ready to compute the partition function. Namely, we choose $\mathcal{L} := \text{Map}(\mathbb{R}^2[1], M)$ as the Lagrangian submanifold of $\text{Map}(\mathbb{R}^2[1], T^*[1]M)$ over which we integrate. Since $\tau_1|_{\mathcal{L}} = \tau_2|_{\mathcal{L}} = 0$, we get

$$Z = \int_{\mathcal{L}} e^{\frac{i}{\hbar} S_{\text{eff}}^{0, t_1, t_2, G}} = \int_{\text{Map}(\mathbb{R}^2[1], M)} e^{\delta_2 \delta_1 G}$$

and we already know that the last integral is independent of G . We only have to make sure that G is chosen in such a way that the integral is well defined (choosing $G = 0$, e.g., would lead to $\infty \cdot 0$). A good choice is $G := g_{\mu\nu}(x) \eta_1^{+\mu} \eta_2^{+\nu}$ where $g^{\mu\nu}$ is a Riemannian metric on target. An explicit computation [3, 4] then shows that $Z = \chi(M)$.

Remark 5.4. Switching ϵ to zero first and then turning on the regularizing term in G is a bit formal since we pass through the solution to the QME where both terms are absent. This solution has a singular integral (of the type $0 \cdot \infty$) on $\mathcal{L}$. In order to find a non formal regularization it is necessary to have additional structure on the Poisson manifold. For instance let us look for G such that $S_{\text{eff}}^{\epsilon, t_1, t_2, G}$ satisfies the

QME for any ϵ , preserving the property that the variation of G produces a quantum canonical transformation. Indeed, let us assume G as above but let g be possibly degenerate. If $(S_{\text{eff}}, G) = 0$ then $S_{\text{eff}}^{\epsilon, t_1, t_2, G}$ satisfies the QME and the change of G is a quantum canonical transformation. This property is equivalent to require that $\pi \circ g = 0$ and $L_V g = 0$ for every vector field V tangent to the symplectic leaves. In special cases we may find such a g and in addition a Kähler structure on the leaves, compatible with the symplectic structure, such that a gauge fixing given by a mixture of what we discussed in this subsection and the Kähler one is available. We plan to investigate the geometrical conditions needed for this gauge fixing in the future.

5.3. Regularization on the space of fields. The argument of the previous subsection may formally be lifted to the space of fields to show that the regularized action is actually independent, up to quantum canonical transformations, of the Poisson structure. Let s^1 and s^2 denote the coordinates on the two S^1 factors of the torus $\mathbb{T}^2$, and let e^1 and e^2 denote the corresponding fiber coordinates on $T[1]\mathbb{T}^2$. We now denote by δ_1 , δ_2 , K_1 and K_2 the lifts of the vector fields $\frac{\partial}{\partial e^1}$, $\frac{\partial}{\partial e^2}$, $e^2 \frac{\partial}{\partial e^1}$ and $e^1 \frac{\partial}{\partial e^2}$ to the space of fields $\mathcal{F} = \text{Map}(T[1]\mathbb{T}^2, T^*[1]M)$. We denote by τ_1 , τ_2 , ρ_1 and ρ_2 their Hamiltonian functions. They satisfy (5.5) and (5.6), and formally also (5.7).

Remark 5.5. For completeness, we give explicit expressions also in this case, even if we are not going to need them. If we write

$$\begin{aligned} \mathbf{X} &= X + \eta_1^+ e^1 + \eta_2^+ e^2 + \beta^+ e^1 e^2, \\ \boldsymbol{\eta} &= \beta + \eta_1 e^1 + \eta_2 e^2 + X^+ e^1 e^2, \end{aligned}$$

we then have

$$\begin{aligned} \delta_1 X &= -\eta_1^+, & \delta_1 \eta_2^+ &= \beta^+, & \delta_1 \beta &= \eta_1, & \delta_1 \eta_2 &= -X^+, \\ \delta_2 X &= -\eta_2^+, & \delta_2 \eta_1^+ &= -\beta^+, & \delta_2 \beta &= \eta_2, & \delta_2 \eta_1 &= X^+, \end{aligned}$$

and

$$\begin{aligned} K_1 \eta_2^+ &= \eta_1^+, & K_2 \eta_1^+ &= \eta_2^+, \\ K_1 \eta_2 &= \eta_1, & K_2 \eta_1 &= \eta_2. \end{aligned}$$

With respect to the symplectic form

$$\Omega = \int_{\mathcal{F}} \delta \mathbf{X} \delta \boldsymbol{\eta} = \int_{\mathbb{T}^2} (\delta X^\mu \delta X_\mu^+ - \delta \eta_1^{+\mu} \delta \eta_{2\mu} + \delta \eta_2^{+\mu} \delta \eta_{1\mu} + \delta \beta^{+\mu} \delta \beta_\mu) ds^1 ds^2,$$

the corresponding Hamiltonian functions are

$$\begin{aligned}\tau_1 &= \int_{\mathbb{T}^2} (-\eta_1^{+\mu} X_\mu^+ + \beta^{+\mu} \eta_{1\mu}) ds^1 ds^2, \\ \tau_2 &= \int_{\mathbb{T}^2} (-\eta_2^{+\mu} X_\mu^+ + \beta^{+\mu} \eta_{2\mu}) ds^1 ds^2, \\ \rho_1 &= \int_{\mathbb{T}^2} \eta_1^{+\mu} \eta_{1\mu} ds^1 ds^2, \\ \rho_2 &= - \int_{\mathbb{T}^2} \eta_2^{+\mu} \eta_{2\mu} ds^1 ds^2.\end{aligned}$$

Notice that despite their non covariant look the above formulae are actually globally well defined.

The action $S = S_0 + S_\pi$ is δ_1 - and δ_2 -closed; it turns out that the interaction part S_π is actually exact:

$$S_\pi = \delta_2 \delta_1 \sigma_\pi, \quad \sigma_\pi := \int_{\mathbb{T}^2} \frac{1}{2} \pi^{\mu\nu}(X) \beta_\mu \beta_\nu ds^1 ds^2.$$

From all the above it follows that S Poisson commutes not only with τ_1 and τ_2 , but also with ρ_1 and ρ_2 . Notice that the Jacobi identity for π implies $(S, \sigma_\pi) = 0$.

Now consider the regularized action

$$S^{\epsilon, t_1, t_2} := S_0 + \epsilon S_\pi - i\hbar(t_1 \tau_1 + t_2 \tau_2),$$

which satisfies the CME and formally also the QME for all ϵ, t_1, t_2 . By all the above it follows that, formally,

$$\begin{aligned}\frac{\partial}{\partial t_1} e^{\frac{i}{\hbar} S^{\epsilon, t_1, t_2}} &= \Delta \left(\frac{1}{t_2} e^{\frac{i}{\hbar} S^{\epsilon, t_1, t_2}} \rho_1 \right), \\ \frac{\partial}{\partial t_2} e^{\frac{i}{\hbar} S^{\epsilon, t_1, t_2}} &= \Delta \left(\frac{1}{t_1} e^{\frac{i}{\hbar} S^{\epsilon, t_1, t_2}} \rho_2 \right),\end{aligned}$$

which means that, as long as the parameters t_1 and t_2 are different from zero, the regularized action is independent of them up to quantum canonical transformations. We also have, again formally,

$$\frac{\partial}{\partial \epsilon} e^{\frac{i}{\hbar} S^{\epsilon, t_1, t_2}} = \Delta \left(\frac{i}{\hbar t_2} e^{\frac{i}{\hbar} S^{\epsilon, t_1, t_2}} \delta_1 \sigma \right) = -\Delta \left(\frac{i}{\hbar t_1} e^{\frac{i}{\hbar} S^{\epsilon, t_1, t_2}} \delta_2 \sigma \right),$$

which implies that, as long as one of the parameters t_1 and t_2 is different from zero, the regularized action is independent of ϵ up to quantum canonical transformations. This in particular means that the partition function is independent of ϵ and that, in order to compute it, we may simply set ϵ to zero. It is now easy to see that, for a reasonable choice of propagators, the effective action for S^{0, t_1, t_2} is simply the restriction

to vacua, that is the the regularized effective action $S_{\text{eff}}^{0,t_1,t_2}$ considered in the previous subsection.

6. GLOBALIZATION OF THE EFFECTIVE ACTION

We now go back to the problem of globalizing $\tilde{S}_{\text{eff}}^{(0)}$ in the general case. Recall that $\tilde{S}_{\text{eff}} \in \Gamma(\Lambda T^*M \otimes \hat{S}\mathcal{H}^*[[\hbar]])$ satisfies the modified QME (3.2). We write $\tilde{S}_{\text{eff}} = \sum_{i=0}^m \tilde{S}_{\text{eff}}^{(i)}$, where $\tilde{S}_{\text{eff}}^{(i)}$ is the i -form component and $m = \dim M$. In form degree zero, we have

$$\frac{1}{2}(\tilde{S}_{\text{eff}}^{(0)}, \tilde{S}_{\text{eff}}^{(0)}) - i\hbar\Delta\tilde{S}_{\text{eff}}^{(0)} = 0,$$

which is the usual QME.

The modified QME is preserved under modified quantum canonical transformations. Namely, $T \in \Gamma(\Lambda T^*M \otimes \hat{S}\mathcal{H}^*[[\hbar]])$ of total degree -1 generates the infinitesimal transformation

$$\delta\tilde{S}_{\text{eff}} = d_x T + (\tilde{S}_{\text{eff}}, T) - i\hbar\Delta T$$

which preserves the modified QME at the infinitesimal level. Notice that, setting $T = \sum_{i=0}^m T^{(i)}$, we get in form degree zero

$$\delta\tilde{S}_{\text{eff}}^{(0)} = (\tilde{S}_{\text{eff}}^{(0)}, T^{(0)}) - i\hbar\Delta T^{(0)},$$

which is a usual infinitesimal quantum canonical transformation. The goal of this Section is to prove the following

Theorem 6.1. *There is a quantum canonical transformation starting at order 1 in $\hbar$ that makes the form degree zero part $\tilde{S}_{\text{eff}}^{(0)}$ of the effective action closed with respect to the induced Grothendieck differential $D = d_x + (S_R|_{\text{vacua}}, \cdot)$ on $\Gamma(\Lambda T^*M \otimes \hat{S}\mathcal{H}^*[[\hbar]])$, where $S_R|_{\text{vacua}}$ denotes the evaluation of S_R on vacua.*

This will ensure that the so obtained effective action, call it $\check{S}_{\text{eff}}^{(0)}$, is the image under $T\phi^*$ of a global effective action S_{eff} . Since $\check{S}_{\text{eff}}^{(0)} \in \Gamma(\hat{S}\mathcal{H}^*[[\hbar]])$, it follows from the discussion just before subsection 2.1 that S_{eff} is a section of $\hat{S}\tilde{\mathcal{H}}^*[[\hbar]]$,¹⁴ i.e., a formal power series in $\hbar$ of functions on $\tilde{\mathcal{H}}$ (formal in the fiber coordinates). Again, we may identify $\tilde{\mathcal{H}}$ with the canonical global moduli space of vacua by using a connection (e.g., the one in ϕ). By Remark 3.1, we conclude that the class of S_{eff} under quantum canonical transformations is a well-defined object independent of all choices.

¹⁴Recall that $\tilde{\mathcal{H}}_x = H^{>0}(\Sigma) \otimes T_x M \oplus H^\bullet(\Sigma) \otimes T_x^* M[1]$.

6.1. Proof of Theorem 6.1. We start with a simple observation:

Lemma 6.2. *Write $\widetilde{S}_{\text{eff}}^{(i)} = \sum_{k=0}^{\infty} \hbar^k S_k^{(i)}$. If the propagator satisfies the properties in (4.3), then $S_0^{(i)} = 0 \ \forall i > 1$, whereas $S_0^{(0)}$ and $S_0^{(1)}$ are obtained by the evaluation on vacua of $\widehat{S}$ and S_R , respectively.*

Proof. The terms for $k = 0$ correspond to trees in the expansion in Feynman diagrams; so, using the notations of Section 4, what we have to prove is that the source part $W_{\Gamma}^{\text{source}}$ vanishes for any tree Γ containing more than one vertex.

This is checked by the following degree counting argument. Consider a tree Γ containing more than one vertex. Let V_k be the number of vertices in Γ of internal valence (i.e., not counting the leaves) equal to $k \geq 1$. Then the total number of vertices is

$$V = \sum_{k \geq 1} V_k,$$

the number of internal edges is

$$(6.1) \quad E = \frac{1}{2} \sum_{k \geq 1} k V_k,$$

and the Euler characteristic of Γ is

$$(6.2) \quad 1 = V - E = \sum_{k \geq 1} \frac{2-k}{2} V_k.$$

Next, the source part $W_{\Gamma}^{\text{source}}$ vanishes automatically due to $K\mathcal{P} = \mathcal{P}K = 0$ and $K^2 = 0$, unless the following two properties hold for the decoration of leaves by cohomology classes $\chi_{\alpha} \in H^{\bullet}(\Sigma)$:

- (i) At every vertex of internal valence 1 there are at least two incident leaves decorated by cohomology classes of non-zero degree. (Otherwise $W_{\Gamma}^{\text{source}}$ vanishes due to $K\mathcal{P} = 0$.)
- (ii) At every vertex of internal valence 2 there is at least one incident leaf decorated by a cohomology class of non-zero degree. (Otherwise $W_{\Gamma}^{\text{source}}$ vanishes due to $K^2 = 0$.)

This gives a lower bound $E + 2V_1 + V_2$ for the form degree of the integrand in (4.5); since it should coincide with the dimension of the space $\Sigma^{\times V}$ it is integrated against, we have the inequality

$$(6.3) \quad E + 2V_1 + V_2 \leq 2V.$$

By (6.1) this is equivalent to

$$\frac{1}{2} V_1 + \sum_{k \geq 3} \frac{k-4}{2} V_k \leq 0.$$

Subtracting (6.2) from this inequality, we get

$$\sum_{k \geq 3} (k-3)V_k \leq -1$$

which is a contradiction. Thus it is impossible to find a decoration of leaves of Γ satisfying properties (i) and (ii) simultaneously. Therefore, W_Γ^{source} vanishes for any decoration. $\square$

We now set $S^{(1)'} = \tilde{S}_{\text{eff}}^{(1)} - S_0^{(1)}$ and $S^{(i)'} = \tilde{S}_{\text{eff}}^{(i)}$ for $i > 1$.

Lemma 6.3. *There is a modified quantum canonical transformation starting at order 1 in $\hbar$ after which all $S^{(i)'}$ for $i \geq 1$ vanish.*

Proof. We work by induction on the order of $\hbar$. At order zero the statement holds by Lemma 6.2. Assume that $S_r^{(i)'} = 0 \forall i \geq 1$ and $\forall r < k$. Then the modified quantum master equation yields the identities

$$DS_k^{(i)'} + \Omega S_k^{(i+1)'} = 0, \quad \forall i \geq 1,$$

with $D := d_x + (S_0^{(1)}, \cdot)$ and $\Omega := (S_0^{(0)}, \cdot)$. We already know that $D^2 = \Omega^2 = 0$ and that D and Ω commute. If $T = \sum_{r=k}^{\infty} \hbar^r T_r$ is a generator starting at the order k , we then have the infinitesimal transformations

$$\delta S_k^{(i)'} = DT_k^{(i-1)} + \Omega T_k^{(i)}, \quad \forall i \geq 1.$$

By dimensional reasons $DS^{(m)'} = 0$, with $m = \dim M$, and since the D -cohomology is concentrated in degree zero, we can find a $\tau \in \Gamma(\Lambda T^*M \otimes \widehat{\mathcal{H}}^*)$ such that $S^{(m)'} = D\tau$. We now consider the transformation with generator $T = -\hbar^k \tau$. Hence we get $\delta S_k^{(m)'} = -D\tau$, $\delta S_k^{(m-1)'} = -\Omega\tau$ and $\delta S_k^{(i)'} = 0$ for $i < m-1$. Integrating this transformation up to time 1, we make $S_k^{(m)'} vanish; as a result the new $S_k^{(m-1)'}$ will be D -closed. We may then proceed like this until we make all the $S^{(i)'}$ vanish. This proves our claim.$

Notice that these transformations may change the $S_r^{(i)'}$ for $r > k$. Moreover, the generator used to kill $S_k^{(1)'}$ will act on $S_r^{(0)}$ for $r \geq k$ by a quantum canonical transformation. $\square$

This completes the proof of Theorem 6.1.

As a final remark, observe that in the case when π is regular and unimodular we start with $S^{(1)'} = 0$, so we have two different but equivalent ways of getting the global action. One consists in taking the original $\tilde{S}_{\text{eff}}^{(0)}$, the other in applying the method described in this Section since nothing guarantees that the remaining $S^{(i)'}$ vanish at the start. After applying the method we get another effective action $\check{S}_{\text{eff}}^{(0)}$ that simply

differs from $\tilde{S}_{\text{eff}}^{(0)}$ by a quantum canonical transformation and is also the image of $T\phi^*$ of a global action which we denote by $\check{S}$. Eventually, the two global effective actions S and $\check{S}$ simply differ by a quantum canonical transformation starting at order $\hbar$.

7. CONCLUSIONS AND PERSPECTIVES

In this paper we have studied the effective action of the Poisson sigma model on a closed surface Σ , where the Poisson structure π on the target M is treated perturbatively and, for consistency, has to be assumed to be unimodular unless Σ is a torus. We have shown how to obtain a global effective action S_{eff} as an $\hbar$ -dependent function on the moduli space of vacua of the theory with zero Poisson structure, around which we are perturbing. Because of the freedom in the choice of gauge fixing and the details of globalization, S_{eff} is, as usual, only well-defined up to quantum canonical transformations. By a reasonable choice of the class of allowed gauge fixings—namely, those for which the propagator enjoys properties (4.3)—we make sure that the order zero $S_{\text{eff},0}$ of the effective action is fixed and equal to the evaluation on vacua of the Poisson-dependent part S_π of the action; moreover, the remaining quantum canonical transformations will start at order 1.

In the cases when Σ is a torus or π is regular and unimodular, we have shown that S_{eff} has (a representative with) no quantum corrections. In the particular case when Σ is a torus, π is nondegenerate and there is a compatible complex structure, we can use the latter to gauge-fix the remaining integration over vacua: the final result is that, as expected from the Hamiltonian formulation and from comparison with the A-model, the partition function is the Euler characteristic of the target. An alternative approach that produces the same result consists in regularizing the effective action by adding the Hamiltonian functions of supersymmetry generators. In general, the effective action modulo quantum canonical transformations is an invariant of the Poisson structure.

Recall that each order in $\hbar$ of S_{eff} is actually a section of a vector bundle $\mathcal{Z}_g := \widehat{S}\widetilde{\mathcal{H}}^*$ over the target M whose structure is fixed by the genus g of the source Σ . These sections are just tensors of a particular sort. The lowest order in the quantum master equation for S_{eff} implies that $S_{\text{eff},0}$ solves the classical master equation, i.e., that it defines a differential on $\Gamma(\mathcal{Z}_g)$, which we call the genus g Poisson complex. Since $S_{\text{eff},0}$ is also Δ -closed, the lowest nonvanishing quantum contribution

to S_{eff} is a cocycle in the genus g Poisson complex¹⁵ and defines an invariant of the Poisson structure, which might be possible to compute in concrete examples.

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¹⁵If this happens at order 1, which can be easily seen not to be the case for $g \leq 1$, this class is also what is left after modding out by quantum canonical transformations at this order.

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CLASSICAL BV THEORIES ON MANIFOLDS WITH BOUNDARY

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ABSTRACT. In this paper we extend the classical BV framework to gauge theories on spacetime manifolds with boundary. In particular, we connect the BV construction in the bulk with the BFV construction on the boundary and we develop its extension to strata of higher codimension in the case of manifolds with corners. We present several examples including electrodynamics, Yang-Mills theory and topological field theories coming from the AKSZ construction, in particular, the Chern-Simons theory, the BF theory, and the Poisson sigma model. This paper is the first step towards developing the perturbative quantization of such theories on manifolds with boundary in a way consistent with gluing.

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1. INTRODUCTION

One of the key features in quantum field theory is locality. Physically, it is based on the concept of an ideal point-like particle. Mathematically, it means that the action in the corresponding classical field theory should be a local functional.

In quantum field theory locality is expected to correspond to the gluing property of the partition function for manifolds with boundary. For a quantum field theory on Minkowski cylinders this implies the usual composition law for evolution operators. In topological and conformal field theories the locality as a gluing property of partition functions was discussed in [4, 51, 56].

In this paper we formulate the classical Batalin-Vilkovisky (BV) framework for theories on manifolds with boundary. This combines the properties of the standard BV [8] and the Batalin-Fradkin-Vilkovisky (BFV) [9] frameworks. Recall that standard BV theories are odd symplectic extensions of classical gauge theories by ghosts, anti-fields and antighosts. The gauge symmetry appears as a cohomological Hamiltonian vector field whose Hamiltonian function is an extension of the classical action. The BFV theories give a similar cohomological description of the gauge symmetries on the boundary. Another relation between BV and BFV theories is described in [30].

This paper should be considered as a first step towards the perturbative quantization of classical BV theories on manifolds with boundary, and possibly with corners.

In case of topological field theories the ultimate goal is to construct perturbative topological invariants of manifolds with boundary which satisfy the cutting-gluing principle. All known classical topological field theories are gauge theories. Examples are the BF theory, the Chern-Simons theory, and the Poisson sigma model (see Sections 5 and 7 for definitions and details).

Recall some basic facts about BV theories. These are classical field theories with $\mathbb{Z}$ -graded spaces of fields which appear naturally in quantization of gauge theories. The BV construction is a generalization of the Faddeev-Popov [25] and BRST [10] methods. The BV construction is particularly important in cases when the gauge symmetry is given by a non-integrable distribution on the space of fields or when the gauge symmetry is reducible with stabilizers of varying rank. Examples of such theories include the Poisson sigma model and the non-abelian BF theory when the dimension of spacetime is greater than 3. Examples of theories where the gauge symmetry is an integrable distribution include Yang-Mills and Chern-Simons. Even in cases when the distribution is integrable, BV formalism is useful because it is compatible with Wilsonian renormalization of Feynman diagrams, see e.g. [23].

Topological field theories such as Chern-Simons, BF theories and the Poisson sigma model are special for several reasons. First, in the perturbative quantization of such theories Feynman diagrams have no ultraviolet divergencies. Another reason is that the natural choice of gauge condition, the Lorenz gauge, is induced by a choice of metric on the spacetime manifold. Thus the gauge independence of the Feynman diagram expansion in such theories implies that the partition function is metric independent and therefore is a topological invariant.

All these three examples, Chern-Simons, BF and the Poisson sigma model, fit into the general AKSZ construction [1]. The AKSZ construction also gives other examples of topological field theories such as the Rozansky-Witten model and others [21].

There is a consensus that perturbative quantization of the classical Chern-Simons theory gives the same asymptotical expansions as the combinatorial topological field theory based on quantized universal enveloping algebras at roots of unity [45], or, equivalently, on the modular category corresponding to the Wess-Zumino-Witten conformal field theory [56, 42] with the first semiclassical computations involving torsion made in [56]. However this conjecture is still open despite a number of important results in this direction, see for example [47, 3].

One of the reasons why the conjecture is still open is that for manifolds with boundary the perturbative quantization of Chern-Simons theory has not been developed yet. On the other hand, for closed manifolds the perturbation theory involving Feynman diagrams was developed in [32, 27, 7] and in [5, 35, 13]. For the latest development see [19]. Closing this gap and developing the perturbative quantization of Chern-Simons theory for manifolds with boundary is one of the main motivations for the project started in this paper.

Having laid the framework of classical BV-BFV theory, in a forthcoming work we plan to discuss the formal perturbative quantization of BV theories on manifolds with boundary. The main difference with respect to the formal perturbative quantization for closed spacetime manifolds is that in addition to a choice of gauge fixing in the bulk, one has to fix a polarization in the space of boundary fields. Given such choices, the partition function is a function on the space of leaves of boundary polarization. Its value on a leaf is the path integral (understood as a series of

Feynman diagrams) over fluctuations lying in the gauge fixing submanifold with boundary values being in a fixed leaf of this polarization. The resulting partition function is a state in the space of boundary states determined by the polarization. We expect that the equation (10) will be replaced by a deformed quantum master equation for this state, as it happens in the example of 1d Chern-Simons theory [2]. Gluing along a boundary component will be simply given by pairing the partition functions in the space of boundary states.

A classical BV-BFV theory is a functor from the spacetime category to the BFV category. One should expect a similar description in the quantum case. Classically it extends to higher categories, and we anticipate that quantum counterparts of such extensions are higher category versions of topological quantum field theories.

In this paper spacetime manifolds are always compact oriented, possibly with boundary and corners.

In this paper we will use freely the notion of a $\mathbb{Z}$ -graded manifold. See [11, 20], and the Appendix B for details. To simplify the terminology we will call $\mathbb{Z}$ -graded manifolds simply graded manifolds. By a point in a graded manifold we understand the application of the functor of points.

This paper is organized in the following manner. In section 2 we recall the basic structure of classical BV theory for closed spacetime manifolds.

Section 3 is the central part of the paper. Here we formulate the BV-BFV framework for gauge theories. In this section we propose the analog of the Classical Master Equation for spacetime manifolds with boundary. We define the relevant moduli spaces and address the problem of composing them when we glue two spacetime manifolds together. We also show that first order BV theories naturally extend to lower dimensional strata.

In section 4 we introduce the notion of BFV category. We also define the classical BV theory as a functor from the spacetime category to the BFV category.

In section 5 we describe several examples of BV-BFV extensions of classical gauge theories: electrodynamics, Yang-Mills theory, scalar field and the abelian BF theory. The latter is in fact topological and is an example of the AKSZ construction.

The AKSZ construction which provides a class of examples of topological BV-BFV theories is described in section 6. This construction is determined by the choice of a target manifold, which should be a Hamiltonian differential graded manifold. An AKSZ theory extends to strata of all codimensions.

Further examples of BV-BFV extensions of AKSZ theories are described in section 7. We start with abelian and non-abelian Chern-Simons theories, then we describe the non-abelian BF theory and the Poisson sigma model.

In Appendix A we review some useful facts about coisotropic submanifolds and reduction. In Appendix B we recall some basic facts on graded manifolds. A brief discussion of smooth points on moduli spaces associated to differential graded manifolds is given in Appendix C. Elements of Cartan calculus for local forms and functionals on mapping spaces are developed in Appendix D and applications to AKSZ theories are presented.

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2. CLASSICAL BV THEORIES FOR CLOSED SPACETIMES

2.1. Non-reduced theory. When the spacetime manifold N is closed, the space of fields $\mathcal{F}_N$ in a BV theory is a graded manifold with a symplectic form ω_N of degree -1 and the action functional S_N of degree zero. Usually fields are sections of or connections on a fiber bundle on N . Some basic facts on graded manifolds are summarized in Appendix B. The degree in the grading on the space of fields is usually called the ghost number. Thus, $gh(\omega_N) = -1$ and $gh(S_N) = 0$. The space of fields is usually infinite dimensional. The BV data should satisfy the following property: the Poisson bracket of the action functional with itself is zero [8, 9] (see also [49, 52]). This property is known as the classical master equation (CME). The action functional generates the Hamiltonian vector field Q_N of degree 1. The CME implies that the Lie derivative of this vector field squares to zero.

Most interesting for applications both in physics and in geometry and topology are local theories. The notion of locality is more transparent for theories on spacetime manifolds with boundary, see section 3. For closed manifolds locality implies natural isomorphisms $\mathcal{F}_{N_1 \sqcup N_2} \simeq \mathcal{F}_{N_1} \times \mathcal{F}_{N_2}$ and the additivity of the symplectic form and of the action: $\omega_{N_1 \sqcup N_2} = \omega_{N_1} + \omega_{N_2}$, $S_{N_1 \sqcup N_2} = S_{N_1} + S_{N_2}$.

We will use the following definition of classical BV theory for closed spacetime manifolds.

Definition 2.1. *For a closed spacetime N a BV theory [8] on the space of fields $\mathcal{F}_N$ is a triple (ω_N, Q_N, S_N) where ω_N is a symplectic form on $\mathcal{F}_N$ with $gh = -1$, Q_N is a vector field on $\mathcal{F}_N$ with $gh = 1$, and S_N is a function on $\mathcal{F}_N$ with $gh = 0$, subject to*

$$\iota_{Q_N} \omega_N = (-1)^n \delta S_N, \quad L_{Q_N}^2 = 0$$

where L_Q is the Lie derivative for the vector field Q .

These two equations imply the *classical master equation*

$$\{S_N, S_N\} = 0,$$

which can also be written as

$$L_{Q_N} S_N = 0$$

The first equation in the definition means that the vector field Q_N is Hamiltonian and therefore it preserves the symplectic form:

$$L_{Q_N} \omega_N = 0,$$

Critical points of the action functional S_N are solutions to Euler-Lagrange (EL) equations

$$(1) \quad \delta S_N = 0,$$

Because the space of fields is an odd-symplectic manifold, solutions to the EL equations are also zeroes of the Hamiltonian vector field generated by S_N , i.e. of the cohomological vector field Q_N . Denote the space of solutions to Euler-Lagrange equations (1) by $\mathcal{EL}_N$.

Because vector field Q_N is Hamiltonian, and Poisson brackets of local functionals are defined, we can write $L_{Q_N}F = \{S_N, F\}$ for the action of the vector field Q_N on the functional F . Denote by V_F the vector field generated by F , then $L_{V_F}G = \{F, G\}$. It is clear that vector fields $[Q_N, V_F]$ annihilate the functional S_N :

$$L_{[Q_N, V_F]}S_N = \{\{S_N, F\}, S_N\} = 0$$

This follows from CME and from the Jacobi identity. Thus, any such vector field can be considered as a local (off-shell) gauge symmetry of S_N .

From now on we will consider only first order BV theories. In such theories the action functional is local and it is linear in derivatives of fields. It is well known that any local theory can be reformulated as a first order theory by adding corresponding “momenta” variables in fields.

2.2. Other degrees. So far we have assumed the symplectic form ω_N to have degree -1 . But everything can be generalized to the case when the degree is some integer k (we fix however the degree of the cohomological vector field Q_N to be 1). In this setting the action S_N will have degree $k + 1$ [46]. Only a few remarks are in order

- (1) Unless $k = 0$, ω_N is automatically exact (so for $k = 0$ we need this extra assumption). In the following, we will be interested in specifying a primitive 1-form of the same degree k .
- (2) Unless $k = -1$, the condition $L_{Q_N}\omega = 0$ implies that Q_N is Hamiltonian and that the Hamiltonian function S_N is uniquely determined.
- (3) Unless $k = -2$, the condition $Q_N^2 = 0$ implies $\{S_N, S_N\} = 0$.

We will mainly be interested in the cases $k = -1$ (BV manifolds) and $k = 0$ (BFV manifolds). For extended BV theories, we will also be interested in k positive. We are not aware of any application of the cases $k < -1$.

2.3. The Q -reduction. The vector field Q_N vanishes on the subspace $\mathcal{EL}_N \subset \mathcal{F}_N$. It also defines the Lie subalgebra $\widetilde{Vect}_Q$ of the Lie algebra of vector fields on $\mathcal{F}_N$ generated by vector fields which are Lie brackets with Q_N , i.e. by vector fields of the form $[V, Q_N]$. This Lie subalgebra in general does not determine a distribution on $\mathcal{F}_N$. However, the restriction of $\widetilde{Vect}_Q$ to $\mathcal{EL}_N$ defines an integrable distribution $Vect_Q$.

Definition 2.2. *The Q -reduction $\mathcal{EL}_N/Q$ of $\mathcal{EL}_N$ is the space of leaves of the distribution $Vect_Q$.*

Assume that X is a zero of Q_N such that the leaf of $Vect_Q$ through X is a smooth point in $\mathcal{EL}_N/Q$. Define the linear operator $\hat{Q}_X : T_X\mathcal{F}_N \rightarrow T_X\mathcal{F}_N$ as the

linearization of Q_N at X . If we write Q_N in local coordinates as $Q_N = \sum_a q^a(X) \partial_a$, the operator $\hat{Q}_X$ acts on the basis ∂_a in $T_X \mathcal{F}_N$ as

$$\hat{Q}_X \partial_a = \sum_b \partial_a q^b(X) \partial_b$$

Because Q_N squares to zero we have $\hat{Q}_X^2 = 0$. By definition $T_X \mathcal{EL}_N = \ker(\hat{Q}_X)$. Also observe that $\text{Vect}_Q|_X = \text{Im}(\hat{Q}_X) \subset T_X \mathcal{EL}_N$. From now on for geometric considerations we will focus only on smooth points as defined in the Appendix C.

Proposition 2.3. *For a smooth point $[X] \in \mathcal{EL}_N/Q$, there is a natural linear isomorphism*

$$T_{[X]} \mathcal{EL}_N/Q = \ker(\hat{Q}_X)/\text{Im}(\hat{Q}_X)$$

2.4. The symplectic reduction. The following Proposition says that if the subspace $\mathcal{EL}_N$ is a submanifold then it is a coisotropic submanifold.

Proposition 2.4. *If $X \in \mathcal{EL}_N$ is a smooth point, then $T_X \mathcal{EL}_N \subset T_X \mathcal{F}_N$ is a coisotropic subspace.*

Proof. The invariance of ω_N with respect to the vector field Q_N means that $L_{Q_N} \omega_N = 0$. If X is a zero of Q_N this implies $\omega_N(\hat{Q}_X \xi, \eta) = \omega_N(\xi, \hat{Q}_X \eta)$.

These properties of $\hat{Q}_X$ imply $\omega_N(\hat{Q}_X \xi, \hat{Q}_X \eta) = \omega_N(\xi, \hat{Q}_X^2 \eta) = 0$. Therefore $\text{Im}(\hat{Q}_X)$ is an isotropic subspace and therefore its symplectic orthogonal $\text{Im}(\hat{Q}_X)^\perp$ is coisotropic.

On the other hand

$$\begin{aligned} (2) \quad \text{Im}(\hat{Q}_X)^\perp &= \{\xi \mid \omega_N(\xi, \hat{Q}_X \eta) = 0, \text{ for any } \eta\} \\ &= \{\xi \mid \omega_N(\hat{Q}_X \xi, \eta) = 0, \text{ for any } \eta\} = \ker(\hat{Q}_X) \end{aligned}$$

This proves that $T_X \mathcal{EL}_N = \ker(\hat{Q}_X)$ is a coisotropic subspace. Because we used only nondegeneracy of the form ω_N on $T_X \mathcal{F}$, the proof works both in finite-dimensional and in the infinite-dimensional case. $\square$

Recall the definition of the symplectic reduction of a coisotropic submanifold of a symplectic manifold. The symplectic reduction $\underline{C}$ of the coisotropic submanifold C in the symplectic manifold S is the space of leaves of the characteristic foliation of C . The characteristic foliation of C is spanned by Hamiltonian vector fields generated by the vanishing ideal I_C (the ideal in the commutative algebra of functions on S generated by functions vanishing on C). When $\underline{C}$ is a smooth manifold, it is a symplectic manifold. Otherwise only at smooth points the tangent space to $\underline{C}$ has a natural symplectic form. If $[X]$ is such a smooth point, the tangent space $T_{[X]} \underline{C}$ is naturally isomorphic to $T_X C / (T_X C)^\perp$.

Remark 2.5. *An alternative proof of the Proposition 2.4 is algebraic. Observe that the vanishing ideal for $\mathcal{EL}_N$ is generated by functionals of the form $\{S_N, T\}$ where $\{.,.\}$ is the Poisson bracket for the form ω_N and T is a local functional. The simple calculations shows that*

$$\{\{S, U\}, \{S, T\}\} = \{S, \{U, \{S, T\}\}\}$$

The bracket $\{U, \{S, T\}\}$ is a local functional if U and T are. Therefore the vanishing ideal is a Poisson algebra and therefore $\mathcal{EL}_N$ is coisotropic in the algebraic sense.

In Proposition 2.4 we proved that $Im(\hat{Q}_X)^\perp = \ker(\hat{Q}_X)$. This implies $Im(\hat{Q}_X) \subset \ker(\hat{Q}_X)^\perp$, which means that the $\hat{Q}$ -distribution on $\mathcal{EL}_N$ is contained in the characteristic distribution. In the finite dimensional case the inclusion becomes an equality.

Assumption 2.6. *In classical field theory we set*

$$(3) \quad Im(\hat{Q}_X) = \ker(\hat{Q}_X)^\perp,$$

as an assumption.

This corresponds to certain ellipticity condition for Q_N . In all our examples this condition follows from the usual Hodge-de Rham decomposition.

Remark 2.7. *As a consequence of condition (3) the characteristic foliation of $\mathcal{EL}_N$ is the same as the foliation by $Vect_Q$, and in particular, if $[X] \in \underline{\mathcal{EL}}_N$ is a smooth point, then*

$$(4) \quad T_{[X]}\underline{\mathcal{EL}}_N = \ker(\hat{Q}_X)/Im(\hat{Q}_X)$$

In other words, locally the symplectic reduction of $\mathcal{EL}_N$ coincides with its Q -reduction: $\underline{\mathcal{EL}}_N = \mathcal{EL}_N/Q$.

We will call $\underline{\mathcal{EL}}_N = \mathcal{EL}_N/Q$ the $\mathcal{EL}$ -moduli space for N . If the cohomology spaces of $\hat{Q}_X$ are finite dimensional at a generic point X , the $\mathcal{EL}$ -moduli space is finite dimensional, but possibly singular.

Remark 2.8. *With the appropriate assumptions the ring of functions on the reduced space $\mathcal{EL}_N/Q$ is isomorphic to the cohomology space of the ring of functions on $\mathcal{F}_N$ with the differential L_{Q_N} (the Lie derivative with respect to Q_N). To be more specific, one should expect that the smooth locus of $\mathcal{EL}/Q$ is isomorphic to the smooth locus of $Spec(H_Q(Fun(\mathcal{F}_N)))$ with the appropriate definition of $Spec$ and of the ring of functionals $Fun(\mathcal{F}_N)$.*

Because $\delta S_N = 0$ on the subspace $\mathcal{EL}_N \subset \mathcal{F}_N$ we have the following Proposition.

Proposition 2.9. *The action is constant on connected components of $\mathcal{EL}_N$.*

3. BV THEORIES FOR SPACETIME MANIFOLDS WITH BOUNDARY

3.1. Non-reduced theory.

3.1.1. Non-reduced BV-BFV theory. In this section, unless otherwise specified, N is an n -dimensional spacetime manifold, with ∂N being its $n - 1$ dimensional boundary. The BV theory for spacetime manifolds with boundary consists of two parts.

First, in the bulk we have the same data as for BV theory, that is: the space of fields $\mathcal{F}_N$ which is a graded (the degree is called ghost number) symplectic manifold with the symplectic form ω_N with $gh = -1$, the cohomological vector field Q_N with $gh = 1$ and the action functional S_N of ghost number zero.

Second, on the boundary we have the BFV data [9] which consist of: the space of boundary fields $\mathcal{F}_{\partial N}$ which is a graded (usually infinite dimensional) symplectic manifold with the symplectic form $\omega_{\partial N}$ which is exact $\omega_{\partial N} = (-1)^{n-1}\delta\alpha_{\partial N}$ with $gh = 0$, and a vector field $Q_{\partial N}$ with $gh = 1$. The sign in $(-1)^{n-1}$ is here because it appears naturally in AKSZ examples of topological gauge theories. The BFV data satisfy the following conditions:

$$(5) \quad Q_{\partial N}^2 = 0, \quad L_{Q_{\partial N}} \omega_{\partial N} = 0$$

These conditions imply that $Q_{\partial N}$ is Hamiltonian vector field [46] with ghost number 1, with the Hamiltonian $S_{\partial N}$, which is by definition the BFV action. In Appendix B we recall this construction.

A BV theory on a spacetime with boundary can be regarded as a pairing of the BV data with the BFV data through the natural mapping $\pi : \mathcal{F}_N \rightarrow \mathcal{F}_{\partial N}$ which is the restriction of fields to the boundary.

We will say that BV theory in the bulk and BFV theory on the boundary agree and form the BV-BFV theory on the manifold with boundary if the restriction mapping is a surjective submersion and if the BV data in the bulk and BFV data on the boundary satisfy the following conditions

$$(6) \quad Q_N^2 = 0, \quad \delta\pi(Q_N) = Q_{\partial N}, \quad L_{Q_{\partial N}} \omega_{\partial N} = 0$$

$$(7) \quad \iota_{Q_N} \omega_N = (-1)^n \delta S_N + \pi^*(\alpha_{\partial N}),$$

The sign is purely conventional, but it fits well with the natural definition of BV data for AKSZ theories.

Note that first two equations imply that

$$(8) \quad Q_{\partial N}^2 = 0$$

which means BFV axioms are satisfied. The equation (7) implies

$$(9) \quad L_{Q_N} \omega_N = (-1)^n \pi^*(\omega_{\partial N})$$

which means, in particular, that in BV-BFV theory the symplectic form is no longer Q -invariant.

The bulk action satisfies an important identity which describes how the action changes with respect to gauge transformations.

Proposition 3.1. *The following identity holds in any BV-BFV theory*

$$(10) \quad L_{Q_N} S_N = (-1)^{\dim(N)} \pi^*(2S_{\partial N} - \iota_{Q_{\partial N}} \alpha_{\partial N})$$

Remark 3.2. *We can remove the signs $(-1)^n$ by redefining vector fields Q_N , $Q_{\partial N}$ and the 1-form $\alpha_{\partial N}$. Set $Q = (-1)^n Q_N$, $Q_{\partial} = (-1)^n Q_{\partial N}$ and $\alpha_{\partial} = (-1)^{n-1} \alpha_{\partial N}$. Then $\delta\pi(Q) = Q_{\partial}$, $\iota_Q \omega_N = S_N - \pi^*(\alpha_{\partial})$ and $L_Q \omega_N = \pi^*(\omega_{\partial N})$.*

Proof. By definition of the Lie derivative

$$L_{Q_N} = \iota_{Q_N} \delta - \delta \iota_{Q_N}$$

and the same formula for the Lie derivative holds for $Q_{\partial N}$. The sign is minus in this formula because ι_Q is an even operation. We will prove the Proposition by applying L_{Q_N} to the equation:

$$\iota_{Q_N} \omega_N = (-1)^n \delta S_N + \pi^*(\alpha_{\partial N})$$

By definition of L_{Q_N} and because of $Q_N^2 = 0$ we have $[L_{Q_N}, \iota_{Q_N}] = L_{Q_N} \iota_{Q_N} - \iota_{Q_N} L_{Q_N} = 0$ and we already know that

$$L_{Q_N} \omega_N = -\pi^* \delta \alpha_{\partial N} = (-1)^{n+1} \pi^* \omega_{\partial N},$$

Therefore we have identities:

$$(11) \quad L_{Q_N} \iota_{Q_N} \omega_N = \iota_{Q_N} L_{Q_N} \omega_N = -\pi^* \iota_{Q_{\partial N}} \delta \alpha_{\partial N},$$

Now we can apply the Lie derivative of Q_N to the classical master equation:

$$(12) \quad L_{Q_N} \iota_{Q_N} \omega_N = (-1)^n L_{Q_N} \delta S_N + L_{Q_N} \pi^* \alpha_{\partial N} = -(-1)^n \delta \iota_{Q_N} \delta S_N + \pi^* (L_{Q_{\partial N}} \alpha_{paN}) = \\ -(-1)^n \delta L_{Q_N} S + \pi^* (\iota_{Q_{\partial N}} \delta \alpha_{\partial N} - \delta \iota_{Q_{\partial N}} \alpha_{\partial N})$$

Here, in the last line, we used the formula for the Lie derivative $L_{Q_{\partial N}}$. Using (11) and the identity $\delta S_{\partial N} = \iota_{Q_{\partial N}} \omega_{\partial N}$ we arrive at:

$$-\pi^* (\delta S_{\partial N}) = -(-1)^n \delta L_{Q_N} S_N + \pi^* (\delta S_{\partial N}) - \pi^* (\delta \iota_{Q_{\partial N}} \alpha_{\partial N})$$

This is equivalent to

$$\delta((-1)^n L_{Q_N} S_N - \pi^* (2S_{\partial N} - \iota_{Q_{\partial N}} \alpha_{\partial N})) = 0$$

We have δ (function of degree 1) = 0, which means the function must vanish. This proves the Proposition. $\square$

Remark 3.3. In the definition of BV-BFV theory we made an assumption that ω_{∂} is exact. This assumption can be removed. Two examples, a charged particle in an electro-magnetic field, and the WZW model suggest that a more natural condition is to require that there is a line bundle $\mathfrak{L}_{\partial}$ over $\mathcal{F}_{\partial N}$. The symplectic form is $\omega_{\partial N} = (-1)^{n-1} \delta \alpha_{\partial N}$ where $\alpha_{\partial N}$ is a connection on $\mathfrak{L}_{\partial}$. The condition

$$L_{Q_N} \omega_N = (-1)^n \pi^* \omega_{\partial N}$$

implies that the connection $(-1)^n \frac{i}{\hbar} (-\iota_{Q_N} \omega_N + \pi^* \alpha_{\partial N})$ on the line bundle $\pi^*(\mathfrak{L}_{\partial})$ over $\mathcal{F}_N$ is flat. In this case the action functional S_N should be regarded as a horizontal section $s_N = \exp(\frac{i S_N}{\hbar})$ of $\pi^*(\mathfrak{L}_{\partial})$ and the equation (7) becomes

$$(\delta + (-1)^n \frac{i}{\hbar} (-\iota_{Q_N} \omega_N + \pi^* \alpha_{\partial N})) s_N = 0$$

3.1.2. *Digression: gauge invariance.* Now let us discuss the gauge invariance of BV-BFV theory. When $\partial N \neq \emptyset$, Poisson bracket of local functionals is not defined. In this case we will call the vector field V Hamiltonian if there exists a local functional F such that

$$\iota_V \omega_N = \delta F$$

Recall that a vector field V is called *projectable* if for all $X \in \mathcal{F}_N$

$$\delta \pi_X (V_X) = v_{\pi(X)}$$

for some vector field v on $\mathcal{F}_{\partial N}$. The gauge invariance of the BV-BFV action can be formulated as the following statement:

Proposition 3.4. *If V is a projectable Hamiltonian vector field, then the vector field $[Q_N, V]$ preserves S_N up to a pullback from the boundary:*

$$L_{[Q_N, V]} S_N = (-1)^{\dim N} \pi^* (\iota_v L_{Q_{\partial N}} \alpha_{\partial N} + \iota_{Q_{\partial N}} L_v \alpha_{\partial N})$$

Indeed, we have:

$$(13) \quad L_{[Q_N, V]} S_N = L_{Q_N} L_V S + L_V L_{Q_N} S_N = \\ \iota_{Q_N} \delta \iota_V \delta S_N + (-1)^{\dim N} L_V \pi^* (2S_{\partial N} - \iota_{Q_{\partial N}} \alpha_{\partial N}) = \\ = (-1)^{\dim N} (\iota_{Q_{\partial N}} \delta \iota_{Q_N} \iota_V \omega_N + \pi^* (-\iota_{Q_{\partial N}} \delta \iota_v \alpha_{\partial N} + \\ 2\iota_v \delta S_{\partial N} - \iota_v \delta \iota_{Q_{\partial N}} \alpha_{\partial N})) = (-1)^{\dim N} \pi^* (\iota_v L_{Q_{\partial N}} \alpha_{\partial N} + \iota_{Q_{\partial N}} L_v \alpha_{\partial N})$$

Here we used identities $\delta S_N = (-1)^{\dim N}(\iota_{Q_N}\omega_N - \pi^*\alpha_{\partial N})$, $\iota_V\omega_N = \delta F$, $L_{Q_N}L_{Q_N}F = \frac{1}{2}L_{[Q_N, Q_N]}F = 0$, $\delta S_{\partial N} = (-1)^{\dim N-1}\iota_{Q_{\partial N}}\omega_{\partial N} = \iota_{Q_{\partial N}}\omega_{\partial N} = \iota_{Q_{\partial N}}\delta\alpha_{\partial N}$, and F is the generating function for V and v is the projection of V to the boundary.

As a corollary we also have the formula

$$\delta L_{[Q_N, V]}S = -(-1)^{\dim N}\pi^*(L_{[Q_{\partial N}, v]}\alpha_{\partial N})$$

3.1.3. The boundary structure from the bulk. Let us show that a first order BV theory of spacetime manifolds (on the bulk) induces BFV theory on the boundary. Denote the pullback of fields to the boundary by $\mathcal{F}_N|_{\partial N}$ (we say pullback because fields are usually either sections of a fiber bundle, or connections). The differential of the action can be written as the bulk part plus the boundary contribution from the use of Stokes theorem:

$$\delta S_N[X] = \int_N A \wedge \delta X - (-1)^n \pi^* \tilde{\alpha}_{\partial N}[X]$$

where $\tilde{\alpha}_{\partial N}$ is a one-form on the space $\mathcal{F}_N|_{\partial N}$ and A defines the Euler-Lagrange equation. The kernel of $\delta\tilde{\alpha}_{\partial N}$ forms a distribution on the space $\mathcal{F}_N|_{\partial N}$. Denote by $\mathcal{F}_{\partial N}$ the space of leaves of this distribution. We have a natural projection $\pi : \mathcal{F}_N \rightarrow \mathcal{F}_{\partial N}$.

Denote by $\alpha_{\partial N}$ the form on $\mathcal{F}_{\partial N}$ corresponding to the form $\tilde{\alpha}_{\partial N}$ on $\mathcal{F}_N|_{\partial N}$ ¹.

Taking into account that for closed manifolds the vector field Q_N is Hamiltonian, generated by the classical action functional, we conclude that the bulk term in the differential of the action functional is $(-1)^n \iota_{Q_N}\omega_N$. Thus, we have equation (7).

Observe that Q_N is projectable to $\mathcal{F}_{\partial N}$ denote its projection by $Q_{\partial N} = \delta\pi(Q_N)$. For any form θ on boundary fields we will have $L_{Q_N}\pi^*(\theta) = \pi^*(L_{Q_{\partial N}}\theta)$. Equations $Q_N^2 = 0$ and (9) imply that $L_{Q_N}^2\omega_N = \pi^*(L_{Q_{\partial N}}\omega_{\partial N}) = 0$. Therefore $L_{Q_{\partial N}}\omega_{\partial N} = 0$.

It is clear that the BFV action induced by a first order BV theory is also of first order.

3.1.4. Some properties of non-reduced BV-BFV theories. Associated with BV and BFV data we have the following important subspaces in the space of bulk fields and in the spaces of boundary fields:

- The space $\mathcal{EL}_N \subset \mathcal{F}_N$ of zeroes of the vector field Q_N .
- The space $\mathcal{EL}_{\partial N} \subset \mathcal{F}_{\partial N}$ of zeroes of the vector field $Q_{\partial N}$.
- The space $\mathcal{L}_N = \pi(\mathcal{EL}_N) \subset \mathcal{F}_{\partial N}$ of boundary values of solutions of Euler-Lagrange equations. It is clear that $\mathcal{L}_N \subset \mathcal{EL}_{\partial N} \subset \mathcal{EL}_N$.

Proposition 3.5. *a) The subspace $\mathcal{EL}_{\partial N}$ is locally coisotropic in $\mathcal{F}_{\partial N}$ (its tangent space at a smooth point is coisotropic in the tangent space to the space of fields at this point).*

b) $\mathcal{EL}_N$ is still locally coisotropic when $\partial N \neq \emptyset$.

The proof of a) is identical to the proof of Proposition 2.4 for closed manifolds. The same proof carries through for b) by requiring η to vanish on the boundary. The algebraic proof outlined after Proposition 2.4 also works, the only difference is that one should take local functionals with test functions vanishing on the boundary. An example of such functional is $\int_N \alpha_a \wedge X^a$ where X^a are coordinate fields and α_a are test functions which vanish on ∂N .

¹Strictly speaking the form $\tilde{\alpha}_{\partial N}$ becomes a connection on a line bundle over $\mathcal{F}_{\partial N}$ with the curvature $\omega_{\partial N}$. However in our examples this line bundle is trivial. This is why for the time being we will assume this and will regard $\alpha_{\partial N}$ as a 1-form.

Proposition 3.6. *The subspace $\mathcal{L}_N \subset \mathcal{EL}_{\partial N} \subset \mathcal{F}_{\partial N}$ is locally isotropic.*

Proof. Assume X is a smooth point of $\mathcal{EL}_N$. The equation (7) implies that

$$(14) \quad \omega_X(\hat{Q}_X \xi, \eta) - \omega_X(\xi, \hat{Q}_X \eta) = \tilde{\omega}_{\pi(X)}(\delta\pi(\xi), \delta\pi(\eta))$$

where ξ, η are tangent vectors to $\mathcal{F}_N$ at the point X . If ξ and η are tangent to $\mathcal{EL}_N \subset \mathcal{F}_N$ then $\hat{Q}_X \xi = \hat{Q}_X \eta = 0$. Therefore $\tilde{\omega}_{\pi(X)}(\delta\pi(\xi), \delta\pi(\eta)) = 0$ which means that $\delta\pi(T_X \mathcal{EL}_N) = T_{\pi(X)} \mathcal{L}_N$ is an isotropic subspace in $T_{\pi(X)} \mathcal{F}_{\partial N}$. $\square$

We proved that $\mathcal{L}_N$ is always isotropic. We will prove in Section 3.4.3 (Corollary 3.21) that under a natural regularity assumption (see Definition 3.18), $\mathcal{L}_N$ is in fact Lagrangian.

Another natural property of a BV-BFV classical field theory is locality. If two spacetime manifolds N_1 and N_2 have common boundary Σ (say $\Sigma \subset \partial N_1$ and it is identified by an orientation preserving diffeomorphism with part of the boundary of N_2) then

$$\begin{aligned} \mathcal{F}_{N_1 \cup_{\Sigma} N_2} &= \mathcal{F}_{N_1} \times_{\mathcal{F}_{\Sigma}} \mathcal{F}_{N_2} \\ S_{N_1 \cup_{\Sigma} N_2} &= S_{N_1} + S_{N_2} \end{aligned}$$

3.2. The reduction of the boundary BFV theory. The boundary manifold ∂N is closed. This is why both the Q -reduction and the symplectic reduction for the boundary BFV theory work as they do for the BV theory on closed spacetime manifolds. The additional structure in the BV-BFV theory is the reduction of the submanifold $\mathcal{L}_N \subset \mathcal{EL}_{\partial N}$ and the reduction of the fibers of the fiber bundle $\pi : \mathcal{EL}_N \rightarrow \mathcal{L}_N$.

3.2.1. The Q -reduction. The following Proposition is an immediate corollary of the identity $\delta\pi \circ Q_N = Q_{\partial N} \circ \delta\pi$.

Proposition 3.7. *The distribution $\text{Vect}_{Q_{\partial}}$ on $\mathcal{F}_{\partial N}$ generated by Lie brackets of vector fields with $Q_{\partial N}$ is parallel to $\mathcal{L}_N$.*

Definition 3.8. *Define Q -reductions of $\mathcal{L}_N$ and of $\mathcal{EL}_{\partial N}$ as the space of leaves of $\text{Vect}_{Q_{\partial}}$ through $\mathcal{L}_N$ and $\mathcal{EL}_{\partial N}$ respectively.*

In general these spaces are singular. Because our goal here is to develop the set up for the perturbative quantization in a vicinity of a generic classical solution, we will focus on smooth points of $\mathcal{EL}_{\partial N}$ and $\mathcal{L}_N$ and tangent spaces at such points. See Appendix C below for the discussion of smooth points.

3.2.2. The symplectic reduction. Let $\mathcal{EL}_{\partial N}$ be the symplectic reduction of $\mathcal{EL}_{\partial N}$. We make the following assumption, which is the analogue of (3) for the boundary fields:

Assumption 3.9. *We assume that for any $l \in \mathcal{EL}_{\partial N}$,*

$$(15) \quad \ker(\hat{Q}_l)^{\perp} = \text{Im}(\hat{Q}_l)$$

in $T_l \mathcal{F}_{\partial N}$, where $\hat{Q}_l$ is the linearization of $Q_{\partial N}$ at l .

The following Proposition is the counterpart of the Remark 2.7.

Proposition 3.10. *Under the assumption (15), locally, the symplectic reduction of $\mathcal{EL}_{\partial N}$ is equal to its Q -reduction.*

The difference from Remark 2.7 is just in the shift of gradings.

Denote by $\underline{\mathcal{L}}_N \subset \underline{\mathcal{EL}}_{\partial N}$ the image of $\mathcal{L}_N$ with respect to this reduction. That is $\underline{\mathcal{L}}_N$ is the space of leaves of the characteristic foliation of $\mathcal{EL}_{\partial N}$ through $\mathcal{L}_N$. This Proposition implies that locally the reduction $\underline{\mathcal{L}}_N$ of $\mathcal{L}_N$ is equal to its Q -reduction $\mathcal{L}_N/Q$.

Under the regularity assumption (see Definition 3.18, Proposition 3.20), $\underline{\mathcal{L}}_N \subset \underline{\mathcal{EL}}_{\partial N}$ is a Lagrangian submanifold.

3.3. The reduction of the bulk BV theory. The meaning of the Q -reduction is passing from fields to gauge classes of fields. This is why it is natural to reduce not the whole space of solutions to Euler-Lagrange equations but a subspace with fixed boundary conditions. Points of this reduced space correspond to gauge classes of fields with boundary values in a given boundary gauge class.

The space $\mathcal{EL}_N$ is naturally fibered over $\mathcal{L}_N = \pi(\mathcal{EL}_N)$ where $\pi : \mathcal{F}_N \rightarrow \mathcal{F}_{\partial N}$ is the restriction mapping.

For $l \in \mathcal{L}_N$ denote by $[l]$ the leaf of $Vect_{Q_{\partial N}}$ through l , i.e. the image of l in the Q -reduced space $\mathcal{L}_N/Q$. Denote by $\mathcal{EL}(N, [l])$ the fiber of π over $[l]$,

$$\mathcal{EL}(N, [l]) = \pi^{-1}([l]) \cap \mathcal{EL}_N$$

Because the restriction mapping π intertwines vector fields Q_N and $Q_{\partial N}$, the vector field Q_N is parallel to $\mathcal{EL}(N, [l])$. It also induces the projection of reduced spaces $p : \mathcal{EL}_N/Q \rightarrow \mathcal{L}_N/Q$ where $\mathcal{EL}_N/Q$ is the space of leaves of $Vect_Q$ in $\mathcal{EL}_N$.

Definition 3.11. *The reduced space $\mathcal{EL}(N, [l])/Q$ is the space of leaves of $Vect_Q$ on $\mathcal{EL}(N, [l])$.*

Because the projection π intertwines vector fields Q_N and $Q_{\partial N}$ the reduced space $\mathcal{EL}(N, [l])/Q$ is the fiber of p over $[l]$: $\mathcal{EL}(N, [l])/Q = p^{-1}([l])$.

Assume that the image of $X \in \mathcal{EL}_N$ in $\mathcal{EL}(N, [\pi(X)])/Q$ is a smooth point. We have the natural mappings of tangent spaces:

$$(16) \quad \begin{array}{ccc} T_{[X]} \mathcal{EL}_N/Q & \xleftarrow{\iota} & T_{[X]} \mathcal{EL}(N, [l])/Q \\ \delta p \downarrow & & \\ T_{[l]} \mathcal{L}_N/Q \subset T_{[l]} \mathcal{EL}_{\partial N}/Q & & \end{array}$$

where $l = \pi(X)$, and the horizontal mapping is the natural inclusion of fibers, $Im(\iota) = \ker(\delta p)$. We know that $T_{[X]} \mathcal{EL}_N/Q \simeq H(T_X \mathcal{F}_N; \hat{Q}_X)$, and that $T_{[l]} \mathcal{EL}(\partial N)/Q \simeq H(T_l \mathcal{F}_{\partial N}; \hat{Q}_l)^2$.

We also know that the short exact sequence

$$\begin{array}{ccc} T_X \mathcal{F}_N & \xleftarrow{i} & \ker(\delta \pi)_X \\ (\delta \pi)_X \downarrow & & \\ T_l \mathcal{F}_{\partial N} & & \end{array}$$

gives the long exact sequence

$$\cdots \rightarrow H_{\hat{Q}_X}^\bullet(\ker(\delta \pi)_X) \xrightarrow{[i]} H_{\hat{Q}_X}^\bullet(T_X \mathcal{F}_N) \xrightarrow{[\delta \pi_X]} H_{\hat{Q}_l}^\bullet(T_l \mathcal{F}_{\partial N}) \xrightarrow{\phi} H_{\hat{Q}_X}^{\bullet+1}(\ker(\delta \pi)_X) \rightarrow \cdots$$

²Recall that $\hat{Q}_X$ is the linearization of Q_N at $X \in \mathcal{EL}_N$ and $\hat{Q}_l$ is the linearization of $Q_{\partial N}$ at $l = \pi(X) \in \mathcal{EL}_{\partial N}$.

on the $\hat{Q}$ -cohomology spaces.

Truncating this long exact sequence we obtain the short exact sequence

(17)

$$\begin{array}{ccc} H_{\hat{Q}_X}(T_X \mathcal{F}_N) & \xleftarrow{[i]} & H_{\hat{Q}_X}(\ker(\delta\pi)_X)/\phi(H_{\hat{Q}_l}(T_l \mathcal{F}_{\partial N})) \\ \downarrow [\delta\pi_X] & & \\ [\delta\pi_X](H_{\hat{Q}_X}(T_X \mathcal{F}_N)) & \subset & H_{\hat{Q}_l}(T_l \mathcal{F}_{\partial N}) \end{array}$$

Because we have natural isomorphisms $H_{\hat{Q}_X}(T_X \mathcal{F}_N) \cong T_{[X]} \mathcal{EL}_N/Q$, $H_{\hat{Q}_l}(T_l \mathcal{F}_{\partial N}) \cong T_{[l]} \mathcal{EL}_{\partial N}/Q$ and $[\delta\pi_X](H_{\hat{Q}_X}(T_X \mathcal{F}_N)) = T_{[l]} \mathcal{EL}_N/Q \subset T_{[l]} \mathcal{EL}_{\partial N}/Q$ we can identify $T_{[X]} \mathcal{EL}(N, [l])/Q$ with the fiber of $[\delta\pi_X]$. We proved the following:

Proposition 3.12. *If $[\pi(X)] \in \mathcal{EL}_N/Q$ is a smooth point, then*

$$T_{[X]} \mathcal{EL}(N, [l])/Q \cong H_{\hat{Q}_X}(\ker(\delta\pi_X))/\phi(H_{\hat{Q}_l}(T_l \mathcal{F}_{\partial N}))$$

Here is another, equivalent characterization of this space. By definition

$$T_X \mathcal{EL}(N, [l]) = \{\xi \in \ker(\hat{Q}_X) \mid \delta\pi_X(\xi) \in \text{Im}(\hat{Q}_l)\}$$

On the other hand $(\text{Vect}_Q)_X = \text{Im}(\hat{Q}_X)$. Therefore the tangent space to the reduced space is the quotient

$$T_X \mathcal{EL}(N, [l])/Q = \{\xi \in \ker(\hat{Q}_X) \mid \delta\pi_X(\xi) \in \text{Im}(\hat{Q}_l)\} / \text{Im}(\hat{Q}_X)$$

This implies the following.

Proposition 3.13. *We have*

$$T_X \mathcal{EL}(N, [l])/Q \cong \{\xi \in \ker(\hat{Q}_X) \mid \delta\pi_X(\xi) = 0\} / \{\xi \in \text{Im}(\hat{Q}_X) \mid \delta\pi_X(\xi) = 0\}$$

3.4. Symplectic $\mathcal{EL}$ -moduli spaces. In previous sections we introduced $\mathcal{EL}$ -moduli space as the Q -reduction of the space of solutions to Euler-Lagrange equations. In case when $\partial N = \emptyset$ we proved that they carry a symplectic structure coming from symplectic reduction, but this is no longer true in the case with boundary.

In this subsection, for the case with boundary we will introduce a different reduction which is symplectic (see subsection 3.4.3), fibers over the Q -reduction and has simple gluing properties. We call it the symplectic $\mathcal{EL}$ -moduli space.

Denote the bulk and boundary $\mathcal{EL}$ -moduli spaces by

$$\mathcal{M}_N = \mathcal{EL}_N/Q_N, \quad \mathcal{M}_{\partial N} = \mathcal{EL}_{\partial N}/Q_{\partial N}$$

When N is fixed, we will usually suppress the subscript N for brevity.

3.4.1. The main construction. Denote by $\widetilde{\text{Vect}}_Q^{\text{rel}}$ the space of vector fields of the form $[Q_N, v]$ where v is a *vertical* vector field, i.e. a vector field tangent to the fibers of π . This space is also a Lie subalgebra of $\widetilde{\text{Vect}}_Q$. The restriction of $\widetilde{\text{Vect}}_Q^{\text{rel}}$ to $\mathcal{EL}$ gives a distribution which we denote by $\text{Vect}_Q^{\text{rel}}$.

Define the *symplectic* $\mathcal{EL}$ -moduli space $\mathcal{M}^{\text{symp}}$ as the space of leaves of $\text{Vect}_Q^{\text{rel}}$ on $\mathcal{EL}$:

$$\mathcal{M}^{\text{symp}} = \mathcal{EL} / \text{Vect}_Q^{\text{rel}}$$

It is clear that

$$T_{[X]} \mathcal{M}^{\text{symp}} = \ker(\hat{Q}_X) / \hat{Q}_X(\ker(\delta\pi_X))$$

The restriction to the boundary $\pi : \mathcal{F} \rightarrow \mathcal{F}_\partial$ induces the projection $\pi_* : \mathcal{M}^{\text{symp}} \rightarrow \mathcal{EL}_\partial$.

The distribution Vect_Q on $\mathcal{EL}$ induces the distribution b on $\mathcal{M}^{\text{symp}}$ which projects to the distribution Vect_{Q_∂} on $\mathcal{EL}_\partial$ by $\delta\pi_*$. It is easy to see that b is involutive and that

$$b_{[X]} \simeq \text{Im}(\hat{Q}_X)/\hat{Q}_X(\ker(\delta\pi_X))$$

where $[X]$ is the leaf of $\text{Vect}_Q^{\text{rel}}$ through X . The image of this projection can be thought of as a quotient distribution $\text{Vect}_Q/\text{Vect}_Q^{\text{rel}}$.

Notice that there is a natural isomorphism $T_X\mathcal{F}/\ker\delta\pi_X \simeq T_l\mathcal{F}_\partial$ where $l = \pi(X)$. We also have a natural degree 1 surjective map

$$\beta : T\mathcal{F}_\partial \rightarrow b, \quad \beta_X : T_l\mathcal{F}_\partial \rightarrow b_{[X]}$$

defined as follows: for $\xi \in T_X\mathcal{F}$ and $\xi + \ker(\delta\pi)$ being identified with an element in $T_l\mathcal{F}_\partial$, we set

$$\beta_X(\xi + \ker(\delta\pi)) = \hat{Q}_X(\xi) + \hat{Q}_X(\ker(\delta\pi))$$

The following statement follows immediately from $\hat{Q}_X^2 = 0$.

Lemma 3.14. *β vanishes on $\text{Im}(\hat{Q}_l)$.*

The usual $\mathcal{EL}$ -moduli space, i.e. space of leaves of Vect_Q on $\mathcal{EL}$, is naturally isomorphic to the space of leaves of b on $\mathcal{M}^{\text{symp}}$. This follows immediately from the definition of $\mathcal{M}^{\text{symp}}$.

Finally, from the definitions above, it is clear that the following diagram is commutative:

$$(18) \quad \begin{array}{ccccc} b_{[X]} & \xlongequal{\quad} & b_{[X]} & \xrightarrow{\quad \subset \quad} & T_{[X]}\mathcal{M}^{\text{symp}} \\ \beta_{[X]} \uparrow & & (\delta\pi_*)_{[X]} \downarrow & & (\delta\pi_*)_{[X]} \downarrow \\ T_l[-1]\mathcal{F}_\partial & \xrightarrow{\hat{Q}_l} & (\text{Vect}_{Q_\partial})_l & \xrightarrow{\quad \subset \quad} & T_l\mathcal{EL}_\partial \end{array}$$

3.4.2. More on tangent spaces. Define the vertical component $T_{[X]}^{\text{vert}}\mathcal{M}^{\text{symp}}$ of the tangent space $T_{[X]}\mathcal{M}^{\text{symp}}$ as

$$T_{[X]}^{\text{vert}}\mathcal{M}^{\text{symp}} = \ker((\delta\pi_*)_X) \subset T_{[X]}\mathcal{M}^{\text{symp}}$$

and denote the quotient map by $\chi : T_{[X]}^{\text{vert}}\mathcal{M}^{\text{symp}} \rightarrow T_{[X]^Q}\mathcal{M}$ where $\mathcal{M}$ is the usual $\mathcal{EL}$ -moduli space for N , i.e. the space of leaves of Vect_Q on $\mathcal{EL}$, and $[X]^Q$ is the leaf of Vect_Q through X .

The projection $\pi : \mathcal{F} \rightarrow \mathcal{F}_\partial$ restricted to $\mathcal{EL}$ induces the natural projection $\pi_*^Q : \mathcal{M} \rightarrow \mathcal{M}_\partial$. Recall that the fiber of π_*^Q over $[l] \in \mathcal{M}_\partial$ is the space $\mathcal{EL}(N, [l])/Q$ discussed in Section 3.3; the image of π_*^Q is the Q -reduction of the space $\mathcal{L} \subset \mathcal{EL}$ discussed in Section 3.2. Denote by $\psi = \delta\pi_*^Q$ the corresponding mapping of tangent bundles, $\psi_{[X]^Q} : T_{[X]^Q}\mathcal{M} \rightarrow T_{[l]}\mathcal{M}_\partial$.

The restriction of the mapping $\beta : T\mathcal{F}_\partial \rightarrow b$ defined earlier to $\ker(\hat{Q}_l)$ vanishes on $\text{Im}(\hat{Q}_l)$ and therefore induces a linear mapping $(\beta_*)_{[X]} : T_l[-1]\mathcal{M}_\partial \rightarrow T_{[X]}^{\text{vert}}\mathcal{M}^{\text{symp}}$.

We have a sequence of linear maps:

$$(19) \quad \cdots \rightarrow T_{[l]}[-1]\mathcal{M}_\partial \xrightarrow{\beta_*} T_{[X]}^{\text{vert}}\mathcal{M}^{\text{symp}} \xrightarrow{\chi} T_{[X]^Q}\mathcal{M} \xrightarrow{\psi} T_{[l]}\mathcal{M}_\partial \rightarrow \cdots$$

Proposition 3.15. *This is an exact sequence.*

Proof. Sequence (19) can be written as

$$\cdots \rightarrow H_{\hat{Q}_l}^{\bullet-1}(T_l \mathcal{F}_\partial) \rightarrow H_{\hat{Q}_X}^{\bullet}(T_X^{\text{vert}} \mathcal{F}) \rightarrow H_{\hat{Q}_X}^{\bullet}(T_X \mathcal{F}) \rightarrow H_{\hat{Q}_l}^{\bullet}(T_l \mathcal{F}_\partial) \rightarrow \cdots$$

which is induced from the short exact sequence in the non-reduced picture

$$0 \rightarrow T_X^{\text{vert}} \mathcal{F} \rightarrow T_X \mathcal{F} \rightarrow T_l \mathcal{F}_\partial \rightarrow 0$$

by passing to $\hat{Q}$ -cohomology and so is exact by snake lemma. $\square$

3.4.3. Symplectic structure, Lefschetz duality. Denote by $\hat{Q}_X^{\text{vert}}$ the restriction of $\hat{Q}_X$ to $T_X^{\text{vert}} \mathcal{F}$.

We know (cf. the proof of Proposition 3.5) that in $T_X \mathcal{F}$ we have

$$\text{Im}(\hat{Q}_X^{\text{vert}})^\perp = \ker(\hat{Q}_X), \quad \text{Im}(\hat{Q}_X)^\perp = \ker(\hat{Q}_X^{\text{vert}})$$

Assumption 3.16. *We will assume that also the opposite holds:*

$$(20) \quad \ker(\hat{Q}_X)^\perp = \text{Im}(\hat{Q}_X^{\text{vert}}), \quad \ker(\hat{Q}_X^{\text{vert}})^\perp = \text{Im}(\hat{Q}_X)$$

This is a stronger version of assumption (3); in our examples it follows from Hodge-Morrey decomposition theorem for manifolds with boundary [14]. It immediately implies that $\text{Vect}_Q^{\text{rel}}$ coincides with the characteristic distribution on $\mathcal{EL}$, thus we have the following.

Proposition 3.17. *Assuming (20), $\mathcal{M}^{\text{symp}}$ is the symplectic reduction of $\mathcal{EL}$ and carries a degree -1 symplectic structure $\underline{\omega}$ coming from $\mathcal{F}$.*

Definition 3.18. *We call a BV-BFV theory **regular** if the assumption (20) holds for any $X \in \mathcal{EL}$, together with the assumption (15) for any $l \in \mathcal{EL}_\partial$.*

Symplectic structure ω on $\mathcal{F}$ induces a bilinear pairing $H_{\hat{Q}_X^{\text{vert}}} \otimes H_{\hat{Q}_X}[1] \rightarrow \mathbb{R}$ or equivalently $T_{[X]}^{\text{vert}} \mathcal{M}^{\text{symp}} \otimes T_{[X]Q}[1] \mathcal{M} \rightarrow \mathbb{R}$ which is well-defined due to (14) and non-degenerate due to (20). Together with the symplectic structure on $\mathcal{M}_\partial = \mathcal{EL}_\partial$ it gives the Lefschetz duality for the long exact sequence (19), i.e. a non-degenerate pairing on $T_{[X]}^{\text{vert}} \mathcal{M}^{\text{symp}} \oplus T_{[X]Q}[1] \mathcal{M} \oplus T_{[l]} \mathcal{M}_\partial$:

$$(21) \quad \langle \bullet, \bullet \rangle : \begin{cases} T_{[X]}^{\text{vert}} \mathcal{M}^{\text{symp}} \otimes T_{[X]Q}[1] \mathcal{M} & \rightarrow \mathbb{R} \\ T_{[X]Q}[1] \mathcal{M} \otimes T_{[X]}^{\text{vert}} \mathcal{M}^{\text{symp}} & \rightarrow \mathbb{R} \\ T_{[l]} \mathcal{M}_\partial \otimes T_{[l]} \mathcal{M}_\partial & \rightarrow \mathbb{R} \end{cases}$$

Observe that (14) implies that the map χ is self-adjoint with respect to $\langle \bullet, \bullet \rangle$, and ψ and β_* are adjoint to each other, therefore Lefschetz duality can be stated as an injective chain map between the complex (19) and the dual one:

$$\begin{array}{ccccccccccc} \cdots & \longrightarrow & T_{[l]}[-1] \mathcal{M}_\partial & \xrightarrow{\beta_*} & T_{[X]}^{\text{vert}} \mathcal{M}^{\text{symp}} & \xrightarrow{\chi} & T_{[X]Q} \mathcal{M} & \xrightarrow{\psi} & T_{[l]} \mathcal{M}_\partial & \longrightarrow & \cdots \\ & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \\ \cdots & \longrightarrow & T_{[l]}^*[-1] \mathcal{M}_\partial & \xrightarrow{\psi^*} & T_{[X]Q}^*[-1] \mathcal{M} & \xrightarrow{\chi^*} & (T_{[X]}^{\text{vert}})^*[-1] \mathcal{M}^{\text{symp}} & \xrightarrow{(\beta_*)^*} & T_{[l]}^* \mathcal{M}_\partial & \longrightarrow & \cdots \end{array}$$

In case of a topological field theory, the complex (19) consists of finite dimensional vector spaces and vertical arrows in the diagram above become isomorphisms.

The form $\underline{\omega}$ induces presymplectic structures $\omega^{\text{symp,vert}}$ on the fibers of $\pi_* : \mathcal{M}^{\text{symp}} \rightarrow \mathcal{EL}_\partial$. The form $\omega^{\text{symp,vert}}$ can be written in terms of Lefschetz duality as:

$$\omega_{[X]}^{\text{symp,vert}} = \langle \bullet, \chi(\bullet) \rangle : \quad T_{[X]}^{\text{vert}} \mathcal{M}^{\text{symp}} \otimes T_{[X]}^{\text{vert}} [1] \mathcal{M}^{\text{symp}} \rightarrow \mathbb{R}$$

Proposition 3.19. *Fibers of $\pi_*^Q : \mathcal{M} \rightarrow \mathcal{M}_\partial$ carry a natural degree -1 symplectic structure coming from $\mathcal{F}$.*

Proof. By non-degeneracy of (21), the kernel of $\omega^{\text{symp,vert}}$ is exactly the kernel of χ . Thus $\omega^{\text{symp,vert}}$ induces a non-degenerate degree -1 symplectic structure on $\text{Im}(\chi) = \ker(\psi) \subset T_{[X]Q}\mathcal{M}$ and hence on fibers of $\pi_*^Q : \mathcal{M} \rightarrow \mathcal{M}_\partial$. $\square$

Thus, the Q -reduced fibers $\mathcal{EL}(N, [l])/Q$ have natural symplectic structure. We call these reduced fibers *the moduli spaces of vacua*.

Proposition 3.20. *The image of $\pi_*^Q : \mathcal{M} \rightarrow \mathcal{M}_\partial$ is locally Lagrangian in $\mathcal{M}_\partial$.*

Proof. Indeed, for a smooth point $[l] = [\pi(X)] \in \mathcal{M}_\partial$, tangent space to the image of π_*^Q is $\text{Im}(\psi : H_{\hat{Q}_X}^\bullet(T_X\mathcal{F}) \rightarrow H_{\hat{Q}_l}^\bullet(T_l\mathcal{F}_\partial))$. Lagrangianity is proven as follows:

$$\begin{aligned} \text{Im}(\psi)^\perp &= \{a \in H_{\hat{Q}_l}(T_l\mathcal{F}_\partial) \mid \langle a, \psi(b) \rangle = 0 \ \forall b \in H_{\hat{Q}_X}(T_X\mathcal{F})\} \\ &= \{a \in H_{\hat{Q}_l} \mid \langle \beta_*(a), b \rangle = 0 \ \forall b \in H_{\hat{Q}_X}\} \\ &= \ker(\beta_*) \\ &= \text{Im}(\psi) \end{aligned}$$

where in the third line we used the non-degeneracy of Lefschetz pairing. $\square$

The following is a corollary of the above, using also Propositions A.1 and 3.7:

Corollary 3.21. *The space $\mathcal{L} = \pi(\mathcal{EL}) \subset \mathcal{F}_\partial$ is locally Lagrangian.*

3.4.4. Digression: fibers of π_*^Q via symplectic reduction. Let $\Lambda \subset \mathcal{F}_\partial$ be a Lagrangian submanifold which is transversal to $\mathcal{L} = \pi(\mathcal{EL})$. Assume that Λ intersects $\mathcal{L}$ at the point l . The subspace $\pi^{-1}(\Lambda) \subset \mathcal{F}$ is the space of fields with boundary values on Λ . We will assume that the subspace $\pi^{-1}(\Lambda)$ is symplectic.

Let us prove that the subspace $\pi^{-1}(\Lambda) \cap \mathcal{EL}$ is locally coisotropic in $\pi^{-1}(\Lambda)$. For this we need to prove that at a smooth point X the tangent space $\mathcal{S}_X = T_X\pi^{-1}(\Lambda) \cap \mathcal{EL}$ is a coisotropic subspace in $T_X\mathcal{F}$.

Because Lagrangian subspaces $\mathcal{L}$ and Λ are transversal, the intersection of their tangent spaces is trivial, $T_X\Lambda \cap T_X\mathcal{L} = \{0\}$, and therefore for each $\xi \in T_X\pi^{-1}(\Lambda) \cap \mathcal{EL}$ we have $\delta\pi_X(\xi) = 0$. Thus

$$\mathcal{S}_X = \{\xi \in \ker(\hat{Q}_X) \mid \delta\pi(\xi) = 0\}$$

Denote

$$I_X = \{\xi \in \text{Im}(\hat{Q}_X) \mid \delta\pi(\xi) = 0\}$$

Observe that $I_X \subset \mathcal{S}_X$ and that

$$(22) \quad \mathcal{S}_X/I_X \cong T_X\mathcal{EL}(N, [l])/Q$$

by Proposition 3.13.

Lemma 3.22. *For a smooth point X , the symplectic orthogonal subspace to I_X in $T_X\pi^{-1}(\Lambda)$ is $\mathcal{S}_X$.*

Proof. The symplectic orthogonal of I_X in $T_X\pi^{-1}(\Lambda)$ is:

$$I_X^\perp = \{\eta \in T_X\pi^{-1}(\Lambda) \mid \omega_N(\eta, \xi) = 0, \text{ for any } \xi \in I_X\}$$

That is for any $\xi = \hat{Q}_X\lambda$ such that $\delta\pi(\lambda) \in \ker(\hat{Q}_\partial)$. The orthogonality of ξ and η is equivalent to

$$\omega(\hat{Q}_X\eta, \lambda) + \omega_\partial(\delta\pi(\eta), \delta\pi(\lambda)) = 0$$

for any λ such that $\delta\pi(\lambda) \in \ker(\hat{Q}_l)$. Because the first term should vanish for all λ , we have $\eta \in \ker(\hat{Q}_X)$. Thus the last term should vanish separately for any such λ . This implies that $\delta\pi(\eta)$ is symplectic orthogonal to $\ker(\hat{Q}_l)$. By condition (15) this implies that $\delta\pi(\eta) \in \text{Im}(\hat{Q}_l)$. Because $\text{Im}(\hat{Q}_l) \subset T_l\mathcal{L}$ and because $\mathcal{L}$ and Λ are transversal, we have $\delta\pi(\eta) = 0$. Therefore $I_X^\perp = \mathcal{S}_X$. $\square$

Corollary 3.23. *The subspace $\mathcal{S}_X \subset T_X\pi^{-1}(\Lambda)$ is coisotropic. Thus $\pi^{-1}(\Lambda) \cap \mathcal{EL}$ is locally coisotropic in $\pi^{-1}(\Lambda)$.*

Lemma 3.24. *The symplectic orthogonal subspace to $\mathcal{S}_X$ in $T_X\pi^{-1}(\Lambda)$ is I_X .*

Proof. By Lemma 3.22, $I_X^\perp = \mathcal{S}_X$, which implies $I_X \subseteq \mathcal{S}_X^\perp$. On the other hand, due to (22) and to Proposition 3.19, $\mathcal{S}_X/I_X$ is symplectic and thus I_X has to coincide with $\mathcal{S}_X^\perp$. $\square$

The following is the immediate corollary of the Lemma above.

Proposition 3.25. *The symplectic reduction of $\mathcal{S}_X$ is*

$$\underline{\mathcal{S}}_X = \mathcal{S}_X/I_X$$

Comparing with the Q -reduction of $T_X\mathcal{EL}(N, [l])$ we see that two reductions are naturally isomorphic:

$$\underline{T_X\pi^{-1}(\Lambda) \cap \mathcal{EL}} = T_X\mathcal{EL}(N, [l])/Q$$

Thus the fiber of $\pi_^Q : \mathcal{M} \rightarrow \mathcal{M}_\partial$ over $[l]$ coincides with the symplectic reduction of $\pi^{-1}(\Lambda) \cap \mathcal{EL}$ in $\pi^{-1}(\Lambda)$.*

3.4.5. *Example: symplectic $\mathcal{EL}$ -moduli space for the abelian Chern-Simons theory.* In the abelian Chern-Simons theory, see section 7.1 for details, the following occurs:

- The symplectic $\mathcal{EL}$ -moduli space is

$$\begin{aligned} \mathcal{M}^{\text{symp}} &= \mathcal{EL}/\text{Vect}_Q^{\text{rel}} = \\ &= \{A \in \Omega^{\bullet+1}(N) \mid dA = 0\} / (A \sim A + d\alpha \text{ for any } \alpha \in \Omega^\bullet(N) \text{ s.t. } \alpha|_{\partial N} = 0) \end{aligned}$$

- The restriction map $\pi_* : \mathcal{M}^{\text{symp}} \rightarrow \mathcal{EL}_\partial = \Omega_{\text{closed}}^{\bullet+1}(\partial N)$ sends the class $[A] \in \mathcal{M}^{\text{symp}}$ to $A|_{\partial N} \in \mathcal{EL}_\partial$ (which is well defined).
- Fibers of π_* are isomorphic to the relative cohomology $H^{\bullet+1}(N, \partial N)$.
- The map $\beta_{[A]} : T_{A|_{\partial N}}\mathcal{F}_{\partial N} \rightarrow T_{[A]}\mathcal{M}^{\text{symp}}$ sends $\alpha_\partial \in \Omega^\bullet(\partial N)$ to $[d\tilde{\alpha}]$ for arbitrary $\tilde{\alpha} \in \Omega^\bullet(N)$ such that $\tilde{\alpha}|_{\partial N} = \alpha_\partial$.
- Taking the quotient of $\mathcal{M}^{\text{symp}}$ by the distribution $b = \text{im}(\beta)$ gives the usual $\mathcal{EL}$ -moduli space

$$\mathcal{M}^{\text{symp}}/b = \mathcal{M} = H^{\bullet+1}(N)$$

which is the absolute de Rham cohomology of N .

- Exact sequence (19) is the usual long exact sequence of relative cohomology

$$\cdots \rightarrow H^\bullet(\partial N) \rightarrow H^{\bullet+1}(N, \partial N) \rightarrow H^{\bullet+1}(N) \rightarrow H^{\bullet+1}(\partial N) \rightarrow \cdots$$

- The symplectic form ω on $\Omega^\bullet(N)$ descends to

$$\underline{\omega}([A], [B]) = \int_N A \wedge B$$

on $\mathcal{M}^{\text{symp}}$. Lefschetz duality (21) is the usual Lefschetz duality between absolute and relative cohomology, plus Poincaré duality for the cohomology of the boundary.

3.5. The gluing (cutting) of symplectic $\mathcal{EL}$ -moduli spaces.

3.5.1. *The non-reduced case.* Assume that a manifold N is cut into two pieces N_1, N_2 along a codimension 1 submanifold N_2^∂ such that $\partial N_1 = N_1^\partial \sqcup N_2^\partial$ and $\partial N_2 = N_2^\partial \sqcup N_3^\partial$. We have natural commutative diagram:

$$(23) \quad \begin{array}{ccc} \mathcal{F}_N & \longrightarrow & \mathcal{F}_{N_1} \\ \downarrow & & \downarrow \\ \mathcal{F}_{N_2} & \longrightarrow & \mathcal{F}_{N_2^\partial} \end{array}$$

The space of fields $\mathcal{F}_N$ is the subspace in the fiber product of spaces $\mathcal{F}_{N_1}$ and $\mathcal{F}_{N_2}$ over $\mathcal{F}_{N_2^\partial}$ consisting of fields which are smooth at N_2^∂ .

The Euler-Lagrange space for N is the fiber product of Euler-Lagrange spaces for N_1 and for N_2 :

$$\mathcal{EL}_N = \mathcal{EL}_{N_1} \times_{\mathcal{EL}_{N_2^\partial}} \mathcal{EL}_{N_2}$$

3.5.2. *The gluing.* The gluing for symplectic $\mathcal{EL}$ -moduli spaces goes as follows. Let N_1, N_2 be two spacetime manifolds with boundaries $\partial N_1 = N_1^\partial \sqcup N_2^\partial$ and $\partial N_2 = (N_2^\partial)' \sqcup N_3^\partial$ respectively. Assume that N_2^∂ diffeomorphic to $(N_2^\partial)'$, and denote by N the result of gluing N_1 and N_2 along the common boundary component:

$$N = (N_1 \sqcup N_2) / (N_2^\partial \sim (N_2^\partial)')$$

The space $\mathcal{M}_N^{\text{symp}}$ can be constructed intrinsically in terms of $\mathcal{M}_{N_1}^{\text{symp}}$ and $\mathcal{M}_{N_2}^{\text{symp}}$ as follows:

(i) First consider the fiber product

$$(24) \quad \widetilde{\mathcal{M}} = \mathcal{M}_{N_1}^{\text{symp}} \times_{\mathcal{EL}_{N_2^\partial}} \mathcal{M}_{N_2}^{\text{symp}}$$

For any point $\tilde{X} \in \widetilde{\mathcal{M}}$, we have a map

$$(25) \quad \tilde{\beta}_{\tilde{X}} : T_{\pi_1(\tilde{X})} \mathcal{F}_{N_1^\partial} \times T_{\pi_2(\tilde{X})} \mathcal{F}_{N_2^\partial} \times T_{\pi_3(\tilde{X})} \mathcal{F}_{N_3^\partial} \rightarrow T_{\tilde{X}} \widetilde{\mathcal{M}}$$

induced from the two maps

$$\begin{aligned} (\beta_1 \times \beta_2)_{[X_1]} : T_{\pi_1(X_1)} \mathcal{F}_{N_1^\partial} \times T_{\pi_2(X_1)} \mathcal{F}_{N_2^\partial} &\rightarrow T_{[X_1]} \mathcal{M}_{N_1}^{\text{symp}} \\ (\beta_{2'} \times \beta_3)_{[X_2]} : T_{\pi_{2'}(X_2)} \mathcal{F}_{(N_2^\partial)'} \times T_{\pi_3(X_2)} \mathcal{F}_{N_3^\partial} &\rightarrow T_{[X_2]} \mathcal{M}_{N_2}^{\text{symp}} \end{aligned}$$

(ii) The space $\mathcal{M}_N^{\text{symp}}$ now can be identified with the leaf space of the distribution $\tilde{\beta}(0 \times T\mathcal{F}_{N_2^\partial} \times 0)$ on $\widetilde{\mathcal{M}}$:

$$\mathcal{M}_N^{\text{symp}} = \widetilde{\mathcal{M}} / \tilde{\beta}(0 \times T\mathcal{F}_{N_2^\partial} \times 0)$$

It inherits the quotient distribution

$$b_N = b_N^1 \times b_N^3 = \text{im}(\tilde{\beta}) / \tilde{\beta}(0 \times T\mathcal{F}_{N_2^\partial} \times 0)$$

parameterized by $T\mathcal{F}_{N_1^\partial} \times T\mathcal{F}_{N_3^\partial}$.

The important point here is that the gluing of symplectic $\mathcal{EL}$ -moduli spaces is done in intrinsic terms of the symplectically reduced picture, i.e. in terms of the ingredients of diagram (18): $(\mathcal{M}^{\text{symp}}, \mathcal{F}_\partial, Q_\partial, \pi_*, \beta)$.

3.5.3. Gluing tangent spaces. The construction of gluing of symplectic $\mathcal{EL}$ -moduli spaces described above implies the following Mayer-Vietoris type long exact sequence for $T_{[X]}\mathcal{M}^{\text{symp}}$:

$$(26) \quad \cdots \rightarrow T_{[\pi_2(X)]}[-1]\mathcal{M}_{N_2^\partial} \xrightarrow{(\beta_2 - \beta_{2'})_*} T_{[X]}\mathcal{M}_N^{\text{symp}} \rightarrow \\ \rightarrow T_{[X_1]}\mathcal{M}_{N_1}^{\text{symp}, Q_2} \oplus T_{[X_2]}\mathcal{M}_{N_2}^{\text{symp}, Q_{2'}} \rightarrow T_{[\pi_2(X)]}\mathcal{M}_{N_2^\partial} \rightarrow \cdots$$

Here $\mathcal{M}_{N_1}^{\text{symp}, Q_2}$ is the quotient of $\mathcal{M}_{N_1}^{\text{symp}}$ by the distribution b_2 . Similarly, $\mathcal{M}_{N_2}^{\text{symp}, Q_{2'}}$ is the quotient of $\mathcal{M}_{N_2}^{\text{symp}}$ by the distribution $b_{2'}$.³

We have a similar long exact sequence for the tangent space to the usual $\mathcal{EL}$ -moduli space:

$$(27) \quad \cdots \rightarrow T_{[\pi_2(X)]}[-1]\mathcal{M}_{N_2^\partial} \rightarrow T_{[X]}\mathcal{M}_N \rightarrow T_{[X_1]}\mathcal{M}_{N_1} \oplus T_{[X_2]}\mathcal{M}_{N_2} \rightarrow T_{[\pi_2(X)]}\mathcal{M}_{N_2^\partial} \rightarrow \cdots$$

Here $[X]$ is the space of leaves of Vect_Q through X .

3.5.4. Example: gluing in abelian Chern-Simons theory. Here we will give an example of gluing symplectic $\mathcal{EL}$ -moduli spaces. Let $N = N_1 \cup_{N_2^\partial} N_2$ as before. Then

$$\widetilde{\mathcal{M}} = \frac{\{A \in \Omega^{\bullet+1}(N) \mid dA = 0\}}{A \sim A + d\alpha \text{ for any } \alpha \in \Omega^\bullet(N) \text{ s.t. } \alpha|_{N_1^\partial} = \alpha|_{N_2^\partial} = \alpha|_{N_3^\partial} = 0}$$

Mapping (25) acts as

$$\tilde{\beta}_{[A]} : (\alpha_1^\partial, \alpha_2^\partial, \alpha_3^\partial) \mapsto [d\tilde{\alpha}]$$

Here $(\alpha_1^\partial, \alpha_2^\partial, \alpha_3^\partial) \in T_{A|_{N_1^\partial}}\mathcal{F}_{N_1^\partial} \times T_{A|_{N_2^\partial}}\mathcal{F}_{N_2^\partial} \times T_{A|_{N_3^\partial}}\mathcal{F}_{N_3^\partial}$ and $\tilde{\alpha} \in \Omega^\bullet(N)$ is any form such that $\tilde{\alpha}|_{N_k^\partial} = \alpha_k^\partial$ for $k = 1, 2, 3$. The class $[d\tilde{\alpha}] \in T_{[A]}\widetilde{\mathcal{M}}$ does not depend on the choice of $\tilde{\alpha}$. To pass from $\widetilde{\mathcal{M}}$ to $\mathcal{M}_N^{\text{symp}}$ we should mod out differentials of forms on the interface N_2^∂ extended to N . Now we can write the symplectic $\mathcal{EL}$ -moduli space as the quotient

$$\mathcal{M}_N^{\text{symp}} = \widetilde{\mathcal{M}} / \tilde{\beta}(0 \times \Omega^\bullet(N_2^\partial) \times 0) \simeq \\ \simeq \frac{\{A \in \Omega^{\bullet+1}(N) \mid dA = 0\}}{A \sim A + d\alpha \text{ for any } \alpha \in \Omega^\bullet(N) \text{ s.t. } \alpha|_{N_1^\partial} = \alpha|_{N_3^\partial} = 0}$$

It is equipped with two commuting distributions b_N^1, b_N^3 parameterized by $\Omega^\bullet(N_1^\partial), \Omega^\bullet(N_3^\partial)$ respectively.

Mayer-Vietoris sequence (26) becomes the following:

$$\cdots \rightarrow H^\bullet(N_2^\partial) \rightarrow H^{\bullet+1}(N, N_1^\partial \sqcup N_3^\partial) \rightarrow \\ \rightarrow H^{\bullet+1}(N_1, N_1^\partial) \oplus H^{\bullet+1}(N_2, N_3^\partial) \rightarrow H^{\bullet+1}(N_2^\partial) \rightarrow \cdots$$

³An equivalent description: $\mathcal{M}_{N_1}^{\text{symp}, Q_2} = \mathcal{EL}_{N_1} / \text{Vect}_Q^{\text{rel}1}$ where $\text{Vect}_Q^{\text{rel}1}$ is the distribution generated by vector fields of the form $[Q_{N_1}, v]$ with v tangent to the fibers of π_1 . Likewise, $\mathcal{M}_{N_2}^{\text{symp}, Q_{2'}} = \mathcal{EL}_{N_2} / \text{Vect}_Q^{\text{rel}3}$.

whereas the version for usual $\mathcal{EL}$ -moduli spaces (27) reads

$$\cdots \rightarrow H^\bullet(N_2^\partial) \rightarrow H^{\bullet+1}(N) \rightarrow H^{\bullet+1}(N_1) \oplus H^{\bullet+1}(N_2) \rightarrow H^{\bullet+1}(N_2^\partial) \rightarrow \cdots$$

3.6. Higher codimensions. The BV-BFV theory is the extension of the BV theory to manifolds with boundary. In a similar way the BV theory extends to higher codimension submanifolds.

Definition 3.26. *The following collection of data is called a k -extended BV theory in dimension n .*

For each $i = 0, \dots, k-1$, a k -extended BV theory assigns to every $(n-i)$ -dimensional manifold N_i with boundary N_{i+1} :

- (1) *a space of fields $\mathcal{F}_{N_i}$ which is a graded manifold with exact symplectic form $\omega_{N_i} = (-1)^{n-i} \delta \alpha_{N_i}$ with $gh(\omega_{N_i}) = gh(\alpha_{N_i}) = i-1$, and with cohomological vector field Q_{N_i} ,*
- (2) *a projection $\pi_i : \mathcal{F}_{N_i} \rightarrow \mathcal{F}_{N_{i+1}}$,*
- (3) *an action functional S_{N_i} on $\mathcal{F}_{N_i}$ with $gh(S_{N_i}) = i$.*

These data should satisfy the following axioms:

- $\delta \pi_{i-1}(Q_{N_{i-1}}) = Q_{N_i}$,
- $\iota_{Q_{N_{i-1}}} \omega_{N_{i-1}} = (-1)^{n-i+1} \delta S_{N_{i-1}} + \pi_i^*(\alpha_{N_i})$

We call k -extension of a given BV theory in dimension n a k -extended theory with the original data for $i = 0$.

Definition 3.27. *A BV field theory has length k if k is the maximal number such that in its k -extension for $i = 1, \dots, k$, all $\pi_{i-1}(\mathcal{EL}_{N_{i-1}}) = \mathcal{L}_{N_{i-1}} \subset \mathcal{EL}_{N_i} \subset \mathcal{F}_{N_i}$ are Lagrangian and all Q -reduced fibers are finite dimensional. Here $\mathcal{EL}_{N_i} \subset \mathcal{F}_{N_i}$ is the set of zeroes of the vector field Q_{N_i} . If k is equal to n , we say that the theory is **maximally extended**.*

Usually BV field theories have length 1. For example, scalar field theory in dimension greater than one or Yang-Mills theory in dimension greater than two. We never consider theories of length zero. In the rest of the paper we will show that scalar field theory in dimension 1, Yang-Mills theory in dimension 2 and all AKSZ theories are maximally extended (this includes BF , Chern-Simons theories and the Poisson sigma model). Notice that scalar theory in dimension one is just quantum mechanics and the Yang-Mills theory in dimension 2 is known to be almost topological [57] (meaning that it only depends on the topology of the spacetime manifold and on its volume).

Remark 3.28 (Extended BV theories on manifolds with corners). *A k -extended BV theory in dimension n naturally leads to an associated theory on n -dimensional manifolds with corners up to codimension k . Namely, to such a manifold N we associate the data for $N_0 = N$ and N_1 the union of the codimension one⁴ strata in ∂N . To each such stratum $N' \subset \partial N$ we associate the data for $N_1 = N'$ and N_2 the union of the codimension one strata in $\partial N'$ (notice that this is the union of only some codimension two strata in ∂N). To each such stratum $N'' \subset \partial N'$ we associate the data for $N_2 = N''$ and N_3 the union of the codimension one strata in $\partial N''$ (notice that this is the union of only some codimension two strata in $\partial N'$ and in turn of some codimension three strata in ∂N), and so on.*

⁴Here and in the following, the codimension of a boundary stratum is computed in terms of the bulk manifold.

3.7. Boundary conditions. In the current paper we insist on having free boundary conditions. An alternative approach, first explored in [18, 43], fixes boundary conditions in a way compatible with the BFV structure on the boundary. This corresponds to choosing a Lagrangian submanifold $\mathcal{L}$ of the BFV space of boundary fields to which the boundary cohomological vector field is tangent and on which the boundary 1-form vanishes. Equivalently, the boundary action should vanish on $\mathcal{L}$. We call such Lagrangian submanifolds *adapted*.

At first sight, it would also seem natural to impose $\mathcal{L}$ to be transversal to the Lagrangian submanifold L_M . However, this condition is too restrictive and rules out a lot of interesting boundary conditions. Instead, a better assumption would be that the intersection of $\mathcal{L}$ with L_N should be finite dimensional after reduction.

As shown in [18], in the case of AKSZ theories, one can obtain an adapted Lagrangian submanifold $\mathcal{L}$ by choosing an adapted Lagrangian submanifold L_i of the target manifold $\mathcal{M}$ for each boundary component $\partial_i N$ of the spacetime N . The submanifold $\mathcal{L} = \prod_i \text{Map}(T[1]\partial_i N, L_i)$ in this case is an adapted Lagrangian in $\mathcal{F}_{\partial N}$. Notice that usually one does not require the decomposition $\partial M = \cup_i \partial_i M$ to be disjoint, rather the pairwise intersections are assumed to be of lower dimension. The adapted Lagrangian submanifolds of the target are usually referred to as *branes*.

In the case when the target is $T^*[1]P$, with P a Poisson manifold, one can easily show that a brane is necessarily of the form $N^*[1]C$, where C is a coisotropic submanifold of P and N^*C denotes its conormal bundle. When the target is a differential graded symplectic manifold (with symplectic form of degree 2) associated to a Courant algebroid over a manifold N , then a brane covering N is the same as a Dirac structure.

An intermediate type of boundary condition consists in splitting the boundary into two components and in choosing an adapted Lagrangian submanifold of the BFV space of fields on the first component while keeping free boundary conditions for the second component. The analysis performed in this paper still holds for boundary fields of the second component. This mixed approach might have several applications. For example [15], the study of the Poisson Sigma Model on the disk whose boundary is split into an even number of ordered intervals I_i with boundary conditions on I_{2s} determined by the $C = P \forall s$ leads to the construction of what is known as the relative symplectic groupoid integrating P .

3.8. $gh = 0$ part of the BV-BFV theory.

3.8.1. Non reduced theory. The $gh = 0$ part of the BV-BFV theory is a first order classical field theory with the space of fields $F_N = \mathcal{F}_N^{(0)}$ (the $gh = 0$ part of $\mathcal{F}_N$), the classical action $S_N^{(0)} = S_N|_{\mathcal{F}_N^{(0)}}$.

The $gh = 0$ part of the space of boundary fields $F_{\partial N} = \mathcal{F}_{\partial N}^{(0)}$ is a symplectic manifold with the symplectic form $\omega_{\partial N}^{(0)} = \delta\alpha_{\partial N}^{(0)}$ which is the $gh = 0$ part of the symplectic form $\omega_{\partial N}$ on boundary BFV fields. The one-form $\delta\alpha_{\partial N}^{(0)}$ is the $gh = 0$ part of the form $\alpha_{\partial N}$ and it is determined by the $gh = 0$ part of the classical action.

Gauge transformations are $gh = 0$ part of the BV-BFV gauge transformations generated by vector fields $[V, Q]$ and they are Hamiltonian on the boundary.

The $gh = 0$ part of the coisotropic submanifold $\mathcal{EL}_{\partial N} \subset \mathcal{F}_N$ is a coisotropic submanifold $C_{\partial N} = \mathcal{EL}_{\partial N}^{(0)} \subset F_N$. It consists of boundary fields which can be

extended to a solution to Euler-Lagrange equations in the vicinity of the boundary in N .

In regular BV theories the space of solutions to the Euler-Lagrange equations $EL_N = \mathcal{EL}_N^{(0)}$ projects to the Lagrangian subspace $L_N = \mathcal{L}_N^{(0)} = \pi(EL_N) \subset C_{\partial N} \subset F_{\partial N}$.

3.8.2. The reduction. The classical moduli space EL_N/G_N is the $gh = 0$ part of the $\mathcal{EL}$ -moduli space $\mathcal{EL}_N/Q_N$ and maps to the reduced phase space $C_{\partial N}/G_{\partial N} = \underline{C}_{\partial N}$, which is in turn the $gh = 0$ part of the boundary $\mathcal{EL}$ -moduli space $\mathcal{EL}_{\partial N}$. Here G_N denotes the distribution on EL_N induced by Vect_{Q_N} , likewise for $G_{\partial N}$; also $G_{\partial N}$ can be seen as the coisotropic distribution on $C_{\partial N}$.

In the regular case, the image of

$$(28) \quad \pi_* : EL_N/G_N \rightarrow \underline{C}_{\partial N}$$

is the reduced Lagrangian $L_N/G_{\partial N} = \underline{L}_N$.

Remark 3.29. In regular case, there is also the following relation between the smooth loci (cf. Appendix C) of moduli spaces:

$$(29) \quad \begin{array}{ccc} T_{\text{vert}}^*[-1](EL_N/G_N)^{\text{smooth}} & \xrightarrow{\subseteq} & (\mathcal{EL}_N/Q_N)^{\text{smooth}} \\ \pi_* \downarrow & & \pi_* \downarrow \\ (\underline{C}_{\partial N})^{\text{smooth}} & \xrightarrow{\subseteq} & (\mathcal{EL}_{\partial N})^{\text{smooth}} \end{array}$$

Here T_{vert}^* denotes the dual of the vertical tangent bundle of the fibration (28). Horizontal arrows in (29) may be strict inclusions (e.g. abelian Chern-Simons theory where $\mathcal{EL}$ -moduli spaces contain additional smooth pieces: $H^0(N)$ and $H^3(N)$, cf. section 7.1.3), or may be equalities (e.g. non-abelian Chern-Simons theory with a simple gauge group G).

4. THE BFV CATEGORY

4.1. Spacetime categories. Recall that an n -dimensional spacetime category is the category of n -dimensional cobordisms which may have additional structure (smooth, Riemannian etc.). See [4, 51, 54] for the discussion of various examples.

In most general terms *objects* of a d -dimensional spacetime category are $(d-1)$ -dimensional manifolds (space manifolds). In specific examples of spacetime categories, space manifolds are equipped with a structure (orientation, symplectic structure, Riemannian metric, etc.).

A *morphism* between two space manifolds Σ_1 and Σ_2 is a d -dimensional manifold M , possibly with a structure (orientation, symplectic, Riemannian metric, etc.), together with the identification of $\Sigma_1 \sqcup \overline{\Sigma_2}$ with the boundary of M . Here $\overline{\Sigma}$ is the manifold Σ with reversed orientation.

Composition of morphisms is the gluing along the common boundary. Here are examples of spacetime categories.

The d -dimensional topological category. Objects are smooth, compact, oriented $(d-1)$ -dimensional manifolds. A morphism between Σ_1 and Σ_2 is a d -dimensional smooth compact oriented manifold with $\partial M = \Sigma_1 \sqcup \overline{\Sigma_2}$. The orientation on M should agree with the orientations of Σ_i in a natural way. The composition consists of gluing two morphisms along the common boundary.

The d -dimensional Riemannian category. Objects are oriented $(d - 1)$ Riemannian manifolds with collars. Morphisms between two objects N_1 and N_2 are oriented d -dimensional Riemannian manifolds M , such that $\partial M = N_1 \sqcup \overline{N_2}$. The orientation on all three manifolds should naturally agree, and the metric on M agrees with the metric on N_1 and N_2 on a collar of the boundary. The composition is the gluing of such Riemannian cobordisms. For details see [54].

The d -dimensional metrized cell complexes. Objects are $(d-1)$ -dimensional oriented metrized cell complexes (edges have length, 2-cells have area, etc.). A morphism between two such complexes C_1 and C_2 is an oriented metrized d -dimensional cell complex C together with two embeddings of metrized cell complexes $i : C_1 \hookrightarrow C$, $j : \overline{C_2} \hookrightarrow C$ where i is orientation reversing and j is orientation preserving. The composition is the gluing of such triples along the common $(d-1)$ -dimensional subcomplex.

This is the underlying category for all lattice models in statistical mechanics.

The Pseudo-Riemannian category The difference between this category and the Riemannian category is that morphisms are pseudo-Riemannian with the signature $(d - 1, 1)$. This is the most interesting category for physics. When $d = 4$ it represents the spacetime structure of our universe.

4.2. The BFV category. The category BFV has the following objects, morphisms and compositions of morphisms.

Objects of BFV are triples $(\mathcal{F}, \alpha, Q)$ where $\mathcal{F}$ is an exact graded symplectic manifold with the symplectic form $\omega = d\alpha$ with ghost number 0, and Q is a cohomological vector field (i.e. its Lie derivative squares to zero) with ghost number 1. The symplectic form should be preserved by Q :

$$L_Q \omega = 0$$

Let $\mathcal{EL}$ be the space of zeroes of the vector field Q . The restriction of the Lie subalgebra $Vect_Q = [Vect(\mathcal{F}), Q] \subset Vect(\mathcal{F})$ defines an involutive⁵ distribution on $\mathcal{EL}$.

Morphisms between $(\mathcal{F}_1, \alpha_1, Q_1)$ and $(\mathcal{F}_2, \alpha_2, Q_2)$ are differential graded manifolds $\mathcal{F}$ with symplectic form $\omega^\mathcal{F}$ with ghost number -1 , with cohomological vector field $Q^\mathcal{F}$ with ghost number 1, with the function $S^\mathcal{F}$ (action function) on $\mathcal{F}$ with $gh(S^\mathcal{F}) = 0$, and with two projection mappings $\pi_i : \mathcal{F} \rightarrow \mathcal{F}_i$. These data should satisfy the following conditions:

- Projections π_i are mappings of differential graded manifolds, i.e. $\delta\pi_i(Q^\mathcal{F}) = Q_i$.
- The following identity should hold

$$\iota_{Q^\mathcal{F}} \omega^\mathcal{F} = (-1)^n \delta S^\mathcal{F} - \pi_1^*(\alpha_1) + \pi_2^*(\alpha_2),$$

- Let $\mathcal{EL}^\mathcal{F}$ be the zero-locus of the vector field $Q^\mathcal{F}$, then $\mathcal{L}^\mathcal{F} = (\pi_1 \times \pi_2)(\mathcal{EL}^\mathcal{F}) \subset \overline{\mathcal{F}_1} \times \mathcal{F}_2$ should be Lagrangian. Here $\overline{\mathcal{F}_1}$ is the dg manifold $\mathcal{F}_1$ with symplectic form $-\omega_1$.

⁵We will always assume that this distribution is actually integrable, which is not automatic in the infinite dimensional case.

- For each Lagrangian submanifold $\mathcal{L}$ which is generic, relative to $\mathcal{L}^\mathcal{F}$ (the intersection is transversal), the preimage $(\pi_1 \times \pi_2)^{-1}(\mathcal{L}) \subset \mathcal{F}$ should be symplectic.

Composition of morphisms Let $\mathcal{F} : \mathcal{F}_1 \rightarrow \mathcal{F}_2$ and $\mathcal{F}' : \mathcal{F}_2 \rightarrow \mathcal{F}_3$ be two morphisms in BFV. The composition $\mathcal{F}' \cdot \mathcal{F}$ is the fiber product $\mathcal{F} \times_{\mathcal{F}_2} \mathcal{F}' = \{(x, x') \in \mathcal{F} \times \mathcal{F}' \mid \pi_2(x) = \pi_2(x')\}$. It is a submanifold in $\mathcal{F}' \times \mathcal{F}$.

The symplectic form on $\mathcal{F}' \cdot \mathcal{F}$ is the pullback of the symplectic form on $\mathcal{F}' \times \mathcal{F}$. The vector field $Q + Q'$ on $\mathcal{F}' \times \mathcal{F}$ is tangent to $\mathcal{F}' \cdot \mathcal{F} \subset \mathcal{F}' \times \mathcal{F}$ and it induced the vector field Q^{comp} on the fiber product. The action function is additive: $S^{\mathcal{F}' \cdot \mathcal{F}}(x, x') = S^\mathcal{F}(x) + S^{\mathcal{F}'}(x')$.

Let f be a mapping which assigns to each object $\mathcal{F}$ of BFV the functional $f^\mathcal{F}$ on $\mathcal{F}$. Define the mapping $F_f : \text{BFV} \rightarrow \text{BFV}$ as follows. It acts trivially on $\mathcal{F}$. It acts on one forms α_Σ as

$$\alpha^\mathcal{F} \rightarrow \alpha^\mathcal{F} + \delta f^\mathcal{F}$$

and does not change $Q^\mathcal{F}$. On the morphism $(\mathcal{F}, \omega^\mathcal{F}, Q^\mathcal{F}, S^\mathcal{F}) : \mathcal{F}_1 \rightarrow \mathcal{F}_2$ this mapping acts it as follows. It acts trivially on $\omega^\mathcal{F}$, on $Q^\mathcal{F}$, and on $\mathcal{F}$ while on the action $S^\mathcal{F}$ it acts as

$$S^\mathcal{F} \rightarrow S^\mathcal{F} + \pi_1^*(f^{\mathcal{F}_1}) - \pi_2^*(f^{\mathcal{F}_2})$$

It is easy to see that F is a covariant endofunctor for BFV.

Remark 4.1. *The notion of BV-BFV category introduced above has a natural generalization where the objects are quadruples $(\mathcal{F}, \mathcal{L}, \alpha, Q)$. Here $\mathcal{F}$ is a $\mathbb{Z}$ -graded manifold, $\mathcal{L}$ is a line bundle over it, α is a connection on $\mathcal{L}$ and Q is a cohomological degree 1 vector field on $\mathcal{F}$.*

A morphism between $(\mathcal{F}_1, \mathcal{L}_1, \alpha_1, Q_1)$ and $(\mathcal{F}_2, \mathcal{L}_2, \alpha_2, Q_2)$ is a $\mathbb{Z}$ -graded manifold with the same data as before but instead of $S^\mathcal{F}$ being a function on $\mathcal{F}$ we have a section of the line bundle $\mathcal{L}^\mathcal{F} = \pi^(\mathcal{L}_1 \times \mathcal{L}_2)$ over $\mathcal{F}$ which is horizontal for the flat connection $(-1)^{n-1} \frac{i}{\hbar} (\iota_{Q^\mathcal{F}} \omega^\mathcal{F} + \pi_1^*(\alpha_1) - \pi_2^*(\alpha_2))$ on $\mathcal{L}^\mathcal{F}$, cf. Remark 3.3.*

4.3. The BV functor. Now classical BV-BFV theories can be regarded as functors from spacetime categories to BFV category.

Fix a spacetime category. A classical BV-BFV field theory for a spacetime from this category defines the covariant functor, the *BV functor*, from the spacetime category to the BFV category.

On objects: The BV functor assigns the space of fields $\mathcal{F}_\Sigma$ to the object Σ of the spacetime category with the BFV data α_Σ and Q_Σ .

On morphisms: The BV functor assigns the space of fields $\mathcal{F}_N$ for a morphism $N : \Sigma_1 \rightarrow \Sigma_2$ with the BV data ω_N, Q_N, S_N , and mappings $\pi_i : \mathcal{F}_N \rightarrow \mathcal{F}_{\Sigma_i}$.

The properties of the BV-BFV field theories guarantee that this mapping is a covariant functor.

Remark 4.2. *The BFV category is 1-category. The corresponding BV functor is a functor from 1-category of cobordisms to the BFV category. This can be extended to higher categories. The natural target structure is a k -extended BFV category. It is a k -category. The k -extended classical BV field theory is the k -functor from the k -category of k -cobordisms to the k -extended BFV category, similar to k -extended topological quantum field theories [6, 39]. We will discuss extended theories in another publication.*

5. EXAMPLES OF BV-BFV THEORIES

5.1. Electrodynamics. Here we will consider the BV-extended classical Euclidean electrodynamics in the trivial $U(1)$ -bundle. Spacetime manifolds in Euclidean electrodynamics are smooth oriented n -dimensional Riemannian manifolds. By $*$: $\Omega^i(N) \rightarrow \Omega^{n-i}(N)$ we denote the Hodge operation induced by the metric on N .

5.1.1. The BV-BFV structure for classical electrodynamics. The space of fields in the BV-extended classical electrodynamics on the spacetime manifold N is $T^*[-1]E_N$. Here $E_N = \Omega^1(N) \oplus \Omega^{n-2}(N) \oplus \Omega^0(N)[1]$ where the first summand is the space of connections A in the trivial $U(1)$ bundle over N , the second summand is the space of fields B , the Hamiltonian counterpart (“momentum”) of A , the third summand is the space of ghost fields. The total space of fields in BV classical electrodynamics is

$$\mathcal{F}_N = \Omega^1(N) \oplus \Omega^{n-2}(N) \oplus \Omega^0(N)[1] \oplus \Omega^{n-1}(N)[-1] \oplus \Omega^2(N)[-1] \oplus \Omega^n(N)[-2]$$

We will use notations $A, B, c, A^\dagger, B^\dagger, c^\dagger$ for the fields from corresponding summands⁶. Here we regard $\Omega^k(N)$ as a vector space concentrated in degree zero.

The BV-symplectic form on the space of fields is the canonical symplectic form on $T^*[-1]E_N$:

$$\omega_N = \int_N (\delta A \wedge \delta A^\dagger + \delta B \wedge \delta B^\dagger + \delta c \wedge \delta c^\dagger)$$

It has degree -1 .

The BV-extended action of the classical electrodynamics on N has degree zero:

$$S_N = \int_N \left(B \wedge F(A) + \frac{1}{2} B \wedge *B + A^\dagger \wedge dc \right),$$

where $F(A) = dA$ is the curvature of the connection A . The vector field Q_N is

$$Q_N = \int_N \left(dc \wedge \frac{\delta}{\delta A} + dB \wedge \frac{\delta}{\delta A^\dagger} + (*B + dA) \wedge \frac{\delta}{\delta B^\dagger} + dA^\dagger \wedge \frac{\delta}{\delta c^\dagger} \right)$$

and it has $gh = 1$.

It acts on coordinate fields as

$$Q_N A = dc, \quad Q_N A^\dagger = dB, \quad Q_N B^\dagger = *B + dA, \quad Q_N c^\dagger = dA^\dagger$$

On other coordinate fields Q_N acts trivially. Here and below we are using the same notation for the vector field Q_N and for its Lie derivative.

The boundary BFV theory has the space of fields

$$\mathcal{F}_{\partial N} = \Omega^1(\partial N) \oplus \Omega^{n-2}(\partial N) \oplus \Omega^0(\partial N)[1] \oplus \Omega^{n-1}(\partial N)[-1]$$

We will denote corresponding fields by $A, B, c, A^\dagger$ respectively. The projection $\pi : \mathcal{F}_N \rightarrow \mathcal{F}_{\partial N}$ acts as

$$\pi(A) = i^*(A), \quad \pi(B) = i^*(B), \quad \pi(c) = i^*(c), \quad \pi(A^\dagger) = i^*(A^\dagger), \quad \pi(B^\dagger) = 0, \quad \pi(c^\dagger) = 0$$

The boundary symplectic form is the differential of the form

$$\begin{aligned} \alpha_{\partial N} &= \int_{\partial N} (B \wedge \delta A + A^\dagger \wedge \delta c) \\ \omega_{\partial N} &= \delta \alpha_{\partial N} = \int_{\partial N} (\delta B \wedge \delta A + \delta A^\dagger \wedge \delta c) \end{aligned}$$

⁶Here we discuss the *minimal* BV extension of Hamiltonian classical electrodynamics.

The boundary vector field $Q_{\partial N} = \delta\pi Q_N$ is

$$(30) \quad Q_{\partial N} = \int_{\partial N} \left(dB \wedge \frac{\delta}{\delta A^\dagger} + dc \wedge \frac{\delta}{\delta A} \right)$$

The boundary action is

$$S_{\partial N} = \int_{\partial N} c \wedge dB$$

Proposition 5.1. *The data described above satisfy the BV-BFV axioms.*

Proof. The only non-trivial computation is to check the classical master equation (7) when $\partial N \neq \emptyset$. Contracting the vector field Q_N with the symplectic form ω_N we obtain

$$\iota_{Q_N} \omega_N = \int_N (dc \wedge \delta A^\dagger + \delta A \wedge dB + \delta B \wedge (*B + dA) + dA^\dagger \wedge \delta c)$$

The differential of the action is easy to compute:

$$\delta S_N = \int_N (\delta B \wedge dA + B \wedge d\delta A + \delta B \wedge *B + \delta A^\dagger \wedge dc + A^\dagger \wedge d\delta c)$$

Comparing these two formulae and using the Stokes formula we obtain the classical master equation . $\square$

Remark 5.2. *The connection field A in electrodynamics is called the vector potential. When $n = 4$ and $N = [t_1, t_2] \times M$ is equipped with Minkowsky metric choose a basis e_0, e_1, e_2, e_3 in the tangent space where e_0 is a time direction. On EL_N the components B_{0i} , $i = 1, 2, 3$ give the magnetic field on M and the components B_{ij} give the electric field.*

5.1.2. *The Q -reduction of $\mathcal{EL}_N$.* The Euler-Lagrange equations in the bulk are:

$$dB = 0, \quad B = -*dA, \quad dA^\dagger = 0, \quad dc = 0$$

Note that this implies the usual Maxwell's equation for the vector potential $d*dA = 0$.

This defines the subspace $\mathcal{EL}_N \subset \mathcal{F}_N$:

$$(31) \quad \mathcal{EL}_N = \{(A, B) \in \Omega^1(N) \oplus \Omega^{n-2}(N) \mid d*dA = 0, B = -*dA\} \oplus \\ \oplus \Omega_{closed}^0(N)[1] \oplus \Omega_{closed}^{n-1}(N)[-1] \oplus \Omega^2(N)[-1] \oplus \Omega^n(N)[-2]$$

The summands correspond to fields $A, B, c, A^\dagger, B^\dagger, c^\dagger$ respectively.

Proposition 5.3. *The Q -reduced space of solutions to the Euler-Lagrange equations is*

$$(32) \quad \mathcal{EL}_N/Q \simeq \Omega_{Maxw}^1(N)/\Omega_{exact}^1(N) \oplus H^0(N)[1] \oplus H^{n-1}(N)[-1] \oplus H^n(N)[-2]$$

Here $\Omega_{Maxw}^1(N)$ is the space of solutions to Maxwell's equations, i.e. 1-forms A , such that $d*dA = 0$. The summands represent quotient spaces in fields $A, c, A^\dagger, c^\dagger$ respectively.

Proof. Let us first find the Q -reduced tangent space to $\mathcal{EL}_N$:

$$T_X \mathcal{EL}_N/Q = \ker(\hat{Q}_X)/\text{Im}(\hat{Q}_X)$$

Because $\mathcal{EL}_N$ is a vector space its tangent space, which is isomorphic to $\ker(\hat{Q}_X)$, is given by (31). The image of $\hat{Q}_X$ is easy to compute:

$$(33) \quad \text{Im}(\hat{Q}_X) = \Omega_{exact}^1(N) \oplus \{(A^\dagger, B^\dagger) \in \Omega^{n-1}(N)[-1] \oplus \Omega^2(N)[-1] \mid A^\dagger = d\beta, B^\dagger = *\beta + \text{exact}, \beta \in \Omega^{n-2}(N)\} \oplus \Omega_{exact}^n(N)[-2]$$

Here the components correspond to fields $A, A^\dagger, B^\dagger, c^\dagger$ respectively. Let us prove now that

$$(34) \quad \{(A^\dagger, B^\dagger) \in \Omega^{n-1}(N) \oplus \Omega^2(N) \mid A^\dagger = d\beta, B^\dagger = *\beta + \text{exact}, \beta \in \Omega^{n-2}(N)\} = \Omega_{exact}^{n-1}(N) \oplus \Omega^2(N)$$

By the Hodge-Morrey decomposition (see for example [14] and references therein) we can write $\Omega(N) = \Omega_{coclosed}(N) + \Omega_{exact}(N)$. Using this decomposition, for an exact $(n-1)$ -form $A^\dagger = d\gamma$ and an arbitrary 2-form $B^\dagger$ we can write $B^\dagger - *\gamma = *\theta + d\eta$ with θ closed. Then $A^\dagger = d\beta$, $B^\dagger = *\beta + d\eta$ for $\beta = \gamma + \theta$ and therefore the r.h.s. of (34) is the subspace of l.h.s. and since to opposite inclusion is obvious we proved (34).

Now we can write

$$\text{Im}(\hat{Q}_X) = \Omega_{exact}^1(N) \oplus \Omega_{exact}^{n-1}(N)[-1] \oplus \Omega^2(N)[-1] \oplus \Omega_{exact}^n(N)[-2]$$

Together with the formula (31) this proves the Proposition. $\square$

Proposition 5.4. *When $\partial N = \emptyset$*

$$\mathcal{EL}_N/Q = H^1(N) \oplus H^0(N)[1] \oplus H^{n-1}(N)[-1] \oplus H^n(N)[-2]$$

The summands correspond to fields $A, c, A^\dagger, c^\dagger$ respectively.

Proof. A 1-form A satisfies the Maxwell's equation $d * dA = 0$ if and only if $dA \in \ker(d^*)$. The metric on N gives the Hodge decomposition:

$$\Omega(N) = H(N) \oplus \Omega_{d-exact}(N) \oplus \Omega_{d^*-exact}(N)$$

where $H(N)$ are harmonic forms representing cohomology classes of N . It is clear from this decomposition that the first summand in (32) is $H^1(N)$. $\square$

When N does not have a boundary the space $\mathcal{EL}_N/Q$ described above has a natural symplectic structure given by the Poincaré pairing between $H^1(N)$ and $H^{n-1}(N)$ and between $H^0(N)$ and $H^n(N)$.

5.1.3. The reduction of boundary structures. Recall that the space of boundary fields is

$$\mathcal{F}_{\partial N} = \Omega^1(\partial N) \oplus \Omega^{n-2}(\partial N) \oplus \Omega^0(\partial N)[1] \oplus \Omega^{n-1}(\partial N)[-1]$$

Where the summands correspond to pullbacks of fields $A, B, c, A^\dagger$ respectively. The Euler-Lagrange equations on the boundary (equations for zeroes of the vector field $Q_{\partial N}$) are

$$dB = 0, \quad dc = 0$$

Thus, the space of solutions to boundary Euler-Lagrange equations is

$$\mathcal{EL}_{\partial N} = \Omega^1(\partial N) \oplus \Omega_{closed}^{n-2}(\partial N) \oplus \Omega_{closed}^0(\partial N)[1] \oplus \Omega^{n-1}(\partial N)[-1]$$

Because it is a vector space, it is isomorphic to its tangent space at every point.

Similarly to the discussion for the bulk, for $l \in \mathcal{EL}_{\partial N}$ we get

$$(35) \quad \text{Im}(\hat{Q}_l) = \Omega_{exact}^1(\partial N) \oplus \Omega_{exact}^{n-1}(\partial N)[-1] \subset T_l \mathcal{EL}_{\partial N}$$

Taking the quotient space $\ker(\hat{Q}_{\partial N})/\text{Im}(\hat{Q}_{\partial N})$ and identifying the tangent space with the space itself we prove the following.

Proposition 5.5. *The Q -reduced space of boundary fields is*

$$\mathcal{EL}(\partial N)/Q = \Omega^1(\partial N)/\{exact\} \oplus \Omega_{closed}^{n-2}(\partial N) \oplus H^0(\partial N)[1] \oplus H^{n-1}(\partial N)[-1]$$

Here summands correspond to fields $A, B, c, A^\dagger$ respectively.

This space is clearly infinite-dimensional. It coincides with the symplectic reduction of $\mathcal{EL}_{\partial N}$ with symplectic structure given by the natural pairing between the first and the second and between the third and the fourth summands.

Proposition 5.6. *BV-extended classical electrodynamics is a regular theory (in the sense of Definition 3.18).*

Proof. By direct calculation, we have

$$(36) \quad \ker(\hat{Q}_X) = \underbrace{\{(a, b) \in \Omega^1 \oplus \Omega^{n-2} \mid b = - * da, db = 0\}}_{\cong \Omega_{cl}^1 \oplus \Omega_{coex, cl}^{n-2}} \oplus \Omega_{cl}^0[1] \oplus \Omega_{cl}^{n-1}[-1] \oplus \Omega^2[-1] \oplus \Omega^n[-2],$$

$$(37) \quad \text{Im}(\hat{Q}_X) = \Omega_{ex}^1 \oplus \Omega_{ex}^{n-1}[-1] \oplus \Omega^2[-1] \oplus \Omega_{ex}^n[-2],$$

$$(38) \quad \ker(\hat{Q}_X^{\text{vert}}) = \Omega_{cl, D}^1 \oplus \Omega_{cl, D}^0[1] \oplus \Omega_{cl, D}^{n-1}[-1] \oplus \Omega^2[-1] \oplus \Omega^n[-2],$$

$$(39) \quad \text{Im}(\hat{Q}_X^{\text{vert}}) = \Omega_{ex, D}^1 \oplus \Omega_{ex, D}^n[-2] \oplus \underbrace{\{(db, da + *b) \in \Omega^{n-1} \oplus \Omega^2 \mid a \in \Omega_D^1, b \in \Omega_D^{n-2}\}}_{\cong \Omega_{ex, D}^{n-1} \oplus (\Omega_{ex, D}^2 \oplus \Omega_{cocl, N}^2)}[-1]$$

Here we use a shorthand notation with $ex, cl, coex, cocl, D, N$ standing for exact, closed, coexact, coclosed, Dirichlet, Neumann respectively (the last two indicating the imposed boundary condition; $\Omega_{ex, D}$ means exact, with the *primitive* being subject to Dirichlet condition). Hodge-Morrey decomposition theorem implies that subspaces (36) and (39) are mutually orthogonal in $T_X \mathcal{F}_N$, and (38) and (37) are mutually orthogonal too. Thus the assumption (20) holds.

Also,

$$\ker(\hat{Q}_l) = \Omega_{\partial}^1 \oplus \Omega_{cl, \partial}^{n-2} \oplus \Omega_{cl, \partial}^0[1] \oplus \Omega_{\partial}^{n-1}[-1]$$

and

$$\text{Im}(\hat{Q}_l) = \Omega_{ex, \partial}^1 \oplus \Omega_{ex, \partial}^{n-1}[-1]$$

(where ∂ stands for forms on the boundary) are mutually orthogonal in $T_l \mathcal{F}_{\partial N}$ due to Hodge decomposition on ∂N , thus the assumption (15) also holds. Therefore electrodynamics is a regular theory. $\square$

5.1.4. The Lagrangian subspace $\mathcal{L}_N$ and its reduction. Now let us describe the evolution relation $\mathcal{L}_N = \pi(\mathcal{EL}_N) \subset \mathcal{F}_{\partial N}$ and its reduction. Due to regularity, $\mathcal{L}_N$

is Lagrangian. This subspace consists of pullbacks of fields $A, B, c, A^\dagger$ satisfying Euler-Lagrange equation in N :

$$(40) \quad \mathcal{L}_N = \{(i^*(A), i^*(- * dA)) \in \Omega^1(\partial N) \oplus \Omega^{n-2}(\partial N) \mid d * dA = 0, A \in \Omega^1(N)\} \\ \oplus i^*(\Omega_{closed}^0(N))[1] \oplus i^*(\Omega_{closed}^{n-1}(N))[-1]$$

The Q -reduction of the Lagrangian subspace $\mathcal{L}_N$, or equivalently, its symplectic reduction is:

$$(41) \quad \mathcal{L}_N/Q = \{(i^*(A), i^*(- * dA)) \in \Omega^1(\partial N) \oplus \Omega^{n-2}(\partial N) \mid \\ d * dA = 0, A \in \Omega^1(N)\} / \{\text{exact } A\} \oplus \tilde{H}^0(\partial N)[1] \oplus \tilde{H}^{n-1}(\partial N)[-1]$$

Here $\tilde{H}^i(\partial N)$ are cohomology classes of ∂N which are pullbacks of cohomology classes on N .

5.1.5. *Gauge classes of solutions to Euler-Lagrange equations with fixed gauge classes of boundary values.* It is clear that the tangent space $T_X \mathcal{EL}(N, [l])$ depends neither on X nor on l and is

$$(42) \quad \{(A, - * dA) \in \Omega^1(\partial N) \oplus \Omega^{n-2}(\partial N) \mid d * dA = 0, i^*(A) \text{ exact}, A \in \Omega^1(N)\} \oplus \\ \oplus \Omega_{closed}^0(N, \partial N)[1] \oplus \{A^\dagger \in \Omega_{closed}^{n-1}(N)[-1] \mid i^*(A^\dagger) \text{ exact}\} \oplus \\ \oplus \Omega^2(N)[-1] \oplus \Omega^n(N)[-2]$$

For the quotient space we have:

$$(43) \quad T_X \mathcal{EL}(N, [l]) / \text{Im}(\hat{Q}_X) = \\ = \frac{H^1(N, \partial N)}{H^0(\partial N)} \oplus \frac{H^{n-1}(N, \partial N)}{H^{n-2}(\partial N)}[-1] \oplus H^0(N, \partial N)[1] \oplus H^n(N)[-2]$$

It is the Q -reduction of the tangent space $T_X \mathcal{EL}(N, [l])$. It is a finite dimensional symplectic space. Since we are in a linear case, $\mathcal{EL}(N, [l])/Q$ is isomorphic to (43).

5.1.6. *Codimension 2 BV structure.* Here we will describe the extension of the BV electrodynamics to codimension 2 strata. Let Σ be an $(n-1)$ -dimensional manifold with the boundary $\partial\Sigma$. The boundary $\partial\Sigma$ is closed and the space of boundary fields is

$$\mathcal{F}_{\partial\Sigma} = \Omega^{n-2}(\partial\Sigma) \oplus \Omega^0(\partial\Sigma)[1]$$

where the summands correspond to pullbacks of fields B and c from Σ to the boundary. We will denote the pullbacks of B and c by the same letters.

The restriction mapping $\pi : \mathcal{F}_\Sigma \rightarrow \mathcal{F}_{\partial\Sigma}$ acts as $\pi(A) = \pi(A^\dagger) = 0$ and $\pi(B) = B$, $\pi(c) = c$.

The symplectic structure on this space is exact

$$\omega_{\partial\Sigma} = \int_{\partial\Sigma} \delta B \wedge \delta c = \delta \alpha_{\partial\Sigma}$$

where

$$\alpha_{\partial\Sigma} = \int_{\partial\Sigma} \delta B \wedge c$$

The action and the vector field Q on $\mathcal{F}_{\partial\Sigma}$ are trivial:

$$S_{\partial\Sigma} = 0, \quad Q_{\partial\Sigma} = 0$$

Proposition 5.7. *The action $S_\Sigma = \int_\Sigma c \wedge dB$ satisfies the equation (7).*

The proof is a straightforward computation.

It is clear that

$$\mathcal{EL}_{\partial\Sigma} = \mathcal{F}_{\partial\Sigma},$$

The evolution relation $\mathcal{L}_\Sigma = \pi(\mathcal{EL}_\Sigma)$

$$\mathcal{L}_\Sigma = i^*(\Omega_{closed}^{n-2}(\Sigma)) \oplus i^*(\Omega_{closed}^0(\Sigma))[1]$$

When there is only one connected component of the boundary

$$\mathcal{L}_\Sigma = \Omega_{exact}^{n-2}(\partial\Sigma) \oplus \Omega_{closed}^0(\partial\Sigma)[1]$$

Because $Q_{\partial\Sigma} = 0$ the reduction is trivial: the reduced structures are the same as non-reduced ones. The reduced fiber over $l = (B, c)$ is

$$\mathcal{EL}(\Sigma, [l])/Q = \Omega^1(\Sigma)/\{exact\} \oplus \Omega_{closed}^{n-2}(\Sigma) \oplus H^0(\Sigma, \partial\Sigma)[1] \oplus H^{n-1}(\Sigma)[-1]$$

This space is infinite dimensional for $n > 2$ and it is finite dimensional when $n = 2$.

Remark 5.8. *The reduced space $\mathcal{L}_\Sigma/Q$ and fibers of $\pi : \mathcal{EL}_\Sigma/Q \rightarrow \mathcal{L}_\Sigma/Q$ are infinite dimensional when $n > 2$. When $n = 2$ they are finite dimensional. This corresponds to two dimensional electrodynamics which is an almost topological field theory [57].*

5.2. Yang-Mills theory. As in the classical Euclidean electrodynamics spacetime manifolds in the Yang-Mills theory are oriented compact smooth, possibly with boundary (and with corners), Riemannian manifolds. Let $\mathfrak{g}$ be the Lie algebra of a finite dimensional simply connected Lie group G with $\mathfrak{g}$ -invariant scalar product. To simplify notations we assume that $\mathfrak{g}$ is a matrix algebra and that the scalar product is given by the trace: $\langle a, b \rangle = \text{tr}(ab)$.

5.2.1. The non-reduced theory. In the first order formulation of Yang-Mills theory fields are connections A in a principal G -bundle P over N and $(n-2)$ -forms B with coefficients in the associated adjoint bundle. For simplicity, we assume that the principal bundle is trivial and consider connections as 1-forms with coefficients in $\mathfrak{g}$ and B fields as $(n-2)$ -forms with coefficients in $\mathfrak{g}$. The ghost fields c are 0-forms with coefficients in $\mathfrak{g}$. The BV extension includes anti-fields $A^\dagger, B^\dagger, c^\dagger$. The total space of BV extended Yang-Mills theory is⁷

$$\begin{aligned} \mathcal{F}_N = & \underbrace{\mathfrak{g} \otimes \Omega^1(N)}_A \oplus \underbrace{\mathfrak{g} \otimes \Omega^{n-2}(N)}_B \oplus \underbrace{\mathfrak{g} \otimes \Omega^0(N)[1]}_c \oplus \\ & \oplus \underbrace{\mathfrak{g} \otimes \Omega^{n-1}(N)[-1]}_{A^\dagger} \oplus \underbrace{\mathfrak{g} \otimes \Omega^2(N)[-1]}_{B^\dagger} \oplus \underbrace{\mathfrak{g} \otimes \Omega^n(N)[-2]}_{c^\dagger} \end{aligned}$$

This is a graded infinite dimensional vector space with the symplectic form

$$\omega_N = \int_N \text{tr} (\delta A \wedge \delta A^\dagger + \delta B \wedge \delta B^\dagger + \delta c \wedge \delta c^\dagger)$$

The action functional is

$$S_N = \int_N \text{tr} \left(B \wedge F_A + \frac{1}{2} B \wedge *B + A^\dagger \wedge d_A c + B^\dagger \wedge [B, c] + \frac{1}{2} c^\dagger \wedge [c, c] \right)$$

⁷Here, as in the case of classical electrodynamics we only discuss the minimal BV extension. When $n = 4$ the BV extension of Yang-Mills theory can also be presented in a different way using the decomposition of 2-forms into self-dual and anti-self-dual parts, see [16] and [22].

where $F_A = dA + \frac{1}{2}[A, A]$, and the cohomological vector field is

$$\begin{aligned} Q_N = \int_N \text{tr} \left(d_A c \wedge \frac{\delta}{\delta A} + [B, c] \wedge \frac{\delta}{\delta B} + \frac{1}{2}[c, c] \wedge \frac{\delta}{\delta c} + (d_A B + [A^\dagger, c]) \wedge \frac{\delta}{\delta A^\dagger} + \right. \\ \left. + (F_A + *B + [B^\dagger, c]) \wedge \frac{\delta}{\delta B^\dagger} + (d_A A^\dagger + [B, B^\dagger] + [c, c^\dagger]) \wedge \frac{\delta}{\delta c^\dagger} \right) \end{aligned}$$

The boundary structure. We will denote the pullback to the boundary of forms $A, B, A^\dagger, c$ by the same letters. The space of boundary fields is the quotient space of the pullback of $\mathcal{F}_N$ to the boundary over the kernel of the form $\delta\tilde{\alpha}_{\partial N}$, as it is explained in section 3.1:

$$\begin{aligned} \mathcal{F}_{\partial N} = & \underbrace{\mathfrak{g} \otimes \Omega^1(\partial N)[1]}_{A, \text{ gh}=0} \oplus \underbrace{\mathfrak{g} \otimes \Omega^{n-2}(\partial N)[n-2]}_{B, \text{ gh}=0} \oplus \\ & \oplus \underbrace{\mathfrak{g} \otimes \Omega^0(\partial N)[1]}_{c, \text{ gh}=1} \oplus \underbrace{\mathfrak{g} \otimes \Omega^{n-1}(\partial N)[n-2]}_{A^\dagger, \text{ gh}=-1} \end{aligned}$$

The structure of an exact symplectic manifold on $\mathcal{F}_{\partial N}$ is given by

$$\begin{aligned} \alpha_{\partial N} &= \int_{\partial N} \text{tr} (B \wedge \delta A + A^\dagger \wedge \delta c), \\ \omega_{\partial N} &= \int_{\partial N} \text{tr} (\delta B \wedge \delta A + \delta A^\dagger \wedge \delta c) \end{aligned}$$

The vector field $Q_{\partial N}$ and the action $S_{\partial N}$ for the boundary BFV theory are

$$\begin{aligned} Q_{\partial N} &= \int_{\partial N} \text{tr} \left(d_A c \wedge \frac{\delta}{\delta A} + [B, c] \wedge \frac{\delta}{\delta B} + \right. \\ &\quad \left. + (d_A B + [A^\dagger, c]) \wedge \frac{\delta}{\delta A^\dagger} + \frac{1}{2}[c, c] \wedge \frac{\delta}{\delta c} \right), \\ S_{\partial N} &= \int_{\partial N} \text{tr} \left(B \wedge d_A c + \frac{1}{2} A^\dagger \wedge [c, c] \right) \end{aligned}$$

The codimension 2 structure. Let Σ be a stratum of codimension 2. The BV-BFV theory on N and on ∂N induce the following data associated on Σ (see section 3.1. The space of fields:

$$\mathcal{F}_\Sigma = \underbrace{\mathfrak{g} \otimes \Omega^{n-2}(\Sigma)[n-2]}_{B, \text{ gh}=0} \oplus \underbrace{\mathfrak{g} \otimes \Omega^0(\Sigma)[1]}_{c, \text{ gh}=1},$$

Here we denoted the pullback of B and of c to Σ by the same letters. The rest of the BFV data, the exact symplectic form, the vector field Q and the action S (which can be obtained from Q by the Roytenberg's construction) are:

$$\begin{aligned} \alpha_\Sigma &= \int_\Sigma \text{tr} (B \wedge \delta c), \\ \omega_\Sigma &= \int_\Sigma \text{tr} (\delta B \wedge \delta c), \\ Q_\Sigma &= \int_\Sigma \text{tr} \left([B, c] \wedge \frac{\delta}{\delta B} + \frac{1}{2}[c, c] \wedge \frac{\delta}{\delta c} \right), \\ S_\Sigma &= \int_\Sigma \text{tr} \left(\frac{1}{2} B \wedge [c, c] \right) \end{aligned}$$

5.2.2. *The non-reduced $gh = 0$ part.*

The bulk. The fields A and its Hamiltonian counterpart remain, so the space of fields is

$$F_N = \underbrace{\mathfrak{g} \otimes \Omega^1(\partial N)}_A \oplus \underbrace{\mathfrak{g} \otimes \Omega^{n-2}(\partial N)}_B$$

The classical action is

$$S_N^{cl} = \int_N \text{tr} \left(B \wedge F_A + \frac{1}{2} B \wedge *B \right)$$

Its critical points are solutions to Euler-Lagrange equations

$$d_A B = 0, \quad F_A + *B = 0$$

or, equivalently

$$d_A * F_A = 0, \quad B = -* F_A$$

Infinitesimal gauge transformations are infinitesimal automorphisms of the trivial G -bundle over N , i.e. elements of the Lie algebra $\text{Map}(N, \mathfrak{g})$. They act as

$$A \mapsto A + d_A \alpha, \quad B \mapsto B + [B, \alpha]$$

where $\alpha \in \Omega^0(N, \mathfrak{g})$.

The boundary. Boundary fields in the Yang-Mills theory are pullbacks of fields A and B . We will denote them by the same letter:

$$F_{\partial N} = \underbrace{\mathfrak{g} \otimes \Omega^1(\partial N)}_A \oplus \underbrace{\mathfrak{g} \otimes \Omega^{n-2}(\partial N)}_B$$

The exact symplectic structure on $\mathcal{F}_{\partial N}$ is given by

$$\begin{aligned} \alpha_{\partial N}^{cl} &= \int_{\partial N} \text{tr}(B \wedge \delta A) \\ \omega_{\partial N}^{cl} &= \int_{\partial N} \text{tr}(\delta B \wedge \delta A) = \delta \alpha_{\partial N}^{cl} \end{aligned}$$

Euler-Lagrange subspace, $gh = 0$ part of $\mathcal{EL}_{\partial N}$, is defined by the constraint

$$d_A B = 0$$

Boundary gauge transformations are:

$$A \mapsto A + d_A \alpha, \quad B \mapsto B + [B, \alpha]$$

The action of the Lie algebra of gauge transformations is Hamiltonian with the moment map $\mu : F_{\partial N} \longrightarrow \mathfrak{g} \otimes \Omega^{n-1}(\partial N)$:

$$(A, B) \mapsto d_A B$$

Codimension 2 part. The $gh = 0$ part of the codimension 2 structure is given by the pullback of B to the boundary:

$$F_{\Sigma} = \mathfrak{g} \otimes \Omega^{n-2}(\Sigma)$$

with no constraints and the gauge transformations given by

$$B \mapsto B + [B, \alpha]$$

where $\alpha \in \Omega^0(\Sigma, \mathfrak{g})$.

5.2.3. *The reduction in the $gh = 0$ part.* The space EL_N is naturally isomorphic to the space of solutions to the Yang-Mills equation $d_A * F(A) = 0$.

The classical moduli space is naturally isomorphic to the space of gauge classes of solutions to the YM equation:

$$EL_N/G_N = \{(A, B) \mid A \in \mathfrak{g} \otimes \Omega^1(N), d_A * F_A = 0, B = -*F_A\} / \{(A, B) \sim (A + d_A \alpha, B + [B, \alpha])\}$$

The reduced phase space is naturally isomorphic to the cotangent bundle of the space of gauge classes of all connections:

$$EL_{\partial N}/G_{\partial N} = \{(A, B) \mid A \in \mathfrak{g} \otimes \Omega^1(\partial N), B \in \mathfrak{g} \otimes \Omega^{n-2}(\partial N), d_A B = 0\} / \{(A, B) \sim (A + d_A \alpha, B + [B, \alpha])\}$$

The $gh = 0$ part of the $\mathcal{EL}$ -moduli space for a codimension 2 stratum is simply

$$EL_{\Sigma}/G_{\Sigma} \cong (\mathfrak{g} \otimes \Omega^{n-2}(\Sigma))/G$$

where Σ is $(n-2)$ -dimensional and the quotient is by the adjoint action of G .

The restriction map

$$\pi_* : EL_N/G_N \rightarrow EL_{\partial N}/G_{\partial N}$$

is the pullback of the connection A and of the form B to the boundary. The fibers are finite-dimensional. One way to see this is by homological perturbation theory around electrodynamics.

The reduced phase space $EL_{\partial N}/G_{\partial N}$ is infinite-dimensional for $n \geq 3$. The case $n = 2$ is special as for $n = 2$ we have

$$EL_{\partial N}/G_{\partial N} \cong T^* \mathcal{M}_{\partial N}^G$$

where $\mathcal{M}_{\partial N}^G$ denotes the moduli space of flat G -connections on ∂N which is finite-dimensional.

Under the regularity assumption, the image of π_* is Lagrangian. Smooth loci of $\mathcal{EL}$ -moduli spaces for the bulk and the boundary are described by diagram (29) with horizontal arrows being equalities (we assume the group G to be simple for this to hold).

5.3. Scalar field.

5.3.1. *The non-reduced picture.* For the massless free n -dimensional scalar field the bulk BV data is:

$$\begin{aligned} \mathcal{F}_N &= \underbrace{\Omega^0(N)}_{\phi} \oplus \underbrace{\Omega^{n-1}(N)}_p \oplus \underbrace{\Omega^n(N)[-1]}_{\phi^\dagger} \oplus \underbrace{\Omega^1(N)[-1]}_{p^\dagger} \\ \omega_N &= \int_N (\delta\phi \wedge \delta\phi^\dagger + \delta p \wedge \delta p^\dagger) \\ S_N &= \int_N (p \wedge d\phi + \frac{1}{2} p \wedge *p) \\ Q_N &= \int_N \left(dp \wedge \frac{\delta}{\delta\phi^\dagger} + (d\phi + *p) \wedge \frac{\delta}{\delta p^\dagger} \right) \end{aligned}$$

The boundary BFV data in this theory is:

$$\begin{aligned}\mathcal{F}_{\partial N} &= \underbrace{\Omega^0(\partial N)}_{\phi} \oplus \underbrace{\Omega^{n-1}(\partial N)}_p \\ \alpha_{\partial N} &= \int_{\partial N} p \wedge \delta\phi \\ \omega_{\partial N} &= \delta\alpha_{\partial N} = \int_{\partial N} \delta p \wedge \delta\phi \\ Q_{\partial N} &= 0 \\ S_{\partial N} &= 0\end{aligned}$$

It is easy to derive them from the BV data in the bulk as it is explained in section 3.1. It is clear that the theory has length one.

For the Euler-Lagrange space we have

$$(44) \quad \mathcal{EL}_N = \{(\phi, p, \phi^\dagger, p^\dagger) \mid dp = 0, p = - * d\phi\}$$

At the boundary $\mathcal{EL}_{\partial N} = \mathcal{F}_{\partial N}$.

5.3.2. *The reduction.* Similarly to the case of electrodynamics we have,

$$\ker(\hat{Q})/Im(\hat{Q}) \simeq \Omega_{Harm}^0(N) \oplus H^n(N)[-1]$$

where $\Omega_{Harm}^0(N)$ are harmonic zero-forms on N ; the first summand correspond to the field ϕ and the second to $\phi^\dagger$. Because of the linearity, the moduli space $\mathcal{EL}_N/Q_N$ is given by the same formula.

Remark 5.9. *If $\partial N = \emptyset$ then $\mathcal{EL}_N/Q_N \simeq H^0(N) \oplus H^n(N)[-1]$ is a finite-dimensional odd-symplectic space.*

Because $Q_{\partial N} = 0$, the reduced boundary Euler-Lagrange space is the space of boundary fields:

$$(45) \quad \mathcal{EL}_{\partial N}/Q_{\partial N} = \mathcal{F}_{\partial N} = \Omega^0(\partial N) \oplus \Omega^{n-1}(\partial N)$$

The reduced evolution relation is the same as the non-reduced one:

$$\begin{aligned}(46) \quad \underline{\mathcal{L}}_N &= \\ &= \{(\phi, p) \in \mathcal{F}_{\partial N} \mid p = -(*d\tilde{\phi})|_{\partial N} \text{ for } \tilde{\phi} \text{ a harmonic continuation of } \phi \text{ into } N\} \simeq \\ &\simeq \Omega^0(\partial N)\end{aligned}$$

Because the harmonic continuation $\tilde{\phi}$ exists and is unique, the subspace $\underline{\mathcal{L}}_N$ is Lagrangian.

The restriction (the pullback to the boundary) mapping is surjective over $\underline{\mathcal{L}}_N$

$$(47) \quad \begin{array}{c} \mathcal{EL}_N/Q_N \\ \downarrow \\ \underline{\mathcal{L}}_N \subset \mathcal{EL}_{\partial N}/Q_{\partial N} \end{array}$$

Its fibers are finite-dimensional odd-symplectic spaces canonically isomorphic to

$$(48) \quad H^0(N, \partial N) \oplus H^n(N)[-1] \simeq T^*[-1]\mathbb{R}^k$$

Where $k \geq 0$ is the number of closed connected components of N .

5.3.3. *Massive scalar field.* The BV extension of the massive scalar field is similar to the massless scalar field. The space of fields and the symplectic structure are the same. The action and the vector field Q_N are

$$(49) \quad S_N = \int_N \left(p \wedge d\phi + \frac{1}{2} p \wedge *p - \frac{m^2}{2} \phi \wedge *\phi \right)$$

$$(50) \quad Q_N = \int_N \left((dp - m^2 * \phi) \wedge \frac{\partial}{\partial \phi^\dagger} + (d\phi + *p) \wedge \frac{\partial}{\partial p^\dagger} \right)$$

The boundary data are unchanged. The reduced bulk Euler-Lagrange space is

$$(51) \quad \mathcal{EL}_N / Q_N \simeq \Omega_{Klein-Gordon}^0(N)$$

Here $\Omega_{Klein-Gordon}^0(N)$ is the space of functions satisfying the equation $\Delta\phi - m^2\phi = 0$. It fibers over $\mathcal{L} \simeq \Omega^0(\partial N)$ with zero fibers. That is, when $m \neq 0$ the fibers of the $\mathcal{EL}$ -moduli spaces over given boundary values are trivial.

It is also easy to construct BV extensions of scalar fields interacting with the Yang-Mills theory, and to generalize it to Dirac and Majorana fermions.

5.4. Abelian BF theory.

5.4.1. *The BV-BFV structure of BF theory.* Here we will focus on the BV-BFV extension of the classical abelian BF gauge theory. Let us start with the description of the corresponding BV theory.

The space of fields in this theory is:

$$\mathcal{F}_N = \Omega^\bullet(N)[1] \oplus \Omega^\bullet(N)[n-2]$$

For coordinate fields we will write $\mathcal{A} \in \Omega^\bullet(N)[1]$ and $\mathcal{B} \in \Omega^\bullet(N)[n-2]$. Here $\Omega^\bullet(N)$ is regarded as a $\mathbb{Z}$ -graded vector space with $\Omega^k(N)$ being its component of degree $-k$.

Remark 5.10. *The BV-BF theory is an example of the AKSZ theory (cf. section 6) with the target manifold $T^*[1]\mathbb{R}$ and with $\Theta = 0$.*

Remark 5.11. *Dimensions $n = 2, 3$ are special. When $n = 2$ the theory is equivalent to the topological sector of electrodynamics. When $n = 3$ it is equivalent to two copies of abelian Chern-Simons theories.*

The BV symplectic form

$$\omega_N = \int_N \delta\mathcal{A} \wedge \delta\mathcal{B}$$

The total degree (ghost number) is $gh(\omega_N) = -1$.

The action functional is:

$$S_N = \int_N \mathcal{B} \wedge d\mathcal{A}$$

The vector field Q_N is

$$Q_N = \int_N \left(d\mathcal{A} \wedge \frac{\delta}{\delta\mathcal{A}} + d\mathcal{B} \wedge \frac{\delta}{\delta\mathcal{B}} \right)$$

On “coordinate functions” it acts as

$$(52) \quad Q_N \mathcal{A} = d\mathcal{A}, \quad Q_N \mathcal{B} = d\mathcal{B}$$

The space $\mathcal{EL}_N$ of solutions to Euler-Lagrange equations, i.e. the space of zeroes of the vector field Q_N , is the space of closed forms:

$$\mathcal{EL}_N = \Omega_{\text{closed}}^\bullet(N)[1] \oplus \Omega_{\text{closed}}^\bullet(N)[n-2]$$

Clearly this space is coisotropic in $\mathcal{F}_N$.

The space of boundary fields is:

$$\mathcal{F}_{\partial N} = \Omega^\bullet(\partial N)[1] \oplus \Omega^\bullet(\partial N)[n-2]$$

and the natural restriction map $\pi : \mathcal{F}_N \rightarrow \mathcal{F}_{\partial N}$ is the pullback of forms to the boundary.

The space of boundary fields is a symplectic space with the exact symplectic form of degree 0:

$$\omega_{\partial N} = \delta\alpha_{\partial N}, \quad \alpha_{\partial N} = \int_{\partial N} \mathcal{A} \wedge \delta\mathcal{B}$$

The boundary cohomological vector field $Q_{\partial N} = \delta\pi Q_N$ is

$$Q_{\partial N}\mathcal{A} = d\mathcal{A}, \quad Q_{\partial N}\mathcal{B} = d\mathcal{B}$$

The Euler-Lagrange subspace $\mathcal{EL}_{\partial N} \subset \mathcal{F}_{\partial N}$ is the subspace of closed forms on ∂N .

Proposition 5.12. *The data described above is an example of the BV-BFV theory.*

The proof is a straightforward computation which proves identities from section 3.1.

The evolution relation $\mathcal{L}_N \subset \mathcal{EL}_{\partial N}$ consists of closed forms on ∂N which extend to closed forms on N .

Proposition 5.13. *The subspace $\mathcal{L}_N \subset \mathcal{EL}_{\partial N} \subset \mathcal{F}_{\partial N}$ is Lagrangian.*

To prove this Proposition we need the following lemma.

Lemma 5.14. *Denote by $\iota_* : H_\bullet(\partial N) \rightarrow H_\bullet(N)$ the pushforward by $\iota : \partial N \hookrightarrow N$ in homology. The subspace $\ker(\iota_*)$ is isotropic in $H_\bullet(\partial N)$ with respect to the bilinear form given by the intersection pairing.*

Proof. Let U and V be two cycles in N relative to the boundary, with $\dim(U) + \dim(V) = \dim(N) + 1$, and let $u = \partial U$, $v = \partial V$ be their boundaries in ∂N . By general position argument we can assume that $U \cap V$ is a one dimensional chain. This chain gives a cobordism between $u \cap v$ and the empty set. Hence the sum of coefficients in $u \cap v$, which is the intersection pairing, vanishes. $\square$

Proof of Proposition 5.13. By the natural identification of $H_\bullet(\partial N)^*$ with $H^\bullet(\partial N)$, the annihilator of $\ker(\iota_*)$ is $\text{Im}(\iota^*)$. Therefore, by Lemma 5.14, the subspace $\text{Im}(\iota^*)$ is coisotropic in $H^\bullet(\partial N)$.

Because $\mathcal{L}_N \subset \mathcal{EL}_N$ is isotropic by Proposition 3.6, the reduced space $\mathcal{L}_N$ is also isotropic. Since we just proved that it is coisotropic, we conclude that it is Lagrangian. By Proposition A.1 the preimage of a Lagrangian subspace with respect to the symplectic reduction is Lagrangian if it contains the kernel of the presymplectic form. In our case this kernel consists of exact forms on ∂N , and exact forms are clearly in $\mathcal{L}_N$. $\square$

Proposition 5.15. *Abelian BF theory is regular.*

Proof. We have

$$(53) \quad \ker(\hat{Q}_X) = \Omega_{closed}^\bullet(N)[1] \oplus \Omega_{closed}^\bullet(N)[n-2],$$

$$(54) \quad \text{Im}(\hat{Q}_X) = \Omega_{exact}^\bullet(N)[1] \oplus \Omega_{exact}^\bullet(N)[n-2],$$

$$(55) \quad \ker(\hat{Q}_X^{\text{vert}}) = \Omega_{closed}^\bullet(N, \partial N)[1] \oplus \Omega_{closed}^\bullet(N, \partial N)[n-2],$$

$$(56) \quad \text{Im}(\hat{Q}_X^{\text{vert}}) = d(\Omega^\bullet(N, \partial N))[0] \oplus d(\Omega^\bullet(N, \partial N))[n-3],$$

$$(57) \quad \ker(\hat{Q}_l) = \Omega_{closed}^\bullet(\partial N)[1] \oplus \Omega_{closed}^\bullet(\partial N)[n-2],$$

$$(58) \quad \text{Im}(\hat{Q}_l) = \Omega_{exact}^\bullet(\partial N)[1] \oplus \Omega_{exact}^\bullet(\partial N)[n-1]$$

Due to Hodge-Morrey decomposition for forms on N , pairs of subspaces (53), (56) and (55), (54) are mutually orthogonal. Due to Hodge decomposition on ∂N , subspaces (57), (58) are also mutually orthogonal. Thus the theory is regular. $\square$

5.4.2. *The reduction of boundary structures.* From section 2.4 we know that the Q -reduced Euler-Lagrange space coincides with its symplectic reduction and is

$$\mathcal{EL}_{\partial N}/Q = \underline{\mathcal{EL}}_{\partial N} = H^\bullet(\partial N)[1] \oplus H^\bullet(\partial N)[n-2]$$

The reduced space $\mathcal{L}_N/Q$ is the space of cohomology classes of closed forms on ∂N which continue to closed forms on N , and we already proved that it is Lagrangian.

5.4.3. *The reduction of bulk fields.* The Q -reduced space of solutions to the EL equations is:

$$\mathcal{EL}_N/Q = H^\bullet(N)[1] \oplus H^\bullet(N)[n-2]$$

To compute the space $\mathcal{EL}(N, [l])/Q$, for $l \in \mathcal{L}_N$ consider the natural exact sequence:

$$0 \rightarrow \Omega^\bullet(N, \partial N) \rightarrow \Omega^\bullet(N) \xrightarrow{\iota^*} \Omega^\bullet(\partial N) \rightarrow 0$$

where ι^* is the pullback to the boundary corresponding to the inclusion mapping $\iota : \partial N \hookrightarrow N$, and $\Omega^\bullet(N, \partial N)$ are forms vanishing on the boundary (their pullback to the boundary is zero). It induces the standard long exact sequence for cohomology spaces:

$$\begin{aligned} \dots &\rightarrow H^\bullet(N, \partial N) \xrightarrow{\chi} H^\bullet(N) \xrightarrow{\iota^*} H^\bullet(\partial N) \xrightarrow{\beta} \\ &\rightarrow H^{\bullet+1}(N, \partial N) \xrightarrow{\chi} H^{\bullet+1}(N) \xrightarrow{\iota^*} H^{\bullet+1}(\partial N) \xrightarrow{\beta} \dots \end{aligned}$$

For $l \in \mathcal{L}_N$ the Q -reduction of the space $\mathcal{EL}(N, [l]) = \pi^{-1}([l]) \cap \mathcal{EL}_N$ is the Q -reduced space $\mathcal{EL}(N, [l])/Q = (\pi^{-1}([l]) \cap \mathcal{EL}_N)/Q$

$$(59) \quad \begin{aligned} \mathcal{EL}(N, [l])/Q &= \text{Im}(\chi)[1] \oplus \text{Im}(\chi)[n-2] \simeq \\ &\simeq H^\bullet(N, \partial N)/\text{Im}(\beta)[1] \oplus H^\bullet(N, \partial N)/\text{Im}(\beta)[n-2] \end{aligned}$$

Recall that $[l]$ is the leaf of Vect_Q through l . Due to regularity, (59) is symplectic, with the symplectic structure coming from the Lefschetz duality between $H^\bullet(N)$ and $H^\bullet(N, \partial N)$.

In terms of the long exact sequence of the pair $(N, \partial N)$, the reduced space $\mathcal{L}_N$ is:

$$\mathcal{L}_N/Q = \text{Im}(\iota^*)[1] \oplus \text{Im}(\iota^*)[n-2] \subset H^\bullet(\partial N)[1] \oplus H^\bullet(\partial N)[n-2]$$

Remark 5.16. *The symplectic reduction of $\mathcal{EL}_N$ is infinite dimensional and, as a vector space,*

$$\mathcal{EL}_N = H^\bullet(N, \partial N)[1] \oplus H^\bullet(N, \partial N)[n-2] \oplus \Omega_{closed}^\bullet(\partial N)[1] \oplus \Omega_{closed}^\bullet(\partial N)[n-2]$$

Indeed, it is easy to check that the symplectic orthogonal subspace $\mathcal{EL}_N^{\perp, symp}$ to $\mathcal{EL}_N$ is

$$\Omega_{exact}(N, \partial N)[1] \oplus \Omega_{exact}(N, \partial N)[n-2]$$

where $\Omega_{exact}(N, \partial N)$ is the space of exact forms with the pullback to the boundary being zero. The symplectic reduction is the quotient space $\mathcal{EL}_N / \mathcal{EL}_N^{\perp, symp}$. It is the symplectic $\mathcal{EL}$ -moduli space discussed in section 3.4 and comes with residual gauge symmetry data, which in particular allow for a simple gluing formula.

5.4.4. *BV extensions to strata of codimension k .* The BF theory can be maximally extended. On an $n-k$ dimensional stratum N_k the space of fields is

$$\mathcal{F}_{N_k} = \Omega^\bullet(N_k)[1] \oplus \Omega^\bullet(N_k)[n-2]$$

The symplectic form is

$$\omega_{N_k} = \int_{N_k} \delta \mathcal{A} \wedge \delta \mathcal{B}$$

The cohomological vector field Q_{N_k} is

$$Q_{N_k} = \int_{N_k} \left(d\mathcal{A} \wedge \frac{\delta}{\delta \mathcal{A}} + d\mathcal{B} \wedge \frac{\delta}{\delta \mathcal{B}} \right)$$

The action is

$$S_{N_k} = \int_{N_k} \mathcal{B} \wedge d\mathcal{A}$$

Formulae for α, Q, S are structurally the same for all k . This is a general feature of AKSZ theories of which abelian BF theory is an example.

5.4.5. *The $gh = 0$ part of the abelian BF theory.* In this section we will consider the restriction of the abelian BF theory to the fields with $gh = 0$.

The $gh = 0$ part of the space of fields is $F_N = \Omega^1(N) \oplus \Omega^{n-2}(N)$. The action is the restriction of the BV action to the $gh = 0$ sector:

$$S_N = \int_N B \wedge dA$$

where $A \in \Omega^1(N)$ and $B \in \Omega^{n-2}(N)$.

Euler-Lagrange equations are:

$$dA = 0, \quad dB = 0,$$

Solutions are closed forms

$$EL_N = \Omega_{closed}^1(N) \oplus \Omega_{closed}^{n-2}(N)$$

This is the degree zero part of the space $\mathcal{EL}_N$ in the BF -BV theory.

The gauge group $G_N = \Omega^0(N) \oplus \Omega^{n-3}(N)$ acts on fields as

$$A \mapsto A + d\alpha, \quad B \mapsto B + d\beta$$

Vector fields generated by these transformations are degree zero parts of vector fields $Vect_Q$. The action of the abelian BF theory is invariant with respect to these transformations.

The set of gauge classes of solutions is the degree zero part of the Q -reduced Euler-Lagrange space:

$$EL_N/G_N = H^1(N) \oplus H^{n-2}(N)$$

Non-reduced boundary theory. The boundary fields are:

$$F_{\partial N} = \Omega^1(\partial N) \oplus \Omega^{n-2}(\partial N)$$

This is a symplectic manifold with the symplectic form induced by the intersection pairing:

$$\omega_{\partial N} = \int_N \delta A \wedge \delta B$$

This form is restriction of the symplectic form in the BF theory to the $gh = 0$ subspace in the space of fields.

The boundary Euler-Lagrange subspace consists of closed 1- and $(n-2)$ -forms on ∂N :

$$EL_{\partial N} = \Omega_{closed}^1(\partial N) \oplus \Omega_{closed}^{n-2}(\partial N)$$

It is a coisotropic subspace of $F_{\partial N}$.

The evolution relation $L_N = \pi(EL_N)$ is the space of closed 1- and $(n-2)$ -forms on ∂N which continue to closed forms on N . As follows from Lemma 5.14, L_N is a Lagrangian subspace in $EL_{\partial N}$.

Reduced boundary theory. Boundary gauge group $G_{\partial N} = \Omega^0(\partial N) \oplus \Omega^{n-3}(\partial N)$ is the pullback of the group of gauge transformations on N to the boundary.

The symplectically reduced Euler-Lagrange subspace $\underline{EL}_{\partial N}$ is the set of leaves of the characteristic foliation of $EL_{\partial N}$ (the foliation of $EL_{\partial N}$ by Hamiltonian vector fields generated by the ideal $I_{EL_{\partial N}}$ of functions on $F_{\partial N}$ vanishing on $EL_{\partial N}$).

Remark 5.17. *Because the action of the gauge group is Hamiltonian with*

$$H_\alpha = \int_{\partial N} \alpha \wedge dB, \quad H_\beta = \int_{\partial N} \beta \wedge dA$$

The reduced space $\underline{EL}_{\partial N}$ is also the result of Hamiltonian reduction: $\overline{EL}_{\partial N} = J^{-1}(0)/G_{\partial N}$, where $J : F_{\partial N} \rightarrow [\Omega^0(\partial N) \oplus \Omega^{n-3}(\partial N)]^$ is the moment map $(A, B) \mapsto (H_\alpha, H_\beta)$. It is also clear that $EL_{\partial N} = J^{-1}(0)$.*

The Q -reduced Euler-Lagrange space

$$EL_{\partial N}/G_{\partial N} = H^1(\partial N) \oplus H^{n-2}(\partial N)$$

has the natural symplectic structure given by the Poincaré pairing.

The reduced evolution relation $L_N/G_{\partial N}$ consists of cohomology classes of closed forms on ∂N which continue to closed forms on N . In other words, this is the image of the restriction mapping:

$$H^1(N) \oplus H^{n-2}(N) \xrightarrow{\iota^* \oplus \iota^*} H^1(\partial N) \oplus H^{n-2}(\partial N)$$

where $\iota : \partial N \hookrightarrow N$ is the inclusion of the boundary mapping. As follows from the Proposition 5.13, it is a Lagrangian subspace.

The kernel of the restriction map is isomorphic to

$$H^1(N)/Im(\beta_0) \oplus H^{n-2}(N)/Im(\beta_{n-3})$$

Here $\beta_i : H^i(\partial N) \rightarrow H^{i+1}(N, \partial N)$ is the mapping in the long exact sequence of the pair $(N, \partial N)$.

6. THE AKSZ CONSTRUCTION OF CLASSICAL TOPOLOGICAL GAUGE THEORIES

In this section we will recall the construction of classical topological field theories for closed manifolds known as the AKSZ construction [1] and we will extend this construction to manifolds with boundary. We will show that this extension gives an example of a BV theory for spacetimes with boundary. The AKSZ construction generalizes the BV extension of the BF gauge theory.

6.1. The target manifold.

6.1.1. Hamiltonian dg manifolds. The AKSZ construction requires the choice of a Hamiltonian differential graded manifold as the target space.

Recall that a *differential graded (dg) manifold* is a pair $(\mathcal{M}, Q)$ where $\mathcal{M}$ is a graded manifold and Q is a cohomological vector field of degree one. A vector field is cohomological if its Lie derivative squares to zero.

A dg manifold is a *dg symplectic manifold* of degree m if it has a symplectic form ω of degree m which is Q -invariant (i.e. $L_Q\omega = 0$ where L_Q is the Lie derivative with respect to Q). We denote the degree by $\deg$.

Definition 6.1. A dg symplectic manifold $(\mathcal{M}, \omega, Q)$ of degree m is *Hamiltonian* if there exists an element $\Theta \in \text{Fun}(\mathcal{M})$ with $\deg(\Theta) = m + 1$ such that

$$(60) \quad \{\Theta, \Theta\} = 0$$

and Q is the Hamiltonian vector field of Θ .

Remark 6.2. A graded symplectic manifold is always exact when $\deg(\omega) \neq 0$: $\omega = d\alpha$. If $\deg(\omega) = 0$ we will require it to be exact. A dg symplectic manifold is automatically Hamiltonian when $\deg(\omega) \neq -1, -2$. See section 2.2 and [46].

Notice that a Hamiltonian dg manifold is actually defined by the symplectic form and by the function Θ satisfying (60). The vector field Q is defined as the Hamiltonian vector field of Θ and acts on functions as on functions on $\mathcal{M}$ as:

$$Qf = \{\Theta, f\}$$

In local coordinates

$$Qf = \sum_{ab} \Theta \overleftarrow{\frac{\partial}{\partial x^a}} \omega^{ab} \overrightarrow{\frac{\partial}{\partial x^b}} f$$

For polynomial functions f we have

$$\frac{df}{dt}(x + t\epsilon) = \sum_a \epsilon^a \overrightarrow{\frac{\partial}{\partial x^a}} f = \sum_a f \overleftarrow{\frac{\partial}{\partial x^a}} \epsilon^a$$

where $\deg(t) = 0$.

The condition (60) implies

$$Q^2 = 0$$

Because $\deg(\omega) = m$, we have $\deg(\omega^{-1}) = -m$ and because $\deg(\Theta) = m + 1$, we have $\deg(Q) = 1$.

6.1.2. *Examples of Hamiltonian dg manifolds.* Here we will give few examples of Hamiltonian dg manifolds.

Example 1. Let $m = 2$, $\mathfrak{g}$ be a finite dimensional Lie algebra with an invariant inner product, and x^a be coordinates on $\mathfrak{g}$ in an orthonormal linear basis e_a ⁸. Choose $\mathcal{M} = \mathfrak{g}[1]$ with $\deg(x^a) = 1$ and define

$$\omega = \frac{1}{2} \sum_a dx^a \wedge dx^a$$

and

$$\Theta = \frac{1}{6} \sum_{abc} f_{abc} x^a x^b x^c$$

where f_{abc} are structure constants of $\mathfrak{g}$ in the basis e_a . Clearly $\deg(\omega) = 2$, and $\deg(\Theta) = 3$, which agrees with $m = 2$.

The differential Q in this example is given by

$$Q = \frac{1}{2} \sum_{abc} f_{ab}^c x^a x^b \frac{\partial}{\partial x^c}$$

which corresponds to the Chevalley-Eilenberg differential for the Lie algebra $\mathfrak{g}$.

This example describes the target space for the Chern-Simons theory in the BV formalism.

Example 2 Let n be any integer and $\mathfrak{g}$ be a finite dimensional Lie algebra. Define

$$\mathcal{M} = \mathfrak{g}[1] \oplus \mathfrak{g}^*[n-2] = T^*[n-1](\mathfrak{g}[1])$$

Let x^a be coordinates on $\mathfrak{g}$, and p_a be corresponding coordinates on the dual space $\mathfrak{g}^*$, $gh(x^a) = 1$, $gh(p_a) = n-2$. Define

$$\omega = \sum_a dp_a \wedge dx^a, \quad \alpha = \sum_a p_a dx^a$$

$$\Theta = \frac{1}{2} \sum_{abc} f_{bc}^a p_a x^b x^c$$

It is clear that $\deg(\omega) = n-1$ and $\deg(\Theta) = n$.

This is the target space for BF models. When $n = 2$ this space is the same as in the next example (Poisson sigma model) for $M = g^*$ with the Kirillov-Kostant Poisson structure.

Example 3 Let $m = 1$ and $\mathcal{M} = T^*[1]M$, where M is a finite dimensional Poisson manifold with the Poisson tensor π . Let x^i be local coordinates on M , and p_i be corresponding coordinates on the cotangent space T_x^*M . The grading is such that $\deg(x^i) = 0$ and $\deg(p_i) = 1$

Define

$$\omega = \sum_i dp_i \wedge dx^i, \quad \alpha = \sum_i p_i dx^i$$

and

$$\Theta = \frac{1}{2} \sum_{ij} \pi^{ij}(x) p_i p_j$$

Condition (60) is equivalent to the fact that π is Poisson. Clearly $\deg(\omega) = 1$, and $\deg(\Theta) = 2$ which agrees with $m = 1$.

This example describes the target space in the Poisson sigma model.

⁸ $\mathfrak{g}$ can be also a Lie superalgebra

Example 4 When $m = 2$ and the Lie algebra has an invariant bilinear form, the Example 2 can be modified by adding a cubic term in p to Θ . For this we identify $\mathfrak{g}$ and $\mathfrak{g}^*$ using the Killing form and assume x^a, p_a are coordinates in an orthonormal basis. The new potential is

$$\Theta = \frac{1}{2} \sum_{abc} f_{bc}^a p_a x^b x^c \pm \frac{1}{6} f^{abc} p_a p_b p_c$$

This Hamiltonian dg manifold is isomorphic to two copies of the Hamiltonian dg manifold from the first example.

Example 5 When $m = 3$ and $\mathfrak{g}$ has an invariant bilinear form, we can add a quadratic term to Θ . The new potential is

$$\Theta = \frac{1}{2} \sum_{abc} f_{bc}^a p_a x^b x^c + \frac{1}{2} \sum_a p_a^2$$

6.2. The space of fields. Fix a Hamiltonian dg manifold $(\mathcal{M}, \omega, \Theta)$ of degree $n - 1$. The classical n -dimensional AKSZ field theory with the target manifold $\mathcal{M}$ on the spacetime manifold N has $\mathcal{F}_N = \text{Map}(T[1]N, \mathcal{M})$ as the space of fields⁹. We assume that the spacetime is a compact oriented smooth manifold. See Appendix B for basic facts on graded manifolds and their mapping spaces. As in the previous sections we will say that the grading on the space of fields is given by ghost numbers gh .

Let f be a smooth function of fixed degree on $\mathcal{M}$ and $X \in \mathcal{F}_N$, then the composition $X_f = f \circ X$ is a smooth function on $T[1]N$. Let ξ^i be local coordinates on the fiber $T_u[1]N$, $u \in N$. Then X_f at (ξ, u) can be written as

$$X_f(\xi, u) = \sum_{k=0}^n X_f(u)_{i_1, \dots, i_k} \xi^{i_1} \dots \xi^{i_k}$$

with $gh(X_f(u)_{i_1, \dots, i_k}) = \deg(f) - k$.

We have natural identification $\phi : C^\infty(T[1]N) \simeq \Omega(N)$. In coordinates this mapping brings the function X_f to the form

$$\phi(X_f)(u) = \sum_{k=0}^n X_f(u)_{i_1, \dots, i_k} du^{i_1} \wedge \dots \wedge du^{i_k}$$

Let x^a , $a = 1, \dots, n$ be homogeneous local coordinates on $\mathcal{M}$. Denote by X^a the composition of the field $X \in \mathcal{F}_N$ and the coordinate function x^a . Component fields X^a can be regarded as forms $\phi(X^a)(u)$ on N , $a = 1, \dots, n$:

$$(61) \quad \phi(X^a)(u) = \sum_{k=0}^d X^a(u)_{i_1, \dots, i_k} du^{i_1} \wedge \dots \wedge du^{i_k}$$

We will denote the component of degree k by $X_k^a(u)$. We will also call forms $X^a(u)$ coordinate fields, or superfields, when it will not cause a confusion.

As it follows from the definition of $X^a(u)_{i_1, \dots, i_k}$ the ghost number of this field is $gh(X_k^a) = \deg(x^a) - k$.

⁹The AKSZ construction does not have to have $T[1]N$ as the source graded manifold. However we will consider only these cases, as they seem to be more important in field theory

6.3. The non-reduced AKSZ theory for spacetime manifolds with boundary.

6.3.1. *The AKSZ theory in terms of coordinate fields.* The AKSZ action consists of “kinetic” and “interaction” parts:

$$S_N[X] = S_N^{kin}[X] + S_N^{int}[X]$$

Let x^a be local coordinates on $\mathcal{M}$ and $\alpha(x) = \sum_a \alpha_a(x) dx^a$. In local coordinates the “kinetic” part of the AKSZ action is

$$S_N^{kin}[X] = \int_N \sum_a \alpha_a(X(u)) \wedge dX^a(u)$$

Here and below $X^a(u)$ are coordinate components of fields as in (61). It is easy to see that $gh(S_N^{kin}) = 0$.

The “interaction” part of the AKSZ action is the functional

$$S_N^{int}[X] = \int_N \Theta(X)$$

where Θ is the potential function for the target manifold. Because $deg(\Theta) = n$, the action has the zero grading, i.e. $gh(S_N^{int}) = 0$. Here and below the expression $\int_N \Theta(X)$ for a homogeneous polynomial $\Theta(x) = \sum_{\{a\}} \Theta_{\{a\}} x^{a_1} \dots x^{a_k}$ of degree k on $\mathcal{M}$ means $\sum_{\{a\}} \Theta_{\{a\}} \int_N X^{a_1} \wedge \dots \wedge X^{a_k}$.

Assume that in local coordinates $\omega_{\mathcal{M}} = \frac{1}{2} \sum_{ab} \omega_{ab} dx^a \wedge dx^b$, then

$$\omega_N = \frac{1}{2} \int_N \sum_{ab} \omega_{ab}(X(u)) \wedge \delta X^a(u) \wedge \delta X^b(u)$$

where $\delta X^a(x)$ is the de Rham differential on the space of fields.

In local coordinates, the action of Q_N on local functionals is:

$$Q_N F = \int_N \left(dX^a + \omega^{ab}(X) \wedge \frac{\partial \Theta}{\partial x^b}(X) \right) \wedge \frac{\delta F}{\delta X^a}$$

Here ω^{ab} are components of the Poisson bivector field corresponding to the symplectic form ω . Because $deg(\omega) = n - 1$ and $deg(\Theta) = n$, we have $gh(Q_N) = 1$.

It acts on coordinate fields as:

$$(62) \quad Q_N X^a = dX^a + \omega^{ab}(X) \wedge \frac{\partial \Theta}{\partial x^b}(X)$$

The variation of S_N :

$$\delta S_N[X] = \int_N (\delta X^a(u) \wedge \partial_a \alpha_b(X(u)) \wedge dX^b(u) + \alpha_a(X(u)) \wedge \delta dX^a(u) + \delta \Theta(X))$$

After integration by parts this expression becomes

$$\int_{\partial N} \alpha_a(X(u)) \wedge \delta X^a(u) + \int_N \left(\omega_{ab}(X(u)) \wedge dX^a(u) + \frac{\partial \Theta}{\partial X^b}(X(u)) \right) \wedge \delta X^b(u)$$

In this section we assume that $\partial N = \emptyset$, which means the first term is absent. The Euler-Lagrange equations are:

$$(63) \quad dX^a(u) + \sum_b \omega^{ab}(X(u)) \wedge \frac{\partial \Theta}{\partial x^a}(X(u)) = 0$$

The same formulae hold for the boundary BFM action $S_{\partial N}$, for the boundary symplectic form $\omega_{\partial N}$, and for the boundary cohomological vector field $Q_{\partial N}$. One

should simply substitute ∂N instead of N . The only difference is the degree change. Because $\dim(\partial N) = n - 1$, $gh(S_{\partial N}) = 1$, $gh(\omega_{\partial N}) = -1$ and $gh(Q_{\partial N}) = -1$ as in the bulk N .

The BV and BFV data described above have the right grading and give an example of a BV-BFV theory:

Proposition 6.3. *The cohomological vector field Q_N is Hamiltonian, up to a boundary term:*

$$(64) \quad \iota_{Q_N} \omega_N = (-1)^{\dim(N)} \delta S_N + \pi^* \alpha_{\partial N}$$

We defer the proof to Appendix D.

Remark 6.4. *When N has a non-empty boundary, the AKSZ action depends of the choice of the form α , not only on its cohomology class as in case of closed manifolds. Let $\tilde{S}_N$ be the action corresponding to $\tilde{\alpha} = \alpha + df$, then*

$$\tilde{S}_N[X] = S_N[X] + \int_N df(X) = S_N + \int_{\partial N} f(X)$$

Any AKSZ theory can be maximally extended. On an $n - k$ dimensional stratum Σ the space of fields is the space of maps $\mathcal{F}_\Sigma = \text{Map}(T[1]\Sigma, \mathcal{M})$. The symplectic form ω_Σ , the form α_Σ , the action functional S_Σ and the vector field Q_Σ are all given by the same formulae as above. The difference from the n -dimensional stratum is only in the grading:

$$gh(\omega_\Sigma) = k - 1, \quad gh(Q_\Sigma) = 1, \quad gh(S_\Sigma) = k$$

7. EXAMPLES OF AKSZ THEORIES

7.1. Abelian Chern-Simons theory. The target space for this AKSZ theory is $\mathbb{R}[1]$ with symplectic structure $\omega = da \wedge da$, where a is the coordinate on $\mathbb{R}[1]$ and $\Theta = 0$.

7.1.1. The bulk BV theory. The space of fields on the 3-dimensional spacetime manifold N is :

$$\mathcal{F}_N = \Omega^\bullet(N)[1]$$

The fields corresponding to forms of degree 0, 1, 2, 3 will be denoted by $c, A, A^\dagger, c^\dagger$ respectively. The ghost numbers are 1, 0, -1, -2. We will write $\mathcal{A} = c + A + A^\dagger + c^\dagger$ for the BV superfield.

The symplectic form, the vector field Q , and the classical action are:

$$\begin{aligned} \omega_N &= \frac{1}{2} \int_N \delta \mathcal{A} \wedge \delta \mathcal{A} = \int_N (\delta c \wedge \delta c^\dagger + \delta A \wedge \delta A^\dagger) \\ Q_N &= \int_N d\mathcal{A} \wedge \frac{\delta}{\delta \mathcal{A}} = \int_N (dc \wedge \frac{\delta}{\delta A} + dA \wedge \frac{\delta}{\delta A^\dagger} + dA^\dagger \wedge \frac{\delta}{\delta c^\dagger}) \\ S_N &= \frac{1}{2} \int_N \mathcal{A} \wedge d\mathcal{A} = \frac{1}{2} \int_N (A \wedge dA + A^\dagger \wedge dc + c \wedge dA^\dagger) \end{aligned}$$

The Euler-Lagrange space is

$$\mathcal{EL}_N = \Omega_{closed}^\bullet(N)[1]$$

7.1.2. *The boundary BFV theory.* Boundary fields are pullbacks of the bulk fields to the boundary.

$$\mathcal{F}_{\partial N} = \Omega^\bullet(N)[1]$$

We will use the same notation for pullbacks as for bulk fields. This means 0, 1, 2 forms will be denoted by $c, A, A^\dagger$ respectively. They have ghost numbers 1, 0, -1 .

The one form $\alpha_{\partial N}$, the symplectic structure, the vector field Q and the action for the boundary BFV theory are:

$$\begin{aligned}\alpha_{\partial N} &= \frac{1}{2} \int_{\partial N} \mathcal{A} \wedge \delta \mathcal{A} = \frac{1}{2} \int_{\partial N} (A \wedge \delta A + c \wedge \delta A^\dagger + A^\dagger \wedge \delta c) \\ \omega_{\partial N} &= \frac{1}{2} \int_{\partial N} \delta \mathcal{A} \wedge \delta \mathcal{A} = \int_{\partial N} \left(\frac{1}{2} \delta A \wedge \delta A + \delta c \wedge \delta A^\dagger \right) \\ Q_{\partial N} &= \int_{\partial N} d\mathcal{A} \wedge \frac{\delta}{\delta \mathcal{A}} = \int_{\partial N} \left(dc \wedge \frac{\delta}{\delta A} + dA \wedge \frac{\delta}{\delta A^\dagger} \right) \\ S_{\partial N} &= \frac{1}{2} \int_{\partial N} \mathcal{A} \wedge \delta \mathcal{A} = \int_{\partial N} c \wedge dA\end{aligned}$$

The boundary Euler-Lagrange space is

$$\mathcal{EL}_{\partial N} = \Omega_{closed}^\bullet(\partial N)[1]$$

The evolution relation $\mathcal{L}_N \subset \mathcal{EL}_{\partial N}$ is the subspace of forms in $\Omega_{closed}^\bullet(\partial N)[1]$ which continue to closed forms on N .

Abelian Chern-Simons theory is regular, which is proven similarly to Proposition 5.15.

7.1.3. *Reduced BV-BFV theory.* Reduced bulk and boundary Euler-Lagrange spaces are

$$\mathcal{EL}_N/Q \simeq H^\bullet(N)[1], \quad \mathcal{EL}_{\partial N}/Q \simeq H^\bullet(\partial N)[1]$$

respectively. The symplectic form on $\mathcal{EL}_{\partial N}/Q$ is given by the Poincaré duality.

The pullback mapping $\pi : \mathcal{F}_N \rightarrow \mathcal{F}_{\partial N}$ induces the mapping of reduced spaces:

$$\pi_* : \mathcal{EL}_N/Q \rightarrow \mathcal{EL}_{\partial N}/Q$$

The reduced evolution relation $\mathcal{L}_N/Q = \text{Im}(\pi_*) \subset \mathcal{EL}_{\partial N}/Q$ is a Lagrangian subspace.

Because π_* is a linear mapping, its fiber over any point of $\mathcal{EL}_{\partial N}/Q$ is simply $\ker(\pi_*)$. This space

$$\ker(\pi_*) \simeq H^\bullet(N, \partial N)/H^{\bullet-1}(\partial N)[1]$$

has a natural symplectic structure of degree -1 coming from the Lefschetz duality.

7.1.4. *The $gh = 0$ part of the theory.* The $gh = 0$ part of the space of fields is $F_N = \Omega^1(N)$. The $gh = 0$ part of Vect_Q gives the gauge action of $\Omega^0(N)$ on the space of fields: $A \rightarrow A + d\beta$ where $A \in \mathcal{F}_N$ and $\beta \in \Omega^0(N)$. The abelian Chern-Simons action

$$S_N^{cl} = \frac{1}{2} \int_N A \wedge dA$$

is gauge invariant when $\partial N = \emptyset$. When N has non-empty boundary the gauge transformation generated by β changes the action by $\int_{\partial N} \beta \wedge A$.

The space EL_N is the space of closed 1-forms on N . The space of boundary fields $F_{\partial N} = \Omega^1(\partial N)$ is exact symplectic with

$$\alpha_{\partial N} = \frac{1}{2} \int_{\partial N} A \wedge \delta A, \quad \omega_{\partial N} = \delta \alpha_{\partial N} = \frac{1}{2} \int_{\partial N} \delta A \wedge \delta A$$

The $gh = 0$ part of the boundary Euler-Lagrange space, the space $C_{\partial N}$, is the space of closed 1-forms on ∂N . It is clearly coisotropic. The gauge action is Hamiltonian with the momentum map $\mu = d : \Omega^1(\partial N) \rightarrow \Omega^2(\partial N)$, where we consider $\Omega^2(N)$ as the dual space to the abelian Lie algebra of 0-forms.

The $gh = 0$ part of the moduli space $\mathcal{EL}_N/Q$ is the space of gauge orbits EL_N/G_N , and we have the natural isomorphism $EL_N/G_N \simeq H^1(N)$. The $gh = 0$ part of the moduli space $\mathcal{EL}_{\partial N}/Q$ is isomorphic to the space $C_{\partial N}/G_{\partial N}$ of gauge orbits through $C_{\partial N}$, or equivalently, since the action is Hamiltonian, this space is the symplectic reduction of $C_{\partial N}$. It is clear that we have the natural isomorphism $C_{\partial N}/G_{\partial N} \simeq H^1(\partial N)$. This space is symplectic with the symplectic structure given by the Poincaré duality.

The pullback to the boundary $\pi : F_N \rightarrow F_{\partial N}$ induces the restriction mapping $\pi_* : H^1(N) \rightarrow H^1(\partial N)$. The subspace

$$L_N = \text{Im}(\pi_*) \subset H^1(\partial N)$$

is Lagrangian. The fiber over any point of $C_{\partial N}/G_{\partial N}$ is $\ker(\pi_*)$.

7.2. Non-abelian Chern-Simons theory. In this case the target manifold is constructed from a Lie algebra $\mathfrak{g}$ with an invariant scalar product (for example a simple Lie algebra). The target manifold is described in details in section 6.1.2. We assume that the Lie algebra and the corresponding simply connected Lie group are matrix groups and will write $\text{tr}(ab)$ for the Killing form evaluated on two Lie algebra elements.

7.2.1. The bulk BV theory. Fields are graded connections in a principal G -bundle over the spacetime N . We assume the bundle is trivial, so the space of fields is

$$\mathcal{F}_N = \Omega^\bullet(N, \mathfrak{g})[1]$$

Here $\Omega^\bullet(N, \mathfrak{g})[1] = \Omega^\bullet(N) \otimes \mathfrak{g}[1]$. The fields corresponding to forms of degree 0, 1, 2, 3 will be denoted by $c, A, A^\dagger, c^\dagger$ respectively. The ghost numbers are 1, 0, -1 , -2 . We will write $\mathcal{A} = c + A + A^\dagger + c^\dagger$ for the BV superfield.

The symplectic form, the vector field Q , and the classical action are:

$$\omega_N = \frac{1}{2} \int_N \text{tr}(\delta \mathcal{A} \wedge \delta \mathcal{A}) = \int_N \text{tr}(\delta c \wedge \delta c^\dagger + \delta A \wedge \delta A^\dagger),$$

$$\begin{aligned} (65) \quad Q_N &= \int_N \text{tr} \left((d\mathcal{A} + \frac{1}{2}[\mathcal{A}, \mathcal{A}]) \wedge \frac{\delta}{\delta \mathcal{A}} \right) = \\ &= \int_N \text{tr} \left(d_A c \wedge \frac{\delta}{\delta A} + (F(A) + [c, A^\dagger]) \wedge \frac{\delta}{\delta A^\dagger} + (d_A A^\dagger + [c, c^\dagger]) \wedge \frac{\delta}{\delta c^\dagger} + \frac{1}{2}[c, c] \wedge \frac{\delta}{\delta c} \right), \end{aligned}$$

$$\begin{aligned} (66) \quad S_N &= \int_N \text{tr} \left(\frac{1}{2} \mathcal{A} \wedge d\mathcal{A} + \frac{1}{6} \mathcal{A} \wedge [\mathcal{A}, \mathcal{A}] \right) = \\ &= \int_N \text{tr} \left(\frac{1}{2} A \wedge dA + \frac{1}{6} A \wedge [A, A] + \frac{1}{2} A^\dagger \wedge d_A c + \frac{1}{2} c \wedge d_A A^\dagger + \frac{1}{2} c^\dagger \wedge [c, c] \right) \end{aligned}$$

The Euler-Lagrange space is the space of flat graded connections

$$\mathcal{EL}_N = \{\mathcal{A} \in \mathcal{F}_N \mid d\mathcal{A} + \frac{1}{2}[\mathcal{A}, \mathcal{A}] = 0\}$$

In coordinate fields the Euler-Lagrange space consists of $c, A, A^\dagger, c^\dagger$ which satisfy

$$[c, c] = 0, \quad d_A c = 0, \quad F(A) + [c, A^\dagger] = 0, \quad d_A A^\dagger + [c, c^\dagger] = 0$$

The tangent space at $\mathcal{A} \in \mathcal{EL}_N$ is kernel of $\hat{Q}_{\mathcal{A}}$. It consists of elements $(\gamma, \alpha, \alpha^\dagger, \gamma^\dagger) \in \Omega^\bullet(N)[1]$ such that

$$(67) \quad \begin{aligned} [\gamma, c] = 0, \quad d_A \gamma + [\alpha, c] = 0, \quad d_A \alpha + [c, \alpha^\dagger] + [\gamma, A^\dagger] = 0, \\ d_A \alpha^\dagger + [c, \gamma^\dagger] + [A^\dagger, \alpha] + [c^\dagger, \gamma] = 0, \end{aligned}$$

The point $\mathcal{A}$ belongs to the $gh = 0$ part of $\mathcal{EL}_N$ if and only if $c = c^\dagger = A^\dagger = 0$ and $F(A) = 0$. The $gh = 0$ part of $\mathcal{EL}_N$ is the space flat connections. The tangent space to $\mathcal{EL}_N$ at such point is naturally isomorphic to

$$(68) \quad \Omega_{d_A - \text{closed}}^\bullet(N, \mathfrak{g})$$

where forms are closed with respect to the differential $d_A = d + [A, \cdot]$.

In this case the reduced tangent space is

$$\ker(\hat{Q}_{\mathcal{A}})/\text{Im}(\hat{Q}_{\mathcal{A}}) \simeq H_{d_A}^\bullet(N, \mathfrak{g})$$

7.2.2. The boundary BFV theory. Boundary fields in the non-abelian Chern-Simons theory are pullbacks of the bulk fields to the boundary.

$$\mathcal{F}_{\partial N} = \Omega^\bullet(N, \mathfrak{g})[1]$$

We will use the same notation for pullbacks as for bulk fields. This means 0, 1, 2 forms will be denoted by $c, A, A^\dagger$ respectively. They have ghost numbers 1, 0, -1.

The one form $\alpha_{\partial N}$, the symplectic structure, the vector field Q and the action for the boundary BFV theory are:

$$\alpha_{\partial N} = \frac{1}{2} \int_{\partial N} \text{tr}(\mathcal{A} \wedge \delta \mathcal{A}) = \frac{1}{2} \int_{\partial N} \text{tr}(A \wedge \delta A + c \wedge \delta A^\dagger + A^\dagger \wedge \delta c),$$

$$\omega_{\partial N} = \frac{1}{2} \int_{\partial N} \text{tr}(\delta \mathcal{A} \wedge \delta \mathcal{A}) = \int_{\partial N} \text{tr} \left(\frac{1}{2} \delta A \wedge \delta A + \delta c \wedge \delta A^\dagger \right),$$

$$\begin{aligned} Q_{\partial N} &= \int_{\partial N} \text{tr} \left(\left(d\mathcal{A} + \frac{1}{2}[\mathcal{A}, \mathcal{A}] \right) \wedge \frac{\delta}{\delta \mathcal{A}} \right) = \\ &= \int_{\partial N} \text{tr} \left(d_A c \wedge \frac{\delta}{\delta A} + (F(A) + [c, A^\dagger]) \wedge \frac{\delta}{\delta A^\dagger} + \frac{1}{2}[c, c] \wedge \frac{\delta}{\delta c} \right), \end{aligned}$$

$$S_{\partial N} = \int_{\partial N} \text{tr} \left(\frac{1}{2} \mathcal{A} \wedge d\mathcal{A} + \frac{1}{6} \mathcal{A} \wedge [\mathcal{A}, \mathcal{A}] \right) = \int_{\partial N} \text{tr} \left(c \wedge F(A) + \frac{1}{2}[c, c] \wedge A^\dagger \right)$$

The boundary Euler-Lagrange space is the space of graded flat $\mathfrak{g}$ -connections on ∂N :

$$\mathcal{EL}_{\partial N} = \{\mathcal{A} \in \mathcal{F}_{\partial N} \mid d\mathcal{A} + \frac{1}{2}[\mathcal{A}, \mathcal{A}] = 0\}$$

The evolution relation $\mathcal{L}_N \subset \mathcal{EL}_{\partial N}$ is the subspace of graded flat $\mathfrak{g}$ -connections on $G \times \partial N$ which continue to flat graded $\mathfrak{g}$ -connections on $G \times N$.

The flatness of the graded connection $\mathcal{A}$ means its graded components satisfy

$$[c, c] = 0, \quad d_A c = 0, \quad F(A) + [c, A^\dagger] = 0,$$

The tangent space $T_{\mathcal{A}}\mathcal{EL}_{\partial N} \subset T_{\mathcal{A}}\mathcal{F}_{\partial N}$ is the kernel of $\hat{Q}_{\mathcal{A}}$:

$$(69) \quad \ker(\hat{Q}_{\mathcal{A}}) = \{(\gamma, \alpha, \alpha^\dagger) | [\gamma, c] = 0, \quad d_A \gamma + [\alpha, c] = 0, \quad d_A \alpha + [c, \alpha^\dagger] + [\gamma, A^\dagger] = 0\}$$

In $gh = 0$ part of $\mathcal{EL}_{\partial N}$ we have $c = A^\dagger = 0$ and $F(A) = 0$

$$(70) \quad \ker(\hat{Q}_{\mathcal{A}}) = \{(\gamma, \alpha, \alpha^\dagger) | \quad d_A \gamma = 0, \quad d_A \alpha = 0\} \simeq \Omega_{d_A-closed}^\bullet(\partial N, \mathfrak{g})[1]$$

$$Im(\hat{Q}_{\mathcal{A}}) = \{(0, d_A \tilde{\gamma}, d_A \tilde{\alpha})\}$$

Thus, in this case the reduced tangent space is

$$\ker(\hat{Q}_{\mathcal{A}})/Im(\hat{Q}_{\mathcal{A}}) \simeq H_{d_A}^\bullet(\partial N, \mathfrak{g})[1]$$

The smooth locus of the Euler-Lagrange space $\mathcal{EL}_{\partial N}$ is the vector bundle over the space of flat connections on ∂N with fibers $\oplus_{i \neq 1} \Omega_{d_A-closed}^i(\partial N)$. The smooth locus of the boundary $\mathcal{EL}$ -moduli space for the non-abelian Chern-Simons theory is isomorphic to a vector bundle over the smooth locus of the representation variety $\text{Hom}(\pi_1(\partial N), G)/G$ with fiber $H_{d_A}^0(\partial N, \mathfrak{g}) \oplus H_{d_A}^2(\partial N, \mathfrak{g})$ with the symplectic structure naturally extending the Atiyah-Bott symplectic form on the representation variety.

Regularity for non-abelian Chern-Simons theory can be proven using homological perturbation theory around the abelian Chern-Simons theory. This argument extends to general AKSZ theories.

7.2.3. Reduced BV-BFV theory. Reduced Euler-Lagrange spaces for the bulk and for the boundary are the spaces of leaves of the foliation $Vect_Q$ on $\mathcal{EL}_N$ and $\mathcal{EL}_{\partial N}$ respectively. We will write

$$\mathcal{EL}_N/Q_N = \{\mathcal{A} \in \mathcal{F}_N | d\mathcal{A} + \frac{1}{2}[\mathcal{A}, \mathcal{A}] = 0\} / \{\mathcal{A} \mapsto \mathcal{A} + d\lambda + [\mathcal{A}, \lambda] | \lambda \in \Omega^\bullet(N)\}$$

and similarly for ∂N . These notations indicate that leaves of $Vect_Q$ should be considered as gauge orbits on $\mathcal{EL}_N$.

The projection $\pi : \mathcal{F}_N \rightarrow \mathcal{F}_{\partial N}$ (the pullback to the boundary) defines the mapping

$$\pi_* : \mathcal{EL}_N/Q_N \rightarrow \mathcal{EL}_{\partial N}/Q_{\partial N}$$

Its image $\mathcal{L}_N/Q \subset \mathcal{EL}_{\partial N}/Q$ is Lagrangian.

Tangent spaces to $\mathcal{EL}_N$ and $\mathcal{EL}_{\partial N}$ are kernels of corresponding operators $\hat{Q}$ and are described in components in (67), (69). The image of $\hat{Q}$ is also easy to compute. For the bulk we have

$$Im(\hat{Q}_{\mathcal{A}}) = \{[\tilde{\gamma}, c], d_A \tilde{\gamma} + [\tilde{\alpha}, c], d_A \tilde{\alpha} + [c, \tilde{\alpha}^\dagger] + [\tilde{\gamma}, A^\dagger], d_A \tilde{\alpha}^\dagger + [c, \gamma^\dagger] + [A^\dagger, \tilde{\alpha}] + [c^\dagger, \gamma]\}$$

for the boundary:

$$Im(\hat{Q}_{\mathcal{A}}) = \{([\tilde{\gamma}, c], d_A \tilde{\gamma} + [\tilde{\alpha}, c], d_A \tilde{\alpha} + [c, \tilde{\alpha}^\dagger] + [\tilde{\gamma}, A^\dagger])\}$$

The reduced spaces are quotient spaces $\ker(\hat{Q})/Im(\hat{Q})$. In other words these are cohomology spaces of the cochain complexes $\Omega^\bullet(N)[1]$ and $\Omega^\bullet(\partial N)[1]$ respectively with respect to the differential $d_A \omega = [c, \omega] + d_A \omega + [A^\dagger, \omega]$.

When $\mathcal{A}$ is of degree zero, i.e. $c = c^\dagger = A^\dagger = 0$ and $F(A) = 0$ we have

$$T_{\mathcal{A}}\mathcal{EL}_N/Q_N = H_{d_A}^\bullet(N, \mathfrak{g})[1], \quad T_{\mathcal{A}}\mathcal{EL}_{\partial N}/Q_{\partial N} = H_{d_A}^\bullet(\partial N, \mathfrak{g})[1]$$

7.2.4. The $gh = 0$ part of the theory. The non-reduced $gh = 0$ part of the theory has the space of fields $F_N = \Omega^1(N, \mathfrak{g})$. In case of a general principal G -bundle, this is the space of connections. The classical action is the Chern-Simons functional:

$$S_N^{cl} = \int_N \text{tr} \left(\frac{1}{2} A \wedge dA + \frac{1}{6} A \wedge [A, A] \right)$$

Its critical points are flat connections: $F(A) = dA + \frac{1}{2}[A, A] = 0$. The $\mathfrak{g}$ -valued zero-forms act by infinitesimal gauge transformations $A \mapsto A + d_A \alpha$, $\alpha \in \omega^0(N, \mathfrak{g})$.

The $gh = 0$ part of the space of boundary fields is $F_{\partial N} = \Omega^1(N, \mathfrak{g})$. This is an exact symplectic space with

$$\alpha_{\partial N} = \frac{1}{2} \int_{\partial N} \text{tr}(A \wedge \delta A), \quad \omega_{\partial N} = \delta \alpha_{\partial N} = \frac{1}{2} \int_{\partial N} \text{tr}(\delta A \wedge \delta A)$$

The degree zero part $C_{\partial N}$ of the boundary Euler-Lagrange space is the space of flat connections in a trivial G -bundle over ∂N . It is coisotropic in the space of all connections.

The gauge group $G_{\partial N} = \text{Map}(\partial N, G)$ acts on $F_{\partial N}$ by Hamiltonian transformations. Infinitesimally, the action is $A \rightarrow A + d_A \alpha$. The momentum map $\mu : \Omega^1(N, \mathfrak{g}) \rightarrow \Omega^2(N, \mathfrak{g})$ is the curvature. The symplectic reduction of $C_{\partial N}$ coincides with the Hamiltonian reduction. The reduced space $C_{\partial N}/G_{\partial N}$ is the moduli space of flat G -connections on the trivial G -bundle over ∂N .

The reduced $gh = 0$ part of the Euler-Lagrange space is the moduli space of flat connections in the trivial G -bundle over N . The mapping $\pi_* : \mathcal{M}_N^G \rightarrow \mathcal{M}_{\partial N}^G$ is the natural restriction mapping of representations of $\pi_1(N)$ to representations of $\pi_1(\partial N) \subset \pi_1(N)$. The image of this mapping is the Lagrangian subvariety in $\mathcal{M}_{\partial N}^G$, the fibers are those representations of $\pi_1(N)$ which restrict trivially to $\pi_1(\partial N)$. For detailed exposition of the global aspects of the classical Chern-Simons theory see [26].

7.3. Non-abelian BF theory. The target space for the non-abelian BF theory is described in section 6.1.2.

7.3.1. The bulk BV theory. The space of fields in the BV-extended non-abelian BF theory is

$$\mathcal{F}_N = \Omega^\bullet(N, \mathfrak{g})[1] \oplus \Omega^\bullet(N, \mathfrak{g})[n-2]$$

$$\begin{aligned} \omega_N &= \int_N \text{tr}(\delta \mathcal{B} \wedge \delta \mathcal{A}) \\ Q_N &= \int_N \text{tr} \left(\left(d\mathcal{A} + \frac{1}{2}[\mathcal{A}, \mathcal{A}] \right) \wedge \frac{\delta}{\delta \mathcal{A}} + d\mathcal{A} \mathcal{B} \wedge \frac{\delta}{\delta \mathcal{B}} \right) \\ S_N &= \int_N \text{tr} \left(\mathcal{B} \wedge \left(d\mathcal{A} + \frac{1}{2}[\mathcal{A}, \mathcal{A}] \right) \right) \end{aligned}$$

The Euler-Lagrange space:

$$\mathcal{EL}_N = \{(\mathcal{A}, \mathcal{B}) \in \mathcal{F}_N \mid d\mathcal{A} + \frac{1}{2}[\mathcal{A}, \mathcal{A}] = 0, \quad d\mathcal{A} \mathcal{B} = 0\}$$

As in the Chern-Simons theory we are interested in the smooth locus of the Euler-Lagrange space which is, in this case, a vector bundle over the space of flat connections on $G \times N$ with fiber $(\oplus_{i \neq 1} \Omega_{d_A - \text{closed}}^i(N, \mathfrak{g})[1]) \oplus \Omega_{d_A - \text{closed}}^\bullet(N, \mathfrak{g})[n-2]$ over a flat connection A .

7.3.2. *The boundary BFV theory.* The space of boundary fields is

$$\mathcal{F}_{\partial N} = \Omega^\bullet(\partial N, \mathfrak{g})[1] \oplus \Omega^\bullet(\partial N, \mathfrak{g})[n-2]$$

The BFV structure on it is given by the AKSZ construction:

$$\begin{aligned} \alpha_{\partial N} &= \int_{\partial N} \text{tr}(\mathcal{B} \wedge \delta \mathcal{A}), \quad \omega_{\partial N} = \delta \alpha_{\partial N} = \int_{\partial N} \text{tr}(\delta \mathcal{A} \wedge \delta \mathcal{B}) \\ Q_{\partial N} &= \int_{\partial N} \text{tr} \left(\left(d\mathcal{A} + \frac{1}{2}[\mathcal{A}, \mathcal{A}] \right) \wedge \frac{\delta}{\delta \mathcal{A}} + d\mathcal{A} \mathcal{B} \wedge \frac{\delta}{\delta \mathcal{B}} \right) \\ S_{\partial N} &= \int_{\partial N} \text{tr} \left(\mathcal{B} \wedge \left(d\mathcal{A} + \frac{1}{2}[\mathcal{A}, \mathcal{A}] \right) \right) \end{aligned}$$

The boundary Euler-Lagrange space is

$$\mathcal{EL}_{\partial N} = \{(\mathcal{A}, \mathcal{B}) \in \mathcal{F}_{\partial N} \mid d\mathcal{A} + \frac{1}{2}[\mathcal{A}, \mathcal{A}] = 0, \quad d\mathcal{A} \mathcal{B} = 0\}$$

7.3.3. *The $gh = 0$ part of the theory.* The degree zero gauge theory has the space of fields $F_N = \Omega^1(N, \mathfrak{g}) \oplus \Omega^{n-2}(N, \mathfrak{g})$ we will denote 1-forms A and $(n-2)$ -forms B . The action is

$$S_N^{\text{cl}} = \int_N \text{tr}(B \wedge F(A))$$

Euler-Lagrange equations are

$$F(A) = 0, \quad d_A B = 0$$

Infinitesimal gauge transformations act as

$$A \rightarrow A + d_A \mu, \quad B \rightarrow B + [B, \mu] + d_A \lambda$$

The space of boundary fields is the pullback of bulk fields. The restriction mapping is the pullback. The symplectic structure on the space of bulk fields is exact with $\omega_N = \delta \alpha_N$ where

$$\alpha_N = \int_{\partial N} \text{tr}(B \wedge \delta A), \quad \omega_N = \int_{\partial N} \text{tr}(\delta A \wedge \delta B)$$

The coisotropic submanifold $C_{\partial N}$ is the $gh = 0$ part of $\mathcal{EL}_{\partial N}$. The Lagrangian submanifold $L_N = \pi(EL_N)$ consists of pairs (A, B) where A is a flat connection on ∂N which continues to a flat connection on N and $B \in \Omega^{n-2}(\partial N, \mathfrak{g})$ is horizontal with respect to the flat connection A and extends to a horizontal $(n-2)$ -form on N .

The reduced space EL_N/G_N is a vector bundle over the moduli space of flat connections on the trivial G -bundle on N with fiber $H_A^{n-2}(N, \mathfrak{g})$ over a gauge class $[A]$.

The moduli space $EL_{\partial N}/G_{\partial N}$ has the same structure and can be identified with $T^*M_{\partial N}^G$.

7.4. **$BF + B^2$ theory.** In this theory the space is 4-dimensional. The target space is the same as the one for the BF theory.

7.4.1. *The bulk BV theory.* The space of fields is the same as in the 4-dimensional BV-extended BF-theory with the same symplectic structure.

The action and the vector field Q_N are:

$$Q_N = tr \int_N \left(\left(d\mathcal{A} + \frac{1}{2}[\mathcal{A}, \mathcal{A}] + \mathcal{B} \right) \wedge \frac{\delta}{\delta \mathcal{A}} + d_{\mathcal{A}}\mathcal{B} \wedge \frac{\delta}{\delta \mathcal{A}} \right)$$

$$S_N = tr \int_N \left(\mathcal{B} \wedge \left(d\mathcal{A} + \frac{1}{2}[\mathcal{A}, \mathcal{A}] \right) + \frac{1}{2}\mathcal{B} \wedge \mathcal{B} \right)$$

The Euler-Lagrange space:

$$\mathcal{EL}_N = \{(\mathcal{A}, \mathcal{B}) \in \mathcal{F}_N \mid d\mathcal{A} + \frac{1}{2}[\mathcal{A}, \mathcal{A}] + \mathcal{B} = 0, d_{\mathcal{A}}\mathcal{B} = 0\} \simeq \Omega^\bullet(N, \mathfrak{g})[1]$$

Indeed, the first condition $F(\mathcal{A}) + \mathcal{B} = 0$ gives no restriction on $\mathcal{A}$, and the second condition $d_{\mathcal{A}}\mathcal{B} = 0$ follows from the first one and from the Bianchi identity.

7.4.2. *The boundary BFV theory.* The space of boundary fields is the same as for the BF theory with the same symplectic structure of degree 0.

The boundary vector field $Q_{\partial N}$ and the boundary action are

$$Q_{\partial N} = tr \int_{\partial N} \left(\left(d\mathcal{A} + \frac{1}{2}[\mathcal{A}, \mathcal{A}] + \mathcal{B} \right) \wedge \frac{\delta}{\delta \mathcal{A}} + d_{\mathcal{A}}\mathcal{B} \wedge \frac{\delta}{\delta \mathcal{A}} \right)$$

$$S_{\partial N} = tr \int_{\partial N} \left(\mathcal{B} \wedge \left(d\mathcal{A} + \frac{1}{2}[\mathcal{A}, \mathcal{A}] \right) + \frac{1}{2}\mathcal{B} \wedge \mathcal{B} \right)$$

The boundary Euler-Lagrange space:

$$\mathcal{EL}_{\partial N} \simeq \Omega^\bullet(\partial N)[1]$$

7.4.3. *The $gh = 0$ part of the theory.* The space of fields is the same as the space of fields of the non-abelian BF theory in four dimensions. The action is

$$S_N^{cl} = tr \int_N \left(B \wedge F(A) + \frac{1}{2}B \wedge B \right)$$

Euler-Lagrange equations are

$$B + F(A) = 0, \quad d_A B = 0$$

The second equation follows from the first and from the Bianchi identity. Infinitesimal gauge transformations are

$$A \mapsto A + d_A \alpha - \beta, \quad B \mapsto B + [B, \alpha] + d_A \beta$$

The reduced theory is trivial:

$$\mathcal{EL}_N/Q = \{0\}, \quad \mathcal{EL}_{\partial N}/Q = \{0\}$$

so the $gh = 0$ part of it also trivial.

7.5. **The Poisson sigma model.** The target space for the Poisson sigma model is described in section 6.1.2. In this section M is a Poisson manifold with the Poisson tensor $\pi(x)$. In local coordinates $\pi(x) = \sum_{ab} \pi^{ab}(x) \partial_a \partial_b$.

7.5.1. *The bulk BV theory.* In case of the Poisson sigma model the spacetime is two dimensional. The space of fields in the models is $\mathcal{F}_N = \text{Map}(T[1]N, T^*[1]M)$. We will use coordinate field $\tilde{X}^a = X^a + \eta^{+,a} + \beta^{+,a}$ for coordinates q^a on M and $\tilde{\eta}_a = \beta_a + \eta_a + X_a^\dagger$ for coordinates p_a in the cotangent directions. Components of fields $\tilde{X}^a$ and $\tilde{\eta}_a$ are forms of degree 0, 1, 2 respectively. They have ghost numbers $gh(X) = 0$, $gh(\eta^\dagger) = -1$, $gh(\beta^\dagger) = -2$, $gh(\beta) = 1$, $gh(\eta) = 0$, $gh(X^\dagger) = -1$. The symplectic form and the action functional are given by the AKSZ construction:

$$\omega_N = \int_N \sum_a \delta \tilde{\eta}_a \wedge \delta \tilde{X}^a = \int_N \sum_a (\delta X^a \wedge \delta X_a^\dagger + \delta \eta^{+,a} \wedge \delta \eta_a + \delta \beta^{+,a} \wedge X_a^\dagger)$$

$$S_N = \int_N \sum_a \left(\tilde{\eta}_a \wedge d\tilde{X}^a + \frac{1}{2} \int_N \sum_{ab} \pi^{ab}(\tilde{X}) \wedge \tilde{\eta}_a \wedge \tilde{\eta}_b \right)$$

In field components this action is

$$(71) \quad S_N = \int_N \left(\eta_a \wedge dX^a - \eta^{+,a} \wedge \beta_a + \frac{1}{2} \pi^{ab}(X) \wedge \eta_a \wedge \eta_b + \pi^{ab}(X) \wedge \beta_a \wedge X_b^\dagger + \right. \\ \left. \eta^{+,a} \wedge \partial_a \pi^{bc}(X) \wedge \beta_b \wedge \eta_c + \frac{1}{2} \beta^{+,c} \wedge \partial_c \pi^{ab}(X) \wedge \beta_b \wedge \beta_c + \frac{1}{4} \eta^{+,c} \wedge \eta^{+,d} \wedge \partial_c \partial_d \pi^{ab}(X) \wedge \beta_a \wedge \beta_b \right)$$

The vector field Q is

$$Q_N = \int_N \left((d\tilde{X}^a + \pi^{ab}(\tilde{X}) \wedge \tilde{\eta}_a \wedge \tilde{\eta}_b) \wedge \frac{\delta}{\delta \tilde{X}^a} + (d\tilde{\eta}_a + \frac{1}{2} \partial_a \pi^{bc}(\tilde{X}) \wedge \tilde{\eta}_b \wedge \tilde{\eta}_c) \wedge \frac{\delta}{\delta \tilde{\eta}_a} \right)$$

The Euler-Lagrange space $\mathcal{EL}_N$ is the space of zeroes of Q (the space of critical points of S_N), or the space of solutions to

$$d\tilde{X}^a + \pi^{ab}(\tilde{X}) \wedge \tilde{\eta}_a \wedge \tilde{\eta}_b = 0, \quad d\tilde{\eta}_a + \frac{1}{2} \partial_a \pi^{bc}(\tilde{X}) \wedge \tilde{\eta}_b \wedge \tilde{\eta}_c$$

7.5.2. *The boundary BFV theory.* We will use the same notations for coordinate fields in the space of boundary fields $\mathcal{F}_{\partial N} = \text{Map}(T[1]\partial N, T^*[1]M)$. In terms of this coordinate fields the BFV data for the Poisson sigma model are:

$$\alpha_{\partial N} = \int_{\partial N} \tilde{\eta}_a \wedge \delta \tilde{X}^a$$

$$\omega_{\partial N} = \int_{\partial N} \delta \tilde{\eta}_a \wedge \delta \tilde{X}^a$$

$$Q_{\partial N} = \int_{\partial N} \left((d\tilde{X}^a + \pi^{ab}(\tilde{X}) \wedge \tilde{\eta}_a \wedge \tilde{\eta}_b) \wedge \frac{\delta}{\delta \tilde{X}^a} + (d\tilde{\eta}_a + \frac{1}{2} \partial_a \pi^{bc}(\tilde{X}) \wedge \tilde{\eta}_b \wedge \tilde{\eta}_c) \wedge \frac{\delta}{\delta \tilde{\eta}_a} \right)$$

$$S_{\partial N} = \int_{\partial N} \left(\sum_a \tilde{\eta}_a \wedge d\tilde{X}^a + \frac{1}{2} \sum_{ab} \pi^{ab}(\tilde{X}) \wedge \tilde{\eta}_a \wedge \tilde{\eta}_b \right)$$

APPENDIX A. COISOTROPIC SUBMANIFOLDS AND REDUCTION

In this paper we use the definition of Lagrangian submanifolds as submanifolds which are both isotropic and coisotropic. This is because many of the symplectic spaces we work with are infinite dimensional. In finite dimensional symplectic manifolds, Lagrangian submanifolds have half-dimension of the total manifold. In the infinite dimensional case this property becomes meaningless.

Let W be a presymplectic space (possible infinite-dimensional) and $K \subset W$ be the kernel of its presymplectic form. Then $W' = W/K$ is symplectic. Denote by $p : W \rightarrow W'$ the natural projection map. For a subspace $L \subset W$ define $L' \subset W'$ as $L' = p(L) = L/(L \cap K)$.

It is clear that $p^{-1}(L') = L + K$, and it is easy to see that $p^{-1}((L')^\perp) = L^\perp$. Here $A^\perp$ is the orthogonal space with respect to the symplectic (presymplectic) form.

Proposition A.1. *The subspaces L and L' have the following properties:*

- (1) *L is isotropic if and only if L' is isotropic.*
- (2) *If L is Lagrangian, then L' is also Lagrangian.*
- (3) *If L' is Lagrangian and $K \subset L$, then L is Lagrangian.*

Proof. For $\xi, \eta \in W$, denote by $[\xi], [\eta] \in W'$ the equivalence classes of these elements. By definition of the symplectic structure on W' :

$$([\xi], [\eta])_{W'} = (\xi, \eta)$$

The first statement is obvious from this definition.

Now assume that L is coisotropic, i.e. if $(\xi, \eta) = 0$ for any $\eta \in L$, then $\xi \in L$. This implies that also $(\xi + \kappa_1, \eta + \kappa_2) = 0$ for each $\kappa_i \in K$. But this means that if $([\xi], [\eta])_{W'} = 0$ for each $[\eta] \in L'$ then $[\xi] \in L'$. This means L' contains its symplectic orthogonal and therefore is coisotropic. We proved the second statement.

To prove the last statement, we have to prove that if L' is coisotropic and $K \subset L$ then L is coisotropic. Then the first statement implies the third one.

Assume L' is coisotropic, i.e. that if $([\xi], [\eta])_{W'} = 0$ for each $[\eta] \in L'$ then $[\xi] \in L'$. But this means that if $(\xi + \kappa_1, \eta + \kappa_2) = 0$ for each $\eta \in L$ and $\kappa_1, \kappa_2 \in K$ then $\xi \in L + K$. If $K \subset L$, this means that if $(\xi, \eta) = 0$ for each $\eta \in L$ then $\xi \in L$. That is that $L \subset W$ is coisotropic. □

An important particular case is when W is a coisotropic subspace in a bigger symplectic space. In this case $K = W^\perp$. The space W' is the Hamiltonian reduction of the space W .

Remark A.2. *If $L \subset V$, the space V is symplectic and $W \subset V$ is coisotropic but we do not assume that $L \subset W$, then the first statement holds when $L' = p(L \cap W)$ but the second and the third statements do not hold in general unless V is finite dimensional.*

Now, let W be a presymplectic manifold and $K \subset TW$ be the integrable distribution which is the kernel of the presymplectic form. Assume the space of leaves of K is a smooth manifold W' . Let $L \subset W$ be a submanifold, and L' be the space of leaves of K which passes through L . Assume it is also smooth. Then Proposition A.1 holds but the condition in the last statement should be replaced with $K|_L \subset TL$.

APPENDIX B. SOME FACTS ON GRADED MANIFOLDS

B.1. Graded manifolds. Recall that a smooth super manifold M with body M_e is a sheaf of super algebras over M_e locally isomorphic to the tensor product of the algebra of smooth functions on the body with an exterior algebra. Namely, there is an atlas $\{(U_\alpha, \phi_\alpha)\}$ of M_e with super algebra isomorphisms

$$\Phi_\alpha: M|_{U_\alpha} \rightarrow C^\infty(\phi_\alpha(U_\alpha)) \otimes \bigwedge V^* =: \mathcal{A}_\alpha$$

for a fixed vector space V . The coordinate map ϕ_α s take values in some given vector space W . For infinite dimensional supermanifolds one needs more structure: W is assumed to be a Banach or Fréchet space and, if V is infinite dimensional, the tensor product has to be completed (the dual has also to be defined properly).

If $V = \oplus_{k \in \mathbb{Z}} V_k$ and $W = \oplus_{k \in \mathbb{Z}} W_k$ are $\mathbb{Z}$ -graded vector spaces (it is safer to assume they have only finitely many nontrivial, but possibly infinite dimensional, summands), then $\mathcal{A}_\alpha$ gets additional structure as it contains the $\mathbb{Z}$ -graded subalgebra of polynomial functions where by definition linear functions on W_k or V_k have degree $-k$. To extend the notion of grading to nonpolynomial functions, we introduce the local graded Euler vector field E as follows: Pick graded bases (i.e., bases adapted to the decompositions) $\{x^i\}$ and $\{y^i\}$ of W^* and V^* , respectively; then define

$$E := \sum_i |x^i| x^i \frac{\partial}{\partial x^i} + \sum_i |y^i| y^i \frac{\partial}{\partial y^i},$$

where $|x^i| := -k$ for $x^i \in (W_k)^*$ and $|y^i| := -k$ for $y^i \in (V_k)^*$. Notice that this definition is independent of the choice of graded bases. We then say that a function f is of degree k if it satisfies $E(f) = kf$.

If it is possible to choose an atlas of M as above such that all transition functions are compatible with the local graded Euler vector fields on charts,¹⁰ then we say that M is a graded manifold. Notice that a graded manifold is a super manifold with additional structure; namely, that of a globally well-defined graded Euler vector field, which we will keep denoting by E . A global function is said to have degree k if it satisfies

$$E(f) = kf.$$

One may let E act on vector fields and differential forms by Lie derivative and define degree accordingly. Namely, a vector field X has degree k if satisfies $[E, X] = kX$ and a differential form α has degree k if $L_E \alpha := i_E d\alpha + di_E \alpha = k\alpha$. Notice that the graded Euler vector field has degree zero. We recall from [46] some useful facts whose proof is a straightforward computation:

- (1) Let ω be a closed form of degree $m \neq 0$. Then $\omega = d\theta$ with $\theta = \frac{1}{m} \iota_E \omega$.
- (2) Let X be a vector field of degree l and ω a closed X -invariant form of degree m with $m + l \neq 0$. Then $\iota_X \omega = dS$ with $S = \frac{1}{m+l} \iota_E \iota_X \omega$.

In particular, this implies that symplectic forms of degree different from zero are automatically exact and that a symplectic cohomological vector field is automatically Hamiltonian if the degree of the symplectic form is different from -1 .

Definition B.1. *Morphisms of graded manifolds are morphisms of the underlying supermanifolds that respect the graded Euler vector fields.*

¹⁰That is, for any two charts U_α and U_β we have $E_\beta(\Phi_\beta \circ \Phi_\alpha^{-1})^* f = (\Phi_\beta \circ \Phi_\alpha^{-1})^* E_\alpha f$ for all $f \in \mathcal{A}_\alpha$.

Remark B.2. A different definition of graded manifolds commonly used in the literature is that of a sheaf M of graded algebras over the degree zero body M_0 locally isomorphic to the tensor product of smooth functions on the degree zero body with the graded symmetric algebra of a graded vector space. In this setting functions (and transition functions) are polynomial in the coordinates of degree different from zero. To distinguish graded manifolds defined this way from the ones defined above we will call them polygraded manifolds. Notice that a polygraded manifold is not a super manifold with additional structure. On the other hand, if a graded manifold happens to have an atlas for which all transition functions are polynomial in the coordinates of degree different from zero, then it can be given the structure of a polygraded manifold simply by restricting the sheaf. In all the examples discussed in this paper we could work with polygraded manifolds, but this would be problematic for functional integral quantization where one exponentiates the action and wants to consider integration.

Remark B.3. A common notion in the literature is that of an N -manifold. This is a graded manifold with no coordinates of negative degree. They are commonly used as targets of the AKSZ construction. Notice that by degree reasons all transition functions are polynomial in the coordinates of degree different from zero.

Remark B.4. A special class of (poly)graded manifolds are those in which the $\mathbb{Z}$ and $\mathbb{Z}_2$ gradings agree. This means that $W_{2k+1} = 0 = V_{2k}$ for all k or, equivalently, that for all k homogenous sections of degree $2k$ ($2k+1$) are even (odd) with respect to the original super algebra grading. These graded manifolds occur in the BV formalism whenever no fermionic physical fields are present. In this paper we will restrict to this case throughout.

B.2. Mapping spaces. Let M and N be finite dimensional super manifolds. Then the set of morphisms $\text{Mor}(M, N)$ can naturally be given the structure of a (usually infinite dimensional) manifold (non super) as follows: If the target is a super vector space Z , then $\text{Mor}(M, Z)$ is the (usually infinite dimensional) vector space

$$(72) \quad (C^\infty(M) \otimes Z)_e = C^\infty(M)_e \otimes Z_e \oplus C^\infty(M)_o \otimes Z_o$$

For the general case, one applies this construction to local charts of the target to define local charts of the manifold of morphisms.

It is important to extend this construction to define the mapping space $\text{Map}(M, N)$ as a (usually infinite dimensional) super manifold with body $\text{Mor}(M, N)$. Again this is done in terms of local charts, so it is enough to define $\text{Map}(M, Z)$. This is just the super space $C^\infty(M) \otimes Z$ (of which (72) is the even part). The main property of the mapping space is

$$\text{Mor}(X \times M, N) = \text{Mor}(X, \text{Map}(M, N)) \quad \forall X.$$

Vector fields on the source and on the target can naturally be lifted to the mapping space. If X_M (Y_N) is a vector field on M (N), we denote by $\check{X}_M$ ($\hat{Y}_N$) its lift. Notice that the map $Y_N \mapsto \hat{Y}_N$ is a morphism of Lie algebras, whereas the map $X_M \mapsto \check{X}_M$ is an antimorphism.

If M and N are graded manifolds, with graded Euler vector fields E_M and E_N , then one can give $\text{Map}(M, N)$ the structure of a (usually infinite dimensional) graded manifold with graded Euler vector field

$$E_{\text{Map}(M, N)} = \hat{E}_N - \check{E}_M.$$

The set of morphisms from M to N in the category of graded manifolds can then be regarded as the degree zero submanifold of $\text{Map}(M, N)$.

Remark B.5. *In the category of polygraded manifolds, one can also define the mapping space as a polygraded manifold. In this case the local data are given by $\text{Map}(M, Z)$ for Z a graded vector space. Here $\text{Map}(M, Z)$ is defined as the graded tensor product*

$$C^\infty(M) \otimes Z = \bigoplus_k (C^\infty(M) \otimes Z)_k$$

with

$$(C^\infty(M) \otimes Z)_k = \bigoplus_j C^\infty(M)_j \otimes Z_{k-j}.$$

APPENDIX C. ON SMOOTH POINTS OF THE MODULI SPACE $\mathcal{EL}/Q$

Here we will discuss the notion of smooth points in the $\mathcal{EL}$ -moduli spaces which we use throughout the paper. We will use notations: $\mathcal{F}$ for the space of BV fields, F for its $\text{gh} = 0$ part, $\mathcal{EL}$ for BV Euler-Lagrange space and EL for its $\text{gh} = 0$ part. We assume $\mathcal{F}$ is given together with the cohomological vector field Q . Denote by $Vect_Q$ the Lie subalgebra of the Lie algebra of vector fields on $\mathcal{F}$ formed by Lie brackets with Q and by G the distribution on the body F of $\mathcal{F}$ induced by the span of $Vect_Q$. The distribution G on F should be regarded as infinitesimal gauge transformations.

For a $x \in F$ the Taylor expansion of a vector field Q in the formal neighborhood of x is:

$$Q_x^{\text{formal}} = Q_x^{(0)} + Q_x^{(1)} + Q_x^{(2)} + \dots$$

where each

$$Q_x^{(k)} \in \text{Coder}(\hat{S}T_x\mathcal{F})$$

is the extension of a k -linear map $\hat{Q}_x^{(k)} : S^k(T_x\mathcal{F}) \rightarrow T_x\mathcal{F}$ to a coderivation of $\hat{S}T_x\mathcal{F}$ by co-Leibniz identity. The linear maps $\hat{Q}_x^{(k)}$ arise from the Taylor expansion of Q at x . By the natural inclusion $\text{Coder}(\hat{S}T_x\mathcal{F}) \hookrightarrow \text{Der}(\hat{S}T_x^*\mathcal{F})$, Q_x^{formal} acts on $\hat{S}T_x^*\mathcal{F}$ as a derivation.

If $x \in EL \subset F$, then $\hat{Q}_x^{(0)} = 0$ and Q_x^{formal} endows $T_x[-1]\mathcal{F}$ with the structure of L_∞ algebra with differential $\hat{Q}_x^{(1)}$ and higher polylinear operations $\hat{Q}_x^{(k)}$, $k \geq 2$.

We define the formal neighborhood of $[x] \in EL/G$ in $\mathcal{EL}/Q$ as

$$(73) \quad U_{[x]}^{\text{formal}}(\mathcal{EL}/Q) := \text{Spec}(H_{Q_x^{\text{formal}}}(\hat{S}T_x^*\mathcal{F}))$$

It can be regarded as the “BV” (or “stable”) version of the Maurer-Cartan set for the L_∞ structure on $T_x[-1]\mathcal{F}$. When operations $\hat{Q}_x^{(k)}$ for $k \geq 2$ are identically zero, the spectrum (73) is

$$H_{\hat{Q}_x^{(1)}}(T_x\mathcal{F})$$

which means that the formal neighborhood of $[x]$ in $\mathcal{EL}/Q$ is the graded vector space $H_{\hat{Q}_x^{(1)}}(T_x\mathcal{F})$.

When operations $\hat{Q}_x^{(k)}$ do not vanish for $k \geq 2$ they induce operations $\hat{Q}_x'^{(k)}$ on the cohomology space $V_x = H_{\hat{Q}_x^{(1)}}(T_x\mathcal{F})$. These operations endow the graded space $V_x[-1]$ with the structure of minimal L_∞ algebra. In other words the formal series

$$Q'_x = Q_x'^{(0)} + Q_x'^{(1)} + Q_x'^{(2)} + \dots \in \text{Coder}(\hat{S}(V_x))$$

is a (formal) cohomological vector field on V_x . Note that first two terms vanish by our assumptions.

This construction is known as homological perturbation theory [33] (see also [31] for a more concise exposition), more specifically, as the homotopy transfer of L_∞ algebras [37, 38]. By the same homological perturbation theory the formal neighborhood of $[x] \in EL/G \subset \mathcal{EL}/Q$ is

$$U_{[x]}^{\text{formal}}(\mathcal{EL}/Q) = \text{Spec}(H_{Q'_x}(\hat{S}(V_x)^*))$$

This is a singular variety unless Q'_x vanishes.

This is why we define a *smooth point* $[x] \in EL/G$ as a point for which all induced L_∞ operations $\hat{Q}_x^{(k)}$ vanish. The set of such points we will call the smooth locus of the body of the $\mathcal{EL}$ -moduli space and will denote it $(EL/G)^{\text{smooth}}$.

The smooth locus in the $\mathcal{EL}$ -moduli space is a graded vector bundle over $(EL/G)^{\text{smooth}}$ with fiber $H_{\hat{Q}_x^{(1)}}^{\neq 0}(T_x \mathcal{F})$ over $[x]$. Here $H^{\neq 0}$ means that we do not count the cohomology in ghost number 0 to avoid double counting of geometric ($gh = 0$) directions for the tangent space.

APPENDIX D. CARTAN CALCULUS OF LOCAL DIFFERENTIAL FORMS

D.1. Transgression map from forms on the target to forms on the mapping space. Recall that the space of fields in the AKSZ theory is $\mathcal{F}_N = \text{Map}(T[1]N, \mathcal{M})$ with $\mathcal{M}$ a Hamiltonian dg manifold.

Consider natural projections

$$\begin{array}{ccc} \mathcal{F}_N \times T[1]N & \xrightarrow{ev} & \mathcal{M} \\ p \downarrow & & \\ \mathcal{F}_N & & \end{array}$$

Here $ev(f, x) = f(x)$ and $p(f, x) = f$. The pullback ev^* of a form $\mathcal{M}$ gives a form on $\mathcal{F}_N \times T[1]N$. Let Ω be a form on $\mathcal{F}_N \times T[1]N$ of type $(k, 0)$. The pushforward p_* of a form on $\mathcal{F}_N \times T[1]N$ gives a form on $\mathcal{F}_N$.

$$p_*(\Omega)(X) = \int_{T[1]N} i_X^*(\Omega)$$

Here i_X is the composition of the natural isomorphism of $T[1]N$ with the fiber $(X, T[1]N) = p^{-1}(X)$ and the embedding of this fiber into $\mathcal{F}_N \times T[1]N$. The integration over $T[1]N$ is defined as usual taking into account natural isomorphism $\phi : C^\infty(T[1]N) \simeq \Omega^\bullet(N)$. If $f \in C^\infty(T[1]N)$

$$\int_{T[1]N} f \stackrel{\text{def}}{=} \int_N \phi(f)$$

Having in mind this formula we will write for p_*

$$p_*(\Omega)(X) = \int_N \Omega$$

The transgression map is defined as

$$T_N := p_* ev^* : \quad \Omega^\bullet(\mathcal{M}) \rightarrow \Omega^\bullet(\mathcal{F}_N)$$

Given $\alpha \in \Omega^\bullet(\mathcal{M})$ which is $\alpha = \sum_{\{a\}} \alpha_{a_1, \dots, a_k} dx^{a_1} \dots dx^{a_k}$ in local coordinates, its transgression is

$$(74) \quad (T_N \alpha)(X) = \int_N \sum_{a_1, \dots, a_k} \alpha_{a_1, \dots, a_k}(X(u)) \delta X^{a_1}(u) \dots \delta X^{a_k}(u)$$

where $X^a(u)$ are coordinate fields and $\delta X^a(u)$ are their de Rham differentials (on $\mathcal{F}_N$).

Define the space of *ultra local forms* $\Omega_{uloc}^\bullet(\mathcal{F}_N)$ as the image of the transgression map.

D.2. De Rham and lifted vector fields. Recall that the de Rham differential for N can be regarded as a vector field D_N on $T[1]N$ of degree 1. The identification of forms on N with functions on $T[1]N$ identifies the action of the de Rham differential on forms with the action of the vector field D_N on functions on $T[1]N$. In local coordinates $\{u^i\}$ on N and $\xi^i = du^i$ on $T_u[1]N$ the Lie derivative of a function along the de Rham vector field on $T[1]N$ is $D_N f = \sum_i \xi^i \frac{\partial f}{\partial u^i}$.

The *de Rham vector field* $\check{D}_N$ is defined as the lift of de Rham vector field $\xi^i \frac{\partial}{\partial u^i}$ on N to the mapping space $\mathcal{F}_N$.

Indeed, the tangent space $T_X \mathcal{F}_N$ is the space of mappings $Y : T[1]N \rightarrow T\mathcal{M}$ such that $(\xi, u) \mapsto Y(u, \xi) \in T_{X(u, \xi)} \mathcal{M}$. A vector field V on $\mathcal{F}_N$ is a section of the tangent bundle, i.e. it assigns a vector $V_X : \mathcal{F}_N \rightarrow T_X \mathcal{F}_N$ to each $X \in \mathcal{F}_N$.

For de Rham vector field $\check{D}_N$ on $\mathcal{F}_N$ the vector $(\check{D}_N)_X \in T_X \mathcal{F}_N$ is the mapping

$$(\check{D}_N)_X : T[1]N \xrightarrow{D_N} T(T[1]N) \xrightarrow{dX} T\mathcal{M}$$

Here $dX : T(T[1]N) \rightarrow T\mathcal{M}$ is the differential of X . In local coordinates:

$$(\check{D}_N)_X(\xi, u) = \sum_a \sum_{\{i\}, j} \frac{\partial X_{i_1, \dots, i_k}^a}{\partial u^j}(u) \xi^j \xi^{i_1} \dots \xi^{i_k} \frac{\partial}{\partial x^a}$$

The de Rham vector field on $\mathcal{F}_N$ has ghost number $+1$.

Another important class of vector fields on $\mathcal{F}_N$ comes from lifting vector fields on $\mathcal{M}$ to the mapping space $\mathcal{F}_N$. If $v : \mathcal{M} \rightarrow T\mathcal{M}$ is a vector field on $\mathcal{M}$, its lifting is the vector field $\hat{v}$ on $\mathcal{F}_N$ with $\hat{v}_X \in T_X \mathcal{F}_N$ given by the mapping

$$T[1]N \xrightarrow{X} \mathcal{M} \xrightarrow{v} T\mathcal{M}$$

D.3. Local differential forms. *Local differential forms*¹¹ are defined as substitutions of several copies of $\check{D}_N$ into an ultra-local form:

$$(75) \quad T_N^k \chi := (\iota_{\check{D}_N})^k T_N \chi, \quad k \geq 0$$

The space of forms $\Omega^\bullet(\mathcal{M})$ is naturally bi-graded with (de Rham degree of forms on $\mathcal{M}$, $\mathbb{Z}$ -grading on $\mathcal{M}$). The space $\Omega^\bullet(\mathcal{F}_N)$ is also naturally bi-graded with (de Rham degree of forms on $\mathcal{F}_N$, ghost number). Transgression mappings $T_N^k : \Omega^\bullet(\mathcal{M}) \rightarrow \Omega^\bullet(\mathcal{F}_N)$ are bi-graded with

$$\deg(T_N^k) = (-k, -\dim(N) + k)$$

Also note that if $\chi \in \Omega^n(\mathcal{M})$ and $k > n$, by degree reasons we have $T_N^k \chi = 0$.

Let $\{x^a\}$ be local coordinates on $\mathcal{M}$ and $X^a(u)$ be corresponding coordinate fields. If $\alpha = \sum_{\{a\}} \alpha_{a_1, \dots, a_k} dx^{a_1} \wedge \dots \wedge dx^{a_k}$ is a differential form on $\mathcal{M}$ written in

¹¹Note that, by our definition, local 0-forms are special cases of local functionals.

local coordinates, the corresponding local forms on $\mathcal{F}_N$ can be written in coordinate fields as

$$T_N^l(\alpha) = \int_N \sum_{\{a\}=\{a_I\} \sqcup \{a_{II}\}} \alpha_{\{a_I\} \sqcup \{a_{II}\}}(X(u)) \wedge dX^{\{a_I\}}(u) \wedge \delta X^{\{a_{II}\}}(u)$$

Here the sum is taken over partitions of the set $\{a\}$ into two subsets, so that $\{a\}$ is a shuffle of $\{a_I\}$ and $\{a_{II}\}$. We denote $dX^{\{a\}} = dX^{a_1} \wedge \dots \wedge dX^{a_k}$.

We will use the notation $\Omega_{loc}^\bullet(\mathcal{F}_N)$ for the space of such forms.

D.4. The Cartan calculus. Let us denote $\pi : \mathcal{F}_N \rightarrow \mathcal{F}_{\partial N}$ the restriction map (pullback by the natural inclusion $\partial N \hookrightarrow N$) and let

$$(76) \quad \pi^* : \Omega^\bullet(\mathcal{F}_{\partial N}) \rightarrow \Omega^\bullet(\mathcal{F}_N)$$

be the pullback by π for differential forms on the space of fields.

Objects we introduced satisfy the following properties:

(i) De Rham differential δ on $\mathcal{F}_N$ acts on local forms by

$$(77) \quad \delta(T_N^k \chi) = (-1)^{\dim(N)} (T_N^k d\chi - k \cdot \pi^*(T_{\partial N}^{k-1} \chi))$$

where d is the de Rham differential on target.

(ii) “Total derivative property”, which is a special case of (77) for $k = \deg \chi + 1$:

$$(78) \quad \chi \in \Omega^n(\mathcal{M}) \Rightarrow T_N^{n+1}(d\chi) = (n+1) \cdot \pi^*(T_{\partial N}^n \chi)$$

(iii) Substitution of $\check{D}_N$ into a local form:

$$(79) \quad \iota_{\check{D}_N}(T_N^k \chi) = T_N^{k+1} \chi$$

by definition (75).

(iv) Substitution of a lifted vector field into a local form:

$$(80) \quad \iota_{\tilde{v}}(T_N^k \chi) = (-1)^{(\text{gh}(v)+1) \cdot \dim(N)} T_N^k(\iota_v \chi)$$

for any vector field $v \in \text{Vect}(\mathcal{M})$ of ghost number $\text{gh}(v)$ on target.

(v) Rules for commuting the substitution of a vector field on $\mathcal{F}_N$ with the map π^* (76):

$$(81) \quad \iota_{\check{D}_N} \pi^* = \pi^* \iota_{\check{D}_{\partial N}}$$

$$(82) \quad \iota_{\tilde{v}} \pi^* = \pi^* \iota_{\tilde{v}}$$

Here $\tilde{v}$ is the lifting of the vector field v on $\mathcal{M}$ to a vector field on $\mathcal{F}_{\partial N}$. This is a manifestation of the fact that vector fields $\check{D}_N, \tilde{v} \in \text{Vect}(\mathcal{F}_N)$ are projectable by the restriction map $\pi : \mathcal{F}_N \rightarrow \mathcal{F}_{\partial N}$ with projections being $\check{D}_{\partial N}, \tilde{v} \in \text{Vect}(\mathcal{F}_{\partial N})$ respectively.

(vi) The Lie derivative of a local form along $\check{D}_N$:

$$(83) \quad L_{\check{D}_N}(T_N^k \chi) = (-1)^{\dim(N)} \pi^*(T_{\partial N}^k \chi)$$

We define the Lie derivative by Cartan formula $L_V := [\iota_V, \delta] = \iota_V \circ \delta + (-1)^{\text{gh}(V)} \delta \circ \iota_V$ for any vector field $V \in \text{Vect}(\mathcal{F}_N)$. With this definition the formula (83) comes as a consequence of (77), (79), (81).

(vii) Lie derivative of a local form along a lifted vector field:

$$(84) \quad L_{\tilde{v}}(T_N^k \chi) = (-1)^{\text{gh}(v) \cdot \dim(N)} T_N^k(L_v \chi)$$

(this is a consequence of (77), (80), (82)).

(viii) Commutators of vector fields:

$$(85) \quad [\check{D}_N, \check{D}_N] = 0$$

This is because $d^2 = 0$ on the source N . Also

$$(86) \quad [\check{D}_N, \hat{v}] = 0$$

and

$$(87) \quad [\hat{u}, \hat{v}] = \widehat{[u, v]}$$

D.5. Applications to AKSZ theories.

D.5.1. *Invariant definition of AKSZ theories.* Recall that the target manifold in an n -dimensional AKSZ theory is a Hamiltonian dg manifold of degree $n - 1$. In other words, it is symplectic with an exact symplectic form $\omega = d\alpha \in \Omega^2(\mathcal{M})$ and $\deg(\omega) = n - 1$ and with a potential function Θ of degree n such that $\{\Theta, \Theta\} = 0$. The potential function generates the cohomological vector field Q of degree 1:

$$\iota_Q \omega = d\Theta$$

By definition of Q , the symplectic form ω is Q -invariant, i.e.

$$L_Q \omega = 0$$

where $L_Q \eta$ is the Lie derivative of η . The condition $\{\Theta, \Theta\} = 0$ implies that

$$L_Q^2 = 0$$

which we will write as $Q^2 = 0$.

The AKSZ action is the following local functional on the space $\text{Map}(T[1]N \rightarrow \mathcal{M})$

$$S_N = S_N^{kin} + S_N^{int}$$

where

$$\begin{aligned} S_N^{kin} &= T_N^1 \alpha \\ S_N^{int} &= T_N^0 \Theta \end{aligned}$$

are kinetic and interaction parts of the action respectively. Here T_N^k are the transgression maps defined above.

The BV cohomological vector field Q_N in this theory is defined as

$$Q_N = Q_N^{kin} + Q_N^{int} \in \text{Vect}(\mathcal{F}_N)$$

with

$$\begin{aligned} Q_N^{kin} &= \check{D}_N \\ Q_N^{int} &= \hat{Q} \end{aligned}$$

Here $\hat{Q}$ is the lift of the vector field Q on $\mathcal{M}$ to a vector field on $\mathcal{F}_N$, see subsection D.2.

The BV 2-form and its primitive 1-form:

$$\begin{aligned} \omega_N &= T_N^0 \omega \\ \alpha_N &= T_N^0 \alpha \end{aligned}$$

Note that with our conventions we have an extra sign: $\omega_N = (-1)^{\dim(N)} \delta \alpha_N$.

These formulae applied to the boundary ∂N of N also define the BFV action $S_{\partial N}$, the cohomological vector field $Q_{\partial N}$, BFV 2-form $\omega_{\partial N}$ and its primitive 1-form $\alpha_{\partial N}$.

D.5.2. *Proof of Proposition 6.3.* Using the definition of Q_N and the Cartan calculus we can compute the contraction of kinetic and interaction parts of Q_N with ω_N :

$$\iota_{Q_N^{kin}} \omega_N = \iota_{\tilde{D}_N} (T_N^0 \omega) = T_N^1 \omega = T_N^1 \delta \alpha$$

The rule (77) from Cartan calculus on local forms implies that this is equal to

$$(88) \quad (-1)^{\dim(N)} \delta T_N^1 \alpha + \pi^*(T_{\partial N}^0 \alpha) = (-1)^{\dim(N)} \delta S_N^{kin} + \pi^* \alpha_{\partial N}$$

Contracting Q_N^{int} with ω_N we obtain

$$\iota_{Q_N^{int}} \omega_N = \iota_{\tilde{Q}} (T_N^0 \omega) = T_N^0 (\iota_Q \omega)$$

Here in the last equality we used (80). Because Q is a Hamiltonian vector field generated by Θ , $\iota_Q \omega = d\Theta$ and for the above expression we obtain

$$(89) \quad T_N^0 (d\Theta) = (-1)^{\dim(N)} \delta (T_N^0 \Theta) = (-1)^{\dim(N)} \delta S_N^{int}$$

Here the first equality follows from (77).

Equations (88), (89) together give (64).

D.5.3. *Further applications to the AKSZ formalism.* Here we will reprove the Proposition 3.1 using the Cartan calculus.

Proposition D.1. *The following identity holds:*

$$(90) \quad L_{Q_N} S_N = (-1)^{\dim(N)} \pi^* (2S_{\partial N} - \iota_{Q_{\partial N}} \alpha_{\partial N})$$

Proof. The proof is computational:

$$(91) \quad \begin{aligned} L_{Q_N^{kin}} S_N^{kin} &= L_{\tilde{D}_N} T_N^1 \alpha = (-1)^{\dim(N)} \pi^* (T_{\partial N}^1 \alpha) = \\ &= (-1)^{\dim(N)} \pi^* (\iota_{Q_{\partial N}^{kin}} \alpha_{\partial N}) = (-1)^{\dim(N)} \pi^* (2S_{\partial N}^{kin} - \iota_{Q_{\partial N}^{kin}} \alpha_{\partial N}) \end{aligned}$$

Here we used $\iota_{Q_{\partial N}^{kin}} \alpha_{\partial N} = S_{\partial N}^{kin}$.

$$(92) \quad \begin{aligned} L_{Q_N^{kin}} S_N^{int} &= L_{\tilde{D}_N} (T_N^0 \Theta) = (-1)^{\dim(N)} \pi^* (T_{\partial N}^0 \Theta) = \\ &= (-1)^{\dim(N)} \pi^* (S_{\partial N}^{int}) \end{aligned}$$

$$(93) \quad \begin{aligned} L_{Q_N^{int}} S_N^{kin} &= L_{\tilde{Q}} (T_N^1 \alpha) = (-1)^{\dim(N)} T_N^1 (L_Q \alpha) \quad \underbrace{\quad}_{\text{Cartan formula}} \\ &= (-1)^{\dim(N)} T_N^1 (\iota_Q d\alpha - d\iota_Q \alpha) = (-1)^{\dim(N)} T_N^1 d(\Theta - \iota_Q \alpha) \end{aligned}$$

Here we used the Cartan formula and the exactness of the symplectic form on the target, $\omega = d\alpha$. Now applying (78) we obtain

$$(94) \quad = (-1)^{\dim(N)} \pi^* T_{\partial N}^0 (\Theta - \iota_Q \alpha) = (-1)^{\dim(N)} \pi^* (S_{\partial N}^{int} - \iota_{Q_{\partial N}^{int}} \alpha_{\partial N})$$

$$(95) \quad L_{Q_N^{int}} S_N^{int} = L_{\tilde{Q}} T_N^0 \Theta = (-1)^{\dim(N)} T_N^0 (L_Q \Theta) = 0$$

Collecting (91)–(95), we obtain (90). □

The following is a corollary of Proposition 6.3 but here we give an independent proof.

Proposition D.2. *The following holds:*

$$(96) \quad L_{Q_N} \omega_N = (-1)^{\dim(N)} \pi^* \omega_{\partial N}$$

Proof. Indeed:

$$(97) \quad L_{Q_N^{kin}} \omega_N = L_{\tilde{D}_N} (T_N^0 \omega) = (-1)^{\dim(N)} \pi^* \underbrace{T_{\partial N}^0 \omega}_{\omega_{\partial N}}$$

where we used the rule (83). Also,

$$(98) \quad L_{Q_N^{int}} \omega_N = L_{\hat{Q}} (T_N^0 \omega) = (-1)^{\dim(N)} T_N^0 (L_Q \omega) = 0$$

where we used the rule (84) and the fact that the target cohomological vector field is symplectic. Putting (97) and (98) together we get (96). $\square$

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CLASSICAL AND QUANTUM LAGRANGIAN FIELD THEORIES WITH BOUNDARY

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ABSTRACT. This note gives an introduction to Lagrangian field theories in the presence of boundaries. After an overview of the classical aspects, the cohomological formalisms to resolve singularities in the bulk and in the boundary theories (the BV and the BFV formalisms, respectively) are recalled. One of the goals here (and in [7]) is to show how the latter two formalisms can be put together in a consistent way, also in view of perturbative quantization.

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1. INTRODUCTION

Lagrangian field theories are used to describe physical models. Their quantization is somehow expected to satisfy Segal's [25] axioms (which generalize in higher dimensions our understanding of quantum mechanics). Roughly speaking they say

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that a d -dimensional quantum field theory should provide a functor from the category of d -dimensional cobordisms (with possible extra structures) to the category of vector spaces and linear maps. This program has been completed in conformal field theories. Atiyah proposed a stricter version of these axioms in the case of topological field theories, which was first realized by the Reshetikhin–Turaev invariant [20] which may be thought of as some nonperturbative quantization of Chern–Simons theory [28, 11]. Another example is implicitly present in an old work by Migdal [19].

It would be interesting to understand how this picture emerges from the perturbative functional integral quantization of Lagrangian fields theories. An important application would be the construction of perturbative quantum field theories on manifolds out of topologically simple or geometrically small pieces, where the computations might be more tractable. Even in the case of topological field theories this would produce new insight (and might possibly lead to a full understanding of the relation between the perturbative expansion of Chern–Simons theory and the asymptotics of the Reshetikhin–Turaev invariant).

The first step in developing this program, performed in [7], consists in developing the analogous pictures in the classical formalism and in the BV formalism [5] which is the main tool for the perturbative quantization of theories with symmetries. In principle, this already yields the possibility of constructing moduli spaces of solutions to variational problems out of computations on topologically simple or geometrically small pieces.

This note reviews some results of [7], see Section 5, with a didactical introduction through classical Lagrangian field theory given in Sections 2 and 3, and with the BFV formalism [4], being outlined in Section 4. Notice that Section 5 is self contained, so the hasty reader who does not need a motivation or an introduction might well jump directly there.

As a final remark, notice that in this paper every manifold is assumed to be compact, though possibly with boundary.

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2. LAGRANGIAN FIELD THEORY I: OVERVIEW

We start reviewing classical Lagrangian mechanics. This is usually defined by specifying a Lagrangian function L on the tangent bundle TN of some manifold N . The action $S_{[t_0, t_1]}$ corresponding to an interval $[t_0, t_1]$ is a function on the path space $N^{[t_0, t_1]}$ defined by

$$(2.1) \quad S_{[t_0, t_1]}[x] = \int_{t_0}^{t_1} L(\dot{x}(t), x(t)) \, dt.$$

The Euler–Lagrange (EL) equations describe the critical points of the action. As this requires integration by parts, one usually puts appropriate boundary conditions for the boundary terms to vanish, e.g., one fixes the initial and final values $x(t_0)$ and $x(t_1)$. In the sequel we will want to avoid this.

2.1. Symplectic formulation. The typical example is Newtonian mechanics on a Riemannian manifold. In this case N is a Riemannian manifold and the Lagrangian is $L(q, v) = \frac{1}{2}m\|v\|^2 - V(q)$, where $\| \cdot \|$ is the norm induced by a metric on N , V is a function on N and m is a parameter (usually assumed to be strictly positive). The EL equations in this case are the Newton equations with force given by $-\nabla V$. They admit, locally, a unique solution if initial position and velocity are specified. We denote by $C = TN$ the space of such initial conditions. Here C stands here for Cauchy, but we will see later it can also stand for coisotropic. Notice the peculiar coincidence that C is the same as the space on which L is defined. This will not be the case in further examples.

One usually reformulates the problem in symplectic terms using the Legendre mapping $\phi_L: TN \rightarrow T^*N$, $(v, q) \mapsto (p(v, q), q)$, with $p_i = \frac{\partial L}{\partial v^i}$. The Newton equations of motions now become first order and their solution yields the symplectic flow Φ_{H_L} with respect to the canonical symplectic form ω_{can} on T^*M and Hamiltonian function H_L given by the Legendre transform of L : $H_L(p, q) = p_i v^i(p, q) - L(v(p, q), q)$. Here we have used the inverse of the Legendre mapping $\phi_L^{-1}: T^*N \rightarrow TN$, $(p, q) \mapsto (v(p, q), q)$, where $v(p, q)$ is the inverse to $p(v, q)$. Since in the following we will also be interested in degenerate Lagrangians, for which the Legendre mapping is not a local diffeomorphism, we now recall how to reformulate things without going to the cotangent bundle. The first simple fact is that

$$(2.2) \quad \alpha = \frac{\partial L}{\partial v^i} dq^i$$

is a well-defined one-form on $C = TN$ (it would be more precise to write α_L instead of α to stress the dependency on L , but we regard L as given for the whole discussion). Moreover, $\omega := d\alpha$ is non degenerate precisely when the Lagrangian is regular, i.e. when the Legendre mapping is a local diffeomorphism. In this case, one can also easily show that $\omega = \phi_L^* \omega_{\text{can}}$. We can now formulate the Hamiltonian evolution directly on C . For later considerations, however, it is better to consider the graph of the Hamiltonian flow instead of the flow itself. Borrowing notations from the cotangent bundle, we then consider $L_{[t_0, t_1]} = (\phi_L^{-1} \times \phi_L^{-1})(\text{graph}(\Phi_{H_L}|_{t_0}^{t_1}))$. As a Hamiltonian flow is a symplectomorphism and as the graph of a symplectomorphism is a Lagrangian submanifold of the Cartesian product with reversed sign of the symplectic form on the first factor, we have that $L_{[t_0, t_1]}$ is Lagrangian submanifold in $\bar{C} \times C$. The fact that it comes from the graph of a flow yields the property

$$(2.3) \quad L_{[t_1, t_2]} \circ L_{[t_0, t_1]} = L_{[t_0, t_2]}, \quad \lim_{t_1 \rightarrow t_0} L_{[t_0, t_1]} = \text{graph of the Id map}.$$

The limit has to be understood by putting an appropriate topology on the space of submanifolds of C .

The crucial point now is that $L_{[t_0, t_1]}$ may be defined directly without making reference to the Hamiltonian flow. Let

$$\begin{aligned} \pi_{[t_0, t_1]}: N^{[t_0, t_1]} &\rightarrow C \times C \\ \{x(t)\} &\mapsto ((\dot{x}(t_0), x(t_0)), (\dot{x}(t_1), x(t_1))) \end{aligned}$$

and let $EL_{[t_0, t_1]} \subset N^{[t_0, t_1]}$ be the space of solutions to the EL equations. Then we simply have

$$(2.4) \quad L_{[t_0, t_1]} = \pi_{[t_0, t_1]}(EL_{[t_0, t_1]}).$$

This is the fundamental equation we are going to use. Notice that (2.3) immediately follow from this new definition (the limiting property follows from the fact that C is the space of initial conditions that guarantee local existence and uniqueness).

To insure that $L_{[t_0, t_1]}$ is Lagrangian we have to make some further observations. First we observe that, if we do not impose boundary conditions, the variational calculus yields

$$(2.5) \quad \delta S_{[t_0, t_1]} = \text{EL}_{[t_0, t_1]} + \pi_{[t_0, t_1]}^* \alpha,$$

where $\text{EL}_{[t_0, t_1]}$ is the term containing the EL equations that we reinterpret as a one-form on $N^{[t_0, t_1]}$. Notice that we also interpret the variation symbol δ as the de Rham differential on $\Omega^\bullet(N^{[t_0, t_1]})$. The appearance of the same α here and in (2.2) is crucial. The equation yields in fact, after differentiation, $\pi_{[t_0, t_1]}^* \omega = -\delta \text{EL}_{[t_0, t_1]}$, which in turn implies that the restriction of ω to $L_{[t_0, t_1]}$ vanishes (observe here that the space of solutions to the EL equations on $M^{[t_0, t_1]}$ is the zero locus of $\text{EL}_{[t_0, t_1]}$ and that $\pi_{[t_0, t_1]}$ is a surjective submersion). This amounts to saying that $L_{[t_0, t_1]}$ is isotropic.

In addition we know that a unique solution is given, locally, once we specify initial conditions in TN or, equivalently, if we specify initial and final position for a short enough time interval $[t_0, t_1]$. Hence, if we fix initial and final positions Q_0 and Q_1 and let $L_{Q_0, Q_1} := \{(v, q, v', q') \in C \times C \mid q = Q_0, q' = Q_1\}$, we get that, for t_1 sufficiently close to t_0 , $L_{Q_0, Q_1} \cap L_{[t_0, t_1]}$ consists of one point; by dimension counting, assuming the intersection is transversal, this implies that $L_{[t_0, t_1]}$ has half dimension than C and so is Lagrangian¹. Finally, if t_0 and t_1 are not close, we can decompose the interval into short ones on which we can use the previous argument and recover $L_{[t_0, t_1]}$ as the composition of the canonical relations corresponding to the subintervals. Hence $L_{[t_0, t_1]}$ is also a canonical relation.

After understanding this, we can also think of more general boundary conditions by replacing L_{Q_0, Q_1} with another submanifold L of $\overline{C} \times C$ on which α vanishes. The latter condition ensures that the variational problem has no boundary contributions. It also implies that L is isotropic. In order to have, generically, intersection points of L with $L_{[t_0, t_1]}$, one has to require L to have maximal dimension, and hence to be Lagrangian². Now, the intersection $L \cap L_{[t_0, t_1]}$ can be considered as the space of solutions to the EL equations. Notice however that this intersection might as well be empty or contain (infinitely) many points, though generically, it will be a discrete set.

2.2. A degenerate example: geodesics on the Euclidean plane. We now consider the non-regular Lagrangian $L(v, q) := \|v\|$, where $\|\cdot\|$ is the Euclidean norm and $v, q \in T\mathbb{R}^2$. The action is still given by (2.1) which we now define only on the space $N_0^{[t_0, t_1]}$ of immersed paths (i.e., we impose the condition $\dot{\gamma}(t) \neq 0 \forall t \in [t_0, t_1]$). The EL equations have as solutions parameterized segments of straight lines in $\mathbb{R}^2$. By analogy with Newtonian mechanics the Cauchy data include initial position and initial velocity, but these data do not give uniqueness: they define one end of the segment and its slope but do not define a parametrization of the

¹Here is another argument. For a short interval, the initial and final positions specify a unique solution. Hence initial and final positions determine initial and final velocities, which implies the L is a graph, hence Lagrangian.

²If L is Lagrangian but α does not vanish on it, we can modify the action by adding boundary terms such that the modified one-form α is vanishing on L

segment uniquely. Nevertheless, let us consider what will happen if we take the “wrong” space of Cauchy data $C = T\mathbb{R}^2 \setminus \{\text{zero section}\}$. We defer to Remark 2.3 the discussion of the “true” space of Cauchy data.

Using polar coordinates for the velocities, we can write $C = \mathbb{R}^2 \times S^1 \times \mathbb{R}_{>0}$ with coordinates the position $\mathbf{q}$, the normalized velocity $\mathbf{v}$ (i.e. $\|\mathbf{v}\| = 1$) and the speed ρ . Then (2.2) yields $\alpha = \mathbf{v} \cdot d\mathbf{q}$, so that $\omega = d\mathbf{v} \cdot d\mathbf{q}$ is clearly degenerate (as it does not have a $d\rho$ component). On the other hand, (2.4) (with π the evaluation of position and velocity at the initial and at the final point) yields

$$L_{[t_0, t_1]} = \{(\mathbf{q}_0, \mathbf{v}, \rho_0), (\mathbf{q}_1, \mathbf{v}, \rho_1) | (\mathbf{q}_1 - \mathbf{q}_0) \text{ parallel to } \mathbf{v}\}$$

which is clearly not a graph but can easily be checked to be Lagrangian in $\overline{C} \times C$ (see Appendix A for the definition of Lagrangianity in the case of a degenerate two-form).

In this example one can easily get rid of the degeneracy of ω by taking the quotient by its kernel which is the span of the vector fields $\mathbf{v} \cdot \frac{\partial}{\partial \mathbf{q}}$ and $\frac{\partial}{\partial \rho}$ (geometrically, these vector fields represent the space directions parallel to the velocity and the rescalings of the velocity). The quotient turns out to be TS^1 with symplectic structure given by pullback of the canonical one on T^*S^1 by the induced metric on S^1 . Notice that the base S^1 here is the space of normalized velocities, whereas the tangent fiber can be thought of as the space direction orthogonal to the given velocity. One can also project $L_{[t_0, t_1]}$ down to the quotient $TS^1 \times TS^1$. The result is just the graph of the identity map on TS^1 . This is a consequence of the fact that the action is invariant under reparametrization. Notice that this is an example of topological theory: the action does not depend of the metric on $[t_0, t_1]$.

In this example we passed to the quotient space which appeared to be smooth. However that in general reduction may produce very singular quotients, so passing directly to the reduced space had better be avoided. Instead, as we will see, it is better to use the BV-BFV approach.

Remark 2.1. Notice that the one-form $\alpha = \mathbf{v} \cdot d\mathbf{q}$ is not horizontal with respect to the kernel of ω , so it cannot be reduced to TS^1 . On the other hand, we may regard α as a connection one-form on the trivial line bundle on C . We can then reduce this line bundle to a line bundle over TS^1 and reduce α as a connection.

Also notice that evaluating the action on a solution yields a well-defined function S_{HJ} (the Hamilton–Jacobi action) on $L_{[t_0, t_1]}$ which is just the length of the path. Again, S_{HJ} cannot be reduced to a well-defined function on $\overline{TS^1} \times TS^1$, but $\exp \frac{1}{\hbar} S_{HJ}$ can be reduced to a section of the reduced line bundle.

Remark 2.2. If one considers the same example but now with the Minkowski metric, also in higher dimensions, and considers only timelike velocities (i.e., with $ds^2 = dt^2 - d\mathbf{x}^2$, one assumes $\|v\| > 0$), the reduction yields TH , where H is the upper hyperboloid $v_0^2 - \mathbf{v}^2 = 1$, and the symplectic structure is obtained by pullback of the canonical one on the cotangent bundle by the hyperbolic metric on H (which is induced by the Minkowski metric). The reduced Lagrangian is again just the graph of the identity.

Remark 2.3. We mentioned above that $T\mathbb{R}^2 \setminus \{\text{zero section}\}$ is not the true space of initial conditions because giving initial position and velocity does not select a unique parametrized segment. In order to obtain uniqueness in a formal neighborhood of the initial point we can enlarge C by setting $\tilde{C} = T\mathbb{R}^2 \setminus \{\text{zero section}\} \times \mathbb{R}^\infty$, where

the coordinates on $\mathbb{R}^\infty$ are all higher derivatives of the path. In other words here we work with the space of jets. Define the one form $\tilde{\alpha}$ as the pull-back of α with respect to the natural projection $\tilde{C} \rightarrow C$. It is clear the $\mathbb{R}^\infty$ factor is in the kernel of $\tilde{\alpha}$ and of $\tilde{\omega} = \omega$. This is the reason why this factor can be completely neglected just as we neglected the space $\mathbb{R} > 0$ of speeds in the discussion above. The reduction of $\tilde{C}$ is the same as that of C . Boundary values (2.4) of solutions to the Euler-Lagrange equations define a Lagrangian subspace in $(\tilde{C} \times \tilde{C})$. Its reduction is again the graph of the identity.

2.3. Example: Free scalar field theory. We now describe an example of a field theory in dimension d . The space time in such theory is a Riemannian d -manifold (M, g_M) . The space of fields is $\mathbb{R}^M$ (functions on M) and the action is

$$S_{(M, g_M)}[\phi] = \frac{1}{2} \int_M g_M^{\mu\nu} \partial_\mu \phi \partial_\nu \phi \, \text{dvol}_{g_M} = \frac{1}{2} \int_M (d\phi, d\phi) \, \text{dvol}_{g_M}$$

where $\phi \in \mathbb{R}^M$. Solutions to the EL equations are harmonic functions on M .

In order to understand the boundary structure for an arbitrary space time manifold M consider first a thin neighborhood of its boundary $\Sigma = \partial M$. That is consider a short cylinder $\Sigma \times [0, \epsilon]$ where (Σ, g_Σ) is a $(d-1)$ -dimensional Riemannian manifold and $g_M = g_\Sigma + ds^2$, $s \in [0, \epsilon]$. A unique solution to the EL equation is obtained if one specifies the values of ϕ and of its normal derivative on $\Sigma \times \{0\}$. This gives the natural space of Cauchy data associated to Σ , $C_\Sigma = \mathbb{R}^\Sigma \times \mathbb{R}^\Sigma$. Similarly to (2.2) the boundary term in the variation of $S_{\Sigma \times [0, \epsilon]}$ gives the one-form

$$\alpha_{(\Sigma, g_\Sigma)} = \int_\Sigma \chi \delta \phi \, \text{dvol}_{g_\Sigma}$$

with $(\chi, \phi) \in C_\Sigma$. Here χ should be thought of as the restriction to $\Sigma \times \{0\}$ of the normal derivative of the bulk field ϕ . Notice that $\omega_{(\Sigma, g_\Sigma)} = \delta \alpha_{(\Sigma, g_\Sigma)}$ is (weakly) nondegenerate.

For a general Riemannian d -manifold M with boundary ∂M , we have the surjective submersion $\pi_M: \mathbb{R}^M \rightarrow C_{\partial M}$ obtained by evaluating the field ϕ and its normal derivative on the boundary. Formula (2.5) still holds (with a little change in notation):

$$\delta S_{(M, g_M)} = \text{EL}_{(M, g_M)} + \pi_M^* \alpha_{(\partial M, g_M|_{\partial M})}.$$

Moreover, in the spirit of (2.4), define $L_{(M, g_M)} \subset C_{\partial M}$ as $\pi_M(EL_{(M, g_M)})$.

It is easy to see that $L_{(M, g_M)}$ is Lagrangian. Indeed, the Dirichlet problem for ϕ has unique solution on M . Thus, if $\phi \in EL_M$, its boundary values define the the normal derivative of ϕ at the boundary. This map from the Dirichlet data to Neumann data is known as the Dirichlet-to-Neumann mapping. Thus, the submanifold $L_M \in C_{\partial M}$ is the graph of the Dirichlet-to-Neumann mapping $\mathbb{R}^{\partial M} \rightarrow \mathbb{R}^{\partial M}$.

Notice that one may distinguish the connected components of ∂M into incoming and outgoing: $\partial M = \partial_{\text{in}} M \sqcup \partial_{\text{out}} M$. Denoting $\partial_{\text{in}} M$ with opposite orientation by $\partial_{\text{in}} M^{\text{opp}}$, we may then view $L_{(M, g_M)} \subset \overline{C}_{\partial_{\text{in}} M^{\text{opp}}} \times C_{\partial_{\text{out}} M}$ as a canonical relation from $C_{\partial_{\text{in}} M^{\text{opp}}}$ to $C_{\partial_{\text{out}} M}$.

2.4. Conclusions. From these examples we see that the Hamiltonian framework for non-regular Lagrangians has to be replaced by its weaker version. However, certain important patterns remain. We have seen in these examples that that *i*) in all cases we were able to derive a, possibly degenerate, two-form on the space

of initial conditions associated to the boundary of the space time and *ii*) we were able to assign to the bulk of the space time a Lagrangian/isotropic submanifold (not necessarily a graph) in such spaces. In the next Section we will see that this is a quite general fact.

3. LAGRANGIAN FIELD THEORY II (AFTER V. FOCK)

In an unpublished account [10], V. Fock has considered the general structure of Lagrangian field theories on manifolds with boundary. We give here our recapitulation of this account. (To different levels of generality, this structure has been rediscovered many times, see e.g. [13, 24] and references therein.)

Notice that this Section is rather about a philosophical account leading to a concrete construction than a precise mathematical formulation. For simplicity, we also assume that all the "spaces" occurring in the following are actually (possibly infinite dimensional) manifolds.

A Lagrangian field theory is specified by

- (1) fixing the dimension d of the source manifolds;
- (2) fixing a class of d -manifolds, possibly with extra structure, such as a metric in the example of subsection 2.3;
- (3) associating a space of fields F_M (functions, maps to a fixed target manifold, sections of bundles, connections, ...) to every d -manifold M in the class;
- (4) defining a density, the Lagrangian L , of the fields and finitely many of their derivatives.³

The action functional associated to a manifold M , as a function on F_M , is then given by $S_M = \int_M L$. The variation of the action, neglecting the boundary terms, yields the EL equations. We denote by $EL_M \subset F_M$ the space of solutions to the EL equations on a given manifold M .

Let now Σ be a $(d-1)$ -manifold. We extend it to a d -manifold $M := \Sigma \times [0, \epsilon]$ (taking care of the possible additional structure). The variational calculus on this particular M produces two new pieces of data:

- (1) The space C_Σ of Cauchy data consisting on the information on the fields (and their derivatives) that one has to specify on Σ so that there is a unique solution to EL equations on $\Sigma \times [0, \epsilon]$ for ϵ small enough (possibly, one might have to work with a formal neighborhood of 0 like in the example of subsection 2.2).
- (2) A one-form α_Σ on C_Σ arising from the $\Sigma \times \{0\}$ -boundary contribution to the variation of $S_{\Sigma \times [0, \epsilon]}$.

One can see that the Lagrangian is regular iff $\omega_\Sigma := d\alpha_\Sigma$ is non degenerate.

Using these data, one can further develop the induced structure. Namely, for every M in the class we now have a surjective submersion

$$\pi_M: F_M \rightarrow C_{\partial M}$$

and the variation of the action leads to the fundamental equation

$$(3.1) \quad \delta S_M = EL_M + \pi_M^* \alpha_{\partial M}.$$

Again we define $L_M := \pi_M(EL_M) \subset C_{\partial M}$. It follows from (3.1) that L_M is isotropic (i.e., the restriction of $\omega_{\partial M}$ vanishes). In most examples L_M is actually Lagrangian.

³For a precise definition, see, e.g., [9]

Remark 3.1. From now on for simplicity we are going to assume that all F_M and C_Σ can be given a manifold structure and that EL_M and L_M are smooth submanifolds and, apart from the following counterexample, we are always going to assume that L_M is Lagrangian.

Example 3.2 (L_M is not Lagrangian). Consider a one-dimensional example with target $\mathbb{R}^2$, space of fields $F_{[t_0, t_1]} = (\mathbb{R}^2)^{[t_0, t_1]}$, with Lagrangian $L \in C^\infty(T\mathbb{R}^2)$ given by $L(v_x, v_y, x, y) = \frac{1}{2}y v_x^2$ and with the action $S_{[t_0, t_1]} = \int_{t_0}^{t_1} \frac{1}{2}y(t) \dot{x}(t)^2 dt$. This example is related to the one given in [16](section 1.2.2) where it is used to disprove a related conjecture by Dirac. Note also that this is a 1-dimensional version of the Polyakov string action.

The EL equations are $\dot{x}^2 = 0$ and $\frac{d}{dt}(y \dot{x}) = 0$, the latter being trivially implied from the first. The x -component of a solution is then completely determined by its initial value, whereas the y -component is completely free. To get formal uniqueness of solutions to the EL equations define $C = \mathbb{R} \times \mathbb{R}^\infty$. Here the second factor contains the information about y and all its derivatives at the initial time. The variation of the action is $\int_{t_0}^{t_1} (\frac{1}{2}\delta y \dot{x}^2 + y \dot{x} \delta \dot{x}) dt$. However the boundary term here is absent because we have to assume $\dot{x} = 0$ on the boundary in order to have a solution and therefore $\alpha = 0$. The projection $\pi_{[t_0, t_1]}: F_{[t_0, t_1]} \rightarrow C \times C$ is then simply given by

$$\pi(x(\cdot), y(\cdot)) = (x(t_0), y(t_0), \dot{y}(t_0), \ddot{y}(t_0), \dots, x(t_1), y(t_1), \dot{y}(t_1), \ddot{y}(t_1), \dots)$$

so that $L_{[t_0, t_1]} = \{(x, y_0, y_1, y_2, \dots, x, \tilde{y}_0, \tilde{y}_1, \tilde{y}_2, \dots), x, y_i, \tilde{y}_i \in \mathbb{R} \forall i\}$. Now $L_{[t_0, t_1]}$ is obviously isotropic since $\omega = 0$. On the other hand, since $\omega = 0$, $L_{[t_0, t_1]}^\perp = C \neq L$. Hence $L_{[t_0, t_1]}$ is not Lagrangian.

As in subsection 2.3, we can decide to split the boundary of a given d -manifold M into incoming and outgoing boundary components, $\partial M = \partial M_{\text{in}} \sqcup \partial M_{\text{out}}$, and regard L_M as a canonical relation from $C_{\partial M_{\text{in}}^{\text{opp}}}$ to $C_{\partial M_{\text{out}}}$, which we will call the evolution relation since it generalizes the evolution flow.

Suppose we cut a manifold M along a submanifold Σ into two manifolds M_1 and M_2 in such a way that $\partial_{\text{in}} M \subset \partial M_1$ and $\partial_{\text{out}} M \subset \partial M_2$. Then we set $\partial_{\text{in}} M_1 = \partial_{\text{in}} M$, $\partial_{\text{out}} M_1 = \Sigma$, $\partial_{\text{in}} M_2 = \Sigma^{\text{opp}}$ and $\partial_{\text{out}} M_2 = \partial_{\text{out}} M$. We then have

$$L_M = L_{M_2} \circ L_{M_1}$$

since a solution on M corresponds to solutions on M_1 and M_2 that match on Σ . This composition of canonical relations replaces the usual composition of flows.

In particular, in the case of a cylinder $\Sigma \times [t_0, t_1]$ we have

$$L_{\Sigma \times [t_0, t_1]} = L_{\Sigma \times [t, t_1]} \circ L_{\Sigma \times [t_0, t]}$$

for all $t \in (t_0, t_1)$. On the other hand, in general we cannot expect $L_\Sigma := \lim_{t_1 \rightarrow t_0} L_{\Sigma \times [t_0, t_1]}$ to be the graph of the identity on C_Σ as this will happen only in regular theories. We call gauge theories those for which L_Σ , which can be seen to be an equivalence relation, is not the graph of the identity.

Remark 3.3 (Evolution correspondences). Recall that the Cauchy space C_Σ determines uniqueness only locally (or just formally, like in example 3.2 or in remark 2.3) on cylinders. Therefore, we may miss some important information by looking only at L_M . The information about nonuniqueness is contained in the fibers of $\pi_M: EL_M \rightarrow L_M$. Notice that by cutting M along some Σ as above we get $EL_M = EL_{M_2} \times_{C_\Sigma} EL_{M_1}$, so we may also interpret EL_M as a canonical correspondence, the evolution correspondence.

It is tempting to think in terms of gluing manifolds along boundary components instead of cutting them (even though this might require some extra pieces of data, like collars, or to work up to homotopy as in [18]). From this perspective we could think of Lagrangian field theories as inducing a functor from the cobordism category (of manifolds with appropriate structure) to the extended presymplectic category. We may think of this as the classical version of the Segal–Atiyah [25, 3] axioms for quantum field theory.

Remark 3.4. Recall that a closed two-form on a manifold P is called presymplectic if its kernel is a subbundle of TP (in finitely many dimensions this is equivalent to requiring ω to have constant rank). There is no reason why the two-forms ω_Σ we obtained above should be presymplectic, but this is a fundamental requirement for making sense of the rest of this program. This requirement puts some constraints on the theories one can write down.

Remark 3.5. In the above description the two-form is always exact. On the other hand, physical examples with non exact symplectic forms abound. One source for them is reduction, see the next subsection, others arise from dropping the restrictive condition that the action is a (well defined) function. More generally, one should think of the action S , or rather of the Gibbs weight $\exp \frac{i}{\hbar} S$, as the section of a line bundle over F_M . In this more general setting, $\alpha_\Sigma/\hbar$ is no longer a one-form on C_Σ , but a connection one-form on a line bundle.

A simple example where this occurs is that of a charged particle moving in a magnetic field on a manifold N . The action contains the term $\int_{t_0}^{t_1} A_i(x(t)) \dot{x}^i(t) dt$, where $A = A_i dx^i$ is the vector potential regarded as a one-form. This term is also equal to $\int_\gamma A$, where γ is the image of the path. If we make a gauge transformation, the action then changes by boundary terms. Such action is not a function on the space of paths when A is a connection on some nontrivial line bundle E over N . Using the evaluation map at the endpoints, we can pullback this line bundle to $F_{[t_0, t_1]}$. Namely, we define $E_{[t_0, t_1]} = \text{ev}_{t_0}^* E^* \otimes \text{ev}_{t_1}^* E$. We can then see the Gibbs weight as a section of $(E_{[t_0, t_1]})^{\otimes k}$ (where $k = 1/\hbar$ is an integer). The boundary one-form α has the term A as a contribution from the magnetic term in the action and therefore $\alpha/\hbar$ is defined only as a connection on $p^* E^{\otimes k}$, where p is the projection from $C = TN$ to N . The symplectic form ω is the canonical one for a particle on N plus the curvature of A .

Another example is the WZW model, as discussed in [12] [14]. In this paper, for simplicity, we will assume that the action is defined as a function.

3.1. Reduction. If the two-form ω_Σ is degenerate, one may perform reduction by its kernel. If the leaf space $\underline{C}_\Sigma$ is smooth it inherits a symplectic structure $\underline{\omega}_\Sigma$. We may also project an evolution relation L_M to the reduction and denote this by $\underline{L}_M$. If it is a smooth submanifold of $\underline{C}_\Sigma$, it is automatically isotropic. Actually, in all the examples at hand it is Lagrangian. (In example 3.2, this is trivial since the reduction of C is zero-dimensional.) Finally, notice that at the reduced level $\lim_{t_1 \rightarrow t_0} \underline{L}_{\Sigma \times [t_0, t_1]}$ is the graph of the identity on $\underline{C}_\Sigma$. In TFTs, this is so even without taking the limit.

Remark 3.6. In general we cannot expect α_Σ to be horizontal with respect to the kernel of ω_Σ , even though this happens in most examples discussed in this paper (with the notable exception of subsection 2.2). If this is not the case, we should

regard α_Σ as a connection one-form on the trivial line bundle $E_\Sigma := C_\Sigma \times \mathbb{C}$. Since, by definition, the restriction of this line bundle to each leaf is flat, we may reduce to a line bundle with connection $(\underline{E}_\Sigma, \underline{\alpha}_\Sigma)$ if the holonomy of α_Σ is trivial on each leaf. Equivalently, we may think of α_Σ as a contact form on the total space of E_Σ . Its reduction, if smooth, will be a contact manifold $\underline{E}_\Sigma$. Under the same conditions as above it will be the total space of a line bundle with connection over $\underline{C}_\Sigma$.

Remark 3.7. In general the reduced space $\underline{C}_\Sigma$ is singular and we want to avoid reduction. We will see in the next Sections how to give a good cohomological replacement for it. However, some partial reduction is very often possible and useful. See for example Remark 2.3. We will see several other examples in the following.

3.2. Axiomatization. By the above discussion we see that a Lagrangian field theory in d dimensions induces the following "categorical" description:

- The source category is a category of cobordisms: objects are $(d-1)$ -manifolds and morphisms are d -manifolds with boundary. Depending on the theory there might be restriction or additional data (e.g., a metric). Composition of morphisms is given by gluing along boundary components; one way to make sense of this consists in putting a choice of collar of the boundary in the additional data.
- The target "category" has (usually infinite-dimensional) presymplectic manifolds as its objects and correspondences with Lagrangian (or just isotropic) image as morphisms.

A few comments are in order.

- (1) What is actually important is not really gluing manifolds along common boundary, but cutting manifolds along submanifolds. This structure is more relevant than the categorical structure and much less problematic.
- (2) In the case of a regular field theory, the dynamics of the problem on the space time M may be recovered by choosing boundary conditions—viz., the choice of a submanifold L of the symplectic manifold $C_{\partial M}$ —and take the fiber of the evolution correspondence EL_M over the intersection points between L and the evolution relation L_M as the space of solutions for these boundary conditions. We might require for a field theory to be good that these fibers should be generically finite dimensional; we call elements of these fibers the (classical) vacua of the theory.

In order for the variational problem to be well-defined, we have to avoid boundary terms and, as a consequence, require L to be such that the restriction of $\alpha_{\partial M}$ to it vanishes. This in particular implies that L must be isotropic. In order to have, generically, solutions, we should also require L to be maximally isotropic, i.e., Lagrangian.

- (3) In a non regular theory, boundary conditions are also given by the choice of a Lagrangian submanifold L on which the one-form α vanishes. In the reduced theory, one considers the intersection between the reduction of L and that of the evolution relation L_M and the fibers over them. In addition, one also has to consider a reduction of these fibers. This is an additional piece of data (not contained in the Lagrangian function defining the field theory). A more refined definition of the target category would then require endowing the evolution correspondences EL_M with an integrable distribution—the

gauge symmetries of the theory—with the consistency requirement that its image in the evolution relation L_M should coincide with the restriction of the characteristic distribution of the presymplectic manifold.

- (4) The above setting might be too rigid as the one-form $\alpha_{\partial M}$ might not restrict to zero on the Lagrangian submanifold L one wishes to consider. To add more flexibility, one can allow changing $\alpha_{\partial M}$ by an exact term df . By consistency with (3.1) we see that in the Lagrangian field theory we started with we have to change the action by $\pi_M^* f$. To preserve locality we might want f itself to be a local functional. By using Stokes Theorem, we may also write this as a bulk term. The original Lagrangian is changed by a total derivative (which is hence invisible on a manifold without boundary).
- (5) In the target “category,” we might also work in the more refined setting where objects are endowed with a line bundle with connection whose curvature is the presymplectic form (a prequantization bundle). The shift of $\alpha_{\partial M}$ by an exact form in the previous comment should now be replaced by a gauge transformation for the connection one-form. In addition, we may take care of the Hamilton–Jacobi action as a covariantly closed section of the pullback of the flat line bundle from the evolution relation to the evolution correspondence. If the fibers of the correspondence over the relation are connected, this defines a section of the flat line bundle over the relation. In many relevant examples in addition the line bundle over the presymplectic manifold is trivial; in these cases, the presymplectic form is exact and the Hamilton–Jacobi action is a function.
- (6) It might also make sense to allow for singular presymplectic manifolds or for singular relations/correspondences.

3.3. Perturbative quantization. The perturbative functional integral may be extended in the presence of boundary.⁴ Assume first that the theory is regular. For simplicity, assume that the symplectic manifold $C_{\partial M}$ is endowed with a Lagrangian foliation along which $\alpha_{\partial M}$ vanishes and with a smooth leaf space $B_{\partial M}$.⁵ Denote by $p_{\partial M}$ the projection $C_{\partial M} \rightarrow B_{\partial M}$. We then define the boundary vector space $H_{\partial M}$ as the space of functions on $B_{\partial M}$ and the state ψ_M associated to the bulk M as

$$\psi_M(\phi) = \int_{\Phi \in \pi_M^{-1}(p_{\partial M}^{-1}(\phi))} e^{\frac{i}{\hbar} S_M(\Phi)} [D\Phi].$$

The integral is defined by the formal saddle point approximation around critical points. As explained in the previous subsection, we may allow for some finite-dimensional degeneracy. In this case, we should think of ψ_M as a function on the total space of the bundle of vacua over $B_{\partial M}$. If this makes sense, one could also eventually perform the remaining finite-dimensional fiber integration.

In non-regular theories we have two (related) problems. The first is that on the boundary space of fields we only have a presymplectic structure. The second is that the critical points are degenerate (with infinite-dimensional fibers). To approach this problem we have to require as an additional piece of data the choice of gauge symmetries. The idea is that the situation reduces to the regular one if we mod

⁴What we call here perturbative perhaps should be called semiclassical. Strictly speaking the perturbative expansion would be taking a formal power series expansion in coupling constants of the action.

⁵A more general setting would require a discussion of geometric quantization.

out by gauge symmetries in the bulk and by the characteristic foliation on the boundary. However, this usually leads to singular spaces and, even when it is not the case, one should make sense of the functional integral on the quotient. The way out is to replace reduction by its cohomological version. When $\partial M = \emptyset$ this goes under the name of BV formalisms [5] and it is known as BFV formalism [4] in the case of boundary reduction by the characteristic foliation. The goal of this note (and of [7]) is to show how the two formalisms fit together in a consistent way.

3.4. An alternative approach. Instead of introducing the space C_Σ of Cauchy data directly, one can “derive” it from the following construction which is somehow more natural and better fitted to the BFV formalism which will be discussed in Section 4. The space of Cauchy data obtained in this way may not coincide with the one introduced previously, but the two construction agree after reduction.

The main idea is to associate to a $(d-1)$ -manifold Σ the space $\tilde{F}_\Sigma$ of germs of fields at $\Sigma \times \{0\}$ on $\Sigma \times [0, \epsilon]$. We will call it the space of preboundary fields. The boundary term in the variational calculus yields, as above, a one-form $\tilde{\alpha}_\Sigma$ on $\tilde{F}_\Sigma$ and the fundamental equation (3.1) now reads

$$(3.2) \quad \delta S_M = EL_M + \tilde{\pi}_M^* \tilde{\alpha}_{\partial M},$$

where $\tilde{\pi}_M$ is the natural surjective submersion from F_M to $\tilde{F}_{\partial M}$.

We then introduce $\tilde{\omega}_\Sigma := d\tilde{\alpha}_\Sigma$. This two-form will have a huge kernel but is assumed to be presymplectic. We denote by $(F_\Sigma^\partial, \omega_\Sigma^\partial)$ the reduced space, which we will call the space of boundary fields. For the rest of the discussion, we are going to assume F_Σ^∂ is a smooth manifold. In general, we do not require $\tilde{\alpha}_\Sigma$ to be basic. If the trivial line bundle with connection $\tilde{\alpha}_\Sigma$ on $\tilde{F}_\Sigma$ may be reduced to a smooth line bundle on F_Σ^∂ , we will denote by α_Σ^∂ the induced connection one-form.

For simplicity, we are now going to assume that $\tilde{\alpha}_\Sigma$ is indeed horizontal, so α_Σ^∂ is a one-form on F_Σ^∂ , and leave the general case to the reader. If we denote by π_M the composition of $\tilde{\pi}_M$ with the natural projection from $\tilde{F}_{\partial M}$ to $F_{\partial M}^\partial$, we get

$$\delta S_M = EL_M + \pi_M^* \alpha_{\partial M}^\partial.$$

Out of it we get that $L_M := \pi_M(EL_M)$ is isotropic. We finally define the (new version of the) space of Cauchy data C_Σ as the space of points of F_Σ^∂ that can be completed to a pair belonging to $L_{\Sigma \times [0, \epsilon]}$ for some ϵ . Notice that we think of F_Σ^∂ as a relation from the one-point manifold to F_Σ^∂ and of $L_{\Sigma \times [0, \epsilon]}$ as a relation from F_Σ^∂ to itself, we may write

$$C_\Sigma = \bigcup_{\epsilon \in (0, +\infty)} L_{\Sigma \times [0, \epsilon]} \circ F_\Sigma^\partial.$$

Notice that F_Σ^∂ is coisotropic in itself. If $\forall \epsilon$ we assume $L_{\Sigma \times [0, \epsilon]}$ also to be so—and hence to be Lagrangian—, then each composition is coisotropic (up to some infinite-dimensional subtleties), and so will be the union. However, it may happen that C_Σ is coisotropic even if the L s are not Lagrangian.

3.4.1. Examples.

Example 3.8. Consider a nondegenerate Lagrangian function on TN as at the beginning of Section 2. The space $\tilde{F}_{pt}$ is just the infinite jet bundle over N . The one-form is given in (2.2). The kernel of the corresponding two-form consists of all

jets higher than the first, so $F_{pt}^\partial = TN$. Since every point in it can be completed to a pair in $L_{[0,\epsilon]}$, for ϵ small enough, we recover $C_{pt} = TN$.

Example 3.9. In the case discussed in subsection 2.2, $\tilde{F}_{pt}$ is the open submanifold in the infinite jet bundle over $\mathbb{R}^2$ obtained by requiring the first jet to be different from zero. The reduced space F_{pt}^∂ is TS^1 with symplectic form obtained by pullback from the cotangent bundle using the metric and L_I is the graph of the identity for every interval I ; hence, $C_{pt} = F_{pt}^\partial = TS^1$. Notice that in this example the space of Cauchy data given by this construction is different from the previous one, though their reductions are obviously the same. Moreover, in this example the one-form $\tilde{\alpha}_{pt}$ is not basic. The induced one-form connection α_{pt}^∂ is the one discussed in remark 2.1.

Example 3.10. We now work out the new description of example 3.2. Here $\tilde{F}_{pt}$ is the infinite jet bundle over $\mathbb{R}^2$ and $\tilde{\alpha}_{pt} = yv_x dx$. The kernel of the two-form is given by all jets higher than the first for x , by all jets higher than the zero jet for y , and by $X := y\frac{\partial}{\partial y} - v_x\frac{\partial}{\partial v_x}$. So in this case the form is not presymplectic. To solve this problem we assume $(v_x, y) \neq (0, 0)$. This means that the original space of fields F_I has to be defined as paths in $\mathbb{R}^2$ that can hit the x -axis only with non zero x -velocity. The reduction is then $F_{pt}^\partial = \mathbb{R}^2$. If we denote by (p, q) its coordinates we have $\alpha_{pt}^\partial = pdq$. Moreover, $\pi_{[t_0, t_1]}(x(\cdot), y(\cdot)) = (y(t_0)\dot{x}(t_0), x(t_0), y(t_1)\dot{x}(t_1), x(t_1))$. Since EL_I consists of paths that are constant in the x -direction, we get $L_{[t_0, t_1]} = \{(0, q, 0, q), q \in \mathbb{R}\}$ which is clearly not Lagrangian. On the other hand, we have $C_{pt} = \{(0, q), q \in \mathbb{R}\}$ which is coisotropic.

Example 3.11 (Electrodynamics). We now discuss the case of electrodynamics (we leave to the reader the generalization to nonabelian Yang–Mills theory). The space of fields on a manifold M is the space $\mathcal{A}_M$ of connection one-forms for a fixed line bundle over M . The action is $S_M(A) = (dA, dA) = \int_M dA \wedge *dA$, where $(\cdot, \cdot)$ is the Hodge pairing of forms for a fixed metric on M and $*$ is the Hodge $*$ -operation. The EL equations are $d^*dA = 0$, where d^* is the formal adjoint of d with respect to the Hodge pairing.

For simplicity of exposition, we now reformulate electrodynamics in the first-order formalism (and leave to the reader its study in the usual second-order formalism). Namely, we enlarge the space of fields to $F_M := \mathcal{A}_M \times \Omega^{d-2}(M)$ and extend the action to

$$S_M(A, B) = \int_M B \wedge dA + \frac{1}{2} B \wedge *B$$

The EL equations are $dA + *B = 0$ and $dB = 0$. Hence, B is completely determined by A and then A must satisfy $d^*dA = 0$.

On the space of preboundary fields $\tilde{F}_\Sigma$ we have the one-form $\tilde{\alpha}_\Sigma = \int_\Sigma B \wedge \delta A$. It then follows that the space of boundary fields is $F_\Sigma^\partial = \mathcal{A}_\Sigma \times \Omega^{d-2}(\Sigma)$ with one-form $\alpha_\Sigma^\partial = \int_\Sigma B \delta A$ and symplectic form $\omega_\Sigma^\partial = d\alpha_\Sigma^\partial$.

Let us now consider $L_{\Sigma \times [0, \epsilon]}$. The equation $dB = 0$ restricts to the boundary, so it has to be satisfied by a field in $L_{\Sigma \times [0, \epsilon]}$. We will call the direction along $[0, \epsilon]$ vertical. The evolution will impose some other conditions, but we claim that $C_\Sigma = \mathcal{A}_\Sigma \times \Omega_{closed}^{d-2}(\Sigma)$, namely that no other conditions have to be imposed on the first boundary component. The reason is that A on the first boundary can always be extended to a solution. In the axial gauge (i.e., when we require that the vertical component of A vanishes), the solution is unique once we specify the

vertical derivative of A at the first boundary. But this first derivative may be chosen, actually uniquely, so as to yield the given closed B on the boundary.

Notice that B —despite the notation—is the electric field on the boundary (or, better, the $(d-1)$ -form corresponding to the electric vector field using the metric) and the equation $dB = 0$ is just the Gauss law (i.e., divergence of electric field equal to zero). The characteristic distribution on C_Σ consists of just the gauge transformations on A .

Example 3.12 (Abelian BF theories). We may consider the “topological” limit of the first-order formulation of electrodynamics, i.e., drop the term with the Hodge $*$ operator. This way we get the action $S_M = \int_M B \wedge dA$ on the same space of fields $F_M = \mathcal{A}_M \times \Omega^{d-2}(M)$. This theory is called abelian BF theory.

The space of boundary fields is the same as for electrodynamics. What changes are the Lagrangian submanifolds L_M . Since the EL equations are just $dA = 0$ and $dB = 0$, we see that L_M consists closed A and B on the boundary that can be extended to closed A and B in the bulk. As a result $C_\Sigma = \mathcal{A}_\Sigma^{flat} \times \Omega_{closed}^{d-2}(\Sigma)$. The characteristic distribution consists of gauge transformations for A and shifts of B by exact forms.

4. THE BFV FORMALISM

In this Section we address the problem of reformulating the reduction of a presymplectic manifold C cohomologically.

If we work in the setting of subsection 3.4, our presymplectic submanifold is actually given as a coisotropic submanifold of a symplectic manifold. Otherwise, we first recall that Gotay [15] proved that every presymplectic manifold (C, ω_C) may be embedded into a symplectic manifold (F, ω_F) as a coisotropic submanifold such that ω_C is the restriction of ω_F to C . Moreover, such an embedding is unique up to neighborhood equivalence. The existence part is simply proven by taking $F = D^*$, where D is the kernel of ω_C , and $\omega_F = p^*\omega_C + \sigma^*\omega_{can}$, where p is the projection $D^* \rightarrow C$, ω_{can} is the canonical symplectic form on T^*C and σ is a splitting of $T^*C \rightarrow D^*$. Notice that ω_{can} is exact; so an exact presymplectic manifold can be coisotropically embedded into an exact symplectic manifold⁶.

We are then led to consider the problem of how to describe symplectic reduction of a coisotropic submanifold cohomologically. This goes under the name of BFV formalism [4].

We start with the finite dimensional setting. Locally, a coisotropic submanifold C of F can be described as the common zero locus of some differentiably independent functions ϕ_i on F . The characteristic foliation is then the span of the Hamiltonian vector fields X_i of the ϕ_i s. The space of functions on the quotient $\underline{C}$, if it is smooth, is the same as $(C^\infty(F)/\langle \phi_1, \phi_2, \dots \rangle)^{(X_1, X_2, \dots)}$, where $\langle \phi_1, \phi_2, \dots \rangle$ denotes the ideal generated by the ϕ_i s and the exponent denotes taking the subalgebra invariant under all the X_i s. The goal is to describe this space (actually this Poisson algebra) as the zeroth cohomology of a complex (actually the differential graded Poisson algebra of functions on a graded symplectic manifold).

To do this we add to F new odd coordinates b_i of degree -1 and define a vector field Q on the supermanifold so obtained by imposing $Q(f) = 0$ for any

⁶ In this note we will not focus on subtleties of this statement in the infinite dimensional setting.

$f \in C^\infty(F)$ and $Q(b_i) = \phi_i$. The cohomology is concentrated in degree zero and yields $C^\infty(F)/\langle \phi_1, \phi_2, \dots \rangle$. To select the invariant part, we add more odd coordinates c^i of degree +1 and extend Q to the supermanifold so obtained $\mathcal{F}$ by requiring $Q(b_i) = \phi_i$, $Q(c^i) = 0$ and $Q(f) = c^i X_i(f)$. This Q is a Hamiltonian vector field on $\mathcal{F}$ with respect to the symplectic form $\omega_F + db_i dc^i$ and Hamiltonian function $c^i \phi_i$; yet, in general it is not a differential on $C^\infty(\mathcal{F})$. However, using cohomological perturbation theory one can prove [26] that the Hamiltonian function may be deformed in such a way this occurs. The construction may also be globalized [6, 17, 22]. To summarize, we have the

Theorem 4.1. *Let C be a coisotropic submanifold of a finite-dimensional symplectic manifold F . Then one can embed F as the body of a supermanifold $\mathcal{F}$ with an additional $\mathbb{Z}$ -grading endowed with an even symplectic form $\omega_{\mathcal{F}}$ of degree zero and an even function S of degree +1 such that its Hamiltonian vector field Q squares to zero and its cohomology in degree zero is isomorphic as a Poisson algebra to the algebra of functions on C that are invariant under its characteristic distribution. This construction is unique up to symplectomorphisms of $\mathcal{F}$ if one requires it to be minimal (in terms of the newly added coordinates).*

In the case of field theory, the analogous result—with the additional condition that S and $\omega_{\mathcal{F}}$ are local—was proved long ago by [4] (in the description above the index i is now replaced by a worldsheet coordinate and the sum over i by an integral). Notice however that, in order to get S as a local functional, one often has to add extra fields of degree greater than +1 (and consequently extra fields of degree less than −1). In any case, the final result is what we will call a BFV manifold.

Definition 4.2. A BFV manifold is a triple (F, ω, Q) where F is a supermanifold with additional $\mathbb{Z}$ -grading, ω is an even symplectic form of degree zero, and Q is an odd symplectic vector field of degree +1 satisfying $[Q, Q] = 0$.

Remark 4.3. Recall that Q symplectic means $L_Q \omega = 0$. On the other hand the $\mathbb{Z}$ -grading amounts to the existence of an even vector field E of degree zero (the graded Euler vector field) such that the grading on functions, forms and vector fields is given by the eigenvalues of the Lie derivative L_E . We then have $L_E \omega = 0$ and $[E, Q] = Q$. This then implies that Q is automatically Hamiltonian, $\iota_Q \omega = dS$, with $S = \iota_E \iota_Q \omega$ (this remark is due to Roytenberg [21]). Notice that the condition $[Q, Q] = 0$ implies the “classical master equation” (CME) in the BFV formalism:

$$\{S, S\} = 0,$$

where $\{ , \}$ denotes the Poisson bracket induced by ω_F .

The coisotropic submanifold C can also be recovered geometrically. Namely, one defines $\mathcal{EL}$ as the zero locus of Q . More precisely (when EL is singular as is often the case), one considers the ideal $I_{\mathcal{EL}}$ generated by functions of the form $\{S, f\}$ with $f \in C^\infty(\mathcal{F})$. This ideal is clearly a Lie subalgebra (with respect to the Poisson bracket) thanks to the CME. This amounts to saying that $\mathcal{EL}$ is a coisotropic submanifold. The original C is just its body.

Remark 4.4 (Quantization). The (geometric) quantization of F is in this setting replaced by a quantization of $\mathcal{F}$ with a compatible quantization of S . Namely, one has to produce a graded vector space $\mathcal{H}_{\mathcal{F}}$ quantizing $(\mathcal{F}, \omega_{\mathcal{F}})$ together with an

odd operator Ω of degree 1 quantizing S and satisfying $\Omega^2 = 0$. Notice that the CME is the classical counterpart of the last equality and there might be obstruction (“anomalies”) in finding such an Ω . If everything works, however, one can consider the cohomology of Ω . Its degree zero component may be thought of the quantization of the reduction of C .

4.1. BFV as a boundary theory. In Section 3 we saw that a d -dimensional Lagrangian field theory associates a space C_Σ with a closed (often exact) two-form ω_Σ to a $(d-1)$ -dimensional manifold Σ . It was part of the assumptions that C_Σ is a manifold and that ω_Σ is presymplectic. Following the description above, we now associate to Σ a BFV manifold $(\mathcal{F}_\Sigma^\partial, \omega_\Sigma^\partial, Q_\Sigma^\partial)$. (The upper symbol ∂ is a reminder that this is the boundary construction as in the following we will have a similar construction, with similar notations, for the bulk.) If we work in the settings of subsection 3.4, then we take F_Σ^∂ to be the degree zero part of $\mathcal{F}_\Sigma^\partial$.

An important remark is that other cohomology groups may turn out to be non-trivial. As an example, consider first-order electrodynamics as in example 3.11. Recall that the space of boundary fields on Σ is $\mathcal{A}_\Sigma \times \Omega^{d-2}(\Sigma)$ whereas $C_\Sigma = \mathcal{A}_\Sigma \times \Omega_{cl}^{d-2}(\Sigma)$. To implement the BFV construction we add odd fields $c \in \Omega^0(\Sigma)$ of degree +1 and $b \in \Omega^{d-1}(\Sigma)$ of degree -1 and consider the BFV action $\mathcal{S}_\Sigma = \int_\Sigma c dB$. The Hamiltonian vector field Q acts trivially on c and B . On the other hand $Qb = dB$ and $QA = dc$. So the BFV cohomology yields functions on $H^0(\Sigma)[1] \times \mathcal{A}_\Sigma^{flat}/gauge \times \Omega_{cl}^{d-2}(\Sigma) \times H^{d-1}(\Sigma)[-1]$. The extra factors, in degree 0 and $d-1$, express the stacky nature of the reduction and become even more important in the nonabelian Yang–Mills case.

5. THE BV FORMALISM FOR MANIFOLDS WITH BOUNDARY

The BV formalism [5] deals with the degeneracy problem for an action in the bulk. In the BV case, as in Sections 2 and 3, we have a d -dimensional Lagrangian field theory, i.e. the assignment of a space of fields F_M and an action $S_M = \int_M L$ over F_M to a d -manifold M . But in addition we have a distribution $D_M \subset TF_M$ on F_M which describe the “symmetries”. This distribution does not have to be given by an action of a Lie group. It is involutive and of finite codimension when restricted to EL_M . The construction aims at cohomologically resolving the quotient of EL_M by the symmetries. Let us sketch the last point assuming at the beginning that our manifold M has no boundary. The BV construction proceeds by

- (1) first extending the space of fields F_M to the supermanifold $D_M[1] \subset T[1]F_M$ (i.e., one assigns degree 1 and the according Grassmann parity to coordinates on fibers of D_M),
- (2) extending the action S_M to a new local functional $\mathcal{S}_M$ on $\mathcal{F}_M := T^*[-1]D_M[1]$ which has the two following properties:
 - (a) It satisfies the classical master equation (CME) $\{\mathcal{S}_M, \mathcal{S}_M\} = 0$, where $\{ , \}$ is the (degree +1) Poisson bracket associated to the canonical symplectic form (of degree -1) on $\mathcal{F}_M$, and
 - (b) the restriction of D_M to EL_M is the same as the restriction of the characteristic distribution of the coisotropic submanifold $\mathcal{EL}_M$ of critical points of $\mathcal{S}_M$ to its degree zero part.

The solution can be found by cohomological perturbation theory. In order to preserve locality, it is often necessary to extend the above procedure by allowing dependent symmetries and resolving their relations by adding new variables of degree 2 (ghosts for ghosts) and so on. The final result is anyway a supermanifold with odd symplectic form of degree -1 and a solution of the CME.

Remark 5.1. The CME is also the starting point for making sense of the integral of $e^{\frac{i}{\hbar}S_M}$ over F_M . In the saddle point approximation, one expands around critical points, i.e., points of EL_M . If the action is degenerate—namely, its Hessian at a critical point is degenerate—one cannot even begin the perturbative expansion. However, if one quotients by a distribution as above, one saves the game (or at least reduces the problem to a residual finite dimensional integration). This quotient might be very singular; also notice that in general situations the distribution is not even involutive outside of EL_M ; and even if everything worked out properly, it might be difficult to define the perturbative functional integral on the quotient, which might have a much more involved manifold structure. The way out is to extend S_M to a (possibly $\hbar$ -dependent) solution $\tilde{S}_M$ of the quantum master equation (QME) on $\mathcal{F}_M$. Namely, one picks a Berezinian ρ (formally, since we are working in an infinite dimensional context) on $\mathcal{F}_M$ and defines the BV Laplacian Δ by $\Delta f = \frac{1}{2} \operatorname{div}_\rho X_f$, where X_f is the Hamiltonian vector field of a function f and div_ρ is the divergence operator with respect to ρ . One requires ρ to restrict to the original measure on F_M and to be compatible with the symplectic structure: namely, one requires $\Delta^2 = 0$. The QME then reads $\frac{1}{2}\{\tilde{S}_M, \tilde{S}_M\} - i\hbar\Delta\tilde{S}_M = 0$. The limit of $\tilde{S}_M$ for $\hbar \rightarrow 0$ solves the CME and is taken to be S_M . One actually starts with S_M and tries to extend it to a formal power series in $\hbar$ that solves the QME if there are no obstructions (“anomalies”). A consequence of the QME is that the integral of $e^{\frac{i}{\hbar}S_M}$ on a Lagrangian submanifold is invariant under deformations of the Lagrangian submanifold. One then replaces the originally ill-defined integral over F_M by the integral over a deformation of the Lagrangian submanifold $D_M[1]$ where it is well-defined. We refer to [23] for a good introduction.

Definition 5.2. A BV manifold is a triple given by a supermanifold with additional $\mathbb{Z}$ -grading, an odd symplectic form of degree -1 and a function of degree 0 that satisfies the CME, i.e. Poisson commutes with itself.

We may then formulate the result of the BV construction in d -dimensional Lagrangian field theory as the assignment of a BV manifold $(\mathcal{F}_M, \omega_M, S_M)$ to a d -manifold M . Notice that as a consequence of the CME the Hamiltonian vector field Q_M of S_M ,

$$(5.1) \quad \iota_{Q_M} \omega_M = \delta S_M,$$

is cohomological, i.e., it satisfies $[Q_M, Q_M] = 0$.

5.1. The case with boundary. Now let us allow M to have a nonempty boundary. Since the BV construction is local it still assigns to M a quadruple $(\mathcal{F}_M, \omega_M, S_M, Q_M)$. It is still true that ω_M is symplectic and that Q_M is cohomological. On the other hand, S_M is no longer its Hamiltonian. The problem is that (5.1) involves integration by parts. We may overcome this problem working as in the previous Sections (in particular, subsection 3.4).

Namely, we define the space $\tilde{\mathcal{F}}_\Sigma$ of preboundary fields on a $(d-1)$ -manifold Σ as the germs at $\Sigma \times \{0\}$ of $\mathcal{F}_{\Sigma \times [0, \epsilon]}$. Integration by parts in the computation of

$\delta\mathcal{S}_{\Sigma \times [0, \epsilon]}$ yields a one-form $\tilde{\alpha}_\Sigma$ on $\tilde{\mathcal{F}}_\Sigma$. We denote by $\tilde{\omega}_\Sigma$ its differential—which we assume to be presymplectic—and by $(\mathcal{F}_\Sigma^\partial, \omega_\Sigma^\partial)$ its reduction. We also assume that $\tilde{\alpha}_\Sigma$ reduces to a connection one-form α_Σ^∂ on $\mathcal{F}_\Sigma^\partial$. In most examples, α_Σ^∂ will be an actual one-form.

If we take care of boundary terms, instead of (5.1) we now get

$$(5.2) \quad \iota_{Q_M} \omega_M = \delta\mathcal{S}_M + \tilde{\pi}_M^* \tilde{\alpha}_{\partial M},$$

where $\tilde{\pi}_M$ is the natural surjective submersion from $\mathcal{F}_M$ to $\tilde{\mathcal{F}}_{\partial M}$. If we denote by π_M the composition of $\tilde{\pi}_M$ with the natural surjective submersion from $\tilde{\mathcal{F}}_{\partial M}$ to $\mathcal{F}_{\partial M}^\partial$, we finally get the fundamental equation of the BV theory for manifolds with boundary [7]:

$$(5.3) \quad \iota_{Q_M} \omega_M = \delta\mathcal{S}_M + \pi_M^* \alpha_{\partial M}^\partial,$$

To complete the description of the theory with boundary, we still have to study Q_M . The first obvious remark is that it is $\tilde{\pi}_M$ -projectable. More precisely, for every Σ , there is a uniquely defined vector field $\tilde{Q}_\Sigma$ (automatically cohomological) on $\tilde{\mathcal{F}}_\Sigma$ such that for every M the vector field Q_M projects to $\tilde{Q}_{\partial M}$: namely, $\tilde{Q}_{\partial M}(\phi) = d_{\hat{\phi}} \tilde{\pi}_M(Q_M(\hat{\phi}))$, $\forall \phi \in \tilde{\mathcal{F}}_{\partial M}$ and $\forall \hat{\phi} \in \tilde{\pi}_M^{-1}(\phi)$.

Let us now differentiate (5.2). Using the fact that ω_M is closed, we get $L_{Q_M} \omega_M = \tilde{\pi}_M^* \tilde{\omega}_{\partial M}$ (which by the way proves that Q_M is not even symplectic). We now apply L_{Q_M} to this equation. Using the fact Q_M is cohomological and projectable, we get $\tilde{\pi}_M^* L_{\tilde{Q}_{\partial M}} \tilde{\omega}_{\partial M} = 0$. Since $\tilde{\pi}_M$ is a surjective submersion, we conclude that $L_{\tilde{Q}_{\partial M}} \tilde{\omega}_{\partial M} = 0$.

Actually, this proves that, for every Σ , $\tilde{\omega}_\Sigma$ is $\tilde{Q}_\Sigma$ -invariant. This implies that $\tilde{Q}_\Sigma$ is projectable to the reduction. To show this, we have just to check that $[\tilde{Q}_\Sigma, X]$ belongs to the kernel of $\tilde{\omega}_\Sigma$ for every X in the kernel. This follows from the identities $\iota_{[\tilde{Q}_\Sigma, X]} \tilde{\omega}_\Sigma = [L_{\tilde{Q}_\Sigma}, \iota_X] \tilde{\omega}_\Sigma = 0$.

We conclude that, for every Σ , there is a uniquely defined vector field Q_Σ^∂ on $\mathcal{F}_\Sigma^\partial$ (automatically cohomological and symplectic) to which $\tilde{Q}_\Sigma$ projects. This has two fundamental consequences:

- (1) To each $(d-1)$ -dimensional manifold Σ , we now associate a BFV manifold $(\mathcal{F}_\Sigma^\partial, \omega_\Sigma^\partial, Q_\Sigma^\partial)$, see definition 4.2.
- (2) For each d -manifold M , Q_M π_M -projects to Q_M^∂ .

These two final observations together with (5.3) constitute the framework of the BV formalism extended to manifolds with boundaries [7], which we call the BV-BFV formalism.

Definition 5.3. We define a BV-BFV manifold over a given exact BFV manifold $(\mathcal{F}^\partial, \omega^\partial = d\alpha^\partial, Q^\partial)$ as a quintuple $(\mathcal{F}, \omega, \mathcal{S}, Q, \pi)$ where $\mathcal{F}$ is a supermanifold with additional $\mathbb{Z}$ -grading, ω is an odd symplectic form of degree -1 , $\mathcal{S}$ is an even function of degree 0, Q is a cohomological vector field and $\pi: \mathcal{F} \rightarrow \mathcal{F}^\partial$ is a surjective submersion such that

- (1) $\iota_Q \omega = d\mathcal{S} + \pi^* \alpha^\partial$,
- (2) $Q^\partial = d\pi Q$.

The definition may be extended to BV-BFV manifolds over a BFV manifold with connection α^∂ . This requires introducing a line bundle over $\mathcal{F}^\partial$ and viewing $\exp(\frac{i}{\hbar} \mathcal{S})$ as a section of the pulled-back bundle.

Remark 5.4 (Axiomatization). We may now reformulate Lagrangian field theories axiomatically as a "functor" from some "cobordisms category" to a category where the objects are BFV manifold with connection and the morphisms are BV-BFV manifolds over the Cartesian product of the objects.

Remark 5.5. Notice that using this method the BFV construction associated to the boundaries is obtained from the BV construction in the bulk and does not have to be done independently. Also recall that, by general principles [21], Q_Σ^∂ is Hamiltonian with a uniquely defined odd Hamiltonian function S_Σ^∂ of degree +1. This yields as a consequence the following generalization of the CME

$$(5.4) \quad Q_M(\mathcal{S}_M) = \pi_M^*(2\mathcal{S}_{\partial M}^\partial - \iota_{Q_{\partial M}^\partial} \alpha_{\partial M}^\partial),$$

which can be proved as follows. First differentiate (5.3) to obtain $L_{Q_M} \omega_M = \pi_M^* \omega_{\partial M}^\partial$. Then apply ι_{Q_M} to (5.3) and use the obtained equation and the fact that Q_M is cohomological and projects to $Q_{\partial M}^\partial$ to obtain the differential of (5.4). Then observe that the differential of a function of degree 1 vanishes if and only if the function itself vanishes (we have no constants in degree +1).

Example 5.6 (First-order electrodynamics). We return to example 3.11 (first-order YM is explained in details in [7]; we leave the usual second-order formulation as an exercise to the reader). Since we want to implement gauge transformations for A , we define $D_M[1]$ by adding the "ghost field" $c \in \Omega^0(M)$ which is odd and of degree +1. Gauge transformations are given by the vector field dc in the A -direction (here d denotes the de Rham differential on M). The BV space of fields $\mathcal{F}_M = T^*[-1]D_M[1]$ is then

$$\Omega^0(M)[1] \times \mathcal{A}_M \times \Omega^2(M)[-1] \times \Omega^{d-2}(M) \times \Omega^{d-1}(M)[-1] \times \Omega^d(M)[-2].$$

We add a superscript $+$ to denote the canonically conjugate coordinates (a.k.a. antifields) to the fields: namely, $B^+ \in \Omega^2(M)[-1]$, $A^+ \in \Omega^{d-1}(M)[-1]$, $c^+ \in \Omega^d(M)[-2]$. The BV action is just

$$\mathcal{S}_M = \int_M B \wedge dA + \frac{1}{2} B \wedge *B + A^+ \wedge dc.$$

The cohomological vector field Q_M acts as follows (we omit the terms where the action is zero):

$$Q_M A = dc, \quad Q_M A^+ = dB, \quad Q_M B^+ = *B + dA, \quad Q_M c^+ = dA^+.$$

On the space of preboundary fields on a $(d-1)$ -manifold Σ , we get $\tilde{\alpha}_\Sigma = \int_\Sigma B \wedge \delta A + A^+ \delta c$. We immediately see that the kernel of $\tilde{\omega}_\Sigma$ consists of all jets of B^+ and c^+ and all jets higher then the zeroth of A, B, A^+, c . Moreover, $\tilde{\alpha}_\Sigma$ is also basic. We then get

$$\mathcal{F}_\Sigma^\partial = \Omega^0(\Sigma)[1] \times \mathcal{A}_\Sigma \times \Omega^{d-2}(\Sigma) \times \Omega^{d-1}(\Sigma)[-1].$$

Projecting the cohomological vector field, we get the cohomological vector field Q_Σ^∂ , which acts by $Q^\partial A^+ = dB$ and $Q^\partial A = dc$ and has Hamiltonian function $S_\Sigma^\partial = \int_\Sigma c dB$.

5.2. EL correspondences. We now define the space $\mathcal{EL}_M$ as the space of zeros of Q_M and $\mathcal{L}_{\partial M}$ as its image under π_M . This generalizes the classical story of evolution correspondences and evolution relations. Notice that (5.3) implies that $\mathcal{L}_{\partial M}$ is isotropic, and we are going to assume that it is actually Lagrangian. There are two problems to be tackled though: the first is that $\mathcal{EL}_M$ is not smooth in general, the second is that we are usually interested in reduction, which is even more singular in general.

One way to avoid the first problem is by working with an algebraic description only, but we will often pretend that we are dealing with smooth manifolds. Namely, instead of $\mathcal{EL}_M$ we consider its vanishing ideal $I_{\mathcal{EL}_M}$, i.e., the ideal generated by functions of the form $Q_M f$, with $f \in C^\infty(\mathcal{F}_M)$. This ideal is a Lie algebra with respect to the Poisson bracket, which amounts to saying that $\mathcal{EL}_M$ is coisotropic. If M has no boundary, this is obvious since Q_M is symplectic and squares to zero. If M has a boundary, this is still true since, to generate it, it is enough to consider functions f that “vanish near the boundary” (namely, functions in $\pi_{M,U}^* \mathcal{F}_U$ where U is compact in the interior of M and $\pi_{M,U}$ is the restriction map). The characteristic distribution $\mathcal{D}_M$ is generated by the Hamiltonian vector fields of functions of the form Qf . If f is as above, we have $Qf = \{S, f\}$ and hence the characteristic distribution is generated by vector fields of the form $[Q, X]$ where X is a Hamiltonian vector field vanishing near the boundary (we assume here that components of Q on EL are differentiably independent). The reduction $\underline{\mathcal{EL}}_M$ of $\mathcal{EL}_M$ by $\mathcal{D}_M$ carries again a symplectic form of degree -1 (if it is singular, this has to make sense of; e.g., by considering the open smooth locus or using the language of derived algebraic geometry [27]).

If M has a boundary, it makes sense to consider another reduction, namely by the distribution $\mathcal{D}_M^Q \supset \mathcal{D}_M$ generated by vector fields of the form $[Q, X]$ where X is Hamiltonian (but with no vanishing condition). More precisely, observe that, since Q_M projects to $Q_{\partial M}^\partial$, we have that $\mathcal{L}_{\partial M}$ is contained in $\mathcal{EL}_{\partial M}^\partial$, the space of zeros of $Q_{\partial M}^\partial$, which is also coisotropic. Hence its characteristic distribution, generated by vector fields of the form $[Q_{\partial M}^\partial, X]$ with X Hamiltonian, is tangent to $\mathcal{L}_{\partial M}$. Now let ℓ be a point in $\mathcal{L}_{\partial M}$ and let $[\ell]$ denote its orbit. Then $\mathcal{D}_M^Q$ restricts to $\pi_M^{-1}([\ell]) \cap \mathcal{EL}_M$ and we denote by $\underline{\mathcal{EL}}_{[\ell]}$ its quotient. The union of the $\underline{\mathcal{EL}}_{[\ell]}$ s over $[\ell] \in \mathcal{L}_{\partial M}$ is the quotient $\underline{\mathcal{EL}}_{M,Q}$ of $\mathcal{EL}_M$ by $\mathcal{D}_M^Q$, which is by itself a quotient of the symplectic reduction $\underline{\mathcal{EL}}_M$.

In [7] it is shown that each fiber $\underline{\mathcal{EL}}_{[\ell]}$ carries a symplectic form of degree -1 . This follows from a different but equivalent description of this quotient. Namely, pick a Lagrangian submanifold $\mathcal{L}$ of $\mathcal{F}_{\partial M}^\partial$ transversal to $\mathcal{L}_{\partial M}$ at ℓ . Then one shows that $\pi_M^{-1}(\ell) \cap \mathcal{EL}_M$ is coisotropic in $\pi_M^{-1}(\mathcal{L})$ and that its reduction is $\underline{\mathcal{EL}}_{[\ell]}$.

We call $\underline{\mathcal{EL}}_{[\ell]}$ the moduli space of vacua at $[\ell]$ and we assume it to be finite dimensional (if this is not the case, it means that we have not considered enough symmetries).

Notice that there is in principle a second (usually coisotropic) submanifold $\mathcal{C}'_\Sigma$ of $\mathcal{F}_\Sigma$. Namely, the elements of $\mathcal{F}_\Sigma$ are zeroes of Q'_Σ which can be extended as zeroes of $Q_{\Sigma \times [0, \epsilon]}$ for some $\epsilon > 0$. It is meaningful to require $\mathcal{C}'_\Sigma = \mathcal{C}_\Sigma$. Otherwise, it again means that we have not taken enough symmetries into account.

Remark 5.7 (Axiomatization). If reduction were always nice, we could get the following induced axiomatization of a d -dimensional Lagrangian field theory in the

BV-BFV formalism. To a $(d-1)$ -manifold Σ we associate a symplectic supermanifold $\underline{\mathcal{EL}}_{\partial M}^{\partial}$ and to a d -manifold M we associate the “evolution correspondence” $\underline{\mathcal{EL}}_{M_Q} \rightarrow \underline{\mathcal{EL}}_{\Sigma}^{\partial}$ which has a Lagrangian image and whose fibers are finite dimensional symplectic manifolds in degree -1 . If we cut a manifold M along a submanifold Σ , we may try to recover $\underline{\mathcal{EL}}_M$ out of the composition of the evolution correspondences, and some more data, for the two halves. This problem started to be addressed in [7].

Example 5.8 (Electrodynamics—continued). In [7], to which we refer for details, it is shown that, in the case of first-order electrodynamics, for any ℓ we have

$$\underline{\mathcal{E}}_{[\ell]} \simeq H^1(M, \partial M) \oplus H^{n-1}(M)[-1] \oplus H^0(M, \partial M)[1] \oplus H^n(M)[-2],$$

which is indeed finite-dimensional. (Here $H^*(M, \partial M)$ denotes cohomology relative to the boundary.)

5.3. Extended theories. The construction in subsection 5.1 may be applied iteratively to go to lower and lower dimension. Namely, there we have obtained a BFV structure $(\mathcal{F}_{\Sigma}^{\partial}, \omega_{\Sigma}^{\partial}, Q_{\Sigma}^{\partial})$, to which we canonically associate a function $\mathcal{S}_{\Sigma}^{\partial}$, for every $(d-1)$ -dimensional manifold Σ without boundary; yet, since the construction is local, we can use these data on a $(d-1)$ -dimensional manifold Σ with boundary. Again what is not going to work is the condition that $\mathcal{S}_{\Sigma}^{\partial}$ is the Hamiltonian function of Q_{Σ}^{∂} . We correct this equation using the induced one-form on the space of preboundary fields on $\partial\Sigma$ and reduce by the kernel of the two form. Since $\mathcal{S}_{\Sigma}^{\partial}$ has degree 1, this will also be the degree of the induced symplectic form.

As a result, to a $(d-2)$ -manifold γ we associate a triple $(\mathcal{F}_{\gamma}^{\partial\partial}, \omega_{\gamma}^{\partial\partial} = d\alpha_{\gamma}^{\partial\partial}, Q_{\gamma}^{\partial\partial})$, where $\omega_{\gamma}^{\partial\partial}$ is an odd symplectic form of degree $+1$ and $Q_{\gamma}^{\partial\partial}$ is a cohomological, symplectic vector field (hence automatically Hamiltonian with a uniquely defined even Hamiltonian function $\mathcal{S}_{\gamma}^{\partial\partial}$ of degree $+2$). To a $(d-1)$ -manifold Σ with boundary we now associate a quintuple $(\mathcal{F}_{\Sigma}^{\partial}, \omega_{\Sigma}^{\partial}, Q_{\Sigma}^{\partial}, \mathcal{S}_{\Sigma}^{\partial}, \pi_{\Sigma}^{\partial})$, where $\omega_{\Sigma}^{\partial}$ is an even symplectic form of degree 0, Q_{Σ}^{∂} is a cohomological vector field, $\mathcal{S}_{\Sigma}^{\partial}$ is an odd function of degree $+1$ and $\pi_{\Sigma}^{\partial}: \mathcal{F}_{\Sigma}^{\partial} \rightarrow \mathcal{F}_{\partial\Sigma}^{\partial\partial}$ is a surjective submersion such that

- (1) $\iota_{Q_{\Sigma}^{\partial}} \omega_{\Sigma}^{\partial} = d\mathcal{S}_{\Sigma}^{\partial} + (\pi_{\Sigma}^{\partial})^* \alpha_{\partial\Sigma}^{\partial\partial}$,
- (2) $Q_{\partial\Sigma}^{\partial\partial} = d\pi_{\Sigma}^{\partial}$.

We can now consider the zero locus $\underline{\mathcal{EL}}_{\partial\Sigma}^{\partial\partial}$ of $Q_{\partial\Sigma}^{\partial\partial}$, which is coisotropic and contains $L_{\Sigma}^{\partial} := \pi_{\Sigma}^{\partial}(\text{EL}_{\Sigma}^{\partial})$. Repeating the same analysis as above, we conclude that $\underline{\mathcal{EL}}_{\Sigma_Q}^{\partial} \rightarrow \underline{\mathcal{EL}}_{\partial\Sigma}^{\partial\partial}$ has fibers with a symplectic structure in degree zero. Notice that in general these fibers will not be finite dimensional (it would be too restrictive to ask for that).

The construction may now be iterated to $(d-2)$ -manifolds with boundaries. Every time the degree of the symplectic form and of the action increase by 1. However, it probably makes sense to continue this construction only as long as the $\underline{\mathcal{E}}^{\partial\partial\ldots\partial}_{[\ell]}$ spaces are finite dimensional.

Typically, at some point we get $\mathcal{S}^{\partial\partial\ldots\partial} = 0$, so that $\underline{\mathcal{E}}^{\partial\partial\ldots\partial}_{[\ell]}$ is the whole space of fields in the bulk over a point on the boundary and this will usually be infinite dimensional.

On the other hand, in topological field theories of the AKSZ type [2], this construction can be iterated down to dimension 0 always with finite dimensional

$\underline{\mathcal{E}}^{\partial\partial\ldots\partial}_{[\ell]}$ fibers; see [7] for details. Hence we can speak of fully extended field theories. These look like the BV-BFV version of [18].

Example 5.9 (Electrodynamics—continued). In example 5.6, we got the BFV structure for first-order electrodynamics. Applying the above reasoning, we first consider the space of preboundary fields with one-form $\tilde{\alpha}_\gamma^{\partial\partial} = \int_\gamma B \delta c$. The kernel of its differential consists of all jets for A and A^+ and all jets higher than the zeroth for B and c , and $\tilde{\alpha}_\gamma^{\partial\partial}$ is basic. Hence we get

$$\mathcal{F}_\gamma^{\partial\partial} = \Omega^0(\gamma)[1] \times \Omega^{d-2}(\gamma).$$

One can also easily realize that $Q_\gamma^{\partial\partial} = 0$. Moreover, one can also compute, see [7],

$$\underline{\mathcal{E}}\mathcal{L}_{\Sigma_Q}^{\partial} = \Omega^1(\Sigma)/\Omega^1(\Sigma)_{\text{exact}} \oplus \Omega_{\text{closed}}^{d-2}(\Sigma, \partial\Sigma) \oplus H^0(\Sigma, \partial\Sigma)[1] \oplus H^{d-1}(\Sigma)[-1],$$

which is infinite dimensional for $d > 2$. If $d = 2$, this space is finite dimensional, so it makes sense to extend the theory down to codimension two. This is another way of observing that two-dimensional electrodynamics is almost topological (this holds also for nonabelian Yang Mills theories).

5.4. Perturbative quantization. We may finally present the generalization of the formalism discussed in subsection 3.3 to the case of degenerate Lagrangians in the BV-BFV formalism. For simplicity, we assume that the boundary one-form $\alpha_{\partial M}^{\partial}$ is globally well-defined and that $\mathcal{F}_{\partial M}^{\partial}$ is endowed with a Lagrangian foliation on which $\alpha_{\partial M}^{\partial}$ vanishes and which has a smooth leaf space $\mathcal{B}_{\partial M}$. The space of functions on $\mathcal{B}_{\partial M}$ defines the boundary graded vector space $\mathcal{H}_{\partial M}$. Let $p_{\partial M}$ be the projection $\mathcal{F}_{\partial M}^{\partial} \rightarrow \mathcal{B}_{\partial M}$. To produce a state ψ_M associated to the bulk M we first have to choose an embedding of $\underline{\mathcal{E}}\mathcal{L}_{[p_{\partial M}^{-1}(\phi) \cap \mathcal{L}_M]}$ into $\pi_M^{-1}(p_{\partial M}^{-1}(\phi))$ and a tubular neighborhood thereof. Then we have to pick a Lagrangian submanifold $\mathcal{L}_\phi$ in the fiber of this tubular neighborhood. Finally,

$$\psi_M(\phi) = \int_{\mathcal{L}_\phi} e^{\frac{i}{\hbar} S_M(\Phi)} [D\Phi].$$

Notice that $\psi_M(\phi)$ is also a function on the moduli space of vacua $\underline{\mathcal{E}}\mathcal{L}_{[p_{\partial M}^{-1}(\phi) \cap \mathcal{L}_M]}$. As already observed, each of these spaces carries a symplectic structure of degree -1 and is by assumption finite dimensional. The integral has to be computed perturbatively. One can then define a BV operator Δ on the moduli spaces of vacua and (if the theory is not anomalous) a coboundary operator Ω on $\mathcal{H}_{\partial M}$. By general BV arguments we expect that ψ_M satisfies the following generalization of the QME

$$(5.5) \quad \hbar^2 \Delta \psi_M + \Omega \psi_M = 0,$$

whose classical limit should correspond to (5.4).

An example where this kind of quantization has been performed and a solution to (5.5) has been explicitly obtained is described in [1]. Other examples are currently being studied [8].

APPENDIX A. SOME USEFUL FACTS

A relation from a set X to a set Y is just a subset of $X \times Y$. If R_1 is a relation from X to Y and R_2 is a relation from Y to Z , the composition $R_2 \circ R_1$ from X to

Z is defined as

$$R_2 \circ R_1 = \{(x, z) \in X \times Z : \exists y \in Y (x, y) \in R_1, (y, z) \in R_2\}.$$

The composition is associative. If $\phi: X \rightarrow Y$ and $\psi: Y \rightarrow Z$ are maps, then $\text{graph}(\psi) \circ \text{graph}(\phi) = \text{graph}(\psi \circ \phi)$.

If X and Y are symplectic manifolds, a relation from X to Y is called canonical if it is a Lagrangian submanifold. A map $\phi: X \rightarrow Y$ is a symplectomorphism iff $\text{graph}(\phi)$ is a canonical relation. The composition of two canonical relations in general is not a submanifold. On the other hand, being Lagrangian is preserved if X and Y are finite dimensional; otherwise one can only ensure being isotropic in general. A composition of isotropic relations is again isotropic.

In this paper, we often work with presymplectic and weakly symplectic forms. Recall that a closed two-form ω is presymplectic if it has constant rank and is weakly symplectic if it defines an injective linear map from the tangent to the cotangent bundle (in finitely many dimensions, this implies that the form is also symplectic).

The notion of Lagrangian submanifold naturally extends to presymplectic and weakly symplectic manifolds. A submanifold L of (M, ω) is called Lagrangian, if $T_x L^\perp = T_x L \forall x \in L$. Here

$$T_x L^\perp := \{v \in T_x M : \omega_x(v, w) = 0 \forall w \in T_x L\},$$

which makes sense also if ω_x is degenerate. Similarly, $L \subset M$ is coisotropic if $TL^\perp \subset TL$ and it is isotropic when $TL \subset TL^\perp$. Here $\perp$ means orthogonal subbundle with respect to the two-form ω .

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A CONSTRUCTION OF OBSERVABLES FOR AKSZ SIGMA MODELS

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ABSTRACT. A construction of gauge-invariant observables is suggested for a class of topological field theories, the AKSZ sigma-models. The observables are associated to extensions of the target Q -manifold of the sigma model to a Q -bundle over it with additional Hamiltonian structure in fibers.

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1. INTRODUCTION

The interest in observables in topological field theories is largely due to the applications in algebraic topology, where expectation values of observables are known to yield (at least in some examples) invariants of knots and links under ambient isotopy or, more generally, cohomology classes of spaces of embeddings. The seminal example here was given in the work of Witten [16], where the expectation value of the Wilson loop observable in Chern-Simons theory with gauge group $SU(2)$, associated to a knot in the 3-sphere, was found out to yield the Jones polynomial of the knot.

Given a diffeomorphism-invariant action S_Σ of a topological field theory on a manifold Σ with space of fields F_Σ , one is particularly interested in observables $O \in C^\infty(F_\Sigma)$ associated to embedded submanifolds $i : \gamma \hookrightarrow \Sigma$ which depend on fields only via their pull-back from Σ to γ :

$$(1) \quad O_{\gamma,i}(X) = O(i^*X)$$

where X is the field. Such observables are automatically invariant with respect to diffeomorphisms $\phi : \Sigma \rightarrow \Sigma$ in the sense that

$$(2) \quad O_{\gamma,\phi \circ i}((\phi^{-1})^*X) = O_{\gamma,i}(X)$$

where $(\phi^{-1})^*X$ is the pull-back of the field by ϕ^{-1} .

On a formal level, the expectation value of the observable

$$(3) \quad \langle O_{\gamma,i} \rangle = \int_{F_\Sigma} \mathcal{D}X \, O_{\gamma,i}(X) e^{iS_\Sigma(X)}$$

is diffeomorphism-invariant

$$(4) \quad \langle O_{\gamma,\phi \circ i} \rangle = \langle O_{\gamma,i} \rangle$$

due to (2), diffeomorphism-invariance of the action S_Σ and of the path integral measure $\mathcal{D}X$. This means in particular that the expectation value is an invariant of the embedding $i : \gamma \hookrightarrow \Sigma$ under ambient isotopy.

Since topological field theories possess gauge symmetry, one also requires that the observable is gauge-invariant, so that in the path integral (3) one could pass to integration over gauge equivalence classes of fields.

In this paper we employ Batalin-Vilkovisky (BV) formalism to treat systems with gauge symmetry (cf. e.g. [15] for details). In particular the space of fields F_Σ becomes extended to an odd-symplectic Q -manifold $\mathcal{F}_\Sigma$ with the action S_Σ extended to a function on $\mathcal{F}_\Sigma$, satisfying the master equation¹ $\{S_\Sigma, S_\Sigma\} = 0$ and generating the cohomological vector field Q_Σ as its Hamiltonian vector field. In this formalism the gauge-invariance of the observable is expressed as

$$(5) \quad Q_\Sigma O = 0$$

and the path integral (3) for the expectation value is traded for an integral over a Lagrangian submanifold $\mathcal{L}$ in $\mathcal{F}_\Sigma$, with $\mathcal{L}$ playing the role of the gauge-fixing condition.

One idea how to construct an observable for a gauge theory in BV formalism is to consider an extension of the BV theory $(\mathcal{F}_\Sigma, S_\Sigma)$ to a BV theory on a larger space $(\mathcal{F}_\Sigma \times \mathcal{F}^{\text{aux}}, S_\Sigma + S^{\text{aux}})$ with $\mathcal{F}^{\text{aux}}$ the space of “auxiliary fields” and S^{aux}

¹In this outline we do not make explicit distinction between classical and quantum master equation. We write the classical one which also coincides with the quantum one if $\Delta_\Sigma S = 0$ for Δ_Σ the BV Laplacian. Likewise, we write the classical gauge-invariance condition on the observable $Q_\Sigma O = 0$ which coincides with the quantum one if $\Delta_\Sigma O = 0$, cf. section 4 for details.

the action for auxiliary fields Y which is also allowed to depend on the “ambient” fields $X \in \mathcal{F}_\Sigma$. Then one uses the BV push-forward construction

$$(6) \quad O(X) = \int_{\mathcal{L}^{\text{aux}} \subset \mathcal{F}^{\text{aux}}} \mathcal{D}Y e^{iS^{\text{aux}}(X,Y)}$$

(propositions 1, 3 in section 4, cf. also [13]) to integrate out the auxiliary fields and produce an observable for the ambient theory $(\mathcal{F}_\Sigma, S_\Sigma)$. In this paper we call such extensions by auxiliary fields “pre-observables” (to be more precise, we use a slightly better definition with equation (40) on S^{aux} instead of the master equation on $S_\Sigma + S^{\text{aux}}$, which still suffices to produce an observable, cf. definition 6 and remark 5).

In this paper we consider observables obtained by the construction outlined above for topological sigma models coming from AKSZ construction [1] (cf. a brief reminder in section 2), where $\mathcal{F}_\Sigma$ is the mapping space from the tangent bundle of Σ with shifted parity of fibers to a target Q -manifold $\mathcal{M}$ with additional Hamiltonian structure (cf. definition 1). In this setting we give a construction of pre-observables (section 5), which associates a pre-observable for the AKSZ sigma model to an embedded submanifold $i : \gamma \hookrightarrow \Sigma$ and an extension of the target $\mathcal{M}$ to a Q -bundle over $\mathcal{M}$ with additional Hamiltonian structure in fibers (the “Hamiltonian Q -bundle”, cf. section 3). For the ambient and auxiliary actions to have degree zero, degrees of Hamiltonian structures in the base and the fiber of the target Q -bundle have to match² dimensions of Σ and γ respectively.

Having a pre-observable for an AKSZ sigma model, we construct the corresponding observable by the BV push-forward (6). By construction, this observable is gauge-invariant in the sense of (5) and satisfies (1), and so formally produces expectation values which are invariant under ambient isotopy. Of course, the problem with this definition of the observable is that (6) is generally a path integral. In section 6 we consider several simple cases when this integral can be made sense of and its expected properties can be rigorously checked:

- case when the fiber in the target is a point (which gives rise to observables given by an exponential of an integral of a local expression),
- case of one-dimensional observables (when the path integral becomes a path integral of quantum mechanical type and can be regularized via geometric quantization, which gives rise to certain generalization of Wilson loops),
- case when the integral (6) is Gaussian which gives rise to a class of observables which we call “torsion-like” for their similarity to Ray-Singer torsion.

In section 7 we give explicit examples of observables falling within one of the three classes above. In particular, we recover the usual Wilson loop observable in Chern-Simons theory, with the corresponding path integral expression being the Alekseev-Faddeev-Shatashvili path integral formula for the Wilson loop [2]. We also recover the Cattaneo-Rossi “Wilson surface” observable for BF theory [9].

In this paper we concentrate only on the construction of observables, we are not calculating their expectation values.

1.1. Logic of the construction.

- (i) For an AKSZ sigma model on manifold Σ with target $\mathcal{M}$, find a Hamiltonian Q -bundle $\mathcal{E}$ over $\mathcal{M}$. Then for every embedded submanifold $i : \gamma \hookrightarrow \Sigma$ of fixed dimension matching the degree of Hamiltonian structure in fibers of $\mathcal{E}$, we construct a pre-observable.

²More precisely, degrees of Hamiltonians (not of symplectic structures) have to match the dimensions.

- (ii) Take the fiber BV integral (6) over the auxiliary fields of the pre-observable constructed in step (i) to obtain an observable for the AKSZ sigma model.

Unlike step (i), step (ii) is not canonical, in the sense that it is a quantization problem: the path integral has to be made sense of (and the gauge-invariance of the result has to be checked) which can be done for certain classes of target Q -bundles, but it is not clear whether it is possible to do in greater generality³.

1.2. Plan of the paper. The logical organization of the exposition is as follows.

- Sections 2, 3.1, bits of 4 concerning BV observables — background reminders.
- Section 5 — step (i) of the main construction.
- Section 3 — auxiliary construction for step (i) of the main construction.
- Section 4 — motivation for step (ii) of the main construction.
- Section 6 — examples for step (ii) of the main construction.
- Section 7 — fully explicit examples.

Section 2 is a short reminder of the AKSZ construction of topological sigma models in Batalin-Vilkovisky formalism. We also recall how some well-known topological field theories fit into the construction: Chern-Simons theory, BF theory, Poisson sigma model.

In section 3 we first briefly recall the standard notion of a Q -bundle and then define a “Hamiltonian” Q -bundle, preparing the grounds for the construction of pre-observables in section 5. We start with the definition 2 of a trivial Hamiltonian Q -bundle (“trivial” here means trivial as a fiber bundle; the cohomological vector field is *not* required to be a sum of a cohomological vector field on the base and another one on the fiber). All our examples are of this type, but for the completeness of the exposition, we also give a slightly more general definition 3, which does not explicitly rely on the total space being a direct product. However in a local trivialization definition 3 boils down to definition 2.

In section 4 we recall the standard notions of classical and quantum observables in BV formalism and introduce the notion of a pre-observable, which comes in three modifications:

- (i) Classical pre-observable, definition 6: essentially, an extension of the action of the ambient classical BV theory to a solution of classical master equation on the space of fields extended by auxiliary fields. We give a technically more convenient definition with equation (40) required instead of the classical master equation on extended space of fields (cf. remark 5 for the relation between the two).
- (ii) Semi-quantum pre-observable, definition 7, suited for integrating out auxiliary fields to obtain an observable for the ambient theory (proposition 1).
- (iii) Quantum pre-observable for a quantum ambient BV theory (i.e. one with the action satisfying the quantum master equation), definition 9: this is also an extension of the ambient theory by auxiliary fields, plus an extension of the action to a solution of quantum master equation on the extended space. From a quantum pre-observable one can induce a quantum observable for the ambient theory, by integrating out auxiliary fields (proposition 3).

³One can indeed try to define the path integral perturbatively, as sum of Feynman diagrams represented by integrals over compactified configuration spaces of tuples of points on γ . However, for the formal argument of proposition 1 that the result is gauge-invariant to become rigorous, one has to prove that the hidden boundary strata of the configuration spaces appearing in the calculation of $Q_{\Sigma}O$ (cf. the proof of proposition 6) do not contribute, which is not true in general case.

In this section in the discussion of quantization we always work with spaces of fields as with finite dimensional spaces and the integrals are over finite-dimensional super-manifolds. In the context of local field theory, the BV push-forward becomes a path integral, so the proofs given in section 4 stop to work and the propositions become conjectures that have to be proven by more delicate means (cf. the proofs of propositions 5, 6 in section 6).

Section 5 is the logical core of the paper. Here we give the construction of a classical pre-observable for an AKSZ theory out of an extension of the target to a Hamiltonian Q -bundle.

In section 6 we treat several classes of situations when the BV push-forward yielding an observable out of a pre-observable constructed in section 5 can be performed rigorously: case of target fiber being a point, case of 1-dimensional observables (via geometric quantization), case when the BV push-forward is given by a Gaussian integral.

In section 7 we specialize the constructions of section 6 to present several explicit examples, including the Wilson loop together with its path integral representation known from [2], Cattaneo-Rossi codimension 2 observable for BF theory, torsion observables in Chern-Simons and BF theories, etc.

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TERMINOLOGY, NOTATIONS

Terminology.

- Q -manifold (or bundle) = differential graded manifold (or bundle).
- Ghost number = internal degree (to distinguish from de Rham degree of a differential form) = $\mathbb{Z}$ -grading on functions on $\mathbb{Z}$ -graded manifolds.
- Observable = gauge-invariant functional on the space of fields (cf. section 4 for definitions).
- Expectation value = correlator.

Conventions. We set the Planck's constant $\hbar = 1$ (cf. remark 6 on how to re-introduce $\hbar$).

Notations. We use $\mathcal{M}, \mathcal{N}, \mathcal{E}, \mathcal{F}$ etc. for $\mathbb{Z}$ -graded manifolds, M, N etc. for ordinary (non-graded) manifolds; Σ always denotes the source (spacetime) manifold of the sigma model, γ is typically a submanifold of Σ on which the observable is supported. We use $\mathcal{L}$ for a Lagrangian submanifold of a degree -1 symplectic graded manifold.

We denote by $\mathfrak{X}(\mathcal{M})$ the Lie algebra of vector fields on $\mathcal{M}$. For a fiber bundle $\pi : \mathcal{E} \rightarrow \mathcal{M}$, we denote by $\mathfrak{X}^{\text{vert}}(\mathcal{E})$ the Lie algebra of vertical vector fields on the total space $\mathcal{E}$, i.e. the space of sections $\Gamma(\mathcal{E}, T^{\text{vert}}\mathcal{E})$ of the vertical distribution on $\mathcal{E}$, $T^{\text{vert}}\mathcal{E} = \ker(d\pi) \subset T\mathcal{E}$.

We denote the degree (the ghost number) of functions/differential forms/vector fields on a graded manifold by $|\dots|$.

2. AKSZ REMINDER

What follows is a very short reminder of the AKSZ construction of topological sigma models in Batalin-Vilkovisky formalism, to fix the terminology and notation. We refer the reader to the original paper [1] and the later expositions in [7], [14] for details.

2.1. Target data. Let $\mathcal{M}$ be a degree n symplectic Q -manifold, i.e. a $\mathbb{Z}$ -graded manifold endowed with a degree 1 vector field Q satisfying $Q^2 = 0$ (the cohomological vector field) and with a degree⁴ n symplectic form $\omega \in \Omega^2(\mathcal{M})$ which is compatible with Q , i.e. $L_Q\omega = 0$.

Assume⁵ that Q has a Hamiltonian function $\Theta \in C^\infty(\mathcal{M})$ with $|\Theta| = n + 1$, $\{\Theta, \bullet\}_\omega = Q$ and satisfying

$$(7) \quad \{\Theta, \Theta\}_\omega = 0$$

Also assume that ω is exact, with $\alpha \in \Omega^1(\mathcal{M})$ a primitive.

Definition 1. We call the set of data $(\mathcal{M}, Q, \omega = \delta\alpha, \Theta)$ a *Hamiltonian Q -manifold of degree n* .

2.2. The AKSZ sigma model. Fix a Hamiltonian Q -manifold $\mathcal{M}$ of degree $n \geq -1$ and let Σ be an oriented closed manifold, $\dim \Sigma = n + 1$. Then one constructs the space of fields as the space of graded maps between graded manifolds from the degree-shifted tangent bundle $T[1]\Sigma$ to $\mathcal{M}$:

$$(8) \quad \mathcal{F}_\Sigma = \text{Map}(T[1]\Sigma, \mathcal{M})$$

It is a Q -manifold with the cohomological vector field coming from the lifting of Q on the target and of the de Rham operator d_Σ on Σ (viewed as a cohomological vector field on $T[1]\Sigma$) to the mapping space:

$$(9) \quad Q_\Sigma = (d_\Sigma)^{\text{lifted}} + (Q)^{\text{lifted}} \in \mathfrak{X}(\mathcal{F}_\Sigma)$$

Transgression map. The following natural maps

$$(10) \quad \begin{array}{ccc} \mathcal{F}_\Sigma \times T[1]\Sigma & \xrightarrow{\text{ev}} & \mathcal{M} \\ p \downarrow & & \\ & \mathcal{F}_\Sigma & \end{array}$$

(where p is the projection to the first factor) allow us to define the transgression map

$$(11) \quad \tau_\Sigma = p_* \text{ev}^* : \Omega^\bullet(\mathcal{M}) \rightarrow \Omega^\bullet(\mathcal{F}_\Sigma)$$

Here p_* is the fiber integration over $T[1]\Sigma$ (with the canonical integration measure). Map τ_Σ preserves the de Rham degree of a form, but changes the internal grading (the ghost number) by $-\dim \Sigma$.

If $u \in \mathfrak{X}(T[1]\Sigma)$, $v \in \mathfrak{X}(\mathcal{M})$ are any vector fields on the source and the target and $\psi \in \Omega^\bullet(\mathcal{M})$ is a form on the target, then for the Lie derivatives of the transgressed form along the lifted vector fields we have

$$(12) \quad L_{u^{\text{lifted}}} \tau_\Sigma(\psi) = 0$$

$$(13) \quad L_{v^{\text{lifted}}} \tau_\Sigma(\psi) = (-1)^{|v| \dim \Sigma} \tau_\Sigma(L_v \psi)$$

The finite version of (12) is that for $\Phi : T[1]\Sigma \rightarrow T[1]\Sigma$ a diffeomorphism of the source, we have

$$(14) \quad (\Phi^*)^* \tau_\Sigma(\psi) = \tau_\Sigma(\psi)$$

where $\Phi^* : \mathcal{F}_\Sigma \rightarrow \mathcal{F}_\Sigma$ is the lifting of Φ to the mapping space and $(\Phi^*)^* : \Omega^\bullet(\mathcal{F}_\Sigma) \rightarrow \Omega^\bullet(\mathcal{F}_\Sigma)$ is the pull-back by Φ^* .

⁴By “degree” here we mean the internal $\mathbb{Z}$ -grading coming from the grading on $\mathcal{M}$.

⁵In fact (cf. [14]), for $n \neq -1$ the symplectic property of the cohomological vector field ($L_Q\omega = 0$) implies existence and uniqueness of the Hamiltonian $\Theta = \frac{1}{n+1} \iota_E \iota_Q \omega$ where E is the Euler vector field. Maurer-Cartan equation (7) follows from $Q^2 = 0$ for $n \neq -2$. Also, for $n \neq 0$, a closed form ω is automatically exact.

The BV 2-form and the master action. One obtains the degree -1 symplectic form (the “BV 2-form”) on $\mathcal{F}_\Sigma$ from the target by transgression⁶:

$$(15) \quad \Omega_\Sigma = (-1)^{\dim \Sigma} \tau_\Sigma(\omega) \in \Omega^2(\mathcal{F}_\Sigma)$$

The Hamiltonian function for Q_Σ (the master action) is constructed as

$$(16) \quad S_\Sigma = \underbrace{\iota_{d_\Sigma^{\text{lifted}}} \tau_\Sigma(\alpha)}_{S_\Sigma^{\text{kin}}} + \underbrace{\tau_\Sigma(\Theta)}_{S_\Sigma^{\text{target}}} \in C^\infty(\mathcal{F}_\Sigma)$$

It automatically satisfies the classical master equation

$$\{S_\Sigma, S_\Sigma\}_{\Omega_\Sigma} = 0$$

One can summarize the construction above by saying that we have a degree -1 Hamiltonian Q -manifold structure on the mapping space (8): $(\mathcal{F}_\Sigma, Q_\Sigma, \Omega_\Sigma, S_\Sigma)$. The primitive 1-form for the BV 2-form can also be constructed by transgression as $\tau_\Sigma(\alpha)$.

Note that exact shifts of the target 1-form, $\alpha \mapsto \alpha + \delta f$, with $f \in C^\infty(M)$ leave S_Σ unchanged.

Why AKSZ theory is topological. For $\phi \in \text{Diff}(\Sigma)$ a diffeomorphism of Σ , denote $\tilde{\phi} \in \text{Diff}(T[1]\Sigma)$ the tangent lift of ϕ to $T[1]\Sigma$. Then

$$(17) \quad (\tilde{\phi}^*)^* S_\Sigma = S_\Sigma, \quad (\tilde{\phi}^*)^* \Omega_\Sigma = \Omega_\Sigma$$

because of (14) and because $d_\Sigma \in \mathfrak{X}(T[1]\Sigma)$ commutes with $\tilde{\phi}$, since the latter is a tangent lift.

In coordinates. Let x^a be local homogeneous coordinates on the target $\mathcal{M}$, let u^μ be local coordinates on Σ and $\theta^\mu = du^\mu$ be the associated degree 1 fiber coordinates on $T[1]\Sigma$. Then locally an element of $\mathcal{F}_\Sigma$ is parameterized by

$$(18) \quad X^a(u, \theta) = \sum_{k=0}^{\dim \Sigma} \underbrace{\sum_{1 \leq \mu_1 < \dots < \mu_k \leq \dim \Sigma} X_{\mu_1 \dots \mu_k}^a(u) \theta^{\mu_1} \dots \theta^{\mu_k}}_{X_{(k)}^a(u, \theta)}$$

Coefficient functions $X_{\mu_1 \dots \mu_k}^a(u)$ are local coordinates of degree $|x^a| - k$ on the mapping space $\mathcal{F}_\Sigma$. Expression (18) is known as (the component of) the *superfield*, and it can be regarded as a generating function for the coordinates on the mapping space $\mathcal{F}_\Sigma$.

For any function $f \in C^\infty(\mathcal{M})$, we have

$$(19) \quad \begin{aligned} \text{ev}^* f &= f(X) = \\ &= f(X_{(0)}) + X_{\geq 1}^a \cdot (\partial_a f)(X_{(0)}) + \frac{1}{2} X_{\geq 1}^a X_{\geq 1}^b \cdot (\partial_b \partial_a f)(X_{(0)}) + \dots \\ &\in C^\infty(\mathcal{F}_\Sigma \times T[1]\Sigma) \end{aligned}$$

where we denoted $X_{\geq 1}^a = \sum_{k \geq 1} X_{(k)}^a$ the part of the superfield of positive de Rham degree with respect to Σ ; ev is the horizontal arrow in (10).

Let α and ω be locally given as $\alpha = \alpha_a(x) \delta x^a$ and $\omega = \frac{1}{2} \omega_{ab}(x) \delta x^a \wedge \delta x^b$. Then the BV 2-form and its primitive are:

$$\Omega_\Sigma = (-1)^{\dim \Sigma} \int_\Sigma \frac{1}{2} \omega_{ab}(X) \delta X^a \wedge \delta X^b, \quad \alpha_\Sigma = \int_\Sigma \alpha_a(X) \delta X^a$$

⁶We introduce the sign $(-1)^{\dim \Sigma}$ in this definition to avoid signs in the formula for the action below. The reader may encounter different sign conventions for the AKSZ construction in the literature.

(Note that we use δ to denote the de Rham differential on the target $\mathcal{M}$ and on the mapping space $\mathcal{F}_\Sigma$. We reserve symbol d for the de Rham differential on the source Σ .)

The master action is:

$$S_\Sigma(X) = \underbrace{\int_\Sigma \alpha_a(X) dX^a}_{S_\Sigma^{\text{kin}}} + \underbrace{\int_\Sigma \Theta(X)}_{S_\Sigma^{\text{target}}}$$

If the cohomological vector field on the target is locally written as $Q = Q^a(x) \frac{\partial}{\partial x^a}$ then the cohomological vector field (9) is determined by its action on the components of the superfield:

$$Q_\Sigma X^a = dX^a + Q^a(X)$$

Critical points of S_Σ are (with our sign conventions) Q -anti-morphisms between $T[1]\Sigma$ and $\mathcal{M}$, i.e. Q -morphisms between $(T[1]\Sigma, d)$ and $(\mathcal{M}, -Q)$.

2.3. Examples. Here we recall some of the standard examples of the AKSZ construction.

Chern-Simons theory [1]. Let $\mathfrak{g}$ be a quadratic Lie algebra, i.e. a Lie algebra with a non-degenerate invariant pairing $\langle, \rangle$. Denote by $\psi : \mathfrak{g}[1] \rightarrow \mathfrak{g}$ the degree 1 $\mathfrak{g}$ -valued coordinate on $\mathfrak{g}[1]$. We choose the target Hamiltonian Q -manifold of degree 2 as

$$(20) \quad \mathcal{M} = \mathfrak{g}[1], \quad Q = \left\langle \frac{1}{2}[\psi, \psi], \frac{\partial}{\partial \psi} \right\rangle, \\ \omega = \frac{1}{2}(\delta\psi, \delta\psi), \quad \alpha = \frac{1}{2}(\psi, \delta\psi), \quad \Theta = \frac{1}{6}(\psi, [\psi, \psi])$$

where $\langle, \rangle$ is the canonical pairing⁷ between $\mathfrak{g}$ and $\mathfrak{g}^*$. The associated AKSZ sigma model on a closed oriented 3-manifold Σ has the space of fields

$$\mathcal{F}_\Sigma = \text{Map}(T[1]\Sigma, \mathfrak{g}[1]) \cong \mathfrak{g}[1] \otimes \Omega^\bullet(\Sigma)$$

The superfield is

$$(21) \quad A = A_{(0)} + A_{(1)} + A_{(2)} + A_{(3)}$$

with $A_{(k)}$ a coordinate on $\mathcal{F}_\Sigma$ with values in $\mathfrak{g}$ -valued k -forms on Σ , with internal degree (ghost number) $1 - k$, for $k = 0, 1, 2, 3$. The BV 2-form is

$$\Omega_\Sigma = -\frac{1}{2} \int_\Sigma (\delta A, \delta A)$$

and the action is

$$S_\Sigma = \int_\Sigma \frac{1}{2}(A, dA) + \frac{1}{6}(A, [A, A])$$

This is the action of Chern-Simons theory in Batalin-Vilkovisky formalism.

In case $\mathfrak{g} = \mathbb{R}$ with abelian Lie algebra structure, we have $Q = \Theta = 0$ on the target and

$$(22) \quad \mathcal{F}_\Sigma = \Omega^\bullet(\Sigma)[1], \quad \Omega_\Sigma = -\frac{1}{2} \int_\Sigma \delta A \wedge \delta A, \quad S_\Sigma = \frac{1}{2} \int_\Sigma A \wedge dA$$

This is the *abelian* Chern-Simons theory in BV formalism.

BF theory. For $\mathfrak{g}$ a Lie algebra (not necessarily quadratic⁸) and D a non-negative integer, we define the target Hamiltonian Q -manifold of degree $D - 1$ as

⁷We will generally be using notation $\langle, \rangle$ for the canonical pairing $V \otimes V^* \rightarrow \mathbb{R}$ between a vector space V and its dual; sometimes we will indicate the respective V as a subscript: $\langle, \rangle_V$.

⁸However, for the consistency of quantization, in particular for the quantum master equation (48), one has to require that $\mathfrak{g}$ is unimodular.

$$(23) \quad \mathcal{M} = \mathfrak{g}[1] \oplus \mathfrak{g}^*[D-2], \quad Q = \left\langle \frac{1}{2}[\psi, \psi], \frac{\partial}{\partial \psi} \right\rangle + \left\langle \text{ad}^*_{\psi} \xi, \frac{\partial}{\partial \xi} \right\rangle,$$

$$\omega = \langle \delta \xi, \delta \psi \rangle, \quad \alpha = \langle \xi, \delta \psi \rangle, \quad \Theta = \frac{1}{2} \langle \xi, [\psi, \psi] \rangle$$

Here $\psi : \mathfrak{g}[1] \rightarrow \mathfrak{g}$ is as before and $\xi : \mathfrak{g}^*[D-2] \rightarrow \mathfrak{g}^*$ is the $\mathfrak{g}^*$ -valued coordinate on $\mathfrak{g}^*[D-2]$ of degree $D-2$; ad^* is the coadjoint action of $\mathfrak{g}$ on $\mathfrak{g}^*$.

The associated AKSZ sigma model on a closed oriented D -manifold Σ has the space of fields

$$(24) \quad \mathcal{F}_{\Sigma} = \mathfrak{g}[1] \otimes \Omega^{\bullet}(\Sigma) \oplus \mathfrak{g}^*[D-2] \otimes \Omega^{\bullet}(\Sigma)$$

The superfields associated to ψ and ξ are respectively

$$A = \sum_{k=0}^D A_{(k)}, \quad B = \sum_{k=0}^D B_{(k)}$$

with $A_{(k)}$ a $\mathfrak{g}$ -valued k -form on Σ of internal degree $1-k$; $B_{(k)}$ is a $\mathfrak{g}^*$ -valued k -form of internal degree $D-2-k$. The BV 2-form and the action are:

$$(25) \quad \Omega_{\Sigma} = (-1)^D \int_{\Sigma} \langle \delta B, \delta A \rangle, \quad S_{\Sigma} = \int_{\Sigma} \left\langle B, dA + \frac{1}{2}[A, A] \right\rangle$$

This is the BF theory in BV formalism.

In abelian case, $\mathfrak{g} = \mathbb{R}$, we have $Q = \Theta = 0$ on the target and

$$(26) \quad \mathcal{F}_{\Sigma} = \Omega^{\bullet}(\Sigma)[1] \oplus \Omega^{\bullet}(\Sigma)[D-2],$$

$$\Omega_{\Sigma} = (-1)^D \int_{\Sigma} \delta B \wedge \delta A, \quad S_{\Sigma} = \int_{\Sigma} B \wedge dA$$

Poisson sigma model [7]. Let M be a manifold endowed with a Poisson bivector $\pi \in \Gamma(M, \wedge^2 TM)$. We construct the target Hamiltonian Q -manifold of degree 1 as

$$(27) \quad \mathcal{M} = T^*[1]M,$$

$$Q = \left\langle \pi(x), p \wedge \frac{\partial}{\partial x} \right\rangle_{\wedge^2 T_x M} + \frac{1}{2} \left\langle \frac{\partial}{\partial x} \pi(x), (p \wedge p) \otimes \frac{\partial}{\partial p} \right\rangle_{(\wedge^2 T \otimes T^*)_x M},$$

$$\omega = \langle \delta p, \delta x \rangle, \quad \alpha = \langle p, \delta x \rangle, \quad \Theta = \frac{1}{2} \langle \pi(x), p \wedge p \rangle_{\wedge^2 T_x M}$$

Here x and p stand for the local base and fiber coordinates on $T^*[1]M$ respectively. Note that all objects in (27) are globally well defined.

The corresponding AKSZ sigma model on an oriented closed surface Σ has the space of fields

$$\mathcal{F}_{\Sigma} = \text{Map}(T[1]\Sigma, T^*[1]M)$$

with superfields

$$X = X_{(0)} + X_{(1)} + X_{(2)}, \quad \eta = \eta_{(0)} + \eta_{(1)} + \eta_{(2)}$$

associated to local coordinates x and p on the target, respectively. Here $X_{(k)}$ and $\eta_{(k)}$ are k -forms on Σ with internal degrees $-k$ and $1-k$ respectively. The BV 2-form and the action are:

$$\Omega_{\Sigma} = \int_{\Sigma} \langle \delta \eta, \delta X \rangle, \quad S_{\Sigma} = \int_{\Sigma} \langle \eta, dX \rangle + \frac{1}{2} \langle \pi(X), \eta \wedge \eta \rangle$$

3. HAMILTONIAN Q -BUNDLES

In this section we briefly recall the standard notion of a Q -bundle and then introduce “Hamiltonian Q -bundles”, an auxiliary notion necessary for our construction of observables.

3.1. Q -bundles reminder. Recall that a “ Q -bundle” [11] is a fiber bundle in the category of Q -manifolds.

In particular, a *trivial* Q -bundle is trivial bundle of graded manifolds

$$\pi : \underbrace{\mathcal{M} \times \mathcal{N}}_{\mathcal{E}} \rightarrow \mathcal{M}$$

where the base $\mathcal{M}$ is endowed with a cohomological vector field Q and the total space is endowed with a cohomological vector field Q^{tot} in such a way that π is a Q -morphism, i.e. $d\pi(Q^{\text{tot}}) = Q$. This implies the following ansatz for Q^{tot} :

$$(28) \quad Q^{\text{tot}} = Q + \mathcal{A}$$

where $\mathcal{A} \in \mathfrak{X}^{\text{vert}}(\mathcal{E}) \cong \mathfrak{X}(\mathcal{N}) \hat{\otimes} C^\infty(\mathcal{M})$ is the vertical part of Q^{tot} . Cohomological property for Q^{tot} is equivalent to

$$(29) \quad Q\mathcal{A} + \frac{1}{2}[\mathcal{A}, \mathcal{A}] = 0$$

plus the cohomological property for Q . Note that in the first term on the l.h.s. of (29) we are thinking of Q as acting on $C^\infty(\mathcal{M})$ part of $\mathcal{A}$. Equivalently, we can lift Q to a horizontal vector field on $\mathcal{E}$ and write the first term as $[Q, \mathcal{A}]$.

3.2. Trivial Hamiltonian Q -bundles.

Definition 2. We define a trivial Hamiltonian Q -bundle of degree $n' \in \mathbb{Z}$ as the following collection of data.

- (i) A trivial Q -bundle

$$\pi : \mathcal{E} = \mathcal{M} \times \mathcal{N} \rightarrow \mathcal{M}$$

with $Q^{\text{tot}} = Q + \mathcal{A}$ as in (28).

- (ii) The fiber $\mathcal{N}$ is endowed with a degree n' exact symplectic form $\omega' = \delta\alpha'$ and a degree $(n' + 1)$ Hamiltonian $\Theta' \in C^\infty(\mathcal{E})$ satisfying $\mathcal{A} = \{\Theta', \bullet\}_{\omega'}$ and

$$(30) \quad Q\Theta' + \frac{1}{2}\{\Theta', \Theta'\}_{\omega'} = 0$$

Remark 1. (Roytenberg) Symplectic structure ω' is automatically exact for $n' \neq 0$, since $\omega' = \delta\left(\frac{1}{n'}\iota_E\omega'\right)$ where E is the Euler vector field on $\mathcal{N}$.

Remark 2. If $n' = 0$, then ω' is not automatically exact. In this situation, exactness condition may be relaxed⁹ to the Bohr-Sommerfeld condition that $\omega'/2\pi$ defines an integral cohomology class in $H^2(\mathcal{N})$. Then the role of primitive 1-form α' is taken by a Hermitian line bundle L' over $\mathcal{N}$ equipped with a $U(1)$ -connection ∇' of curvature ω' .

Remark 3. For a Hamiltonian Q -manifold of degree $n \neq -2$, equation $\{\Theta, \Theta\} = 0$ follows from the fact that Θ is the Hamiltonian for a cohomological vector field (Jacobi identity implies $\frac{1}{2}\{\Theta, \Theta\}, \bullet\} = Q^2(\bullet) = 0$ which implies that $\{\Theta, \Theta\}$ is a constant of degree $n + 2$ and thus vanishes unless $n = -2$).

Observe that this argument fails for the fiber $\mathcal{N}$: applying the same reasoning to derive the equation (30) from (29), we obtain that l.h.s. of (30) is a pull-back of some function on the base of degree $n' + 2$, which does not imply that it is zero.

⁹Cf. remark 8 for the motivation: primitive α' (or connection ∇' in the relaxed version) will be needed in the construction of section 5 to define the kinetic part of the auxiliary action.

3.3. General Hamiltonian Q -bundles. The assumption of triviality of the bundle $\mathcal{E}$, used in the definition 2, can be relaxed as follows.

Definition 3. A Hamiltonian Q -bundle is the following set of data:

- (i) A Q -bundle $\pi : \mathcal{E} \rightarrow \mathcal{M}$.
- (ii) A degree n' exact pre-symplectic form $\omega' = \delta\alpha'$ on the total space $\mathcal{E}$, with the property that the distribution $\ker \omega' \subset T\mathcal{E}$ is transversal to the vertical distribution $T^{\text{vert}}\mathcal{E}$; thus $\ker \omega'$ defines a flat Ehresmann connection $\nabla_{\omega'}$ on $\mathcal{E}$.
- (iii) A Hamiltonian function $\Theta' \in C^\infty(\mathcal{E})$ satisfying

$$\iota_{Q^{\text{tot}}}\omega' = \delta^{\text{vert}}\Theta'$$

where $\delta^{\text{vert}} = (i^{\text{vert}})^* \circ \delta$ is the vertical part of de Rham differential on $\mathcal{E}$; we denoted $i^{\text{vert}} : T^{\text{vert}}\mathcal{E} \hookrightarrow T\mathcal{E}$ the natural inclusion. We also require that

$$(Q^{\text{hor}} + \frac{1}{2}Q^{\text{vert}})\Theta' = 0$$

where the splitting $Q^{\text{tot}} = Q^{\text{hor}} + Q^{\text{vert}}$ of the cohomological vector field on $\mathcal{E}$ into the horizontal and the vertical part is defined by connection $\nabla_{\omega'}$.

Remark 4. Note that one can introduce a local trivialization of $\mathcal{E}$ consistent with the flat connection $\nabla_{\omega'}$ (i.e. where the horizontal distribution $\ker \omega' \subset T\mathcal{E}$ is defined by the direct product structure on $\mathcal{E}|_U$ coming from the local trivialization, where $U \subset \mathcal{M}$ is a trivializing neighborhood). In such a trivialization, we recover back the definition 2. In this sense, definition 3 does not bring in a drastic increase of generality.

3.4. Examples.

- (i) Let $\mathfrak{g}$ be a Lie algebra and let $\rho : \mathfrak{g} \rightarrow \mathfrak{so}(R) \subset \text{End}(R)$ be a representation of $\mathfrak{g}$ on by anti-symmetric matrices on a Euclidean vector space $R, (\cdot, \cdot)$. Then for any $k \in \mathbb{Z}$ we can define a degree $4k + 2$ trivial Hamiltonian Q -bundle with base $\mathcal{M} = \mathfrak{g}[1]$ and fiber $\mathcal{N} = R[2k + 1]$. Let ψ be the degree 1 $\mathfrak{g}$ -valued coordinate on $\mathfrak{g}[1]$ and let x be the degree $2k + 1$ R -valued coordinate on $R[2k + 1]$. Then we set

$$(31) \quad Q = \frac{1}{2} \left\langle [\psi, \psi], \frac{\partial}{\partial \psi} \right\rangle$$

– the Chevalley-Eilenberg differential, and

$$(32) \quad \mathcal{A} = \left\langle \rho(\psi)x, \frac{\partial}{\partial x} \right\rangle, \quad \omega' = \frac{1}{2}(\delta x, \delta x),$$

$$\alpha' = \frac{1}{2}(x, \delta x), \quad \Theta' = \frac{1}{2}(x, \rho(\psi)x)$$

where $\langle \cdot, \cdot \rangle$ denotes the canonical pairing between a vector space and its dual. It is a straightforward check that this comprises the data of a trivial Hamiltonian Q -bundle of degree $4k + 2$, according to definition 2.

As a variation of this example, we may require that the pairing $(\cdot, \cdot)$ is anti-symmetric (so that R is a symplectic space instead of Euclidean) and that $\rho : \mathfrak{g} \rightarrow \mathfrak{sp}(R) \subset \text{End}(R)$ is a symplectic representation. Then formulae (31,32) define a Hamiltonian Q -bundle structure of degree $4k$ on $\mathfrak{g}[1] \times R[2k]$ (note that here we choose an even degree shift for the fiber).

- (ii) Example above generalizes straightforwardly to the setting of L_∞ algebras and L_∞ modules. Namely, let $\mathfrak{g}$ be an L_∞ algebra with operations $\{l_j : \wedge^j \mathfrak{g} \rightarrow \mathfrak{g}\}_{j \geq 1}$. Also, let R be graded vector space with graded symmetric pairing $(\cdot, \cdot)$ of degree q and with the structure of an L_∞ module over $\mathfrak{g}$ with

the module operations $\{\rho_j : \wedge^j \mathfrak{g} \otimes R \rightarrow R\}_{j \geq 0}$, where we additionally require that operations $(\bullet, \rho_j(\cdots; \bullet))$ are graded anti-symmetric w.r.t. the two entries in the module. For ψ a $\mathfrak{g}$ -valued degree 1 coordinate on $\mathfrak{g}[1]$ and x a R -valued degree $2k + 1$ coordinate on $R[2k + 1]$, we construct a trivial Hamiltonian Q -bundle with $\mathcal{M} = \mathfrak{g}[1]$, $\mathcal{N} = R[2k + 1]$ and the data are:

$$(33) \quad Q = \sum_{j \geq 1} \frac{1}{j!} \left\langle l_j(\psi, \dots, \psi), \frac{\partial}{\partial \psi} \right\rangle$$

for the base and

$$(34) \quad \mathcal{A} = \sum_{j \geq 0} \frac{1}{j!} \left\langle \rho_j(\psi, \dots, \psi; x), \frac{\partial}{\partial x} \right\rangle, \quad \omega' = \frac{1}{2}(\delta x, \delta x),$$

$$\alpha' = \frac{1}{2}(x, \delta x), \quad \Theta' = \sum_{j \geq 0} \frac{1}{2j!}(x, \rho_j(\psi, \dots, \psi; x))$$

This comprises the data of a trivial Hamiltonian Q -bundle of degree $4k + 2 + q$.

- (iii) Example (ii) admits the following variation. If $\mathfrak{h}$ is an L_∞ algebra with $I \subset \mathfrak{h}$ an L_∞ ideal¹⁰, then there is a natural Q -bundle

$$(35) \quad \pi : \mathfrak{h}[1] \rightarrow (\mathfrak{h}/I)[1]$$

with bundle projection being the quotient map and with the fiber I . If in addition we fix a section of the quotient map $\mathfrak{h} \rightarrow \mathfrak{h}/I$, i.e. a splitting of the short exact sequence

$$(36) \quad I \rightarrow \mathfrak{h} \rightarrow \mathfrak{h}/I$$

and if I carries a degree q cyclic¹¹ inner product, then (35) becomes a Hamiltonian Q -bundle of degree $2 + q$. Explicitly, the structure on the base $\mathcal{M} = (\mathfrak{h}/I)[1]$ is again given by (33) with $\{l_j\}$ the quotient L_∞ operations on $\mathfrak{h}/I$, ψ the degree 1 $(\mathfrak{h}/I)$ -valued coordinate on $(\mathfrak{h}/I)[1]$. The structure on the fiber $\mathcal{N} = I[1]$ is:

$$(37) \quad \mathcal{A} = \sum_{j \geq 0, k \geq 0, j+k \geq 1} \frac{1}{j!k!} \left\langle P_I \lambda_{j+k}(\underbrace{\psi, \dots, \psi}_j, \underbrace{x, \dots, x}_k), \frac{\partial}{\partial x} \right\rangle,$$

$$\omega' = \frac{1}{2}(\delta x, \delta x), \quad \alpha' = \frac{1}{2}(x, \delta x),$$

$$\Theta' = \sum_{j \geq 0, k \geq 0, j+k \geq 1} \frac{1}{j!(k+1)!} (x, P_I \lambda_{j+k}(\underbrace{\psi, \dots, \psi}_j, \underbrace{x, \dots, x}_k))$$

where $P_I : \mathfrak{h} \rightarrow I$ is the projection to the ideal fixed by the choice of splitting of (36), $\{\lambda_j\}$ are the L_∞ operations on $\mathfrak{h}$, x is the degree 1 I -valued coordinate on $I[1]$ and $(,)$ is the cyclic pairing on I .

- (iv) Let $\mathfrak{g}$ be a Lie algebra and suppose there is a Hamiltonian action of $\mathfrak{g}$ on an exact symplectic manifold $(M, \omega_M = \delta \alpha_M)$ with equivariant moment map $\mu : M \rightarrow \mathfrak{g}^*$. Then we set $\mathcal{M} = \mathfrak{g}[1]$ as in (i), with the structure (31). For the fiber, we set $\mathcal{N} = M$ and

$$(38) \quad \mathcal{A} = \{\langle \psi, \mu \rangle, \bullet\}, \quad \omega' = \omega_M, \quad \alpha' = \alpha_M, \quad \Theta' = \langle \psi, \mu \rangle$$

¹⁰Recall that a subspace $I \subset \mathfrak{h}$ is called an L_∞ ideal if the L_∞ operations on $\mathfrak{h}$ take values in I if at least one argument is in I .

¹¹By cyclic property for the pairing on the ideal we mean that extending the pairing on I to $\mathfrak{h}$ by zero on $\mathfrak{h}/I$, we obtain a degenerate cyclic pairing for the L_∞ algebra $\mathfrak{h}$.

This is the data of a trivial Hamiltonian Q -bundle of degree 0 on $\mathfrak{g}[1] \times M$. Exactness condition for ω_M can be relaxed to the Bohr-Sommerfeld integrality condition, cf. remark 2.

- (v) For any Q -manifold $(\mathcal{M}, Q)$, given a function $\theta \in C^\infty(\mathcal{M})$ satisfying

$$Q\theta = 0, \quad |\theta| = p$$

we can construct a trivial Hamiltonian Q -bundle over $\mathcal{M}$ of degree $p - 1$ with fiber $\mathcal{N}$ a point, $\mathcal{A} = 0$, $\omega' = 0$, $\Theta' = \theta$ (and we formally prescribe degree $p - 1$ to ω').

4. BV PRE-OBSERVABLES AND OBSERVABLES

In this section we recall some basic structures of BV formalism: classical and quantum BV theory (cf. [15]), observables in classical and quantum setting. We also introduce the notion of a *pre-observable*¹²: an extension of a BV theory by auxiliary fields endowed with an odd-symplectic structure and an action (which is also dependent on the fields of the ambient theory). Given a pre-observable, one can construct an observable by integrating out the auxiliary fields (propositions 1, 3). Pre-observables come in three versions: purely classical, *semi-quantum* – suited for integrating out auxiliary fields to produce a classical observable for the ambient theory, and *quantum* – suited for integrating out auxiliary fields to produce a quantum observable for the ambient theory.

Throughout this section in our treatment of quantum objects we are assuming that the spaces of fields are finite-dimensional. This is almost never the case in the interesting examples. In the setting of local quantum field theory, when the spaces of fields are infinite-dimensional, measures on fields are problematic to define; however the BV Laplacians and the integrals over auxiliary fields (which are perturbative path integrals now) can be made sense of with an appropriate regularization/renormalization procedure (one systematic approach is via Wilson's renormalization flow, cf. [10]). Proofs of propositions 1, 3 cannot be taken for granted in the infinite-dimensional setting, and should be redone within the framework of perturbative path integrals (cf. e.g. the proof of proposition 6 in section 6.3).

In view of this, our treatment of quantum objects in this section should be viewed as a simplified discussion, to motivate the interest in pre-observables and argue that given a pre-observable one can expect a path integral expression for an observable.

4.1. Pre-observables and observables for a classical ambient BV theory.

Definition 4. We will call a (*classical*) *BV theory* a Hamiltonian Q -manifold of degree -1 , i.e. a quadruple $(\mathcal{F}, Q, \Omega, S)$ consisting of a Q -manifold $(\mathcal{F}, Q)$ (the *space of fields* with the *BRST operator*), the degree -1 symplectic structure (the *BV 2-form*) Ω and the degree 0 action S satisfying $\{S, \bullet\}_\Omega = Q$ and the *classical master equation*:

$$(39) \quad \{S, S\}_\Omega = 0$$

Definition 5. A (*classical*) *observable* for a classical BV theory $(\mathcal{F}, Q, \Omega, S)$ is a degree 0 function¹³ $O \in C^\infty(\mathcal{F})$ satisfying

$$Q(O) = 0$$

¹²This is not a part of the standard BV lore; the term is sometimes used in a totally different sense in the literature.

¹³We will be assuming degree 0 condition for observables by default, but sometimes it is interesting to relax it. In such cases we will indicate the degree explicitly.

Observables O and O' are called *equivalent* if $O' - O = Q(\Psi)$ for some $\Psi \in C^\infty(\mathcal{F})$, i.e. O' and O give the same class in Q -cohomology.

Definition 6. For a classical BV theory $(\mathcal{F}, Q, \Omega, S)$ (which we will call the *ambient theory*), a *pre-observable* is a degree -1 Hamiltonian Q -bundle over $\mathcal{F}$. We will call the fiber the *space of auxiliary fields* $\mathcal{F}^{\text{aux}}$, which comes with its own degree -1 symplectic structure Ω^{aux} and the *action for auxiliary fields* $S^{\text{aux}} \in C^\infty(\mathcal{F} \times \mathcal{F}^{\text{aux}})$ satisfying

$$(40) \quad QS^{\text{aux}} + \frac{1}{2}\{S^{\text{aux}}, S^{\text{aux}}\}_{\Omega^{\text{aux}}} = 0$$

Remark 5. Note that if in addition to (40) we have the property

$$\{S^{\text{aux}}, S^{\text{aux}}\}_\Omega = 0$$

then the pre-observable gives a new classical BV theory

$$(\mathcal{F} \times \mathcal{F}^{\text{aux}}, Q + \{S^{\text{aux}}, \bullet\}_{\Omega + \Omega^{\text{aux}}}, \Omega + \Omega^{\text{aux}}, S + S^{\text{aux}})$$

Note also that this cohomological vector field on $\mathcal{F} \times \mathcal{F}^{\text{aux}}$ differs from the one arising from the Q -bundle structure by the term $\{S^{\text{aux}}, \bullet\}_\Omega$ and is in general not projectable to $\mathcal{F}$.

In the case when $\mathcal{F}^{\text{aux}}$ is finite-dimensional, it makes sense to introduce the following notion.

Definition 7. For a classical BV-theory $(\mathcal{F}, Q, \Omega, S)$, a *semi-quantum* pre-observable is a quadruple $(\mathcal{F}^{\text{aux}}, \Omega^{\text{aux}}, \mu^{\text{aux}}, S^{\text{aux}})$ consisting of a $\mathbb{Z}$ -graded manifold $\mathcal{F}^{\text{aux}}$ and a degree -1 symplectic form Ω^{aux} on it. Further,

- $\mathcal{F}^{\text{aux}}$ is endowed with a volume element μ^{aux} compatible with Ω^{aux} in the sense that the associated BV Laplacian on $C^\infty(\mathcal{F}^{\text{aux}})$,

$$\Delta^{\text{aux}} : f \mapsto \frac{1}{2} \operatorname{div}_{\mu^{\text{aux}}} \{f, \bullet\}_{\Omega^{\text{aux}}}$$

satisfies

$$(\Delta^{\text{aux}})^2 = 0$$

- Degree 0 action $S^{\text{aux}} \in C^\infty(\mathcal{F} \times \mathcal{F}^{\text{aux}})$ satisfies

$$(41) \quad QS^{\text{aux}} + \frac{1}{2}\{S^{\text{aux}}, S^{\text{aux}}\}_{\Omega^{\text{aux}}} - i\Delta^{\text{aux}}S^{\text{aux}} = 0$$

or, equivalently,

$$(42) \quad (Q - i\Delta^{\text{aux}})e^{iS^{\text{aux}}} = 0$$

We call two semi-quantum pre-observables $(\mathcal{F}^{\text{aux}}, \Omega^{\text{aux}}, \mu^{\text{aux}}, S^{\text{aux}})$ and $(\mathcal{F}^{\text{aux}}, \Omega^{\text{aux}}, \mu^{\text{aux}}, \tilde{S}^{\text{aux}})$ *equivalent*, if there exists a degree -1 function $R^{\text{aux}} \in C^\infty(\mathcal{F} \times \mathcal{F}^{\text{aux}})$ such that

$$(43) \quad e^{i\tilde{S}^{\text{aux}}} - e^{iS^{\text{aux}}} = (Q - i\Delta^{\text{aux}})(e^{iS^{\text{aux}}}R^{\text{aux}})$$

Infinitesimally, this equivalence condition can be written as

$$(44) \quad \tilde{S}^{\text{aux}} - S^{\text{aux}} = QR^{\text{aux}} + \{S^{\text{aux}}, R^{\text{aux}}\}_{\Omega^{\text{aux}}} - i\Delta^{\text{aux}}R^{\text{aux}} + \mathcal{O}(R^2)$$

In particular, a classical pre-observable can be promoted to a semi-quantum pre-observable if there exists a volume element μ^{aux} on $\mathcal{F}^{\text{aux}}$ compatible with Ω^{aux} , such that in addition to (40) we have $\Delta^{\text{aux}}S^{\text{aux}} = 0$.

Given a semi-quantum pre-observable, one can construct an observable for the ambient classical BV theory using fiber BV integrals [13] as follows.

Proposition 1. *Given a semi-quantum pre-observable $(\mathcal{F}^{\text{aux}}, \Omega^{\text{aux}}, \mu^{\text{aux}}, S^{\text{aux}})$, set*

$$(45) \quad O_{\mathcal{L}} = \int_{\mathcal{L} \subset \mathcal{F}^{\text{aux}}} e^{iS^{\text{aux}}} \sqrt{\mu^{\text{aux}}} |_{\mathcal{L}} \in C^\infty(\mathcal{F})$$

for $\mathcal{L} \subset \mathcal{F}^{\text{aux}}$ a Lagrangian submanifold. Then:

- (i) $O_{\mathcal{L}}$ is an observable, i.e. $Q(O_{\mathcal{L}}) = 0$.
- (ii) If Lagrangian submanifolds $\mathcal{L}$ and $\mathcal{L}'$ can be connected by a Lagrangian homotopy, then the difference $O_{\mathcal{L}'} - O_{\mathcal{L}}$ is Q -exact, i.e. observables $O_{\mathcal{L}}$ and $O_{\mathcal{L}'}$ are equivalent.
- (iii) Given two equivalent semi-quantum pre-observables S^{aux} and $\tilde{S}^{\text{aux}}$, the associated observables $O_{\mathcal{L}}$ and $\tilde{O}_{\mathcal{L}}$ are equivalent.

Proof. Applying Q to the integral (45), we obtain

$$(46) \quad Q(O_{\mathcal{L}}) = Q \int_{\mathcal{L}} e^{iS^{\text{aux}}} = \int_{\mathcal{L}} (Q - i\Delta^{\text{aux}}) e^{iS^{\text{aux}}} = 0$$

where we used (42) and the Stokes' theorem for BV integrals [15], that the BV integral of a Δ -coboundary vanishes. We suppress the measure in the short-hand notation for integrals here. This proves (i).

Now let $\{\mathcal{L}_t\}_{t \in [0,1]}$ be a smooth family of Lagrangian submanifolds of $(\mathcal{F}^{\text{aux}}, \Omega^{\text{aux}})$ connecting $\mathcal{L} = \mathcal{L}_0$ and $\mathcal{L}' = \mathcal{L}_1$. An infinitesimal deformation of a Lagrangian submanifold $\mathcal{L}_t$ is given by a graph of a Hamiltonian vector field¹⁴

$$(47) \quad \frac{d}{dt} \mathcal{L}_t = \text{graph}(\{\bullet, \Psi_t\}_{\Omega^{\text{aux}}}) \in \Gamma(\mathcal{L}_t, N\mathcal{L}_t)$$

viewed as a section of the normal bundle of $\mathcal{L}_t$, with $\Psi_t \in C^\infty(\mathcal{L}_t)$, $|\Psi_t| = -1$ the generator. Hence,

$$\begin{aligned} \frac{d}{dt} O_{\mathcal{L}_t} &= \frac{d}{dt} \int_{\mathcal{L}_t} e^{iS^{\text{aux}}} |_{\mathcal{L}_t} \\ &= \int_{\mathcal{L}_t} \{e^{iS^{\text{aux}}}, \Psi_t\}_{\Omega^{\text{aux}}} + e^{iS^{\text{aux}}} \cdot \Delta^{\text{aux}}(\Psi_t) \\ &= \int_{\mathcal{L}_t} \Delta^{\text{aux}}(e^{iS^{\text{aux}}} \Psi_t) - \Delta^{\text{aux}}(e^{iS^{\text{aux}}}) \cdot \Psi_t \\ &= \int_{\mathcal{L}_t} \Delta^{\text{aux}}(e^{iS^{\text{aux}}} \Psi_t) + iQ(e^{iS^{\text{aux}}}) \cdot \Psi_t \\ &= Q \left(i \int_{\mathcal{L}_t} e^{iS^{\text{aux}}} \Psi_t \right) \end{aligned}$$

The term $\Delta^{\text{aux}}(\Psi_t)$ in the second line comes from the transformation of measure $\sqrt{\mu^{\text{aux}}} |_{\mathcal{L}_t}$ by the infinitesimal shift of Lagrangian (47). This calculation implies that

$$O_{\mathcal{L}'} - O_{\mathcal{L}} = Q \left(\int_0^1 dt \, i \int_{\mathcal{L}_t} e^{iS^{\text{aux}}} \Psi_t \right)$$

and thus proves (ii).

Item (iii) follows immediately from the construction (45), our definition of equivalence (43) and the Stokes' theorem for BV integrals:

$$\tilde{O}_{\mathcal{L}} - O_{\mathcal{L}} = \int_{\mathcal{L}} e^{i\tilde{S}^{\text{aux}}} - e^{iS^{\text{aux}}}$$

¹⁴A general infinitesimal Lagrangian deformation is given by $\text{graph}(\iota_{(\Omega^{\text{aux}})^{-1}\chi_t})$ for any degree -1 closed 1-form $\chi_t \in \Omega^1(\mathcal{L}_t)$. Since in non-zero degree a closed form is automatically *exact* (cf. remark 1), we have (47) with the generator $\Psi_t = \iota_E \chi_t$.

$$= \int_{\mathcal{L}} (Q - i\Delta^{\text{aux}}) \left(e^{iS^{\text{aux}}} R^{\text{aux}} \right) = Q \left(\int_{\mathcal{L}} e^{iS^{\text{aux}}} R^{\text{aux}} \right)$$

□

Here we are implicitly assuming convergence of the integral (45), and for the proof of (ii) the integral over $\mathcal{L}_t$ should converge for every $t \in [0, 1]$.

The following construction allows one to produce new semi-quantum pre-observables out of old ones by partially integrating out the auxiliary fields.

Proposition 2. *Given a semi-quantum pre-observable $(\mathcal{F}^{\text{aux}}, \Omega^{\text{aux}}, \mu^{\text{aux}}, S^{\text{aux}})$, assume that $\mathcal{F}^{\text{aux}}$ is given as a product¹⁵, $\mathcal{F}^{\text{aux}} = \mathcal{F}_z^{\text{aux}} \times \tilde{\mathcal{F}}^{\text{aux}}$, so that Ω^{aux} and μ^{aux} split:*

$$\Omega^{\text{aux}} = \Omega_z^{\text{aux}} + \tilde{\Omega}^{\text{aux}}, \quad \mu^{\text{aux}} = \mu_z^{\text{aux}} \times \tilde{\mu}^{\text{aux}}$$

Define $S_z^{\text{aux}} \in C^\infty(\mathcal{F} \times \mathcal{F}_z^{\text{aux}})$ by

$$e^{iS_z^{\text{aux}}} = \int_{\tilde{\mathcal{L}} \subset \tilde{\mathcal{F}}^{\text{aux}}} e^{iS^{\text{aux}}} \sqrt{\tilde{\mu}^{\text{aux}}} |_{\mathcal{L}}$$

with $\tilde{\mathcal{L}}$ a Lagrangian submanifold of $(\tilde{\mathcal{F}}^{\text{aux}}, \tilde{\Omega}^{\text{aux}})$. Then

- (i) $(\mathcal{F}_z^{\text{aux}}, \Omega_z^{\text{aux}}, \mu_z^{\text{aux}}, S_z^{\text{aux}})$ is a semi-quantum pre-observable for the same ambient classical BV theory.
- (ii) The observable for the ambient theory induced from S_z^{aux} using (45) with Lagrangian $\mathcal{L}_z \subset \mathcal{F}_z^{\text{aux}}$ is equivalent to the one induced directly from S^{aux} using Lagrangian $\mathcal{L} \subset \mathcal{F}^{\text{aux}}$, provided that $\mathcal{L}$ can be connected with $\mathcal{L}_z \times \tilde{\mathcal{L}}$ by a Lagrangian homotopy in $\mathcal{F}^{\text{aux}}$.

Proof. Similarly to (46), we have

$$(Q - i\Delta_z^{\text{aux}}) e^{iS_z^{\text{aux}}} = \int_{\tilde{\mathcal{L}}} (Q - \underbrace{-i\Delta_z^{\text{aux}} - i\tilde{\Delta}^{\text{aux}}}_{-i\Delta^{\text{aux}}}) e^{iS^{\text{aux}}} = 0$$

which proves (i). Item (ii) is an immediate corollary of (ii) of proposition 1. □

4.2. Pre-observables and observables for a quantum ambient BV theory.

In the case when the space of fields of the ambient theory $\mathcal{F}$ is finite-dimensional, it makes sense to introduce the following notions.

Definition 8 (cf. [15]). A *quantum BV theory* is a quadruple $(\mathcal{F}, \Omega, \mu, S)$ where $(\mathcal{F}, \Omega)$ is a $\mathbb{Z}$ -graded manifold with a degree -1 symplectic form Ω ; μ is a volume element on $\mathcal{F}$, compatible with Ω as in definition 7, i.e. the BV Laplacian $\Delta : f \mapsto \frac{1}{2} \text{div}_\mu \{f, \bullet\}_\Omega$ satisfies $\Delta^2 = 0$; the degree 0 action $S \in C^\infty(\mathcal{F})$ is required to satisfy the *quantum master equation*

$$(48) \quad \frac{1}{2} \{S, S\}_\Omega - i\Delta S = 0$$

or, equivalently,

$$\Delta e^{iS} = 0$$

For a quantum BV theory it is convenient to introduce a degree 1 second order differential operator

$$\delta_{\text{BV}} = \{S, \bullet\}_\Omega - i\Delta = e^{-iS} (-i\Delta) e^{iS}$$

which satisfies $(\delta_{\text{BV}})^2 = 0$.

¹⁵Subscript “z” refers to auxiliary zero-modes.

Definition 9. A *quantum observable*¹⁶ for a quantum BV theory is a degree 0 function $O \in C^\infty(\mathcal{F})$ satisfying

$$\delta_{\text{BV}} O = 0$$

which is equivalent to

$$\Delta(O e^{iS}) = 0$$

Observables O and O' are called *equivalent* if $O' - O = \delta_{\text{BV}}(\Psi)$ for some $\Psi \in C^\infty(\mathcal{F})$, i.e. O' and O give the same class in δ_{BV} -cohomology.

Given a quantum observable, one can define its *correlator* as

$$(49) \quad \langle O \rangle_{\mathcal{L}} = \frac{\int_{\mathcal{L} \subset \mathcal{F}} O e^{iS} \sqrt{\mu}|_{\mathcal{L}}}{\int_{\mathcal{L} \subset \mathcal{F}} e^{iS} \sqrt{\mu}|_{\mathcal{L}}} \in \mathbb{C}$$

where $\mathcal{L}$ is a Lagrangian submanifold of $(\mathcal{F}, \Omega)$ and we are assuming convergence. By the Stokes' theorem for BV integrals, this expression does not change with Lagrangian homotopy of $\mathcal{L}$: $\langle O \rangle_{\mathcal{L}} = \langle O \rangle_{\mathcal{L}'}$, and also correlators of equivalent observables coincide: $\langle O \rangle_{\mathcal{L}} = \langle O' \rangle_{\mathcal{L}}$, cf. [15].

Finally, if both $\mathcal{F}$ and $\mathcal{F}^{\text{aux}}$ are finite-dimensional, the following definition makes sense.

Definition 10. A *quantum pre-observable* for a quantum BV theory is a semi-quantum pre-observable $(\mathcal{F}^{\text{aux}}, \Omega^{\text{aux}}, \mu^{\text{aux}}, S^{\text{aux}})$ where instead of equation (41) we require that $S + S^{\text{aux}}$ satisfies the quantum master equation on $\mathcal{F} \times \mathcal{F}^{\text{aux}}$:

$$(\Delta + \Delta^{\text{aux}}) e^{i(S + S^{\text{aux}})} = 0$$

The analog of proposition 1 in the setting of quantum BV theory is as follows.

Proposition 3. Given a quantum pre-observable $(\mathcal{F}^{\text{aux}}, \Omega^{\text{aux}}, \mu^{\text{aux}}, S^{\text{aux}})$, define again $O_{\mathcal{L}}$ as

$$O_{\mathcal{L}} = \int_{\mathcal{L} \subset \mathcal{F}^{\text{aux}}} e^{iS^{\text{aux}}} \sqrt{\mu^{\text{aux}}}|_{\mathcal{L}} \in C^\infty(\mathcal{F})$$

for $\mathcal{L} \subset \mathcal{F}^{\text{aux}}$ a Lagrangian submanifold. Then:

- (i) $O_{\mathcal{L}}$ is a quantum observable, i.e. $\delta_{\text{BV}} O_{\mathcal{L}} = 0$.
- (ii) If Lagrangian submanifolds $\mathcal{L}$ and $\mathcal{L}'$ can be connected by a Lagrangian homotopy, then the difference $O_{\mathcal{L}'} - O_{\mathcal{L}}$ is δ_{BV} -exact.

Proof. The proof is obtained from the proof of proposition 1 by replacing Q by $-i\Delta, S^{\text{aux}}$ by $S + S^{\text{aux}}$ and O by $O e^{iS}$ everywhere. $\square$

Similarly, proposition 2 holds in the context of quantum pre-observables.

Remark 6. It is customary for perturbative quantization to introduce a formal parameter (the “Planck’s constant”) $\hbar$ by making a rescaling

$$(50) \quad \Omega \rightarrow \hbar^{-1} \Omega, \quad S \rightarrow \hbar^{-1} S$$

(which means $\Omega^{\text{old}} = \hbar^{-1} \Omega^{\text{new}}$ etc.) where the new action is a formal power series in $\hbar$, $S \in C^\infty(\mathcal{F})[[\hbar]]$. Hamiltonian vector field $\{S, \bullet\}_\Omega$ does not rescale and the

¹⁶Here we define a quantum observable in the framework of Lagrangian field theory in BV formalism, as something that can be averaged over the space of fields to yield a correlator (49). In other contexts one has quite different definitions of quantum observables: e.g. in the framework of Atiyah’s topological quantum field theory [12], an observable is a pair consisting of a submanifold $\gamma \subset \Sigma$ of the spacetime manifold and a vector $\hat{O}_\gamma$ in the space of states associated to the boundary of a tubular neighborhood of γ in Σ . The passage from the first picture to the second one goes through path integral quantization of the ambient topological theory on the tubular neighborhood (as on a manifold with boundary) with the insertion of observable.

BV Laplacian rescales as $\Delta \rightarrow \hbar \Delta$. For the structure on auxiliary fields, one does the same:

$$(51) \quad \Omega^{\text{aux}} \rightarrow \hbar^{-1} \Omega^{\text{aux}}, \quad S^{\text{aux}} \rightarrow \hbar^{-1} S^{\text{aux}}$$

With these redefinitions the equation (41) on semi-quantum pre-observables becomes

$$Q S^{\text{aux}} + \frac{1}{2} \{S^{\text{aux}}, S^{\text{aux}}\}_{\Omega^{\text{aux}}} - i \hbar \Delta^{\text{aux}} S^{\text{aux}} = 0$$

and can be solved by obstruction theory order by order in $\hbar$ for $S^{\text{aux}} = S_{(0)}^{\text{aux}} + \hbar S_{(1)}^{\text{aux}} + \hbar^2 S_{(2)}^{\text{aux}} + \dots$ starting from $S_{(0)}^{\text{aux}}$ a solution of (40). Likewise, the BV push-forward formula (45) becomes

$$O = \int_{\mathcal{L} \subset \mathcal{F}^{\text{aux}}} e^{\frac{i}{\hbar} S^{\text{aux}}} \sqrt{\mu^{\text{aux}}} |_{\mathcal{L}} \in C^\infty(\mathcal{F})[[\hbar]]$$

and can be evaluated by the stationary phase formula.

Note that one can also introduce two independent Planck's constants $\hbar, \hbar^{\text{aux}}$ for the ambient (50) and auxiliary (51) theory respectively.

5. AKSZ PRE-OBSERVABLE ASSOCIATED TO A HAMILTONIAN Q -BUNDLE

Let $(\mathcal{M}, Q, \omega = \delta\alpha, \Theta)$ be a Hamiltonian Q -manifold and Σ a closed oriented spacetime manifold, $\dim \Sigma = |\omega| + 1$. Then we have the AKSZ theory with

$$(52) \quad \mathcal{F}_\Sigma = \text{Map}(T[1]\Sigma, \mathcal{M}), \quad Q_\Sigma = (d_\Sigma)^{\text{lifted}} + Q^{\text{lifted}}, \\ \Omega_\Sigma = (-1)^{\dim \Sigma} \tau_\Sigma(\omega), \quad S_\Sigma = \iota_{d_\Sigma^{\text{lifted}}} \tau_\Sigma(\alpha) + \tau_\Sigma(\Theta)$$

as in section 2.

Given a trivial Hamiltonian Q -bundle over $\mathcal{M}$ with fiber data $(\mathcal{N}, \mathcal{A}, \omega' = \delta\alpha', \Theta')$, a closed oriented manifold γ of dimension $\dim \gamma = |\omega'| + 1$ and a smooth map¹⁷ $i : \gamma \rightarrow \Sigma$, we can construct a pre-observable for the AKSZ theory (52) with target $\mathcal{M}$ as follows. Set

$$(53) \quad \mathcal{F}_\gamma = \text{Map}(T[1]\gamma, \mathcal{N}), \quad \mathcal{A}_\gamma = (d_\gamma)^{\text{lifted}} + p^* \mathcal{A}^{\text{lifted}}, \\ \Omega_\gamma = (-1)^{\dim \gamma} \tau_\gamma(\omega'), \quad S_\gamma = \iota_{(d_\gamma)^{\text{lifted}}} \tau_\gamma(\alpha') + p^* \tau_\gamma^{\text{tot}}(\Theta')$$

Here the notations are as follows.

- $\tau_\gamma : \Omega^\bullet(\mathcal{N}) \rightarrow \Omega^\bullet(\mathcal{F}_\gamma)$ and $\tau_\gamma^{\text{tot}} : \Omega^\bullet(\mathcal{E}) \rightarrow \Omega^\bullet(\text{Map}(T[1]\gamma, \mathcal{E}))$ are the transgression maps, defined as in (11); $\mathcal{E} = \mathcal{M} \times \mathcal{N}$ is the total space of the target Hamiltonian Q -bundle.
- $\text{Map } p = i^* : \mathcal{F}_\Sigma \rightarrow \text{Map}(T[1]\gamma, \mathcal{M})$ is the pull-back of ambient fields by $i : \gamma \rightarrow \Sigma$ and $p^* : C^\infty(\text{Map}(T[1]\gamma, \mathcal{M})) \rightarrow C^\infty(\mathcal{F}_\Sigma)$ is the pull-back by p . By the trivial extension to auxiliary fields, $p = p \times \text{id}_{\mathcal{F}_\gamma}$ also maps $\mathcal{F}_\Sigma \times \mathcal{F}_\gamma$ to $\text{Map}(T[1]\gamma, \mathcal{E})$.
- $(d_\gamma)^{\text{lifted}} \in \mathfrak{X}(\mathcal{F}_\gamma) \subset \mathfrak{X}^{\text{vert}}(\mathcal{F}_\Sigma \times \mathcal{F}_\gamma)$ is the lifting of the de Rham differential on γ to a vector field on the mapping space $\text{Map}(T[1]\gamma, \mathcal{N})$.
- $\mathcal{A}^{\text{lifted}}$ is the lifting of the vertical vector field $\mathcal{A} \in \mathfrak{X}^{\text{vert}}(\mathcal{E})$ to a vertical vector field on the mapping space $\text{Map}(T[1]\gamma, \mathcal{E})$, so that $p^* \mathcal{A}^{\text{lifted}}$ becomes a vertical vector field on $\mathcal{F}_\Sigma \times \mathcal{F}_\gamma$.

Proposition 4. *The data (53) defines a classical pre-observable for the AKSZ theory (52), in particular*

$$(54) \quad Q_\Sigma S_\gamma + \frac{1}{2} \{S_\gamma, S_\gamma\}_{\Omega_\gamma} = 0$$

¹⁷By default, we assume that i is an embedding, cf. remark 10 below.

Proof. First let us check that $\mathcal{A}_\gamma$ as defined in (53) is the Hamiltonian vector field for S_γ . Introduce the notations for the source (kinetic) and target parts of $\mathcal{A}_\gamma$ and S_γ :

$$\mathcal{A}_\gamma = \underbrace{(d_\gamma)^{\text{lifted}}}_{\mathcal{A}_\gamma^{\text{kin}}} + \underbrace{p^* \mathcal{A}^{\text{lifted}}}_{\mathcal{A}_\gamma^{\text{target}}}, \quad S_\gamma = \underbrace{\iota_{(d_\gamma)^{\text{lifted}}} \tau_\gamma(\alpha')}_{S_\gamma^{\text{kin}}} + \underbrace{p^* \tau_\gamma^{\text{tot}}(\Theta')}_{S_\gamma^{\text{target}}}$$

Then we have

$$(55) \quad \begin{aligned} \iota_{\mathcal{A}_\gamma^{\text{kin}}} \Omega_\gamma &= \iota_{(d_\gamma)^{\text{lifted}}} (-1)^{\dim \gamma} \tau_\gamma(\omega') = \iota_{(d_\gamma)^{\text{lifted}}} \delta \tau_\gamma(\alpha') = \\ &= \underbrace{L_{(d_\gamma)^{\text{lifted}}} \tau_\gamma(\alpha') + \delta \iota_{(d_\gamma)^{\text{lifted}}} \tau_\gamma(\alpha')}_{=0 \text{ by (12)}} = \delta S_\gamma^{\text{kin}} = \delta^{\text{vert}} S_\gamma^{\text{kin}} \end{aligned}$$

and

$$(56) \quad \begin{aligned} \iota_{\mathcal{A}_\gamma^{\text{target}}} \Omega_\gamma &= p^* \iota_{\mathcal{A}^{\text{lifted}}} (-1)^{\dim \gamma} \tau_\gamma^{\text{tot}}(\omega') = \\ &= (-1)^{\dim \gamma} p^* \tau_\gamma^{\text{tot}}(\underbrace{\iota_{\mathcal{A}} \omega'}_{=\delta^{\text{vert}} \Theta'}) = \delta^{\text{vert}} p^* \tau_\gamma^{\text{tot}}(\Theta') = \delta^{\text{vert}} S_\gamma^{\text{target}} \end{aligned}$$

Here δ^{vert} stands for the de Rham differential in the fiber direction, as in section 3.3. Collecting (55,56), we get

$$\{S_\gamma, \bullet\}_{\Omega_\gamma} = \mathcal{A}_\gamma$$

Next, let us prove (54). Note that

$$Q_\Sigma S_\gamma^{\text{kin}} = 0$$

since S_γ^{kin} does not depend on the ambient fields (i.e. is a function on $\mathcal{F}_\Sigma \times \mathcal{F}_\gamma$, constant in the direction of $\mathcal{F}_\Sigma$); also

$$Q_\Sigma^{\text{kin}} S_\gamma^{\text{target}} = p^*(Q_\Sigma^{\text{kin}}|_\gamma \tau_\gamma^{\text{tot}}(\Theta')) = 0$$

as in (12), where $Q_\Sigma^{\text{kin}}|_\gamma$ is the lifting of d_γ to $\text{Map}(T[1]\gamma, \mathcal{M})$, extended to a horizontal vector field on $\text{Map}(T[1]\gamma, \mathcal{E})$. Thus

$$(57) \quad \begin{aligned} Q_\Sigma S_\gamma + \frac{1}{2} \{S_\gamma, S_\gamma\}_{\Omega_\gamma} &= \\ &= Q_\Sigma^{\text{target}} S_\gamma^{\text{target}} + \frac{1}{2} \{S_\gamma^{\text{kin}}, S_\gamma^{\text{kin}}\}_{\Omega_\gamma} + \{S_\gamma^{\text{kin}}, S_\gamma^{\text{target}}\} + \frac{1}{2} \{S_\gamma^{\text{target}}, S_\gamma^{\text{target}}\}_{\Omega_\gamma} \\ &= \frac{1}{2} \{S_\gamma^{\text{kin}}, S_\gamma^{\text{kin}}\}_{\Omega_\gamma} + \underbrace{\mathcal{A}_\gamma^{\text{kin}} S_\gamma^{\text{target}}}_{0 \text{ by (12)}} + (Q_\Sigma^{\text{target}} + \frac{1}{2} \mathcal{A}_\gamma^{\text{target}}) S_\gamma^{\text{target}} \end{aligned}$$

Here the first term on the r.h.s. has $((d_\gamma)^{\text{lifted}})^2 = 0$ as its Hamiltonian vector field on $\mathcal{F}_\gamma$, thus it is a degree 1 constant function on $\mathcal{F}_\gamma$, and hence vanishes. Therefore, continuing the calculation (57) and using (13), we have

$$(58) \quad \begin{aligned} Q_\Sigma S_\gamma + \frac{1}{2} \{S_\gamma, S_\gamma\}_{\Omega_\gamma} &= (-1)^{\dim \gamma} p^* \tau_\gamma^{\text{tot}}(Q\Theta' + \frac{1}{2} \mathcal{A}\Theta') = \\ &= (-1)^{\dim \gamma} p^* \tau_\gamma^{\text{tot}}(Q\Theta' + \frac{1}{2} \{\Theta', \Theta'\}_{\omega'}) = 0 \end{aligned}$$

since the target is a Hamiltonian Q -bundle and equation (30) holds there. This finishes the proof. $\square$

In coordinates. Alongside with the superfield (18) for the ambient theory on Σ , we now have a superfield for the auxiliary theory on γ :

$$(59) \quad Y^i(v^1, \dots, v^{\dim \gamma}, \xi^1, \dots, \xi^{\dim \gamma}) =$$

$$= \sum_{k=1}^{\dim \gamma} \underbrace{\sum_{1 \leq \nu_1 < \dots < \nu_k \leq \dim \gamma} Y_{\nu_1 \dots \nu_k}^i(v^1, \dots, v^{\dim \gamma}) \xi^{\nu_1} \dots \xi^{\nu_k}}_{Y_{(k)}^i(v, \xi)}$$

associated to local homogeneous coordinates y^i on $\mathcal{N}$. Here $v^1, \dots, v^{\dim \gamma}$ are local coordinates on γ and ξ^ν stands for dv^ν , $\nu = 1, \dots, \dim \gamma$.

Let α' and ω' be locally given on $\mathcal{N}$ as $\alpha' = \alpha'_i(y) \delta y^i$, $\omega' = \frac{1}{2} \omega'_{ij}(y) \delta y^i \wedge \delta y^j$. Then the BV 2-form for auxiliary fields is:

$$\Omega_\gamma = (-1)^{\dim \gamma} \int_\gamma \frac{1}{2} \omega'_{ij}(Y) \delta Y^i \wedge \delta Y^j$$

Its primitive is:

$$\alpha_\gamma = \int_\gamma \alpha'_i(Y) \delta Y^i$$

The action for auxiliary fields is:

$$S_\gamma = \int_\gamma \alpha'_i(Y) dY^i + \int_\gamma \Theta'(i^* X, Y)$$

Vertical part of the cohomological vector field $\mathcal{A}_\gamma$ acts on the auxiliary superfield by

$$\mathcal{A}_\gamma Y^i = dY^i + \mathcal{A}^i(i^* X, Y)$$

if on the target we have $\mathcal{A} = \mathcal{A}^i(x, y) \frac{\partial}{\partial y^i}$.

Remark 7. In the case when $i(\gamma) \subset \Sigma$ is an embedded submanifold of positive codimension, the expression

$$(60) \quad \{S_\gamma, S_\gamma\}_{\Omega_\Sigma}$$

is ill-defined, because it formally contains $\int_\Sigma (\delta_\gamma)^2 \dots$ where δ_γ is the delta function supported on $i(\gamma) \subset \Sigma$. On the other hand, if we use adapted local coordinates $u^1, \dots, u^{\dim \Sigma}$ on Σ in which $i(\gamma)$ is given by $u^{\dim \gamma + 1} = \dots = u^{\dim \Sigma} = 0$, the auxiliary action S_γ only depends on the components of the ambient superfield $X_{\mu_1 \dots \mu_k}^a$ with $\mu_1, \dots, \mu_k \leq \dim \gamma$, whereas the BV 2-form Ω_Σ couples the components $X_{\mu_1 \dots \mu_k}^a$ and $X_{\nu_1 \dots \nu_l}^b$ if and only if $\{\mu_1, \dots, \mu_k\} \sqcup \{\nu_1, \dots, \nu_l\} = \{1, \dots, \dim \Sigma\}$ (for the sake of this argument we choose local Darboux coordinates on $\mathcal{M}$ in which $\omega_{ab}(x)$ does not depend on x). So, formally (60) is $\infty \cdot 0$, and we naively regularize it to zero. Thus, by remark 5, we have (in the sense of our regularization) the classical master equation on $\mathcal{F}_\Sigma \times \mathcal{F}_\gamma$:

$$(61) \quad \{S_\Sigma + S_\gamma, S_\Sigma + S_\gamma\}_{\Omega_\Sigma + \Omega_\gamma} = 0$$

In the case when $i : \gamma \rightarrow \Sigma$ is a diffeomorphism, we have

$$\{S_\gamma, S_\gamma\}_{\Omega_\Sigma} = \{S_\gamma^{\text{target}}, S_\gamma^{\text{target}}\}_{\Omega_\Sigma} = (-1)^{\dim \Sigma} p^* \tau_\gamma^{\text{tot}}(\{\Theta', \Theta'\}_\omega)$$

Thus, (61) holds if and only if on the target we have

$$(62) \quad \{\Theta', \Theta'\}_\omega = 0$$

in addition to (30).

Remark 8. If $\dim \gamma = 1$, ω' is not automatically exact. The interesting case is when ω' is not exact but satisfies Bohr-Sommerfeld integrality condition, see remark 2. We assume that γ is a circle; the discussion extends to a disjoint union of circles trivially. The kinetic part of the action for auxiliary fields S_γ^{kin} cannot be defined as in (53), because the primitive α' does not exist globally on $\mathcal{N}$. Instead one can define

$$(63) \quad e^{i S_\gamma^{\text{kin}}(Y)} := \text{Hol}_{Y_{(0)}^*} \nabla'(\gamma) \in U(1)$$

– the holonomy around γ of the pull-back of the connection ∇' (cf. remark 2) by $Y_{(0)} : \gamma \rightarrow \mathcal{N}$. Here the l.h.s. should be viewed as one symbol. Target part of the action is defined as before, so we have

$$e^{iS_\gamma(X,Y)} = e^{i\oint_\gamma \Theta'(i^*X,Y)} \text{Hol}_{Y_{(0)}}^* \nabla'(\gamma)$$

Otherwise, using Stokes' theorem one can rewrite the kinetic part of the auxiliary action as

$$(64) \quad S_\gamma^{\text{kin}}(Y) = \int_D \tilde{Y}_{(0)}^* \omega'$$

for D a disc bounded by γ and $\tilde{Y}_{(0)}$ an arbitrary extension of $Y_{(0)}$ from γ to D . Due to integrality of $\omega'/2\pi$, arbitrariness of $\tilde{Y}_{(0)}$ results in (64) being defined modulo multiples of 2π :

$$S_\gamma^{\text{kin}}(Y) \in \mathbb{R}/2\pi\mathbb{Z}$$

From this point of view, the l.h.s. of (63) is well-defined.

It is easy to see that the proof of proposition 4 still works in this case (e.g. because one can find a primitive of ω' in the tubular neighborhood of the loop $Y_{(0)}(\gamma) \subset \mathcal{N}$).

Remark 9. Pre-observable (53) is invariant w.r.t. $\text{Diff}(\gamma)$ (reparametrizations) and $\text{Diff}(\Sigma)$ (ambient diffeomorphisms) in the following sense. Let us denote explicitly the dependence of S_γ on $i : \gamma \rightarrow \Sigma$ (through $p = i^*$ in (53)) as $S_\gamma(X, Y; i)$. Then for an ambient diffeomorphism $\Phi \in \text{Diff}(\Sigma)$ and a reparametrization $\phi \in \text{Diff}(\gamma)$ we have

$$(65) \quad S_\gamma(X, Y; \Phi \circ i \circ \phi) = S_\gamma(\Phi^*X, (\phi^{-1})^*Y; i)$$

which can be seen directly from the definition (53).

6. FROM PRE-OBSERVABLES TO OBSERVABLES

The general idea of passage from pre-observables constructed in section 5 to observables for the underlying AKSZ theory is by means of integrating out the auxiliary fields as in proposition 1, i.e.

$$(66) \quad O_\gamma = \int_{\mathcal{L} \subset \mathcal{F}_\gamma} e^{iS_\gamma} \in C^\infty(\mathcal{F}_\Sigma)$$

where $\mathcal{L}$ is a Lagrangian submanifold of $(\mathcal{F}_\gamma, \Omega_\gamma)$. When we want to emphasize the dependence of O_γ on the map $i : \gamma \rightarrow \Sigma$, we will write $O_{\gamma,i}$.

Assuming that we can make sense of the path integral (66), the observable O_γ is expected to have the following properties:

- (i) O_γ depends on the fields of the ambient AKSZ theory only via pull-back by $i : \gamma \rightarrow \Sigma$.
- (ii) Gauge invariance: $Q_\Sigma O_\gamma = 0$ (which is, indeed, our definition of an observable).
- (iii) The class of O_γ in Q_Σ -cohomology is independent of deformations of the gauge fixing $\mathcal{L}$ (as a Lagrangian submanifold of $\mathcal{F}_\gamma$).
- (iv) The class of O_γ in Q_Σ -cohomology is invariant under isotopy of γ (reparametrizations of γ homotopic to identity): for $\phi \in \text{Diff}_0(\gamma)$, we have

$$O_{\gamma, i \circ \phi} = O_{\gamma, i} + Q_\Sigma(\cdots)$$

- (v) Invariance under ambient diffeomorphisms: for $\Phi \in \text{Diff}(\Sigma)$, we have

$$O_{\gamma, \Phi \circ i}(X) = O_{\gamma, i}(\Phi^*X)$$

(vi) Invariance of the correlator $\langle O_\gamma \rangle$ under ambient isotopy: for $\Phi \in \text{Diff}_0(\Sigma)$, we have

$$\langle O_{\gamma, \Phi \circ i} \rangle = \langle O_{\gamma, i} \rangle$$

Here (i) follows from our construction of the pre-observable (53) and (v) follows from (i). Properties (ii) and (iii) are expected to hold in view of the proposition 1 for finite-dimensional fiber BV integrals. Property (iv) formally follows from (65), (iii) and $\text{Diff}(\gamma)$ -invariance of the measure on $\mathcal{F}_\gamma$:

$$\begin{aligned} (67) \quad O_{\gamma, i \circ \phi} &= \int_{\mathcal{L} \subset \mathcal{F}_\gamma} \mathcal{D}Y e^{iS_\gamma(X, Y; i \circ \phi)} = \int_{\mathcal{L}} \mathcal{D}Y e^{iS_\gamma(X, (\phi^{-1})^* Y; i)} \\ &= \int_{(\phi^{-1})^* \mathcal{L}} (\phi^{-1})^* (\mathcal{D}Y) e^{iS_\gamma(X, Y; i)} = \int_{(\phi^{-1})^* \mathcal{L}} \mathcal{D}Y e^{iS_\gamma(X, Y; i)} \\ &= \int_{\mathcal{L}} \mathcal{D}Y e^{iS_\gamma(X, Y; i)} + Q_\Sigma(\cdots) = O_{\gamma, i} + Q_\Sigma(\cdots) \end{aligned}$$

Here $\int_{\mathcal{L}} \mathcal{D}Y \cdots$ should be understood as $\int_{\mathcal{L}} \sqrt{\mu}|_{\mathcal{L}} \cdots$ where μ is the path integral measure on $\mathcal{F}_\gamma$. Note that we do use the fact that ϕ is an isotopy rather than a general diffeomorphism, since otherwise Lagrangians $\mathcal{L}$ and $(\phi^{-1})^* \mathcal{L}$ are not guaranteed to be homotopic. Property (vi) formally follows from (v), $\text{Diff}(\Sigma)$ -invariance of the AKSZ action (17) and of the path integral measure on $\mathcal{F}_\Sigma$, and from the BV-Stokes' theorem:

$$\begin{aligned} (68) \quad \langle O_{\gamma, \Phi \circ i} \rangle &= \int_{\mathcal{L} \subset \mathcal{F}_\Sigma} \mathcal{D}X O_{\gamma, \Phi \circ i}(X) e^{iS_\Sigma(X)} = \int_{\mathcal{L}} \mathcal{D}X O_{\gamma, i}(\Phi^* X) e^{iS_\Sigma(X)} \\ &= \int_{\Phi^* \mathcal{L}} (\Phi^*)^* (\mathcal{D}X) O_{\gamma, i}(X) e^{iS_\Sigma((\Phi^{-1})^* X)} = \int_{\Phi^* \mathcal{L}} \mathcal{D}X O_{\gamma, i}(X) e^{iS_\Sigma(X)} \\ &= \int_{\mathcal{L}} \mathcal{D}X O_{\gamma, i}(X) e^{iS_\Sigma(X)} = \langle O_{\gamma, i} \rangle \end{aligned}$$

Remark 10. For our construction of observable O_γ it is not a priori necessary to impose any restrictions on the map $i : \gamma \rightarrow \Sigma$. However, if we want to make sense of the correlator $\langle O_\gamma \rangle$ via perturbation theory for the path integral (which is outside of the scope of this paper), we have to require that i is an embedding. The technical reason is that if $i : \gamma \rightarrow \Sigma$ is not an embedding, it does not lift to a map between compactified configuration spaces of pairs of points on γ and Σ , $\text{Conf}_2(\gamma) \rightarrow \text{Conf}_2(\Sigma)$, which prevents one from defining the pull-back of the propagator of the ambient theory to γ , which in turn leads to Feynman diagrams for the correlator being ill-defined.

We will not try to give meaning to the path integral (66) in the most general situation here, but rather will discuss several special cases.

6.1. Case $\mathcal{N} = \text{point}$. In the case when $\mathcal{N}$ is a point the target Hamiltonian Q -bundle is described in example (v) in section 3.4: $\mathcal{A} = 0$, $\omega' = 0$, $\Theta' = \theta \in C^\infty(\mathcal{M})$ — a Q -cocycle of degree p for the target $\mathcal{M}$. The associated pre-observable (53) for a p -dimensional closed manifold γ and a map $i : \gamma \rightarrow \Sigma$ is:

$$\mathcal{F}_\gamma = \text{point}, \quad \mathcal{A}_\gamma = 0, \quad \Omega_\gamma = 0, \quad S_\gamma(X) = \int_\gamma \theta(i^* X)$$

The passage to the observable (66) is trivial in this setting, since there are no auxiliary fields:

$$(69) \quad O_\gamma = e^{i \int_\gamma \theta(i^* X)}$$

This observable obviously satisfies properties (i–v) above.

6.2. Case of 1-dimensional observables. Let $i : \gamma = S^1 \rightarrow \Sigma$ be a circle mapped into Σ and suppose we have a Hamiltonian Q -bundle of degree 0 over the AKSZ target $(\mathcal{M}, Q, \omega = \delta\alpha, \Theta)$ with fiber data $(\mathcal{N}, \mathcal{A}, \omega' = \delta\alpha', \Theta')$. We will assume for simplicity that $\mathcal{N}$ is concentrated in degree 0, i.e. $(\mathcal{N}, \omega')$ is an ordinary (non-graded) symplectic manifold.

In local coordinates, the auxiliary superfield (59) is:

$$Y^i(u, du) = Y_{(0)}^i(u) + Y_{(1)}^i(u) du$$

where u is the coordinate on $\gamma = S^1$ and index i corresponds to local coordinates on $\mathcal{N}$. For the pre-observable (53) we have

$$(70) \quad \mathcal{F}_\gamma = \text{Map}(T[1]\gamma, \mathcal{N}) \\ = \left\{ (Y_{(0)}, Y_{(1)}) \mid Y_{(0)} : \gamma \rightarrow \mathcal{N}, Y_{(1)} \in \Gamma(\gamma, T^*\gamma \otimes Y_{(0)}^* T\mathcal{N})[-1] \right\}, \\ \Omega_\gamma = - \oint_\gamma \omega'_{ij}(Y_{(0)}) \delta Y_{(0)}^i \wedge \delta Y_{(1)}^j + \frac{1}{2} Y_{(1)}^k \partial_k \omega'_{ij}(Y_{(0)}) \delta Y_{(0)}^i \wedge \delta Y_{(0)}^j, \\ S_\gamma = \oint_\gamma \alpha'_i(Y_{(0)}) dY_{(0)}^i + \Theta'(i^* X, Y)$$

One natural choice of Lagrangian in (66) is to set

$$(71) \quad \mathcal{L} = \text{Map}(\gamma, \mathcal{N}) \subset \text{Map}(T[1]\gamma, \mathcal{N})$$

I.e. the 1-form component $Y_{(1)}$ of the auxiliary superfield vanishes on $\mathcal{L}$. With this choice (66) becomes

$$(72) \quad O_\gamma = \int_{\text{Map}(\gamma, \mathcal{N})} \mathcal{D}Y_{(0)} e^{i \oint_\gamma \alpha'_i(Y_{(0)}) dY_{(0)}^i + \Theta'(i^* X, Y_{(0)})} \\ = \int_{\text{Map}(\gamma, \mathcal{N})} \mathcal{D}Y_{(0)} e^{i \oint_\gamma \alpha'_i(Y_{(0)}) dY_{(0)}^i + i^* X_{(1)}^a \partial_a \Theta'(i^* X_{(0)}, Y_{(0)})}$$

Note that with this gauge fixing, one formally has a strict version of property (iv), namely that O_γ is invariant under isotopy of γ (instead of only the class of O_γ in Q_Σ -cohomology being invariant). This follows from invariance of $\mathcal{L}$ under $\text{Diff}(\gamma)$, together with $\text{Diff}(\gamma)$ -invariance of the path integral measure $\mathcal{D}Y_{(0)}$.

The quantum mechanical path integral (72) can be understood in the Hamiltonian formalism, using the fiber geometric quantization of the target Hamiltonian Q -bundle, as follows.

Proposition 5. *Assume that the symplectic manifold $(\mathcal{N}, \omega')$ can be geometrically quantized to a complex vector space of states $\mathcal{H}$ and the Hamiltonian $\Theta' \in C^\infty(\mathcal{M} \times \mathcal{N})$ can be quantized (viewing coordinates on $\mathcal{M}$ as external parameters) to an operator-valued function on $\mathcal{M}$, $\hat{\Theta}' \in C^\infty(\mathcal{M}) \otimes \text{End}(\mathcal{H})$ satisfying*

$$(73) \quad Q\hat{\Theta}' + i(\hat{\Theta}')^2 = 0$$

(the quantum counterpart of the equation (30)). Then define O_γ as the trace of a path ordered exponential¹⁸

$$(74) \quad O_\gamma = \text{tr}_{\mathcal{H}} \mathcal{P} \exp \left(i \oint_\gamma \hat{\Theta}'(i^* X) \right)$$

Then

$$Q_\Sigma O_\gamma = 0$$

i.e. O_γ is an observable.

¹⁸To define the path ordered exponential, we need to choose a starting point $p \in \gamma$, but the dependence on p is cancelled by taking the trace.

Proof. Let $j : [0, 1] \rightarrow \Sigma$ be a path in Σ parameterized by $t \in [0, 1]$. Let us denote $\psi = \hat{\Theta}'(j^* X) \in \Omega^\bullet([0, 1]) \hat{\otimes} C^\infty(\mathcal{F}_\Sigma) \otimes \text{End}(\mathcal{H})$, also let $\psi_{(0)}(t)$ and $\psi_{(1)}(t, dt)$ be the 0- and 1-form components of ψ . Consider the path ordered exponential

$$(75) \quad W_j = \mathcal{P} \exp \left(i \int_0^1 \psi_{(1)} \right) \\ = \lim_{N \rightarrow \infty} \overleftarrow{\prod}_{0 \leq k < N} \left(\text{id}_{\mathcal{H}} + i \iota_{\frac{1}{N} \partial_t} \psi_{(1)}(k/N, dt) \right) \in C^\infty(\mathcal{F}_\Sigma) \otimes \text{End}(\mathcal{H})$$

Then

$$(76) \quad Q_\Sigma W_j = -i \int_0^1 \mathcal{P} \exp \left(i \int_t^1 \psi_{(1)} \right) \cdot Q_\Sigma \psi(t, dt) \cdot \mathcal{P} \exp \left(i \int_0^t \psi_{(1)} \right) \\ = -i \int_0^1 \mathcal{P} \exp \left(i \int_t^1 \psi_{(1)} \right) \cdot \left(dt \frac{\partial}{\partial t} \psi_{(0)}(t) - i [\psi_{(0)}(t), \psi_{(1)}(t, dt)] \right) \cdot \mathcal{P} \exp \left(i \int_0^t \psi_{(1)} \right) \\ = -i \lim_{N \rightarrow \infty} \sum_{l=0}^{N-1} \overleftarrow{\prod}_{l < k < N} \left(\text{id}_{\mathcal{H}} + i \iota_{\frac{1}{N} \partial_t} \psi_{(1)} \left(\frac{k}{N}, dt \right) \right) \cdot \\ \cdot \left(\psi_{(0)} \left(\frac{l+1}{N} \right) \left(\text{id}_{\mathcal{H}} + i \iota_{\frac{1}{N} \partial_t} \psi_{(1)} \left(\frac{l}{N}, dt \right) \right) - \left(\text{id}_{\mathcal{H}} + i \iota_{\frac{1}{N} \partial_t} \psi_{(1)} \left(\frac{l}{N}, dt \right) \right) \psi_{(0)} \left(\frac{l}{N} \right) \right) \\ \cdot \overleftarrow{\prod}_{0 \leq k < l} \left(\text{id}_{\mathcal{H}} + i \iota_{\frac{1}{N} \partial_t} \psi_{(1)} \left(\frac{k}{N}, dt \right) \right) \\ = -i (\psi_{(0)}(1) \cdot W_j - W_j \cdot \psi_{(0)}(0))$$

where we used that due to (73), we have $Q_\Sigma \psi = d\psi - \frac{i}{2}[\psi, \psi]$. For $j = i : S^1 \rightarrow \Sigma$, we have $j(0) = j(1)$ and applying (76), we get

$$Q_\Sigma O_\gamma = \text{tr}_{\mathcal{H}} Q_\Sigma W_j = -i \text{tr}_{\mathcal{H}} [\hat{\Theta}'(X_{(0)}(j(0))), W_j] = 0$$

□

Note that the construction of proposition 5 does not require ω' to be exact, it is enough to require that ω' satisfies Bohr-Sommerfeld integrality condition.

6.3. Torsion-like observables. Now let again $i : \gamma \rightarrow \Sigma$ be an oriented closed manifold of arbitrary dimension mapped into Σ and suppose that $\mathcal{N}$ is a finite-dimensional graded vector space, ω' is a graded-antisymmetric pairing on $\mathcal{N}$ of degree $\dim \Sigma - 1$ and Θ' is quadratic in $\mathcal{N}$ directions:

$$\Theta'(x, y) = \frac{1}{2} \omega'(y, \theta(x)y)$$

with $\theta \in C^\infty(\mathcal{M}) \otimes \text{End}(\mathcal{N})$ of degree 1. Equation (30) is then equivalent to

$$(77) \quad Q\theta - \theta^2 = 0$$

We also assume that

$$(78) \quad \text{Str}_{\mathcal{N}} \theta(x) = 0$$

where $\text{Str}_{\mathcal{N}}$ stands for the super-trace over $\mathcal{N}$.

One way to make sense of the expression (66) is to choose a Riemannian metric on γ . Then we have the Hodge decomposition for differential forms on γ with values in $\mathcal{N}$

$$(79) \quad \underbrace{\mathcal{N} \otimes \Omega^\bullet(\gamma)}_{\mathcal{F}_\gamma} = \underbrace{\mathcal{N} \otimes \Omega_{\text{harm}}^\bullet(\gamma)}_{\mathcal{F}_\gamma^z} \oplus \underbrace{\mathcal{N} \otimes \Omega_{\text{exact}}^\bullet(\gamma)}_{\mathcal{F}_\gamma} \oplus \overbrace{\mathcal{N} \otimes \Omega_{\text{co-exact}}^\bullet(\gamma)}^{\tilde{\mathcal{L}}}$$

which can be used to first take the BV integral over the complement of harmonic forms, as in proposition 2, and then take the BV integral over the zero-modes (harmonic forms):

$$(80) \quad O_\gamma = \int_{\mathcal{L}_z \subset \mathcal{F}_\gamma^z} \mathcal{D}Y_z \int_{\tilde{\mathcal{L}} \subset \tilde{\mathcal{F}}_\gamma} \mathcal{D}\tilde{Y} e^{\frac{i}{2} \int_\gamma \omega'(Y, dY + \theta(i^* X)Y)}$$

where $Y = Y_z + \tilde{Y}$ is the splitting of the auxiliary superfield into the harmonic part and the part in the complement; $\mathcal{F}_\gamma^z$ comes with the degree -1 symplectic structure Ω_γ^z induced from ω' and the Poincaré pairing on cohomology of γ ; $\mathcal{L}_z \subset \mathcal{F}_\gamma^z$ is a Lagrangian vector subspace w.r.t. Ω_γ^z .

The internal integral in (80) is a path integral and can be made sense of using perturbation theory¹⁹:

$$(81) \quad O_\gamma^z(X, Y_z) = \exp \left(\frac{i}{2} \int_\gamma \omega'(Y_z, \theta(i^* X)) (\text{id} + G \theta(i^* X))^{-1} Y_z + \right. \\ \left. + \text{Str}_{\mathcal{N}} \sum_{k \geq 2} \frac{(-1)^k}{2k} \int_{\text{Conf}_k(\gamma)} \overleftarrow{\prod}_{1 \leq j \leq k} \pi_{j+1,j}^* \eta \cdot \pi_j^* \theta(i^* X) \right)$$

The notations here are:

- $\text{Conf}_k(\gamma)$ is the Fulton-Macpherson-Axelrod-Singer compactification of the configuration space of k points on γ , cf. [5].
- $G = d^*(\Delta + \mathcal{P}_{\text{harm}})^{-1} : \Omega^\bullet(\gamma) \rightarrow \Omega^{\bullet-1}(\gamma)$ is the Hodge-theoretic inverse of d ; its integral kernel extends to a smooth $(\dim \gamma - 1)$ -form η on $\text{Conf}_2(\gamma)$ (the *Hodge propagator*), cf. [5]. By $\mathcal{P}_{\text{harm}}$ we denote the orthogonal projection to harmonic forms, using the Hodge inner product.
- $\pi_{j+1,j} : \text{Conf}_k \rightarrow \text{Conf}_2$ is the map associated to forgetting all points except points j and $j+1$; also $\pi_j : \text{Conf}_k \rightarrow \text{Conf}_1 = \gamma$ is forgetting all points except the point j . By convention, $\pi_{k+1,k} = \pi_{1,k}$.
- In the first term in the exponential in (81), $\theta(i^* X)$ is understood as a multiplication operator on $\Omega^\bullet(\gamma)$ and an endomorphism of $\mathcal{N}$, depending on ambient fields. Expression $(\text{id} + G \theta(i^* X))^{-1}$ is understood as $\sum_{k=0}^{\infty} (-1)^k (G \theta(i^* X))^k$.

The BV integral over zero-modes in (80) then is:

$$(82) \quad O_\gamma = \int_{\mathcal{L}_z \subset \mathcal{F}_\gamma^z} e^{\frac{i}{2} \int_\gamma \omega'(Y_z, \theta(i^* X)) (\text{id} + G \theta(i^* X))^{-1} Y_z} \sqrt{\mu_z|_{\mathcal{L}_z}} \cdot \\ \cdot \exp \left(\text{Str}_{\mathcal{N}} \sum_{k \geq 2} \frac{(-1)^k}{2k} \int_{\text{Conf}_k(\gamma)} \overleftarrow{\prod}_{1 \leq j \leq k} \pi_{j+1,j}^* \eta \cdot \pi_j^* \theta(i^* X) \right)$$

where μ_z is some fixed translation-invariant volume element on $\mathcal{F}_z$.

Remark 11. The requirement that the integral over zero-modes in (82) should converge imposes the following restriction on $\mathcal{L}_z$. If $\{Y_{\mathcal{L}_z}^I\}$ are coordinates on $\mathcal{L}_z$ and if we write

$$\int_\gamma \omega'(Y_z, \theta(i^* X)) (\text{id} + G \theta(i^* X))^{-1} Y_z = Y_{\mathcal{L}_z}^I H_{IJ} (i^* X) Y_{\mathcal{L}_z}^J$$

then we require the even-even block (i.e. the block pairing coordinates $Y_{\mathcal{L}_z}^I$ of even degree with coordinates $Y_{\mathcal{L}_z}^J$ of even degree) of the matrix (H_{IJ}) to be invertible. One universal way to satisfy this condition is to choose $\mathcal{L}_z \subset \mathcal{F}_\gamma^z$ by setting all even

¹⁹We are working modulo constants here. We are also implicitly assuming convergence of the sum over k in (81).

coordinates on $\mathcal{F}_\gamma^z$ to zero. With this choice, the integral over $\mathcal{L}_z$ in (82) is (up to normalization) the Pfaffian of (H_{IJ}) .

Proposition 6. (i) Expression (82) is an observable, i.e.

$$Q_\Sigma O_\gamma = 0$$

(ii) Dependence of O_γ on deformations of Riemannian metric on γ and on deformations of $\mathcal{L}_z$ is Q_Σ -exact.

Proof. Define $S_\gamma^z \in C^\infty(\mathcal{F}_\Sigma \times \mathcal{F}_\gamma^z)$ by (81) and $O_\gamma^z = e^{iS_\gamma^z}$:

$$(83) \quad S_\gamma^z = \frac{1}{2} \int_\gamma \omega'(Y_z, \theta (\text{id} + G \theta)^{-1} Y_z) - \\ - i \text{Str}_\mathcal{N} \sum_{k \geq 2} \frac{(-1)^k}{2k} \int_{\text{Conf}_k(\gamma)} \overleftarrow{\prod}_{1 \leq j \leq k} \pi_{j+1,j}^* \eta \cdot \pi_j^* \theta$$

where θ actually means $\theta(i^* X)$.

First let us check that S_γ^z is a semi-quantum pre-observable, i.e. satisfies the equation (41):

$$(84) \quad Q_\Sigma S_\gamma^z + \frac{1}{2} \{S_\gamma^z, S_\gamma^z\}_{\Omega_\gamma^z} - i \Delta_z S_\gamma^z = 0$$

Indeed, we have

$$(85) \quad \frac{1}{2} \{S_\gamma^z, S_\gamma^z\}_{\Omega_\gamma^z} = \frac{1}{2} \int_\gamma \omega'(Y_z, (\text{id} + \theta G)^{-1} \theta \mathcal{P}_{\text{harm}} \theta (\text{id} + G \theta)^{-1} Y_z)$$

$$(86) \quad -i \Delta_z S_\gamma^z = -\frac{i}{2} \text{Str}_{\mathcal{F}_\gamma^z} (\mathcal{P}_{\text{harm}} \theta (\text{id} + G \theta)^{-1})$$

$$(87) \quad Q_\Sigma S_\gamma^z = -\frac{1}{2} \int_\gamma \omega'(Y_z, (\text{id} + \theta G)^{-1} \underbrace{(d\theta + \theta^2)}_{Q_\Sigma \theta(i^* X)} (\text{id} + G \theta)^{-1} Y_z) - \\ + \frac{i}{2} \text{Str}_\mathcal{N} \sum_{k \geq 2} (-1)^k \int_{\text{Conf}_k(\gamma)} \left(\overleftarrow{\prod}_{2 \leq j \leq k} \pi_{j+1,j}^* \eta \cdot \pi_j^* \theta \right) \cdot \pi_{2,1}^* \eta \cdot \pi_1^* (d\theta + \theta^2)$$

Using Stokes' theorem, (87) can be written as

$$(88) \quad Q_\Sigma S_\gamma^z = \frac{1}{2} \int_\gamma \omega'(Y_z, (\text{id} + \theta G)^{-1} \theta ([d, G] - \text{id}) \theta (\text{id} + G \theta)^{-1} Y_z) + \\ + \frac{i}{2} \text{Str}_\mathcal{N} \sum_{k \geq 2} (-1)^{k+\dim \gamma} \int_{\text{Conf}_k(\gamma)} \left(\overleftarrow{\prod}_{2 \leq j \leq k} \pi_{j+1,j}^* \eta \cdot \pi_j^* \theta \right) \cdot \pi_{2,1}^* d\eta \cdot \pi_1^* \theta + \\ + \frac{i}{2} \text{Str}_\mathcal{N} \sum_{k \geq 2} (-1)^k \int_{\text{Conf}_k(\gamma)} \left(\overleftarrow{\prod}_{2 \leq j \leq k} \pi_{j+1,j}^* \eta \cdot \pi_j^* \theta \right) \cdot \pi_{2,1}^* \eta \cdot \pi_1^* \theta^2 - \\ - \frac{i}{2} \text{Str}_\mathcal{N} \sum_{k \geq 2} \frac{(-1)^{k+k \dim \gamma}}{k} \int_{\partial \text{Conf}_k(\gamma)} \overleftarrow{\prod}_{1 \leq j \leq k} \pi_{j+1,j}^* \eta \cdot \pi_j^* \theta$$

Two last terms here cancel due to the contribution of principal boundary strata of $\text{Conf}_k(\gamma)$, where points j and $j+1$ collapse (and we use the property of the propagator that the fiber integral of η over fibers of $\partial \text{Conf}_2(\gamma) \rightarrow \gamma$ is the constant function $-(-1)^{\dim \gamma} \mathbf{1} \in C^\infty(\gamma)$). Hidden boundary strata (i.e. those corresponding to the collapse of ≥ 3 points) do not contribute due to the fact that the collapsed subgraph of the “wheel” graph either has a vertex of valence 2, and then the integral

vanishes by (anti-) symmetry of η on $\partial\text{Conf}_2(\gamma)$ w.r.t. the antipodal involution, or the collapsed subgraph has an isolated vertex, and then the integral vanishes for degree reasons, cf. [6].

Since $[d, G] = \text{id} - \mathcal{P}_{\text{harm}}$ and since $d\eta = -(-1)^{\dim \gamma} \mathbf{P}_{\text{harm}}$ where $\mathbf{P}_{\text{harm}} \in \Omega^{\dim \gamma}(\gamma \times \gamma)$ is the integral kernel of $\mathcal{P}_{\text{harm}}$, collecting (85,86,88) we get the semi-quantum master equation (84).

Next, knowing that (84) holds, we infer by proposition 1 that $O_\gamma = \int_{\mathcal{L}_z \subset \mathcal{F}_\gamma^z} e^{iS_\gamma^z}$ given by (82) satisfies $Q_\Sigma O_\gamma = 0$. Thus we proved (i).

To prove (ii) we first note that deformations of $\mathcal{L}_z$ induce Q_Σ -exact shifts in O_γ by proposition 1.

Next we check the dependence of S_γ^z on metric on γ . Let us now be more careful with the space of zero-modes and define it as $\mathcal{F}_\gamma^z = \mathcal{N} \otimes H^\bullet(\gamma)$ – the $\mathcal{N}$ -valued de Rham cohomology of γ ; it comes with a metric-dependent embedding of cohomology into differential forms $i : H^\bullet(\gamma) \hookrightarrow \Omega^\bullet(\gamma)$ which realizes cohomology classes by harmonic forms. In view of this redefinition we should now write $i(Y_z)$ instead of Y_z in (83).

Now assume that we have a path in the space of Riemannian metrics on γ , $g_t \in \text{Met}(\gamma)$ with $t \in [0, 1]$ the parameter. Then we have the t -dependent Hodge-theoretic inverse G_t for de Rham operator, the t -dependent propagator η_t and the t -dependent embedding of cohomology into differential forms i_t with exact time-derivatives:

$$\frac{d}{dt} G_t = [d, H_t], \quad \frac{\partial}{\partial t} \eta_t = d\zeta_t, \quad \frac{\partial}{\partial t} i_t = dj_t$$

for some linear operator $H_t : \Omega^\bullet(\gamma) \rightarrow \Omega^{\bullet-2}(\gamma)$ with smooth integral kernel $\zeta_t \in \Omega^{\dim \gamma - 2}(\gamma \times \gamma)$ and some $j_t : H^\bullet(\gamma) \rightarrow \Omega^{\bullet-1}(\gamma)$; H_t , ζ_t and j_t depend on g_t and $\frac{\partial}{\partial t} g_t$. Then one can introduce the “extended” versions of G , η and i :

$$\begin{aligned} \tilde{G} = G_t + dt H_t &\in \Omega^\bullet([0, 1]) \hat{\otimes} \text{End}(\Omega^\bullet(\gamma)) \\ \tilde{\eta} = \eta_t + dt \zeta_t &\in \Omega^{\dim \Sigma - 1}([0, 1] \times \text{Conf}_2(\gamma)) \\ \tilde{i} = i_t + dt j_t &\in \Omega^\bullet([0, 1]) \hat{\otimes} \text{Hom}(H^\bullet(\gamma), \Omega^\bullet(\gamma)) \end{aligned}$$

They satisfy respectively

$$\begin{aligned} \left(dt \frac{\partial}{\partial t} + [d_\gamma, \bullet] \right) \tilde{G} &= \text{id} - \mathcal{P}_{\text{harm}} \\ \left(dt \frac{\partial}{\partial t} + d_{\text{Conf}_2(\gamma)} \right) \tilde{\eta} &= -(-1)^{\dim \gamma} \mathbf{P}_{\text{harm}} \\ \left(dt \frac{\partial}{\partial t} + d_\gamma \right) \tilde{i} &= 0 \end{aligned}$$

Next, we define the extended version of S_γ^z , $\tilde{S}_\gamma^z \in \Omega^\bullet([0, 1]) \hat{\otimes} C^\infty(\mathcal{F}_\Sigma \times \mathcal{F}_\gamma^z)$ by substituting $\tilde{G}$ instead of G , $\tilde{\eta}$ instead of η and $\tilde{i}(Y_z)$ instead of Y_z into (83). Repeating our proof of (84) in the extended case we find that

$$\left(dt \frac{\partial}{\partial t} + Q_\Sigma \right) \tilde{S}_\gamma^z + \frac{1}{2} \{ \tilde{S}_\gamma^z, \tilde{S}_\gamma^z \}_{\Omega_\gamma^z} - i \Delta_z \tilde{S}_\gamma^z = 0$$

which, when we decompose $\tilde{S}_\gamma^z$ into the 0-form and 1-form parts w.r.t. t , $\tilde{S}_\gamma^z = S_\gamma^z(t) + dt R_\gamma^z(t)$, is equivalent to semi-quantum master equation (84) for $S_\gamma^z(t)$ plus the equation

$$(89) \quad \frac{\partial}{\partial t} S_\gamma^z(t) = Q_\Sigma R_\gamma^z(t) + \{ S_\gamma^z(t), R_\gamma^z(t) \}_{\Omega_\gamma^z} - i \Delta_z R_\gamma^z(t)$$

Therefore (cf. (44)) semi-quantum pre-observables $S_\gamma^z(t)$ are equivalent for all values of the parameter t and so by item (iii) of proposition 1, dependence of the induced

observable $O_\gamma(t) = \int_{\mathcal{L}_z \subset \mathcal{F}_\gamma^z} e^{iS_\gamma^z(t)}$ on t is Q_Σ -exact. This finishes the proof of (ii). $\square$

7. EXAMPLES OF OBSERVABLES

Here we will illustrate the construction of AKSZ pre-observables/observables of sections 5, 6 by several explicit examples.

7.1. Observables with $\mathcal{N}$ a point. First we give some examples of the situation described in section 6.1.

- For abelian Chern-Simons theory (22) we may take $\theta = c\psi \in C^\infty(\mathbb{R}[1])$ with $c \in \mathbb{R}$ any constant. This yields by (69) a 1-dimensional observable for $i : \gamma = S^1 \hookrightarrow \Sigma$

$$O_\gamma = e^{ic \oint_\gamma i^* A_{(1)}}$$

— the abelian Wilson loop; parameter c can be viewed as the weight of a one-dimensional representation of the gauge group $\mathbb{R}$. Note that this is also an observable for the abelian BF theory (26).

- For D -dimensional abelian BF theory, take $\theta = c\xi$ with $c \in \mathbb{R}$ a parameter. The associated observable for a closed oriented submanifold $i : \gamma \hookrightarrow \Sigma$ of codimension 2 is

$$(90) \quad O_\gamma = e^{ic \int_\gamma i^* B_{(D-2)}}$$

- For 2-dimensional BF theory assume that $\mathfrak{g}$ is quadratic so that we can identify $\mathfrak{g}^*$ with $\mathfrak{g}$ and let $\rho : \mathfrak{g} \rightarrow \text{End}(R)$ be a representation. Set $\theta = -i \log \text{tr}_R f(\rho(\xi))$ for $f \in C^\infty(\mathbb{R})$ a function. The corresponding 0-dimensional observable for $\gamma \in \Sigma$ a point is

$$(91) \quad O_\gamma = \text{tr}_R f(\rho(B_{(0)}|_\gamma))$$

- For 2-dimensional abelian BF theory take $\theta = f(\xi)\psi$ for $f \in C^\infty(\mathbb{R})$. The corresponding 1-dimensional observable for $i : \gamma = S^1 \hookrightarrow \Sigma$ is

$$(92) \quad O_\gamma = e^{i \oint_\gamma i^* (A \cdot f(B))}$$

Note that for f non-constant, O_γ depends non-trivially on components of the superfields of ghost number $\neq 0$, namely on $A_{(0)}$ and $B_{(1)}$:

$$O_\gamma = e^{i \oint_\gamma i^* (A_{(1)} \cdot f(B_{(0)}) + A_{(0)} \cdot B_{(1)} \cdot f'(B_{(0)}))}$$

- For Poisson sigma model and $\theta = f \in C^\infty(M)$ a Casimir of the Poisson structure (i.e. $[\pi, f] = 0$, with $[\cdot, \cdot]$ the Schouten-Nijenhuis bracket on polyvector fields on M), we get a 0-dimensional observable for γ a point in Σ :

$$(93) \quad O_\gamma = e^{i \int_\gamma f(X_{(0)}|_\gamma)}$$

- For Poisson sigma model and $v \in \mathfrak{X}(M)$ a Poisson vector field (i.e. $[\pi, v] = 0$), set $\theta = \langle p, v(x) \rangle$. The associated 1-dimensional observable is:

$$(94) \quad O_\gamma = e^{i \oint_\gamma i^* \langle \eta, v(X) \rangle}$$

Note that if $\tilde{v} = v + [\pi, f]$ — another Poisson vector field giving the same class in Poisson cohomology of M as v , then the observable $\tilde{O}_\gamma$ constructed from $\tilde{v}$ by (94) is equivalent to O_γ : $\langle p, \tilde{v}(x) \rangle - \langle p, v(x) \rangle = Qf(x)$ on the target implies $\oint_\gamma i^* \langle \eta, \tilde{v}(X) \rangle - \oint_\gamma i^* \langle \eta, v(X) \rangle = -Q_\Sigma \oint_\gamma i^* f(X)$ on the mapping space $\mathcal{F}_\Sigma$. This implies in turn

$$\tilde{O}_\gamma = O_\gamma + Q_\Sigma \left(-i \int_0^1 dt e^{i \oint_\gamma i^* \langle \eta, (1-t)v(X) + t\tilde{v}(X) \rangle} \oint_\gamma i^* f(X) \right)$$

Thus (94) defines a map from Poisson cohomology of M in degree one to Q_Σ -cohomology in degree zero.

Note that the observable (91) is a special case of (93) when we take M to be $\mathfrak{g}^*$ with Kirillov-Kostant Poisson structure. Also, (92) is a special case of (94) for $M = \mathbb{R}$ with $\pi = 0$.

7.2. Wilson loop observable in Chern-Simons theory. Here we will combine the discussion of section 6.2 with the example (iv) of section 3.4.

Let $\mathfrak{g}$ be a quadratic Lie algebra and let (M, ω_M) be a symplectic manifold with a Hamiltonian action of $\mathfrak{g}$ with equivariant moment map $\mu : M \rightarrow \mathfrak{g}^*$. Further, we assume that ω_M satisfies Bohr-Sommerfeld integrality condition and is the curvature of a $U(1)$ -connection ∇ on a line bundle L over M .

Then we have a degree 0 Hamiltonian Q -bundle structure on $\mathfrak{g}[1] \times M \rightarrow \mathfrak{g}[1]$, with base structure given by (20) and fiber structure (38), with the correction that now we relaxed the exactness condition on ω_M . Applying the construction of section 5 to this target data we get the following 1-dimensional pre-observable for Chern-Simons theory:

$$(95) \quad \begin{aligned} \mathcal{F}_\gamma &= \text{Map}(T[1]\gamma, M) \\ &= \{(Y_{(0)}, Y_{(1)}) \mid Y_{(0)} : \gamma \rightarrow M, Y_{(1)} \in \Gamma(\gamma, T^*\gamma \otimes Y_{(0)}^* TM)[-1]\}, \\ \Omega_\gamma &= - \oint_\gamma \omega_M(Y) = - \oint_\gamma \omega_{ij}(Y_{(0)}) \delta Y_{(0)}^i \wedge \delta Y_{(1)}^j + \frac{1}{2} Y_{(1)}^k \partial_k \omega_{ij}(Y_{(0)}) \delta Y_{(0)}^i \wedge \delta Y_{(0)}^j, \\ e^{iS_\gamma} &= \text{Hol}_{Y_{(0)}^* \nabla}(\gamma) e^{i \oint_\gamma \langle i^* A, \mu(Y) \rangle} = \text{Hol}_{Y_{(0)}^* \nabla}(\gamma) e^{i \oint_\gamma \langle i^* A_{(1)}, \mu(Y_{(0)}) \rangle + \langle i^* A_{(0)}, Y_{(1)}^i \partial_i \mu(Y_{(0)}) \rangle} \end{aligned}$$

where $i : \gamma = S^1 \hookrightarrow \Sigma$ is a circle embedded into 3-manifold Σ , $A = \sum_{k=0}^3 A_{(k)}$ is the superfield of the ambient Chern-Simons theory, $Y = Y_{(0)} + Y_{(1)}$ is the auxiliary superfield (21), with components $Y_{(0)}, Y_{(1)}$ having internal degrees 0 and -1 respectively; indices i, j, k refer to local coordinates on M .

To construct an observable out of the pre-observable (95), we impose the gauge fixing (71) which consists in setting $Y_{(1)} = 0$. The path integral (72) reads

$$(96) \quad O_\gamma = \int_{\text{Map}(\gamma, M)} \mathcal{D}Y_{(0)} \text{Hol}_{Y_{(0)}^* \nabla}(\gamma) e^{i \oint_\gamma \langle i^* A_{(1)}, \mu(Y_{(0)}) \rangle}$$

Assume that there exists a Lagrangian polarization P of (M, ω_M) such that the geometric quantization of $(M, \omega, L, \nabla, P)$ yields a space of states $\mathcal{H}$ and components $\mu_a \in C^\infty(M)$ of the moment map are quantized into operators $\hat{\mu}_a \in \text{End}(\mathcal{H})$ such that the map $\hat{\mu} : \mathfrak{g} \rightarrow \text{End}(\mathcal{H})$ sending generator $T_a \in \mathfrak{g}$ into $\hat{\mu}_a$ is a Lie algebra homomorphism (i.e. $\hat{\mu}$ is a representation of $\mathfrak{g}$ on $\mathcal{H}$). Then

$$\hat{\Theta}' = -i \langle \psi, \hat{\mu} \rangle \in C^\infty(\mathfrak{g}[1]) \otimes \text{End}(\mathcal{H})$$

(where we view $\hat{\mu}$ as an element of $\mathfrak{g}^* \otimes \text{End}(\mathcal{H})$) is the quantization of $\Theta' = \langle \psi, \mu(x) \rangle$ satisfying (73). Then the construction of proposition 5 produces out of the pre-observable (95) the usual Wilson loop of the connection component $A_{(1)}$ of the ambient superfield in representation $\hat{\mu}$ along γ :

$$(97) \quad O_\gamma = \text{tr}_{\mathcal{H}} \mathcal{P} \exp \oint_\gamma \hat{\mu}(i^* A_{(1)})$$

which is indeed a gauge invariant observable (i.e. $Q_\Sigma O_\gamma = 0$) for Chern-Simons theory on Σ .

Expression (97) can be viewed as a regularization of the path integral (96).

Coadjoint orbit case. In case when (M, ω_M) is an integral coadjoint orbit of a compact Lie group G such that $\mathfrak{g} = \text{Lie}(G)$, with ω_M the Kirillov symplectic structure and $\mu : M \hookrightarrow \mathfrak{g}^*$ the embedding of M into $\mathfrak{g}^*$ (which corresponds to

the coadjoint Hamiltonian action of G on M), formula (96) becomes the Alekseev-Faddeev-Shatashvili path integral formula for the Wilson loop [2]. The respective $\hat{\mu}$ is the irreducible representation of G associated by Kirillov's orbit method to the coadjoint orbit M , and (97) is just the Wilson loop in this irreducible representation.

7.3. “Wilson loops” in Poisson sigma model. Let (M, π) be a Poisson manifold, $(N, \omega_N = \delta\alpha_N)$ an exact symplectic manifold and $F : N \rightarrow \mathfrak{X}(M)$ a map²⁰ with the property

$$(98) \quad \frac{1}{2}\{F, F\}_{\omega_N} + [\pi, F] = 0$$

where $[\cdot, \cdot]$ is the Schouten-Nijenhuis bracket on polyvector fields on M ; expression $\frac{1}{2}\{F, F\}$ is a map from N to bivectors on M , given in local coordinates as $\frac{1}{2}(\omega^{-1}(y))^{ab} \frac{\partial}{\partial y^a} F^i(x, y) \frac{\partial}{\partial y^b} F^j(x, y) \frac{\partial}{\partial x^i} \wedge \frac{\partial}{\partial x^j}$ where $\{x^i\}$ are local coordinates on M and $\{y^a\}$ are local coordinates on N . Then we have a degree 0 Hamiltonian Q -bundle structure on $T^*[1]M \times N \rightarrow T^*[1]M$ with base structure (27) and fiber structure

$$(99) \quad \mathcal{N} = N, \quad \mathcal{A} = \langle p, \{F, \bullet\}_{\omega_N} \rangle, \\ \omega' = \omega_N, \quad \alpha' = \alpha_N, \quad \Theta' = \langle p, F \rangle \in C^\infty(T^*[1]M \times N)$$

The corresponding 1-dimensional pre-observable for Poisson sigma model associated to (M, π) is given by $\mathcal{F}_\gamma$, Ω_γ as in (95), changing M to N , and the auxiliary action is

$$S_\gamma = \oint_\gamma \alpha_a(Y) dY^a + \langle i^* \eta, F(i^* X, Y) \rangle$$

The corresponding observable is formally given (in the gauge (71)) by the path integral (72):

$$(100) \quad O_\gamma = \int_{\text{Map}(\gamma, N)} \mathcal{D}Y_{(0)} e^{i \oint_\gamma \alpha_a(Y_{(0)}) dY_{(0)}^a + \langle i^* \eta, F(i^* X, Y_{(0)}) \rangle}$$

Assume that (N, ω_N) can be geometrically quantized to a space of states $\mathcal{H}$ and F is quantized to an operator-valued vector field $\hat{F} \in \text{End}(\mathcal{H}) \hat{\otimes} \mathfrak{X}(M)$ satisfying

$$(101) \quad [\pi, \hat{F}] + i\hat{F} \wedge \hat{F} = 0$$

Then (74) gives

$$(102) \quad O_\gamma = \text{tr}_{\mathcal{H}} \mathcal{P} \exp \left(i \oint_\gamma \langle i^* \eta, \hat{F}(i^* X) \rangle \right)$$

which can be regarded as the evaluation of the path integral (100) via Hamiltonian formalism. By proposition 5, O_γ is indeed an observable for the Poisson sigma model.

Remark 12. Note that in case $N = \text{point}$ we get back the observable (94). Also, in case $M = \mathfrak{g}^*$ with Kirillov-Kostant Poisson structure and requiring that F takes values in constant vector fields on F , equation (98) becomes the equation on equivariant moment map and the corresponding observable (102) is the usual Wilson loop (97) for 2-dimensional BF theory.

²⁰Alternatively, one can view F as a vertical vector field on the trivial fiber bundle $N \times M \rightarrow N$.

7.4. Torsion observables in Chern-Simons theory. Fix $n' \in \{-1, 0, 1, 2\}$ and $m \in \mathbb{Z}$ (only the parity of m will matter for our discussion). Let again $\mathfrak{g}$ be a quadratic Lie algebra and let $\rho : \mathfrak{g} \rightarrow \text{End}(R)$ be a representation on a vector space R with values in traceless matrices (cf. (78)). Then we have a degree n' Hamiltonian Q -bundle structure on $\mathfrak{g}[1] \oplus R[m] \oplus R^*[n' - m] \rightarrow \mathfrak{g}[1]$ with base structure (20) and fiber structure

$$(103) \quad \mathcal{N} = R[m] \oplus R^*[n' - m], \quad \mathcal{A} = \left\langle \rho(\psi)q, \frac{\partial}{\partial q} \right\rangle + \left\langle \rho^*(\psi)p, \frac{\partial}{\partial p} \right\rangle, \\ \omega' = \langle \delta p, \delta q \rangle, \quad \alpha' = \langle p, \delta q \rangle, \quad \Theta' = \langle p, \rho(\psi)q \rangle$$

where $\rho^* : \mathfrak{g} \rightarrow \text{End}(R^*)$ is the representation dual to ρ ; q is the R -valued coordinate of degree m on $R[m]$ and p is the R^* -valued coordinate of degree $n' - m$ on $R^*[n' - m]$.

The corresponding pre-observable for Chern-Simons theory on a closed oriented 3-manifold Σ , for $i : \gamma \hookrightarrow \Sigma$ an embedded closed oriented $(n' + 1)$ -manifold, reads:

$$(104) \quad \mathcal{F}_\gamma = R[m] \otimes \Omega^\bullet(\gamma) \oplus R^*[n' - m] \otimes \Omega^\bullet(\gamma), \quad \Omega_\gamma = \int_\gamma \langle \delta \mathbf{p}, \delta \mathbf{q} \rangle, \\ S_\gamma = \int_\gamma \langle \mathbf{p}, d\mathbf{q} \rangle + \langle \mathbf{p}, \rho(i^* A) \mathbf{q} \rangle$$

where $\mathbf{q} = \sum_{k=0}^{n'+1} \mathbf{q}_{(k)}$ and $\mathbf{p} = \sum_{k=0}^{n'+1} \mathbf{p}_{(k)}$ are the auxiliary superfields corresponding to q and p ; component $\mathbf{q}_{(k)}$ is a R -valued k -form on γ with internal degree $m - k$ and $\mathbf{p}_{(k)}$ is a R^* -valued k -form on γ of internal degree $n' - m - k$.

Pre-observable (104) can be pushed forward to zero-modes (in the sense of proposition 2) as in section 6.3:

$$(105) \quad \mathcal{F}_\gamma^z = R[m] \otimes H^\bullet(\gamma) \oplus R^*[n' - m] \otimes H^\bullet(\gamma), \quad \Omega_\gamma^z = \int_\gamma \langle \delta \mathbf{p}_z, \delta \mathbf{q}_z \rangle, \\ S_\gamma^z = \int_\gamma \langle \mathbf{p}_z, \rho(i^* A)(\text{id} + G \rho(i^* A))^{-1} \mathbf{q}_z \rangle - i \log \text{tor}(\gamma, i^* A, \rho)$$

where we introduced the notation

$$(106) \quad \text{tor}(\gamma, i^* A, \rho) = \exp \text{tr}_R \sum_{k \geq 2} \frac{(-1)^k}{k} \int_{\text{Conf}_k(\gamma)} \overleftarrow{\prod}_{1 \leq j \leq k} \pi_{j+1, j}^* \eta \cdot \pi_j^* \rho(i^* A)$$

Notations G, η, π are the same as in section 6.3. We use harmonic representatives for zero-modes $\mathbf{q}_z, \mathbf{p}_z$ in $\mathcal{F}_\gamma$.

Taking the BV integral over zero-modes in (105) we obtain the observable for Chern-Simons theory on Σ :

$$(107) \quad O_\gamma = \int_{\mathcal{L}_z \subset \mathcal{F}_\gamma^z} \sqrt{\mu_z}|_{\mathcal{L}_z} e^{i \int_\gamma \langle \mathbf{p}_z, \rho(i^* A)(\text{id} + G \rho(i^* A))^{-1} \mathbf{q}_z \rangle} \cdot \text{tor}(\gamma, i^* A, \rho)$$

with $\mathcal{L}_z \subset \mathcal{F}_\gamma^z$ a Lagrangian subspace and μ_z a translation invariant volume element on $\mathcal{F}_\gamma^z$. Proposition 6 implies that O_γ is Q_Σ -closed and that deformations of gauge-fixing induce Q_Σ -exact shifts in O_γ .

In case $n' = 0, m = 1$, with $\gamma = S^1$, (107) can be evaluated explicitly²¹:

$$(108) \quad O_\gamma = \int_{R[1] \oplus R^*[-1]} d\mathbf{q}_{(0)}^z d\mathbf{p}_{(0)}^z e^{i \oint_\gamma \langle \mathbf{p}_{(0)}^z, \rho(\bar{A}) \mathbf{q}_{(0)}^z \rangle} \cdot \det_R \left(\frac{\sinh \rho(\bar{A}/2)}{\rho(\bar{A}/2)} \right) \\ = \det_R \left(\rho(W)^{1/2} - \rho(W)^{-1/2} \right) = \det_R(\rho(W) - \text{id}_R)$$

²¹One trick to simplify the calculation is to use gauge-invariance of O_γ to pick the constant representative of the gauge equivalence class of connection $i^* A$.

where $W = \mathcal{P} \exp \oint_{\gamma} i^* A_{(1)}$ is the holonomy of the ambient connection around γ (since this requires the choice of a starting point on γ , one can think of W as a group element defined modulo conjugation). We also denote $\bar{A} = \log W$.

In (108) we used the gauge-fixing Lagrangian (71). In particular, the gauge-fixing for the integral over zero-modes is

$$\underbrace{(R[1] \oplus R^*[-1]) \otimes H^0(\gamma)}_{\mathcal{L}_z} \subset \underbrace{(R[1] \oplus R^*[-1]) \otimes H^\bullet(\gamma)}_{\mathcal{F}_\gamma^z}$$

Remark 13. Observable (108) is only interesting (not identically zero) for representations ρ such that $\det \rho \neq 0$, i.e. $\rho(x)$ does not have zero eigenvalue for x in an open dense subset of $\mathfrak{g}$. In particular, (108) is identically zero for $\rho = \text{ad}$ the adjoint representation.

Remark 14. Observable (108) has the property that it only depends on $i^* A_{(1)}$ and not on the other components of the ambient field. Also, it is the only representative of its class in Q_Σ -cohomology which depends only on $i^* A$, for degree reasons (since $C^\infty(\mathcal{F}_\Sigma|_\gamma)$ is non-negatively graded; we denote $\mathcal{F}_\Sigma|_\gamma = \text{Map}(T[1]\gamma, \mathcal{M})$ the space of pull-backs of ambient fields to γ).

In case $n' > 0$, O_γ defined by (107) generally depends on $i^* A_{(k)}$ for $0 \leq k \leq n' + 1$. Also, there is no canonical representative of the class of O_γ in Q_Σ -cohomology and no canonical choice for the gauge-fixing Lagrangian $\mathcal{L} \subset \mathcal{F}_\gamma$.

For $n' = -1$, convergence requirement for the integral over zero-modes (107) forces (cf. remark 11) $\mathcal{L}_z = (R[m] \oplus R^*[-1-m])^{\text{odd}}$ which yields $O_\gamma = 0$.

For $n' = 2$, for γ a connected closed oriented 3-manifold with non-zero first Betti number, we may take $m = 1$ and choose the gauge-fixing for zero-modes as

$$(109) \quad \mathcal{L}_z = (R[1] \oplus R^*[1]) \otimes (H^0(\gamma) \oplus \underbrace{\lambda^\perp}_{\subset H^1(\gamma)} \oplus \underbrace{\lambda}_{\subset H^2(\gamma)})$$

where λ is any line in $H^1(\gamma)$ and $\lambda^\perp$ is its orthogonal complement in $H^2(\gamma)$ w.r.t. the Poincaré duality.

If the condition $\det \rho \neq 0$ as in remark 13 holds, O_γ defined with zero-mode gauge-fixing (109) is well-defined on an open subset of $\mathcal{F}_\Sigma|_\gamma$ and is not identically zero.

In case $\dim H^1(\gamma) = 1$, Lagrangian (109) is the same as in remark 11, i.e. $\mathcal{L}_z = (\mathcal{F}_\gamma^z)^{\text{odd}}$. In case $\dim H^1(\gamma) > 1$, the Lagrangian $(\mathcal{F}_\gamma^z)^{\text{odd}}$ yields O_γ which is identically zero.

In case of γ a rational homology sphere, $H^1(\gamma) = 0$, construction (109) is not applicable. However, one can take

$$\mathcal{L}_z = (R[1] \oplus R^*[1]) \otimes H^0(\gamma)$$

which produces an observable O_γ of degree $-2 \dim R$.

Remark 15. In case $n' = 2$ and $\gamma = \Sigma$, $i = \text{id}$, property (62) does not generally hold, so (104) is a pre-observable in the sense of definition 6, but $S_\Sigma + S_\gamma$ does not satisfy the classical master equation (61) on the space $\mathcal{F}_\Sigma \times \mathcal{F}_\gamma$. On the level of observable O_γ this means that $Q_\Sigma O_\gamma = 0$, but $S_\Sigma - i \log O_\gamma$ does not satisfy the classical master equation on $\mathcal{F}_\Sigma$.

In case $n' = 1$, for γ a connected closed oriented surface of genus $g > 0$, we may take $m = 0$ and set

$$\mathcal{L}_z = R \otimes (H^0(\gamma) \oplus \lambda^\perp) \oplus R^*[1] \otimes (H^0(\gamma) \oplus \lambda)$$

where λ is again any line $H^1(\gamma)$ and $\lambda^\perp$ is its orthogonal in $H^1(\gamma)$ w.r.t. the Poincaré duality. Assuming $\det \rho \neq 0$, with this choice of $\mathcal{L}_z$ we produce an observable O_γ of degree $(2g-2)\dim R$ which is well-defined and non-zero on an open subset of $\mathcal{F}_\Sigma|_\gamma$.

Expression (107) also is an observable for BF theory in dimension D . In this case n' should satisfy $-1 \leq n' \leq D-1$. For $n' = 2k$ and γ a closed oriented submanifold of Σ of dimension $2k+1$, we may take $m = 2k+1$ and choose the gauge-fixing $\mathcal{L}_z$ as

$$\mathcal{L}_z = (R[2k+1] \oplus R^*[-1]) \otimes \mathcal{L}'$$

where $\mathcal{L}' \subset H^\bullet(\gamma)$ a Lagrangian subspace with the property

$$(110) \quad \dim \mathcal{L}'_j = \dim \mathcal{L}'_{2k-j}$$

where $\mathcal{L}'_j = \mathcal{L}' \cap H^j(\gamma)$, which is equivalent to

$$(111) \quad \dim \mathcal{L}'_j = B_j - B_{j-1} + \cdots + (-1)^j B_0$$

where $B_j = \dim H^j(\gamma)$. For $\mathcal{L}'$ to exist, we require that for γ the combinations of Betti numbers on the r.h.s. of (111) are non-negative. Assuming $\det \rho \neq 0$, property (110) ensures that the integral over zero-modes in (107) exists and is non-zero in an open subset of $\mathcal{F}_\Sigma|_\gamma$.

7.5. A codimension 2 (pre-)observable in BF theory. The following example is due to Cattaneo-Rossi [9]. Let $\mathcal{M} = \mathfrak{g}[1] \oplus \mathfrak{g}^*[D-2]$ with the structure of degree $D-1$ Hamiltonian Q -manifold as in (23). One constructs a trivial degree $D-3$ Hamiltonian Q -bundle over $\mathcal{M}$ with fiber data

$$(112) \quad \mathcal{N} = \mathfrak{g} \oplus \mathfrak{g}^*[D-3], \quad \mathcal{A} = \langle [\psi, q], \frac{\partial}{\partial q} \rangle + \langle \text{ad}_\psi^* p, \frac{\partial}{\partial p} \rangle + (-1)^D \langle \xi, \frac{\partial}{\partial p} \rangle,$$

$$\omega' = \langle \delta p, \delta q \rangle, \quad \alpha' = \langle p, \delta q \rangle, \quad \Theta' = \langle p, [\psi, q] \rangle + \langle \xi, q \rangle$$

Here q is the $\mathfrak{g}$ -valued coordinate of degree 0 on $\mathfrak{g}$ and p is the $\mathfrak{g}^*$ -valued coordinate of degree $D-3$ on $\mathfrak{g}^*[D-3]$; ψ and ξ are coordinates on $\mathcal{M}$ as in section 2.3.

Given a closed oriented D -manifold Σ and a closed oriented submanifold $i: \gamma \hookrightarrow \Sigma$ of codimension 2, by the construction of section 5 we get a pre-observable for the BF theory on Σ associated to the Hamiltonian Q -bundle (23,112):

$$(113) \quad \mathcal{F}_\gamma = \mathfrak{g} \otimes \Omega^\bullet(\gamma) \oplus \mathfrak{g}^*[D-3] \otimes \Omega^\bullet(\gamma),$$

$$(114) \quad \Omega_\gamma = (-1)^D \int_\gamma \langle \delta \mathbf{p}, \delta \mathbf{q} \rangle,$$

$$(115) \quad S_\gamma = \int_\gamma \langle \mathbf{p}, d\mathbf{q} \rangle + \langle \mathbf{p}, [i^* A, \mathbf{q}] \rangle + \langle i^* B, \mathbf{q} \rangle$$

where $\mathbf{q} = \sum_{k=0}^{D-2} \mathbf{q}_{(k)}$ and $\mathbf{p} = \sum_{k=0}^{D-2} \mathbf{p}_{(k)}$ are the auxiliary superfields corresponding to q and p ; $\mathbf{q}_{(k)}$ and $\mathbf{p}_{(k)}$ are k -forms on γ with values in $\mathfrak{g}$ and $\mathfrak{g}^*$ respectively, having internal degrees $-k$ and $D-3-k$ respectively.

Push-forward of the pre-observable (113–115) to zero-modes (in the sense of proposition 2) is evaluated as in section 6.3:

$$(116) \quad \mathcal{F}_\gamma^z = \mathfrak{g} \otimes H^\bullet(\gamma) \oplus \mathfrak{g}^*[D-3] \otimes H^\bullet(\gamma), \quad \Omega_\gamma^z = \int_\gamma \langle \delta \mathbf{p}_z, \delta \mathbf{q}_z \rangle,$$

$$S_\gamma^z = \int_\gamma \langle i^* B + (-1)^D \text{ad}_{i^* A}^* \mathbf{p}_z, (\text{id} + G \text{ad}_{i^* A})^{-1} \mathbf{q}_z \rangle - i \log \text{tor}(\gamma, i^* A, \text{ad})$$

where tor is as in (106), for $\rho = \text{ad}$ the adjoint representation of $\mathfrak{g}$.

In the case of abelian BF theory, $\mathfrak{g} = \mathbb{R}$, (116) yields

$$(117) \quad e^{iS_\gamma^z} = e^{i \int_\gamma \langle i^* B, \mathbf{q}_z \rangle}$$

which is an observable if we understand $\mathbf{q}_z \in H^\bullet(\gamma)$ as an external parameter (the crucial point here is that by construction (117) satisfies the semi-quantum master equation (42), but the last term in (42) vanishes since S_γ^z depends only on $\mathbf{q}_z$). Taking $\mathbf{q}_z = c \cdot 1 \in H^0(\gamma)$, we obtain the observable (90).

In [9] it is shown that in the case of “long knots” $\mathbb{R}^{D-2} \sim \gamma \subset \Sigma = \mathbb{R}^D$, embeddings of $\mathbb{R}^{D-2}$ into $\mathbb{R}^D$ with prescribed linear asymptotics, the push-forward of pre-observable (113–115) to zero-modes yields, upon fixing the values of zero-modes to $\mathbf{q}_z = c \otimes 1 \in \mathfrak{g}^* \otimes \Omega^0(\gamma)$ and $\mathbf{p}_z = 0$, an observable, generalizing (90) in non-abelian case.

8. FINAL REMARKS

In this paper we presented a two-step construction of classical observables in AKSZ sigma models, which consists of: (i) constructing an extension of the AKSZ theory to a BV theory on a larger space of fields using an extension of the target to a Hamiltonian Q -bundle, (ii) using the BV push-forward to the original space of fields to produce an observable. We also provided some examples, and in particular recovered the well-known Wilson loop observable in Chern-Simons theory together with its path integral representation [2] and the Cattaneo-Rossi “Wilson surface” observable in BF theory.

There are natural questions to this construction not answered here, to which we hope to return in a future publication:

- Extend the zoo of explicit examples. In particular,
 - give more explicit examples of “Wilson loops” in Poisson sigma model, section 7.3 (which requires a solution of (98) as the input for step (i) of the construction and its quantization satisfying (101) for step (ii)) other than those mentioned in remark 12;
 - give an example corresponding to a *non-trivial* Hamiltonian Q -bundle over the target of the AKSZ sigma model.
- Calculate the isotopy invariants of embeddings given by expectation values of our observables.

8.1. Extension to source manifolds with boundary. Our construction of observables possesses an extension to the case when the source manifold Σ has a boundary $\partial\Sigma$, and the submanifold $\gamma \subset \Sigma$ on which the observable is supported is also allowed to have boundary, $\partial\gamma \subset \partial\Sigma$. We will outline this extension here, treating gauge theories with boundary in “BV-BFV formalism” [8]. For a more detailed example, Chern-Simons theory on a manifold with boundary with Wilson lines ending on the boundary, the reader is referred to [4].

Recall [8] that a gauge theory on manifold Σ with boundary $\partial\Sigma$ in BV-BFV formalism is described by the following data:

- A degree 0 Hamiltonian Q -manifold $(\mathcal{F}_{\partial\Sigma}, Q_{\partial\Sigma}, \Omega_{\partial\Sigma} = \delta\alpha_{\partial\Sigma}, S_{\partial\Sigma})$ — “the BFV phase space” (or “the space of boundary fields”) associated to the boundary $\partial\Sigma$.
- A Q -manifold $(\mathcal{F}_\Sigma, Q_\Sigma)$ (“the space of bulk fields”) endowed with a Q -morphism $\pi : \mathcal{F}_\Sigma \rightarrow \mathcal{F}_{\partial\Sigma}$ (restriction of fields to the boundary), a degree -1 symplectic form Ω_Σ and a degree 0 action $S_\Sigma \in C^\infty(\mathcal{F}_\Sigma)$ satisfying

$$(118) \quad \delta S_\Sigma = \iota_{Q_\Sigma} \Omega_\Sigma + \pi^* \alpha_{\partial\Sigma}$$

Equation (118) replaces the condition that S_Σ is the Hamiltonian function for Q_Σ in definition 4, and its consequence $Q_\Sigma S_\Sigma = \pi^*(\iota_{Q_\Sigma} \alpha_{\partial\Sigma} - 2S_{\partial\Sigma})$ or, equivalently,

$$\frac{1}{2} \underbrace{\{S_\Sigma, S_\Sigma\}_{\Omega_\Sigma}}_{:= \iota_{Q_\Sigma} \iota_{Q_\Sigma} \Omega_\Sigma} + \pi^* S_{\partial\Sigma} = 0$$

replaces the classical master equation.

AKSZ sigma models on manifolds with boundary fit naturally into BV-BFV formalism: one takes the standard construction (8,9,15,16) for the bulk and sets

$$\begin{aligned} \mathcal{F}_{\partial\Sigma} &= \text{Map}(T[1]\partial\Sigma, \mathcal{M}), \quad Q_{\partial\Sigma} = d_{\partial\Sigma}^{\text{lifted}} + Q^{\text{lifted}}, \quad \Omega_{\partial\Sigma} = \tau_{\partial\Sigma}(\omega), \\ \alpha_{\partial\Sigma} &= (-1)^{\dim \Sigma - 1} \tau_{\partial\Sigma}(\alpha), \quad S_{\partial\Sigma} = (-1)^{\dim \Sigma - 1} \left(\iota_{d_{\partial\Sigma}^{\text{lifted}}} \tau_{\partial\Sigma}(\alpha) + \tau_{\partial\Sigma}(\Theta) \right) \end{aligned}$$

for the boundary; Q -morphism $\pi : \text{Map}(T[1]\Sigma, \mathcal{M}) \rightarrow \text{Map}(T[1]\partial\Sigma, \mathcal{M})$ is the pull-back by the inclusion $\partial\Sigma \hookrightarrow \Sigma$.

A classical observable for a gauge theory in BV-BFV formalism (generalizing definition 5) can be defined as the following collection of data.

- A graded vector space (the auxiliary space of states) $\mathcal{H}_{\partial\gamma}$ and a degree 1 element

$$\delta_{\partial\gamma} \in C^\infty(\mathcal{F}_{\partial\Sigma}) \otimes \text{End}(\mathcal{H}_{\partial\gamma})$$

satisfying

$$Q_{\partial\Sigma} \delta_{\partial\gamma} + \delta_{\partial\gamma}^2 = 0$$

- A degree 0 $\mathcal{H}_{\partial\gamma}$ -valued function of bulk fields

$$O_\gamma \in C^\infty(\mathcal{F}_\Sigma) \otimes \mathcal{H}_{\partial\gamma}$$

satisfying

$$(Q_\Sigma + \pi^* \delta_{\partial\gamma}) O_\gamma = 0$$

The locality requirement is that O_γ only depends on the restrictions of bulk fields to γ ; likewise, $\delta_{\partial\gamma}$ should only depend on the restrictions of boundary fields to $\partial\gamma$.

We can also define a pre-observable in the BV-BFV context as the following data.

- A degree 0 Hamiltonian Q -bundle over $(\mathcal{F}_{\partial\Sigma}, Q_{\partial\Sigma})$ with fiber data $(\mathcal{F}_{\partial\gamma}, \mathcal{A}_{\partial\gamma}, \Omega_{\partial\gamma} = \delta\alpha_{\partial\gamma}, S_{\partial\gamma})$.
- A trivial Q -bundle over $(\mathcal{F}_\Sigma, Q_\Sigma)$ with fiber data $(\mathcal{F}_\gamma, \mathcal{A}_\gamma)$, with projection $\bar{\pi} : \mathcal{F}_\gamma \rightarrow \mathcal{F}_{\partial\gamma}$ such that $d\bar{\pi}(\mathcal{A}_\gamma) = \mathcal{A}_{\partial\gamma}$, with fiber degree -1 symplectic structure Ω_γ and with degree 0 auxiliary action $S_\gamma \in C^\infty(\mathcal{F}_\Sigma \times \mathcal{F}_\gamma)$ satisfying

$$\delta^{\text{vert}} S_\gamma = \iota_{\mathcal{A}_\gamma} \Omega_\gamma + \bar{\pi}^* \alpha_{\partial\gamma}, \quad Q_\Sigma S_\gamma + \frac{1}{2} \underbrace{\{S_\gamma, S_\gamma\}_{\Omega_\gamma}}_{:= \iota_{\mathcal{A}_\gamma} \iota_{\mathcal{A}_\gamma} \Omega_\gamma} + \pi_{\text{tot}}^* S_{\partial\gamma} = 0$$

where δ^{vert} is the de Rham differential on $\mathcal{F}_\gamma$ and $\pi_{\text{tot}} = \pi \times \bar{\pi} : \mathcal{F}_\Sigma \times \mathcal{F}_\gamma \rightarrow \mathcal{F}_{\partial\Sigma} \times \mathcal{F}_{\partial\gamma}$.

Passing from a pre-observable to observable is a quantization problem. We construct $\mathcal{H}_{\partial\gamma}$ as the geometric quantization of the symplectic manifold $(\mathcal{F}_{\partial\gamma}, \Omega_{\partial\gamma})$ and $\delta_{\partial\gamma}$ as the geometric quantization of $S_{\partial\gamma}$:

$$(119) \quad \mathcal{H}_{\partial\gamma} = \text{GeomQuant}(\mathcal{F}_{\partial\gamma}, \Omega_{\partial\gamma}), \quad \delta_{\partial\gamma} = i\hat{S}_{\partial\gamma}$$

For simplicity, assume that the Lagrangian polarization of $\mathcal{F}_{\partial\Sigma}$ used for the geometric quantization is the vertical polarization of a fibration $\mathbf{p} : \mathcal{F}_{\partial\Sigma} \rightarrow \mathcal{B}$. Then O_γ is given by the following modification of the fiber BV integral:

$$(120) \quad O_\gamma(X, b) = \int_{\mathcal{L}_b \subset \bar{\pi}^{-1} \mathbf{p}^{-1} b} \mathcal{D}Y e^{iS_\gamma(X, Y)}$$

for $b \in \mathcal{B}$, $X \in \mathcal{F}_\Sigma$; Y runs over $\mathcal{F}_\gamma$; $\mathcal{L}_b$ is a Lagrangian in $\bar{\pi}^{-1}\mathfrak{p}^{-1}b \subset \mathcal{F}_\gamma$.

Given an AKSZ theory on Σ with target $\mathcal{M}$ and a Hamiltonian Q -bundle over $\mathcal{M}$, as a first step, we construct a pre-observable with the bulk data given by old formulae (53) and boundary data given by the same formulae with γ replaced by $\partial\gamma$:

$$\begin{aligned}\mathcal{F}_{\partial\gamma} &= \text{Map}(T[1]\partial\gamma, \mathcal{N}), \quad \mathcal{A}_{\partial\gamma} = d_{\partial\gamma}^{\text{lifted}} + p_{\partial}^* \mathcal{A}^{\text{lifted}}, \quad \Omega_{\partial\gamma} = \tau_{\partial\gamma}(\omega'), \\ \alpha_{\partial\gamma} &= (-1)^{\dim \gamma - 1} \tau_{\partial\gamma}(\alpha'), \quad S_{\partial\gamma} = (-1)^{\dim \gamma - 1} \left(\iota_{d_{\partial\gamma}^{\text{lifted}}} \alpha_{\partial\gamma} + p_{\partial}^* \tau_{\partial\gamma}^{\text{tot}}(\Theta') \right)\end{aligned}$$

where $p_{\partial} : \text{Map}(T[1]\partial\Sigma, \mathcal{M}) \rightarrow \text{Map}(T[1]\partial\gamma, \mathcal{M})$ is the restriction of ambient boundary fields to $\partial\gamma$.

As the second step, we pass from this pre-observable to an observable using construction (119,120).

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Chern-Simons Theory with Wilson Lines and Boundary in the BV-BFV Formalism

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Abstract

We consider the Chern-Simons theory with Wilson lines in 3D and in 1D in the BV-BFV formalism of Cattaneo-Mnev-Reshetikhin. In particular, we allow for Wilson lines to end on the boundary of the space-time manifold. In the toy model of 1D Chern-Simons theory, the quantized BFV boundary action coincides with the Kostant cubic Dirac operator which plays an important role in representation theory. In the case of 3D Chern-Simons theory, the boundary action turns out to be the odd (degree 1) version of the BF model with source terms for the B field at the points where the Wilson lines meet the boundary. The boundary space of states arising as the cohomology of the quantized BFV action coincides with the space of conformal blocks of the corresponding WZW model.

1 Introduction

In the case of complicated space-time topology, a promising approach to quantization of general field theories involves cutting the space-time manifold into simple pieces, where the problem is more easily solved, and then gluing back the individual elements to obtain the final answer. This method was proposed by Atiyah and successfully applied by Witten in [15] to study the quantization of the Chern-Simons theory with Wilson lines (cf. also [9]). Following this idea, a systematic program to understand quantization in the Batalin-Vilkovisky formalism for field theories with degeneracies on manifolds with boundaries has been initiated in [6]. As a part of the construction, a Batalin-Fradkin-Vilkovisky model is associated to the boundary of the space-time manifold. The canonical quantization of this boundary BFV model provides a space of boundary states, together with a cohomological invariance condition that defines the admissible quantum states of the theory among all boundary states. In the case of quantum field theories on manifolds without boundary, the partition function and

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other correlation functions are complex-valued. In the presence of a boundary, the correlators of the bulk theory take values in this boundary space of states.

The aim of this paper is to apply the BV-BFV formalism of [6] to the Chern-Simons theory on manifolds with boundary, with Wilson lines ending on the boundary. The BV formulation of this theory on closed manifolds is well understood (and served as the motivating example for the AKSZ construction). We will also consider the one-dimensional Chern-Simons model, obtained when the AKSZ construction is carried out in one dimension.

We include in our construction Wilson lines which may end on the boundary of the manifold. This requires some extra work in BV-BFV formalism. Our treatment is based on the path integral representation for Wilson loops suggested in [1], [7]. This is also an example of a more general construction of observables for AKSZ sigma models proposed in [12].

We compare our answers with those obtained using the geometric quantization for the boundary [15] and using canonical quantization [10], [8].

One of our main results is the boundary BFV action for Chern-Simons theory with Wilson lines, which has the form of an odd (degree 1) version of BF action modified by source terms for the B field at points where the Wilson lines meet the boundary. We also consider the toy model of one-dimensional Chern-Simons theory and derive the corresponding boundary action. Its quantization coincides with Kostant cubic Dirac operator. We compare the BV-BFV results for the one-dimensional model with the ones obtained in [3] on segments, and also see how Wilson lines can be added to the one-dimensional model. In the three-dimensional case, the boundary space of states, arising as the cohomology of the quantized BFV action, coincides with the space of conformal blocks of the WZW model on the boundary (in the picture of [10]).

We begin in section 2 with the treatment of Wilson lines in the BV formalism. We also provide a short introduction to the relevant aspects of the BV formalism. In section 3 we describe the BV formulation of the Chern-Simons theory with Wilson lines, applying the AKSZ construction to this special setting. Then we proceed to explain how the bulk model gets supplemented with a boundary BFV theory if the underlying manifold has a boundary. We repeat this procedure in section 4 for the one-dimensional Chern-Simons model. The $\mathbb{Z}_2$ -grading that replaces the usual $\mathbb{Z}$ -grading in this case leads to certain subtleties with the master equation. In section 5 we present the quantization of the boundary BFV models and describe the arising spaces of quantum states, that we compare with known results for the quantization of the involved models.

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2 Wilson lines in the BV formalism

In this section, we start with a brief introduction to the BV formalism. The main point is to incorporate the Wilson line observables in this approach.

2.1 A short introduction to the BV formalism

We know that the path integral in quantum field theories is not well defined if the classical action S_{cl} defined over the space of classical fields $\mathcal{F}_{cl}$ is degenerate, for instance due to gauge symmetries. The Batalin-Vilkovisky formalism provides a general method for the perturbative calculation of partition functions and correlators.

In the BV formalism, the space of fields is augmented to a BV space of fields $\mathcal{F}_{BV}$, a graded infinite-dimensional manifold equipped with a symplectic structure Ω_{BV} of degree -1 called the BV structure. The grading (usually $\mathbb{Z}$, sometimes $\mathbb{Z}_2$) is commonly referred to as “ghost number”, in relation with the Faddeev-Popov prescription. The BV bracket is defined as the Poisson bracket obtained by inverting the BV structure,

$$\{F, G\} = \Omega_{BV}^{-1}(\delta F, \delta G),$$

and obviously has a ghost number 1. Note that the variational operator δ can be interpreted as a de Rham differential in the space of fields. In many cases of interest, the BV space of fields is a cotangent bundle where the degree of the fibers is shifted to -1 , which ensures its canonical symplectic form has the proper degree. Coordinates along the cotangent fibers are then called antifields. In the case of gauge theories, where the degeneracy arises under the action of a gauge group, the BV space of fields is simply the shifted cotangent bundle of the BRST space of fields which contains all classical fields as well as the ghosts parametrizing the gauge symmetries (basically the infinitesimal gauge parameters with a ghost number shifted by one), $\mathcal{F}_{BV} = T^*[-1]\mathcal{F}_{BRST}$.

At the classical level, infinitesimal gauge transformations and the classical action can be used to construct a differential acting on the functionals on the BV space of fields. Geometrically, this differential corresponds to a cohomological vector field Q on $\mathcal{F}_{BV}$. Moreover, Q is a Hamiltonian vector field,

$$\iota_Q \Omega_{BV} = \delta S_{BV},$$

with the Hamiltonian function being the BV action S_{BV} that reduces to the classical action when all antifields are set to zero. The condition $Q^2 = 0$ follows from the classical master equation $\{S_{BV}, S_{BV}\} = 0$, and determining the BV formulation of a given theory amounts to determining an extension of the classical action to the BV space of fields that satisfies this classical master equation.

2.2 The AKSZ construction

While it is usually difficult to find the BV formulation of a given field theory with a degenerate action, the study of the geometric interpretation of the clas-

sical master equation in [4] led to an insightful procedure to construct solutions thereof, called the AKSZ construction after its authors. A formalized more recent treatment can be found in [5].

In this construction, the target space of the theory is a graded manifold Y equipped with a symplectic structure ω_Y of degree $n - 1$ and a compatible cohomological vector field Q_Y of degree 1, in the sense that it preserves the symplectic structure, $\mathcal{L}_{Q_Y}\omega_Y = 0$. We also want ω_Y to be associated to a Liouville one-form α_Y (of degree $n - 1$ as well), namely $\omega_Y = \delta\alpha_Y$. Here δ denotes the de Rham exterior derivative on Y , while we keep the usual d for the one on the source manifold N of the model. For $n \neq 0$ (see for instance [13] for details), Q_Y can be shown to be Hamiltonian, i.e. there exists a function Θ_Y of degree n on Y such that $\iota_{Q_Y}\omega_Y = \delta\Theta_Y$. Like in the BV formalism, the nilpotency of Q_Y follows from the condition $\{\Theta_Y, \Theta_Y\}_Y = 0$, where the curly braces with a subscript Y denote the Poisson bracket on Y associated to its symplectic structure.

The BV space of fields is then given by maps between the odd tangent bundle of some n -dimensional manifold N and the graded manifold Y ,

$$\mathcal{F}_{AKSZ} = \text{Map}(T[1]N, Y).$$

The odd tangent bundle is naturally equipped with a cohomological vector field, the de Rham vector field, which can be expressed as

$$D = \theta^\mu \frac{\partial}{\partial x^\mu}$$

in coordinates x^μ of the base manifold N and θ^μ of the odd fibers. Notice also that real-valued functions on $T[1]N$ can be interpreted as differential forms on N (by expanding the function in powers of θ^μ),

$$C^\infty(T[1]N, \mathbb{R}) \simeq \Omega^\bullet(N),$$

which allows to define a canonical measure μ on $T[1]N$: the Berezinian integration along all odd fibers simply extracts the top-form out of this expansion and it remains to integrate it over the base N .

Roughly, the idea behind the AKSZ construction involves lifting the symplectic structure ω_Y from the target space to define the BV structure Ω_{AKSZ} on the space of fields. In effect, we replace functions and differential forms on Y by functionals on the space of fields with values in differential forms on N and their variations (which also explains the choice of δ to denote the exterior derivative on Y), and we integrate over the source space $T[1]N$ using its canonical measure μ ,

$$\Omega_{AKSZ} = \int_{T[1]N} \mu \tilde{\omega}_Y. \quad (1)$$

The tilde denotes the extension from function on Y to functional on the space of fields. The Berezinian integration along the fibers will lower the ghost number of ω_Y from $n - 1$ to -1 as required.

In a second stage we need to lift the cohomological vector field Q_Y on the target-space Y as well as the de Rham vector field D on the source-space $T[1]N$ to the space of fields, and combine them to form the BV cohomological vector field Q discussed above that will happen to be Hamiltonian. Its generating functional is nothing but the BV-AKSZ action

$$S_{\text{AKSZ}} = \int_{T[1]N} \mu \left(\iota_{Q_D} \tilde{\alpha}_Y + \tilde{\Theta}_Y \right), \quad (2)$$

where $Q_D = \Sigma_i D\phi^i \frac{\delta}{\delta\phi^i}$ is the lift of the de Rham vector field (ϕ^i denotes generic coordinates on the space of fields). This AKSZ action automatically solves the classical master equation as a consequence of the integrability condition on Θ_Y and the fact that the integral of exact forms vanishes provided $\partial N = \emptyset$.

As an example of the AKSZ construction, we derive here the BV formulation of the Chern-Simons theory, which corresponds to the special case $n = 3$ with $Y = \mathfrak{g}[1]$, where $\mathfrak{g}$ is a Lie algebra equipped with an invariant scalar product. As required, the target space $\mathfrak{g}[1]$ supports a symplectic structure of degree 2 and a Hamiltonian cohomological vector field of degree 1 (sometimes called a Q -structure). If we denote with δ the exterior derivative on $\mathfrak{g}[1]$ and ψ a generic element, the symplectic form, its Liouville potential and the Hamiltonian of the cohomological vector field respectively can be written as

$$\omega_{\mathfrak{g}[1]} = -\frac{1}{2} (\delta\psi, \delta\psi), \quad (3)$$

$$\alpha_{\mathfrak{g}[1]} = -\frac{1}{2} (\psi, \delta\psi), \quad (4)$$

$$\Theta_{\mathfrak{g}[1]} = -\frac{1}{6} (\psi, [\psi, \psi]). \quad (5)$$

Note that Grassmanian variables ψ anticommute, but so do differential forms of odd degree, which explains why $\omega_{\mathfrak{g}[1]}$ may be built out of a symmetric product.

If we use coordinates $x^\mu, \mu = 1, 2, 3$ on N and corresponding Grassmanian coordinates θ^μ on the odd fibres of $T[1]N$, we can decompose the fields $\mathbf{A} \in \text{Map}(T[1]N, \mathfrak{g}[1])$ into $\mathfrak{g}$ -valued differential forms of various degrees and grading,

$$\mathbf{A} = \gamma + A_\mu \theta^\mu + \frac{1}{2} A_{\mu\nu}^+ \theta^\mu \theta^\nu + \frac{1}{6} \gamma_{\mu\nu\sigma}^+ \theta^\mu \theta^\nu \theta^\sigma, \quad (6)$$

specifically

$$\begin{aligned} \gamma &\in \text{Map}(N, \mathfrak{g}[1]), \\ A &\in \Gamma(T^*N \otimes \mathfrak{g}), \\ A^+ &\in \Gamma(\bigwedge^2 T^*N \otimes \mathfrak{g}[-1]), \\ \gamma^+ &\in \Gamma(\bigwedge^3 T^*N \otimes \mathfrak{g}[-2]). \end{aligned}$$

These fields are endowed with two gradings, namely the ghost-grading (that stands in square brackets when non-zero) and the degree as a differential form. Their sum, the total degree, should amount to 1, since each θ^μ has a ghost number 1, and all terms in the decomposition (6) should have the same total

ghost number of 1. As usual, the fields of ghost number 0 are the classical fields, here a $\mathfrak{g}$ -valued connection A , and the fields of ghost number 1 are simply called ghosts. The other two fields are their antifields (which in the BV formalism means canonically conjugated), as is clear when one computes the BV structure,

$$\begin{aligned}\Omega_{\text{BV}}^{\text{CS}} &= \int_{T[1]N} \mu \tilde{\omega}_{\mathfrak{g}[1]} = - \int_{T[1]N} \mu (\delta \mathbf{A}, \delta \mathbf{A}) \\ &= \int_N ((\delta \gamma^+, \delta \gamma) - (\delta A^+, \delta A)) = \int_N (-(\delta \gamma, \delta \gamma^+) + (\delta A, \delta A^+)).\end{aligned}\tag{7}$$

Note that the commutation rules for the fields (which are simultaneously functions on $\mathcal{F}$ and differential forms on N) are determined by the total degree (the de Rham degree of the differential form plus the ghost number): two fields of odd total degree anti-commute, and commute if at least one field has even total degree.

It remains to compute the BV action, which is straightforward in the AKSZ scheme,

$$\begin{aligned}S_{\text{BV}}^{\text{CS}} &= \int_{T[1]N} \mu \left(\iota_{Q_D} \tilde{\alpha}_{\mathfrak{g}[1]} + \tilde{\Theta}_{\mathfrak{g}[1]} \right) \\ &= \int_{T[1]N} \mu \left((\mathbf{A}, D\mathbf{A}) + \frac{1}{6} (\mathbf{A}, [\mathbf{A}, \mathbf{A}]) \right) \\ &= \int_N \left(\frac{1}{2} (A, dA) + \frac{1}{6} (A, [A, A]) - (A^+, d\gamma + [A, \gamma]) + \left(\gamma^+, \frac{1}{2} [\gamma, \gamma] \right) \right).\end{aligned}\tag{8}$$

In the two terms involving only physical fields, we recognize the classical action of the Chern-Simons theory. The other terms complete the BV action, and by construction, it is clear that it satisfies the classical master equation. Nevertheless, we will show it explicitly, mainly to present an example of calculations in the space of fields, on which we will rely in the rest of this paper.

2.3 Calculations in the BV formalism

First of all, we need to find the BV bracket, simply by inverting the BV structure (7), without forgetting that the product between two differential forms in the space of fields really means the exterior product,

$$\delta \phi^+ \delta \phi = \delta \phi^+ \wedge \delta \phi = \delta \phi^+ \otimes \delta \phi \pm \delta \phi \otimes \delta \phi^+,$$

where the sign depends on the commutation rules between ϕ and ϕ^+ . We find the following expression for the BV bracket of two functionals F_1 and F_2 ,

$$\begin{aligned}\{F_1, F_2\} &= \int_N \left(\left(\frac{F_1 \overleftarrow{\delta}}{\delta \gamma}, \frac{\overrightarrow{\delta} F_2}{\delta \gamma^+} \right) - \left(\frac{F_1 \overleftarrow{\delta}}{\delta \gamma^+}, \frac{\overrightarrow{\delta} F_2}{\delta \gamma} \right) \right. \\ &\quad \left. - \left(\frac{F_1 \overleftarrow{\delta}}{\delta A}, \frac{\overrightarrow{\delta} F_2}{\delta A^+} \right) + \left(\frac{F_1 \overleftarrow{\delta}}{\delta A^+}, \frac{\overrightarrow{\delta} F_2}{\delta A} \right) \right)\end{aligned}\tag{9}$$

where the functional derivatives $\overrightarrow{\frac{\delta}{\delta\phi}}$ and $\overleftarrow{\frac{\delta}{\delta\phi}}$ for $\phi \in \{A, A^+, \gamma, \gamma^+, g^+\}$ are the duals of the differentials $\delta\phi$ in the space of fields (which can be interpreted as variations of fields in the framework of variational calculus). We need to make the difference between right- and left-derivatives due to the commutation rules that depend on ghost numbers and degrees of differential forms on N .

All the fields of the Chern-Simons model are $\mathfrak{g}$ -valued, so taking the functional derivative of a real-valued functional F on $\mathcal{F}$ by one of these fields should produce a $\mathfrak{g}^*$ -valued result, but we can use the non-degenerate scalar product $(\cdot, \cdot)$ to identify $\mathfrak{g}$ with its dual. If F is constructed as an integral, like an action, the left- and right-derivatives by a field $\phi \in \{A, A^+, \gamma, \gamma^+, g^+\}$ can be defined as the components of the exterior derivative with respect to the local frame induced by these coordinate-fields of $\mathcal{F}$,

$$\delta F(\phi_1, \dots, \phi_n) = \int_{\partial N} \sum_{j=1}^n \left(\delta\phi_j, \frac{\overrightarrow{\delta} F}{\delta\phi_j} \right) = \int_{\partial N} \sum_{j=1}^n \left(\frac{F \overleftarrow{\delta}}{\delta\phi_j}, \delta\phi_j \right).$$

If on the other hand F is a local functional, we may still express it as an integral provided we filter its position with a Dirac distribution, a distribution that will stick to the functional derivative.

At a later stage, we will need to consider Lie group-valued fields of the form $g \in \text{Map}(N, G)$. The problem with such a field is that its variation does not take value in $\mathfrak{g}$, but rather in the tangent space at g of the Lie group G . Natural coupling with the other $\mathfrak{g}$ -valued fields via the invariant scalar product involves the right multiplication by g^{-1} to bring it back to the Lie algebra, explicitly $\delta g g^{-1}$. The dual derivative $\frac{\delta}{\delta g}$ assumes its value in $T_{g^{-1}}^* G$, which is isomorphic to $T_{g^{-1}} G$ thanks to the invariant non-degenerate scalar product, and we need to apply this time left-multiplication by g to get back to the Lie algebra. If the functional F depends also on g , we find for the derivative

$$\delta F(\phi_1, \dots, \phi_n, g) = \int_{\partial N} \sum_{j=1}^n \left(\delta\phi_j, \frac{\overrightarrow{\delta} F}{\delta\phi_j} \right) + \left(\delta g g^{-1}, g \frac{\overrightarrow{\delta} F}{\delta g} \right).$$

To compute $\{S_{\text{BV}}^{\text{CS}}, S_{\text{BV}}^{\text{CS}}\}$, we need the derivatives of the Chern-Simons BV action. We find

$$\begin{aligned} \delta S_{\text{BV}}^{\text{CS}} = \int_N & \left(\left(\delta A, dA + \frac{1}{2} [A, A] + [A^+, \gamma] \right) \right. \\ & + (\delta\gamma, -d_A A^+ - [\gamma^+, \gamma]) \\ & \left. + (\delta A^+, -d_A \gamma) + \left(\delta\gamma^+, \frac{1}{2} [\gamma, \gamma] \right) \right), \end{aligned} \quad (10)$$

where we introduced the covariant derivative $d_A = d + [A, \cdot]$. Note that we need to integrate by parts to find the contribution of the exterior derivatives, such as

$$\int_N (A, \delta dA) = \int_N (A, d\delta A) = - \int_{\partial N} (A, \delta A) + \int_N (dA, \delta A),$$

where the boundary term vanishes for a closed manifold N . When we consider source spaces with boundaries, these terms will no longer vanish, and they will contribute to a one-form in the boundary space of fields. For now we can check the classical master equation,

$$\begin{aligned} \frac{1}{2} \{S_{\text{BV}}^{\text{CS}}, S_{\text{BV}}^{\text{CS}}\} &= \int_N \left(\left(\frac{S_{\text{BV}}^{\text{CS}} \overleftarrow{\delta}}{\delta\gamma}, \frac{\overrightarrow{\delta} S_{\text{BV}}^{\text{CS}}}{\delta\gamma^+} \right) - \left(\frac{S_{\text{BV}}^{\text{CS}} \overleftarrow{\delta}}{\delta A}, \frac{\overrightarrow{\delta} S_{\text{BV}}^{\text{CS}}}{\delta A^+} \right) \right) \\ &= \int_N \left(\left(d_A A^+ + [\gamma^+, \gamma], \frac{1}{2} [\gamma, \gamma] \right) \right. \\ &\quad \left. - \left(dA + \frac{1}{2} [A, A] + [A^+, \gamma], -d_A \gamma \right) \right) \\ &= 0. \end{aligned}$$

In the last step we make repeated use of the invariance of the scalar product, the Jacobi identity for $\mathfrak{g}$, and the Stokes theorem.

We are now fully prepared to describe Wilson lines in the BV formalism.

2.4 Wilson Lines

In gauge theories, a degeneracy arises under the local action of a Lie group on the space of fields, the gauge group. In what follows, we will denote by G the gauge group, and $\mathfrak{g}$ its associated Lie algebra. Their so-called gauge field is a connection A in a principal G -bundle over some manifold N . The gauge symmetry is parametrized in the BV (and BRST) formalism by a ghost field $\gamma \in \text{Map}(N, \mathfrak{g}[1])$. The BV variation of these two fields depends only on their behavior under gauge transformations and not on the specific type of the underlying ambient theory. We assume that the dynamics and the gauge structure of this ambient theory is encoded in the BV action S^{amb} and the corresponding BV structure Ω^{amb} (both defined as integrals over N), and of course that S^{amb} solves the ambient classical master equation $\{S^{\text{amb}}, S^{\text{amb}}\}_{\text{amb}} = 0$. The part of this ambient BV bracket involving the gauge connection and the ghost field relevant for our further investigation is defined by the BV variation of these fields, namely

$$\begin{aligned} \{S^{\text{amb}}, A\} &= Q A = d_A \gamma, \\ \{S^{\text{amb}}, \gamma\} &= Q \gamma = \frac{1}{2} [\gamma, \gamma]. \end{aligned} \tag{11}$$

Natural non-local observables to consider in gauge theories are given by Wilson-loops, traces of the holonomy of the connection A along a curve Γ embedded in N in given representations of the Lie algebra,

$$W_{\Gamma, R}[A] = \text{Tr}_R \text{Pexp} \left(\int_{\Gamma} A \right),$$

where P stands for the path-ordering and R labels the representation of $\mathfrak{g}$.

This cumbersome path-ordering can be removed at the price of integrating over all inequivalent gauge transformations along the loop [1],

$$W_{\Gamma,R}[A] = \int \mathcal{D}g \exp \left(\int_{\Gamma} \langle T_0, g^{-1} A g + g^{-1} dg \rangle \right). \quad (12)$$

The dual algebra element $T_0 \in \mathfrak{g}^*$ encodes the representation R , along the lines of the orbit method [11] that links unitary irreducible representations of Lie groups and their coadjoint orbits. This expression for the Wilson loop can be absorbed into an extended action by adding the auxiliary term

$$S_{\text{Wilson}} = \int_{\Gamma} \langle \text{Ad}_g^*(T_0), A + dg g^{-1} \rangle \quad (13)$$

to the ambient action S^{amb} of the model under consideration. In this last step we replaced the adjoint action on the second factor of the product by the coadjoint action on the first factor, to emphasize the role of the coadjoint orbit $\mathcal{O}$ of T_0 .

Now we would like to find a BV formulation of this contribution, so as to obtain a BV action of the full model with Wilson loops. The partition function of such a model with an action extended to take into account a Wilson line as an auxiliary term actually corresponds to the expectation value of this Wilson line in the pure theory,

$$Z_{S^{\text{amb}}+S^{\text{aux}}} = \langle W_{\Gamma,R} \rangle_{S^{\text{amb}}}.$$

We note that the coadjoint orbit $\mathcal{O}$ supports the Kirillov symplectic structure $\omega_{\mathcal{O}}$, of ghost number 0, and that the curve Γ carrying the Wilson line has dimension 1, the first two main ingredients for the AKSZ construction for $n = 1$. It is thus tempting to try to apply the prescription proposed in [12] to construct observables within the AKSZ formalism. Nonetheless, [12] treats exclusively the case of an ambient theory of the AKSZ type, whereas we want to consider gauge theories, with the sole requirement that their space of fields contains a gauge connection and an associated ghost field obeying the relations (11). The obvious solution is to study a gauge theory of the AKSZ type, a condition fulfilled by the Chern-Simons model, the main subject of this paper. The BV formulation of the Wilson line contribution will happen to remain valid for other gauge theories.

So following [12], the auxiliary fields are the maps between the odd tangent bundle of the curve Γ and the coadjoint orbit,

$$\mathcal{F}^{\text{aux}} = \text{Map}(T[1]\Gamma, \mathcal{O}).$$

This auxiliary space of fields needs to be equipped with its own BV structure, Ω^{aux} , that once added to the BV structure Ω^{amb} of the ambient theory will provide the BV structure $\Omega = \Omega^{\text{amb}} + \Omega^{\text{aux}}$ of the full model with space of fields $\mathcal{F} = \mathcal{F}^{\text{amb}} \oplus \mathcal{F}^{\text{aux}}$. Then it will be possible to add to the ambient action S^{amb} an auxiliary term S^{aux} that obeys certain constraints to obtain a solution of the master equation of the full model.

The definition of Ω^{aux} is similar to the one of the AKSZ-BV structure (1), we just need to change the source space and the symplectic structure of the target,

$$\Omega^{\text{aux}} = \int_{T[1]\Gamma} \mu_\Gamma \tilde{\omega}_\mathcal{O}.$$

Here μ_Γ is obviously the canonical measure on $T[1]\Gamma$.

Unfortunately, the Kirillov symplectic form on the coadjoint orbit is in general not exact, so it is in general not possible to find a Liouville one-form, which we would normally use to construct the kinetic term of the auxiliary action. However, in the case of integrable orbits, we may pick a line bundle (the pre-quantum line bundle in the language of geometric quantization) and a connection $\alpha_\mathcal{O}$ thereon with curvature $\omega_\mathcal{O}$,

$$\delta\alpha_\mathcal{O} = \omega_\mathcal{O}$$

(we recall that in the target spaces of AKSZ theories, we denote by δ the exterior derivative), and we can simply use this connection to construct the kinetic term of the auxiliary action.

This formulation is not very practical to carry out calculations. To find expressions easier to deal with, we apply the defining property of the Kirillov symplectic form, that the pullback by the projection map

$$\pi : G \rightarrow \mathcal{O} \simeq G/\text{Stab}(T_0) \quad (14)$$

brings it to an explicit presymplectic form ω_G on G ,

$$\pi^*(\omega_\mathcal{O}) = \omega_G = -\langle \text{Ad}_g^*(T_0), \frac{1}{2} [\delta g g^{-1}, \delta g g^{-1}] \rangle. \quad (15)$$

This two-form is the contraction of T_0 with the exterior derivative of the Maurer-Cartan one-form on G . It thus admits a potential

$$\alpha_G = -\langle \text{Ad}_g^*(T_0), \delta g g^{-1} \rangle. \quad (16)$$

As it happens, the pullback by the projection map π brings the connection $\alpha_\mathcal{O}$ over to the one-form α_G ,

$$\pi^*(\alpha_\mathcal{O}) = \alpha_G,$$

that we will use in the place of the more cumbersome connection to compute certain quantities. Since ω_G is degenerate, it is not possible to construct a Poisson bracket out of it, unless we restrict it to invariant functions on G , such as the ones obtained by pullback of functions on the coadjoint orbit $\mathcal{O}$ by the projection map π .

It remains to define the interaction term. The idea of [12] is to construct a function $\Theta_\mathcal{O}$ on $\mathfrak{g}[1] \times \mathcal{O}$ that will generate together with $\Theta_{\mathfrak{g}[1]}$ a Hamiltonian cohomological vector field on $\mathfrak{g}[1] \times \mathcal{O}$. While $\Theta_{\mathfrak{g}[1]}$ already satisfies an integrability condition on its own and generates a cohomological vector field $\mathcal{Q}_{\mathfrak{g}[1]}$, the

integrability condition for $\Theta_{\mathcal{O}}$ needs to be slightly adapted to account for the mixed term, namely

$$\mathcal{Q}_{\mathfrak{g}[1]}\Theta_{\mathcal{O}} + \frac{1}{2}\{\Theta_{\mathcal{O}}, \Theta_{\mathcal{O}}\}_{\mathcal{O}} = 0.$$

As it happens, the function

$$\Theta_{\mathcal{O}} = \langle \text{Ad}_g^*(T_0), \psi \rangle \quad (17)$$

satisfies this requirement and naturally extends the term $\langle \text{Ad}_g^*(T_0), A \rangle$ that already appeared in the classical part (13). This integrability condition is most easily checked by pulling it back by π to a function on $\mathfrak{g}[1] \times G$, where the Poisson bracket $\{\cdot, \cdot\}_{\mathcal{O}}$ becomes $\{\cdot, \cdot\}_G$, which can be explicitly determined by inverting ω_G .

We now have all the ingredients to construct the auxiliary BV action, that we just need to combine into a formula similar as the usual AKSZ action (2),

$$S^{\text{aux}} = \int_{T[1]\Gamma} \mu_{\Gamma} \left(\iota_{Q_D} \tilde{\alpha}_{\mathcal{O}} + \tilde{\Theta}_{\mathcal{O}} \right).$$

By construction, if S^{amb} is the AKSZ action of the Chern-Simons model, the total action

$$S = S^{\text{amb}} + S^{\text{aux}}$$

automatically satisfies the classical master equation generated by the total BV structure

$$\Omega = \Omega^{\text{amb}} + \Omega^{\text{aux}}.$$

We claimed that it remains true when S^{amb} is the BV action of a generic gauge theory with gauge group G . To verify this assertion, we need to compute

$$\frac{1}{2}\{S, S\} = \frac{1}{2}\{S^{\text{amb}}, S^{\text{amb}}\} + \{S^{\text{amb}}, S^{\text{aux}}\} + \frac{1}{2}\{S^{\text{aux}}, S^{\text{aux}}\}. \quad (18)$$

The first two terms involve only the ambient BV structure, since S^{amb} does not depend on the auxiliary fields. The first one vanishes due to the master equation of the BV ambient model. To compute the second term, we should know the exact dependence of the auxiliary term S^{aux} on the ambient fields A and γ , and to compute the last one, we need an expression of the auxiliary BV structure Ω^{aux} that we know how to invert.

These two issues can be addressed by using the projection map (14) to define an extended space of fields,

$$\hat{\mathcal{F}}_G^{\text{aux}} = \pi^*(\mathcal{F}^{\text{aux}}) = \{(g, g^+) | g \in \text{Map}(\Gamma, G), g^+ \in \Omega^1(\Gamma) \otimes g^*(T\mathcal{O})[-1]\}.$$

The subscript G emphasizes the fact that the coadjoint orbit is replaced by the whole group.

This projection map, now seen as a map between spaces of fields,

$$\pi : \hat{\mathcal{F}}_G^{\text{aux}} \rightarrow \mathcal{F}^{\text{aux}},$$

acts on the group-valued component g by sending it to its image $\text{Ad}_g^*(T_0)$ in the coadjoint orbit of T_0 . It can be used to pull back differential forms on the auxiliary space of fields (such as the auxiliary BV structure, a two-form, or the auxiliary BV action, a zero-form) to this extended space of fields, where it is easier to compute BV brackets of G -invariant functionals given the explicit formulas for the pullbacks of the auxiliary BV structure and of the auxiliary action.

In $\hat{\mathcal{F}}_G^{\text{aux}}$, both fields g and g^+ can be combined into a superfield of total degree 0 that we can use to express the pullback of the auxiliary BV structure and action,

$$\mathbf{H}(x, \theta) = \text{Ad}_{g(x)}^*(T_0) - \theta g^+(x).$$

Here x is a coordinate of Γ and θ a Grassmanian coordinate on the odd fibers of $T[1]\Gamma$, and $g^+(x)$ is the component of the one-form g^+ expressed in this coordinate system, $g^+ = g^+(x)dx$.

We now have all the tools to compute the pullback of the auxiliary BV structure,

$$\begin{aligned} \hat{\Omega}_G^{\text{aux}} &= \pi^*(\Omega^{\text{aux}}) = \int_{T[1]\Gamma} \mu_\Gamma \pi^*(\tilde{\omega}_\mathcal{O}) = \int_{T[1]\Gamma} \mu_\Gamma \tilde{\omega}_G \\ &= - \int_{T[1]\Gamma} \mu \delta \langle \mathbf{H}, \delta g g^{-1} \rangle = \int_\Gamma \delta \langle g^+, \delta g g^{-1} \rangle \\ &= \int_\Gamma \langle \delta g^+, \delta g g^{-1} \rangle + \langle g^+, \frac{1}{2} [\delta g g^{-1}, \delta g g^{-1}] \rangle, \end{aligned} \quad (19)$$

and of the auxiliary BV action,

$$\begin{aligned} \hat{S}_G^{\text{aux}} &= \pi^*(S^{\text{aux}}) = \int_{T[1]\Gamma} \mu_\Gamma \left(\iota_{Q_D} \tilde{\alpha}_G + \pi^*(\tilde{\Theta}_\mathcal{O}) \right) \\ &= \int_{T[1]\Gamma} \mu_\Gamma \left(\langle \mathbf{H}, Dg g^{-1} \rangle + \langle \mathbf{H}, \mathbf{A} \rangle \right) \\ &= \int_\Gamma \left(\langle \text{Ad}_g^*(T_0), A + dg g^{-1} \rangle - \langle g^+, \gamma \rangle \right). \end{aligned} \quad (20)$$

The additional term $\langle g^+, \gamma \rangle$ encodes the action of gauge transformations of the ambient model on the auxiliary classical field $\text{Ad}_g^*(T_0)$.

We insist on the fact that $\hat{\Omega}_G^{\text{aux}}$ is only a pre-BV structure in $\hat{\mathcal{F}}_G^{\text{aux}}$, it is closed but degenerate. Nevertheless, like its finite dimensional counterpart ω_G , it can be used to define a BV bracket on invariant functionals, such as the ones obtained by pullback from $\mathcal{F}^{\text{aux}}$, for instance $\hat{S}_G^{\text{aux}}$.

We can turn our attention to the pullback of the last two terms in (18). To compute the second one, we notice that $\{S^{\text{amb}}, \cdot\}$ acts as a differential (namely Q) on the fields of the ambient model, of which only A and γ appear in the

auxiliary action (20), so that we may simply use the relations (11) to find

$$\begin{aligned}\{S^{\text{amb}}, \hat{S}_G^{\text{aux}}\} &= \int_{\Gamma} (\langle \text{Ad}_g^*(T_0), Q(A) \rangle - \langle g^+, Q(\gamma) \rangle) \\ &= \int_{\Gamma} \left(\langle \text{Ad}_g^*(T_0), d_A \gamma \rangle - \langle g^+, \frac{1}{2} [\gamma, \gamma] \rangle \right).\end{aligned}\quad (21)$$

Next, the bracket of the third term contains only contributions from the auxiliary structure, and we may compute

$$\begin{aligned}\frac{1}{2} \{ \hat{S}_G^{\text{aux}}, \hat{S}_G^{\text{aux}} \} &= \int_{\Gamma} \left(\langle g \frac{\hat{S}_G^{\text{aux}} \overleftarrow{\delta}}{\delta g}, \frac{\overrightarrow{\delta} \hat{S}_G^{\text{aux}}}{\delta g^+} \rangle + \langle g^+, \frac{1}{2} \left[\frac{\hat{S}_G^{\text{aux}} \overleftarrow{\delta}}{\delta g^+}, \frac{\overrightarrow{\delta} \hat{S}_G^{\text{aux}}}{\delta g^+} \right] \rangle \right) \\ &= \int_{\Gamma} \left(\langle (d + \text{ad}_A^*) \text{Ad}_{g^{-1}}^* T_0, \gamma \rangle + \langle g^+, \frac{1}{2} [\gamma, \gamma] \rangle \right).\end{aligned}\quad (22)$$

The first line displays the BV bracket of invariant functionals on $\hat{\mathcal{F}}_G^{\text{aux}}$ constructed out of the pre-BV structure $\hat{\Omega}_G^{\text{aux}}$.

The sum of these two terms yields the integral of an exact term that vanishes since Γ is closed (for now), and the pullback of the classical master equation is satisfied,

$$\pi^* \left(\frac{1}{2} \{ S^{\text{amb}} + S^{\text{aux}}, S^{\text{amb}} + S^{\text{aux}} \} \right) = 0.$$

Furthermore, since the left-hand side of this last equality is G -invariant, it behaves nicely enough under the projection π so that $S^{\text{amb}} + S^{\text{aux}}$ still solves the classical master equation at the level of $\mathcal{F} = \mathcal{F}^{\text{amb}} \oplus \text{Map}(T[1]\Gamma, \mathcal{O})$, also for generic gauge theories.

2.5 Quadratic Lie algebras

In many cases of interest, the Lie algebra $\mathfrak{g}$ is equipped with a non-degenerate scalar product $(\cdot, \cdot)$, which we can use to define an isomorphism $\beta : \mathfrak{g}^* \rightarrow \mathfrak{g}$, that we can apply to T_0 , H and g^+ . The relation $\beta(\text{Ad}_g^*(T_0)) = \text{Ad}_g(\beta(T_0))$ will be very useful, in particular we can replace the canonical pairing between $\mathfrak{g}$ and $\mathfrak{g}^*$ with the scalar product,

$$\langle \text{Ad}_g^*(T_0), \cdot \rangle = (\text{Ad}_g(\beta(T_0)), \cdot),$$

and thus identify coadjoint orbits with adjoint orbits.

In the rest of this article, we will assume that $\mathfrak{g}$ admits such a non-degenerate scalar product. We will make use of it to write down all actions and BV structures, and we will simply consider T_0 , H and g^+ to be elements of $\mathfrak{g}$ instead of its dual, $\mathfrak{g}^*$ (we drop the β for simplicity). Coadjoint orbits will therefore be identified with adjoint orbits. To summarize the main results of this section with this new convention, we can re-write the auxiliary BV structure (19) as

$$\hat{\Omega}_G^{\text{aux}} = \int_{\Gamma} (\delta g^+, \delta g g^{-1}) + (g^+, \frac{1}{2} [\delta g g^{-1}, \delta g g^{-1}]) \quad (23)$$

and the auxiliary BV action (20) as

$$\hat{S}_G^{\text{aux}} = \int_{\Gamma} ((\text{Ad}_g(T_0), A + dg g^{-1}) - (g^+, \gamma)). \quad (24)$$

3 3D Chern-Simons Theory with a Wilson line

Our main goal in this paper is to study the behaviour of Chern-Simons models on a manifold N with boundaries and Wilson lines ending on these boundaries, both in three and one dimension, in the BV formalism. The presence of a boundary requires either a careful choice in the boundary conditions for the fields, so as to keep the classical master equation under control, or the application of the recently developed BV-BFV formalism for gauge theories with boundaries [6], which presents the big advantage that it allows to glue pieces together along their boundary.

In order to obtain the BV-BFV formulation of the three-dimensional Chern-Simons model with a boundary supporting some Wilson lines, we first need to determine the BV theory of the bulk. Actually we already know all its ingredients. Obviously, we will treat the Wilson lines as an auxiliary part of the action, as described in section 2, added to the ambient Chern-Simons action (8),

$$S^{\text{amb}} = S_{BV}^{CS}.$$

To include Wilson lines to our model, say n of them, we need to extend the ambient space of fields

$$\mathcal{F}^{\text{amb}} = \text{Map}(T[1]N, \mathfrak{g}[1])$$

carrying the ambient BV structure $\Omega^{\text{amb}} = \Omega_{BV}^{CS}$ with an auxiliary part

$$\mathcal{F}^{\text{aux}} = \bigoplus_{k=1}^n \text{Map}(T[1]\Gamma_k, \mathcal{O}_k)$$

made of n components, one for each Wilson line labeled by k . We recall that we denote by $\mathcal{O}_k$ the (co)adjoint orbit of a Lie algebra element $T_{0,k}$ encoding the representation in which the k -th Wilson line is computed and by Γ_k the curve embedded in N supporting this Wilson line. The BV structure of this auxiliary space of fields is the sum of N copies of the auxiliary BV structure of a single Wilson line,

$$\Omega^{\text{aux}} = \sum_{k=1}^n \int_{T[1]\Gamma_k} \mu_{\Gamma_k} \tilde{\omega}_{\mathcal{O}_k}.$$

The auxiliary BV action is similarly constructed as a sum,

$$S^{\text{aux}} = \sum_{k=1}^n \int_{T[1]\Gamma_k} \mu_{\Gamma_k} \left(\iota_{Q_D} \tilde{\alpha}_{\mathcal{O}_k} + \tilde{\Theta}_{\mathcal{O}_k} \right).$$

From now on, unless specified otherwise, a superscript “amb” will always describe a quantity associated to the BV formulation of the bare Chern-Simons model, be it a BV structure, a BV action or a BV space of fields, “aux” will always describe a quantity associated to the BV formulation of the auxiliary contribution of n Wilson lines, and no superscript will mean a BV quantity of the full model, namely $\mathcal{F} = \mathcal{F}^{\text{amb}} \oplus \mathcal{F}^{\text{aux}}$, $\Omega = \Omega^{\text{amb}} + \Omega^{\text{aux}}$ and $S = S^{\text{amb}} + S^{\text{aux}}$.

Before we consider the case of a source manifold with boundary, we need to compute the Hamiltonian vector field Q generated by S , i.e. satisfying

$$\iota_Q \Omega = \delta S. \quad (25)$$

Once the BV structure Ω is inverted to form the BV bracket $\{\cdot, \cdot\}$, this is equivalent to

$$Q = \{S, \cdot\}.$$

This relation is linear in Q and S , namely $\iota_{Q^{\text{amb}}} \Omega = \delta S^{\text{amb}}$ and $\iota_{Q^{\text{aux}}} \Omega = \delta S^{\text{aux}}$. We will again use the projection maps $\pi_k : G \rightarrow \mathcal{O}_k$ to pull back the auxiliary action and the auxiliary BV structure to the space of fields $\bigoplus_{k=1}^n \hat{\mathcal{F}}_G^{\text{aux}}$ where the calculations are easier. Note that we need one copy of $\hat{\mathcal{F}}_G^{\text{aux}}$ for each Wilson line. The results can then be easily brought over to the actual auxiliary space of fields by the n projections π_k .

For the ambient Hamiltonian vector field we obtain

$$\begin{aligned} Q^{\text{amb}} = \{S^{\text{amb}}, \cdot\} = & \left((d_A A^+ + [\gamma^+, \gamma]), \frac{\vec{\delta}}{\delta \gamma^+} \right) - \frac{1}{2} \left([\gamma, \gamma], \frac{\vec{\delta}}{\delta \gamma} \right) \\ & - \left(\left(dA + \frac{1}{2} [A, A] + [A^+, \gamma] \right), \frac{\vec{\delta}}{\delta A^+} \right) + \left(d_A \gamma, \frac{\vec{\delta}}{\delta A} \right), \end{aligned} \quad (26)$$

and for its auxiliary counterpart

$$\begin{aligned} \hat{Q}_G^{\text{aux}} = \{\hat{S}_G^{\text{aux}}, \cdot\} = & - \sum_k \left(\text{Ad}_g(T_{0,k}) \delta(\Gamma_k), \frac{\vec{\delta}}{\delta A^+} \right) - \left(d_A(\text{Ad}_g(T_{0,k})), \frac{\vec{\delta}}{\delta g^+} \right) \\ & - \left(\gamma, g \frac{\vec{\delta}}{\delta g} \right) - \sum_k \left(g^+ \delta(\Gamma_k), \frac{\vec{\delta}}{\delta \gamma^+} \right). \end{aligned} \quad (27)$$

Here $\delta(\Gamma_k)$ denotes a Dirac distribution two-form centred on Γ_k to filter the curve out of the whole manifold N . The fields g and g^+ that appear in front of these Dirac two-forms are defined only on the Wilson lines. The functional derivatives appearing right after them act on functionals in the bulk, but their results are zero- and one-forms that make sense on the curves Γ_k . Since the ambient and the total actions solve classical master equations, we know that Q^{amb} and $Q = Q^{\text{amb}} + Q^{\text{aux}}$ are cohomological, but not Q^{aux} .

If the source space has a boundary, on which might end an open Wilson line along one or more curves Γ_k (for simplicity, we will assume all of them), these

cohomological vector fields cease to be Hamiltonian due to boundary effects affecting the variation of the differentiated terms in the action. The integration by part required to compute the contribution of these terms to the variation of the action now contains a surface integral. The BV-BFV formalism is based on the observation that this correction can be seen as a one-form in the boundary space of fields $\mathcal{F}_\partial$. This boundary space of fields contains the restriction of the fields of the bulk BV theory to their value on the boundary of the source manifold, ∂N in the case of our ambient theory, $\bigcup_{k=1}^n \partial\Gamma_k$ for the auxiliary model describing the Wilson lines. We denote by

$$\pi_\partial : \mathcal{F} \rightarrow \mathcal{F}_\partial$$

the projection corresponding to this restriction. The correction to the Hamiltonian condition (25) can be expressed as

$$\delta S = \iota_Q \Omega + \pi_\partial^*(\alpha_\partial). \quad (28)$$

The exterior derivative of this one-form,

$$\Omega_\partial = \delta \alpha_\partial,$$

happens to be symplectic and is called the BFV structure. It is a two-form of ghost number 0 in the boundary space of fields. The corrected Hamiltonian condition (28) is linear in α_∂ , too, so we may decompose the boundary BFV structure

$$\Omega_\partial = \Omega_\partial^{\text{amb}} + \Omega_\partial^{\text{aux}}$$

and compute it in two parts. The ambient one corresponds to the Chern-Simons model,

$$\Omega_\partial^{\text{amb}} = \int_{\partial N} \left(\frac{1}{2} (\delta A, \delta A) + (\delta \gamma, \delta A^+) \right), \quad (29)$$

and could actually be derived in a two-dimensional adaptation of the AKSZ construction. To compute the auxiliary part, we make use of the usual trick to do calculations in the augmented space of fields. The result

$$\hat{\Omega}_{\partial, G}^{\text{aux}} = \sum_k \int_{\partial\Gamma_k} \left(\text{Ad}_g(T_{0,k}), \frac{1}{2} [\delta g g^{-1}, \delta g g^{-1}] \right) \quad (30)$$

is the sum of $2k$ copies of the symplectic form ω_G of the target space of the augmented space of fields $\hat{\mathcal{F}}_{\partial, G}^{\text{aux}}$, one carried by each extremity of every Wilson line. Using the relation (15), we immediately find the BFV structure on the actual auxiliary boundary space of fields $\mathcal{F}_\partial^{\text{aux}} = \bigoplus_{k=1}^n \text{Map}(\partial\Gamma_k, \mathcal{O}_K)$,

$$\Omega_\partial^{\text{aux}} = \sum_k \int_{\partial\Gamma_k} \tilde{\omega}_{\mathcal{O}_k}, \quad (31)$$

where $\tilde{\omega}_{\mathcal{O}_k}$ is evidently the Kirillov symplectic form on the k -th (co)adjoint orbit $\mathcal{O}_k$ lifted to the space of fields.

The curve Γ_k being one dimensional, we have $\partial\Gamma_k = \{z_k, z'_k\} \subset \partial N$, and what the last integral really means is $\int_{\partial\Gamma_k} \tilde{\omega}_{\mathcal{O}_k} = \tilde{\omega}_{\mathcal{O}_k}(z_k) - \tilde{\omega}_{\mathcal{O}_k}(z'_k)$.

In the last step of the construction of the boundary BFV model, we know that the restriction of Q to the boundary surface ∂N is Hamiltonian with respect to the BFV structure, and the boundary BFV action is defined as its generating functional,

$$\iota_{Q_\partial} \Omega_\partial = \delta S_\partial.$$

Ghost number counting shows that the BFV action has ghost number 1.

In the case of the Chern-Simons model with Wilson lines, we calculate the restriction of $\hat{Q}_G = Q^{\text{amb}} + \hat{Q}_G^{\text{aux}}$ in the extended space of fields,

$$\begin{aligned} \hat{Q}_{\partial, G} = & - \left(\frac{1}{2} [\gamma, \gamma], \frac{\vec{\delta}}{\delta\gamma} \right) - \left(\left(dA + \frac{1}{2} [A, A] + [A^+, \gamma] \right), \frac{\vec{\delta}}{\delta A^+} \right) \\ & + \left(d_A \gamma, \frac{\vec{\delta}}{\delta A} \right) - \sum_k \left(\text{Ad}_g(T_{0,k}) \delta(\Gamma_k), \frac{\vec{\delta}}{\delta A^+} \right) - \left(\gamma, g \frac{\vec{\delta}}{\delta g} \right), \end{aligned} \quad (32)$$

which leads to the two contributions

$$S_\partial^{\text{amb}} = - \int_{\partial N} \left(\left(dA + \frac{1}{2} [A, A], \gamma \right) + \left(A^+, \frac{1}{2} [\gamma, \gamma] \right) \right) \quad (33)$$

and

$$\hat{S}_{\partial, G}^{\text{aux}} = - \sum_k \int_{\partial\Gamma_k} (\text{Ad}_g(T_{0,k}), \gamma) \quad (34)$$

to the boundary BFV action. As expected, the G -valued field g appears only in a (co)adjoint action, so the projection to the auxiliary space of fields is straightforward, and we obtain the BFV action

$$\begin{aligned} S_\partial = & - \int_{\partial N} \left(\left(dA + \frac{1}{2} [A, A], \gamma \right) + \left(A^+, \frac{1}{2} [\gamma, \gamma] \right) \right. \\ & \left. + \sum_k (\text{Ad}_g(T_{0,k}), \gamma) (\delta^{(2)}(z_k) - \delta^{(2)}(z'_k)) \right). \end{aligned} \quad (35)$$

In the last line, we have cast everything into the integral over ∂N by making use of Dirac distributions centered on the extremities of the Wilson lines.

We recognize in the boundary BFV action of the Chern-Simons model with Wilson lines an odd version of the two-dimensional BF model with sources, where the role of the B field is taken over by the restriction to the boundary of the ghost field γ of the bulk theory.

We conclude this section with a short remark regarding the insertions (labeled by z_k and z'_k) of the boundary model. In our setting, with Wilson lines ending on the boundary, these insertions always come in pairs of points carrying the same representation, one insertion at each end of a Wilson line. If we consider Wilson graphs in the bulk model, which are a natural generalizations of

the Wilson lines, we can obtain any configuration of points and representations as insertions. Wilson graphs are observables modeled after Wilson lines, but based on oriented graphs instead of curves. Each edge carries a representation of $\mathfrak{g}$ and contributes with a similar term as a Wilson line to the total action, while each vertex carries an intertwining operator between the representations of the attached edges. If the formulation of these intertwining operators is straightforward in the operator formalism, their description is more involved in the path-integral formalism and goes beyond the scope of this paper, where we will for simplicity consider only Wilson loops and open Wilson lines.

4 1D Chern-Simons Theory with a Wilson line

The AKSZ construction for the Chern-Simons model can also be carried out in one dimension [3]. In this section, we will see how to add a Wilson line to this model, by following the same procedure as in the previous section. The main difference comes from the fact that the Wilson line is now a space-filling observable, and that the BV bracket of the auxiliary term with itself will pick up terms from the ambient part of the BV structure. Moreover, as stated before, we will now use a $\mathbb{Z}_2$ -grading, since a $\mathbb{Z}$ -grading is not possible in one dimension, so instead of denoting the ghost number in square brackets, we will use the parity-reversing operator Π .

Given a one-dimensional manifold Γ (in general a disjoint union of circles and open segments), the space of fields is

$$\mathcal{F}^{\text{amb}} = \text{Map}(\Pi T\Gamma, \Pi \mathfrak{g}), \quad (36)$$

where $\mathfrak{g}$ is again assumed to be equipped with an invariant scalar product $(\cdot, \cdot)$.

The target space $\Pi \mathfrak{g}$ supports the same geometric structures as before, that we may again transpose to the space of fields.

If x is a coordinate on Γ and θ a Grassmannian coordinate on the odd fibers of $\Pi T\Gamma$, we can decompose the fields $\Psi \in \text{Map}(\Pi T\Gamma, \Pi \mathfrak{g})$ into a $\mathfrak{g}$ -valued fermion ψ and a $\mathfrak{g}$ -valued one-form $A = A(x)dx$,

$$\Psi = \psi + \theta A(x). \quad (37)$$

We repeat the same procedure to find the BV structure

$$\Omega^{\text{amb}} = - \int_{\Pi \Gamma} \mu(\delta \Psi, \delta \Psi) = \int_{\Gamma} (\delta \psi, \delta A) \quad (38)$$

and the BV action

$$S^{\text{amb}} = \int_{\Pi \Gamma} \mu \left(\frac{1}{2} (\Psi, D\Psi) + \frac{1}{6} (\Psi, [\Psi, \Psi]) \right) = \int_{\Gamma} \frac{1}{2} (\psi, d_A \psi). \quad (39)$$

The $\mathfrak{g}$ -valued one-form A can be interpreted as a connection for some principal G -bundle over Γ , where G is a Lie group integrating $\mathfrak{g}$. The odd $\mathfrak{g}$ -valued scalar

ψ serves simultaneously as a ghost for the gauge symmetry and an antifield for A .

In order to add Wilson lines to this model, we need to extend the space of fields, the BV structure and the action with precisely the same auxiliary structure Ω^{aux} and action S^{aux} as in the three-dimensional case, except that they are now supported directly by the base manifold of the ambient source space Γ . We consider a single Wilson line for simplicity, it is easy to add similar terms for additional lines. Furthermore we assume it covers the whole source space Γ . Actually it could involve only some of the connected components of Γ , and the other ones would support a bare (in the sense that there are no Wilson lines) one-dimensional Chern-Simons model. Notice also that instead of γ we write ψ to emphasize the fact that it plays simultaneously the role of γ and A^+ of the previous model.

We insist once more that since the Wilson line is a space-filling observable, we need to check that the classical master equation is solved, a result which is not guaranteed by the AKSZ construction due to the term $\{S^{\text{aux}}, S^{\text{aux}}\}_{\text{ambient}}$ coming from the auxiliary part of the action and the ambient part of the BV structure. If we use again the projection map $\pi : G \rightarrow \mathcal{O}$ to pull differential forms from the auxiliary space of fields $\mathcal{F}^{\text{aux}}$ to the extended one $\hat{\mathcal{F}}_G^{\text{aux}}$, we can calculate

$$\begin{aligned}
\frac{1}{2} \{ \hat{S}_G, \hat{S}_G \} &= \int_{\Gamma} \left(\left(\frac{\hat{S}_G \overleftarrow{\delta}}{\delta A}, \frac{\overrightarrow{\delta} \hat{S}_G}{\delta \psi} \right) + \left(g \frac{\hat{S}_G \overleftarrow{\delta}}{\delta g}, \frac{\overrightarrow{\delta} \hat{S}_G}{\delta g^+} \right) \right. \\
&\quad \left. - \frac{1}{2} \left(\frac{\hat{S}_G \overleftarrow{\delta}}{\delta g^+}, \left[g^+, \frac{\overrightarrow{\delta} \hat{S}_G}{\delta g^+} \right] \right) \right) \\
&= \int_{\Gamma} \left(-\frac{1}{2} [\psi, \psi] + \text{Ad}_g T_0, d_A \psi + g^+ \right) + (-d_A(\text{Ad}_g T_0), -\psi) \\
&\quad - \frac{1}{2} (\psi, [g^+, -\psi]) \\
&= \int_{\Gamma} (\text{Ad}_g T_0, g^+) \\
&= -\frac{1}{2} \int_{\Pi T \Gamma} \mu(\mathbf{H}, \mathbf{H}) \\
&= -\frac{1}{2} \int_{\Pi T \Gamma} \mu \left(\frac{S^{\text{aux}} \overleftarrow{\delta}}{\delta \Psi}, \frac{\overrightarrow{\delta} S^{\text{aux}}}{\delta \Psi} \right),
\end{aligned} \tag{40}$$

and the last line shows it explicitly. Nevertheless, this term vanishes, since g^+ takes value in the tangent space $T_H \mathcal{O}$ at $H = \text{Ad}_g T_0$ to the adjoint orbit, which is easily seen to be orthogonal to H with respect to the invariant scalar product on $\mathfrak{g}$.

Again, before we turn to the case of a source space with a boundary, we need to compute the Hamiltonian cohomological vector field Q generated by S ,

or more accurately its counterpart in the extended space of fields, namely

$$\begin{aligned}\hat{Q}_G = & \left((d_A \psi + g^+), \frac{\delta}{\delta A} \right) + \left(\left(-\frac{1}{2} [\psi, \psi] + \text{Ad}_g(T_0) \right), \frac{\delta}{\delta \psi} \right) \\ & - \left(\psi, g \frac{\delta}{\delta g} \right) - \left(([\psi, g^+] + d_A(\text{Ad}_g(T_0))), \frac{\delta}{\delta g^+} \right).\end{aligned}\quad (41)$$

If the source space has a boundary, in other words if some of its components are segments, we can repeat the procedure to construct the BFV boundary model. We first calculate the image of the symplectic potential of the boundary BFV structure in the augmented space of fields from the variation of the BV action,

$$\hat{\alpha}_{\partial, G} = \int_{\partial\Gamma} \left(\frac{1}{2} (\psi, \delta\psi) + (T_0, g^{-1} \delta g) \right), \quad (42)$$

and the corresponding pre-BFV structure,

$$\hat{\Omega}_{\partial, G} = \int_{\partial\Gamma} \left(\frac{1}{2} (\delta\psi, \delta\psi) - \frac{1}{2} (\text{Ad}_g(T_0), [\delta g g^{-1}, \delta g g^{-1}]) \right). \quad (43)$$

We see that the second term is connected to the pullback by the projection map $\pi : G \rightarrow \mathcal{O}$ of the Kirillov-Kostant-Souriau symplectic structure, and we obtain as a BFV structure in the proper boundary space of fields

$$\Omega_{\partial} = \int_{\partial\Gamma} \left(\frac{1}{2} (\delta\psi, \delta\psi) + \tilde{\omega}_{\mathcal{O}} \right). \quad (44)$$

In all these expressions, the integral over $\partial\Gamma$ is nothing but a sum over the boundary points, with each term carrying a sign given by the orientation of its segment.

The restriction of the cohomological vector field $\hat{Q}_G$ to the boundary,

$$\hat{Q}_{\partial, G} = \left(\left(-\frac{1}{2} [\psi, \psi] + \text{Ad}_g(T_0) \right), \frac{\delta}{\delta \psi} \right) - \left(\psi, g \frac{\delta}{\delta g} \right), \quad (45)$$

is Hamiltonian with respect to the BFV structure, and it is generated by the BFV action of the boundary model,

$$S_{\partial} = \int_{\partial\Gamma} \left(-\frac{1}{6} (\psi, [\psi, \psi]) + (\text{Ad}_g(T_0), \psi) \right). \quad (46)$$

Finally we show that S_{∂} solves the master equation of the BFV model,

$$\{S_{\partial}, S_{\partial}\} = Q_{\partial} S_{\partial} = \int_{\partial\Gamma} (T_0, T_0) = 0. \quad (47)$$

As stated before, the last integral is really a sum over the boundary elements of $\partial\Gamma$ with a sign assigned to their orientation, and since they come in pairs, at each end of every segment, the overall sum vanishes.

5 Boundary Quantum States

Upon quantization, the partition function and correlators of a field theory defined on a manifold N without boundary are complex numbers. In the presence of a boundary, one should rather expect quantum states, elements of a Hilbert space associated to each component of the boundary ∂N , according to the Atiyah-Segal picture of quantum field theory. The disjoint union of boundary components corresponds to the tensor product of the associated Hilbert spaces. Then gluing together a pair of components of ∂N corresponds to taking the scalar product of the two corresponding factors of the tensor product.

For instance, if $N = [0, 1]$ is an interval, the partition function of the BV-BFV model should take value in some Hilbert space of the form $\mathcal{H} \otimes \mathcal{H}$, with one factor for each component of the boundary $\partial N = \{0, 1\}$, and upon gluing the two ends, contracting this tensor product using the scalar product on $\mathcal{H}$ should yield the BV partition function of the same model constructed on the circle S^1 .

If the bulk theory is studied in the BV formalism, the boundary information is encoded in the associated BFV model, at least at the classical level, as we saw in the particular cases of the Chern-Simons theory in one and three dimensions, possibly with Wilson lines.

To pass to the quantum level, we first observe that a BFV boundary model can be canonically quantized. The BFV structure, a symplectic structure of ghost number 0 in the space of fields, is used to define the (anti)commutation rules for the quantized fields. These act on the Hilbert space $\mathcal{H}_\partial^{\text{BFV}}$ associated to the boundary where the partition function of the bulk takes value. This Hilbert space inherits a grading from the ghost number of the classical fields. In this picture, the BFV action S_∂ , which was the generator of the cohomological vector field on the boundary Q_∂ , can be quantized by replacing the classical fields with their quantized counterparts, and we obtain the quantized BFV charge $\hat{S}_\partial$. Its action on the boundary space of states squares to zero and it roughly encodes the gauge transformations. At the classical level, a physical observable is a functional annihilated by the cohomological vector field Q generated by the BV action in the bulk and the BFV action S_∂ on the boundary, and two observables are gauge equivalent if they differ by a Q -exact term. At the quantum level, the role of Q is taken over by the BFV charge $\hat{S}_\partial$: a gauge invariant boundary state should be annihilated by the BFV charge, and two states are gauge-equivalent if they differ by a BFV-exact term. Moreover, we require physical states to depend only on physical quantum fields, and not on the ghosts or the antifields. In other words, the space of boundary quantum states should correspond to the BFV-cohomology at ghost number zero $H_{\hat{S}_\partial}^0(\mathcal{H}_\partial^{\text{BFV}})$.

The relation with the quantized bulk theory is that the partition function (and all the other correlators) should obviously be gauge invariant and therefore belong to this cohomology $H_{\hat{S}_\partial}^0(\mathcal{H}_\partial^{\text{BFV}})$. Its determination thus becomes a subject of interest.

We will start with the one-dimensional Chern-Simons theory, a simpler model where all calculations can be done until the end, before we study the

more interesting three-dimensional model.

5.1 1D Chern-Simons Theory

The zero-dimensional boundary model of the one-dimensional Chern-Simons theory contains $\mathfrak{g}$ -valued fermions and bosonic fields $H = \text{Ad}_g(T_0)$ which take value in the (co)adjoint orbit $\mathcal{O}$. Once quantized, the fermions form a Clifford algebra $Cl(\mathfrak{g})$. If $(t^a)_{a=1}^{\dim \mathfrak{g}}$ is an orthonormal basis of $\mathfrak{g}$ with structure constants f_{abc} , we obtain the anticommutation rules

$$[\hat{\psi}_a, \hat{\psi}_b] = \hbar \delta_{ab} \quad (48)$$

for the quantized fermions.

For the bosonic content of the model, the Kirillov symplectic form on the (co)adjoint orbits is the inverse of the restriction from $\mathfrak{g}^* \simeq \mathfrak{g}$ to $\mathcal{O}$ of the Kirillov-Kostant-Souriau Poisson structure, so that the commutator of two $\mathcal{O}$ -valued quantized fields is simply given by their Lie bracket. If we use the basis (t_a) of the Lie algebra to write

$$H = \text{Ad}_g(T_0) = X_a t_a,$$

we can express the commutation rules with the structure constants of the Lie algebra,

$$[\hat{X}_a, \hat{X}_b] = \hbar f_{abc} \hat{X}_c. \quad (49)$$

The corresponding sector of the algebra of quantum operators is a representation of the enveloping algebra $\mathcal{U}(\mathfrak{g})$ of the Lie algebra $\mathfrak{g}$, namely $\rho_R(\mathcal{U}(\mathfrak{g})) \subset \text{End}(V_R)$. This representation is simply the representation R in which we computed the Wilson loops in the previous sections.

We can use these operators $\hat{\psi}$ and $\hat{X}$ to construct the expectation value of the Wilson line $\langle W_{\Gamma, R} \rangle$ in the operator formalism, such as in [3], where the partition function for the one-dimensional Chern-Simons model is derived in both the path-integral and the operator formalism. If the curve Γ is open, this expectation value maps the space of fields to the boundary space of quantum states which is the cohomology at level 0 of the quantum BFV charge,

$$\langle W_{I, R} \rangle \in H_{\hat{S}_\partial}^0(\mathcal{H}_\partial).$$

We need to find this cohomology.

The BFV charge

$$\hat{S}_\partial = \int_{\partial \Gamma} \hat{X}_a \hat{\psi}_a - \frac{1}{6} f_{abc} \hat{\psi}_a \hat{\psi}_b \hat{\psi}_c \quad (50)$$

carries one copy of the cubic Dirac operator [2]

$$\mathfrak{D} = \hat{X}_a \hat{\psi}_a - \frac{1}{6} f_{abc} \hat{\psi}_a \hat{\psi}_b \hat{\psi}_c$$

at each boundary point of Γ . This operator squares to

$$\mathfrak{D}^2 = \frac{1}{2} [\mathfrak{D}, \mathfrak{D}] = \frac{1}{2} \hat{X}_a \hat{X}_a - \frac{1}{48} f_{abc} f_{abc},$$

a central element in the quantum Weil algebra $\mathcal{U}(\mathfrak{g}) \otimes Cl(\mathfrak{g})$, which guarantees that the action of the BFV charge squares to zero.

It is known that this cohomology is trivial ([2],[3]). The resulting quantized BV-BFV is therefore not very interesting, so we should turn to the more involved problem of the three-dimensional Chern-Simons theory.

5.2 3D Chern-Simons Theory

We may now repeat the same procedure for the three-dimensional model. The first observation is that the treatment of the part coming from the extremities of the Wilson lines, namely the terms in the insertion points labeled by z_k and z'_k , is essentially the same as in the one-dimensional model. Each insertion contributes to the overall BFV structure with a term in (31), that when canonically quantized gives the algebra of operators (49) we encountered in the quantization of the one-dimensional model. We can formally express the quantization map

$$\text{Ad}_{g(z_k)}(T_{k,0}) = X_{k,a}(z_k)t_a \mapsto \rho_k(\hat{X}_a(z_k))t^a.$$

Even though the orbits might be different for different insertions, the commutation rules (49) are identical for all of them, only the representation ρ_k differs, as we emphasized on the right-hand side.

In the next step, if we choose a complex structure on the boundary surface $\Sigma = \partial N$, we get a polarization of the connection

$$A = A_z dz + A_{\bar{z}} d\bar{z}, \quad (51)$$

which allows us to rewrite the ambient part of the BFV structure (29) in Darboux coordinates of the corresponding sector $\mathcal{F}_{\partial}^{\text{amb}}$ of the BFV space of boundary fields,

$$\Omega_{\partial}^{\text{amb}} = \int_{\partial N} dz d\bar{z} ((\delta A_z \delta A_{\bar{z}}) + (\delta \gamma, \delta A^+)). \quad (52)$$

Consequently, we may perform the canonical quantization by choosing among each pair of conjugated fields one quantum field and replace the other one by the corresponding functional differential, for instance

$$\begin{aligned} A_{\bar{z}} &\rightarrow a, \\ A_z &\rightarrow -\frac{\delta}{\delta a}, \\ \gamma &\rightarrow \gamma, \\ A^+ &\rightarrow \frac{\delta}{\delta \gamma}, \end{aligned} \quad (53)$$

so as to obtain canonical (anti)commutation rules. Note that a is a boson and γ a fermion.

The Hilbert space $\mathcal{H}_\partial^{\text{BFV}}$ of boundary states on which act all these operators is therefore the space of functionals in a and γ with value in a tensor product of all the representation space associated to each insertion,

$$\mathcal{H}_\partial^{\text{BFV}} = \text{Fun} \left(a, \gamma; \bigotimes_k V_{\rho_k} \otimes V_{\rho_k} \right).$$

We recall it is graded by the ghost number.

Among these states, we want to determine the cohomology $H_{\hat{S}_\partial}^0(\mathcal{H}_\partial^{\text{BFV}})$ of the BFV charge at ghost number zero, made up of the quantum states of the BV-BFV model. At degree zero, we are considering functionals ψ of the $\mathfrak{g}$ -valued $(0, 1)$ -form a , independent of the ghosts γ , which take value in the tensor product of all representation spaces associated to the extremities of the Wilson lines of the models.

The BFV charge

$$\begin{aligned} \hat{S}_\partial = & - \int_{\partial N} dz d\bar{z} \left(\left(\partial a + \bar{\partial} \frac{\delta}{\delta a} + \left[a, \frac{\delta}{\delta a} \right], \gamma \right) \right. \\ & - \sum_k \left(\rho_k(\hat{X}_a(z_k)) \delta(z - z_k) - \rho_k(\hat{X}_a(z_{k'})) \delta(z - z_{k'}) \right) (t^a, \gamma) \\ & \left. + \left(\frac{1}{2} [\gamma, \gamma], \frac{\delta}{\delta \gamma} \right) \right) \end{aligned} \quad (54)$$

acts on the Hilbert space $\mathcal{H}_\partial^{\text{BFV}}$ via multiplication and differentiation by the quantum fields a and γ and via the obvious action of the representation ρ_k on its representation space V_{ρ_k} .

At ghost number zero, BFV quantum states ψ are therefore subject to the condition

$$\left(\partial a + \bar{\partial} \frac{\delta}{\delta a} + \left[a, \frac{\delta}{\delta a} \right] - \sum_k \left(\rho_k(\hat{X}_a(z_k)) \delta_{z_k}(z) - \rho_k(\hat{X}_a(z_{k'})) \delta_{z_k}(z) \right) t^a \right) \psi = 0. \quad (55)$$

This actually coincides with the constraint (1) in [10] imposed to the Schrödinger picture states in the canonical quantization of the Chern-Simons model on $\Sigma \times \mathbb{R}$ at genus 0, or the constraint (2.2) in [8] in the same situation at genus 1, where it is found that the cohomology $H_{\hat{S}_\partial}^0(\mathcal{H}_\partial^{\text{BFV}})$ coincides with the space of conformal blocks in the WZW model for a correlator of fields inserted at the extremities of the Wilson lines.

The condition (55) for quantum states also appears in the geometric quantization framework, see for instance constraint (3.4) in [15], therefore the space of states in geometric quantization coincides with the space of quantum boundary states in BV-BFV quantization.

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Order theory for discrete gradient methods

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Abstract

The discrete gradient methods are integrators designed to preserve invariants of ordinary differential equations. From a formal series expansion of a subclass of these methods, we derive conditions for arbitrarily high order. We derive specific results for the average vector field discrete gradient, from which we get P-series methods in the general case, and B-series methods for canonical Hamiltonian systems. Higher order schemes are presented, and their applications are demonstrated on the Hénon–Heiles system and a Lotka–Volterra system, and on both the training and integration of a pendulum system learned from data by a neural network.

Keywords Geometric integration · Energy preservation · B-series · Neural networks · Order of accuracy · High-order methods

Mathematics Subject Classification 65L05 · 65P10 · 37M15 · 05C05

1 Energy preservation and discrete gradient methods

For an ordinary differential equation (ODE)

$$\dot{x} = f(x), \quad x \in \mathbb{R}^d, \quad f : \mathbb{R}^d \rightarrow \mathbb{R}^d, \quad (1)$$

a first integral, or invariant, is a function $H : \mathbb{R}^d \rightarrow \mathbb{R}$ such that $H(x(t)) = H(x(t_0))$ along the solution curves of (1). If we can write

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$$f(x) = S(x)\nabla H(x), \quad (2)$$

where $S(x) : \mathbb{R}^{d \times d} \rightarrow \mathbb{R}^d$ is a skew-symmetric matrix, then (1) preserves H : this follows from the skew-symmetry of $S(x)$, which yields

$$\frac{d}{dt}H(x) = \nabla H(x)^T \dot{x} = \nabla H(x)^T S(x)\nabla H(x) = 0. \quad (3)$$

The converse is also true: McLachlan et al. showed in [30] that, whenever (1) has a first integral H , there exists a skew-symmetric matrix $S(x)$, bounded near every non-degenerate critical point of H , such that (1) can be written on what is called the *skew-gradient form*:

$$\dot{x} = S(x)\nabla H(x). \quad (4)$$

The proof provided in [30] for this is based on presenting a general form of one such $S(x)$, the so-called default formula

$$S(x) = \frac{f(x)\nabla H(x)^T - \nabla H(x)f(x)^T}{\nabla H(x)^T \nabla H(x)}. \quad (5)$$

Unless $d = 2$, this is generally not a unique choice of $S(x)$, as e.g.

$$S(x) = \frac{f(x)g(x)^T - g(x)f(x)^T}{g(x)^T \nabla H(x)}$$

will satisfy (2) for any non-vanishing function $g : \mathbb{R}^d \rightarrow \mathbb{R}^d$. Many ODEs with first integrals have a well-known skew-gradient form (4). This includes Poisson systems, and the important class consisting of canonical Hamiltonian ODEs. For the latter, S will be constant, so that we may write

$$\dot{x} = S\nabla H(x). \quad (6)$$

A numerical integrator preserving a first integral H exactly is called an integral-preserving, or *energy-preserving*, method. Starting in the late 1970s, a few energy-preserving methods were proposed which relied on some discrete analogue of the property (3), see e.g. [9, 23, 25, 26]. Most prominent among these is the class of methods called discrete gradient methods, defined formally by Gonzalez in [18] and given their current name in [30].

Given the first integral H , a discrete gradient $\bar{\nabla}H : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ is a function satisfying the conditions

$$\bar{\nabla}H(x, y)^T(y - x) = H(y) - H(x), \quad (7)$$

$$\bar{\nabla}H(x, x) = \nabla H(x), \quad (8)$$

for all $x, y \in \mathbb{R}^d$. Introducing also the discrete approximation $\bar{S}(x, y, h)$ to $S(x)$, skew-symmetric and satisfying $\bar{S}(x, x, 0) = S(x)$, the corresponding discrete gradient

method is given by

$$\frac{\hat{x} - x}{h} = \bar{S}(x, \hat{x}, h) \bar{\nabla} H(x, \hat{x}). \quad (9)$$

This scheme satisfies a discrete analogue to (3):

$$H(\hat{x}) - H(x) = h \bar{\nabla} H(x, \hat{x})^T \bar{S}(x, \hat{x}, h) \bar{\nabla} H(x, \hat{x}) = 0.$$

We say that (9) is consistent to the skew-gradient system (4), since $\bar{S}(x, \hat{x}, h)$ is a consistent approximation of $S(x)$ and $\bar{\nabla} H(x, \hat{x})$ is a consistent approximation of $\nabla H(x)$.

If $d \geq 2$, there are in general infinitely many functions satisfying (7)–(8). Many explicit definitions of concrete discrete gradients have been suggested, and we will discuss the most prominent among them in Sect. 2.1. One of these is the *average vector field (AVF) discrete gradient*, first introduced in [22] and sometimes called the mean value discrete gradient [30]. For a given H , it is given by the average of ∇H on the segment $[x, y]$:

$$\bar{\nabla}_{\text{AVF}} H(x, y) = \int_0^1 \nabla H((1 - \xi)x + \xi y) d\xi. \quad (10)$$

When applied to the constant S system (6), the discrete gradient method with $\bar{S}(x, y, h) = S$ and $\bar{\nabla} H = \bar{\nabla}_{\text{AVF}} H$ coincides with the scheme

$$\frac{\hat{x} - x}{h} = \int_0^1 f((1 - \xi)x + \xi \hat{x}) d\xi. \quad (11)$$

This is sometimes viewed as a method by itself, applicable to any system (1), in which case it is called the average vector field (AVF) method [37]. This was shown in [6] to be a B-series method.

As pointed out in [30], the discrete gradient is restricted by its definition to be at best a second order approximation to point values of ∇H . In much of the literature on discrete gradient methods, see e.g. [18, 21], the approximation $\bar{S}$ is defined as being independent of h . In that case, the discrete gradient scheme (9) can at best guarantee second order convergence towards the exact solution. Over the last two decades, there have been published some notable papers on higher order discrete gradient methods. McLaren and Quispel were first out with their bootstrapping technique derived in [31, 32]. Given any discrete gradient $\bar{\nabla} H$ and an approximation to $S(x)$ given by $\bar{S}(x, y, h)$, they compare the Taylor expansion of the corresponding discrete gradient scheme to that of the exact solution, and thus find a new approximation $\tilde{S}(x, y, h)$ to $S(x)$ which yields higher order. This quickly becomes a very involved procedure, but by using a symmetric discrete gradient, they derive fourth order methods. A downside of this method is that the schemes of order higher than two require the calculation of tensors of order three or higher at every time step.

A fourth order generalization of the AVF method is proposed by the same authors in [37]. This can be viewed as a fourth-order discrete gradient method for all skew-gradient systems where S is constant. Also worth mentioning in this setting is the collocation-like method introduced by Hairer [20] and then generalized to Poisson

systems by Cohen and Hairer [10]. This is a multi-stage extension of the AVF discrete gradient method. To get higher than second order, more than one stage is required. In that case the method is not a discrete gradient method, although it is energy-preserving.

Norton et al. show in [35] that linear projection methods can be viewed as a class of discrete gradient methods for skew-gradient systems with $S(x)$ given by the default formula (5). In connection to this, Norton and Quispel suggest in [36] the class of approximations to (5) given by

$$\bar{S}(x, y, h) = \frac{\tilde{f}(x, y, h)\tilde{g}(x, y, h)^T - \tilde{g}(x, y, h)\tilde{f}(x, y, h)^T}{\hat{g}(x, y, h)^T \check{g}(x, y, h)}, \quad (12)$$

where $\tilde{f}(x, y, h)$ is a consistent approximation to $f(x)$, and $\tilde{g}(x, y, h)$, $\hat{g}(x, y, h)$ and $\check{g}(x, y, h)$ are all consistent approximations to $\nabla H(x)$. The corresponding discrete gradient method then inherits the order of the method $\hat{x} = x + h\tilde{f}(x, \hat{x}, h)$.

The use of the discrete gradient method expands beyond the numerical integration of ODEs. It can be applied to the time-integration of partial differential equations (PDEs) which has constants of motion, guaranteeing preservation of a discrete approximation of an integral [5, 14, 16]. Higher-order methods have been studied in this context, but then only schemes that are expanding on the AVF method [3, 24]. Furthermore, building on the recently introduced Hamiltonian neural networks [8, 19], Matsubara et al. have shown how discrete gradients can be used to learn energy-preserving systems from data [27]. Since the energy is given by a neural network, none of the higher-order methods reviewed above are applicable in that case, and the authors suggest to use multi-step methods to get higher than second order.

To the best of our knowledge, no one has so far suggested higher than fourth order discrete gradient methods for a general skew-gradient system (4). Furthermore, for this general case, all discrete gradient methods suggested of higher than second order involve tensors of order three or higher. In this paper we present general theory for achieving methods of arbitrary order, using any discrete gradient and not depending on tensors of order higher than two. Largely inspired by the above mentioned references, especially [31, 32, 37], we present here a general form giving a class of approximations $\bar{S}(x, y, h)$ to any $S(x)$ in (4), with corresponding conditions for achieving an arbitrary order of the discrete gradient method (9). We do this step by step. In the next section, we derive some useful properties of a general discrete gradient and discuss the most common specific discrete gradients. Then we consider the AVF method and use order theory for B-series methods to obtain a generalization of this, with corresponding order conditions. In Sect. 4, we build on this to develop higher order discrete gradient methods for a general skew-gradient system, using the AVF discrete gradient. Then, in Sect. 5, we generalize this further to allow for a free choice of the discrete gradient, thus arriving at the general form $\bar{S}(x, y, h)$ mentioned above, and a formal series expansion of the corresponding discrete gradient methods. Throughout the paper, we present several examples of higher order schemes for the different cases. In the final section, we apply some of these schemes to numerical examples: the Hénon–Heiles system with a constant S , a Lotka–Volterra system with a non-constant S , and a pendulum system learned by a neural network.

2 A preliminary analysis of discrete gradients

To simplify notation in the following derivations, we define $g := \nabla H$. Furthermore, we suppress the first argument of $\bar{\nabla} H$ and define $\bar{g}(y) := \bar{\nabla} H(x, y)$. We use Einstein summation convention, with a comma in the subscript to differentiate between components of covectors and derivatives. That is, we write $\bar{g}(y)_{,j}^i := \frac{\partial \bar{g}(y)^i}{\partial y^j}$, $\bar{g}(y)_{i,j} := \frac{\partial \bar{g}(y)_i}{\partial y^j}$ and so forth. Taylor expanding $\bar{g}(y)$ around x , we get

$$\begin{aligned} \bar{g}(y)^i &= \bar{g}(x)^i + \bar{g}(x)_{,j}^i (y^j - x^j) + \frac{1}{2} \bar{g}(x)_{,jk}^i (y^j - x^j)(y^k - x^k) \\ &\quad + \frac{1}{6} \bar{g}(x)_{,jkl}^i (y^j - x^j)(y^k - x^k)(y^l - x^l) + \mathcal{O}(|y - x|^4), \end{aligned} \quad (13)$$

or

$$\bar{g}(y) = \sum_{\kappa=0}^{\infty} \frac{1}{\kappa!} \bar{g}^{(\kappa)}(x) (y - x)^\kappa. \quad (14)$$

By the consistency criterion (8), we have $\bar{g}(x) = g(x)$. However, if we require the discrete gradient to be a differentiable function in its second argument, (8) follows directly from (7). To see this, we write (7) as

$$H(y) - H(x) = \bar{g}(y)_i (y^i - x^i). \quad (15)$$

Differentiating this with respect to y^j , we get

$$g(y)_j = H(y)_{,j} = \bar{g}(y)_{i,j} (y^i - x^i) + \bar{g}(y)_j, \quad (16)$$

The case $y = x$ immediately gives $g(x)_j = \bar{g}(x)_j$, or (8). Assuming further that $\bar{\nabla} H \in C^2(\mathbb{R}^d \times \mathbb{R}^d, \mathbb{R}^d)$, we can differentiate once more to get

$$g(y)_{j,k} = H(y)_{,jk} = \bar{g}(y)_{i,jk} (y^i - x^i) + \bar{g}(y)_{j,k} + \bar{g}(y)_{k,j}, \quad (17)$$

which means that

$$g(x)_{j,k} = H(x)_{,jk} = \bar{g}(x)_{j,k} + \bar{g}(x)_{k,j},$$

or

$$D^2 H(x) = D_2 \bar{\nabla} H(x, x) + (D_2 \bar{\nabla} H(x, x))^T, \quad (18)$$

where $D^2 H := D \nabla H$ denotes the Hessian of H , and $D_2 \bar{\nabla} H$ denotes the Jacobian of $\bar{\nabla} H$ with respect to its second argument.

Lemma 1 *If the discrete gradient $\bar{\nabla} H$ is symmetric, i.e. $\bar{\nabla} H(x, y) = \bar{\nabla} H(y, x)$ for all $x, y \in \mathbb{R}^d$, then*

$$D_2 \bar{\nabla} H(x, x) = \frac{1}{2} D^2 H(x). \quad (19)$$

Proof Disclosing the suppressed argument x in (16), we have

$$g(y)_j = \frac{\partial}{\partial y^j} (\bar{g}(x, y)_i)(y^i - x^i) + \bar{g}(x, y)_j,$$

which we can differentiate by x^k to get

$$0 = \frac{\partial^2}{\partial x^k \partial y^j} (\bar{g}(x, y)_i)(y^i - x^i) - \frac{\partial}{\partial y^j} \bar{g}(x, y)_k + \frac{\partial}{\partial x^k} \bar{g}(x, y)_j.$$

If $\bar{\nabla} H$ is symmetric,

$$\frac{\partial}{\partial x^k} \bar{g}(x, y)_j = \frac{\partial}{\partial x^k} \bar{g}(y, x)_j.$$

Thus, for $y = x$ we get $\bar{g}(x)_{k,j} = \bar{g}(x)_{j,k}$, or $(D_2 \bar{\nabla} H(x, x))^T = D_2 \bar{\nabla} H(x, x)$. Inserting that in (18), we obtain (19).

Definition 1 Given a discrete gradient $\bar{\nabla} H \in C^1(\mathbb{R}^d \times \mathbb{R}^d, \mathbb{R}^d)$, we define the function $Q : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}^{d \times d}$ by

$$Q(x, y) := \frac{1}{2} \left((D_2 \bar{\nabla} H(x, y))^T - D_2 \bar{\nabla} H(x, y) \right). \quad (20)$$

Note that $Q(x, y)$ is a skew-symmetric matrix. From (18), we see that $Q(x, x) = \frac{1}{2} D^2 H(x) - D_2 \bar{\nabla} H(x, x)$. The following result is crucial for the the main result of this paper: the development of a general theory for higher order discrete gradient methods.

Lemma 2 For a discrete gradient $\bar{\nabla} H \in C^p(\mathbb{R}^d \times \mathbb{R}^d, \mathbb{R})$ and the corresponding Q given by (20),

$$\begin{aligned} D_2^\kappa \bar{\nabla} H(x, x) v^\kappa &= \frac{1}{\kappa + 1} D^\kappa \nabla H(x) v^\kappa \\ &\quad - \frac{2\kappa}{\kappa + 1} D_2^{\kappa-1} Q(x, x) v^\kappa \quad \text{for any } \kappa \in [1, p], v \in \mathbb{R}^d. \end{aligned}$$

Proof Differentiating (17) $\kappa - 1$ times by y and setting $y = x$, we find that the κ th derivatives of $g(x)$ can be expressed by the κ th derivatives of $\bar{g}(x)$ through the relation

$$g(x)_{j,I} = \bar{g}(x)_{j,I} + \sum_{m=1}^{\kappa} \bar{g}(x)_{i_m, \{j, I_m\}}, \quad \text{for all } j, I, \kappa, \quad (21)$$

where $I = \{i_1, i_2, \dots, i_\kappa\}$ is an ordered set of κ indices, and

$$I_m = I \setminus \{i_m\} = \{i_1, i_2, \dots, i_{m-1}, i_{m+1}, \dots, i_\kappa\},$$

i.e. I with the m th element excluded. Similarly, by continued differentiation of (20), we obtain

$$(D_2^{\kappa-1} Q(x, x))_{j,I} = \frac{1}{2} \bar{g}(x)_{i_1, \{j, I_1\}} - \frac{1}{2} \bar{g}(x)_{j,I}.$$

Thus

$$\begin{aligned} (D_2^{\kappa-1} Q(x, x) v^\kappa)_j &= (D_2^{\kappa-1} Q(x, x))_{j,I} v^I = \frac{1}{2} \bar{g}(x)_{i_1, \{j, I_1\}} v^{\{j, I_1\}} - \frac{1}{2} \bar{g}(x)_{j,I} v^I \\ &= \frac{1}{2\kappa} \sum_{m=1}^{\kappa} \bar{g}(x)_{i_m, \{j, I_m\}} v^{\{j, I_m\}} + \frac{1}{2\kappa} \bar{g}(x)_{j,I} v^I - \frac{1}{2\kappa} \bar{g}(x)_{j,I} v^I - \frac{1}{2} \bar{g}(x)_{j,I} v^I \\ &= \frac{1}{2\kappa} g(x)_{j,I} v^I - \left(\frac{1}{2\kappa} + \frac{1}{2}\right) \bar{g}(x)_{j,I} v^I = \frac{1}{2\kappa} g(x)_{j,I} v^I - \frac{\kappa+1}{2\kappa} \bar{g}(x)_{j,I} v^I. \end{aligned}$$

2.1 Properties of different discrete gradients

While introducing the discrete gradient methods in [18], Gonzalez also gave an example of a discrete gradient satisfying (7)–(8): the *midpoint discrete gradient* is given by

$$\bar{\nabla}_M H(x, y) := \nabla H\left(\frac{x+y}{2}\right) + \frac{H(y) - H(x) - \nabla H\left(\frac{x+y}{2}\right)^T (y-x)}{(y-x)^T (y-x)} (y-x).$$

Even when H is analytic, this discrete gradient is often not; the second order partial derivatives are in general singular in $y = x$. For that reason, it is not suited for achieving higher order methods by the techniques we consider in this paper.

The *Itoh–Abe discrete gradient method*, introduced in [23], notably does not require evaluation of the gradient. The corresponding discrete gradient, which has also been called the coordinate increment discrete gradient [30], is defined by

$$\bar{\nabla}_{IA} H(x, y) := \sum_{j=1}^d \alpha_j e_j, \quad (22)$$

where e_j is the j^{th} canonical unit vector and

$$\begin{aligned} \alpha_j &= \begin{cases} \frac{H(w_j) - H(w_{j-1})}{y^j - x^j} & \text{if } y^j \neq x^j, \\ \frac{\partial H}{\partial x^j}(w_{j-1}) & \text{if } y^j = x^j, \end{cases} \\ w_j &= \sum_{i=1}^j y^i e_i + \sum_{i=j+1}^n x^i e_i. \end{aligned}$$

While the other discrete gradients we consider in this paper are symmetric and thus second order approximations to ∇H , the Itoh–Abe discrete gradient is only of first

order. However, a second order discrete gradient, which we call the *symmetrized Itoh–Abe (SIA) discrete gradient*, is given by

$$\bar{\nabla}_{\text{SIA}} H(x, y) := \frac{1}{2} (\bar{\nabla}_{\text{IA}} H(x, y) + \bar{\nabla}_{\text{IA}} H(y, x)). \quad (23)$$

Furihata presented the discrete variational derivative method for a class of PDEs in [16], a method which has been developed further by Furihata, Matsuo and co-authors in a series of papers, e.g. [29, 38], as well as the monograph [17]. As shown in [14], these schemes can also be obtained by semi-discretizing the PDE in space and then applying a discrete gradient method on the resulting system of ODEs. We give here the definition of the specific discrete gradient that gives the schemes of Furihata and co-author, defined for a class of invariants that includes all polynomial functions:

Definition 2 Assume that we can write the first integral as

$$H(x) = \sum_l c_l \prod_{j=1}^d f_j^l(x^j), \quad (24)$$

for functions $f_j^l : \mathbb{R} \rightarrow \mathbb{R}$. The *Furihata discrete gradient* $\bar{\nabla}_{\text{F}} H(x, y)$ is defined by

$$\bar{\nabla}_{\text{F}} H(x, y) := \sum_{j=1}^d \alpha_j e_j, \quad (25)$$

where e_j is the j^{th} canonical unit vector and

$$\alpha_j = \begin{cases} \sum_l \frac{c_l}{2} \frac{f_j^l(y^j) - f_j^l(x^j)}{y^j - x^j} \left(\prod_{k=1}^{j-1} f_k^l(x^k) + \prod_{k=1}^{j-1} f_k^l(y^k) \right) \prod_{k=j+1}^d \frac{f_k^l(x^k) + f_k^l(y^k)}{2} & \text{if } y^j \neq x^j, \\ \sum_l \frac{c_l}{2} \frac{df_j^l(x^j)}{dx^j} \left(\prod_{k=1}^{j-1} f_k^l(x^k) + \prod_{k=1}^{j-1} f_k^l(y^k) \right) \prod_{k=j+1}^d \frac{f_k^l(x^k) + f_k^l(y^k)}{2} & \text{if } y^j = x^j. \end{cases}$$

The discrete gradient introduced by Matsubara and co-authors in [27] shares some relation to the Furihata discrete gradient, in that they are both relying on a discrete analogue to the product rule for derivatives. But the former discrete gradient, obtained by what the authors call the *automatic discrete differentiation algorithm*, more importantly depends on an analogue to the chain rule. By using this algorithm instead of standard automatic differentiation in a neural network that learns a dynamical system from data, the discrete gradient is obtained together with the preserved energy. Thus the learned system can be integrated in an energy-preserving manner, which distinguishes the approach of Matsubara et al. from comparable studies [8, 19, 39].

Lastly we consider the AVF discrete gradient (10), which has several noteworthy characteristics.

Lemma 3 The $Q(x, y)$ corresponding to the AVF discrete gradient is the zero matrix, since $(D_2 \bar{\nabla}_{\text{AVF}} H(x, y))^T = D_2 \bar{\nabla}_{\text{AVF}} H(x, y)$.

Proof For $\bar{g}(y) := \bar{\nabla}_{\text{AVF}} H(x, y)$, we have

$$\begin{aligned}\bar{g}(y)_{i,j} &= \frac{\partial}{\partial y^j} \int_0^1 g((1-\xi)x + \xi y)_i \, d\xi = \int_0^1 \frac{\partial}{\partial y^j} g((1-\xi)x + \xi y)_i \, d\xi \\ &= \int_0^1 \xi g((1-\xi)x + \xi y)_{i,j} \, d\xi = \int_0^1 \xi g((1-\xi)x + \xi y)_{j,i} \, d\xi = \bar{g}(y)_{j,i}.\end{aligned}$$

From the Integrability Lemma (see e.g. [21, Lemma VI.2.7]) and the above, we have that the AVF discrete gradient defines a gradient vector field:

Corollary 1 *The AVF discrete gradient is the gradient with respect to the second argument of a function $\tilde{H}(x, y)$. That is,*

$$\bar{\nabla}_{\text{AVF}} H(x, y) = \nabla_2 \tilde{H}(x, y),$$

for some $\tilde{H} : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ and all $x, y \in \mathbb{R}^d$.

Any other discrete gradient will have these properties only for functions H for which it coincides with the AVF discrete gradient:

Proposition 1 *The AVF discrete gradient is the unique discrete gradient satisfying $(D_2 \bar{\nabla} H(x, y))^T = D_2 \bar{\nabla} H(x, y)$ for all H, x and y , and it has the formal expansion*

$$\bar{\nabla}_{\text{AVF}} H(x, y) = \sum_{\kappa=0}^{\infty} \frac{1}{(\kappa+1)!} D^{\kappa} \nabla H(x)(y-x)^{\kappa}. \quad (26)$$

Proof Assume that ∇H is an analytic function. As in the proof of Lemma 2, let $I = \{i_1, i_2, \dots, i_{\kappa}\}$ be an ordered set of κ indices, and let I_m be I with the m^{th} element excluded. If $\bar{g}(y)_{i,j} = \bar{g}(y)_{j,i}$ for all i, j , then also

$$\bar{g}(y)_{i,I} = \bar{g}(y)_{i_m, \{i, I_m\}} \quad \text{for all } i, I, m. \quad (27)$$

Inserting (27) in (21) we get $g^{(\kappa)}(x) = (1+\kappa)\bar{g}^{(\kappa)}(x)$. Then inserting this for $\bar{g}^{(\kappa)}(x)$ in (14), we get (26), which uniquely defines the AVF discrete gradient.

A consequence of the above result is that the AVF discrete gradient is the unique discrete gradient for which the scheme (9) with $\bar{S}(x, \hat{x}, h) = S$ is a B-series method when applied to the system (6).

As we see from the above definitions and discussion, each of the discrete gradients have their advantages and disadvantages. Gonzalez' midpoint discrete gradient is easily calculated from the energy H and the gradient ∇H , but it is in general only once differentiable. The Itoh–Abe and Furihata discrete gradient methods do not require knowledge of the gradient, but the former is only a first order method and the latter is only defined for H of the form (24). The AVF discrete gradient is the unique discrete gradient whose series expansion is given by the differentials of the gradient. It does however require an integral to be calculated. The discrete gradient of Matsubara and

co-authors has a different area of use than the others: it is only defined when H is a neural network. In that case neither Gonzelez' midpoint, the AVF or the Furihata discrete gradients can be obtained.

In the next section we consider the specific case where the AVF discrete gradient is chosen and S is constant, so that we can build on established results for B-series methods to develop new theory.

3 A generalization of the AVF method

Let us recall the concept of B-series. Referring to the definitions in [21, Section III.1], we let T be the set of rooted trees, built recursively from starting with $\tau = \bullet$ and letting $\tau = [\tau_1, \dots, \tau_m]$ be the tree obtained by grafting the roots of the trees $\tau_1, \dots, \tau_m$ to a new root. Furthermore, $F(\tau)$ is the elementary differential associated with the tree τ , defined by $F(\bullet)(x) = f(x)$ and

$$F(\tau)(x) = f^{(m)}(x)(F(\tau_1)(x), \dots, F(\tau_m)(x)),$$

and $\sigma(\tau)$ is the symmetry coefficient for τ , defined by $\sigma(\bullet) = 1$ and

$$\sigma(\tau) = \sigma(\tau_1) \cdots \sigma(\tau_m) \cdot \mu_1! \mu_2! \cdots, \quad (28)$$

where the integers $\mu_1, \mu_2, \dots$ count equal trees among $\tau_1, \dots, \tau_m$. Then, if $\phi : T \cup \{\emptyset\} \rightarrow \mathbb{R}$ is an arbitrary map, a *B-series* is a formal series

$$B(\phi, x) = \phi(\emptyset)x + \sum_{\tau \in T} \frac{h^{|\tau|}}{\sigma(\tau)} \phi(\tau) F(\tau)(x). \quad (29)$$

The exact solution of (1) can be written as the B-series $B(\frac{1}{\gamma}, x)$, where the coefficient γ satisfies $\gamma(\emptyset) = \gamma(\bullet) = 1$ and

$$\gamma(\tau) = |\tau| \gamma(\tau_1) \cdots \gamma(\tau_m) \quad \text{for } \tau = [\tau_1, \dots, \tau_m], \quad (30)$$

where $|\tau|$ is the order, i.e. the number of nodes, of τ .

Definition 3 The *generalized AVF method* is given by

$$\begin{aligned} \frac{\hat{x} - x}{h} &= \left(I + \sum_{n=2}^{p-1} h^n \sum_j b_{nj} \left(\prod_{k=1}^n f'(z_{njk}) + (-1)^n \prod_{k=1}^n f'(z_{nj(n-k+1)}) \right) \right) \\ &\quad \cdot \int_0^1 f((1-\xi)x + \xi\hat{x}) d\xi, \end{aligned} \quad (31)$$

where each $z_{njk} := z_{njk}(x, \hat{x}, h) = B(\phi_{njk}, x)$ can be written as a B-series with $\phi(\emptyset) = 1$.

Theorem 1 When applied to (1) with $f(x) = S\nabla H(x)$, where S is a constant skew-symmetric matrix, the scheme (31) preserves H , in that $H(\hat{x}) = H(x)$.

Proof With $f(x) = S\nabla H(x)$, (31) becomes

$$\frac{\hat{x} - x}{h} = \bar{S}(x, \hat{x}, h) \bar{\nabla}_{\text{AVF}} H(x, \hat{x}),$$

with

$$\bar{S}(x, \hat{x}, h) = S + \sum_{n=2}^{p-1} h^n \sum_j b_{nj} \left(\prod_{k=1}^n SD^2 H(z_{njk}) + (-1)^n \prod_{k=1}^n SD^2 H(z_{nj(n-k+1)}) \right) S.$$

We have

$$\begin{aligned} & \left(\prod_{k=1}^n SD^2 H(z_{njk}) \cdot S + (-1)^n \prod_{k=1}^n SD^2 H(z_{nj(n-k+1)}) \cdot S \right)^T \\ &= S^T \prod_{k=1}^n (D^2 H(z_{nj(n-k+1)})^T S^T) + (-1)^n S^T \prod_{k=1}^n (D^2 H(z_{njk})^T S^T) \\ &= (-1)^{i+1} S \prod_{k=1}^n D^2 H(z_{nj(n-k+1)}) S - S \prod_{k=1}^n D^2 H(z_{njk}) S \\ &= - \left(\prod_{k=1}^n SD^2 H(z_{njk}) \cdot S + (-1)^n \prod_{k=1}^n SD^2 H(z_{nj(n-k+1)}) \cdot S \right), \end{aligned}$$

and thus $\bar{S}(x, \hat{x}, h)$ is a skew-symmetric matrix.

Before considering the order conditions of the generalized AVF method, let us recall a couple of results from the literature on B-series.

Lemma 4 ([21, Lemma III.1.9]) Let $B(a, x)$ be a B-series with $a(\emptyset) = 1$. Then $hf(B(a, x)) = B(a', x)$ is also a B-series, with $a'(\emptyset) = 0$, $a'(\bullet) = 1$ and otherwise

$$a'(\tau) = a(\tau_1) \cdots a(\tau_m) \quad \text{for } \tau = [\tau_1, \dots, \tau_m].$$

Lemma 5 ([34, Theorem 2.2]) Let $B(a, x)$ and $B(b, x)$ be two B-series with $a(\emptyset) = 1$ and $b(\emptyset) = 0$. Then $hf'(B(a, x))B(b, x) = B(a \times b, x)$, i.e. a B-series, with $(a \times b)(\emptyset) = (a \times b)(\bullet) = 0$ and otherwise

$$(a \times b)(\tau) = \sum_{i=1}^m \prod_{j=1, j \neq i}^m a(\tau_j) b(\tau_i) \quad \text{for } \tau = [\tau_1, \dots, \tau_m].$$

Proposition 1 in [6] states that the standard AVF method is a B-series method. We build on the proof of that proposition to prove the following result.

Proposition 2 *The generalized AVF method (31) is a B-series method.*

Proof First we define $\hat{e} : T \cup \{\emptyset\} \rightarrow \mathbb{R}$ by $\hat{e}(\emptyset) = 1$ and $\hat{e}(\tau) = 0$ for all $\tau \neq \emptyset$. Then, assuming that the solution $\hat{x}$ of (31) can be written as the B-series $\hat{x} = B(\Phi, x)$, we find the B-series

$$\begin{aligned} h \int_0^1 f((1-\xi)x + \xi\hat{x}) \, d\xi &= h \int_0^1 f(B((1-\xi)\hat{e} + \xi\Phi, x)) \, d\xi \\ &= \int_0^1 B(((1-\xi)\hat{e} + \xi\Phi)', x) \, d\xi \\ &= B\left(\int_0^1 ((1-\xi)\hat{e} + \xi\Phi)' \, d\xi, x\right). \end{aligned}$$

Setting $\theta := \int_0^1 ((1-\xi)\hat{e} + \xi\Phi)' \, d\xi = \int_0^1 ((1-\xi)\hat{e})' \, d\xi + \int_0^1 (\xi\Phi)' \, d\xi = \int_0^1 (\xi\Phi)' \, d\xi$, we get

$$\theta(\emptyset) = 0, \quad \theta(\bullet) = 1, \quad \theta([\tau_1, \dots, \tau_m]) = \frac{1}{m+1} \Phi(\tau_1) \cdots \Phi(\tau_m). \quad (32)$$

Then we may rewrite (31) as

$$\begin{aligned} \hat{x} &= x + \left(I + \sum_{n=2}^{p-1} h^n \sum_j b_{nj} \left(\prod_{k=1}^n f'(B(\phi_{njk}, x)) + (-1)^n \prod_{k=1}^n f'(B(\phi_{nj(n-k+1)}, x)) \right) \right) B(\theta, x) \\ &= x + B(\theta, x) \\ &\quad + \sum_{n=2}^{p-1} \sum_j b_{nj} (B(\phi_{nj1} \times \cdots \times \phi_{njn} \times \theta, x) + (-1)^n B(\phi_{njn} \times \cdots \times \phi_{nj1} \times \theta, x)) \\ &= B(\Phi, x), \end{aligned}$$

with

$$\Phi = \hat{e} + \theta + \sum_{n=2}^{p-1} \sum_j b_{nj} (\phi_{nj1} \times \cdots \times \phi_{njn} \times \theta + (-1)^n \phi_{njn} \times \cdots \times \phi_{nj1} \times \theta). \quad (33)$$

Comparing the B-series of the exact solution and the B-series of the solution of (31), and noting that the elementary differentials are independent, we immediately get the following result.

Theorem 2 *The generalized AVF method (31) is of order p if and only if*

$$\Phi(\tau) = \frac{1}{\gamma(\tau)} \quad \text{for } |\tau| \leq p, \quad (34)$$

Table 1 Elementary differentials and their coefficients in the B-series of the solution of (31), up to fourth order

$ \tau $	$F(\tau)^i$	τ	$\sigma(\tau)$	$\gamma(\tau)$	$\Phi(\tau)$
1	f^i	$\bullet$	1	1	1
2	$f_j^i f^j$	$\vdots$	1	2	$\frac{1}{2}$
3	$f_{jk}^i f^j f^k$	$\vee$	2	3	$\frac{1}{3}$
4	$f_j^i f_k^j f^k$	$\vdots$	1	6	$\frac{1}{4} + 2 \sum_j b_{2j}$
	$f_{jkl}^i f^j f^k f^l$	$\vee$	6	4	$\frac{1}{4}$
	$f_{jk}^i f^j f_l^k f^l$	$\vee$	1	8	$\frac{1}{6} + \sum_{j,k} b_{2j} \phi_{2jk}(\bullet)$
	$f_j^i f_{kl}^j f^k f^l$	$\vee$	2	12	$\frac{1}{6} + 2 \sum_{j,k} b_{2j} \phi_{2jk}(\bullet)$
	$f_j^i f_k^j f_l^k f^l$	$\vdots$	1	24	$\frac{1}{8} + 2 \sum_j b_{2j}$

where Φ is given by (33) and γ is given by (30).

The terms $\Phi(\tau)$ can be found from (33) by applying Lemma 5 recursively, as illustrated by the following example.

Example 1 Consider $\tau = \vee$, and assume we have found Φ for all trees up to and including order four already, as given in Table 1. We have

$$\theta(\vee) = \frac{1}{3} \Phi(\bullet) \Phi(\bullet) = \frac{1}{3} \left(\frac{1}{4} + 2 \sum_j b_{2j} \right) = \frac{1}{12} + \frac{2}{3} \sum_j b_{2j}.$$

Then we calculate

$$\begin{aligned} (\phi_{2j1} \times \phi_{2j2} \times \theta)(\vee) &= \phi_{2j1}(\bullet)(\phi_{2j2} \times \theta)(\bullet) + \phi_{2j1}(\bullet)(\phi_{2j2} \times \theta)(\bullet) \\ &= \phi_{2j1}(\bullet)(\phi_{2j2} \times \theta)(\bullet) = \phi_{2j1}(\bullet) \phi_{2j2}(\emptyset) \theta(\emptyset) = \frac{1}{2} \phi_{2j1}(\bullet), \end{aligned}$$

where we have used in the second equality that $(\phi_{2j2} \times \theta)(\bullet) = \phi_{2j2}(\emptyset) \theta(\emptyset) = 0$.

Similarly we find $(\phi_{2j2} \times \phi_{2j1} \times \theta)(\vee) = \frac{1}{2} \phi_{2j2}(\bullet)$. Furthermore,

$$\begin{aligned} (\phi_{3j1} \times \phi_{3j2} \times \phi_{3j3} \times \theta)(\vee) &= \phi_{3j1}(\bullet)(\phi_{3j2} \times \phi_{3j3} \times \theta)(\bullet) \\ &= \phi_{3j1}(\bullet) \phi_{3j2}(\emptyset) (\phi_{3j3} \times \theta)(\bullet) \\ &= \phi_{3j1}(\bullet) \phi_{3j3}(\emptyset) \theta(\bullet) = \phi_{3j1}(\bullet), \end{aligned}$$

and $(\phi_{3j3} \times \phi_{3j2} \times \phi_{3j1} \times \theta)(\mathbf{V}) = \phi_{3j3}(\bullet)$. Hence,

$$\begin{aligned} \Phi(\mathbf{V}) &= \frac{1}{12} + \frac{2}{3} \sum_j b_{2j} + \frac{1}{2} \sum_j b_{2j}(\phi_{2j1}(\bullet) + \phi_{2j2}(\bullet)) \\ &\quad + \sum_j b_{3j}(\phi_{3j1}(\bullet) - \phi_{3j3}(\bullet)). \end{aligned}$$

Now, if we assume the order condition (34) to be satisfied for all trees up to and including order four, we can replace $\sum_j b_{2j} = -\frac{1}{24}$ and $\sum_j b_{2j}(\phi_{2j1}(\bullet) + \phi_{2j2}(\bullet)) = -\frac{1}{24}$ in the above expression, use that $\gamma(\mathbf{V}) = 30$, and get that (34) is satisfied for $\mathbf{V}$ if and only if

$$\sum_j b_{3j}(\phi_{3j1}(\bullet) - \phi_{3j3}(\bullet)) = -\frac{1}{720}. \quad (35)$$

3.1 Construction of higher order schemes

As the size of the trees grows, finding $\Phi(\tau)$ from (33) can become quite a cumbersome operation. Furthermore, we observe from Table 1 that there are some equivalent order conditions for different trees. Before presenting more convenient techniques for finding order conditions for the generalized AVF method, let us define some more concepts related to B-series and trees.

First, recall that the Butcher product of two trees $u = [u_1, \dots, u_m]$ and $v = [v_1, \dots, v_n]$ is given by $u \circ v = [u_1, u_2, \dots, u_m, v]$. This operation is neither associative nor commutative, and in contrast to the practice in [21], we here take the product of several factors without parentheses to mean evaluation from right to left:

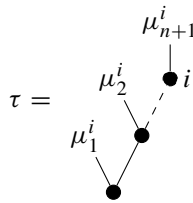
$$u_1 \circ u_2 \circ \dots \circ u_k := u \circ (u_2 \circ (\dots \circ u_k)).$$

Given a forest $\mu = (\tau_1, \dots, \tau_m)$, the tree obtained by grafting the roots of every tree in μ to a new root is denoted by $[\mu] = [\tau_1, \dots, \tau_m]$. Moreover, $\mu^{-1}(\tau)$ denotes the forest such that $[\mu^{-1}(\tau)] = \tau$. We extend the maps $\phi : T \cup \{\emptyset\} \rightarrow \mathbb{R}$ and $\gamma : T \cup \{\emptyset\} \rightarrow \mathbb{R}$ to forests by the letting $\phi(\mu) = \prod_{i=1}^m \phi(\tau_i)$ and $\gamma(\mu) = \prod_{i=1}^m \gamma(\tau_i)$ for $\mu = (\tau_1, \dots, \tau_m)$.

Consider now a tree τ consisting of $|\tau|$ nodes. We may number every tree from 1 to $|\tau|$, starting at the root and going from left to right on the increasing levels above. For a given node $i \in \{1, \dots, |\tau|\}$ on level $n+1$, there exists a unique set of forests $\hat{\tau}^i = \{\mu_1^i, \dots, \mu_{n+1}^i\}$ such that

$$\tau = [\mu_1^i] \circ [\mu_2^i] \circ \dots \circ [\mu_{n+1}^i].$$

That is, labeling node i ,



Proposition 3 The Φ of (34) can alternatively be found by

$$\Phi(\tau) = \hat{e}(\tau) + \theta(\tau) + \sum_{i \text{ s.t. } n \geq 2} \Lambda(\hat{\tau}^i) \quad (36)$$

where $\hat{e}(\emptyset) = 1$ and $\hat{e}(\tau) = 0$ for all $\tau \neq \emptyset$, $\theta(\emptyset) = 0$, $\theta(\bullet) = 1$,

$$\theta([\tau_1, \dots, \tau_m]) = \frac{1}{m+1} \Phi(\tau_1) \cdots \Phi(\tau_m),$$

and

$$\Lambda(\hat{\tau}^i) = \theta([\mu_{n+1}^i]) \sum_j b_{nj} \left(\phi_{nj1}(\mu_1^i) \cdots \phi_{njn}(\mu_n^i) + (-1)^n \phi_{njn}(\mu_1^i) \cdots \phi_{nj1}(\mu_n^i) \right). \quad (37)$$

Proof Define n_i so that $n_i + 1$ is the level of node i . Collect the children of node i in the set C_i . We have

$$[\mu_{n_i+1}^i] = [\mu_{n_k}^k] \circ [\mu_{n_k+1}^k] \quad \text{for all } k \in C_i,$$

and thus

$$(a \times b)([\mu_{n_i+1}^i]) = \sum_{k \in C_i} a(\mu_{n_k}^k) b([\mu_{n_k+1}^k]).$$

Note also that $\mu_{n_i}^i = \mu_{n_i}^k = \mu_{n_k-1}^k$ if $k \in C_i$. Then we get

$$\begin{aligned} (\phi_{nj1} \times \cdots \times \phi_{njn} \times \theta)(\tau) &= (\phi_{nj1} \times \cdots \times \phi_{njn} \times \theta)([\mu_1^1]) \\ &= \sum_{i_1 \in C_1} \phi_{nj1}(\mu_1^{i_1}) (\phi_{nj2} \times \cdots \times \phi_{njn} \times \theta)([\mu_2^{i_1}]) \\ &= \sum_{i_1 \in C_1} \phi_{nj1}(\mu_1^{i_1}) \sum_{i_2 \in C_{i_1}} \phi_{nj2}(\mu_2^{i_2}) (\phi_{nj3} \times \cdots \times \phi_{njn} \times \theta)([\mu_3^{i_2}]) \\ &= \sum_{i_1 \in C_1} \sum_{i_2 \in C_{i_1}} \phi_{nj1}(\mu_1^{i_2}) \phi_{nj2}(\mu_2^{i_2}) (\phi_{nj3} \times \cdots \times \phi_{njn} \times \theta)([\mu_3^{i_2}]) \\ &\vdots \\ &= \sum_{i_1 \in C_1} \sum_{i_2 \in C_{i_1}} \cdots \sum_{i_n \in C_{i_{n-1}}} \phi_{nj1}(\mu_1^{i_n}) \cdots \phi_{njn}(\mu_n^{i_n}) \theta([\mu_{n+1}^{i_n}]) \end{aligned}$$

$$= \sum_{i \text{ on level } n+1} \phi_{nj1}(\mu_1^i) \cdots \phi_{njn}(\mu_n^i) \theta([\mu_{n+1}^i]).$$

Inserting this and the corresponding result for $(\phi_{njn} \times \cdots \times \phi_{nj1} \times \theta)(\tau)$ in (33), we get (37).

In [7, 15], conditions are derived for a B-series method to be energy preserving when applied to the system (6). In [37], while giving the AVF method as one such method, Quispel and McLaren present a general form of what they call energy-preserving linear combinations of rooted trees:

Here we give their result as a lemma, which is proved later by the proof of the more general Theorem 5.

Lemma 6 *Let $\mu_1, \dots, \mu_n$ be n arbitrary forests. Then, if $f(x) = S \nabla H(x)$ for some skew-symmetric constant matrix S , we have that $F(\omega)(x) \cdot \nabla H(x) = 0$ for*

$$\omega = [\mu_1] \circ [\mu_2] \circ \cdots [\mu_n] \circ [\emptyset] + (-1)^n [\mu_n] \circ [\mu_{n-1}] \circ \cdots [\mu_1] \circ [\emptyset]. \quad (38)$$

There is a connection between (36) and Lemma 6 such that instead of order conditions for every tree, we can calculate order conditions for every energy-preserving linear combination. To see this we start by collecting the leaf nodes, i.e. nodes with no children, of the tree τ in a set I_l and the other nodes in the set I_n . If node $i \in I_n$, we may then use the relation

$$\Lambda(\{\mu_1^i, \dots, \mu_n^i, \mu_{n+1}^i\}) = \theta([\mu_{n+1}^i]) \Lambda(\{\mu_1^i, \dots, \mu_n^i, \emptyset\})$$

to find $\Lambda(\hat{\tau}^i)$ from the previously calculated Λ for a smaller tree. Then if lower order conditions are satisfied, we have numerical values for these Λ . The leaf nodes on the other hand, with their corresponding $\hat{\tau}^i = \{\mu_1^i, \dots, \mu_n^i, \emptyset\}$, gives an energy-preserving linear combination (38) which τ belongs to. If i is on level two, this combination is simply $\tau - \tau = 0$, and accordingly Λ is not calculated for these nodes in (36). Moreover, leaves on the same level have identical $\hat{\tau}^i$. Thus, a tree with leaves on m different levels above level two will belong to at most m non-zero energy-preserving linear combinations (38). For each of these combinations there is a corresponding order condition, with the left hand side given by (37). The right hand side can be found by considering the individual trees.

If we assume the conditions for order $< p$ to be satisfied, we may replace (34) by

$$\sum_{i \in I_l} \Lambda(\hat{\tau}^i) = \frac{1}{\gamma(\tau)} - \hat{e}(\tau) - \sum_{i \in I_n} \frac{\Lambda(\{\mu_1^i, \dots, \mu_n^i, \emptyset\})}{(|\mu_{n+1}^i| + 1) \gamma(\mu_{n+1}^i)}, \quad (39)$$

where $|\mu|$ denotes the number of trees in the forest μ . Note that $\Lambda(\{\emptyset\}) = 1$ and hence $\Lambda(\hat{\tau}^1) = \theta(\tau)$. Then we can calculate the numerical value for the right hand side and, if τ has leaves on only one level > 2 , find an order condition for both τ and the other tree in the combination (38). This warrants an example.

Example 2 Consider again the tree $\tau = \text{V}$, which is part of the energy-preserving linear combination $\omega = \text{V} - \text{Y}$. Ignoring the two nodes on level 2, there are three nodes to calculate Λ for: $i = 1$, $i = 4$ and $i = 5$. We find

$$\begin{aligned}\Lambda(\hat{\tau}^1) &= \frac{1}{(2+1)\gamma(\bullet)\gamma(\text{V})} = \frac{1}{3 \cdot 1 \cdot 6} = \frac{1}{18} \\ \Lambda(\hat{\tau}^4) &= \frac{1}{(1+1)\gamma(\bullet)} \Lambda(\{\bullet, \emptyset, \emptyset\}) \\ &= \frac{1}{2\gamma(\bullet)} \left(\frac{1}{\gamma(\text{V})} - \frac{1}{3\gamma(\bullet)\gamma(\text{V})} \right) = \frac{1}{2} \left(\frac{1}{8} - \frac{1}{6} \right) = -\frac{1}{48}, \\ \Lambda(\hat{\tau}^5) &= \Lambda(\{\bullet, \emptyset, \emptyset\}) = \sum_j b_{3j}(\phi_{3j1}(\bullet) - \phi_{3j3}(\bullet)).\end{aligned}$$

The right hand side of (39) becomes

$$\frac{1}{\gamma(\tau)} - \frac{1}{18} - \left(-\frac{1}{48}\right) = \frac{1}{30} - \frac{1}{18} + \frac{1}{48} = -\frac{1}{720},$$

and we have the order condition (35) for the linear combination $\text{V} - \text{Y}$.

If there are leaves on $r > 1$ different levels above level two, things get slightly more complicated. Then we get r different terms on the left hand side of (39) and we need to consider the order condition for τ and the r trees it forms energy-preserving linear combinations with, so that we get an equation for every energy-preserving combination of these trees, also those not including τ . This is illustrated by the following example.

Example 3 The tree V forms energy-preserving combinations with both Y and W . Thus we have to calculate (39) for all three trees to find order conditions for the corresponding linear combinations. Starting with $\tau = \text{V}$, which has three nodes above level two, two leaves and one non-leaf, we get

$$\begin{aligned}\Lambda(\hat{\tau}^4) &= \Lambda(\{\bullet, \text{V}, \emptyset\}) = \sum_j b_{2j}(\phi_{2j1}(\bullet)\phi_{2j2}(\text{V}) + \phi_{2j2}(\bullet)\phi_{2j1}(\text{V})), \\ \Lambda(\hat{\tau}^5) &= \frac{1}{(1+1)\gamma(\bullet)} \Lambda(\{\bullet, \bullet, \emptyset\}) = \frac{1}{2\gamma(\bullet)} \frac{1}{2} \left(\frac{1}{\gamma(\text{V})} - \frac{1}{3\gamma(\bullet)\gamma(\text{V})} \right)\end{aligned}$$

$$\begin{aligned}
&= \frac{1}{2} \frac{1}{2} \left(\frac{1}{15} - \frac{1}{9} \right) = -\frac{1}{90}, \\
\Lambda(\hat{\tau}^6) &= \Lambda(\{\bullet, \bullet, \emptyset, \emptyset\}) = \sum_j b_{3j} (\phi_{3j1}(\bullet) \phi_{3j2}(\bullet) - \phi_{3j3}(\bullet) \phi_{3j2}(\bullet)) \\
&= \sum_j b_{3j} \phi_{3j2}(\bullet) (\phi_{3j1} - \phi_{3j3})(\bullet).
\end{aligned}$$

For the right hand side of (39), we get

$$\frac{1}{\gamma(\tau)} - \frac{1}{(2+1)\gamma(\bullet)\gamma(\mathfrak{V})} - \left(-\frac{1}{90}\right) = \frac{1}{48} - \frac{1}{3 \cdot 1 \cdot 8} + \frac{1}{90} = -\frac{7}{720},$$

and hence the order condition for $\mathfrak{V}$ is

$$\sum_j b_{2j} (\phi_{2j1}(\bullet) \phi_{2j2}(\mathfrak{I}) + \phi_{2j2}(\bullet) \phi_{2j1}(\mathfrak{I})) + \sum_j b_{3j} \phi_{3j2}(\bullet) (\phi_{3j1} - \phi_{3j3})(\bullet) = -\frac{7}{720}. \quad (40)$$

Similarly we calculate (39) for $\mathfrak{V}$,

$$\sum_{jk} b_{2j} \phi_{2jk}(\mathfrak{V}) - 2 \sum_j b_{3j} \phi_{3j2}(\bullet) (\phi_{3j1} - \phi_{3j3})(\bullet) = -\frac{1}{120}, \quad (41)$$

and for $\mathfrak{V}$,

$$\sum_{jk} b_{2j} \phi_{2jk}(\mathfrak{V}) + 2 \sum_j b_{2j} (\phi_{2j1}(\bullet) \phi_{2j2}(\mathfrak{I}) + \phi_{2j2}(\bullet) \phi_{2j1}(\mathfrak{I})) = -\frac{1}{36}. \quad (42)$$

Combining (40), (41) and (42), we get the equivalent system of equations

$$\sum_j b_{3j} \phi_{3j2}(\bullet) (\phi_{3j1}(\bullet) - \phi_{3j3}(\bullet)) = \frac{1}{240} + \alpha, \quad (43)$$

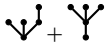
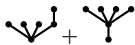
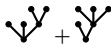
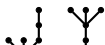
$$\sum_j b_{2j} (\phi_{2j1}(\bullet) \phi_{2j2}(\mathfrak{I}) + \phi_{2j2}(\bullet) \phi_{2j1}(\mathfrak{I})) = -\frac{1}{72} - \alpha, \quad (44)$$

$$\sum_{j,k} b_{2j} \phi_{2jk}(\mathfrak{V}) = 2\alpha, \quad (45)$$

where the choice of $\alpha \in \mathbb{R}$ is arbitrary. The order conditions (43)–(45) can be associated

to the linear combinations $\mathfrak{V} - \mathfrak{V}$, $\mathfrak{V} + \mathfrak{V}$ and $\mathfrak{V} + \mathfrak{V}$, respectively.

Table 2 Energy-preserving linear combinations of elementary differentials, and their associated order conditions for the scheme (31), up to sixth order. The coefficients $\alpha_1, \alpha_2 \in \mathbb{R}$ are arbitrary

$ \tau $	ω	Order condition
1	$\bullet$	—
2	—	—
3		$\sum_j b_{2j} = -\frac{1}{24}$
4		$\sum_{j,k} b_{2j} b_{2jk} \phi_{2jk}(\bullet) = -\frac{1}{24}$
5		$\sum_{j,k} b_{2j} b_{2jk} \phi_{2jk}(\bullet)^2 = -\frac{1}{40}$
		$\sum_j b_{2j} \phi_{2j1}(\bullet) \phi_{2j2}(\bullet) = -\frac{1}{90}$
		$\sum_j b_{3j} (\phi_{3j1}(\bullet) - \phi_{3j3}(\bullet)) = -\frac{1}{720}$
		$\sum_{j,k} b_{2j} b_{2jk} \phi_{2jk}(\mathfrak{I}) = -\frac{1}{60}$
		$\sum_j b_{4j} = \frac{1}{240}$
6		$\sum_{j,k} b_{2j} b_{2jk} \phi_{2jk}(\bullet)^3 = -\frac{1}{60}$
		$\sum_j b_{2j} (\phi_{2j1}(\bullet)^2 \phi_{2j2}(\bullet) + \phi_{2j1}(\bullet) \phi_{2j2}(\bullet)^2) = -\frac{1}{72}$
		$\sum_j b_{3j} (\phi_{3j1}(\bullet)^2 - \phi_{3j3}(\bullet)^2) = -\frac{1}{720}$
		$\sum_{j,k} b_{2j} b_{2jk} \phi_{2jk}(\bullet) \phi_{2jk}(\mathfrak{I}) = -\frac{1}{96}$
		$\sum_j b_{3j} \phi_{3j2}(\bullet) (\phi_{3j1}(\bullet) - \phi_{3j3}(\bullet)) = \frac{1}{240} + \alpha_1$
		$\sum_j b_{4j} (\phi_{4j1}(\bullet) + \phi_{4j4}(\bullet)) = \frac{1}{240}$
		$\sum_j b_{2j} (\phi_{2j1}(\bullet) \phi_{2j2}(\mathfrak{I}) + \phi_{2j1}(\mathfrak{I}) \phi_{2j2}(\bullet)) = -\frac{1}{72} - \alpha_1$
		$\sum_j b_{3j} (\phi_{3j1}(\mathfrak{I}) - \phi_{3j3}(\mathfrak{I})) = -\frac{1}{180} - \alpha_2$
		$\sum_{j,k} b_{2j} b_{2jk} \phi_{2jk}(\mathfrak{V}) = 2\alpha_1$
		$\sum_{j,k} b_{2j} b_{2jk} \phi_{2jk}(\mathfrak{I}) = \alpha_2$
		$\sum_j b_{4j} (\phi_{4j2}(\bullet) + \phi_{4j3}(\bullet)) = -\frac{1}{1440} - \alpha_2$

By considering the order conditions in Table 2, we find a fifth order scheme of the form (31) given by

$$\begin{aligned} \frac{\hat{x} - x}{h} = & \left(I - \frac{5}{136} h^2 (f'(z_2) f'(z_3) + f'(z_3) f'(z_2)) - \frac{1}{102} h^2 f'(x) f'(x) \right. \\ & + \frac{1}{288} h^3 (f'(x) f'(x) f'(z_1) + f'(z_1) f'(x) f'(x)) \\ & \left. + \frac{1}{120} h^4 f'(x) f'(x) f'(x) f'(x) \right) \int_0^1 f((1 - \xi)x + \xi \hat{x}) d\xi, \end{aligned} \quad (46)$$

where

$$z_1 = x + \frac{2}{5} h f(x), \quad z_2 = x + \frac{17 + \sqrt{17}}{30} h f(z_1), \quad z_3 = x + \frac{17 - \sqrt{17}}{30} h f(z_1).$$

A symmetric sixth order scheme is given by

$$\begin{aligned} \frac{\hat{x} - x}{h} = & \left(I - \frac{13}{360} h^2 f'(\bar{x} + \frac{\sqrt{13}}{26} h f(\bar{x} - \frac{3\sqrt{13}}{26} h f(\bar{x}))) f'(\bar{x} - \frac{\sqrt{13}}{26} h f(\bar{x} + \frac{3\sqrt{13}}{26} h f(\bar{x}))) \right. \\ & - \frac{13}{360} h^2 f'(\bar{x} - \frac{\sqrt{13}}{26} h f(\bar{x} + \frac{3\sqrt{13}}{26} h f(\bar{x}))) f'(\bar{x} + \frac{\sqrt{13}}{26} h f(\bar{x} - \frac{3\sqrt{13}}{26} h f(\bar{x}))) \\ & - \frac{1}{180} h^2 f'(x) f'(x) - \frac{1}{180} h^2 f'(\hat{x}) f'(\hat{x}) \\ & + \frac{1}{720} h^3 f'(\bar{x} - \frac{1}{2} h f(\bar{x})) f'(\bar{x}) f'(\bar{x} + \frac{1}{2} h f(\bar{x})) \\ & - \frac{1}{720} h^3 f'(\bar{x} + \frac{1}{2} h f(\bar{x})) f'(\bar{x}) f'(\bar{x} - \frac{1}{2} h f(\bar{x})) \\ & \left. + \frac{1}{120} h^4 f'(\bar{x}) f'(\bar{x}) f'(\bar{x}) f'(\bar{x}) \right) \int_0^1 f((1 - \xi)x + \xi \hat{x}) d\xi, \end{aligned} \quad (47)$$

where $\bar{x} = \frac{x + \hat{x}}{2}$. If we wish to calculate the matrix in front of the integral explicitly, we have a non-symmetric sixth order scheme given by

$$\begin{aligned} \frac{\hat{x} - x}{h} = & \left(I - \frac{13}{360} h^2 (f'(z_6) f'(z_7) + f'(z_7) f'(z_6)) \right. \\ & - \frac{1}{180} h^2 (f'(x) f'(x) + f'(z_1) f'(z_1)) \\ & + \frac{1}{720} h^3 (f'(x) f'(z_2) f'(z_3) - f'(z_3) f'(z_2) f'(x)) \\ & \left. + \frac{1}{120} h^4 f'(z_2) f'(z_2) f'(z_2) f'(z_2) \right) \int_0^1 f((1 - \xi)x + \xi \hat{x}) d\xi, \end{aligned} \quad (48)$$

with

$$\begin{aligned} z_1 &= x + \frac{1}{4}hf(x) + \frac{3}{4}hf\left(x + \frac{2}{3}hf\left(x + \frac{1}{3}hf(x)\right)\right), \quad z_2 = x + \frac{1}{2}hf(x), \\ z_3 &= x + hf(z_2), \quad z_4 = \frac{1}{2}(x + z_3) - \frac{3\sqrt{13}}{26}hf(z_2), \\ z_5 &= \frac{1}{2}(x + z_3) + \frac{3\sqrt{13}}{26}hf(z_2), \\ z_6 &= \frac{1}{2}(x + z_1) + \frac{\sqrt{13}}{26}hf(z_4), \quad z_7 = \frac{1}{2}(x + z_1) - \frac{\sqrt{13}}{26}hf(z_5). \end{aligned}$$

4 AVF discrete gradient methods for general skew-gradient systems

We will now build on the results of the previous section by generalizing the results to the situation where $S(x)$ in the skew-gradient system (4) is not necessarily constant. Consider therefore now an ODE of the form (4), and set again $g := \nabla H$. By Taylor expansion of x around $t = t_0$ we get

$$\begin{aligned} x(t_0 + h) &= x + hSg + \frac{h^2}{2}(S'gSg + Sg'Sg) + \frac{h^3}{6}(S''g(Sg, Sg) + 2S'g'(Sg, Sg) \\ &\quad + Sg''(Sg, Sg) + S'gS'gSg + S'gSg'Sg \\ &\quad + Sg'S'gSg + Sg'Sg'Sg) + \mathcal{O}(h^4), \end{aligned}$$

where $x := x(t_0)$, and S, g and their derivatives are evaluated in x . Introducing the notation $f^\circ := S'g$ and $f^\bullet := Sg'$, we can write this in the abbreviated form

$$\begin{aligned} x(t_0 + h) &= x + hf + \frac{h^2}{2}(f^\circ f + f^\bullet f) + \frac{h^3}{6}(f^{\circ\circ}(f, f) + 2f^{\circ\bullet}(f, f) + f^{\bullet\bullet}(f, f) \\ &\quad + f^\circ f^\circ f + f^\circ f^\bullet f + f^\bullet f^\circ f + f^\bullet f^\bullet f) + \mathcal{O}(h^4). \end{aligned} \quad (49)$$

4.1 Skew-gradient systems and P-series

A *P-series* is given by

$$P(\phi, (x, y)) = \left(\phi(\emptyset)x + \sum_{\tau \in TP_\bullet} \frac{h^{|\tau|}}{\sigma(\tau)} \phi(\tau) F(\tau)(x, y), \right. \\ \left. \phi(\emptyset)y + \sum_{\tau \in TP_\circ} \frac{h^{|\tau|}}{\sigma(\tau)} \phi(\tau) F(\tau)(x, y) \right), \quad (50)$$

where TP is the set of rooted bi-colored trees and $TP_\bullet$ and $TP_\circ$ are the subsets of TP whose roots are black and white, respectively [21, Section III.2]. The bi-colored trees are built recursively; starting with $\bullet$ and $\circ$, we let $\tau = [\tau_1, \dots, \tau_m]_\bullet$ be the tree you get by grafting the roots of $\tau_1, \dots, \tau_m$ to a black root and $\tau = [\tau_1, \dots, \tau_m]_\circ$ the tree you get by grafting $\tau_1, \dots, \tau_m$ to a white root. No subscript, i.e. $\tau = [\tau_1, \dots, \tau_m]$, means grafting to a black root.

The exact solution of a partitioned system

$$\begin{aligned}\dot{x} &= f(x, y), & x(t_0) &= x_0, \\ \dot{y} &= g(x, y), & y(t_0) &= y_0,\end{aligned}\tag{51}$$

can be written as $(x(t_0 + h), y(t_0 + h)) = P(1/\gamma, (x_0, y_0))$, where the coefficient γ is given by $\gamma(\emptyset) = \gamma(\bullet) = \gamma(\circ) = 1$ and (30). As noted in [10], setting $f(x, y) := S(y)\nabla H(x)$, the skew-gradient system (4) can be written as (51) with $g = f$. When $g = f$, all coefficients and the elementary differentials $F(\tau)$ in (50) are given independently of the color of the root. Thus for the system (4), it suffices to consider

$$P(\phi, x) = \phi(\emptyset)x + \sum_{\tau \in TP_\bullet} \frac{h^{|\tau|}}{\sigma(\tau)} \phi(\tau) F(\tau)(x),\tag{52}$$

and we have that the exact solution of (4) can be written as $x(t_0 + h) = P(1/\gamma, x_0)$. Breaking slightly with convention, we define a P-series to be the single row version (52) in the remainder of this paper. Denoting black-rooted subtrees by τ_i and white-rooted subtrees by $\bar{\tau}_i$, the elementary differentials $F(\tau)$ for the skew-gradient system are given recursively by $F(\bullet)(x) = F(\circ)(x) = S(x)\nabla H(x)$, and

$$F(\tau)(x) = S^{(l)} D^m \nabla H(F(\tau_1)(x), \dots, F(\tau_m)(x), F(\bar{\tau}_1)(x), \dots, F(\bar{\tau}_l)(x))\tag{53}$$

for both $\tau = [\tau_1, \dots, \tau_m, \bar{\tau}_1, \dots, \bar{\tau}_l]_\bullet$ and $\tau = [\tau_1, \dots, \tau_m, \bar{\tau}_1, \dots, \bar{\tau}_l]_\circ$. The bi-colored trees in $TP_\bullet$ and their corresponding elementary differentials F are given up to order three in Table 3. The number of trees grows very quickly with the order; see <https://oeis.org/A000151>.

The following lemma is Lemma III.2.2 in [21] amended to fit our setting.

Lemma 7 *Let $P(a, x)$ and $P(b, x)$ be two P-series with $a(\emptyset) = b(\emptyset) = 1$. Then*

$$hS(P(a, x))\nabla H(P(b, x)) = P(a \vee b, x),$$

where $(a \vee b)(\emptyset) = 0$, $(a \vee b)(\bullet) = 1$, and

$$(a \vee b)(\tau) = a(\tau_1) \cdots a(\tau_m) b(\bar{\tau}_1) \cdots b(\bar{\tau}_l) \quad \text{for } \tau = [\tau_1, \dots, \tau_m, \bar{\tau}_1, \dots, \bar{\tau}_l]_\bullet.$$

Proposition 4 *The AVF discrete gradient scheme*

$$\frac{\hat{x} - x}{h} = S\left(\frac{x + \hat{x}}{2}\right) \int_0^1 \nabla H((1 - \xi)x + \xi \hat{x}) d\xi\tag{54}$$

is a second order P-series method.

Table 3 Bi-colored trees and their elementary differentials up to third order

$ \tau $	$F(\tau)^i$	$F(\tau)$	τ	$\alpha(\tau)$	$\gamma(\tau)$	$\sigma(\tau)$
1	$S_j^i g^j$	f	$\bullet$	1	1	1
2	$S_{jk}^i g^j S_l^k g^l$	$f^\circ f$	$\begin{array}{c} \circ \\ \\ \bullet \end{array}$	1	2	1
	$S_j^i g_k^j S_l^k g^l$	$f^\bullet f$	$\begin{array}{c} \bullet \\ \\ \bullet \end{array}$	1	2	1
3	$S_{jkm}^i g^j S_l^k g^l S_n^m g^n$	$f^{\circ\circ}(f, f)$	$\begin{array}{c} \circ \\ \vee \\ \bullet \end{array}$	1	3	2
	$S_{jk}^i g_m^j S_l^k g^l S_n^m g^n$	$f^{\circ\bullet}(f, f)$	$\begin{array}{c} \circ \\ \vee \\ \bullet \end{array}$	2	3	1
	$S_j^i g_{km}^j S_l^k g^l S_n^m g^n$	$f^{\bullet\bullet}(f, f)$	$\begin{array}{c} \bullet \\ \vee \\ \bullet \end{array}$	1	3	2
	$S_{jk}^i g^j S_{lm}^k g^l S_n^m g^n$	$f^\circ f^\circ f$	$\begin{array}{c} \circ \\ \\ \circ \\ \\ \bullet \end{array}$	1	6	1
	$S_j^i g_k^j S_{lm}^k g^l S_n^m g^n$	$f^\bullet f^\circ f$	$\begin{array}{c} \bullet \\ \\ \circ \\ \\ \bullet \end{array}$	1	6	1
	$S_{jk}^i g^j S_l^k g_m^l S_n^m g^n$	$f^\circ f^\bullet f$	$\begin{array}{c} \circ \\ \\ \bullet \\ \\ \bullet \end{array}$	1	6	1
	$S_j^i g_k^j S_l^k g_m^l S_n^m g^n$	$f^\bullet f^\bullet f$	$\begin{array}{c} \bullet \\ \\ \bullet \\ \\ \bullet \end{array}$	1	6	1

Proof As in the proof of Proposition 2, we define $\hat{e}$ by $\hat{e}(\emptyset) = 1$ and $\hat{e}(\tau) = 0$ for all $\tau \neq \emptyset$. Now, assume that the solution $\hat{x}$ of (54) can be written as the P-series $\hat{x} = P(\Phi, x)$. Then, using Lemma 7, we find the P-series

$$\begin{aligned}
 & h S\left(\frac{x + \hat{x}}{2}\right) \int_0^1 \nabla H((1 - \xi)x + \xi \hat{x}) d\xi \\
 &= h S\left(P\left(\frac{1}{2}\hat{e} + \frac{1}{2}\Phi, x\right)\right) \int_0^1 \nabla H(P((1 - \xi)\hat{e} + \xi\Phi, x)) d\xi \\
 &= \int_0^1 h S\left(P\left(\frac{1}{2}\hat{e} + \frac{1}{2}\Phi, x\right)\right) \nabla H(P((1 - \xi)\hat{e} + \xi\Phi, x)) d\xi \\
 &= P\left(\int_0^1 \left(\left(\frac{1}{2}\hat{e} + \frac{1}{2}\Phi\right) \vee ((1 - \xi)\hat{e} + \xi\Phi)\right) d\xi, x\right).
 \end{aligned}$$

Thus we get $\Phi = \hat{e} + \int_0^1 \left(\left(\frac{1}{2}\hat{e} + \frac{1}{2}\Phi\right) \vee ((1 - \xi)\hat{e} + \xi\Phi)\right) d\xi = \hat{e} + \int_0^1 \left(\left(\frac{1}{2}\Phi\right) \vee (\xi\Phi)\right) d\xi$. That is, $\Phi(\emptyset) = 1$, $\Phi(\bullet) = 1$, and

$$\Phi([\tau_1, \dots, \tau_m, \bar{\tau}_1, \dots, \bar{\tau}_l]) = \frac{1}{(m+1)2^l} \Phi(\tau_1) \cdots \Phi(\tau_m) \Phi(\bar{\tau}_1) \cdots \Phi(\bar{\tau}_l).$$

Writing out the first few terms of the series, we have

$$\begin{aligned}
 \hat{x} &= x + hf + \frac{h^2}{2}(f^\circ f + f^\bullet f) + h^3\left(\frac{1}{8}f^{\circ\circ}(f, f) + \frac{1}{4}f^{\circ\bullet}(f, f) + \frac{1}{6}f^{\bullet\bullet}(f, f)\right) \\
 &\quad + \frac{1}{4}f^\circ f^\circ f + \frac{1}{4}f^\circ f^\bullet f + \frac{1}{4}f^\bullet f^\circ f + \frac{1}{4}f^\bullet f^\bullet f + \mathcal{O}(h^4),
 \end{aligned}$$

which, after comparing with the expanded exact solution (49), we see is of order two.

The following lemma is obtained in a manner similar to Lemma 5, i.e. Theorem 2.2 in [34], and hence we present it without its proof.

Lemma 8 *Let $P(a, x)$, $P(b, x)$ and $P(c, x)$ be three P -series with $a(\emptyset) = b(\emptyset) = 1$ and $c(\emptyset) = 0$. Then*

$$h S(P(a, x)) D^2 H(P(b, x)) P(c, x) = P((a, b) \times c, x)$$

with $((a, b) \times c)(\emptyset) = ((a, b) \times c)(\bullet) = 0$ and otherwise

$$((a, b) \times c)(\tau) = \sum_{i=1}^m \prod_{j=1, j \neq i}^m \prod_{k=1}^l a(\bar{\tau}_k) b(\tau_j) c(\tau_i) \quad \text{for } \tau = [\tau_1, \dots, \tau_m, \bar{\tau}_1, \dots, \bar{\tau}_l]. \quad (55)$$

Note that $\{\emptyset\}$ counts as both a black-rooted and a white-rooted tree. Hence we have e.g.

$$((a, b) \times c)(\heartsuit) = a(\circ) b(\emptyset) c(\bullet) = a(\bullet) c(\bullet),$$

where we also use that $a(\circ) = a(\bullet)$.

We now present a subclass of the AVF discrete gradient method, for which we will find order conditions using Lemma 7 and Lemma 8. This subclass is every AVF discrete gradient method for which the approximation of $S(x)$ can be written in the form

$$\begin{aligned} \bar{S}(x, \hat{x}, h) = & \sum_{n=0}^{p-1} h^n \sum_j b_{nj} \left(\prod_{k=1}^n S(\bar{z}_{nj k}) D^2 H(z_{nj k}) \cdot S(\bar{z}_{nj(n+1)}) \right. \\ & \left. + (-1)^n S(\bar{z}_{nj(n+1)}) \prod_{k=1}^n D^2 H(z_{nj(n-k+1)}) S(\bar{z}_{nj(n-k+1)}) \right), \end{aligned} \quad (56)$$

where, if $\hat{x}$ is the solution of

$$\frac{\hat{x} - x}{h} = \bar{S}(x, \hat{x}, h) \bar{\nabla}_{\text{AVF}} H(x, \hat{x}),$$

each $z_{nj k} := z_{nj k}(x, \hat{x}, h) = P(\phi_{nj k}, x)$ and each $\bar{z}_{nj k} := \bar{z}_{nj k}(x, \hat{x}, h) = P(\psi_{nj k}, x)$ can be written as a P -series with $\phi_{nj k}(\emptyset) = \psi_{nj k}(\emptyset) = 1$ for all n, j, k . We require that $\sum_j b_{0j} = \frac{1}{2}$, which ensures that (56) is a consistent approximation of $S(x)$.

Theorem 3 *The discrete gradient scheme (9) with the AVF discrete gradient (10) and the approximation of $S(x)$ given by (56) is a P -series method.*

Proof Generalizing the argument in the proof of Proposition 4, we find the P -series

$$h S(P(a, x)) \int_0^1 \nabla H((1 - \xi)x + \xi \hat{x}) d\xi = P\left(\int_0^1 (a \vee ((1 - \xi)\hat{e} + \xi \Phi)) d\xi, x\right),$$

where $\bar{\theta}(a) := \int_0^1 (a \vee ((1 - \xi)\hat{e} + \xi\Phi)) d\xi = \int_0^1 (a \vee \xi\Phi) d\xi$, so that $\bar{\theta}(a)(\emptyset) = 0$, $\bar{\theta}(a)(\bullet) = 1$, and

$$\bar{\theta}(a)([\tau_1, \dots, \tau_m, \bar{\tau}_1, \dots, \bar{\tau}_l]) = \frac{1}{m+1} \Phi(\tau_1) \cdots \Phi(\tau_m) a(\bar{\tau}_1) \cdots a(\bar{\tau}_l). \quad (57)$$

Thus we may write the solution $\hat{x}$ found from applying the scheme (9) with the AVF discrete gradient (10) and $\bar{S}(x, \hat{x}, h)$ given by (56) as

$$\begin{aligned} \hat{x} &= x + \sum_{n=0}^{p-1} h^n \sum_j b_{nj} \left(\prod_{k=1}^n S(P(\psi_{njk}, x)) D^2 H(P(\phi_{njk}, x)) \cdot P(\bar{\theta}(\psi_{nj(n+1)}), x) \right. \\ &\quad \left. + (-1)^n \prod_{k=1}^n S(P(\psi_{nj(n-k+2)}, x)) D^2 H(P(\phi_{nj(n-k+1)}, x)) \cdot P(\bar{\theta}(\psi_{nj1}), x) \right) \\ &= x + \sum_{n=0}^{p-1} \sum_j b_{nj} (P((\psi_{nj1}, \phi_{nj1}) \times (\psi_{nj2}, \phi_{nj2}) \times \cdots \times (\psi_{njn}, \phi_{njn}) \times \bar{\theta}(\psi_{nj(n+1)}), x) \\ &\quad + (-1)^n P((\psi_{nj(n+1)}, \phi_{njn}) \times (\psi_{njn}, \phi_{nj(n-1)}) \times \cdots \times (\psi_{nj2}, \phi_{nj1}) \times \bar{\theta}(\psi_{nj1}), x)) \\ &= P(\Phi, x), \end{aligned} \quad (58)$$

with

$$\begin{aligned} \Phi &= \hat{e} + \sum_{n=0}^{p-1} \sum_j b_{nj} ((\psi_{nj1}, \phi_{nj1}) \times \cdots \times (\psi_{njn}, \phi_{njn}) \times \bar{\theta}(\psi_{nj(n+1)}) \\ &\quad + (-1)^n (\psi_{nj(n+1)}, \phi_{njn}) \times \cdots \times (\psi_{nj2}, \phi_{nj1}) \times \bar{\theta}(\psi_{nj1})). \end{aligned} \quad (59)$$

Theorem 4 *The AVF discrete gradient method with $\bar{S}$ given by (56) is of order p if and only if*

$$\Phi(\tau) = \frac{1}{\gamma(\tau)} \quad \text{for } |\tau| \leq p. \quad (60)$$

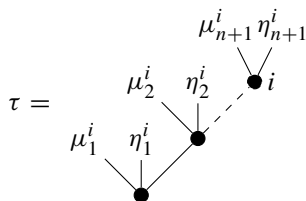
The values $\Phi(\tau)$ can be found from (59) using (55) recursively and then (57). However, a more convenient approach is derived in the next subsection.

4.2 Order conditions

This subsection is devoted to generalizations of the results in Sect. 3.1 to the cases where $S(x)$ is not necessarily constant. To that end, for a tree $\tau \in TP_\bullet$, we cut off all branches between black and white nodes and denote the mono-colored tree we are left with by τ^b . We number the nodes in that tree as before, from 1 to $|\tau^b|$, and reattach the cut-off parts to the tree to get τ again. Let μ denote a forest of black-rooted trees and η a forest of white-rooted trees. Then, for a given node $i \in [1, \dots, |\tau^b|]$ on level $n+1$, there exists a unique set of forests $\hat{\tau}^i = \{(\mu_1^i, \eta_1^i), \dots, (\mu_{n+1}^i, \eta_{n+1}^i)\}$ such that

$$\tau = [(\mu_1^i, \eta_1^i)] \circ [(\mu_2^i, \eta_2^i)] \circ \cdots \circ [(\mu_{n+1}^i, \eta_{n+1}^i)].$$

That is,



Now we can generalize Proposition 3 as follows.

Proposition 5 *The Φ of (59) can be found by*

$$\Phi(\tau) = \hat{e}(\tau) + \sum_{i=1}^{|\tau^b|} \Lambda(\hat{\tau}^i) \quad (61)$$

where $\hat{e}(\emptyset) = 1$ and $\hat{e}(\tau) = 0$ for all $\tau \neq \emptyset$, and

$$\begin{aligned} \Lambda(\hat{\tau}^i) = & \theta([\mu_{n+1}^i]) \sum_j b_{nj} \left(\psi_{nj1}(\eta_1^i) \phi_{nj1}(\mu_1^i) \cdots \psi_{njn}(\eta_n^i) \phi_{njn}(\mu_n^i) \psi_{nj(n+1)}(\eta_{n+1}^i) \right. \\ & \left. + (-1)^n \psi_{nj(n+1)}(\eta_1^i) \phi_{njn}(\mu_1^i) \psi_{njn}(\eta_2^i) \cdots \phi_{nj1}(\mu_n^i) \psi_{nj1}(\eta_{n+1}^i) \right), \end{aligned} \quad (62)$$

with

$$\theta([\tau_1, \dots, \tau_m]) = \frac{1}{m+1} \Phi(\tau_1) \cdots \Phi(\tau_m).$$

Proof Defining n_i and C_i as in the proof of Proposition 3, we have

$$\begin{aligned} [(\mu_{n_i+1}^i, \eta_{n_i+1}^i)] &= [(\mu_{n_k}^k, \eta_{n_k}^k)] \circ [(\mu_{n_k+1}^k, \eta_{n_k+1}^k)] \quad \text{for all } k \in C_i, \\ ((a, b) \times c)([\mu_{n_i+1}^i, \eta_{n_i+1}^i]) &= \sum_{k \in C_i} a(\eta_{n_k}^k) b(\mu_{n_k}^k) c([\mu_{n_k+1}^k, \eta_{n_k+1}^k]). \end{aligned}$$

Observe that $\bar{\theta}(a)([\mu, \eta]) = a(\eta)\theta([\mu])$. For $n = 0$ we have

$$\bar{\theta}(\psi_{0j1})(\tau) = \bar{\theta}(\psi_{0j1})([\mu_1^1, \eta_1^1]) = \psi_{0j1}(\eta_1^1)\theta([\mu_1^1]),$$

and for $n > 0$ we get

$$\begin{aligned} & ((\psi_{nj1}, \phi_{nj1}) \times \cdots \times (\psi_{njn}, \phi_{njn}) \times \bar{\theta}(\psi_{nj(n+1)}))(\tau) \\ &= ((\psi_{nj1}, \phi_{nj1}) \times \cdots \times (\psi_{njn}, \phi_{njn}) \times \bar{\theta}(\psi_{nj(n+1)}))([\mu_1^1, \eta_1^1]) \\ &= \sum_{i_1 \in C_1} \psi_{nj1}(\eta_1^{i_1}) \phi_{nj1}(\mu_1^{i_1}) (\psi_{nj2}, \phi_{nj2}) \\ & \quad \times \cdots \times (\psi_{njn}, \phi_{njn}) \times \bar{\theta}(\psi_{nj(n+1)})([\mu_2^{i_1}, \eta_2^{i_1}]) \\ & \quad \vdots \end{aligned}$$

$$\begin{aligned}
&= \sum_{i_1 \in C_1} \cdots \sum_{i_n \in C_{i_{n-1}}} \psi_{nj1}(\eta_1^{i_n}) \phi_{nj1}(\mu_1^{i_n}) \cdots \psi_{njn}(\eta_n^{i_n}) \phi_{njn}(\mu_n^{i_n}) \bar{\theta}(\psi_{nj(n+1)})([\mu_{n+1}^{i_n}, \eta_{n+1}^{i_n}]) \\
&= \sum_{i \text{ on level } n+1} \psi_{nj1}(\eta_1^i) \phi_{nj1}(\mu_1^i) \cdots \psi_{njn}(\eta_n^i) \phi_{njn}(\mu_n^i) \psi_{nj(n+1)}(\eta_{n+1}^i) \theta([\mu_{n+1}^i]).
\end{aligned}$$

Inserting this and the corresponding result for $((\psi_{nj(n+1)}, \phi_{njn}) \times \cdots \times (\psi_{nj2}, \phi_{nj1}) \times \bar{\theta}(\psi_{nj1}))(\tau)$ in (59), we get (62).

Note that if τ only has black nodes, we have $\Lambda(\hat{\tau}^1) = \theta(\tau) \sum_j b_{0j}(\psi_{0j1}(\emptyset) + \psi_{0j1}(\emptyset)) = \theta(\tau)$, and also $\Lambda(\hat{\tau}^i) = 0$ for all nodes i on level 2. Thus (61) simplifies to (36).

Like for the constant S case, the order conditions can be given for energy-preserving linear combinations of elementary differentials instead for each elementary differential. In the following generalization of Lemma 6, we state that the energy-preserving linear combinations of bi-colored rooted trees are given by

$$\omega = \text{left tree} + (-1)^n \text{right tree}$$

Theorem 5 Let $\mu_1, \mu_2, \dots, \mu_n$ be arbitrary forests of black-rooted trees and $\eta_1, \eta_2, \dots, \eta_{n+1}$ arbitrary forests of white-rooted trees. Given $f(x) = S(x) \nabla H(x)$, where $S(x)$ is a skew-symmetric matrix, and elementary differentials defined by (53), the linear combinations of trees given by

$$\omega = [(\mu_1, \eta_1)] \circ \cdots \circ [(\mu_n, \eta_n)] \circ [\eta_{n+1}] + (-1)^n [(\mu_n, \eta_{n+1})] \circ \cdots \circ [(\mu_1, \eta_2)] \circ [\eta_1] \quad (63)$$

are energy-preserving in the sense that $F(\omega)(x) \cdot \nabla H(x) = 0$.

Proof For any forest of black-rooted trees μ_j , we have $F([\mu_j] \circ [\emptyset]) = S B_j S \nabla H$ for some symmetric matrix B_j , suppressing the argument x . Similarly, for a forest of white-rooted trees η_j , we have $F([\eta_j]) = W_j \nabla H$ for some skew-symmetric matrix W_j . Note that the empty forest is considered both a black-rooted and a white-rooted forest, and accordingly we have $F([\emptyset] \circ [\emptyset]) = F(\mathbf{1}) = S(D^2 H) S \nabla H$ and $F([\emptyset]) = F(\bullet) = S \nabla H$. For these matrices B_j and W_j corresponding to the forests μ_j and η_j , we get

$$F([(\mu_1, \eta_1)] \circ \cdots \circ [(\mu_n, \eta_n)] \circ [\eta_{n+1}]) = W_1 B_1 W_2 B_2 \cdots B_n W_{n+1} \nabla H.$$

We have

$$(W_1 B_1 W_2 B_2 \cdots B_n W_{n+1})^T = \begin{cases} -W_{n+1} B_n W_n B_{n-1} \cdots B_1 W_1 & \text{if } n \text{ even,} \\ W_{n+1} B_n W_n B_{n-1} \cdots B_1 W_1 & \text{if } n \text{ odd.} \end{cases}$$

Thus $F(\omega)(x)$ is a skew-symmetric matrix times $\nabla H(x)$, and the statement in the above theorem follows directly.

For bi-colored trees, we define a node on the tree τ to be a leaf if it is a leaf on the corresponding cut tree τ^b by the definition of leaves given in the previous section. We let I_l be the set of leaves and I_n the set of non-leaf nodes which are also in τ^b , so that $I_l \cup I_n = [1, \dots, |\tau^b|]$. In contrast to the case with mono-colored trees, a leaf i on level one or two of a bi-colored tree may give rise to a non-zero energy-preserving linear combination; it does so if and only if $\eta_k^i \neq \emptyset$ for any $k = 1, 2$. Accordingly, $\Lambda(\hat{\tau}^i)$ is calculated in (61) also when $n = 0, 1$. Furthermore, two leaves i and j on the same level will belong to two different energy-preserving combinations if $\eta_{n+1}^i \neq \eta_{n+1}^j$. Therefore we now simply state that a tree with r leaves, also including the lower two levels, belong to at most r non-zero linear combinations. We thus get r terms on the left hand side of

$$\sum_{i \in I_l} \Lambda(\hat{\tau}^i) = \frac{1}{\gamma(\tau)} - \hat{e}(\tau) - \sum_{i \in I_n} \frac{\Lambda(\{(\mu_1^i, \eta_1^i), \dots, (\mu_n^i, \eta_n^i), (\emptyset, \eta_{n+1}^i)\})}{(|\mu_{n+1}^i| + 1)\gamma(\mu_{n+1}^i)}, \quad (64)$$

which is equivalent to (60) if we assume the conditions for lower order to be satisfied.

Example 4 Consider the tree $\tau = \mathbf{V}^{\circ}$, which is part of the energy-preserving linear combination $\mathbf{V}^{\circ} - \mathbf{V}^{\circ}$. Assume that the order conditions up to and including order three are all satisfied. The cut tree $\tau^b = \mathbf{V}$ has three nodes of which two are leaves. Node number 2 is a leaf on level 2 with $\eta_1^2 = \eta_2^2 = \emptyset$, and thus gives $\Lambda(\hat{\tau}^2) = 0$. We find for the other two,

$$\begin{aligned} \Lambda(\hat{\tau}^1) &= \Lambda(\{((\bullet, \mathbf{V}^{\circ}), \emptyset)\}) = \frac{1}{(|\mu_1^1| + 1)\gamma(\mu_1^1)} \Lambda(\{(\emptyset, \emptyset)\}) = \frac{1}{(2 + 1)\gamma(\bullet)\gamma(\mathbf{V}^{\circ})} \frac{1}{\gamma(\bullet)} = \frac{1}{6}, \\ \Lambda(\hat{\tau}^3) &= \Lambda(\{(\bullet, \emptyset), (\emptyset, \circ)\}) = \sum_j b_{1j} \phi_{1j1}(\bullet) \psi_{1j2}(\circ) - \psi_{1j1}(\circ) \phi_{1j1}(\bullet) \\ &= \sum_j b_{1j} \phi_{1j1}(\bullet) (\psi_{1j2} - \psi_{1j1})(\circ). \end{aligned}$$

For the right hand side of (64) we get

$$\frac{1}{\gamma(\tau)} - \Lambda(\hat{\tau}^1) = \frac{1}{8} - \frac{1}{6} = -\frac{1}{24},$$

and thus the order condition

$$\sum_j b_{1j} \phi_{1j1}(\bullet) (\psi_{1j2} - \psi_{1j1})(\circ) = -\frac{1}{24}$$

for the energy-preserving linear combination $\mathbf{V}^{\circ} - \mathbf{V}^{\circ}$.

Even though the number of black-rooted bi-colored trees grows very quickly, e.g. to 26 for $|\tau| = 4$ and 107 for $|\tau| = 5$, finding and satisfying the order conditions is not as daunting a task as it might first appear. First of all, it suffices to find order conditions for the non-zero linear combinations given by (63). Moreover, a couple key observations simplifies the process further:

- The large number of trees τ for which $\tau^b = \bullet$, i.e. trees with no black nodes on level 2, are all energy-preserving. They can be written $\tau = [\eta_1^1]$, and their order condition is given by

$$2 \sum_j b_{0j} \psi_{0j1}(\eta_1^1) = \frac{1}{\gamma(\tau)}.$$

- For trees that are identical except for the colors of the descendants of white nodes, it suffices to calculate one order condition. E.g. for  we have the order condition $2b_{0j} \psi_{0j1}(\mathfrak{V}) = \frac{1}{12}$, where each of the gray nodes may be black or white. To satisfy these conditions, it is natural to require that $\bar{z}_{0j1}$ in (56) is a B-series up to order $p - 1$.

From the order conditions displayed in Table 4 we find that one second order scheme is given by (9) using the AVF discrete gradient (10) and an explicit skew-symmetric approximation of S given by $\bar{S}(x, \cdot, h) = S(x + \frac{1}{2}hf(x))$. A third order scheme is obtained if we instead use the skew-symmetric approximation of S explicitly given by

$$\begin{aligned} \bar{S}(x, \cdot, h) = & \frac{1}{4}S(x) + \frac{3}{4}S(z_2) + \frac{1}{4}h \left(S(z_1)D^2H(x)S(x) - S(x)D^2H(x)S(z_1) \right) \\ & - \frac{1}{12}h^2 S(x)D^2H(x)S(x)D^2H(x)S(x), \end{aligned} \quad (65)$$

where $z_1 = x + \frac{1}{3}hf(x)$, $z_2 = x + \frac{2}{3}hf(z_1)$.

A symmetric fourth order scheme is given by (9) using the AVF discrete gradient (10) and the skew-symmetric approximation of S

$$\begin{aligned} \bar{S}(x, \hat{x}, h) = & \frac{1}{2}S\left(\bar{x} - \frac{1}{\sqrt{12}}hf(\bar{x}) + \frac{1}{\sqrt{12}}hf(\bar{x})\right) \\ & + \frac{1}{2}S\left(\bar{x} + \frac{1}{\sqrt{12}}hf(\bar{x}) - \frac{1}{\sqrt{12}}hf(\bar{x})\right) \\ & + \frac{1}{2}h S\left(\bar{x} + \frac{1}{12}hf(\bar{x})\right)D^2H(\bar{x})S\left(\bar{x} - \frac{1}{12}hf(\bar{x})\right) \\ & - \frac{1}{2}h S\left(\bar{x} - \frac{1}{12}hf(\bar{x})\right)D^2H(\bar{x})S\left(\bar{x} + \frac{1}{12}hf(\bar{x})\right) \\ & - \frac{1}{12}h^2 S(\bar{x})D^2H(\bar{x})S(\bar{x})D^2H(\bar{x})S(\bar{x}), \end{aligned} \quad (66)$$

Table 4 Linear combinations ω of bi-colored black-rooted trees corresponding to energy-preserving elementary differentials of $f(x) = S(x)\nabla H(x)$, where $S(x)$ is a skew-symmetric matrix, as well as their associated order conditions for the discrete gradient method (9) with the AVF discrete gradient (10) and $\bar{S}(x, \hat{x}, h)$ given by (56)

$ \tau $	ω	Order condition
1		$2 \sum_j b_{0j} = 1$
2		$2 \sum_j b_{0j} \psi_{0j1}(\bullet) = \frac{1}{2}$
3		$2 \sum_j b_{0j} \psi_{0j1}(\bullet)^2 = \frac{1}{3}$
4		$\sum_j b_{2j} = -\frac{1}{24}$
		$2 \sum_j b_{0j} \psi_{0j1}(\mathbf{1}) = \frac{1}{6}$
		$2 \sum_j b_{0j} \psi_{0j1}(\mathbf{1}) = \frac{1}{6}$
		$\sum_j b_{1j} (\psi_{1j2} - \psi_{1j1})(\bullet) = -\frac{1}{12}$
		$2 \sum_j b_{0j} \psi_{0j1}(\bullet)^3 = \frac{1}{4}$
		$2 \sum_j b_{2j} \psi_{2j2}(\bullet) = -\frac{1}{24}$
		$2 \sum_j b_{0j} \psi_{0j1}(\mathbf{V}) = \frac{1}{12}$
		$2 \sum_j b_{0j} \psi_{0j1}(\mathbf{V}) = \frac{1}{12}$
		$2 \sum_j b_{0j} \psi_{0j1}(\mathbf{V}) = \frac{1}{12}$
		$2 \sum_j b_{0j} \psi_{0j1}(\bullet) \psi_{0j1}(\mathbf{1}) = \frac{1}{8}$
		$2 \sum_j b_{0j} \psi_{0j1}(\bullet) \psi_{0j1}(\mathbf{1}) = \frac{1}{8}$
		$\sum_j b_{2j} (\phi_{2j1} + \phi_{2j2})(\bullet) = -\frac{1}{24}$
		$\sum_j b_{1j} (\psi_{1j2}(\bullet)^2 - \psi_{1j1}(\bullet)^2) = -\frac{1}{12}$
		$\sum_j b_{1j} \phi_{1j1}(\bullet) (\psi_{1j1} - \psi_{1j0})(\bullet) = -\frac{1}{24}$
		$2 \sum_j b_{0j} \psi_{0j1}(\mathbf{1}) = \frac{1}{24}$
		$2 \sum_j b_{0j} \psi_{0j1}(\mathbf{1}) = \frac{1}{24}$
		$2 \sum_j b_{0j} \psi_{0j1}(\mathbf{1}) = \frac{1}{24}$
		$2 \sum_j b_{0j} \psi_{0j1}(\mathbf{1}) = \frac{1}{24}$
		$2 \sum_j b_{0j} \psi_{0j1}(\mathbf{1}) = \frac{1}{24}$
		$\sum_j b_{2j} (\psi_{2j1} + \psi_{2j3})(\bullet) = -\frac{1}{24}$
		$\sum_j b_{1j} (\psi_{1j2} - \psi_{1j1})(\mathbf{1}) = -\frac{1}{24}$
		$\sum_j b_{1j} (\psi_{1j2} - \psi_{1j1})(\mathbf{1}) = -\frac{1}{24}$

where $\bar{x} = \frac{x+\hat{x}}{2}$. Another fourth order scheme is obtained if we instead use the explicit skew-symmetric approximation of S found by

$$\begin{aligned}\bar{S}(x, \cdot, h) &= \frac{1}{2}(S(z_5 + z_6) + S(z_5 - z_6)) \\ &\quad + \frac{1}{12}h(S(z_2)D^2H(z_1)S(x) - S(x)D^2H(z_1)S(z_2)) \\ &\quad - \frac{1}{12}h^2S(z_1)D^2H(z_1)S(z_1)D^2H(z_1)S(z_1),\end{aligned}\quad (67)$$

where

$$\begin{aligned}z_1 &= x + \frac{1}{2}hf(x), \quad z_3 = x + hf(z_2), \quad z_5 = \frac{1}{3}(x + z_1 + z_2) + \frac{1}{12}(-z_3 + z_4), \\ z_2 &= x + hf(z_1), \quad z_4 = x + hf(z_3), \quad z_6 = \frac{\sqrt{3}}{36}(7x - 2z_1 - 4z_2 + z_3 - 2z_4).\end{aligned}$$

5 Order conditions for general discrete gradient methods

We will now generalize the results of the two previous sections to discrete gradient methods with a general discrete gradient, as defined by (7)–(8). To that end, we introduce two new series in the vein of B- and P-series, as well as related tree structures.

5.1 The constant S case

Consider mono-colored rooted trees whose nodes can have two different shapes: the circle shape of the nodes in trees of B-series, but also a triangle shape. Let TG be the set of such trees whose leaves are always circles. That is, from the first tree $\bullet$, every tree $\tau \in TG$ can be built recursively through

$$[\tau_1, \dots, \tau_m]_{\bullet}, \quad [\tau_1, \dots, \tau_m]_{\blacktriangle}, \quad \tau_1, \dots, \tau_m \in TG,$$

which denotes the grafting of the trees $\tau_1, \dots, \tau_m$ to a root $\bullet$ or $\blacktriangle$, respectively. The elementary differentials $F(\tau)$ corresponding to a tree $\tau \in TG$ are likewise defined recursively by $F(\bullet)(x) = f(x) = S\nabla H(x)$ and

$$F(\tau)(x) = \begin{cases} SD^m \nabla H(x)(F(\tau_1)(x), \dots, F(\tau_m)(x)) & \text{for } \tau = [\tau_1, \dots, \tau_m]_{\bullet}, \\ SD_2^{m-1} Q(x, x)(F(\tau_1)(x), \dots, F(\tau_m)(x)) & \text{for } \tau = [\tau_1, \dots, \tau_m]_{\blacktriangle}. \end{cases}$$

We can then define a generalization of B-series which includes these elementary differentials.

Definition 4 A G-series is a formal series of the form

$$G(\phi, x) = \phi(\emptyset)x + \sum_{\tau \in TG} \frac{h^{|\tau|}}{\sigma(\tau)} \phi(\tau) F(\tau)(x), \quad (68)$$

where $\phi : TG \cup \{\emptyset\} \rightarrow \mathbb{R}$ is an arbitrary mapping, and the symmetry coefficient σ is given by (28).

The G-series of the exact solution is given by $x(t_0 + h) = G(\xi, x(t_0))$, with

$$\xi(\tau) = \begin{cases} \frac{1}{\gamma(\tau)} & \text{if } \tau \in T, \\ 0 & \text{otherwise.} \end{cases} \quad (69)$$

For use in the remainder of this paper, we generalize the Butcher product by the definition

$$u \circ v = [u_1, \dots, u_m, v]_{\star}, \quad \text{for } u = [u_1, \dots, u_m]_{\star}, \quad \star \in \{\bullet, \blacktriangle\}.$$

Furthermore, we let $|\tau|$ denote the total number of nodes in τ , and $|\tau|_{\star}$ the number of nodes of type $\star$. Let SG be the set of tall trees in TG ; that is, the set of trees with only one node on each level. For a tree $\tau \in TG$, number every tree from 1 to $|\tau|$, as before. For any node i on level $n + 1$, we define the stem $s^i \in SG$ to be the tall tree consisting of the nodes connecting the root to node i , including the root and node i . Denote the j^{th} node of s^i by s^i_j , so that s^i_1 is the root and $s^i_{n+1} = i$. Then we have a unique set of forests $\hat{\tau}^i = \{\mu^i_1, \dots, \mu^i_{n+1}\}$ such that

$$\tau = [\mu^i_1]_{s^i_1} \circ [\mu^i_2]_{s^i_2} \circ \dots \circ [\mu^i_{n+1}]_{s^i_{n+1}}.$$

That is,

$$\tau = \begin{array}{c} \mu^i_{n+1} \\ | \\ \mu^i_2 \quad s^i_{n+1} \\ \diagdown \quad \diagup \\ \mu^i_1 \quad s^i_2 \\ \diagdown \quad \diagup \\ s^i_1 \end{array}$$

The following lemma is a generalization of Lemma 5 to G-series. Its proof is very similar to the proof of [34, Theorem 2.2], and hence omitted.

Lemma 9 *Let $G(a, x)$ and $G(b, x)$ be two G-series with $a(\emptyset) = 1$ and $b(\emptyset) = 0$. Then the G-series $hSD^2H(G(a, x))G(b, x) = G(a \times b, x)$ is given by $(a \times b)(\emptyset) = (a \times b)(\bullet) = 0$ and otherwise*

$$(a \times b)(\tau) = \begin{cases} \sum_{i=1}^m \prod_{j=1, j \neq i}^m a(\tau_j) b(\tau_i) & \text{for } \tau = [\tau_1, \dots, \tau_m]_{\bullet}, \\ 0 & \text{for } \tau = [\tau_1, \dots, \tau_m]_{\blacktriangle}. \end{cases}$$

Moreover, $hSQ(x, G(a, x))G(b, x) = G(a \otimes b, x)$, with $(a \otimes b)(\emptyset) = (a \otimes b)(\bullet) = 0$ and otherwise

$$(a \otimes b)(\tau) = \begin{cases} 0 & \text{for } \tau = [\tau_1, \dots, \tau_m]_{\bullet}, \\ \sum_{i=1}^m \prod_{j=1, j \neq i}^m a(\tau_j) b(\tau_i) & \text{for } \tau = [\tau_1, \dots, \tau_m]_{\blacktriangle}. \end{cases}$$

To every stem $s \in SG$ of height $n + 1 = |s|$, we associate coefficients b_{sj} and ϕ_{sjk} . Letting s_k be the k^{th} node of s , we define the function

$$R(\phi_{sjk}, x) := \begin{cases} D^2 H(G(\phi_{sjk}, x)) & \text{if } s_k = \bullet, \\ Q(x, G(\phi_{sjk}, x)) & \text{if } s_k = \blacktriangle. \end{cases}$$

Then we have $hSR(\phi_{sjk}, x)G(b, x) = G(\phi_{sjk} \diamond b)$, with $(\phi_{sjk} \diamond b)(\emptyset) = (\phi_{sjk} \diamond b)(\bullet) = 0$ and

$$(\phi_{sjk} \diamond b)(\tau) = \begin{cases} \sum_{i=1}^m \prod_{j=1, j \neq i}^m \phi_{sjk}(\tau_j) b(\tau_i) & \text{for } \tau = [\tau_1, \dots, \tau_m]_{s_k}, \\ 0 & \text{if root of } \tau \neq s_k. \end{cases}$$

Consider now the class of skew-symmetric and consistent approximations to S that can be written in the form

$$\bar{S}(x, \hat{x}, h) = \sum_{s \in SG} h^n \sum_j b_{sj} \left(\prod_{k=1}^n SR(\phi_{sjk}, x) + (-1)^{|s| \bullet - 1} \prod_{k=1}^n SR(\phi_{sj(n-k+1)}, x) \right) S \quad (70)$$

whenever $\hat{x}$ is the solution of

$$\frac{\hat{x} - x}{h} = \bar{S}(x, \hat{x}, h) \bar{\nabla} H(x, \hat{x}),$$

with $\phi_{sjk}(\emptyset) = 1$ for every s, j, k , and with $\sum_j b_{\bullet j} = \frac{1}{2}$.

Lemma 10 *The discrete gradient method (9) with $\bar{S}(x, \hat{x}, h)$ given by (70) and $\bar{\nabla} H \in C^\infty(\mathbb{R}^d \times \mathbb{R}^d, \mathbb{R}^d)$ is a G-series method when applied to a constant S skew-gradient system (6).*

Proof Assume that the solution $\hat{x}$ of (9) with $\bar{S}(x, \hat{x}, h)$ given by (70) can be written as the G-series $\hat{x} = G(\Phi, x)$. Then, using Lemma 2 and $\bar{\nabla} H(x, x) = \nabla H(x)$,

$$\begin{aligned} hS\bar{\nabla} H(x, \hat{x}) &= hS \sum_{m=0}^{\infty} \frac{1}{m!} D_2^m \bar{\nabla} H(x, x) (G(\Phi, x) - x)^m \\ &= hS \sum_{m=0}^{\infty} \frac{1}{(m+1)!} D^m \nabla H(x) (G(\Phi, x) - x)^m \\ &\quad - hS \sum_{m=1}^{\infty} \frac{2m}{(m+1)!} D_2^{m-1} Q(x, x) (G(\Phi, x) - x)^m. \end{aligned}$$

Arguing as in the proof of Lemma III.1.9 in [21], we get $h S \bar{\nabla} H(x, \hat{x}) = G(\theta, x)$, with $\theta(\emptyset) = 0$, $\theta(\bullet) = 1$, and

$$\begin{aligned}\theta([\tau_1, \dots, \tau_m]_{\bullet}) &= \frac{1}{m+1} \Phi(\tau_1) \cdots \Phi(\tau_m), \\ \theta([\tau_1, \dots, \tau_m]_{\blacktriangle}) &= \frac{-2m}{m+1} \Phi(\tau_1) \cdots \Phi(\tau_m).\end{aligned}\quad (71)$$

Then we can write (9) with $\bar{S}(x, \hat{x}, h)$ given by (70) as

$$\begin{aligned}\hat{x} &= x + \sum_{s \in SG} h^n \sum_j b_{sj} \left(\prod_{k=1}^n SR(\phi_{sjk}, x) + (-1)^{|s|\bullet-1} \prod_{k=1}^n SR(\phi_{sj(n-k+1)}, x) \right) G(\theta, x) \\ &= x + G(\theta, x) \\ &\quad + \sum_{s \in SG, n>0} \sum_j b_j^s (G(\phi_{sj1} \diamond \cdots \diamond \phi_{sjn} \diamond \theta, x) \\ &\quad + (-1)^{|s|\bullet-1} G(\phi_{sjn} \diamond \cdots \diamond \phi_{sj1} \diamond \theta, x)) \\ &= G(\Phi, x),\end{aligned}$$

with

$$\Phi = \hat{e} + \theta + \sum_{s \in SG, n>0} \sum_j b_{sj} \left(\phi_{sj1} \diamond \cdots \diamond \phi_{sjn} \diamond \theta + (-1)^{|s|\bullet-1} \phi_{sjn} \diamond \cdots \diamond \phi_{sj1} \diamond \theta \right). \quad (72)$$

Theorem 6 *The discrete gradient method (9) with $\bar{S}(x, \hat{x}, h)$ given by (70) and $\bar{\nabla} H \in C^\infty(\mathbb{R}^d \times \mathbb{R}^d, \mathbb{R}^d)$ is of order p if and only if*

$$\Phi(\tau) = \xi(\tau) \quad \text{for } |\tau| \leq p, \quad (73)$$

where Φ is given by (72) and the ξ is given by (69).

We remark that $\bar{\nabla} H \in C^\infty(\mathbb{R}^d \times \mathbb{R}^d, \mathbb{R}^d)$ is a necessary condition for the method to be a G-series method for all S and H , but not for its order; $\bar{\nabla} H \in C^{p-1}(\mathbb{R}^d \times \mathbb{R}^d, \mathbb{R}^d)$ is sufficient for the scheme to be of order p . The following proposition is presented without its proof, which follows along the lines of the proof of Proposition 3.

Proposition 6 *The Φ of (73) satisfies*

$$\Phi(\tau) = \hat{e}(\tau) + \theta(\tau) + \sum_{i \text{ s.t. } n \geq 1} \Lambda(\hat{\tau}^i, s^i) \quad (74)$$

where $\hat{e}(\emptyset) = 1$ and $\hat{e}(\tau) = 0$ for all $\tau \neq \emptyset$, θ is given by (71), and

$$\begin{aligned}\Lambda(\hat{\tau}^i, s^i) &= \theta([\mu_{n+1}^i]_{s_{n+1}^i}) \left(\sum_j b_{sj} \phi_{sj1}(\mu_1^i) \cdots \phi_{sjn}(\mu_n^i) \right. \\ &\quad \left. + (-1)^{|s|\bullet-1} \sum_j b_{\hat{s}^i j} \phi_{\hat{s}^i jn}(\mu_1^i) \cdots \phi_{\hat{s}^i j1}(\mu_n^i) \right),\end{aligned}\quad (75)$$

with $\hat{s}^i$ given by $\hat{s}_k^i = s_{n-k+1}^i$ for $k = 1, \dots, n$, and $\hat{s}_{n+1}^i = s_{n+1}^i$.

As for the AVF method, one does not need to find the order conditions for every tree; it suffices to find the order condition for each energy-preserving linear combination of the form

$$\omega = [\mu_1]_{s_1} \circ [\mu_2]_{s_2} \circ \dots \circ [\mu_n]_{s_n} \circ [\emptyset]_{\bullet} + (-1)^n [\mu_n]_{s_n} \circ [\mu_{n-1}]_{s_{n-1}} \circ \dots \circ [\mu_1]_{s_1} \circ [\emptyset]_{\bullet}. \quad (76)$$

The above does not give every energy-preserving linear combination of the elementary differentials of G-series; it gives the combinations one gets in the scheme (9) with $\bar{S}(x, \hat{x}, h)$ given by (70). Now, let again I_l and I_n denote the sets of leaf nodes and non-leaf nodes, respectively. If we assume the conditions for order $< p$ to be satisfied, we have an equivalent order condition to (73) by

$$\sum_{i \in I_l} \Lambda(\hat{\tau}^i, s^i) = \xi(\tau) - \hat{e}(\tau) - \sum_{i \in I_n} \Lambda(\hat{\tau}^i, s^i), \quad (77)$$

where we may use the relation

$$\Lambda(\{\mu_1^i, \dots, \mu_n^i, \mu_{n+1}^i\}, s^i) = \hat{\theta}([\mu_{n+1}^i]_{s_{n+1}^i}) \Lambda(\{\mu_1^i, \dots, \mu_n^i, \emptyset\}, \bar{s}^i)$$

to calculate $\Lambda(\hat{\tau}^i)$ for $i \in I_n$. Here $\bar{s}^i$ is s^i with s_{n+1}^i replaced by $\bullet$, and $\hat{\theta}(\emptyset) = 0$, $\hat{\theta}(\bullet) = 1$, and

$$\begin{aligned} \hat{\theta}([\tau_1, \dots, \tau_m]_{\bullet}) &= \frac{1}{m+1} \xi(\tau_1) \cdots \xi(\tau_m), \\ \hat{\theta}([\tau_1, \dots, \tau_m]_{\blacktriangle}) &= \frac{-2m}{m+1} \xi(\tau_1) \cdots \xi(\tau_m). \end{aligned} \quad (78)$$

Note that $\Lambda(\hat{\tau}^1, s^1) = \hat{\theta}(\tau)$.

From the order conditions in Table 5, we can find an $\bar{S}(x, \hat{x}, h)$ so that (9) becomes a fourth order scheme for any $\bar{\nabla}H \in C^3(\mathbb{R}^d \times \mathbb{R}^d, \mathbb{R}^d)$. For instance, the stem $s = \mathbf{1}$ has the related order conditions $\sum_j b_{sj} = \frac{1}{2}$ and $\sum_{j,k} b_{sj} \phi_{sjk}(\bullet) = \frac{2}{3}$, which sets the requirements for the term

$$h^2 \sum_j b_{sj} (SQ(x, z_{1j}) SQ(x, z_{2j}) + SQ(x, z_{2j}) SQ(x, z_{1j})) S.$$

Table 5 Energy-preserving linear combinations of the form (76) and their associated order conditions for the discrete gradient method (9) with $\bar{S}(x, \hat{x}, h)$ given by (70)

$ \tau $	ω	s	Order condition
1	$\bullet$	$\bullet$	$\sum_j b_{sj} = \frac{1}{2}$
2	$\mathbf{I}$	$\mathbf{I}$	$\sum_j b_{sj} = \frac{1}{2}$
3	$\mathbf{V}$	$\mathbf{I}$	$\sum_j b_{sj} \phi_{sj1}(\bullet) = \frac{1}{3}$
	$\mathbf{I}$	$\mathbf{I}$	$\sum_j b_{sj} = \frac{1}{2}$
	$\mathbf{I} - \mathbf{I}$	$\mathbf{I}$	$\sum_j b_{sj} - \sum_j b_{\bar{s}j} = 0$
	$\mathbf{I}$	$\mathbf{I}$	$\sum_j b_{sj} = -\frac{1}{24}$
4	$\mathbf{V}$	$\mathbf{I}$	$\sum_j b_{sj} \phi_{sj1}(\bullet)^2 = \frac{1}{4}$
	$\mathbf{V}$	$\mathbf{I}$	$\sum_j b_{sj} \phi_{sj1}(\mathbf{I}) = 0$
	$\mathbf{V}$	$\mathbf{I}$	$\sum_j b_{sj} \phi_{sj1}(\mathbf{I}) = \frac{1}{6}$
	$\mathbf{V} + \mathbf{V}$	$\mathbf{I}$	$\sum_{j,k} b_{sj} \phi_{sjk}(\bullet) = \frac{2}{3}$
	$\mathbf{V} - \mathbf{V}$	$\mathbf{I}$	$\sum_j b_{sj} \phi_{sj2}(\bullet) - \sum_j b_{\bar{s}j} \phi_{\bar{s}j1}(\bullet) = 0$
	$\mathbf{V} - \mathbf{V}$	$\mathbf{I}$	$\sum_j b_{sj} \phi_{sj2}(\bullet) - \sum_j b_{\bar{s}j} \phi_{\bar{s}j1}(\bullet) = 0$
	$\mathbf{V} + \mathbf{V}$	$\mathbf{I}$	$\sum_{j,k} b_{sj} \phi_{sjk}(\bullet) = -\frac{1}{24}$
	$\mathbf{I}$	$\mathbf{I}$	$\sum_j b_{sj} = \frac{1}{2}$
	$\mathbf{I} - \mathbf{I}$	$\mathbf{I}$	$\sum_j b_{sj} - \sum_j b_{\bar{s}j} = 0$
	$\mathbf{I} + \mathbf{I}$	$\mathbf{I}$	$\sum_j b_{sj} - \sum_j b_{\bar{s}j} = 0$
	$\mathbf{I}$	$\mathbf{I}$	$\sum_j b_{sj} = 0$

Choosing $b_{s1} = \frac{1}{2}$ and $z_{11} = z_{21} = x + \frac{2}{3}hf(x)$, we have fulfilled these conditions. Likewise, finding terms that satisfy the other order conditions, we get e.g.

$$\begin{aligned}
 \bar{S}(x, h) = & S + hS\left(\frac{8}{9}Q(x, z_3) + \frac{1}{9}Q(x, x)\right)S \\
 & + h^2S\left(Q(x, z_2)SQ(x, z_2) - \frac{1}{12}D^2H(z_1)SD^2H(z_1)\right)S \\
 & + h^3S\left(Q(x, x)SQ(x, x)SQ(x, x) \right. \\
 & \left. - \frac{1}{12}D^2H(x)SD^2H(x)SQ(x, x) - \frac{1}{12}Q(x, x)SD^2H(x)SD^2H(x)\right)S,
 \end{aligned} \tag{79}$$

where

$$z_1 = x + \frac{1}{2}hf(x), \quad z_2 = x + \frac{2}{3}hf(x), \quad z_3 = x + \frac{3}{4}hf(z_1).$$

5.2 The general case

Allowing for S to be a function of the solution, we define now the set TV of bi-colored trees whose nodes are either circles or triangles, and whose leaves on the cut tree τ^b , defined as the mono-colored tree left when all branches between black and white nodes are cut off, are always circles. Denoting as before black-rooted subtrees by τ_i and white-rooted subtrees by $\bar{\tau}_i$, the elementary differentials of trees $\tau \in TV$ are given by $F(\bullet)(x) = F(\circ)(x) = f(x) = S\nabla H(x)$ and

$$F(\tau)(x) = \begin{cases} S^{(l)} D^m \nabla H(x)(F(\tau_1)(x), \dots, F(\bar{\tau}_l)(x)) & \text{for } \tau = [\tau_1, \dots, \tau_m, \bar{\tau}_1, \dots, \bar{\tau}_l]_{\circ}, \\ S^{(l)} D_2^{m-1} Q(x, x)(F(\tau_1)(x), \dots, F(\bar{\tau}_l)(x)) & \text{for } \tau = [\tau_1, \dots, \tau_m, \bar{\tau}_1, \dots, \bar{\tau}_l]_{\Delta}, \end{cases}$$

where $\circ$ can be either $\bullet$ or $\circ$ and Δ can be either $\blacktriangle$ or $\triangle$. Let $TV_{\bullet}$ denote the set of trees in TV with black roots, either of the shape $\bullet$ or $\blacktriangle$.

Definition 5 A V -series is a formal series of the form

$$V(\phi, x) = \phi(\emptyset)x + \sum_{\tau \in TV_{\bullet}} \frac{h^{|\tau|}}{\sigma(\tau)} \phi(\tau) F(\tau)(x), \quad (80)$$

where $\phi : TV_{\bullet} \cup \{\emptyset\} \rightarrow \mathbb{R}$ is an arbitrary mapping, and the symmetry coefficient σ is given by (28).

Proofs of the theorems in this subsection can be obtained similarly to the proofs in Sects. 4 and 5.1, and are therefore omitted.

We redefine

$$R(\phi_{sjk}, x) := \begin{cases} D^2 H(G(\phi_{sjk}, x)) & \text{if } s_k = \bullet, \\ Q(x, V, (\phi_{sjk}, x)) & \text{if } s_k = \blacktriangle. \end{cases}$$

and consider now approximations of $S(x)$ that can be written as

$$\begin{aligned} \bar{S}(x, \hat{x}, h) = & \sum_{s \in SG} h^n \sum_j b_{sj} \left(\prod_{k=1}^n S(V(\psi_{sjk}, x)) R(\phi_{sjk}, x) \cdot S(V(\psi_{sj(n+1)}, x)) \right. \\ & \left. + (-1)^{|s|_{\bullet}-1} S(V(\psi_{sj(n+1)}, x)) \prod_{k=1}^n R(\phi_{sj(n-k+1)}, x) S(V(\psi_{sj(n-k+1)}, x)) \right) \end{aligned} \quad (81)$$

whenever $\hat{x}$ is the solution of

$$\frac{\hat{x} - x}{h} = \bar{S}(x, \hat{x}, h) \bar{\nabla} H(x, \hat{x}),$$

with $\phi_{sjk}(\emptyset) = \psi_{sjk}(\emptyset) = 1$ for every s, j, k , and with $\sum_j b_{\bullet j} = \frac{1}{2}$.

Theorem 7 *The discrete gradient scheme (9) with the approximation of $S(x)$ given by (81) and $\bar{\nabla}H \in C^\infty(\mathbb{R}^d \times \mathbb{R}^d, \mathbb{R}^d)$ is a V-series method. It can be written $\hat{x} = V(\Phi, x)$, with*

$$\begin{aligned} \Phi = \hat{e} + \sum_{s \in SG} \sum_j b_{sj} & ((\psi_{sj1}, \phi_{sj1}) \diamond \cdots \diamond (\psi_{sjn}, \phi_{sjn}) \diamond \hat{\theta}(\psi_{sj(n+1)}) \\ & + (-1)^n (\psi_{sj(n+1)}, \phi_{sjn}) \diamond \cdots \diamond (\psi_{sj2}, \phi_{sj1}) \diamond \hat{\theta}(\psi_{sj1})). \end{aligned} \quad (82)$$

where

$$\begin{aligned} \hat{\theta}(a)([\tau_1, \dots, \tau_m, \bar{\tau}_1, \dots, \bar{\tau}_l]_{\mathbf{A}}) &= \frac{1}{m+1} \Phi(\tau_1) \cdots \Phi(\tau_m) a(\bar{\tau}_1) \cdots a(\bar{\tau}_l), \\ \hat{\theta}(a)([\tau_1, \dots, \tau_m, \bar{\tau}_1, \dots, \bar{\tau}_l]_{\mathbf{A}}) &= \frac{-2m}{m+1} \Phi(\tau_1) \cdots \Phi(\tau_m) a(\bar{\tau}_1) \cdots a(\bar{\tau}_l). \end{aligned} \quad (83)$$

The scheme is of order p if and only if

$$\Phi(\tau) = \xi(\tau) \quad \text{for } |\tau| \leq p, \quad (84)$$

where

$$\xi(\tau) = \begin{cases} \frac{1}{\gamma(\tau)} & \text{if } \tau \in TP, \\ 0 & \text{otherwise.} \end{cases} \quad (85)$$

As in Sect. 4.2, we cut the branches between black and white nodes, regardless of the shape of the nodes, and denote this tree by τ^b . Number the nodes and reattach the cut-off parts. For the node i and the corresponding stem s^i , there exists a unique set of forests $\hat{\tau}^i = \{(\mu_1^i, \eta_1^i), \dots, (\mu_{n+1}^i, \eta_{n+1}^i)\}$ such that

$$\tau = [(\mu_1^i, \eta_1^i)]_{s_1^i} \circ \cdots \circ [(\mu_n^i, \eta_n^i)]_{s_n^i} \circ [(\mu_{n+1}^i, \eta_{n+1}^i)]_{s_{n+1}^i}$$

Proposition 7 *The Φ of (82) satisfies*

$$\Phi(\tau) = \hat{e}(\tau) + \sum_{i=1}^{|\tau^b|} \Lambda(\hat{\tau}^i, s^i) \quad (86)$$

where $\hat{e}(\emptyset) = 1$ and $\hat{e}(\tau) = 0$ for all $\tau \neq \emptyset$, and

$$\begin{aligned} \Lambda(\hat{\tau}^i, s^i) &= \theta([(\mu_{n+1}^i, \eta_{n+1}^i)]_{s_{n+1}^i}) \left(\sum_j b_{s^i j} \psi_{s^i j 1}(\eta_1^i) \phi_{s^i j 1}(\mu_1^i) \cdots \phi_{s^i j n}(\mu_n^i) \psi_{s^i j(n+1)}(\eta_{n+1}^i) \right. \\ &\quad \left. + (-1)^{|s^i| \bullet - 1} \sum_j b_{\hat{s}^i j} \psi_{\hat{s}^i j(n+1)}(\eta_1^i) \phi_{\hat{s}^i j n}(\mu_1^i) \cdots \phi_{\hat{s}^i j 1}(\mu_n^i) \psi_{\hat{s}^i j 1}(\eta_n^i) \right), \end{aligned} \quad (87)$$

with θ given by (71) and $\hat{s}^i$ given by $\hat{s}_k^i = s_{n-k+1}^i$ for $k = 1, \dots, n$, and $\hat{s}_{n+1}^i = s_{n+1}^i$.

The number of trees in TV grows very quickly. However, in our task of finding higher order schemes we may use the lessons of the previous sections, and require that the arguments of S , D^2H and Q in (81) are B-series up to order $p - 1$. Then we only need to find order conditions for energy-preserving linear combinations of the form

$$\begin{aligned}\omega &= [(\mu_1, \eta_1)]_{s_1} \circ \cdots \circ [(\mu_n, \eta_n)]_{s_n} \circ [\eta_{n+1}]_{\bullet} \\ &+ (-1)^n [(\mu_n, \eta_{n+1})]_{s_n} \circ \cdots \circ [(\mu_1, \eta_2)]_{s_1} \circ [\eta_1]_{\bullet},\end{aligned}\quad (88)$$

where μ_i and η_i are forests of trees in $TP_{\bullet}$ and $TP_{\circ}$ respectively, for $i = 1, \dots, n + 1$. Thus we can disregard any tree with $\triangle$ in it. Furthermore, we may color all nodes of the trees in μ_i and η_i except the roots gray, and let the elementary differentials corresponding to these trees be the same as the elementary differentials of B-trees.

We find the order conditions

$$\sum_{i \in I_l} \Lambda(\hat{\tau}^i, s^i) = \xi(\tau) - \hat{e}(\tau) - \sum_{i \in I_n} \Lambda(\hat{\tau}^i, s^i),$$

by using the relation

$$\begin{aligned}\Lambda(\{(\mu_1^i, \eta_1^i), \dots, (\mu_{n+1}^i, \eta_{n+1}^i)\}, s^i) \\ = \hat{\theta}([\mu_{n+1}^i]_{s_{n+1}^i}) \Lambda(\{(\mu_1^i, \eta_1^i), \dots, (\emptyset, \eta_{n+1}^i)\}, \bar{s}^i)\end{aligned}$$

to calculate $\Lambda(\hat{\tau}^i)$ for $i \in I_n$. The $\hat{\theta}$ is given by (78), and $\bar{s}^i$ is s^i with s_{n+1}^i replaced by $\bullet$.

Example 5 Consider $\mathfrak{Y}$, which is part of the energy-preserving linear combination $\mathfrak{I} + \mathfrak{Y}$. We have two black nodes, and calculate

$$\begin{aligned}\Lambda(\hat{\tau}^1, s^1) &= \Lambda(\{(\bullet, \mathfrak{I})\}, \triangle) = \hat{\theta}(\mathfrak{I}) \Lambda(\{(\emptyset, \mathfrak{I})\}, \bullet) = -\xi(\bullet) \xi(\mathfrak{I}) = -\frac{1}{6}, \\ \Lambda(\hat{\tau}^2, s^2) &= \Lambda(\{(\emptyset, \mathfrak{I}), (\emptyset, \emptyset)\}, \mathfrak{I}) = \sum_j b_{s^2 j} \psi_{s^2 j 1}(\mathfrak{I}) + \sum_j b_{\bar{s}^2 j} \psi_{\bar{s}^2 j 2}(\mathfrak{I}) \\ &= \sum_j b_{s^2 j} (\psi_{s^2 j 1} + \psi_{s^2 j 2})(\mathfrak{I}).\end{aligned}$$

Hence the order condition associated to this linear combination is

$$\sum_j b_{s^2 j} (\psi_{s^2 j 1} + \psi_{s^2 j 2})(\mathfrak{I}) = \frac{1}{6}.$$

An approximation of $S(x)$ satisfying all the order conditions in Tables 5 and 6 is given by

$$\begin{aligned}
 \bar{S}(x, \cdot, h) = & \frac{1}{2}(S(z_{11} + z_{12}) + S(z_{11} - z_{12})) \\
 & + \frac{1}{12}h(S(z_6)D^2H(z_2)S(x) - S(x)D^2H(z_2)S(z_6)) \\
 & + \frac{3}{7}h(S(z_3)Q(x, z_5)S(z_4) + S(z_4)Q(x, z_5)S(x, z_3)) \\
 & + \frac{8}{105}hS(x)Q(x, z_7)S(x) + \frac{1}{15}hS(x)Q(x, x)S(x) \\
 & + h^2S(z_2)Q(x, z_5)S(z_8)Q(x, z_5)S(z_2) \\
 & - \frac{1}{12}h^2S(z_2)D^2H(z_2)S(z_2)D^2H(z_2)S(z_2) \\
 & + \frac{1}{6}h^2(S(z_2) - S(x))D^2H(x)S(x)Q(x, x)S(x) \\
 & - \frac{1}{6}h^2S(x)Q(x, x)S(x)D^2H(x)(S(z_2) - S(x)) \\
 & + h^3S(x)Q(x, x)S(x)Q(x, x)S(x)Q(x, x) \\
 & - \frac{1}{12}h^3S(x)D^2H(x)S(x)D^2H(x)S(x)Q(x, x)S(x) \\
 & - \frac{1}{12}h^3S(x)Q(x, x)S(x)D^2H(x)S(x)D^2H(x)S(x),
 \end{aligned} \tag{89}$$

with

$$\begin{aligned}
 z_1 &= x + \frac{1}{3}hf(x), & z_2 &= x + \frac{1}{2}hf(x), & z_3 &= x + \frac{7-\sqrt{7}}{12}hf(z_1), \\
 z_4 &= x + \frac{7+\sqrt{7}}{12}hf(z_1), & z_5 &= x + \frac{2}{3}hf(z_2), & z_6 &= x + hf(z_2), \\
 z_7 &= x + \frac{5}{4}hf(z_2), & z_8 &= x + \frac{4}{3}hf(z_2), & z_9 &= x + hf(z_6), & z_{10} &= x + hf(z_9), \\
 z_{11} &= \frac{1}{3}(x + z_2 + z_6) + \frac{1}{12}(-z_9 + z_{10}), & z_{12} &= \frac{\sqrt{3}}{36}(7x - 2z_2 - 4z_6 + z_9 - 2z_{10}),
 \end{aligned}$$

and hence a discrete gradient scheme with this $\bar{S}(x, \cdot, h)$ and any $\bar{\nabla} \in C^3(\mathbb{R}^d \times \mathbb{R}^d, \mathbb{R}^d)$ will be of fourth order.

One advantage of choosing the AVF discrete gradient is that the resulting scheme generally requires fewer computations at each time step. This is clearly evident in the above example: if $\bar{\nabla} = \bar{\nabla}_{\text{AVF}}$, then (89) collapses to (67). However, if the AVF discrete gradient is difficult to calculate, there can also be much to gain in computational cost by choosing a symmetric discrete gradient, like the symmetrized Itoh–Abe discrete gradient (23) or the Furihata discrete gradient (25). Then one can ignore the order condition for any combination (88) for which $s_j = \blacktriangle$ and $\mu_j = \emptyset$ for some $j \in [1, n]$,

$ \tau $	ω	s	Order condition
2		$\bullet$	$2 \sum_j b_{sj} \psi_{sj1}(\bullet) = \frac{1}{2}$
3		$\bullet$	$2 \sum_j b_{sj} \psi_{sj1}(\bullet)^2 = \frac{1}{3}$
		$\bullet$	$2 \sum_j b_{sj} \psi_{sj1}(\mathring{\bullet}) = \frac{1}{6}$
		$\mathbf{I}$	$\sum_j b_{sj} (\psi_{sj1} + \psi_{sj2})(\bullet) = \frac{1}{2}$
		$\mathbf{I}$	$\sum_j b_{sj} (\psi_{sj2} - \psi_{sj1})(\bullet) = -\frac{1}{12}$
4		$\bullet$	$2 \sum_j b_{sj} \psi_{sj1}(\bullet)^3 = \frac{1}{4}$
		$\bullet$	$2 \sum_j b_{sj} \psi_{sj1}(\bullet) \psi_{0j1}(\mathring{\bullet}) = \frac{1}{8}$
		$\bullet$	$2 \sum_j b_{sj} \psi_{sj1}(\mathring{\bullet}) = \frac{1}{12}$
		$\bullet$	$2 \sum_j b_{sj} \psi_{sj1}(\mathring{\bullet}) = \frac{1}{24}$
		$\mathbf{I}$	$\sum_j b_{sj} \phi_{sj1}(\mathring{\bullet}) = \frac{1}{6}$
		$\mathbf{I}$	$\sum_j b_{sj} \psi_{sj1}(\bullet) \psi_{sj2}(\bullet) = \frac{1}{8}$
		$\mathbf{I}$	$\sum_j b_{sj} \phi_{sj1}(\bullet) (\psi_{sj1} + \psi_{sj2})(\bullet) = \frac{1}{3}$
		$\mathbf{I}$	$\sum_j b_{sj} (\psi_{sj1}(\bullet)^2 + \psi_{sj2}(\bullet)^2) = \frac{1}{3}$
		$\mathbf{I}$	$\sum_j b_{sj} (\psi_{sj1} + \psi_{sj2})(\mathring{\bullet}) = \frac{1}{6}$
		$\mathbf{I}$	$\sum_j b_{sj} \phi_{sj1}(\bullet) (\psi_{sj2} - \psi_{sj1})(\bullet) = -\frac{1}{24}$
		$\mathbf{I}$	$\sum_j b_{sj} (\psi_{sj2}(\bullet)^2 - \psi_{sj1}(\bullet)^2) = -\frac{1}{12}$
		$\mathbf{I}$	$\sum_j b_{sj} (\psi_{sj2} - \psi_{sj1})(\mathring{\bullet}) = -\frac{1}{24}$
		$\mathbf{I}$	$\sum_j b_{sj} \psi_{sj2}(\bullet) = \frac{2}{3}$
		$\mathbf{I}$	$\sum_j b_{sj} (\psi_{sj1} + \psi_{sj3})(\bullet) = \frac{1}{2}$
		$\mathbf{I}$	$\sum_j b_{sj} (\psi_{sj1} - \psi_{sj3})(\bullet) = \frac{1}{12}$
		$\mathbf{I}$	$\sum_j b_{sj} \psi_{sj3}(\bullet) - \sum_j b_{\bar{s}j} \psi_{\bar{s}j1}(\bullet) = 0$
		$\mathbf{I}$	$2 \sum_j b_{2j} \psi_{2j1}(\bullet) = -\frac{1}{24}$
		$\mathbf{I}$	$\sum_j b_{sj} (\psi_{sj1} + \psi_{sj3})(\bullet) = -\frac{1}{24}$

since this corresponds to elementary differentials involving $Q(x, x)$, which we recall is zero when the discrete gradient is symmetric. If we consider the conditions for fourth order presented in Tables 5 and 6, this eliminates 17 of the 22 conditions for trees with $\blacktriangle$ in the stem. By considering the remaining order conditions we get that, if $\bar{\nabla}H \in C^3(\mathbb{R}^d \times \mathbb{R}^d, \mathbb{R}^d)$ and $\bar{\nabla}H(x, y) = \bar{\nabla}H(y, x)$, the discrete gradient scheme (9) is of fourth order if

$$\begin{aligned} \bar{S}(x, \cdot, h) = & \frac{1}{2}(S(z_5 + z_6) + S(z_5 - z_6)) \\ & + \frac{1}{12}h(S(z_2)D^2H(z_1)S(x) - S(x)D^2H(z_1)S(z_2)) \\ & + \frac{8}{9}hS(z_1)Q(x, z_7)S(z_1) - \frac{1}{12}h^2S(z_1)D^2H(z_1)S(z_1)D^2H(z_1)S(z_1), \end{aligned} \quad (90)$$

with

$$\begin{aligned} z_1 &= x + \frac{1}{2}hf(x), & z_2 &= x + hf(z_1), & z_3 &= x + hf(z_2), \\ z_4 &= x + hf(z_3), & z_5 &= \frac{1}{3}(x + z_1 + z_2) + \frac{1}{12}(-z_3 + z_4), \\ z_6 &= \frac{\sqrt{3}}{36}(7x - 2z_1 - 4z_2 + z_3 - 2z_4), & z_7 &= x + \frac{3}{4}hf(z_1). \end{aligned}$$

If S is constant, (90) simplifies to

$$\bar{S}(x, \cdot, h) = S + \frac{8}{9}hSQ(x, z_7)S - \frac{1}{12}h^2SD^2H(z_1)SD^2H(z_1)S. \quad (91)$$

6 Numerical experiments

For all the examples considered in the following, the resulting discrete gradient schemes are nonlinearly implicit systems, which we here solve using Newton's method. Note that whenever $\bar{S}(x, \hat{x}, h)$ is independent of $\hat{x}$, the Jacobian of

$$F(\hat{x}) = \hat{x} - x - h\bar{S}(x, \hat{x}, h)\bar{\nabla}H(x, \hat{x})$$

is given by

$$J(\hat{x}) = I - h\bar{S}(x, \hat{x}, h)D_2\bar{\nabla}H(x, \hat{x}), \quad (92)$$

where $D_2\bar{\nabla}H(x, \hat{x})$ may also be used for the calculation of $Q(x, \hat{x})$. The extra computational cost of using a higher-order scheme with an explicit $\bar{S}$ thus only lies in the computation of this $\bar{S}$ once each time step. A scheme with an implicitly defined $\bar{S}(x, \hat{x}, h)$ can give a much more complicated Jacobian, but a quasi-Newton method using the approximate Jacobian given by (92) may still be very efficient.

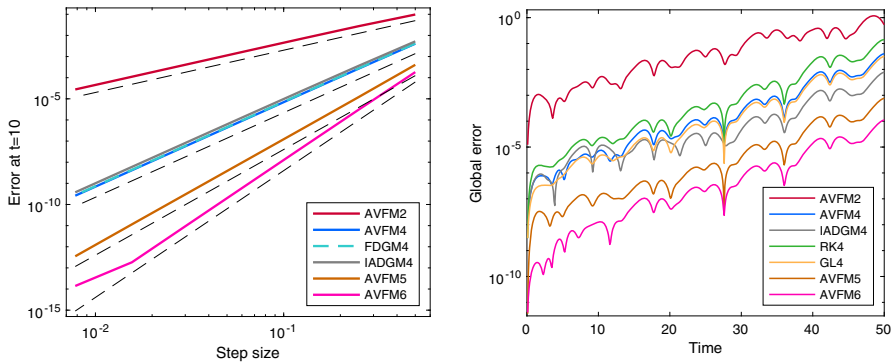


Fig. 1 Error plots for the Hénon–Heiles system (93) solved by various discrete gradient methods: AVFM2/4/5/6 are AVF discrete gradient methods of different orders; FDGM4 is a fourth order Furihata discrete gradient method; IADGM4 is a fourth order Itoh–Abe discrete gradient method. RK4 is the classic Runge–Kutta method and GL4 is the fourth order Gauss–Legendre method, included for comparison. The black dashed lines in the order plot are reference lines of order two, four, five and six. The step size in the right plot is $h = 0.1$

6.1 Hénon–Heiles system

The Hénon–Heiles system can be written in the form (6) with

$$S = \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}, \quad H(q, p) = \frac{1}{2}(q_1^2 + q_2^2 + p_1^2 + p_2^2) + q_1^2 q_2 - \frac{1}{3} q_2^3, \quad (93)$$

where I is the 2×2 identity matrix. We use here the same initial conditions used in [37]: $q_1 = \frac{1}{10}$, $q_2 = -\frac{1}{2}$, $p_1 = p_2 = 0$. The order of some of the energy-preserving methods proposed in this paper are confirmed by the left plot in Fig. 1. We compare the performance of the fourth order discrete gradient methods obtained by using the $\bar{S}$ given by (79) coupled with three different discrete gradients: the Itoh–Abe discrete gradient (22), the Furihata discrete gradient (25), and the AVF discrete gradient (10) (right plot of Fig. 1). The symmetrized Itoh–Abe discrete gradient (23) is for this H identical to the Furihata discrete gradient. The AVF and Furihata discrete gradient methods perform in this case very similarly, and thus the error from the Furihata discrete gradient method is excluded from the right plot in Fig. 1. We observe that, although it initially performs on par with the AVF method, the Itoh–Abe discrete gradient method gives a lower global error than the other fourth order methods as time goes on. Note however that this method requires the most computations at every time step.

6.2 Lotka–Volterra

The methods should also be tested on a skew-gradient system with non-constant S . We choose the Lotka–Volterra system also used for numerical experiments in [10]. It is given by

Fig. 2 Order of discrete gradient methods applied to the Lotka–Volterra system (94), with different $\bar{S}$: $\bar{S}(x, \hat{x}) = S(\frac{x+\hat{x}}{2})$ for DGM2, (65) for DGM3, (67) for DGM4-exp, (66) for DGM4-imp. The dashed lines are reference lines of order two, three and four

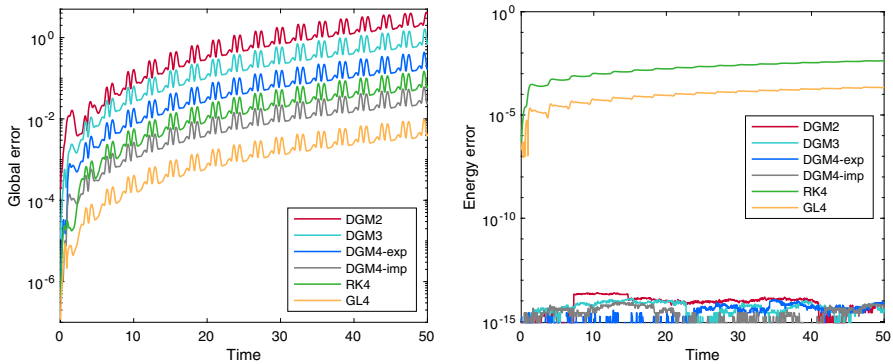
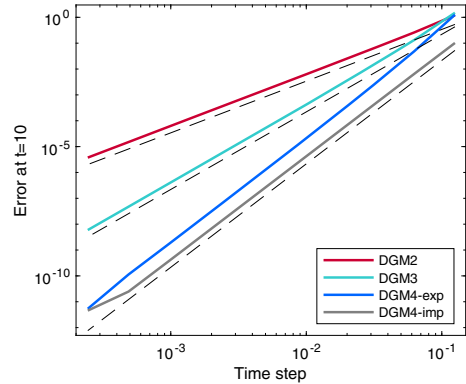


Fig. 3 Error in the solution and in the energy for discrete gradient methods with $\bar{S}$ given by $\bar{S}(x, \hat{x}) = S(\frac{x+\hat{x}}{2})$ for DGM2, (65) for DGM3, (67) for DGM4-exp and (66) for DGM4-imp, applied to the Lotka–Volterra system, with step size $h = 0.05$. For comparison, errors from using the standard fourth order Runge–Kutta (RK4) and Gauss–Legendre (GL4) methods are also included

$$S = \frac{1}{2} \begin{pmatrix} 0 & -x_1 x_2 & x_1 x_3 \\ x_1 x_2 & 0 & -2x_2 x_3 \\ -x_1 x_3 & 2x_2 x_3 & 0 \end{pmatrix}, \quad H(x) = 2x_1 + x_2 + 2x_3 + \ln(x_2) - 2\ln(x_3), \quad (94)$$

and initial conditions $x_1 = 1$, $x_2 = \frac{19}{10}$, $x_3 = \frac{1}{2}$. For this H , the Itoh–Abe, Furihata and AVF discrete gradients are all equivalent. Order plots are given in Fig. 2. We consider fourth order discrete gradient methods where ∇S is given either dependent on or independent of $\hat{x}$; that is, (66) or (67). The implicitly given (67) yields a significantly lower error in the solution of the corresponding discrete gradient method, as can be witnessed from the left plot in Fig. 3. In contrast to what we observed for the canonical Hamiltonian system studied above, none of the discrete gradient methods give a global error lower than that of the fourth order Gauss–Legendre method.

6.3 Hamiltonian neural networks

We present here results on the pendulum problem also considered in [19, 27]. Training data is first generated by adding noise to solutions of the system (6) with

$$S = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad H(q, p) = 2mgl(1 - \cos q) + \frac{l^2}{2m}p^2,$$

with $l = m = 1$ and $g = 3$, for various times $t \in [0, 20]$ and 50 randomly sampled initial coordinates. Then a Hamiltonian neural network (HNN) is used to learn the Hamiltonian. If the automatic discrete differentiation algorithm is used for the training, the corresponding discrete gradient is learned simultaneously. To compute $Q(x, y)$, we require that the network also learns the Jacobian of the discrete gradient with respect to the second argument. This necessitates a modification of the network developed by Matsubara et al [27], which we have done using the autograd class of PyTorch. The resulting Jacobian is in any case very useful for integrating the system by any discrete gradient method, since we then can employ a root-finding algorithm that use (92), e.g. Newton's method.

We have tested second, third and fourth order discrete gradient methods both for training the network and for integrating the obtained dynamical system. For the fourth order scheme, we have used the $\bar{S}$ given by (91) and for the third order scheme we have used

$$\bar{S}(x, \cdot, h) = S + hS Q(x, z)S - \frac{1}{12}h^2 SD^2H(x)SD^2H(x)S, \quad (95)$$

where $z = x + \frac{2}{3}hS\nabla H(x)$. Note that ∇H and D^2H can be obtained from the network through the relations $\nabla H(x) = \bar{\nabla}H(x, x)$ and $D^2H(x) = 2D_2\bar{\nabla}H(x, x)$.

As is also noted in [27], energy preservation seems to be more important than high order for the method used to train the system. In Fig. 4 we see an example of this: the second order discrete gradient method outperforms the fourth order Runge–Kutta method. Whether or not higher-order discrete gradient methods give an improved performance that justifies the extra computational cost is not clear on this example. A thorough investigation of this for more complex problems is a matter for future studies.

For energy-preserving integration of the learned system, much can be gained in accuracy by using a higher-order discrete gradient method, at the expense of not much additional computational cost. The order plots in Fig. 5 are obtained by comparing to a solution found by the fourth order discrete gradient method with a smaller step size. Python code for the neural networks used to produce the plots in Figs. 4 and 5, and additional results on a mass-spring system, is available at <https://github.com/solveeid/dgnet4>.

Fig. 4 The trajectories that goes through $(q, p) = (2, 0)$ for the dynamical systems learned by an HNN with the Euler method, the classic Runge–Kutta method and discrete gradient methods with $\bar{S}$ given by S (DGM2), (95) (DGM3) and (91) (DGM4). The plots for DGM2 and DGM3 are almost entirely hidden behind the plot for DGM4. The step size used in the training is $h = 0.5$

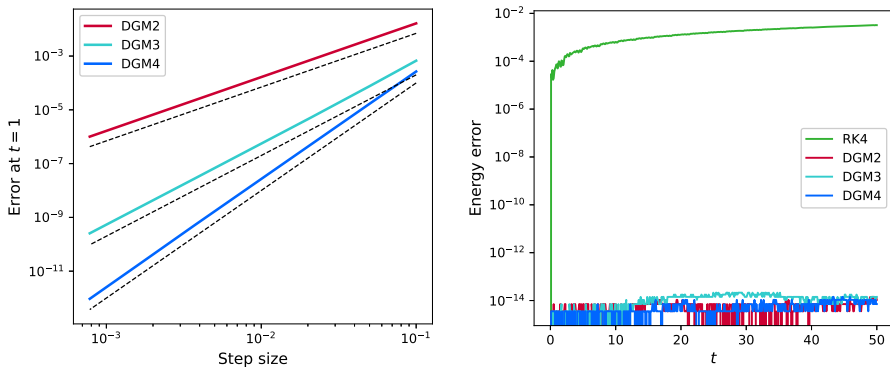
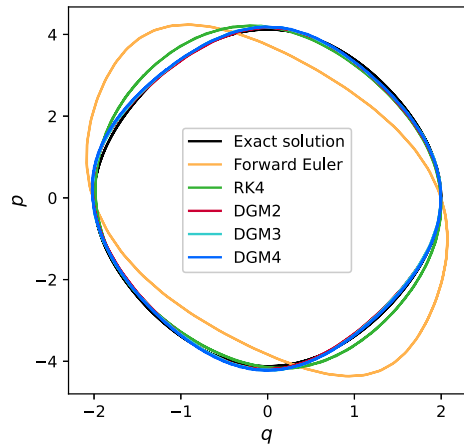


Fig. 5 Comparison of different integrators for the pendulum system found using an HNN with the second order discrete gradient method. *Left:* Order plots for discrete gradient methods with $\bar{S}$ given by S (DGM2), (95) (DGM3) and (91) (DGM4). The dashed lines are reference lines of order two, three and four. *Right:* Energy errors for the same methods, and the fourth-order Runge–Kutta method. Step size $h = 0.1$

7 Conclusions and future work

The main purpose of this paper has been to develop order theory for discrete gradient methods. This was achieved through the introduction of the function Q , Lemma 2 and a generalization of B- and P-series results. Propositions 3, 5, 6 and 7 present results which simplifies the derivation of the conditions from the general theory. The techniques introduced in this paper for building on B- and P-series methods can possibly be used on more methods than the discrete gradient methods. Future research may utilize this.

We have proposed some higher order schemes satisfying the derived order conditions. The development of specific schemes has however not been the main focus of this paper; analysis to find more optimal schemes is something we leave for the future. After such an analysis is performed, the methods could be tested on more advanced problems than those considered above, e.g. for the temporal discretization of Hamil-

tonian partial differential equations, and their performance as measured by accuracy relative to computational cost could be compared to existing methods.

The use of neural networks to train dynamical systems with preservation properties is an emerging field of study where new developments are coming with a high frequency. To our knowledge, the use of higher-order one-step energy-preserving integrators in this setting has not previously been studied, and may not even have been known to be possible until the novel results of this paper. The potential utility of the methods remain largely unexplored and will be considered in future work. This will include using methods of order higher than four and analysis of the methods compared to and in connection with other recent developments, both in the training of the network and for the integration of the resulting system. It would also be interesting to investigate how the methods perform on more advanced examples, including Hamiltonian PDEs, and for Lagrangian neural networks [1, 11].

The order theory presented in this paper can also be developed further in several different directions. The schemes given with $\bar{S}$ independent of $\hat{x}$ are linearly implicit when H is quadratic; if the order theory is extended to the polarized discrete gradient methods of [12, 13, 28], we could get higher order linearly implicit multi-step schemes for systems with polynomial first integrals of any degree. Another avenue could be to consider order conditions for the discrete Riemannian gradient methods presented in [4]. Then the results in Sect. 5 are especially interesting, since the integral in the AVF discrete Riemannian gradient can be challenging to compute analytically. Lastly, building on [2], order conditions for the exponential AVF method is given in [33]. This could be extended to exponential integrators using any discrete gradient by the theory presented in this paper.

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Ollivier-Ricci curvature convergence in random geometric graphs

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Connections between continuous and discrete worlds tend to be elusive. One example is curvature. Even though there exist numerous nonequivalent definitions of graph curvature, none is known to converge in any limit to any traditional definition of curvature of a Riemannian manifold. Here we show that Ollivier curvature of random geometric graphs in any Riemannian manifold converges in the continuum limit to Ricci curvature of the underlying manifold, but only if the definition of Ollivier graph curvature is properly generalized to apply to mesoscopic graph neighborhoods. This result establishes a rigorous link between a definition of curvature applicable to networks and a traditional definition of curvature of smooth spaces.

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I. INTRODUCTION

Curvature is one of the most basic geometric characteristics of space. The original definitions of curvature apply only to smooth Riemannian or Lorentzian manifolds, but there exist numerous extensions of curvature definitions applicable to graphs, simplicial complexes, and other discrete structures. For a variety of reasons, these extensions have recently seen a surge of interest in areas as diverse as network or data science and quantum gravity.

In network science, graph curvature is interesting in general since many real-world networks were found to possess different flavors of geometry [1]. Measuring network curvature is then a way to learn what this geometry really is [2]. On the application side, graph curvature was used to characterize congestion in telecommunication networks [3,4], detect cancer cells [5], predict COVID-19 spreading [6] and stock market fluctuations [7], analyze robustness and other properties of the Internet, financial, and brain networks [8–12], and in a number of classic applications such as community inference [13] and network embedding in machine learning [14]. In a majority of these applications, graph curvature appears as a signature of a (pathological) event or property of interest. How reliable such a signature is for a task at hand depends on whether the used definition of graph curvature does what is intended—that it is indeed curvature, and not something else.

In general relativity, scalar Ricci curvature is a key player since it appears in the Einstein-Hilbert action whose least-action extremization leads to Einstein's equations [15]. Therefore, various extensions of Ricci curvature to discrete structures representing “quantum spacetimes” have been considered. The first step in that direction was based on replacing continuum manifolds with simplicial complexes [16]. To that end, variations of the seminal definition of discrete curvature by Regge [17] were developed and utilized. More recently, a collection of more minimalistic structures was considered, such as causal sets [18–27] and random graphs [28–33]. Yet another class of approaches aimed at explaining spacetime emergence from “quantum bits” appears in space-from-entanglement proposals such as ER=EPR or tensor networks [34–40]. In all these examples, one can hope that a particular discrete structure represents Planck-scale gravitational physics, only if discrete curvature of this structure converges to Ricci curvature of classical spacetime in the continuum limit. If such convergence does not hold in a particular approach, then the approach does not agree with general relativity, so it cannot be considered realistic, regardless of how attractive it is in other respects.

Quite contrary to the continuum world of Riemannian and Lorentzian manifolds where there is no real ambiguity regarding the definition of curvature, there exist not one or two but very many nonequivalent definitions of curvature applicable to graphs. We briefly review the most prominent definitions in the next section. Unfortunately, none of these definitions are known to converge in the continuum limit to any traditional curvature of any Riemannian manifold. Therefore, it is often unclear how one should interpret different measurements of different curvatures in different graphs and networks, raising question concerning reliability and generalizability of

predictions based on such measurements. Similarly, the evasiveness of classical limit in some approaches to quantum gravity can be often linked to unclarities concerning whether discrete “quantum” curvature converges to a continuous classical one. These and other problems related to curvature convergence become particularly acute in view of growing evidence that different graph curvature definitions may disagree even about the sign of curvature in some paradigmatic graphs and networks [41,42].

Here we show that Ollivier curvature [43–45] of random geometric graphs [46–48] in any Riemannian manifold converges to Ricci curvature of the manifold in the continuum limit. Unfortunately, this convergence does not and cannot hold for the standard definition of Ollivier curvature of microscopic one-link neighborhoods in sparse unweighted graphs [49]. Fortunately, we find a natural generalization of Ollivier curvature applicable to mesoscopic neighborhoods in unweighted and weighted graphs. This generalization is a not-previously-considered key ingredient that allows us to prove the curvature convergence. Our proofs hold only when the sizes of mesoscopic graph neighborhoods fall within certain windows that depend on graph density and on how graph edges are weighted. We also perform large-scale simulations that agree with our proofs and suggest that the convergence windows are actually much wider than the limits imposed by our proof techniques.

These results link rigorously a definition of graph curvature to the traditional Ricci curvature of a Riemannian manifold. To the best of our knowledge, the closest, in spirit, previous results are Cheeger *et al.*’s proof [50] of Regge’s seminal observation [17] that an angle-defect-based curvature of increasingly finer-grained simplicial triangulations of a manifold converges to its Ricci curvature, and the more recent demonstration [21,22] that the Benincasa-Dowker action of causal sets converges to its classical counterpart in the continuum limit.

To proceed, we first recall the Ollivier curvature definition in perspective of other curvature definitions applicable to graphs and networks, Sec. II. We then describe our generalization of the Ollivier curvature definition to mesoscopic neighborhoods in graphs in general and in random geometric graphs in particular, Sec. III. We then state our main results in Sec. IV, and conclude with a discussion of their implications and caveats in Sec. V.

II. GRAPH CURVATURE DEFINITIONS

There are many nonequivalent definitions of curvature applicable to graphs. It is impossible to list them all. Here we mention some notable ones due to Steiner [51], Regge [17,50], Bakry-Émery [52], Gromov [53,54], Higuchi [55], Eckmann-Moses [56], Forman [57–59], Lin-Yau [60,61], Knill [62,63], Keller [64,65], and hybrids thereof [66]. We first outline the taxonomy of these curvatures and basic ideas behind their definitions to put Ollivier curvature [43–45] in perspective, and then recall the detailed definition of the latter for completeness.

With a few exceptions, the discrete curvature definitions above fall into the following classes.

A. Direct approaches

The first class of definitions comes from relatively direct extensions of continuous curvature to nonsmooth objects such as simplices, simplicial complexes, or more general complexes. Since graphs are 1-dimensional simplicial complexes, any notion of curvature applicable to complexes applies to graphs as well. Casting cliques in graphs as higher-dimensional simplices turns graphs into higher-dimensional complexes, so that curvature definitions applicable to higher-dimensional complexes are also applicable to graphs.

One subclass of such approaches first defines curvature of polyhedra using the Gauss-Bonnet theorem that relates the curvature of a surface to its Euler’s characteristic. This definition is then extended to general simplicial complexes. Curvatures defined by Steiner [51], Regge [17,50], Knill [62,63], and Keller [64,65] all fall into this category.

Forman curvature [57–59] follows similar ideas, but is slightly different in that it uses the Bochner-Weitzenböck formula that relates the Laplacian of a Riemannian manifold to its discrete approximation, which is then used to derive a discrete version of curvature applicable to a very general class of complexes. The definition of Forman curvature for unweighted graphs is one of the simplest. If a graph is considered as a 1-complex, so that only nodes (0-simplices) and links (1-simplices) are considered, then the Forman curvature $F_1(i, j)$ of a link between nodes i and j of degrees k_i and k_j is $F_1(i, j) = 4 - (k_i + k_j)$. If the graph is considered as a 2-complex, so that triangles are also considered, and $\Delta_{i,j}$ is the number of triangles containing the link, then the definition becomes $F_2(i, j) = F_1(i, j) + 3\Delta_{i,j}$.

B. Indirect approaches

Another class of ideas behind discrete curvature is less direct as these are based on more sophisticated curvature-related properties of smooth spaces.

One such idea, pioneered by Bakry and Emery [52], relies on the characterization of Ricci curvature via curvature-dimension inequalities. The curvature-dimension inequality establishes a lower bound for the Ricci curvature of a Riemannian manifold based on the gradient of harmonic functions on it. The inequality is an immediate consequence of Bochner’s identity that relates Ricci curvature to the gradient of harmonic functions. It was an important insight by Bakry and Emery that the curvature-dimension inequality can be lifted from manifolds and considered as a substitute of the lower bound for Ricci curvature in spaces where direct discretizations of Ricci curvature are not possible. In addition to the Bakry-Emery definition [52], the definitions of Lin-Yau [60,61] also fall into this category.

The key point behind this type of approaches to discrete curvature is to base its definitions on some fundamental non-trivial curvature-related properties that transcend well beyond the limits of smooth spaces.

C. Ollivier curvature

Ollivier curvature belongs to the second class of approaches outlined above. The fundamental aspect of Ricci curvature that Ollivier curvature embarks on to sail to the

discrete world is somewhat different from—although related to—the curvature-dimension inequalities. This aspect is of a less analytic and more geometric nature. It deals with how balls shrink or expand under parallel transport. Recall that if Ricci curvature of a space is negative, zero, or positive, then under parallel transport, balls in this space expand, stay the same, or shrink, respectively.

To capture this property, one needs to consider the Wasserstein, also known as transportation, distance $W(\mu_1, \mu_2)$ between two probability distributions μ_1, μ_2 in *any* nice (Polish to be exact) metric space:

$$W(\mu_1, \mu_2) = \inf_{\mu \in \Gamma(\mu_1, \mu_2)} \int d(x, y) \mu(x, y) dx dy, \quad (1)$$

where $d(x, y)$ is the distance between points x and y in the space, and the infimum is taken over all possible transportation plans which are joint probability distributions μ whose marginals are μ_1 and μ_2 . If μ_1, μ_2 are represented by sand piles, then $W(\mu_1, \mu_2)$ is the minimum cost to transport pile μ_1 into pile μ_2 , where the grain of sand at x gets transported to y incurring cost $d(x, y)$.

Let x and y be now two points at a small distance $d_M(x, y) = \delta$ in a Riemannian manifold M , and let $\mu_1 = \mu_x$ and $\mu_2 = \mu_y$ be the normalized restrictions of the volume form in M onto the balls $B_M(x, \delta), B_M(y, \delta)$ of radius δ centered at x, y , i.e.,

$$\mu_x(z) = \frac{\text{vol}(z)}{\text{vol}[B_M(x, \delta)]} \quad \text{if } z \in B_M(x, \delta), \quad (2)$$

$$\mu_y(z) = \frac{\text{vol}(z)}{\text{vol}[B_M(y, \delta)]} \quad \text{if } z \in B_M(y, \delta), \quad (3)$$

where $\text{vol}(z) dz$ is the volume element in M . Ollivier curvature between x and y is then defined by

$$\kappa_M(x, y) = 1 - \frac{W_M(\mu_x, \mu_y)}{\delta}. \quad (4)$$

We note that this curvature definition depends on the ball radius δ which can be any positive real number. It was shown in [44] that

$$\lim_{\delta \rightarrow 0} \frac{\kappa_M(x, y)}{\delta^2} = \frac{\text{Ric}(v, v)}{2(D+2)}, \quad (5)$$

where D is M 's dimension, and $\text{Ric}(v, v)$ is the Ricci curvature at x along v which is the unit tangent vector at x pointing along the geodesic from x to y . Ricci curvature $\text{Ric}(v, v)$ is equal to the average of sectional curvatures at x over all tangent planes containing v .

In words, the important convergence result in (5) says that in Riemannian manifolds, the rescaled Ollivier curvature converges to Ricci curvature in the limit of small ball sizes $\delta \rightarrow 0$.

It is relatively straightforward to extend the definition of Ollivier curvature from manifolds M to simple unweighted graphs G , but there are important differences. The most crucial one is that in smooth spaces M the Ollivier ball radius δ appearing in the Ollivier curvature definition (4) can be any positive real number, which can tend to zero to prove the convergence in (5). In unweighted graphs G , however, the possible nonzero values of the distance—and so of the δ —are positive integers, the hop lengths of shortest paths

between pairs of nodes. The smallest possible nonzero distance is $\delta = 1$, the graph distance $d_G(x, y)$ between directly connected nodes x and y . Presumably because of the “ δ better be small” logic, the Ollivier ball radius δ was routinely set to its smallest possible value $\delta = 1$ in the definition of Ollivier graph curvature appearing in the past literature, even though δ in graphs can certainly be set to any positive integer value.

If $\delta = 1$, then the balls $B_G(x, 1)$ and $B_G(y, 1)$ of radius 1 around nodes x and y are the sets of the neighbors of x and y , while the probability distributions μ_x and μ_y become the uniform distributions over these sets: $\mu_x = 1/k_x$ and $\mu_y = 1/k_y$, where k_x and k_y are the degrees of x and y . These distributions govern the standard random walk in the graph. Thus defined, Ollivier graph curvature was shown to be bounded by -2 and 1 , with the bounds achieved on infinite double stars and complete graphs [49]. The Ollivier graph curvature in this definition was measured in a great variety of synthetic and real-world networks [5,8,9,13,41,42,49,67]. It was also investigated at great depths in connection to quantum gravity [28–33].

However, it is evident that Ollivier curvature of graphs as defined above cannot converge to Ricci curvature of manifolds under any circumstances, simply because δ is no longer real but integer, so that $\lim_{\delta \rightarrow 0}$ makes no sense, and because Ollivier graph curvature is always between -2 and 1 , while Ricci curvature of a Riemannian manifold can be any real number. Therefore, in the next section, we generalize the Ollivier graph curvature definition to make it much more versatile, capturing a much wider spectrum of graph properties that are not necessarily microscopic.

III. MESOSCOPIC GRAPH NEIGHBORHOODS

A. General idea

The main limitation of the Ollivier graph curvature definition in the previous section is that it is “too microscopic.” It limits the curvature-related considerations to graph balls or neighborhoods of the radius of one hop, $\delta = 1$. Such neighborhoods may be too small to “feel” any curvature in a general case. Recall that microscopically, any smooth curved space is locally flat. This observation instructs us to consider much larger mesoscopic graph neighborhoods with $\delta \gg 1$. Unfortunately, this instruction cannot be correct either, because to talk about any curvature convergence, we really need $\delta \ll 1$ since δ must go to zero in the key convergence equation (5), which is the main motivation to set δ to its smallest possible value 1 in graphs.

The combination of the calls for $\delta \ll 1$ and $\delta \gg 1$ looks like a clear sign of unhealthy pathology, suggesting abandoning any hopes for Ollivier-to-Ricci curvature convergence in graphs. And indeed we believe it is obvious that there cannot be any such convergence in unweighted graphs, simply because δ cannot be less than 1 in them.

However, these observations do not preclude the curvature convergence in weighted graphs with positive real weights. In such graphs, the Ollivier ball radius δ is back to being a positive real number, which can tend to zero because link weights can tend to zero with some characteristic rate ε . Suppose also that ε and δ tend to zero such that $\varepsilon \ll \delta$. Then, on

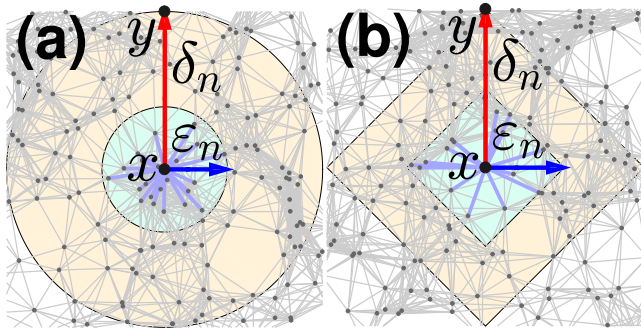


FIG. 1. Random geometric graphs in (a) Riemannian space and (b) Lorentzian spacetime. The figure visualizes the flat two-dimensional Euclidian plane (a) and Minkowski spacetime (b), where distances are the ℓ_2 and ℓ_1 norms, respectively. Nodes are sprinkled randomly via the Poisson point process with rate $r = n$. The blue regions are the balls $B_M(x, \varepsilon_n)$ of radius $t = \varepsilon_n$ around node x located at the origin. In (b) this ball is the intersection of the past light cone centered at time $t = \varepsilon_n$ with the future light cone centered at time $t = -\varepsilon_n$. Node x is connected to all other nodes in the graph that happen to lie within these blue balls. The yellow regions are the Ollivier balls $B_M(x, \delta_n)$ of radius $\delta_n \geq \varepsilon_n$. Node y is at distance δ_n from x . Vector v (not shown) is the tangent vector at x toward y . The probability distribution μ_x is the uniform distribution over all the nodes z that happen to lie within distance δ_n from x in the graph, $z \in B_G(x, \delta_n)$. These are not exactly the nodes lying with the yellow Ollivier balls, but at large n the difference is negligible, Appendix A 2.

the one hand, we have that $\delta \rightarrow 0$ as needed for convergence (5), while on the other hand, the hopwise lengths of the radii of Ollivier's balls are $\delta/\varepsilon \gg 1$, so that their sizes actually grow in terms of number of nodes. This way we simultaneously satisfy the two requirements above: the Ollivier balls are small ($\delta \ll 1$) and large ($\delta/\varepsilon \gg 1$) at the same time, and hence we call them *mesoscopic*. It is intuitive to think of ε and δ as the microscopic and mesoscopic scales of the system.

The outlined approach is very general and can be implemented in a variety of situations. In particular, the graphs do not really have to be weighted, but δ must necessarily play a role of a properly rescaled graph distance. In the next section, we make these ideas concrete in application to random geometric graphs.

B. Random geometric graphs

To describe our implementation of the general ideas from the previous subsection in application to random geometric graphs (RGGs), we first recall what RGGs are.

Given any Riemannian manifold, which henceforth we will often call just *space*, the RGGs in it are defined constructively via the following two-step procedure: (1) sprinkle points uniformly at random in the space via a Poisson point process of rate $r > 0$ driven by the volume form defined by the metric in the space [68], and (2) connect by edges all pairs of points whose pairwise distances in the space are smaller than threshold $t > 0$, Fig. 1(a). In topology, RGGs are a very fundamental object because they are 1-skeletons of Vietoris-Rips complexes [69] whose topology was proven to converge to the space topology under very mild assumptions [70]. In

that context, our results below show that in a proper sense, the geometry of RGGs also converges to the space geometry.

We are now ready to describe our ε, δ settings with RGGs in the continuum limit $n \rightarrow \infty$ suitable for curvature convergence investigations.

Given any nice D -dimensional Riemannian manifold M , we first fix any point x in it and any unit tangent vector v at x . We are concerned with Ricci curvature at x in the v direction. For a given n , we add the second point y at distance δ_n from x , $d_M(x, y) = \delta_n$, along the geodesic from x in the v direction. The distance δ_n is also the radius of the balls in the Ollivier curvature definition above, so that it goes to zero in the limit, $\delta_n \rightarrow 0$.

We then add the Poisson point process (PPP) of rate $r \sim n$ in M to the two nonrandom points x, y . Henceforth, $f(n) \sim g(n)$ means $f(n)/g(n) \rightarrow c \in (0, \infty)$, and $f(n) \approx g(n)$ means $c = 1$. The connection distance threshold t in RGGs defined on top of this PPP + $\{x, y\}$ is then set to $t = \varepsilon_n \rightarrow 0$. That is, all pairs of points are linked in RGG G if the distance between them in M does not exceed ε_n . In the continuum limit $n \rightarrow \infty$ of this setup, our PPP samples the manifold increasingly densely (as there are on average $r \sim n$ vertices in G per unit volume in M), while the RGGs become increasingly “microscopic” (as their edges span M 's distances no greater than $\varepsilon_n \rightarrow 0$).

Every created edge is then weighted as described in the next section. For any two vertices z_1 and z_2 in RGG G , we thus have two distances defined: the manifold distance $d_M(z_1, z_2)$ and the graph distance $d_G(z_1, z_2)$, which is the length of the shortest path between z_1 and z_2 in the *weighted graph* G .

The Ollivier balls centered at x, y are then the sets of G 's vertices z lying within *weighted graph* distances $d_G \leq \delta_n$ from vertices x, y : $B_G(x, \delta_n) = \{z \in G : d_G(x, z) \leq \delta_n\}$, $B_G(y, \delta_n) = \{z \in G : d_G(y, z) \leq \delta_n\}$. The probability distributions μ_x, μ_y are the uniform distributions on the vertex sets $B_G(x, \delta_n), B_G(y, \delta_n)$:

$$\mu_x(z) = \frac{1}{|B_G(x, \delta_n)|} \quad \text{if } z \in B_G(x, \delta_n), \quad (6)$$

$$\mu_y(z) = \frac{1}{|B_G(y, \delta_n)|} \quad \text{if } z \in B_G(y, \delta_n). \quad (7)$$

Finally, Ollivier *graph* curvature $\kappa_G(x, y)$ is defined by the same equation (4), except that probability distributions μ_x, μ_y are given by Eqs. (6) and (7), Fig. 1(a):

$$\kappa_G(x, y) = 1 - \frac{W_G(\mu_x, \mu_y)}{\delta_n}. \quad (8)$$

We reiterate that as opposed to previous literature on Ollivier graph curvature, we do not require the RGG connection radius ε_n to be equal to the Ollivier ball radius δ_n , Fig. 1(a). However, we do not exclude this possibility either. In fact, we will see in the next section that the Ollivier-to-Ricci curvature convergence does hold even with $\varepsilon_n = \delta_n$ if links are weighted by the manifold distances between the linked nodes, and if graphs are sufficiently dense. That is, our only requirements to ε_n and δ_n are that they both must go zero, and that $\varepsilon_n \leq \delta_n$.

IV. RESULTS

We obtain curvature convergence results for two different schemes of edge weighting in our random geometric graphs. We consider them separately in the next two sections.

A. Space distance weights

We first state our results for the settings in the previous section with the weights w_{ij} of all edges i, j in RGGs G set equal to the distances between i and j in manifold M :

$$w_{ij} = d_M(i, j). \quad (9)$$

With these settings, we can show that in the continuum limit, rescaled Ollivier curvature in random graph G converges to Ricci curvature in manifold M ,

$$\lim_{n \rightarrow \infty} \left\langle \left| \frac{\kappa_G(x, y)}{\delta_n^2} - \frac{\text{Ric}(v, v)}{2(D+2)} \right| \right\rangle = 0, \quad (10)$$

if the RGG connectivity radius ε_n and Ollivier ball radius δ_n shrink to zero with n as

$$\varepsilon_n \sim n^{-\alpha}, \quad \delta_n \sim n^{-\beta}, \quad (11)$$

with exponents α, β satisfying

$$0 < \beta \leq \alpha, \quad \alpha + 2\beta < \frac{1}{D}. \quad (12)$$

The connectivity and Ollivier balls can be equal and can shrink at the same rate,

$$\varepsilon_n \sim \delta_n \sim n^{-\alpha}, \quad (13)$$

in which case (12) becomes

$$\alpha < \frac{1}{3D}. \quad (14)$$

The expectation $\langle \cdot \rangle$ in (10) is with respect to the RGG ensemble, in which $\kappa_G(x, y)$ is random. The result in (10) is strong in the sense that it implies not only the convergence of the expected value of Ollivier graph curvature to Ricci curvature generalizing (5) to graphs,

$$\lim_{n \rightarrow \infty} \frac{\langle \kappa_G(x, y) \rangle}{\delta_n^2} = \frac{\text{Ric}(v, v)}{2(D+2)}, \quad (15)$$

but also the concentration of random $\kappa_G(x, y)$ around its expected value,

$$\lim_{n \rightarrow \infty} \text{Prob} \left[\left| \frac{\kappa_G(x, y)}{\delta_n^2} - \frac{\text{Ric}(v, v)}{2(D+2)} \right| > \varepsilon \right] = 0 \quad (16)$$

for any $\varepsilon > 0$.

We note that the conditions (12) allow graphs to be arbitrarily sparse, but not exactly ultrasparse. We call graphs *dense*, *sparse*, and *ultrasparse* (also *truly sparse*) if their expected average degree

$$\bar{k}_n \sim n \text{vol}[B_M(\cdot, \varepsilon_n)] \sim n \varepsilon_n^D \sim n^{1-\alpha D} \quad (17)$$

is $O(n)$, $o(n)$, and $\sim \text{constant}$, respectively. Observe that the exponent β in (12) cannot be exactly zero because the Ollivier balls must shrink for the sake of convergence, but it can be arbitrarily close to zero. The closer the β to zero, the closer the α can be to $1/D$ according to (12), while $\alpha = 1/D$ corresponds

to the ultrasparse case with $\bar{k}_n \sim \text{constant}$. We note though that the closer the β to zero, the slower the convergence, simply because the Ollivier balls shrink too slowly in this case.

Another important observation is that the case with $\varepsilon_n = \delta_n$ corresponds to the traditional definition of Ollivier graph curvature with microscopic one-hop neighborhoods. In this case, Eq. (14) says that we cannot prove the convergence anywhere close to the ultrasparse limit with $\alpha = 1/D$, and our simulations suggest that there is no convergence there indeed.

In general, the sparser the graphs, the more problematic the convergence becomes. This is because the sparser the graphs, the smaller the connection radius ε_n . As discussed in Sec. III, if ε_n is too small, and if $\delta_n = \varepsilon_n$, then all balls are of the radius of one hop, and such microscopic neighborhoods may not “feel” any curvature; all they can “feel” is locally Euclidean flatness.

1. Proofs

Our proof of (10), outlined in Appendix A, is not trivial. It consists of three major steps.

First, we extend the weighted shortest path distance in RGGs G to space M . We do so by defining the following auxiliary distance in M . For any pair of points $x, y \in M$ we add x and y as nodes to the graph G , and connect them to all other nodes in G within some distance $\lambda_n \ll \varepsilon_n$ from x, y in M . The auxiliary manifold distance between x, y is then the distance between them in the extended graph. We must select λ_n very carefully, to make sure the auxiliary distance is a sufficiently good approximation of the true distance in M .

Once we have this auxiliary distance in M , we need to approximate the Ollivier balls in RGGs G with the corresponding balls in M . That is, on the one hand we have sets of nodes within graph distance δ_n from node x , while on the other hand we have sets of nodes within manifold distance δ_n from x . We work out this approximation by deriving an upper bound on the Wasserstein distance between the uniform probability distributions on these two balls. This allows us to work with balls defined by manifold distances, instead of graph distances.

Finally, the most nontrivial step, we must relate the discrete uniform probability distribution defined on finite sets of nodes in balls in G to the corresponding nonuniform continuous distributions defined on the continuum of points in balls in M . To this end, using a result on matching two Poisson point processes, we first bound the Wasserstein distance between these two distributions in Euclidean space. We finally extend this result to non-Euclidean M , using the local flatness of the space together with the fact that the neighborhoods that we consider shrink as $\delta_n \rightarrow 0$.

Overall, at each step of this proof, we introduce an approximation error to the relation between the Wasserstein distances in graphs G and manifold M . The crux of each step is that it must be done with care to ensure that the introduced error is upper bounded by $\delta_n^3 \rightarrow 0$. This is because if the error is that small, then (10) follows from (5). The further details behind how we achieve this at each step of the proof can be found in Appendix A.

2. Simulations

The fact that we can prove the Ollivier $\rightarrow$ Ricci convergence only under the conditions (12) does not mean that there is no convergence outside of this parameter region, and this is indeed what we observe in simulations.

For simulations we select the sphere, torus, and the Bolza surface as the three 2D manifolds of constant Ricci curvature $+1$, 0 , and -1 . The Bolza surface [71,72] is the simplest hyperbolic 2D manifold of genus 2 with no boundaries. We consider the Bolza surface, versus something simpler of negative curvature, such as a hyperbolic disk, because we want our manifolds to have no boundaries. We want this, because computing Ollivier curvature in simulations is a major challenge at large n , so that we do not want any boundary effects at any n .

We then fix x and y as described above, sprinkle n points on the three manifolds uniformly at random according to the manifold volume form, and link all pairs of points at distances $\leq \varepsilon_n$ on the manifold. We then find the Ollivier balls $B_G(x, \delta_n)$, $B_G(y, \delta_n)$, set up the uniform probability distributions $\mu_x = 1/|B_G(x, \delta_n)|$, $\mu_y = 1/|B_G(y, \delta_n)|$ on them, and compute all the pairwise distances $d_G(x_i, y_j)$ in graph G between nodes $i \in B_G(x, \delta_n)$ and $j \in B_G(y, \delta_n)$ whose coordinates are x_i and y_j . All these data allow us to compute the Wasserstein distance $W_G(\mu_x, \mu_y)$ by solving the linear program

$$W_G(\mu_x, \mu_y) = \min_{\rho_{ij}} \sum_{i,j} d_G(x_i, y_j) \rho_{ij} \mu_x, \text{ such that}$$

$$\sum_i \rho_{ij} \mu_x = \mu_y, \quad 0 \leq \rho_{ij} \leq 1, \quad \sum_j \rho_{ij} = 1, \quad (18)$$

where the minimization is over transportation plans ρ_{ij} whose entries describe pairwise movements of probability masses. Having computed this $W_G(\mu_x, \mu_y)$, Ollivier curvature is finally given by (8). Further details on the simulations are in Appendix C.

The results are shown in Fig. 2. Remarkably, we observe that rescaled Ollivier curvature in our RGGs converges to Ricci curvature of the underlying manifold not only in the parameter regime accessible to our proofs, Fig. 2(a), but also well outside of this regime, for much sparser graphs, Fig. 2(b). In particular, we observe that the convergence is as good for truly sparse graphs with constant average degree as for denser graphs.

B. ε_n weights

On the one hand, we may be not too happy that the weighted RGGs considered in the previous section contain too much information about the manifold in their edge weights $w_{ij} = d_M(i, j)$, helping the convergence “too much.” On the other hand, we cannot be happy at all if edges are unweighted, $w_{ij} = 1$, because in this case there cannot be any curvature convergence whatsoever for the reasons discussed in Sec. III. The maximum-happiness point appears to be the middle-ground possibility discussed in Sec. III—all edges are weighted by the same weight decaying to zero and playing the role of a system scale, à la the Planck scale. This “Planck scale” in our case is the RGG connectivity radius ε_n . That is,

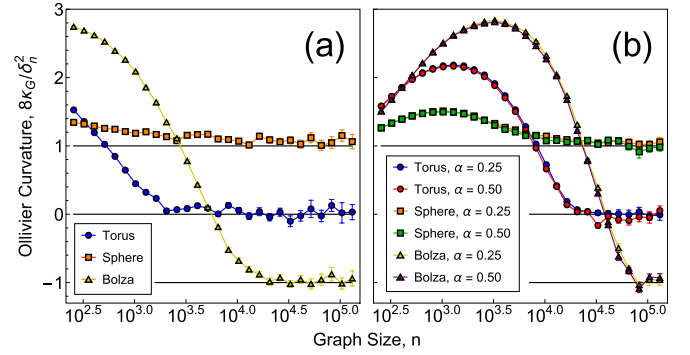


FIG. 2. Ollivier-Ricci curvature convergence in constant-curvature random geometric graphs. Panel (a) shows the simulation data for rescaled Ollivier curvature (8) in (9)-weighted random geometric graphs on the ($D = 2$)-dimensional torus, sphere, and Bolza surface. The scaling exponents of the connectivity and the Ollivier ball radii (11) in the graphs are set to $\alpha = \beta = 0.16 < 1/6$ lying within the proof-accessible regime (14). Panel (b) shows apparent convergence for $\alpha = \{1/4, 1/2\}$ and $\beta = 1/4$. Both of these two settings lie well outside of the proof-accessible regime (12), while $\alpha = 1/2$ corresponds to truly sparse graphs with constant average degree. The error bars represent the standard error $\sigma/\sqrt{n_s}$, where σ is the standard deviation and $n_s = 10 \times 2^{17}/n$ is the number of sampled random graphs of size n . The smallest graph size n is chosen to ensure the graphs are connected, while the largest size $n = 2^{17}$ is bounded by memory constraints of the MOSEK package.

the weights w_{ij} of all edges i, j in our RGGs in this section are

$$w_{ij} = \varepsilon_n. \quad (19)$$

This weighting scheme is equivalent to saying that we actually work with *unweighted* graphs, except that we multiply by ε_n all distances in them, which are hop lengths of shortest paths. If $d_G(z_1, z_2)$ is the shortest path hop distance between vertices z_1 and z_2 in the unweighted RGG G , then the rescaled graphs distance $d'_G(z_1, z_2) = \varepsilon_n d_G(z_1, z_2)$ is actually a good approximation to the manifold distance $d_M(z_1, z_2)$. Unfortunately, how good this approximation is (characterized by *stretch* d'_G/d_M) has been rigorously quantified only for the RGGs in the Euclidean plane [73]. Since the distances we approximate are bounded by the Ollivier ball radius δ_n that tends to zero, and since any Riemannian manifold is locally Euclidean, our results below relying on [73] hold for any 2D manifold. In fact, it turns out that the results in [73] are strong enough for our proof strategy in the weighted case to work essentially without a change in this unweighted case as well, Appendix B. While our result below holds only for $D = 2$, analogous results for higher dimensions can be worked out as soon as stretch results for higher-dimensional RGGs are obtained.

The result in Appendix B is that with weights in (19), the Ollivier $\rightarrow$ Ricci convergence holds in the same strong sense (10) if

$$0 < \beta < \frac{1}{9}, \quad 3\beta < \alpha < \frac{1 - 3\beta}{2}. \quad (20)$$

Similarly to the weighted case, these conditions allow for arbitrarily sparse but not ultrasparse graphs ($\beta \rightarrow 0$, $\alpha \rightarrow$

1/2). Unlike the weighted case, these conditions imply that, as expected, the convergence is slower: $\beta < 1/9$ ($2D$ unweighted) versus $\beta < 1/6$ ($2D$ weighted). Another difference with the weighted case is that these conditions do not allow for the connectivity radius ε_n to be the same as the Ollivier ball radius δ_n : $\alpha \geq 3\beta \Rightarrow \delta_n \gg \varepsilon_n$. We believe this is not a deficiency in our proof techniques, but a reflection of reality. It seems obvious that one must consider mesoscopic neighborhoods of size $\delta_n \gg \varepsilon_n$ in unweighted graphs since one-hop microscopic ε_n neighborhoods in these graphs are too small to feel any curvature.

V. DISCUSSION

We have established a rigorous connection between curvatures of discrete and continuous objects—random graphs and Riemannian manifolds. Anecdotally, this connection is reminiscent of what Riemann had in mind working on the foundations of Riemannian geometry [74]. The closest result to ours, which deals with convergence of curvature of simplicial triangulations of manifolds in Regge calculus [17,50] and its many derivatives [75], is very different. Informally, while any simplex in any simplicial triangulation of any manifold is a chunk of space, graphs are much more primordial objects in that they do not have any space attached to them whatsoever.

We saw why the convergence of discrete curvature to a continuous one is such a delicate matter. If we use the standard simple definition of Ollivier graph curvature, then the immediately obvious observation is that there cannot be any convergence. We really have to consider mesoscopic regions in graphs with properly rescaled distances for the convergence to take place, an observation that has been entirely overlooked in the past. Even then the graph density and mesoscopic region sizes cannot be arbitrary, but should enter in a delicate interplay. These nontrivial delicate points are one reason why there are so few results on discrete $\rightarrow$ continuous curvature convergence.

Our convergence proofs can likely be improved since our simulations suggest that the convergence holds well outside of the parameter regimes accessible by our proof techniques. Our simulations can be improved as well. We could not simulate graphs large enough to see any convergence in ε_n -weighted graphs, where the convergence is slow. In any case, the investigation of the exact boundaries between regimes where the convergence does and does not take place is a wide open problem, both for Ollivier curvature and for all other discrete curvature definitions mentioned in Sec. II, for most of which it is not even known under what circumstances, if any, they do converge.

If some discrete curvature does not converge to a continuous one for some discrete structure, then it may very well happen that this discrete curvature tells us something very different about the structure from what is intended. In such a case, why should we call this “curvature” *curvature*? This problem is not semantic but practical. If we are sure that graph curvature converges to manifold curvature, then we can be also sure that our graph curvature can be used to learn reliably manifold geometry—an exciting bit of news for manifold learning and many other network or data science applications mentioned in the introduction. As an example of the other

extreme, if we are sure that curvature of some tensor network does not converge to Ricci curvature of classical spacetime, then we can be also sure that this tensor network is not a good model of quantum spacetime.

As far as possible applications to quantum gravity are concerned, our results are an important but very high-level step that does not touch on many critical aspects. We comment on some of them.

First of all, we proved the convergence of curvature only for graph links and paths. Their curvatures first need to be averaged to result in scalar Ollivier-Ricci curvature at a vertex, and then integrated over vertex sets to yield a discrete version of the Einstein-Hilbert action. Is there any convergence in this case, and if so, then under what circumstances? Such convergence results would indicate unambiguously that the discrete Einstein-Hilbert action can be constructed from Ollivier curvature.

A much more difficult problem is the following. What we have solved is the direct problem of graph curvature convergence to space curvature when space is there. The problem in quantum gravity is actually the inverse problem of geometrogenesis [28,29,76]—there is no space, only a discrete quantum structure, which must “look like” space in the classical limit. That is, an illusion of continuous space must somehow emerge in the limit.

One strategy to address this problem is conceptually similar to [77]. One can consider a canonical maximum-entropy Gibbs ensemble of random graphs with fixed values of Ollivier curvature on links, paths, or vertices. The questions then are as follows: is this ensemble equivalent (in the continuum limit) to the ensemble of random geometric graphs on a Riemannian manifold with the corresponding values of Ricci curvature, and do we have any (second-order) phase transitions in this ensemble? If the answer to the first question is *yes*, then the graphs in this ensemble do look like spatial graphs, even though there is no space in their definition. If the connection between Ollivier curvature and the Einstein-Hilbert action is there, then the temperature parameter in this ensemble must be analogous to the gravitational coupling constant. In that context, the settings considered in this paper are particularly interesting. Indeed, since Ollivier curvature (8) is well defined for any shortest path between any pairs of nodes x, y located at any distance in the graph, one can probe graph curvature at *any* scale for renormalization and other purposes.

Finally, yet another critical issue is the issue of time, causality, and Lorentzian metric signature. There are two schools of thought concerning the issue. The first school believes that time and Lorentzian signature must be also emergent. This route is followed, for example, in the combinatorial quantum gravity program [28,29], where time and causality are expected to appear only in the continuum limit above a certain scale, below which spacetime is Riemannian, turning into a random graph at a second-order phase transition defining quantum gravity nonperturbatively.

The second school builds in time and causality *ab initio*, as in causal sets or causal dynamical triangulation. For our results to be applicable in these settings, one needs first to extend them from Riemannian spaces to Lorentzian spacetimes. To the best of our knowledge, no notion of random geometric

graphs or Rips complexes has been defined for Lorentzian manifolds. Since intersections of past and future light cones, Fig. 1(b), known as *causal diamonds* or *Alexandroff sets*, play the role of balls in Lorentzian geometry—specifically, the topology defined by Alexandroff sets agrees with the base topology in any nice Lorentzian spacetime [78]—we propose to define Lorentzian random geometric graphs as shown in Fig. 1(b). This definition is very different from causal sets, where any two time-like-separated nodes are linked, and there are no other links. In the Fig. 1(b) definition of Lorentzian random geometric graphs, there are both time-like and space-like links but only between nodes within causal diamonds of a finite size. Can Ollivier curvature be defined for these graphs and shown to converge to Ricci curvature of the underlying Lorentzian manifold?

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APPENDIX A: OLLIVIER $\rightarrow$ RICCI CONVERGENCE IN DISTANCE-WEIGHTED RGGs

Here we outline the main steps and ingredients in our curvature convergence proof. All further lower-level details can be found in [79].

Let μ_x^M and μ_y^M denote the normalized restrictions of the volume form in manifold M onto the balls $B_M(x, \delta_n)$, $B_M(y, \delta_n)$ in M , while μ_x^G and μ_y^G denote the uniform probability distribution over nodes lying within the balls $B_G(x, \delta_n)$, $B_G(y, \delta_n)$ in weighted RGG G . Observe that if

$$|W_G(\mu_x^G, \mu_y^G) - W_M(\mu_x^M, \mu_y^M)| \ll \delta_n^3, \quad (\text{A1})$$

then (10) follows from (5). Indeed, if (A1) holds, then

$$|\kappa_G(x, y) - \kappa_M(x, y)| = \frac{|W_G(\mu_x^G, \mu_y^G) - W_M(\mu_x^M, \mu_y^M)|}{\delta_n} \ll \delta_n^2,$$

so that

$$\left| \frac{\kappa_G(x, y)}{\delta_n^2} - \frac{\kappa_M(x, y)}{\delta_n^2} \right| \ll 1.$$

The main idea of the proof is thus to show (A1). We proceed in three steps.

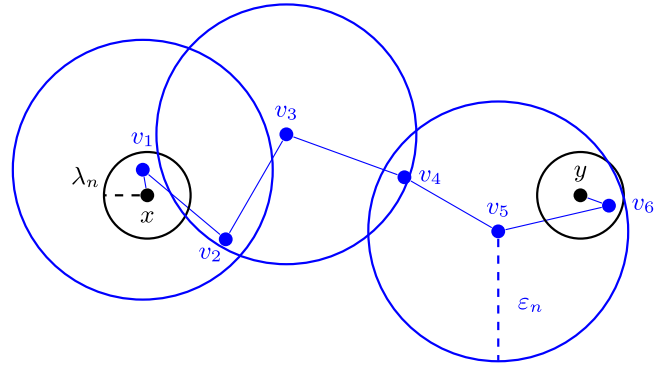


FIG. 3. Illustration of the construction of the extended graph distance $\tilde{d}_M$. The blue points are nodes of the RGG and the two black points are the selected points on the manifold. The blue circles indicate the connection radius ε_n , while the black circles have radius λ_n . In this example x is connected to node v_1 and y to node v_6 .

1. Approximating graph distances by manifold distances

The first difficulty we face is that the Wasserstein distances W_G and W_M are in different spaces with different distances: on the one hand we have RGG G with weighted shortest path distance d_G , and on the other hand we have the manifold M with distances d_M . How can we compare the two then? We simply extend d_G to a new distance $\tilde{d}_M$ on M , such that the difference between the new and original Wasserstein distances $\tilde{W}_M$ and W_M is $\ll \delta_n^3$.

This extension is accomplished by the following procedure. Let

$$\lambda_n = \ln(n)^{2/D} n^{-1/D}, \quad (\text{A2})$$

so that

$$n\lambda_n^D \rightarrow \infty. \quad (\text{A3})$$

If the conditions (12) hold, then we have

$$\lambda_n \ll \varepsilon_n, \quad (\text{A4})$$

$$\lambda_n \ll \delta_n^3. \quad (\text{A5})$$

Given any two points x, y in M we extend the RGG G by adding x and y to the set of G 's nodes, and connecting both x and y to all other nodes in G whose distance from x and y , respectively, in the manifold is less than λ_n , Fig. 3. We then define the new manifold distance $\tilde{d}_M(x, y)$ as the weighted shortest path distance in the extended graph. This new manifold distance $\tilde{d}_M$ is well defined, with probability converging to 1, because the expected number of points of a PPP of rate n in a ball of radius λ_n is $n \text{vol}[B_M(x, \lambda_n)] \sim n\lambda_n^D$, so that the probability to find a graph vertex within distance λ_n from any point x in M is given by $1 - e^{-n \text{vol}[B_M(x, \lambda_n)]}$ which converges to 1 thanks to (A3). Observe that if x and y happen to be already nodes in G , then the new manifold distance between them is the graph distance, $\tilde{d}_M(x, y) = d_G(x, y)$.

We now have to show that for any two points x and y in M , the distance $\tilde{d}_M(x, y) = d_M(x, y) + o(\delta_n^3)$. As evident from Fig. 3, $\tilde{d}_M(x, y) \geq d_M(x, y)$. For the upper bound, let us partition the geodesic between x and y into $m = \lceil 3d_M(x, y)/\varepsilon_n \rceil$ sections of equal length, and let $c_1 = x, c_2, \dots, c_m, c_{m+1} = y$

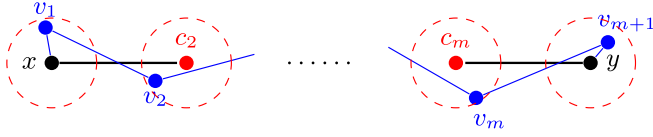


FIG. 4. Approximating the manifold distance by the extended weighted shortest path graph distance.

denote the $m + 1$ end points of this partition, Fig. 4. Consider the $m + 1$ balls $B_M(c_i, \lambda_n/4)$. The probability that every ball contains a graph vertex converges to 1. Denote those $m + 1$ vertices by $v_1, \dots, v_{m+1}$. We then observe the following:

(1) the node v_1 is at distance at most $\lambda_n/2$ from x , and so is v_m from y ;

(2) the distance between each consecutive pair of nodes v is bounded by

$$d_M(v_i, v_{i+1}) \leq d_M(x, y)/m + \lambda_n \leq \varepsilon_n/3 + \lambda_n;$$

(3) $d_G(v_i, v_{i+1}) \leq d_M(c_i, c_{i+1}) + \lambda_n$.

From these observations and (A4), we conclude that each consecutive pair of nodes v is connected in the RGG and

$$\begin{aligned} \tilde{d}_M(x, y) &\leq \lambda_n + \sum_{i=1}^m d_M(v_i, v_{i+1}) \\ &\leq \lambda_n + d_M(x, y)(1 + 3\lambda_n/\varepsilon_n). \end{aligned}$$

If $d_M(x, y) \leq C\delta_n$, for some $C > 0$, then

$$|d_M(x, y) - \tilde{d}_M(x, y)| \leq d_M(x, y) \frac{3\lambda_n}{\varepsilon_n} + \lambda_n \quad (\text{A6})$$

$$\leq 3C\lambda_n\delta_n\varepsilon_n^{-1} + \lambda_n \quad (\text{A7})$$

$$\sim \ln(n)^{\frac{2}{D}} n^{-\frac{1}{D}-\beta+\alpha} + \lambda_n. \quad (\text{A8})$$

The second term in the last equation is $\ll \delta_n^3$ by (A5), and so is the first term if (12) holds. We therefore conclude that

$$\tilde{d}_M(x, y) = d_M(x, y) + o(\delta_n^3).$$

Finally, the Wasserstein distances W_M and $\tilde{W}_M$ are defined by distances d_M and $\tilde{d}_M$, respectively. Therefore we immediately conclude that

$$|W_M(\mu_1, \mu_2) - \tilde{W}_M(\mu_1, \mu_2)| \ll \delta_n^3$$

for any two probability distributions μ_1 and μ_2 on M . If now x, y are also any two nodes of our RGG G , then $d_G(x, y) = d_M(x, y)$, so that it follows that if μ_1 and μ_2 are defined on G 's nodes, then

$$|W_M(\mu_1, \mu_2) - W_G(\mu_1, \mu_2)| \ll \delta_n^3. \quad (\text{A9})$$

The calculations above thus show that from now on we can always work only with the Wasserstein distance W_M in the space M , and this indeed is what we do in the next steps.

2. Approximating probability distributions in graphs

The next hurdle we need to overcome is comparing the uniform probability distribution μ_x^G on $B_G(x, \delta_n)$ with μ_x^M , the normalized restriction of the volume form to $B_M(x, \delta_n)$. The main difficulty here lies in that $d_M(x, z) \leq \delta_n$ does not necessarily imply that $d_G(x, z) \leq \delta_n$. In general,

the shape of the intersection $B_G(x, \delta_n) \cap B_M(x, \delta_n)$ can be highly nontrivial. We tame this shape by introducing a new probability distribution $\hat{\mu}_x^G$ which is the uniform probability distribution on the set $B_G := B_M(x, \delta_n) \cap G$ which is the set of G 's nodes that happen to lie within distance δ_n from x in the space, i.e., all nodes in the rose ball in Fig. 1(a). Observe that $B_G(x, \delta_n) \subseteq B_G$ since $d_G(x, y) \geq d_M(x, y)$. The main goal of this step is to show that $\langle W_M(\mu_x^G, \hat{\mu}_x^G) \rangle \ll \delta_n^3$.

First, consider (A6). If the conditions (12) hold, then $\lambda_n \ll \delta_n^3$ and $\lambda_n/\varepsilon_n \ll \delta_n^2$. It follows then that there exists a $\xi_n \ll \delta_n$ such that

$$|d_M(x, y) - d_G(x, y)| \leq d_M(x, y)\xi_n^2 + \xi_n^3. \quad (\text{A10})$$

For instance, we can take the ξ_n to be

$$\xi_n = \max \left\{ \sqrt{3\lambda_n/\varepsilon_n}, \lambda_n^{1/3} \right\}.$$

Define now the new radii $\delta_n^\pm = (\delta_n \pm \xi_n^3)/(1 \mp \xi_n^2)$, and note that $\delta_n^- < \delta_n < \delta_n^+$, so that

$$B_M(x, \delta_n^-) \subset B_M(x, \delta_n) \subset B_M(x, \delta_n^+).$$

Let $B_G^\pm := B_M(x, \delta_n^\pm) \cap G$ and $\mu_x^\pm$ denote the uniform probability distribution on $B_G^\pm$. We then obtain an upper bound for $W_M(\hat{\mu}_x^G, \mu_x^+)$ by considering the following joint probability distribution, also known as transport plan, μ for $(u, v) \in B_G \times B_G^+$:

$$\mu(u, v) = \begin{cases} \frac{1}{|B_G^+|}, & \text{if } u = v \in B_G, \\ \frac{1}{|B_G||B_G^+|}, & \text{if } u \in B_G, v \in B_G^+ \setminus B_G, \\ 0, & \text{otherwise.} \end{cases}$$

Since the Wasserstein distance $W_M(\hat{\mu}_x^G, \mu_x^+)$ is the infimum over all joint distributions (1), and since $\hat{\mu}_x^G$ and μ_x^+ are discrete, it follows that

$$\begin{aligned} W_M(\hat{\mu}_x^G, \mu_x^+) &\leq \sum_{u \in B_G, v \in B_G^+} d_M(u, v) \mu(u, v) \\ &= \sum_{u \in B_G, v \in B_G^+ \setminus B_G} \frac{d_M(u, v)}{|B_G||B_G^+|} \\ &\leq 2\delta_n^+ \frac{|B_G^+ \setminus B_G|}{|B_G^+|}. \end{aligned}$$

Since we are dealing with a PPP, both $|B_G^+ \setminus B_G|$ and $|B_G^+|$ are Poisson distributed with diverging means. Although they are not completely independent, it can be shown that

$$\left\langle \frac{|B_G^+ \setminus B_G|}{|B_G^+|} \right\rangle \sim \frac{\langle |B_G^+ \setminus B_G| \rangle}{\langle |B_G^+| \rangle + \langle |B_G| \rangle} \sim \frac{(\delta_n^+)^D - (\delta_n^-)^D}{\delta_n^D}.$$

This fraction is of the order $\xi_n^2 \ll \delta_n^2$, so that we conclude

$$\langle W_M(\hat{\mu}_x^G, \mu_x^+) \rangle \ll \delta_n^+ \delta_n^2 \sim \delta_n^3.$$

Finally, let node z lie in the manifold ball of radius δ_n^- around node x , $d_M(x, z) \leq \delta_n^-$. Then (A10) implies that the graph distance between x and z is upper bounded by δ_n :

$$d_G(x, z) \leq d_M(x, z)(1 + \xi_n^2) + \xi_n^3 \leq \frac{\delta_n - \xi_n^3}{1 + \xi_n^2} (1 + \xi_n^2) + \xi_n^3 = \delta_n.$$

This means that $z \in B_G(x, \delta_n)$, so that we have the following sandwich:

$$B_G^-(x, \delta_n) \subseteq B_G(x, \delta_n) \subseteq B_G^+(x, \delta_n).$$

Using calculations similar to the ones above for this sandwich, we obtain

$$\langle W_M(\mu_x^G, \mu_x^+) \rangle \ll \delta_n^3.$$

Therefore, by the triangle inequality of the W_M distance,

$$\langle W_M(\mu_x^G, \hat{\mu}_x^G) \rangle \leq \langle W_M(\mu_x^G, \mu_x^+) \rangle + \langle W_M(\mu_x^+, \hat{\mu}_x^G) \rangle \ll \delta_n^3. \quad (\text{A11})$$

The calculations above thus show that from now on we can work with the probability distribution $\hat{\mu}_x^G$ instead of μ_x^G . The former is more convenient to work with than the latter because it is over G 's nodes that happen to lie within manifold ball $B_M(x, \delta_n)$, the rose ball in Fig. 1(a).

3. Going from discrete to continuous probability distributions

The final step is to go from the discrete probability distribution $\hat{\mu}_x^G$ to the continuous distribution μ_x^M . Similarly to the goal of the previous section, the task is to show that

$$\langle W_M(\hat{\mu}_x^G, \mu_x^M) \rangle \ll \delta_n^3. \quad (\text{A12})$$

This is done by applying results on the matching distance between a PPP and points of a grid on the same space [80–83].

Note that both $\hat{\mu}_x^G$ and μ_x^M are now defined on $B_M(x, \delta_n)$. We now need to devise a transport plan from $\hat{\mu}_x^G$ to μ_x^M that assigns to any (measurable) set $A \subseteq B_M(x, \delta_n)$ how much mass from each PPP point in $B_G = G \cap B_M(x, \delta_n)$ is used to make up $\mu_x^M(A)$. We do so as follows.

First, suppose that space M is flat, the D -dimensional Euclidean space. We construct our transport plan in three steps:

- (1) we first place a grid on the space;
- (2) we next find a minimal matching between this grid and the PPP points in B_G ; and
- (3) finally, for each (measurable) set A in $B_M(x, \delta_n)$, we find all the PPP points in B_G that are matched to the grid points that lie in A .

Observe that this transport plan is such that every PPP point, i.e., graph vertex in B_G contributes an equal fraction of its mass to make up $\mu_x^M(A)$. This plan is also such that the largest distance any amount of mass has to move is given by the largest matching distance. Relying on the results from [80–83], it can then be shown that

$$\langle W_M(\hat{\mu}_x^G, \mu_x^M) \rangle \sim \ln(n)n^{-1/D} \ll \delta_n^3, \quad (\text{A13})$$

where the last inequality holds if the conditions (12) are satisfied.

Next, we relax the condition that the space M is flat. Indeed, it can be any nice Riemannian manifolds because $\delta_n \rightarrow 0$, so that we can map the ball $B_M(x, \delta_n)$ to the flat D -dimensional tangent space at x using the exponential map. We have to be careful here though, because the exponential map does not preserve distances. Still, by fixing a sufficiently small neighborhood U around the origin of the tangent space, we can ensure that for large enough n , $\exp^{-1} B_M(x, \delta_n) \subset U$,

and the distances are distorted by a fixed small amount:

$$B_D\left(0, \frac{\delta_n}{1+\xi}\right) \subseteq \exp^{-1} B_M(x, \delta_n) \subseteq B_D\left(0, \frac{\delta_n}{1-\xi}\right) \subset U,$$

where $B_D(0, \delta)$ is the D -dimensional Euclidean ball of small radius δ around the origin in the tangent space.

Finally, relying on this mapping and on the PPP mapping theorem [68] that applies to U , we see that the mapped PPP is still a PPP with intensity $\sim n$, now relative to the Euclidean volume form. Moreover, since the mapped ball is sandwiched between two balls whose radii scale as δ_n , the results for the D -dimensional Euclidean space above yield matching lower and upper bounds for the Wasserstein distance, from which (A12) follows.

The combination of (A12), (A11), and (A9) yields (A1), completing the proof.

APPENDIX B: OLLIVIER $\rightarrow$ RICCI CONVERGENCE IN ε_n -WEIGHTED RGGs

We first observe that the last step in the proof outlined in Appendix A does not rely on any graph distances at all. Further, the key element in the second step is (A10), which is also the fundamental ingredient in the first step in showing that distance $\tilde{d}_M$ is a good approximation to distance d_M , leading to (A9). Therefore, the convergence proof for ε_n -weighted graphs is complete as soon as (A10) is established for the new graph distance $d'_G = \varepsilon_n d_G$, where d_G is now the vanilla shortest path (hop count) distance in graph G . We also note that we have to do this only for the 2-dimensional Euclidean space, since the neighborhoods shrink and we can apply techniques similar to the ones in Appendix A3 to extend the result to any curved manifold.

We first observe that the conditions (20) imply that $\varepsilon_n \gg \sqrt{\ln(n)/n}$. In this case, the main stretch result from [73] reads

$$|d'_G(x, y) - d_M(x, y)| \leq d_M(x, y)\gamma_n + \varepsilon_n, \quad (\text{B1})$$

where $\gamma_n = \max\{\gamma_n^{(1)}, \gamma_n^{(2)}, \gamma_n^{(3)}\}$, and where the three γ_n 's have somewhat complicated explicit expressions with different constants and different scalings with n which can be shown to be

$$\gamma_n^{(1)} \sim \left(\frac{\ln n}{n\varepsilon_n^2}\right)^{\frac{2}{3}}, \quad \gamma_n^{(2)} \sim \left(\frac{\ln n}{n\varepsilon_n^2}\right)^2, \quad \text{and} \quad \gamma_n^{(3)} \sim \left(\frac{1}{n\varepsilon_n^2}\right)^{\frac{2}{3}},$$

if the conditions (20) hold. Since these conditions also imply that $n\varepsilon_n^2 \rightarrow \infty$, it follows that

$$\gamma_n \sim \left(\frac{\ln n}{n\varepsilon_n^2}\right)^{\frac{2}{3}} \quad (\text{B2})$$

in our case.

The ε_n in (B1) is $\ll \delta_n^3$ if the lower bound in the second condition in (20) holds. Therefore, to translate (B1) to (A10), all we need to do is to show that $\gamma_n \ll \delta_n^2$. If this holds, then we can simply take the ξ_n in (A10) to be $\xi_n = \max\{\sqrt{\gamma_n}, \varepsilon_n^{1/3}\}$. The requirement $\gamma_n \ll \delta_n^2$ is satisfied if

$$\frac{\ln n}{n\varepsilon_n^2\delta_n^3} \sim \ln(n)n^{2\alpha+3\beta-1} \ll 1, \quad (\text{B3})$$

which is indeed true if the upper bound in the second condition in (20) holds.

APPENDIX C: NUMERICAL METHODS

The computation of Ollivier graph curvature in simulations consists of three tasks: graph construction, distance matrix computation, and Wasserstein distance computation. We rely on a hybrid multicore solution which parallelizes each of these tasks, with the first two utilizing custom CUDA methods split among multiple GPUs, and the last utilizing multiple CPU cores via the MOSEK software package.

1. Graph construction

We construct random geometric graphs of size n by first sampling n points uniformly at random, according to the manifold volume form, on the surface of a unit torus, sphere, and Bolza surface. This is trivial for the first two manifolds, and less so for the Bolza surface.

The simplest representation of the Bolza surface [71,72] is the hyperbolic octagon with vertices at complex coordinates

$$o_k = R \exp \left[\frac{i\pi}{4} \left(\frac{1}{2} + k \right) \right], \quad k = 0, 1, \dots, 7, \quad (\text{C1})$$

where $R = 2^{-1/4}$ in the Poincaré disk model of the hyperbolic plane. When the opposite sides of this octagon are glued, the Bolza surface of constant curvature -1 is formed. To sprinkle points uniformly at random onto this octagon, we first sprinkle them uniformly at random, according to the volume form in the Poincaré model, which in the polar coordinates is

$$dV = \frac{4r dr d\theta}{(1 - r^2)^2}, \quad (\text{C2})$$

onto the Poincaré disk of radius R , and then remove those points that lie in this disk but do not lie in the octagon. We do this removal in the Klein disk model of the hyperbolic plane, because geodesics—octagon sides, in particular—are straight lines there. The map to go from the Poincaré (r, θ) to Klein (r_K, θ) polar coordinates is $(r_K, \theta) = (2r/(1 + r^2), \theta)$, and in the latter coordinates, the coordinates of points that lie in the octagon meet the condition $r_K < r_c[\phi(\theta)]$, where

$$r_c(\phi) = R_K \frac{\cos(\pi/8)}{\cos(\pi/8 - \phi)}, \quad \phi \in [0, \pi/4], \quad (\text{C3})$$

$$\phi(\theta) = \left(\theta - \frac{\pi}{8} [1 + 2k(\theta)] \right) \bmod 2\pi, \quad (\text{C4})$$

$$k(\theta) = \left\lfloor \frac{4}{\pi} \left(\theta - \frac{\pi}{8} \right) \right\rfloor \bmod 8, \quad (\text{C5})$$

where $R_K = 2R/(1 + R^2) = 2^{5/4}/(1 + \sqrt{2})$ is the octagon radius in the Klein model. The nodes that do not pass this test are thrown out. The remaining nodes lie in the octagon, and we work back with their Poincaré coordinates, $(r, \theta) = (1/r_K - \sqrt{1/r_K^2 - 1}, \theta)$.

After all the n nodes are sprinkled, we add two additional nodes x, y separated by distance δ_n , which are the centers of the two balls $B_G(x, \delta_n), B_G(y, \delta_n)$. Given the coordinates of all the $n + 2$ nodes, all node pairs are linked whenever

their pairwise distance on the surface is below the connection threshold ε_n . The distances on the torus and the sphere are

$$d_T(x_1, y_1; x_2, y_2) = \left[\left(\frac{1}{2} - \left| \frac{1}{2} - |x_1 - x_2| \right| \right)^2 + \left(\frac{1}{2} - \left| \frac{1}{2} - |y_1 - y_2| \right| \right)^2 \right]^{1/2}, \quad (\text{C6})$$

$$d_S(\theta_1, \phi_1; \theta_2, \phi_2) = \arccos [\cos \theta_1 \cos \theta_2 + \sin \theta_1 \sin \theta_2 \cos(\phi_1 - \phi_2)]. \quad (\text{C7})$$

On the Bolza surface, the distance formula is more complicated. We describe it next.

The Bolza surface can also be considered as the factor space $\mathbb{H}^2/\mathcal{F}$ [72], where $\mathbb{H}^2$ is the hyperbolic plane, and the Fuchsian group $\mathcal{F}$ is defined by its eight generators

$$g_k = \begin{pmatrix} a & be^{ik\pi/4} \\ be^{-ik\pi/4} & a \end{pmatrix}, \quad k = 0, 1, \dots, 7, \quad (\text{C8})$$

where $a = 1 + \sqrt{2}$ and $b = \sqrt{a^2 - 1}$. This group acts on the hyperbolic plane by linear fractional transformations. Let $\mathcal{N}$ be the set of the 48 elements of $\mathcal{F}$ given by

$$\hat{g}_{k,\ell} = \prod_{t=0}^{\ell} g_{k+3t}, \quad (\text{C9})$$

where $k = 0, 1, \dots, 7$ and $\ell = 0, 1, \dots, 5$. One can check that these elements map the original octagon to its 48 either side- or vertex-adjacent octagons in the tessellation of the hyperbolic plane induced by these identical octagons. Define $\mathcal{N}_0 := \mathcal{N} \cup I$ to be these 48 elements plus the identity element I . One can show that the distance on the Bolza surface between two points with complex coordinates z_1, z_2 ($z = re^{i\theta}$) is given by

$$d_B(z_1, z_2) = \min_{\hat{g} \in \mathcal{N}_0} d_H(z_1, \hat{g}z_2), \quad \text{where} \quad (\text{C10})$$

$$d_H(z_1, z_2) = 2 \operatorname{arctanh} \left| \frac{z_1 - z_2}{1 - z_1^* z_2} \right| \quad (\text{C11})$$

is the hyperbolic distance between the points in the Poincaré model.

We compute what nodes are linked in parallel using a multi-GPU solution, where each thread works on one pair of nodes, except in the case of the Bolza surface where each thread computes one or two of 49 distances and each thread block works on two node pairs. The adjacency matrix is first tiled such that (1) the data required to generate each tile fit on a single GPU and (2) the total number of tiles is a multiple of the number of GPUs available. This decomposition provides a scalable solution independent of the number or type of GPUs present in the system. The algorithm is improved by using the shared L1 memory cache to store the Bolza generators and partial results, warp shuffling to accumulate results, function templating to eliminate kernel branches, and asynchronous CUDA calls to pipeline data transfers across different GPUs.

2. Distance matrix computation

After constructing the graph, we identify the balls $B_G(x, \delta_n)$ and $B_G(y, \delta_n)$. When $\varepsilon_n = \delta_n$, these are simply the nearest neighbors of x and y , respectively; otherwise, they

are calculated using Dijkstra's algorithm [84]. For the largest graphs we consider, the size of these balls can be up to tens of thousands of nodes, resulting in over billions of pairwise distances between the nodes in the two balls. The overall simulation bottleneck is the computation of the matrix of the weighted shortest-path distances between these large sets of nodes.

We employ a custom multi-GPU A^* search algorithm using the methods described in [85]. The standard A^* algorithm works by constructing a priority queue of visited nodes z using a binary heap. In computing the shortest path distance between a source node x_i and destination node y_j , priorities are assigned to node z in the queue according to the heuristic function

$$f(z) = d_G(x_i, z) + h(z, y_j), \quad (\text{C12})$$

where $d_G(x_i, z)$ is the weighted graph distance between z and source $x_i \in B_G(x, \delta_n)$, while $h(z, y_j)$ is the lower-bound estimate of the distance between z and destination $y_j \in B_G(y, \delta_n)$. As soon as new node $z' \in B_G(y, \delta_n)$ is added to the priority queue, its weighted graph distance $d_G(x_i, z')$ is added to the distance matrix.

The A^* algorithm is implemented on a single GPU by utilizing multiple priority queues, one per CUDA thread, so

that it is efficient for graphs with large average degrees of the order of hundreds. Each priority queue extracts multiple states, after which we detect duplicates using a technique called parallel hashing with replacement, which is a modification of the cuckoo hashing scheme that avoids hash conflicts by allowing some duplicates to remain [85,86]. During this step, the heuristic for extracted nodes is updated, and they are re-added to the priority queues using a parallel heap insertion. This procedure continues until the destination node y_j has been extracted by at least one of the priority queues.

3. Wasserstein distance computation

Having in place the distance matrix between the sets of nodes in the two balls $B_G(x, \delta_n)$ and $B_G(y, \delta_n)$, we compute the Wasserstein distance between the uniform probability distributions on these balls by solving the linear program (18). For the simulations presented here, we found it sufficient to use the MOSEK package [87] as long as the number of variables, given by $|B_G(x, \delta)| |B_G(y, \delta)|$, is roughly less than 1.3×10^9 , past which we run out of memory. In most cases with smaller balls, however, we solve the linear program quite quickly using the standard primal-dual interior point method [88].

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Maple Computation of the Ollivier-Ricci Curvature of Graphs

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1 Background

Ricci curvature, originally defined in the setting of differential geometry, has recently attracted significant attention in the areas of graph theory, network science, and machine learning[4]. Departing from the original setting of Ricci Curvature, the *Ollivier-Ricci Curvature on Graphs* adapts the concept to discrete spaces, enabling the analysis of geometric and structural properties of networks.

Whereas the original definition of Ricci curvature refers to the inherent geometric properties of a Riemannian manifold, the Ollivier-Ricci Curvature on Graphs assigns a probability distribution to each vertex and takes the *Wasserstein/Earth Mover's metric* as its notion of distance ([3]). Put simply, the probability distribution assigned to a vertex v encodes the likelihood of moving to any one of the neighbors of v . In the context of “earth-moving,” it represents the fraction of the total mass of the neighbors of v which is assigned to each of the individual neighbors. (*Note:* From this point forward, *Ollivier-Ricci Curvature* refers to the Ollivier-Ricci Curvature on Graphs.) In addition to this, we permit edge weights, which often represent cost or distance. Following [2], the Ollivier-Ricci curvature of an edge xy is

$$\kappa(x, y) := 1 - \frac{W_1(\mu_x, \mu_y)}{d(x, y)},$$

where $W_1(\mu_x, \mu_y)$ is the Wasserstein-1 metric taking the probability distributions on μ_x and μ_y as its inputs and $d(x, y)$ is the weight of the edge xy .

1.1 Examples:

1.1.1 Cycles:

The Ollivier-Ricci curvature of C_5 : Assuming each vertex is assigned a uniform distribution, it suffices to compute the Ollivier-Ricci Curvature of an arbitrary edge, say $\{v_1, v_2\}$.

1. **Establishing the Distributions:** We chose to use a uniform distribution. Each vertex of C_5 has exactly two neighbors, so the distribution vector assigned to each vertex is $(1/2, 1/2)$.
2. **Cost Matrix:** The cost matrix represents the graph distance from the neighbors of v_1 , which are v_2 and v_5 , to the neighbors of v_2 , which are v_1 and v_3 .

	to v_1	to v_3
from v_2	1	1
from v_5	1	2

Table 1: Cost Matrix for the edge $\{v_1, v_2\}$ in C_5

3. **Finding the Wasserstein Distance:** This amounts to solving an LP. We want to move mass from μ_1 , the distribution vector of v_1 , to μ_2 , the distribution vector of v_2 .

	to v_1	to v_3	Total Supply
from v_2	a	b	1/2
from v_5	c	d	1/2
Total Demand	1/2	1/2	

Table 2: Transport Plan for $\{v_1, v_2\}$ in C_5

We seek to minimize $C = 1 \cdot a + 1 \cdot b + 1 \cdot c + 2 \cdot d$, under the constraints that $a + b = 1/2$, $c + d = 1/2$, $a + c = 1/2$, $b + d = 1/2$. The solution to this LP is $a = 0$, $b = 1/2$, $c = 1/2$, $d = 0$, so the minimal cost is $C = 1$.

4. **Computing Edge Curvature:**

$$\kappa(v_1, v_2) = 1 - \frac{W_1(\mu_1, \mu_2)}{d(v_1, v_2)} = 1 - 1 = 0$$

The edge curvatures for all edges is 0, so the Ollivier-Ricci curvature of C_n , which is the average over all edge curvatures, is 0. This is an opportune time for the interested reader to note how a weighted edge ($d(v_1, v_2) \neq 1$) would affect the curvature.

1.1.2 Complete Graphs:

Given uniform probability distributions at each vertex, the Ollivier-Ricci Curvature of a complete graph is easy to compute because of its substantial vertex transitivity. With this in mind, we compute the Ollivier-Ricci Curvature for a general complete graph, rather than a complete graph of fixed order, below.

1. **Establishing the Distributions:** In K_n , the uniform distribution vector has length $n - 1$ and entries all equal to $(n - 1)^{-1}$.
2. **Conceptualizing the Cost Matrix:** The neighborhoods of any two vertices u, v in K_n have substantial overlap. The only non-overlap in the neighborhoods is the two endvertices themselves: the only neighbor of u which is not also a neighbor of v is v itself. The Wasserstein distance between u and v is therefore $1/(n - 1)$.
3. **Computing Edge Curvature:** Assuming the edges are all unweighted,

$$\kappa(u, v) = 1 - \frac{(n - 1)^{-1}}{d(u, v)} = 1 - \frac{1}{n - 1}.$$

4. **Graph Curvature:** Since all the edge curvatures are the same, the Ollivier-Ricci Curvature of the graph is simply

$$1 - \frac{1}{n - 1}.$$

However, if one of the edges had weight 100, for example, the curvature of that edge would be $1 - (100n - 100)^{-1}$. In this case, the Ollivier-Ricci curvature of the graph would be

$$\frac{\left(\binom{n}{2} - 1 \right) \left(1 - \frac{1}{n-1} + \left[1 - \frac{1}{100n-100} \right] \right)}{\binom{n}{2}}.$$

1.1.3 A Negatively Curved Edge

Consider the graph G with 5 vertices and edge set

$$\{\{v_1, v_2\}, \{v_1, v_3\}, \{v_2, v_3\}, \{v_3, v_5\}, \{v_5, v_4\}\}.$$

We compute the curvature of the edge $\{1, 3\}$ by hand and leave the remaining edges as an exercise to the interested reader.

1. **Determining the Distribution:** Assuming a uniform distribution, the distribution vector at v_3 is $(1/3, 1/3, 1/3)$ corresponding to the three neighbors of v_3 . The distribution vector at v_5 is $(1/2, 1/2)$.
2. **Constructing the Cost Matrix:**

	to v_3	to v_4
from v_1	1	3
from v_2	1	3
from v_5	1	1

Table 3: Cost Matrix for the edge $\{v_3, v_5\}$ in G

3. **Finding the Wasserstein Distance:**

	to v_3	to v_4	Total Supply
from v_1	a	b	$1/3$
from v_2	c	d	$1/3$
from v_5	e	f	$1/3$
Total Demand	$1/2$	$1/2$	

Table 4: Transport Plan for edge $\{v_3, v_5\}$ in G

We seek to minimize $C = 1a + 3b + 1c + 3d + 1e + 1f$, subject to the following constraints: $a + b = 1/3, c + d = 1/3, e + f = 1/3, a + c + e = 1/2, b + d + f = 1/2$. The solution to this LP is $a = 1/3, b = 0, c = 1/6, d = 1/6, e = 0, f = 1/3$, which corresponds to $C = 4/3$.

4. **Computing the Curvature:** The curvature of the unweighted edge v_3, v_5 in G is therefore $1 - 4/3 = -1/3$.

Note: In the accompanying Maple file, you can verify this example by running `EdgeCurvsList(G)` where

$$G = \text{Graph}([1, 2, 3, 4, 5], \{\{1, 2\}, \{1, 3\}, \{2, 3\}, \{3, 5\}, \{4, 5\}\}).$$

1.2 Community Detection

We can, of course, compute the Ollivier-Ricci curvature of graphs much larger and/or denser than those shown above. In practice, graphs of interest are quite large; for example, a protein-protein interaction network can have thousands of nodes, as could the network representing researchers and their coauthors. Methods relying on Ollivier-Ricci curvature have generated substantial interest in the area of community detection, where these methods often outperform standard connectivity- and eigenvalue-based techniques (see [4]).

The connection between Ollivier-Ricci Curvature and Community Detection is as follows: edges with negative Ollivier-Ricci Curvature represent bridges between communities, and edges with positive curvature tend to hold a community together. The community detection algorithm given in [4] iteratively removes the negatively curved edge of greatest magnitude until no negatively curved edges remain, leaving a set of likely communities. We implement this algorithm at the end of the following section.

2 Maple Code

There are three key pieces of information involved in the computation of Ricci Curvature: a graph, edge weights, and the distributions associated with each vertex. In the code that follows, we handle each of these components, while giving increasing flexibility to the user in terms of input parameters.

We begin with the case of Ollivier-Ricci Curvature for unweighted graphs with uniform distributions assigned to each vertex. The following is a simple procedure to create the uniform distribution vector and list of Neighbors for a vertex x in a graph G . Graphs are to be given as a standard Maple graph object unless otherwise noted.

```
distributionx:=proc(G,x) local Neis, n, distvec, i:
  Neis:=Neighbors(G,x):
  n:=nops(Neis):
  distvec:= Vector(n):
```



```

    distvec:=Vector([seq(1/n, i=1..n)]):
return([Neis, distvec]):
end:

```

Given two vertices x, y , we observe that the associated distributions μ_x, μ_y are supported on the set of neighbors of x and y , respectively. We denote these sets by $\{x_1, \dots, x_n\}$ and $\{y_1, \dots, y_m\}$, respectively. Let $d(x_i, y_i)$ denote the graph shortest path distance between x_i and y_i . The cost matrix procedure in `OlliRicci.txt` inputs G and two vertices x, y and outputs the $n \times m$ matrix C such that $C(i, j) = d(x_i, y_j)$. We refer to this matrix in the procedure `wasserstein`, in which we compute the Wasserstein/Earth Mover's distance using the following LP (see [1]):

$$\begin{aligned}
& \text{minimize} \sum_{i=1}^n \sum_{j=1}^m C_{ij} z_{ij} : \\
& \text{subject to} \sum_{j=1}^m z_{ij} = \text{mass at } i \text{ in } \mu_x \\
& \sum_{i=1}^n z_{ij} = \text{mass at } j \text{ in } \mu_y \\
& z_{ij} \geq 0 \text{ for all } i, j.
\end{aligned}$$

The implementation is as follows:

```

wasserstein:=proc(x, y, G) local n,m, obj, distx, disty,
Neisx, Neisy, C, flovars, constr,
i, j, flovarslist, LPsol, wassersteindist:
distx:=distributionx(G, x):
disty:=distributionx(G, y):

Neisx:=distx[1]:
Neisy:=disty[1]:
n:=nops(Neisx):
m:=nops(Neisy):

```

```

C:=costmatrix(G, x, y):

if n=0 or m=0 then return("ERROR: isolated vertex"):
fi:

flovars:=Matrix(n,m,(i,j) -> Z[i,j]):
obj:=add(add(C[i,j]*flovars[i,j], j=1..m),i=1..n):
constr:={}:

#outflow constraints
for i from 1 to n do
  constr:=constr union {add(flovars[i,j], j=1..m)=distx[2][i]}:
od:
#inflow constraints
for j from 1 to m do
  constr:=constr union {add(flovars[i,j], i=1..n)=disty[2][j]}:
od:
#nonnegativity
for i from 1 to n do
  for j from 1 to m do
    constr:=constr union {flovars[i,j]>=0}:
  od:
od:
#formatting for LP solver
flovarslist:=[seq(seq(flovars[i,j], j=1..m), i=1..n)]:

LPsol:=LPSolve(obj, constr):
wassersteindist:=LPsol[1]:
RETURN(wassersteindist);

end:

```

We also include in `OlliRicci.txt` the procedure `wassermanualdist`, whose inputs are $(x, \text{distx}, y, \text{disty}, G)$, where x and y are vertices, G is a graph, and, distributions `distx` and `disty` as lists.

Synthesizing the aforementioned procedures, `OlliRicciEdge` computes the Ollivier-Ricci curvature of an edge. This procedure allows the user to input the weight of the edge, allowing for use with weighted graphs. Extend-

ing this, `OlliRicciEdgeD` calls `wassermanualdist`, accommodating graphs with non-uniform distribution vectors.

Finally, the collection of procedures `OlliRicci`, `OlliRicciW`, and `OlliRicciWD`, compute the Ollivier-Ricci Curvature of unweighted graphs with uniform distributions, weighted graphs with uniform distributions, and weighted graphs with non-uniform manually entered distributions. In the case of weighted graphs, we depart from the standard Maple graph notation and use the format $G = E$, where each $e \in E$ is of the form `{endvertex1, endvertex2, weight}`. For convenience, `OlliRicciWD` is included below:

```
OlliRicciWD:=proc(G1, D) local G, verts, wedges, edgecurvs,
    edges,i, f, e:
    verts:=[seq(i, i=1..G1[1])]:
    wedges:=G1[2]:
    edgecurvs:=[]:
#need to process G
    edges:={}:
    for f in wedges do
        edges:= edges union {{f[1], f[2]}}:
    od:
    G:=Graph(verts, edges):
#build list of weighted curvatures
    for e in wedges do
        dist(e[1]):=[Neighbors(G, e[1]), Vector(D[e[1]])]:
        print(e, e[1], dist(e[1]));
        dist(e[2]):=[Neighbors(G, e[2]), Vector(D[e[2]])]:
        print(e, e[2], dist(e[2]));
        edgecurvs:=[op(edgecurvs), OlliRicciEdgeMD(G, e[1],
            dist(e[1]), e[2], dist(e[2]), e[3])]:
    od:
    return(add(edgecurvs)/nops(wedges)):
end:
```

As discussed in 1 community detection is an application of Ricci Curvature that has garnered recent interest. The authors of [4] supply an algorithm for detecting communities in graphs using Ricci Curvature. Their implementation scales well (much better than what we show below) and has yielded novel insights for real-world problems. In the following, we implement their algorithm in Maple.

The first step involves “pruning” the given graph by removing all edges with negative Ricci curvatures. We recall that negative curvatures tend to indicate FILL IN. It is worth noting that this is the slowest step in our implementation.

```

Prune:=proc(G) local Gcopy, ecurvs, x, y, edge,
                minEdge, minCurv, aff, u,v, Neis, e:
Gcopy:=CopyGraph(G):
ecurvs:=table():
for edge in Edges(Gcopy) do
    x:=edge[1]:
    y:=edge[2]:
    ecurvs[[x,y]]:=OlliRicciEdge(Gcopy, x, y, 1):
od:

while true do
    minEdge:=NULL:
    minCurv:=0:

    for edge in Edges(Gcopy) do
        x:=edge[1]:
        y:=edge[2]:
        if assigned(ecurvs[[x,y]]) and ecurvs[[x,y]]< minCurv then
            minCurv:=ecurvs[[x,y]]:
            minEdge:=[x,y]:
        fi:
    od:

    if minEdge =NULL then break: fi:

    x:=minEdge[1]:
    y:=minEdge[2]:
    RemoveEdge(Gcopy, x, y):

    ecurvs[[x,y]]:='undef':

    #optimizing: recomputing curvatures only for
    #affected edges, not for all except removed

```

```

for u in [x,y] do
  Neis:=Neighbors(Gcopy, u):
  for v in Neis do
    if v<u then
      e:=[v,u]:
    else
      e:=[u,v]:
    fi:
    ecurvs[[op(e)]]:=OlliRicciEdge(Gcopy, e[1], e[2], 1):
  od:
od:
RETURN(Gcopy):
end:

```

After the graph is “pruned” we consider each of the remaining connected components as communities. In the Maple package accompanying this paper, the community detection procedure `RicciCommDetect` takes two arguments, (G, s) , where s is the smallest community size we consider.

3 Future Work

As with any computer program, there is room for optimization. Particularly, the shortest path computations in the cost matrix could be optimized for large graphs potentially using a breadth-first search or other algorithms. The current computation is slow for dense graphs. In the table below, we show the computing time on a standard laptop for the unweighted Ollivier-Ricci Curvature of each of the complete graphs on 2 to 20 vertices:

2	3	4	5	6	7	8	9	10	
0.001	0.003	0.005	0.012	0.025	0.044	0.081	0.238	0.241	
11	12	13	14	15	16	17	18	19	20
1.41	1.77	3.56	5.57	8.63	13.0	19.1	25.3	41.5	52.4

Table 5: Computation times for OlliRicci for Complete Graphs

Additionally, for any graph with negative edge curvatures, the **Prune** step is very slow. Optimizing that step would make the community detection

algorithm more viable.

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SPECTRAL GRAPH THEORY

NICHOLAS PURPLE

ABSTRACT. This paper is an introduction to certain topics in graph theory, spectral graph theory, and random walks. We begin with basic definitions in graph theory, moving then to topics in linear algebra that are necessary to study the spectra of graphs. Then, finally, to basic results of the graph's matrices and an introduction to random walks.

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1. INTRODUCTION

While graphs are typically represented as a set of vertices and a set of edges, they can also be represented by several different matrices. This naturally motivates the inquiry as to the nature of a graph's spectrum. The spectrum of a graph has proven itself to be surprisingly informative about that graph's characteristics, such as the diameter of a graph and its connectivity. We will explore some of these results within this paper as well as discuss the basics of random walks on graphs.

We begin with basic but necessary definitions in graph theory that are important to both describe and prove results in spectral graph theory. Then, we move to topics in linear algebra that are required to understand spectral graph theory, such as the Rayleigh quotient and spectral theorem. Next, we discuss basic results of a graph's matrices and characteristics of a graph that can be read from the matrices and their corresponding eigenvalues. Finally, we begin a very basic introduction to random walks on graphs with a discussion of the transition matrix.

2. BASIC DEFINITIONS

Definition 2.1. A *graph* is a pair $G = (V, E)$, where E is a multiset whose elements are 2-subsets of V .

Elements of V are called the vertices of a graph G and the elements of E are called the edges of a graph G . The above definition is sometimes referred to as an undirected graph. Edges can be thought of as lines connecting two vertices or as loops starting and ending at the same vertex.

Definition 2.2. An edge $e = (v_i, v_j)$ is *incident to* a vertex v if $v = v_i$ or $v = v_j$.

Definition 2.3. An edge $e = (v_i, v_j)$ *joins* the two vertices v_i and v_j .

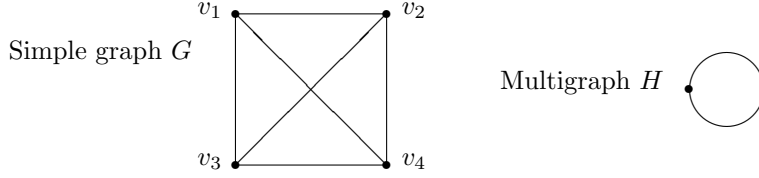
An edge that joins a vertex v to itself is called a *loop*, and if there exists more than one edge that is incident to a pair of vertices u and v , then there are *multiple edges* joining those u and v .

Definition 2.4. Two vertices v and u are *adjacent* if there exists an edge $e \in E$ where e joins v and u . If v and u are adjacent, then $v \sim u$.

Definition 2.5. A *simple* graph G is a graph that contains no loops or multiple edges.

Unless stated otherwise, in this paper graphs will be *simple* and finite. This means that the set of vertices is finite.

Definition 2.6. The *degree* of a vertex v , denoted d_v , is the number of edges incident to v .



Example 2.7. The diagram above on the left is a simple graph G with vertex set $V = \{v_1, v_2, v_3, v_4\}$ and edge set $E = \{\{v_1, v_2\}, \{v_1, v_3\}, \{v_1, v_4\}, \{v_2, v_3\}, \{v_3, v_4\}, \{v_2, v_4\}\}$. A special property of the above graph is that every pair of vertices is adjacent, forming a *complete graph*. Complete graphs are denoted by K_n , with n being the number of vertices in the graph, meaning the above graph is a K_4 . It should also be noted that all vertices are incident to the same number of edges. Equivalently, for all $v \in V$, $d_v = 3$. We call a graph where d_v is constant a *regular* graph. Therefore, all complete graphs are regular but not all regular graphs are complete.

The graph on the right, H , is the simplest example of a multigraph: a graph with one vertex and a loop.

Definition 2.8. A *walk* on a graph $G = (V, E)$ is a sequence of vertices $(v_0, \dots, v_{n-1})$ where $\{v_{i-1}, v_i\} \in E$ for $1 \leq i \leq n-1$. The *length* of the walk is $n-1$.

Definition 2.9. A *path* on a graph $G = (V, E)$ is a walk where all vertices and edges are distinct.

Definition 2.10. If there exists a path in G from vertex v_1 to vertex v_2 , then v_1 and v_2 are *connected*. A graph G is *connected* if every pair of vertices in G is connected.

Remark 2.11. We will assume in this paper that graphs are connected unless stated otherwise.

Other than representing graphs visually with vertices and edges, one can also represent them in terms of matrices. Three matrices that can be used to study graphs are the adjacency matrix, the Laplacian, and the normalized Laplacian. While all three matrices have different constructions and eigenvalues, they all can indicate important features of a graph, including regularity, connectedness, and completeness.

Definition 2.12. The *adjacency matrix* A of a graph G is an $n \times n$ matrix indexed by V , where $n = |V|$, defined as:

$$A_{i,j} = \begin{cases} 1 & v_i \sim v_j \\ 0 & \text{otherwise} \end{cases}$$

Definition 2.13. The *Laplacian matrix* L of a graph G is also an $n \times n$ matrix indexed by V , where $n = |V|$, defined as:

$$L_{i,j} = \begin{cases} d_{v_i} & v_i = v_j \\ -1 & v_i \sim v_j \\ 0 & \text{otherwise} \end{cases}$$

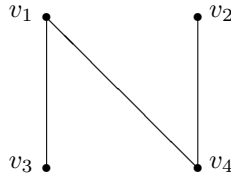
Remark 2.14. An equivalent definition of the Laplacian would be $L = D - A$, where D is a diagonal matrix with the degrees of the vertices down the diagonal and A is the adjacency matrix.

Definition 2.15. The *normalized Laplacian* $\mathcal{L}$ is an $n \times n$ matrix defined as:

$$\mathcal{L}_{i,j} = \begin{cases} 1 & v_i = v_j \\ -\frac{1}{\sqrt{d_{v_i}d_{v_j}}} & v_i \sim v_j \\ 0 & \text{otherwise} \end{cases}$$

Remark 2.16. Again, the normalized Laplacian can be defined with another, equivalent definition. It can be seen as the product of both the diagonal matrix D and the Laplacian L . More specifically, it can be defined as $\mathcal{L} = D^{-1/2}LD^{-1/2}$.

All of the matrices that were defined depend on a numbering of the vertices in V . However, different numberings can be thought of as simple permutations of the vertices. This means that the different matrices are unique up to conjugation by permutation matrices. So then, no matter the numbering placed on the vertices of a graph G , the spectral properties of all three matrices defined will be consistent.



Example 2.17. The vertices of the above graph have been assigned an order, so the adjacency, Laplacian, and normalized Laplacian matrices can be expressed as:

$$A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

$$L = \begin{pmatrix} 1 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix}$$

$$\mathcal{L} = \begin{pmatrix} 1 & -\frac{1}{\sqrt{2}} & 0 & 0 \\ -\frac{1}{\sqrt{2}} & 1 & -\frac{1}{2} & 0 \\ 0 & -\frac{1}{2} & 1 & -\frac{1}{2} \\ 0 & 0 & -\frac{1}{\sqrt{2}} & 1 \end{pmatrix}$$

3. RELEVANT LINEAR ALGEBRA

Spectral graph theory, as implied by the name, relies heavily on linear algebra. Before the results of the eigenvalues of the various matrices can be discussed, certain features of linear algebra must be defined and proven. By definition and seen in Example 2.14, all three matrices are symmetric, and therefore all of their respective eigenvalues, are real.

Definition 3.1. The standard inner product $\langle \cdot, \cdot \rangle$ of vectors $a, b \in \mathbb{C}^n$ is defined to be

$$\langle a, b \rangle = a^T \bar{b}$$

Theorem 3.2. *If A is a symmetric matrix with all real entries, then all eigenvalues of A are real. There also exists a basis for $\mathbb{R}^n$ comprised of orthonormal eigenvectors of A .*

Proof. Let $\lambda \in \mathbb{C}$ be an eigenvalue of A with corresponding eigenvector v . Note that since A is a symmetric matrix, we know that $A = \bar{A} = A^T$. Then:

$$\begin{aligned} \lambda \langle v, v \rangle &= \langle Av, v \rangle \\ &= \langle v, Av \rangle \\ &= \langle v, \lambda v \rangle \\ &= \bar{\lambda} \langle v, v \rangle \end{aligned}$$

Because $\bar{\lambda} = \lambda$, it must be true that $\lambda \in \mathbb{R}$. So if λ is an eigenvalue of A , then λ must be real.

A proof for the basis of orthonormal eigenvectors can be found [4]. □

Definition 3.3. A matrix A is called *Hermitian* if $A = \bar{A}^T$.

Because the adjacency matrix, Laplacian matrix, and normalized Laplacian matrix are all real, symmetric matrices, it is clear that all three of them are Hermitian matrices. A useful property of Hermitian matrices is that the Rayleigh quotient

can be used to find the eigenvalues, and by extension the eigenvectors, of Hermitian matrices.

Definition 3.4. Let A be an $n \times n$, Hermitian matrix. The *Rayleigh quotient* R_A is a function $\mathbb{R}^n \setminus \{0\} \rightarrow \mathbb{R}$ where

$$R_A(x) = \frac{x^T A x}{x^T x}$$

Quickly note that if x is an eigenvector with corresponding eigenvalue λ , then $R_A(x) = \lambda$ because

$$R_A(x) = \frac{x^T A x}{x^T x} = \frac{x^T \lambda x}{x^T x} = \lambda \frac{x^T x}{x^T x} = \lambda$$

The Rayleigh quotient can be used to find the minimum and maximum eigenvalues of a Hermitian matrix. Using those corresponding eigenvectors, all of the intermediate eigenvalues can be found.

So starting with a symmetric, $n \times n$ matrix A , there is a basis for $\mathbb{R}^n$ of orthonormal eigenvectors $v_1, \dots, v_n$ with corresponding eigenvalues $\lambda_1, \dots, \lambda_n$ where $\lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n$. Then, because $v_1, \dots, v_n$ forms a basis for $\mathbb{R}^n$, for any vector $x \in \mathbb{R}^n$, we can express x as a linear combination of those eigenvectors

$$x = \sum_i \alpha_i v_i \quad (\alpha_i \in \mathbb{R}).$$

First looking at the denominator of the Rayleigh quotient:

$$x^T x = \sum_{i=1, j=1}^n \alpha_i \alpha_j v_i^T v_j = \sum_i \alpha_i^2$$

Note that the j term is eliminated because when $i \neq j$, $v_i^T v_j = 0$ because they are orthogonal. Now, the numerator of Rayleigh's quotient:

$$x^T A x = x^T \left(\sum_{i=1}^n \alpha_i A v_i \right) = \sum_{i=1}^n \alpha_i x^T (\lambda_i v_i) = \sum_{i=1}^n \lambda_i \alpha_i^2$$

The entirety of Rayleigh's quotient can then be expressed as:

$$R_A(x) = \frac{\sum_{i=1}^n \alpha_i^2 \lambda_i}{\sum_{i=1}^n \alpha_i^2}$$

Then, because of the ordering placed on the eigenvalues:

$$R_A(x) = \frac{\sum_{i=1}^n \alpha_i^2 \lambda_i}{\sum_{i=1}^n \alpha_i^2} \leq \frac{\sum_{i=1}^n \alpha_i^2}{\sum_{i=1}^n \alpha_i^2} \lambda_n = \lambda_n$$

Because $R_A(x)$ has a maximum value of λ_n and we know that $R_A(x)$ achieves its maximum when $x = v_n$, we have that $\max_{x \neq 0} R_A(x) = \lambda_n$. A similar argument follows for $\min_{x \neq 0} R_A(x) = \lambda_1$.

We can expand this method to include all eigenvalues rather than just λ_1 and λ_n . For any k where $1 \leq k \leq n$, let $W = \text{span}(v_1, \dots, v_k)$. Then, for any $x \in W$,

we can represent x as $x = \sum_{i=1}^k \alpha_i v_i$. Therefore, the Rayleigh quotient for $x \in W$ will have similar bounds as the Rayleigh quotient for $x \in \mathbb{R}^n$:

$$R_A(x) = \frac{\sum_{i=1}^k \alpha_i^2 \lambda_i}{\sum_{i=1}^k \alpha_i^2} \leq \frac{\sum_{i=1}^k \alpha_i^2}{\sum_{i=1}^k \alpha_i^2} \lambda_k = \lambda_k$$

So we have that

$$\lambda_k = \max_{x \in W, x \neq 0} R_A(x)$$

For finding the k th eigenvalue with the minimum, we let $W = \text{span}(v_k, \dots, v_n)$ and the bounds on the Rayleigh quotient are reversed

$$R_A(x) = \frac{\sum_{i=1}^k \alpha_i^2 \lambda_i}{\sum_{i=1}^k \alpha_i^2} \geq \frac{\sum_{i=1}^k \alpha_i^2}{\sum_{i=1}^k \alpha_i^2} \lambda_k = \lambda_k$$

So we have that

$$\lambda_k = \min_{x \in W, x \neq 0} R_A(x)$$

4. RESULTS OF THE LAPLACIAN AND ADJACENCY MATRICES

As stated previously, different properties of a graph can be identified through the adjacency matrix, Laplacian matrix, and normalized Laplacian matrix. All three can display the regularity of a graph, whether a graph is connected, and if a graph is complete.

Theorem 4.1. *Let $G = (V, E)$ be a graph, where $|V| = n$, and let A be the adjacency matrix of G . If A is raised to the m -th power, then the entry $A_{i,j}^m$ is the number of walks from v_i to v_j of length m .*

Proof. We induct on m .

Base Case: Let $m = 1$. The $A_{i,j}$ entry of the adjacency matrix is by definition 1 if $v_i \sim v_j$ and 0 otherwise. Therefore, the $A_{i,j}$ entry is the number of walks from v_i to v_j .

Inductive Step: We assume the theorem is true for $k \geq 1$. Now let $m = k + 1$. We begin with A^k , where $A_{i,l}^k$ is the number of length k walks from v_i to v_l . Then, the number of walks from v_i to v_j is the sum of the number of walks from v_i to v_l multiplied by the number of walks from v_l to v_j for all $v_l \sim v_j$. So the desired number of walks can be represented as

$$\sum_{l=1}^n A_{i,l}^k \cdot A_{l,j}$$

This is the formula for matrix multiplication so $A_{i,j}^{k+1} = \sum_{l=1}^n A_{i,l}^k \cdot A_{l,j}$. Therefore,

$A_{i,j}^{k+1}$ is the number of walks of length $k + 1$ from v_i to v_j . $\square$

Remark 4.2. Notice that the diagonal entries of A^{k+1} will be the number of closed walks, or the number of walks of length $k + 1$ from a vertex w to w , which leads to an interesting property of the trace of A^{k+1} .

Lemma 4.3. *Let M be an $n \times n$ matrix. If β is an eigenvalue of M , then β^n is an eigenvalue of M^n .*

Proof. We induct on n .

Base case: Let $n = 1$. Then $\beta^1 = \beta$ is the definition of eigenvalue of M .

Now we assume that the statement holds for $k \geq 1$. Let $n = k + 1$. Because β is an eigenvalue of M , there is a corresponding eigenvector v . So then $\beta^{k+1}v = \beta \cdot \beta^k v = \beta A^k v$ by the Inductive hypothesis. Then

$$\begin{aligned}\beta A^k v &= A^k \beta v \\ &= A^k A v \\ &= A^{k+1} v\end{aligned}$$

Therefore, if β is an eigenvalue of M , then β^n is an eigenvalue of M^n . $\square$

We denote the eigenvalues of A as $a_1 \leq \dots \leq a_n$, the eigenvalues of L as $\lambda_1 \leq \dots \leq \lambda_n$, and the eigenvalues of $\mathcal{L}$ as $\gamma_1 \leq \dots \leq \gamma_n$.

Theorem 4.4. *Let $G = (V, E)$ be a graph with adjacency matrix A . Then $\sum_{i=1}^n \alpha_i^k = \text{tr}(A^k)$ is the number of closed paths on G .*

Proof. Let $k \in \mathbb{N}$. By Theorem 4.1, $\sum_{i=1}^n A_{i,i}^k$ is the total number of closed walks of length k on G . By Lemma 4.3, A^k has eigenvalues $\alpha_1^k, \dots, \alpha_n^k$. The trace of A^k is $\sum_{i=1}^n \alpha_i^k$, so $\sum_{i=1}^n A_{i,i}^k = \sum_{i=1}^n \alpha_i^k$. Therefore, $\sum_{i=1}^n \alpha_i^k$ is the number of closed paths on G . $\square$

Now, shifting focus to the Laplacian, we can use the Rayleigh quotient to observe an interesting property of the Laplacian's eigenvalues. Let $G = (V, E)$ be a graph with vertices $u_1, \dots, u_n$. Looking just at the numerator of the quotient $\frac{f^T L f}{f^T f}$, for any $f \in \mathbb{R}^n$:

$$\begin{aligned}f^T L f &= f^T (D - A) f \\ &= (f_1, \dots, f_n) \cdot \begin{pmatrix} d_1 f_1 - \sum_{u_i \sim u_1} f_i \\ \vdots \\ d_n f_n - \sum_{u_i \sim u_n} f_i \end{pmatrix} \\ &= \sum_{i=1}^n f_i \left(d_i f_i - \sum_{u_j \sim u_i} f_j \right) \\ &= \sum_{i=1}^n \left(d_i f_i^2 - \sum_{u_j \sim u_i} f_j \cdot f_i \right)\end{aligned}$$

$$\begin{aligned}
&= \sum_{i=1}^n \left(d_i f_i^2 - \frac{1}{2} \sum_{u_j \sim u_i} (f_j - f_i)^2 - \frac{1}{2} \sum_{u_j \sim u_i} f_j^2 - \frac{1}{2} d_i f_i^2 \right) \\
&= \sum_{i=1}^n d_i f_i^2 + \frac{1}{2} \left(2 \cdot \sum_{u_i \sim u_j} (f_i - f_j)^2 \right) - \frac{1}{2} \sum_{\substack{(u_i, u_j) \\ u_i \sim u_j}} f_i^2 - \frac{1}{2} \sum_{i=1}^n d_i f_i^2 \\
&= \sum_{u_i \sim u_j} (f_i - f_j)^2
\end{aligned}$$

Therefore, $f^T L f \geq 0$ for all $f \in \mathbb{R}^n$, which means that $\lambda_i \geq 0$ for all $1 \leq i \leq n$. Also, this means that $\gamma_i \geq 0$ for all $1 \leq i \leq n$.

Theorem 4.5. *Let $G = (V, E)$ be a graph where d is the maximal degree of the vertices in G . The interval $[-d, d]$ contains the adjacency eigenvalues. The interval $[0, 2d]$ contains the Laplacian eigenvalues. The interval $[0, 2]$ contains the normalized Laplacian eigenvalues.*

Proof. Let α be an adjacency eigenvalue with eigenvector a having entries a_i for $1 \leq i \leq n$. Then for vertices v_i for $1 \leq i \leq n$:

$$A \cdot a = \alpha a = \begin{pmatrix} \sum_{v_i \sim v_1} a_i \\ \vdots \\ \sum_{v_i \sim v_n} a_i \end{pmatrix}$$

Then, $\alpha a_i = \sum_{v_j \sim v_i} a_j$. Let $v_k \in V$ be a vertex in G where $|a_k|$ is maximal. Then:

$$\begin{aligned}
|\alpha| |a_k| &\leq \sum_{v_i \sim v_k} |a_i| \leq \sum_{v_i \sim v_k} |a_k| \\
&= d_v \cdot |a_k| \leq d \cdot |a_k|
\end{aligned}$$

Therefore, $|\alpha| \leq d$, implying that $-d \leq \alpha \leq d$.

Next, let λ be a Laplacian eigenvalue with eigenvector b . Then for vertices v_i for $1 \leq i \leq n$:

$$L \cdot b = \lambda b = \begin{pmatrix} d_{v_1} b_1 - \sum_{v_i \sim v_1} b_i \\ \vdots \\ d_{v_n} b_n - \sum_{v_i \sim v_n} b_i \end{pmatrix}$$

Then, $\lambda b_i = d_{v_i} b_i - \sum_{v_j \sim v_i} b_j$. Let $v_k \in V$ be a vertex in G where $|b_k|$ is maximal.

Then:

$$|d_{v_k} - \lambda| |b_k| \leq \sum_{v_i \sim v_k} |b_i| \leq |b_k| d_{v_k}$$

This implies that $|d_{v_k} - \lambda| \leq d_{v_k}$. Therefore, $0 \leq \lambda \leq 2d_{v_k} \leq 2d$.

Lastly, let γ be a normalized Laplacian eigenvalue with eigenvector c . Then for vertices v_i for $1 \leq i \leq n$:

$$\mathcal{L} \cdot c = \gamma c = \begin{pmatrix} c_1 - \sum_{v_i \sim v_1} \frac{c_i}{\sqrt{d_{v_1} d_{v_i}}} \\ \vdots \\ c_n - \sum_{v_i \sim v_n} \frac{c_i}{\sqrt{d_{v_n} d_{v_i}}} \end{pmatrix}$$

Then, $\gamma c_i = c_i - \sum_{v_j \sim v_i} \frac{c_j}{\sqrt{d_{v_j} d_{v_i}}}$. Let $v_k \in V$ be a vertex in G where $|c_k|$ is maximal.

Then:

$$|1 - \gamma| |c_k| \leq \sum_{v_j \sim v_k} \frac{c_j}{\sqrt{d_{v_j} d_{v_k}}} \leq \sum_{v_j \sim v_k} \frac{c_j}{d_{v_j}} \leq c_k$$

This implies that $|1 - \gamma| \leq 1$. Therefore, $0 \leq \gamma \leq 2$. $\square$

Definition 4.6. A graph $G = (V, E)$ is called *bipartite* if V is the union of two disjoint, independent sets V_1 and V_2 where every edge connects a vertex in V_1 to a vertex in V_2 . We call V_1 and V_2 *partite sets*.

Lemma 4.7. A graph is bipartite if and only if it only contains even closed walks.

Proof. First, let us prove the forward direction. Let G be a graph with partite sets V_1 and V_2 . Let $v \in V$, and without loss of generality let $v \in V_1$. A closed walk from v to v has to alternate between V_1 and V_2 , starting and ending in V_1 . Therefore, the closed walk must have an even length, or it would never return to v .

Now we prove the reverse. Assume that G has no odd closed walks. Then let $u \in V$ and divide the remaining vertices into two sets:

$$A = \{v \in V | \text{the shortest path between } u \text{ and } v \text{ has an odd length}\}$$

$$B = \{v \in V | \text{the shortest path between } u \text{ and } v \text{ has an even length}\}$$

Then let $u \in B$. Suppose that there exists $a, b \in A$ such that $a \sim b$. Then there is an odd path from u to a , an odd path from u to b , and a path of length 1 from a to b . Combined, this would form an odd closed walk, so no such a, b exist. A similar argument follows for adjacent vertices in B . Therefore, G is bipartite. $\square$

Theorem 4.8. Consider a graph $G = (V, E)$. The following statements are equivalent:

- (1) G is bipartite.
- (2) If α is an eigenvalue of A , then $-\alpha$ is also an eigenvalue.
- (3) $\sum_{i=1}^n \alpha_i^r = \text{tr}(A^r) = 0$ for all odd r .

Proof. First we prove that (1) implies (2): Let G be a bipartite graph. Then, if G is bipartite, disconnected vertices can be added to both partite sets in order to make them the same size. Each disconnected vertex will add a row of zeroes to

the adjacency matrix, meaning the rank of the matrix will not change. Then, the matrix can be permuted to achieve a matrix that resembles this form:

$$A = \begin{pmatrix} 0 & M \\ M^T & 0 \end{pmatrix}$$

Now, consider α to be an eigenvalue of A with a corresponding eigenvector $x = \begin{pmatrix} u \\ v \end{pmatrix}$. Then $Ax = \alpha x = \begin{pmatrix} Mv \\ M^T u \end{pmatrix}$. This implies that $Mv = \alpha u$ and $M^T u = \alpha v$.

Consider then vector $y = \begin{pmatrix} u \\ -v \end{pmatrix}$. So then $Ay = \begin{pmatrix} -Mv \\ Mu \end{pmatrix} = \begin{pmatrix} -\alpha u \\ \alpha v \end{pmatrix}$, which implies that $-\alpha$ is an eigenvalue of A .

Now we prove that (2) implies (3): For each eigenvalue α_i there is a corresponding eigenvalue α_j where $\alpha_j = -\alpha_i$. Then, for all odd r , $\alpha_j^r = -\alpha_i^r$. Therefore,

$$\sum_{i=1}^n \alpha_i^r = 0 \text{ for all odd } r.$$

Finally, we prove that (3) implies (1): This implies that there are no closed walks of odd length in G . So by Lemma 4.7, G is bipartite. $\square$

5. BASICS OF RANDOM WALKS

Random walks on a graph can be thought of as choosing, at random, a vertex of the graph. Then taking a step away from the chosen vertex simply entails moving randomly to one of its neighbors. The sequence of randomly chosen vertices can then be referred to as a *random walk*. When discussing random walks on graphs, the graphs discussed will typically be connected.

There are several types of graphs random walks can take place on, namely unweighted, undirected graphs and weighted, undirected graphs. A common feature of both is that the transition probabilities, which are the probabilities from one vertex to another, are described in a matrix P called the *transition matrix*. It should be noted that unlike other matrices discussed in this paper, the transition matrix is not symmetric.

Definition 5.1. For an unweighted graph, the i, j th entry of the transition matrix P is described by

$$P_{i,j} = \begin{cases} \frac{1}{d_i} & \text{if } v_i \sim v_j \\ 0 & \text{else} \end{cases}$$

Definition 5.2. An *edge weight* $w(u, v)$ is a weight on moving from vertex u to vertex v where $w(u, v) = w(v, u)$ and $d_u = \sum_x w(u, x)$.

Definition 5.3. The i, j th entry for the transition matrix for a weighted graph is then described by

$$P_{i,j} = \begin{cases} \frac{w(v_i, v_j)}{d_i} & \text{if } v_i \sim v_j \\ 0 & \text{else} \end{cases}$$

For an unweighted graph, it is fairly clear that for a vertex u , $\sum_v P_{u,v} = 1$, showing that it is a probability distribution. For a weighted graph, it is still true that $\sum_v P_{u,v} = 1$ by how the edge weights are defined, so it is also a probability

distribution. For a graph $G = (V, E)$ where $|V| = n$, the initial distribution can be determined by a $1 \times n$ vector f where $\sum_{i=1}^n f_i = 1$.

Lemma 5.4. *Let G be a graph and P be the corresponding transition matrix. Then the rows of P^k are distributions.*

Proof. We induct on k .

Base case: Let $k = 1$. By definition the rows of P are distributions.

Now we suppose the statement is true for $l \geq 1$. Let $k = l + 1$. We have that $P^{l+1} = P^l P$ and by the inductive hypothesis the rows of P^l are distributions. Let $P_{i,j}^l$ denote the i, j th element of P^l . The product $P^l P$ gives us

$$\begin{pmatrix} P_{1,1}^l & \cdots & P_{1,n}^l \\ \vdots & \ddots & \vdots \\ P_{n,1}^l & \cdots & P_{n,n}^l \end{pmatrix} \begin{pmatrix} P_{1,1} & \cdots & P_{1,n} \\ \vdots & \ddots & \vdots \\ P_{n,1} & \cdots & P_{n,n} \end{pmatrix} = \begin{pmatrix} \sum_i P_{1,i}^l P_{i,1} & \cdots & \sum_i P_{1,i}^l P_{i,n} \\ \vdots & \ddots & \vdots \\ \sum_i P_{n,i}^l P_{i,1} & \cdots & \sum_i P_{n,i}^l P_{i,n} \end{pmatrix}$$

Without loss of generality, let us look at the first row. The sum of all the elements in the first row can be expressed as $\sum_j \sum_i P_{1,i}^l P_{i,j}$. However this can be rearranged to be:

$$\begin{aligned} \sum_j \sum_i P_{1,i}^l P_{i,j} &= \sum_i P_{1,i}^l \sum_j P_{i,j} \\ &= \sum_i P_{1,i}^l \cdot 1 \\ &= \sum_i P_{1,i}^l \\ &= 1 \end{aligned}$$

Therefore, the rows of P^k are distributions. $\square$

Theorem 5.5. *Let $G = (V, E)$ be a graph where $|V| = n$, P is the transition matrix, and f is the initial distribution. Then for any $k \in \mathbb{N}$, fP^k is the distribution after k steps.*

Proof. We induct on k .

Base case: Let $k = 1$. By definition, P is the transition matrix containing the probabilities of taking one step from one vertex to another. The product of f and P is then

$$fP = \begin{pmatrix} \sum_i f_i P_{i,1} \\ \vdots \\ \sum_i f_i P_{i,n} \end{pmatrix}$$

To prove this is a distribution we take the sum $\sum_j \sum_i f_i P_{i,j}$. This sum can be reordered so that

$$\begin{aligned} \sum_j \sum_i f_i P_{i,j} &= \sum_j f_j \sum_i P_{j,i} \\ &= \sum_j f_j \cdot 1 \\ &= \sum_j f_j \\ &= 1 \end{aligned}$$

Therefore, fP is the distribution after one step.

Now we suppose the statement is true for $l \geq 1$. Let $k = l + 1$. We begin with P^l , where $P_{i,j}^l$ is the probability of moving from v_i to v_j in l steps. Then the probability of moving from v_i to v_m in $l + 1$ steps is the sum of the probability of moving from v_i to v_j in l steps multiplied by the probability of moving from v_j to v_m in one step. This can be represented as

$$\sum_j P_{i,j}^k P_{j,l}$$

This is the formula for matrix multiplication so $P_{i,m}^{l+1} = \sum_j P_{i,j}^k P_{j,l}$. Therefore $P_{i,m}^{l+1}$ is the probability of moving from v_i to v_m in $l + 1$ steps. The proof of fP^{l+1} being a distribution is similar to the proof in the base case. $\square$

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A CHARACTERIZATION OF THE CLIFFORD TORUS

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ABSTRACT. In this paper, we prove that an n -dimensional closed minimal hypersurface M with Ricci curvature $Ric(M) \geq \frac{n}{2}$ of a unit sphere $S^{n+1}(1)$ is isometric to a Clifford torus if $n \leq S \leq n + \frac{14(n+4)}{9n+30}$, where S is the squared norm of the second fundamental form of M .

1. INTRODUCTION

Let M be an n -dimensional closed minimal hypersurface in a unit sphere $S^{n+1}(1)$ of dimension $n + 1$. Let S denote the squared norm of the second fundamental form of M . From the Gauss equation (see section 2), we know that S , which is extrinsic by definition, is actually an intrinsic quantity. It is well-known that Chern, do Carmo and Kobayashi [3] and Lawson [4] obtained independently that Clifford tori are the only closed minimal hypersurfaces of the unit sphere with $S = n$. When the scalar curvature of M is constant, Yang and the first named author proved in [6] and [7] that if $n \leq S \leq n + \frac{n}{3}$, then M is isometric to a Clifford torus $S^m(\sqrt{\frac{m}{n}}) \times S^{n-m}(\sqrt{\frac{n-m}{n}})$. A natural problem is that, for a closed minimal hypersurface M of a unit sphere, whether there exists a constant $\epsilon(n) > 0$ such that if $n \leq S \leq n + \epsilon(n)$, then $S = n$ and M is isometric to a Clifford torus $S^m(\sqrt{\frac{m}{n}}) \times S^{n-m}(\sqrt{\frac{n-m}{n}})$. The first named author [2] gave a positive answer under the additional condition that M has only two distinct principal curvatures. In general, it still remains open and it is a very hard problem. On the other hand, the Clifford torus $S^m(\sqrt{\frac{m}{n}}) \times S^{n-m}(\sqrt{\frac{n-m}{n}})$ is a closed minimal hypersurface in $S^{n+1}(1)$ with $S = n$ and its Ricci curvature varies between $\frac{n(m-1)}{m}$ and $\frac{n(n-m-1)}{n-m}$. If $2 \leq m \leq n-2$, then $Ric(M) \geq \frac{n}{2}$. Hence it is natural to ask

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whether there exists a constant $\epsilon(n) > 0$ such that if M is a closed minimal hypersurface with $\text{Ric}(M) \geq \frac{n}{2}$ and $n \leq S \leq n + \epsilon(n)$, then $S = n$ and M is isometric to a Clifford torus $S^m(\sqrt{\frac{m}{n}}) \times S^{n-m}(\sqrt{\frac{n-m}{n}})$ ($1 < m < n-1$). In this paper, we give an affirmative answer for the above problem.

Theorem 1. *Let M be an n -dimensional closed minimal hypersurface of a unit sphere $S^{n+1}(1)$ with Ricci curvature $\text{Ric}(M) \geq \frac{n}{2}$. If*

$$n \leq S \leq n + \frac{14(n+4)}{9n+30},$$

then $S = n$ and M is isometric to a Clifford torus $S^m(\sqrt{\frac{m}{n}}) \times S^{n-m}(\sqrt{\frac{n-m}{n}})$ ($1 < m < n-1$).

In particular, if $n \leq 5$, we obtain the following

Theorem 2. *Let M be an n -dimensional ($n \leq 5$) closed minimal hypersurface of a unit sphere $S^{n+1}(1)$. If*

$$n \leq S \leq n + \epsilon(n),$$

then $S = n$ and M is isometric to a Clifford torus $S^m(\sqrt{\frac{m}{n}}) \times S^{n-m}(\sqrt{\frac{n-m}{n}})$, where $\epsilon(3) = \frac{42}{85}$, $\epsilon(4) = \frac{8}{31}$ and $\epsilon(5) = \frac{3(21-5\sqrt{17})}{28+3\sqrt{17}}$.

Remark. For $n \leq 5$, Peng-Terng [5] proved the following: Let M be an n -dimensional ($n \leq 5$) closed minimal hypersurface of a unit sphere $S^{n+1}(1)$. If

$$n \leq S \leq n + \epsilon_1(n),$$

then $S = n$, where $\epsilon_1(n) = \frac{6 - 1.13n}{5 + \sqrt{17}}$. It is obvious that our pinching constant in Theorem 2 is larger than theirs.

2. LOCAL FORMULAE

Let M be an n -dimensional hypersurface in a unit sphere $S^{n+1}(1)$. We choose a local orthonormal frame field $\{e_1, \dots, e_{n+1}\}$ in $S^{n+1}(1)$, restricted to M , so that $e_1, \dots, e_n$ are tangent to M . Let $\omega_1, \dots, \omega_{n+1}$ denote the dual coframe field in $S^{n+1}(1)$. Then, in M ,

$$\omega_{n+1} = 0.$$

It follows from Cartan's Lemma that

$$(2.0) \quad \omega_{n+1i} = \sum_j h_{ij} \omega_j, \quad h_{ij} = h_{ji}.$$

The second fundamental form α and the mean curvature of M are defined by

$$(2.1) \quad \alpha = \sum_{i,j} h_{ij} \omega_i \omega_j e_{n+1} \quad \text{and} \quad nH = \sum_i h_{ii},$$

respectively. We recall that M is by definition a minimal hypersurface if the mean curvature of M is identically zero. The connection form ω_{ij} is characterized by the

structure equations

$$(2.2) \quad \begin{cases} d\omega_i + \sum_j \omega_{ij} \wedge \omega_j = 0, & \omega_{ij} + \omega_{ji} = 0, \\ d\omega_{ij} + \sum_k \omega_{ik} \wedge \omega_{kj} = \Omega_{ij}, \\ \Omega_{ij} = \frac{1}{2} \sum_{k,l} R_{ijkl} \omega_k \wedge \omega_l, \end{cases}$$

where Ω_{ij} (resp. R_{ijkl}) denotes the curvature form (resp. the components of the curvature tensor) of M . The Gauss equation is given by

$$(2.3) \quad R_{ijkl} = (\delta_{ik}\delta_{jl} - \delta_{il}\delta_{jk}) + (h_{ik}h_{jl} - h_{il}h_{jk}).$$

The covariant derivative $\nabla\alpha$ of the second fundamental form α of M with components h_{ijk} is given by

$$\sum_k h_{ijk} \omega_k = dh_{ij} + \sum_k h_{jk} \omega_{ik} + \sum_k h_{ik} \omega_{jk}.$$

Then the exterior derivative of (2.0) together with the structure equation yields the following Codazzi equation:

$$(2.4) \quad h_{ijk} = h_{ikj} = h_{jik}.$$

From the Codazzi equation, we know that h_{ijk} is symmetric in the indices i, j and k . Similarly, we have the covariant derivative $\nabla^2\alpha$ of $\nabla\alpha$ with components h_{ijkl} as follows:

$$\sum_l h_{ijkl} \omega_l = dh_{ijk} + \sum_l h_{ljk} \omega_{il} + \sum_l h_{ilk} \omega_{jl} + \sum_l h_{ijl} \omega_{kl},$$

and it is easy to get the following Ricci formula:

$$(2.5) \quad h_{ijkl} - h_{ijlk} = \sum_m h_{im} R_{mjkl} + \sum_m h_{mj} R_{mikl}.$$

Similarly, we also have

$$(2.6) \quad h_{ijklm} - h_{ijkml} = \sum_r h_{rjk} R_{ril m} + \sum_r h_{irk} R_{rjlm} + \sum_r h_{ijr} R_{rklm},$$

where the h_{ijklm} 's are the components of the covariant derivative $\nabla^3\alpha$ of $\nabla^2\alpha$. We should remark that h_{ijkl} and h_{ijklm} are symmetric in the first three indices i, j and k and generally not symmetric in the other ones. The components of the Ricci curvature and the scalar curvature are given by

$$(2.7) \quad R_{ij} = (n-1)\delta_{ij} - \sum_k h_{ik}h_{jk},$$

$$(2.8) \quad R = n(n-1) - \sum_{i,j} h_{ij}^2.$$

Now we compute certain local formulae. For any fixed point p in M , we can choose a local orthonormal frame field $e_1, \dots, e_n$ such that

$$(2.9) \quad h_{ij} = \begin{cases} 0 & \text{if } i \neq j, \\ \lambda_i & \text{if } i = j. \end{cases}$$

The following formulas can be obtained by a direct computation (cf. [1]). Let

$$S := \sum_{i,j} h_{ij}^2 = \sum_i \lambda_i^2,$$

$$(2.10) \quad \frac{1}{2}\Delta S = \sum_{i,j,k} h_{ijk}^2 - S(S-n),$$

$$(2.11) \quad \begin{aligned} \frac{1}{2}\Delta \sum_{i,j,k} h_{ijk}^2 &= \sum_{i,j,k,l} h_{ijkl}^2 + (2n+3-S) \sum_{i,j,k} h_{ijk}^2 \\ &\quad + 3(2B-A) - \frac{3}{2}|\nabla S|^2, \end{aligned}$$

where $A = \sum_{i,j,k} \lambda_i^2 h_{ijk}^2$ and $B = \sum_{i,j,k} \lambda_i \lambda_j h_{ijk}^2$.

3. PROOFS OF THE THEOREMS

At first we give two algebraic lemmas which will play a crucial role in the proofs of our theorems.

Lemma 1. *Let a_i ($i = 1, 2, 3, 4$) be real numbers satisfying $\sum_i a_i = 0$ and $\sum_i a_i^2 = a$. Then $\sum_i a_i^4 \leq \frac{7}{12}a^2$.*

Proof. We maximize the function $\sum_i a_i^4$ with the constraints $\sum_i a_i = 0$ and $\sum_i a_i^2 = a$. By means of the method of the Lagrange multiplier, we solve the following problem:

$$f = \sum_i a_i^4 + \lambda \sum_i a_i + \mu \left(\sum_i a_i^2 - a \right),$$

where λ and μ are the Lagrange multipliers. The maximum point of $\sum_i a_i^4$ is a critical point of f . Taking the derivative of f with respect to a_i , we have

$$f_{a_i} = 4a_i^3 + \lambda + 2\mu a_i = 0.$$

Hence, at most three of the a_i 's are distinct with each other at a critical point of f . We consider the following three cases.

- (1) Three of the a_i 's are distinct with each other. Without loss of generality, we denote them by a_1, a_2, a_3 and assume $a_1 = a_4$; then

$$2a_1 + a_2 + a_3 = 0, \quad 2a_1^2 + a_2^2 + a_3^2 = a.$$

Hence,

$$\begin{aligned} \sum_i a_i^4 &= 2a_1^4 + a_2^4 + a_3^4 = \frac{(a + 2a_1^2)^2}{2} - 14a_1^4 \\ &= \frac{a^2}{2} + 2aa_1^2 - 12a_1^4 \leq \frac{7}{12}a^2, \end{aligned}$$

i.e.,

$$\sum_i a_i^4 \leq \frac{7}{12}a^2.$$

- (2) Two of the a_i 's are distinct with each other. Without loss of generality, we denote them by a_1, a_2 and assume $a_1 = a_4$ and $a_2 = a_3$ or $a_1 = a_3 = a_4$; then $\sum_i a_i^4 \leq \frac{7}{12}a^2$.
- (3) If all of the a_i 's are the same, then $\sum_i a_i^4 = 0$.

Therefore, we conclude

$$\sum_i a_i^4 \leq \frac{7}{12} a^2.$$

This completes the proof of Lemma 1. $\square$

Lemma 2. *Let a_{ij} and b_i ($i, j = 1, \dots, n$) be real numbers satisfying $\sum_i b_i = 0$, $\sum_i b_i^2 = b > 0$, $\sum_{i,j} b_i a_{ij} = \frac{1}{2}b(n-b)$ and $\sum_{i,j} b_j a_{ij} = \frac{1}{2}b(n-b)$. Then*

$$\sum_i a_{ii}^2 + 3 \sum_{i \neq j} a_{ij}^2 \geq \frac{3b(n-b)^2}{2(n+4)}.$$

Proof. We consider $F = \sum_i a_{ii}^2 + 3 \sum_{i \neq j} a_{ij}^2$ as a function of a_{ij} with constraints $\sum_{i,j} b_i a_{ij} = \frac{1}{2}b(n-b)$ and $\sum_{i,j} b_j a_{ij} = \frac{1}{2}b(n-b)$. Let

$$f := \sum_i a_{ii}^2 + 3 \sum_{i \neq j} a_{ij}^2 + \lambda \left[\sum_{i,j} b_i a_{ij} - \frac{1}{2}b(n-b) \right] + \mu \left[\sum_{i,j} b_j a_{ij} - \frac{1}{2}b(n-b) \right],$$

where λ and μ are the Lagrange multipliers. It is obvious that the minimum point of F is a critical point of f . Taking the derivative of f with respect to a_{ij} , we get

$$(3.1) \quad f_{a_{ii}} = 2a_{ii} + \lambda b_i + \mu b_i = 0, \quad \text{for } i,$$

$$(3.2) \quad f_{a_{ij}} = 6a_{ij} + \lambda b_i + \mu b_j = 0, \quad \text{for } i \neq j.$$

Hence

$$\sum_i a_{ii} f_{a_{ii}} = 2 \sum_i a_{ii}^2 + \lambda \sum_i a_{ii} b_i + \mu \sum_i a_{ii} b_i = 0$$

and

$$\sum_{i \neq j} a_{ij} f_{a_{ij}} = 6 \sum_{i \neq j} a_{ij}^2 + \lambda \sum_{i \neq j} a_{ij} b_i + \mu \sum_{i \neq j} a_{ij} b_j = 0.$$

Therefore,

$$(3.3) \quad 2 \left[\sum_i a_{ii}^2 + 3 \sum_{i \neq j} a_{ij}^2 \right] = \lambda \frac{1}{2} b(b-n) + \mu \frac{1}{2} b(b-n).$$

From (3.1) and (3.2), we have

$$(3.4) \quad 2 \sum_i b_i a_{ii} + (\lambda + \mu) \sum_i b_i^2 = 0,$$

$$6 \sum_{i \neq j} b_i a_{ij} + \lambda \sum_{i \neq j} b_i^2 + \mu \sum_{i \neq j} b_i b_j = 0$$

and

$$6 \sum_{i \neq j} b_j a_{ij} + \lambda \sum_{i \neq j} b_i b_j + \mu \sum_{i \neq j} b_j^2 = 0.$$

From (3.4) and the two equalities above, we get

$$-4 \sum_i b_i a_{ii} + 3b(n-b) + \lambda n b = 0$$

and

$$-4 \sum_i b_i a_{ii} + 3b(n-b) + \mu nb = 0,$$

$$\lambda + \mu = \frac{6(b-n)}{(n+4)}.$$

According to (3.3), we obtain

$$f_{min} = \frac{3b(n-b)^2}{2(n+4)}.$$

Thus we have finished the proof of Lemma 2. $\square$

For any fixed point p in M , we can choose a local frame field $e_1, \dots, e_n$ such that

$$(3.5) \quad h_{ij} = \lambda_i \delta_{ij}.$$

Defining $f_3 = \sum_i \lambda_i^3$ and $f_4 = \sum_i \lambda_i^4$, then f_3 and f_4 are functions defined globally on M .

Proposition 1. *Let M be a minimal hypersurface in $S^{n+1}(1)$. Then*

$$\sum_{i,j,k,l} h_{ijkl}^2 \geq \frac{3}{2}(Sf_4 - f_3^2 - 2S^2 + nS) + \frac{3S(S-n)^2}{2(n+4)}$$

holds.

Proof. From the Ricci formula (2.5) and the Gauss equation (2.3), we have

$$(3.6) \quad \begin{aligned} h_{iijj} - h_{jjii} &= h_{ijij} - h_{ijji} = \sum_m h_{im} R_{mjij} + \sum_m h_{mj} R_{mii j} \\ &= \lambda_i R_{ijij} + \lambda_j R_{jii j} = (\lambda_i - \lambda_j) R_{ijij} \\ &= (\lambda_i - \lambda_j)(1 + \lambda_i \lambda_j). \end{aligned}$$

We define

$$(3.7) \quad u_{ijkl} = \frac{1}{4}(h_{ijkl} + h_{lij k} + h_{kl i j} + h_{j k l i}).$$

Since h_{ijkl} is symmetric in the indices i, j, k , from formula (3.6), we obtain

$$(3.8) \quad \sum_{i,j,k,l} h_{ijkl}^2 \geq \sum_{i,j,k,l} u_{ijkl}^2 + \frac{3}{2}[Sf_4 - f_3^2 - 2S^2 + nS].$$

Since $\Delta h_{ij} = (n-S)h_{ij}$ and $\sum_i h_{iikl} = 0$, we have

$$\sum_{i,j} u_{iijj} \lambda_i = \sum_{i,j} u_{iijj} \lambda_j = \frac{1}{2}S(n-S).$$

From $\sum_i \lambda_i = 0$ and $\sum_i \lambda_i^2 = S$ and defining $a_{ij} := u_{iijj}$ and $b_i := \lambda_i$, then a_{ij} and b_i satisfy the conditions in Lemma 2. From the definition of u_{ijkl} , we know that u_{ijkl} is symmetric in the indices i, j, k, l . From Lemma 2, we infer

$$(3.9) \quad \sum_{i,j,k,l} u_{ijkl}^2 \geq \sum_i u_{iiii}^2 + 3 \sum_{i \neq j} u_{iijj}^2 \geq \frac{3S(S-n)^2}{2(n+4)}.$$

Hence, from (3.8) and (3.9), we obtain

$$\sum_{i,j,k,l} h_{ijkl}^2 \geq \frac{3}{2}(Sf_4 - f_3^2 - 2S^2 + nS) + \frac{3S(S-n)^2}{2(n+4)}.$$

This completes the proof of Proposition 1. $\square$

Proposition 2. *Let M be a closed minimal hypersurface in $S^{n+1}(1)$. Then*

$$\int_M [(S - 2n - \frac{3}{2}) \sum_{i,j,k} h_{ijk}^2 + 2(S - n)f_4 - \frac{3S(S-n)^2}{2(n+4)} + \frac{9}{8}|\nabla S|^2] dM \geq 0$$

holds.

Proof. The following integral formula (3.10) can be found in [2]:

$$(3.10) \quad \int_M (A - 2B) dM = \int_M [Sf_4 - S^2 - f_3^2 - \frac{1}{4}|\nabla S|^2] dM.$$

From the Ricci formula (2.5), by a direct computation, we obtain

$$\frac{1}{4}\Delta f_4 = (n - S)f_4 + 2A + B.$$

Integrating both sides of the above equality, we have

$$(3.11) \quad \int_M (S - n)f_4 dM = \int_M (2A + B) dM.$$

Formulas (3.10) and (3.11) yield

$$(3.12) \quad \int_M [(S - 4n)f_4 + 3f_3^2 + 3S^2 + \frac{3}{4}|\nabla S|^2] dM \geq 0.$$

According to Stokes' formula, we integrate the formula (2.11) and obtain

$$(3.13) \quad \begin{aligned} & \int_M \sum_{i,j,k,l} h_{ijkl}^2 dM \\ &= \int_M [-(2n + 3 - S) \sum_{i,j,k} h_{ijk}^2 - 3(2B - A) + \frac{3}{2}|\nabla S|^2] dM. \end{aligned}$$

From Proposition 1, (3.10) and (3.13), we infer

$$(3.14) \quad \int_M \{(S - 2n - \frac{3}{2}) \sum_{i,j,k} h_{ijk}^2 + \frac{3}{4}|\nabla S|^2 + \frac{3}{2}[Sf_4 - f_3^2 - S^2] - \frac{3S(n-S)^2}{2(n+4)}\} dM \geq 0.$$

(3.12) + 2 $\times$ (3.14) yields

$$\int_M [(S - 2n - \frac{3}{2}) \sum_{i,j,k} h_{ijk}^2 + 2(S - n)f_4 - \frac{3S(S-n)^2}{2(n+4)} + \frac{9}{8}|\nabla S|^2] dM \geq 0.$$

Thus Proposition 2 is valid. $\square$

Proof of Theorem 1. According to (2.10) and Stokes' Theorem, we obtain

$$(3.15) \quad \int_M \sum_{i,j,k} h_{ijk}^2 dM = \int_M [S(S - n)] dM$$

and

$$(3.16) \quad - \int_M \frac{1}{2} |\nabla S|^2 = \int_M [S \sum_{i,j,k} h_{ijk}^2 + (n-S)S^2] dM.$$

From formula (2.7) and the assumption in Theorem 1, we have

$$R_{ii} = n - 1 - \lambda_i^2 \geq \frac{n}{2}.$$

Therefore,

$$\lambda_i^2 \leq \frac{n-2}{2},$$

$$\sum_i \lambda_i^4 \leq \frac{n-2}{2} \sum_i \lambda_i^2,$$

that is,

$$(3.17) \quad f_4 \leq \frac{n-2}{2} S.$$

From Proposition 2 and (3.17), we have

$$\int_M \left\{ (S - 2n - \frac{3}{2}) \sum_{i,j,k} h_{ijk}^2 + (n-2)S(S-n) - \frac{3S(S-n)^2}{2(n+4)} + \frac{9}{8} |\nabla S|^2 \right\} dM \geq 0.$$

From (3.15), (3.16) and the above inequality, we infer

$$\int_M \left\{ (-\frac{5}{4}S - n - \frac{7}{2}) \sum_{i,j,k} h_{ijk}^2 + [\frac{9}{4}S - \frac{3(S-n)}{2(n+4)}] S(S-n) \right\} dM \geq 0.$$

Since

$$n \leq S \leq n + \frac{14(n+4)}{9n+30},$$

we have

$$\int_M \left\{ (-\frac{5}{4}S - n - \frac{7}{2}) \sum_{i,j,k} h_{ijk}^2 + (\frac{9n}{4} + \frac{7}{2}) S(S-n) \right\} dM \geq 0.$$

Hence

$$\int_M \frac{5}{4} (S-n) \sum_{i,j,k} h_{ijk}^2 dM = 0.$$

Since S and $\sum_{i,j,k} h_{ijk}^2$ are continuous functions, we have $S = n$. Thus, M is isometric to a Clifford torus $S^m(\sqrt{\frac{m}{n}}) \times S^{n-m}(\sqrt{\frac{n-m}{n}})$ ($1 < m < n-1$) from the result due to Chern, do Carmo and Kobayashi [3] or Lawson [4]. This completes the proof of Theorem 1. $\square$

Proof of Theorem 2. In the case $n = 3$, because $\sum_i \lambda_i = 0$, we have $f_4 = \sum_i \lambda_i^4 = \frac{S^2}{2}$. From Proposition 2, we have

$$\int_M \left\{ (S - 2n - \frac{3}{2}) \sum_{i,j,k} h_{ijk}^2 + S^2(S-n) - \frac{3S(S-n)^2}{2(n+4)} + \frac{9}{8} |\nabla S|^2 \right\} dM \geq 0.$$

From (3.16) and the above inequality, we have

$$\int_M \left\{ \left(-\frac{5}{4}S - 2n - \frac{3}{2} \right) \sum_{i,j,k} h_{ijk}^2 + \frac{13}{4}S^2(S-n) - \frac{3S(S-n)^2}{2(n+4)} \right\} dM \geq 0.$$

Since

$$n \leq S \leq n + \frac{42}{85},$$

we have

$$\int_M \left\{ -\frac{5}{4}(S-n) \sum_{i,j,k} h_{ijk}^2 \right\} dM \geq 0.$$

Hence

$$\int_M \frac{5}{4}(S-n) \sum_{i,j,k} h_{ijk}^2 dM = 0.$$

By making use of the same proof as in Theorem 1, we know that Theorem 2 is true in the case $n = 3$.

In the case $n = 4$, from Lemma 1, we have $f_4 \leq \frac{7}{12}S^2$. By using this inequality, we obtain, from Proposition 2,

$$\int_M \left\{ \left(S - 2n - \frac{3}{2} \right) \sum_{i,j,k} h_{ijk}^2 + \frac{7}{6}S^2(S-n) - \frac{3S(S-n)^2}{2(n+4)} + \frac{9}{8}|\nabla S|^2 \right\} dM \geq 0.$$

By the same proof as in the case $n = 3$, we know that Theorem 2 is also valid in the case $n = 4$.

In the case $n = 5$, from Proposition 1, (3.10) and (3.13), we have

$$(3.18) \quad \int_M \left\{ \left(S - 2n - \frac{3}{2} \right) \sum_{i,j,k} h_{ijk}^2 + \frac{3}{2}(A - 2B) - \frac{3S(S-n)^2}{2(n+4)} + \frac{9}{8}|\nabla S|^2 \right\} dM \geq 0.$$

Since

$$\begin{aligned} 3(A - 2B) &= \sum_{i,j,k} (\lambda_i^2 + \lambda_j^2 + \lambda_k^2 - 2\lambda_i\lambda_j - 2\lambda_j\lambda_k - 2\lambda_k\lambda_i) h_{ijk}^2 \\ &= \sum_{i \neq j \neq k \neq i} [2(\lambda_i^2 + \lambda_j^2 + \lambda_k^2) - (\lambda_i + \lambda_j + \lambda_k)^2] h_{ijk}^2 \\ &\quad + 3 \sum_{i \neq k} (\lambda_k^2 - 4\lambda_i\lambda_k) h_{iik}^2 - 3 \sum_i \lambda_i^2 h_{iii}^2 \\ &\leq \frac{\sqrt{17}+1}{2} S \sum_{i,j,k} h_{ijk}^2, \end{aligned}$$

by making use of this inequality and (3.18), a similar proof as in the case $n = 3$ yields that Theorem 2 is also valid in this case. We have finished the proof of Theorem 2. $\square$

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OTHER QUANTUM RELATIVES OF THE ALEXANDER POLYNOMIAL THROUGH THE LINKS-GOULD INVARIANTS

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ABSTRACT. In 2006, Oleg Viro studied two interpretations of the (multivariable) Alexander polynomial understood as a quantum link invariant: either by considering the quasitriangular Hopf algebra associated to $U_q\mathfrak{sl}(2)$ at fourth roots of unity, or by considering the super Hopf algebra $U_q\mathfrak{gl}(1|1)$. In this paper, we show these Hopf algebras share properties with the -1 specialization of $U_q\mathfrak{gl}(n|1)$ leading to the proof of a conjecture by David De Wit, Atsushi Ishii and Jon Links on the Links-Gould invariants.

1. INTRODUCTION

The Links-Gould invariants of links $LG^{n,m}$ are two variable quantum link invariants. They are derived from super Hopf algebras $U_q\mathfrak{gl}(n|m)$. David De Wit, Atsushi Ishii and Jon Links conjectured in [3] that for any link L

$$LG^{n,m}(L; \tau, e^{i\pi/m}) = \Delta_L(\tau^{2m})^n,$$

where Δ_L is the Alexander-Conway polynomial of L . They showed that equality when $(n, m) = (1, m)$ and when $(n, m) = (2, 1)$ for a particular class of braids. A complete proof of the $(n, 1)$ case for $n = 2, 3$ can be found in [9]. However this is achieved by studying the invariants at hand at the level of *representations*, which requires computation of an explicit R -matrix for each $LG^{n,1}$, making that method hard to implement as n grows.

Here we prove the $(n, 1)$ case of the conjecture for any n :

$$LG^{n,1}(L; \tau, -1) = \Delta_L(\tau^2)^n.$$

To do so we study the structure of the *universal* objects directly, and in particular the (super) Hopf algebras and universal R -matrices that are involved.

However, the strong version of the conjecture is still open.

2. HOPF ALGEBRAS FOR THE ALEXANDER POLYNOMIAL

We first define a Hopf algebra U which is an essential ingredient in the study of quantum relatives of the Alexander polynomial. Unfortunately, this algebra is only braided in a weak sense. Then we recall the definition of two quantum groups which can be seen as central extensions of U . One was first used by Murakami [11], the second by Kauffman and Saleur [7], and both were studied by Viro in [15]. Finally we compare the braidings of the two Hopf algebras.

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2.1. A braided Hopf algebra U . The following Hopf algebra U is a version of quantum $\mathfrak{sl}(2)$ when the quantum parameter q is chosen to be a fourth root $\mathbf{i}$ of 1. This complex algebra U is finitely presented by generators $k^{\pm 1}, e, f$ and relations

$$ke + ek = kf + fk = e^2 = f^2 = 0 \quad \text{and} \quad ef - fe = k - k^{-1}.$$

The coproduct, counity and antipode of U are given by

$$\begin{aligned} \Delta(e) &= 1 \otimes e + e \otimes k, & \varepsilon(e) &= 0, & S(e) &= -ek^{-1}, \\ \Delta(f) &= k^{-1} \otimes f + f \otimes 1, & \varepsilon(f) &= 0, & S(f) &= -kf, \\ \Delta(k) &= k \otimes k, & \varepsilon(k) &= 1, & S(k) &= k^{-1}. \end{aligned}$$

This Hopf algebra can be seen in some sense as a “double” of Bodo Pareigis’ Hopf algebra [12] that would be $\langle k, f \rangle$ with our notation. A pivotal structure is a group like element ϕ that induces a conjugation map equal to the square of the antipode. There is a better choice which can be made by setting $\phi = k^{-1}$.

Let $\tau : x \otimes y \mapsto y \otimes x$ be the switch of factors. Hopf algebra U is not quasi-triangular but it is braided as defined in [13]: there exists an (outer) algebra automorphism $\mathcal{R} : U \otimes U \rightarrow U \otimes U$ different from τ that satisfies

$$\begin{aligned} (1) \quad & \mathcal{R} \circ \Delta = \tau \circ \Delta, \\ (2) \quad & \Delta_1 \circ \mathcal{R} = \mathcal{R}_{13} \mathcal{R}_{23}, \\ (3) \quad & \Delta_2 \circ \mathcal{R} = \mathcal{R}_{13} \mathcal{R}_{12}. \end{aligned}$$

Automorphism $\mathcal{R}$ admits a regular splitting (see [13]) $\mathcal{R} = \mathcal{D} \circ \text{Ad}_{\check{R}}$ where $\text{Ad}_{\check{R}}$ is the conjugation by the invertible element

$$\check{R} = 1 + e \otimes f$$

and $\mathcal{D}$ is an outer automorphism satisfying equations similar to (2) and (3) and defined by:

$$\mathcal{D} \circ \tau = \tau \circ \mathcal{D}, \quad \mathcal{D}(e \otimes 1) = e \otimes k, \quad \mathcal{D}(f \otimes 1) = f \otimes k^{-1} \quad \text{and} \quad \mathcal{D}(k \otimes 1) = k \otimes 1.$$

Elements $k^{\pm 2}$ generate a central Hopf subalgebra and for any $g \in \mathbb{C} \setminus \{0, 1\}$, the quotient $U/(k^2 - g)$ is an 8-dimensional semi-simple Hopf algebra with two isomorphism classes of irreducible representations $V_{\pm a}$ where $a^2 = g$. The representation V_a is 2-dimensional and can be written as follows in a certain basis (e_0, e_1) :

$$(4) \quad k = \begin{pmatrix} a & 0 \\ 0 & -a \end{pmatrix}, \quad e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad f = \begin{pmatrix} 0 & 0 \\ a - \frac{1}{a} & 0 \end{pmatrix}.$$

Then the central element $ef + fe$ acts by $(a - a^{-1})I_2$.

2.2. The $\mathfrak{sl}(2)$ model and the Alexander polynomial. From [1, 15] the $\mathfrak{sl}(2)$ model is the unrolled version of quantum $\mathfrak{sl}(2)$ at $q = \mathbf{i} = \exp(\mathbf{i}\pi/2)$. It is an algebra $U_{\mathbf{i}}^H \mathfrak{sl}(2)$ generated by $K^{\pm 1}, E, F, H$. Its presentation is obtained from that of U ($U_0 = \langle K^{\pm 1}, E, F \rangle \simeq U$) by adding a generator H and the following relations:

$$[H, K] = 0, \quad [H, E] = 2E, \quad [H, F] = -2F.$$

We will consider the category $\mathcal{C}$ of weight modules, that is, finite dimensional vector spaces where element H acts diagonally and

$$(5) \quad K = \mathbf{i}^H = \exp(\mathbf{i}\pi H/2).$$

The pivotal Hopf algebra structure on U is extended to $U_{\mathbf{i}}^H \mathfrak{sl}(2)$ using the following relations:¹

$$\Delta(H) = 1 \otimes H + H \otimes 1, \quad \varepsilon(H) = 0, \quad S(H) = -H.$$

Like in the case of U , the pivotal element is $\Phi = K^{-1}$ so that $S^2(\cdot) = \Phi \cdot \Phi^{-1}$. With this pivotal structure, category $\mathcal{C}$ is ribbon with braiding given by the switch $\tau : x \otimes y \mapsto y \otimes x$ composed with the action of the universal R -matrix:

$$R^H = \mathbf{i}^{H \otimes H/2} (1 + E \otimes F).$$

Lemma 2.1. *For any two representations $V, W \in \mathcal{C}$, the conjugation by $D^H := \mathbf{i}^{H \otimes H/2}$ in $V \otimes W$ induces an automorphism $\mathcal{D}^H$ of $\text{End}_{\mathbb{C}}(V \otimes W)$ that satisfies*

$$\rho_{V \otimes W} \circ \mathcal{D} = \mathcal{D}^H \circ \rho_{V \otimes W} : U \otimes U \rightarrow \text{End}_{\mathbb{C}}(V \otimes W).$$

Proof. This is an easy consequence of equation (5). More generally, if $x, y \in U$ satisfy $[H, x] = 2mx$ and $[H, y] = 2m'y$, then $H \otimes H.x \otimes y = x \otimes y.(H + 2m) \otimes (H + 2m')$ so

$$\begin{aligned} & \mathbf{i}^{\rho_{V \otimes W}(H \otimes H/2)} \rho_{V \otimes W}(x \otimes y) \\ &= \rho_{V \otimes W}(x \otimes y) \mathbf{i}^{\rho_{V \otimes W}((H+2m) \otimes (H+2m')/2)} \\ &= \rho_{V \otimes W}((x \otimes K^m)(K^{m'} \otimes y)) \mathbf{i}^{\rho_{V \otimes W}(H \otimes H/2)} \\ &= \rho_{V \otimes W}(\mathcal{D}(x \otimes y)) \mathbf{i}^{\rho_{V \otimes W}(H \otimes H/2)}. \end{aligned}$$

2.1

For any complex number α that is not an odd integer, $U_{\mathbf{i}}^H \mathfrak{sl}(2)$ possesses up to isomorphism a unique 2-dimensional irreducible representation V_{α} with $\text{Spec}(H) = \{\alpha+1, \alpha-1\}$. When restricted to U , this is the representation V_a where $a = \mathbf{i}^{\alpha+1}$ and the action of H is given by $H = \begin{pmatrix} \alpha+1 & 0 \\ 0 & \alpha-1 \end{pmatrix}$.

In the representation $V_{\alpha} \otimes V_{\beta}$ with respect to basis $(e_0 \otimes e_0, e_0 \otimes e_1, e_1 \otimes e_0, e_1 \otimes e_1)$, the braiding is:

$$\mathbf{i}^{\frac{\alpha\beta-1}{2}} \begin{pmatrix} \mathbf{i}^{\frac{\alpha+\beta+2}{2}} & 0 & 0 & 0 \\ 0 & 0 & \mathbf{i}^{\frac{\alpha-\beta}{2}} & 0 \\ 0 & \mathbf{i}^{\frac{-\alpha+\beta}{2}} & \mathbf{i}^{\frac{-\alpha+\beta}{2}} (\mathbf{i}^{\beta+1} - \mathbf{i}^{-\beta-1}) & 0 \\ 0 & 0 & 0 & \mathbf{i}^{\frac{-\alpha-\beta+2}{2}} \end{pmatrix}.$$

In the case where $\alpha = \beta$, the R -matrix then takes the particular form

$$\tau R^H = \mathbf{i}^{\frac{\alpha^2-1}{2}} \begin{pmatrix} t^{-1/2} & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & (t^{-1/2} - t^{1/2}) & 0 \\ 0 & 0 & 0 & -t^{1/2} \end{pmatrix}$$

where we set $t^{1/2} = \mathbf{i}^{-\alpha-1}$.

The ribbon category we consider here allows us to apply the Reshetikhin-Turaev theory [14] to construct a framed link isotopy invariant in S^3 . It becomes an unframed link isotopy invariant if one divides the above R -matrix on $V_{\alpha} \otimes V_{\alpha}$ by the value of the twist $\theta_{\alpha} = \mathbf{i}^{\frac{\alpha^2-1}{2}}$. In this particular case, the invariant we find is the

¹Compared to Viro, we use the opposite coproduct here.

Conway normalization of the classical Alexander polynomial; see [15]. The Links-Gould invariants $LG^{n,1}$ that will interest us in the following are obtained by the same general construction using other Hopf algebras. Explicitly, the Reshetikhin-Turaev functor gives representations of braid groups B_ℓ

$$\begin{aligned}\Psi_{V_\alpha^{\otimes \ell}} : B_\ell &\longrightarrow GL(V_\alpha^{\otimes \ell}) \\ \sigma_i &\mapsto \text{Id}_{V_\alpha}^{\otimes i-1} \otimes \theta_\alpha^{-1} \tau R^H \otimes \text{Id}_{V_\alpha}^{\otimes \ell-i-1},\end{aligned}$$

where σ_i is the i^{th} standard Artin generator of braid group B_ℓ .

Let L be an oriented link in S^3 obtained as closure of a braid in ℓ strands $b \in B_\ell$. Then:

1) There exists a scalar c such that $\text{trace}_{2,3,\dots,\ell}((\text{Id}_{V_\alpha} \otimes (K^{-1})^{\otimes \ell-1}) \circ \Psi_{V_\alpha^{\otimes \ell}}(b)) = c \cdot \text{Id}_{V_\alpha}$.

2) $L \mapsto c$ is a link invariant and is equal to the Alexander polynomial of L , $\Delta_L(t)$.

Remark 2.2. Identifying algebras $\text{End}_{\mathbb{C}}(V_\alpha^{\otimes \ell})$ and $\text{End}_{\mathbb{C}}(V_\alpha)^{\otimes \ell}$, the partial trace operator is defined by $\text{trace}_{2,3,\dots,\ell}(f_1 \otimes \dots \otimes f_\ell) := \text{trace}(f_2) \text{trace}(f_3) \dots \text{trace}(f_\ell) f_1 \in \text{End}_{\mathbb{C}}(V_\alpha)$ for any $f_1, \dots, f_\ell \in \text{End}_{\mathbb{C}}(V_\alpha)$.

2.3. An example of bosonization: The $\mathfrak{gl}(1|1)$ model.

2.3.1. Bosonization. Here we recall Majid's trick [10] to transform a super Hopf algebra into an ordinary one.

Let H be a pivotal super Hopf algebra and $\mathcal{C}$ be its even monoidal category of representations (morphisms are formed by even H -linear maps). Let H^σ be the bosonization of H : as an algebra, H^σ is the semi-direct product of H with $\mathbb{Z}/2\mathbb{Z} = \{1, \sigma\}$ where the action of σ or equivalently the commutation relations in H^σ are given by

$$\forall x \in H, \sigma x = (-1)^{|x|} x \sigma.$$

The coproduct Δ^σ on H^σ is given by $\Delta^\sigma \sigma = \sigma \otimes \sigma$ and

$$\forall x \in H, \Delta^\sigma(x) = \sum_i x_i \sigma^{|x'_i|} \otimes x'_i \text{ where } \Delta(x) = \sum_i x_i \otimes x'_i.$$

If $R = \sum_i R_i^{(1)} \otimes R_i^{(2)}$ is the universal R -matrix in H , then the following formula defines a universal R -matrix in H^σ :

$$R^\sigma = R_1 \sum_i R_i^{(1)} \sigma^{|R_i^{(2)}|} \otimes R_i^{(2)}, \text{ where } R_1 = \frac{1}{2}(1 \otimes 1 + \sigma \otimes 1 + 1 \otimes \sigma - \sigma \otimes \sigma).$$

Given a super representation $V = V_0 \oplus V_1$ of H we get a representation of H^σ by setting $\sigma|_V = \text{Id}_{V_0} - \text{Id}_{V_1}$. On the other hand, since $\sigma^2 = 1$, every H^σ -module inherits a natural $\mathbb{Z}/2\mathbb{Z}$ grading: W splits into $W = W_0 \oplus W_1$ where we define $W_0 = \ker(\sigma - 1)$ and $W_1 = \ker(\sigma + 1)$.

Theorem 2.3 (Majid Theorem 4.2). *The even category of super H -modules can be identified with the category of H^σ -modules.*

Note that the antipode of H^σ is given by $x \mapsto \sigma^{|x|} S(x)$. Also, if H has a pivot ϕ , then one can choose $\phi^\sigma = \sigma \phi$ as a pivot in H^σ .

2.3.2. *The $\mathfrak{gl}(1|1)$ model.* Using the same notation as Viro: $U_q\mathfrak{gl}(1|1)$ is the pivotal super Hopf algebra generated by two odd generators X, Y , two even generators I, G satisfying the relations

$$XY + YX = \frac{C - C^{-1}}{q - q^{-1}}, \quad X^2 = Y^2 = 0,$$

$$[I, X] = [I, Y] = [I, G] = 0,$$

$$[G, X] = X, \quad [G, Y] = -Y,$$

where $C = q^I$, with coproduct

$$\Delta(I) = 1 \otimes I + I \otimes 1, \quad \Delta(G) = 1 \otimes G + G \otimes 1,$$

$$\Delta(X) = X \otimes C^{-1} + 1 \otimes X, \quad \Delta(Y) = Y \otimes 1 + C \otimes Y,$$

counit

$$\varepsilon(X) = \varepsilon(Y) = \varepsilon(I) = \varepsilon(G) = 0,$$

antipode

$$S(I) = -I, \quad S(G) = -G, \quad S(X) = -XC, \quad S(Y) = -YC^{-1},$$

pivot

$$\phi = K$$

and universal R -matrix

$$R = (1 + (q - q^{-1})(X \otimes Y)(C \otimes C^{-1}))q^{-I \otimes G - G \otimes I}.$$

Its bosonization $U_q\mathfrak{gl}(1|1)^\sigma$ contains a Hopf subalgebra U_1 isomorphic to U defined by setting

$$e = (q - q^{-1})X\sigma, \quad f = Y \quad \text{and} \quad k = C^{-1}\sigma.$$

Indeed, these elements satisfy the following:

$$ef - fe = (q - q^{-1})(X\sigma Y - YX\sigma) = (q - q^{-1})(-XY - YX)\sigma = k - k^{-1},$$

$$ke + ek = kf + fk = 0,$$

$$\Delta^\sigma(e) = (q - q^{-1})\Delta^\sigma(X\sigma) = (q - q^{-1})(X \otimes C^{-1} + \sigma \otimes X)(\sigma \otimes \sigma) = e \otimes k + 1 \otimes e,$$

$$\Delta^\sigma(f) = \Delta^\sigma(Y) = Y \otimes 1 + C\sigma \otimes Y = f \otimes 1 + k^{-1} \otimes f,$$

$$\Delta^\sigma(k) = k \otimes k.$$

In the bosonization, the universal R -matrix is

$$R^\sigma = R_1 q^{-(I \otimes G + G \otimes I)}(1 + e \otimes f), \quad \text{where } R_1 = \frac{1}{2}(1 \otimes 1 + \sigma \otimes 1 + 1 \otimes \sigma - \sigma \otimes \sigma).$$

Lemma 2.4. Denoting $D' = q^{-I \otimes G - G \otimes I}$ and $D^\sigma = R_1 D'$ we have that for any $x, y \in U = U_1$

$$R_1(x \otimes y)R_1^{-1} = \sigma^{|y|}x \otimes y\sigma^{|x|}, \quad D'(x \otimes y)(D')^{-1} = xC^{-d_G(y)} \otimes yC^{-d_G(x)},$$

$$D^\sigma(x \otimes y)(D^\sigma)^{-1} = (C^{-1}\sigma)^{d_G(y)}x \otimes y(C^{-1}\sigma)^{d_G(x)} = \mathscr{D}(x \otimes y),$$

where $d_G(x) \in \mathbb{Z}$ is defined by $[G, x] = d_G(x)x$.

Remark 2.5. For a homogeneous $a \in U_0$, $|a| = d_G(a)$ modulo 2.

Let us recall the expression of a family of 2-dimensional $U_q\mathfrak{gl}(1|1)^\sigma$ -modules. This family is parametrized by two complex numbers (j, J) and $\varepsilon \in \{0, 1\}$; see [15]. It extends the representation V_a of U_1 where $a = (-1)^\varepsilon q^{-2j}$. Written in matrix form,

$$I = \begin{pmatrix} 2j & 0 \\ 0 & 2j \end{pmatrix}, G = \begin{pmatrix} \frac{J+1}{2} & 0 \\ 0 & \frac{J-1}{2} \end{pmatrix},$$

$$X = \begin{pmatrix} 0 & \frac{q^{2j}-q^{-2j}}{q-q^{-1}} \\ 0 & 0 \end{pmatrix}, Y = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \sigma = \begin{pmatrix} (-1)^\varepsilon & 0 \\ 0 & -(-1)^\varepsilon \end{pmatrix}.$$

2.4. Comparing the actions of R^σ and R^H . $U_0 \subset U_{\mathbf{i}}^H \mathfrak{sl}(2)$ and $U_1 \subset U_q \mathfrak{gl}(1|1)^\sigma$ are two isomorphic Hopf algebras. The goal of this paragraph is to show the action of

$$R^H = \mathbf{i}^{H \otimes H/2} (1 + E \otimes F) \in U_{\mathbf{i}}^H \mathfrak{sl}(2) \otimes U_{\mathbf{i}}^H \mathfrak{sl}(2)$$

and that of

$$R^\sigma = R_1 q^{-(I \otimes G + G \otimes I)} (1 + e \otimes f) \in U_q \mathfrak{gl}(1|1)^\sigma \otimes U_q \mathfrak{gl}(1|1)^\sigma$$

on two representations $V_1^H \otimes V_2^H$ of $U_{\mathbf{i}}^H \mathfrak{sl}(2)$ and $V_1^\sigma \otimes V_2^\sigma$ of $U_q \mathfrak{gl}(1|1)^\sigma$ are identical up to a scalar multiple of the identity, when V_i^H and V_i^σ have the same underlying $U_0 = U_1$ -module structure.

We recall conjugations by $D^H = \mathbf{i}^{H \otimes H/2}$ in $V_1^H \otimes V_2^H$ and D^σ in $V_1^\sigma \otimes V_2^\sigma$ induce the same automorphism $\mathscr{D}$ of $U \otimes U$.

Proposition 2.6. *For $i = 1, 2$ let V_i^H be a representation of $U_{\mathbf{i}}^H \mathfrak{sl}(2)$ and V_i^σ be a representation of $U_q \mathfrak{gl}(1|1)^\sigma$ that both restrict to the same irreducible representation of $U = U_0 = U_1$. Then $D^H (D^\sigma)^{-1} \in \text{End}_{\mathbb{C}}(V_1 \otimes V_2)$ is a scalar multiple of the identity.*

Proof. The density theorem states that if V is a finite dimensional irreducible representation of an algebra A over an algebraically closed field, then $A \rightarrow \text{End}(V)$ is surjective. Denote the representations at hand $\rho_{V_i^H}, \rho_{V_i^\sigma}$ for $i = 1, 2$. We supposed

$$\rho_{V_i^H}|_U = \rho_{V_i^\sigma}|_U.$$

So if $\rho_H = \rho_{V_1^H} \otimes \rho_{V_2^H}$ and $\rho_\sigma = \rho_{V_1^\sigma} \otimes \rho_{V_2^\sigma}$ we define $\rho := \rho_H|_{U \otimes U} = \rho_\sigma|_{U \otimes U}$. Using Lemma 2.1 and Lemma 2.4, for any $x, y \in U$:

$$\rho_H(D^H) \rho(x \otimes y) \rho_H((D^H)^{-1}) = \rho(\mathscr{D}(x \otimes y)) = \rho_\sigma(D^\sigma) \rho(x \otimes y) \rho_\sigma((D^\sigma)^{-1}).$$

Which means

$$\rho_H(D^H)^{-1} \rho_\sigma(D^\sigma) \rho(x \otimes y) = \rho(x \otimes y) \rho_H(D^H)^{-1} \rho_\sigma(D^\sigma).$$

Using the density theorem, $\rho_H(D^H)^{-1} \rho_\sigma(D^\sigma)$ commutes with any element in

$$\text{End}_{\mathbb{C}}(V_1) \otimes \text{End}_{\mathbb{C}}(V_2) = \text{End}_{\mathbb{C}}(V_1 \otimes V_2).$$

So this linear map is a scalar multiple of the identity. 2.6

Let us now study Hopf algebra $A = U_{\mathbf{i}}^H \mathfrak{sl}(2) \otimes_U U_q \mathfrak{gl}(1|1)^\sigma$. This algebra contains both algebras $U_{\mathbf{i}}^H \mathfrak{sl}(2)$ and $U_q \mathfrak{gl}(1|1)^\sigma$ as subalgebras.

Formally, setting $q = e^h$, $q^T := e^{hT}$ and $\mathbf{i}^\alpha = e^{i\frac{\pi}{2}\alpha}$, we also consider that

$$\mathbf{i}^H = k = q^{-I} \sigma$$

which means that we will only study representations of A that satisfy this relation. Recall from equations (4) the representation of U with parameter a . We can look

for the representations of A that simultaneously extend to the representations of $U_{\mathbf{i}}^H \mathfrak{sl}(2)$ and $U_q \mathfrak{gl}(1|1)^\sigma$ we already described. If $\varepsilon \in \{0, 1\}$ is the degree of the first vector e_0 of the basis (e_0, e_1) we choose, direct computation of such a representation $V(\alpha, a, 2j, \varepsilon, J)$ shows it is well defined if and only if:

$$(6) \quad \begin{cases} (-1)^\varepsilon q^{-2j} = a, \\ a = e^{\mathbf{i}\frac{\pi}{2}(\alpha+1)} = \mathbf{i}^{\alpha+1}. \end{cases}$$

Setting $s = q^j \mathbf{i}^{\frac{\alpha-3-2\varepsilon}{2}} = \pm 1$, we can compute the coefficient $R^H/R^\sigma = D^H/D^\sigma$ given by Proposition 2.6 in our case.

Proposition 2.7. $R^H/R^\sigma = D^H/D^\sigma = ss'(-1)^{\varepsilon\varepsilon'} \mathbf{i}^{\varepsilon+\varepsilon'} \mathbf{i}^{\frac{\alpha\alpha'-1}{2}} q^{jJ'+j'J}$.

Proof. Using representation $V \otimes V' = V(\alpha, a, 2j, \varepsilon, J) \otimes V(\alpha', a', 2j', \varepsilon', J')$ in basis $(e_0 \otimes e_0, e_0 \otimes e_1, e_1 \otimes e_0, e_1 \otimes e_1)$, we can write:

$$D^H = \mathbf{i}^{\alpha\alpha'/2} \begin{pmatrix} \mathbf{i}^{\frac{\alpha+\alpha'+1}{2}} & 0 & 0 & 0 \\ 0 & \mathbf{i}^{\frac{-\alpha+\alpha'-1}{2}} & 0 & 0 \\ 0 & 0 & \mathbf{i}^{\frac{\alpha-\alpha'-1}{2}} & 0 \\ 0 & 0 & 0 & \mathbf{i}^{\frac{-\alpha-\alpha'+1}{2}} \end{pmatrix}.$$

Moreover, $D^\sigma = R_1 D'$ and

$$R_1 = (-1)^{\varepsilon\varepsilon'} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & (-1)^\varepsilon & 0 & 0 \\ 0 & 0 & (-1)^{\varepsilon'} & 0 \\ 0 & 0 & 0 & (-1)^{\varepsilon+\varepsilon'+1} \end{pmatrix},$$

$$D' = q^{-jJ'-j'J} \begin{pmatrix} q^{-j-j'} & 0 & 0 & 0 \\ 0 & q^{j-j'} & 0 & 0 \\ 0 & 0 & q^{-j+j'} & 0 \\ 0 & 0 & 0 & q^{j+j'} \end{pmatrix}.$$

Since $a = \mathbf{i}^\alpha = (-1)^{\varepsilon+1} \mathbf{i} q^{-2j}$, the formulas make two square roots of a appear:

$$\sqrt[j]{a} = \mathbf{i}^{\alpha/2} \text{ and } \sqrt[\varepsilon]{a} = \mathbf{i}^{\varepsilon+\frac{3}{2}} q^{-j} = s \sqrt[\varepsilon]{a}.$$

That way, one can compute any of the diagonal coefficients of $D^H(D^\sigma)^{-1}$ to end the proof. 2.7

3. AN INTEGRAL FORM OF $U_q \mathfrak{gl}(n|1)$ AND ITS SPECIALIZATION

3.1. Quasitriangular Hopf superalgebra $U_q \mathfrak{gl}(n|1)$. Here we define the h -adic quasitriangular Hopf superalgebra $U_q \mathfrak{gl}(n|1)$ that we will use to construct the Links-Gould invariant $LG^{n,1}$. The conventions we use for generators and relations are those chosen by Zhang and De Wit in [2, 16]. We use the set of indices $\mathcal{I} = \{1, 2, \dots, n+1\}$. We also introduce a grading $[a] \in \mathbb{Z}/2\mathbb{Z}$ for any $a \in \mathcal{I}$ by setting

$$[a] = 0 \text{ if } a \leq n \text{ and } [a] = 1 \text{ when } a = n+1.$$

The superalgebra has $(n+1)^2$ generators divided into three families. There are $n+1$ even Cartan generators E_a^a . There are $\frac{1}{2}n(n+1)$ lowering generators E_a^b parametrized by $a < b$. Finally there are $\frac{1}{2}n(n+1)$ raising generators E_b^a , with $a < b$. The degree of E_a^b is given by $[a] + [b]$.

For $a \in \mathcal{I}$, $a \neq n+1$, set $K_a = q^{E_a}$, and set $K_{n+1} = q^{-E_{n+1}^{n+1}}$. In the following $[X, Y]$ denotes the super commutator $[X, Y] = XY - (-1)^{|X||Y|}YX$.

Now let us present the relations there are between elements of $U_q \mathfrak{gl}(n|1)$.

For any, $a, b \in \mathcal{I}$ with $|a - b| \geq 2$ and for any c in the interval between a and b ,

$$E_b^a = E_c^a E_b^c - q^{\text{sign}(a-b)} E_b^c E_c^a.$$

For any $a, b \in \mathcal{I}$,

$$E_a^a E_b^b = E_b^b E_a^a, \quad E_a^a E_{b\pm 1}^b = E_{b\pm 1}^b (E_a^a + \delta_b^a - \delta_{b\pm 1}^a),$$

$$[E_{a+1}^a, E_b^{b+1}] = \delta_b^a \frac{K_a K_{a+1}^{-1} - K_a^{-1} K_{a+1}}{q - q^{-1}},$$

$$\text{which generalizes for } a < b \text{ to } [E_b^a, E_a^b] = \frac{K_a K_b^{-1} - K_a^{-1} K_b}{q - q^{-1}},$$

$$(E_{n+1}^n)^2 = (E_n^{n+1})^2 = 0, \text{ which implies } (E_{n+1}^i)^2 = (E_i^{n+1})^2 = 0 \text{ for } i < n+1.$$

The Serre relations: for any $a, b \in \mathcal{I}$ with $|a - b| \geq 2$,

$$E_a^{a+1} E_b^{b+1} = E_b^{b+1} E_a^{a+1}, \quad E_{a+1}^a E_{b+1}^b = E_{b+1}^b E_{a+1}^a,$$

and for $a \leq n-1$,

$$E_{a+1}^a E_{a+2}^a = q E_{a+2}^a E_{a+1}^a, \quad E_a^{a+1} E_a^{a+2} = q E_a^{a+2} E_a^{a+1},$$

$$E_{a+2}^a E_{a+2}^{a+1} = q E_{a+2}^{a+1} E_{a+2}^a \quad \text{and} \quad E_a^{a+2} E_{a+1}^{a+2} = q E_{a+1}^{a+2} E_a^{a+2}.$$

These relations can be completed into a set of “quasicommutation” relations indexed by pairs of root vectors (see [2, Lemma 1] where a reordering algorithm gives a constructive proof of the Poincaré-Birkhoff-Witt theorem) but these relations are redundant over the field $\mathbb{C}(q)$.

We consider the Hopf algebra structure given by the coproduct

$$\Delta(E_{a+1}^a) = E_{a+1}^a \otimes K_a K_{a+1}^{-1} + 1 \otimes E_{a+1}^a, \quad \Delta(E_a^{a+1}) = K_a^{-1} K_{a+1} \otimes E_a^{a+1} + E_a^{a+1} \otimes 1$$

$$\Delta(K_a) = K_a \otimes K_a \quad \text{and} \quad \Delta(E_a^a) = E_a^a \otimes 1 + 1 \otimes E_a^a$$

which admits² the universal R -matrix $R^{\mathfrak{gl}} = D^{\mathfrak{gl}} \check{R}^{\mathfrak{gl}}$ with

$$D^{\mathfrak{gl}} = q^{\sum_{i \leq n} E_i^i \otimes E_i^i - E_{n+1}^{n+1} \otimes E_{n+1}^{n+1}}$$

and

$$\check{R}^{\mathfrak{gl}} = \prod_{i=1}^n \left(\prod_{j=i+1}^n e_q((q - q^{-1})E_j^i \otimes E_i^j) \right) e'_q(E_{n+1}^i \otimes E_i^{n+1}),$$

where $e'_q(x) = (1 - (q - q^{-1})x)$, $e_q(x) = \sum_{k=0}^{+\infty} \frac{x^k}{(k)_q!}$, $(k)_q = \frac{1-q^k}{1-q}$ and $(k)_q! = (1)_q(2)_q \dots (k)_q$. Do note that the order in which the factors are written in $\check{R}^{\mathfrak{gl}}$ matters.

²We use here the coproduct and R -matrix of [8] conjugated by $D^{\mathfrak{gl}}$.

3.2. Integral form and interesting subalgebras. We now give an integral form of $U_q\mathfrak{gl}(n|1)$ that supports evaluation at $q = -1$. Let $\mathcal{A}_q$ be the $\mathbb{Z}[q, q^{-1}]$ -subalgebra of $U_q\mathfrak{gl}(n|1)$ generated by elements K_a , $\mathcal{E}_b^a := (q - q^{-1})E_b^a$ when $a < b$ and $\mathcal{E}_b^a := E_b^a$ when $a > b$. The relations of $U_q\mathfrak{gl}(n|1)$

$$[E_b^a, E_a^b] = \frac{K_a K_b^{-1} - K_a^{-1} K_b}{q - q^{-1}}$$

for $a < b$, are replaced in algebra $\mathcal{A}_q$ by

$$[\mathcal{E}_b^a, \mathcal{E}_a^b] = K_a K_b^{-1} - K_a^{-1} K_b.$$

Still, $\mathcal{A}_q$ admits a presentation similar to that of $U_q\mathfrak{gl}(n|1)$. No additional relations are needed because the analog of the above commutation relations are enough to express any element in the Poincaré-Birkhoff-Witt basis.

In the bosonization $\mathcal{A}_q^\sigma$ of $\mathcal{A}_q$, define for $i = 1, \dots, n$ the algebra

$$A_i = \langle e_i = -\mathcal{E}_{n+1}^i \sigma, f_i = \mathcal{E}_i^{n+1}, k_i = K_i K_{n+1}^{-1} \sigma \rangle \subset \mathcal{A}_q^\sigma.$$

Proposition 3.1. *Algebra A_i is isomorphic to U . Indeed:*

$$e_i f_i - f_i e_i = k_i - k_i^{-1},$$

$$k_i e_i + e_i k_i = k_i f_i + f_i k_i = 0.$$

Proof. Direct computations from the defining relations of $\mathcal{A}_q$ and Lemma 1 of [2]. In particular, $e_i f_i - f_i e_i = -\mathcal{E}_{n+1}^i \sigma \mathcal{E}_i^{n+1} + \mathcal{E}_i^{n+1} \mathcal{E}_{n+1}^i \sigma = [\mathcal{E}_{n+1}^i, \mathcal{E}_i^{n+1}] \sigma = k_i - k_i^{-1}$. 3.1

Remark 3.2. However, A_i is not isomorphic to U as a Hopf algebra (except for A_n), which can be seen by looking at the coproduct of elements of A_i in $\mathcal{A}_q$. This will not be a problem for us.

Set $1 \leq i \neq j \leq n$. Using [2] Lemma 1 once again, we want to see at what conditions any $x \in A_i$ and $y \in A_j$ commute.

Lemma 3.3. *We have the following commutations:*

$$e_i e_j = -q^{-1} e_j e_i, f_i f_j = -q^{-1} f_j f_i, k_i k_j = k_j k_i,$$

$$\text{if } i < j, e_i f_j - f_j e_i = \sigma K_j K_{n+1}^{-1} \mathcal{E}_j^i, \text{ otherwise } e_i f_j - f_j e_i = \sigma (q - q^{-1}) \mathcal{E}_j^i K_{n+1} K_i^{-1},$$

$$k_j e_i = -q^{-1} e_i k_j, k_j f_i = -q f_i k_j.$$

Proof. The first two equalities correspond to [2, Eq. (38) and (39)] and the two brackets $[e_i, f_j]$ correspond to [2, Eq. (36) (c) and (d)]. 3.3

Corollary 3.4. *Setting $q = -1$, in any quotient of $\mathcal{A}_{-1}^\sigma$ where for any $1 \leq i < j \leq n$, $\mathcal{E}_j^i = 0$, the elements of two distinct A_i commute.*

3.3. Highest weight representation $V(0^n, \alpha)$. Let $V(0^n, \alpha)$ be the highest weight irreducible 2^n -dimensional representation of $U_q \mathfrak{gl}(n|1)$ of weight $(0^n, \alpha)$, with $\alpha \notin \mathbb{Z}$. So E_i^i is represented by 0, except for E_{n+1}^{n+1} that is represented by α . Set v_0 a highest weight vector in $V(0^n, \alpha)$ and let $V_q(0^n, \alpha) = \mathcal{A}_q v_0$. The Poincaré-Birkhoff-Witt theorem proves that

$$\left(\prod_{i=1}^n f_i^{m_i} v_0 \right)_{m_i \in \{0,1\}}$$

is a basis for vector space $V(0^n, \alpha)$ and for the free $\mathbb{Z}[q, q^{-1}]$ -module $V_q(0^n, \alpha)$. Set $\mathcal{A}_{-1}^\sigma = \mathcal{A}_q^\sigma \otimes_{q=-1} \mathbb{C}$ and $V_{-1}(0^n, \alpha) = V_q(0^n, \alpha) \otimes_{q=-1} \mathbb{C}$.

Proposition 3.5. *In the representation $V_{-1}(0^n, \alpha)$, $\mathcal{E}_j^i = 0$ for any $1 \leq i < j \leq n$. So $\mathcal{E}_j^i$ belongs to the kernel I of the representation $\mathcal{A}_{-1}^\sigma \rightarrow \text{End}(V_{-1}(0^n, \alpha))$. As a consequence, the following map is well defined:*

$$\begin{aligned} \Theta : \bigotimes_{i=1}^n A_i &\longrightarrow \mathcal{A}_{-1}^\sigma / I \\ \otimes_i x_i &\mapsto \prod_i x_i \end{aligned}$$

Proof. We want to show that for any basis vector $v \in V_{-1}(0^n, \alpha)$ and for $1 \leq i < j \leq n$, $\mathcal{E}_j^i v = 0$. We can write $v = f_1^{i_1} \dots f_n^{i_n} v_0$ where $i_k = 0, 1$.

Using [2] Lemma 1 once more, if

$$\begin{aligned} c < i, & \text{ then } [E_j^i, E_c^{n+1}] = 0 \text{ by [2, Eq. (40)]} \\ c = i, & \text{ then } [E_j^i, E_c^{n+1}] = -K_i K_j^{-1} E_j^{n+1} \text{ by [2, Eq. (36)(a)]} \\ i < c < j, & \text{ then } [E_j^i, E_c^{n+1}] = -(q - q^{-1}) K_c K_j^{-1} E_c^i E_j^{n+1} \text{ by [2, Eq. (43)(a)]} \\ j \leq c, & \text{ then } [E_j^i, E_c^{n+1}] = 0 \text{ by [2, Eq. (37),(40)].} \end{aligned}$$

In all cases, $[\mathcal{E}_j^i, f_c] = [\mathcal{E}_j^i, \mathcal{E}_c^{n+1}] = (q - q^{-1})[E_j^i, E_c^{n+1}] = 0$ in $\mathcal{A}_{-1}^\sigma$. So $\mathcal{E}_j^i v = f_1^{i_1} \dots f_n^{i_n} (\mathcal{E}_j^i v_0)$. But $\mathcal{E}_j^i$ is a raising generator, so $\mathcal{E}_j^i v_0 = 0$. Using Corollary 3.4, for $i \neq j$ A_i and A_j commute in that representation. 3.5

3.4. $\check{R}^{\mathfrak{gl}}$ makes sense when $q = -1$. Here we intend to show that the nondiagonal part $\check{R}^{\mathfrak{gl}}$ of the universal R -matrix of $U_q \mathfrak{gl}(n|1)$ supports evaluation at $q = -1$, which is not obvious given the formula defining $\check{R}^{\mathfrak{gl}}$. In the bosonization $U_q \mathfrak{gl}(n|1)^\sigma$, the universal R -matrix is given by

$$(R^{\mathfrak{gl}})^\sigma = D^{\mathfrak{gl}} (\check{R}^{\mathfrak{gl}})^\sigma = D^{\mathfrak{gl}} \prod_{i=1}^n \left(\prod_{j=i+1}^n e_q(\mathcal{E}_j^i \otimes \mathcal{E}_i^j) \right) (1 + e_i \otimes f_i).$$

Proposition 3.6. *For any $1 \leq i < j \leq n$,*

$$\left(e_q(\mathcal{E}_j^i \otimes \mathcal{E}_i^j) - 1 \right) V_q(0^n, \alpha) \otimes V_q(0^n, \alpha) \subset (q+1) \mathbb{Z}[q, q^{-1}]_{loc} V_q(0^n, \alpha) \otimes V_q(0^n, \alpha)$$

where $\mathbb{Z}[q, q^{-1}]_{loc}$ is the localization of $\mathbb{Z}[q, q^{-1}]$ at $(q+1)$. Hence $(R^{\mathfrak{gl}})^\sigma$ induces a well-defined automorphism of $V_{-1}(0^n, \alpha) \otimes V_{-1}(0^n, \alpha)$ where the action of $(\check{R}^{\mathfrak{gl}})^\sigma$ is given by

$$(\check{R}^{\mathfrak{gl}})^\sigma = \prod_{i=1}^n (1 + e_i \otimes f_i).$$

Proof. Define $V = \mathbb{Z}[q, q^{-1}]_{loc} V_q(0^n, \alpha) \subset V(0^n, \alpha)$ so that $V_{-1}(0^n, \alpha) \cong V \otimes_{q=-1} \mathbb{C}$. We wish to prove that for $1 \leq i < j \leq n$, in the representation $V \otimes V$, $e_q(\mathcal{E}_j^i \otimes \mathcal{E}_i^j) = 1$

mod $(q+1)$. Set $1 \leq i < j \leq n$. We show by induction on $k \geq 1$, that

$$\frac{(\mathcal{E}_j^i)^k}{(k)_q!} V \subset (q+1)V.$$

For $k=1$, it follows from $\mathcal{E}_j^i \in I$ (see Proposition 3.5). Now let us suppose that the result holds for any $l \in \{1, \dots, k-1\}$. Since $\frac{(\mathcal{E}_j^i)^k}{(k)_q!} = \frac{(\mathcal{E}_j^i)^{k-1}}{(k-1)_q!} \frac{\mathcal{E}_j^i}{(k)_q}$ it is enough to show that $\frac{\mathcal{E}_j^i}{(k)_q} V \subset V$.

We know that $\mathcal{E}_j^i V \subset (q+1)V$, so $\frac{\mathcal{E}_j^i}{(k)_q} V \subset \frac{q+1}{(k)_q} V$.

If k is even,

$$(k)_q = (q+1)\left(\frac{k}{2}\right)_{q^2} \text{ with } \left(\frac{k}{2}\right)_{q^2} = \frac{k}{2} \bmod (q+1) \text{ so } \frac{\mathcal{E}_j^i}{(k)_q} V \subset \frac{1}{\left(\frac{k}{2}\right)_{q^2}} V = V.$$

If k is odd, $(k)_q = 1 \bmod (q+1)$ and therefore $\frac{\mathcal{E}_j^i}{(k)_q} V \subset (q+1)V$. This concludes the proof. 3.6

4. LINKS-GOULD INVARIANTS AND THE CONJECTURE

4.1. Links-Gould invariants $LG^{n,1}$. The Links-Gould invariants $LG^{n,1}$ are the framed link invariants obtained by applying the modified (one has to use a modified trace; see [4]) Reshetikhin-Turaev construction to the ribbon Hopf algebras $U_q \mathfrak{gl}(n|1)^\sigma$ we just studied. Like in the Alexander case, the R -matrix can be divided by the value of the twist so that $LG^{n,1}$ becomes an *unframed* link invariant. Note that this definition and Viro's work [15] show that the first LG invariant $LG^{1,1}$ coincides with the Alexander-Conway polynomial Δ .

There are several sets of variables used in papers studying LG invariants. Three of them appear regularly: (t_0, t_1) , (τ, q) and (q^α, q) . Each set can be expressed in terms of the others using the following defining relations:

$$t_0 = q^{-2\alpha}, \quad t_1 = q^{2\alpha+2},$$

$$\tau = t_0^{1/2} = q^{-\alpha}.$$

In the case of $LG^{2,1}$, variables (t_0, t_1) nicely lead to a symmetric Laurent polynomial that has all sorts of Alexander-type properties [6].

Here we are interested in what happens to $LG^{n,1}$ when you evaluate q at -1 , or in other words when you set $t_0 t_1 = 1$.

4.2. Proof of the conjecture. Our study of ribbon Hopf algebra $U_q \mathfrak{gl}(n|1)^\sigma$ allows us to prove the following, that was conjectured in [3]:

Theorem 4.1. *For any link L in S^3 , $LG^{n,1}(L; \tau, -1) = \Delta_L(\tau^2)^n$. This can be translated in variables (t_0, t_1) :*

$$LG^{n,1}(L; t_0, t_0^{-1}) = \Delta_L(t_0)^n.$$

The rest of the section is devoted to proving this identity. First we identify $V_{-1}(0^n, \alpha)$ as an $\otimes_i A_i$ -module:

Proposition 4.2. *Let us denote by V^i an A_i -module isomorphic to the 2-dimensional U -module $V_{q^{-\alpha}}$. Equipped with the action of $\otimes_i A_i$ induced by $\Theta : \otimes_i A_i \rightarrow \mathcal{A}_{-1}^\sigma/I$, representation $V_{-1}(0^n, \alpha)$ is isomorphic to the irreducible representation $\otimes_i V^i$.*

Proof. By $\otimes_i V^i$, we mean the representation

$$\otimes_i \rho_i : \otimes_i A_i \rightarrow \otimes_i \text{End}_{\mathbb{C}}(V^i) \cong \text{End}_{\mathbb{C}}(\otimes_i V^i) \text{ where } \rho_i : A_i \rightarrow \text{End}_{\mathbb{C}}(V^i).$$

Set $a = q^{-\alpha}$. For each i , k_i^2 acts by a^2 on $V_{-1}(0^n, \alpha)$. Thus $V_{-1}(0^n, \alpha)$ is a representation of the 8^n -dimensional semi-simple algebra $\bigotimes_{i=1}^n (A_i/(k_i^2 - a^2))$. But for each A_i , v_0 is a highest weight vector of weight a . So it belongs to a summand of the $\bigotimes_{i=1}^n (A_i/(k_i^2 - a^2))$ -module $V_{-1}(0^n, \alpha)$ of the form $\otimes_i V^i$. Comparing the dimensions which are equal to 2^n for both vector spaces, we have that $V_{-1}(0^n, \alpha) \simeq \otimes_i V^i$. 4.2

Now we study the action of the pivotal element of $\mathcal{A}_q^\sigma$ in the representation at $q = -1$.

Proposition 4.3. *If $K_{2\rho}^\sigma$ is the pivotal element of $\mathcal{A}_q^\sigma$, in the representation $V_{-1}(0^n, \alpha)$,*

$$K_{2\rho}^\sigma = \Theta(\otimes_i \phi_i)$$

where $\phi_i = k_i^{-1} \in A_i$.

Proof. The antipode of $U_q \mathfrak{gl}(n|1)$ satisfies $S(E_{i+1}^i) = -E_{i+1}^i K_{i+1} K_i^{-1}$ and $S^2(E_{i+1}^i) = K_i K_{i+1}^{-1} E_{i+1}^i K_{i+1} K_i^{-1} = K_{2\rho} E_{i+1}^i K_{2\rho}^{-1}$. We can write $K_{2\rho}$ in terms of Cartan generators:

$$K_{2\rho} = K_{n+1}^n \prod_{i=1}^n K_i^{n-2i}.$$

Denoting $\langle a|b \rangle := \sum_{i=1}^n a_i b_i - a_{n+1} b_{n+1}$, and ρ the graded half sum of all positive roots, we find:

$$2\rho = \sum_{i=1}^n (n-2i)\varepsilon_i + n\varepsilon_{n+1},$$

where ε_i is the i^{th} basis vector of $\mathbb{C}^{n+1}$ and we write any vector $x = \sum_{i=1}^{n+1} x_i \varepsilon_i$ in this basis. Element $K_{2\rho}$ conjugates $e_i \in A_i$ as follows:

$$\begin{aligned} K_{2\rho} e_i K_{2\rho}^{-1} &= q^{\langle 2\rho | \varepsilon_i - \varepsilon_{n+1} \rangle} e_i \\ &= q^{(n-2i+n)} e_i \\ &= q^{2n-2i} e_i. \end{aligned}$$

So if $q = -1$,

$$\begin{aligned} \sigma K_{2\rho} e_i K_{2\rho}^{-1} \sigma &= -e_i \\ &= \phi_i e_i \phi_i^{-1} \\ &= \Theta(\otimes_j \phi_j) e_i \Theta(\otimes_j \phi_j^{-1}). \end{aligned}$$

Similarly to Proposition 2.6, we therefore can say that in the irreducible $\otimes_i A_i$ -module $V_{-1}(0^n, \alpha)$, $K_{2\rho}^\sigma$ is a scalar multiple of $\Theta(\otimes_j \phi_j)$. We call this element λ .

Since the two maps both act by $q^{n\alpha}$ on the highest weight vector, we conclude that $\lambda = 1$. 4.3

Proposition 4.4. *For any $x \in A_i \otimes A_i \subset \mathcal{A}_q \otimes \mathcal{A}_q$, we have*

$$D^{\mathfrak{gl}} x (D^{\mathfrak{gl}})^{-1} = \mathcal{D}(x)$$

where we identified $A_i \otimes A_i \cong U \otimes U$.

Proof. By a direct computation,

$$D^{\mathfrak{gl}} E_{n+1}^j \otimes 1 = E_{n+1}^j \otimes 1 q^{\sum_{i \leq n} ((E_i^i + \delta_j^i) \otimes E_i^i - (E_{n+1}^{n+1} - 1) \otimes E_{n+1}^{n+1})} = E_{n+1}^j \otimes K_j K_{n+1}^{-1} D^{\mathfrak{gl}}.$$

Thus $D^{\mathfrak{gl}} e_j \otimes 1 (D^{\mathfrak{gl}})^{-1} = e_j \otimes k_j$. Similarly $D^{\mathfrak{gl}} f_j \otimes 1 (D^{\mathfrak{gl}})^{-1} = f_j \otimes k_j^{-1}$. Finally $k_i \otimes 1$ clearly commutes with $D^{\mathfrak{gl}}$ and we can conclude using $\tau \circ D^{\mathfrak{gl}} = D^{\mathfrak{gl}} \circ \tau$. 4.4

Proof of Theorem 4.1. Let us sum up what we proved up to now to obtain Theorem 4.1. Let $V_{-1}(0^n, \alpha) \simeq \bigotimes_{i=1}^n V^i$ be the isomorphic representations of Proposition 4.2. In the following we fix such an isomorphism. Let V_H^i be a $U_1^H \mathfrak{sl}(2)$ -module structure on V^i extending the representation of A_i . We therefore obtain n commuting R -matrices $R^i = D^i \check{R}^i$ in $\text{End}_{\mathbb{C}}(V^i \otimes V^i) \hookrightarrow \text{End}_{\mathbb{C}}(V_{-1}(0^n, \alpha) \otimes V_{-1}(0^n, \alpha))$, where the explicit inclusion maps are given by $\iota_i : v \otimes w \mapsto (id^{\otimes i-1} \otimes v \otimes id^{\otimes n-i}) \otimes (id^{\otimes i-1} \otimes w \otimes id^{\otimes n-i})$. By Proposition 3.6,

$$\check{R}_{|q=-1}^{\mathfrak{gl}} = \prod_i \iota_i(\check{R}^i) \in \text{End}_{\mathbb{C}}(V_{-1}(0^n, \alpha) \otimes V_{-1}(0^n, \alpha)).$$

Moreover, using Lemma 2.1, Proposition 4.4, and the density lemma, the conjugation by $\prod_i \iota_i(D^i)$ is equal to the conjugation by $D^{\mathfrak{gl}}$ in $\text{End}_{\mathbb{C}}(V_{-1}(0^n, \alpha) \otimes V_{-1}(0^n, \alpha))$. Hence the braidings on $(\bigotimes_{i=1}^n V_H^i) \otimes (\bigotimes_{i=1}^n V_H^i)$ and on $V_{-1}(0^n, \alpha) \otimes V_{-1}(0^n, \alpha)$ are proportional. Now in the process of computing both the Links-Gould invariant and the Alexander polynomial, the R -matrices are rescaled by the inverse of their twist θ^{-1} so that the invariants become framing independent:

$$\text{trace}_2(\theta^{-1}(\text{Id} \otimes \phi) \tau R) = \text{Id}_{V_{-1}(0^n, \alpha)}$$

(here ϕ denotes any of the pivotal structures which are equal by Proposition 4.3). Hence the rescaled R -matrices $R_{|q=-1}^{\mathfrak{gl}} = \prod_i \iota_i(R_{V^i \otimes V^i}^H)$ and $\bigotimes_i R_{V^i \otimes V^i}^H$ are equal up to reordering factors. Finally, for any braid $\beta \in B_\ell$, the associated operators by the Reshetikhin-Turaev construction correspond up to reordering as well:

$$\Psi_{V_{-1}(0^n, \alpha) \otimes \ell}^{\mathfrak{gl}}(\beta) = \left(\Psi_{V_{-\alpha}^{\otimes \ell}}^{U_1^H \mathfrak{sl}(2)}(\beta) \right)^{\otimes n}.$$

At the end, if $\text{trace}_{2,3,\dots,\ell} \left((\text{Id}_{V_{-1}(0^n, \alpha)} \otimes \phi^{\otimes \ell-1}) \circ \Psi_{V_{-1}(0^n, \alpha) \otimes \ell}^{\mathfrak{gl}}(\beta) \right) = d \cdot \text{Id}_{V_{-1}(0^n, \alpha)}$ when

$$\text{trace}_{2,3,\dots,\ell} \left((\text{Id}_{V_\alpha} \otimes \phi_U^{\otimes \ell-1}) \circ \Psi_{V_\alpha^{\otimes \ell}}^{U_1^H \mathfrak{sl}(2)}(\beta) \right) = c \cdot \text{Id}_{V_\alpha},$$

we obtain

$$d = c^n$$

by considering the trace of these two maps. Indeed, the trace is blind to reordering factors. 4.1

Remark 4.5. In [4], the LG invariant is extended to a multivariable link invariant $M(L; q, q_1, \dots, q_c)$ for links with $c \geq 2$ ordered components, taking its values in Laurent polynomials $\mathbb{Z}[q^\pm, q_1^\pm, \dots, q_c^\pm]$. It is shown in [5] that

$$LG^{n,1}(\tau, q) = \left(\prod_{i=0}^{n-1} \frac{q^i}{\tau} - \frac{\tau}{q^i} \right) M(L; q, \tau^{-1}, \dots, \tau^{-1}).$$

The proof in this paper should adapt to show that

$$M(L; -1, q_1, \dots, q_c) = \nabla(q_1, \dots, q_c)^n$$

where ∇ is the Conway potential function, a version of the multivariable Alexander polynomial.

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Regular graphs with positive Ollivier–Ricci curvature

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Abstract

We study the Ollivier–Ricci curvature and its modification introduced by Lin, Lu, and Yau on regular graphs. We derive precise formulas for the Ollivier–Ricci curvature and the Lin–Lu–Yau curvature on regular graphs, expressed in terms of graph parameters and an optimal assignment between neighborhoods. Using these precise formulas, we provide an efficient implementation for computing the Ollivier–Ricci curvature and the Lin–Lu–Yau curvature on regular graphs, enabling us to explore the distribution of regular graphs with positive Lin–Lu–Yau curvature. Our main contribution is establishing a lower bound on the degree of a regular graph that guarantees positive Lin–Lu–Yau curvature and non-negative Ollivier–Ricci curvature, along with an analysis of the sharpness of this bound. Additionally, we study the maximal number of vertices that an d -regular graph with positive Lin–Lu–Yau curvature can have.

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1 Introduction and statement of results

Since Gauss and Riemann introduced the geometric notion of curvature over 150 years ago, it played a central role in differential geometry. Among the various types of curvature, Ricci curvature is of particular importance, serving as a fundamental tool in the study of Riemannian manifolds. Given the important role of Ricci curvature in differential geometry, there is a natural interest in extending this concept to metric spaces more general than Riemannian manifolds. This has led to the proposal of various generalized notions of curvature for non-smooth or discrete structures (see, e.g., [1–4]). Each approach captures a different aspect of the Riemannian notion, and each is thus suited for a different purpose.

Von Renesse and Sturm [5] established a connection between Ricci curvature and optimal transport on smooth Riemannian manifolds. Building on their results, Ollivier [6] introduced a discrete notion of Ricci curvature on metric spaces equipped with Markov chains or a measure. His approach leverages optimal transport theory, as his

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definition of curvature is based on the Wasserstein distance. For a precise definition a detailed introduction is required, which is provided in Sect. 2.1.

In the context of locally finite graphs, Ollivier's notion of Ricci curvature has recently received considerable attention. In this setting, a modification of Ollivier's notion was introduced by Lin, Lu, and Yau in [7], the *Lin–Lu–Yau curvature*. The first contribution of this work is the derivation of a precise formula for the Lin–Lu–Yau curvature on regular graphs using elementary means.

Theorem 1.1 *Let $G = (V, E)$ be a locally finite graph. Let $x, y \in V$ be of equal degree d with $x \sim y$. Then the Lin–Lu–Yau curvature*

$$\kappa(x, y) = \frac{1}{d} \left(d + 1 - \inf_{\phi \in \mathcal{A}_{xy}} \sum_{z \in S_1(x) \setminus B_1(y)} d(z, \phi(z)) \right),$$

where $\mathcal{A}_{xy}$ denotes the set of assignments between $S_1(x) \setminus B_1(y)$ and $S_1(y) \setminus B_1(x)$.

Bonini et al. [8, Theorem 4.6] derived an exact formula for the Lin–Lu–Yau curvature on strongly regular graphs in terms of graph parameters and the size of a maximal matching in the core neighborhood. In [9], Münch et al. provided an exact formula for a modified curvature notion on combinatorial graphs. To the best of my knowledge, precise expressions for the Lin–Lu–Yau curvature on arbitrary regular graphs have not yet been established.

We then use the exact formula for the Lin–Lu–Yau curvature to provide an efficient implementation for computing the Lin–Lu–Yau curvature on regular graphs. This allows us to investigate the distribution of regular graphs with positive Lin–Lu–Yau curvature, leading to the main result of this work: The establishment of a lower bound on the degree of a regular graph that ensures positive Lin–Lu–Yau curvature.

Theorem 1.2 *Let $G = (V, E)$ be a finite graph. If G is d -regular with*

$$d > \frac{2|V|}{3} - 2,$$

then $\kappa(x, y) > 0$ for every edge $x \sim y$.

A similar result holds true for the Ollivier–Ricci curvature.

Theorem 1.3 *Let $G = (V, E)$ be a finite graph. If G is d -regular with*

$$d > \frac{2|V|}{3} - 2,$$

then $\kappa_0(x, y) \geq 0$ for every edge $x \sim y$.

After deriving this lower bound on the degree of regular graphs that guarantees positive Lin–Lu–Yau curvature and non-negative Ollivier–Ricci curvature, we proceed to discuss the sharpness of the bound.

Theorem 1.4 *Let $n \in \mathbb{N}$ such that*

$$d = \frac{2n - 8}{3} \in 2\mathbb{N}$$

and $d \geq 12$. Then, there exists a d -regular graph on n vertices such that $\kappa(x, y) = 0$ and $\kappa_0(x, y) < 0$ for an edge $x \sim y$.

Finally, we study the maximal number of vertices that a d -regular graph G with positive Lin–Lu–Yau curvature can have,

$$\mathcal{M}_d = \max_{\substack{G \text{ } d\text{-regular,} \\ \text{Ric}(G) > 0}} |V(G)|.$$

We prove that $\mathcal{M}_d$ is finite for every degree d and we obtain precise values for d equal to two and three.

Theorem 1.5 *Let $G = (V, E)$ be a d -regular graph, satisfying $\text{Ric}(G) > 0$. If $d = 2$, then $|V| \leq 5$. If $d = 3$, then $|V| \leq 10$. Consequently, we have $\mathcal{M}_2 = 5$ and $\mathcal{M}_3 = 10$.*

We conclude this introduction by providing an outline of the remainder of the paper. In Sect. 2 we review the relevant concepts of graph theory and optimal transport theory, and introduce the Ollivier–Ricci curvature, as well as its modification by Lin, Lu, and Yau. In Sect. 3 we derive exact formulas for the different curvature notions. In Sect. 4 we present our efficient implementation for the computation of the Lin–Lu–Yau curvature and explore the distribution of regular graphs with positive Lin–Lu–Yau curvature. In Sect. 5, we establish our main result. Namely, we derive a lower bound on the degree of a regular graph that ensures positive Lin–Lu–Yau curvature. Finally, in Sect. 6, we investigate the maximal number of vertices that a d -regular graph with positive Lin–Lu–Yau curvature can have and present some open questions.

2 Definitions and notations

We begin by reviewing some fundamental concepts of graph theory and optimal transport theory. We then introduce Ollivier’s discrete notion of Ricci curvature on graphs, as well as its modification by Lin, Lu, and Yau.

2.1 Graph theory

A *simple graph* $G = (V, E)$ is an unweighted, undirected graph that contains no multiple edges or self-loops. For two vertices $x, y \in V$ we denote the existence of an edge between x and y by $x \sim y$. For any two vertices $x, y \in V$, the *shortest-path distance* $d(x, y)$ is the number of edges in a shortest path connecting x and y . If no such path exists, $d(x, y)$ is defined to be infinity. For two nonempty sets $A, B \subseteq V$, we define

$$\text{dist}(A, B) = \inf \{d(x, y) : x \in A, y \in B\}.$$

The *diameter* of G is denoted by $\text{diam}(G) = \max_{x,y \in V} d(x, y)$. For $x \in V$ and $r \in \mathbb{N}$ we define the r -sphere centered at x as $S_r(x) = \{y \in V : d(x, y) = r\}$ and the r -ball centered at x as $B_r(x) = \{y \in V : d(x, y) \leq r\}$. For an edge $x \sim y$, we denote by $\Delta(x, y) = S_1(x) \cap S_1(y)$ the set of common neighbors of x and y .

The *degree* of a vertex $x \in V$ is denoted by $d_x = |S_1(x)|$. The graph G is called *locally finite* if every vertex has finite degree. The graph is said to be d -regular if every vertex has the same degree d .

The Moore bound provides an upper bound on the number of vertices a regular graph with given diameter can have.

Theorem 2.1 (Moore bound) *Let $G = (V, E)$ be a d -regular graph with diameter k . Then*

$$|V| \leq 1 + d \sum_{i=0}^{k-1} (d-1)^i.$$

A simple graph $G = (V, E)$ is called *bipartite*, if there is a partition of V into two disjoint sets $V = A \cup B$, such that every edge connects a vertex in A to a vertex in B . If the partition $V = A \cup B$ is specified, we denote the bipartite graph by $G = (A, B, E)$.

In what follows, the famous *Hall's marriage theorem* [10] will be of importance. Before stating the theorem, we must first introduce the concept of a perfect matching. For a simple graph $G = (V, E)$, we call a subset of edges $S \subseteq E$ a *matching* of G if no vertex in V is incident to more than one edge in S . For a bipartite graph $G = (A, B, E)$, we call a matching S a *perfect matching of A into B* if every vertex in A is incident to exactly one edge in S .

Theorem 2.2 (Hall's marriage theorem) *Let $G = (A, B, E)$ be a bipartite graph. Then there exists a perfect matching of A into B if and only if $|N(X)| \geq |X|$ for every $X \subseteq A$, where $N(X) = \bigcup_{x \in X} S_1(x)$.*

As a consequence of Hall's marriage theorem, we obtain the following Corollary.

Corollary 2.3 *Let $G = (A, B, E)$ be a bipartite graph. If G is regular, then there exists a perfect matching of A into B .*

In what follows, $G = (V, E)$ is always assumed to be a locally finite, connected, and simple graph.

2.2 Optimal transport theory

A probability measure on the vertex set V is a function $\mu : V \rightarrow [0, 1]$ that satisfies $\sum_{x \in V} \mu(x) = 1$. We denote by $\mathcal{P}(V)$ the set of probability measures on V . The Wasserstein distance, a metric on the set $\mathcal{P}(V)$, is a fundamental concept in optimal transport theory.

Definition 2.4 (*Wasserstein distance*) Let $G = (V, E)$ be a locally finite graph. Let μ_1, μ_2 be two probability measures on V . The *Wasserstein distance* between μ_1 and μ_2 is defined as

$$W_1(\mu_1, \mu_2) = \inf_{\pi \in \Pi(\mu_1, \mu_2)} \sum_{x \in V} \sum_{y \in V} d(x, y) \pi(x, y), \quad (2.1)$$

where

$$\Pi(\mu_1, \mu_2) = \left\{ \pi : V \times V \rightarrow [0, 1] : \sum_{y \in V} \pi(x, y) = \mu_1(x), \sum_{x \in V} \pi(x, y) = \mu_2(y) \right\}.$$

Intuitively, imagine two distributions given by μ_1 and μ_2 as piles of earth. The Wasserstein distance measures the minimal effort required to transform one pile of earth into another. We call $\pi \in \Pi(\mu_1, \mu_2)$ a *transport plan* and if the infimum in 2.1 is attained, we call π an *optimal transport plan* transporting μ_1 to μ_2 .

The following Lemma allows us to impose some assumptions on an optimal transport plan. It states that it is not necessary to move any mass that is shared between the two probability measures.

Lemma 2.5 *Let $G = (V, E)$ be a locally finite graph. Let μ_1, μ_2 be two probability measures on V . Then there exists an optimal transport plan π transporting μ_1 to μ_2 satisfying*

$$\pi(x, x) = \min\{\mu_1(x), \mu_2(x)\}$$

for all $x \in V$.

The proof is an immediate consequence of the triangle inequality. For more details, we refer to [12, Lemma 4.1]

In what follows, probability measures under which all points in their support carry equal mass will be of importance, i.e.,

$$\mu_1 = \frac{1}{n} \sum_{i=1}^n \delta_{x_i}, \quad \mu_2 = \frac{1}{n} \sum_{j=1}^n \delta_{y_j}.$$

Here, δ_x denotes the Dirac measure centered on some fixed vertex $x \in V$:

$$\delta_x(y) = \begin{cases} 1, & \text{if } x = y; \\ 0, & \text{otherwise.} \end{cases}$$

In this case, any transport plan $\pi \in \Pi(\mu_1, \mu_2)$ can be represented as a bistochastic $n \times n$ matrix $\pi = (\pi_{ij})_{i,j=1,\dots,n}$. Here bistochasticity means that $\pi_{ij} \geq 0$ and that

$$\sum_{k=1}^n \pi_{ik} = \sum_{k=1}^n \pi_{kj} = 1$$

for any $i, j \in \{1, \dots, n\}$. We denote the set of all bistochastic $n \times n$ matrices by $\mathcal{B}_n$. Thus, the Wasserstein distance reduces to

$$W_1(\mu_1, \mu_2) = \inf_{\pi \in \mathcal{B}_n} \frac{1}{n} \sum_{i,j=1}^n \pi_{ij} d(x_i, y_j).$$

Therefore, calculating the Wasserstein distance is a linear optimization problem on the bounded and convex set $\mathcal{B}_n$. *Choquet's minimization theorem* states that some extremal points of $\mathcal{B}_n$ are optimal. *Birkhoff's theorem* states that these extremal points are permutation matrices, i.e.,

$$\pi_{ij} = \delta_{\sigma(i),j} = \begin{cases} 1, & \text{if } \sigma(i) = j; \\ 0, & \text{otherwise.} \end{cases}$$

for some permutation σ of $\{1, \dots, n\}$. Here δ_{ij} is the Kronecker symbol. We denote by S_n the set of all permutations of $\{1, \dots, n\}$. Thus, we obtain the following Lemma [11].

Lemma 2.6 *Let $G = (V, E)$ be a locally finite graph. Let $\mu_1 = \frac{1}{n} \sum_{i=1}^n \delta_{x_i}$ and $\mu_2 = \frac{1}{n} \sum_{i=1}^n \delta_{y_i}$, where $\{x_1, \dots, x_n\} \subset V$ and $\{y_1, \dots, y_n\} \subset V$. Then the Wasserstein distance between μ_1 and μ_2 reduces to*

$$W_1(\mu_1, \mu_2) = \inf_{\sigma \in S_n} \frac{1}{n} \sum_{i=1}^n d(x_i, y_{\sigma(i)}).$$

In other words, there exists an optimal transport plan that corresponds to a one-to-one assignment between the source vertices $x_1, \dots, x_n$ and the target vertices $y_1, \dots, y_n$. We refer to such an optimal transport plan as an *optimal assignment*. In other words, the Kantorovich problem coincides with the Monge problem in this special case.

2.3 Ollivier–Ricci curvature and its modification

To introduce Ollivier's notion of Ricci curvature on graphs, we define the probability measures μ_x^α for $x \in V$ and $\alpha \in [0, 1]$ by

$$\mu_x^\alpha(y) = \begin{cases} \alpha, & \text{if } y = x; \\ \frac{1-\alpha}{d_x}, & \text{if } y \sim x; \\ 0, & \text{otherwise.} \end{cases}$$

Then, the Ollivier–Ricci curvature is defined as follows.

Definition 2.7 (*Ollivier–Ricci curvature*). Let $G = (V, E)$ be a locally finite graph. We define the α -Ollivier–Ricci curvature of an edge $x \sim y$ by

$$\kappa_\alpha^G(x, y) = 1 - W_1^G(\mu_x^\alpha, \mu_y^\alpha).$$

The parameter α is called the *idleness*.

Ollivier considered idleness parameters $\alpha = 0$ and $\alpha = \frac{1}{2}$. In [12], the authors study the Ollivier–Ricci curvature as a function of the idleness parameter. To this end, they introduced the *Ollivier–Ricci idleness function* $\alpha \rightarrow \kappa_\alpha(x, y)$. It was first shown by Lin, Lu, and Yau in [7], that the idleness function is concave. Using that $\kappa_1(x, y) = 0$, this implies that the function $h(\alpha) = \frac{\kappa_\alpha(x, y)}{1 - \alpha}$ is increasing over the interval $[0, 1)$. They further showed that $h(\alpha)$ is bounded and thus, the limit $\lim_{\alpha \rightarrow 1} h(\alpha)$ exists. Lin, Lu, and Yau used this result to introduce a modification of the Ollivier–Ricci curvature.

Definition 2.8 (*Lin–Lu–Yau curvature*). Let $G = (V, E)$ be a locally finite graph. The *Lin–Lu–Yau curvature* of an edge $x \sim y$ is defined as

$$\kappa(x, y) = \lim_{\alpha \rightarrow 1} \frac{\kappa_\alpha(x, y)}{1 - \alpha}.$$

Remark 2.9 Observe that $\kappa_1(x, y) = 0$ for any edge $x \sim y$. Thus, $\kappa(x, y)$ is the negative of the derivative of $\kappa_\alpha(x, y)$ with respect to the idleness parameter in $\alpha = 1$.

In what follows, we write $\text{Ric}(G) > 0$ if $\kappa(x, y) > 0$ for all edges $x \sim y$ in G .

Bourne et al. [12] showed that the idleness function is piecewise linear with at most 3 linear parts. They also derived the length of the last linear part.

Theorem 2.10 [12, Theorem 4.4]. *Let $G = (V, E)$ be a locally finite graph and let $x, y \in V$ with $x \sim y$ and $d_x \geq d_y$. Then $\alpha \rightarrow \kappa_\alpha(x, y)$ is linear over $\left[\frac{1}{d_x+1}, 1\right]$.*

Thus, we obtain the following relation between the α -Ollivier–Ricci curvature and its modification by Lin, Lu, and Yau as an immediate consequence of the mean value theorem.

Theorem 2.11 *Let $G = (V, E)$ be a locally finite graph and let $x, y \in V$ with $x \sim y$ and $d_x \geq d_y$. Then*

$$\kappa_\alpha(x, y) = (1 - \alpha)\kappa(x, y)$$

for $\alpha \in \left[\frac{1}{d_x+1}, 1\right]$.

Hence, the Lin–Lu–Yau curvature coincides up to a scaling factor with the α -Ollivier–Ricci curvature for large values of α .

3 Exact formulas for the Ollivier–Ricci curvature

In this section, we shall derive precise formulas for the Lin–Lu–Yau curvature and the α -Ollivier–Ricci curvature of an edge $x \sim y$ with $d_x = d_y$.

3.1 Decomposing the neighborhood

To derive our precise formulas, we decompose the 1-spheres centered at the vertices x and y into disjoint sets. We denote by $\Delta(x, y)$ the set of common neighbors of both x and y , namely,

$$\Delta(x, y) = S_1(x) \cap S_1(y).$$

Furthermore, we define $R_x(x, y) = S_1(x) \setminus B_1(y)$ and $R_y(x, y) = S_1(y) \setminus B_1(x)$, giving rise to the following disjoint decompositions of $S_1(x)$ and $S_1(y)$:

$$S_1(x) = \{y\} \cup \Delta(x, y) \cup R_x(x, y) \quad \text{and} \quad S_1(y) = \{x\} \cup \Delta(x, y) \cup R_y(x, y).$$

Using these decompositions, we show that the computation of the Wasserstein distance reduces to an assignment problem between $R_x(x, y)$ and $R_y(x, y)$. Therefore, we introduce the concept of optimal assignments.

Definition 3.1 (*Optimal assignment*). Let $G = (V, E)$ be a locally finite graph. Let $U, W \subseteq V$ with $|U| = |W| < \infty$. We call a bijection $\phi : U \rightarrow W$ an *assignment* between U and W . Denote by $\mathcal{A}_{UW}$ the set of all assignments between U and W . We call $\phi \in \mathcal{A}_{UW}$ an *optimal assignment* between U and W if

$$\sum_{z \in U} d(z, \phi(z)) = \inf_{\psi \in \mathcal{A}_{UW}} \sum_{z \in U} d(z, \psi(z)).$$

Denote the set of all optimal assignments between U and W by $\mathcal{O}_{UW}$.

Note that for an edge $x \sim y$ with $d_x = d_y$, we have $|R_x(x, y)| = |R_y(x, y)|$. For the sake of simplicity, we will denote by $\mathcal{A}_{xy}$ and $\mathcal{O}_{xy}$ the sets $\mathcal{A}_{R_x(x, y)R_y(x, y)}$ and $\mathcal{O}_{R_x(x, y)R_y(x, y)}$, respectively.

3.2 Ollivier–Ricci curvature on the last linear part

To derive our formula for the Lin–Lu–Yau curvature, we begin by deriving an expression for the Ollivier–Ricci curvature on the last linear part. We then utilize the fact that on this segment, the Lin–Lu–Yau curvature coincides with the Ollivier–Ricci curvature, differing only by a scaling factor.

We first need the following Lemma about some of the assumptions we can impose on an optimal transport plan transporting μ_x^α to μ_y^α .

Lemma 3.2 *Let $G = (V, E)$ be a locally finite graph. Let $x, y \in V$ be of equal degree d with $x \sim y$. Furthermore, let $\alpha \in \left[\frac{1}{d+1}, 1\right]$. Then there exists an optimal transport plan π transporting μ_x^α to μ_y^α with the following properties:*

- (i) $\forall i \in \Delta(x, y) : \pi(i, i) = \mu_x^\alpha(i)$;
- (ii) $\pi(x, x) = \pi(y, y) = \mu_y^\alpha(x)$;
- (iii) $\pi(x, y) = \mu_y^\alpha(y) - \mu_y^\alpha(x)$.

Remark 3.3 Observe that $\mu_y^\alpha(y) - \mu_y^\alpha(x) \geq 0$ if and only if $\alpha \geq \frac{1}{d+1}$. Hence, the subsequent discussion pertains only to the last linear part of κ_α .

Proof Properties (i) and (ii) are immediate consequence of Lemma 2.5.

For the proof of (iii), let π be an optimal transport plan transporting μ_x^α to μ_y^α while satisfying properties (i) and (ii). Since π satisfies property (ii), we must have

$$\begin{aligned}\pi(x, y) &= \mu_y^\alpha(y) - \sum_{j \in V \setminus \{x\}} \pi(j, y) \\ &\leq \mu_y^\alpha(y) - \pi(y, y) \\ &= \mu_y^\alpha(y) - \mu_y^\alpha(x).\end{aligned}$$

Now, assume that $\pi(x, y) < \mu_y^\alpha(y) - \mu_y^\alpha(x)$. Thus, the sets

$$I = \{z \in S_1(x) \setminus \{y\} : \pi(z, y) > 0\} \quad \text{and} \quad O = \{z \in S_1(y) \setminus \{x\} : \pi(x, z) > 0\}$$

are non-empty. Due to property (i), we must have $I \cap O = \emptyset$. Since π satisfies property (ii), we obtain:

$$\begin{aligned}\sum_{z \in I} \pi(z, y) &= \mu_y^\alpha(y) - \pi(y, y) - \pi(x, y) \\ &= \mu_x^\alpha(x) - \pi(x, x) - \pi(x, y) \\ &= \sum_{z \in O} \pi(x, z).\end{aligned}$$

In other words, the mass transported from I to y equals the mass transported from x to O under π . We denote this mass by m , namely $m = \sum_{z \in I} \pi(z, y)$.

We now obtain a new transport plan π' from modifying π as follows:

- Set $\pi'(x, y) = \mu_y^\alpha(y) - \mu_y^\alpha(x)$, $\pi'(x, z) = 0 \forall z \in O$ and $\pi'(z, y) = 0 \forall z \in I$
- Move the mass m directly from I to O .

See Fig. 1 for an illustration of this modification.

Due to property (i), we must have $O \cap \Delta(x, y) = I \cap \Delta(x, y) = \emptyset$. Therefore, $d(i, y) = 2$ for any $i \in I$ and $d(x, j) = 2$ for any $j \in O$. Thus, π moves mass

- m at distance two from I to y ,
- m at distance two from x to O ,
- $\pi(x, y) = \mu_y^\alpha(y) - \mu_y^\alpha(x) - m$ at distance one from x to y ,

resulting in a cost of:

$$3 \cdot m + \mu_y^\alpha(y) - \mu_y^\alpha(x).$$

On the other hand, observe that $d(i, j) \leq 3$ for any $i \in I$ and $j \in O$. Thus, π' moves the mass

- $\mu_y^\alpha(y) - \mu_y^\alpha(x)$ at distance one from x to y ,
- m from I to O at distance less than or equal to three,

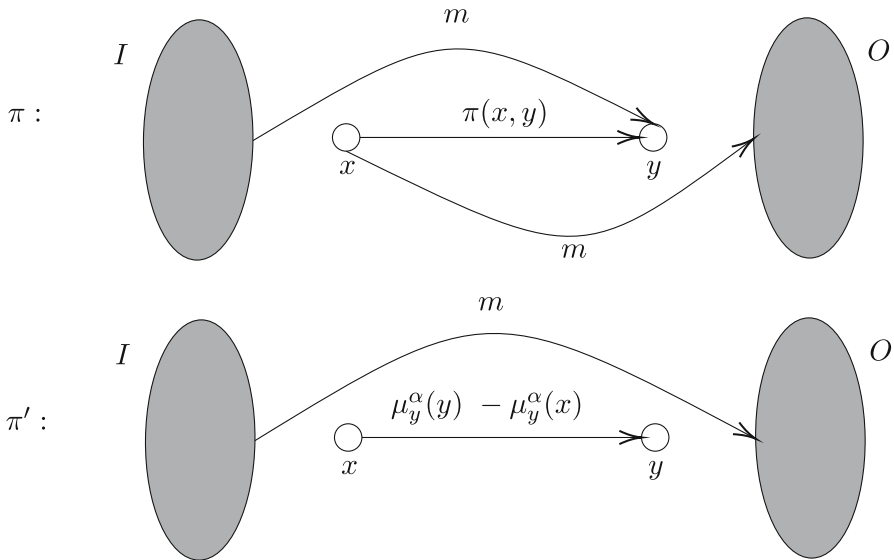


Fig. 1 Illustration of the optimal transport plan π and its modification π' for the proof of Lemma 3.2

resulting in a cost less than or equal to:

$$3 \cdot m + \mu_y^\alpha(y) - \mu_y^\alpha(x).$$

This is exactly the cost of moving this part of the mass under the optimal transport plan π . As π' coincides with π everywhere else, π' is an optimal transport plan as well. Finally, observe that by construction,

$$\pi'(j, j) = \pi(j, j) \quad \forall j \in \Delta(x, y), \quad \pi'(x, x) = \pi(x, x) \quad \text{and} \quad \pi'(y, y) = \pi(y, y).$$

Hence, we have constructed an optimal transport plan satisfying properties (i)–(iii). $\square$

We are now ready to derive a precise expression for the Ollivier–Ricci curvature on the last linear part.

Theorem 3.4 *Let $G = (V, E)$ be a locally finite graph. Let $x, y \in V$ be of equal degree d with $x \sim y$. Furthermore, let $\alpha \in \left[\frac{1}{d+1}, 1\right]$. Then the α -Ollivier–Ricci curvature*

$$\kappa_\alpha(x, y) = \frac{1 - \alpha}{d} \left(d + 1 - \inf_{\phi \in \mathcal{A}_{xy}} \sum_{z \in R_x(x, y)} d(z, \phi(z)) \right).$$

Remark 3.5 This result can be derived from the linearity of the idleness function together with properties (i), (ii), and (iii) in the proof of Theorem 1.1 in [19]. For the reader's convenience, we provide a full proof here.

Proof To demonstrate this, we proceed with a proof by cases.

Case 1: $R_x(x, y) = \emptyset$. In this case, $\mathcal{A}_{xy}$ only contains the empty function. Thus,

$$\frac{1-\alpha}{d} \left(d + 1 - \inf_{\phi \in \mathcal{A}_{xy}} \sum_{z \in R_x(x, y)} d(z, \phi(z)) \right) = \frac{1-\alpha}{d} (d + 1).$$

On the other hand, the transport plan π transporting μ_x^α to μ_y^α , defined by

$$\pi(i, j) = \begin{cases} \frac{1-\alpha}{d}, & \text{if } i = j \in \{x, y\} \cup \Delta(x, y); \\ \alpha - \frac{1-\alpha}{d}, & \text{if } i = x \text{ and } j = y; \\ 0, & \text{otherwise.} \end{cases}$$

is optimal, according to Lemma 3.2. Therefore,

$$W_1(\mu_x^\alpha, \mu_y^\alpha) = \sum_{i \in B_1(x)} \sum_{j \in B_1(y)} d(i, j) \pi(i, j) = \alpha - \frac{1-\alpha}{d},$$

and hence,

$$\kappa_\alpha(x, y) = \frac{1-\alpha}{d} (d + 1).$$

Case 2: $R_x(x, y) \neq \emptyset$. Let $\phi \in \mathcal{O}_{xy}$ be an optimal assignment between $R_x(x, y)$ and $R_y(x, y)$. Define the transport plan π transporting μ_x^α to μ_y^α by

$$\pi(i, j) = \begin{cases} \frac{1-\alpha}{d}, & \text{if } i = j \in \{x, y\} \cup \Delta(x, y); \\ \frac{1-\alpha}{d}, & \text{if } i \in R_x(x, y) \text{ and } j = \phi(i); \\ \alpha - \frac{1-\alpha}{d}, & \text{if } i = x \text{ and } j = y; \\ 0, & \text{otherwise.} \end{cases}$$

We claim that this transport plan is optimal. Assume on the contrary that is is not optimal. Then let π' be an optimal transport plan transporting μ_x^α to μ_y^α , satisfying properties (i)–(iii) stated in Lemma 3.2. Then,

$$\sum_{i \in B_1(x)} \sum_{j \in B_1(y)} \pi'(i, j) d(i, j) = \alpha - \frac{1-\alpha}{d} + \sum_{i \in R_x(x, y)} \sum_{j \in R_x(x, y)} \pi'(i, j) d(i, j). \quad (3.1)$$

On the other hand, we obtain by construction of π

$$\sum_{i \in B_1(x)} \sum_{j \in B_1(y)} \pi(i, j) d(i, j) = \alpha - \frac{1-\alpha}{d} + \sum_{i \in R_x(x, y)} \sum_{j \in R_x(x, y)} \pi(i, j) d(i, j). \quad (3.2)$$

Due to the non-optimality of π , we must have:

$$\sum_{i \in B_1(x)} \sum_{j \in B_1(y)} \pi'(i, j) d(i, j) < \sum_{i \in B_1(x)} \sum_{j \in B_1(y)} \pi(i, j) d(i, j). \quad (3.3)$$

Combining Eqs. 3.1, 3.2 and 3.3 leads to

$$\sum_{i \in R_x(x, y)} \sum_{j \in R_x(x, y)} \pi'(i, j) d(i, j) < \sum_{i \in R_x(x, y)} \sum_{j \in R_x(x, y)} \pi(i, j) d(i, j). \quad (3.4)$$

Now, observe that $\pi|_{R_x(x, y) \times R_y(x, y)}$ and $\pi'|_{R_x(x, y) \times R_y(x, y)}$ are transport plans transporting $\mu_x^\alpha|_{R_x(x, y)}$ to $\mu_y^\alpha|_{R_y(x, y)}$. Furthermore, $\mu_x^\alpha|_{R_x(x, y)}$ and $\mu_y^\alpha|_{R_y(x, y)}$ are measures under which all points in their support carry equal mass $\frac{1-\alpha}{d}$.

Thus, we can apply Lemma 2.6 (after multiplication with a scaling factor) and conclude that $\pi|_{R_x(x, y) \times R_y(x, y)}$ is an optimal transport plan. This contradicts Eq. 3.4.

Therefore, our assumption was incorrect and π is an optimal transport plan. The rest of the proof is just a straightforward calculation. Namely,

$$\begin{aligned} W_1(\mu_x^\alpha, \mu_y^\alpha) &= \sum_{i \in B_1(x)} \sum_{j \in B_1(y)} d(i, j) \pi(i, j) \\ &= \alpha - \frac{1-\alpha}{d} + \frac{1-\alpha}{d} \sum_{z \in R_x(x, y)} d(z, \phi(z)), \end{aligned}$$

which leads to

$$\kappa_\alpha(x, y) = \frac{1-\alpha}{d} \left(d + 1 - \sum_{z \in R_x(x, y)} d(z, \phi(z)) \right).$$

The fact that ϕ is an optimal assignment concludes the proof. $\square$

3.3 Lin–Lu–Yau curvature

According to Theorem 2.11, the identity

$$\kappa(x, y) = \frac{\kappa_\alpha(x, y)}{1-\alpha}$$

holds true on the last linear part of the idleness function. Therefore, our formula for the Lin–Lu–Yau curvature is an immediate consequence of Theorem 3.4.

Theorem 1.1 *Let $G = (V, E)$ be a locally finite graph. Let $x, y \in V$ be of equal degree d with $x \sim y$. Then the Lin–Lu–Yau curvature*

$$\kappa(x, y) = \frac{1}{d} \left(d + 1 - \inf_{\phi \in \mathcal{A}_{xy}} \sum_{z \in R_x(x, y)} d(z, \phi(z)) \right). \quad (3.5)$$

Let us now state some consequence of our formula.

Corollary 3.6 *Let $G = (V, E)$ be a locally finite graph. Let $x, y \in V$ be of equal degree d with $x \sim y$. Then $\kappa(x, y) \in \mathbb{Z}/d$. Furthermore,*

$$-2 + \frac{4}{d} + 3 \frac{|\Delta(x, y)|}{d} \leq \kappa(x, y) \leq \frac{2 + |\Delta(x, y)|}{d}.$$

Proof The fact that $\kappa(x, y) \in \mathbb{Z}/d$ follows directly from formula 3.5.

Using that $d(z, \phi(z)) \geq 1$ for all $z \in R_x(x, y)$ and any assignment $\phi \in \mathcal{A}_{xy}$, we obtain

$$\begin{aligned} \kappa(x, y) &= \frac{1}{d} \left(d + 1 - \inf_{\phi \in \mathcal{A}_{xy}} \sum_{z \in R_x(x, y)} d(z, \phi(z)) \right) \\ &\leq \frac{1}{d} \left(d + 1 - |R_x(x, y)| \right) \\ &= \frac{1}{d} \left(d + 1 - (d - 1 - |\Delta(x, y)|) \right) \\ &= \frac{2 + |\Delta(x, y)|}{d}. \end{aligned}$$

Similarly, using that $d(z, \phi(z)) \leq 3$, we obtain

$$\begin{aligned} \kappa(x, y) &= \frac{1}{d} \left(d + 1 - \inf_{\phi \in \mathcal{A}_{xy}} \sum_{z \in R_x(x, y)} d(z, \phi(z)) \right) \\ &\geq \frac{1}{d} \left(d + 1 - 3|R_x(x, y)| \right) \\ &= \frac{1}{d} \left(-2d + 4 + 3|\Delta(x, y)| \right). \end{aligned}$$

□

3.4 Ollivier–Ricci curvature for vanishing idleness

This section aims to derive an exact formula for the α -Ollivier–Ricci curvature of an edge $x \sim y$ with $d_x = d_y$, in the case of vanishing idleness, i.e., $\alpha = 0$.

Recall that for $\alpha = 0$, μ_x^α simplifies to

$$\mu_x^\alpha(y) = \begin{cases} \frac{1}{d_x}, & \text{if } y \in S_1(x); \\ 0, & \text{otherwise.} \end{cases}$$

As per Lemma 2.6, the Kantorovich problem coincides with the Monge problem in this case. The following Lemma states some of the assumptions we can impose on an optimal assignment.

Lemma 3.7 *Let $G = (V, E)$ be a locally finite graph. Let $x, y \in V$ be of equal degree d with $x \sim y$. Then there exists an optimal assignment ϕ between $S_1(x)$ and $S_1(y)$, such that $\phi(i) = i$ for all $i \in \Delta(x, y)$.*

Using analogous reasoning as in the case of the Lin–Lu–Yau curvature, we obtain a precise expression for the 0-Ollivier–Ricci curvature.

Theorem 3.8 *Let $G = (V, E)$ be a locally finite graph. Let $x, y \in V$ be of equal degree d with $x \sim y$. Then*

$$\kappa_0(x, y) = \frac{1}{d} \left(d - \inf_{\phi} \sum_{z \in S_1(x) \setminus S_1(y)} d(z, \phi(z)) \right),$$

where the infimum is taken over all assignments ϕ between $S_1(x) \setminus S_1(y)$ and $S_1(y) \setminus S_1(x)$.

4 Distribution of positive Lin–Lu–Yau curvature in regular graphs

In this section, we wish to study the distribution of regular graphs G satisfying $\text{Ric}(G) > 0$. To do so, we are faced with the problem of generating exhaustive lists of all d -regular graphs with a given number of vertices, up to isomorphism. This is one of the oldest problems in the field of constructive combinatorics [13]. Already determining the number of d -regular graphs on n vertices, denoted by $|\mathcal{G}_{n,d}|$, proves to be an exceedingly difficult problem for which no closed formula is currently known. But there are some interesting asymptotic results. Wormald and McKay [14] conjectured that asymptotically

$$|\mathcal{G}_{n,d(n)}| \sim \sqrt{2} \binom{n-1}{d(n)}^n (\lambda^\lambda (1-\lambda)^{(1-\lambda)})^{\binom{n}{2}} e^{\frac{1}{4}},$$

where $\lambda = \frac{d(n)}{n-1}$ and $d(n)$ satisfying $1 \leq d(n) \leq n-2$ with $nd(n)$ always an even integer. A complete proof of this conjecture was only presented in 2017 [15].

For the asymptotic number of regular graphs up to isomorphism, the study of the automorphism groups of regular graphs is necessary. We refer the interested reader to the work by Wormald and McKay [16] for more details.

While no closed formula is available, algorithms designed to generate all d -regular graphs on n vertices have determined the number of such graphs, up to isomorphism, for small values of n . In the following, we use the algorithm *GENREG*, proposed by Meringer in [13]. Refer to Table 1 for the total number of connected d -regular graphs on n vertices.

Table 1 Number of connected d -regular graphs with given number of vertices n , up to isomorphism; empty fields indicate values that are not yet known [13]

d	3	4	5	6	7	8
n						
4	1	0	0	0	0	0
5	0	1	0	0	0	0
6	2	1	1	0	0	0
7	0	2	0	1	0	0
8	5	6	3	1	1	0
9	0	16	0	4	0	1
10	19	59	60	21	5	1
11	0	265	0	266	0	6
12	85	1,544	7,848	7,849	1,547	94
13	0	10,778	0	367,860	0	10,786
14	509	88,168	3,458,383	21,609,300	21,609,301	3,459,386
15	0	805,491	0	1,470,293,675	0	1,470,293,676
16	4060	8,037,418	2,585,136,675	113,314,233,808	733,351,105,934	733,351,105,935
17	0	86,221,634	0		0	
18	41301	985,870,522				
19	0	11,946,487,647	0		0	
20	510,489					

Given the rapid growth in the number of regular graphs with specified degree and vertex count, an efficient implementation for the computation of the Lin–Lu–Yau curvature is necessary to quantify the number of regular graphs G satisfying $Ric(G) > 0$. Current implementations do not provide sufficient efficiency for this task at scale.

Building on the precise formula derived in Theorem 1.1, we provide a new, efficient implementation for the computation of the Lin–Lu–Yau curvature on regular graphs. The source code is available in a public repository.¹ See Appendix A for a comparative analysis between our implementation and the existing implementations by Ni et al. [17], and by Cushing et al. [20]. On smaller graphs ($n \leq 20$) our implementation achieves a speedup of 20–50 times, depending on the exact values of d and n .

Using our efficient implementation for the computation of the Lin–Lu–Yau curvature, we are able to calculate the number of d -regular graphs G on n vertices that satisfy $Ric(G) > 0$ for $n < 15$. Due to the rapid growth in the number of regular graphs, it becomes computationally infeasible to determine the exact number of regular graphs with positive Lin–Lu–Yau curvature for larger values of n . Instead, we approximate this by randomly generating one million d -regular graphs on n vertices and calculating their Lin–Lu–Yau curvature. For the generation of the random regular graphs, we use the algorithm proposed by Steger and Wormald in [18], generating the regular graphs approximately uniformly at random.

The results of our analysis are shown in Table 2. If only a percentage (and not an exact count) is provided, the value was approximated by random sampling.

One notable observation is that regular graphs with relatively large degree always satisfy $Ric(G) > 0$. This observation will be discussed in the following section.

Additionally, observe that for a fixed degree d , there exists a threshold $\mathcal{M}_d$, such that no d -regular graph $G = (V, E)$ satisfies $|V| > \mathcal{M}_d$ and $Ric(G) > 0$. An investigation of this is conducted in Sect. 6.

5 Condition for positive Lin–Lu–Yau curvature

In this section, we present our main result: a condition on the degree of a regular graph G , ensuring that G satisfies $Ric(G) > 0$. We then discuss the sharpness of this lower bound.

Theorem 1.2 *Let $G = (V, E)$ be a finite graph. If G is d -regular with*

$$d > \frac{2|V|}{3} - 2,$$

then $Ric(G) > 0$.

To prove this, we need the following lemmata.

Lemma 5.1 *Let $G = (V, E)$ be a finite graph. If G is d -regular with*

¹ <https://github.com/moritzmath/Regular-Graph-Curvature>.

Table 2 Number of connected d -regular graphs on n vertices, up to isomorphism, that satisfy $\text{Ric}(G) > 0$, followed by the percentage of the total number of d -regular graphs on n vertices in parentheses

d	3	4	5	6	7	8
n						
4	1 (100%)	–	–	–	–	–
5	–	1 (100%)	–	–	–	–
6	2 (100%)	1 (100%)	1 (100%)	–	–	–
7	–	2 (100%)	–	1 (100%)	–	–
8	2 (40%)	6 (100%)	3 (100%)	1 (100%)	1 (100%)	–
9	–	15 (93.8%)	–	4 (100%)	–	1 (100%)
10	1 (5.3%)	32 (54%)	60 (100%)	21 (100%)	5 (100%)	1 (100%)
11	–	68 (25.7%)	–	266 (100%)	–	6 (100%)
12	0 (0%)	44 (2.9%)	6569 (83.7%)	7,849 (100%)	1,547 (100%)	94 (100%)
13	–	23 (0.2%)	–	366,804 (99.7%)	–	10,786 (100%)
14	0 (0%)	9 (0.01%)	719,285 (20.8%)	20,998,083 (97.2%)	21,609,251 (99.9%)	3,459,385 (100%)
15	–	12 (0.0015%)	–	$\approx 86.5\%$	–	$\approx 100\%$
16	0 (0%)	4 (0.00005%)	$\approx 0.23\%$	$\approx 62.6\%$	$\approx 99.7\%$	$\approx 99.9\%$
17	–	3 (0.000004%)	–	$\approx 31.8\%$	–	$\approx 99.9\%$

If no exact count is provided, the values are estimated by random sampling; a dash indicates that no d -regular graph on n vertices exists

$$d \geq \frac{|V| - 1}{2},$$

then $\text{diam}(G) \leq 2$.

Proof Assume that $\text{diam}(G) > 2$. Hence, there exist vertices $x, y \in V$ such that $d(x, y) > 2$, and therefore $S_1(x) \cap S_1(y) = \emptyset$. As $x, y \notin S_1(x) \cup S_1(y)$, we have

$$|S_1(x) \cup S_1(y)| \leq |V| - 2.$$

This contradicts

$$|S_1(x) \cup S_1(y)| = 2d \geq |V| - 1.$$

Therefore, our assumption was incorrect, and it follows that $\text{diam}(G) \leq 2$ must hold. $\square$

Lemma 5.2 *Let $G = (V, E)$ be a finite graph. Assume that G is d -regular with*

$$d > \frac{2|V|}{3} - 2.$$

If an edge $x \sim y$ satisfies $|\Delta(x, y)| > 2d - |V|$, then $\kappa(x, y) > 0$.

Proof Using the integrality of d together with Lemma 5.1, we conclude that $\text{diam}(G) \leq 2$. If $|\Delta(x, y)| \geq 2d - |V| + 2$, we obtain

$$\begin{aligned} \kappa(x, y) &\geq \frac{1}{d} \left(d + 1 - 2|R_x(x, y)| \right) \\ &= \frac{1}{d} \left(d + 1 - 2(d - 1 - |\Delta(x, y)|) \right) \\ &\geq \frac{1}{d} \left(3d + 7 - 2|V| \right). \end{aligned}$$

Utilizing the lower bound on d , we conclude that $\kappa(x, y) > 0$.

Now, assume $|\Delta(x, y)| = 2d - |V| + 1$ and $\kappa(x, y) \leq 0$. As $\text{diam}(G) \leq 2$, we have

$$\kappa(x, y) \geq \frac{1}{d} \left(d + 1 - 2|R_x(x, y)| \right),$$

and therefore we obtain

$$d + 1 \leq 2|R_x(x, y)| = |R_x(x, y)| + |R_y(x, y)|. \quad (5.1)$$

Let $j \in \Delta(x, y)$ be arbitrary. According to inequality 5.1, j cannot be adjacent to every vertex in $R_x(x, y) \cup R_y(x, y)$. Assume, without loss of generality, that there

exists an $i \in R_x(x, y)$ such that $i \approx j$. Thus, i has at most $d - 3$ neighbors in $S_1(x)$, as it is not adjacent to y , j and itself. Now, observe that

$$|V \setminus (S_1(x) \cup S_1(y))| = |V| - 2d + |\Delta(x, y)| = 1$$

by assumption. Therefore, there exist a $k \in R_y(x, y)$ such that $i \sim k$. Hence, an optimal assignment $\phi \in \mathcal{O}_{xy}$ must satisfy $|\{z \in R_x(x, y) : d(z, \phi(z)) = 1\}| \geq 1$, leading to

$$\begin{aligned} \kappa(x, y) &= \frac{1}{d} \left(d + 1 - \inf_{\phi \in \mathcal{A}_{xy}} \sum_{z \in R_x(x, y)} d(z, \phi(z)) \right) \\ &\geq \frac{1}{d} \left(d + 1 - 2(|R_x(x, y)| - 1) - 1 \right) \\ &= \frac{1}{d} \left(3d + 6 - 2|V| \right). \end{aligned}$$

Using the lower bound on d , we conclude that $\kappa(x, y) > 0$, contradicting our assumption. $\square$

Lemma 5.3 *Let $G = (V, E)$ be a finite graph. Assume that G is d -regular with*

$$d > \frac{2|V|}{3} - 2.$$

If an edge $x \sim y$ satisfies $|\Delta(x, y)| = 2d - |V|$, then $\kappa(x, y) > 0$.

Proof Using the integrality of d together with Lemma 5.1, we conclude that $\text{diam}(G) \leq 2$. If $R_x(x, y) = \emptyset$, then $\kappa(x, y) = \frac{d+1}{d} > 0$. Thus, we now assume $R_x(x, y) \neq \emptyset$.

Let $i \in R_x(x, y)$ be arbitrary. Since $i \approx y$, it is adjacent to at most $d - 2$ vertices in $S_1(x)$. By assumption, we have $V = S_1(x) \cup S_1(y)$, so there must exist a vertex $j \in R_y(x, y)$ such that $i \sim j$. Similarly, every vertex in $R_y(x, y)$ is adjacent to a vertex in $R_x(x, y)$.

Now, denote by H the bipartite subgraph with vertex set $R_x(x, y) \cup R_y(x, y)$ and containing all edges connecting a vertex in $R_x(x, y)$ with a vertex in $R_y(x, y)$. According to the preceding argument, every vertex in H is of degree greater than or equal to one in H . Denote by $\Delta(H)$ the maximal degree of H . We proceed with a proof by cases.

Case 1: $\Delta(H) = 1$. The induced bipartite subgraph is 1-regular. According to Corollary 2.3, there exists a perfect matching of $R_x(x, y)$ into $R_y(x, y)$. Hence, we obtain

$$\kappa(x, y) = \frac{1}{d} \left(d + 1 - \inf_{\phi \in \mathcal{A}_{xy}} \sum_{z \in R_x(x, y)} d(z, \phi(z)) \right)$$

$$= \frac{1}{d} \left(d + 1 - |R_x(x, y)| \right) > 0.$$

Case 2: $1 < \Delta(H) \leq |R_x(x, y)| - 2$. Let $z_1 \in R_x(x, y)$ be, without loss of generality, a vertex of degree $\Delta(H)$ in H . There exist distinct vertices $i, j \in R_y(x, y)$ such that z_1 is not adjacent to both i and j . Choose $z_2 \in R_x(x, y)$ such that $z_2 \sim i$. If the degree of z_2 in H is greater than one, choose $z_3 \in R_x(x, y) \setminus \{z_1, z_2\}$ arbitrary. otherwise, choose $z_3 \in R_x(x, y)$ such that $z_3 \sim j$. By construction, every $X \subseteq \{z_1, z_2, z_3\}$ satisfies $|N(X)| \geq |X|$ in H . Hall's marriage theorem implies that an optimal assignment $\phi \in \mathcal{O}_{xy}$ must satisfy $|\{z \in R_x(x, y) : d(z, \phi(z)) = 1\}| \geq 3$, leading to

$$\begin{aligned} \kappa(x, y) &= \frac{1}{d} \left(d + 1 - \inf_{\phi \in \mathcal{A}_{xy}} \sum_{z \in R_x(x, y)} d(z, \phi(z)) \right) \\ &\geq \frac{1}{d} \left(d + 1 - 2(|R_x(x, y)| - 3) - 3 \right) \\ &= \frac{1}{d} (3d + 6 - 2|V|). \end{aligned}$$

Using the lower bound on d , we conclude that $\kappa(x, y) > 0$.

Case 3: $\Delta(H) \geq |R_y(x, y)| - 1$. Let $z_1 \in R_x(x, y)$ be, without loss of generality, a vertex of degree $\Delta(H)$ in H . Recall that $\text{diam}(G) \leq 2$ and that z_1 can be matched with a vertex at distance one in $R_y(x, y)$, since the degree of z_1 is at least one in H . Therefore, we obtain

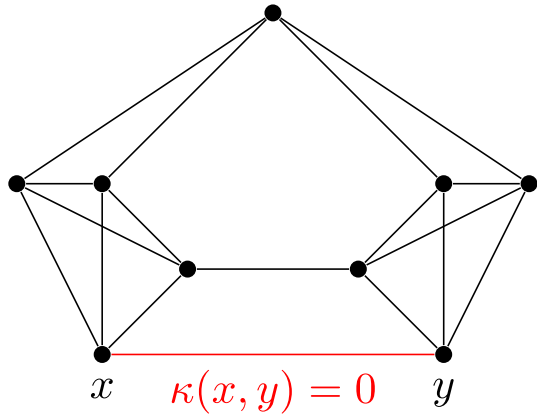
$$\kappa(x, y) \geq \frac{1}{d} (d + 2 - 2|R_x(x, y)|).$$

If $|R_x(x, y)| + |R_y(x, y)| \leq d + 1$, we conclude that $\kappa(x, y) > 0$. Thus, we now assume

$$d < |R_x(x, y)| + |R_y(x, y)| - 1. \quad (5.2)$$

Note that this implies $|R_x(x, y)| \geq 3$, since $d \geq |R_x(x, y)| + 1$. Since z_1 is adjacent to x and to at least $|R_y(x, y)| - 1$ vertices in $R_y(x, y)$, it cannot be adjacent to every vertex in $R_x(x, y)$ according to inequality 5.2. Thus, we can choose $z_2 \in R_x(x, y)$ such that $z_1 \not\sim z_2$ in G . Hence, z_2 is adjacent to at most $d - 3$ vertices in $S_1(x)$. Recall that, by assumption, $V = S_1(x) \cup S_1(y)$, implying that the degree of z_2 in H is greater than or equal to two. Choose $z_3 \in R_x(x, y) \setminus \{z_1, z_2\}$ arbitrary. Then, by construction, every $X \subseteq \{z_1, z_2, z_3\}$ satisfies $|N(X)| \geq |X|$ in H . Hall's marriage theorem implies that an optimal assignment $\phi \in \mathcal{O}_{xy}$ between $R_x(x, y)$ and $R_y(x, y)$ satisfies $|\{z \in R_x(x, y) : d(z, \phi(z)) = 1\}| \geq 3$, leading to $\kappa(x, y) > 0$, as in *Case 2*. $\square$

Fig. 2 Illustration of a 4-regular graph G on 9 vertices that does not satisfy $\text{Ric}(G) > 0$



Proof of Theorem 1.2 Since $S_1(x) \cup S_1(y) \subseteq V$, the inequality

$$2d - |V| \leq |\Delta(x, y)|$$

always holds true. If $2d - |V| < |\Delta(x, y)|$, the claim follows from Lemma 5.2. If $2d - |V| = |\Delta(x, y)|$, the claim follows from Lemma 5.3. This concludes the proof of Theorem 1.2. $\square$

Remark 5.4 For $d = \frac{2|V|}{3} - 2$, it is possible for edges $x \sim y$ with $\kappa(x, y) \leq 0$ to occur. An example is illustrated in Fig. 2.

We can generalize Theorem 1.2 to the 0-Ollivier–Ricci-curvature, using the following result by Cushing et al.

Theorem 5.5 [19, Theorem 1.1]. *Let $G = (V, E)$ be a regular graph. Then we have the following implication for all edges $x \sim y$:*

$$\kappa(x, y) > 0 \implies \kappa_0(x, y) \geq 0.$$

Hence, we obtain the following result for the Ollivier–Ricci curvature.

Theorem 1.3 *Let $G = (V, E)$ be a finite graph. If G is d -regular with*

$$d > \frac{2|V|}{3} - 2,$$

then $\kappa_0(x, y) \geq 0$ for every edge $x \sim y$.

Next, we discuss the asymptotic sharpness of the lower bound.

Theorem 1.4 *Let $n \in \mathbb{N}$ such that*

$$d = \frac{2n - 8}{3} \in 2\mathbb{N}.$$

and $d \geq 12$. Then there exists a d -regular graph on n vertices such that $\kappa(x, y) = 0$ and $\kappa_0(x, y) < 0$ for an edge $x \sim y$.

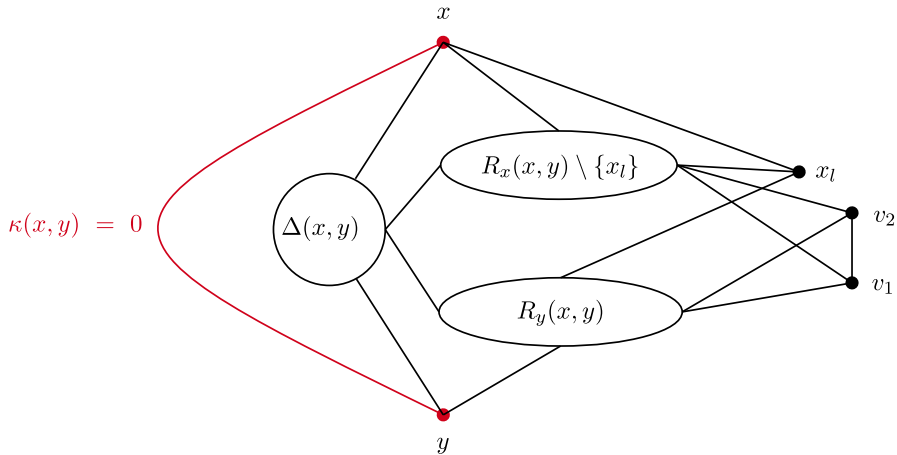


Fig. 3 Illustration of the graph constructed in Theorem 1.4

Proof Let the assumptions of Theorem 1.4 hold. We construct a d -regular graph on n vertices that contains an edge $x \sim y$ such that $\kappa(x, y) = 0$. To this end, define the vertex set

$$V = \{x, y, z_0, \dots, z_{l-3}, x_0, \dots, x_l, y_0, \dots, y_l, v_1, v_2\},$$

where $l = \frac{d}{2}$. Next, we add the edges

- $x \sim y$,
- $x \sim z_i$ for $i = 0, \dots, l-3$,
- $x \sim x_i$ for $i = 0, \dots, l$,
- $y \sim z_i$ for $i = 0, \dots, l-3$,
- $y \sim y_i$ for $i = 0, \dots, l$.

to the set of edges E . Therefore, x and y are of degree d , and

$$\Delta(x, y) = \{z_0, \dots, z_{l-3}\}, \quad R_x(x, y) = \{x_0, \dots, x_l\}, \quad R_y(x, y) = \{y_0, \dots, y_l\}.$$

Next, we add for every $z_i \in \Delta(x, y)$ the edges $z_i \sim x_j$ for $j = 0, \dots, l-1$ if $j \neq i$ as well as the edges $z_i \sim y_j$ for $j = 0, \dots, l$ if

$$j \notin \{2i \pmod{l+1}, 2i+1 \pmod{l+1}\}.$$

Thus, every $z_j \in \Delta(x, y)$ is of degree d , as it has $d-2$ neighbors in $R_x(x, y) \cup R_y(x, y)$ and is adjacent to both x and y .

On the other hand, every x_i is adjacent to $|\Delta(x, y)| - 1$ vertices in $\Delta(x, y)$ if $i \leq l-3$, and to all vertices in $\Delta(x, y)$ if $i \in \{l-2, l-1\}$. Every y_i is adjacent to $|\Delta(x, y)| - 2$ vertices in $\Delta(x, y)$ if $i \leq l-6$ and to $|\Delta(x, y)| - 1$ otherwise.

Next, we add the edges $x_i \sim x_j$ for $i \neq j$, except $x_{l-2} \sim x_{l-1}$. Analogously, we add the edges $y_i \sim y_j$ for $i \neq j$, as well as $x_l \sim y_j$ for $j = 0, \dots, l-2$.

Thus, every $x_i \in R_x(x, y) \setminus \{x_l\}$ has degree $d-2$ and x_l has degree d . On the other hand, y_i has degree $d-1$ if $i \in \{l-5, l-4, l-3, l-2\}$ and $d-2$ otherwise.

Finally, we add the edges

- $v_1 \sim x_i$ and $v_2 \sim x_i$ for $i = 0, \dots, l - 1$.
- $v_1 \sim y_i$ and $v_2 \sim y_i$ for $i = 0, \dots, l - 6$ and $i = l - 1, l$.
- $v_1 \sim y_{l-5}, v_1 \sim y_{l-4}, v_2 \sim y_{l-3}, v_2 \sim y_{l-2}$ and $v_1 \sim v_2$.

The constructed graph $G = (V, E)$ is d -regular. Now, denote by H the bipartite subgraph with vertex set $R_x(x, y) \cup R_y(x, y)$ and containing all edges connecting a vertex in $R_x(x, y)$ with a vertex in $R_y(x, y)$. By construction, every x_j for $0 \leq j \leq l - 1$ is of degree zero in H . Therefore, according to Hall's marriage theorem, an optimal assignment $\phi \in \mathcal{O}_{xy}$ must satisfy $|\{z \in R_x(x, y) : d(z, \phi(z)) = 1\}| = 1$. Thus, we obtain

$$\begin{aligned} \kappa(x, y) &= \frac{1}{d} \left(d + 1 - \inf_{\phi \in \mathcal{A}_{xy}} \sum_{z \in R_x(x, y)} d(z, \phi(z)) \right) \\ &= \frac{1}{d} \left(d + 1 - 2(|R_x(x, y)| - 1) - 1 \right) \\ &= 0. \end{aligned}$$

Furthermore, every vertex in $R_x(x, y)$ can be assigned to x at distance one and y can be assigned to every vertex in $R_y(x, y)$ at distance one. Therefore, we obtain

$$\begin{aligned} \kappa_0(x, y) &= \frac{1}{d} \left(d - \inf_{\phi} \sum_{z \in S_1(x) \setminus S_1(y)} d(z, \phi(z)) \right) \\ &= \frac{1}{d} \left(d - 2(|S_1(x) \setminus S_1(y)| - 3) - 3 \right) \\ &= -\frac{1}{d}. \end{aligned}$$

□

6 Maximal regular graphs with positive Ricci curvature

We finish this article with an investigation of the maximal number of vertices that an d -regular graph G satisfying $\text{Ric}(G) > 0$ can have,

$$\mathcal{M}_d = \max_{\substack{G \text{ } d\text{-regular,} \\ \text{Ric}(G) > 0}} |V(G)|.$$

The following Proposition shows that $\mathcal{M}_d$ is always finite.

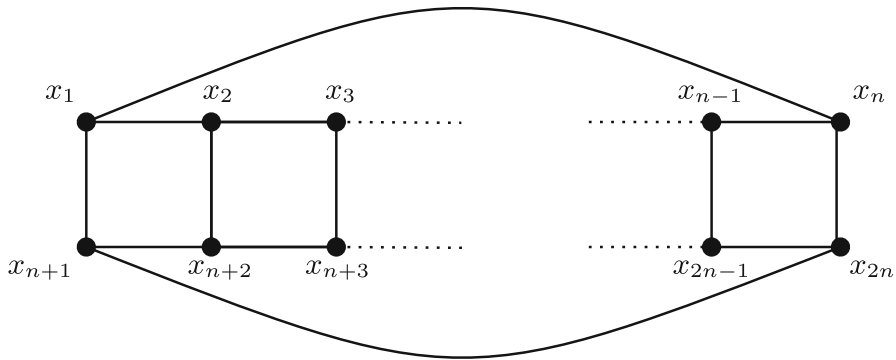


Fig. 4 The prism graph Y_n

Proposition 6.1 *Let $d \in \mathbb{N}$. Then*

$$\mathcal{M}_d \leq 1 + d \sum_{i=0}^{2d-1} (d-1)^i.$$

For the proof of Proposition 6.1, we make use of the following Bonnet–Myers type theorem on graphs by Lin, Lu, and Yau.

Theorem 6.2 (Discrete Bonnet–Myers Theorem [7]). *Let $G = (V, E)$ be a locally finite graph. If $\text{Ric}(G) \geq k > 0$, then the diameter of graph G is bounded as follows:*

$$\text{diam}(G) \leq \frac{2}{k}.$$

Proof of Proposition 6.1 Let G be an arbitrary d -regular graph, satisfying $\text{Ric}(G) > 0$. According to Corollary 3.6, we have $\kappa(x, y) \in \mathbb{Z}/d$ for every edge $x \sim y$. Therefore, $\text{Ric}(G) \geq \frac{1}{d}$ must hold. Using Theorem 6.2, we obtain

$$\text{diam}(G) \leq 2d.$$

The proposition is now an immediate consequence of the Moore bound. $\square$

The following theorem quantifies $\mathcal{M}_d$ for d equal to two and three.

Theorem 1.5 *Let $G = (V, E)$ be a d -regular graph, satisfying $\text{Ric}(G) > 0$. If $d = 2$, then $|V| \leq 5$. If $d = 3$, then $|V| \leq 10$. Consequently, we have $\mathcal{M}_2 = 5$ and $\mathcal{M}_3 = 10$.*

For the proof of Theorem 1.5, we need the following classification result by Cushing et al. [20].

Theorem 6.3 *Let $G = (V, E)$ be a 3-regular graph. Then $\kappa_0(x, y) \geq 0$ for all edges $x \sim y$ if and only if G is a prism graph Y_n for some $n \geq 3$ or a Möbius ladder M_n for some $n \geq 2$.*

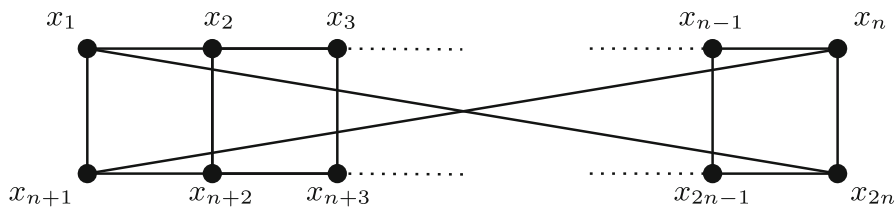


Fig. 5 The Möbius Ladder M_n

Figure 4 illustrates the prism graph Y_n , while Fig. 5 provides a visualization of the Möbius ladder M_n . With this classification result established, we are now prepared to prove Theorem 1.5.

Proof of Theorem 1.5 The only 2-regular, connected graphs are the cycle graphs C_n . It is straightforward to verify that $Ric(C_n) = 0$ for $n \geq 6$. Hence, if $G = (V, E)$ is 2-regular satisfying $Ric(G) > 0$, then $G = C_n$ for some $2 \leq n \leq 5$ and therefore $|V| \leq 5$ holds true.

We now turn to the case where $d = 3$. Let $x \sim y$ be an arbitrary edge in G . By assumption, we have $\kappa(x, y) > 0$. Since G is regular, it follows from Theorem 5.5 that $\kappa_0(x, y) \geq 0$. Thus, according to Theorem 6.3, G is a prism graph Y_n for some $n \geq 3$ or a Möbius ladder M_n for some $n \geq 2$. The prism graph Y_n does not satisfy $Ric(Y_n) > 0$ for $n \geq 6$ and the Möbius ladder M_n does not satisfy $Ric(M_n) > 0$ for $n \geq 5$. Therefore, G has at most 10 vertices. $\square$

The values $\mathcal{M}_d$ for $d \geq 4$ are still unknown. The Cartesian product of n 5-cycles $(C_5)^{\square n}$ is $2n$ -regular, has 5^n vertices and satisfies

$$Ric((C_5)^{\square n}) = \frac{1}{2n} > 0.$$

This is an immediate consequence of $Ric(C_5) = \frac{1}{2}$ and Corollary 3.2 in [7]. Thus, $\mathcal{M}_d$ exhibits exponential growth, satisfying

$$\mathcal{M}_d \geq \sqrt{5}^d$$

for d even.

Recall that $\kappa(x, y) \in \mathbb{Z}/d$ for every edge $x \sim y$ of an d -regular graph. Hence $(C_5)^{\square n}$ exhibits constant, minimal positive Lin–Lu–Yau curvature for an $2n$ -regular graph. This led us to formulate the following conjecture.

Conjecture 6.4 For $d \in 2\mathbb{N}$,

$$\mathcal{M}_d = \sqrt{5}^d.$$

Moreover, for odd d , $\mathcal{M}_d$ also exhibits exponential growth, as e.g. the n -dimensional hypercube $\mathcal{Q}_n$ for odd n shows.

Appendix

A Comparative analysis of implementations

In this section, we present a comparative analysis of our implementation for computing the Lin–Lu–Yau curvature against existing implementations. We compare our program with those by Ni et al., presented in [17], and by Cushing et al., presented in [20]. For this comparison, we generate random d -regular graphs on n vertices, employing the algorithm proposed by Steger and Wormald in [18]. We then calculate the Lin–Lu–Yau curvature of these graphs using the different implementations. To ensure robustness, computation times are averaged over 100 graphs for each combination of parameters n and d . The computations are performed on an Apple M2 chip (8-core CPU at up to 3.5 GHz) with 16 GB RAM. Due to the limited efficiency of the implementation by Cushing et al., we exclude it from the analysis on larger graphs, as the computation time is prohibitively high. For smaller graphs ($n \leq 20$) our implementation achieves a speedup of 20–50 times over the other implementations. The results of our comparative analysis are presented in Fig. 6.

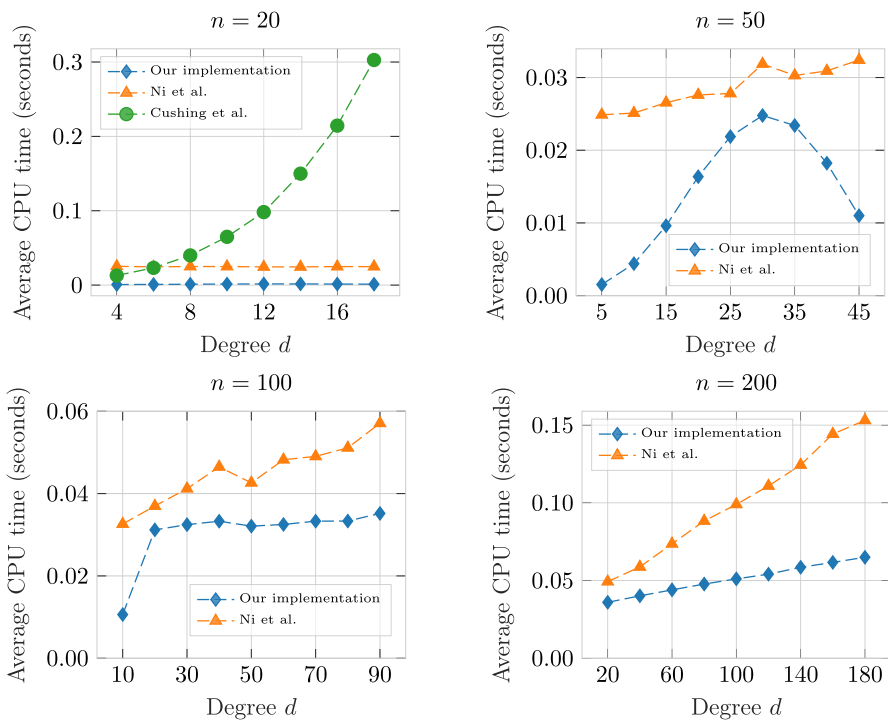


Fig. 6 A comparative analysis of the different implementations for computing the Lin–Lu–Yau curvature on regular graphs. We display the average CPU time of each implementation for computing the Lin–Lu–Yau curvature of an d -regular graph on n vertices

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Data Availability The source code for the implementation presented in Sect. 4 is publicly available at <https://github.com/moritzmath/Regular-Graph-Curvature>.

Declarations

Conflict of interest The author certifies that he has no affiliations with or involvement in any organization or entity with any financial interest or non-financial interest in the subject matter or materials discussed in this manuscript.

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Assembly Theory: What It Does and What It Does Not Do

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Abstract

A recent publication in *Nature* has generated much heated discussion about evolution, its tendency towards increasing diversity and complexity, and its potential status above and beyond the known laws of fundamental physics. The argument at the heart of this controversy concerns assembly theory, a method to detect and quantify the influence of higher-level emergent causal constraints in computational worlds made of basic objects and their combinations. In this short essay, I briefly review the theory, its basic principles and potential applications. I then go on to critically examine its authors' assertions, concluding that assembly theory has merit but is not nearly as novel or revolutionary as claimed. It certainly does not provide any new explanation of biological evolution or natural selection, or a new grounding of biology in physics. In this regard, the presentation of the paper is starkly distorted by hype, which may explain some of the outrage it created.

Keywords Assembly theory · Complexification · Evolutionary innovation · Natural selection · Darwinian evolution

A commotion

There has been a lot of agitation over the past few weeks about a paper that recently appeared in *Nature* (Sharma et al. 2023; see also the news & views feature by Ellis 2023). Many of my fellow evolutionary biologists felt compelled to express their disgust and outrage on social media. Sadly, there is much shouting but little serious argumentation on display.¹ Evolutionary biology is being grossly misrepresented, they say. Even worse: the perpetrators are chemists and physicists who are claiming that there is an explanatory gap between physics and evolution. Accusations of “non-sense,” “mumbo jumbo,” “word salad,” creationist intent, and (perhaps most surprising of all) injecting wokeness into evolutionary theory soon started flying from all kinds of directions.

But what is all the fuss about? It's about something called *assembly theory*, which was created by Sara Walker, Lee Cronin, and colleagues. It has been around for a few years (Marshall et al. 2017, 2021, 2022; Liu et al. 2021), and has ruffled some feathers previously within the communities of combinatorial chemistry² and complexity science³ (Uthamacumaran et al. 2022). So what exactly is going on?

To get straight to the point: *I find assembly theory intriguing and worth considering*. Originally developed to detect signatures of alien life in the atmosphere of faraway exoplanets⁴ (Marshall et al. 2017, 2021), it is a neat and simple model for the combinatorial generation of innovation in rule-based abstract “worlds” of recombining objects. Arguably, it is a prime example of classic computational

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¹ The replies to this post on X/Twitter <https://twitter.com/Nature/status/1709800557257306572?s=20>, present a fairly representative sample of the kinds of reactions the paper generated online.

² See this thread on X/Twitter, for example: <https://x.com/mitomaths/status/1711320060197736918?s=20>.

³ A blog post by Hector Zenil (<https://hectorzenil.medium.com/test-8f0be54817c4>) identifies no less than eight fallacies of assembly theory. Scroll to the section “Original Post” for the actual beginning of the article. A video essay by the same author (<https://www.youtube.com/watch?v=078EXZeS8Y0>) summarizes these fallacies, and highlights conceptual/methodological limitations, and the pervasive failure by the proponents of assembly theory to acknowledge relevant previous work in the field of complexity science.

⁴ The original aim was to classify molecules present in an exoplanet atmosphere into those that are likely to be of organic and biological origin. See also Zenil et al (2018).

complexity theory. Applied to the natural world, the metrics it proposes may allow us to detect whether levels of organization above the basic laws of physics have emerged in an observed system, and to estimate the causal influence of these higher levels on the underlying dynamics. As an added bonus, assembly theory manages all this without having to assume anything specific about what those higher levels of organization actually *are*. To me, this sounds interesting and potentially useful. We should not dismiss the work outright and, despite our skepticism, give the theory a fair chance to prove its worth.

On the downside, there are several serious problems that stem from the framing and presentation of the argument in the paper, the hype the authors have generated around it (online and in peer-reviewed print), and their problematic and overextended interpretation of the model.

The first major issue is that the paper is entitled “*Assembly theory explains and quantifies selection and evolution*” (Sharma et al. 2023) when it does absolutely no such thing. In fact, assembly theory is not specifically about Darwinian evolution by natural selection and, besides, the paper uses the term “selection” in a very broad sense that no biologist would recognize. This leads to unnecessary confusion, as is evident from the angry online comments. To make things worse, the abstract is a textbook example of how *not* to write an abstract. It is phrased in vague and hyperbolic terms that are misleading. We will come back to that later.

At this point, let me make one point very clear: it is not fair or proper to dismiss a paper based on its abstract alone, which is exactly what a surprising number of online commenters did who obviously (and sometimes admittedly) did not bother to read the whole paper.

That’s why I *did* read the paper carefully (plus some of its predecessors and criticisms as well). This helped me clarify what assembly theory is good for, and what its limitations are. Let me share these insights with you, and then quickly review the entire controversy within the context of contemporary scientific publishing and career structures.

Some Assembly Required

The fundamental idea behind constructing an assembly theory model is that you define an “*assembly universe*,” which consists of a finite set of distinct basic building blocks (colored bricks in Fig. 1) and another finite set of rules that allow you to assemble them into more complex composite objects. This universe, in principle, can contain all possible combinations of building blocks that do not violate your basic assembly rules. If you want to simulate real-world chemistry, for example, you can derive all kinds of composite molecules by recombining atoms with their corresponding chemical bonds.



Fig. 1 Building blocks of an assembly universe

So far so good. Now you introduce a dimension of time to the model, which is implemented by *recursivity*. In other words, at each step of the assembly process, you can use all objects that are already assembled (not just the basic building blocks you started with) for further assembly. Thus, at each step, you get a bigger choice of objects to build with. In fact, the number of possible rule-based combinations will increase hyper-exponentially with each step. This way, you can see what kinds of composites emerge over time. An example of such a process is depicted in Fig. 2.

Recursivity makes the dynamics of the model *historically contingent*. In the end, the kinds of objects that you actually *can* assemble are not only restricted by the rules of your universe, but also by the starting point and trajectory you actually chose to take. This renders the whole model computationally tractable, because it greatly reduces the combinations of possible composites you can get. You don’t have to deal with all possible combinations of building blocks, just the ones that are actually present in whatever “world” (a mix or “ensemble” of different objects) that you are simulating or observing. For instance, you can start with a given set of substrates, and simulate what kind of products you can get from there, given all possible chemical reactions plus the relative abundances of the substrates.

You can consider such a system a kind of a null model: given your basic building blocks and the assembly rules you defined, you can calculate how many steps you need to take before a specific composite object can arise. That minimal path to construct a composite object is called its *assembly index*. If your rules have specific weights or probabilities of application, you can also predict the expected abundance (copy number) for any composite object in your mix. This may not be easy to do in practice, but the authors at least show that it is possible in principle for any well-defined world with finite sets of building blocks and rules.

Fig. 2 An assembly process (adapted from Sharma et al. 2023)

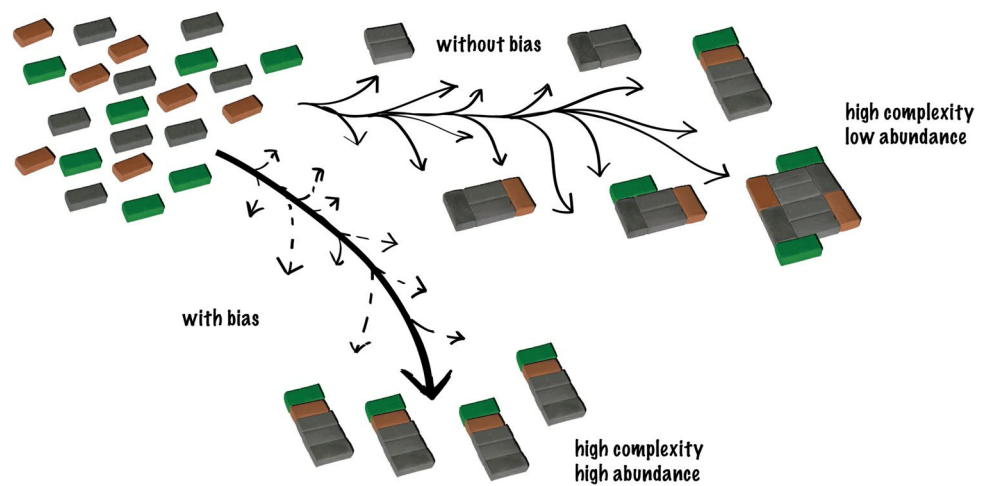
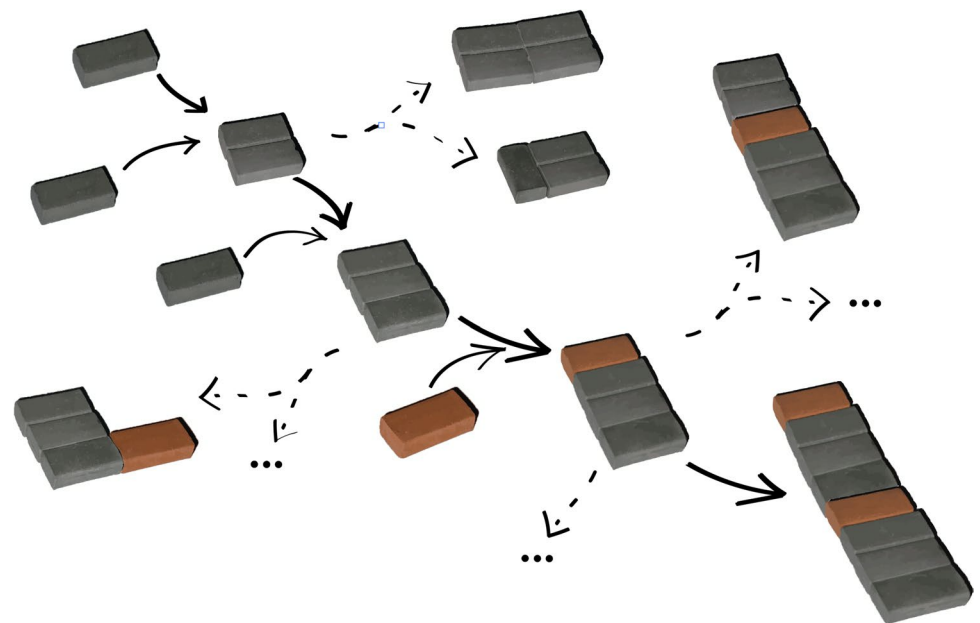


Fig. 3 “Selection” in assembly theory means bias in outcome (adapted from Sharma et al. 2023)



By the way, these metrics are not particularly novel. They are special cases of classical measures of *algorithmic* or *computational complexity* (especially Huffman 1952; Bennett 1988), which are used in a peculiar combination here, tailored to the context of assembly theory. We will come back to this point later.

Things become truly interesting once your model produces very complex composites. The higher the complexity of a composite, the longer it will take to appear, and the more unlikely it is to appear by chance, especially after just a minimal number of steps. Based on this, you would generally expect many different complex composites to be present at very low abundance at later steps. Yet, *if you find certain complex composites enriched, especially early on, that’s a sign that things are not just based on the random interplay of the basic rules in your system.* Figure 3 illustrates the point.

Put simply: finding composites with high complexity at high abundance means the basic rules of your “world” have probably been skewed in some way that is not built into the basic rules. That’s what the authors mean by “*selection*” (Fig. 3). This concept is, of course, a whole lot broader and more generic than what evolutionary biologists mean by the term. And the bias that assembly theory calls “*selection*” could be caused by processes that are very different from Darwinian evolution (many of them not selective at all). We’ll come back to that shortly. For now, let me reiterate that “*selection*” in the context of assembly theory simply means that the basic rules of the world you have defined have been constrained or channeled in some way to lead to an unexpected outcome.

This conception of bias is not mysterious at all. It simply means that *you can use assembly theory to check whether*

something unexpected is going on in a very broad range of computational model “worlds” or “universes” defined by different building blocks and rules. If that “something” is present, then more than just your basic rules must be at work. As a practical example, the authors suggest that assembly theory can be used to analyze mass spectrographs of exoplanet atmospheres in search for complex molecules at high abundance, which would mean something more than just the laws of chemistry are at work there (Marshall et al. 2017, 2021). Likewise, you could monitor the complexity and abundance of some technology, let’s say Schnitzel hammers⁵ in Austria, to infer that this technology must have been selected for in a particular environment. It did not just pop up randomly, but co-evolves with its substrate, the Viennese Schnitzel. So far, so good.

At this point it’s important to reiterate that *assembly theory does not (and need not) make any assumptions as to what is being “selected,” and in what way*. On the one hand, this is bad: it is one of the things that is confusing to evolutionary biologists complaining about the model. Assembly theory is not specifically about Darwinian evolution. It does not care about populations, individuals, genes, and so on. It does not fit into the conceptual framework of traditional evolutionary theory, that is for sure. And it almost certainly won’t help you to measure selective pressures in the wild. On the other hand, its lack of specificity is also a strength of assembly theory: it is very broadly applicable to all kinds of “worlds” and “selection” biases (if you can formulate an appropriate “world” and figure out how to measure the assembly index, which is far from trivial in most cases).

In other words, *assembly theory is a tool to detect the emergence of new levels of organization and their causal influence on lower-level phenomena in the world you are observing*. If the outcome you are detecting is biased, the underlying rules must have been constrained or channeled in some way to generate that bias. Philosophers call this “downward causation,” and keep on arguing about it. In particular, some reductionists find it objectionable, but for reasons that really don’t hold up to closer scrutiny. To explain why would take another paper-length argument. Suffice it to say that the problem goes away if you consider that processes and relations (*i.e.*, the rules that are affecting your objects) are fundamental, and not only the objects with their intrinsic properties themselves.

All you need to know about downward causation in this context is that it does not change the underlying rules. Instead, it constrains and channels the underlying processes in unexpected ways. And, if you think about it, that’s exactly

what evolution does with the laws of physics: natural selection never alters the rules of physics and chemistry underlying the processes that compose your body, but constrains and channels the direction of these processes in ways that you fundamentally cannot predict from the underlying physics or chemistry alone. This is why (evolutionary) biology is (and will always remain) an independent science after all. And in this sense, there undoubtedly *is* an explanatory gap between physics and biology, as the authors rightly point out.

It is in this very general and roundabout way that *assembly theory can indeed be useful for evolutionary biology*. It may allow us to establish, once and for all, that biology is more than just chemistry and physics. More specifically, it provides us with a potentially useful perspective on what may be driving innovation and complexification in evolution (and other higher-level processes) at a very fundamental level: it’s the emergence of ever new combinations of (chemical and higher-level cell-, tissue-level, or organismic) components. It’s *the constant growth of a co-evolving possibility space*, something which biologist Stuart Kauffman (2000) has called *the adjacent possible*. This should make us reconsider how we formally describe what is possible in evolution, and it adds a temporal directionality to the random effects of mutations: for a mutation to produce a complex outcome, more time will be required than for simple ones.

Arguably, *this is not exactly rocket science* and assembly theory oversimplifies the complexity of real-world evolutionary innovation to the point of potentially losing its connection with reality altogether. Yet, I would say it *is* a step forward compared to accounts of evolution that simply take the emergence of complexity for granted or assume a global preexisting space of possibilities for evolution. Such views are outdated and unhelpful, but still surprisingly common among reductionist evolutionary biologists and traditional complexity scientists, and that *is* a problem, in my opinion.

For this reason, *it would definitely not hurt if some of the people disparaging assembly theory as a matter of principle would profit from engaging with it in a more good-faith and open-minded manner*. A new perspective, even if controversial and not yet thoroughly worked out, can be inspiring and useful. Give it some slack! Personally, I find it enriching to engage with a different point of view every once in a while, because it allows me to see the advantages and limitations of my own approach more clearly. Judging from what is going on online or when I get my papers or grants reviewed, this is not a common attitude in our field. And that is a real shame, in my opinion. It clearly limits the ways we think about the incredible richness of evolutionary phenomena. And this, I firmly believe, is hampering conceptual progress in our field right now (Fig. 4).

⁵ Because of their locally restricted occurrence, the Wikipedia page for Schnitzel hammers only exists in German: <https://de.wikipedia.org/wiki/Fleischklopfen>.



Fig. 4 Assembly theory is surrounded by hype

A Flurry of Hype

Having said all this, it is time to turn to some of the issues surrounding the paper. The main problem is that it is wrapped in a misleading package, while much of the actual substance of the work is hidden in the technical appendices. Let me emphasize again that assembly theory is *not* a new theory of Darwinian evolution, nor is it a new interpretation of the concept of “selection” used in Darwinian theory. The title raises expectations the paper simply cannot meet. The authors do not even bother to explain how assembly theory could be applied in the context of genes, organisms, or populations. Granted, assembly theory *can* be applied to detect the signature of Darwinian evolution, but that requires all sorts of auxiliary hypotheses. For instance, the presence of specific chemical signatures in an exoplanet’s atmosphere only indicate the presence of alien life if we can confidently exclude potential sources for the detected bias other than biological evolution.

What are such alternative higher-level constraints? They can arise through self-organization, such as that observed in far-from-equilibrium (dissipative) systems, *e.g.*, hurricanes, eddies, and candle flames, which can form highly complex and improbable structures. It can also arise through the peculiar self-referential organization of living matter, which goes beyond mere self-organization in non-living systems (see Jaeger 2024a, 2024b, and references therein). Yet, deviations from expected outcomes can also occur for much more mundane reasons. For instance, kinetic energy barriers can heavily alter the rates of chemical reactions influencing what steady states will be reached. Furthermore, stochastic phenomena such as founder effects and random drift will play a role at early stages, especially in relatively small worlds. Or bias can be due to some hidden or neglected causal factor that was omitted from the basic rules when formulating the model.

Therefore, *how we interpret bias in outcomes largely depends on auxiliary hypotheses and the very assumptions we put into the model of our rule-based world in the first place.* This kind of circularity is not necessarily vicious, and is common in scientific modeling practice, but it clearly begs the question the paper is claiming to address. To say it again: assembly theory cannot tell us whether some bias is due to natural selection or not. It only tells us whether the bias is there, and how much of it is there, given the basic assumptions underlying the rule-based world we are modeling with assembly theory.

Of course, this also means that all the talk about *biological function* in the paper remains completely vacuous. If you cannot tell whether some bias in your outcome is the product of actual natural selection, you cannot tell whether it is functional in a sense of the term any biologist would recognize. As a matter of fact, *assembly theory does not deal with biological function at all.* So why do the authors bring it up?

Are these various blind spots of assembly theory known to the authors? I would strongly hope so. Otherwise, they would be rather fundamentally misinterpreting the nature and reach of their own model. I choose to give them the benefit of doubt, and will assume that they realize they are using the term “selection” in a broad and vague sense, including biases that are not selective at all. But if they realize this, how can they not see that their use of the term “selection” is grossly misleading?

Based on the evidence presented so far, I cannot help but conclude that we are dealing with a rather disingenuous presentation of what the paper actually achieves. *What assembly theory really does is to detect and quantify bias caused by higher-level constraints* in some well-defined rule-based worlds. That’s it! Even if Darwinian selection may contribute to such bias, assembly theory cannot tell you if it does, how much it does, or if other factors play a role as well.

Why this kind of packaging? It seems to do so much more harm than good. In fact, there is much food for thought hidden behind a thick smokescreen of impenetrable language and implausible claims. But it is probably exactly the glitzy packaging that got the paper into *Nature* in the first place. This highlights well-known weaknesses and failures in the editorial and peer-review processes of high-profile scientific journals. Exposure in high-profile journals still counts for far too much social capital in today's scientific career and funding market, and selling assembly theory for what it really is was evidently not sexy enough to get that kind of exposure.

All this reflects rather unfavorably both on the authors and the scientific publication system in general. Unfortunately, failures like this one abound in our field and beyond. What we need more of are substantive theoretical discussions, but they keep on getting overshadowed by false or shallow controversies, as well as self-serving public-relations campaigns without much intellectual depth or substance. It is high time we focus on what really matters again. Conceptual progress is at stake.

But let me finish on a more positive note: shortly after the publication of the paper discussed here, another publication came out in *PNAS*, which is similar in spirit, tackling the problem of increasing complexification in general evolutionary processes (Wong et al. 2023). Just like assembly theory, it takes a very broad picture of evolution and “selection,” going far beyond the biological realm of Darwinian evolution by natural selection. In contrast to the paper discussed here, it does not overstate its case, defines its terms precisely (especially when concepts are employed in unusual ways), and frames its discussion in terms of a specific notion of “function.” I have some criticisms regarding these authors' approach as well, but reading this paper will be a good use of your time. Please engage its ideas with an open mind! They are well worth considering, and may even inspire you to think differently, or reflect on some of the aspects of your own approach to evolution.

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Deformation of framed curves with boundary conditions

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Abstract

We provide a general approach to deform framed curves while preserving their clamped boundary conditions (this includes closed framed curves) as well as properties of their curvatures. We apply this to director theories, which involve a curve $\gamma : (0, 1) \rightarrow \mathbb{R}^3$ and orthonormal directors $d_1, d_2, d_3 : (0, 1) \rightarrow \mathbb{S}^2$ with $d_1 = \gamma'$. We show that γ and the d_i can be approximated smoothly while preserving clamped boundary conditions at both ends. The approximation process also preserves conditions of the form $d_i \cdot d'_j = 0$. Moreover, it is continuous with respect to natural functionals on framed curves. In the context of Γ -convergence, our approach allows to construct recovery sequences for director theories with prescribed clamped boundary conditions. We provide one simple application of this kind. Finally, we use similar ideas to derive Euler–Lagrange equations for functionals on framed curves satisfying clamped boundary conditions.

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1 Introduction

An elastic inextensible rod of length ℓ can be modelled by an arclength parametrized curve $\gamma : (0, \ell) \rightarrow \mathbb{R}^3$ describing the deformation of the center line, and a field of orthonormal directors $d_2, d_3 : (0, \ell) \rightarrow \mathbb{S}^2$ which are normal to the (unit) velocity vector $d_1 = \gamma'$ of the curve. The triple (d_1, d_2, d_3) is called a frame (or framing) adapted to the curve γ' . We assume that $(d_1, d_2, d_3) \in SO(3)$ almost everywhere. There are generalizations of this setting, e.g., to non-orthonormal frames and to models for extensible rods; we refer to [2, 7] and the references therein.

Rod-like theories also arise as asymptotic theories for narrow plates. One such instance are theories for ribbons [10, 26]. In such instances, there can be a natural choice of directors, which is characterized by the vanishing of curvature or torsion, i.e., of expressions of the form $d_i \cdot d'_j$. A common choice of directors gives rise to the Frenet frame, cf. [2, 8]. In [6] alternative choices of frames are discussed. One such alternative is what is called a relatively parallel

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adapted frame in [6]. The Frénet frame is determined by the condition $d_1' \cdot d_3 = 0$, while the relatively parallel frame in [6] is determined by the condition $d_2' \cdot d_3 = 0$. In the theory derived in [10], the natural choice of directors instead satisfies the constraint $d_1 \cdot d_2' = 0$.

Similar objects and constraints arise in the context of curves which lie within a surface $\Sigma \subset \mathbb{R}^3$. A common choice of directors in this case is to take $d_3(t)$ to equal the normal to the surface Σ at the point $\gamma(t)$. The curvature constraints then correspond to special curves on Σ such as geodesics or lines of curvature, and the frame is called the Darboux frame.

In hyperelasticity, the behaviour of rod-like objects is governed by a suitable energy functional which assigns an elastic energy to each triple (γ, d_2, d_3) resp. (d_1, d_2, d_3) .

A common mechanical situation is that of a rod (or ribbon) whose ends are clamped (or welded to some planes), cf. [2,4]. That is, one prescribes (γ, d_1, d_2, d_3) at both ends of the rod:

$$\begin{aligned}\gamma(0) &= \bar{\gamma}(0), \quad \gamma(\ell) = \bar{\gamma}(\ell) \\ d_i(0) &= \bar{d}_i(0), \quad d_i(\ell) = \bar{d}_i(\ell) \text{ for } i = 1, 2, 3.\end{aligned}\tag{1}$$

Here $\bar{\gamma}$ and $\bar{d}_i$ are given boundary data defined on $\{0, \ell\}$. Observe that closed framed curves (e.g. infinitesimally narrow ribbons in the shape of a Möbius band [25,27]) are particular cases of framed curves with clamped ends.

In order to study variational problems in such a setting, it is useful to be able to modify a given framed curve (γ, d_2, d_3) in a way that preserves both the boundary conditions and curvature conditions, e.g., those of the form $d_i \cdot d_j' = 0$. The purpose of this article is to describe a general process to achieve such modifications. An important feature of this process is that the required modifications can be chosen to enjoy additional properties, such as smoothness or small support.

One class of applications arises in the derivation of asymptotic theories. In the context of Γ -convergence, the need arises to construct, for a given limiting framed curve (γ, d_2, d_3) (which satisfies the boundary conditions), sequences $(\gamma^{(n)}, d_2^{(n)}, d_3^{(n)})$ of framed curves which are asymptotically optimal, i.e., so-called recovery sequences. Using the modification process, one can first construct these recovery sequences without taking the boundary conditions into account, and subsequently modify the resulting $(\gamma^{(n)}, d_2^{(n)}, d_3^{(n)})$ to become admissible for the prescribed boundary conditions. In such a context, it is essential to find well-controlled modifications even when the sequence $(\gamma^{(n)}, d_2^{(n)}, d_3^{(n)})$ obtained in the first step is only weakly compact in $W^{2,2} \times W^{1,2}$.

In the present article we only give a simple application of the general results in this context. It is related to the existence of generalized optimal designs for rods with clamped ends via Γ -convergence, see Proposition 4.4 below. A more sophisticated application to such a question in the context of ribbons will be given elsewhere.

Another area of applications of the modification procedure is the approximation of a given framed curve by smooth ones. Such density results are useful in the context of Γ -convergence, as well. They allow to construct recovery sequences $(\gamma^{(n)}, d_2^{(n)}, d_3^{(n)})$ starting from smooth limiting objects (γ, d_2, d_3) . This can be much easier than having to base the construction on an irregular object.

They are also an important tool to establish the convergence of discrete approximations in [4], as they permit to remove artificial regularity hypotheses. This is particularly useful in the case of ribbons, since the constraints are not of lower order.

We will obtain the following density result; cf. Theorem 4.1 below for its precise statement.

Let Q be continuous and define $\tilde{\mathcal{F}}_Q$ by

$$\tilde{\mathcal{F}}_Q(\gamma, d_2, d_3) = \int_0^\ell Q(d_1 \cdot d'_2, d_2 \cdot d'_3, d_1 \cdot d'_3).$$

Then for all framed curves

$$(\gamma, d_2, d_3) \in W^{2,2}(0, \ell) \times W^{1,2}(0, \ell) \times W^{1,2}(0, \ell)$$

with $\tilde{\mathcal{F}}_Q(\gamma, d_2, d_3) < \infty$ there exist $(\gamma^{(n)}, d_2^{(n)}, d_3^{(n)})$, which are C^∞ on $[0, \ell]$ and which satisfy the following:

- *convergence*: $(\gamma^{(n)}, d_2^{(n)}, d_3^{(n)}) \rightarrow (\gamma, d_2, d_3)$ strongly in $W^{2,2} \times W^{1,2} \times W^{1,2}$ as $n \rightarrow \infty$.
- *convergence of energies*: $\tilde{\mathcal{F}}_Q(\gamma^{(n)}, d_2^{(n)}, d_3^{(n)}) \rightarrow \tilde{\mathcal{F}}_Q(\gamma, d_2, d_3)$ as $n \rightarrow \infty$.
- *boundary conditions*: $(\gamma^{(n)}, d_2^{(n)}, d_3^{(n)})$ satisfy (1) for all $n \in \mathbb{N}$.
- *structure*: if $d_i \cdot d'_j = 0$ on $(0, \ell)$, then $d_i^{(n)} \cdot (d_j^{(n)})' = 0$ on $(0, \ell)$ for all $n \in \mathbb{N}$.

The above theorem applies to the standard bending-torsion theory of rods, for which a similar result has been obtained recently, by a different approach, in [5]; see also [19] for a related result. However, it also applies to curves endowed with the Frénet frame, or with the almost parallel frame in [6], or to the frame used in the context of ribbons in [10]. For more details we refer to Sect. 3.

Both Theorems 4.1 and 4.4 are applications of the following more general result; we refer to Theorem 3.2 for the precise statement and to Definition 3.1 for the notion of degeneracy:

Let (γ, d_2, d_3) be a framed curve satisfying (1). Assume that there is a set $J \subset I$ of positive measure on which (γ, d_2, d_3) is nondegenerate. Let $(\gamma^{(n)}, d_2^{(n)}, d_3^{(n)})$ be a sequence of framed curves converging weakly to (γ, d_2, d_3) in $W^{2,2} \times W^{1,2}$.

Then there exist $(\hat{\gamma}^{(n)}, \hat{d}_2^{(n)}, \hat{d}_3^{(n)})$, still converging weakly to (γ, d_2, d_3) , which satisfy (1) and whose curvatures differ from those of $(\gamma^{(n)}, d_2^{(n)}, d_3^{(n)})$ only by well-behaved functions supported on J .

Roughly speaking, by well-behaved functions we mean those which preserve the conditions $d_i \cdot d'_j = 0$ on the curvatures, and which moreover can be chosen to belong to any space with is typically dense in L^2 .

Clearly, this last result is only interesting if nondegeneracy is a very weak condition. This is indeed the case, as shown in Proposition 3.3 below. In fact, the notion of degeneracy turns out to be strong enough to see why the result cannot be true if (γ, d_2, d_3) is degenerate on $(0, \ell)$.

In Sect. 3.5 we derive the Euler–Lagrange equations for stationary points of functionals on framed curves satisfying the boundary conditions (1). In Sect. 4.3 we apply this to a ribbon model.

A more general notion of framed curves (allowing for zero velocity) is studied, e.g., in [3, 11, 15]. In [12] conditions are obtained under which a smooth curve with points of zero velocity nevertheless admits a smooth adapted framing.

In [9, 20–22] equilibrium states of clamped rods were studied under various kinds of boundary conditions; we also refer to [13]. A formalism to study such variational problems on framed curves was developed in [14] and applied for fixed boundary conditions e.g. in [18].

Notation. We tacitly sum over repeated indices, unless stated otherwise. All sets and functions are measurable. The letter I will denote the identity matrix, but it will also denote

the open interval $(0, 1)$. We denote by $\mathfrak{so}(n)$ the space of skew-symmetric $n \times n$ -matrices and by $SO(n)$ the set of rotation matrices. The scalar product on $\mathbb{R}^{n \times n}$ will be denoted by $A : B = \text{Tr}(A^T B)$, and we use the norm $|A|^2 = A : A$. We use the notation $\wedge : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathfrak{so}(n)$ given by $a \wedge b = a \otimes b - b \otimes a$. We denote by (e_1, e_2, e_3) the canonical basis of $\mathbb{R}^3$.

2 Abstract results

2.1 An implicit function theorem

In what follows, for a normed space $(H, \|\cdot\|_H)$, we will denote by $B_R^{\|\cdot\|_H}(x)$ the open ball of radius $R > 0$ with respect to the norm $\|\cdot\|_H$ centered at $x \in H$; we will omit the superscript when the norm is evident from the context. For a given linear map T we will denote its kernel by $N(T)$. By $L(E, G)$ we denote the space of linear bounded operators from E to G .

Theorem 2.1 *Let $K, k \in \mathbb{N}$, let $\mathcal{M} \subset \mathbb{R}^K$ be a smooth k -dimensional submanifold, let H be a Banach space, let $V \subset H$ be open in the strong topology of H . Let $\mathcal{P} : V \rightarrow \mathcal{M}$ be continuously Fréchet differentiable and let $x \in V$ be such that $D\mathcal{P}(x)$ maps some subspace $\tilde{E} \subset H$ onto the tangent space to $\mathcal{M}$ at $\mathcal{P}(x)$. Then there exists a subspace $E \subset \tilde{E}$ such that*

$$H = E \oplus N(D\mathcal{P}(x)). \quad (2)$$

Moreover, for any subspace $E \subset H$ satisfying (2) the following is true:

- (i) *E is k -dimensional, and there exists an $r > 0$ and a continuously Fréchet differentiable map*

$$Y : B_r^{\|\cdot\|_H}(x) \rightarrow E$$

satisfying $Y(x) = 0$ and

$$\mathcal{P}(z + Y(z)) = \mathcal{P}(x) \text{ for all } z \in B_r^{\|\cdot\|_H}(x). \quad (3)$$

- (ii) *If both $\mathcal{P}$ and its Fréchet derivative $D\mathcal{P} : V \rightarrow L(H, \mathbb{R}^K)$ are sequentially continuous with respect to weak convergence in H and if $R > 0$ is such that*

$$B_{2R}^{\|\cdot\|_H}(x) \subset V \quad (4)$$

and $(x_n) \subset B_R^{\|\cdot\|_H}$ is such that $x_n \rightarrow x$ weakly in H as $n \rightarrow \infty$, then there exist $y_n \in E$ converging to zero in E and satisfying $\mathcal{P}(x_n + y_n) = \mathcal{P}(x)$ for all $n \in \mathbb{N}$ large enough.

Remarks (i) Theorem 2.1 (i) is a common version of the implicit function theorem. Theorem 2.1 (ii) will be relevant for the derivation of our main result, Theorem 3.2 below.

- (ii) If $V = H$ then there always exists an $R > 0$ satisfying the hypotheses of the second part, since in this case condition (4) is void and the sequence (x_n) is bounded in H because it converges weakly.

- (iii) In applications (such as in [4, 10, 16] and Sect. 4) one needs the freedom to choose E in order to ensure that the corrections y_n enjoy suitable properties.

- (iv) A consequence of Theorem 2.1 (i) is that

$$N(D\mathcal{P}(x)) \subset N(DY(x)). \quad (5)$$

This follows at once by applying the chain rule to (3).

The proof of Theorem 2.1 (ii) uses the following implicit function theorem.

Lemma 2.2 *Let X be a topological space, let E and G be Banach spaces and let $U \subset E$ be open in the strong topology. Let*

$$F : X \times U \rightarrow G \quad (6)$$

be continuous. Assume, moreover, that the partial Fréchet derivative

$$D_2 F : X \times U \rightarrow L(E, G) \quad (7)$$

of F with respect to its second argument exists and is continuous, and let $(a, b) \in X \times U$ be such that

$$D_2 F(a, b) : E \rightarrow G \text{ is bijective.} \quad (8)$$

Then there exists a neighbourhood $W \subset X$ of a and a map $Y : W \rightarrow U$ such that

$$F(x, Y(x)) = F(a, b) \text{ for all } x \in W.$$

Moreover, Y is continuous in a with $Y(a) = b$.

Proof The proof closely follows the usual proof of the implicit function theorem. Without loss of generality we assume that $b = 0$, that U is an open ball centered at 0 and that $F(a, 0) = 0$. By the closed graph theorem, the inverse of $D_2 F(a, 0)$ is bounded.

Set $A = D_2 F(a, 0)$ and define, for every $x \in X$ the map $T_x : U \rightarrow E$ by setting

$$T_x y = A^{-1} F(x, y) - y \text{ for all } y \in U.$$

Then for all $y, z \in U$:

$$\begin{aligned} \|T_x y - T_x z\| &\leq \|A^{-1}\|_{L(G, E)} \cdot \|F(x, y) - F(x, z) - A(y - z)\|_G \\ &= \|A^{-1}\|_{L(G, E)} \cdot \left\| \int_0^1 \left(D_2 F(x, ty + (1-t)z) - A \right) dt (y - z) \right\|_G \\ &\leq \|A^{-1}\|_{L(G, E)} \cdot \sup_{t \in [0, 1]} \|D_2 F(x, z + t(y - z)) - A\|_{L(E, G)} \|y - z\|_E \end{aligned}$$

By continuity of $D_2 F$, there exist $R_0 > 0$ and a neighbourhood W_0 of a such that $\overline{B}_{R_0}(0) \subset U$ and such that for all $(x, \tilde{y}) \in W_0 \times \overline{B}_{R_0}(0)$ we have

$$\|A^{-1}\|_{L(G, E)} \|D_2 F(x, \tilde{y}) - A\|_{L(E, G)} \leq \frac{1}{2}.$$

Thus

$$\|T_x y - T_x z\|_E \leq \frac{1}{2} \|y - z\|_E \text{ for all } x \in W_0, y, z \in \overline{B}_{R_0}(0). \quad (9)$$

Now let $R \in (0, R_0)$ and $y \in \overline{B}_R(0)$. Then for all $x \in W_0$ we have by (9) that

$$\begin{aligned} \|T_x y\|_E &\leq \|T_x y - T_x(0)\|_E + \|T_x(0)\|_E \\ &= \frac{1}{2} \|y\|_E + \|A^{-1} F(x, 0)\|_E \\ &\leq \frac{R}{2} + \|A^{-1}\|_{L(G, E)} \|F(x, 0)\|_G. \end{aligned}$$

Since $F(a, 0) = 0$ and since F is continuous in $(a, 0)$, there exists a neighbourhood $W \subset W_0$ of a such that

$$\|A^{-1}\|_{L(G, E)} \|F(x, 0)\|_G \leq \frac{R}{2} \text{ for all } x \in W.$$

Summarizing, for every $R \in (0, R_0)$ there exists a neighbourhood W of a such that the map $T_x : \overline{B}_R(0) \subset E \rightarrow \overline{B}_R(0) \subset E$ is a contraction for each $x \in W$. Hence for each $x \in W$ the map T_x has a unique fix point $Y(x) \in \overline{B}_R(0)$.

Since, moreover, $R \in (0, R_0)$ was arbitrarily small, we conclude that Y is continuous in a with $Y(a) = 0$. $\square$

Remark 2.3 Let $K, k \in \mathbb{N}$ and let $\mathcal{M}$ be a smooth k -dimensional submanifold of $\mathbb{R}^K$. Then Lemma 2.2 remains true if G is replaced by $\mathcal{M}$ in (6), by $\mathbb{R}^K$ in formula (7) and by $T_{F(a,b)}\mathcal{M}$ (i.e., the tangent space to $\mathcal{M}$ at the point $F(a, b)$) in (8).

Proof Let $\tilde{\mathcal{M}} \subset \mathcal{M}$ be relatively open with $F(a, b) \in \tilde{\mathcal{M}}$ and such that there exists a chart $\Psi : \tilde{\mathcal{M}} \rightarrow \mathbb{R}^k$. By continuity of F there exist open neighbourhoods $\tilde{X} \subset X$ of a and $\tilde{U} \subset U$ of b such that $F(\tilde{X} \times \tilde{U}) \subset \tilde{\mathcal{M}}$. The conclusion now follows by applying Lemma 2.2 with $\tilde{X}, \tilde{U}$ and $\Psi \circ F$ instead of X, U and F . $\square$

Since Remark 2.3 is such an immediate consequence of the lemma, we will at times refer to Lemma 2.2 even when we should, more precisely, be referring to Remark 2.3.

Proof of Theorem 2.1 The existence of E is standard. In fact, since $D\mathcal{P}(x)$ maps H onto the tangent space to $\mathcal{M}$ at $\mathcal{P}(x)$, there exist $y^{(1)}, \dots, y^{(k)} \in H$ such that

$$D\mathcal{P}(x)(y^{(1)}), \dots, D\mathcal{P}(x)(y^{(k)})$$

are linearly independent. We can take E to be the span of the $y^{(j)}$.

If E is any subspace satisfying (2) then $\dim E \leq k$. In fact, since $E \cap N(D\mathcal{P}(x)) = \{0\}$, we see that the restriction of $D\mathcal{P}(x)$ to E is injective. Hence $\dim E \leq \dim D\mathcal{P}(x)(E)$. On the other hand, the range of $D\mathcal{P}(x)$ equals the range of the restriction $D\mathcal{P}(x)|_E$. Since $D\mathcal{P}(x)$ is surjective, we conclude that $\dim E \geq k$.

Now let $R > 0$ satisfy (4), let $X = B_R^{\|\cdot\|_H}(x)$ and $U = E \cap B_{R/2}^{\|\cdot\|_H}(0)$. Then $z + y \in V$ for all $(z, y) \in X \times U$ and therefore we can define the map $F : X \times U \rightarrow \mathcal{M}$ by setting

$$F(z, y) = \mathcal{P}(z + y).$$

For every fixed $z \in X$ we see that $F(z, \cdot)$ is differentiable with differential

$$D_2F(z, y) = D\mathcal{P}(z + y) \text{ for all } y \in U.$$

In particular, $D_2F(x, 0) = D\mathcal{P}(x)|_E$ is injective. Since $\mathcal{P}$ and its Fréchet derivative $D\mathcal{P}$ are strongly continuous, the existence of $r > 0$ and Y follows from Lemma 2.2 and Remark 2.3 if we endow X with the relative topology inherited from the strong topology on H . The continuous Fréchet differentiability of Y is standard as well.

In order to prove Theorem 2.1 (ii), we endow X with the relative topology inherited from the weak topology on H . Then both F and D_2F are continuous on $X \times U$ because $\mathcal{P}$ and $D\mathcal{P}$ and the operation of addition are sequentially weakly continuous and the relative weak topology on X is first countable. Hence the claim follows from Lemma 2.2 and Remark 2.3 by setting $y_n = Y(x_n)$ for large enough n . $\square$

Corollary 2.4 Assume that the hypotheses of Theorem 2.1 are satisfied and let $E \subset H$ be a subspace satisfying (2). Then there exists a neighbourhood $\hat{E} \subset E$ of the origin such that $\mathcal{P}$ maps $x + \hat{E}$ homeomorphically onto a neighbourhood of $\mathcal{P}(x)$ in $\mathcal{M}$.

Proof We define a C^1 -map $\tilde{F}$ from a neighbourhood of the origin in E into $\mathcal{M}$ by setting $\tilde{F}(y) = \mathcal{P}(x + y)$. Then $D\tilde{F}(0) = D\mathcal{P}(x)|_E$ is a bijection from the k -dimensional space E onto the k -dimensional tangent space to $\mathcal{M}$ at $\mathcal{P}(x)$. Hence the claim follows from the inverse function theorem. $\square$

In Sect. 3.5 we will use the following Lagrange multiplier rule.

Corollary 2.5 *Assume that the hypotheses of Theorem 2.1 are satisfied. Assume, moreover, that $\mathcal{F} : V \rightarrow \mathbb{R}$ is Fréchet differentiable in x and that $\mathcal{F}(x) \leq \mathcal{F}(z)$ for all $z \in V$ with $\mathcal{P}(z) = \mathcal{P}(x)$. Then there exists a unique Λ in the tangent space to $\mathcal{M}$ at $\mathcal{P}(x)$ such that*

$$D\mathcal{F}(x) + \Lambda \cdot D\mathcal{P}(x) = 0 \text{ on } H.$$

Proof Since $D\mathcal{P}(x)$ maps H onto the tangent space to $\mathcal{M}$ at $\mathcal{P}(x)$, existence and uniqueness of Λ follow once we prove that

$$D\mathcal{F}(x)y = 0 \text{ for all } y \in N(D\mathcal{P}(x)). \quad (10)$$

By Theorem 2.1 (i) there exists a finite dimensional subspace $E \subset H$, an open ball $B \subset V$ centered at x and a continuously Fréchet differentiable map $Y : B \rightarrow E$ such that $\mathcal{P}(z + Y(z)) = \mathcal{P}(x)$ for all $z \in B$. In particular, for any $y \in N(D\mathcal{P}(x))$ there is an $r > 0$ such that

$$\mathcal{P}(x + ty + Y(x + ty)) = \mathcal{P}(x) \text{ for all } t \in (-r, r).$$

Hence

$$\mathcal{F}(x + ty + Y(x + ty)) - \mathcal{F}(x) \geq 0 \text{ for all } t \in (-r, r). \quad (11)$$

On the other hand, (5) implies that $\frac{1}{t}Y(x + ty)$ converges to zero in E as $t \rightarrow 0$. Dividing (11) by t and letting $t \rightarrow 0$ we therefore deduce (10). $\square$

3 Framed curves with boundary conditions

Throughout this chapter I denotes the open interval $(0, 1)$. Any other interval would be as good. The same letter I will denote the identity matrix.

For given $A \in L^2(I, \mathbb{R}^{n \times n})$ the initial value problem

$$\begin{aligned} R' &= AR \\ R(0) &= R_0 \end{aligned}$$

has a unique solution $R \in W^{1,2}(I, \mathbb{R}^{n \times n})$ for any choice of initial condition $R_0 \in \mathbb{R}^{n \times n}$.

From now on, a framed curve is a pair consisting of a frame field $r : I \rightarrow SO(3)$ and an associated curve $\gamma : I \rightarrow \mathbb{R}^3$ satisfying $\gamma' = r^T e_1$ on I . For every $A \in L^2(I, \mathfrak{so}(3))$ let us denote by $r_A \in W^{1,2}(I, SO(3))$ the unique solution to the initial value problem

$$\begin{aligned} r'_A &= Ar_A \\ r_A(0) &= I \end{aligned} \quad (12)$$

In what follows we will at times write r instead of r_A and r_i instead of $r_A^T e_i$. Notice that $r_2 = d_2$ and $r_3 = d_3$ are the directors mentioned in the introduction. The differential equation (12) can be spelled out as follows:

$$\begin{pmatrix} r_1 \\ r_2 \\ r_3 \end{pmatrix}' = \begin{pmatrix} 0 & A_{12} & A_{13} \\ -A_{12} & 0 & A_{23} \\ -A_{13} & -A_{23} & 0 \end{pmatrix} \begin{pmatrix} r_1 \\ r_2 \\ r_3 \end{pmatrix}. \quad (13)$$

If γ is a curve lying in a surface and r_3 is the normal to the surface along γ , then A_{13} is the normal curvature, A_{12} the geodesic curvature and A_{23} the geodesic torsion of γ . One can

also regard r_A as a curve in $SO(3)$, in which case A is the pull-back of the Maurer-Cartan form.

The standard bending-torsion theory of rods [5] corresponds to the case when A in (12) is allowed to take arbitrary values in $\mathfrak{so}(3)$, i.e., all entries A_{12} , A_{13} and A_{23} can be nonzero in (13).

The Frénet frame r_A corresponds to the case when $A_{13} \equiv 0$ in (13); the almost parallel frame in [6], which is also relevant in [16], corresponds to the case when $A_{23} \equiv 0$; whereas in [10] the case $A_{12} \equiv 0$ is relevant.

The cases when only one of the A_{ij} may differ from zero have obvious geometric interpretations:

- If $A_{13} = A_{23} = 0$ everywhere or $A_{12} = A_{23} = 0$ everywhere, then one director is constant and the curve $t \mapsto \int_0^t r_A^T e_1$ lies in a plane.
These cases are equivalent to planar framed curves, because one component of r_A is constant. Therefore, the results obtained later imply similar results for planar framed curves.
- If $A_{12} = A_{13} = 0$, then the curve $t \mapsto \int_0^t r_A^T e_1$ is a straight line and the frame spins around this line.

As mentioned in the introduction, the boundary conditions we address here amount to clamped ends, i.e., boundary conditions prescribing $\gamma(0)$, $r_A(0)$ and $\gamma(1)$, $r_A(1)$. Theorem 3.2 below shows that, for a given generic framed curve (γ, r) and any sequence (γ_n, r_n) converging to (γ, r) weakly in $W^{2,2} \times W^{1,2}$, it is possible to perturb (γ_n, r_n) in a well-controlled way such that each of the modified framed curves satisfies the same clamped boundary conditions as (γ, r) . By a well-controlled perturbation we mean, firstly, that it converges to zero as $n \rightarrow \infty$; secondly, that it is supported on a small set; thirdly, that it preserves the structure of A . By the latter we simply mean that if A_{ij} vanishes identically then the same should remain true for modified framed curve.

3.1 Main results

Let $i, j, k \in \{1, 2, 3\}$ and set

$$\mathfrak{A} = \text{span}\{e_i \wedge e_j, e_i \wedge e_k, e_j \wedge e_k\}. \quad (14)$$

In this definition i, j, k are not necessarily distinct, so that $\mathfrak{A}$ can be one, two or three dimensional; let us exclude the trivial case $\mathfrak{A} = \{0\}$. For a given choice of $\mathfrak{A}$ we define the following subspace $\tilde{\mathfrak{A}}$ of $\mathfrak{so}(3)$:

$$\tilde{\mathfrak{A}} = \begin{cases} \mathfrak{so}(3) & \text{if } \dim \mathfrak{A} \geq 2 \\ \mathfrak{A} & \text{if } \dim \mathfrak{A} = 1. \end{cases}$$

and the following subspace Γ of $\mathbb{R}^3$:

$$\Gamma = \begin{cases} \mathbb{R}^3 & \text{if } \dim \mathfrak{A} \geq 2 \\ \text{span}\{e_1, e_k\} & \text{if } \mathfrak{A} = \text{span}\{e_1 \wedge e_k\} \text{ for some } k = 2, 3 \\ \{0\} & \text{if } \mathfrak{A} = \text{span}\{e_2 \wedge e_3\}, \end{cases}$$

as well as the following subgroup of $SO(3)$:

$$G = \begin{cases} SO(3) & \text{if } \dim \mathfrak{A} \geq 2 \\ \{R \in SO(3) : R(e_1 \times e_k) = e_1 \times e_k\} & \text{if } \mathfrak{A} = \text{span}\{e_1 \wedge e_k\} \text{ for some } k = 2, 3 \\ \{R \in SO(3) : Re_1 = e_1\} & \text{if } \mathfrak{A} = \text{span}\{e_2 \wedge e_3\}, \end{cases}$$

where the symbol $\times$ denotes the cross product in $\mathbb{R}^3$.

From now on, the symbols $\mathfrak{A}$, $\tilde{\mathfrak{A}}$, Γ and G will always be as just defined, i.e., $\mathfrak{A}$ will be a space of the form (14) and $\tilde{\mathfrak{A}}$, Γ and G will denote the sets determined by this choice of $\mathfrak{A}$.

These definitions and many of the results extend to higher dimensions and to other subspaces $\mathfrak{A} \subset \mathfrak{so}(n)$ as well. The natural definition of $\tilde{\mathfrak{A}}$ is that it is the smallest Lie subalgebra of $\mathfrak{so}(3)$ containing $\mathfrak{A}$, and $G = \exp(\tilde{\mathfrak{A}})$.

The following technical definition is to be viewed in conjunction with Proposition 3.3 below. That proposition shows that, in most cases, degeneracy of A in the sense of Definition 3.1 in fact implies that A_{12} and A_{13} vanish identically, which essentially means that the curve $t \mapsto \int_0^t r_A^T e_1$ is a straight line. A partial converse to Proposition 3.3 is also true, see Proposition 3.9. Summarizing, the following degeneracy condition turns out to be very restrictive, i.e., most $A \in L^2(I, \mathfrak{A})$ are nondegenerate.

Definition 3.1 A map $A \in L^2(I, \tilde{\mathfrak{A}})$ is said to be *degenerate with respect to $\mathfrak{A}$ on a set $J \subset I$* if there exist $\lambda \in \Gamma$ and $M \in \tilde{\mathfrak{A}}$, not both zero, such that

$$M - \lambda \wedge \left(\int_t^1 r_A^T e_1 \right) \in (r_A^T(t) \mathfrak{A} r_A(t))^\perp \text{ for all } t \in J. \quad (15)$$

In (15) we can also take $\int_0^t r_A^T e_1$ instead of $-\int_t^1 r_A^T e_1$. In fact, one can readily verify that $\lambda \wedge \int_0^1 r_A^T e_1 \in \tilde{\mathfrak{A}}$ which can therefore be absorbed into M .

The following theorem is our main result regarding the deformation of framed curves while preserving prescribed boundary conditions.

Theorem 3.2 Let $A \in L^2(I, \tilde{\mathfrak{A}})$ and assume that there is a measurable set $J \subset I$ of positive measure such that A is not degenerate with respect to $\mathfrak{A}$ on J . Then every L^2 -dense subspace of

$$\{\hat{A} \in L^2(I, \mathfrak{A}) : \text{spt } \hat{A} \subset J\} \quad (16)$$

contains a finite dimensional subspace E such that, whenever $A_n \in L^2(I, \tilde{\mathfrak{A}})$ converge weakly in L^2 to A , then there exist $\hat{A}_n \in E$ converging to zero in E and such that

$$r_{A_n + \hat{A}_n}(1) = r_A(1) \text{ and} \quad (17)$$

$$\int_0^1 r_{A_n + \hat{A}_n}^T(t) e_1 dt = \int_0^1 r_A^T(t) e_1 dt \quad (18)$$

for all n .

Remarks.

- (i) This result should be viewed in conjunction with Proposition 3.3, which provides simple sufficient criteria for nondegeneracy.
- (ii) Equations (17) and (18) only refer to the boundary conditions at 1. The boundary conditions at 0 are determined by the initial values of the framed curves and are 0 (for the curve) resp. I (for the frame r_A). This entails no loss of generality, as it is always possible to arrange these conditions in 0 by means of a rigid motion.

- (iii) Observe that A_n and A are allowed to attain values in all of $\widetilde{\mathfrak{A}}$, while $\widehat{A}_n$ takes only values in $\mathfrak{A}$. Clearly, if A_n takes values in $\mathfrak{A}$, then so does $A_n + \widehat{A}_n$.
- (iv) In the cases when $\dim \mathfrak{A} = 1$, we have $\widetilde{\mathfrak{A}} = \mathfrak{A}$ and therefore the hypotheses require that $A_n \in \mathfrak{A}$ almost everywhere. This hypothesis cannot be dropped. In fact, if $A_n \notin \mathfrak{A}$ on a set of positive measure, then it is impossible to find $\widehat{A}_n : I \rightarrow \mathfrak{A}$ such that (18) is satisfied.

For instance, consider the case $\mathfrak{A} = \text{span}\{e_2 \wedge e_3\}$ and let $\widetilde{A} \in L^2(I, \mathfrak{so}(3))$. Then we have $\widetilde{A} \in \mathfrak{A}$ almost everywhere on I if, and only if, $\int_0^1 r_{\widetilde{A}}^T e_1 = 1$. In fact, both assertions are equivalent to $t \mapsto \int_0^t r_{\widetilde{A}}^T e_1$ being a straight line.

Therefore, if A takes values in $\mathfrak{A}$ but $A_n \notin \mathfrak{A}$ on a set of positive measure, then there exist no $\widehat{A}_n : I \rightarrow \mathfrak{A}$ which lead to (18), because $A_n + \widehat{A}_n \notin \mathfrak{A}$ on a set of positive measure.

- (v) By adapting the proofs, similar results can be derived for other boundary conditions. If, for instance, we were only interested in preserving the boundary conditions (17), then there would exist no degenerate A at all; this follows from the assertion, in Proposition 3.3 below, that $\lambda \neq 0$ whenever A is degenerate.
- (vi) A variant of Theorem 3.2 in the particular case $\mathfrak{A} = \{\widehat{A} \in \mathfrak{so}(3) : \widehat{A}_{23} = 0\}$ is used in [16]. The assertion for the case $\mathfrak{A} = \{\widehat{A} \in \mathfrak{so}(3) : \widehat{A}_{12} = 0\}$ is relevant in ribbon models [10,26]. Both in [16] and in Sect. 4.1 we choose J to be an interval and the dense subspace to consist of smooth functions. But we will also need to apply Theorem 3.2 for J which are not intervals.

The following result provides some natural sufficient criteria for nondegeneracy.

Proposition 3.3 *Let $A \in L^2(I, \widetilde{\mathfrak{A}})$, let $J \subset I$ have positive measure and assume that A is degenerate with respect to $\mathfrak{A}$ on J . Let M and λ be as in (15). Then $\lambda \neq 0$ and the following are true:*

- (i) *If $\dim \mathfrak{A} = 1$ then $e_2 \wedge e_3 \notin \mathfrak{A}$ and $A = 0$ almost everywhere on J .*
- (ii) *If $\dim \mathfrak{A} \geq 2$ and $e_2 \wedge e_3 \in \mathfrak{A}$, then $A_{12} = A_{13} = 0$ almost everywhere on J .*
- (iii) *If $\mathfrak{A} = \text{span}\{e_1 \wedge e_2, e_1 \wedge e_3\}$, and $J \subset I$ is an interval and $A \in \mathfrak{A}$ almost everywhere on J , then there exists a nonzero constant multiple Λ of λ such that, with $r_i = r_A^T e_i$, we have*

$$r'_1 = \Lambda \times r_1 \text{ on } J, \quad (19)$$

and there is an $a \in [0, 1]$ and a constant c_1 such that

$$\begin{aligned} \frac{\Lambda}{|\Lambda|} \cdot r_1 &= a \\ -A_{13}(t) &= \Lambda \cdot r_2(t) = |\Lambda| \sqrt{1-a^2} \cdot \cos(a|\Lambda|t - c_1) \\ A_{12}(t) &= \Lambda \cdot r_3(t) = |\Lambda| \sqrt{1-a^2} \cdot \sin(a|\Lambda|t - c_1) \end{aligned} \quad (20)$$

for almost every $t \in J$.

Remarks.

- (i) The conclusions of Proposition 3.3 are not exhausting. We refer to Sect. 3.4 for details.
- (ii) The assertion $e_2 \wedge e_3 \notin \mathfrak{A}$ in Proposition 3.3 (i) means that in the case $\mathfrak{A} = \text{span}\{e_2 \wedge e_3\}$ every $A \in L^2(I, \mathfrak{A})$ is nondegenerate on any $J \subset I$ of positive measure.

- (iii) Equation (19) asserts that the so-called Darboux vector Λ of r_1 is constant. It amounts to the existence of a constant rotation $R_\Lambda \in SO(3)$ taking e_1 into $\frac{\Lambda}{|\Lambda|}$ such that

$$R_\Lambda^T r_1(t) = \begin{pmatrix} a \\ \sqrt{1-a^2} \cos(|\Lambda|t) \\ \sqrt{1-a^2} \sin(|\Lambda|t) \end{pmatrix}. \quad (21)$$

Hence with $t_0 = \inf J$ and setting $\tilde{\gamma}(t) = \int_{t_0}^t r_1$, we have

$$R_\Lambda^T \tilde{\gamma}(t) = \begin{pmatrix} a(t-t_0) \\ \frac{\sqrt{1-a^2}}{|\Lambda|} (\sin(|\Lambda|t) - \sin(|\Lambda|t_0)) \\ -\frac{\sqrt{1-a^2}}{|\Lambda|} (\cos(|\Lambda|t) - \cos(|\Lambda|t_0)) \end{pmatrix},$$

which is a circular helix. Observe that the equations for $\Lambda \cdot r_2$ and $\Lambda \cdot r_3$ are in fact consequences of the equations for the A_{ij} , up to additive constants.

- (iv) Summarizing, for most choices of $\mathfrak{A}$ degeneracy implies that $(r_A^T e_1)' = 0$. The case $\mathfrak{A} = \text{span}\{e_1 \wedge e_2, e_1 \wedge e_3\}$ is an exception. Here $r_A(1)$ cannot be changed without modifying $r_A^T e_1$. Intuitively, if we wish to impose torsion on a narrow strip of paper, then the ends of the strip must be moved closer to each other, cf. Remark 3.8 below.

In the following sections we prove Theorem 3.2 and Proposition 3.3.

3.2 Boundary conditions as a level set

The boundary conditions (17), (18) can be regarded as a level set of a well-behaved functional $\mathcal{G}$ on the space of connections A . In fact, introducing

$$\begin{aligned} \mathcal{G} : L^2(I, \mathfrak{so}(3)) &\rightarrow SO(3) \times \mathbb{R}^3 \\ A &\mapsto \left(r_A(1), \int_0^1 r_A^T e_1 \right), \end{aligned} \quad (22)$$

equations (17), (18) become

$$\mathcal{G}(A_n + \hat{A}_n) = \mathcal{G}(A).$$

In what follows, the letter $\mathcal{G}$ will always refer to this map. The regularity of $\mathcal{G}$ depends on the dependence of r_A from A . Therefore, the following results will be useful.

Lemma 3.4 *Assume that $A_n \rightharpoonup A$ weakly in $L^2(I, \mathfrak{so}(3))$. Then*

$$r_{A_n} \rightharpoonup r_A \text{ weakly in } W^{1,2}(I, \mathbb{R}^{3 \times 3}). \quad (23)$$

If $A_n \rightarrow A$ strongly in $L^2(I, \mathfrak{so}(3))$, then $r_{A_n} \rightarrow r_A$ strongly in $W^{1,2}(I, \mathbb{R}^{3 \times 3})$.

Proof We define the C^1 -functions $\Phi_n : I \rightarrow [0, \infty)$ by setting

$$\Phi_n(t) = \int_0^t |r_{A_n} - r_A|^2.$$

Recall that

$$(r_{A_n} - r_A)' = A_n(r_{A_n} - r_A) + (A_n - A)r_A \quad (24)$$

and that $r_{A_n}(0) = r_A(0)$. Introducing

$$\Psi_n(t) = \left| \int_0^t (A_n - A)r_A \right|^2$$

we can therefore estimate, for all $t \in I$,

$$\begin{aligned} \Phi'_n(t) &\leq \left(\int_0^t |A_n| |r_{A_n} - r_A| \right)^2 + \Psi_n(t) \\ &\leq C \Phi_n(t) + \Psi_n(t), \end{aligned}$$

because (A_n) is bounded in $L^2(I)$. Since $\Phi_n(0) = 0$, we deduce from Gronwall's inequality that $\Phi_n(1) \leq C \int_0^1 \Psi_n$. But $\Psi_n \rightarrow 0$ pointwise on I , because $A_n \rightarrow A$ weakly in $L^2(I)$. Moreover, $|\Psi_n| \leq C$ on I because (A_n) is bounded in $L^2(I)$. Hence dominated convergence ensures that $\Psi_n \rightarrow 0$ in $L^1(I)$. Thus $\Phi_n(1) \rightarrow 0$ as $n \rightarrow \infty$. This means that

$$r_{A_n} \rightarrow r_A \text{ strongly in } L^2(I, \mathbb{R}^{3 \times 3}). \quad (25)$$

On the other hand, we see from $r'_{A_n} = A_n r_{A_n}$ and the fact that r_{A_n} takes values in $SO(3)$, that the sequence (r'_{A_n}) is bounded in $L^2(I)$. Combining this with (25) we deduce (23).

Finally, if $A_n \rightarrow A$ strongly in L^2 , then $A_n r_{A_n} \rightarrow A r_A$ strongly in L^2 because $r_{A_n} \rightarrow r_A$ uniformly on I by (23). Therefore, $r'_{A_n} \rightarrow r'_A$ strongly in L^2 . $\square$

Lemma 3.5 *Let $t_0 \in [0, 1]$. Then the map*

$$\begin{aligned} \vec{r} : L^2(I, \mathfrak{so}(3)) &\rightarrow SO(3) \\ A &\mapsto r_A(t_0) \end{aligned}$$

is Fréchet differentiable with

$$D\vec{r}(A)(B) = r_A(t_0) \int_0^{t_0} (r_A^T B r_A) \text{ for all } B \in L^2(I, \mathfrak{so}(3)).$$

Moreover, $\vec{r}$ and its Fréchet derivative $D\vec{r}$ are sequentially continuous with respect to weak convergence in $L^2(I, \mathfrak{so}(3))$.

In particular, $\mathcal{G}$ is Fréchet differentiable and, moreover, $\mathcal{G}$ and $D\mathcal{G}$ are sequentially continuous with respect to weak convergence in L^2 . We have

$$D\mathcal{G}(A)(B) = \left(r_A(1) \int_0^1 (r_A^T B r_A), - \int_0^1 \int_0^t (r_A^T B r) r_A^T(t) e_1 dt \right) \quad (26)$$

for all $B \in L^2(I, \mathfrak{so}(3))$.

Proof We may assume without loss of generality that $t_0 = 1$. For $A, B \in L^2(I, \mathfrak{so}(3))$ set

$$W_A(B) = \int_0^1 (r_A^T B r_A). \quad (27)$$

Then

$$D\vec{r}(A)(B) = \vec{r}(A) W_A(B) \quad (28)$$

is the derivative of $\vec{r}$ at A in the direction B . In fact, let $B \in L^2(I, \mathfrak{so}(3))$. Since

$$\begin{aligned} r_{A+B}(1) &= r_A(1) + r_A(1) \int_0^1 r_A^T B r_{A+B} \\ &= r_A(1) + r_A(1) W_A(B) + r_A(1) \int_0^1 r_A^T B (r_{A+B} - r_A), \end{aligned}$$

we see that

$$|\vec{r}(A+B) - \vec{r}(A) - \vec{r}(A)W_A(B)| \leq \|B\|_{L^2(I)} \|r_{A+B} - r_A\|_{L^2(I)}.$$

By Lemma 3.4 the second factor on the right-hand side converges to zero as $B \rightarrow 0$ in L^2 . This proves the Fréchet differentiability and completes the proof of (28).

Now let $A_n \rightarrow A$ weakly in L^2 . We claim that then $D\vec{r}(A_n) \rightarrow D\vec{r}(A)$ in the operator norm on $L(L^2, \mathbb{R}^{3 \times 3})$. In fact, by (28),

$$\begin{aligned} |D\vec{r}(A_n)(B) - D\vec{r}(A)(B)| &= |\vec{r}(A_n)W_{A_n}(B) - \vec{r}(A)W_A(B)| \\ &\leq |W_{A_n}(B) - W_A(B)| + |\vec{r}(A_n) - \vec{r}(A)| |W_A(B)|. \end{aligned}$$

From (27) we deduce that

$$|W_{A_n}(B) - W_A(B)| \leq 2\|B\|_{L^1(I)} \|r_{A_n} - r_A\|_{L^\infty(I)}.$$

Since $|W_A(B)| \leq \|B\|_{L^1(I)}$, we conclude that

$$|D\vec{r}(A_n)(B) - D\vec{r}(A)(B)| \leq C\|r_{A_n} - r_A\|_{L^\infty(I)} \|B\|_{L^1(I)}.$$

So by Lemma 3.4 we have $D\vec{r}(A_n) \rightarrow D\vec{r}(A)$ in the operator norm on $L(L^1, \mathbb{R}^{3 \times 3})$.

Finally, the second component of (26) follows by integration from the corresponding formulae for r_A . $\square$

3.3 Surjectivity

The following observation motivates Definition 3.1. Notice that $(r_A(1)\tilde{\mathfrak{A}}) \times \Gamma$ is the tangent space of $G \times \Gamma$ at $\mathcal{G}(A)$.

Lemma 3.6 *Let $A \in L^2(I, \tilde{\mathfrak{A}})$, let $J \subset I$ have positive measure and assume that A is not degenerate with respect to $\mathfrak{A}$ on J . Then $D\mathcal{G}(A)$ maps any L^2 -dense subspace of*

$$\{\hat{A} \in L^2(I, \mathfrak{A}) : \text{spt } \hat{A} \subset J\}. \quad (29)$$

onto $(r_A(1)\tilde{\mathfrak{A}}) \times \Gamma$.

Proof As usual, we write r instead of r_A . Lemma 3.5 implies that $\mathcal{G}$ is Fréchet differentiable with derivative given by (26). If the restriction of $D\mathcal{G}(A)$ to a dense subspace of (29) does not map onto $r(1)\tilde{\mathfrak{A}} \times \Gamma$, then there exist $M \in \tilde{\mathfrak{A}}$ and $\lambda \in \Gamma$, not both zero, such that, setting $\gamma(t) = -\int_t^1 r^T e_1$, we have

$$\begin{aligned} 0 &= r(1)M : r(1) \int_0^1 (r^T \hat{A} r) - \lambda \cdot \int_0^1 \int_0^t (r^T \hat{A} r) r^T(t) e_1 dt \\ &= \int_0^1 (r M r^T) : \hat{A} + \lambda \cdot \int_0^1 (r^T \hat{A} r) \gamma \\ &= \int_0^1 r(M + \lambda \otimes \gamma) r^T : \hat{A} \end{aligned}$$

for all $\widehat{A}$ in a dense subspace of (29). Thus on J the matrix-valued map $r(M + \lambda \otimes \gamma)r^T$ takes values in the orthogonal complement of $\mathfrak{A} \subset \mathbb{R}^{3 \times 3}$. This implies (15) because $r^T \mathfrak{A}^\perp r = (r^T \mathfrak{A} r)^\perp$. $\square$

Combining Lemma 3.6 with Theorem 2.1 we can prove Theorem 3.2:

Proof of Theorem 3.2 Let $\widetilde{E}$ be a dense subspace of (16). According to Lemma 3.6 the hypotheses of Theorem 2.1 are satisfied with $V = H = L^2(I, \widetilde{\mathfrak{A}})$, $\mathcal{M} = G \times \Gamma$ and $\mathcal{P} = \mathcal{G}$. Hence the assertions follow from Theorem 2.1, taking into account Remark (ii) following its statement. $\square$

Another consequence of Lemma 3.6 is the following result about the relative openness of the set $\mathcal{G}(L^2(I, \mathfrak{A}))$.

Corollary 3.7 *Let $A \in L^2(I, \mathfrak{A})$ and let $J \subset I$ have positive measure. Assume that A is not degenerate with respect to $\mathfrak{A}$ on J . Then there exists a nonempty open neighbourhood $W \subset G \times \Gamma$ of $\mathcal{G}(A)$ such that for all $(\bar{\gamma}, \bar{r}) \in W$ there is some $\widetilde{A} \in L^2(I, \mathfrak{A})$ with $\mathcal{G}(\widetilde{A}) = (\bar{\gamma}, \bar{r})$.*

Proof Let $\widetilde{E}$ be an L^2 -dense subspace of (29). Combining Lemma 3.6 with Corollary 2.4, we see that there exists a finite dimensional subspace $E \subset \widetilde{E}$ and an open neighbourhood $\widehat{E} \subset E$ of the origin such that $\mathcal{G}$ maps $A + \widehat{E}$ onto a relatively open neighbourhood $W \subset G \times \Gamma$ of $\mathcal{G}(A)$. $\square$

The following consequence of Corollary 3.7 asserts that the maximal distance $\left| \int_0^1 r_A^T e_1 \right|$ of the endpoints of the curve for a prescribed value of $r_A(1)$ can be attained only when A is degenerate.

Remark 3.8 Let $\widetilde{E}$ be an L^2 -dense subspace of $L^2(I, \mathfrak{A})$ and assume that $A \in L^2(I, \mathfrak{A})$ maximizes the functional $L^2(I, \mathfrak{A}) \rightarrow \mathbb{R}$ given by

$$\widetilde{A} \mapsto \left| \int_0^1 r_{\widetilde{A}}^T e_1 \right|$$

among all $\widetilde{A} \in A + \widetilde{E}$ with $r_{\widetilde{A}}(1) = r_A(1)$. Then A is degenerate with respect to $\mathfrak{A}$ on I .

Proof If A is not degenerate with respect to $\mathfrak{A}$ on I , then by (the proof of) Corollary 3.7 there exists a finite dimensional subspace E of $\widetilde{E}$ and a neighbourhood $\widehat{E} \subset E$ of the origin such that $\mathcal{G}(A + \widehat{E})$ contains the set $\{r_A(1)\} \times W$, where W is some neighbourhood of $\int_0^1 r_A^T e_1$ in Γ . Hence A cannot be a maximizer. $\square$

For most choices of $\mathfrak{A}$, for global maximizers Remark 3.8 and its converse are immediate consequences of Proposition 3.3 and Proposition 3.9 below, because they show that $r_A^T e_1$ is constant if and only if A is degenerate.

3.4 Degeneracy

Proof of Proposition 3.3 We write r instead of r_A and $r_i = r^T e_i$. We set $\gamma(t) = -\int_t^1 r_1$.

We begin with the proof of part (ii). Let $(j, k) = (2, 3)$ or $(j, k) = (3, 2)$. From the definition of $\mathfrak{A}$ and (15) we deduce that there exist $M \in \mathfrak{so}(3)$ and $\lambda \in \mathbb{R}^3$, not both zero, such that

$$M + \lambda \wedge \gamma \text{ is parallel to } r_1 \wedge r_j \text{ on } J.$$

Denote by $f \in W^{1,2}(I)$ the coefficient of the projection of $M + \lambda \wedge \gamma$ onto $r_1 \wedge r_j$. Then

$$M + \lambda \wedge \gamma = f r_1 \wedge r_j \text{ on } J. \quad (30)$$

Taking derivatives in (30) we find

$$\lambda \wedge r_1 = f' r_1 \wedge r_j + f A_{1k} r_k \wedge r_j + f A_{jk} r_1 \wedge r_k \text{ (no sum) on } J. \quad (31)$$

We conclude that $f A_{1k} = 0$. Hence

$$f = 0 \text{ on } J \cap \{A_{1k} \neq 0\}. \quad (32)$$

Therefore also $f' = 0$ on this set. Hence

$$\lambda \wedge r_1 = 0 \text{ on } J \cap \{A_{1k} \neq 0\}.$$

Taking the derivative we find

$$A_{1m} \lambda \wedge r_m = 0 \text{ on } J \cap \{A_{1k} \neq 0\}, \quad (33)$$

where, as usual, we sum over the repeated index m . Since $\lambda \wedge r_1 = 0$ on this set, if $\lambda \neq 0$ we conclude in particular $A_{1k} = 0$ almost everywhere on $J \cap \{A_{1k} \neq 0\}$. Thus we have shown:

$$\lambda \neq 0 \implies A_{1k} = 0 \text{ almost everywhere on } J. \quad (34)$$

But if $\lambda = 0$, then we would deduce $M = 0$ by combining (32) with (30).

Let us prove part (iii). With f defined in analogy to part (ii), there exist $M \in \mathfrak{so}(3)$ and $\lambda \in \mathbb{R}^3$, not both zero, such that

$$M + \lambda \wedge \gamma = f r_2 \wedge r_3 \text{ on } J. \quad (35)$$

If $f = 0$ almost everywhere on J , then $\lambda \neq 0$ (since otherwise $M = 0$) and $\lambda \wedge r_1 = 0$ on J . This implies that $A_{12} = A_{13} = 0$ almost everywhere on J , which is the conclusion with $a = 1$.

So it remains to consider the case when f does not vanish identically. Taking the derivative in (35), we deduce

$$\lambda \wedge r_1 = f' r_2 \wedge r_3 + f (A_{13} r_1 \wedge r_2 - A_{12} r_1 \wedge r_3) \text{ on } J. \quad (36)$$

Hence $f' = 0$ on J (hence f is a nonzero constant on J) and

$$\begin{aligned} \lambda \cdot r_2 &= -f A_{13} \\ \lambda \cdot r_3 &= f A_{12} \end{aligned} \quad (37)$$

on J . Since f is a nonzero constant on J , the vector $\Lambda = \frac{\lambda}{f}$ is nonzero and constant, and (37) is satisfied with $f = 1$ and Λ instead of λ . If $\lambda = 0$ then $A_{12} = A_{13} = 0$ by (37) on J , which as we saw above is the conclusion.

Let us therefore assume that $\lambda \neq 0$, so $\Lambda \neq 0$. Returning to (36), we see that

$$\Lambda \times r_1 = A_{13} r_3 + A_{12} r_2 = r_1' \quad (38)$$

where $\times$ denotes the cross product in $\mathbb{R}^3$. We conclude that r_1 satisfies the equation (19) for a precession about the axis Λ . Formula (21) follows by applying first a rotation taking e_1 into $\Lambda/|\Lambda|$ and afterwards possibly rotating about Λ .

Equation (19) implies that

$$a = \frac{\Lambda}{|\Lambda|} \cdot r_1$$

is constant on J , and after possibly reflecting Λ we may assume that $a \geq 0$.

Projecting (38) onto r_2 and r_3 we recover (37). Using (37) we find

$$\begin{aligned}(\Lambda \cdot r_2)' &= -|\Lambda|aA_{12} = -|\Lambda|a(\Lambda \cdot r_3) \\ (\Lambda \cdot r_3)' &= -|\Lambda|aA_{13} = |\Lambda|a(\Lambda \cdot r_2).\end{aligned}\quad (39)$$

This implies that there are (say, positive) constants $c_1, c_2 \in \mathbb{R}$ such that

$$-A_{13} + iA_{12} = \Lambda \cdot (r_2 + ir_3) = c_2 e^{i(|\Lambda|at - c_1)}.$$

Finally, notice that $|\Lambda|^2 = \sum_i (\Lambda \cdot r_i)^2 = |\Lambda|^2 a^2 + c_2^2$, so that $c_2 = |\Lambda| \sqrt{1 - a^2}$.

Let us prove part (i). First let $(j, k) = (2, 3)$ or $(j, k) = (3, 2)$ and $\mathfrak{A} = \text{span}\{e_1 \wedge e_k\}$.

Recall that in this case r_j is constant and $\lambda \perp r_j$ and $M \in \mathfrak{A} = \text{span}\{r_1 \wedge r_k\}$. For suitable $f_1, f_2 \in W^{1,2}(J)$ we have

$$M + \lambda \wedge \gamma = f_1 r_1 \wedge r_j + f_2 r_k \wedge r_j \text{ on } J. \quad (40)$$

This equation shows that $M = 0$ if $\lambda = 0$. Hence $\lambda \neq 0$. Taking derivatives, we find

$$\lambda \wedge r_1 = f_1'(r_1 \wedge r_j) + f_2'(r_k \wedge r_j) + A_{1k} f_1 r_k \wedge r_j - A_{1k} f_2 r_1 \wedge r_j \text{ (no sum) on } J,$$

because r_j is constant. Thus $\lambda \cdot r_k = 0$ on J and therefore r_1 is parallel to λ on J . Moreover, taking derivatives we find $A_{1k} \lambda \cdot r_1 = 0$ on J . Since $\lambda \neq 0$ we conclude that $A_{1k} = 0$ on J .

Let us now address the case $\mathfrak{A} = \text{span}\{e_2 \wedge e_3\}$. Then r_1 is constant and so $\mathfrak{A} = \text{span}\{r_2 \wedge r_3\}$. Since in this case degeneracy implies $\lambda = 0$, we must have $M \neq 0$. Since by degeneracy $M \in \{r_2 \wedge r_3\}^\perp$, we obtain a contradiction (to our assumption that J has positive measure). $\square$

From the proof we can in fact draw more conclusions than those stated in Proposition 3.3 (with notation as in its proof):

For $\mathfrak{A} = \text{span}\{e_1 \wedge e_k\}$ in fact we have shown that $r_1 \parallel \lambda$ on J . Recalling that $\gamma \cdot r_1 \times r_k = 0$ we can insert this into (40) to conclude that $M + \lambda \wedge \gamma = 0$ on J ; in particular, $\lambda \cdot r_k$ is constant and $\gamma - \gamma(t_0)$ is parallel to λ on J , for any $t_0 \in J$. Similarly, in the case $\mathfrak{A} = \mathfrak{so}(3)$ we can use (30) (which for $\mathfrak{A} = \mathfrak{so}(3)$ holds with $f = 0$) to deduce that $r_1 \parallel \lambda$ on J and, from (30), that $\gamma - \gamma(t_0)$ is parallel to λ on J for any $t_0 \in J$.

In the case $\mathfrak{A} = \text{span}\{e_1 \wedge e_k, e_2 \wedge e_3\}$ we can use $A_{1k} = 0$ in (31) to obtain the system

$$\begin{aligned}f' &= -\lambda \cdot r_j \\ A_{jk} f &= -\lambda \cdot r_k\end{aligned}$$

almost everywhere on J . So for instance we see that A_{jk} is not arbitrary; in particular, it agrees with a continuous function on the relatively open set $J \cap \{f \neq 0\}$.

In view of these remarks only a partial converse to Proposition 3.3 can be expected. In particular, the last observation shows that in Proposition 3.9 below we cannot allow A to take arbitrary values in $\mathfrak{so}(3)$ in the cases when $\dim \mathfrak{A} = 2$.

Proposition 3.9 *Let $J \subset I$ be an interval, let $A \in L^2(I, \mathfrak{A})$ and assume that one of the following conditions is true:*

- (i) $\dim \mathfrak{A} \geq 2$, $e_2 \wedge e_3 \in \mathfrak{A}$ and $A_{12} = A_{13} = 0$ almost everywhere on J .
- (ii) $\mathfrak{A} = \text{span}\{e_1 \wedge e_2, e_1 \wedge e_3\}$ and there is a constant vector $\Lambda \in \mathbb{R}^3$ such that (19) is satisfied almost everywhere on J .
- (iii) $k \in \{2, 3\}$ and $\mathfrak{A} = \text{span}\{e_1 \wedge e_k\}$ and $A = 0$ almost everywhere on J .

Then A is degenerate with respect to $\mathfrak{A}$ on J .

Proof We use the same notation as in the proof of Proposition 3.3. Set $t_0 = \inf J$.

We begin with part (i). Since we assumed that A takes values in $\mathfrak{A}$, we have $A_{12} = A_{13} = 0$ on J . Since J is an interval, this implies that r_1 is constant on J . We set $\lambda = r_1(t_0)$. Then

$$(\lambda \wedge \gamma)' = \lambda \wedge r_1 = \lambda \wedge r_1(t_0) = 0 \text{ on } J.$$

Hence there is an $M \in \mathfrak{so}(3)$ such that $M + \lambda \wedge \gamma = 0$ on J .

In case (ii) notice that (19) implies that $\Lambda \wedge r_1 = (r_2 \wedge r_3)'$. Hence there is an $M \in \mathfrak{so}(3)$ such that $M + \Lambda \wedge \gamma = r_2 \wedge r_3$ on J ; in particular, we cannot have $M = 0$ and $\Lambda = 0$ because the right-hand side is not zero. Thus A is degenerate.

In case (iii) notice that r is constant on J and therefore all $r_i \wedge r_j$ are constant. Setting $\lambda = r_1(t_0)$ we have $\lambda \in \Gamma$. Moreover, $\lambda \wedge r_1 = 0$ on J and therefore by integration $\lambda \wedge (\gamma - \gamma(t_0)) = 0$ on J . As in Remark (i) to Definition 3.1 we see that $\lambda \wedge \gamma(t_0) \in \mathfrak{A}$. $\square$

3.5 Euler–Lagrange equations

In this section we derive Euler–Lagrange equations for minimizers of functionals on $L^2(I, \mathfrak{A})$ under the boundary conditions considered in earlier sections.

In order to define a suitable notion of minimizers, let $\mathfrak{A}$ as in Definition 3.1. We will say that A is a minimizer of some functional $\mathcal{F} : L^2(I, \mathfrak{A}) \rightarrow (-\infty, \infty]$ under its own boundary conditions on $(t_0, t_1) \subset I$ if $\mathcal{F}(A) \leq \mathcal{F}(\tilde{A})$ for all $\tilde{A} \in L^2(I, \mathfrak{A})$ satisfying $\tilde{A} = A$ on $I \setminus (t_0, t_1)$ as well as

$$\begin{aligned} r_{\tilde{A}}(t_1) &= r_A(t_1) \\ \int_{t_0}^{t_1} r_{\tilde{A}}^T e_1 &= \int_{t_0}^{t_1} r_A^T e_1. \end{aligned}$$

The main result of this section is as follows.

Proposition 3.10 *Let $Q \in C^1(\mathfrak{A})$ be a nonnegative function satisfying*

$$|\nabla Q(A)| \leq C(1 + |A|^2) \text{ for all } A \in \mathfrak{A}, \quad (41)$$

and define $\mathcal{F} : L^2(I, \mathfrak{A}) \rightarrow [0, \infty]$ by

$$\mathcal{F}(\tilde{A}) = \int_I Q(\tilde{A}).$$

Let $A \in L^2(I, \mathfrak{A})$ be such that $\mathcal{F}(A) < \infty$. Assume, moreover, that there is a nonempty open interval $J \subset I$ such that A is a nondegenerate minimizer of $\mathcal{F}$ under its own boundary conditions on J . Then $\nabla Q(A) \in W^{1,2}(J, \mathfrak{A})$ and there exist $M \in \mathfrak{A}$ and $\lambda \in \Gamma$ such that

$$r_A^T \nabla Q(A) r_A + M + \lambda \wedge \int_t^1 r_A^T e_1 \in (r_A^T \mathfrak{A} r_A)^\perp \text{ on } J. \quad (42)$$

Remarks.

- (i) Proposition 3.10 is a Lagrange multiplier rule. The rule fails when A is degenerate. However, we understand this situation well due to Proposition 3.3.

(ii) Let

$$\mathcal{I}_{\mathfrak{A}} = \{(i, j) \in \{1, 2, 3\}^2 : i < j \text{ and } e_i \wedge e_j \in \mathfrak{A}\}$$

and let f be the function defined by

$$f(A_{12}, A_{13}, A_{23}) = Q\left(\sum_{(i,j) \in \mathcal{I}_{\mathfrak{A}}} A_{ij} e_i \wedge e_j\right).$$

We will abuse notation by writing $f(A)$ instead of $f(A_{12}, A_{13}, A_{23})$. Writing $r_i = r_A^T e_i$ and $\gamma(t) = \int_t^1 r_1$, the conclusion of Proposition 3.10 can be stated as follows:

There exist $M \in \widetilde{\mathfrak{A}}$, $\lambda \in \Gamma$ and $H_{ij} \in W^{1,2}(I)$ such that on J we have

$$M + \lambda \wedge \gamma = - \sum_{(i,j) \in \mathcal{I}_{\mathfrak{A}}} \frac{\partial f}{\partial A_{ij}}(A) r_i \wedge r_j + \sum_{(i,j) \in \mathcal{I}_{\widetilde{\mathfrak{A}}} \setminus \mathcal{I}_{\mathfrak{A}}} H_{ij} r_i \wedge r_j. \quad (43)$$

In fact, (42) implies that

$$\sum_{\mathcal{I}_{\mathfrak{A}}} \frac{\partial f}{\partial A_{ij}}(A) r_i \wedge r_j + M + \lambda \wedge \gamma \perp r_k \wedge r_\ell \text{ for all } (k, \ell) \in \mathcal{I}_{\mathfrak{A}} \quad (44)$$

on J ; notice that we can absorb any nonzero constant factor into M and λ . Since $M + \lambda \wedge \gamma$ takes values in $\widetilde{\mathfrak{A}}$, there exist coefficients $H_{ij} \in W^{1,2}(I)$ such that

$$M + \lambda \wedge \gamma = \sum_{\mathcal{I}_{\mathfrak{A}}} H_{ij} r_i \wedge r_j. \quad (45)$$

Equation (44) can be restated as

$$\frac{\partial f}{\partial A_{ij}}(A) = -H_{ij} \text{ for all } (i, j) \in \mathcal{I}_{\mathfrak{A}}. \quad (46)$$

In particular $\frac{\partial f}{\partial A_{ij}}(A) \in W^{1,2}(I)$, hence $\nabla Q(A) \in W^{1,2}(I)$, and equation (43) also follows at once.

Proof of Proposition 3.10 We may assume without loss of generality that $J = I$. Let $A \in L^2(I, \mathfrak{A})$ be such that $\mathcal{F}(A) < \infty$. For every $\widehat{A} \in L^\infty(I, \mathfrak{A})$ we have $\mathcal{F}(A + \widehat{A}) < \infty$, due to the growth condition on ∇Q . Moreover, since $Q \in C^1$, for each $R > 0$ the expression

$$\eta_R(\delta) = \sup_{|\widehat{A}| \leq R, |B| \leq \delta} |\nabla Q(\widetilde{A} + B) - \nabla Q(\widetilde{A})|$$

converges to zero as $\delta \downarrow 0$. For every $\widehat{A} \in L^\infty(I, \mathfrak{A})$ with $\|\widehat{A}\|_{L^\infty} \leq \delta \leq 1$ and every $R > 0$ we have

$$\left| \int_I Q(A + \widehat{A}) - \int_I Q(A) - \int_I \nabla Q(A) : \widehat{A} \right| \leq \left(\eta_R(\delta) + C \int_{|A| > R} (1 + |A|^2) \right) \delta,$$

due to (41). Choosing R large enough and then δ small enough, the first factor on the right-hand side can be made arbitrarily small.

Hence the function $\widetilde{\mathcal{F}} : L^\infty(I, \mathfrak{A}) \rightarrow \mathbb{R}$ defined by $\widetilde{\mathcal{F}}(\widehat{A}) = \mathcal{F}(A + \widehat{A})$ is Fréchet differentiable at 0, with

$$D\widetilde{\mathcal{F}}(0)\widehat{A} = \int_I \nabla Q(A) : \widehat{A}. \quad (47)$$

Define $\tilde{\mathcal{G}} : L^\infty(I, \mathfrak{A}) \rightarrow G \times \Gamma$ by setting $\tilde{\mathcal{G}}(\hat{A}) = \mathcal{G}(A + \hat{A})$. Then $\tilde{\mathcal{F}}$ and $\mathcal{P} = \tilde{\mathcal{G}}$ satisfy the hypotheses of Corollary 2.5, with $H = L^\infty(I, \mathfrak{A})$ and V some neighbourhood of $x = 0$ in H . The surjectivity of $D\tilde{\mathcal{G}}(0)$ follows from Lemma 3.6.

We conclude that there exists a unique Λ in the tangent space to $G \times \Gamma$ at $(r_A(1), \int_0^1 r_A^T e_1)$ such that $D\tilde{\mathcal{F}}(0) + \Lambda \cdot D\tilde{\mathcal{G}}(0) = 0$. By (47) and since $D\tilde{\mathcal{G}}(0) = D\mathcal{G}(A)$, this amounts to

$$\int_I \nabla Q(A) : \hat{A} + \Lambda \cdot D\mathcal{G}(A)\hat{A} = 0 \quad \text{for all } \hat{A} \in L^\infty(I, \mathfrak{A}). \quad (48)$$

The existence of M and λ satisfying (42) now follows from (48) as in the proof of Lemma 3.6. $\square$

4 Applications

4.1 Smooth approximation of framed curves with clamped ends

In this section we construct smooth approximations for framed curves which preserve clamped boundary conditions. Results of this kind can be useful in proving the convergence of numerical schemes, cf. [4]. We will prove the following result.

Theorem 4.1 *Let $\mathfrak{A}$ be as in Definition 3.1. Assume that $Q : \mathfrak{A} \rightarrow [0, \infty)$ is continuous and define $\mathcal{F}_Q : L^2(I, \mathfrak{A}) \rightarrow [0, \infty]$ by*

$$\mathcal{F}_Q(A) = \int_I Q(A).$$

Then for all $A \in L^2(I, \mathfrak{A})$ with $\mathcal{F}_Q(A) < \infty$ there exist $A_n \in C^\infty(\bar{I}, \mathfrak{A})$ such that $A_n \rightarrow A$ strongly in $L^2(I, \mathbb{R}^{3 \times 3})$ and $\mathcal{F}_Q(A_n) \rightarrow \mathcal{F}_Q(A)$ and, moreover,

$$r_{A_n}(1) = r_A(1) \text{ and } \int_0^1 r_{A_n}^T e_1 = \int_0^1 r_A^T e_1 \quad \text{for all } n \in \mathbb{N}.$$

Remarks.

- (i) According to Lemma 3.4, strong convergence of (A_n) in L^2 implies strong convergence of (r_{A_n}) in $W^{1,2}$.
- (ii) The case $\mathfrak{A} = \mathfrak{so}(2)$ is implicit in the degenerate cases when $\dim \mathfrak{A} = 1$.

In [10] a model for ribbons was obtained which generalises the model from [25]. Both models are governed by an energy functional of the form considered in Proposition 4.1, with $\mathfrak{A} = \text{span}\{e_1 \wedge e_3, e_2 \wedge e_3\}$. Proposition 4.1 therefore is a density result for ribbons with prescribed boundary conditions.

In [16] framed curves with $\mathfrak{A} = \text{span}\{e_1 \wedge e_2, e_1 \wedge e_3\}$ were used to describe developable surfaces and a smooth approximation result was obtained using similar ideas. However, the boundary conditions considered there were different and the length of the curves was not fixed.

Lemma 4.2 *Let L be an index set and let $\{J_i\}_{i \in L}$ be a family of open subintervals of I such that $I = \bigcup_{i \in L} J_i$. Let $A \in L^2(I, \mathfrak{A})$ and assume that A is degenerate with respect to $\mathfrak{A}$ on J_i , for each $i \in L$. Then A is degenerate for $\mathfrak{A}$ on I .*

Proof As usual, we write $r_i = r_A^T e_i$. Since the J_i cover I , the claim follows if we show that whenever $J_1, J_2 \in \{J_i\}_{i \in L}$ are such that $J = J_1 \cap J_2$ is nonempty, then A is degenerate on $J_1 \cup J_2$.

Let us first consider the case when for a $k \in \{2, 3\}$ we have $e_1 \wedge e_k, e_2 \wedge e_3 \in \mathfrak{A}$. Then Proposition 3.3 implies that $A_{12} = A_{13} = 0$ on $J_1 \cup J_2$. Proposition 3.9 then implies that A is degenerate with respect to $\mathfrak{A}$ on the interval $J_1 \cup J_2$.

Now let us consider the case when $\mathfrak{A}$ is as in Proposition 3.3 (iii). By definition of $\mathfrak{A}$ we have $A_{23} = 0$ on $J_1 \cup J_2$. By Proposition 3.3 (iii) there exist $\Lambda^{(1)}, \Lambda^{(2)} \in \mathbb{R}^3 \setminus \{0\}$ such that $r'_1 = \Lambda^{(i)} \times r_1$ on J_i . If $\Lambda^{(1)} = \Lambda^{(2)}$ then Proposition 3.9 implies that A is degenerate with respect to $\mathfrak{A}$ on $J_1 \cup J_2$. If $\Lambda^{(1)} \neq \Lambda^{(2)}$ then $(\Lambda^{(1)} - \Lambda^{(2)}) \times r_1 = 0$ on J implies that r_1 is constant on J . Since J is a nontrivial subinterval of J_1 and of J_2 and since (20) applies on each of them, we conclude that r_1 is constant on $J_1 \cup J_2$. Hence A is degenerate with respect to $\mathfrak{A}$ on $J_1 \cup J_2$, by Proposition 3.9.

Finally, if $\dim \mathfrak{A} = 1$ then Proposition 3.3 implies that $e_2 \wedge e_3 \notin \mathfrak{A}$ and $A = 0$ on $J_1 \cup J_2$. And then Proposition 3.9 implies that A is degenerate with respect to $\mathfrak{A}$ on $J_1 \cup J_2$. $\square$

Lemma 4.3 *Let $J_0 \subset J_1 \subset J_2 \subset \dots$ be measurable subsets of I with $I = \bigcup_{n \in \mathbb{N}} J_n$. Let $A \in L^2(I, \mathfrak{A})$ be degenerate with respect to $\mathfrak{A}$ on J_n , for each $n \in \mathbb{N}$. Then A is degenerate with respect to $\mathfrak{A}$ on I .*

Proof Set $r = r_A$ and define $\gamma(t) = -\int_t^1 r^T e_1$. We may assume that each J_n has positive measure. As usual, Proposition 3.3 shows that $\mathfrak{A} \neq \text{span}\{e_2 \wedge e_3\}$.

Since A is degenerate with respect to $\mathfrak{A}$ on each J_n , for each n there exist $M_n \in \tilde{\mathfrak{A}}$ and $\lambda_n \in \Gamma$, not both zero, such that the map $F_n : I \rightarrow \mathfrak{so}(3)$ given by $F_n = M_n + \lambda_n \wedge \gamma$ satisfies

$$F_n \in \left(r^T \mathfrak{A} r\right)^\perp \text{ on } J_n. \quad (49)$$

According to Proposition 3.3 we have $\lambda_n \neq 0$, so after dividing M_n and λ_n by $|\lambda_n|$ we may assume that $\lambda_n \in \mathbb{S}^2$. Hence there exists $\lambda \in \mathbb{S}^2 \cap \Gamma$ such that, after passing to a subsequence (not relabelled), we have $\lambda_n \rightarrow \lambda$ as $n \rightarrow \infty$.

Let us first consider the case when (M_n) has a bounded subsequence. After possibly taking a further subsequence we may assume that there exists $M \in \tilde{\mathfrak{A}}$ such that $M_n \rightarrow M$ as $n \rightarrow \infty$.

Define $F = M + \lambda \wedge \gamma$. Then $F_n \rightarrow F$ uniformly on I . Hence (49) implies that $F \in (r^T \mathfrak{A} r)^\perp$ almost everywhere on I . Since $\lambda \in \Gamma \setminus \{0\}$ and $M \in \tilde{\mathfrak{A}}$, this means that A is degenerate with respect to $\mathfrak{A}$ on all of I .

It remains to consider the case when (M_n) does not admit a bounded subsequence. Then we may assume that $M_n \neq 0$ and we define

$$F_n = \frac{M_n}{|M_n|} + \frac{\lambda_n \wedge \gamma}{|M_n|}.$$

Then F_n satisfies (49) and there exists an $M \in \tilde{\mathfrak{A}}$ such that, after passing to a subsequence,

$$\lim_{n \rightarrow \infty} \frac{M_n}{|M_n|} = M.$$

Since $\lambda_n \in \mathbb{S}^2$ and γ is bounded, we conclude that $F_n \rightarrow M$ uniformly. As before, we conclude that A is degenerate with respect to $\mathfrak{A}$ on all of I . (As $\lambda = 0$, Proposition 3.3 leads to a contradiction, showing that in fact (M_n) always admits a bounded subsequence.) $\square$

Proof of Theorem 4.1 Let $A \in L^2(I, \mathfrak{A})$. We may assume without loss of generality that $\mathfrak{A}$ the smallest among the spaces allowed on the right-hand side of (14) with the property that $A \in \mathfrak{A}$ almost everywhere on I .

Let us first consider the case when A is not degenerate with respect to $\mathfrak{A}$ on all of I .

Step 1 (Truncation). Set $J_n = \{|A| \leq n\}$. Since A is not degenerate on all of I , Lemma 4.3 implies that there exists an $N \in \mathbb{N}$ such that A is not degenerate with respect to $\mathfrak{A}$ on

$$J = \{|A| \leq N\};$$

we may also assume that J has positive measure. Denote by $\tilde{E}_0$ the space of measurable functions $I \rightarrow \mathfrak{A}$ supported on J and attaining only finitely many values.

For all $m \in \mathbb{N}$ and all $t \in I$ define

$$\tilde{A}_m(t) = \begin{cases} A(t) & \text{if } |A(t)| \leq m \\ 0 & \text{if } |A(t)| > m. \end{cases}$$

By Theorem 3.2 there is a finite dimensional subspace E_0 of $\tilde{E}_0$ and there are $\hat{A}_m \in E_0$ converging to zero uniformly as $m \uparrow \infty$ and such that $A_m = \tilde{A}_m + \hat{A}_m$ satisfies $\mathcal{G}(A_m) = \mathcal{G}(A)$ for all m . Clearly $A_m \rightarrow A$ strongly in $L^2(I)$. After taking a subsequence, we may also assume that A_m is not degenerate with respect to $\mathfrak{A}$ on I for any m . Moreover, for all $m \geq N$ we have $\tilde{A}_m = A$ on J and therefore

$$\begin{aligned} \left| \int_I Q(A) - \int_I Q(A_m) \right| &\leq \int_I |Q(A) - Q(\tilde{A}_m)| + \int_I |Q(\tilde{A}_m) - Q(A_m)| \\ &= \int_{\{A > m\}} |Q(A) - Q(0)| + \int_J |Q(A) - Q(A + \hat{A}_m)|. \end{aligned}$$

As $m \rightarrow \infty$ the first term converges to zero because the measure of $\{A > m\}$ converges to zero and $Q(A) \in L^1(I)$ because $\mathcal{F}_Q(A) < \infty$. The second term converges to zero because Q is uniformly continuous on the ball of radius $2N$ around the origin and $\hat{A}_m \rightarrow 0$ uniformly. Thus $Q(A + \hat{A}_m) \rightarrow Q(A)$ uniformly on J .

Step 2 (Mollification). In view of Step 1, we may assume from the outset and without loss of generality that $A \in L^\infty(I, \mathfrak{A})$. Let $A_n \in C^\infty(\bar{I}, \mathfrak{A})$ be such that $|A_n| \leq 2\|A\|_{L^\infty(I)}$ on I and such that $A_n \rightarrow A$ strongly in all $L^p(I)$ with $p \geq 2$.

Let $r > 0$ be small. By Lemma 4.2 there exists an open interval $J \subset I$ of length r such that A is not degenerate with respect to $\mathfrak{A}$ on J . The space $\tilde{E}_1 = C_0^\infty(J, \mathfrak{A})$ is L^2 -dense in (16). Hence by Theorem 3.2 there exist $\hat{A}_n$ belonging to a finite dimensional subspace E_1 of $\tilde{E}_1$ such that $\hat{A}_n \rightarrow 0$ uniformly and such that $A_n = \tilde{A}_n + \hat{A}_n$ satisfies $\mathcal{G}(A_n) = \mathcal{G}(A)$ for all n . Clearly $A_n \rightarrow A$ strongly in $L^2(I)$. Since Q is locally bounded, $Q(A_n)$ is dominated by a constant and therefore $\int_I Q(A_n) \rightarrow \int_I Q(A)$ by dominated convergence.

Step 3 (Degenerate cases). Let us now consider the case when A is degenerate with respect to $\mathfrak{A}$ on all of I . If $\dim \mathfrak{A} \geq 2$ then

$$\mathfrak{A} = \text{span}\{e_1 \wedge e_2, e_1 \wedge e_3\}. \quad (50)$$

In fact, otherwise Proposition 3.3 together with the fact that $A \in \mathfrak{A}$ almost everywhere would imply that $A_{12} = A_{13} = 0$, contradicting the minimality of $\mathfrak{A}$.

If A is degenerate with respect to (50), then Proposition 3.3 implies that (20) holds on all of I . If, on the other hand, $\dim \mathfrak{A} = 1$ then Proposition 3.3 implies that $A = 0$ almost everywhere on I . In both cases we already had $A \in C^\infty(\bar{I}, \mathfrak{A})$ to begin with. $\square$

4.2 Shape design for rods with clamped boundary conditions

As a simple application of Theorem 3.2, in this section we consider an optimal design problem for rods. In [17] the relaxed problem was derived by considering the Euler–Lagrange equations; the proof was similar to the one given in [1] for the linear case.

Here, instead, we prove Γ -convergence of the functionals and, in addition, we impose clamped boundary conditions at both ends of the rod. More precisely, let $r^{(1)} \in G$ and $\gamma^{(1)} \in \Gamma$ be given. For simplicity let us assume that, in the case when $\mathfrak{A} = \text{span}\{e_1 \wedge e_2, e_1 \wedge e_3\}$, these prescribed boundary conditions $r^{(1)}$ and $\gamma^{(1)}$ are not satisfied by any helix (20) with the initial data $r(0) = I$ and $\gamma(0) = 0$.

Let $0 < a < b$, let $\mathcal{L} : \mathfrak{so}(3) \rightarrow \mathfrak{so}(3)$ be linear and positive definite and for a given $B \in L^\infty(I; [a, b])$ define $\mathcal{F}^{(B)} : L^2(I, \mathfrak{A}) \rightarrow [0, \infty]$ by

$$\mathcal{F}^{(B)}(A) = \begin{cases} \int_I BA : \mathcal{L}A & \text{if } r_A(1) = r^{(1)} \text{ and } \int_0^1 r_A^T e_1 = \gamma^{(1)} \\ +\infty & \text{otherwise.} \end{cases}$$

The number $B(t) \in [a, b]$ models the ‘hardness’ of the material at the point $t \in I$. The aim is to find the material distribution B which is optimal in the sense that it minimizes some compliance functional

$$B \mapsto J(B, A_B). \quad (51)$$

Here A_B is a suitable minimizer of $\mathcal{F}^{(B)}$ among all admissible $A \in L^2(I, \mathfrak{A})$, i.e., those satisfying $r_A(1) = r^{(1)}$ as well as $\int_0^1 r_A^T e_1 = \gamma^{(1)}$.

The minimization of (51) commonly requires considering sequences $(B_n) \subset L^\infty(I; [a, b])$ and understanding the asymptotic behaviour of the corresponding A_{B_n} . This motivates the following proposition, which addresses the asymptotic behaviour of the functionals $\mathcal{F}^{(B_n)}$. Since it is a Γ -convergence result, it remains true if we add a lower order functional to $\mathcal{F}^{(B)}$. Its proof is closely related to an argument for a similar problem without clamped boundary conditions; see e.g. [24].

Proposition 4.4 *Let $B_n : I \rightarrow [a, b]$ be measurable and assume that $B_n^{-1} \xrightarrow{*} (B^*)^{-1}$ weakly- $*$ in $L^\infty(I)$. Then the Γ -limit of the functionals $\mathcal{F}^{(B_n)} : L^2(I, \mathfrak{A}) \rightarrow [0, \infty]$ with respect to weak convergence in $L^2(I, \mathfrak{A})$ agrees with the functional $\mathcal{F}^{(B^*)} : L^2(I, \mathfrak{A}) \rightarrow [0, \infty]$.*

Proof As usual, we define $\mathcal{G}$ as in (22) and write the boundary conditions as $\mathcal{G}(A) = \mathcal{G}_0$, where $\mathcal{G}_0 = (r^{(1)}, \gamma^{(1)})$.

Let us begin by proving the upper bound. Let A be such that $\mathcal{F}^{(B^*)}(A) < \infty$, so in particular $\mathcal{G}(A) = \mathcal{G}_0$. Let $\hat{A}_n \in L^\infty(I, \mathfrak{A})$ be such that $\hat{A}_n \rightarrow 0$ strongly in L^∞ , and define

$$A_n = \frac{B^*}{B_n} A + \hat{A}_n.$$

Clearly $A_n \in L^2(I, \mathfrak{A})$ and $A_n \rightharpoonup A$ weakly in L^2 . Moreover,

$$\mathcal{F}^{(B_n)}(A_n) = \int (B^* A + B_n \hat{A}_n) : \mathcal{L}A_n \rightarrow \mathcal{F}^{(B^*)}(A),$$

because $B_n \hat{A}_n \rightarrow 0$ strongly in L^∞ and $\mathcal{L}A_n \rightharpoonup \mathcal{L}A$ weakly in L^2 .

It remains to show that the $\hat{A}_n$ can be chosen in such a way that $\mathcal{G}(A_n) = \mathcal{G}_0$ for all n .

If A is not degenerate with respect to $\mathfrak{A}$ on I , then the existence of such $\hat{A}_n$ follows from Theorem 3.2, because $L^\infty(I, \mathfrak{A})$ is dense in $L^2(I, \mathfrak{A})$.

If $e_2 \wedge e_3 \in \mathfrak{A}$ and A is degenerate with respect to $\mathfrak{A}$ on I , then Proposition 3.3 implies that $A_{12} = A_{13} = 0$ on I . Hence $A \in \mathfrak{A}_0 = \text{span}\{e_2 \wedge e_3\}$ almost everywhere. Proposition 3.3 shows that such A are never degenerate with respect to $\mathfrak{A}_0$. Hence Theorem 3.2 again implies the existence of suitable $\hat{A}_n$ (taking values only in $\mathfrak{A}_0$).

If $\mathfrak{A} = \text{span}\{e_1 \wedge e_k\}$ and A is degenerate with respect to $\mathfrak{A}$ on I , then Proposition 3.3 implies that $A = 0$ on I . Then we can simply set $\hat{A}_n = 0$ as well.

If $\mathfrak{A} = \text{span}\{e_1 \wedge e_2, e_1 \wedge e_3\}$ and A were degenerate with respect to $\mathfrak{A}$ on I , then by Proposition 3.3 it would have to define a helix, which we have excluded by our choice of $r^{(1)}$ and $\gamma^{(1)}$.

It remains to prove the lower bound, which is standard: assume that $A_n \rightarrow A$ weakly in $L^2(I, \mathfrak{A})$ and that $\mathcal{F}^{(B_n)}(A_n) < \infty$. In particular, $\mathcal{G}(A_n) = \mathcal{G}_0$ for all n . Hence $\mathcal{G}(A) = \mathcal{G}_0$ because $\mathcal{G}$ is weakly continuous. Set

$$q_n = A_n - \frac{B^*}{B_n} A.$$

Then $q_n \rightarrow 0$ weakly in L^2 because both terms on the right converge to A weakly in L^2 . Hence

$$\begin{aligned} \mathcal{F}^{(B_n)}(A_n) &= \int (B^* A + B_n q_n) : \mathcal{L} A_n \\ &= \int B^* A : \mathcal{L} A_n + \int B_n q_n : \mathcal{L} q_n + \int q_n : \mathcal{L} B^* A \\ &\geq \int B^* A : \mathcal{L} A_n + \int q_n : \mathcal{L} B^* A. \end{aligned}$$

The right-hand side converges to $\mathcal{F}^{(B^*)}(A)$ because $q_n \rightarrow 0$ and $\mathcal{L} A_n \rightarrow \mathcal{L} A$ weakly in L^2 . $\square$

The relaxation for planar rods with clamped boundary conditions is a particular case of Proposition 4.4. In order to relate to the setting in [17], we introduce the space

$$X = \{K \in W^{1,2}(I) : K(0) = 0\}$$

and for any $B : I \rightarrow [a, b]$, $g^{(1)} \in B_1(0)$ and $K^{(1)} \in \mathbb{R}$ we define the functionals

$$\mathcal{F}_{2d}^{(B)}(K) = \begin{cases} \int_I B(K')^2 & \text{if } K(1) = K^{(1)} \text{ and } \int_0^1 e^{iK} = g^{(1)} \\ +\infty & \text{otherwise.} \end{cases}$$

Corollary 4.5 *Let B^* and B_n be as in Proposition 4.4. Then $\mathcal{F}_{2d}^{(B_n)}$ Γ -converge to $\mathcal{F}_{2d}^{(B^*)}$ with respect to weak $W^{1,2}$ -convergence.*

Proof Set

$$r^{(1)} = \begin{pmatrix} \cos K^{(1)} & 0 & -\sin K^{(1)} \\ 0 & 1 & 0 \\ \sin K^{(1)} & 0 & \cos K^{(1)} \end{pmatrix}$$

and $\gamma^{(1)} = (g^{(1)} \cdot e_1, 0, g^{(1)} \cdot e_2)^T$ and let $\mathcal{L}$ denote the identity. Then

$$\mathcal{F}_{2d}^{(B)}(K) = \frac{1}{2} \mathcal{F}^{(B)}(K' \cdot e_1 \wedge e_2).$$

Hence the claim follows from Proposition 4.4. $\square$

4.3 Euler–Lagrange equation for clamped ribbons

The functional for ribbons derived in [10] is defined on $L^2(I, \mathfrak{A})$ with $\mathfrak{A} = \text{span}\{e_1 \wedge e_3, e_2 \wedge e_3\}$. For isotropic materials it is, up to a constant prefactor, of the form $\mathcal{F}(A) = \int_I Q(A)$, with $Q(A) = f(A_{13}, A_{23})$ and $f : \mathbb{R}^2 \rightarrow [0, \infty)$ defined by

$$f(x_1, x_2) = \begin{cases} \frac{|x_1|^4}{x_1^2} & \text{if } |x_1| > |x_2| \\ 4x_2^2 & \text{if } |x_1| \leq |x_2|. \end{cases} \quad (52)$$

A formal derivation of the Euler–Lagrange equation for this functional can be found in [23]. Applying Proposition 3.10, we obtain the following result.

Corollary 4.6 *Let $\mathfrak{A} = \text{span}\{e_1 \wedge e_3, e_2 \wedge e_3\}$ and let $\mathcal{F} : L^2(I, \mathfrak{A}) \rightarrow \mathbb{R}$ be defined by $\mathcal{F}(A) = \int_I f(A_{13}, A_{23})$, with f as in (52). Let $A \in L^2(I, \mathfrak{A})$ be a nondegenerate minimizer for $\mathcal{F}$ under its own boundary conditions on I . Then there exist $M \in \mathfrak{so}(3)$ and $\lambda \in \mathbb{R}^3$ such that, with $H_{ij} : I \rightarrow \mathbb{R}$ defined by*

$$M + \lambda \wedge \left(\int_t^1 r_A^T e_1 \right) = \sum_{i < j} H_{ij} (r_A^T e_i) \wedge (r_A^T e_j),$$

and setting

$$\widehat{\theta} = \min \left\{ \frac{|A_{23}|}{|A_{13}|}, 1 \right\},$$

we have

$$(1 - \widehat{\theta}^4)A_{13} = -H_{13} \quad (53)$$

$$2(1 + \widehat{\theta}^2)A_{23} = -H_{23} \quad (54)$$

Proof We have $f \in C^1(\mathbb{R}^2)$ with

$$(\partial_1 f)(A_{13}, A_{23}) = \begin{cases} 2A_{13} \left(1 - \frac{A_{23}^4}{A_{13}^4} \right) & \text{if } |A_{13}| > |A_{23}| \\ 0 & \text{if } |A_{13}| \leq |A_{23}| \end{cases}$$

and

$$(\partial_2 f)(A_{13}, A_{23}) = \begin{cases} 4A_{23} \left(1 + \frac{A_{23}^2}{A_{13}^2} \right) & \text{if } |A_{13}| > |A_{23}| \\ 8A_{23} & \text{if } |A_{13}| \leq |A_{23}|. \end{cases}$$

Hence we can apply Proposition 3.10. Inserting these expressions into (46) we obtain (53) and (54) after absorbing a constant into M and λ . $\square$

A regularity analysis for solutions of the Euler–Lagrange equations is not the topic of the present paper. We only note one immediate consequence of (53) and (54).

Remarks.

- (i) The set $J^+ = \{t \in I : |A_{13}(t)| > |A_{23}(t)|\}$ is open and $A \in C^\infty(J^+)$. The proof is given below.
- (ii) On $J^- = \{t \in I : |A_{13}(t)| \leq |A_{23}(t)|\}$ we have $A_{23} = -\frac{H_{23}}{4}$. The right-hand side is a continuous function on I .

Proof of Remark (i) On J^+ we have $A_{13} \neq 0$; so $\theta = A_{23}/A_{13}$ is well-defined on this set. Observe that $\theta^2 = \widehat{\theta}^2$ on J^+ . Equation (53) implies that

$$J^+ = \{H_{13} \neq 0\}, \quad (55)$$

because on J^+ we have $A_{13} \neq 0$ and $\theta \neq 1$, so the left-hand side of (53) is nonzero. Since H_{13} is continuous, up to a null set J^+ agrees with an open set.

The function $\Psi : (-1, 1) \rightarrow \mathbb{R}$ given by $\Psi(s) = \frac{2s}{1-s^2}$ is strictly increasing and smooth. Hence it has a smooth inverse $\Psi^{-1} : \mathbb{R} \rightarrow (-1, 1)$. Dividing (54) by (53) we conclude that

$$\theta = \Psi^{-1}\left(\frac{H_{23}}{H_{13}}\right) \text{ on } J^+. \quad (56)$$

In particular, $\theta \in C^0(J^+)$. But then (53) and (54) imply that $A \in C^0(J^+)$. Hence $r_A \in C^1(J^+)$. Thus $H_{ij} \in C^1(J^+)$. Hence (56) implies that $\theta \in C^1(J^+)$ and then (53) and (54) imply that $A \in C^1(J^+)$. Continuing to bootstrap, we conclude that $A \in C^\infty(J^+)$. $\square$

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A note on the fractional Hardy inequality

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Abstract

We give a direct proof of fractional Hardy inequality by means of Littlewood–Paley decomposition and properties of singular homogeneous kernels of degree $-d$. A refinement when $q > 2$ is proved.

Keywords Hardy inequality · Littlewood–Paley decomposition · Fractional Sobolev spaces

Mathematics Subject Classification 46E35 · 39B62

The classical Hardy inequality states that when $d \geq 3$

$$\int_{\mathbb{R}^d} \frac{|u|^2}{|x|^2} dx \leq \frac{4}{(d-2)^2} \int_{\mathbb{R}^d} |\nabla u|^2 dx \quad (0.1)$$

and it is clearly of fundamental importance in analysis. There are of course many different proofs of (0.1), the simplest one consists in restrict by density to $D(\mathbb{R}^d \setminus \{0\})$, to observe that $\frac{1}{|x|^2} = -\frac{1}{2}x \cdot \nabla(\frac{1}{|x|^2})$, then to integrate by parts and eventually to apply Cauchy–Schwarz inequality.

A natural extension of (0.1) is in the framework of fractional Sobolev spaces $\dot{H}^s(\mathbb{R}^d)$. In this setting the following Hardy-type inequality holds

$$\int_{\mathbb{R}^d} \frac{|u|^2}{|x|^{2s}} dx \leq C \|f\|_{\dot{H}^s(\mathbb{R}^d)}^2, \quad (0.2)$$

provided that $0 \leq s < \frac{d}{2}$. For a compact and nice proof of (0.2) we quote Theorem 2.57 in [1] and the proof given by Tao in the Appendix of [16] while for an improvement involving Besov spaces we quote [2].

If one is interested in proving an L^q estimate for $\frac{|f|}{|x|^s}$ we need to recall the definition of the homogeneous Sobolev norm $\|f\|_{\dot{W}^{s,q}(\mathbb{R}^d)}$ which is defined as $\| |D|^s f \|_{L^q(\mathbb{R}^d)}$ where

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$(\widehat{|D|^s f})(\xi) = |2\pi\xi|^s \widehat{u}(\xi)$. In this note we give a direct proof and a refinement when $q > 2$ for the following class of Hardy-type inequalities that generalize the fractional Hardy inequality (0.2).

Theorem 0.1 (Fractional Hardy inequality) *Let $0 < s < \frac{d}{q}$, $1 < q < \infty$ and $f \in \dot{W}^{s,q}(\mathbb{R}^d)$, then*

$$\left\| \frac{f}{|x|^s} \right\|_{L^q(\mathbb{R}^d)} \leq C(d, s, q) \|f\|_{\dot{W}^{s,q}(\mathbb{R}^d)}. \quad (0.3)$$

The explicit value of the constant $C(d, s, q)$ in (0.3) is due to Herbst [11]. The proof of (0.3) goes back to the end of the fifties of the last century thanks to the work of Stein and Weiss [15] who proved an even more general version of (0.3) called Stein–Weiss inequality given by

$$\left(\int_{\mathbb{R}^d} (|T_\lambda f(x)| |x|^{-\beta})^q dx \right)^{\frac{1}{q}} \leq C(d, q, p, \lambda) \left(\int_{\mathbb{R}^d} (|f(x)| |x|^\alpha)^p dx \right)^{\frac{1}{p}} \quad (0.4)$$

where

$$T_\lambda f(x) = \int_{\mathbb{R}^d} \frac{f(y)}{|x-y|^\lambda} dy \quad 0 < \lambda < d,$$

and

$$0 < \lambda < d, 1 < p < \infty, \alpha < \frac{d}{p'}, p \leq q < \infty, \beta < \frac{d}{q}, \alpha + \beta \geq 0, \\ \frac{1}{q} = \frac{1}{p} + \left(\frac{\lambda + \alpha + \beta}{d} \right) - 1.$$

The fact that (0.4) implies (0.3) follows by the fact that $T_\lambda f = c|D|^{-s}f$, with $\lambda = d - s$, $c = \frac{\pi^{d/2} \Gamma((d-\lambda)/2)}{\Gamma(\lambda/2)}$ and choosing $p = q$ and $\alpha = 0$, $\beta = s$.

In order to state our result we recall the standard definition for Homogeneous Besov norm $\|\cdot\|_{\dot{B}_{p,q}^s}$ and Triebel–Lizorkin norm $\|\cdot\|_{\dot{F}_{p,q}^s}$ (see e.g. [8] for general references). Let f be a tempered distribution such that $\hat{f} \in L_{loc}^1$ and $P_N(f)$ the Littlewood–Paley projector on the dyadic frequency N , i.e. $\widehat{P_N(f)}(\xi) = \psi_N(\xi) \hat{f}(\xi)$ where $\psi_N(\xi) = \psi(\frac{\xi}{N})$ and $\sum_{N \in 2^{\mathbb{Z}}} \psi_N = 1$, then we define

$$\|f\|_{\dot{B}_{p,q}^s} = \left(\sum_{N \in 2^{\mathbb{Z}}} \|N^s P_N(f)\|_{L^p}^q \right)^{\frac{1}{q}}, \\ \|f\|_{\dot{F}_{p,q}^s} = \left\| \left(\sum_{N \in 2^{\mathbb{Z}}} |N^s P_N(f)(x)|^q \right)^{\frac{1}{q}} \right\|_{L^p}.$$

Our result is a direct proof of the following

Theorem 0.2 *Let $0 < s < \frac{d}{q}$, $1 < q < \infty$ then*

$$\left\| \frac{f}{|x|^s} \right\|_{L^q(\mathbb{R}^d)} \leq C(d, s, q) \|f\|_{\dot{B}_{q,q}^s(\mathbb{R}^d)}, \quad (0.5)$$

with the following corollary

Corollary 0.1 *Let $0 < s < \frac{d}{q}$, if $1 < q \leq 2$ then*

$$\left\| \frac{f}{|x|^s} \right\|_{L^q(\mathbb{R}^d)} \leq C(d, s, q) \|f\|_{\dot{W}^{s,q}(\mathbb{R}^d)}, \quad (0.6)$$

if $q > 2$

$$\left\| \frac{f}{|x|^s} \right\|_{L^q(\mathbb{R}^d)} \leq C(d, s, q) \|f\|_{\dot{W}^{s,q}(\mathbb{R}^d)}^{\frac{1}{q}} \|f\|_{\dot{F}^{s,2(q-1)}(\mathbb{R}^d)}^{\frac{q-1}{q}}. \quad (0.7)$$

The fact that $\left\| \frac{f}{|x|^s} \right\|_{L^q(\mathbb{R}^d)}$ can be controlled by homogeneous Besov norms is not a novelty, a proof of Theorem 0.2 can be found in [18], see also [19]. Here we present a direct proof using the Schur test. We shall remark that our corollary when $q > 2$ is a refinement of Hardy inequality (0.3). Indeed we have when $2(q-1) > 2$

$$\|f\|_{\dot{F}^{s,2(q-1)}(\mathbb{R}^d)}^{\frac{q-1}{q}} \leq \|f\|_{\dot{F}^{s,2}(\mathbb{R}^d)}^{\frac{q-1}{q}} \sim \|f\|_{\dot{W}^{s,q}(\mathbb{R}^d)}^{\frac{q-1}{q}}$$

thanks to square function estimate

$$\|f\|_{\dot{F}^{s,2}} = \left\| \left(\sum_{N \in 2\mathbb{Z}} |N^s P_N(f)(x)|^2 \right)^{\frac{1}{2}} \right\|_{L^q} \sim \| |D|^s f \|_{L^q(\mathbb{R}^d)}.$$

The case $1 < q < 2$ is proved by duality and it requires proving the L^q continuity for singular homogeneous kernels of degree- d . This fact is well known and is Lemma 2.1 in [15]. We underline however that our strategy in proving Theorem 0.2 permits to skip the more delicate lemmas in the Stein and Weiss paper [15] that are needed to prove (0.3).

As a final comment, recalling that $|D|f = \sum_{j=1}^d R_j(\partial_{x_j} f)$ with R_j the Riesz transform defined as $(\widehat{R_j f})(\xi) = -i \frac{\xi_j}{|\xi|} \widehat{u}(\xi)$ and that hence $\| |D|f \|_{L^q(\mathbb{R}^d)} \lesssim \| \nabla f \|_{L^q(\mathbb{R}^d)}$ when $1 < q < \infty$, we get

Corollary 0.2 *Let $2 < q < d$ then*

$$\left\| \frac{f}{|x|} \right\|_{L^q(\mathbb{R}^d)} \leq C(d, s, q) \| \nabla f \|_{L^q(\mathbb{R}^d)}^{\frac{1}{q}} \|f\|_{\dot{F}^{s,2(q-1)}(\mathbb{R}^d)}^{\frac{q-1}{q}}. \quad (0.8)$$

We underline that Corollary 0.2 is a refinement of the classical Hardy inequality involving ∇f

$$\left\| \frac{f}{|x|} \right\|_{L^q(\mathbb{R}^d)} \leq \left(\frac{q}{d-q} \right) \| \nabla f \|_{L^q(\mathbb{R}^d)}. \quad (0.9)$$

by the fact that $\|f\|_{\dot{F}^{s,2(q-1)}(\mathbb{R}^d)} \leq \|f\|_{\dot{F}^{s,2}(\mathbb{R}^d)} \lesssim \| \nabla f \|_{L^q(\mathbb{R}^d)}$. In the literature there is a lot of interest in proving improvements for (0.9), typically such improvement (in bounded or unbounded domains) are in the direction to add a negative term in r.h.s of (0.9), see e.g. [3–7, 9, 10, 13]. Our refinement, although obtained with different techniques, is more in the spirit of [2, 17], i.e. to control r.h.s. of (0.9) with terms that are smaller (up to a multiplicative constant) than the Sobolev norms.

1 Proof of Theorem 0.2

A key argument in our proof is given by the following well known version of Schur test

Proposition 1.1 *Let $\alpha_{N,R} \geq 0$, with $N, R \in 2^{\mathbb{Z}}$, $1 < q < \infty$, then*

$$\sum_R \left(\sum_N \alpha_{N,R} C_N \right)^q \lesssim \sum_N (C_N)^q$$

provided there exists a sequence of positive numbers p_N such that

$$\left(\sum_N \alpha_{N,R} p_N^{\frac{q'}{q}} \right)^{\frac{q}{q'}} \lesssim p_R \quad (1.1)$$

$$\sum_R \alpha_{N,R} p_R \lesssim p_N. \quad (1.2)$$

Proof By Holder's inequality with conjugated exponent (q, q')

$$\sum_N \alpha_{N,R} C_N = \sum_N \alpha_{N,R}^{\frac{1}{q}} \alpha_{N,R}^{\frac{1}{q'}} p_N^{\frac{1}{q}} \frac{C_N}{p_N^{\frac{1}{q}}} \leq \left(\sum_N \alpha_{N,R} p_N^{\frac{q'}{q}} \right)^{\frac{1}{q'}} \left(\sum_N \alpha_{N,R} \frac{C_N^q}{p_N} \right)^{\frac{1}{q}}$$

we get

$$\sum_R \left(\sum_N \alpha_{N,R} C_N \right)^q \leq \sum_R \left(\sum_N \alpha_{N,R} p_N^{\frac{q'}{q}} \right)^{\frac{q}{q'}} \left(\sum_N \alpha_{N,R} \frac{C_N^q}{p_N} \right)$$

that, thanks to (1.1) and Fubini, implies

$$\sum_R \left(\sum_N \alpha_{N,R} C_N \right)^q \lesssim \sum_R p_R \left(\sum_N \alpha_{N,R} \frac{C_N^q}{p_N} \right) = \sum_N \frac{C_N^q}{p_N} \left(\sum_R \alpha_{N,R} p_R \right).$$

Now by (1.2) we conclude

$$\sum_R \left(\sum_N \alpha_{N,R} C_N \right)^q \lesssim \sum_N \frac{C_N^q}{p_N} p_N = \sum_N C_N^q.$$

□

The strategy of the proof for is an adaptation of proof of Hardy inequality in the case $q = 2$ given by Tao [16], i.e. to prove the following estimate

$$\int_{\mathbb{R}^d} \frac{|f(x)|^q}{|x|^{sq}} dx \lesssim \sum_N N^{qs} \|P_N f\|_{L^q(\mathbb{R}^d)}^q \quad (1.3)$$

where $P_N f$ are the classical Littlewood–Paley projectors with N a dyadic number.

We divide $\mathbb{R}^d$ in dyadic shells obtaining

$$\int_{\mathbb{R}^d} \frac{|f(x)|^q}{|x|^{qs}} dx = \sum_{R \in 2^{\mathbb{Z}}} \int_{\frac{R}{2} \leq |x| \leq R} \frac{|f(x)|^q}{|x|^{qs}} dx \lesssim \sum_{R \in 2^{\mathbb{Z}}} \frac{1}{R^{sq}} \int_{\{\frac{R}{2} \leq |x| \leq R\}} |f|^q dx. \quad (1.4)$$

such that using the Littlewood-Paley decomposition we get

$$\sum_{R \in 2^{\mathbb{Z}}} \frac{1}{R^{sq}} \int_{\{\frac{R}{2} \leq |x| \leq R\}} |f|^q dx \leq \sum_{R \in 2^{\mathbb{Z}}} R^{-sq} \left(\sum_{N \in 2^{\mathbb{Z}}} \left(\int_{\{\frac{R}{2} \leq |x| \leq R\}} |P_N(f)|^q \right)^{\frac{1}{q}} \right)^q. \quad (1.5)$$

By the Bernstein inequality $\|P_N(f)\|_{L^\infty(\mathbb{R}^d)} \leq N^{\frac{d}{q}} \|P_N(f)\|_{L^q(\mathbb{R}^d)}$ it follows that

$$\begin{aligned} \left(\int_{\frac{R}{2} < |x| < R} |P_N(f)|^q \right)^{\frac{1}{q}} &\leq R^{\frac{d}{q}} \|P_N(f)\|_{L^\infty} \\ &\leq (NR)^{\frac{d}{q}} \|P_N(f)\|_{L^q}, \end{aligned} \quad (1.6)$$

and clearly

$$\left(\int_{\frac{R}{2} < |x| < R} |P_N(f)|^q \right)^{\frac{1}{q}} \leq \|P_N f\|_{L^q},$$

such that we get

$$\begin{aligned} \int_{\mathbb{R}^d} \frac{|f(x)|^q}{|x|^{qs}} dx &\lesssim \sum_R R^{-qs} \left(\sum_N \min\{1, (NR)^{\frac{d}{q}}\} \|P_N f\|_{L^q} \right)^q \\ &= \sum_R \left(\sum_N \min\{(NR)^{-s}, (NR)^{\frac{d}{q}-s}\} N^s \|P_N f\|_{L^q} \right)^q. \end{aligned}$$

The last step is to apply the Schur test given by Proposition 1.1 in order to conclude that

$$\begin{aligned} \sum_R \left(\sum_N \min\{(NR)^{-s}, (NR)^{\frac{d}{q}-s}\} N^s \|P_N f\|_{L^q} \right)^q &\leq \sum_{N \in 2^{\mathbb{Z}}} N^{sq} \|P_N(f)\|_{L^q}^q \\ &= \sum_{N \in 2^{\mathbb{Z}}} N^{sq} \int_{\mathbb{R}^d} |P_N(f)|^q = \int_{\mathbb{R}^d} \sum_{N \in 2^{\mathbb{Z}}} N^{sq} |P_N(f)|^q. \end{aligned}$$

Notice that

$$\begin{aligned} \sum_{N > \frac{1}{R}} \min\{(NR)^{-s}, (NR)^{\frac{d}{q}-s}\} + \sum_{N \leq \frac{1}{R}} \min\{(NR)^{-s}, (NR)^{\frac{d}{q}-s}\} \\ = R^{-s} \sum_{N > \frac{1}{R}} N^{-s} + R^{\frac{d}{q}-s} \sum_{N \leq \frac{1}{R}} N^{\frac{d}{q}-s} \lesssim 1 \end{aligned}$$

such that (arguing in the same way when summing over R)

$$\sum_N \min\{(NR)^{-s}, (NR)^{\frac{d}{q}-s}\} \lesssim 1 \quad (1.7)$$

$$\sum_R \min\{(NR)^{-s}, (NR)^{\frac{d}{q}-s}\} \lesssim 1. \quad (1.8)$$

The hypotheses for Schur test given by Proposition 1.1 are hence fulfilled by choosing $\alpha_{N,R} = \min\{(NR)^{-s}, (NR)^{\frac{d}{q}-s}\}$ and $p_N = 1$ in Proposition 1.1. This proves (0.3).

2 Proof of Corollary 0.1

In Theorem 0.2 we proved the following estimate

$$\int_{\mathbb{R}^d} \frac{|f(x)|^q}{|x|^{sq}} dx \lesssim \sum_N N^{qs} \|P_N f\|_{L^q(\mathbb{R}^d)}^q \quad (2.1)$$

where $P_N f$ are the classical Littlewood–Paley projectors with N a dyadic number. First we prove that (2.1) implies the Fractional Hardy inequality. We have two cases: $q \geq 2$, $q < 2$.

Case $q \geq 2$:

Thanks to (2.1) we derive

$$\sum_N N^{qs} \|P_N f\|_{L^q(\mathbb{R}^d)}^q = \int_{\mathbb{R}^d} \sum_N N^{sq} |P_N f(x)|^q dx \leq \int_{\mathbb{R}^d} \left(\sum |N^s P_N f(x)|^2 \right)^{\frac{q}{2}} dx$$

from the elementary inequality $(\sum_i a_i^{p_1})^{\frac{1}{p_1}} \leq (\sum_i a_i^{p_2})^{\frac{1}{p_2}}$ with $p_1 \geq p_2$, obtaining

$$\begin{aligned} \int_{\mathbb{R}^d} \frac{|f(x)|^q}{|x|^{sq}} dx &\lesssim \sum_N N^{qs} \|P_N f\|_{L^q(\mathbb{R}^d)}^q \\ &\leq \int_{\mathbb{R}^d} \left(\sum_N |N^s P_N f(x)|^2 \right)^{\frac{q}{2}} dx \sim \| |D|^s f \|_{L^q(\mathbb{R}^d)}^q \end{aligned}$$

where the last equivalence is nothing but the classical square function estimate, see for instance [14].

To prove (0.7) we notice that

$$\begin{aligned} &\int_{\mathbb{R}^d} \sum_N N^{sq} |P_N f(x)|^q dx \\ &\leq \int_{\mathbb{R}^d} \left(\sum_N N^{2s} |P_N f(x)|^2 \right)^{\frac{1}{2}} \left(\sum_N N^{2s(q-1)} |P_N f(x)|^{2(q-1)} \right)^{\frac{1}{2}} dx \\ &\leq \left(\int_{\mathbb{R}^d} \left(\sum_N N^{2s} |P_N f(x)|^2 \right)^{\frac{q}{2}} dx \right)^{\frac{1}{q}} \left(\int_{\mathbb{R}^d} \left(\sum_N N^{2s(q-1)} |P_N f(x)|^{2(q-1)} \right)^{\frac{q}{2(q-1)}} dx \right)^{\frac{q-1}{q}} \end{aligned}$$

by applying twice the Holder's inequality, first in the serie with conjugated exponent $(2, 2)$ and then in the integral with conjugated exponent $(q, \frac{q}{q-1})$. By definition

$$\left(\int_{\mathbb{R}^d} \left(\sum_N N^{2s(q-1)} |P_N f(x)|^{2(q-1)} \right)^{\frac{q}{2(q-1)}} dx \right)^{\frac{q-1}{q}} = \|f\|_{\dot{F}_{q, 2(q-1)}^s}^{q-1}.$$

Case $q < 2$:

For the case $q < 2$ we use the dual characterization of L^q norms, i.e.

$$\begin{aligned} \left\| \frac{f}{|x|^s} \right\|_{L^q} &= \sup_{\|g\|_{q'}=1} \left\langle \frac{f(x)}{|x|^s}, g \right\rangle = \sup_{\|g\|_{q'}=1} \left\langle f(x), \frac{g(x)}{|x|^s} \right\rangle \\ &= \sup_{\|g\|_{q'}=1} \left\langle |D|^{-s}(|D|^s f(x)), \frac{g(x)}{|x|^s} \right\rangle = \sup_{\|g\|_{q'}=1} \left\langle |D|^s f, |D|^{-s} \left(\frac{g(x)}{|x|^s} \right) \right\rangle \\ &\leq \left\| |D|^s f \right\|_{L^q} \left\| |D|^{-s} \left(\frac{g(x)}{|x|^s} \right) \right\|_{L^{q'}}. \end{aligned}$$

Now we aim to prove that

$$\left\| |D|^{-s} \left(\frac{g(x)}{|x|^s} \right) \right\|_{L^{q'}(\mathbb{R}^d)} \lesssim \|g\|_{L^{q'}(\mathbb{R}^d)}, \quad (2.2)$$

for all $g \in L^{q'}$ with $q' > 2$ such that we could conclude that

$$\left\| \frac{f}{|x|^s} \right\|_{L^q(\mathbb{R}^d)} = \sup_{\|g\|_{q'}=1} \left\langle \frac{|f(x)|}{|x|^s}, g \right\rangle \lesssim \| |D|^s f \|_{L^q(\mathbb{R}^d)}.$$

Now we prove (2.2). We recall that $|D|^{-s} f \sim \int_{\mathbb{R}^d} \frac{f(y)}{|x-y|^{d-s}} dy$, see Theorem 5.9 in [12], such that we have (renaming q' by q to simplify the notation)

$$\begin{aligned} |D|^{-s} \left(\frac{g(x)}{|x|^s} \right) &\sim \left| \int_{\mathbb{R}^d} \frac{g(y)}{|x-y|^{d-s} |y|^s} dy \right|^q \leq \left| \int_{\mathbb{R}^d} \frac{|g(y)|}{|y|^s |x-y|^{d-s}} dy \right|^q \\ &\lesssim \left| \int_{\mathbb{R}^d} \frac{|g(y)| \mathbb{1}_{\{|y| > \frac{|x|}{2}\}}(y)}{|y|^s |x-y|^{d-s}} dy \right|^q + \left| \int_{\mathbb{R}^d} \frac{|g(y)| \mathbb{1}_{\{|y| \leq \frac{|x|}{2}\}}(y)}{|y|^s |x-y|^{d-s}} dy \right|^q \\ &\lesssim \frac{1}{|x|^{qs}} \left| \int_{\mathbb{R}^d} \frac{|g(y)| \mathbb{1}_{\{|y| > \frac{|x|}{2}\}}(y)}{|x-y|^{d-s}} dy \right|^q + \left| \int_{\mathbb{R}^d} \frac{|g(y)| \mathbb{1}_{\{|y| \leq \frac{|x|}{2}\}}(y)}{|y|^s |x-y|^{d-s}} dy \right|^q \\ &\lesssim \frac{1}{|x|^{qs}} \left| \int_{\mathbb{R}^d} \frac{|g(y)|}{|x-y|^{d-s}} dy \right|^q + \left| \int_{\mathbb{R}^d} \frac{|g(y)| \mathbb{1}_{\{|y| \leq \frac{|x|}{2}\}}(y)}{|y|^s |x-y|^{d-s}} dy \right|^q \\ &:= |S_1(g)|^q + |S_2(g)|^q \end{aligned}$$

By previous estimates using Paley–Littlewood decomposition and the square function equivalence we get when $q > 2$

$$\int_{\mathbb{R}^d} |S_1(g)|^q dx \sim \int_{\mathbb{R}^d} \left| \frac{|D|^{-s} |g(x)|}{|x|^s} \right|^q dx \lesssim \| |D|^s (|D|^{-s} |g|) \|_{L^q(\mathbb{R}^d)}^q = \|g\|_{L^q(\mathbb{R}^d)}^q.$$

Concerning $\|S_2(g)\|_{L^q}$ we follow the strategy of Stein and Weiss [15] proving the L^q continuity for singular homogeneous kernels of degree- d . The proof of this fact is Lemma 2.1 in [15] that we show for reader convenience. First notice that $\frac{|y|}{|x|} \leq \frac{1}{2}$ implies

$$|x-y| \geq |x| - |y| \geq \frac{|x|}{2},$$

such that

$$\int_{|y| \leq \frac{|x|}{2}} \frac{|g(y)|}{|x-y|^{d-s} |y|^s} dy \lesssim \int_{|y| \leq \frac{|x|}{2}} \frac{|g(y)|}{|y|^s |x|^{d-s}} dy. \quad (2.3)$$

Now we introduce following [15] the function,

$$K(x, y) = \begin{cases} |y|^s |x|^{d-s} & |y| \leq \frac{|x|}{2} \\ 0 & \text{otherwise} \end{cases}$$

and

$$Ug(x) := \int_{|y| \leq \frac{|x|}{2}} \frac{|g(y)|}{|y|^s |x|^{d-s}} dy = \int_{\mathbb{R}^d} K(|x|, |y|) |g(y)| dy.$$

To conclude the proof it suffices hence to show that

$$\int_{\mathbb{R}^d} |Ug|^q dx \lesssim \int |g|^q dx.$$

Fixing $\eta \in S^{d-1}$ and calling $|x| = R$ we define

$$U_\eta g(R) := \int_0^{+\infty} r^{d-1} K(R, r) \cdot |g(r\eta)| dr,$$

such that

$$\begin{aligned} Ug(x) &= \int_{\mathbb{R}} K(|x|, |y|) |g(y)| dy = \int_0^{+\infty} \left(\int_{S^{d-1}} K(R, r) |g(r\eta)| d\sigma_\eta \right) r^{d-1} dr \\ &= \int_{S^{d-1}} \int_0^{+\infty} K(R, r) |g(r\eta)| r^{d-1} dr d\sigma_\eta = \int_{S^{d-1}} U_\eta g(R) d\sigma_\eta. \end{aligned}$$

By the substitution $r = tR$ we obtain

$$\begin{aligned} U_\eta g(R) &= \int_0^{+\infty} K(R, Rt) |g(tR\eta)| R^{d-1} t^{d-1} R dt \\ &= \int_0^{+\infty} K(1, t) |g(tR\eta)| t^{d-1} dt, \end{aligned}$$

thanks to the fact that K is homogeneous of degree $-d$, i.e. that

$$K(\lambda x, \lambda y) = |\lambda|^{-d} K(|x|, |y|).$$

Let h be the function in $L^{q'}((0, +\infty); R^{d-1} dR)$ of unitary norm such that

$$\begin{aligned} \left(\int_0^{+\infty} |U_\eta g(R)|^q R^{d-1} dR \right)^{\frac{1}{q}} &= \int_0^{+\infty} U_\eta g(R) h(R) R^{d-1} dR \\ &= \int_0^{+\infty} \left\{ \int_0^{+\infty} K(1, t) |g(tR\eta)| t^{d-1} dt \right\} R^{d-1} h(R) dR \\ &= \int_0^{+\infty} K(1, t) t^{d-1} \left\{ \int_0^{+\infty} |g(tR\eta)| h(R) R^{d-1} dR \right\} dt \\ &\leq \int_0^{+\infty} K(1, t) t^{d-1} \left\{ \int_0^{+\infty} |g(tR\eta)|^q R^{d-1} dR \right\}^{\frac{1}{q}} dt \\ &= \left(\int_0^{+\infty} K(1, t) t^{d-1-\frac{d}{q}} dt \right) \cdot \left\{ \int_0^{+\infty} |g(R\eta)|^q R^{d-1} dR \right\}^{\frac{1}{q}} \\ &= \left(\int_0^1 t^{d-\frac{d}{q}-1-s} dt \right) \cdot \left\{ \int_0^{+\infty} |g(R\eta)|^q R^{d-1} dR \right\}^{\frac{1}{q}} =: J \cdot \left\{ \int_0^{+\infty} |g(R\eta)|^q R^{d-1} dR \right\}^{\frac{1}{q}}, \end{aligned}$$

where the last integral J converges due to the fact that by our assumptions $s < \frac{d}{q'}$ (remember that we skipped q' with q).

Now we estimate $L^q(\mathbb{R}^d)$ norm of Ug . By Jensen inequality

$$|Ug(R)|^q = \left| \int_{S^{d-1}} |U_\eta g(R)| d\sigma_\eta \right|^q \leq \{|S^{d-1}|\}^{q-1} \int_{S^{d-1}} |U_\eta g|^q d\sigma_\eta,$$

such that integrating with respect to the measure $R^{d-1} dR$ we get

$$\begin{aligned} & \int_0^{+\infty} |Ug(R)|^q R^{d-1} dR \\ & \leq J^q |S^{d-1}|^{q-1} \left(\int_0^{+\infty} \left\{ \int_{S^{d-1}} |U_\eta g(R)|^q d\sigma_\eta \right\} R^{d-1} dR \right) \\ & = J^q |S^{d-1}|^{q-1} \int_{S^{d-1}} \int_0^{+\infty} |U_\eta g(R)|^q R^{d-1} dR d\sigma_\eta \\ & \leq J^q |S^{d-1}|^{q-1} \int_{S^{d-1}} \int_0^{+\infty} |g(R\eta)|^q R^{d-1} dR d\sigma = J^q |S^{d-1}|^{q-1} \int_{\mathbb{R}^d} |g(x)|^q dx. \end{aligned}$$

By the fact that $Uf(x)$ is radial we can conclude that

$$\int_{\mathbb{R}^d} |Ug(x)|^q dx = |S^{d-1}| \cdot \int_0^{+\infty} |Ug(R)|^q R^{d-1} dR \leq J^q |S^{d-1}|^q \int_{\mathbb{R}^d} |g(x)|^q dx.$$

This concludes the proof in the case $q < 2$.

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Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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Assembly theory explains and quantifies selection and evolution

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Scientists have grappled with reconciling biological evolution^{1,2} with the immutable laws of the Universe defined by physics. These laws underpin life's origin, evolution and the development of human culture and technology, yet they do not predict the emergence of these phenomena. Evolutionary theory explains why some things exist and others do not through the lens of selection. To comprehend how diverse, open-ended forms can emerge from physics without an inherent design blueprint, a new approach to understanding and quantifying selection is necessary^{3–5}. We present assembly theory (AT) as a framework that does not alter the laws of physics, but redefines the concept of an 'object' on which these laws act. AT conceptualizes objects not as point particles, but as entities defined by their possible formation histories. This allows objects to show evidence of selection, within well-defined boundaries of individuals or selected units. We introduce a measure called assembly (*A*), capturing the degree of causation required to produce a given ensemble of objects. This approach enables us to incorporate novelty generation and selection into the physics of complex objects. It explains how these objects can be characterized through a forward dynamical process considering their assembly. By reimagining the concept of matter within assembly spaces, AT provides a powerful interface between physics and biology. It discloses a new aspect of physics emerging at the chemical scale, whereby history and causal contingency influence what exists.

In evolutionary theory, natural selection¹ describes why some things exist and others do not². Darwin's theory of evolution and its modern synthesis point out how selection among variants in the past generates current functionality³, as well as a forward-looking process⁴. Neither addresses the space in which new phenotypic variants are generated. Physics can, in theory, take us from past initial conditions to current and future states. However, because physics has no functional view of the Universe, it cannot distinguish novel functional features from random fluctuations, which means that talking about true novelty is impossible in physical reductionism. Thus, the open-ended generation of novelty⁵ does not fit cleanly in the paradigmatic frameworks of either biology⁶ or physics⁷, and so must resort ultimately to randomness⁸. There have been several efforts to explore the gap between physics and evolution^{9,10}. This is because a growing state space over time requires the exploration of a large combinatorial set of possibilities¹¹, such as in the theory of the adjacent possible¹². However, the search generates an unsustainable expansion in the number of configurations possible in a finite universe in finite time, and does not include selection. In addition, this approach has limited predictive power with respect to why only some evolutionary innovations happen and not others. Other efforts have studied the evolution of rules acting on other rules¹³; however, these models are abstract so it is difficult to see how they can describe—and predict—the evolution of physical objects.

Here, we introduce AT, which addresses these challenges by describing how novelty generation and selection can operate in forward-evolving processes. The framework of AT allows us to predict features of new discoveries during selection, and to quantify how much selection was necessary to produce observed objects^{14,15} without having to prespecify individuals or units of selection. In AT, objects are not considered as point particles (as in most physics), but are defined by the histories of their formation as an intrinsic property, mapped as an assembly space. The assembly space is defined as the pathway by which a given object can be built from elementary building blocks, using only recursive operations. For the shortest path, the assembly space captures the minimal memory, in terms of the minimal number of operations necessary to construct an observed object based on objects that could have existed in its past¹⁶. One feature of biological assemblies of objects is multiple realizability wherein biological evolution can produce functionally equivalent classes of objects with modular use of units in many different contexts. For each unit, the minimal assembly is unique and independent of its formation, and therefore accounts for multiple realizability in how it could be constructed^{17,18}.

We introduce the foundations of AT and its implementation to quantify the degree of selection and evolution found in a collection of objects. Assembly is a function of two quantities: the number of copies of the observed objects and the objects' assembly indices (an assembly

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index is the number of steps on a minimal path producing the object). Assembly captures the amount of memory necessary to produce a selected configuration of historically contingent objects in a manner similar to how entropy quantifies the information (or lack thereof) necessary to specify the configuration of an ensemble of point particles, but assembly differs from entropy because of its explicit dependence on the contingency in construction paths intrinsic to complex objects. We demonstrate how AT leads to a unified language for describing selection and the generation of novelty, and thereby produce a framework to unify descriptions of selection across physics and biology.

Assembly theory

The concept of an object in AT is simple and rigorously defined. An object is finite, is distinguishable, persists over time and is breakable such that the set of constraints to construct it from elementary building blocks is quantifiable. This definition is, in some sense, opposite to standard physics, which treats objects of interest as fundamental and unbreakable (for example, the concept of ‘atoms’ as indivisible, which now applies to elementary particles). In AT, we recognize that the smallest unit of matter is typically defined by the limits of observational measurements and may not itself be fundamental. A more universal concept is to treat objects as anything that can be broken and built. This allows us to naturally account for the emergent objects produced by evolution and selection as fundamental to the theory. The concept of copy number is of foundational importance in defining a theory that accounts for selection. The more complex a given object, the less likely an identical copy can exist without selection of some information-driven mechanism that generates that object. An object that exists in multiple copies allows the signatures describing the set of constraints that built it to be measured experimentally. For example, mass spectrometry can be used to measure assembly for molecules, because it can measure how molecules are built by making bonds¹⁹.

Assembly index and copy number

To construct an assembly space for an object, one starts from elementary building blocks comprising that object and recursively joins these to form new structures, whereby, at each recursive step, the objects formed are added back to the assembly pool and are available for subsequent steps (Supplementary Information Sections 1 and 2). AT captures symmetry breaking arising along construction paths due to recursive use of past objects that can be combined in different ways to make new things. For any given object i , we can define its assembly space as all recursively assembled pathways that produce it. For each object, the most important feature is the assembly index a_i , which corresponds to the shortest number of steps required to generate the object from basic building blocks. This can be quantified as the length of the shortest assembly pathway that can generate the object (Fig. 1).

In chemical systems, molecular assembly theory treats bonds as the elementary operations from which molecules are constructed. The shortest path to build a given molecule can be found by breaking its bonds and then ordering its motifs in order of size, starting from atoms and moving to larger motifs by adding bonds in sequence. Given a motif generated on the path, the motif remains available for reuse. The recursivity allows identifying the shortest construction path with parts already built on that path, allowing us to quantify the minimum number of constraints, or memory size, to construct the molecule. The assembly index can be estimated from any complex discrete object with well-defined building blocks, which can be broken apart, as shown in Fig. 1. At every step, the size of the object increases by at least one. The number of total possible steps, although potentially large, is always finite for any finite object and thus the assembly index is computable in finite time. For molecules, the assembly index can be determined experimentally.

A hallmark feature of life is how complex objects are generated by evolution, of which many are functional. For example, a DNA molecule holds genetic information reliably and can be copied easily. By contrast, a random string of letters requires much information to describe it, but is not normally seen as very complex or useful. Thus far, science has not been able to find a measure that quantifies the complexity of functionality to distinguish these two cases. Here we overcome this inherent problem by pointing out another feature of the evolutionary process: the complex and functional objects it generates take many steps to make, and selection allows many identical copies of these objects. Therefore, an evolutionary process can be identified by the production of many identical, or near-identical, multistep objects. The assembly index on its own cannot detect selection, but copy number combined with the assembly index can. This approach defines a new way to measure complexity in terms of the hierarchy of causation stemming from selection at different levels.

Because we do not typically know the full assembly trajectory of an object, we instead adopt a conservative alternative. AT finds the minimal number of steps to produce the object. We assume that every subobject, once available, can be used as often as needed to generate the object. A different approach would be to use Kolmogorov complexity^{20,21} applied to a given molecule, but this requires starting with a graphical representation, and a program to compute the graph of that molecule. The Kolmogorov complexity of a string is the shortest program that will output that string for a programming language capable of universal computation. This measure cannot be easily computed, because checking whether any single program will output the string is uncomputable, as it involves, at least, deciding whether the program stops. Running this program reflects nothing of the underlying process of how the molecule was constructed. Only late in the evolutionary process will molecules be produced by anything starting to resemble Turing machines, loops, stacks, tapes and so on²². Thus, using universal computation to assess molecules adds unrealistic dynamics, making the answer uncomputable. The assembly measure that we have presented here both uses realistic dynamics for molecules, using bonds as building blocks, and is computable for any molecule. The main work for detecting evolution and memory is done here by combining the assembly index and copy number of the objects.

The aim of AT is to develop a new understanding of the evolution of complex matter that naturally accounts for selection and history in terms of what operations are physically possible in constructing an object^{23,24}. We will discuss AT as applied to chemical systems as the main application in this manuscript because their assembly index has been experimentally measured. For molecules, assembly index has a clear physical interpretation and has been validated as quantifying evidence of selection in its application to the detection of molecular signatures of life. However, we anticipate the theory to be sufficiently general to apply to a wide variety of other systems including polymers, cell morphology, graphs, images, computer programs, human languages and memes, as well as many others. The challenge in each case will be to construct an assembly space that has a clear physical meaning in terms of what operations can be caused to occur to make the object²³ (Fig. 1).

In AT there are two important features of the context the object is found in. First, there must be objects in its environment that can constrain the steps to assemble the object and second these objects themselves have been selected because they must be retained over subsequent steps to physically instantiate the memory needed to build the target object. Among the most relatable examples are enzyme catalysts in biochemistry, which permit the formation of very unlikely molecules in large numbers because the enzymes themselves are also selected to exist with many copies. We make no distinction between the traditional notion of biological ‘individual’ and objects that are selected in the environment to quantify the selection necessary to produce a given configuration. Thus, our approach naturally accounts

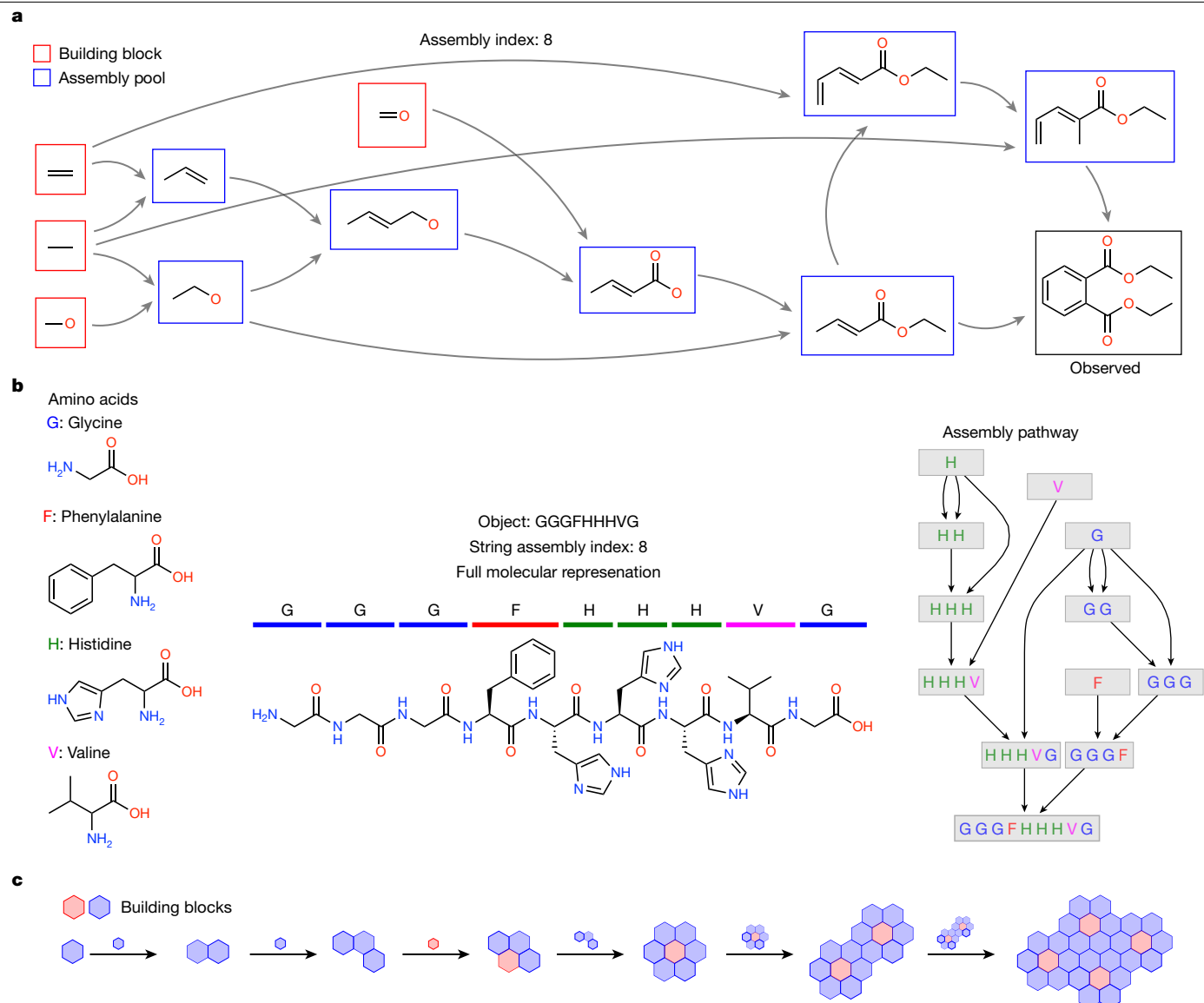


Fig. 1 | Assembly index and shortest path(s). **a–c**, AT is generalizable to different classes of objects, illustrated here for three different general types. **a**, Assembly pathway to construct diethyl phthalate molecule considering molecular bonds as the building blocks. The figure shows the pathway starting with the irreducible constructs to create the molecule with assembly index 8.

b, Assembly pathway of a peptide chain by considering building blocks as strings. Left, four amino acids as building blocks. Middle, the actual object and its representation as a string. Right, assembly pathway to construct the string. **c**, Generalized assembly pathway of an object comprising discrete components.

for well-known phenomena, such as niche construction, whereby organisms and environment are co-constructed and co-selected.

Copy number is important because a single example of a highly complex molecule (with a very high assembly index) could potentially be generated in a series of random events that become increasingly less likely with increasing assembly index. If we consider a forward-building assembly process (see Supplementary Information Sections 1 and 2 for details), without a specific target in mind, the number of possible objects that could be built at each recursive step grows super-exponentially in the absence of any constraints. The likelihood of finding and measuring more than one copy of an object therefore decreases super-exponentially with increasing assembly index in the absence of selection for a specified target. Objects with high assembly index, found in abundance, provide evidence of selection because of the combinatorially growing space of possible objects at each recursive assembly step (Fig. 2). Finding more than one identical copy indicates the presence of a non-random process generating the object.

The assembly equation

We define assembly as the total amount of selection necessary to produce an ensemble of observed objects, quantified using equation (1):

$$A = \sum_{i=1}^N e^{a_i} \left(\frac{n_i - 1}{N_T} \right) \quad (1)$$

where A is the assembly of the ensemble, a_i is the assembly index of object i , n_i is its copy number, N is the total number of unique objects, e is Euler's number and N_T is the total number of objects in the ensemble. Normalizing by the number of objects in the ensemble allows assembly to be compared between ensembles with different numbers of objects.

Assembly quantifies two competing effects, the difficulty of discovering new objects, but, once discovered, some objects become easier to make; this is indicative of how selection was required to discover

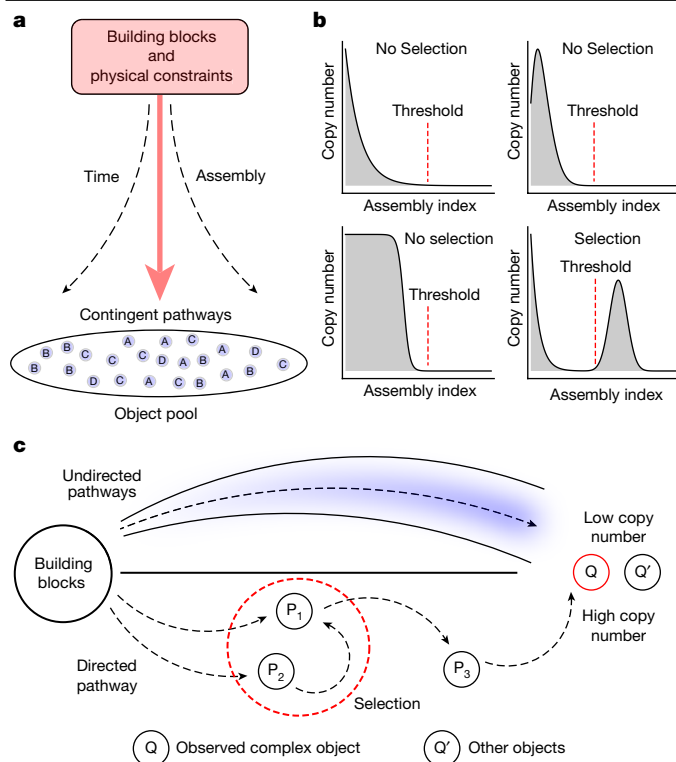


Fig. 2 | Selection in assembly space. **a**, Pictorial representation of the assembly space representing the formation of combinatorial object space from building blocks and physical constraints. **b**, Observed copy number distributions of objects at different assembly indices as an outcome of selection or no selection. **c**, Representation of physical pathways to construct objects with undirected and directed pathways (selected) leading to the low and high copy numbers of the observed object.

and make them. The exponential growth of assembly with depth in assembly space, as quantified by assembly index, is derived by considering a linearly expanding assembly pool that has objects that combine at step $a \rightarrow a + 1$, whereby an object at the assembly index a combines with another object from the assembly pool. Discovering new objects at increasing depth in an assembly space gets increasingly harder with depth because the space of possibilities expands exponentially. Once the pathway for a new object has been discovered, the production of an object (copy number greater than 1) gets easier as the copy number increases because a high copy number implies that an object can be produced readily in a given context. Thus, the hardest innovation is making an object for the first time, which is equivalent to its discovery, followed by making the first copy of that object, but once an object exists in very high abundance it must already be relatively easy to make. Hence, assembly (A) scales linearly with copy number for more than one object for a fixed cost per object once a process has been discovered (see Supplementary Information Section 3 for additional details).

Increasing assembly (A) results from increasing copy numbers n and increasing assembly indices a . If high values of assembly can be shown to capture cases in which selection has occurred, it implies that finding high assembly index objects in high abundance is a signature of selection. In AT, the information required at each step to construct the object is 'stored' within the object (Fig. 2). Each time two objects are combined from an assembly pool, the specificity of the combination process constitutes selection. As we will show, randomly combining objects within the assembly pool at each step does not constitute selection because no combinations exist in memory to be used again for building the same object. If, instead, certain combinations are preferentially used, it implies that a mechanism exists that selects the specific

operations and, by extension, specific target objects to be generated. Later we will quantify the degree of selectivity by parameter α in the growth dynamics, which allows parameterizing selection in an empirically observable manner by parameterizing reuses of specific sets of operations (see Supplementary Information Section 3 for example).

Assembly as given in equation (1) is determined for identified finite and distinguishable objects (with copy number greater than 1) and their distinct assembly spaces. However, in real samples, there are almost always several different coexisting objects, which will include a common history for their formation. Transistors, for example, are used across several different technologies, suggesting a common subspace in the assembly spaces of many modern technologies that includes transistor-like objects. This common subspace, constituting the overlap in the assembly paths of distinct structures, is called a co-assembly space. By contrast, a joint assembly space of several objects is the combined assembly space required to generate those objects. As a potential extension of the assembly equation, to account for the joint assembly of objects, we expand the formulation of the assembly equation that includes the quantification of shared pathways to construct objects to determine the assembly (A) of an ensemble with different objects that share common history (Supplementary Information Section 3).

Selection within assembly spaces

The concept of the assembly space allows us to understand how selection and historical contingency impose constraints on what can be made in the future. By aiming to detect 'selection', we mean a process similar to selection in Darwinian evolution. We do not, however, model functional differences that selection might act on. Instead, we account only for the specificity of selection—that some objects are more likely to be used to make new things and some are less likely. The only functionality we want to detect or describe is in the memory of the process to generate the object, with examples including a metabolic reaction network or a genome. This allows the three Lewontin conditions for evolution to hold²⁵. A key feature of assembly spaces is that they are combinatorial, with objects combined at every step. Combinatorial spaces do not play a prominent role in current physics, because their objects are modelled as point particles and not as combinatorial objects (with limited exceptions). However, combinatorial objects are important in chemistry, biology and technology, in which most objects of interest (if not all) are hierarchical modular structures. More objects exist in assembly space than can be built in finite time with finite resources because the space of possibilities grows super-exponentially with the assembly index. To tame this explosive growth, in AT historical contingency is intrinsic with the space built compositionally, where items are combined recursively (accounting for hierarchical modularity) and this substantially constrains the number of possible objects. It is the combination of this compositionality with combinatorics that allows us to describe selection (Fig. 3).

To produce an assembly space, an observed object is broken down recursively to generate a set of elementary building units. These units can be used to then recursively construct the assembly pathways of the original object(s) to build what we call assembly observed, A_o . A_o captures all histories for the construction of the observed object(s) from elementary building blocks, consistent with what physical operations are possible. Because objects in AT are compositional, they contain information about the larger space of possible objects from which they were selected. To see how, we first build an assembly space from the same building blocks in A_o , which include all possible pathways for assembling any object composed of the same set elementary building blocks as our target object. The space so constructed is the assembly universe (A_u).

In the assembly universe, all objects are possible with no rules, yielding a combinatorial explosion and with double exponential growth in

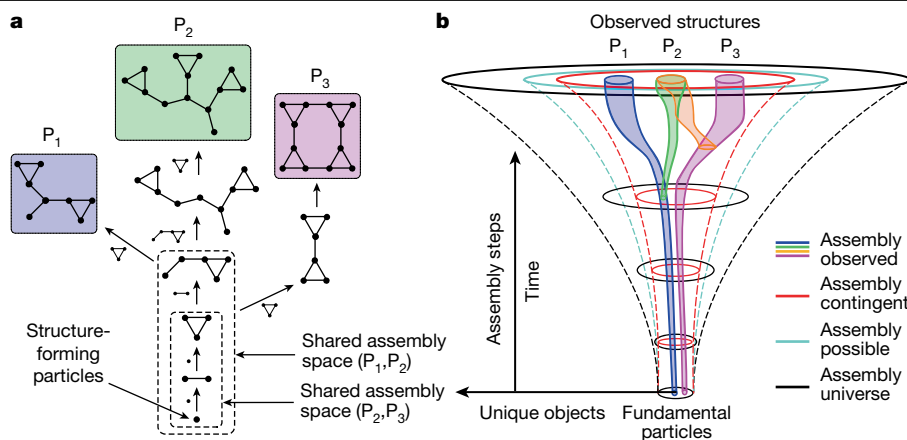


Fig. 3 | Assembly spaces. **a**, Assembly observed of the three objects shown as graphs (P_1 , P_2 and P_3) with their shared minimal construction process called their ‘joint assembly space’. **b**, Illustration of the expansion of the assembly universe, assembly possible, assembly contingent and assembly observed

(see text for details). Assembly universe has no dynamics and is displayed with assembly steps as the time axis. Note that the figure illustrates their nested structure only, not the relative size of the spaces where each set is typically exponentially larger than the subset.

the number of objects, as is characteristic of exploding state spaces and the adjacent possible (see Supplementary Information Section 4 for details). Although mathematically well defined, this double exponential growth is unphysical because the physical processes place restrictions on what is possible (in the case of molecules, an example is how quantum mechanics constrains the numbers of bonds per atom). The assembly universe also has no concept of directionality in time, as there is no ordering to construction processes. Because everything can exist, there is an implication that objects can be constructed independently of what has existed in the past and of resource or time constraints, which is not what we observe in the real universe. For most systems of interest, including in molecular assembly spaces, the number of molecules in the assembly universe is orders of magnitude larger than the amount of matter available in the cosmologically observable universe. There is no way to computationally build and exhaust the entire space, even for objects with relatively low assembly indices. For larger objects, such as proteins, this can be truly gigantic²⁶. In AT, we do not observe all possible objects at a given depth in the assembly space because of selection, more reflective of what we see in the real universe. We next show how taking account of memory and resource limitation severely restricts the size of the space of what can be built, but also allows higher-assembly objects to be built before exhausting resources constructing all the possible lower-assembly objects. AT can account for selection precisely because of the historical contingency in the recursive construction of objects along assembly paths.

Assembly possible (A_p) is the space of physically possible objects, which can be generated by means of the combinatorial expansion of all the known physical rules of object construction and allowing all rules to be available at every step to every object. This can be described by a dynamical model representing undirected forward dynamics in AT. When an object with assembly index a combines with its own history, its assembly index increases by one, $a \rightarrow a + 1$. If the resulting object can be made by means of other, shorter path(s), its assembly index will be smaller than $a + 1$ or even a . Another assumption behind the dynamical model of undirected dynamics is a microscopically driven stochastic rule that uses existing objects uniformly: the probability of choosing an object with assembly index a to be combined with any other object is proportional to N_a , the number of objects with assembly index a (see Supplementary Information Section 5 for further details).

Within assembly possible, assembly contingent (A_c) describes the possible space of objects where history, and selection on that history, matter. Historical contingency is introduced by assuming that only the knowledge or constraints built on a given path can be used in the

future, or with different paths interacting in cases in which selected objects that had not interacted previously now interact. We define the probability P_a of an object being selected with assembly index (a) as $P_a \propto (N_a)^\alpha$, where N_a is the number of objects with assembly index a . Here, α parameterizes the degree of selection: for $\alpha = 1$ all objects that have been assembled in the past are available for reuse, and for $0 \leq \alpha < 1$, only a subset (that grows non-linearly with assembly index) are available for reuse, indicating that selection has occurred. This leads to the growth dynamics:

$$\frac{dN_{a+1}}{dt} = k_d(N_a)^\alpha \quad (2)$$

where k_d represents the rate of discovery (expansion rate) of new objects. For $\alpha = 1$, there is historical dependence without selection. We build assembly paths by taking two randomly chosen objects from the assembly pool and combining them; if a new object is formed, it is added back into the pool. Here we are building random objects, but these are fundamentally different from random combinatorial objects because the randomness we implement is distributed across the recursive construction steps leading to an object (see Supplementary Information Section 5 for solutions). The case of $\alpha = 1$, in which there is historical dependence but no selection, defines the boundary of assembly possible.

Within assembly possible, the assembly contingent (A_c) is the space of possible configurations of objects where $0 \leq \alpha < 1$, that is, where selection is possible, and the objects found in the space are controlled by a path-dependency contingent on each object that has already been built. The growth of the assembly contingent is much slower than exponential; indeed, not all possible paths are explored equally. Instead, the dynamics are channelled by constraints imposed by the selectivity emerging along specific paths. Indeed, a signature of selection in assembly spaces is a slower-than-exponential growth of the number of unique objects. To show this, we use a simple phenomenological model of linear polymers to demonstrate how assembly differentiates cases when selection happens. Starting with a single monomer in the assembly pool, the undirected exploration process combines two randomly selected polymers and adds them back to the assembly pool. In the case of directed exploration with selection, the polymer that has been created most recently is selected to join a randomly selected polymer from the assembly pool. For both directed and undirected exploration, this process was iterated up to 10^4 steps and repeated 25 times. For each observed polymer in the assembly pool, the shortest pathway was generated. For

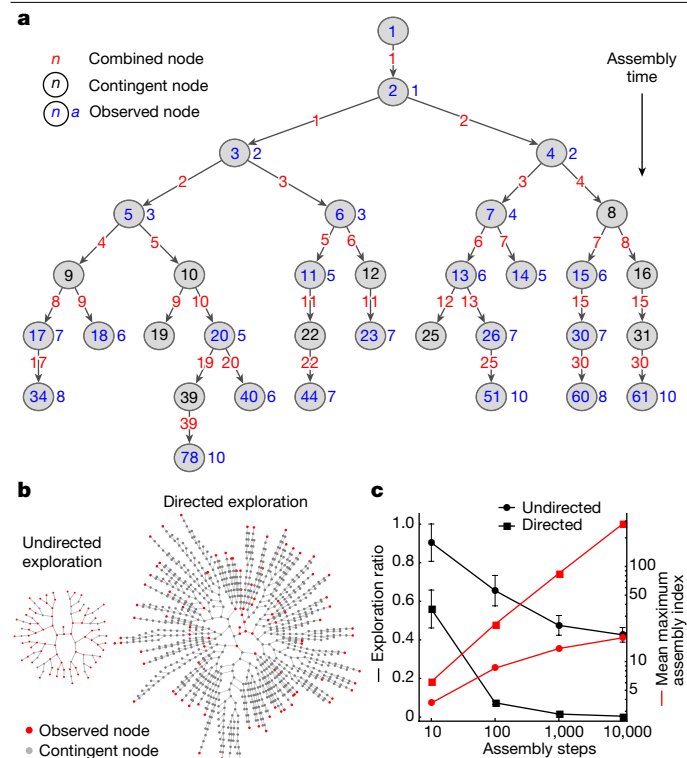


Fig. 4 | Undirected and directed exploration in a forward assembly process. **a**, The joint assembly space of polymeric chains (with their lengths indicated) after 30 steps created by combining randomly selected polymers from the assembly pool. The length of the realized polymers is shown in blue (observed nodes), whereas nodes shown in black represent polymers that have not been realized but are part of the joint assembly space of all realized objects (contingent nodes). For simplicity of representing the joint assembly space, the edge nodes (shown in red) represent the combined node along the directed graph. **b**, The comparison between undirected and directed exploration after 100 assembly steps using a graph with radial embedding (observed and contingent nodes shown in red and grey, respectively). **c**, The mean and standard deviation of the exploration ratio (defined by the ratio of the number of observed nodes and the number of total nodes, which includes observed and contingent nodes) and mean maximum assembly index. n is 25 runs all averaged up to 10^4 assembly steps.

each run, the assembly space of multiple coexisting polymers, their joint assembly space, was approximated by the union of the shortest pathways of all observed polymers. An example of joint assembly space in an undirected exploration up to 30 steps is shown in Fig. 4a.

Comparison between the explored joint assembly space in undirected and directed exploration up to 100 steps is shown in Fig. 4b (see Supplementary Information Section 6 for details). To quantify the degree of exploration at a given assembly step, we calculated the exploration ratio, defined by the ratio of observed nodes to total number of nodes present in the joint assembly space. Figure 4c shows the exploration ratio and the mean maximum assembly index observed, approximated by $\log_2(n)$, where n is the length of the polymer for the undirected and directed exploration processes (both upper and lower bounds scale as $\log_2(n)$ in leading order). Here, the mean maximum assembly index was estimated by calculating the assembly index of the mean value of the longest observed polymeric chains over 25 runs. Comparing the directed process to the undirected exploration illustrates a central principle: the signal of selection is simply a lower exploration ratio and higher complexity (as defined by the maximum assembly index). The observation of a lower exploration ratio in the directed process than in the undirected process is the evidence of the presence of selectivity in the combination process between the

polymers existing in the assembly pool. The process representing sorting and selecting chains within the assembly pool represents an outcome of a physical process leading to selection (see Supplementary Information Section 7 for an additional model).

We conjecture that, the ‘more assembled’ an ensemble of objects, the more selection is required for it to come into existence. The historical contingency in AT means that assembly dynamics explores higher-assembly objects before exhausting all lower-assembly objects, leading to a vast separation in scales separating the number of objects that could have been explored versus those that are actually constructed following a particular path. For example, proteins built both from D and L amino acids and their pathways are part of assembly possible, but, within an assembly contingent trajectory, only proteins constructed out of L amino acids might be present, because of early selection events. This early symmetry breaking along historically contingent paths is a fundamental property of all assembly processes. It introduces an ‘assembly time’ that ticks at each object being made: assembly physics includes an explicit arrow of time intrinsic to the structure of objects.

Assembly unifies selection with physics

In the real universe, objects can be built only from parts that already exist. The discovery of new objects is therefore historically contingent. The rate of discovery of new objects can be defined by the expansion rate (k_d) from equation (2), introducing a characteristic timescale $\tau_d \approx \frac{1}{k_d}$, defined as the discovery time. In addition, once a pathway to build an object is discovered, the object can be reproduced if the mechanism in its environment is selected to build it again. Thus far, we have considered discovery dynamics within the assembly spaces and did not account for the abundance or copy number of the observed objects when discovered. To include copy number in the dynamics of AT, we must introduce a second timescale, the rate of production (k_p) of a specific object, with a characteristic production timescale $\tau_p \approx \frac{1}{k_p}$ (Fig. 5). For simplicity, we assume that selectivity and interaction among emerging objects are similar across assembled objects. Defining these two distinct timescales for initial discovery of an object and making copies of existing objects allows us to determine the regimes in which selection is possible (Fig. 5).

For $\frac{\tau_p}{\tau_d} \gg 1$, whereby objects are discovered quickly but reproduced slowly, the expansion of assembly space is too fast under mass constraints to accumulate a high abundance of any distinguishable objects, leading to a combinatorial explosion of unique objects with low copy numbers. This is consistent with how some unconstrained prebiotic synthesis reactions, such as the formose reaction, end up producing tar, which is composed of a large number of molecules with too low a copy number to be individually identifiable^{27,28}. Selection and evolution cannot emerge if new objects are generated on timescales so fast that resources are not available for making more copies of those objects that already exist. For $\frac{\tau_p}{\tau_d} \ll 1$, objects are reproduced quickly but new ones are discovered slowly. Here resources are primarily consumed in producing additional copies of objects that already exist. Typically, new objects are discovered infrequently. This leads to a high abundance of objects produced by extreme constraints, which could limit the further growth of assembly space. This illustrates how exploration versus exploitation can play out in AT. Significant separation of the two timescales of discovery of new objects and (re)production of selected objects results in either a combinatorial explosion of objects with low copy numbers or, conversely, high copy numbers of low assembly objects. In both cases, we will not observe trajectories that grow more complex structures.

The emergence of selection and open-ended evolution in a physical system should occur in the transition regime where there is only a small separation in the timescales between discovering new objects and reproducing ones that are selected, for example the region located

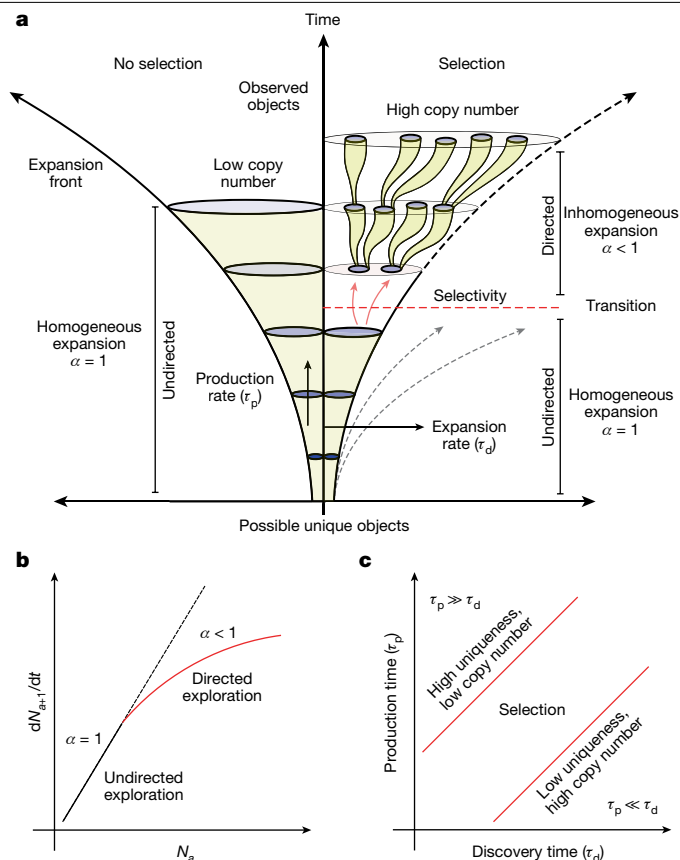


Fig. 5 | Selection and evolution in assembly space. a, Assembly processes with and without selection. The selection process is defined by a transition from undirected to directed exploration. The parameter α represents the selectivity of the assembly process ($\alpha = 1$: undirected/random expansion, $\alpha < 1$: directed expansion). Undirected exploration leads to the fast homogeneous expansion of discovered objects in the assembly space, whereas directed exploration leads to a process that is more like a depth-first search. Here, τ_d is the characteristic timescale of discovery, determining the growth of the expansion front, and τ_p is the characteristic timescale of production that determines the rate of formation of objects (increasing copy number). **b**, Rate of discovery of unique objects at assembly $a + 1$ versus number of objects at assembly a . The transition of $\alpha = 1$ to $\alpha < 1$ represents the emergence of selectivity limiting the discovery of new objects. **c**, Phase space defined by the production (τ_p) and discovery (τ_d) timescales. The figure shows three different regimes: (1) $\tau_d \ll \tau_p$, (2) $\tau_d \gg \tau_p$, and (3) $\tau_d \approx \tau_p$. Selection is unlikely to emerge in regimes 1 and 2, and is possible in regime 3.

between $\tau_d \ll \tau_p$ and $\tau_d \gg \tau_p$ (Fig. 5). To investigate discovery and production dynamics simultaneously, we introduce mass action kinetics in the framework of AT. Our aim is to demonstrate how the generation of novelty can be described alongside selection in a forward process (thus unifying key features of life with physics) and how measuring assembly identifies how much selection occurred. We do so by studying phenomenological models, with the understanding that we are putting selection in by hand in our examples to demonstrate foundational principles of how assembly quantifies selection. To explore this, we consider a forward assembly process whereby the copy numbers of emerging objects follow homogeneous kinetics, together with the discovery dynamics as given by equation (2). With the discovery of new unique objects over time, symmetry breaking in the construction of contingent assembly paths will create a network of growing branches within the assembly possible. In principle, interactions among existing objects and external factors lead to discovery of new objects, expanding the space of possible future objects. Such events can drastically

change the copy number distribution of objects at various assembly indices, depending on the emerging kinetics in the formation of new objects. By combining discovery and production kinetics in a simplified formulation, we estimate copy numbers of objects at different assembly indices and show assembly of the ensemble over time in the forward process at different degrees of selection (see Supplementary Information Section 8 for an example).

The interplay between the two characteristic timescales describes how discovery dynamics ($\tau_d \approx 1/k_d$) and forward kinetics ($\tau_p \approx 1/k_p$), together with selection (characterized by the selection parameter α), are essential for driving processes towards creating higher-assembly objects. This is characteristic of trajectories within assembly contingent. Assembly captures key features of how the open-ended growth of complexity can occur within a restricted space only by generating new objects with increasing assembly indices, while also producing them with a high copy number. Selectivity ($\alpha < 1$) together with comparable production timescales ($\tau_d \approx \tau_p$) is essential for the production of high assembly ensembles. This suggests that selectivity in an unknown physical process can be explained by experimentally detecting the number of objects, their assembly index and copy number as a function of time. Considering molecules as objects and assuming that molecules observed using analytical techniques such as mass spectrometry implies a high copy number, the discovery rate and the selection index (α) can be computed from the temporal data of observed molecules at all assembly indices.

Conclusions

We have introduced the foundations of AT and how it can be implemented to quantify the degree of selection found in an ensemble of evolved objects, agnostic to the detailed formation mechanisms of the objects or knowing a priori which objects are products of units of selection. To do so, we introduced a quantity, assembly, built from two quantities: the number of copies of an object and its assembly index, where the assembly index is the minimal number of recursive steps necessary to build the object (its size). We demonstrated how AT allows a unified language for describing selection and the generation of novelty by showing how it quantifies the discovery and production of selected objects in a forward process described by mass action kinetics. AT provides a framework to unify descriptions of selection across physics and biology, with the potential to build a new physics that emerges in chemistry in which history and causal contingency through selection must start to play a prominent role in our descriptions of matter. For molecules, computing the assembly index is not explicitly necessary, because the assembly index can be probed directly experimentally with high accuracy with spectroscopy techniques including mass spectrometry, infrared and nuclear magnetic resonance spectroscopy²⁹.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41586-023-06600-9>.

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Methods

All the calculations were performed using Mathematica 13 (Wolfram Ltd). In addition, assembly index calculations on polymeric strings in the Supplementary Information were performed using a string assembly calculator previously developed using Python and C++.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All Mathematica Notebooks used to perform the calculations are available at <https://github.com/croningp/assemblyphysics>. The string assembly calculator and the dataset of assembly index calculations is available from the Zenodo repository <https://doi.org/10.5281/zenodo.8017327>.

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Ollivier Ricci curvature of directed hypergraphs

Marzieh Eidi^{1✉} & Jürgen Jost^{1,2}

Many empirical networks incorporate higher order relations between elements and therefore are naturally modelled as, possibly directed and/or weighted, hypergraphs, rather than merely as graphs. In order to develop a systematic tool for the statistical analysis of such hypergraph, we propose a general definition of Ricci curvature on directed hypergraphs and explore the consequences of that definition. The definition generalizes Ollivier's definition for graphs. It involves a carefully designed optimal transport problem between sets of vertices. While the definition looks somewhat complex, in the end we shall be able to express our curvature in a very simple formula, $\kappa = \mu_0 - \mu_2 - 2\mu_3$. This formula simply counts the fraction of vertices that have to be moved by distances 0, 2 or 3 in an optimal transport plan. We can then characterize various classes of hypergraphs by their curvature.

Principles of network analysis. Network analysis constitutes one of the success stories in the study of complex systems^{3,6,11}. For the mathematical analysis, a network is modelled as a (perhaps weighted and/or directed) graph. One can then look at certain graph theoretical properties of an empirical network, like its degree or motif distribution, its assortativity or clustering coefficient, the spectrum of its Laplacian, and so on. One can also compare an empirical network with certain deterministic or random theoretical models. Successful as this analysis clearly is, we nevertheless see two important limitations. One is that many of the prominent concepts and quantities used in the analysis of empirical networks are node based, like the degree sequence. The structure of a network is encoded, however, not in its vertices or nodes, but rather in its edges, that is, the relations between the nodes. Therefore, here and in other works of our group, we wish to advocate and pursue an analysis whose fundamental ingredients are edge based quantities. Secondly, many real data sets are naturally modelled by structures that are somewhat more general than graphs, because they may contain relations involving more than two elements. For instance, chemical reactions typically involve more than two substances. This leads to hypergraphs, a subject that is currently gaining much momentum, see for instance⁴. In an undirected graph, an edge is given by an unordered pair of vertices, that is, by a two-element subset of the set of all vertices. For an undirected hypergraph, a hyperedge is given by any non-empty subset of the vertex set. In a directed graph, an edge is an ordered pair of vertices, that is, it connects two vertices, its tail and its head. Analogously, in a directed hypergraph, a hyperedge connects two non-empty sets of vertices, a tail set and a head set. Directed graphs thus are special directed hypergraphs, where each such set contains a single vertex. Like graphs, hypergraphs can also be weighted. Let us look at some examples. In a coauthorship network, the authors are the vertices, and a set of authors constitutes a hyperedge when they coauthor a paper. Thus, coauthorship networks are naturally modelled as undirected hypergraphs. Modelling them as a graph would connect two vertices when they are part of the set of authors of a paper. But already for three authors A, B, C , when they are pairwise connected in a graph, this cannot distinguish between the case where we have just three papers with two authors each, or whether there is a joint paper by all three of them, and perhaps in addition also some two or single author papers. Hypergraphs modelling empirical networks can also be directed. Taking the example of chemical reactions, they are typically not reversible, but rather transform a set of educts into a set of products. Thus, the task we set ourselves here is to forge tools that are powerful in the analysis of empirical networks modelled as hypergraphs. And as advocated above, such a tool should primarily evaluate properties of hyperedges rather than of vertices. Of course, once we have suitable quantities associated to hyperedges, quantities for vertices can be then also be derived, for instance by averaging over all hyperedges incident to or emanating from a vertex. So, how to go about this task? We take a conceptual approach and consider a hypergraph as a geometric object and then look for mathematical strategies for identifying invariants that can characterize geometric structures. And that leads us to one of the greatest successes in mathematics, Riemannian geometry. There, the fundamental invariants are curvatures. It

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might now seem that curvature is a concept only suitable for smooth structures, like smooth curves or surfaces, and therefore ill-suited for discrete structures like hypergraphs. But as it turns out, curvature concepts can be formulated more abstractly than taking second derivatives of some smooth objects. We shall now explain this in more detail, in order to motivate the approach taken in this paper.

Ricci curvature: from Riemannian geometry to network analysis. In Riemannian geometry (see for instance⁷ as a reference), the curvature of a space quantifies its non-flatness. Among the various curvature notions that are of importance in Riemannian geometry, Ricci curvature quantifies this deviation by comparing the average distance between two sufficiently close points and the distance between two small balls around them. Bounds on curvatures can be used to connect the geometry of a Riemannian manifold with its topology, or to control stochastic processes on it. More precisely, a positive lower bound for the Ricci curvature yields the Bonnet–Myers theorem, which bounds the diameter of the space in terms of such a lower Ricci bound, the Lichnerowicz theorem for the spectral gap of the Laplacian, a control on mixing properties of Brownian motion and the Levy–Gromov theorem for isoperimetric inequalities and concentration of measures. In view of these strong implications, it is desirable to extend this to metric spaces that are more general than Riemannian manifolds. With this motivation, several generalized curvature notions have been proposed for non-smooth or discrete structures. In particular, Ollivier¹² defined a notion of Ricci curvature on metric spaces equipped with a Markov chain, and extended some of the mentioned results for positively curved manifolds. His definition compares the Wasserstein distance between probability measures supported in the neighborhoods of two given points with the distance between these points. The Wasserstein distance between two probability measures is defined as the minimal cost needed for transporting one into the other. That is, an optimal transport problem has to be solved. On Riemannian manifolds, this recovers the original notion of Ricci curvature (up to some scaling factor), and at the same time, it naturally applies to discrete metric spaces like graphs. Recently, this curvature has been applied in network analysis, to determine spreading or local clustering in networks modelled by undirected or directed graphs, see for instance¹⁴. It is therefore desirable to have such a tool also for hypergraphs. In this paper, we shall develop a notion of an Ollivier-type Ricci curvature for, possibly directed and/or weighted, hypergraphs. There have been some prior proposals for extensions of Ollivier–Ricci curvature in such a direction (see for instance^{1,2,15}), but our approach is more general and, as we argue, also more natural both in terms of its conceptual motivation and its range of applicability to empirical networks. From a geometric perspective, the fundamental principle that curvature characterizes types of spaces also applies here as we can distinguish and classify particular classes of directed hypergraphs in terms of their curvature. A definition of the Ollivier Ricci curvature of directed graphs was firstly proposed and investigated in¹⁵ where out-out directions for assigning measures are used. For that, however, one needs to assume strong connectivity of the underlying directed graphs in order to find transportation plans with finite cost, but this does not hold in many real directed networks. Therefore, in this paper, we work with in-out directions, which does not require such a strong assumption. The resulting theory is rather different from that of¹⁵. The first extension of the notion of Ollivier Ricci curvature to hypergraphs was proposed in¹, using a multi-marginal optimal transport problem to define curvature. Because of that, the resulting curvature in the end is an analogue of Riemannian scalar rather than Ricci curvature. Also, it does not directly apply to directed hypergraphs. In this paper, we therefore propose a notion of directed hypergraph curvature that extends Ricci curvature rather than scalar curvature. Since in our setting, hyperedges are directed and each direction separates the vertices of the hyperedge into two classes, similar to directed graphs, we consider a double marginal optimal transport problem. We study some implications of our definition and then take a closer look at hypergraphs of constant Ricci curvature. In⁹, our notion of Ricci curvature for hypergraphs is systematically applied to empirical networks, leading to insight not readily available through other methods of network analysis.

Ricci curvature. In order to proceed, we first need to explain the geometric meaning of Ricci curvature. Ricci curvature is a fundamental concept from Riemannian Geometry (see for instance⁷) that more recently has been extended to a discrete setting.

For a Riemannian manifold M of dimension N , Ricci curvature can be defined in several equivalent ways. What is relevant for the extension to the discrete setting is that it measures the local amount of non-flatness of the manifold by comparing the distance between two small balls with the distance of their centers when these centers are sufficiently close to each other. To be more precise, consider a unit tangent vector w at a point x in a Riemannian manifold M and let $\varepsilon, \delta > 0$ be smaller than the injectivity radius of M . Suppose $\exp_x(\cdot) : T_x M \rightarrow M$ denotes the exponential map and y is the endpoint of $\exp_x \delta w$ and hence at distance δ from x . Let S_x be the sphere of radius ε in the tangent space at x (and hence $\exp_x S_x$ is the sphere of radius ε around x in the manifold). Then if S_x is mapped to S_y using parallel transport, the average distance between a point of $\exp_x S_x$ and its image in $\exp_y S_y$ is

$$\delta \left(1 - \frac{\varepsilon^2}{2N} \text{Ric}(w, w) + O(\varepsilon^3 + \varepsilon^2 \delta) \right)$$

when $(\varepsilon, \delta) \rightarrow 0$. If balls are used instead of spheres, the scaling factor is $\frac{\varepsilon^2}{2(N+2)}$ instead of $\frac{\varepsilon^2}{2N}$ ¹³. This follows from standard Jacobi field estimates. These estimates involve the sectional curvature, but summing over all directions orthogonal to the geodesic connecting x and y results in a Ricci curvature term. Here, one should think of ε as being smaller than δ , and $O(\varepsilon^3)$ then simply indicates a higher term, whereas $O(\varepsilon^2 \delta)$ is needed when the Ricci curvature is not constant. In Riemannian geometry, one can then average the Ricci curvatures of the directions at a point. This then yields the scalar curvature, which thus is a quantity naturally associated to points. The scalar

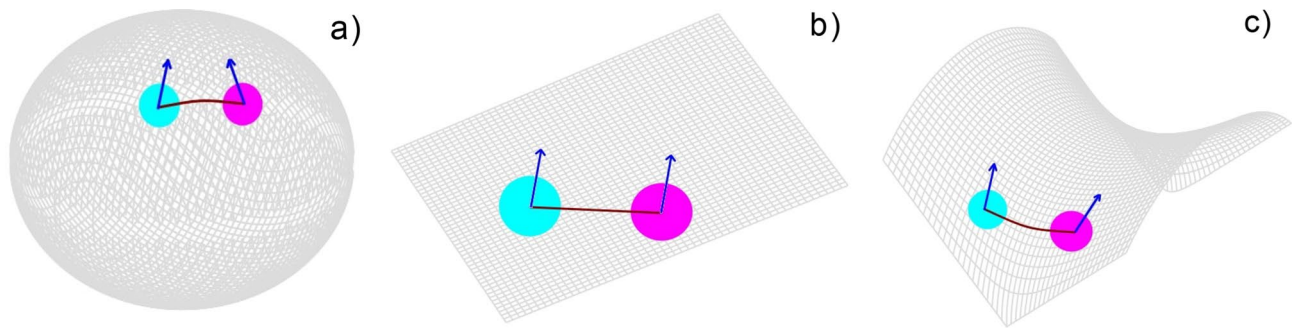


Figure 1. Manifolds with constant Ricci curvatures; (a) positive, (b) zero and (c) negative.

curvature is a much weaker geometric invariant than the Ricci curvature. Returning to the latter, if balls in average are closer than their centers (Fig. 1a), Ricci curvature in the direction of xy is positive. If the manifold is locally flat, Euclidian (Fig. 1b), then the two distances coincide. Most manifolds, however, are locally negatively curved (Fig. 1c)¹⁰.

This local characterization is the key property for defining Ricci curvature notions in more general settings than smooth manifolds. In 2007, Ollivier defined a notion of Ricci curvature, called Ollivier (coarse) Ricci curvature, on metric spaces equipped with a random walk m ; Recall that if (X, d) is a Polish metric space equipped with its Borel σ -algebra, a random walk m on X is a family of probability measures $\{m_x | x \in X\}$ satisfying the following conditions¹²:

- The map $x \rightarrow m_x$ is measurable.
- Each m_x has finite first moment, i.e., for some (hence any) $z \in X$ one has $\int d(z, y) dm_x(y) < \infty$.

Also in the following definition, we let $d(x, y)$ be the distance from x to y obtained from the metric.

Definition 1.1¹² Let (X, d) be a metric space with a random walk m , let $x, y \in X$ be two distinct points. The Ricci curvature of (X, d, m) in the direction (x, y) is

$$\kappa(x, y) := 1 - \frac{W_1(m_x, m_y)}{d(x, y)}$$

where W_1 is the 1 – Wasserstein distance between m_x and m_y on X :

$$W_1(m_x, m_y) := \inf_{\mathcal{E} \in \Pi(m_x, m_y)} \int_{(x, y) \in X \times X} d(x, y) d\mathcal{E}(x, y)$$

where $\Pi(m_x, m_y)$ is the set of measures on $X \times X$ whose first (second) marginal is m_x (m_y) (thus, each $\mathcal{E}(x, y)$ is a coupling between random walks projecting to m_x and m_y).

Here instead of taking metric balls around two close enough points we consider the Wasserstein distance (Transportation or Earthmover distance), between two probability measures m_x and m_y corresponding to two random walks which are starting at x and y respectively. When (X, d, m) is Riemannian manifold equipped with Riemannian volume measure, as it is shown in¹², this notion coincides with the Riemannian Ricci curvature in the direction of xy (up to some scaling factor).

In network analysis, and particularly for networks that are modeled by undirected graphs, this notion has proved itself as a very useful tool to determine clustering and coherence in the network^{8,14}, and since it is based on Markov chains, it is very well suited for capturing diffusion and stochastic processes in the network. One can see this as a motivation for the definition of¹ where multi-marginal optimal transport was used to define a notion of curvature for collections of points in metric spaces. Since this is defined for points rather than for directions, from our perspective, this should be considered as a version of scalar curvature, rather than of Ricci curvature. However, the definition we shall present in the next section will be different.

Results

Transport plans and curvature of directed hypergraphs. As already mentioned, similar to directed graphs, in directed hypergraphs, every hyperedge e in E represents a directional relation between two non-empty subsets A_e (tail), B_e (head) of the vertex set V . In this paper we often write A instead of A_e , and B instead of B_e when the hyperedge e is specified and fixed. Note that the sets A_e , B_e and V can be equal, namely when the hypergraph has just one hyperedge e and its tail set and head set coincide. However if for a hyperedge e , A_e or B_e is a proper subset of V , then since we consider non-empty subsets of V , the other one should also be proper. Similarly, for any vertex $x \in V$, d_x^{in} is the number of incoming hyperedges to x (those hyperedges which include x in their head set), and d_x^{out} is the number of outgoing hyperedges from x (those hyperedges which have x in their tail set). Also a directed path between the vertices in a directed hypergraph G is an alternating sequence of

distinct vertices and directed hyperedges $(v_1, e_1, \dots, v_k, e_k, v_{k+1})$ such that for each i , v_i and v_{i+1} are in the tail and head sets of e_i respectively. If $k \geq 1$ and v_1 and v_{k+1} are the same vertices, the path is called a directed k -cycle. From now on we consider directed hypergraphs to be weakly connected, meaning that the underling undirected hypergraph is connected.

Definition 2.1 Let $H = (V, E)$ be an unweighted directed hypergraph and $e \in E$ be an arbitrary directed hyperedge such that $A = \{x_1, \dots, x_n\} \xrightarrow{e} B = \{y_1, \dots, y_m\}$ ($n, m \leq |V|$). We define the Ollivier–Ricci curvature of this hyperedge as

$$\kappa(e) := 1 - W(\mu_{A^{in}}, \mu_{B^{out}})$$

where the probability measures $\mu_{A^{in}}$ (called mass) and $\mu_{B^{out}}$ (called hole), are defined on V as follows:

$$\mu_{A^{in}} = \sum_{i=1}^n \mu_{x_i}^{in} \text{ where } \forall 1 \leq i \leq n \text{ and } \forall z \in V(H)$$

$$\mu_{x_i}^{in}(z) = \begin{cases} 0 & z = x_i \text{ \& } d_{x_i}^{in} \neq 0 \\ \frac{1}{n} & z = x_i \text{ \& } d_{x_i}^{in} = 0 \\ \sum_{e': x_i \in B_{e'}, z \in A_{e'}} \frac{1}{n \times d_{x_i}^{in} \times |A_{e'}|} & z \neq x_i \text{ \& } z \in A_{e'} \\ 0 & \text{otherwise} \end{cases}$$

and likewise

$$\mu_{B^{out}} = \sum_{j=1}^m \mu_{y_j}^{out} \text{ where } \forall 1 \leq j \leq m, z \in V(H) :$$

$$\mu_{y_j}^{out}(z) = \begin{cases} 0 & z = y_j \text{ \& } d_{y_j}^{out} \neq 0 \\ \frac{1}{m} & z = y_j \text{ \& } d_{y_j}^{out} = 0 \\ \sum_{e': y_j \in A_{e'}, z \in B_{e'}} \frac{1}{m \times d_{y_j}^{out} \times |B_{e'}|} & z \neq y_j \text{ \& } z \in B_{e'} \\ 0 & \text{otherwise} \end{cases}$$

and $W(\mu_{A^{in}}, \mu_{B^{out}})$ is the 1–Wasserstein distance between these two discrete measures defined as follows:

$$W(\mu_{A^{in}}, \mu_{B^{out}}) = \min_{\mathcal{E} \in \Pi(\mu_{A^{in}}, \mu_{B^{out}})} \sum_{u \rightarrow A} \sum_{B \rightarrow v} d(u, v) \mathcal{E}(u, v)$$

where $d(u, v)$ is the length of a shortest path from u to v and for a transport plan $\mathcal{E}$, $\mathcal{E}(u, v)$ represents the amount of mass moved from vertex u to vertex v and $\Pi(\mu_{A^{in}}, \mu_{B^{out}})$ is the set of transport plans, that is, probability measures on $V \times V$ that have $\mu_{A^{in}}$ and $\mu_{B^{out}}$ as their marginals. In other words, the minimum is taken over all couplings $\mathcal{E}$ between $\mu_{A^{in}}$ and $\mu_{B^{out}}$ which satisfy

$$\sum_{u \rightarrow A} \mathcal{E}(u, v) = \sum_{j=1}^m \mu_{y_j}^{out}(v) \text{ and } \sum_{B \rightarrow v} \mathcal{E}(u, v) = \sum_{i=1}^n \mu_{x_i}^{in}(u)$$

and by $A^{in}(u \rightarrow A)$ we mean the vertices $u \in V$ with $\mu_{A^{in}}(u) \neq 0$. Similarly $B^{out}(B \rightarrow v)$ refers to the vertices $v \in V$ with $\mu_{B^{out}}(v) \neq 0$.

Remark 2.1 Since we are working with directed objects, in this definition we had to provide for the situation where a vertex in the tail does not have incoming connections, or where a vertex in the head does not have any outgoing ones. In undirected networks, this is not an issue, because then the hyperedge in question itself provides both.

We can construct for each directed hypergraph a corresponding directed graph. That graph has the same set of vertices as the directed hypergraph and for each hyperedge, we draw an edge from each vertex in its tail to every vertex in its head. Thus, a directed hyperedge $A = \{x_1, \dots, x_n\} \xrightarrow{e} B = \{y_1, \dots, y_m\}$ corresponds to a set with nm elements of directed edges. Note, however, that there might be directed graphs that correspond to more than one directed hypergraph.

Proposition 2.1 The curvature of a hyperedge $e : A = \{x_1, \dots, x_n\} \rightarrow B = \{y_1, \dots, y_m\}$ is bounded from below by the minimum of the Ricci curvatures of directed edges in its corresponding directed graph.

Proof Let $\mathcal{E}_{ij}$ be the optimal transport plan for the edge $e_{ij} : x_i \rightarrow y_j$, i.e.

$$W(\mu_{x_i}^{in}, \mu_{y_j}^{out}) = \sum_{u, v \in V} d(u, v) \mathcal{E}_{ij}(u, v)$$

Then

$$\mathcal{E} := \frac{1}{mn} \sum_{i=1}^n \sum_{j=1}^m \mathcal{E}_{ij}$$

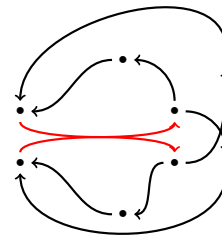
has the marginal distributions $\mu_{A^{in}}$ and $\mu_{B^{out}}$. Therefore:

$$W(\mu_{A^{in}}, \mu_{B^{out}}) \leq \sum_{u,v \in V} d(u,v) \mathcal{E}(u,v) = \frac{1}{mn} \sum_{i=1}^n \sum_{j=1}^m W(\mu_{x_i}^{in}, \mu_{y_j}^{out}) \leq \max_{\substack{1 \leq i \leq n \\ 1 \leq j \leq m}} W(\mu_{x_i}^{in}, \mu_{y_j}^{out})$$

and therefore $\kappa(e) \geq \min_{\substack{1 \leq i \leq n \\ 1 \leq j \leq m}} \kappa(e_{ij})$. $\square$

However, the corresponding directed edges of a directed hyperedge do not fully represent the geometric structure of that hyperedge as shown in the following.

Remark 2.2 The maximum of the Ricci curvatures of directed edges corresponding to a directed hyperedge is not necessarily an upper bound for its Ricci curvature, as one can see from the example where the curvature of the hyperedge with red colour is one and the curvature of all its four corresponding directed edges is $-1/2$.



Lemma 2.1 For a directed hyperedge $e : A \rightarrow B$ we have

$$W(\mu_{A^{in}}, \mu_{B^{out}}) \geq \sup \left(\sum_{u \rightarrow A} f(u) \mu_{A^{in}}(u) - \sum_{B \rightarrow v} f(v) \mu_{B^{out}}(v) \right)$$

where the supremum is taken over all functions on $V(H)$ with $f(u) - f(v) \leq d(u,v)$.

Proof The proof is similar to the proof of proposition 2.10 in¹⁵ where another measure (out-out) for defining Ricci curvature of a directed edge is considered. Here we show that the same result holds for directed hypergraphs by assuming other directions (in-out) for defining measures. We have:

$$\begin{aligned} \sum_{u \rightarrow A} \sum_{B \rightarrow v} d(u,v) \mathcal{E}(u,v) &\geq \sum_{u \rightarrow A} \sum_{B \rightarrow v} (f(u) - f(v)) \mathcal{E}(u,v) \\ &= \sum_{u \rightarrow A} f(u) \sum_{B \rightarrow v} \mathcal{E}(u,v) - \sum_{B \rightarrow v} f(v) \sum_{u \rightarrow A} \mathcal{E}(u,v) \end{aligned}$$

thus:

$$= \sum_{u \rightarrow A} f(u) \mu_{A^{in}}(u) - \sum_{B \rightarrow v} f(v) \mu_{B^{out}}(v)$$

and since for all Lipschitz functions on hypergraphs this inequality holds and the left hand side is independent of f , we obtain

$$W(\mu_{A^{in}}, \mu_{B^{out}}) \geq \sup \left(\sum_{u \rightarrow A} f(u) \mu_{A^{in}}(u) - \sum_{B \rightarrow v} f(v) \mu_{B^{out}}(v) \right).$$

$\square$

Remark 2.3 In general, we do not get equality in this lemma; equality holds for undirected hypergraphs (see¹⁵), or more generally, if for every directed hyperedge e from A_e to B_e , we have a directed hyperedge e' in the reverse direction, from B_e to A_e (i.e. $A_{e'} = B_e$, $B_{e'} = A_e$). In that case, all distances $d(u,v)$ become symmetric.

We now propose another formula for the curvature of a hyperedge which is more intuitive and in some cases much easier to work with.

For defining the Ricci curvature of a hyperedge e , we use incoming hyperedges to its tail set (A_e) and outgoing hyperedges from its head set (B_e). If u and v are in the support of $\mu_{A^{in}}$ and $\mu_{B^{out}}$ respectively, then $d(u, v) \leq 3$. Recall that for any real valued function f on V , the support of f (denoted by $\text{supp}(f)$) is the set of all vertices in V where f is non-zero, i.e., $\text{supp}(f) = \{u \in V \mid f(u) \neq 0\}$.

If μ_i is the amount of mass that is moved with distance i ($i \leq 3$) in an optimal transport plan, then we have:

$$\sum_{i=0}^3 \mu_i = 1, \quad \sum_{i=1}^3 i\mu_i = W. \quad (1)$$

If $\kappa = 0$ then $W = 1$, and we thus have $\mu_0 = \mu_2 + 2\mu_3$. More generally, we obtain

Theorem 2.1 *The curvature of a hyperedge is given by*

$$\kappa = \mu_0 - \mu_2 - 2\mu_3. \quad (2)$$

In fact, we could simply use (2) as a *definition* of Ricci curvature. The formula (2) for the curvature of a directed hyperedge, also works for curvature of edges in undirected graphs. As in the (undirected) graph case, μ_0 represents the amount of mass which is not moved in an optimal plan, i.e., the amount of the stable mass in directed 3-cycles ($u \rightarrow x_i \rightarrow y_j \rightarrow u$) (where u is simultaneously in the support of $\mu_{A^{in}}$ and $\mu_{B^{out}}$) or directed loops emerging from any of the x_i s. Although μ_1 (the mass moved with distance one, possibly through directed quadrangles ($u \rightarrow x_i \rightarrow y_j \rightarrow v$, while u is directly connected to v , $u \rightarrow v$) does not appear in the formula for the curvature, computing μ_1 is an intermediate step for the computations of μ_2 and μ_3 where μ_2 is the amount of mass that should be moved with distance 2 (possibly through directed pentagons including x_i and y_j) and μ_3 is the amount of the mass that is moved with distance 3 in an optimal plan.

Remark 2.4 While finding the general formula for μ_1 (and μ_2) may be difficult, any lower bound for μ_1 (after simply knowing the exact amount of μ_0) helps us to derive some upper bounds for W and therefore a lower bound for the curvature. Also the μ_i can differ between different optimal transport plans, but equalities (1) and (2) will always hold.

We also point out that while optimal transport plans always exist in our finite setting, they need not be unique.

As an example, in Fig. 2, for computing the curvature of the green hyperedge which connects three vertices in its left(A) to two vertices in its right(B), separated by dots, we assign masses (red bullets) and holes (empty squares) respectively to the incoming neighbours of A and outgoing neighbours of B . Note that since for a vertex x in A (the lowest vertex in A), $d_x^{in} = 0$, we put all its assigned mass at x which is equal to $1/3$. Similarly there is a vertex y in B such that $d_y^{out} = 0$. Therefore we put its whole assigned measure at y itself and it is equal to $1/2$. Also for the other vertices in A (respectively B), their assigned masses (holes) are divided between the vertices in the tails of incoming hyperedges to A (the vertices in the heads of outgoing hyperedges from B) based on Definition 2.1. It is straightforward to check that in (any) optimal transport plan, $1/12$ of the mass need not be moved. This is the amount of the mass which coincides with one of the holes. Also $1/3$ of the mass at x is moved with distance one to the hole at y . $1/6$ is moved with distance two and the remained part is moved with distance three. Hence the curvature of the green hyperedge is $-11/12$.

Bounds for the curvature. For obtaining an upper bound for the curvature of a hyperedge we need to control μ_0 which corresponds to the stable mass at directed 3 cycles (triangles in the undirected graph case) and those vertices which are in the intersection of A and B .

In what follows, we shall see that increasing the number of vertices in this intersection will make the curvature more positive. Here, a directed hyperloop is a directed hyperedge $e: A = \{x_1, \dots, x_n\} \rightarrow B = \{y_1, \dots, y_m\}$ for which $A \cap B$ is non-empty. Specifically where $A \cap B = A = B$, we shall have $\kappa(e) = 1$. Directed k -cycles where $k \geq 4$ do not affect the curvature of directed hyperedges since they can not make short-cuts for moving any of the masses to any of the holes.

Proposition 2.2 *For a directed hyperedge $e: A = \{x_1, \dots, x_n\} \rightarrow B = \{y_1, \dots, y_m\}$ we have*

$$\sum_{u \in \text{supp } \mu_{A^{in}}(u) \cup \text{supp } \mu_{B^{out}}(u)} \mu_{A^{in}}(u) \wedge \mu_{B^{out}}(u) \geq \kappa(e).$$

where $\alpha \wedge \beta := \min\{\alpha, \beta\}$.

Proof Jost and Liu have established this theorem in undirected graphs (theorem 7 in⁸). Here, we simply notice that the number of non-zero elements in this summation coincides with the number of vertices u belonging to a directed 3-cycle ($u \rightarrow x_i \rightarrow y_j \rightarrow u$) or $A \cap B$. For other vertices u either of $\mu_{A^{in}}(u)$ or $\mu_{B^{out}}(u)$ is zero.

Also the equality holds when for a hyperedge e , all the masses coincide with all the holes with the same size. For instance in directed hyperloops when $A \cap B = A = B$ both sides of the above formula are equal to one. As another general example see Lemma 2.2. $\square$

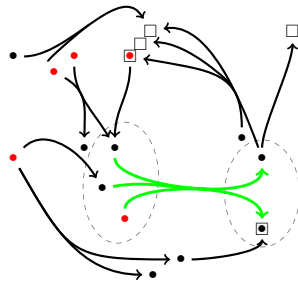


Figure 2. The green hyperedge is negatively curved.

As already mentioned in Remark 2.4 after computing μ_0 , any non-zero amount for μ_1 would give us an upper bound for W . For that, at least one incoming neighbour of A should be at distance one from some outgoing neighbour of B . For example, when for the hyperedge $e, e: A = \{x_1, \dots, x_n\} \rightarrow B = \{y_1, \dots, y_m\}$, there is at least one hyperedge e' from any y_j to any x_i ($e': y_j \rightarrow x_i$) or/ and when there is at least one x_i with $d_{x_i}^{in} = 0$ and at least one y_j with $d_{y_j}^{out} = 0$, the required condition is satisfied and there is at least one mass which is in distance one from at least one hole and therefore we can present a transfer plan (similar to that in Theorem 3 in⁸) to obtain a positive lower bound for μ_1 . Specifically, in the same way that trees reach the smallest possible amount of curvature in undirected graphs, here hyperedges in directed hypertrees get the lowest possible number.

Definition 2.2 A directed loopless hypergraph is a hypertree if

- (1) There is at most one directed path between any two vertices and
- (2) It does not contain any directed cycle.

We note that although these two conditions are equivalent in undirected (hyper)graphs, they do not coincide in the directed case.

Theorem 2.2 Let $A = \{x_1, \dots, x_n\}$ and $B = \{y_1, \dots, y_m\}$ be two subsets of vertices V of a hypertree H with respectively n and m elements and $A \rightarrow B$ be a hyperedge in this hypertree. If k elements in A have no incoming hyperedge, namely $\#\{x_i \in A, d_{x_i}^{in} = 0\} = k$ and k' element in B have no outgoing hyperedge, i.e. $\#\{y_j \in B, d_{y_j}^{out} = 0\} = k'$, we have: $\kappa(e) = -2 + \frac{k}{n} + \frac{k'}{m}$.

Proof Note that since e is in a hypertree, according to the definition $\mu_0 = 0$. Hence in a hypertree, the curvature of any hyperedge is non-positive ($\kappa(e) \leq 0$). We shall propose a transfer plan, which gives us an upper bound for W , and we will obtain a lower bound for W based on a single Lipschitz function (defined on the support of $\mu_{A^{in}}$ and $\mu_{B^{out}}$). We shall see that these two bounds coincide.

First we move $\left(\frac{k}{n} \wedge \frac{k'}{m}\right)$ of the mass from k x_i 's to k' y_j 's with distance one. Then if $\frac{k}{n} \geq \frac{k'}{m}$ we move $\frac{k}{n} - \frac{k'}{m}$ of the mass from x_i 's with no incoming hyperedges to outgoing neighbours of the y_j 's with distance 2 and if $\frac{k'}{m} > \frac{k}{n}$ we move $\frac{k'}{m} - \frac{k}{n}$ of the mass at incoming neighbours of x_i 's to those y_j 's with no outgoing hyperedges with distance 2. Then we move the remaining part of the mass with distant 3. Therefore $W \leq 3 - \frac{k}{n} - \frac{k'}{m}$.

On the other hand, for all z in $V(H)$ we define

$$f(z) = \begin{cases} 3 & \exists 1 \leq i \leq n, \exists e: z \rightarrow x_i \\ 2 & \exists 1 \leq i \leq n, z = x_i \\ 1 & \exists 1 \leq j \leq m, z = y_j \\ 0 & \text{otherwise} \end{cases}$$

It is straightforward to check that f is a Lipschitz function on $A \cup B \cup \text{supp } \mu_{A^{in}} \cup \text{supp } \mu_{B^{out}}$, hence according to the Theorem 3.5, we have

$$\begin{aligned} W(\mu_{A^{in}}, \mu_{B^{out}}) &\geq \sup_{z \rightarrow A} \sum \mu_{A^{in}}(z) f(z) - \sum_{B \rightarrow z'} \mu_{B^{out}}(z') f(z') \\ &\geq 3 \left(1 - \frac{k}{n}\right) + 2 \left(\frac{k}{n}\right) - 1 \times \frac{k'}{m} - 0 \times \left(1 - \frac{k'}{m}\right) \\ &= 3 - \frac{k}{n} - \frac{k'}{m}. \end{aligned}$$

$$\text{and } \kappa(e) = -2 + \frac{k}{n} + \frac{k'}{m}.$$

□

Extension and reduction.

- **Removing vertices from a hyperedge (Reduction)** In this part we want to investigate what happens to the curvature of an edge $e : A = \{x_1, \dots, x_n\} \rightarrow B = \{y_1, \dots, y_m\}$ if we remove a number (l, l') of vertices from A ($l \leq n$) and/or from B ($l' \leq m$). Curvature depends on the connections between elements of $\text{supp } \mu_{A^{\text{in}}}$ and $\text{supp } \mu_{B^{\text{out}}}$ and removing different vertices from A (and/or B) might have different effects on the curvature. However since the amount of the masses (size of holes) which is assigned to any x_i (and y_j) is already determined and is equal to respectively $1/n$ (and $1/m$), we can give a bound for such changes.

Proposition 2.3 Let $e : A = \{x_1, \dots, x_n\} \rightarrow B = \{y_1, \dots, y_m\}$. By removing a vertex x_i from A we get $e' : A - \{x_i\} \rightarrow B = \{y_1, \dots, y_m\}$ and we have

$$|\kappa(e') - \kappa(e)| \leq \frac{3}{n}.$$

Similarly, by removing l vertices from A ($l < n$) we have

$$|\kappa(e') - \kappa(e)| \leq \frac{3l}{n}$$

Proof The two bounds for the curvature of e' arise from two extreme scenarios which might happen while removing vertex x_i (or l vertices) from A .

1. If the whole mass which is assigned to x_i is in directed loops or directed 3-cycles including x_i and any of y_j s, and after removing x_i , its corresponding mass has to be moved with distance 3 in an optimal plan, then

$$\begin{cases} \kappa(e) = \mu_0 - \mu_2 - 2\mu_3 \\ \kappa(e') = (\mu_0 - \frac{1}{n}) - \mu_2 - 2(\mu_3 + \frac{1}{n}). \end{cases}$$

2. If the whole mass around x_i was transported with distance 3 in an optimal plan and after removing this vertex, the corresponding mass is in the place of directed loops or directed 3-cycles including vertices of $A - \{x_i\}$ and B , then

$$\kappa(e') = \left(\mu_0 + \frac{1}{n}\right) - \mu_2 - 2\left(\mu_3 - \frac{1}{n}\right) = \kappa(e) + \frac{3}{n}.$$

Therefore we have

$$\kappa(e) + \frac{3}{n} \geq \kappa(e') \geq \kappa(e) - \frac{3}{n}.$$

The same argument works for removing l vertices from A and the proof is complete. $\square$

Analogously, by removing l' vertices from B ($l' < m$) and using the same argument as before for the holes assigned to B , we have

$$|\kappa(e') - \kappa(e)| \leq \frac{3l'}{m}.$$

Therefore:

Proposition 2.4 By removing l vertices from the set A and l' vertices from B ($e : A \rightarrow B$) the following relation holds between the curvature of the resulting hyperedge (e') and the old one :

$$|\kappa(e') - \kappa(e)| \leq 3\left(\frac{l}{n} + \frac{l'}{m}\right) \wedge 3$$

- **Adding vertices to a hyperedge (Extension)** Here we want to obtain bounds for the curvature of a hyperedge obtained by adding some new vertices to the set A and/or to B and possibly adding new connections between them.

Proposition 2.5 Let $e : A = \{x_1, \dots, x_n\} \rightarrow B = \{y_1, \dots, y_m\}$. By adding l vertices to A and l' vertices to B we get a hyperedge $e' : A' = \{x_1, \dots, x_{n+l}\} \rightarrow B' = \{y_1, \dots, y_{m+l'}\}$ with

$$|\kappa(e') - \kappa(e)| \leq 3\left(\frac{l}{l+n} + \frac{l'}{l'+m}\right) \wedge 3$$

Proof Here, since to each x_i in A' we assign $\frac{1}{l+n}$ of the total mass ($= 1$) and $\frac{1}{l'+m}$ of the total hole ($= 1$) is assigned to each y_j in B' , by considering the two extreme scenarios as before we have:

$$\mu_0(e) \rightarrow \mu_0(e') \pm \left(\frac{l}{l+n} + \frac{l'}{l'+m} \right) \wedge 1$$

and therefore:

$$\mu_3(e) \rightarrow \mu_3(e') \mp \left(\frac{l}{l+n} + \frac{l'}{l'+m} \right) \wedge 1$$

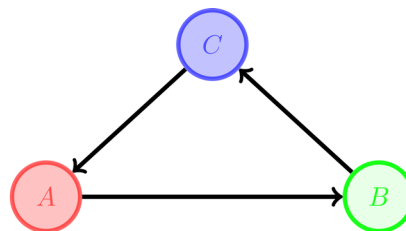
and according to the formula 2 the proof is complete. $\square$

Remark 2.5 Note that in Propositions 2.3, 2.4, 2.5, the equality holds when any of the mentioned extreme scenarios happen.

Hypergraphs with constant Ricci curvature. We want to construct examples of directed hypergraphs in which the curvature of the hyperedges is constant ($\kappa = 1, \kappa = 0, \kappa = -2$). In the case of $\kappa = 0$ these (hyper) graphs are called Ricci flat. For brevity, we also call the others Ricci 1 and Ricci -2 directed hypergraphs.

- Ricci 1 directed hypergraphs

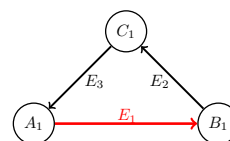
Theorem 2.3 The vertices of a Ricci 1 directed loopless hypergraph which for every hyperedge $e : \{x_1, \dots, x_n\} \rightarrow \{y_1, \dots, y_m\}$ does not have any hyperedge in the reverse direction ($\nexists e' : y_j \rightarrow x_i$), can be divided into 3 subsets A, B, C such that $A \rightarrow B \rightarrow C \rightarrow A$. This means that some (not necessarily all) vertices in A are connected to vertices in B via a non-empty collection of directed hyperedges and similarly for the other connections as in the following diagram.



Proof Consider a hyperedge $e_1 : A_1 \rightarrow B_1$. Since $\kappa(e_1) = 1$

$$\text{supp } \mu_{A_1^{\text{in}}} = \text{supp } \mu_{B_1^{\text{out}}} =: C_1 \quad \text{and} \quad \forall z \in C_1 : \mu_{A_1^{\text{in}}}(z) = \mu_{B_1^{\text{out}}}(z) \quad (3)$$

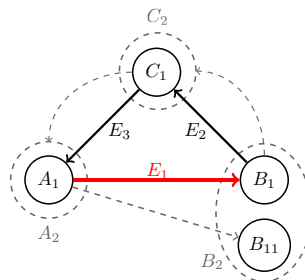
So the diagram related to e_1 looks like



where E_2 and E_3 represent collections of directed hyperedges. Now, if there is no outgoing hyperedge from C_1 other than elements in E_3 and there is no incoming hyperedge to C_1 other than elements of E_2 and there is no outgoing hyperedge other than e_1 from A_1 and there is no incoming hyperedge to B_1 other than e_1 , then A_1, B_1 and C_1 would be the desired partitioning set; Since the hypergraph is loopless, the intersection of any two of these sets is empty and because it is weakly connected the union of the vertices is the whole vertex set, $V(H)$. If any of the above conditions does not hold, we can extend A_1 and/or B_1 and/or C_1 as follows:

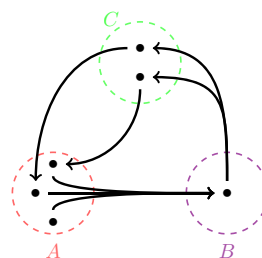
For instance, let there be at least one hyperedge going out of A_1 other than e_1 ; we call it $e_{OA_1} : A_1 \rightarrow B_{11}$ and we put $B_2 = B_1 \cup B_{11}$ where B_1 and B_{11} are not necessarily disjoint. Since $\kappa(e_{OA_1}) = 1$, so $C_2 := \text{supp } \mu_{B_{11}^{\text{out}}} = \text{supp } \mu_{A_1^{\text{in}}} \supseteq C_1$. We next consider edges in E_3 ; If any of them has an endpoint outside A_1 and if the set of endpoints of E_3 is denoted by A_2 , then $A_1 \subseteq A_2$. By repeating this process we obtain an increasing

sequence of A_i 's, B_j 's and C_k 's. We put $A = \cup A_i$, $B = \cup B_j$ and $C = \cup C_k$. Obviously, based on the process, elements in A are connected to B , B to C and C to A and these 3 sets are our desired partitioning.



□

Remark 2.6 The converse of this theorem is not necessarily true. For instance, the following hypergraph is not Ricci 1 although there is such a partitioning for the set of vertices of this hypergraph:



Instead we have the following:

Proposition 2.6 If in the corresponding directed graph of a directed (loopless) hypergraph, the set of vertices can be partitioned into three different sets A, B, C such that $A \rightarrow B \rightarrow C \rightarrow A$ and all of the elements in A are connected (via directed edges) to all the elements in B and similarly for the other arrows as above, then the original directed hypergraph is Ricci 1.

Before proving this Proposition we state the next Lemma.

Lemma 2.2 A directed (loopless) graph is Ricci 1 iff its set of vertices can be partitioned into 3 sets A, B, C such that $A \rightarrow B \rightarrow C \rightarrow A$ and all the vertices in A are connected to all the vertices in B and similarly for the other arrows, as in the digram in previous theorem.

Proof $\Rightarrow$ The proof is similar to that of Theorem 2.3. Here, in addition all the vertices of A should be connected to all the vertices of B and so on. The reason is that here, for every edge $e : x \rightarrow y$, $d_x^{\text{in}} = d_y^{\text{out}}$, and the condition that $\text{supp } \mu_x^{\text{in}} = \text{supp } \mu_y^{\text{out}}$ implies that the tails of incoming edges to x coincide with the heads of outgoing edges from y . Hence in the resulted partition every vertex in A is connected to every vertex in B and similarly for the connections between other sets the same situation holds.

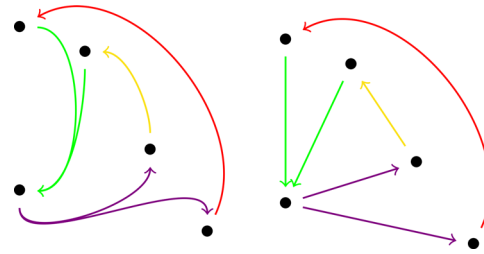
$\Leftarrow$ For proving that every edge has curvature 1, for every edge $e : x \rightarrow y$, we should have $d_x^{\text{in}} = d_y^{\text{out}}$ and $\text{supp } \mu_x^{\text{in}} = \text{supp } \mu_y^{\text{out}}$ and for every z in this support $\mu_x^{\text{in}}(z) = \mu_y^{\text{out}}(z)$. Since in the partition all the vertices in A are connected to all the vertices of B and so on, for every edge the needed conditions obviously hold and the directed graph is Ricci 1. □

Proof of Proposition 2.6 Since we have such a partition for the vertices of the corresponding directed graph of this hypergraph, according to the previous Lemma, the curvature of all the corresponding edges of each directed hyperedge is 1. Therefore their minimum also has curvature 1. On the other hand, according to Proposition 2.1

$$\kappa(\text{every hyperedge}) \geq \min \kappa(\text{edges in the corresponding directed graph})$$

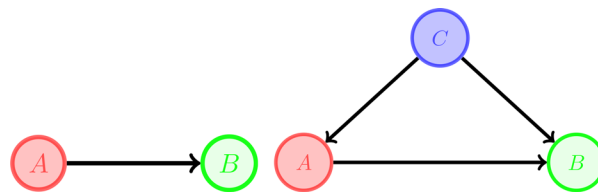
So for all hyperedges e , $\kappa(e) = 1$ and the hypergraph is Ricci 1. □

Remark 2.7 It might be possible that the directed hypergraph is Ricci 1, but as shown in the following example, its corresponding directed graph is not.



Ricci flat directed hypergraphs

Theorem 2.4 If the vertices of a directed hypergraph can be divided, as in the left diagram, into two sets A (source) and B (sink) such that all the vertices in A have outgoing hyperedges and no incoming hyperedges and all the vertices of B have incoming hyperedges and no outgoing hyperedges, then the hypergraph is Ricci flat. Also If the set of vertices can be divided into three sets, A, B, C such that all the vertices in these sets are connected to the vertices of the other sets as indicated in the right diagram, then the hypergraph is Ricci flat.



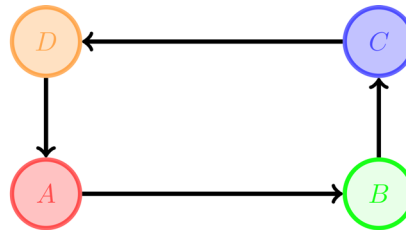
Proof Based on the construction, for the two set partitioning, for every hyperedge, the masses are in the source set (A) which is at distance one from the holes which are in the sink set (B). So μ_1 is equal to 1 and the hypergraph is Ricci flat. For the other case, since we are assuming that all the vertices of A are connected to all the vertices of B and similarly for other connections the same condition holds, for every hyperedge $e : A \rightarrow B$, the distance between any incoming neighbour of A to any outgoing neighbour of B is one. So μ_1 is equal to 1 (and obviously by construction $\mu_0 = 0$). Therefore the hypergraph is Ricci flat. $\square$

Remark 2.8 The converse of the previous theorem is not necessarily true. However, If in a Ricci flat directed hypergraph for every directed hyperedge e there is no incoming hyperedge to its tail set and there is no outgoing hyperedge from its head set, then the set of vertices in this directed hypergraph can be partitioned into two sets A and B as above. That is because we can put all the tail sets of all of directed hyperedges in set A and all the head sets of all of directed hyperedges in set B .

In the above theorem, for 3-set partitioning, in contrast to the 2-set partitioning, the sets are partitioned into source, saddle and sink sets. The vertices in a saddle have both incoming and outgoing hyperedges. Also similar to the Ricci 1 case, connections inside any of these three sets might violate constant curvature along different hyperedges. As the last case, in the following part we introduce a class of directed hypergraphs with the most negative curvature.

Ricci negative (-2) directed hypergraphs

Proposition 2.7 If the set of vertices of a directed hypergraph can be divided into 4 sets, A, B, C and D such that all the vertices in these sets are connected to the vertices of the other sets as indicated, then the hypergraph is Ricci -2 . The presence of internal hyperedges (connections inside each of these sets) might violate constant curvature along different hyperedges.



Proof Here for every hyperedge $e : A \rightarrow B$, the distance between any incoming neighbour of A to any outgoing neighbour of B is 3. Therefore all $\mu_i = 0$, except $\mu_3 = 1$, and every hyperedge has curvature -2 . $\square$

Weighted directed hypergraphs. We can extend our constructions to weighted directed hypergraphs where the vertices and hyperedges may both carry weights. The vertices may carry different weights depending on the hyperedges they are involved in (This can be represented by a vector of the dimension of the hyperedge set with non-negative components. Here, zero means the corresponding hyperedge does not involve that vertex). For a specified hyperedge whose curvature we want to measure, the weights of its vertices need to be fixed, of course. In this case we denote the vertex and hyperedge weights by w_v and w_e , respectively

Definition 2.3

$$\kappa(e) := 1 - W(\mu_{A^{in}}, \mu_{B^{out}})$$

where the probability measures $\mu_{A^{in}}$ (called mass) and $\mu_{B^{out}}$ (called hole), are defined on V as follows: Let $H = (V, E)$ be a weighed directed hypergraph and $e \in E$ an arbitrary directed hyperedge such that $A = \{x_1, \dots, x_n\} \xrightarrow{e} B = \{y_1, \dots, y_m\}$ ($n, m \leq |V|$). We define the Ricci curvature of this hyperedge as

$$\kappa(e) := 1 - W(\mu_{A^{in}}, \mu_{B^{out}})$$

where the probability measures $\mu_{A^{in}}$ and $\mu_{B^{out}}$ are defined on V as follows:

$$\mu_{A^{in}} = \sum_{i=1}^n \mu_{x_i}^{in} \quad \forall 1 \leq i \leq n \quad \text{and} \quad \forall z \in V(H)$$

$$\mu_{x_i}^{in}(z) = \begin{cases} 0 & z = x_i \quad \& \quad d_{x_i}^{in} \neq 0 \\ \frac{w_{x_i}}{\sum_{i=1}^n w_{x_i}} & z = x_i \quad \& \quad d_{x_i}^{in} = 0 \\ \sum_{e' : z \in A_{e'}, x_i \in B_{e'}} \frac{w_{x_i}}{\sum_{x_j \in A_{e'}} w_{x_j}} \times \frac{w_{e'}}{\sum_{e : z \rightarrow A} w_e} \times \frac{w_z}{\sum_{z \in A_{e'}} w_z} & z \neq x_i \quad \& \quad z \in A_{e'} \\ 0 & \text{otherwise} \end{cases}$$

Similarly, $\mu_{B^{out}} = \sum_{j=1}^m \mu_{y_j}^{out}, \forall 1 \leq j \leq m, z \in V(H)$

$$\mu_{y_j}^{out}(z) = \begin{cases} 0 & z = y_j \quad \& \quad d_{y_j}^{out} \neq 0 \\ \frac{w_{y_j}}{\sum_{j=1}^m w_{y_j}} & z = y_j \quad \& \quad d_{y_j}^{out} = 0 \\ \sum_{e' : y_j \in A_{e'}, z \in B_{e'}} \frac{w_{y_j}}{\sum_{y_i \in A_{e'}} w_{y_i}} \times \frac{w_{e'}}{\sum_{e : B \rightarrow z} w_e} \times \frac{w_z}{\sum_{z \in B_{e'}} w_z} & z \neq y_j \quad \& \quad z \in B_{e'} \\ 0 & \text{otherwise} \end{cases}$$

Remark 2.9 In weighted directed graphs, the Proposition 2.6 does not hold because, due to the weights we might have masses which coincide with holes of different sizes and therefore it violates Ricci 1 condition. For instance if we consider two directed 3-cycles which have one edge in common, by considering non-equal weights assigned to two other edges in the two cycles, the curvature of the common edge is not one although we have a three set partitioning in which all the required connections exist.

Further differences between directed and undirected (hyper)graphs.

- In directed (hyper)graphs, lower curvature bounds no longer control random walks.
- Since the Wasserstein distance no longer needs to satisfy a triangle inequality, we cannot define curvatures for vertex sets that are not connected by a hyperedge.

These problems come essentially from the fact that we consider incoming edges at the tail A and outgoing edges at the head B of a hyperedge. In principle, we could of course also consider in-in or out-out relationships instead, but then, we might not always be able to move our masses, and so, curvatures might then become $-\infty$. This can only be avoided if we assume some strong connectedness condition in the directed case (see for instance¹⁵). Such a condition, however, is typically not satisfied in empirical data sets.

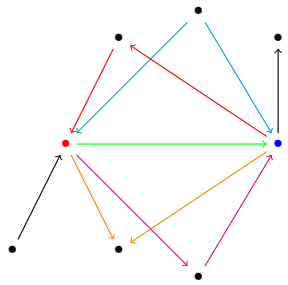


Figure 3. Directed triangles and their different impact on curvature.

- The curvature of a directed (hyper)graph might be rather different from that of the underlying undirected (hyper)graph. For instance, in an undirected graph, every edge in a quadrangle has curvature zero. But a directed k -cycle where $k \geq 4$ is negatively curved.
- Directed triangles; For undirected graphs, it has been proven that the local clustering coefficient at each vertex can control the scalar curvature of any vertex which by definition is obtained by averaging over the Ricci curvature of all the edges connecting to that vertex (see Corollary 1 in⁸). In the undirected graph case, the local clustering coefficient is based on the number of triangles containing that vertex and its neighbours. In the directed case, after fixing the direction of every hyperedge, we encounter with four different types of triangles, bounded by only three edges, which share the property of having a directed edge that goes out from A and enters into the set B . Therefore in contrast to the undirected graph case, not all directed triangles but only some of them affect the curvature; in fact, for μ_0 the presence of directed 3-cycles containing the vertices of A and B and of vertices u where $u \rightarrow x_i \rightarrow y_j \rightarrow u$, $u \in A^{in}(u \rightarrow A)$ and $u \in B^{out}(B \rightarrow u)$ increases the curvature of the corresponding hyperedge since they directly increase the amount of μ_0 . In contrast, those directed triangles which contain vertices of A and B and u in such a way that $x_i \rightarrow u$, $x_i \rightarrow y_j \rightarrow u$, $u \in A^{out}(A \rightarrow u)$ and $u \in B^{out}(B \rightarrow u)$ and those containing A and B and u such that $u \rightarrow x_i \rightarrow y_j$, $u \rightarrow y_j$, $u \in A^{in}(u \rightarrow A)$, $u \in B^{in}(u \rightarrow B)$ might affect on the amount of μ_2 and thereby can make the curvature less negative. The last type of directed triangles are those which include vertices of A and B , outgoing vertices of A and incoming vertices to the set B , but they do not affect any of the μ_i s and therefore the curvature. For instance in the directed graph depicted in Fig. 3, for computing the curvature of the green edge, the triangle consisting of the red and green edges affects μ_0 . Both triangles including orange-green and blue-green edges have an effect on μ_2 and the curvature is not affected by the presence of the triangle of the pink-green edges.

Conclusion

In contrast to ordinary graphs, hypergraphs can faithfully model higher order interactions between data points. Therefore, in empirical network science, nowadays graph representations gradually give way to hypergraphs. Under the label of Petri nets, directed hypergraphs have played an important role in computer science, but with an emphasis rather on processing schemes than on network properties. As systems get larger, however, network aspects are gaining importance⁵. Similarly, with the availability of large scale data sets, hypergraph methods are currently actively developed for the investigation of chemical reaction systems. Since we believe that instead of ad hoc schemes, one should rather develop a systematic mathematical theory, here we generalize concepts and constructions from Riemannian and metric geometry in order to provide such a theory. In particular, we have found that a version of the fundamental notion of Ricci curvature seems a natural tool to probe the local geometry of hypergraphs. Like the Ollivier Ricci curvature in graph theory, this curvature is defined in terms of optimal transport problem between probability distributions assigned to sets of vertices. Our notion can be used to differentiate between positively curved, flat and negatively curved hypergraphs. Consequently, in⁹, applying this tool to metabolic networks we have been able to identify patterns in these networks that can clearly distinguish them from random models that have been proposed as universal models in the literature. Thus, in this paper we demonstrate that a concept inspired from pure mathematics can be fashioned into a novel tool in applied mathematics for analysing systems of interacting elements.

Data availability

The datasets generated and/or analysed during the current study are available from the corresponding author on request.

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Author contributions

This paper is part of the dissertation of M.E., written under the supervision of J.J.

Competing interests

The authors declare no competing interests.

Additional information

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Assembly theory and its relationship with computational complexity



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Assembly theory (AT) quantifies selection using the assembly equation, identifying complex objects through the assembly index, the minimal steps required to build an object from basic parts, and copy number, the observed instances of the object. These measure a quantity called Assembly, capturing causation necessary to produce abundant objects, distinguishing selection-driven complexity from random generation. Unlike computational complexity theory, which often emphasizes minimal description length via compressibility, AT explicitly focuses on the causation captured by selection as the mechanism behind complexity. We illustrate formal distinctions through mathematical examples demonstrating that the assembly index is fundamentally distinct from complexity metrics like Shannon entropy, Huffman encoding, and Lempel–Ziv–Welch compression. We provide proofs showing that the assembly index belongs to a different computational complexity class compared to these measures and compression algorithms. Additionally, we highlight AT's unique ontological grounding as a physically measurable framework, setting it apart from abstract theoretical approaches to formalizing life that lack empirical measurement foundations.

Understanding evolution in the most general terms, including how the process of selection manifested before the genetic code, is of primary interest for uncovering the origin of life and the prospects of other life in the universe^{1–7}. A major challenge is that current theories of evolution typically begin from some intermediate, preexisting biological system and, as such, can only describe the evolutionary dynamics within that system. Assembly theory (AT) was developed to provide a more general framework for understanding evolution and selection, allowing detection of the signatures of any evolutionary process⁸, including probing stages of evolution on Earth predating genomes. By combining a formal theoretical^{8–10} and experimental^{11,12} approach to determine whether or not observed objects are the product of evolution, AT is opening new avenues of research for life detection^{13–16}, the design of synthetic life^{17,18}, the design of experiments to probe the de novo origin of life in the laboratory¹⁹, and opportunities to look deeper into the history of life on Earth (e.g., predating the evolution of the genome)²⁰. In its formalization, AT includes a new, empirically measurable form of complexity, Assembly, that adds unique features to the field of complexity metrics and opens avenues for how complexity can be measured as a physical quantity.

As with any theory that attempts to capture broad regularities in the physical world, there is necessarily a large amount of abstraction in going

from measurement to theory in the development of AT²¹. This has led to some confusion about whether what the theory proposes is genuinely new^{22–25} or is already captured by other theoretical metrics of complexity developed over many decades^{26,27}. Some researchers have claimed that AT is identical to several other complexity measures^{22–24}, but do so by pointing to mathematical formalizations that are not formally equivalent to one another, and, as we show here, are also not formally equivalent to AT. In this work, we hope to clarify how AT differs formally and in practice from currently available complexity measures. Providing clarity to the discussion requires demonstrating formal uniqueness and understanding the connection between different theoretical concepts of complexity, and in the case of AT, the role of its ontological basis in physical measurement. The assembly index (AI), a critical component of AT, has been demonstrated to be measurable as an intrinsic physical observable in molecules. This requires being explicit about differences in the physical basis of computational complexity and AT, because the latter was explicitly developed as an empirically-grounded theory of complexity, not a purely theoretical one.

The theory of computation was not explicitly developed to solve the problem of the origin of life²⁸, nor was it even devised as a formal approach to life or to physical systems. In fact, one of the central goals of theoretical computer science is formal analysis without reference to particular

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hardware or implementations. Nonetheless, the connection between computation and living processes has been noted by many researchers over the years^{29–32}. The key ideas of theoretical computer science were laid out in the seminal work of Turing³³ and Church³⁴. Early work by von Neumann attempted to analogize the concept of universal computation to that of universal construction³⁵, providing a thought experiment of an abstract machine that can construct any buildable physical object (rather than compute any computable function as a universal Turing machine does). This difference is important because one theory, that of computation, deals with the inputs and outputs of programs, how to describe these, the computational resources (time and space) required, and the complexity of algorithms for producing specific outputs. The other, that of construction, was intended to deal with objects constructed by physical processes, not data and algorithms. That is, in the latter, the constraints of physical law are necessarily foundational³⁶. While von Neumann's universal constructor was primarily a theoretical concept, it was later implemented in computational experiments with cellular automata³⁵, and the distinction between computation and construction has, thus far, remained mostly abstract. Self-replicating robotics has extended this idea to the physical world, but so far requires prefabricated building blocks and are far from being universal constructors³⁷. Assembly theory (AT) makes the connection of these ideas to physics and metrology³⁸ concrete, thus providing the first step toward a testable theory of selection as a mechanism for constructing objects.

In what follows, we first introduce the foundations of AT as developed to date. We subsequently establish why AT is not the same as computational approaches to life. (1) We show with explicit examples how a key component of AT, the assembly index (AI), is not formally equivalent to other measures of complexity derived from computer science and information theory. Specifically, we show, with simple but rigorous mathematical counterexamples, why AI yields quantitatively different results when compared to compression algorithms like Huffman coding and Lempel–Ziv–Welch (LZW) compression, which have been argued to be identical to AI. (2) We provide formal proofs for the calculation of the assembly index of strings, showing it is not in the same computational complexity class as these compression algorithms, definitively demonstrating that no formal equivalence is possible. We demonstrate that computation of the assembly index is NP-complete, so its computation is mathematically in an equivalent class to other optimal compression algorithms. However, this similarity does not undermine its ontological differences. We make this comparison not to equate assembly theory with abstract computation, but to lay to rest certain claims made that the assembly index is equivalent to metrics and algorithms in the complexity class P, which it cannot be. We therefore (3) also discuss metrology, and how assembly theory provides a rigorous approach to the physical measurement of complexity and life detection, which has not been shown to be emulated by computational complexity measures. We discuss how empirically, the assembly index is developed as a physical observable, whereas Huffman coding, LZW compression and Kolmogorov complexity are computed from labeled data. (4) Because it is physically relevant, it should be expected that assembly, when used to detect life and evolution, will have lower false positive and false negative rates than purely algorithmic measures. Because it is a context-free approach, it is also a standardized comparison across experiments and planetary environments. Any of these points would be sufficient to distinguish AT from computational approaches to formalizing life, but we also highlight how (5) AT is not based on AI alone and reemphasize the importance of the copy number in AT. Approximating any feature of AT with a mathematical measure does not replace the physical insights the theory aims to provide or invalidate the explanations it might afford.

Any useful approach to quantifying complexity relevant to life detection (in the lab or on other worlds) must bridge the reality gap between theory and empirical testing and validation. That is, it must engage with metrology. This is a primary reason why AT was developed in the context of a modern chemistry lab's capabilities, with its initial development as a rigorous method for laboratory studies aiming to identify molecules as

biosignatures¹¹. A key conjecture of AT is that “life” is the only mechanism for producing highly complex identical objects in abundance, ranging from molecules, to cells, to memes and beyond. This proposal leads to a key, testable prediction of the theory that high AI objects with a high copy number will not be found independently of evolutionary processes³⁹ (see e.g., ref. 40 for a broad discussion on abstracting the living process). While many kinds of measures of molecular complexity computed on graphs have been developed over the years^{41–43}, the key advance offered by AT was to propose and formalize a precise conjecture on the relationship between molecular complexity and life and allow an experimental test. This observation necessarily and intentionally grounds AT in experimental science, since the threshold of the copy number and AI that constitute a biosignature is an essentially empirical one. AT provides a physical measurement of how much selective constraints have been applied to generate an object and, therefore, how likely it is to be associated with a living process.

Assembly theory has the following observables: copy number, n , and the assembly index, a_i ⁸. The copy number, n , of an object refers to how many identical objects are observed. Here, it is important to clarify what an object is in AT, because the definition departs from most definitions in physics and other fields, including as used in other approaches to formalize complexity (we shall return to this point later). In AT, objects have the properties of being finite, distinguishable, breakable (and therefore constructable), and they must be able to persist in multiple identical copies. With this definition of an object, the assembly index a_i quantifies the minimal number of recursive, compositional joining operations necessary to construct an object from its elementary parts. The exact definition of assembly index is restricted to the assembly space of physically meaningful operations, see Fig. 1 for an example of the molecule adenosine. As AT is developed, we expect it will find applications beyond the initial focus on covalent chemistry, including more abstracted spaces like minerals, gene/protein sequences, language, and many types of technology.

For molecules, both the copy number, n (abundance of molecular species) and a (minimal number of bond-making operations, allowing reuse of parts) are measurable in the lab. Assembly index a_i , referred to as the molecular assembly index (MA) for molecules, can be measured using mass spectrometry, NMR, and infrared techniques¹². For an observed configuration of complex matter (e.g., a system that can be broken apart to more basic building blocks), taking account of both the number of copies of distinguishable objects and their assembly indices allows quantifying Assembly, A , which we conjectured captures the total amount of selection required to produce it. Formally, for an ensemble of N total (non-unique) objects, the Assembly is defined⁸ as

$$A = \sum_i e^{a_i} \left(\frac{n_i - 1}{N} \right) \quad (1)$$

where the terms of n_i and a_i are the copy number and assembly index, respectively, of the i th distinguishable object in the ensemble. The assembly equation, Eq. 1, represents a formalization of selection and evolution in a very large combinatorial space, the assembly space, defined as the set of recursive, physically implementable (and thus measurable) operations from which objects can be constructed (Fig. 1).

AT suggests that the combinatorial space of evolutionary objects is best understood in terms of coordinates of assembly index and copy number within the assembly space⁸. Finding objects with high assembly indices in high copy number, yields large values of A , implying substantial selection must have occurred^{11,12}. This is because it becomes super-exponentially more difficult to produce identical copies of increasingly assembled objects with large a_i by chance alone (i.e., there is a super-exponential expansion of the combinatorial space for each possible joining operation). That is, observing $n_i \gg 1$ for a complex object with sufficiently high a_i is nontrivial, and in AT is conjectured to be impossible to happen abiotically. In AT, configurations of matter with large A are unreasonably unlikely to occur outside of selective processes found in what we call “life”. Here, we are using selection in a generalized sense: selection is the mechanism whereby a set of

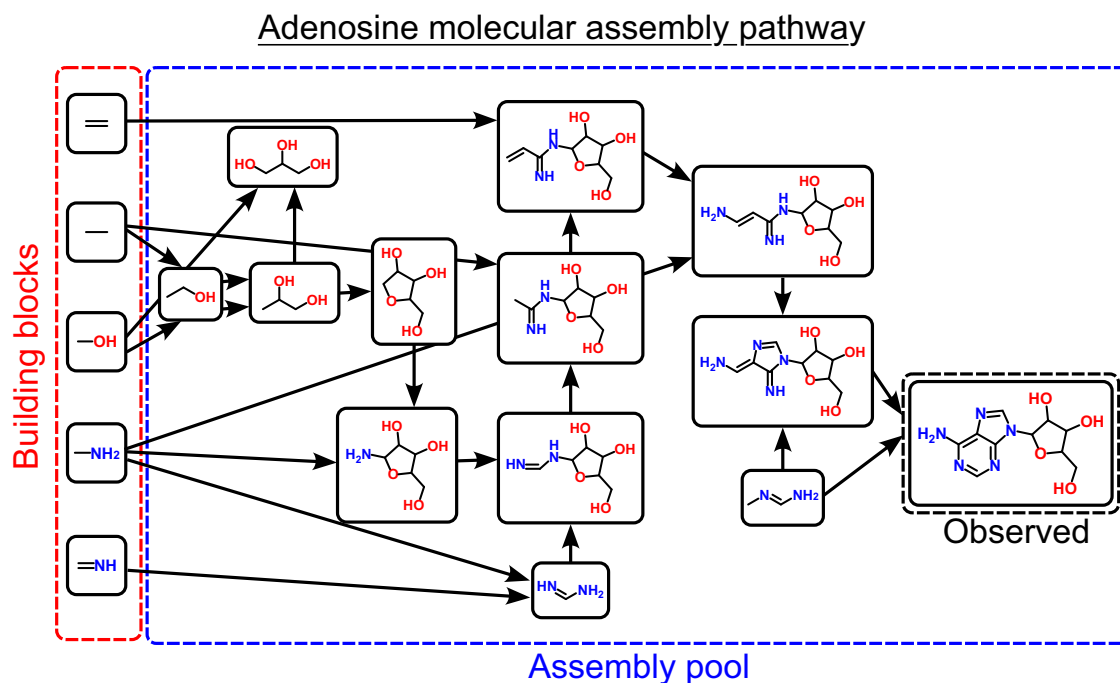


Fig. 1 | An illustration showing how a molecule can be cut into basic units, and recursively assembled from them to construct a minimum assembly path. The molecule's molecular assembly index is the number of steps on the path.

Table 1 | Comparative ontology of complexity measures

Complexity Theory	Operands	Measures	Description/Details
Algorithmic complexity	Classes of mathematical problems	Computational complexity Kolmogorov complexity;	Measures are concerned with optimal (minimal) descriptions of products of computational procedures, or the computational resources required for an algorithm that could produce the given output
Compression and statistical complexity	Data and its representations	Huffman code; Lempel–Ziv–Welch compression; Information theory	Measures provide data transformations to generate reduced representations
Physical complexity	States of physical systems	Logical depth; Thermodynamic depth;	Measures are concerned with the physical resource needed to compute a minimum description, or the probabilities of observed paths to produce a state
Assembly theory	Physical objects	Assembly index	Measure is concerned with the minimum construction as an intrinsic and measurable physical attribute

Examples of different kinds of complexity theories and measures, describing their operands, measures, and conceptual underpinnings.

constructive circumstances yields the formation of some subsets of objects over others. This spans simple reaction rate differences to formal modern genetics, where the strength of selection would also be larger for the latter, thus yielding higher A . Laboratory measurements of molecular assembly indices confirm the conjecture that an observational threshold for the evidence of selection and evolution exists. To date, experimental studies of abiotic, biological, and blinded samples indicate that molecules with $a_i > 15$ are found only in living samples¹¹.

Results

Overview of complexity measures

The ontology underlying AT as a theory of physics is different than that of the theory of computation, see Table 1. In what follows, we compare AT to computational and information-theoretic approaches to quantifying complexity. For some comparisons, we give explicit examples, while for others we do not because the differences in ontology and computability already make clear the most salient distinctions between different approaches, and in some cases these distinctions are so apparent as to make direct comparison not meaningful. For establishing mathematical uniqueness, we take the simplest and most illustrative approach, in the form of basic counterexamples. We consider cases where the assembly index is distinct for two

outputs, but those two outputs yield the same value in another complexity measure; and we show cases where the assembly index is the same for two outputs, but other complexity measures yield distinct values. Either scenario would be enough to show that the assembly index is not one-to-one with the comparison complexity measure.

Before we dive into computational complexity, it is important to note that there are different notions of equivalence. Computational complexity employs a weak notion of equivalence that does not require a strict one-to-one equality for metrics in the same broad complexity class. Physics and the natural sciences often require a stronger notion of equivalence associated with equality and one-to-one correspondence, often rooted in mechanism, and this notion is what we would require for the assembly index. These different disciplinary notions of equivalence have caused a lot of confusion.

The most widely discussed measure of computational complexity is the Kolmogorov–Chaitin complexity^{44,45}, also known as algorithmic complexity. Formally, the algorithmic complexity $K(x)$ of a string x is defined as the length of the shortest computer program, which, when run, will output x (assuming a fixed programming language). That is, $K(x)$ is the length of the minimum description of x , found by searching over all programs that output x in the language x is described in. The value of $K(x)$ for a given string, therefore, depends on the choice of description language (and, for example,

might be different if we wrote the program in C versus Python)⁴⁶. Foundational results in the theory of computation establish there can be no algorithm that determines whether an arbitrary program will produce a given output as this would involve, at least, deciding whether the program will halt, which is uncomputable³³ and therefore $K(x)$ is also uncomputable^{44,45}. For the example above, the two strings have equivalent size, but their complexity is not the same: $K(x_1) > K(x_2)$, and x_2 would thus be described as more compressible. However, there is no way to guarantee x_1 does not have some hidden pattern in it that makes it just as compressible as x_2 (that is, although we have shown $K(x_1) > K(x_2)$ for the two shortest programs we were able to identify, it could still be possible that there exists an algorithm $K'(x_1) \leq K(x_2)$).

In Abrahão et al.²² and Ozelim et al.²⁴, the authors make an argument that since AT does not offer any innovations in approximating K , it is rendered redundant by preexisting methods of lossless compression. However, this represents a fundamental misunderstanding because approximating K is not, and was never, the goal of AT. It is our view that K is not as relevant as a physical measurement of complexity generated by evolutionary processes. The assembly index comes to estimate a lower bound on the chance of the random generation of the molecule from molecular inflow into the system, by formalizing the minimum causation necessary to make an observed configuration. The algorithmic complexity, on the other hand, is the minimal size of the program required to generate a certain output—given a universal Turing machine. The latter introduces a deep philosophical challenge in that the construction of the molecule requires construction of the molecule itself by the UTM (we ignore for a second the fact that a UTM cannot construct molecules), but also the construction of the UTM itself (this in essence was von Neumann's point in developing the idea of a universal constructor). Construction of such a reliable computer will require many more steps than the construction of the molecules that we are interested in when detecting life. Though it is true that very simple physical systems can exhibit universal computation when properly encoded by human intervention⁴⁷, there is no evidence of any non-evolved system that exhibits universal computation, that can reliably control molecular reactions, and can form itself within a small number of steps in finite time. In fact, the only process that we know of anywhere in the universe that can build such a machine is the evolution of life itself. Until we can find such a process, or a reason for it to exist outside of evolution, the assumption that before the origin of life, molecules form in large numbers only through assembly processes in the absence of computers seems well-founded. Thus, the assembly index does not come to approximate K , instead K is a poor approximation for the minimal chance of formation of molecules, a task that the assembly index was specifically designed for. As an example, let us look at a polymer that represents the first 100 digits of the number e in some base, that is present in an abundance such that it can be unambiguously detected. This is a relatively simple computation, and therefore, the program to compute it can be small. However, the length of this program will not represent the chance for the formation of such a molecule. Instead, it will form by assembly of the monomers plus assembly of the 100 digits, very similar to the assembly index of any other random polymer of length 100. Without a universal Turing machine, the fact that there is a small program to generate the sequence is meaningless, and the hard task in the evolution of life is building the machines in the first place.

Motivation for AT and the Assembly index

Assembly index was developed as an exact, objective measure that can be probed by different experimental techniques, capturing symmetries and reuse of fragments in the assembly pathway to construct a molecule (Fig. 1). This distinguishes it from statistical notions of complexity, which rely on computed probabilities from the observed frequency of reuse of segments in a data structure. Nonetheless, the assembly index has been critiqued as both a restricted version of Huffman coding^{22,23} and described as reducible to Shannon entropy²⁴. Formally, a “restricted version” or “reducible” should imply that one formulation is a formal subset of another. For Huffman coding, Shannon entropy and assembly index, it is easy to explicitly show

these are not equivalent and therefore that assembly index is by no formal means a “restricted version” of Huffman coding or reducible to Shannon entropy.

Information theory provides the theoretical underpinnings for efficient data compression and coding techniques, establishing limits on to what degree data can be compressed without losing information (lossless compression), and how information can be transmitted over a noisy channel reliably. Concepts like algorithmic complexity^{44,45}, Huffman coding⁴⁸, arithmetic coding⁴⁹, and channel capacity⁵⁰ are all grounded in information-theoretic principles. Information-theoretic measures like Shannon entropy⁵⁰, mutual information⁵¹, transfer entropy⁵², and synergistic information⁵³, etc., are widely used to characterize the complexity, dependencies, and similarities in data across complex systems^{54,55}.

Shannon entropy is a powerful tool due its generality, but this also means one must exercise caution in its interpretation as it can be applied to any data labeling scheme one might wish to define^{56,57}. Indeed, Shannon entropy provides a good example of how being explicit about the ontology of the abstract variables that appear in our different mathematical concepts of information is critical to understand their utility (see Table 1). For example, if X labels the microstate of a physical system, the mathematical form for Shannon entropy is nearly identical to the thermodynamic entropy⁵⁸ $S(X) = kTH(X)$ where here k is the Boltzmann constant and T the temperature of the system. This allows a physical interpretation that has utility: there are many reasons to think that many complex systems, from languages to computer algorithms, would target distributions of states, and the energetic costs of moving between states, as a key feature for refinement. The formal equivalence of Boltzmann and Shannon entropies has therefore led to interesting work on the connections between information theory, computation, and thermodynamics⁵⁹. Stemming from the work of Landauer⁶⁰, much of this effort has focused on the minimal energy required to perform a computation by casting computations as entropy transformations, and on showing that storing and erasing information comes with certain energetic costs, which includes experimental verification⁶¹. Such connections have even been used to bound the thermodynamic efficiency of the ribosome as a string writing device.

Huffman coding⁴⁸ is among the most widely used lossless data compression algorithms. A Huffman code is constructed by first building a frequency table of characters and their occurrence counts, see Fig. 2. The compression efficacy of Huffman encoding depends on the character frequency skew, where data with highly frequent characters are compressed the most, and data with uniform distributions of characters are compressed the least.

As a simple illustrative example of formal differences between assembly index, Shannon entropy and Huffman coding, we consider the strings:

$$zbzbzca = a \text{ of } 4$$

$$zzzbbca = a \text{ of } 5$$

These strings are identical in length and composition (with the same number of z 's, b 's, and c 's), and thus have the same string entropy and Huffman coding. However, the two strings have different assembly indices (4 and 5, respectively).

The Huffman tree is constructed by creating a node for each character and repeatedly combining the two least frequent nodes/characters into a new parent node, until only one parent node, the root node, remains, see Fig. 2. For the example above, the tree that generates the code alphabet is identical for the two strings (because they have the same character frequencies) with the following code assignments

$$z \rightarrow 1$$

$$b \rightarrow 01$$

$$c \rightarrow 00$$

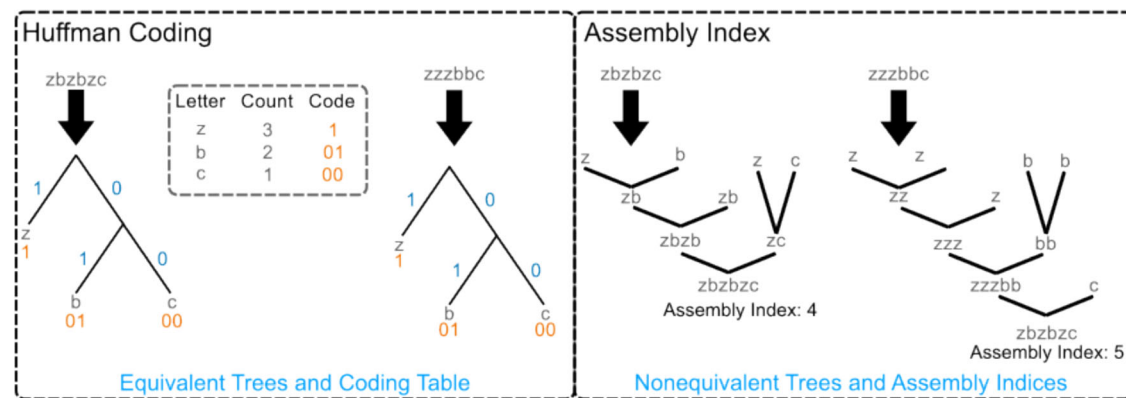


Fig. 2 | Two strings with different assembly index but identical Huffman trees. A single case such as this is sufficient to demonstrate that the assembly index cannot be a “restricted case” of Huffman coding, because it demonstrates that the assembly indices cannot be mapped one-to-one with Huffman trees.

Lempel-Ziv-Welch				Assembly	
Steps	Longest Reachable String	Construction Process	Coding Dictionary	Longest Reachable String	Construction Process
1	“z”	z	1, “z”	“z”	z
2	“zzz”	Z+ZZ	2, “zz”	“zz”	Z+Z
3	“zzzzzz”	Z+ZZ+ZZZ	3, “zzz”	“zzzz”	ZZ+ZZ
4	“zzzzzzzzzz”	Z+ZZ+ZZZ+ZZZZ	4, “zzzz”	“zzzzzzzz”	ZZZZ+ZZZZ
5	“zzzzzzzzzzzzzzzz”	Z+ZZ+ZZZ+ZZZZ+ZZZZZ	5, “zzzzz”	“zzzzzzzzzzzzzzzz”	ZZZZZZZZ+ZZZZZZZZ
String Length = 15				String Length = 16	

Fig. 3 | The most compressible string of a single letter as represented in n steps of Lempel–Ziv–Welch (LZW) compared with n assembly steps, demonstrating that the scaling of the two algorithms is not equivalent and by extension there can be no formal equivalence between the two.

which results in the encoded output strings

$$\begin{aligned} zbzbzc &\rightarrow 101101100 \\ zzzbbc &\rightarrow 111010100 \end{aligned}$$

There are several observations of note here. Two strings with different assembly indices can produce identical Huffman trees and code alphabets. The two representations, Huffman coding and assembly index, are not one-to-one because the Huffman trees are identical for strings that have different assembly spaces, see Fig. 2, providing a counterexample proof that AT cannot be a subset of Huffman coding (contrary to the claims of ref. 22) because there can be no 1:1 map between the two applicable in every case. This establishes that the assembly index is formally unique when compared to Huffman coding.

Assembly index is not a statistical measure and does not rely on the frequencies of characters; instead, it aims to capture features of the recursive composition of objects as they are constructed from a base alphabet. By contrast, Huffman trees are often highly degenerate because they map the specific structure of strings onto character frequencies, which is a many-to-one mapping. In general, the assembly index is more weakly dependent on character frequencies than such statistical measures, and it will yield different values in many cases where Shannon entropy or Huffman coding would not resolve differences between strings.

Some have argued that lossless compression, like that involved in generating a Huffman code, is relevant to evolution, particularly in describing genomes⁶². However, genomes are highly constrained by the evolutionary history of the ribosome, and physical constraints on protein folding⁶³. As such, compression might not yield the most accurate insights into evolutionary refinement of genomic architecture, especially as it does not account for the physical and physiological constraints present in all cellular life^{64–66} (or viruses⁶⁷). Given that compression-based measures are sensitive to the choice of data representation, they must be applied with care in a biological context: the labels humans use for data to describe living

things may not be the same labeling scheme those organisms implement themselves, even if it were the case where an organism was indeed performing data compression.

By contrast, the aim of AT is different because it does not presuppose that a compressed description is necessary, and the theory is agnostic to assumptions that cellular life is doing any kind of computation or algorithmic compression. That is, AT does not rely on a human or computer-based labeling scheme, or any kind of statistical description of data; instead, it aims to identify the underlying structure of the physical space in which selection operates (the assembly space).

The assembly index has also been criticized by the same group of researchers as identical to other compression algorithms, including the Lempel–Ziv–Welch (LZW) for an explicit example⁶⁸. For LZW a string sequence is read from left to right, and a lookup table is built for new substrings. The longest matching substring is chosen, and a new substring which includes that substring and the next letter in the input is added to the lookup table along with a new index. Regular signals can be characterized by a small number of patterns and hence have low LZW complexity. Irregular signals require long dictionaries and hence have a high LZW complexity (most schemes employ a maximum dictionary size based on the number of available character bits). The easiest way to establish that LZW is not formally equivalent to the assembly index is to look at the scaling of the simplest string case, a repeated single letter (Fig. 3). If we use LZW compression, string length, l , will grow following:

$$\begin{aligned} l_{i+1} &= l_i + i + 1 \\ l_1 &= 1 \end{aligned}$$

where i counts the number of compression steps to build up to the next most compressible string of a single letter. This yields a scaling of the longest string length that can be reached by n compression steps (or dictionary

construction steps) as

$$l_n = \binom{n(n+1)}{2}$$

Assembly index looks for the shortest path using parts that have already been constructed, and it has previously been shown that this string will grow with assembly steps like^{8–10}:

$$l_a = 2^a$$

where we have used the relation of assembly index to string length for single character strings, $a = n - 1$, for n assembly steps. It is clear that the scaling of length versus steps for assembly index and LZW are not equivalent: for n steps in LZW compression, string length scales like $l_n \propto n^2$ (in the limit of large strings) compared to the much faster scaling of $l_a \propto 2^a$ for $a + 1$ steps along an assembly path. Inverting these we find the number of compression steps scales with string length as $n \propto \sqrt{l}$ for LZW schemes, and the number of assembly steps scales like $a \propto \log_2 l$.

For LZW compression to be implementable, it requires the development of a rather sophisticated structure of reading, labeling, and storing data⁶⁸. It is not clear how many complex systems will emergently discover such sophisticated algorithms for data processing. It also depends on how sequences of data are read: data with similar character count, repetition of structure etc., can have very different LZW compressions because these are highly dependent on reading the data from left to right, what characters are at the start of the sequence, and therefore are sensitive to how one chooses to read the data⁶⁹.

Comparison of AT with other complexity measures

Other LZ algorithms such as LZ77 have also been claimed by the same group of authors to be equivalent to, or outperform, assembly index^{22–24}. However, these algorithms are always constructed to read data with directionality (e.g., from left to right), with no look-ahead. This feature makes these algorithms efficient—they do not need many passes over the file, or often just one. The Assembly index, on the other hand, looks at global optima. It does not just do a look-ahead; it goes much further and looks over all possible assembly pathways. This is because there is no requirement for speed, but instead to reach an inherent, global, property of the molecule. Sacrificing this feature to make LZ algorithms look more like assembly index calculations therefore removes a central feature of LZ compression schemes, making the direct comparison of these algorithms more of a mathematical exercise in approximating assembly index with a LZ like scheme, then a meaningful statement of exact equivalences between LZ algorithms and assembly index. In the field of computational complexity measures, differences of this degree are sufficient to categorize algorithms as distinct. Indeed, as we will show in the next section, computation of the mathematical metric of the assembly index is NP-complete, which puts it in a different computational complexity class than most compression measures, including LZ, meaning it must be formally distinct from them. We note that for molecules, algorithms to compute exact values of assembly index are nontrivial⁷⁰, and we do not expect any of the approximate approaches presented in ref. 22 to work in many important cases (this again goes back to the issues of representation of data and labeling, the algorithms the authors implement in ref. 22 will not yield the same results as assembly index in all cases, and would require modification for each new data type for encoding molecular data). The mathematical relationships of assembly index to optimal compression are indeed intriguing, but it is important to keep in mind the difference between math and physics, and the assembly index is intended to be a mathematical representation of a physical attribute of evolved objects.

Computational complexity class of AT

In theoretical computer science, a common way to classify algorithms is by their computational complexity class²⁷. If two algorithms are in the same

complexity class, they could be equivalent in the difficulty of their computation, although the form of the metrics may not be one-to-one or indicate the same things about the outputs. However, if two algorithms are found to not be in the same complexity class, it is understood they will never be shown to be formally equivalent (assuming P does not equal NP). In the supplement, we show that the formal definition of assembly spaces, as originally stated by ref. 11 and expanded on in the supplement, is NP-complete. It should be noted that, thus far in this paper, we have focused on strings since they provide simple examples, and while they have not so far been the primary focus of the developers of AT, these are the focus of several works that are critical of the theory. Indeed, AT is more general than the string structures discussed in these criticisms. AT was developed to deal with molecules, and these are much more complicated structures with more complicated attachment rules than strings; thus, we would not expect all properties of string assembly to generalize to the more complex case of molecules. AT, and within it the calculation of the assembly index, covers a wide range of complexity classes. In the proof in the supplement, we give an exact definition of assembly index and show that a set of string assembly problems are NP-complete. The problem of tiling—whether a shape can be fully covered using a set of tiles—can be included in assembly space and is very similar to assembling a molecule using a set of input molecular structures. Tiling of 2D and 3D shapes is known to be NP-complete⁷¹. Answering whether a molecule can be assembled out of a set of tiles is simpler than answering what the minimal number of steps to do so; therefore, we know that molecular assembly must be at least NP-complete. Other assembly problems could be simpler. Thus, string assembly is probably NP-hard, whereas the problem of assembling strings of different lengths out of a single letter is equivalent to the shortest addition chain problem, whose exact complexity is unknown, but possibly lower⁷².

It is significant to our discussion that the assembly index is NP-complete because most compression problems can be solved in polynomial time. Denoted P, this class is defined as the space of all algorithms that can be computed in time bounded by a polynomial function of the size of the input. This makes sense for compression algorithms, because the goal of compression is providing data representations that are easy to compute. Another widely studied class of algorithms lives in the class of non-deterministic polynomial time problems, denoted NP, defined as the space of algorithms where a known output has a proof that can be verified in polynomial time by a deterministic Turing machine, or solved in polynomial time by a non-deterministic one. Non-deterministic Turing machines can be simulated by deterministic Turing machines, but this simulation generally requires exponential time in the worst case. After that, the next class of interest is non-deterministic polynomial time complete problems, denoted NP-complete. NP-complete problems form an equivalence class, defined as the space of algorithms where it is possible to reduce one NP problem to another NP-complete problem in polynomial time. This is relevant because the class of NP-complete problems cannot be mapped to problems in P. Problems in P, however, can be trivially mapped to problems in the NP-complete class, since the problems in P are an easier subset of the NP-complete problems.

The result that computing the assembly index for strings is NP-complete has several important consequences. First, NP-complete problems are expected to have an exponential worst-case runtime²⁷, and so algorithms that find the assembly index could have an exponential runtime. Second, proving assembly index is NP-complete means it cannot be strictly equivalent to LZ algorithms, Huffman coding, or Shannon entropy because these are all in the complexity class P and computing these scales like $O(n)\log(n)$, $O(n)$, and $O(n)$ respectively^{73,74} as opposed to $\exp(n)$. Indeed, any complexity measure that lives in P cannot be strictly equivalent to the assembly index, which is NP-complete. There will be NP-complete problem instances that cannot be mapped to problem instances in P. This result implies our counterexamples are not just specific cases but represent general and fundamental differences between the assembly index and these compression algorithms. Comparisons to Huffman coding, Shannon entropy, LZW, and LZ77 can only be approximate comparisons to assembly index and will never represent exact equivalences.

Purpose of AT and assembly index

A critical divergence between these compression algorithms and AT lies in the interest in optimality. Compression algorithms, while theoretically concerned with finding the shortest possible representation, are more often used in a context where one settles for “good enough” solutions that achieve practical compression ratios quickly. The actual global minimum is not essential for their primary function of efficient data reduction. However, in AT, the global minimum (assembly index) is of central interest because it captures minimum causation as an intrinsic physical feature. Note we have not ruled out that computation of the assembly index is equivalent to some NP-complete compression algorithms; for example, optimal, i.e., minimal output length, text-substitution, and run-count compression algorithms are NP-complete^{74,75}. Shannon entropy is in $O(n)$ where n is the length of the input string, since the algorithm only must iterate over the string once to calculate the frequency of each character. Compression algorithms in the computational complexity class P, such as those reviewed herein, are useful because they are efficient, and in some cases, can be computed with a single pass over the input. Compression algorithms in the NP-complete computation class are intractable for compression, and therefore, efficient heuristic compression is universally preferred. Assembly index might be considered an optimal compression strategy in this sense since it returns the minimum object reconstruction path, but it is important to recognize that the goal of AT is not to provide a formal measure of compression. One should not confuse the mathematical representation of a theory with what it describes about the physical world, and indeed, many mathematical theories of nature can be expressed by equivalent but different representations simply due to the nature of mathematical description. A trivial example is how many mechanistic derivations will lead to a square root of time scaling, but not all of these are a random walk. The assembly index, as far as what it is meant to represent about the physical world, does not describe algorithmic compression, even if its formal representation may be equivalent mathematically to descriptions of some optimal compression algorithms. Finding a formal equivalence with another algorithm that is likewise NP-complete would in no way invalidate the theory, but could be useful in providing other ways to compute the assembly index, which might in turn be useful in testing AT's theoretical predictions against experimental data. Compression algorithms used in practice are heuristics that attempt to approximate optimal compression while being computationally tractable. Similarly, AI requires heuristic approximations for large problems. Heuristics often leverage assumptions about the problem domain to improve their approximations. In the domain of string AI, compression heuristics such as LZ compression may indeed provide useful insights. However, in the case of approximating molecular assembly, relevant to agnostic biosignature detection, the best heuristics will likely leverage domain chemistry knowledge.

Many complexity measures can be correlated while not being formally identical. In fact, it should be expected that most computational complexity measures will be correlated with output size, and indeed, all measures we have discussed herein are (at least weakly) correlated for this reason. In Ozelim et al.²⁴, the authors claimed correlations between assembly index and other complexity metrics invalidate the use of assembly index as a physical observable. This represents a misconception about the differences between correlation and metrology. It may be possible to correlate all sorts of variables, but only the assembly index has been shown to be reconstructable from the physical properties of molecules as probed by mass spectrometry, NMR, and infrared^{11,12}. It is worth pointing out that the assembly index was constructed exactly because the physical size of a molecule does not say much about its complexity. There are many large molecules, e.g., polymers, that easily occur in a chemical sample that are large and can form in the absence of evolution because they are composed of repeating units or other molecular symmetries. Therefore, size and all complexity measures whose only empirical measurement is correlation size are unreliable methods for life detection.

It is widely recognized that correlation neither implies causation nor equivalence. As a simple illustrative example of the category error of confusing correlation with mechanism, consider the two physical laws

mathematically represented by the equations $T_{\text{ps}}^2 = \frac{4\pi^2}{Gm}$ and $E = mc^2$. Both are functions of mass with one effective parameter, but each calculates a different physical quantity and in a different context. They are built on different mechanisms, one describes planetary motion and the other gives the energy of a particle in its rest frame as derived from special relativity. Superficially, the two correlate perfectly because of their linear dependence on m , but this in no way implies the two theories are equivalent or capture the same physics. This is an absurd example aimed at highlighting that trivial correlations are possible even for well-developed theories. Likewise, correlation in complexity measures, especially when controlled by overall output size, does not imply equivalence. For example, many efforts approximate Kolmogorov complexity, which is formally uncomputable, with Lempel–Ziv, or other compression schemes, which are in the class P. This means that Lempel–Ziv will fail at the most interesting cases of Kolmogorov complexity. Furthermore, the point of building measures and theories in physics is to compactly capture mechanism, and so the other complexity measures that do not share the same mechanistic foundations should not be assumed equivalent because of correlations. There is a long history in the physics of measurement about the challenges of developing measurement schemes that accurately capture physics, see e.g., Chang³⁸ for a history of the measurement of temperature.

In Marshall et al.¹¹, assembly index was shown to be reconstructed by counting the number of MS/MS peaks, demonstrating a meaningful measurement of assembly index using mass spectrometry. Fragmentation of molecules via mass spectrometry is expected by the theory to reconstruct the structure of the assembly space for molecules via recursive decomposition of molecular fragments. Thus, the measurement using MS/MS maps directly to the ontology of the theory, rendering the correspondence of assembly index and MS/MS peaks physically meaningful. Subsequently, inference of molecular assembly index was shown to lead to convergent results for mass spectrometry, NMR and infrared measurements, again considering molecules as physical objects in AT, where these techniques reconstruct features of the assembly space¹². This means that the assembly index is invariant with respect to the measurement technique used to read it out, motivating the interpretation that the molecular assembly index is an intrinsic molecular property of a molecule.

The authors in Uthamacumaran et al.²³ and Ozelim et al.²⁴ show a correlation of different compression measures to assembly index and mass spectrometry data, but do so devoid of ontology and without an explanation for why their reported correlations should be metrologically meaningful. Specifically, they show how the length of string-based representations of molecules (e.g., using SMILES) and compression values for these strings are correlated with the MS/MS peaks in molecular data²⁴. These correlations have no basis in chemistry, but they use these kinds of correlations to argue for why the assembly index should not be considered as measurable or any better than any other complexity metric that could be used in a similar way²³. However, their results should be trivially expected based on the scaling cross-correlation of complexity measures with output size alone, and this does not recover what assembly index is capturing. A simple example is to consider oligomers, which by Ozelim et al.'s analyses would have values that scale with their size, but the assembly index would yield much lower values due to recursion in the use of monomer fragments (this example is along similar lines to the scaling differences noted above in our discussion on the differences between assembly index and LZW). It is well documented that assembly index has a weak correlation with molecular weight, which is progressively weaker with increasing molecular weight (e.g. see Fig. 2b in Marshall et al.¹¹ and Figure 5a in Jirasek et al.¹² where it is shown that the upper limit of assembly index scales roughly with molecular weight and the lower limit with the logarithm of the molecular weight). The correlations in Ozelim et al.²⁴ with the size of the molecule do not recover what the assembly index is capturing.

In general, if one is not careful to control for string length, correlations across complexity measures will be artificially inflated. This is the kind of spurious correlation we pointed to in the example of mass above, where correlations are apparent but only due to superficial similarity. Some

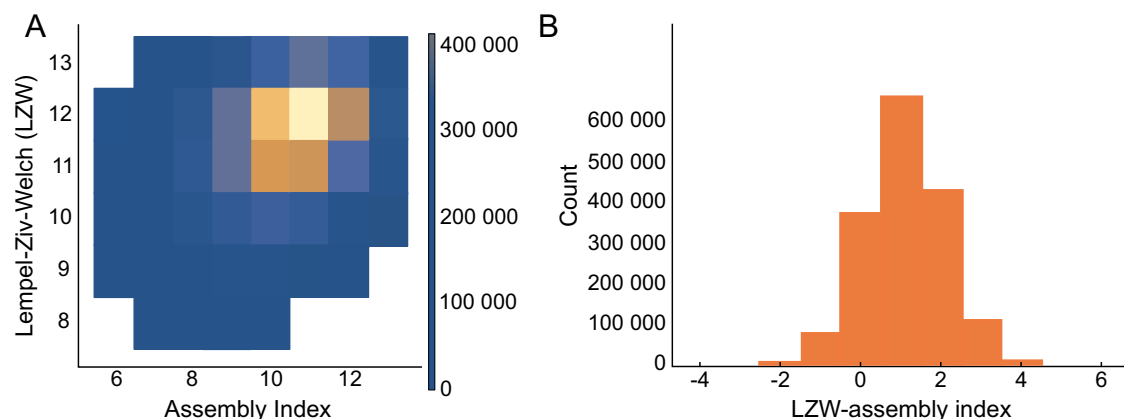


Fig. 4 | Comparison of the assembly index, a, and the Lempel–Ziv–Welch compression for every permutation of “aaaaabbbbbbcccc”. A The density map of a compared with LZW, where the two have a weak Pearson correlation of 0.25. **B** The histogram of the difference between a and the length of LZW (given as LZW

–assembly index) showing an approximate Gaussian distribution. These results confirm that when controlling for size, assembly index and LZW compression are weakly correlated.

correlation is evidenced in ref. 24, where the authors show a plot of correlation between LZW and AI by plotting both scaling with size (Fig. 2 in ref. 24). Such correlations are much weaker if one does account for size and would get weaker with larger objects because the assembly index’s correlation weakens with object size. As an explicit example, we consider LZW compression and assembly index, comparing the ratio of the two values for strings of fixed length and composition. We considered 10,000 random rearrangements of the string “aaaaabbbbbbcccc” and took the ratio of assembly index to the number of bytes in the compressed string as computed using LZW, see Fig. 4. We find that this ratio roughly follows a Gaussian distribution, showing assembly index and LZW are only weakly correlated with each other when controlling for size. The Pearson correlation between the two is very weak at 0.25. The set of permutations of “aaaaabbbbbbcccc” is a small subset of the strings of length 16 constructed from the characters {a,b,c}. We have already shown for the simplest case of strings composed of a single character how LZW scales with length l like $n \propto \sqrt{l}$ while the assembly index scales like $a \propto \log_2 l$. This implies that the worst mismatch between LZW and the assembly index will grow at least as fast as $\sqrt{l}/\log_2 l$, approximated as $\sqrt{l}$ for large l ; that is, while both correlate with size, the worst mismatch between the two increases with size. This is also evident in ref. 24, where the scatter correlating assembly index and LZW increases with string length. Given that the assembly index is constructed from the measurement scheme itself, its correlation is not due to averages over size-correlated effects, which give a superficial appearance of similarity even though cases of significant interest, particularly those for biosignature science, should be expected to yield very different results.

Compression algorithms will assign values to molecules that depend on the choice of representation of the data. Theoretical physics proceeds by using specific language in which all statements are intended to carry a physical meaning. In AT, this is done by selecting a measurement scheme for system representation and specifying a set of operations that correspond to physically realizable transformations. For molecular AT, this choice is well-motivated by knowledge of covalent chemistry⁷⁶. The frontier of the theory is in developing metrology for new domains of application, where work is to be done to ground equally well-motivated choices for construction of the assembly space in empirical measurement⁷⁷. Computational complexity methods face serious ambiguities as tools for physics because they suffer from an arbitrary choice of universal computer, whereas AT resolves this ambiguity through metrology, constructing the assembly space from laboratory-based measurement.

The measures the authors of refs. 22–24 introduce all suffer from ambiguity in the choice of data representation, analogous to the ambiguity in choosing a universal computer for K (see above). They will lack an absolute ordering for complexity that would accurately delineate the boundary

between non-life and life, obfuscating the distinction that AT formalizes. Indeed, different data representations of the same molecule, e.g., in the form of SMILES⁷⁸, or INCHI⁷⁹, graph-based representations, or even mass spectrometry data files, should be expected to yield quantitatively different results for the computed compressibility of the molecular data. Thus, a general correlative trend might be established, but the values assigned to specific molecules will depend on how one chooses to represent the data (e.g., this characterizes what is observed Fig. 7 in ref. 24). Using these correlations would provide a very poor basis for life detection because it should be expected that the same molecule will fall in different locations on such correlation plots depending on choices made representing the molecular data. That is, we should expect weak correlation to be possible, but the structure underlying where specific molecules lie to be representation dependent. This is untenable for life detection, because as we have already pointed out, one could choose any representation of the data and its compression that would represent any claim one might wish to make about the relative complexity of the molecules (Section 1.1), thus rendering the correlations useless without an underlying ontology forcing a choice of representation, which is a key difference between physics and computer science.

Many considerations about the physical nature of the process of computation, as well as the physical instantiation of outputs of algorithmic procedures, are not accounted for by the theory of computation²⁷. This has motivated several researchers to develop related complexity measures that might capture more physically relevant details. We now discuss these measures, and why AT represents a significant advance in the development of a physics of complexity, measurable in the laboratory. Originally developed by Bennett⁸⁰, logical depth was inspired by universal computation, but with a goal to quantify achieved complexity rather than potential complexity. Logical depth is distinct from algorithmic complexity because it quantifies complexity based on the work a computer must do to produce a given output, rather than an abstracted notion of program size. Formally, the logical depth of a string is defined as the execution time, or number of machine cycles, it takes a computer to calculate the output from the shortest possible program. In general, these yield different results: random objects have little depth despite having potentially long descriptions, while structured objects like repeating strings are expected to have a depth more coincident with the difficulty of the process that generated them. Logical depth has an advantage in applications where it is necessary to distinguish complexity due to randomness as distinct from meaningful computational work. However, like algorithmic complexity, it is dependent on the choice of language to describe the output, and in general, the shortest program is not computable.

Thermodynamic depth was introduced by Lloyd and Pagels as a measure of macroscopic complexity⁸¹ and is closely related to, but differs

from, logical depth. Thermodynamic depth captures the number of steps a system passes through to reach the current state. The goal of thermodynamic depth is to quantify complexity as a physical property of a state, noting that complexity should be a function of the process, or assembly routine, that brought the state into existence (and not just a description of the state)³⁸. We have now arrived at the first comparison complexity measure reviewed herein, which deals explicitly with physical states rather than computational programs and their outputs. Thus, this is the first candidate, besides the assembly index, that is explicitly about physics and what can be experimentally determined. Lloyd and Pagels tried to empirically ground their measure of physical complexity by defining it as a continuous function of the probabilities of the experimentally defined trajectories that can result in a macrostate, d , which might be observed in a lab. The thermodynamic depth is thus defined as $D(d) = -k \ln(p_i)$, where p_i is the probability, the system arrived at state d by the i th observed trajectory and k is an arbitrary constant. Thermodynamic depth will depend on what experiment is done, e.g., via empirically observed frequencies, p_i , of the observed paths. Its value can therefore differ for the same macrostate simply because the experiment was prepared differently (or more experiments were done). This introduces subjectivity in its definition⁸².

In a chemical system, the thermodynamic depth would depend on the probability of the reaction pathways that produced it⁸¹. This is untenable as a complexity metric for life detection because the same molecules will have a different thermodynamic depth in different contexts (where the reactions to generate the molecule would differ). This measure is therefore not standardizable across different planetary environments or different experiments. Also, one cannot infer thermodynamic depth from an observation of a complex chemical system at just one point in time, as one needs to know the full history of its formation: this information is almost never available for many scientific questions of interest. These challenges are exacerbated by the astronomical size of chemical space, where counting all possible reactions, pathways, and molecules is not computationally possible^{83–85}. The assembly index, by contrast, is independent of the path that produced the object(s), rather it depends only on the minimum path that could have produced the object, and therefore can be measured without knowing formation history. It will yield the same value for the same physical object(s) anywhere in the universe we might observe them. The length of the minimum recursively constructed path to make an object is a unique value for every object (even in cases where there may be degeneracy in minimum paths for the object's construction, the minimum path size is still a measurable value).

Falsification of any physical or chemical theory should come from experimental and observational tests of the explanations it provides, and the predictions it makes. AT was developed by critically addressing the question of how complex a detectable molecule must be before we know it was not formed by a random or undirected process. Exploring this idea in a laboratory setting allowed the concept of causal or contingent control, at the molecular level, to be formalized as the explanation of how highly complex objects are produced in abundance. AT, as a theory, is still in its infancy, with many open questions remaining about the mechanism for selection, and the transition in Assembly from the abiotic world to the biotic world. What is clear is that AT is poised to help uncover how selection, as a driving force, allowed the evolution of evolution in a manner that is testable with standardized measurements. This is because AT provides a measure that can detect the emergence of evolution in chemical experiments, as well as giving a new approach on which the emergence of evolution can be theoretically studied. As the theory develops, it will be important to explore its foundations. In this regard, it is worth noting that any good scientific theory should be “hard to vary”⁸⁶, especially as concrete predictions are made that can be experimentally or conceptually tested and built on. This is one reason the assembly index having an exact value, not dependent on data representation, is critical in its role as a foundation of the theoretical framework of AT.

Critical to the conceptual shift presented by AT is the central role of copy number, which so far, we have not addressed here. Criticisms of AT as redundant or not new, have not addressed the bulk of the theory, including the structure of the assembly equation and its role in formalizing selection

and the observed assembly space as a necessarily contingent structure^{22–25,87}. In AT, objects must be found in multiple copies. This means that in applications of the theory, it is not meaningful to talk about the assembly index of objects independent of any considerations of how to measure and count identical copies of them (where “identical” is defined within measurement uncertainty). Whilst this may seem an ad hoc requirement, it is a critical feature of AT's ability to uniquely identify objects as products of selection and evolution: the conjecture of the theory is that high Assembly configurations of matter (with abundant, large assembly index objects) only occur via evolution and selection. Note in Eq. 1 if $n_i = 1$, then the contribution of object i to the total value of A is 0. A recent hypothesized falsification of AT did not address the conceptual foundations of the theory accurately⁸⁷, because it applied the theory to idealized mathematical representations of unit cells, where in real materials, these are not possible to be observed in identical copies due to defects⁷⁷.

Discussion

In a recent discussion of AT, it was stated that AT cannot account for unanticipated biases, e.g., that AT cannot distinguish whether the bias is natural selection or something else that might favor production of an object given the “rule-based world” that AT is modeling²⁵. However, this criticism is made without looking at the empirical data and seemingly without interacting with the ontology of the assembly index. AT is a link between the abiotic world and the biological world and shows that a much more generalized mechanism of selection must have acted before genetic selection appeared. That is, AT makes explicit that selection is a more general phenomenon than that found just in biology. Our central claim is that the combinatorial space is so large, that some configurations simply cannot exist without “systematic bias” stored in a generic memory, what we call “selection,” which will include modern genetic evolution, but also many other more general modes of selection. So far, the central claim holds up to empirical tests, as presented previously¹¹ and those are not explored in criticisms of AT. Also, critiques do not provide empirical evidence to support the argument that the mundane deviations from expected outcomes they cite would falsify the central hypotheses of AT by producing arbitrarily high Assembly configurations of matter in abiotic settings. To date, there exists no empirical evidence that natural environments, outside of the processes of life, can produce objects of arbitrarily high complexity.

An interesting critique is related to how AT does not explicitly address function²⁵. However, a goal of AT is to measure evolved complexity without referencing function explicitly. This is intentional, as it avoids the issue of predefining all possible functions of a given object where there will always be some unanticipated context in which the object might acquire a new function⁸⁸. There is some debate as to whether a relevant measure of complexity should include explicit reference to context or be context-free. For example, the environmental context or functional context might inform how one quantifies complexity^{89,90}. This point is similar to early debates on measuring temperature, where standardization of measurement was challenged by different contexts for experimental protocols, such as material structure of the vessel, pressure, purity of sample, etc.³⁸. However, ultimately, a fixed temperature scale was developed, which could provide standardized comparison across all these different experimental protocols. It is our contention that the most significant advances in the science of measuring complexity will come from similarly standardized measurements, and assembly theory has been developed with this goal in mind. It is important to note that there are two types of context dependence: The first is whether a mathematical or physical measure is context-dependent. The second is whether the outcomes, possibilities, states, etc. of a particular physical process are context-dependent, even if the measure describing those features is not, which is the case for temperature. Temperature is a universal measure that we use in diverse situations ranging from the surface temperature of the sun to the internal temperature of the ocean: the scale we use to measure these temperatures is the same and fixed across all environments, but the specific values we observe are context-dependent. The assembly index as a measure is context-free; it is like temperature. However, the values

achievable or observed are context-dependent. Planetary contexts will determine how far selection can go in generating higher AI.

Another reason to focus on constructability rather than functionality is that it is unclear if the possible uses of a complex object are finite or (at least asymptotically) infinite⁹¹. If one has N objects, there are potentially $\sim N^N$ possible combinatorial interactions, meaning the number of possible interactions (functions) trends super-exponentially faster toward infinity than the number of objects does as the system size grows. However, while the number of functions an object can perform is not pre-statable⁹², the number of functions that could have constructed any observed object is always finite and well-defined. This is exactly what underlies the definition of the assembly index and the assembly space: AT explains how an object found in high copy number is constructed by functions, rather than what function it has, specifically because this is finite and measurable, as opposed to undefined. As an explicit example of the utility of this approach, one can consider the long-standing paradoxical question of how to tell if a given sequence of DNA is “functional” or not, without referring to the context of the cell⁹³. AT would instead ask whether the abundant DNA found in the cell was evidence of selection or not. A similar approach is taken in population genetics, where functionality of a sequence can be detected by finding many copies of it across the genome—we can detect functional jumping elements by finding those that have a large number of copies, vs. mutated non-functional ones that will have only a single copy. We can also find functional elements across different species by finding those that appear preserved across the species. Thus, within an evolutionary context, copy number is an indicator of function. In all these cases, the copying of the genome itself is a given—non-functional parts of the genome will replicate just as well as functional parts across generations. But where they fail is in the first to jump across the genome, and in the second to resist mutations through selection over the population.

Well-developed theories often lead to counterintuitive reformulations of our understanding of phenomenon that depart from our colloquial understanding of them, reframing the question being asked. It may be the case that a formal theory of life will require abandoning our biases about the role and nature of function, to learn something deeper by building abstractions that tie more closely to what empirical data we can objectively measure. Whether AT will continue to pass tests of its empirical validation or be falsified will be determined by future work of the scientific community. But one should not claim a theory invalid because it does not conform to prior expectations of what explanations we should find. Instead, we must always be open as scientists to theories we can rigorously test, and the explanations they provide.

We have shown that the assembly index is not identical to measures of computational complexity and compression, and we have discussed how the phenomenology of assembly theory as a theory of the physical world rooted in measurement distinguishes it from computational complexity theory. Assembly theory was designed to explore how the evolution of evolution was possible before formalized genetics. That is, AT aims to elucidate how selection produces the machinery of modern genetic evolution. One of the challenges with traditional computational complexity theory as an approach to physics and biology is that it has been applied to quantify the amount of complexity found in the natural world using an ontology that may not be meaningful in this context, using metrics that cannot be meaningfully mapped directly to physical measurement. Whereas the other measures described herein are in some cases undecidable, and in all cases dependent on choices of data representation, AT provides a formalization constructed from the physics that generated the object with a grounding in metrology. Furthermore, AT makes predictions falsifiable through empirical observations. One prediction is that objects experimentally inferred to have high assembly index and copy number are uniquely associated with life. Of course, falsifiability is not unique to AT as a scientific theory. What is unique is the ability to use readily available biochemistry lab equipment to obtain fragmentation patterns and copy number data that map to the AT framework, making it the first theory tailored to life detection that is amenable to experimental validation.

Methods

Methods for computing the assembly indices of strings were implemented using algorithms developed in Seet et al.⁷⁰. The assembly indices of strings were compared to Lempel–Ziv–Welch (LZW) compression and Huffman coding using standard algorithms for implementing these compression algorithms. For comparison across a larger sample, values of assembly index and LZW were computed across all 1,681,680 permutations of “aaaaabbbbbbbccccc”. For a proof of assembly index calculations being NP-hard, we performed a Karp reduction of the vertex cover algorithm to the calculation of assembly index (see supplement).

Data availability

All data is available in the SI.

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Author contributions

C.P.K., M.L., S.I.W., and L.C. conceived of the project, performed numerical analyses, developed mathematical counterexamples, and wrote the manuscript. A.I., G.M.F., and M.R.C. did the complexity class proof for the assembly index, edited the manuscript, and wrote the supplement.

Competing interests

The authors declare no competing interests.

Additional information

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SYMPLECTIC GEOMETRY: LECTURE 1

LIAT KESSLER

1. SYMPLECTIC LINEAR ALGEBRA

Symplectic vector spaces. Let V be a finite dimensional real vector space and $\omega \in \wedge^2 V^*$, i.e., ω is a bilinear antisymmetric 2-form:

$$\omega: V \times V \rightarrow \mathbb{R}, \omega(u, v) = -\omega(v, u)$$

(hence $\omega(v, v) = 0$ for all $v \in V$). We say that ω is *symplectic* if it is non-degenerate: for every $v \neq 0$ there is u such that $\omega(v, u) \neq 0$. We call (V, ω) a *symplectic vector space*.

Claim 1.1. *Let $\omega \in \wedge^2 V^*$. The following are equivalent.*

- *The form ω is symplectic.*
- *The kernel*

$$\ker \omega := \{v \in V : \omega(v, u) = 0 \text{ for all } u \in V\}$$

is trivial.

- *The map*

$$\omega^\flat: V \rightarrow V^*, \omega^\flat(v)(u) = \omega(v, u)$$

is an isomorphism.

A *symplectomorphism* ϕ between symplectic vector spaces (V, ω) and (V', ω') is a linear isomorphism $\phi: V \rightarrow V'$ such that $\phi^* \omega' = \omega$. (By definition, $(\phi^* \omega')(u, v) = \omega'(\phi(u), \phi(v))$.) If a symplectomorphism exists, (V, ω) and (V', ω') are said to be *symplectomorphic*. Note that being symplectomorphic is an equivalence relation on vector spaces of finite dimension. The group of symplectomorphisms of (V, ω) is denoted $\text{Sp}(V)$.

Example. The standard symplectic vector space. Consider $V = \mathbb{R}^{2n}$ with basis $(e_1, \dots, e_n, f_1, \dots, f_n)$. Then the antisymmetric bilinear form defined by

$$\omega_0(e_i, e_j) = 0, \omega_0(f_i, f_j) = 0, \omega_0(e_i, f_j) = \delta_{i,j}$$

is a symplectic form. Can you give examples of symplectomorphisms? E.g., $A(e_j) = f_j$, $A(f_j) = -e_j$, or $A(e_j) = e_j + f_j$, $A(f_j) = f_j$.

We will show that every symplectic vector space is symplectomorphic to the standard one.

Lemma 1.1. *For (V, ω) , there exists a basis $(e_1, \dots, e_n, f_1, \dots, f_n)$ of V such that $\omega(e_i, f_j) = \delta_{i,j}$ and $\omega(e_i, e_j) = \omega(f_i, f_j) = 0$.*

Such a basis is called a *symplectic basis*.

Remarks. (1) A choice of a symplectic basis for (V, ω) yields a symplectomorphism to $(\mathbb{R}^{2n}, \omega_0)$. Hence the dimension of a symplectic vector space is the only invariant of its isomorphism type.

(2) A symplectic basis is not unique, but is called a “canonical” basis.

(3) Using a symplectic basis, we can write $\omega = e_1^* \wedge f_1^* + \dots + e_n^* \wedge f_n^*$ where $e_1^*, \dots, e_n^*, f_1^*, \dots, f_n^*$ is a basis of V^* dual to the symplectic basis. It is easy to see (e.g., by induction) then that the n -th exterior power

$$\omega^n = \omega \wedge \dots \wedge \omega = \text{alt}(\omega \otimes \dots \otimes \omega) = n! e_1^* \wedge f_1^* \wedge \dots \wedge e_n^* \wedge f_n^*,$$

in particular, $\omega^n(e_1, f_1, \dots, e_n, f_n) = n! \neq 0$. So ω^n is a non-vanishing top degree form.

Before we prove the lemma, we define some of the terms that we will use in the proof and later on.

Subspaces of a symplectic vector space. The *symplectic orthogonal complement* of a linear subspace $W \subseteq V$ is defined as the subspace

$$W^\omega = \{v \in V \mid \omega(v, w) = 0 \text{ for all } w \in W\}.$$

We note that W^ω is the pre-image of the annihilator $\text{ann}(W) \subseteq V^*$ under the isomorphism $\omega^\flat$. Therefore

$$\dim W^\omega + \dim W = \dim V, \quad (W^\omega)^\omega = W.$$

A linear subspace W is called *isotropic* if $W \subseteq W^\omega$; *coisotropic* if $W^\omega \subseteq W$; *symplectic* if $W \cap W^\omega = 0$; *Lagrangian* if $W = W^\omega$. For example, in $(\mathbb{R}^{2n}, \omega_0)$, the span of e_1, e_2 is isotropic; the span of e_1, f_1 is symplectic; the span of $(e_1, \dots, e_n)$ (or of $(f_1, \dots, f_n)$) is Lagrangian.

Notice that every 1-dimensional subspace is isotropic (ask: $\omega(v, v) = 0$), and that W is isotropic iff W^ω is coisotropic. We will use these facts to prove the following lemma.

Lemma 1.2. *In any symplectic vector space (V, ω) there is a Lagrangian subspace.*

Proof. As noted, there exists an isotropic subspace. Let L be an isotropic subspace that is not contained in any isotropic subspace of strictly larger dimension. Then L must be Lagrangian: otherwise, there is $v \in L^\omega \setminus L$ and $L \oplus \text{span}(v) > L$ is isotropic. $\square$

Consequently, the dimension of a symplectic vector space is even: $\dim V = \dim L + \dim L^\omega = 2 \dim L$ for a Lagrangian subspace L . Note that the dimension of an isotropic subspace is $\leq \frac{1}{2} \dim V$.

You will prove a stronger version of the lemma in Problem Set 1:

Lemma 1.3. *Given a Lagrangian subspace M in (V, ω) , there is a Lagrangian subspace L such that $L \cap M = \{0\}$.*

Sketch of proof: Now let L be a maximal isotropic subspace with $L \cap M = \{0\}$. If it is not Lagrangian consider the quotient $\pi: L^\omega \rightarrow L^\omega/L$. The image $\pi(M \cap L^\omega)$ is isotropic hence of positive codimension. Therefore, one can choose an 1-dimensional subspace $F \subset L^\omega/L$ such that $F \cap \pi(M \cap L^\omega) = \{0\}$. Then $L' = \pi^{-1}(F)$ is an isotropic subspace that contains L and whose intersection with M is empty.

Proof of Lemma 1.1. Take two Lagrangian subspaces L, M of (V, ω) such that $L \cap M = \{0\}$. Since $L^\omega = L \cap M = \{0\}$, the composition

$$M \hookrightarrow V \xrightarrow{\omega^\flat} V^* \rightarrow L^*,$$

where the last map is the dual to the inclusion $L \hookrightarrow V$, is an isomorphism. Let $e_1, \dots, e_n$ be a basis for L and $f_1, \dots, f_n$ the dual basis for $L^* \cong M$. $\square$

We will give another proof of the lemma using compatible complex structures.

Compatible complex structures.

Example. Let V be a complex vector space of complex dimension n , with a Hermitian metric (complex positive definite inner product, complex linear with respect to the second entry and complex anti-linear with respect to the first entry) $h: V \times V \rightarrow \mathbb{C}$. Then $\omega = \text{Im}(h)$ is a symplectic form on V (considered as a real vector space) (check). Every unitary map $V \rightarrow V$ is a symplectomorphism. Note that $g = \text{Re}(h)$ is a real positive definite inner product (check).

Can you always find a Hermitian metric on a finite dim complex vector space? (Sure. Since the answer is clearly yes in $\mathbb{C}^n$, e.g., $h(u, v) = \bar{u}^T \text{Id } v$.)

Lemma 1.4. Wirtinger's inequality: *Let V be a complex linear space of dimension n with a positive definite Hermitian form h on V . Let $g = \operatorname{Re}(h)$ and $\omega = \operatorname{Im}(h)$.*

For $X_1, \dots, X_{2k}$ in V that are orthonormal with respect to g , we have

$$|\omega^k(X_1, \dots, X_{2k})| \leq k!,$$

with equality holding precisely when $W = \operatorname{span}(X_1, \dots, X_{2k})$ is a complex k -dimensional subspace of L .

Proof. Note that the value of $|\omega^k(X_1, \dots, X_{2k})|$ does not depend on the choice of an orthonormal basis to W , and that there is an orthonormal basis $X_1, \dots, X_{2k}$ such that

$$\omega|_W = \sum_{j=1}^k \omega(X_{2j-1}, x_{2j})(X_{2j-1}^* \wedge X_{2j}^*),$$

where X_i^* is the dual to X_i . (Check!) So

$$|\omega^k(X_1, \dots, X_{2k})| = k! \pi_{j=1}^k |\omega(X_{2j-1}, X_{2j})|.$$

Therefore it is enough to check the case $k = 1$.

If $k = 1$ then $\omega(X_1, X_1) = 0 = \omega(X_2, X_2)$ (since antisymmetric) hence $h(X_1, X_1) = g(X_1, X_1) = 1 = h(X_2, X_2)$, and $h(X_1, X_2)$ is in $\sqrt{-1}\mathbb{R}$ (since $\operatorname{Re} h(X_1, X_2) = g(X_1, X_2) = 0$ hence $\omega(X_1, X_2) = -\sqrt{-1}h(X_1, X_2) = h(iX_1, X_2)$). Therefore, by Schwartz's inequality (applied to h):

$$|\omega(X_1, X_2)| = |h(\sqrt{-1}X_1, X_2)| \leq h(\sqrt{-1}X_1, \sqrt{-1}X_1)h(X_2, X_2) = 1$$

with equality if and only if $X_2 = cX_1$ for a complex number $c \neq 0$. Since $h(X_1, X_1) = 1 = h(X_2, X_2)$ and $\operatorname{Re}(h(X_1, X_2)) = 0$, we have $c^2 = \pm 1$ and $c \neq \pm 1$, i.e., $c = \pm i$ and $\operatorname{span}(X_1, X_2 = \pm iX_1)$ is a complex 1-dimensional subspace. □

We now go on the reverse direction, starting from a real symplectic vector space. A *complex structure* on a real vector space V is an automorphism $J: V \rightarrow V$ such that $J^2 = -\operatorname{Id}$. A complex structure J on a symplectic vector space (V, ω) is *ω -compatible* if

$$g(u, v) = \omega(u, Jv)$$

defines a positive definite inner product. This implies that

$$\text{for } v \neq 0, \omega(v, Jv) = g(v, v) > 0,$$

and

$$\omega(Ju, Jw) = g(Ju, v) = g(v, Ju) = \omega(v, j^2u) = -\omega(v, u) = \omega(u, v),$$

i.e., J is a symplectomorphism. Moreover, a complex structure J is compatible with ω iff these two conditions hold: since ω is bilinear so is g , the first condition is equivalent to g being positive definite; the second condition is equivalent to g being symmetric.

Example. Consider the standard symplectic vector space $(\mathbb{R}^{2n}, \omega_0)$. The complex structure given by $Je_i = f_i$, $Jf_i = -e_i$ is compatible with ω_0 (we noticed before that it is a symplectomorphism; also, by definition of ω_0 and J , $\omega_0(e_i, Je_i) = \omega_0(e_i, f_i) = 1 > 0$ and $\omega_0(f_i, Jf_i) = \omega_0(f_i, -e_i) = 1 > 0$). This identifies $(\mathbb{R}^{2n}, \omega_0, J)$ with $\mathbb{C}^n$ with $\text{Im}(h)$ for $h(u, v) = \bar{u}^T \text{Id } v$.

In general, a compatible complex structure on (V, ω) makes V a complex vector space with Hermitian metric

$$h(u, v) = g(u, v) + \sqrt{-1}\omega(u, v).$$

(h is complex linear with respect to the second entry, $h(u, Jv) = \sqrt{-1}h(u, v)$, and complex anti-linear with respect to the first entry, $h(Ju, v) = -\sqrt{-1}h(u, v)$, and $h(v, v) > 0$ for $v \neq 0$.)

Lemma 1.5. *Let (V, ω) be a symplectic vector space. Given a positive inner product G on V , there is a linear isomorphism $A: V \rightarrow V$ such that*

$$J = J^G = (\sqrt{AA^*})^{-1}A$$

is well defined and is a compatible complex structure on (V, ω) .

The factorization $A = (\sqrt{AA^*})J^G$ is called the *polar decomposition* of A .

Proof. (As noted before), since ω and G are non-degenerate, the maps $\omega^\flat, G^\flat: V \rightarrow V^*$ are isomorphisms between V and V^* . Hence $\omega(u, v) = G(Au, v)$ for some linear isomorphism $A: V \rightarrow V$. The map A is skew-adjoint (with respect to G) since

$G(A^*u, v) = G(u, Av) = G(Av, u) = \omega(v, u) = -\omega(u, v) = G(-Au, v)$, hence $A^* = -A$. Note that AA^* is symmetric ($(AA^*)^* = AA^*$), and positive: $G(AA^*u, u) = G(A^*u, A^*u) > 0$ for $u \neq 0$. We conclude that AA^* diagonalizes with positive eigenvalues λ_i :

$$AA^* = B \text{diag}(\lambda_1, \dots, \lambda_n) B^{-1}.$$

So we can define

$$\sqrt{AA^*} = B \text{diag}(\sqrt{\lambda_1}, \dots, \sqrt{\lambda_n}) B^{-1};$$

and it is symmetric and positive definite. Let

$$J = J^G := (\sqrt{AA^*})^{-1}A.$$

Since A commutes with $\sqrt{AA^*} = \sqrt{-A^2}$, we get that J commutes with A . Since $A^* = -A$, also $J^* = -J$. Hence, since $JJ^* = \text{Id}$ (check), we have $J^2 = -\text{Id}$, i.e., J is a complex structure. We get

$$\omega(Ju, Jv) = G(AJu, Jv) = G(JAu, Jv) = G(JJ^*Au, v) = G(Au, v) = \omega(u, v),$$

and

$$\omega(u, Ju) = G(Au, Ju) = G(J^*Au, u) = G(\sqrt{AA^*}u, u) > 0$$

for $u \neq 0$ (since $\sqrt{AA^*}$ is positive definite). Hence J is ω -compatible. $\square$

Remarks. (1) In general, the positive inner product defined by $\omega(u, Jv)$ is different from $G(u, v)$. However, if J is given and $G(u, v) = \omega(u, Jv)$ then $J^G = J$. (In that case, $A = J^*$ and $AA^* = \text{Id}$.)
 (2) The construction is canonical. The map $\sqrt{AA^*}$ does not depend on the choice of B nor on the ordering of the eigenvalues in the diagonal map.

Denote by $\mathcal{J}(V, \omega)$ the space of compatible complex structures. We equip it with the subset topology induced from the topology on the space of linear endomorphisms of V . Let $\text{Riem}(V)$ denote the space of positive definite inner products; it is a convex open subset of the space of symmetric bilinear forms.

The previous lemma and Remarks imply the following result.

Corollary 1.1. *The map $G \mapsto J^G$ is a continuous and surjective map $F: \text{Riem}(V) \rightarrow \mathcal{J}(V, \omega)$. Furthermore (by construction) for $H: \mathcal{J}(V, \omega) \rightarrow \text{Riem}(V)$ associating to J the inner product $\omega(u, Jv)$ is a section, i.e., $F \circ H(J) = J$.*

Theorem 1.1. *The space $\mathcal{J}(V, \omega)$ is contractible and not empty.*

Proof. The existence of a compatible complex structure follows directly from the previous Corollary and the fact that $\text{Riem}(V)$ is not empty. Since $\text{Riem}(V)$ is a convex subset of a vector space, it is contractible. If $\Phi: I \times \text{Riem}(V) \rightarrow \text{Riem}(V)$ is a contraction: $\Phi_0 = \text{Id}_{\text{Riem}(V)}$ and Φ_1 is a map onto a point, then $F \circ \Phi \circ (\text{Id} \times H): I \times \mathcal{J}(V, \omega) \rightarrow \mathcal{J}(V, \omega)$ is a contraction of $\mathcal{J}(V, \omega)$ to a point. $\square$

We deduce a second proof for the existence of a symplectic basis for (V, ω) .

Proof of Lemma 1.1. Let J be an ω -compatible complex structure on V (exists by the previous theorem). Let h be the Hermitian form:

$$h(u, v) = \omega(u, Jv) + \sqrt{-1}\omega(u, v).$$

Pick an orthonormal (w.r.t h) basis of (V, J) as a complex space: $(e_1, \dots, e_n)$. Let $f_i = Je_i$. Then $(e_1, \dots, e_n, f_1, \dots, f_n)$ is a symplectic basis:

$$\omega(e_i, f_j) = \operatorname{Im}(h(e_i, Je_j)) = \operatorname{Im}(\sqrt{-1}h(e_i, e_j)) = \operatorname{Im}(\sqrt{-1}\delta_{i,j}) = \delta_{i,j},$$

$$\omega(e_i, e_j) = \operatorname{Im}(h(e_i, e_j)) = \operatorname{Im} \delta_{i,j} = 0,$$

and similarly $\omega(f_i, f_j) = 0$. $\square$

Remark 1.1. The two proofs we gave to Lemma 1.1 are related. The transition from the second proof to the first one is by the following claim: If J is ω -compatible and L is a Lagrangian subspace of (V, ω) then JL is also Lagrangian and $JL = L^\perp$, where $\perp$ denotes orthogonality with respect to the positive inner product $g(u, v) = \omega(u, Jv)$.

The two proofs used two important tools of symplectic geometry: Lagrangian sub-manifolds and compatible almost complex structures, which we will discuss in the next talk, when we move from symplectic vector spaces to symplectic manifolds.

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Everything about Spectral Graph Theory

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Chapter 1

Introduction

The aim of this book is to understand the spectral graph theory. We combine all classic sources, e.g. Fan Chung's book, Dan Spielman and Luca Trevisan's graduate courses. Also some other important sources.

There is not much to say in the introduction. If later I have some insightful ideas about this entire field I would probably write some rubbish here.

Chapter 2

Linear Algebra Preliminaries

2.1 Views about a matrix for a graph

We have two important views about a matrix $\mathbf{A}$. First it is a function that maps a vector $\mathbf{x}$ to the **vector** $\mathbf{Ax}$. Second, it is a function that maps a vector $\mathbf{x}$ to a **number** $\mathbf{x}^T \mathbf{A} \mathbf{x}$, which is a **quadratic** form. Note the name quadratic form is very fancy but it just means

$$x^T A x = \sum_{i,j} x(i) A(i,j) x(j)$$

The adjacency matrix is

$$A_G(u, v) = \begin{cases} 1 & \text{if } (u, v) \in E \\ 0 & \text{otherwise} \end{cases}$$

Though the adjacency matrix is natural to see a graph, it is the least useful. The eigenvectors and eigenvalues are the most meaningful when trying to understand a natural operator or a natural quadratic form.

2.2 Similar matrices and diagonalization

The definition of the similarity between two matrices is:

Definition 2.1. A matrix A is similar to a matrix B if there is a non-singular matrix M such that $M^{-1} A M = B$. In this case, A and B have the same eigenvalues.

Definition 2.2. A square matrix A is **diagonalizable** or nondefective if it is similar to a diagonal matrix, i.e. if there is an invertible matrix P such that $P^{-1} A P$ is a diagonal matrix.

2.3 Spectral Theory

Recall that an eigenvector of a matrix M with eigenvalue λ is defined as

$$Mf = \lambda f$$

where $\lambda I - M$ should be a **singular** matrix. The eigenvalues are the roots of the **characteristic** polynomial of M :

$$\det(xI - M)$$

Theorem 2.3 (The Spectral Theorem). *If M is an $n \times n$, real, symmetric matrix, then there exist real numbers $\lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n$ and n mutually orthogonal unit vectors $f_1, \dots, f_n$ and such that f_i is an eigenvector of M of eigenvalue λ_i for each i .*

This is why we like symmetric matrices. If the matrix is not symmetric it might not have n eigenvalues. Even if it has n eigenvalues, their eigenvectors will not be orthogonal (prove by contradiction).

If the matrix is not symmetric, we may even not be interested in their eigenvalues/eigenvectors.

Definition 2.4 (The Rayleigh quotient). *The Rayleigh quotient of a vector x with respect to a matrix M is the ratio*

$$\frac{x^T M x}{x^T x}$$

Observe that if f is an eigenvector, then

$$\frac{f^T M f}{f^T f} = \frac{\lambda f^T f}{f^T f} = \lambda$$

Theorem 2.5. *Let M be a symmetric matrix and x be a non-zero vector that maximizes the Rayleigh quotient w.r.t. M . Then x is an eigenvector of M with eigenvalue equal to the Rayleigh quotient. Moreover, this eigenvalue is the largest eigenvalue of M .*

Proof. The common trick we would use to prove stuff in spectral graph theory is to decompose the vector into n eigenvectors directions.

$$x = \sum_i (f_i^T x) f_i$$

The intuition here is that, we first compute the projection length of x onto f_i which is just the inner product $x^T f_i$. Then we multiply it with the eigenvector of that direction. Assuming x is a unit vector, then

$$\begin{aligned} \frac{x^T M x}{x^T x} &= x^T M x \\ &= \left(\sum_i (f_i^T x) f_i \right)^T M \left(\sum_j (f_j^T x) f_j \right) \\ &= \left(\sum_i (f_i^T x) f_i \right)^T \left(\sum_j (f_j^T x) \lambda_j f_j \right) \\ &= \sum_{i,j} (f_i^T x) (f_j^T x) \lambda_j f_i^T f_j && \text{because } f_i^T x \text{ and } f_j^T x \text{ are constants} \\ &= \sum_j (f_j^T x)^2 \lambda_j && \text{because the eigenvectors are orthogonal, their products are 0} \\ &\leq \lambda_n \sum_j (f_j^T x)^2 \\ &= \lambda_n && \text{because } x \text{ is assumed to be a unit vector} \end{aligned}$$

It is easy to show that the equality can be reached. Hence the Rayleigh quotient is never greater than λ_n . $\square$

Note that this theorem shows how the Rayleigh quotient plus the spectral information characterize the matrix M . When we want to study a symmetric matrix, we study this quotient and it should reveal some information we need.

Similarly we have a very useful conclusion about the relationship between the eigenvalues and the Rayleigh quotient.

$$\lambda_i = \min_{x \perp f_1, \dots, f_{i-1}} \frac{x^T M x}{x^T x}$$

So the i th smallest eigenvalue of M is the minimum Rayleigh quotient of M when x is a vector from the perpendicular space of the subspace formed by f_1 to f_{i-1} . Note that when $i = n$, we only have one dimension in our matrix, so x is trivial.

Similarly we have

$$f_i = \arg \min_{x \perp f_1, \dots, f_{i-1}} \frac{x^T M x}{x^T x}$$

2.4 Positive (Semi)definite Matrices

For a **symmetric** matrix A , we write

$$A \succcurlyeq 0$$

if A is **positive semidefinite**, which means all the eigenvalues are **nonnegative**. This is equivalent to

$$v^T A v \geq 0$$

for all v .

An extension is

$$A \succcurlyeq B$$

if

$$A - B \succcurlyeq 0$$

which is equivalent to

$$v^T A v \geq v^T B v$$

for all v .

2.5 Matrix Norm

The operator norm of a matrix M also called the 2-norm is defined by

$$\|M\| = \max_v \frac{\|Mv\|}{\|v\|}$$

It measures how much a vector can increase in size when it is multiplied by M . When M is symmetric, the 2-norm is just the largest absolute value of an eigenvalue of M . Also we have

$$\|M_1 M_2\| \leq \|M_1\| \|M_2\|$$

Chapter 3

Graph Preliminaries

Definition 3.1. *A graph is 3-connected if there is no set of two vertices whose removal disconnects the graph.*

Definition 3.2. *A **planar** graph is a special graph that can be drawn in the plane without crossing edges.*

Definition 3.3. *A planar graph divides the plane into connected regions called faces. Each face is identified with the vertices and edges on its boarder.*

Theorem 3.4. *Let $G = (V, E)$ and F be the set of faces. Then*

$$|V| - |E| + |F| = 2$$

Definition 3.5. *A **hypercube** is the graph with vertex set $\{0, 1\}^d$ with edges between vertices whose labels differ in exactly one bit.*

Chapter 4

The Matrices of Interest

We give some basic information about the matrices we like to discuss and some simple facts about their eigenvalues.

4.1 Laplacian matrix

This is the most basic matrix in the spectral graph theory.

$$L := D - A$$

where D is the diagonal matrix in which

$$D(u, u) = \deg(u)$$

and the rest are all 0s.

Why is this design important? If we look at the Laplacian quadratic form, we see its relationship with the Rayleigh quotient

$$x^T L x = \sum_{(u,v) \in E} (x(u) - x(v))^2$$

because we would have a $-x(u)x(v)$ and a $-x(v)x(u)$ and also one $x(u)^2$ and one $x(v)^2$. The thing is we should notice that for one edge (u, v) we only compute it once. We do not consider the edge (u, v) and (v, u) .

This form measures the **smoothness** of the function x . It is small if the function x does not jump too much over any edge. Intuitively, if $x(u) - x(v)$ is large, then it contributes a lot to the quadratic form. Note that if the edge is not in the graph, then it is ok to have large difference between the two nodes.

If the graph is weighted then we can have $G = (V, E, w)$ where $w : E \rightarrow \mathbb{R}^+$. Then the Laplacian quadratic form is

$$x^T L x = \sum_{(u,v) \in E} w_{u,v} (x(u) - x(v))^2$$

For two vertices in the graph G , the corresponding entries of them u, v , in the graph G in the Laplacian is

$$L_{G_{u,v}} = (\delta_u - \delta_v)(\delta_u - \delta_v)^T$$

4.2 Adjacency Matrix

In this part we study the features of the adjacency matrix. We denote its eigenvalues by

$$\mu_1 \geq \mu_2 \geq \dots \geq \mu_n$$

here we make it a bit different from what we do for the Laplacian. First we order them in a decreasing order, and we use μ instead of λ . There is a reason why we do such things. Note that in the Laplacian we use $L = D - A$. Hence μ_i in fact corresponds to λ_i . If the graph is d -regular, then

$$L = Id - A$$

hence

$$\lambda_i = d - \mu_i$$

Note that if it is not regular, then we do not see this relationship immediately. So the largest eigenvalue of the adjacency matrix of a d -regular graph is d because we know the smallest eigenvalue of the Laplacian is 0. The eigenvector is the uniform eigenvector.

4.3 Non-regular graphs

Now we look at a more interesting case, graphs that are not necessarily regular and we study its eigenvalues.

Lemma 4.1. *For a graph G , we let $d_{\max}$ be the maximum degree and let d_{ave} be the average degree. Then we have*

$$d_{ave} \leq \mu_1 \leq d_{\max}$$

Proof. The lower bound proof simply follows the Rayleigh quotient. The upper bound proof requires using the eigenvector of eigenvalue. $\square$

Lemma 4.2. *If G is connected and $\mu_1 = d_{\max}$ then G is $d_{\max}$ -regular.*

The eigenvector corresponding to the largest eigenvalue is usually not a constant vector. It is always a positive vector if the graph is connected.

Theorem 4.3 (Perron-Frobenius). *Let G be a graph and A its adjacency matrix, then*

- $\mu_1 \geq -\mu_n$
- $\mu_1 > \mu_2$
- The eigenvalue μ_1 has a strictly positive eigenvector

Proposition 4.4. *If G is connected, then $\mu_n = -\mu_1$ if and only if G is bipartite.*

Lemma 4.5. *Let A be symmetric and S be a subset of its row and column indices then we have*

$$\lambda_{\max}(A) \geq \lambda_{\max}(A(S)) \geq \lambda_{\min}(A(S)) \geq \lambda_{\min}(A)$$

Chapter 5

Spectral Information

We have seen some interesting facts in the first preliminary chapter. Here we basically focus more on the eigenvalues stuff.

5.1 Eigenvectors of the Laplacian

The matrix M in previous chapter is general. We need the theorems because we want to study the Laplacian matrix L and we want to know its eigenvalues.

By observing the above theorem, we should notice that the quadratic form $x^T L x$ is non-negative. Hence the smallest eigenvalue λ_1 is 0. We can also show that $\lambda_2 > 0$ if and only if the graph is connected. If the graph is disconnected, then we can construct two orthogonal vectors with eigenvalue zero: consider we have two components, then one eigenvector can be constant on one component and 0 everywhere else for the other component.

λ_2 has a name: algebraic connectivity of a graph [1]. When we relate λ_2 to how well a graph is connected, we are converting qualitative statements to quantitative statements.

An interesting thing is that if we compute all the eigenvectors of the Laplacian, and for each node u , we use two of the eigenvectors as its coordinates say $f_i(u), f_j(u)$ we can have very nice embedding drawings.

5.2 The Courant-Fischer Theorem

We have seen this theorem briefly before, now we give the complete version of it.

Theorem 5.1 (Courant-Fischer Theorem). *Let L be a symmetric matrix with eigenvalues $\lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n$. Then,*

$$\lambda_k = \min_{\substack{S \subset \mathbb{R}^n \\ \dim(S)=k}} \max_{x \in S} \frac{x^T L x}{x^T x} = \max_{\substack{T \subset \mathbb{R}^n \\ \dim(T)=n-k+1}} \min_{x \in T} \frac{x^T L x}{x^T x}$$

In previous chapters, we denote S as the eigenspace formed by the first k eigenvectors. Also we denote T as the complement space of the eigenspace formed by the first $k - 1$ eigenvectors.

To see it more intuitively, when $k = 1$, then S is just the span of f_1 and T is all of $\mathbb{R}^n$. For general k , the optima will be achieved when S is the span of $f_1, \dots, f_k$ and T is the span of $f_k, \dots, f_n$.

Proof. We only prove the first equality because they are similar.

For $x \in S_k$ which is the span of $f_1, \dots, f_k$, we decompose x :

$$x = \sum_{i=1}^k c_i f_i$$

so we apply the old trick,

$$\frac{x^T L x}{x^T x} = \frac{\sum_{i=1}^k \lambda_i c_i^2}{\sum_{i=1}^k c_i^2} \leq \frac{\sum_{i=1}^k \lambda_k c_i^2}{\sum_{i=1}^k c_i^2} = \lambda_k$$

Now we should explain the $\min_{\substack{S \subset \mathbb{R}^n \\ \dim(S)=k}}$ part. Essentially this says that we should pick a subspace S of dimension k , among all candidates, we pick the S in which the maximum Rayleigh quotient that $x \in S$ can make is minimized.

We pick T_k be the span of $f_k, \dots, f_n$. Then we know S must have an intersection with T_k because their dimensions sum to $n + 1$.

$$\max_{x \in S} \frac{x^T L x}{x^T x} \geq \max_{y \in S \cap T_k} \frac{y^T L y}{y^T y}$$

We decompose y as

$$y = \sum_{i=k}^n c_i f_i$$

and so

$$\frac{y^T L y}{y^T y} = \frac{\sum_{i=k}^n \lambda_i c_i^2}{\sum_{i=k}^n c_i^2} \geq \frac{\sum_{i=k}^n \lambda_k c_i^2}{\sum_{i=k}^n c_i^2} = \lambda_k$$

This shows that all k -dimensional subspaces should have a lower bound on the Rayleigh quotient λ_k . The equality holds trivially. $\square$

5.3 Bounds on λ_2

We apply the Courant-Fischer theorem then we have

$$\lambda_2 = \min_{v: v^T \mathbf{1} = 0} \frac{v^T L v}{v^T v}$$

Chapter 6

Graph Coloring

Definition 6.1. A *coloring* of a graph is an assignment of one color to every vertex in a graph so that each edge attaches vertices of different colors.

It is easy to see that we are interested in using as few colors as possible.

Definition 6.2. The *chromatic* number of a graph, χ_G , is the least k for which G is k -colorable.

Lemma 6.3. A graph is 2-colorable if and only if it is bipartite.

6.1 Wilf's theorem

Theorem 6.4. $\chi_G \leq \lfloor \mu_1 \rfloor + 1$.

Proof. We order all the vertices. Then we look at the number κ where

$$\forall u, |\{v : v < u, (u, v) \in E\}| \leq \kappa$$

Once we have the smallest κ we can color the graph because for i we can always pick a color different from its previous less than κ neighbours. Now we need to show $\kappa \leq \lfloor \mu_1 \rfloor$.

Since we know the average degree is less than μ_1 we put a node with that degree as n . For the subgraph, meaning we remove the row/column corresponding to that vertex, the largest eigenvalue of the new adjacency matrix is at most μ_1 , so we put another node to $n - 1$ and we keep doing so. $\square$

Note that the graph's max eigenvalue can be smaller than d_{max} so the chromatic number can be smaller than $d_{max} + 1$ where we can easily show that all graphs can be colored by $d_{max} + 1$ colors.

6.2 Hoffman's bound

Hoffman proved a bound on the chromatic number of a graph in terms of its adjacency matrix eigenvalues that is tight for bipartite graphs.

Chapter 7

Conductance

7.1 Isoperimetry and λ_2

Here λ_2 is the second **smallest** eigenvalue of the Laplacian.

Definition 7.1. For a subset S of the vertices of a graph. One way of measuring how well S can be separated from the graph is to count the number of edges connecting S to the rest of the graph. It is the **boundary** of S :

$$\partial(S) = \{(u, v) \in E : u \in S, v \notin S\}$$

Then we can define the **isoperimetric ratio** of S .

$$h(S) = \frac{|\partial(S)|}{|S|}$$

The **isoperimetric number** of a graph is the minimum isoperimetric number over all sets of at most half the vertices:

$$h(G) = \min_{|S| \leq n/2} h(S)$$

Theorem 7.2 (Lower bound of Cheeger).

$$h(S) \geq \lambda_2(1 - s)$$

where $s = |S|/|V|$. In particular

$$h(G) \geq \lambda_2/2$$

Proof. By the min-max theorem, we know for all vectors orthogonal to $\mathbf{1}$, we have

$$x^T L x \geq \lambda_2 x^T x$$

Hence we construct an indicator vector χ_S where

$$\chi_S(u) = \begin{cases} 1 & u \in S \\ 0 & \text{otherwise} \end{cases}$$

We can observe a nice feature:

$$\chi_S^T L \chi_S = \sum_{(u,v) \in E} (\chi_S(u) - \chi_S(v))^2 = |\partial(S)|$$

To make χ_S orthogonal to $\mathbf{1}$, we make $x = \chi_S - s\mathbf{1}$. Then x satisfies everything. To finish the proof we only need to see

$$x^T x = |S|(1 - s)$$

□

The isoperimetric number shows the worst connected part of the graph because it has the least ratio. This is lower bounded by the second-smallest eigenvalue of the Laplacian. Hence λ_2 shows how good a graph is connected: if it is large, then G is well connected. If

7.1.1 Complete Graph K_n

Lemma 7.3. *The Laplacian of K_n has eigenvalue 0 with multiplicity 1 and n with multiplicity $n - 1$.*

7.2 Conductance

In previous section we have seen some relationship between the isoperimetric parameter and the second smallest eigenvalue of the Laplacian. To get tighter bounds and cleaner results, we use the **conductance** and the second smallest eigenvalue of the **normalized** Laplacian.

Definition 7.4 (Conductance). *We define*

$$\Phi(S) = \frac{|\partial(S)|}{\min(\text{vol}(S), \text{vol}(V - S))}$$

where $\text{vol}(S)$ is the sum of the degrees of the vertices in S . Note that many similar, though slightly different definitions appear in the literature, like

$$\frac{\text{vol}(V)\partial(S)}{\text{vol}(S)\text{vol}(V - S)}$$

Also the conductance of a graph is

$$\Phi_G = \min_{S \subset V} \Phi(S)$$

7.3 The Normalized Laplacian

Definition 7.5. *We define the **normalized** Laplacian:*

$$\mathcal{L} = D^{-1/2} L D^{-1/2}$$

7.4 Cheeger's inequality

Theorem 7.6. *Let $0 = \nu_1 \leq \nu_2 \leq \dots \leq \nu_n$ be the eigenvalues of $\mathcal{L}$. Then Cheeger gives us*

$$\frac{\nu_2}{2} \leq \Phi_G \leq \sqrt{2\nu_2}$$

Proof. For now we omit it and if we have time we go back to type in the proof. The harder part, which is the square root part, should be one's favorite theorem in spectral graph theory. □

Chapter 8

Random Walks

The details about random walks can be found in other books on Markov chains. Here we only see its analysis in spectral graph theory.

The matrix form of a lazy random walk is

$$W = \frac{1}{2}(I + D^{-1}A)$$

The distribution of the walk p we use is a row vector, though in many places people use column vectors. The difference is minor. Note that the degrees of different nodes are not necessarily the same, hence the matrix may not be symmetric.

How do we deal with this matrix and how to see its eigenvalues? We borrow the normalized Laplacian:

$$\begin{aligned} W &= I - \frac{1}{2}(I - D^{-1}A) \\ &= I - \frac{1}{2}D^{-1/2}(I - D^{-1/2}AD^{-1/2})D^{1/2} \\ &= I - \frac{1}{2}D^{-1/2}\mathcal{L}D^{1/2} \end{aligned}$$

where $\mathcal{L} = I - D^{-1/2}AD^{-1/2}$. $\mathcal{L}$ is symmetric, hence $D^{-1/2}\mathcal{L}D^{1/2}$ is diagonalizable. Therefore W is also diagonalizable.

Assuming we know f_i and ν_i are the eigenvectors/eigenvalues of $\mathcal{L}$, then we can prove that $f_i D^{1/2}$ is a left-eigenvector of W of eigenvalue $1 - \nu_i/2$.

$$\begin{aligned} (f_i D^{1/2})W &= (f_i D^{1/2}) \left(I - \frac{1}{2}D^{-1/2}\mathcal{L}D^{1/2} \right) \\ &= f_i D^{1/2} - \frac{1}{2}f_i \mathcal{L}D^{1/2} \\ &= f_i D^{1/2} - \frac{\nu_i}{2}f_i D^{1/2} \\ &= (1 - \nu_i/2)f_i D^{1/2} \end{aligned}$$

Note that because in Markov chain theory we usually use row vector. So here we adjust the eigenvector to be left row vector. Essentially it is the same as using right column vector. Also

here we use the advantage that $\mathcal{L}$ is symmetric, so the eigenvectors are the same no matter which form we would use. Also note that since W can be asymmetric, so these eigenvectors of W are not necessarily orthogonal.

Lazy random walk has some good features. One is that the eigenvalues of W are all between 1 and 0. Also from the above relationship, we can see the relationship between the eigenvalues of the normalized Laplacian and the eigenvalues of the random walk matrix: $\omega_i = (1 - \nu_i/2)$. So the second largest eigenvalue of W corresponds to $1 - \nu_2/2$. Then the spectral gap of W is just $\nu_2/2$. So when we use Cheeger to discuss the second smallest eigenvalue of the Laplacian we are talking about the spectral gap of the random walk. They are essentially the **same** thing.

8.1 The rate of convergence

We know the mixing time is related to the relaxation time: $1/\gamma$ where γ is the spectral gap of the random walk matrix.

Theorem 8.1. *For all a, b and t , if $p_0 = e_a$, then*

$$|p_t(b) - \pi(b)| \leq \sqrt{\frac{\deg(b)}{\deg(a)}} \omega_2^t$$

To be consistent with our Markov chain stuff, we use row vectors.

Proof. First of all, we have

$$p_t(b) = p_t e_b^T$$

and we decompose $p_0 D^{-1/2} = \sum_i c_i f_i$.

$$\begin{aligned} p_t &= p_0 W^t \\ &= p_0 \left(D^{-1/2} \left(I - \frac{1}{2} \mathcal{L} \right) D^{1/2} \right)^t \\ &= p_0 \left(D^{-1/2} \left(I - \frac{1}{2} \mathcal{L} \right)^t D^{1/2} \right) \\ &= \left(\sum_i c_i f_i \right) \left(I - \frac{1}{2} \mathcal{L} \right)^t D^{1/2} \\ &= \left(\sum_i c_i f_i \right) \left(\sum_j (1 - \nu_i/2)^t f_i^T f_j \right) D^{1/2} \\ &= \left(\sum_i (1 - \nu_i/2)^t c_i f_i \right) D^{1/2} \\ &= c_1 f_1 D^{1/2} + \left(\sum_{i \geq 2} (1 - \nu_i/2)^t c_i f_i \right) D^{1/2} \end{aligned}$$

The last line is true because $\nu_1 = 0$. Also note that $c_1 f_1 D^{1/2} = \pi$, so we have

$$p_t(b) = \pi(b) + \left(\sum_{i \geq 2} (1 - \nu_i/2)^t c_i f_i \right) D^{1/2} e_b^T$$

Since $p_0 = e_a$, hence

$$c_i = e_a D^{-1/2} f_i^T$$

and

$$\left(\sum_{i \geq 2} (1 - \nu_i/2)^t c_i f_i \right) D^{1/2} e_b^T = \left(\sum_{i \geq 2} \omega_i^t e_a D^{-1/2} f_i^T f_i \right) D^{1/2} e_b^T = \sqrt{\frac{\deg(b)}{\deg(a)}} \left(\sum_{i \geq 2} \omega_i^t e_a f_i^T f_i \right) e_b^T$$

We upper bound the right hand side.

$$\left| \left(\sum_{i \geq 2} \omega_i^t e_a f_i^T f_i \right) e_b^T \right| \leq \sum_{i \geq 2} \omega_i^t |e_a f_i^T| |f_i e_b^T| \leq \omega_2^t \sum_{i \geq 2} |e_a f_i^T| |f_i e_b^T| \leq \omega_2^t$$

We know $\omega_2 = 1 - \nu_2/2$. So when converging $t = O(\log n / \nu_2)$. □

Note that this proof is very classic. We should memorize it. How? Essentially we decompose the product of W^t and notice that the first eigenvector is related to π . Then the rest is to bound the rest of the decomposition using the second largest eigenvalue.

Chapter 9

Expanders

Intuitively expanders are graphs that connect well. For a set of vertices within an expander graph, we can expect it to have a lot of neighbours and that is why people like expanders.

Definition 9.1. *Expander graphs are an incredibly useful family of graphs. An expander is a graph with constant vertex degree and constant conductance.*

An expander family is a sequence of graphs of increasing numbers of vertices if there exist constants d and ϕ such that every graph in the family has degree d and conductance at least ϕ .

The spectral characterization of expanders is:

$$|\lambda_i - d| \leq \epsilon d, \forall i \geq 2$$

Random d -regular graphs are expanders with high probability.

9.1 Approximations of the Complete Graph

By introducing the approximation of graphs, we use this theory to determine good graphs when we cannot achieve complete graphs.

Definition 9.2. *An ϵ -expander is a d -regular graph such that*

$$|\mu_i| \leq \epsilon d$$

for $i \geq 2$ where $\mu_1 \geq \dots \geq \mu_n$ are the eigenvalues of the adjacency matrix. The Laplacian eigenvalues are

$$\lambda_i = d - \mu_i$$

which is equivalent to what we have above.

Note that if the spectral gap is a constant, then the mixing time is also of order $\log n$ which is very small.

Definition 9.3. *A graph G is ϵ -approximation of a graph H if*

$$(1 - \epsilon)H \preceq G \preceq (1 + \epsilon)H$$

where $H \preceq G$ means for all x ,

$$x^T L_H x \leq x^T L_G x$$

9.2 Vertex Expansion

Expanders have a good property: for a small set of vertices, it has unusually large number of neighbours.

Theorem 9.4. *Let $G = (V, E)$ be a d -regular graph on n vertices that ϵ -approximates $\frac{d}{n}K_n$. Then for all $S \subset V$,*

$$|\Gamma(S)| \geq \frac{|S|}{\epsilon^2(1 - \alpha) + \alpha}$$

where $|S| = \alpha n$

9.3 Ramanujan graphs

The Ramanujan graphs constructed by Margulis [3] and Lubotzky, Phillips and Sarnak [2] achieve

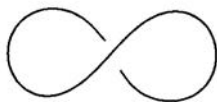
$$\epsilon \leq \frac{2\sqrt{d-1}}{d}$$

What does it mean? It means the Ramanujan graph can be very close to a complete graph in terms of the adjacent matrices.

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- [3] Grigorii Aleksandrovich Margulis. Explicit group-theoretical constructions of combinatorial schemes and their application to the design of expanders and concentrators. *Problemy peredachi informatsii*, 24(1):51–60, 1988.

Chapter 1



Knots and Links

1. Knots and Links

A *knot* is a simple closed curve, where “simple” means the curve does not intersect itself and “closed” means there are no loose ends. We usually think of knots in three-dimensional space since simple closed curves in the line and plane are pretty boring and, perhaps surprisingly, simple closed curves in 4 or more dimensions are also boring, as we will see.

Two knots K_0 and K_1 have the same *knot type* if we can move K_0 around in space in a continuous way, i.e. without cutting or tearing the knot (or the space in which the knot lives!) to match up K_0 with K_1 . Formally, K_0 is *ambient isotopic* to K_1 if there is a continuous map $H : \mathbb{R}^3 \times [0, 1] \rightarrow \mathbb{R}^3$ such that $H(K_0, 0) = K_0$, $H(K_0, 1) = K_1$ and $H(x, t)$ is injective (one-to-one) for every $t \in [0, 1]$. Such a map is called an *ambient isotopy*; if you think of t as a time variable, then H is a movie showing how to continuously deform K_0 onto K_1 . If there exists an ambient isotopy H taking K_0 to K_1 we write $H : K_0 \xrightarrow{\sim} K_1$.

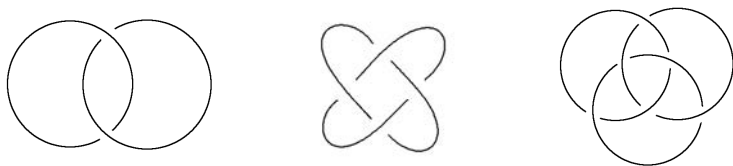
To specify a knot K we could make a physical model by tying the knot in a rope or cord; a nice trick suggested by Colin Adams in [Ada04] is to use an extension cord, so you can join the ends together by plugging the plug into the outlet end.

To specify knots in a more print-friendly format, we could give a parametric function $f(t) = (x(t), y(t), z(t))$ where $0 \leq t \leq 1$ and $f(0) = f(1)$. This approach is required in order to study *geometric knot theory*, where the exact positioning of K in space is important. In *topological knot theory*, however, we only care about the position up to ambient isotopy; thus, a simpler solution is to draw pictures or *knot diagrams*. Formally, a *knot diagram* is a projection or shadow of a knot on a plane where we indicate which strand passes over and which passes under at apparent crossing points by drawing the understrand broken.



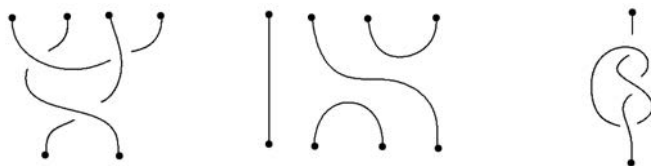
A knot is *tame* if it has a diagram with a finite number of crossing points; knots in which every projection has infinitely many crossing points are called *wild knots*. We will only deal with tame knots in this book.

Links, Tangles and Braids (oh my!) There are many kinds of objects related to knots. A *link* consists of several knots possibly linked together; each individual simple closed curve is a *component* of the link. A knot is a link with only one component.

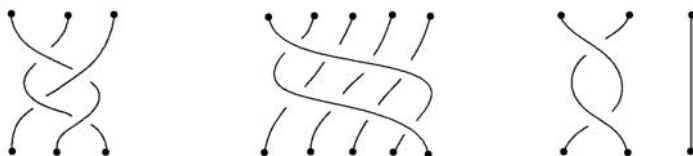


A *tangle* is a portion of a knot or link with fixed endpoints we can think of as inputs and outputs. If there are n inputs and m outputs,

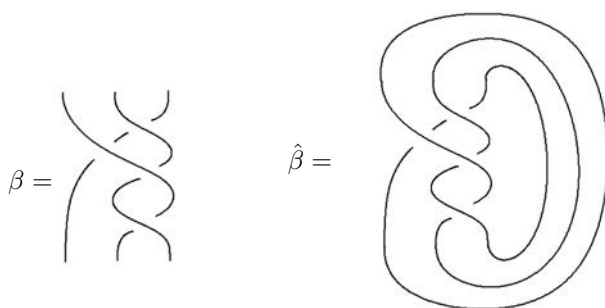
we have an (n, m) -tangle.



A *braid* is a tangle which has no maxima and no minima in the vertical direction, i.e., a tangle whose strands do not turn around. Note that in any braid, the number of inputs must equal the number of outputs, unlike more general tangles.

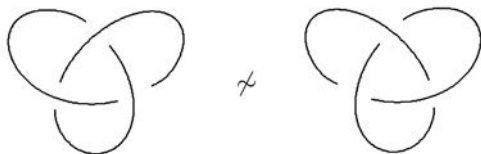


For any braid β there is a knot or link $\hat{\beta}$ called the *closure* of the braid, obtained by joining the top strands to the bottom strands. The converse is also true – every knot or link can be put into braid form, a fact known as *Alexander's Theorem*.

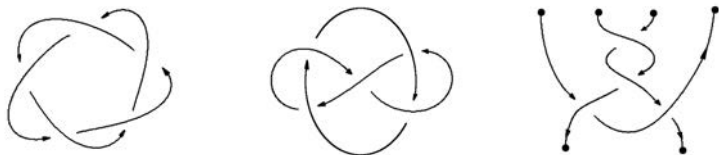


The *obverse* of a knot K is the mirror image of K , denoted $\bar{K}$. A knot may or may not be equivalent to its obverse – the trefoil knot comes in distinct left- and right-handed varieties, for instance. Knots which are different from their obverses are called *chiral*, while knots

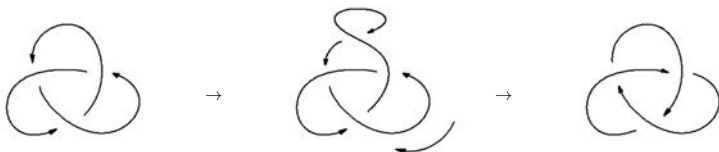
which are ambient isotopic to their obverses are called *amphichiral*.



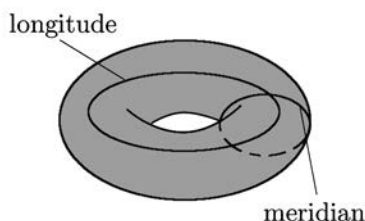
Oriented Knots. For each strand in a knot, link, tangle or braid, we can make a choice of *orientation* or preferred direction of travel. Knots described by a parametrization have an implied orientation in the direction of increasing t value; braids also have an implied orientation of all strands oriented in the same direction (up or down depending on the author's choice of convention). For generic oriented knots, links and tangles, we specify the orientation of each strand with an arrow.



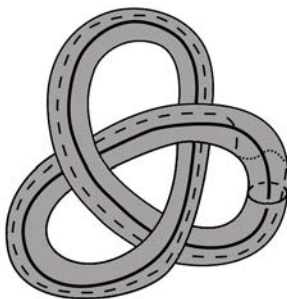
Reversing the orientation of an oriented knot K yields a possibly different oriented knot called the *inverse* or *reverse* of K , denoted $-K$. For two oriented knots K_0 and K_1 to be equivalent, we need an ambient isotopy $H : K_0 \xrightarrow{\sim} K_1$ which respects the orientation of K_1 . For example, the trefoil knot K below is equivalent to its inverse $-K$ as illustrated:

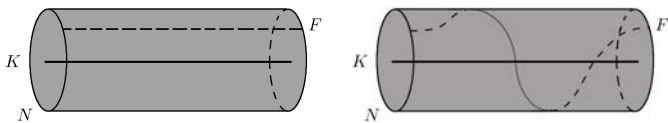


Framed Knots. Like a choice of orientation, a *framing* of a knot is a choice of extra structure we can give to a knot which then must be preserved by an ambient isotopy for two framed knots to be equivalent. Start by inflating the knot K like an inner tube, so we have a knotted solid torus N with K as its core. This solid torus is called a *regular neighborhood* of the knot. A circle on the torus which goes around the torus with the knot is called a *longitude*, while a circle going around a disk slice of the solid torus with the knot at its center is called a *meridian*.



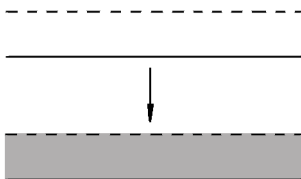
A *framing curve* F is a simple closed curve on the surface of the torus which projects down onto the original knot K in an injective (one-to-one) way, i.e., a longitude of the torus. While F goes around the torus with K exactly once in the longitudinal direction, it can wrap around the meridional direction of the torus any integer number of times.





Let K be a knot and F a framing curve. A *framed isotopy* of (K_0, F_0) to (K_1, F_1) is an ambient isotopy which carries K_0 to K_1 and carries F_0 to F_1 . For a given knot K , with framing curve F , the number of times F wraps meridianally around K (with counterclockwise wraps counted with a positive sign and clockwise twists counted with a minus sign) is called the *framing number* of the framed knot (K, F) . For a fixed knot K , two framed knots (K, F_0) and (K, F_1) are framed isotopic only if the framing numbers are equal.

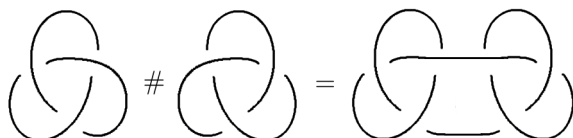
We can think of a framed knot as a 2-component link with the knot and its framing curve forming two sides of a ribbon.



Then, framed isotopy can be understood as movement of the ribbon through space. Similarly a framed link of n components can be understood as an ordinary link of $2n$ components where the components come in parallel pairs forming the two sides of n ribbons, with each of the original n components having its own framing curve and framing number. Similarly, in a framed braid or framed tangle, each strand has its own framing curve and framing number.

We can think of framed isotopy as a mathematical model for knotted 3-dimensional ropes or tori, where ambient isotopy is the model for knotted 1-dimensional curves.

Connected Sums. Knots and links have an operation known as *connected sum* where two knots are joined into a single knot by cutting the knots and joining the loose ends to form a single knot. We write $K_0 \# K_1$ for the connected sum of K_0 and K_1 .

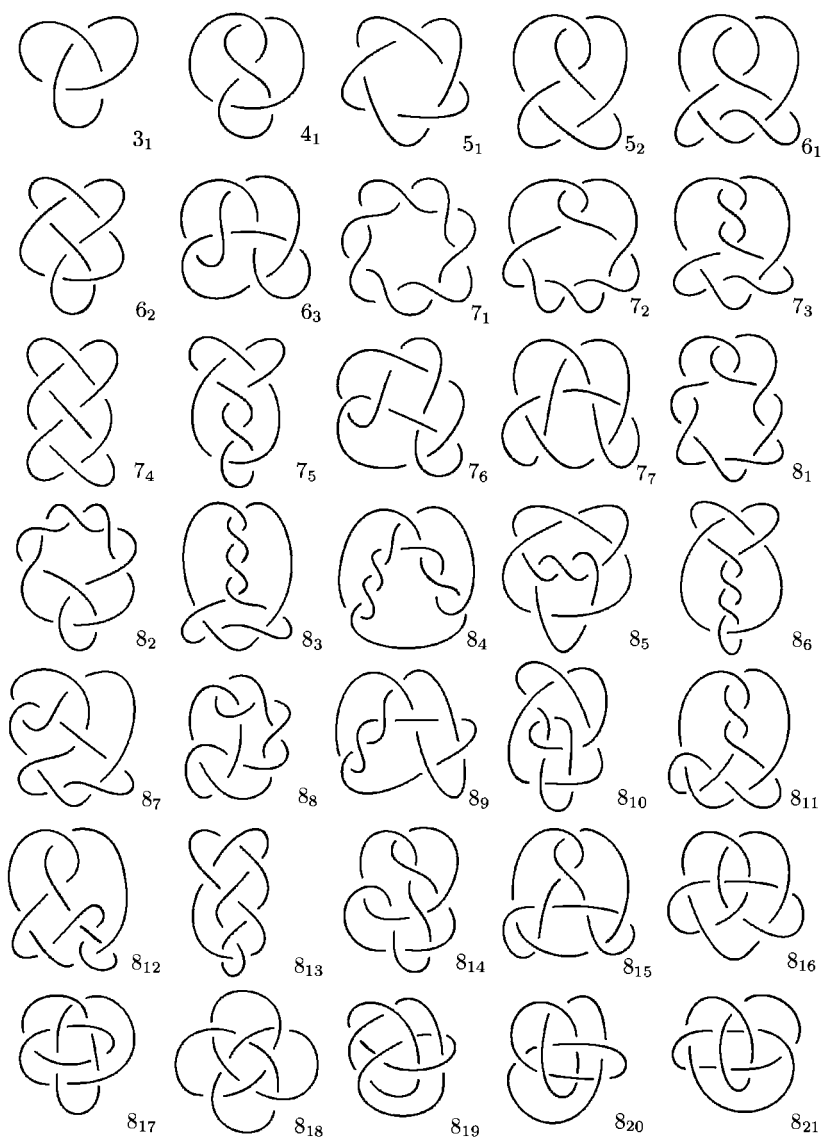


A connected sum $K_0 \# K_1$ can also be understood as the result of first tying K_0 in a piece of string, then tying K_1 before joining the ends.

A knot K is *prime* if the only way to decompose K as a connected sum of two knots is as $K \# 0_1$ where 0_1 is the unknotted circle or *unknot*; that is, K is prime if K does not break down as a connected sum of two nontrivial knots. For any knot, to decide whether the knot is prime, we can look at all the ways of intersecting the knot with a circle which meets the knot at exactly two points; this divides the knot into a connected sum of the portion outside the circle (completed by the arc along the circle) and the portion inside the circle (completed with the arc along the circle). If the knot is prime, then every such division will have one side unknotted.



Below is a table of all prime knots with up to eight crossings. Knots are named according to their crossing number with a subscript indicating their position on the table.

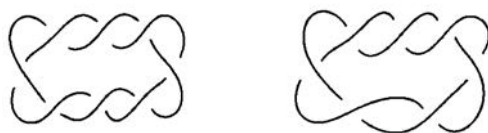


Exercises. 1. Take a piece of rope or an extension cord, tie a knot very loosely, and join the ends together. Lay the knot on a flat surface and draw a knot diagram representing the knot. Now, move the knot around to a new position – don't be afraid to add a few twists, just keep the ends joined. Now, draw a diagram of your knot in its new configuration. Repeat a few times; then repeat with a new knot.

2. Draw all possible knot or link diagrams with exactly two crossings.

3. Draw all possible knot or link diagrams with exactly three crossings.

4. Is it possible, given what you currently know, that the two diagrams below represent the same knot or link?



5. Is it possible, given what you currently know, that the two diagrams below represent the same knot or link?



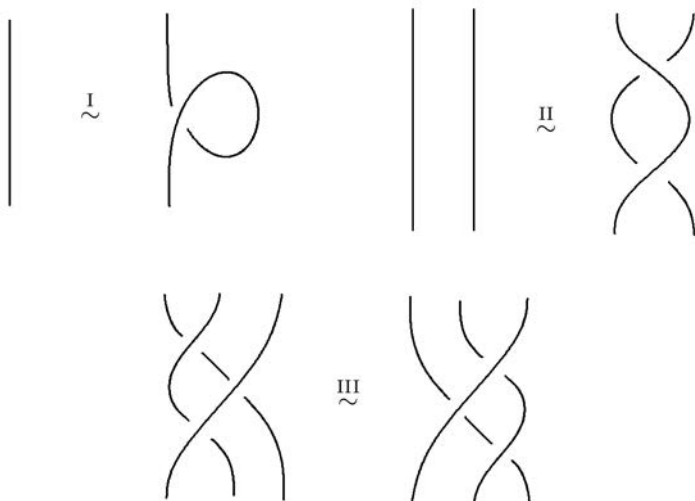
2. Combinatorial Knot Theory

The basic question in knot theory is how to tell when two knot diagrams represent the same knot. This is really two questions: (1) given a knot diagram, what are all possible diagrams which represent the same knot type and (2) how can we prove two diagrams represent different knots. The second question we leave to the next section.

The first question was answered in the 1920s by Kurt Reidemeister. A *local move* on a knot diagram involves replacing one portion of the diagram inside a small disk with something else, while the rest of the diagram outside the disk remains unchanged. A *planar isotopy* is a local move which replaces a strand without crossings with another strand without crossings with the same endpoints.

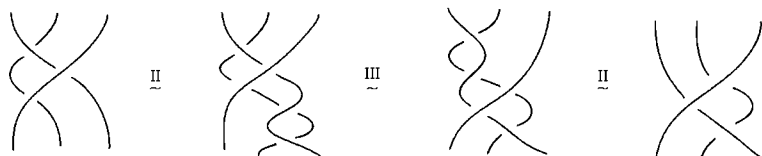


In 1926, Kurt Reidemeister and independently, in 1927, J. W. Alexander and G. B. Briggs proved that two tame knot diagrams, K_0 and K_1 , are ambient isotopic if and only if one can be changed into the other by a finite sequence of planar isotopies and moves of the following three types:



If you look closely, you will find that some other similar moves are implied by the listed moves. For example, move III says you

can move a strand over a crossing where the crossing is the same type (left-over-right) as the other two crossings. We can derive an additional type III move which says you can pass a strand over the other kind of crossing using the given III move and a II move:



The Reidemeister moves let us translate the topological relationship of ambient isotopy into a combinatorial equivalence relation. That is, we started thinking of knots as geometric objects, simple closed curves in space, but we now have a new way to think of knots: as equivalence classes of knot diagrams under the equivalence relation generated by planar isotopy moves and the Reidemeister moves.

Thus, to prove that two knot diagrams, K_0 and K_1 , represent the same knot type, we can identify an explicit sequence of Reidemeister moves taking K_0 to K_1 . For example, the knot below is secretly the unknot, i.e., an unknotted circle. To prove it, we give a sequence of Reidemeister moves taking it to a circle without crossings.

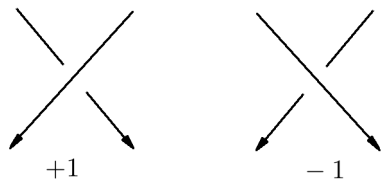


In practice it is often easier to redraw knots using the principle that any portion of a strand with only overcrossings may be replaced with another strand with the same endpoints and all new overcrossings, with the resulting breaks healing. Note that any such “overpass move” can always be broken down into a sequence of Reidemeister

moves and planar isotopies.



Combinatorial Oriented Knots. Introducing an orientation on our knot diagrams gives us two kinds of crossings, which we identify as “positive” and “negative” depending on whether the understrand is directed right-to-left or left-to-right when viewed from the overstrand.



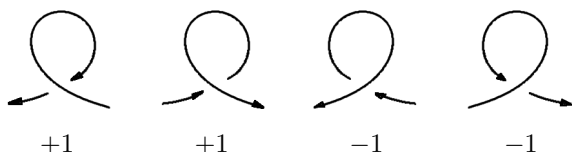
We will denote the sign of a crossing C as $\epsilon(C) = \pm 1$.

Including orientations means we now have more Reidemeister moves than we did before. Instead of two type I moves, we now have four; one type II becomes four – two *direct* moves where the strands are oriented in the same direction, and *reverse* moves where the strands are oriented in opposite directions, and there are eight oriented type III moves.

In practice, many of the moves are implied by the other moves. Indeed, it is an interesting exercise to find a minimal generating set of moves, i.e., a subset containing as few moves as possible from which all of the other oriented moves can be recovered.

The sum of all the crossing signs is a quantity known as the *writhe* of the diagram; writhe is a property of diagrams, not of knots, since starting with a given knot diagram we can adjust the writhe to whatever we want using type I moves. For links, each component has its own writhe determined by counting only crossings where the component crosses itself; multi-component crossings do not contribute to the component writhes.

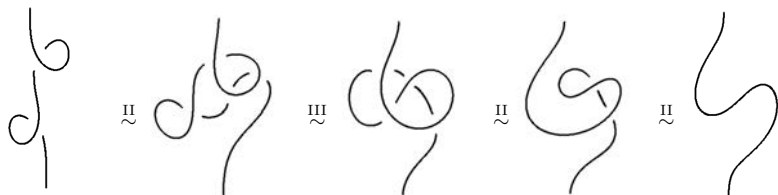
Note that for any single-component crossing both possible choices of orientation determine the same sign for each crossing since switching the orientation of a component reverses the directions of both strands in the crossing. In particular, writhe is well defined even for unoriented diagrams, and kinks have well-defined signs regardless of orientation choice.



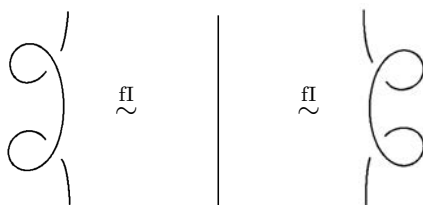
Combinatorial Framed Knots. Given a knot diagram K , there is an easy standard way to choose a framing curve – simply “push off” a copy of K , i.e., draw a framing curve parallel to K . This is traditionally called the *blackboard framing* since it is the easiest framing to draw on a blackboard. More precisely, let F be the knot traced by a normal vector to the knot K . If F is endowed with an orientation parallel to that of K , then the *framing number* of K is the sum of the crossing signs at crossings where F crosses over K . Conversely, if we assign an integer j to a knot K , then we can construct a normal vector to K and a knot F such that j is the framing number. The blackboard framing is then the natural framing with framing number equal to the writhe of K . In particular, every knot or link diagram can be considered as a framed knot or link by using the blackboard framing.

Geometrically, the *framing number* of a framing curve F is the number of times F wraps around the solid torus with K as its core. The framing number of a blackboard-framed knot is equal to its writhe. A little thought reveals that Reidemeister II and III moves do not change the writhe of a diagram, while Reidemeister I moves do. Thus, to preserve the blackboard framing, we must modify the type I move to preserve writhe. In particular, to cancel the $+1$ to writhe from adding a positive kink, we must also add a negative kink. Kinks of both signs come in two versions, clockwise and counterclockwise, also known as kinks of *winding number* -1 and $+1$, respectively.

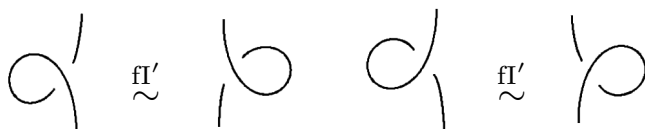
As observed in [FR92], it turns out that if both the winding numbers and crossing signs of the kinks are opposite, we can cancel the kinks using only II and III moves (the following illustration is the simplest “Whitney trick” [Whi44]):



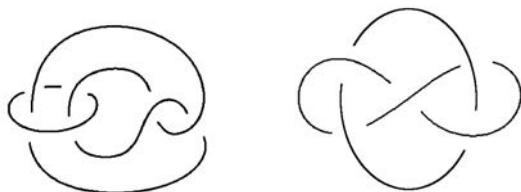
In the case of kinks with equal winding number and opposite writhe, we need an explicit move. These are the *blackboard framed type I moves*:



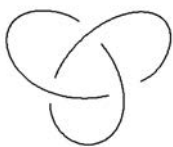
These moves are equivalent to the alternate framed type I moves:



Exercises. 1. Using Reidemeister moves, show that the diagrams below represent the same link. This link is known as the *Whitehead link*.



2. Using Reidemeister moves, determine whether the knot K below is equivalent to the *trefoil* or the *Figure 8*.

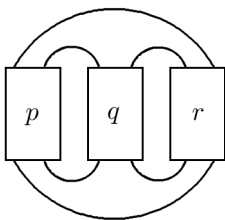


Trefoil

 K 

Figure 8

3. Let p, q, r be three integers. A (p, q, r) -pretzel link is a knot or link of the form



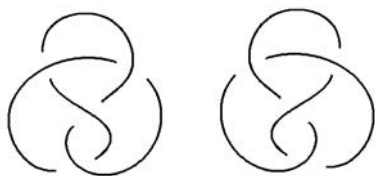
where the boxes are replaced with stacks of p, q and r oriented crossings respectively (a negative value means use negative crossings – also note that the orientation of the crossings in the boxes may not extend to the whole link!) For example, the $(2, 1, -3)$ pretzel link is



How many components are possible in a pretzel link? What conditions on p, q and r ensure that we have a knot? A 2-component link?

4. Show that the figure eight knot 4_1 is ambient isotopic to its mirror image by changing the diagram on the left to the one on the right

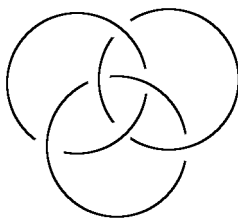
using Reidemeister moves.



5. Using framed Reidemeister moves, show that the knot below is framed isotopic to the unknot with writhe -2 .



6. A link is called *Brunnian* if it is nontrivial, but deleting any component makes the remaining link trivial. Given that the *Borromean rings*



form a nontrivial link, show that the link is Brunnian.

7. Show that the fI and fI' moves are equivalent in the presence of the type II and III moves by deriving the fI' move using only type fI , II, and III moves and then deriving the fI move using only type fI' , II, and III moves.

3. Knot and Link Invariants

Changing K into K' with Reidemeister moves proves that the two diagrams represent the same knot or link. What if we cannot see

a way to change K into K' ? Our inability to change K into K' with Reidemeister moves does not say that K and K' are different; it might be that there is a way, but it's very complicated, perhaps involving hundreds of moves and requiring introducing and removing many crossings. In order to prove that two diagrams represent different knots, we must be more clever.

A *knot invariant* is a function $f : \mathcal{K} \rightarrow X$ from the set of all knot diagrams to a set X such that for each Reidemeister move, we have

$$f(K_1) = f(K_2)$$

where K_1 is the knot diagram before the move and K_2 is the same diagram after the move. If f is a knot invariant, then any two diagrams related by Reidemeister moves must give the same value when we evaluate f .

Knot theory might be described as the search for and the study of knot invariants. Many knot and link invariants have been discovered and studied, mostly in the 20th and 21st centuries. For the remainder of this section we will explore a few well-known knot and link invariants.

Geometric Invariants. One way to define a knot or link invariant is to identify some geometric or topological quantity determined by a knot or link diagram and take the minimum over all diagrams of K . Many examples of this style of invariant have been defined and studied, from basic to more esoteric:

- *Crossing Number* – The minimal number of crossings in any diagram of K .
- *Braid Index* – The smallest number of strands of any braid whose closure is a diagram of K .
- *Bridge Number* – The smallest number of maxima in any diagram of K .
- *Stick Number* – The smallest number of straight line segments needed to form K in $\mathbb{R}^3$.
- *Rope Length* – The minimal length of a rope of radius 1 needed to tie K .

- *Genus* – The minimal number of holes in a surface whose edge is K .
- *Unknotting Number* – The minimal number of crossing changes needed to unknot K .

These invariants are easy to define but generally hard to compute. From a particular diagram of K we can compute an upper bound on the actual value of the invariant for K , but finding a diagram of K which realizes the minimal value is not always easy to do.

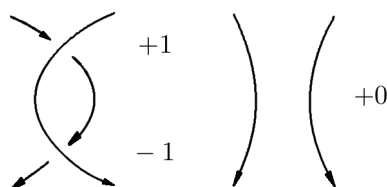
A knot or link invariant f is *computable* if the actual value of $f(K)$ can be determined, not just bounded, using any diagram of K . We will now see three examples of computable knot and link invariants.

Linking Number. Perhaps the easiest example of a computable link invariant is the *linking number*. Let $L = L_1 \cup L_2$ be an oriented link with two components. Let $\mathcal{M}$ be the set of crossings in L with one strand from each component. Then the linking number of L is the sum of the crossing signs of the crossings in $\mathcal{M}$ divided by 2 since this sum is always even:

$$\text{lk}(L) = \frac{1}{2} \sum_{C \in \mathcal{M}} \epsilon(C).$$

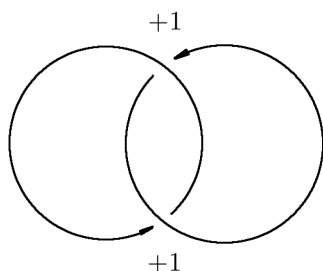
We can verify that the linking number is a link invariant by checking that the contributions match before and after each move. In a type I move, the crossing being introduced or removed is single-component, so it contributes 0 to the linking number, which matches the contribution from the straight strand.

In a type II move, there are two possibilities: either both crossings are multicomponent or both are not. As before, if both crossings are single-component, the contribution of zero matches the zero contribution of the two uncrossed strands. In the multicomponent case, there is always one positive crossing and one negative crossing, so the contribution is $+1 - 1 = 0$.

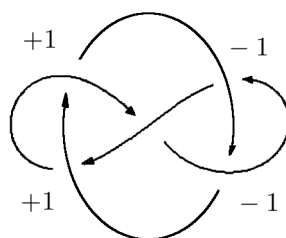


Verifying that $f(L_B) = f(L_A)$ for type III moves is left to the reader as an exercise; see problem 1.

The fact that the linking number is a link invariant lets us distinguish some links from others. For example, the Hopf link below has linking number 1 while the Whitehead link has linking number 0.



$\text{lk} = 1$



$\text{lk} = 0$

The Jones Polynomial. In 1984 knot theory was reinvigorated by the discovery by Vaughan Jones [Jon85] of a powerful knot and link invariant now known as the *Jones polynomial*. The simplest way to define the Jones polynomial is a recursive definition due to Louis Kauffman using the *Kauffman bracket skein relation*. There are several versions of this invariant related by variable substitution; the version we'll use is from [BN02].

Let K be a knot or link diagram. The skein relation can be understood as a way of interpreting a crossing as a linear combination of *smoothings*:

$$\left\langle \begin{array}{c} \diagup \quad \diagdown \\ \diagdown \quad \diagup \end{array} \right\rangle = \left\langle \begin{array}{c} \cup \\ \cup \end{array} \right\rangle - q \left\langle \begin{array}{c} \cup \\ \cap \end{array} \right\rangle \quad \left\langle \begin{array}{c} \diagup \quad \diagdown \\ \diagup \quad \diagdown \end{array} \right\rangle.$$

Recursively applying this relation to each crossing lets us replace a knot or link diagram with n crossings with a sum of polynomials in q times diagrams without crossings. We need a rule for evaluating the bracket of a diagram without crossings. Thus, for a disjoint union of n copies of the diagram of the unknot with no crossings, we define

$$\left\langle \bigcirc \bigcirc \dots \bigcirc \right\rangle = (q + q^{-1})^{n-1}.$$

In particular, we can erase a closed curve without crossings at the cost of multiplying by $(q + q^{-1})$.

The bracket function is unchanged by Reidemeister III moves: The reader is encouraged to verify that both $\left\langle \text{diagram 1} \right\rangle$ and $\left\langle \text{diagram 2} \right\rangle$ are equal to

$$-q \left(\left\langle \text{diagram 3} \right\rangle + \left\langle \text{diagram 4} \right\rangle \right) + q^2 \left(\left\langle \text{diagram 5} \right\rangle + \left\langle \text{diagram 6} \right\rangle \right) - q^3 \left\langle \text{diagram 7} \right\rangle.$$

However, Reidemeister I and II moves do change the value of $\langle K \rangle$, but in a predictable way. More precisely, removing a positive crossing multiplies $\langle K \rangle$ by q^{-1} and removing a negative crossing multiplies $\langle K \rangle$ by $-q^2$:

$$\begin{aligned} \left\langle \text{diagram 8} \right\rangle &= \left\langle \text{diagram 9} \right\rangle - q \left\langle \text{diagram 10} \right\rangle \\ &= (q + q^{-1} - q) \left\langle \text{diagram 11} \right\rangle = q^{-1} \left\langle \text{diagram 12} \right\rangle \end{aligned}$$

and

$$\begin{aligned} \left\langle \text{diagram 13} \right\rangle &= \left\langle \text{diagram 14} \right\rangle - q \left\langle \text{diagram 15} \right\rangle \\ &= (1 - q(q + q^{-1})) \left\langle \text{diagram 16} \right\rangle = -q^2 \left\langle \text{diagram 17} \right\rangle. \end{aligned}$$

Likewise, a crossing-removing type II move multiplies $\langle K \rangle$ by a factor of $(-q^2)q^{-1} = -q$:

$$\begin{aligned}
 \left\langle \begin{array}{c} \diagup \diagdown \\ \diagdown \diagup \end{array} \right\rangle &= \left\langle \begin{array}{c} \frown \\ \smile \end{array} \right\rangle - q \left\langle \begin{array}{c} \bigcirc \\ \bigcirc \end{array} \right\rangle - q \left\langle \begin{array}{c} \diagup \\ \diagdown \end{array} \right\rangle \left\langle \begin{array}{c} \diagdown \\ \diagup \end{array} \right\rangle + q^2 \left\langle \begin{array}{c} \cup \\ \cup \end{array} \right\rangle \\
 &= (1 - q(q + q^{-1}) + q^2) \left\langle \begin{array}{c} \frown \\ \smile \end{array} \right\rangle - q \left\langle \begin{array}{c} \diagup \\ \diagdown \end{array} \right\rangle \left\langle \begin{array}{c} \diagdown \\ \diagup \end{array} \right\rangle \\
 &= -q \left\langle \begin{array}{c} \diagup \\ \diagdown \end{array} \right\rangle \left\langle \begin{array}{c} \diagdown \\ \diagup \end{array} \right\rangle.
 \end{aligned}$$

Thus, to cancel the effects of type I and II moves, we need to multiply by $(-1)^n q^{p-2n}$ where p is the number of positive crossings and n is the number of negative crossings. The *Jones polynomial* of a link L is

$$J(L) = (-1)^n q^{p-2n} \langle L \rangle.$$

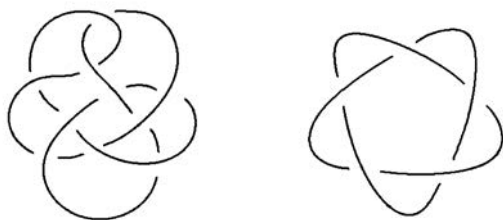
Example 1. Let us compute the Jones polynomial of the Hopf link:

$$\begin{aligned}
 \left\langle \begin{array}{c} \bigcirc \\ \bigcirc \end{array} \right\rangle &= \left\langle \begin{array}{c} \bigcirc \\ \bigcirc \end{array} \right\rangle - q \left\langle \begin{array}{c} \diagup \diagdown \\ \diagdown \diagup \end{array} \right\rangle \\
 &= \left\langle \begin{array}{c} \bigcirc \\ \bigcirc \end{array} \right\rangle - q \left\langle \begin{array}{c} \diagup \diagdown \\ \diagup \diagdown \end{array} \right\rangle \\
 &= -q \left\langle \begin{array}{c} \diagup \diagdown \\ \diagdown \diagup \end{array} \right\rangle + q^2 \left\langle \begin{array}{c} \bigcirc \\ \bigcirc \end{array} \right\rangle \\
 &= q + q^{-1} - q - q + q^2(q + q^{-1}) = q^{-1} + q^3.
 \end{aligned}$$

If we orient the components so that both crossings are positive, we then have $q^2(q^{-1} + q^3) = q + q^5$; if we reverse the orientation of one component while fixing the other, we have two negative crossings and the Jones polynomial becomes $q^{-4}(q^{-1} + q^3) = q^{-1} + q^{-5}$. Thus, the two possible oriented Hopf links have different Jones polynomials and cannot be ambient isotopic to each other.

It turns out that the Jones polynomial of the mirror image of a knot or link K can be obtained from the Jones polynomial of K by

replacing q with q^{-1} . The Jones polynomial is a very powerful invariant, but it is not a complete invariant – there are known examples of pairs of different knots which have the same Jones polynomial, such as the knots below:

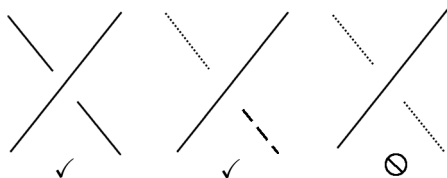


Indeed, one of the more famous unsolved (as of this writing) problems in knot theory is whether the Jones polynomial detects the unknot; that is, is there a nontrivial knot K with Jones polynomial $J(K) = 1$? For links with multiple components, the answer is yes, there are nontrivial links with trivial Jones polynomial. For knots, though, the problem is currently unsolved. Direct computations have shown that no nontrivial knot with fewer than 16 crossings has trivial Jones polynomial.

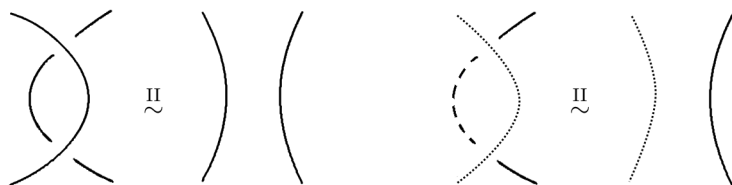
The Jones polynomial is a powerful knot invariant, but computationally it is very intense. The recursive algorithm described above is an *exponential time* algorithm, meaning each additional crossing doubles the number of computations needed to compute $J(K)$.

Tricoloring. For our final example of a computable knot invariant, we will define *Fox tricoloring*, introduced by Ralph Fox in the 1950s. A *tricoloring* of a knot or link diagram is a choice of color for each arc in the diagram from a set of three colors – we’ll use solid, dotted and dashed, but you can use whatever colors you like. A tricoloring is *valid* if at every crossing we either have all three colors the same or all three colors different. A valid tricoloring is *nontrivial* if it uses

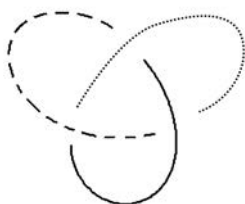
all three colors.



To use tricoloring as a knot invariant, we notice that if we start with a valid tricoloring of a diagram K before doing a Reidemeister move, there is a unique valid tricoloring of the diagram after the move which agrees with the original coloring outside the move area. For example, all strands in a type I move must be the same color. There are two cases for type II moves: both strands the same color before crossing and two different colors before crossing:

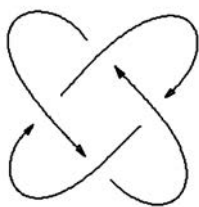


For the type III moves, there are various cases which the reader is encouraged to check. Moreover, if a tricolored diagram is monochrome before a move, the corresponding diagram after the move is also monochrome. Hence, the existence of a nontrivial tricoloring of a knot or link diagram is an invariant of knots and links. For example, the only valid tricolorings of an unknotted diagram are monochrome colorings, while the trefoil has a nontrivial tricoloring; thus there can be no sequence of Reidemeister moves taking the trefoil to the unknot.



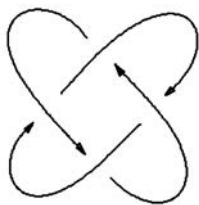
Exercises. 1. Verify that Reidemeister III moves do not change linking number. (Hint: Choose one oriented type III move and consider all cases depending on which strands are from the same component).

2. A link is *split* if it is possible to separate the components so that one component lies entirely on one side of a line and the other component lies entirely on the other side of the line. Prove that the $(4, 2)$ -torus link below is not split.

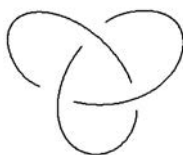
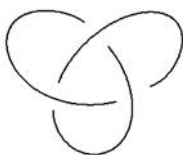


3. Prove that the framing number of a blackboard framed oriented knot is the linking number of the framed knot considered as a 2-component link.

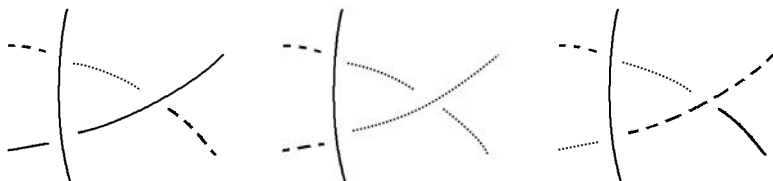
4. Compute the Jones polynomial of the $(4, 2)$ -torus link below.



5. Use the Jones polynomial to prove that the right-handed and left-handed trefoils below are not equivalent.



6. For each of the tricolored tangle diagrams below, find the unique corresponding tricolored tangle diagram after doing a type III move:



7. Show that there is no nontrivial tricoloring of the figure 8 knot below.



8. How many valid tricolorings of the trefoil knot are possible? How many are nontrivial?

A GUIDE TO SYMPLECTIC GEOMETRY

IVO TEREK



These are lecture notes for the SYMPLECTIC GEOMETRY CRASH COURSE held at The Ohio State University during the summer term of 2021, as our first attempt for a series of mini-courses run by graduate students for graduate students. Due to time and space constraints, many things will have to be omitted, but this should serve as a quick introduction to the subject, as courses on Symplectic Geometry are not currently offered at OSU. There will be many exercises scattered throughout these notes, most of them routine ones or just really remarks, not only useful to give the reader a working knowledge about the basic definitions and results, but also to serve as a self-study guide. And as far as references go, arXiv.org links as well as links for authors' webpages were provided whenever possible.

Columbus, May 2021

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1 Symplectic Linear Algebra

1.1 Symplectic spaces and their subspaces

There is nothing more natural than starting a text on *Symplectic Geometry*¹ with the definition of a symplectic vector space. All the vector spaces considered will be real and finite-dimensional unless otherwise specified.

Definition 1

A **symplectic vector space** is a pair (V, Ω) , where:

- V is a vector space, and;
- $\Omega: V \times V \rightarrow \mathbb{R}$ is a non-degenerate^a skew-symmetric bilinear form.

We say that Ω is a **linear symplectic form**.

^aThat is, if $\Omega(v, w) = 0$ for all $w \in V$, then $v = 0$.

Before anything else, a quick observation: every symplectic vector space (V, Ω) is even-dimensional. To wit, let $m = \dim V$ and pick any basis $(v_1, \dots, v_m)$ for V . Identify Ω with the matrix $(\Omega(v_i, v_j))_{i,j=1}^m$. Non-degeneracy of Ω says that $\det \Omega \neq 0$, while skew-symmetry leads to $\det \Omega = (-1)^m \det \Omega$, and thus $(-1)^m = 1$. Thus m is even. Our first example will be manifestly even-dimensional:

Example 1

The mother of all examples (for precise reasons we'll see later in the chapter) is $(\mathbb{R}^{2n}, \Omega_{2n})$, where $\Omega_{2n}: \mathbb{R}^{2n} \times \mathbb{R}^{2n} \rightarrow \mathbb{R}$ is given by

$$\Omega_{2n}((x, y), (x', y')) \doteq \langle x, y' \rangle - \langle x', y \rangle.$$

Here, we write $\mathbb{R}^{2n} = \mathbb{R}^n \oplus \mathbb{R}^n$ and $\langle \cdot, \cdot \rangle$ stands for the standard Euclidean inner product in $\mathbb{R}^n$. Here's a concrete instance (up to perhaps relabeling some coordinates): in $\mathbb{R}^4$, the symplectic form Ω_4 is given by

$$\Omega_4((x, y, z, w), (x', y', z', w')) = xy' - x'y + zw' - z'w.$$

The matrix representing Ω_{2n} relative to the standard basis of $\mathbb{R}^{2n}$, in block form, is simply

$$-J_{2n} \doteq \begin{pmatrix} 0 & \text{Id}_n \\ -\text{Id}_n & 0 \end{pmatrix}.$$

We call $(\mathbb{R}^{2n}, \Omega_{2n})$ the **canonical symplectic prototype**.

¹By the way, the name "symplectic" is due to Hermann Weyl, changing the Latin *com/plex* to the Greek *sym/plectic*.

Here's another example, more abstract in spirit.

Example 2

Let L be a vector space, and consider $V \doteq L \oplus L^*$, where L^* is the dual space of L . We define $\Omega_L: V \times V \rightarrow \mathbb{R}$ by

$$\Omega_L((x, f), (y, g)) \doteq g(x) - f(y).$$

This Ω_L is called the **canonical symplectic structure of $L \oplus L^*$** .

Exercise 1

- (a) Check that the symplectic forms given in the previous two examples are indeed non-degenerate.
- (b) In the previous example, let $(e_1, \dots, e_n)$ be a basis for L , take the dual basis $(e^1, \dots, e^n)$ in L^* , and compute the matrix of Ω relative to the joint basis $((e_1, 0), \dots, (e_n, 0), (0, e^1), \dots, (0, e^n))$ of $L \oplus L^*$. Does the result surprise you?

When dealing with spaces equipped with inner products, we have orthogonal complements. There's a natural analogue here:

Definition 2

Let (V, Ω) be a symplectic vector space and $S \subseteq V$ be a subspace. The **symplectic complement** of S is defined to be $S^\Omega \doteq \{x \in V \mid \Omega(x, y) = 0 \text{ for all } y \in S\}$.

However, the behavior of symplectic complements is not exactly the same as the one for orthogonal complements. The most striking difference is that the sum of S and S^Ω does not need to be direct, nor equal to V , in general. But there's a saving grace:

Proposition 1

Let (V, Ω) be a symplectic vector space, and $S \subseteq V$ a subspace. Then:

- (i) $\dim S + \dim S^\Omega = \dim V$.
- (ii) $(S^\Omega)^\Omega = S$.
- (iii) $S \oplus S^\Omega = V \iff S \cap S^\Omega = \{0\} \iff (S, \Omega|_{S \times S})$ is a symplectic vector space.

Proof:

- (i) The map $V \ni x \mapsto \Omega(x, \cdot)|_S \in S^*$ is surjective with kernel S^Ω , so the formula follows from the rank-nullity theorem, noting that $\dim S = \dim S^*$.
- (ii) Clearly $S \subseteq (S^\Omega)^\Omega$, so equality follows since $\dim S = \dim (S^\Omega)^\Omega$ by the previous item applied twice.

- (iii) Since $\dim(S + S^\Omega) = \dim S + \dim S^\Omega - \dim(S \cap S^\Omega)$, item (i) now says that $\dim(S + S^\Omega) = \dim V$ if and only if $\dim(S \cap S^\Omega) = 0$.

□

In particular, this proposition shows that it is not true that restricting Ω to subspaces of V gives symplectic spaces. This is a very special condition. There are other types of subspaces worthy of attention.

Definition 3

Let (V, Ω) be a symplectic vector space and $S \subseteq V$ be a subspace. We say that S is:

- (i) **symplectic** if $S \cap S^\Omega = \{0\}$.
- (ii) **isotropic** if $S \subseteq S^\Omega$.
- (iii) **coisotropic** if $S^\Omega \subseteq S$.
- (iv) **Lagrangian** if $S = S^\Omega$.

As immediate examples, all lines passing through the origin are isotropic, and all hyperplanes passing through the origin are coisotropic. And Lagrangian subspaces are the isotropic subspaces which are maximal (relative to inclusion) — it is not hard to see that Lagrangian subspaces are always mid-dimensional. Here are more concrete examples:

Example 3

- (1) Let $S \subseteq \mathbb{R}^n$ be any subspace. Then $S \times \{0\}$ and $\{0\} \times S$ are isotropic subspaces of $(\mathbb{R}^{2n}, \Omega_{2n})$, and they're Lagrangian if and only if $S = \mathbb{R}^n$. Moreover, $S \times S^\perp$ is always Lagrangian.
- (2) Let L be a vector space, and consider the canonical symplectic structure Ω_L on $V = L \oplus L^*$. If $S \subseteq L$ and $X \subseteq L^*$ are vector subspaces, then $S \oplus \{0\}$ and $\{0\} \oplus X$ are both isotropic, and Lagrangian if and only if $S = L$ and $X = L^*$, respectively. If $\text{Ann}(S) \subseteq L^*$ is the annihilator of S , then $S \oplus \text{Ann}(S)$ is Lagrangian.
- (3) If (V_1, Ω_1) and (V_2, Ω_2) are symplectic vector spaces, so is their (external) direct sum $(V_1 \oplus V_2, \Omega_1 \oplus \Omega_2)$, where

$$(\Omega_1 \oplus \Omega_2)((v_1, v_2), (v'_1, v'_2)) \doteq \Omega_1(v_1, v'_1) + \Omega_2(v_2, v'_2).$$

If $S_1 \subseteq V_1$ and $S_2 \subseteq V_2$ are subspaces, then $(S_1 \times S_2)^{\Omega_1 \oplus \Omega_2} = S_1^{\Omega_1} \oplus S_2^{\Omega_2}$, so that if S_1 and S_2 are both of the same “type” (i.e., symplectic, isotropic, coisotropic, Lagrangian), then $S_1 \times S_2$ will also have the same type.

Exercise 2

Explore the above example further: if $S_1, S_2 \subseteq \mathbb{R}^n$ are two subspaces, consider $(\mathbb{R}^{2n}, \Omega_{2n})$ and show that $(S_1 \times S_2)^{\Omega_{2n}} = S_2^\perp \times S_1^\perp$. Conclude that $S_1 \times S_2$ is isotropic if and only if $S_1 \subseteq S_2^\perp$, coisotropic if and only if $S_1 \supseteq S_2^\perp$, and Lagrangian if and only if $S_1 = S_2^\perp$. State and prove a similar result replacing $(\mathbb{R}^{2n}, \Omega_{2n})$ with $(L \oplus L^*, \Omega_L)$ and orthogonal complements with annihilators. Check the claims made in item (3).

Here's a less obvious example of a symplectic vector space, coming from a differential equation:

Example 4

Let $(L, \langle \cdot, \cdot \rangle)$ be a pseudo-Euclidean space (i.e., $\langle \cdot, \cdot \rangle$ is a non-degenerate symmetric bilinear form on L), $I \subseteq \mathbb{R}$ be an open interval, and $\Phi: I \rightarrow \text{End}(L)$ be a smooth curve of self-adjoint operators. Let V be the space of smooth solutions $v: I \rightarrow L$ of the ordinary differential equation $\ddot{v}(t) = \Phi(t)v(t)$ (note that $\dim V = 2 \dim L$ is even). Then, define $\Omega: V \times V \rightarrow \mathbb{R}$ by $\Omega(v, w) = \langle \dot{v}, w \rangle - \langle v, \dot{w} \rangle$ (observe that the right side is indeed constant). Clearly Ω is skew-symmetric, but we claim that it is non-degenerate as well: if $v \in V \setminus \{0\}$, there is $t_0 \in I$ such that $v(t_0) \neq 0$, so non-degeneracy of $\langle \cdot, \cdot \rangle$ provides a vector $w_0 \in L$ with $\langle v(t_0), w_0 \rangle \neq 0$. Then the unique $w \in V$ with $w(t_0) = 0$ and $\dot{w}(t_0) = w_0$ satisfies $\Omega(v, w) \neq 0$. For each $t \in I$, we have the operators $\delta_t, \dot{\delta}_t: V \rightarrow L$ given by $\delta_t(v) = v(t)$ and $\dot{\delta}_t(v) = \dot{v}(t)$, and both $\ker \delta_t$ and $\ker \dot{\delta}_t$ are Lagrangian.

In particular, the same idea shows that the space $\mathcal{J}(\gamma)$ of Jacobi fields along a geodesic γ on a pseudo-Riemannian manifold (M, g) has a natural symplectic form, since $J \in \mathcal{J}(\gamma)$ if and only if

$$\frac{D^2 J}{dt}(t) = R(\dot{\gamma}(t), J_t)\dot{\gamma}(t),$$

where D/dt is the covariant derivative along γ induced by the Levi-Civita connection of (M, g) , and each $\Phi(t) = R(\dot{\gamma}(t), \cdot)\dot{\gamma}(t)$ is self-adjoint due to symmetries of the curvature tensor.

The following exercise is also useful to practice using the definitions:

Exercise 3

Let (V, Ω) be a symplectic vector space, and $S_1, S_2 \subseteq V$ be two subspaces. Show that:

- (a) $(S_1 + S_2)^\Omega = S_1^\Omega \cap S_2^\Omega$;
- (b) $(S_1 \cap S_2)^\Omega = S_1^\Omega + S_2^\Omega$.

Conclude that if S is any subspace of V , then $S \cap S^\Omega$ is always isotropic and $S + S^\Omega$ is always coisotropic. **Hint:** how does (b) follow from (a)?

Let's make sure you're comfortable with everything seen so far.

Exercise 4 (Challenge #1)

Let V and W be two vector spaces. A map $B: V \times W \rightarrow \mathbb{R}$ is called a **perfect pairing** if both maps $V \rightarrow W^*$ and $W \rightarrow V^*$ induced by B are isomorphisms. Define $\Omega_B: (V \times W) \times (V \times W) \rightarrow \mathbb{R}$ by $\Omega_B((v, w), (v', w')) = B(v, w') - B(v', w)$.

- (a) Show that $(V \times W, \Omega_B)$ is a symplectic vector space and explain how the two main examples we have been dealing with so far fit here. Namely, what are the perfect pairings?
- (b) Let $T: V \rightarrow W$ be a linear map. Show that the graph $\text{gr}(T)$ is a Lagrangian subspace of $V \times W$ if and only if the diagram

$$\begin{array}{ccc} V & \xrightarrow{T} & W \\ \cong \downarrow & & \downarrow \cong \\ W^* & \xrightarrow{T^*} & V^* \end{array}$$

commutes.

- (c) Conclude that if L is a vector space and $T: L \rightarrow L^*$ is linear, then $\text{gr}(T)$ is a Lagrangian subspace of $(L \oplus L^*, \Omega_L)$ if and only if the bilinear map induced by T , $(x, y) \mapsto T(x)(y)$, is symmetric.

We move on. Spaces with inner products have orthonormal bases, but we have seen that orthogonal complements and symplectic complements are not quite the same. Are there symplectic analogues of orthonormal bases? You shouldn't be surprised that the answer is "yes".

Theorem 1

Let (V, Ω) be a symplectic vector space, with dimension $\dim V = 2n$. There is a basis $\mathcal{B} = (e_1, \dots, e_n, f_1, \dots, f_n)$ for V such that for all $i, j = 1, \dots, n$, we have $\Omega(e_i, e_j) = 0$, $\Omega(f_i, f_j) = 0$ and $\Omega(e_i, f_j) = \delta_{ij}$. Or, in other words, the matrix of Ω relative to $\mathcal{B}$ is $-J_{2n}$. Such a basis $\mathcal{B}$ is called a **Darboux basis** for (V, Ω) .

Proof: The proof goes by (strong) induction on n . If $\dim V = 2$, then pick any $e_1 \in V$, non-zero. Since Ω is non-degenerate, there is $f_1 \in V$ such that $\Omega(e_1, f_1) \neq 0$. Rescaling f_1 , we may assume that $\Omega(e_1, f_1) = 1$, which establishes the base of the induction. For the inductive step, assume that $\dim V = 2n$, take $e_1, f_1 \in V$ just as above, and let $S = \text{span}(e_1, f_1)$. Clearly S is symplectic, so S^Ω is also symplectic and $S \oplus S^\Omega = V$. By the induction assumption, we may take a Darboux basis $(e_2, \dots, e_n, f_2, \dots, f_n)$ for S^Ω . Throw in e_1 and f_1 to get a Darboux basis for V . $\square$

Example 5

Let (V, Ω) be a symplectic vector space and $\mathcal{B} = (e_1, \dots, e_n, f_1, \dots, f_n)$ be a Darboux basis for V . Let $I \subseteq [n]$, where $[n] = \{1, \dots, n\}$. Then $S_{\mathcal{B}, e, I} = \text{span}(e_i : i \in I)$ and $S_{\mathcal{B}, f, I} = \text{span}(f_i : i \in I)$ are isotropic subspaces, Lagrangian if and only if $I = [n]$. And the subspace $S_{\mathcal{B}, e, f, I} = \text{span}(e_i, f_i : i \in I)$ is symplectic, with complement $S_{\mathcal{B}, e, f, I}^\Omega = S_{\mathcal{B}, e, f, [n] \setminus I}$.

Let's conclude this section with a lemma that will be useful later.

Lemma 1

Let (V, Ω) be a symplectic vector space, and $L \subseteq V$ a Lagrangian subspace. Then L has a Lagrangian complement, that is, there is a second Lagrangian subspace $L' \subseteq V$ such that $V = L \oplus L'$.

Proof: Start taking any complementary subspace $W \subseteq V$ for L , i.e., a subspace such that $V = L \oplus W$. There's no reason this W should already be Lagrangian, so we correct it as follows: for any $w \in W$, we look at the linear functional $f_w : V \rightarrow \mathbb{R}$ given by $f_w(x) = \Omega(w, \text{pr}_W(x))/2$, where $\text{pr}_W : V \rightarrow W$ is the projection defined by the direct sum. Note that f_w annihilates L , so we may write $f_w = \Omega(w', \cdot)$ for some $w' \in L^\Omega = L$ which is uniquely determined by w . Now we let $L' = \{w - w' \mid w \in W\}$, and claim that L' is Lagrangian and $V = L \oplus L'$. If $w_1, w_2 \in W$, compute

$$\begin{aligned} \Omega(w_1 - w'_1, w_2 - w'_2) &= \Omega(w_1, w_2) - \Omega(w_1, w'_2) - \Omega(w'_1, w_2) + \Omega(w'_1, w'_2) \\ &= \Omega(w_1, w_2) + f_{w_2}(w_1) - f_{w_1}(w_2) - 0 \\ &= \Omega(w_1, w_2) + \frac{1}{2}\Omega(w_2, w_1) - \frac{1}{2}\Omega(w_1, w_2) \\ &= 0. \end{aligned}$$

So L' is isotropic and $V = L + L'$, hence L' is Lagrangian and $V = L \oplus L'$ by a dimension count. The equality $V = L + L'$ is justified as follows: if $v \in V$, then $v = x + w$ with $x \in L$ and $w \in W$, so we have $v = (x + w') + (w - w')$. $\square$

1.2 Symplectomorphisms

In the previous section, we have seen symplectic analogues of orthogonal complements and of orthonormal bases. Naturally, there are analogues of isometries.

Definition 4

Let (V_1, Ω_1) and (V_2, Ω_2) be two symplectic vector spaces, and $\varphi : V_1 \rightarrow V_2$ be linear. The map φ is called **symplectic** if $\Omega_1(x, y) = \Omega_2(\varphi(x), \varphi(y))$ for all vectors $x, y \in V$. A symplectic isomorphism is then called a **symplectomorphism**.

Symplectic maps should behave well relative to symplectic complements, just like isometries behave well relative to orthogonal complements. There are some minor things one should pay attention to, though. The next exercise will clarify them.

Exercise 5

Let (V_1, Ω_1) and (V_2, Ω_2) be two symplectic vector spaces, and $\varphi: V_1 \rightarrow V_2$ be a symplectic map.

- Show that φ is necessarily injective. Give an example showing that φ does not need to be surjective.
- Show that if $S \subseteq V_1$ is a subspace, then $\varphi[S^{\Omega_1}] \subseteq \varphi[S]^{\Omega_2}$. Give an example showing that the reverse inclusion need not hold.
- With the same notation as in (b), show that $\varphi[S^{\Omega_1}] = \varphi[S]^{\Omega_2} \cap \text{Ran}(\varphi)$.

With this in place, let's see some examples.

Example 6

Let L be a vector space and consider the canonical symplectic space $(L \oplus L^*, \Omega_L)$. If $T \in \text{GL}(L)$, then $\varphi \doteq T \oplus (T^*)^{-1}: L \oplus L^* \rightarrow L \oplus L^*$ is a symplectomorphism, since

$$\begin{aligned} \Omega_L(\varphi(x, f), \varphi(y, g)) &= (T^*)^{-1}(g)(Tx) - (T^*)^{-1}(f)(Ty) \\ &= T^* \circ (T^*)^{-1}(g)(x) - T^* \circ (T^*)^{-1}(f)(y) \\ &= g(x) - f(y) \\ &= \Omega_L((x, f), (y, g)). \end{aligned}$$

As a particular example, we may take the dilations $T(x) = e^a x$, where $a \in \mathbb{R}$ is fixed. This construction gives $\varphi_a(x, f) = (e^a x, e^{-a} f)$.

Exercise 6

Show that the above example is optimal, in the following sense: if $T: L \rightarrow L$ and $\hat{T}: L^* \rightarrow L^*$ are any linear maps, then $T \oplus \hat{T}$ is a symplectomorphism of $L \oplus L^*$ if and only if $\hat{T} = (T^*)^{-1}$.

Example 7

Let (V, Ω) be a symplectic vector space and $\mathcal{B} = (e_1, \dots, e_n, f_1, \dots, f_n)$ be a Darboux basis for V .

- The linear map $J: V \rightarrow V$ defined by setting $J(e_i) = f_i$ and $J(f_i) = -e_i$, for all $i = 1, \dots, n$, is a symplectomorphism. Note that $J^2 = -\text{Id}_V$ (that is, J is a

linear complex structure on V).

- If $\sigma \in S_n$ is a permutation, then $\varphi_\sigma: V \rightarrow V$ given by $\varphi_\sigma(e_i) = e_{\sigma(i)}$ and $\varphi_\sigma(f_i) = f_{\sigma(i)}$, for all $i = 1, \dots, n$, is a symplectomorphism.
- Let's generalize the above item. Let $A \in \text{GL}_n(\mathbb{R})$, and define $\varphi_A: V \rightarrow V$ by

$$\varphi(e_j) = \sum_{i=1}^n (A^\top)^i_j e_i \quad \text{and} \quad \varphi(f_j) = \sum_{i=1}^n (A^{-1})^i_j f_i.$$

Then φ_A is a symplectomorphism. It's clear that $\Omega(\varphi_A(e_i), \varphi_A(e_j)) = 0$ and $\Omega(\varphi_A(f_i), \varphi_A(f_j)) = 0$ for all choices of i and j , so let's only check the last Darboux relation:

$$\begin{aligned} \Omega(\varphi_A(e_i), \varphi_A(e_j)) &= \Omega\left(\sum_{k=1}^n (A^\top)^k_i e_k, \sum_{\ell=1}^n (A^{-1})^\ell_j f_\ell\right) \\ &= \sum_{k,\ell=1}^n (A^\top)^k_i (A^{-1})^\ell_j \Omega(e_k, f_\ell) \\ &= \sum_{k,\ell=1}^n (A^\top)^k_i (A^{-1})^\ell_j \delta_{k\ell} \\ &= \sum_{k=1}^n (A^\top)^k_i (A^{-1})^k_j \\ &= \delta_{ij}. \end{aligned}$$

Exercise 7

Show that the last bullet of the previous example is optimal, in the following sense (with same notations as above and before): if $\varphi: V \rightarrow V$ is a symplectomorphism which leaves $S_{\mathcal{B},e,[n]}$ and $S_{\mathcal{B},f,[n]}$ invariant, there are two matrices $B, C \in \text{GL}_n(\mathbb{R})$, $B = (b^i_j)_{i,j=1}^n$ and $C = (c^i_j)_{i,j=1}^n$, such that

$$\varphi(e_j) = \sum_{i=1}^n b^i_j e_i \quad \text{and} \quad \varphi(f_j) = \sum_{i=1}^n c^i_j f_i.$$

Then there is $A \in \text{GL}_n(\mathbb{R})$ such that $B = A^\top$ and $C = A^{-1}$.

Example 8

As in Example 4 (p. 4), let $(L, \langle \cdot, \cdot \rangle)$ be a pseudo-Euclidean space and (V, Ω) be the vector space of smooth solutions of $\ddot{v}(t) = \Phi(t)v(t)$ on some interval $I \subseteq \mathbb{R}$, for a curve of self-adjoint endomorphisms of L . For every $t \in I$, the evaluation map $V \ni v \mapsto (v(t), \dot{v}(t)) \in L \oplus L$ is a symplectomorphism, where $L \oplus L$ is considered

with the symplectic form from Example 1 (p. 1, with L playing the role of $\mathbb{R}^n$) or, alternatively, Exercise 4 (p. 5) with the perfect pairing $B = \langle \cdot, \cdot \rangle$.

Now we are ready to discuss the symplectic analogue of $O(V, g)$, where (V, g) is an inner product space.

Definition 5

Let (V, Ω) be a symplectic vector space. The **symplectomorphism group** of (V, Ω) , denoted by $\text{Sp}(V, \Omega)$, is the collection of all symplectomorphisms from V to itself.

Clearly $\text{Sp}(V, \Omega)$ is a group when equipped with composition of functions. In particular, we write $\text{Sp}_{2n}(\mathbb{R}) \doteq \text{Sp}(\mathbb{R}^{2n}, \Omega_{2n})$ when discussing things on the matrix level. Namely, we have that

$$\text{Sp}_{2n}(\mathbb{R}) = \{R \in \text{GL}_{2n}(\mathbb{R}) \mid R^\top J_{2n} R = J_{2n}\},$$

and from there we see that $\text{Sp}_{2n}(\mathbb{R})$ is closed under matrix transposition (this is not true for every matrix group). Writing matrices in block-form, we can explicitly write what it means for R to be in $\text{Sp}_{2n}(\mathbb{R})$.

Exercise 8 (The Luneburg Relations)

Write $R \in \text{GL}_{2n}(\mathbb{R})$ as

$$R = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$$

Show that $R \in \text{Sp}_{2n}(\mathbb{R})$ if and only if we have that $A^\top C$ and $B^\top D$ are symmetric and $A^\top D - C^\top B = \text{Id}_n$. In this case, also show that

$$R^{-1} = \begin{pmatrix} D^\top & -B^\top \\ -C^\top & A^\top \end{pmatrix}.$$

We will not be able to go into much detail about $\text{Sp}(V, \Omega)$, so we will avoid technical proofs (references will be given) and settle for a few comments regarding:

- a distinguished set of generators $\langle \tau_{v, \lambda} \mid (v, \lambda) \in (V \setminus \{0\}) \times \mathbb{R} \rangle$.
- the spectrum $\sigma(\varphi)$ of a symplectomorphism $\varphi \in \text{Sp}(V, \Omega)$;
- the fundamental group $\pi_1 \text{Sp}(V, \Omega)$.

We will start on the group-theoretic side of things.

Definition 6

Let (V, Ω) be a symplectic vector space, and fix $v \in V$, $v \neq 0$, and $\lambda \in \mathbb{R}$. The **symplectic transvection** in the direction of v with intensity λ is the linear map $\tau_{v,\lambda}: V \rightarrow V$ given by $\tau_{v,\lambda}(x) \doteq x + \lambda\Omega(x, v)v$.

A symplectic transvection $\tau_{v,\lambda}$ is essentially a “shear-like reflection” about the hyperplane $(\mathbb{R}v)^\Omega$. It is instructive to establish some of their basic properties:

Exercise 9

Let (V, Ω) be a symplectic vector space, and $v \in V$ a non-zero vector. Then:

- (a) $\tau_{v,\lambda} \in \text{Sp}(V, \Omega)$, for all $\lambda \in \mathbb{R}$;
- (b) $\tau_{v,\lambda_1} \circ \tau_{v,\lambda_2} = \tau_{v,\lambda_1+\lambda_2}$, for all $\lambda_1, \lambda_2 \in \mathbb{R}$;
- (c) $\tau_{v,\lambda} = \text{Id}_V$ if and only if $\lambda = 0$;
- (d) if $\varphi \in \text{Sp}(V, \Omega)$, then $\varphi \circ \tau_{v,\lambda} \circ \varphi^{-1} = \tau_{\varphi(v),\lambda}$, for all $\lambda \in \mathbb{R}$;
- (e) $\tau_{av,\lambda} = \tau_{v,a^2\lambda}$, for all $a, \lambda \in \mathbb{R}$, $a \neq 0$.

The takeaway here is that if $v \neq 0$, the curve $\mathbb{R} \ni \lambda \mapsto \tau_{v,\lambda} \in \text{Sp}(V, \Omega)$ is an injective group homomorphism, and the images of any two such curves are conjugate in $\text{Sp}(V, \Omega)$.

The properties given above are the key for proving the big theorem on the group structure of $\text{Sp}(V, \Omega)$:

Theorem 2 (Symplectic “Cartan-Dieudonné”)

For any symplectic vector space (V, Ω) , the group $\text{Sp}(V, \Omega)$ is generated by symplectic transvections.

See [24] for a proof. From the definition of $\text{Sp}_{2n}(\mathbb{R})$ and using Darboux bases, we can see that the determinant of any symplectomorphism equals 1 or -1 . This theorem allows us to obtain the optimal conclusion:

Corollary 1

If (V, Ω) is a symplectic vector space and $\varphi \in \text{Sp}(V, \Omega)$, then $\det \varphi = 1$.

Proof: By the symplectic version of Cartan-Dieudonné, it suffices to check that given a non-zero vector $v \in V$ and $\lambda \in \mathbb{R}$, then $\det \tau_{v,\lambda} = 1$. But it is easy to see that $\det \tau_{v,\lambda} = 1 + \lambda\Omega(v, v) = 1$. $\square$

Now, we change gears. It turns out that knowing that every symplectomorphism is unimodular is exactly what we need to understand its spectrum.

Theorem 3 (The Symplectic Eigenvalue Theorem)

Let (V, Ω) be a symplectic vector space and $\varphi \in \text{Sp}(V, \Omega)$. If $\lambda \in \mathbb{C}$ is an eigenvalue of φ , so will be λ^{-1} , $\bar{\lambda}$ and $\bar{\lambda}^{-1}$. Moreover, the algebraic multiplicities of λ and λ^{-1} are the same, and the multiplicities of 1 and -1 (if they occur at all) are even.

For a proof, see [1]. The key idea, though, is to show that if $p(\lambda)$ is the characteristic polynomial of φ , the relation $p(\lambda) = \lambda^{2n} p(\lambda^{-1})$ holds — and here enters the condition $\det \varphi = 1$ (try to prove it). And to conclude this section, we observe that $\text{Sp}(V, \Omega)$ is (perhaps not surprisingly) a Lie group (with dimension $\dim \text{Sp}(V, \Omega) = n(2n + 1)$, if $2n = \dim V$, by Lunenburg's relations), and its Lie algebra $\mathfrak{sp}(V, \Omega)$ consists of all linear operators with are Ω -skew (those are called **Hamiltonian operators**). There is a version of the above theorem for $\mathfrak{sp}(V, \Omega)$, as well as “infinitesimal Lunenburg relations”. Now, every Lie group has, first and foremost, a topology. And what can we say about the topology of $\text{Sp}(V, \Omega)$? The simplest (functorial) topological invariant is the fundamental group. It turns out that $\pi_1 \text{Sp}(V, \Omega) \cong \mathbb{Z}$, but the surprising fact is that there is a very specific isomorphism, called the **Maslov index** — defined from loops in the Grassmannian of Lagrangian subspaces of (V, Ω) and the usual notion of degree between self-maps of the circle $\mathbb{S}^1$. See [28] or [35] for more details.

1.3 Local linear forms

The reason we're starting these lecture notes discussing linear algebra is not only for pedagogical reasons, but really because symplectic vector spaces are the linear prototypes of symplectic manifolds we'll study in the next chapters. More precisely, the tangent spaces to symplectic manifolds are symplectic vector spaces, so (essentially) all linear algebra tools can be smoothly applied pointwise on a manifold. And frequent questions are “what does a symplectic manifold look like locally” and “how does a given submanifold fit into a symplectic manifold?”. Answering these questions in the linear setting is an easy task, and we'll do this now.

Theorem 4 (Linear Darboux)

Let (V, Ω) be a symplectic vector space with $2n = \dim V$. Then (V, Ω) is symplectomorphic to the canonical symplectic prototype $(\mathbb{R}^{2n}, \Omega_{2n})$.

Proof: Take a Darboux basis $\mathcal{B} = (e_1, \dots, e_n, f_1, \dots, f_n)$ for V and define $\varphi: \mathbb{R}^{2n} \rightarrow V$ by

$$\varphi(x, y) = \sum_{i=1}^n (x^i e_i + y^i f_i).$$

Clearly φ is a symplectomorphism. □

This is underwhelming, and we could have established this theorem in the last section (perhaps the reader even guessed the statement by then). One drawback should be pointed: the symplectomorphism is not natural (in a categorical sense), and depends on a choice of Darboux basis for V — which may be completely arbitrary. Despite this, it has somewhat deep consequences (e.g., the only invariant of a symplectic vector space is its dimension), which will become more evident when we encounter its non-linear version in the next chapter. We will see that every symplectic manifold locally looks like $(\mathbb{R}^{2n}, \Omega_{2n})$, not only at a given point, but at a *neighborhood* of each point. For example, the symplectomorphism $(V, \Omega) \cong (\mathbb{R}^{2n}, \Omega_{2n})$ should be thought of as a *global* Darboux coordinate system for V . More on this later. From the several “types” of subspaces (V, Ω) may have, Lagrangian subspaces seem to be the most important ones, and thus one could ask whether there is a local form for Lagrangian submanifolds of a symplectic manifold. We’ll start to answer this question now, in the linear setting — the proof, as opposed to the previous one, will be slightly more interesting. The naturality problem pointed out for the linear Darboux theorem will not be an issue now due to this simple fact from linear algebra: if V is a vector space, $L \subseteq V$ is a subspace, and $L', L'' \subseteq V$ are two subspaces complementary to L , then the diagram

$$\begin{array}{ccc} & V/L & \\ \widetilde{\text{pr}}_{L'} \swarrow & & \searrow \widetilde{\text{pr}}_{L''} \\ L' & \xrightarrow{\text{pr}_{L''}|_{L'}} & L'' \end{array}$$

commutes where all the pr stand for the corresponding projections given by direct sum decompositions, and the $\widetilde{\text{pr}}$ stand for maps induced on V/L .

Theorem 5 (Linear Weinstein)

Let (V, Ω) be a symplectic vector space, and $L \subseteq V$ a Lagrangian subspace. Then there is a natural symplectomorphism between (V, Ω) and $(L \oplus L^*, \Omega_L)$ such that the diagram

$$\begin{array}{ccc} & & V \\ & \nearrow & \parallel \\ L & & \cong \\ & \searrow & \\ & & L \oplus L^* \end{array}$$

commutes.

Proof: Choose a complementary Lagrangian subspace L' for L , and write $V = L \oplus L'$. We have the following chain of equalities and isomorphisms:

$$V = L \oplus L' \xrightarrow[\cong]{\text{Id}_L \oplus \widetilde{\text{pr}}_{L'}^{-1}} L \oplus \frac{V}{L} = L \oplus \frac{V}{L\Omega} \xrightarrow[\cong]{\text{Id}_L \oplus \Omega} L \oplus L^*$$

Above, $\widetilde{\text{pr}}_{L'}: V/L \rightarrow L'$ is the map induced by the projection $\text{pr}_{L'}: V \rightarrow L'$ given by the direct sum decomposition — which also yields $\text{pr}_L: V \rightarrow L$. The given diagram obviously commutes, so we only have to check that the full composition φ is symplectic. Write $\varphi(x) = (\text{pr}_L(x), f_x)$, where $f_x = \Omega(\cdot, x)|_L$ (be careful here: doing $\Omega(x, \cdot)|_L$ as usual will create a negative sign later). First, note that for all $y \in V$, we have that

$$\begin{aligned} f_x(\text{pr}_L(y)) &= \Omega(\text{pr}_L(y), x) = \Omega(y - \text{pr}_{L'}(y), x) \\ &= \Omega(y, x) - \Omega(\text{pr}_{L'}(y), x) \stackrel{(1)}{=} \Omega(y, x) - \Omega(\text{pr}_{L'}(y), \text{pr}_L(x)) \\ &\stackrel{(2)}{=} \Omega(y, x) - \Omega(y, \text{pr}_L(x)) = \Omega(\text{pr}_L(x) - x, y), \end{aligned}$$

where in (1) we use that L' is Lagrangian, and in (2) that L is Lagrangian. Hence $\Omega(\text{pr}_L(x) - x, \cdot) = f_x \circ \text{pr}_L$. With this in place, for all $x, y \in V$, we may compute

$$\begin{aligned} \Omega_L(\varphi(x), \varphi(y)) &= \Omega_L((\text{pr}_L(x), f_x), (\text{pr}_L(y), f_y)) \\ &= f_y(\text{pr}_L(x)) - f_x(\text{pr}_L(y)) \\ &= \Omega(\text{pr}_L(y) - y, x) - \Omega(\text{pr}_L(x) - x, y) \\ &= \Omega(\text{pr}_L(y), x) - \Omega(y, x) - \Omega(\text{pr}_L(x), y) + \Omega(x, y) \\ &= 2\Omega(x, y) - \Omega(x, \text{pr}_L(y)) - \Omega(\text{pr}_L(x), y) \\ &= \Omega(x, y), \end{aligned}$$

as wanted, since $\Omega(x, y) = \Omega(\text{pr}_L(x), \text{pr}_{L'}(y)) + \Omega(\text{pr}_{L'}(x), \text{pr}_L(y))$, as both L and L' are Lagrangian. $\square$

2 Symplectic Manifolds

2.1 Definitions and examples

The most important definition in this text requires some motivation. Let's think of the canonical symplectic prototype not as a vector space, but as a manifold. Each tangent space to $\mathbb{R}^{2n}$ is naturally isomorphic to $\mathbb{R}^{2n}$ itself, so let $\omega_{2n} \in \Omega^2(\mathbb{R}^{2n})$ assign to each point in $\mathbb{R}^{2n}$, the linear symplectic form Ω_{2n} . If we write the coordinates in $\mathbb{R}^{2n}$ as $(x^1, \dots, x^n, y_1, \dots, y_n)$, then we have that

$$\omega_{2n} = \sum_{k=1}^n dx^k \wedge dy_k = -d \left(\sum_{k=1}^n y_k dx^k \right).$$

Thus ω_{2n} is an exact form (the reason for writing it like this, with the negative sign, will become clear soon), and it is non-degenerate at each point. This would suggest defining a symplectic manifold as a smooth manifold equipped with a non-degenerate exact 2-form. It turns out that this is *not* a good definition. Here's the reason:

Proposition 2

Let M be a smooth manifold and $\omega \in \Omega^2(M)$ be exact and non-degenerate. Then M is not compact.

Proof: Assume by contradiction that M is compact. Since ω is non-degenerate, the dimension of M must be even, say $\dim M = 2n$. And non-degeneracy also gives that $\omega^{\wedge n}$ is a volume form on M . But if ω is exact, write $\omega = d\eta$ for some $\eta \in \Omega^1(M)$. Now Stokes's theorem gives that

$$0 \neq \int_M \omega^{\wedge n} = \int_M d\eta \wedge \omega^{\wedge(n-1)} = \int_M d(\eta \wedge \omega^{\wedge(n-1)}) = 0,$$

as $\partial M = \emptyset$. □

This means that our first guess of what a symplectic manifold should be would exclude all compact manifolds from the theory. Thus, we must move on to the next best thing. If we cannot require ω to be exact, let's settle for locally exact. By Poincaré's lemma, this is equivalent to being closed.

Definition 7

A **symplectic manifold** is a pair (M, ω) , where M is a smooth manifold and ω is a closed and non-degenerate 2-form on M . We'll say that (M, ω) is **exact** if ω is exact, and we'll call ω a **symplectic form**.

Remark. Non-degeneracy is an algebraic condition, while closedness is an analytic condition. It turns out that closedness is crucial for the development of the theory, and we will have several chances to appreciate this as we proceed.

Rephrasing Proposition 2 above, we already have our very first result² about symplectic manifolds:

Proposition 3

If (M, ω) is a compact symplectic manifold, then $H_{\text{dR}}^2(M) \neq 0$.

This means that unlike what happens with Riemannian metrics, for example, it is not true that every manifold has a symplectic form, even if one assumes from the start that it already has even dimension. With a minor modification of what we just did above, we obtain a slightly stronger topological restriction.

Exercise 10

- (a) Let (M^{2n}, ω) be a compact symplectic manifold. Show that $H_{\text{dR}}^{2k}(M) \neq 0$ for all $1 \leq k \leq n$. **Hint:** Show that $[\omega^{\wedge k}] \neq 0$ in $H_{\text{dR}}^{2k}(M)$ using Stokes' theorem.
- (b) Show that $H_{\text{dR}}^{2k}(\mathbb{S}^2 \times \mathbb{S}^4) = \mathbb{R}$ for $k = 0, 1, 2, 3$, but yet $\mathbb{S}^2 \times \mathbb{S}^4$ does not admit any symplectic form. **Hint:** To compute cohomologies, use the Künneth formula. Then assume by contradiction that $\mathbb{S}^2 \times \mathbb{S}^4$ has a symplectic form ω , and write $[\omega] = c[\pi^*\alpha]$, where $c \neq 0$, $[\alpha]$ generates $H_{\text{dR}}^2(\mathbb{S}^2)$, and $\pi: \mathbb{S}^2 \times \mathbb{S}^4 \rightarrow \mathbb{S}^2$ is the projection. What happens next?

Time for some examples.

Example 9

As mentioned in the beginning of this section, if (V, Ω) is a symplectic vector space, we may use that $T_x V \cong V$ to let $\omega_x = \Omega$, for all $x \in V$, so that (V, ω) becomes a symplectic manifold.

Example 10

Let $M \subseteq \mathbb{R}^3$ be an orientable surface, with unit normal field $N: M \rightarrow \mathbb{S}^2$. Define $\omega \in \Omega^2(M)$ by $\omega_x(v, w) = \langle N(x), v \times w \rangle$, where $\langle \cdot, \cdot \rangle$ is the standard Euclidean product and $\times$ is the associated cross product. Clearly ω is skew-symmetric, and it is closed by dimension reasons. Here's a proof for non-degeneracy: fix $x \in M$, $v \in T_x M$, and assume that $\omega_x(v, \cdot) = 0$. So $\langle v, w \times N(x) \rangle = 0$ for all $w \in T_x M$, but the values $N(x) \times w$ fill up $T_x M$, which says that $v \in (T_x M)^\perp$ — hence $v = 0$. Note that this ω is the standard area form of M .

The next example is more of an useful construction.

²Second result, if you consider that: ω being non-degenerate gives that $\omega^{\wedge n}$ is a volume form, so that M must be orientable. Explicitly: every symplectic manifold is orientable.

Example 11

If (M_1, ω_1) and (M_2, ω_2) are symplectic manifolds, so is $(M_1 \times M_2, \omega_1 \oplus \omega_2)$. Here, $\omega_1 \oplus \omega_2$ is, of course, a shorthand for $\pi_1^* \omega_1 + \pi_2^* \omega_2$, where for $i = 1, 2$, $\pi_i: M_1 \times M_2 \rightarrow M_i$ is the projection. Since d commutes with pull-backs, we directly have $d(\omega_1 \oplus \omega_2) = d\omega_1 \oplus d\omega_2 = 0$, and non-degeneracy follows from $\det(\omega_1 \oplus \omega_2) = \det \omega_1 \det \omega_2 \neq 0$, where this equality is to be understood locally and identifying each ω_i with its matrix relative to a coordinate system for M_i .

Exercise 11 (Hinting at future ideas)

Consider the prototype $(\mathbb{R}^{2n}, \omega_{2n})$. There are (at least?) two possible realizations of the torus $\mathbb{T}^{2n}$. The first, as $\mathbb{T}^{2n} = (\mathbb{S}^1 \times \mathbb{S}^1)^n$ — this inherits a product symplectic structure — and the second one as the quotient by a lattice $\mathbb{T}^{2n} = \mathbb{R}^{2n} / \mathbb{Z}^{2n}$ — this also has a symplectic structure, as ω_{2n} is translation-invariant, and thus survives in the quotient. What is the relation between the two structures?

The rest of this section is dedicated to one of the fundamental examples of a symplectic manifold — perhaps even more fundamental than $(\mathbb{R}^{2n}, \omega_{2n})$. Let Q be a smooth manifold (we're using Q instead of M because Q itself will not be the symplectic manifold we're after). Recall that its **cotangent bundle** T^*Q is the disjoint union of all the **cotangent spaces** T_x^*Q , as x ranges over Q , and we denote an element of T^*Q by a pair (x, p) , where $p \in T_x^*Q$. Local coordinates $(q^1, \dots, q^n)$ induce **cotangent coordinates** $(q^1, \dots, q^n, p_1, \dots, p_n)$ for T^*Q , via

$$(x, p) \mapsto \left(q^1(x), \dots, q^n(x), p \left(\frac{\partial}{\partial q^1} \Big|_x \right), \dots, p \left(\frac{\partial}{\partial q^n} \Big|_x \right) \right).$$

In classical mechanics, Q is taken to be the configuration space of a mechanical system, while T^*Q is seen as the phase space of positions and momenta. The tangent bundle TQ , in turn, would be seen as the phase space of positions and velocities. In some sense (which can be made precise), momenta and velocities are the same thing up to isomorphism (once a Riemannian metric or a hyperregular Lagrangian function has been introduced on Q). But T^*Q has some natural structure that TQ does not, and we'll exploit that.

Definition 8

Let Q be a smooth manifold. The **tautological form** (also called the **Liouville form** of T^*Q) is the 1-form $\lambda \in \Omega^1(T^*Q)$ defined by $\lambda_{(x,p)}(X) \doteq p(d\pi_{(x,p)}(X))$, for all $X \in T_{(x,p)}(T^*Q)$, where $\pi: T^*Q \rightarrow Q$ is the natural projection. The **canonical symplectic structure** of T^*Q is $\omega_{\text{can}} \doteq -d\lambda$.

Remark. Note that if $\pi: T^*Q \rightarrow Q$, then $d\pi_{(x,p)}: T_{(x,p)}(T^*Q) \rightarrow T_x Q$, for all elements $(x, p) \in T^*Q$. Some texts define ω_{can} to be $d\lambda$ without the negative sign. You should always pay attention to sign conventions throughout the literature.

Example 12

If Q is a smooth manifold, then $(T^*Q, \omega_{\text{can}})$ is a symplectic manifold. The only non-trivial thing to be checked is that ω_{can} is indeed non-degenerate. This can be done locally, noting that in terms of cotangent coordinates $(q^1, \dots, q^n, p_1, \dots, p_n)$, we have that

$$\lambda = \sum_{k=1}^n p_k dq^k \quad \text{and} \quad \omega_{\text{can}} = \sum_{k=1}^n dq^k \wedge dp_k.$$

To wit, we have that

$$\lambda = \sum_{k=1}^n \lambda \left(\frac{\partial}{\partial q^k} \right) dq^k + \sum_{k=1}^n \lambda \left(\frac{\partial}{\partial p_k} \right) dp_k,$$

but

$$\lambda_{(x,p)} \left(\frac{\partial}{\partial q^k} \Big|_{(x,p)} \right) = p \left(d\pi_{(x,p)} \left(\frac{\partial}{\partial q^k} \Big|_{(x,p)} \right) \right) = p \left(\frac{\partial}{\partial q^k} \Big|_x \right) = p_k(x, p)$$

and

$$\lambda_{(x,p)} \left(\frac{\partial}{\partial p_k} \Big|_{(x,p)} \right) = p \left(d\pi_{(x,p)} \left(\frac{\partial}{\partial p_k} \Big|_{(x,p)} \right) \right) = p(\mathbf{0}) = 0.$$

Exercise 12

Let Q be a smooth manifold. Here's another justificative for the name “tautological”, given to $\lambda \in \Omega^1(T^*Q)$, in two steps:

- Show that for every $\sigma \in \Omega^1(Q)$, we have that $\sigma^* \lambda = \sigma$.
- Conversely, show that if $\alpha \in \Omega^1(T^*Q)$ has the property that $\sigma^* \alpha = \sigma$ for all $\sigma \in \Omega^1(Q)$, then $\alpha = \lambda$.

Hint: Here, you should think of $\sigma \in \Omega^1(Q)$ as a map $\sigma: Q \rightarrow T^*Q$ (which just happens to be a section of $\pi: T^*Q \rightarrow Q$), so pull-backs make sense.

Here's one last example for this section:

Example 13 (Magnetic symplectic forms)

Let Q be a smooth manifold, and let $B \in \Omega^2(Q)$ be a closed. Define $\omega_B \in \Omega^2(T^*Q)$ by $\omega_B = \omega_{\text{can}} + \pi^* B$, where $\pi: T^*Q \rightarrow Q$ is the natural projection. Note that $d(\omega_B) = \pi^*(dB) = 0$ since d commutes with pull-backs and, conversely, that ω_B is not closed if B is not (because since π is a surjective submersion, π^* is injective). Moreover, taking cotangent coordinates and identifying B , ω , and ω_B with their

matrices relative to those coordinates, we have that

$$\omega_B = \begin{pmatrix} B & \text{Id}_n \\ -\text{Id}_n & 0 \end{pmatrix},$$

which has full-rank (no matter what B is), so ω_B is non-degenerate. We call ω_B the **magnetic symplectic form** associated to B . The reason for this name is that the equations which dictate the motion of a particle on Q subject to the action of a magnetic field B are the equations describing the flow of a certain vector field associated to a Hamiltonian function (i.e., the “total energy” classical observable) via ω_B .

Some general advice: you should keep all of those examples in mind as we proceed, as they’re so useful to gain intuition for new definitions and theorems.

2.2 Symplectomorphisms (*redux*)

What is the correct notion of equivalence for symplectic manifolds? Equivalently, what are the symplectic analogues of isometries? Or yet, what are the isomorphisms in the category of symplectic manifolds?

Definition 9

Let (M_1, ω_1) and (M_2, ω_2) be two symplectic manifolds. A **symplectomorphism** is a diffeomorphism $\varphi: M_1 \rightarrow M_2$ such that $\varphi^*\omega_2 = \omega_1$ or, more explicitly, such that

$$(\omega_1)_x(v, w) = (\omega_2)_{\varphi(x)}(d\varphi_x(v), d\varphi_x(w)),$$

for all $x \in M_1$, $v, w \in T_x M_1$. The **symplectomorphism group** of a symplectic manifold (M, ω) , denoted by $\text{Sp}(M, \omega)$, is the collection of all symplectomorphisms from M to itself.

As before, $\text{Sp}(M, \omega)$ is a group when equipped with the composition of functions. Of course that symplectomorphisms between symplectic vector spaces are symplectomorphisms in the above sense (the total derivative of a linear map is itself). But in general, symplectomorphisms may be highly non-linear.

Exercise 13

Consider in $\mathbb{R}^4$, with coordinates (x, y, u, v) , the standard symplectic form

$$\omega = dx \wedge du + dy \wedge dv.$$

Show that $\varphi: \mathbb{R}^4 \rightarrow \mathbb{R}^4$ given by

$$\varphi(x, y, u, v) = (x, y, u + e^x(\cos y + \sin y), v + e^x(\cos y - \sin y))$$

is a symplectomorphism. Note that identifying $\mathbb{R}^4$ with $\mathbb{C}^2$, we have that this map is just $\varphi(z, w) = (z, w + (1 - i)e^{iz})$. The complex Jacobian has determinant 1, so maybe it shouldn't surprise you that φ is a symplectomorphism.

Here's a friendlier example:

Example 14

Consider (S^2, ω) , where $\omega \in \Omega^2(S^2)$ is the standard area form, given (as before) by $\omega_x(v, w) = \langle x, v \times w \rangle$. Restrict a linear isometry $A \in \text{SO}(3)$ to the sphere, $A: S^2 \rightarrow S^2$. Then $A \in \text{Sp}(S^2, \omega)$. Indeed, we have that

$$\langle Ax, Av \times Aw \rangle = \langle Ax, A(v \times w) \rangle = \langle x, v \times w \rangle,$$

since $\det A = 1$ and A preserves $\langle \cdot, \cdot \rangle$. This shows that $\text{SO}(3) \subseteq \text{Sp}(S^2, \omega)$. We will see later (after Theorem 8, p. 29) that the inclusion is strict.

However, the symplectic manifold really rich in “symmetries” are cotangent bundles.

Example 15 (Fiberwise translations)

Let Q be a smooth manifold, and consider $(T^*Q, \omega_{\text{can}})$. Given $\sigma \in \Omega^1(Q)$, define $\tau_\sigma: T^*Q \rightarrow T^*Q$ by $\tau_\sigma(x, p) = (x, p + \sigma_x)$. When is τ_σ a symplectomorphism? A general rule of thumb is that when dealing with cotangent bundles, always look at what happens with the tautological form $\lambda \in \Omega^1(T^*Q)$ first — if λ is preserved, so will be ω_{can} . Note that if $\pi: T^*Q \rightarrow Q$ is the standard projection, then $\pi \circ \tau_\sigma = \pi$. Now take $(x, p) \in T^*Q$ and compute

$$\begin{aligned} (\tau_\sigma^* \lambda)_{(x,p)} &= \lambda_{(x,p+\sigma_x)} \circ d(\tau_\sigma)_{(x,p)} \\ &= (p + \sigma_x) \circ d\pi_{(x,p+\sigma_x)} \circ d(\tau_\sigma)_{(x,p)} \\ &= (p + \sigma_x) \circ d(\pi \circ \tau_\sigma)_{(x,p)} \\ &= (p + \sigma_x) \circ d\pi_{(x,p)} \\ &= p \circ d\pi_{(x,p)} + \sigma_x \circ d\pi_{(x,p)} \\ &= \lambda_{(x,p)} + (\pi^* \sigma)_{(x,p)}. \end{aligned}$$

So $\tau_\sigma^* \lambda = \lambda + \pi^* \sigma$ leads to $\tau_\sigma^* \omega_{\text{can}} = \omega_{\text{can}} - \pi^*(d\sigma)$, and we conclude that τ_σ is a symplectomorphism for $(T^*Q, \omega_{\text{can}})$ if and only if σ is closed. Note that τ_σ is indeed a diffeomorphism because $(\tau_\sigma)^{-1} = \tau_{-\sigma}$. If we choose a closed form $B \in \Omega^2(Q)$ to form the associated magnetic symplectic form ω_B , the same computation shows that τ_σ with σ closed is a symplectomorphism for (T^*Q, ω_B) as well.

You can explore this situation a bit further in the next exercise.

Exercise 14

Let Q be a smooth manifold and take two closed forms $B, B' \in \Omega^2(Q)$. Show that if $[B] = [B']$ in $H_{\text{dR}}^2(Q)$, then (T^*Q, ω_B) and $(T^*Q, \omega_{B'})$ are symplectomorphic under a fiberwise translation. In other words, a magnetic symplectic structure depends only on the cohomology class of the chosen magnetic form.

It turns out that cotangent bundles have another natural family of (this time non-linear) symplectomorphisms. It relies on the following concept:

Definition 10

Let Q_1 and Q_2 be two smooth manifold, and $f: Q_1 \rightarrow Q_2$ be a diffeomorphism. The **cotangent lift** of f is the map $\hat{f}: T^*Q_1 \rightarrow T^*Q_2$ defined by

$$\hat{f}(x, p) \doteq (f(x), p \circ (df_x)^{-1}).$$

At a first sight, the definition of a cotangent lift might seem a bit awkward, due to the presence of the *inverse* of the derivative of f . The reason for that is not only to make the compositions work. The next exercise should give you some more context.

Exercise 15 (The cotangent T^* functor)

Let $\text{Core}(\text{Man})$ be the category whose objects are smooth manifolds, and morphisms are diffeomorphisms^a. Define $T^*: \text{Core}(\text{Man}) \rightarrow \text{Core}(\text{Man})$ as follows:

- on objects: $Q \mapsto T^*Q$;
- on morphisms: $(h: Q_1 \rightarrow Q_2) \mapsto (T^*h: T^*Q_2 \rightarrow T^*Q_1)$, where T^*h is defined by $(T^*h)(y, p) \doteq (h^{-1}(y), (dh_{h^{-1}(y)})^*(p))$.

Show that T^* is a contravariant (faithful) functor. With the notation of the previous definition, we have that $\hat{f} = T^*(f^{-1})$, to make $\hat{}$ a covariant functor.

^aThe *core* of a category $\mathcal{C}$ is the category (in fact, groupoid) $\text{Core}(\mathcal{C})$ whose objects are the same as the objects in $\mathcal{C}$, but the morphisms in $\text{Core}(\mathcal{C})$ are only the morphisms in $\mathcal{C}$ which are isomorphisms. One could (should?) expect $\text{Core}: \text{Cat} \rightarrow \text{Grpd}$ to be a functor too, where Cat and Grpd are exactly what you think. Is it, though? If yes, how? If not, why?

Proposition 4

Let Q_1 and Q_2 be two smooth manifolds, with tautological forms λ_1 and λ_2 on their cotangent bundles. If $f: Q_1 \rightarrow Q_2$ is a diffeomorphism, then $(\hat{f})^*\lambda_2 = \lambda_1$, and thus $\hat{f}: T^*Q_1 \rightarrow T^*Q_2$ is a symplectomorphism.

Proof: Cotangent lifts are natural, in the sense that the diagram

$$\begin{array}{ccc} T^*Q_1 & \xrightarrow{\hat{f}} & T^*Q_2 \\ \pi \downarrow & & \downarrow \pi \\ Q_1 & \xrightarrow{f} & Q_2 \end{array}$$

commutes. Now, take $(x, p) \in T^*Q_1$ and repeatedly use the chain rule to compute:

$$\begin{aligned} ((\hat{f})^*\lambda_2)_{(x,p)} &= (\lambda_2)_{\hat{f}(x,p)} \circ d\hat{f}_{(x,p)} \\ &= (p \circ (df_x)^{-1}) \circ d\pi_{\hat{f}(x,p)} \circ d\hat{f}_{(x,p)} \\ &= p \circ (df_x)^{-1} \circ d(\pi \circ \hat{f})_{(x,p)} \\ &= p \circ (df_x)^{-1} \circ d(f \circ \pi)_{(x,p)} \\ &= p \circ (df_x)^{-1} \circ df_x \circ d\pi_{(x,p)} \\ &= p \circ d\pi_{(x,p)} \\ &= (\lambda_1)_{(x,p)}. \end{aligned}$$

□

Corollary 2

Let Q_1 and Q_2 be smooth manifolds equipped with closed 2-forms $B_1 \in \Omega^2(Q_1)$ and $B_2 \in \Omega^2(Q_2)$, and consider the magnetic cotangent bundles (T^*Q_1, ω_{B_1}) and (T^*Q_2, ω_{B_2}) . If $f: Q_1 \rightarrow Q_2$ is a diffeomorphism, then $\hat{f}: T^*Q_1 \rightarrow T^*Q_2$ is a symplectomorphism if and only if $f^*B_2 = B_1$.

Now, cotangent lifts are a very special type of symplectomorphism, because not only they preserve the canonical symplectic form, they also preserve the tautological form, which is a much stronger condition to require. It turns out that this condition is characteristic of cotangent lifts, in the sense of the following:

Theorem 6

Let Q_1 and Q_2 be smooth manifolds, with tautological forms λ_1 and λ_2 on their cotangent bundles. If $\varphi: T^*Q_1 \rightarrow T^*Q_2$ is a diffeomorphism such that $\varphi^*\lambda_2 = \lambda_1$, then there is a diffeomorphism $f: Q_1 \rightarrow Q_2$ such that $\varphi = \hat{f}$.

The proof essentially consists on showing that the condition $\varphi^*\lambda_2 = \lambda_1$ implies that φ takes fibers to fibers, that is, if $x \in Q_1$, then the image $\varphi[T_x^*Q_1]$ actually equals $T_y^*Q_2$ for some (unique) $y \in Q_2$ — one then sets $f(x) \doteq y$ and checks directly that $\hat{f} = \varphi$, as desired. See [14], [15], or even [43] for more details.

2.3 Hamiltonian fields

On Riemannian manifolds, smooth functions have gradients. Namely, gradients are vector fields equivalent to derivatives under the metric. The key fact here is not that Riemannian metrics are positive-definite, but instead that they are non-degenerate. Well, symplectic forms are non-degenerate. So it should not be surprising that we will also have symplectic analogues of gradient fields.

Definition 11

Let (M, ω) be a symplectic manifold, and $H: M \rightarrow \mathbb{R}$ be a smooth function. The **Hamiltonian field** (or **symplectic gradient**) of H is the vector field $X_H \in \mathfrak{X}(M)$ characterized by the relation $\omega(X_H, \cdot) = dH$.

Remark. Again there's an ambiguity regarding sign conventions here: some texts replace dH with $-dH$, or $\omega(X_H, \cdot)$ with $\omega(\cdot, X_H)$ in the above definition. Our convention should agree with most of the Physics literature.

As usual, let's start with examples.

Example 16

Let $M \subseteq \mathbb{R}^3$ be an orientable surface with unit normal field $N: M \rightarrow S^2$ and area form $\omega \in \Omega^2(M)$. Let $H: M \rightarrow \mathbb{R}$ be smooth. The standard Euclidean inner product of $\mathbb{R}^3$ restricts to a Riemannian metric on M , and so we may consider the gradient field $\text{grad } H \in \mathfrak{X}(M)$. What is the relation between $\text{grad } H$ and X_H ? Using the definitions and invariance under cyclic permutations of the triple product, we see that for each $x \in M$ and $v \in T_x M$, we have

$$\langle (\text{grad } H)_x, v \rangle = dH_x(v) = \omega_x(X_H|_x, v) = \langle N(x), X_H|_x \times v \rangle = \langle v, N(x) \times X_H|_x \rangle,$$

so that $\text{grad } H = N \times X_H$. With this in place, we use double cross product formulas to obtain

$$N \times \text{grad } H = N \times (N \times X_H) = \langle N, X_H \rangle N - \langle N, N \rangle X_H = -X_H,$$

and so $X_H = \text{grad } H \times N$.

Example 17

Let (V, ω) be a symplectic vector space, and let $H \in V^*$. Here are two ways to see X_H , since $dH_x = H$ for all $x \in V$:

- Identify $V \cong V^*$ using ω . So $X_H = H$, simply because $H = \omega(X_H, \cdot)$.
- Identify $V \cong V^*$ using a Darboux basis $\mathcal{B} = (e_1, \dots, e_n, f_1, \dots, f_n)$ (that is, send $\mathcal{B}$ to its dual basis and extend linearly). Also consider the linear complex structure J determined by $\mathcal{B}$, that is, defined by $J(e_i) = f_i$ and

$J(f_i) = -e_i$. Writing H and X_H as linear combinations of elements in $\mathcal{B}$ and evaluating both sides of $H = \omega(X_H, \cdot)$ at e_j and f_j , it will follow that $X_H = -JH$.

The next example has such a fundamental importance in mechanics, that we'll phrase it as a theorem.

Theorem 7 (Hamilton's equations)

Let Q be a smooth manifold, consider its cotangent bundle $(T^*Q, \omega_{\text{can}})$, and let $H: T^*Q \rightarrow \mathbb{R}$ be a smooth function. In T^*Q , take local cotangent coordinates $(q^1, \dots, q^n, p_1, \dots, p_n)$. Then:

- (i) the Hamiltonian field X_H of H is given by

$$X_H = \sum_{k=1}^n \left(\frac{\partial H}{\partial p_k} \frac{\partial}{\partial q^k} - \frac{\partial H}{\partial q^k} \frac{\partial}{\partial p_k} \right)$$

- (ii) if $\gamma: I \rightarrow T^*Q$ is an integral curve of the field X_H , described in coordinates by $\gamma(t) = (q^1(t), \dots, q^n(t), p_1(t), \dots, p_n(t))$, it satisfies **Hamilton's equations**:

$$\begin{cases} \frac{dq^k}{dt}(t) = \frac{\partial H}{\partial p_k}(\gamma(t)) \\ \frac{dp_k}{dt}(t) = -\frac{\partial H}{\partial q^k}(\gamma(t)). \end{cases}$$

Proof: Item (b) follows trivially from (a). So let's check (a). Write

$$X_H = \sum_{k=1}^n \left(a^k \frac{\partial}{\partial q^k} + b_k \frac{\partial}{\partial p_k} \right),$$

where the a^k and b^k are to be determined. We have that

$$\frac{\partial H}{\partial q^k} = dH \left(\frac{\partial}{\partial q^k} \right) = \omega_{\text{can}} \left(X_H, \frac{\partial}{\partial q^k} \right) = -b_k$$

and

$$\frac{\partial H}{\partial p_k} = dH \left(\frac{\partial}{\partial p_k} \right) = \omega_{\text{can}} \left(X_H, \frac{\partial}{\partial p_k} \right) = a^k,$$

since for all choices of i and j we have that

$$\omega_{\text{can}} \left(\frac{\partial}{\partial q^i}, \frac{\partial}{\partial q^j} \right) = 0, \quad \omega_{\text{can}} \left(\frac{\partial}{\partial p_i}, \frac{\partial}{\partial p_j} \right) = 0 \quad \text{and} \quad \omega_{\text{can}} \left(\frac{\partial}{\partial q^i}, \frac{\partial}{\partial p_j} \right) = \delta_i^j.$$

□

Remark. We'll see in the next section that every symplectic manifold admits coordinates for which the symplectic form looks like the canonical symplectic prototype. This means that any local computations we make, for instance, in a cotangent bundle, will be locally true for any symplectic manifold.

Exercise 16

Let Q be a smooth manifold, $B \in \Omega^2(Q)$ a closed form, and consider the magnetic cotangent bundle (T^*Q, ω_B) . Let $H: T^*Q \rightarrow \mathbb{R}$ be a smooth function and denote by X_H^B its Hamiltonian field computed with ω_B . Show that relative to cotangent coordinates $(q^1, \dots, q^n, p_1, \dots, p_n)$, we have that:

(a) the expression for X_H^B is

$$X_H^B = \sum_{k=1}^n \left(\frac{\partial H}{\partial p_k} \frac{\partial}{\partial q^k} + \left(-\frac{\partial H}{\partial q^k} + \sum_{j=1}^n B_{jk} \frac{\partial H}{\partial p_j} \right) \frac{\partial}{\partial p_k} \right),$$

where the functions B_{jk} are the components of B relative to the coordinate system $(q^1, \dots, q^n)$ on Q .

(b) if $\gamma: I \rightarrow T^*Q$ is an integral curve of the field X_H^B , described in coordinates by $\gamma(t) = (q^1(t), \dots, q^n(t), p_1(t), \dots, p_n(t))$, it satisfies the magnetic version of Hamilton's equations:

$$\begin{cases} \frac{dq^k}{dt}(t) = \frac{\partial H}{\partial p_k}(\gamma(t)) \\ \frac{dp_k}{dt}(t) = -\frac{\partial H}{\partial q^k}(\gamma(t)) + \sum_{j=1}^n B_{jk}(\gamma(t)) \frac{\partial H}{\partial p_j}(\gamma(t)). \end{cases}$$

(c) if $\mathbb{R} \times T^*Q$ is the *extended phase space* equipped with the (non-symplectic, for dimensional reasons) 2-form $\eta = \omega_{\text{can}} + dH \wedge dt$ (we identify ω_{can} and dH with their pull-backs under $\mathbb{R} \times T^*Q \rightarrow T^*Q$), then X_H^B is the unique field on T^*Q such that $\eta(X_H^B + \partial_t, \cdot) = 0$.

Hint: Item (c) is independent from (a) and (b) and, again, (b) follows trivially from (a) (why?).

Exercise 17 (Variational characterizations of Hamilton's equations)

Let Q be a smooth manifold and $H: T^*Q \rightarrow \mathbb{R}$ be a smooth function. Define the **action integral** of H by sending every curve $\gamma = (x, p): [a, b] \rightarrow T^*Q$ to the number

$$\mathcal{A}^H[\gamma] = \int_a^b p(t)(\dot{x}(t)) - H(x(t), p(t)) dt.$$

Show that critical points of $\mathcal{A}^H$ appear, in cotangent coordinates, as solutions to Hamilton's equations. **Hint:** see [38] if needed.

Hamilton's equations have far-reaching consequences in Physics and Geometry. We'll illustrate one situation from each area in what follows. Having some more vocabulary before proceeding will be helpful.

Definition 12

An **autonomous Hamiltonian system** is a triple (M, ω, H) , where (M, ω) is a symplectic manifold, and $H: M \rightarrow \mathbb{R}$ is smooth.

- The function H is called the **Hamiltonian** of the system;
- The flow of the vector field X_H is called the **Hamiltonian flow** of the system;
- For each $e \in \mathbb{R}$, the inverse image $\Sigma_e \doteq H^{-1}(e)$ is called an **energy level** (or **energy surface**) of the system.
- A smooth function $f: M \rightarrow \mathbb{R}$ is called a **constant of motion** if it is constant along the integral curves of X_H .

Example 18 (The Harmonic Oscillator)

Consider a unidimensional simple **harmonic oscillator**:

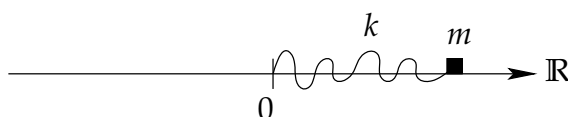


Figure 1: Simple Harmonic Oscillator.

Suppose that the mass of the particle attached to the string is $m > 0$, that the elasticity constant of the string is $k > 0$, and that the particle moves without friction. The configuration space here is $Q = \mathbb{R}$, and the phase space of positions and momenta is $T^*\mathbb{R}$ — so we'll use the position x and the momentum p of the particle as coordinates (in particular, the symplectic form to be used will be simply $dx \wedge dp$). By Hooke's Law, the force needed to extend or compress the spring by some distance x is $F(x) = -kx$. Moreover, since the total energy is the sum of the kinetic and potential energies, and velocity and momentum are linked through $p = mv$, the Hamiltonian function controlling the time evolution of this system is $H: T^*\mathbb{R} \cong \mathbb{R}^2 \rightarrow \mathbb{R}$ given by

$$H(x, p) = \frac{p^2}{2m} + \frac{kx^2}{2}.$$

Thus we have a Hamiltonian system $(T^*\mathbb{R}, dx \wedge dp, H)$. In matrix form, Hamilton's equations become just

$$\begin{pmatrix} \dot{x}(t) \\ \dot{p}(t) \end{pmatrix} = \begin{pmatrix} 0 & 1/m \\ -k & 0 \end{pmatrix} \begin{pmatrix} x(t) \\ p(t) \end{pmatrix}$$

Let's denote by A the constant matrix of coefficients above. A straightforward computation (for instance, using that $A^2 = (-k/m)\text{Id}_2$ and a series expansion) gives that

$$\exp(tA) = \cos\left(t\sqrt{\frac{k}{m}}\right) \text{Id}_2 + \sin\left(t\sqrt{\frac{k}{m}}\right) \sqrt{\frac{m}{k}} A.$$

This motivates us to define the **angular frequency** of the system by $\omega \doteq \sqrt{k/m}$. Thus, it follows that the solution (x, p) to Hamilton's equations with initial condition (x_0, p_0) is given by

$$x(t) = x_0 \cos(\omega t) + \frac{p_0}{\sqrt{km}} \sin(\omega t) \quad \text{and} \quad p(t) = -x_0 \sqrt{km} \sin(\omega t) + p_0 \cos(\omega t).$$

Observe that if we set $H_0 \doteq H(x_0, y_0)$, then $H(x(t), y(t)) = H_0$ for all $t \in \mathbb{R}$. One can see that $H(x, p) = H_0$ is the equation of an ellipse centered at the origin, in the xp -plane, which is then parametrized by $t \mapsto (x(t), p(t))$. Hence, we may draw the phase portrait for the harmonic oscillator:

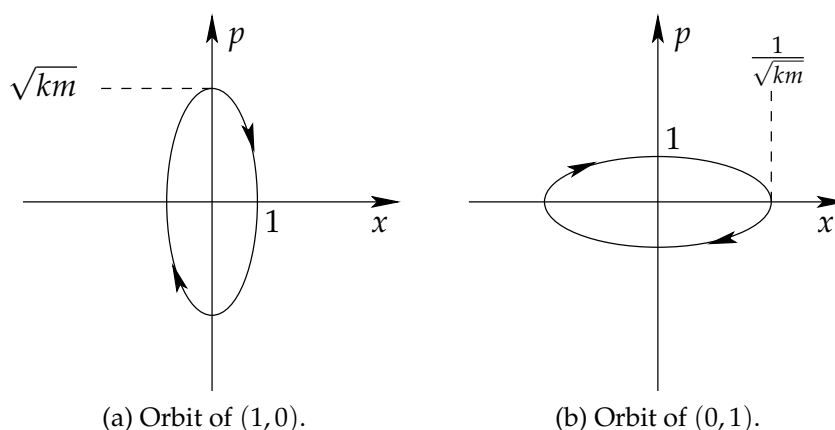


Figure 2: Periodic orbits with time interval $0 \leq t \leq 2\pi/\omega$.

This phenomenon of conservation of energy is something very general: for every Hamiltonian system, the Hamiltonian itself is a constant of motion — simply because skew-symmetry of the symplectic form implies that $\omega(X_H, X_H) = 0$.

Exercise 18 (“Chain” conservation of energy)

Let's expand on the last comment made on the above example. Let (M, ω, H) be a Hamiltonian system. Show that any function of the Hamiltonian itself is also a constant of motion. Namely, if $\varphi: \mathbb{R} \rightarrow \mathbb{R}$ is any smooth function, show that $X_{\varphi \circ H} = (\varphi' \circ H)X_H$ and conclude that $\varphi \circ H$ is a constant of motion.

Example 19 (Cogeodesic flow)

Let (Q, g) be a Riemannian manifold. The isomorphism $TQ \cong T^*Q$ induced by g allows us to define a fiber metric on T^*Q , which we'll also denote by g (note that this is not the same as saying that we have a Riemannian metric on the manifold T^*Q). Consider the Hamiltonian $H: T^*Q \rightarrow \mathbb{R}$ given by

$$H(x, p) \doteq \frac{1}{2} g_x(p, p).$$

Using the canonical symplectic form ω_{can} on T^*Q , we may consider the associated Hamiltonian field $X_H \in \mathfrak{X}(T^*Q)$. The corresponding Hamiltonian flow is called the **cogeodesic flow** on T^*Q , because its integral curves project to geodesics on (Q, g) . Let's understand how this happens, checking that Hamilton's equations are equivalent to the geodesic equations. So, consider tangent and cotangent coordinates for TQ and T^*Q , related under g . Namely, we have

$$(q^1, \dots, q^n, v^1, \dots, v^n, p_1, \dots, p_n),$$

where

$$p_j = \sum_{i=1}^n g_{ij} v^i \quad \text{and} \quad v^j = \sum_{i=1}^n g^{ij} p_i.$$

Here, $(g_{ij})_{i,j=1}^n$ is the matrix of g relative to the chosen coordinate system for Q , and $(g^{ij})_{i,j=1}^n$ is its inverse matrix — such inverse matrix, incidentally, represents the fiber metric induced on T^*Q , so that

$$H = \frac{1}{2} \sum_{i,j=1}^n g^{ij} p_i p_j.$$

The first of Hamilton's equations becomes just $\dot{q}^k = v^k$, as expected. The second equation, in the presence of the first one, will become the geodesic equation. Here's how this happens. First, we compute the derivative

$$\frac{\partial H}{\partial q^k} = \frac{1}{2} \sum_{i,j=1}^n \frac{\partial g^{ij}}{\partial q^k} p_i p_j = -\frac{1}{2} \sum_{i,j,r,s=1}^n g^{ir} \frac{\partial g_{rs}}{\partial q^k} g^{sj} p_i p_j = -\frac{1}{2} \sum_{r,s=1}^n \frac{\partial g_{rs}}{\partial q^k} v^r v^s,$$

and rename back $r, s \rightarrow i, j$. Now,

$$\left(\sum_{i=1}^n g_{ik} v^i \right)' = \frac{1}{2} \sum_{i,j=1}^n \frac{\partial g_{ij}}{\partial q^k} v^i v^j$$

becomes

$$\sum_{i=1}^n g_{ik} \dot{v}^i + \sum_{i,j=1}^n \frac{\partial g_{ik}}{\partial q^j} v^i \dot{q}^j = \frac{1}{2} \sum_{i,j=1}^n \frac{\partial g_{ij}}{\partial q^k} v^i v^j.$$

Splitting the middle term as a sum (since it's symmetric in i and j) and writing everything in terms of derivatives of the q^i 's yields

$$\sum_{i=1}^n g_{ik} \ddot{q}^i + \frac{1}{2} \sum_{i,j=1}^n \left(\frac{\partial g_{ik}}{\partial q^j} + \frac{\partial g_{jk}}{\partial q^i} - \frac{\partial g_{ij}}{\partial q^k} \right) \dot{q}^i \dot{q}^j = 0.$$

Multiply everything by g^{kr} , sum over r , and rename $r \rightarrow k$ to get

$$\ddot{q}^k + \sum_{i,j=1}^n \Gamma_{ij}^k \dot{q}^i \dot{q}^j = 0,$$

where the

$$\Gamma_{ij}^k \doteq \frac{1}{2} \sum_{r=1}^n g^{kr} \left(\frac{\partial g_{ir}}{\partial q^j} + \frac{\partial g_{jr}}{\partial q^i} - \frac{\partial g_{ij}}{\partial q^r} \right)$$

are the **Christoffel symbols** describing the **Levi-Civita connection** of (Q, g) in coordinates. Thus we have obtained the coordinate description of the geodesic equation

$$\frac{D\dot{\gamma}}{dt}(t) = 0,$$

as claimed.

Remark. Using the isomorphism $TQ \cong T^*Q$ given by g to convert X_H into a vector field on TQ , one obtains the standard **geodesic field** $G \in \mathfrak{X}(TQ)$, characterized by

$$G_{(x,v)} = \frac{d}{dt} \Big|_{t=0} (\gamma_{x,v}(t), \dot{\gamma}_{x,v}(t)),$$

where $\gamma_{x,v}$ is the unique maximal geodesic on (Q, g) with $\gamma_{x,v}(0) = x$ and $\dot{\gamma}_{x,v}(0) = v$.

Exercise 19 (Challenge #2)

Let (Q, g) be a Riemannian manifold and $B \in \Omega^2(Q)$ be closed. Consider the (skew-adjoint) bundle morphism $F: TQ \rightarrow TQ$, called the **Lorentz force** associated with B , characterized by the relation $g_x(F_x(v), w) = B_x(v, w)$, for all $x \in Q$ and $v, w \in T_x Q$. Mimic what was done in the previous example and show that integral curves of the Hamiltonian field $X_H^B \in \mathfrak{X}(T^*Q)$ (computed with ω_B) of $H: T^*Q \rightarrow \mathbb{R}$ given by

$$H(x, p) = \frac{1}{2} g_x(p, p)$$

project onto solutions of the **magnetic geodesic equation**

$$\frac{D\dot{\gamma}}{dt}(t) = F_{\gamma(t)}(\dot{\gamma}(t))$$

on Q . **Hint:** You already did some of the hard work on Exercise 16 (p. 24).

Remark. Recall that the geodesics of (Q, g) are also characterized as critical points of the **energy functional** $\mathcal{E}$, defined as

$$\mathcal{E}[\gamma] = \frac{1}{2} \int_a^b g_{\gamma(t)}(\dot{\gamma}(t), \dot{\gamma}(t)) dt,$$

where $\gamma: [a, b] \rightarrow M$ is smooth (actually, one can get away with assuming that γ is just absolutely continuous — smoothness of minimizing geodesics then follows). If we introduce a magnetic form $B \in \Omega^2(Q)$, and B is exact, that is, $B = dA$ for some $A \in \Omega^1(Q)$ (called **magnetic potential**), then it turns out that magnetic geodesics appear as critical points of the **magnetic energy functional** $\mathcal{E}_A$, defined as

$$\mathcal{E}_A[\gamma] = \frac{1}{2} \int_a^b g_{\gamma(t)}(\dot{\gamma}(t), \dot{\gamma}(t)) dt + \int_a^b A_{\gamma(t)}(\dot{\gamma}(t)) dt.$$

Note that the choice of magnetic potential A is not unique, as one could replace A with $A + df$, for any smooth $f: Q \rightarrow \mathbb{R}$. Despite this gauge freedom, there is still some physical interpretation for A , justifying its name. Related to this, there's the so-called **Aharonov-Bohm experiment**. You can read more about it, from a mathematical perspective, in [41]. For a crash review on variational calculus on manifolds, in case you want to try and prove the claims made here, see [38].

We continue to explore the importance of flows.

Theorem 8

Let (M, ω) be a compact symplectic manifold and $H: M \rightarrow \mathbb{R}$ be a smooth function. Then each stage $\Phi_{t, X_H}: M \rightarrow M$ of the Hamiltonian flow is a symplectomorphism of (M, ω) .

Proof: Since M is compact, the field X_H is complete. Noting that the relation characterizing X_H can be rewritten as $\iota_{X_H}\omega = dH$, where ι stands for the interior derivative (or contraction), we may use Cartan's homotopy formula³:

$$\mathcal{L}_{X_H}\omega = \iota_{X_H}(d\omega) + d(\iota_{X_H}\omega) = \iota_{X_H}(0) + d(dH) = 0 + 0 = 0.$$

□

Remark.

- This theorem is, in some sense, the true reason why we require symplectic forms to be closed. Namely, for any $H: M \rightarrow \mathbb{R}$, we have that $\mathcal{L}_{X_H}\omega = \iota_{X_H}(d\omega)$, so requiring that the flows of all X_H preserve ω directly leads to $d\omega = 0$.

³If M is any smooth manifold, $X \in \mathfrak{X}(M)$, and $\alpha \in \Omega^k(M)$, then $\mathcal{L}_X\alpha = \iota_X d\alpha + d(\iota_X\alpha)$. This formula says that the operator $\mathcal{L}_X$ is nullhomotopic, in the following sense: the contraction ι_X defines a (co)chain homotopy between $\mathcal{L}_X$ and the zero map, seen as (co)chain self-maps of the de Rham complex $(\Omega^\bullet(M), d)$ of M . Thus $\mathcal{L}_X$ induces the zero map in cohomology.

- We can also show now that the inclusion $\mathrm{SO}(3) \subseteq \mathrm{Sp}(\mathbb{S}^2, \omega)$ seen in Example 14 (p. 19) is strict: let φ be the time 1 of the Hamiltonian flow of some $H: \mathbb{S}^2 \rightarrow \mathbb{R}$ which vanishes on some proper open subset of $\mathbb{S}^2$. Then φ will act there as the identity, but will do something else on the complement (hence φ is non-linear).

This raises a pertinent question: if (M, ω) is a symplectic manifold, and $X \in \mathfrak{X}(M)$ is any vector field, when does the flow of X consist of symplectomorphisms of (M, ω) ? Again, by Cartan's homotopy formula we have that $\mathcal{L}_X \omega = d(\iota_X \omega)$, as $d\omega = 0$. So if $\iota_X \omega \in \Omega^1(M)$ is closed (which in particular happens if it is exact), ok.

Definition 13

Let (M, ω) be a symplectic manifold and $X \in \mathfrak{X}(M)$. We say that X is:

- **Hamiltonian**, if there is a smooth function $H: M \rightarrow \mathbb{R}$ such that $X = X_H$ (such H is then called an **energy function** for X);
- **symplectic** (or, **locally Hamiltonian**) if $\iota_X \omega$ is closed.

We'll write $\mathcal{S}(M, \omega)$ and $\mathcal{H}(M, \omega)$ for the algebras of symplectic and Hamiltonian fields on (M, ω) .

Clearly every Hamiltonian field is symplectic, but for the converse there is the obvious unique obstruction: $H_{\mathrm{dR}}^1(M)$. Thus, finding symplectic fields which are not Hamiltonian is, in spirit, the same as knowing how to find closed 1-forms which are not exact. The next exercise illustrates this.

Exercise 20

Consider the cylinder $\mathbb{S}^1 \times \mathbb{R} \subseteq \mathbb{R}^3$ equipped with its standard symplectic form $\omega = x \, dy \wedge dz + y \, dz \wedge dx$. Consider the field $\partial/\partial z \in \mathfrak{X}(\mathbb{S}^1 \times \mathbb{R})$.

(a) Show that $\partial/\partial z$ is symplectic.

(b) Compute $\int_{\mathbb{S}^1} \iota_{\partial/\partial z} \omega$ and conclude that $\partial/\partial z$ is not Hamiltonian.

You can also explore more relations between symplectic and Hamiltonian fields here:

Exercise 21

Let (M, ω) be a symplectic manifold. Show that:

(a) for any $X, Y, Z \in \mathcal{S}(M, \omega)$, the Jacobi-type identity

$$\omega(X, [Y, Z]) + \omega(Y, [Z, X]) + \omega(Z, [X, Y]) = 0$$

holds.

- (b) $[\mathcal{S}(M, \omega), \mathcal{S}(M, \omega)] \subseteq \mathcal{H}(M, \omega)$ (in words: the Lie bracket of two symplectic fields is Hamiltonian).

And we'll conclude this section with an instructive exercise:

Exercise 22

Let Q be a smooth manifold, $X \in \mathfrak{X}(Q)$ be a complete vector field, and consider its flow $\Phi_{t,X}: Q \rightarrow Q$, for each $t \in \mathbb{R}$. Equip T^*Q with ω_{can} . Show that:

- (a) there is a complete vector field $\widehat{X} \in \mathfrak{X}(T^*Q)$ such that $\widehat{\Phi_{t,X}} = \Phi_{t,\widehat{X}}$ for all $t \in \mathbb{R}$, where $\widehat{}$ on the left side denotes a cotangent lift, as usual. **Hint:** use the functorial behavior of $\widehat{}$ to show that $\widehat{\Phi_{t,X}}$ defines a flow on T^*Q .
- (b) the field $\widehat{X}$ found in (a) is Hamiltonian and find an energy function for it. **Hint:** Use Cartan's homotopy formula.

Exercise 23

Let Q be a smooth manifold, $f: Q \rightarrow \mathbb{R}$ be a smooth function, and consider the cotangent bundle $(T^*Q, \omega_{\text{can}})$. If $\pi: T^*Q \rightarrow Q$ is the natural projection, we have the pull-back $\pi^*f: T^*Q \rightarrow \mathbb{R}$. Show that the time 1 of the flow of X_{π^*f} maps the zero section $Q \hookrightarrow T^*Q$ to the graph of $-df$. **Hint:** show that X_{π^*f} is vertical and compute $\Phi_{t,X_{\pi^*f}}(x, 0)$ explicitly, for instance, using cotangent coordinates.

2.4 Submanifolds and local forms

This situation will mimic what happened with subspaces of a symplectic vector space (V, Ω) . As usual, we'll smoothly apply all the linear algebra concepts previously seen to all tangent spaces to a manifold. For this, let's register our first definition:

Definition 14

Let (M, ω) be a symplectic manifold and $N \subseteq M$ a submanifold. We'll say that N is **symplectic**, **isotropic**, **coisotropic** or **Lagrangian** if, for every $x \in N$, the tangent space $T_x N$ is a subspace of $(T_x M, \omega_x)$ of the corresponding type.

Remark. More abstractly, one can say that an immersion $\iota: N \hookrightarrow M$ is Lagrangian if $\iota^*\omega = 0$ and $2 \dim N = \dim M$. This is particularly useful to show that certain submanifolds are Lagrangian, when employing convenient identifications.

Two immediate examples: any curve in a symplectic manifold is isotropic, and any hypersurface is coisotropic, for dimensional reasons (of course, these are not all

the examples). Still, the main focus will thus be on Lagrangian submanifolds. Mid-dimensional submanifolds appear naturally in several situations, while symplectic submanifolds generally turn out to be studied as symplectic manifolds on their own right. Before going to examples, the following result should further emphasize why should one care about Lagrangian submanifolds.

Proposition 5

Let (M, ω, H) be a Hamiltonian system, and assume that $N \subseteq M$ is a coisotropic submanifold which is contained in some energy level of H . Then N is invariant under the Hamiltonian flow. More precisely, $\Phi_{t, X_H}[N] = N$ for all t for which the flow is defined.

Proof: It suffices to show that for each $x \in N$, we have $X_H|_x \in T_x N$ — the result then follows by integration. Since we assume that N is coisotropic, we have by definition that $(T_x N)^{\omega_x} \subseteq T_x N$. So, let's show that $X_H|_x \in (T_x N)^{\omega_x}$. For any $v \in T_x N$, we have that

$$\omega_x(X_H|_x, v) = dH_x(v) = 0,$$

using that N is contained in an energy level of H . This concludes the proof. $\square$

Remark. So, coisotropic submanifolds contained in energy levels of a Hamiltonian system are invariant under the dynamics. This holds *a fortiori* for Lagrangian submanifolds. However, in dynamics one often cares about minimal invariant sets. But “Lagrangian” not only means “maximal isotropic”, but also “minimal coisotropic”.

With this in place, here's our first non-trivial example.

Example 20

Let (M_1, ω_1) and (M_2, ω_2) be symplectic manifolds, and let $\varphi: M_1 \rightarrow M_2$ be smooth. Let's see that the graph

$$\text{gr}(\varphi) = \{(x, \varphi(x)) \mid x \in M_1\}$$

is a Lagrangian submanifold of $(M_1 \times M_2, \omega_1 \oplus (-\omega_2))$ if and only if we have $\dim M_1 = \dim M_2$ and $\varphi^* \omega_2 = \omega_1$ (which then implies that φ is a local diffeomorphism). First, note that for every $x \in M_1$ we have that

$$T_{(x, \varphi(x))} \text{gr}(\varphi) = \{(v, d\varphi_x(v)) \mid v \in T_x M_1\}.$$

With this, if $\tilde{v} = (v, d\varphi_x(v))$, $\tilde{w} = (w, d\varphi_x(w)) \in T_{(x, \varphi(x))} \text{gr}(\varphi)$, we may compute

$$\begin{aligned} (\omega_1 \oplus (-\omega_2))_{(x, \varphi(x))}(\tilde{v}, \tilde{w}) &= (\omega_1)_x(v, w) - (\omega_2)_{\varphi(x)}(d\varphi_x(v), d\varphi_x(w)) \\ &= (\omega_1)_x(v, w) - (\varphi^* \omega_2)_x(v, w), \end{aligned}$$

which says that $T_{(x, \varphi(x))} \text{gr}(\varphi)$ is isotropic if and only if $\varphi^* \omega_2 = \omega_1$. As for the

dimension, we have that $\text{gr}(\varphi)$ is diffeomorphic to M_1 , so that

$$\dim \text{gr}(\varphi) = \frac{1}{2}(\dim M_1 + \dim M_2)$$

if and only if $\dim M_1 = \dim M_2$.

Remark. Taking $(M, \omega) \doteq (M_1, \omega_1) = (M_2, \omega_2)$ and $\varphi = \text{Id}_M$ in the above example, we see that the **diagonal of** M , $\Delta(M) \doteq \{(x, x) \mid x \in M\}$, is a Lagrangian submanifold of $(M \times M, \omega \oplus (-\omega))$. In this case, the projection $\Delta(M) \rightarrow M$ is a symplectomorphism.

As usual, cotangent bundles will serve as rich source of examples.

Example 21 (Lagrangian submanifolds of cotangent bundles)

There are three types of submanifolds here which deserve attention:

- (1) Fibers. That is, any fixed cotangent space T_x^*Q . We consider the inclusion $\iota: T_x^*Q \hookrightarrow T^*Q$, given by $\iota(p) = (x, p)$. We have that

$$(\iota^*\lambda)_p = \lambda_{(x,p)} \circ d\iota_p = p \circ d\pi_{(x,p)} \circ d\iota_p = p \circ d(\pi \circ \iota)_p = p \circ 0 = 0,$$

since $(\pi \circ \iota)(p) = x$ is a constant. Hence $\iota^*\lambda = 0$ leads to $\iota^*\omega_{\text{can}} = 0$. If $B \in \Omega^2(Q)$ is closed, the fact that $\pi \circ \iota$ again implies that $\iota^*\pi^*B = 0$. So T_x^*Q is a Lagrangian submanifold of both $(T^*Q, \omega_{\text{can}})$ and (T^*Q, ω_B) , as it already has the correct dimension.

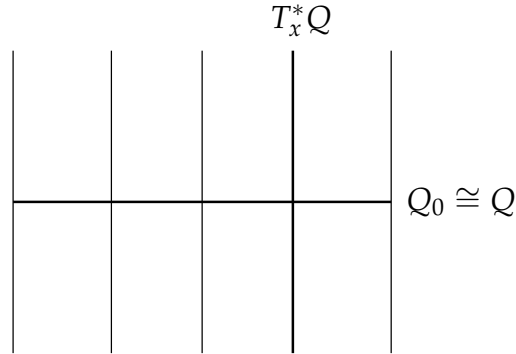


Figure 3: Embedding Q in T^*Q as the zero section.

- (2) Ranges of 1-forms. Let $\sigma \in \Omega^1(Q)$ and let $Q_\sigma \doteq \text{Ran}(\sigma) \subseteq T^*Q$. More precisely, Q_σ is the range of σ regarded as a map $\sigma: Q \rightarrow T^*Q$. Note that Q_σ is automatically mid-dimensional. The inclusion $Q_\sigma \hookrightarrow T^*Q$ is identified with σ itself. By Exercise 12 (p. 17), we have that $\sigma^*\lambda = \sigma$, so that applying $-d$ we have $\sigma^*\omega_{\text{can}} = -d\sigma$. Moreover, since $\pi \circ \sigma = \text{Id}_Q$, we have that if $B \in \Omega^2(Q)$ is closed, then $\sigma^*\pi^*B = B$. So, Q_σ is a Lagrangian submanifold of $(T^*Q, \omega_{\text{can}})$ if and only if σ is closed, and it is a Lagrangian submanifold of (T^*Q, ω_B) if and only if $d\sigma = B$ (i.e., σ is a magnetic potential for B). Note

that $Q_0 = Q$ is always a Lagrangian submanifold of $(T^*Q, \omega_{\text{can}})$, but never for (T^*Q, ω_B) . Broadly speaking, the introduction of a magnetic term eliminates the choice of a distinguished Lagrangian section of T^*Q — this is akin to what happens when one deals with affine spaces in terms of vector spaces (there is no canonical choice of origin).

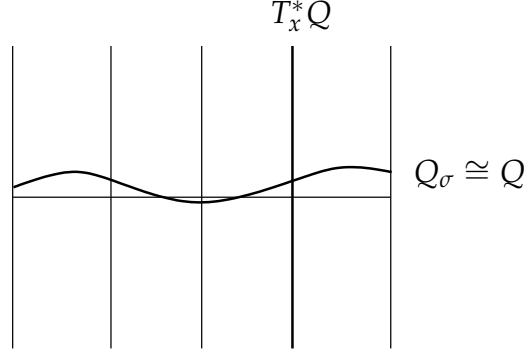


Figure 4: Embedding Q in T^*Q as the graph Q_σ .

- (3) Conormal bundles: Let $P \subseteq Q$ be a submanifold. The **conormal bundle** of P (relative to Q) is the vector bundle $\pi^*: \nu^*P \rightarrow P$ whose fibers are the annihilators of the tangent spaces to P . Namely, for all $x \in P$, we have

$$\nu_x^*P \doteq \text{Ann}(T_xP) = \{p \in T_x^*Q \mid p|_{T_xP} = 0\}.$$

We have that

$$\dim \nu^*P = \dim P + (\dim Q - \dim P) = \dim Q = \frac{1}{2} \dim T^*Q,$$

so the dimension of ν^*P is already correct. Let $\iota: \nu^*P \hookrightarrow T^*Q$ be the inclusion, and write $\pi: T^*Q \rightarrow Q$ for the projection. Note that $\pi^*: \nu^*P \rightarrow P$ is valued on P , so its derivative is valued on the tangent spaces to P , which are annihilated by elements in the fibers of ν^*P . So the relation $\pi \circ \iota = \pi^*$ gives that

$$(\iota^*\lambda)_{(x,p)} = p \circ d\pi^*_{(x,p)} = 0,$$

provided $(x, p) \in \nu^*P$. We conclude that ν^*P is a Lagrangian submanifold of $(T^*Q, \omega_{\text{can}})$. One can also easily see that ν^*P will be a Lagrangian submanifold of (T^*Q, ω_B) , for closed $B \in \Omega^2(Q)$, if and only if B pulls back to the zero 2-form on P .

The main idea we'll discuss here is called **Moser's trick**. It is crucial to the study of deformation of symplectic structures, to establishing local form theorems for submanifolds of symplectic manifolds and, in particular, has Darboux's theorem as a consequence. For that, fix a manifold M and recall that an **isotopy** is a family $(\varphi_t)_{t \in \mathbb{R}}$ of diffeomorphisms of M with φ_0 equal to the identity map, and such that the joint map $\mathbb{R} \times M \ni (t, x) \mapsto \varphi_t(x) \in M$ is smooth. Note that an isotopy does not necessarily satisfy the group-like property that flows of vector fields satisfy, but for each such iso-

topy there is an associated **time-dependent vector field** on M , that is, a family $(X_t)_{t \in \mathbb{R}}$ of vector fields on M such that the joint map $\mathbb{R} \times M \ni (t, x) \mapsto X_{t,x} \in T_x M \subseteq TM$ is smooth. Namely, the isotopy defines the time-dependent vector field with the relation

$$\frac{d}{dt} \varphi_t(x) = X_{t, \varphi_t(x)}.$$

When M is compact, a time-dependent vector field gives rise to an isotopy, via “integration”.

Lemma 2 (Moser’s trick)

Let M be a compact manifold with symplectic forms ω_0 and ω_1 . Assume that $(\omega_t)_{t \in [0,1]}$ is a smooth family of symplectic forms joining ω_0 and ω_1 , such that the cohomology class $[\omega_t]$ is independent of t . Then there is an isotopy $(\varphi_t)_{t \in \mathbb{R}}$ such that $\varphi_t^* \omega_t = \omega_0$ for all $t \in [0, 1]$ — in particular, $\varphi_1^* \omega_1 = \omega_0$.

Proof: The idea essentially consists in reverse engineering what $(\varphi_t)_{t \in \mathbb{R}}$ must be. Since the cohomology class $[\omega_t]$ in $H_{\text{dR}}^2(M)$ is independent of t , we may get a smooth family of 1-forms $(\alpha_t)_{t \in [0,1]}$ such that

$$\frac{d}{dt} \omega_t = d\alpha_t, \quad \text{for all } t \in [0, 1].$$

Now, if the isotopy exists and we denote by $(X_t)_{t \in \mathbb{R}}$ its associated time-dependent vector field, we may take the derivative of $\varphi_t^* \omega_t$ using a “product rule” and Cartan’s homotopy formula to obtain

$$\begin{aligned} 0 &= \frac{d}{dt} \varphi_t^* \omega_t = \varphi_t^* \mathcal{L}_{X_t} \omega_t + \varphi_t^* \left(\frac{d}{dt} \omega_t \right) \\ &= \varphi_t^* \left(d(\iota_{X_t} \omega_t) + \iota_{X_t} (d\omega_t) + \frac{d}{dt} \omega_t \right) \\ &= \varphi_t^* d(\iota_{X_t} \omega_t + \alpha_t). \end{aligned}$$

So, provided we can solve **Moser’s equation** $\iota_{X_t} \omega_t + \alpha_t = 0$ for X_t , we can extend it from $[0, 1]$ to $\mathbb{R}$ and then integrate it to the desired isotopy, by compactness of M . But we can clearly solve Moser’s equation: it suffices to take, for each $(t, x) \in [0, 1] \times M$, the vector $X_{t,x}$ to be the image of $-\alpha_{t,x}$ under the isomorphism $T_x^* M \rightarrow T_x M$ induced by ω_t . Hence $\varphi_t^* \omega_t$ is independent of t , and it equals its value for $t = 0$, namely, $\varphi_0^* \omega_0 = (\text{Id}_M)^* \omega_0 = \omega_0$. $\square$

Compactness of M is a somewhat strong assumption above. But we can work around it, to some extent. One useful tool we’ll need for this is a more powerful version of Poincaré’s Lemma:

Lemma 3 (Poincaré’s Lemma — relative version)

Let M be a smooth manifold, $N \subseteq M$ be a compact submanifold, and $\alpha \in \Omega^k(M)$ a closed k -form such that $\iota^* \alpha = 0$, where $\iota: N \hookrightarrow M$ is the inclusion. Then there is a (tubular) neighborhood of N in M and a $(k-1)$ -form β defined on such neigh-

neighborhood such that $\alpha = d\beta$ holds there, with $\beta_x = 0$ for all $x \in N$.

Remark. If $\alpha_x = 0$ for all $x \in N$, one can also arrange for the first order partial derivatives of β , relative to any adapted coordinate system to N , to also vanish along N . Note that when N is a single point, this is the usual Poincaré’s Lemma. See [26] or [43] for a proof.

With this in place:

Theorem 9 (Moser stability)

Let M be a smooth manifold, $N \subseteq M$ be a compact submanifold, and assume that $\omega_0, \omega_1 \in \Omega^2(M)$ are two symplectic forms such that $(\omega_0)_x = (\omega_1)_x$ for all $x \in N$. Then there are (tubular) neighborhoods U_0 and U_1 of N in M and a diffeomorphism $\varphi: U_0 \rightarrow U_1$ such that $\varphi^*\omega_1 = \omega_0$ and $\varphi(x) = x$ for all $x \in N$.

Proof: (Sketch) By Poincaré’s Lemma there is a 1-form β defined on a tubular neighborhood U_0 of N such that $\omega_1 - \omega_0 = d\beta$ holds there. Define $\omega_t \doteq \omega_0 + t d\beta$ on U_0 , for all $t \in [0, 1]$. By compactness of N and $[0, 1]$, reducing U_0 if necessary, we may assume that all ω_t are symplectic. Now, since

$$\frac{d}{dt}\omega_t = d\beta$$

does not depend on t , by reducing U_0 further, Moser’s trick gives an isotopy φ_t of U_0 such that $\varphi_t^*\omega_t = \omega_0$ for all $t \in [0, 1]$. Let $\varphi = \varphi_1$ and $U_1 = \varphi[U_0]$ — we may arrange for $\varphi(x) = x$ for all $x \in N$ by doing a choice of gauge $\beta \mapsto \beta + df$, with f smooth on U_0 , according to the remark after Poincaré’s Lemma above, before running Moser’s argument. $\square$

Theorem 10 (Darboux)

Let (M, ω) be a symplectic manifold. Around each point in (M, ω) there are coordinates $(x^1, \dots, x^n, y_1, \dots, y_n)$ for which ω is expressed as

$$\omega = \sum_{k=1}^n dx^k \wedge dy_k.$$

We’ll call such coordinates **Darboux coordinates** for (M, ω) .

Proof: Let N be a singleton in Theorem 9 above. More precisely, fix $x_0 \in M$ and pick coordinates $(\tilde{x}^1, \dots, \tilde{x}^n, \tilde{y}_1, \dots, \tilde{y}_n)$ around x_0 such that

$$\left(\frac{\partial}{\partial \tilde{x}^1} \Big|_{x_0}, \dots, \frac{\partial}{\partial \tilde{x}^n} \Big|_{x_0}, \frac{\partial}{\partial \tilde{y}_1} \Big|_{x_0}, \dots, \frac{\partial}{\partial \tilde{y}_n} \Big|_{x_0} \right)$$

is a Darboux basis for $(T_{x_0}M, \omega_{x_0})$. On this coordinate domain, consider ω itself and

$$\tilde{\omega} = \sum_{k=1}^n d\tilde{x}^k \wedge d\tilde{y}_k.$$

Since $\omega_{x_0} = \tilde{\omega}_{x_0}$ by construction, reducing the coordinate domain if necessary, by Moser we obtain a diffeomorphism φ of this domain for which $\varphi^*\tilde{\omega} = \omega$. Then we set $x^i \doteq \tilde{x}^i \circ \varphi$ and $y_j \doteq \tilde{y}_j \circ \varphi$ for all $i, j = 1, \dots, n$. By definition of pull-back, $(x^1, \dots, x^n, y_1, \dots, y_n)$ are the desired Darboux coordinates. $\square$

Note that in the same way that every symplectic vector space (V, Ω) is symplectomorphic to the prototype $(\mathbb{R}^{2n}, \Omega_{2n})$ with the correct dimension (due to the existence of Darboux bases), this theorem is saying that every point in a symplectic manifold (M, ω) has a neighborhood symplectomorphic to an open subset of $(\mathbb{R}^{2n}, \omega_{2n})$ (due to the existence of Darboux coordinates). In particular, we know that cotangent coordinates for cotangent bundles are Darboux coordinates, so pretty much everything we have established for a cotangent bundle holds locally in an arbitrary symplectic manifold — for instance, Theorem 7 (p. 23) tells us how to describe Hamiltonian flows in Darboux coordinates.

More importantly, this theorem says that there are **no local invariants in symplectic geometry**, such as curvature. We know that curvatures come from a connection on the manifold. But even that becomes slightly tricky, as there is no “Koszul formula” in this setting.

Proposition 6 ([34])

Let M be a smooth manifold, and $\omega \in \Omega^2(M)$ be a non-degenerate 2-form. There is a torsionfree connection on M which parallelizes ω if and only if ω is closed. And if this is the case, there are infinitely many such connections.

Proof: If ω is closed, then (M, ω) is symplectic. Cover M with Darboux coordinates, on each such domain define a local connection declaring all the Christoffel symbols to vanish, and then “glue” everything into a single connection using a partition of unity. Conversely, if there is such a torsionfree connection ∇ parallelizing ω , recall that (up to a constant multiple depending on conventions), $d\omega$ is the alternator of $\nabla\omega$, which is zero by assumption. $\square$

We’ll conclude this section by stating local form results for symplectic and Lagrangian submanifolds. They require two brief concepts:

- If M is a smooth manifold and N is a submanifold, we can define the **normal bundle** of N relative to M even without a Riemannian metric on M . Namely, since for all $x \in N$ we may regard $T_x N$ as a subspace of $T_x M$, we can just consider the quotient $\nu_x N = T_x M / T_x N$. This of course defines $\nu N \rightarrow N$, we have an isomorphism $TN \oplus \nu N \cong TM|_N$, and when M does have a Riemannian metric, we have $\nu N \cong TN^\perp$. If we have a smooth map $f: M_1 \rightarrow M_2$ between two

manifolds which restricts to a smooth map $N_1 \rightarrow N_2$ between submanifolds, for each $x \in N_1$ we have a “derivative” $\nu f_x: \nu_x N_1 \rightarrow \nu_{f(x)} N_2$ given by

$$\nu f_x(v + T_x N_1) = df_x(v) + T_{f(x)} N_2.$$

If f is a diffeomorphism, this bundle morphism $\nu N_1 \rightarrow \nu N_2$ is an isomorphism and, conversely, given an isomorphism $\nu N_1 \rightarrow \nu N_2$ covering a diffeomorphism $N_1 \rightarrow N_2$, such diffeomorphism extends to a diffeomorphism on a tubular neighborhood of N_1 which induces the given bundle isomorphism via derivative (the gist of it is to take Riemannian metrics on M_1 and M_2 and use normal exponential maps).

- A **symplectic vector bundle**, denoted by $(E, \omega) \rightarrow M$, is nothing more than vector bundle $E \rightarrow M$ equipped with (smoothly varying) linear symplectic forms ω_x on each fiber E_x . Note that the requirement $d\omega = 0$ is meaningless in this setting. We already have plenty of examples: the tangent bundle of a symplectic manifold, every symplectic vector space is a symplectic vector bundle over a point, the direct sum of any vector bundle with its dual, normal bundles (in the above sense) of symplectic submanifolds of symplectic manifolds, and so on. In particular, a **morphism of symplectic vector bundles** is exactly what you’re thinking — a bundle morphism preserving the symplectic structures.

Theorem 11 (Weinstein’s symplectic normal form theorem)

Let (M_1, ω_1) and (M_2, ω_2) be symplectic manifolds, with symplectic submanifolds N_1 and N_2 , and $\nu N_1 \rightarrow \nu N_2$ be an isomorphism of symplectic vector bundles covering a symplectomorphism $N_0 \rightarrow N_1$. Then such symplectomorphism extends to a symplectomorphism between tubular neighborhoods of N_0 and N_1 which induces the given isomorphism via derivative.

To motivate the next result, consider a symplectic manifold (M, ω) and a Lagrangian submanifold $Q \subseteq M$. We know that for each point $x \in Q$, $T_x Q \cong (T_x Q)^{\omega_x}$ holds, and the isomorphism $T_x M \cong T_x^* M$ induced by ω_x directly restricts to an isomorphism $(T_x Q)^{\omega_x} \cong \text{Ann}(T_x Q)$. Dualizing, it follows that $\nu_x Q \cong T_x^* Q$. Gathering those isomorphisms as x ranges over Q , we have an isomorphism $\nu Q \cong T^* Q$ of symplectic vector bundles.

Theorem 12 (Weinstein’s Lagrangian neighborhood theorem)

Let (M, ω) be a symplectic manifold and $Q \subseteq M$ be a Lagrangian submanifold. Then there is a tubular neighborhood U of Q in M and a neighborhood U_0 of Q as the zero section in $T^* Q$, and a diffeomorphism $\varphi: U_0 \rightarrow U$ such that $\varphi(x, 0) = x$ for all $x \in Q$ and $\varphi^*(\omega) = \omega_{\text{can}}$.

For proofs of the last two results, see [28]. Let’s explore one last consequence of this. We have seen after Example 20 (p. 32) that if (M, ω) is a symplectic manifold,

then the diagonal $\Delta(M)$ is a Lagrangian submanifold of $(M \times M, \omega \oplus (-\omega))$, but now we know that the projection $\Delta(M) \rightarrow M$ actually extends to a symplectomorphism φ_Δ in some neighborhood of $\Delta(M)$, taking values in a neighborhood of M as the zero section of T^*M :

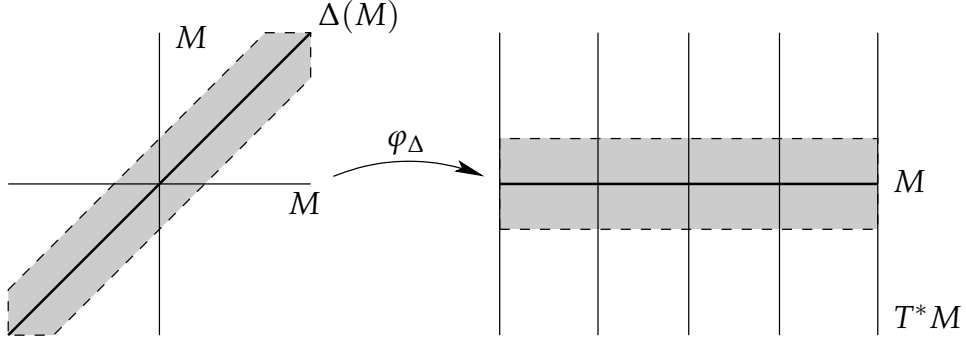


Figure 5: Illustrating the symplectomorphism φ_Δ .

Now, if $\varphi \in \text{Sp}(M, \omega)$, we know by Example 20 (p. 32), again, that the graph $\text{gr}(\varphi)$ is a Lagrangian submanifold of $(M \times M, \omega \oplus (-\omega))$. If φ is $\mathcal{C}^1$ -close to the identity, then $\text{gr}(\varphi)$ is close enough to $\Delta(M)$, and so the image $L \doteq \varphi_\Delta[\text{gr}(\varphi)] \subseteq T^*M$ is close enough to the zero section, thus being equal to M_σ for some closed $\sigma \in \Omega^1(M)$, as in Example 21 (p. 33).

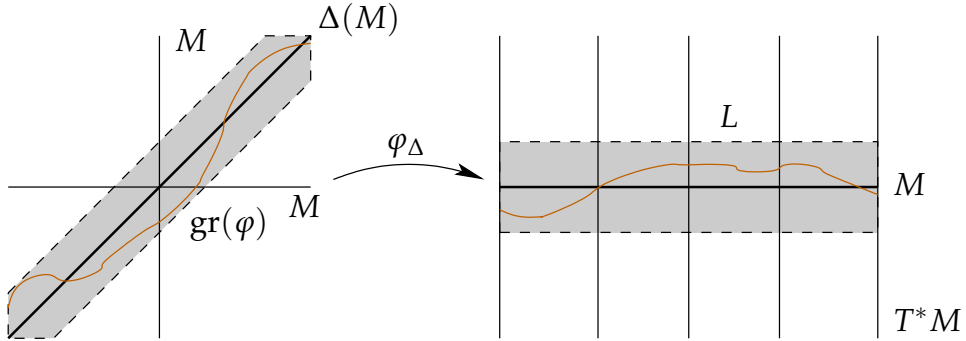


Figure 6: Sending $\text{gr}(\varphi)$ to T^*M as $L = M_\sigma$.

In particular, we see that fixed points of φ are in bijective correspondence with zeros of σ .

Corollary 3

Let (M, ω) be a compact symplectic manifold with $H_{\text{dR}}^1(M) = 0$ and $\varphi \in \text{Sp}(M, \omega)$ be $\mathcal{C}^1$ -close to the identity function. Then φ has at least two fixed points.

Proof: As per the above discussion, consider the closed 1-form σ corresponding to the graph $\text{gr}(\varphi)$. Since $H_{\text{dR}}^1(M) = 0$, write $\sigma = df$ for some $f: M \rightarrow \mathbb{R}$. By compactness of M , f achieves its maximum and minimum values at some points in M — those correspond to zeros of σ and hence to fixed points of φ . $\square$

3 Hamiltonian Actions

3.1 Poisson Manifolds

Let (M, ω, H) be a Hamiltonian system. One usually would like to find constants of motion for the system to understand it better. We know that $f \in \mathcal{C}^\infty(M)$ is a constant of motion if f is constant along integral curves of X_H , and this is equivalent to $df(X_H)$ vanishing. By definition of Hamiltonian field, this is the same as requiring $\omega(X_f, X_H)$ to vanish.

Definition 15

Let (M, ω) be a symplectic manifold. The **Poisson bracket associated with ω** is the operation $\{\cdot, \cdot\}_\omega: \mathcal{C}^\infty(M) \times \mathcal{C}^\infty(M) \rightarrow \mathcal{C}^\infty(M)$ given by $\{f, g\}_\omega \doteq \omega(X_f, X_g)$.

Example 22 ($\{\cdot, \cdot\}_\omega$ in coordinates)

Let (M, ω) be a symplectic manifold, take $f, g \in \mathcal{C}^\infty(M)$, and (x^k, y_k) be Darboux coordinates for M . By Theorem 7 (p. 23), we know that

$$X_f = \sum_{k=1}^n \left(\frac{\partial f}{\partial y_k} \frac{\partial}{\partial x^k} - \frac{\partial f}{\partial x^k} \frac{\partial}{\partial y_k} \right) \quad \text{and} \quad X_g = \sum_{k=1}^n \left(\frac{\partial g}{\partial y_k} \frac{\partial}{\partial x^k} - \frac{\partial g}{\partial x^k} \frac{\partial}{\partial y_k} \right).$$

Hence, we obtain that

$$\{f, g\}_\omega = \sum_{k=1}^n \left(\frac{\partial f}{\partial x^k} \frac{\partial g}{\partial y_k} - \frac{\partial g}{\partial x^k} \frac{\partial f}{\partial y_k} \right),$$

by applying df to X_g . The Hamiltonian fields and Poisson brackets of coordinate functions also make sense where they are defined. So, in particular, we have that

$$X_{x^k} = -\frac{\partial}{\partial y_k} \quad \text{and} \quad X_{y_k} = \frac{\partial}{\partial x^k},$$

as well as the relations

$$\{x^i, x^j\}_\omega = 0, \quad \{x^i, y_j\}_\omega = \delta_j^i, \quad \text{and} \quad \{y_i, y_j\}_\omega = 0,$$

for all $i, j = 1, \dots, n$. Hamilton's equations, in turn, read simply

$$\dot{x}^k = \{x^k, H\}_\omega \quad \text{and} \quad \dot{y}^k = \{y^k, H\}_\omega.$$

In the same way that the symplectic form determines the Poisson bracket, the converse actually holds.

Exercise 24

Let M be a smooth manifold and $\omega, \omega_1, \omega_2 \in \Omega^2(M)$ be symplectic forms.

- (a) Show that if $\{\cdot, \cdot\}_{\omega_1} = \{\cdot, \cdot\}_{\omega_2}$, then $\omega_1 = \omega_2$. **Hint:** Show that for all functions $f, g \in \mathcal{C}^\infty(M)$, we'll have that $X_f^{\omega_1}(g) = X_f^{\omega_2}(g)$, so that $X_f^{\omega_1} = X_f^{\omega_2}$. Go from there.
- (b) Conclude that $\varphi \in \text{Sp}(M, \omega)$ if and only if $\{f \circ \varphi, g \circ \varphi\}_\omega = \{f, g\}_\omega \circ \varphi$ for all $f, g \in \mathcal{C}^\infty(M)$.

However, Poisson brackets are more than just a convenient device to describe Hamiltonian dynamics. It will turn out that the Poisson bracket alone is enough to do Hamiltonian dynamics, even when a symplectic form is not present! Of course, a few definitions are still required to make sense of this, but we'll continue to develop more intuition with examples and basic results first.

Example 23

Let $M \subseteq \mathbb{R}^3$ be an orientable surface, with unit normal field $N: M \rightarrow \mathbb{S}^2$, and standard area form $\omega \in \Omega^2(M)$. We have seen in Example 16 (p. 22) that for any $f \in \mathcal{C}^\infty(M)$, we have that $X_f = \text{grad } f \times N$. With this, we may compute the Poisson bracket on (M, ω) using double cross product identities:

$$\begin{aligned} \{f, g\}_\omega &= \langle N, X_f \times X_g \rangle \\ &= \langle N, (\text{grad } f \times N) \times (\text{grad } g \times N) \rangle \\ &= \langle N \times (\text{grad } f \times N), \text{grad } g \times N \rangle \\ &= \langle \langle N, N \rangle \text{grad } f - \langle N, \text{grad } f \rangle N, \text{grad } g \times N \rangle \\ &= \langle \text{grad } f, \text{grad } g \times N \rangle \\ &= \langle N, \text{grad } f \times \text{grad } g \rangle. \end{aligned}$$

In particular, since $\text{grad } f \times \text{grad } g$ is always normal to the surface, we conclude that $\{f, g\}_\omega = 0$ if and only if $\text{grad } f$ and $\text{grad } g$ are always proportional.

Exercise 25

Let (M, ω) be a symplectic manifold. Show that for any $f, g \in \mathcal{C}^\infty(M)$, we have $X_{\{f, g\}_\omega} = -[X_f, X_g]$. This negative sign is just collateral damage from our choices of sign conventions^a. **Hint:** Use Cartan's homotopy formula together with the general identity $\iota_{[X, Y]} = \mathcal{L}_X \circ \iota_Y - \iota_X \circ \mathcal{L}_Y$ for interior derivatives.

^aPerhaps the "correct" definition of our Poisson bracket should have had the opposite sign instead. Arguably, it wouldn't look so natural. But we would have to pay the price somewhere. To try to avoid this sort of thing, some texts (like [28], for example), even define the Lie bracket with the opposite sign!

Next, let's finally justify the name “bracket” (which should have made you think of a Lie bracket):

Theorem 13

Let (M, ω) be a symplectic manifold. Then the Poisson bracket $\{\cdot, \cdot\}_\omega$ is skew-symmetric, $\mathbb{R}$ -bilinear, satisfies the **Jacobi identity**

$$\{f, \{g, h\}_\omega\}_\omega + \{g, \{h, f\}_\omega\}_\omega + \{h, \{f, g\}_\omega\}_\omega = 0,$$

and the **Leibniz rule**

$$\{f, gh\}_\omega = g\{f, h\}_\omega + h\{f, g\}_\omega,$$

for all $f, g, h \in \mathcal{C}^\infty(M)$.

Proof: Skew-symmetry and bilinearity over real scalars are obvious. For the Jacobi identity, we will use for the second time in this text the condition $d\omega = 0$. First, we have that

$$\begin{aligned} X_f(\omega(X_g, X_h)) &= d\{g, h\}_\omega(X_f) = \omega(X_{\{g, h\}_\omega}, X_f) \\ &= \{\{g, h\}_\omega, f\}_\omega = -\{f, \{g, h\}_\omega\}_\omega. \end{aligned}$$

Next, compute (with the result from the previous exercise) that

$$\omega([X_g, X_h], X_f) = \omega(-X_{\{g, h\}_\omega}, X_f) = \omega(X_f, X_{\{g, h\}_\omega}) = \{f, \{g, h\}_\omega\}_\omega.$$

Finally, we use that

$$\begin{aligned} 0 = d\omega(X_f, X_g, X_h) &= X_f(\omega(X_g, X_h)) - X_g(\omega(X_f, X_h)) + X_h(\omega(X_f, X_g)) \\ &\quad - \omega([X_f, X_g], X_h) + \omega([X_f, X_h], X_g) - \omega([X_g, X_h], X_f). \end{aligned}$$

This becomes exactly the Jacobi identity for $\{\cdot, \cdot\}_\omega$ (in fact, $d\omega = 0$ is equivalent to the Jacobi identity). $\square$

Corollary 4

Let (M, ω, H) be a Hamiltonian system, and $f, g \in \mathcal{C}^\infty(M)$ be constants of motion. Then $\{f, g\}_\omega$ is also a constant of motion.

Being a Lie bracket, it is natural to expect the Poisson bracket to have relations with integrability and submanifolds. The next two results should illustrate this.

Proposition 7

Let (M, ω) be a symplectic manifold, and $N \subseteq M$ be a submanifold. We define the **annihilator ideal** of N to be $\mathcal{C}^\infty(M)_N = \{f \in \mathcal{C}^\infty(M) \mid f|_N = 0\}$. Then, the following are equivalent:

- (i) N is coisotropic.

- (ii) For each $f \in \mathcal{C}^\infty(M)_N$, $\mathbf{X}_f$ is tangent to N along its points.
- (iii) $\mathcal{C}^\infty(M)_N$ is closed under $\{\cdot, \cdot\}_\omega$.

Proof: First, note that if $f \in \mathcal{C}^\infty(M)_N$, then for each $x \in N$ and $v \in T_x N$, we have that $0 = df_x(v) = \omega_x(\mathbf{X}_f|_x, v)$, so $\mathbf{X}_f|_x \in (T_x N)^{\omega_x}$. In fact, we also have that such values fill the space: $(T_x N)^{\omega_x} = \{\mathbf{X}_f|_x \mid f \in \mathcal{C}^\infty(M)_N\}$. With this in place, we move on to the implications.

- (i) $\implies$ (ii): Clear from the above as $(T_x N)^{\omega_x} \subseteq T_x N$ for all $x \in N$ by (i).
- (ii) $\implies$ (iii): If $f, g \in \mathcal{C}^\infty(M)_N$, then $\{f, g\}_\omega|_N = df(\mathbf{X}_g)|_N = 0$ because for all $x \in N$, df_x annihilates $T_x N$ and $\mathbf{X}_g|_x \in T_x N$ by (ii).
- (iii) $\implies$ (i): Take $x \in N$, and let's show that $(T_x N)^{\omega_x} \subseteq (T_x N^{\omega_x})^{\omega_x}$. For that, we may just use Hamiltonian fields of functions coming from $\mathcal{C}^\infty(M)_N$. By (iii), we have that $\omega_x(\mathbf{X}_f|_x, \mathbf{X}_g|_x) = \{f, g\}_\omega(x) = 0$ if $f, g \in \mathcal{C}^\infty(M)_N$.

□

Corollary 5

Let (M, ω) be a symplectic manifold and $f_1, \dots, f_k \in \mathcal{C}^\infty(M)$ be functions in **involution**, that is, $\{f_i, f_j\}_\omega = 0$ for all $i, j = 1, \dots, k$. Gather them into a map $f: M \rightarrow \mathbb{R}^k$ and assume it is a submersion. Then the fibers of f are coisotropic submanifolds of M .

Proof: Since f is a submersion, the fibers of f are submanifolds of M , and the Hamiltonian fields $\{\mathbf{X}_{f_i}|_x\}_{i=1}^k$ form a basis for $(T_x N)^{\omega_x}$, for all $x \in N$. The involution assumption says that these vectors are “orthogonal”, and so all of them are, in fact, in $T_x N$. □

Remark. If $2k = n$, we obtain a foliation by Lagrangian submanifolds, and this is the usual setup to study **integrable systems**: a Hamiltonian system (M, ω, H) with dimension $2n$ admitting n constants of motion $f_1 = H, f_2, \dots, f_n$ which are in involution, whose differentials are linearly independent in some open dense subset of M .

Exercise 26

The converse to the previous corollary holds: if (M, ω) is a symplectic manifold and $N \subseteq M$ is a codimension k coisotropic submanifold, then every point in N has a neighborhood U in M and a submersion $f: U \rightarrow \mathbb{R}^k$ whose coordinate functions are in involution, and $U \cap N = f^{-1}(0)$. **Hint:** Use induction on k and see [30].

Poisson brackets can also be related to a mild generalization of Hamiltonian fields.

Exercise 27 (Challenge #3)

Let (M, ω) be a symplectic manifold. For each $\alpha \in \Omega^1(M)$, let $Z_\alpha \in \mathfrak{X}(M)$ be the unique field characterized by the relation $\omega(Z_\alpha, \cdot) = \alpha$. Now, define a bracket $[[\cdot, \cdot]]: \Omega^1(M) \times \Omega^1(M) \rightarrow \Omega^1(M)$ by the relation $Z_{[[\alpha, \beta]]} \doteq -[Z_\alpha, Z_\beta]$. Fix 1-forms $\alpha, \beta \in \Omega^1(M)$. Show that:

- (a) $[[\alpha, \beta]] = \mathcal{L}_{Z_\beta} \alpha - \mathcal{L}_{Z_\alpha} \beta - d(\omega(Z_\alpha, Z_\beta))$;
- (b) If α and β are closed, so is $[[\alpha, \beta]]$.
- (c) $d\{f, g\}_\omega = [[df, dg]]$.
- (d) $[[\cdot, \cdot]]$ is a Lie bracket on $\Omega^1(M)$.

If Q is a smooth manifold and $B \in \Omega^2(Q)$ is closed, consider (T^*Q, ω_B) .

- (e) Show that for each $H \in \mathcal{C}^\infty(T^*Q)$, $X_H^B = X_H - Z_{\iota_{X_H}(\pi^*B)}$.
- (f) Conclude that for $f, g \in \mathcal{C}^\infty(T^*Q)$, $\{f, g\}_{\omega_B} = \{f, g\}_{\omega_{\text{can}}} - (\pi^*B)(X_f, X_g)$.

Hint: Expect the condition $d\omega = 0$ to appear somewhere between (a) and (d).

Remark. If $(M, -d\lambda)$ is an exact symplectic manifold, $E \doteq Z_\lambda$ is called the **Euler field**. Its flow (compute it in Darboux coordinates and you'll see the reason for the name "Euler" field) is an useful tool in the proof of Theorem 6, mentioned in p. 21.

It will be useful later in this chapter to have an abstraction of all we discussed so far. The inspiration comes from Theorem 13 (p. 42).

Definition 16 (Poisson geometry)

A **Poisson manifold** is a pair $(P, \{\cdot, \cdot\})$, where P is a smooth manifold and $\{\cdot, \cdot\}$ is a Lie bracket (to be called a **Poisson bracket**) on $\mathcal{C}^\infty(P)$ for which $\{f, \cdot\}$ is a derivation for all $f \in \mathcal{C}^\infty(P)$. Then:

- (i) The vector field $X_f \doteq -\{f, \cdot\}$ is called the **Hamiltonian field** of f .
- (ii) The functions $f \in \mathcal{C}^\infty(P)$ for which $X_f = 0$ are called **Casimir functions**.
- (iii) The **Poisson bivector** is the twice-contravariant skew-symmetric tensor field Π defined by linearly extending the prescription $\Pi(df, dg) \doteq \{f, g\}$, for all $f, g \in \mathcal{C}^\infty(P)$.

Remark. The negative sign in (i) is again an attempt to do some damage control: it makes X_f so defined coincide with the Hamiltonian field in a symplectic manifold. In any case, discussing Hamiltonian flows and coisotropic submanifolds now makes sense in any Poisson manifold (the latter in view of Proposition 7, p. 42). Writing (P, Π) instead of $(P, \{\cdot, \cdot\})$ is also usual.

Exercise 28

Check that in the above definition Π is indeed well-defined, e.g., by using coordinates for P . Namely, show that if we take coordinates $(x^1, \dots, x^m)$ for P and write $\alpha, \beta \in \Omega^1(P)$ locally as

$$\alpha = \sum_{i=1}^m \alpha_i dx^i \quad \text{and} \quad \beta = \sum_{j=1}^m \beta_j dx^j,$$

then $\Pi(\alpha, \beta) = \sum_{i,j=1}^m \alpha_i \beta_j \{x^i, x^j\}$, and that the result is coordinate-independent. One usually writes $\Pi^{ij} \doteq \{x^i, x^j\}$ for the components of Π , so that

$$\Pi = \sum_{1 \leq i < j \leq m} \Pi^{ij} \frac{\partial}{\partial x^i} \wedge \frac{\partial}{\partial x^j}.$$

So, if (M, ω) is a symplectic manifold, then $(M, \{\cdot, \cdot\}_\omega)$ is a Poisson manifold, whose Casimir functions are precisely the (locally) constant ones. But not every Poisson manifold arises in this way.

Example 24 (Canonical Poisson prototype)

Consider $\mathbb{R}^{2n+r}$, with coordinates $(x^1, \dots, x^n, y_1, \dots, y_n, z^1, \dots, z^r)$ and just mimic the usual formula seen in Example 22 (p. 40), ignoring the z -coordinates. Namely, define $\{\cdot, \cdot\}: \mathcal{C}^\infty(\mathbb{R}^{2n+r}) \times \mathcal{C}^\infty(\mathbb{R}^{2n+r}) \rightarrow \mathcal{C}^\infty(\mathbb{R}^{2n+r})$ by

$$\{f, g\} \doteq \sum_{k=1}^n \left(\frac{\partial f}{\partial x^k} \frac{\partial g}{\partial y_k} - \frac{\partial g}{\partial x^k} \frac{\partial f}{\partial y_k} \right).$$

Note that the Casimir functions for this bracket are precisely the ones depending only on the z -variables. There's a Darboux-type theorem (actually due to Lie) stating that every Poisson bracket on $\mathbb{R}^{2n+r}$ whose rank is equal to r (in general, it doesn't need to be constant) is locally equivalent to this standard one.

As the above example shows, there is no dimension restriction (dissapointingly, note that one may also always choose $\{\cdot, \cdot\} = 0$ on any manifold) and one has little to no control over the Casimir functions of an arbitrary Poisson bracket. Still, there are very interesting examples of non-trivial Poisson brackets coming from Mechanics.

Example 25 (Nambu mechanics)

Define $\{\cdot, \cdot\}_N: \mathcal{C}^\infty(\mathbb{R}^3) \times \mathcal{C}^\infty(\mathbb{R}^3) \rightarrow \mathcal{C}^\infty(\mathbb{R}^3)$ by

$$\{f, g\}_N(x) \doteq \langle x, \nabla f(x) \times \nabla g(x) \rangle,$$

where ∇ stands for the gradient operator, while $\langle \cdot, \cdot \rangle$ and $\times$ denote the standard Euclidean inner and cross products. This $\{\cdot, \cdot\}_N$ is a Poisson bracket on $\mathbb{R}^3$, called

the **Nambu bracket** (the only non-trivial thing to verify is the Jacobi identity). Here's some intuition: if $x, y, z: \mathbb{R}^3 \rightarrow \mathbb{R}$ denote the coordinate functions on $\mathbb{R}^3$, as a consequence of the cross product relations between vectors in the standard basis of $\mathbb{R}^3$, we have that

$$\{x, y\}_N = z, \quad \{y, z\}_N = x, \quad \text{and} \quad \{z, x\}_N = y,$$

and thus the **Nambu bivector** is given by

$$\Pi_N = x \frac{\partial}{\partial y} \wedge \frac{\partial}{\partial z} + y \frac{\partial}{\partial z} \wedge \frac{\partial}{\partial x} + z \frac{\partial}{\partial x} \wedge \frac{\partial}{\partial y}.$$

So, under the canonical isomorphism $(\mathbb{R}^3)^{\wedge 2} \cong \mathbb{R}^3$, Π_N corresponds to the position vector field on $\mathbb{R}^3$. We now claim that a function $f \in \mathcal{C}^\infty(\mathbb{R}^3)$ is Casimir if and only if it is radial. Assume that f is radial, so that $f(x) = \rho(\|x\|^2)$ for some smooth $\rho: \mathbb{R} \rightarrow \mathbb{R}$. Then we have that $\nabla f(x) = 2\rho'(\|x\|^2)x$, so

$$\{f, g\}_N(x) = 2\rho'(\|x\|^2)\langle x, x \times \nabla g(x) \rangle = 0$$

for all $g \in \mathcal{C}^\infty(\mathbb{R}^3)$. Conversely, assume that f is Casimir. Let's show that for each $r > 0$, the restriction $f|_{rS^2}$ is constant, by showing that its differential vanishes (here, rS^2 denotes the sphere centered at the origin with radius r). The condition $\{f, \cdot\}_N = 0$ implies that for all $x \in \mathbb{R}^3$, we have $x \times \nabla f(x) = \mathbf{0}$, since the values $\nabla g(x)$ fill up $T_x(\mathbb{R}^3)$ as g ranges over $\mathcal{C}^\infty(\mathbb{R}^3)$ (or, more simply, just take $g = x$, $g = y$ and $g = z$). Thus, write $\nabla f(x) = \alpha(x)x$ for some smooth $\alpha: \mathbb{R}^3 \rightarrow \mathbb{R}$. Now assume that $\gamma: I \rightarrow rS^2$ is any smooth curve, and compute

$$\frac{d}{dt}f(\gamma(t)) = \langle \nabla f(\gamma(t)), \dot{\gamma}(t) \rangle = \alpha(\gamma(t))\langle \gamma(t), \dot{\gamma}(t) \rangle = 0,$$

as desired (differentiate $\langle \gamma, \gamma \rangle = r^2$ to obtain $\langle \gamma, \dot{\gamma} \rangle = 0$). In higher dimensions, we have $(\mathbb{R}^n)^{\wedge(n-1)} \cong \mathbb{R}^n$, so multivectors instead of bivectors are required to study Nambu mechanics and Poisson brackets are no longer enough. See [21] for more details.

Remark. More generally, once can fix any smooth function $C: \mathbb{R}^3 \rightarrow \mathbb{R}$ and define a Nambu bracket from C , by $\{f, g\}_{N,C}(x) \doteq \langle \nabla C(x), \nabla f(x) \times \nabla g(x) \rangle$. Then C itself is a Casimir function and, as you may now expect, the associated bivector $\Pi_{N,C}$ corresponds under $(\mathbb{R}^3)^{\wedge 2} \cong \mathbb{R}^3$ to ∇C .

Exercise 29

Verify that $\{\cdot, \cdot\}_N$ satisfies the Jacobi identity. **Hint:** Vector calculus identities might help.

Exercise 30

Let $\mathfrak{g}$ be a Lie algebra. Define $\{\cdot, \cdot\}_{\mathfrak{g}^*}: \mathcal{C}^\infty(\mathfrak{g}^*) \times \mathcal{C}^\infty(\mathfrak{g}^*) \rightarrow \mathcal{C}^\infty(\mathfrak{g}^*)$ by setting

$$\{f, g\}_{\mathfrak{g}^*}(p) = p([df_p, dg_p]),$$

where $df_p, dg_p \in \mathfrak{g}^{**} \cong \mathfrak{g}$, so that taking their Lie bracket makes sense. This is called the **standard Lie-Poisson structure** on $\mathfrak{g}^*$. Show that $\{\cdot, \cdot\}_{\mathfrak{g}^*}$ is a Poisson bracket. **Hint:** You'll need the Jacobi identity for $[\cdot, \cdot]$ somewhere.

Unfortunately, the dimension and the presence of non-constant Casimir functions are not the only obstacle for a Poisson bracket to have been induced by a symplectic form. In $\mathbb{R}^2$, define

$$\{f, g\}(x, y) = y \left(\frac{\partial f}{\partial x} \frac{\partial g}{\partial y} - \frac{\partial f}{\partial y} \frac{\partial g}{\partial x} \right).$$

This is clearly a Poisson bracket on $\mathbb{R}^2$. If f is Casimir, then f is constant on $\mathbb{R} \times \mathbb{R}_{>0}$ and $\mathbb{R} \times \mathbb{R}_{<0}$, hence constant on the entire $\mathbb{R}^2$ by continuity. But $\{\cdot, \cdot\}$ is not symplectic as

$$\det \begin{pmatrix} 0 & y \\ -y & 0 \end{pmatrix} = y^2$$

vanishes along the x -axis. A bit more is required.

Theorem 14 (Pauli-Jost)

Let $(P, \{\cdot, \cdot\})$ be a Poisson manifold and assume that the induced map $T^*P \rightarrow TP$ characterized by $df_x \mapsto X_f|_x$, for all $x \in P$ and $f \in \mathcal{C}^\infty(P)$, is an isomorphism. Then there is a symplectic form $\omega \in \Omega^2(P)$ for which $\{\cdot, \cdot\}_\omega = \{\cdot, \cdot\}$.

Here's a sketch of the proof: use the isomorphism $T^*P \cong TP$ to convert the bivector Π to a 2-form ω — it is non-degenerate by assumption. And it is closed by the Jacobi identity for $\{\cdot, \cdot\}$ (reverse the proof of Theorem 13, p. 42). It satisfies $\{\cdot, \cdot\}_\omega = \{\cdot, \cdot\}$ by construction.

For more on Poisson manifolds, see the delightful article [44], or the book [42] for a more thorough presentation.

3.2 Group actions on manifolds

Throughout the rest of this text, we will explore symmetries of symplectic manifolds. Symmetries are naturally encoded on group actions. We are naturally led to consider actions of Lie groups on manifolds. Thus, we'll start briefly reviewing group actions. Let's fix some notation. Here, G will denote a (connected) Lie group, e its identity element⁴ and, for each $g \in G$, $L_g, R_g, C_g, \text{inv}: G \rightarrow G$ will denote, respectively, left

⁴The letter e comes from German, *einselement*.

translation by g , right translation by g , conjugation by g , and the inversion map. The Lie algebra $\mathfrak{g}$ (also denoted by $\text{Lie}(G)$) of G is the tangent space $T_e G$, and we'll identify it with the algebra $\mathfrak{X}^L(G)$ of left-invariant vector fields on G , via $X \mapsto X^L$, where X^L is the unique left-invariant extension of $X \in \mathfrak{g}$, namely, $(X^L)_g \doteq d(L_g)_e(X)$. Similar comments hold for $\mathfrak{X}^R(G)$ and X^R .

Definition 17

Let G be a Lie group and M be a smooth manifold. A (smooth) **left action** of G on M , denoted $G \curvearrowright M$, is a smooth map $G \times M \ni (g, x) \mapsto g \cdot x \in M$, satisfying:

- (i) $e \cdot x = x$ for all $x \in M$
- (ii) $g \cdot (h \cdot x) = (gh) \cdot x$, for all $g, h \in G, x \in M$.

In other words, $G \curvearrowright M$ is equivalent to a Lie group homomorphism $G \rightarrow \text{Diff}(M)$. We will identify $g \in G$ with the diffeomorphism $g: M \rightarrow M$ given by $x \mapsto g \cdot x$, and given $x \in M$, we'll write $\mathcal{O}_x: G \rightarrow M$ for the **orbit map** of x , $\mathcal{O}_x(g) \doteq g \cdot x$. The **orbit space** of $G \curvearrowright M$, denoted by M/G , is the quotient of M under the equivalence relation " $x \sim y \iff$ there is $g \in G$ such that $y = g \cdot x$ ".

Exercise 31

In the above definition, check that $\sim$ is indeed an equivalence relation.

Here are some frequent and useful concepts.

Definition 18

Let $G \curvearrowright M$ be a Lie group action on a manifold.

- (i) Given $x \in M$, $G \cdot x \doteq \{g \cdot x \mid g \in G\}$ is the **orbit** of x ;
- (ii) Given $x \in M$, $G_x \doteq \{g \in G \mid g \cdot x = x\}$ is the **stabilizer** of x ;
- (iii) The action is **transitive** if there is $x \in M$ such that $G \cdot x = M$.
- (iv) The action is **free** if $G_x = \{e\}$ for all $x \in M$.
- (v) The action is **proper** if the **enriched action** $G \times M \rightarrow M \times M$ (given by $(g, x) \mapsto (x, g \cdot x)$) is a proper map (i.e., whose inverse images of compact sets are compact).

Remark. Given a smooth Lie group action $G \curvearrowright M$, freeness and properness of the action are enough to ensure that M/G has a unique smooth manifold structure for which the quotient projection $M \rightarrow M/G$ is a surjective submersion. In particular, if G is compact, properness is automatic. For more precise conditions ensuring that the quotient $M/\sim$ of a manifold under an equivalence relation $\sim$ has a smooth manifold

structure making $M \rightarrow M/\sim$ a surjective submersion, see **Godement's Theorem**, for example, in [20].

Example 26

- (1) A **real linear Lie group** is a Lie subgroup of $GL_n(\mathbb{R})$ (for example, $SL_n(\mathbb{R})$, $O(n, \mathbb{R})$, $SO(n, \mathbb{R})$, etc.). Each linear Lie group acts on $\mathbb{R}^n$ via evaluation. In particular, restricting we obtain actions such as $O(n, \mathbb{R}) \curvearrowright \mathbb{S}^{n-1}$.
- (2) $\mathbb{S}^1 \curvearrowright \mathbb{S}^2$ via rotations about the z -axis, where elements of $\mathbb{S}^1$ are regarded as angles.

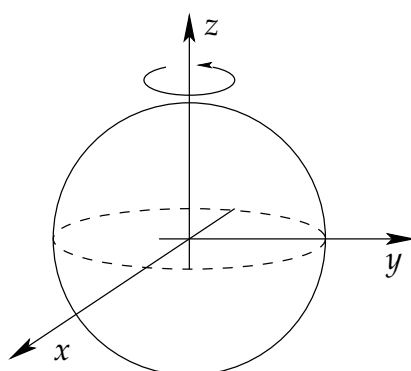


Figure 7: Action of the circle on the sphere.

- (3) Any Lie group G acts on itself, in several ways:

- Via left translations: $g \cdot x \doteq L_g(x) = gx$.
- Via right translations: $g \cdot x \doteq R_{g^{-1}} = xg^{-1}$.
- Via conjugation: $g \cdot x \doteq C_g(x) = gxg^{-1}$.

- (4) Fixed $\alpha \in \mathbb{R}$, we have an action $\mathbb{R} \curvearrowright \mathbb{T}^2 = \mathbb{R}^2/\mathbb{Z}^2$ via

$$t \cdot ((x, y) \bmod 1) \doteq (x + t, y + \alpha t) \bmod 1.$$

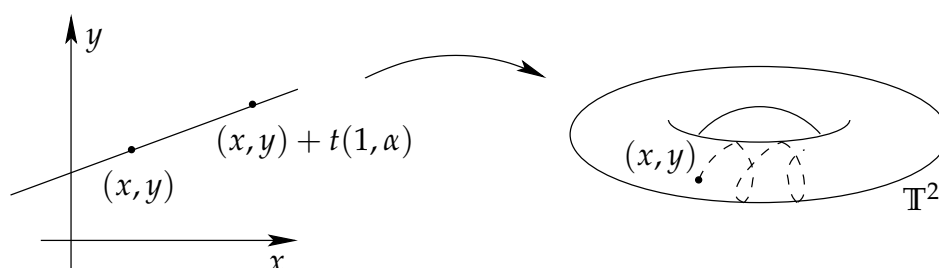


Figure 8: A line action on the torus.

It is possible to show that if $\alpha \in \mathbb{Q}$, the orbits of the action are embedded closed curves in $\mathbb{T}^2$, while if $\alpha \in \mathbb{R} \setminus \mathbb{Q}$, they're dense in $\mathbb{T}^2$ (this is called **Kronecker's flow theorem**).

(5) Let Q be a smooth manifold, and let $G \curvearrowright Q$ be a smooth action. This action induces actions:

- $G \curvearrowright TQ$, via $g \cdot (x, v) \doteq (g \cdot x, dg_x(v))$.
- $G \curvearrowright T^*Q$, via $g \cdot (x, p) \doteq (g \cdot x, p \circ (dg_x)^{-1})$.

Combining the above, G will also act on every tensor bundle over Q .

To complement the examples above, consider the following exercise:

Exercise 32

- (a) Which of the actions given in the above example are transitive? Free? Try to describe the orbit spaces. **Hint:** The orbit-stabilizer theorem may come in handy.
- (b) Show that if the action $G \curvearrowright Q$ is free and proper, so are $G \curvearrowright TQ$ and $G \curvearrowright T^*Q$. **Hint:** Recall that $G \curvearrowright Q$ is proper if and only if given any sequences $(g_n)_{n \geq 1}$ in G and $(x_n)_{n \geq 1}$ in Q such that both $(g_n \cdot x_n)_{n \geq 1}$ and $(x_n)_{n \geq 1}$ converge in Q , then $(g_n)_{n \geq 1}$ has a convergent subsequence in G .

And while we're talking about the orbit-stabilizer theorem, note that each stabilizer G_x is a closed subgroup of G (hence a Lie subgroup — not normal, in general), so the quotient G/G_x is a smooth manifold as well. Then $G/G_x \rightarrow G \cdot x \subseteq M$ says that each orbit $G \cdot x$ is an immersed submanifold of G . If G is compact, the orbits are embedded (simply because a proper injective immersion must be an embedding). To further exploit the fact we're dealing with Lie groups, derivatives must come into play.

Definition 19

Let $G \curvearrowright M$ be a smooth Lie group action. For any $X \in \mathfrak{g}$, the **action field** of X is $X^\# \in \mathfrak{X}(M)$ defined by $(X^\#)_x \doteq d(\mathbb{O}_x)_e(X)$.

Remark. When emphasizing the manifold being acted on is needed, one might write $X_M^\#$ instead. Many texts directly define the action field by explicitly writing

$$X_x^\# = \left. \frac{d}{dt} \right|_{t=0} \exp(tX) \cdot x,$$

which automatically makes the flow relation $\Phi_{t, X^\#}(x) = \exp(tX) \cdot x$ true.

Exercise 33

Let $G \curvearrowright M$ be a smooth Lie group action. Show that:

- (a) the map $\mathfrak{g} \ni X \mapsto X^\# \in \mathfrak{X}(M)$ is an anti-homomorphism of Lie algebras. For right actions, it's a homomorphism.
- (b) given $x \in M$, $(X^\#)_x = \mathbf{0}$ if and only if $X \in \mathfrak{g}_x \doteq \text{Lie}(G_x)$ (it is called the **stabilizer algebra at x**).

Here are quick examples, based on the previous ones.

Example 27

- (1) For the rotation action $S^1 \curvearrowright S^2$, since $\text{Lie}(S^1) = i\mathbb{R} \cong \mathbb{R}$ (the obvious isomorphism, delete i), we have that

$$1^\#_{(x,y,z)} = \left. \frac{d}{d\theta} \right|_{t=0} (x \cos \theta - y \sin \theta, x \sin \theta + y \cos \theta, z) = -y \left. \frac{\partial}{\partial x} \right|_{(x,y,z)} + x \left. \frac{\partial}{\partial y} \right|_{(x,y,z)}.$$

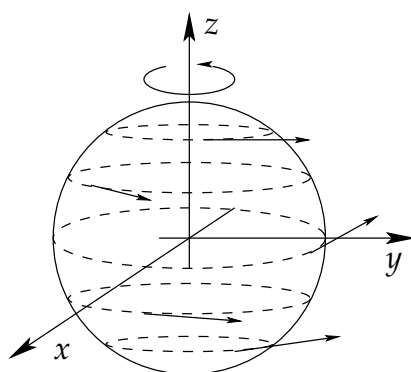
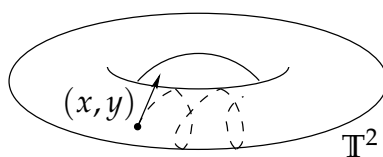


Figure 9: Infinitesimal generator for $S^1 \curvearrowright S^2$.

- (2) Take a Lie group G acting on itself, and $X \in \mathfrak{g}$. For the action given by:
 - left translations, we have $X^\# = X^R$.
 - right translations, we have $X^\# = -X^L$ (here $d(\text{inv})_e = -\text{Id}_{\mathfrak{g}}$ plays a role).
 - conjugation, we have $X^\# = X^R - X^L$.
- (3) For the action $\mathbb{R} \curvearrowright \mathbb{T}^2$ coming from a parameter $\alpha \in \mathbb{R}$, we have that

$$1^\#_{(x,y)} = \left. \frac{d}{dt} \right|_{t=0} (x + t, y + \alpha t) \bmod 1 = \left. \frac{\partial}{\partial \theta_1} \right|_{(x,y)} + \alpha \left. \frac{\partial}{\partial \theta_2} \right|_{(x,y)}.$$


 Figure 10: Infinitesimal generator for $\mathbb{R} \curvearrowright \mathbb{T}^2$.

Exercise 34

- (a) Let $G_1 \curvearrowright M_1$ and $G_2 \curvearrowright M_2$ be two smooth Lie group actions, $\psi: G_1 \rightarrow G_2$ be a Lie group homomorphism, and $F: M_1 \rightarrow M_2$ be ψ -**equivariant**, that is, it satisfies $F(g \cdot x) = \psi(g) \cdot F(x)$, for all $g \in G_1$ and $x \in M_1$. Show that for each $X \in \mathfrak{g}_1$ and $x \in M_1$, we have $dF_x(X^\#_{M_1}|_x) = d\psi_e(X)^\#_{M_2}|_{F(x)}$.
- (b) Conclude that if $G \curvearrowright Q$ is a smooth Lie group action, then for each $X \in \mathfrak{g}$, the fields $X^\#_{T^*Q}$ and $X^\#_{T^*Q}$ project to $X^\#_Q$ under the natural bundle projections.

Now, let's get back to the symplectic category.

Definition 20

Let (M, ω) be a symplectic manifold and G be a Lie group. A smooth action $G \curvearrowright M$ is called a **symplectic action** if for each $g \in G$, the action map $g: M \rightarrow M$ is a symplectomorphism. We'll denote a symplectic action by $G \curvearrowright (M, \omega)$.

Remark. This is equivalent to requiring the homomorphism $G \rightarrow \text{Diff}(M)$ to be actually valued in $\text{Sp}(M, \omega)$.

As an immediate consequence of the definition, since the flows of the action fields are given by $\Phi_{t, X^\#}(x) = \exp(tX) \cdot x$ (which in particular shows that $X^\#$ is complete), we have from Theorem 8 (p. 29) that all the action fields of a symplectic action are... symplectic.

Example 28

- (1) If (M, ω, H) is a compact Hamiltonian system, we obtain $\mathbb{R} \curvearrowright (M, \omega)$ via $t \cdot x \doteq \Phi_{t, X_H}(x)$.
- (2) More generally, if (M, ω) is a compact symplectic manifold and we have functions $f_1, \dots, f_k \in \mathcal{C}^\infty(M)$ in involution, that is, $\{f_i, f_j\}_\omega = 0$ for all indices $i, j = 1, \dots, k$, we obtain an action $\mathbb{R}^k \curvearrowright (M, \omega)$ by

$$(t_1, \dots, t_k) \cdot x \doteq \Phi_{t_1, X_{f_1}} \circ \dots \circ \Phi_{t_k, X_{f_k}}(x).$$

- (3) Identify $\mathbb{R}^{2n} \cong \mathbb{C}^n$, so that we get an action $\mathbb{T}^n \curvearrowright (\mathbb{R}^{2n}, \omega_{2n})$ via

$$(e^{i\theta_1}, \dots, e^{i\theta_n}) \cdot (z^1, \dots, z^n) \doteq (e^{i\theta_1} z^1, \dots, e^{i\theta_n} z^n).$$

By the way, you can check that in complex coordinates, one has

$$\omega_{2n} = \frac{i}{2} \sum_{k=1}^n dz^k \wedge d\bar{z}^k.$$

- (4) If $G \curvearrowright Q$, then $G \curvearrowright (T^*Q, \omega_{\text{can}})$ by Proposition 4 (p. 20). And clearly, if $B \in \Omega^2(Q)$ is closed, we'll have $G \curvearrowright (T^*Q, \omega_B)$ if and only if the original action $G \curvearrowright Q$ preserves B .

With this in place, the next natural question is whether if given $G \curvearrowright (M, \omega)$, free and proper, ω induces a symplectic form in the orbit space M/G (which is a smooth manifold, under the given assumptions). The answer is an unfortunate “no”, as $\dim(M/G) = \dim M - \dim G$ might not be even. Thus we are forced to move on to the next best thing that could survive in the quotient with good properties: the Poisson bracket $\{\cdot, \cdot\}_\omega$.

Theorem 15

Let $G \curvearrowright (M, \omega)$ be given. Then the **algebra** $\mathcal{C}^\infty(M)^G$ of G -invariant functions is closed under $\{\cdot, \cdot\}_\omega$. Hence, if $G \curvearrowright M$ is free and proper, there is a unique Poisson bracket $\{\{\cdot, \cdot\}\}$ on M/G for which the quotient projection $M \rightarrow M/G$ becomes a Poisson morphism.

Proof: By Exercise 24 (p. 41), for all $f_1, f_2 \in \mathcal{C}^\infty(M)^G$ and $g \in G$, we have that

$$\{f_1, f_2\}_\omega \circ g \stackrel{(*)}{=} \{f_1 \circ g, f_2 \circ g\}_\omega \circ g = \{f_1, f_2\}_{g^*\omega} = \{f_1, f_2\}_\omega,$$

where in $(*)$ we use invariance of f_1 and f_2 . The rest is clear once we establish the correspondence $\mathcal{C}^\infty(M)^G \cong \mathcal{C}^\infty(M/G)$. But this is just the characteristic property of surjective submersions. Namely, each $f \in \mathcal{C}^\infty(M)^G$ passes to the quotient giving a function $\tilde{f} \in \mathcal{C}^\infty(M/G)$, while composing any given $\tilde{f} \in \mathcal{C}^\infty(M/G)$ with the quotient projection $M \rightarrow M/G$ gives $f \in \mathcal{C}^\infty(M)^G$ — these two processes are inverses to each other. With this in place, there's nothing else to do: define

$$\{\{\tilde{f}_1, \tilde{f}_2\}\}(G \cdot x) \doteq \{f_1, f_2\}_\omega(x),$$

where the representative x of $G \cdot x$ is chosen at will. □

Remark. The above result should be regarded as a little factory of Poisson manifolds from symplectic manifolds, on its own right: just find the good Lie groups. In fact, nothing changes if one replaces (M, ω) with some Poisson manifold $(P, \{\cdot, \cdot\})$ instead, assuming that the action $G \curvearrowright P$ itself is a **Poisson action**, in the sense that every action map $g: P \rightarrow P$ preserves $\{\cdot, \cdot\}$.

Example 29

Let G be a Lie group acting on itself by left translations (which is clearly free and proper), and also consider the induced action $G \curvearrowright T^*G$ via cotangent lifts. Let's fully understand the quotient T^*G/G . For that, fix the global trivialization $\Phi: G \times \mathfrak{g}^* \rightarrow T^*G$, given by $\Phi(g, p) \doteq (g, p \circ d(L_g)_e^{-1})$. Using Φ to pull-back the action on T^*G to $G \times \mathfrak{g}^*$, we compute

$$\begin{aligned} g \cdot (h, p) &= \Phi^{-1}(g \cdot \Phi(h, p)) = \Phi^{-1}(g \cdot (h, p \circ d(L_h)_e^{-1})) \\ &= \Phi^{-1}(gh, p \circ d(L_h)_e^{-1} \circ d(L_g)_h^{-1}) = \Phi^{-1}(gh, p \circ d(L_{gh})_e^{-1}) \\ &= \Phi^{-1}\Phi(gh, p) = (gh, p). \end{aligned}$$

That is, Φ undoes the twisting of the cotangent spaces. Since the action only happens on the first factor, it follows that

$$T^*G/G \xrightarrow{\Phi^{-1}} (G \times \mathfrak{g}^*)/G \xrightarrow{\text{pr}_2} \mathfrak{g}^*,$$

but here's the catch: this diffeomorphism is Poisson, when we equip $\mathfrak{g}^*$ with its standard Lie-Poisson structure (of course, by Φ^{-1} and pr_2 above we understand the induced maps on the quotients). Note that

$$\mathcal{C}^\infty((G \times \mathfrak{g}^*)/G) \cong \mathcal{C}^\infty(G \times \mathfrak{g}^*)^G \ni f \mapsto f(e, \cdot) \in \mathcal{C}^\infty(\mathfrak{g}^*)$$

is an isomorphism. With this, it would suffice to check that for all $X, Y \in \mathfrak{g}$, one has $\{\hat{X}, \hat{Y}\}_{\Phi^*\omega}(g, p) = p([X, Y])$, where $\hat{X}: G \times \mathfrak{g}^* \rightarrow \mathbb{R}$ is given by $\hat{X}(g, p) = p(X)$, and this can be done in several ways (think about what is the most efficient one).

Exercise 35

In the setting of the above example, with the same notation, show that the “tautological form” $\Phi^*\lambda \in \Omega^1(G \times \mathfrak{g}^*)$ is given by $(\Phi^*\lambda)_{(g,p)} = p \circ (dg_{(e,p)})^{-1}$, where by g we mean, as usual, the action $g: G \times \mathfrak{g}^* \rightarrow G \times \mathfrak{g}^*$ of the element $g \in G$.

3.3 Moment maps and Noether's Theorem

Recall that if a symplectic manifold (M, ω) has $H_{\text{dR}}^1(M) = 0$, then every symplectic field is Hamiltonian. If a Lie group G acts on such (M, ω) , then every action field $X^\#$ is Hamiltonian. It seems natural to try and collect all the Hamiltonian functions realizing these fields into a single map. So, assume that there is a map $\mu_c: \mathfrak{g} \rightarrow \mathcal{C}^\infty(M)$ for which $X_{\mu_c^X} = X^\#$ for all $X \in \mathfrak{g}$, where we write μ_c^X for what would be $\mu_c(X)$ — let's say that if such map μ_c exists, the action is **weakly Hamiltonian**. This implies that for

each $X, Y \in \mathfrak{g}$, the relation

$$X_{\mu_c^{[X,Y]}} = [X, Y]^\# = -[X^\#, Y^\#] = -[X_{\mu_c^X}, X_{\mu_c^Y}] = X_{\{\mu_c^X, \mu_c^Y\}_\omega}$$

holds, by exercises 25 (p. 41) and 33 (p. 51). In a perfect world, this would imply that $\mu_c^{[X,Y]} = \{\mu_c^X, \mu_c^Y\}_\omega$, but we can only say that those functions differ by a constant, which depends on X and Y themselves. This cannot be helped.

Definition 21

A symplectic action $G \curvearrowright (M, \omega)$ is called a **Hamiltonian action** if there exists a linear map $\mu_c: \mathfrak{g} \rightarrow \mathcal{C}^\infty(M)$, to be called **comoment map**, such that the diagram

$$\begin{array}{ccc} (\mathfrak{g}, [\cdot, \cdot]) & \xrightarrow{\mu_c} & (\mathcal{C}^\infty(M), \{\cdot, \cdot\}_\omega) \\ \downarrow \# & & \downarrow X_\bullet \\ (\mathcal{S}(M, \omega), [\cdot, \cdot]) & \longleftrightarrow & (\mathcal{H}(M, \omega), [\cdot, \cdot]) \end{array}$$

commutes, where $\mathcal{S}(M, \omega)$ and $\mathcal{H}(M, \omega)$ stand (as before), respectively, for the algebras of symplectic and Hamiltonian vector fields on (M, ω) , and the vertical maps are anti-homomorphisms. The associated **moment map** is the function $\mu: M \rightarrow \mathfrak{g}^*$ given by $\mu(x)(X) \doteq \mu_c^X(x)$, and μ and μ_c carry the same information. We say that the data (M, ω, G, μ) forms a **Hamiltonian G -space**.

Remark. Digression. Moment/comoment maps need not exist for a given symplectic action and, when they do, they might not be unique. One possible tool to study this question is the so-called **Chevalley-Eilenberg cohomology** of a Lie algebra. Here's the quick gist of it: let $\rho: \mathfrak{g} \rightarrow \mathfrak{gl}(V)$ be a representation of $\mathfrak{g}$ in some vector space V . For each $0 \leq k \leq \dim \mathfrak{g}$, let $\mathcal{A}_\rho^k(\mathfrak{g}; V)$ denote the space of V -valued alternating k -linear maps on $\mathfrak{g}$, and define the **Chevalley-Eilenberg derivative** $d: \mathcal{A}_\rho^k(\mathfrak{g}; V) \rightarrow \mathcal{A}_\rho^{k+1}(\mathfrak{g}; V)$ by mimicking the classical Palais formula:

$$\begin{aligned} d\omega(X_0, \dots, X_k) &= \sum_{i=0}^k (-1)^i \rho(X_i) \omega(X_0, \dots, \widehat{X}_i, \dots, X_k) \\ &\quad + \sum_{0 \leq i < j \leq k} (-1)^{i+j} \omega([X_i, X_j], X_0, \dots, \widehat{X}_i, \dots, \widehat{X}_j, \dots, X_k). \end{aligned}$$

Then we have the complex

$$\dots \xrightarrow{d} \mathcal{A}_\rho^{k-1}(\mathfrak{g}; V) \xrightarrow{d} \mathcal{A}_\rho^k(\mathfrak{g}; V) \xrightarrow{d} \mathcal{A}_\rho^{k+1}(\mathfrak{g}; V) \xrightarrow{d} \dots$$

and $d^2 = 0$, meaning that if we set $Z_\rho^k(\mathfrak{g}; V)$ and $B_\rho^k(\mathfrak{g}; V)$ as the kernel and range of the correct d 's, we can form the cohomology $H_\rho^k(\mathfrak{g}; V) \doteq Z_\rho^k(\mathfrak{g}; V) / B_\rho^k(\mathfrak{g}; V)$.

Exercise 36

Fix a Lie algebra $\mathfrak{g}$ and its adjoint representation $\text{ad}: \mathfrak{g} \rightarrow \mathfrak{gl}(\mathfrak{g})$. We have that $[\cdot, \cdot] \in \mathcal{A}_{\text{ad}}^2(\mathfrak{g}; \mathfrak{g})$. Show that the bracket $[\cdot, \cdot]$ is closed and defines the trivial class in $H_{\text{ad}}^2(\mathfrak{g}; \mathfrak{g})$. **Hint:** there is a (somewhat) obvious $f \in \mathcal{A}_{\text{ad}}^1(\mathfrak{g}; \mathfrak{g})$ for which $df = [\cdot, \cdot]$.

For example, $H_{\rho}^1(\mathfrak{g}; V)$ controls the existence of ρ -invariant complements for ρ -invariant subspaces of V , and $H_{\rho}^2(\mathfrak{g}; V)$ (for a certain ρ) controls whether $\mathfrak{g}$ admits abelian extensions. **Whitehead's Lemmas** state that if $\mathfrak{g}$ is semi-simple (that is, its Killing form is non-degenerate) then both $H_{\rho}^1(\mathfrak{g}; V)$ and $H_{\rho}^2(\mathfrak{g}; V)$ are trivial (in particular, the **Weyl's reducibility theorem** for semi-simple Lie algebras follows from this). As far as moment maps go, assume again that we have $G \curvearrowright (M, \omega)$ and a linear map $\mu_c: \mathfrak{g} \rightarrow \mathcal{C}^{\infty}(M)$ such that for each $X \in \mathfrak{g}$, μ_c^X is a Hamiltonian function for $X^{\#}$, but no more, and indeed write $\mu_c^{[X, Y]} = \{\mu_c^X, \mu_c^Y\}_{\omega} + \zeta_{\mu}(X, Y)$, for some $\zeta_{\mu}(X, Y) \in \mathbb{R}$.

Exercise 37

Show that the map $\zeta_{\mu}: \mathfrak{g} \times \mathfrak{g} \rightarrow \mathbb{R}$ satisfies:

- (a) ζ_{μ} is bilinear;
- (b) $\zeta_{\mu}(Y, X) = -\zeta_{\mu}(X, Y)$ for all $X, Y \in \mathfrak{g}$;
- (c) $\zeta_{\mu}(X, [Y, Z]) + \zeta_{\mu}(Y, [Z, X]) + \zeta_{\mu}(Z, [X, Y]) = 0$ for all $X, Y, Z \in \mathfrak{g}$.

Hence $\zeta_{\mu} \in Z_{\rho_0}^2(\mathfrak{g}; \mathbb{R})$, where $\rho_0: \mathfrak{g} \rightarrow \mathfrak{gl}(\mathbb{R})$ is the trivial representation.

Note: item (c) says precisely that $d(\zeta_{\mu}) = 0$.

It turns out that μ_c can be “corrected” to a legitimate comoment map if and only if the cohomology class $[\zeta_{\mu}]$ is trivial in $H_{\rho_0}^2(\mathfrak{g}; \mathbb{R})$. Namely, we just add a primitive for ζ_{μ} to μ . If $[\zeta_{\mu}]$ is non-trivial, there is a way to make it trivial by passing to a certain central extension of $\mathfrak{g}$ (in other words, one must enlarge $\mathfrak{g}$) — this is related to phenomena of group quantization, see [23] and the other references in [33] for more details. One useful conclusion of all of this is the:

Theorem 16

A symplectic action $G \curvearrowright (M, \omega)$, where $H_{\text{dR}}^1(M) = 0$ and G is semi-simple, is Hamiltonian.

As for the uniqueness, we'll just mention that two “comoment maps” for a weakly Hamiltonian action differ by an element of the annihilator of the commutant ideal $[\mathfrak{g}, \mathfrak{g}]$. End of digression.

With this in place, let's revisit a few of the previous examples of symplectic actions.

Example 30

- (1) Consider a symplectic manifold (M, ω) and a smooth function $H: M \rightarrow \mathbb{R}$ for which the Hamiltonian field X_H is complete, and define the action $\mathbb{R} \curvearrowright (M, \omega)$ via Hamiltonian flow: $t \cdot x = \Phi_{t, X_H}(x)$. We know that $\text{Lie}(\mathbb{R}) = \mathbb{R}$, so given $a \in \mathbb{R}$, we can note that $\Phi_{ta, X_H} = \Phi_{t, aX_H} = \Phi_{t, X_{aH}}$ and compute

$$(a^\#)_x = \frac{d}{dt} \Big|_{t=0} (ta) \cdot x = \frac{d}{dt} \Big|_{t=0} \Phi_{ta, X_H}(x) = \frac{d}{dt} \Big|_{t=0} \Phi_{t, X_{aH}}(x) = X_{aH} \Big|_x,$$

so that $\mu_c: \mathbb{R} \rightarrow \mathcal{C}^\infty(M)$ given by $\mu_c^a = aH$ works. It is a homomorphism since $\{\mu_c^a, \mu_c^b\}_\omega = ab\{H, H\}_\omega = 0$ for all $a, b \in \mathbb{R}$ and $\mathbb{R}$ is abelian. The corresponding moment map $\mu: M \rightarrow \mathbb{R}^*$ takes a point $x \in M$ to the map $\mu(x): \mathbb{R} \rightarrow \mathbb{R}$ given by multiplication by $H(x)$ (so in this sense, $\mu = H$).

- (2) Consider the circle action $S^1 \curvearrowright (\mathbb{C}^n, \omega_{2n})$ given just by scalar multiplication: $e^{i\theta} \cdot (z^1, \dots, z^n) = (e^{i\theta} z^1, \dots, e^{i\theta} z^n)$. In other words, compose the diagonal inclusion $S^1 \hookrightarrow \mathbb{T}^n$ with the action seen in Example 28 (p. 52). Note that $\text{Lie}(S^1) = i\mathbb{R}$:

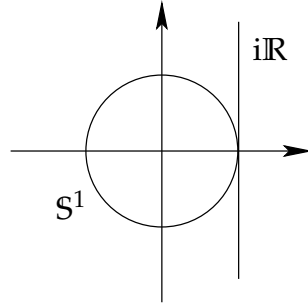


Figure 11: Lie algebra of the circle group.

A direct computation (being careful with the identification $\mathbb{R}^{2n} \cong \mathbb{C}^n$) shows that if $ia \in i\mathbb{R}$, then

$$(ia)^\# = \sum_{k=1}^n \left(ia z^k \frac{\partial}{\partial z^k} - ia \bar{z}^k \frac{\partial}{\partial \bar{z}^k} \right),$$

so that

$$\begin{aligned}\iota_{(ia)^\#} \omega_{2n} &= \omega_{2n}((ia)^\#, \cdot) = \frac{i}{2} \sum_{k=1}^n (dz^k \wedge d\bar{z}^k)((ia)^\#, \cdot) \\ &= \frac{i}{2} \sum_{k=1}^n (iaz^k d\bar{z}^k - (-ia\bar{z}^k dz^k)) = -\frac{a}{2} \sum_{k=1}^n (z^k d\bar{z}^k + \bar{z}^k dz^k) \\ &= -\frac{a}{2} \sum_{k=1}^n d(|z^k|^2) = d\left(-\frac{a}{2} \sum_{k=1}^n |z^k|^2\right).\end{aligned}$$

This suggests that this circle action is indeed Hamiltonian with comoment map $\mu_c: i\mathbb{R} \rightarrow \mathcal{C}^\infty(\mathbb{C}^n)$ given by

$$\mu_c^{ia}(z) = -\frac{a\|z\|^2}{2}.$$

You can directly check that $\{\mu_c^{ia}, \mu_c^{ib}\}_{\omega_{2n}} = 0$ for all $a, b \in \mathbb{R}$.

- (3) Let $(M, -d\lambda)$ be an exact symplectic manifold and assume that $G \curvearrowright (M, -d\lambda)$ is an action not only preserving $\omega = -d\lambda$, but the primitive λ as well (clearly the case of interest is when λ is indeed the tautological form on some cotangent bundle $M = T^*Q$). We claim that this already ensures that the action is Hamiltonian. Namely, given $X \in \mathfrak{g}$, Cartan's homotopy formula comes into play again:

$$0 = \mathcal{L}_{X^\#} \lambda = \iota_{X^\#} d\lambda + d\iota_{X^\#} \lambda \implies \iota_{X^\#} \omega = d(\lambda(X^\#)).$$

This leads us to consider $\mu_c: \mathfrak{g} \rightarrow \mathcal{C}^\infty(M)$ given by $\mu_c^X = \lambda(X^\#)$. Let's verify that this is a comoment map. For that, start noting that for any $X, Y \in \mathfrak{g}$, we have

$$X^\#(\lambda(Y^\#)) = X^\#(\mu_c^Y) = d\mu_c^Y(X^\#) = \omega(X_{\mu_c^Y}, X^\#) = -\omega(X^\#, Y^\#).$$

Now, we use the definition of exterior derivative to compute

$$\begin{aligned}\{\mu_c^X, \mu_c^Y\}_\omega &= \omega(X^\#, Y^\#) \\ &= Y^\#(\lambda(X^\#)) - X^\#(\lambda(Y^\#)) + \lambda([X^\#, Y^\#]) \\ &= \omega(X^\#, Y^\#) - \omega(Y^\#, X^\#) - \lambda([X, Y]^\#),\end{aligned}$$

so that cancelling one $\omega(X^\#, Y^\#)$ on both sides and reusing the very first line of the computation, it follows that $\{\mu_c^X, \mu_c^Y\}_\omega = \mu_c^{[X, Y]}$, as required. Lastly, note that we may write the moment map $\mu: M \rightarrow \mathfrak{g}^*$ as $\mu(x) = (\mathbb{O}_x^* \lambda)_e$, where $\mathbb{O}_x: G \rightarrow M$ is the orbit map of x .

Remark. Note that if a smooth action $G \curvearrowright Q$ preserves some $A \in \Omega^1(Q)$, then the action $G \curvearrowright (T^*Q, \omega_B)$ induced via cotangent lifts, where $B = dA$, is Hamiltonian with comoment map $\mu_c: \mathfrak{g} \rightarrow \mathcal{C}^\infty(T^*Q)$ given by $\mu_c^X(x, p) = (p - A_x)(X_Q^\#|_x)$.

Exercise 38

Let (M, ω) be a compact symplectic manifold, $f_1, \dots, f_k \in \mathcal{C}^\infty(M)$ be in involution, and consider the action $\mathbb{R}^k \curvearrowright (M, \omega)$ given by composition of Hamiltonian flows, as in Example 28 (p. 52). Namely, set

$$(t_1, \dots, t_k) \cdot x = \Phi_{t_1, X_{f_1}} \circ \dots \circ \Phi_{t_k, X_{f_k}}(x).$$

Show that this action is Hamiltonian with comoment map $\mu_c: \mathbb{R}^k \rightarrow \mathcal{C}^\infty(M)$ given by

$$\mu_c^{(a_1, \dots, a_k)} = \sum_{i=1}^k a_i f_i$$

and moment map $\mu: M \rightarrow (\mathbb{R}^k)^*$ given by $\mu(x) = \langle f(x), \cdot \rangle$, where $f = (f_1, \dots, f_k)$ and $\langle \cdot, \cdot \rangle$ is the standard inner product in $\mathbb{R}^k$. **Hint:** to show that μ_c is a Poisson map, the involution assumption is crucial.

Let's look at one last example, of mechanical flavor:

Example 31 (The reason for the name “co/moment” map)

Consider $\mathbb{R}^3$ equipped with its standard Euclidean inner product $\langle \cdot, \cdot \rangle$, and consider the diagonal derivative action $\mathrm{SO}(3) \curvearrowright T\mathbb{R}^3 \cong \mathbb{R}^6$, given by

$$R \cdot (x, y) \doteq (Rx, Ry).$$

Here, $\mathbb{R}^6$ is equipped with the standard symplectic form ω_6 . Recall that we have a natural isomorphism $(\mathfrak{so}(3), [\cdot, \cdot]) \cong (\mathbb{R}^3, \times)$, via the map

$$\mathbb{R}^3 \ni (a, b, c) \mapsto A_{(a,b,c)} = \begin{pmatrix} 0 & -c & b \\ c & 0 & -a \\ -b & a & 0 \end{pmatrix} \in \mathfrak{so}(3)$$

That is, $A_{(a,b,c)}$ is the matrix of $(a, b, c) \times _$ relative to the standard basis of $\mathbb{R}^3$. The inner product on $\mathbb{R}^3$ then induces an isomorphism $\mathfrak{so}(3) \cong \mathfrak{so}(3)^*$. We claim that this action is Hamiltonian with comoment map $\mu_c: \mathbb{R}^3 \rightarrow \mathcal{C}^\infty(\mathbb{R}^6)$ given by

$$\mu_c^v(x, y) = \langle v, x \times y \rangle.$$

A quick way to see this is to compute, for each $v \in \mathbb{R}^3$, the action field $v^\#$ as $v^\#_{(x,y)} = (v \times x, v \times y)$ (the total derivative of a linear map is itself), and to note that for any function $f \in \mathcal{C}^\infty(\mathbb{R}^6)$, if $\nabla f = (\nabla_x f, \nabla_y f)$, then $X_f = (\nabla_y f, -\nabla_x f)$,

so that $(\nabla \mu_c^v)_{(x,y)} = (y \times v, v \times x)$ leads to $X_{\mu_c^v} = v^\#$ as required. So, with all due isomorphisms, the moment map $\mu: \mathbb{R}^3 \times \mathbb{R}^3 \rightarrow \mathbb{R}^3$ appears as $\mu(x, v) = x \times v$, which is the actual **angular momentum**.

Here's one last exercise describing a simple construction providing more examples:

Exercise 39

Let $(M_1, \omega_1, G, \mu_1)$ and $(M_2, \omega_2, G, \mu_2)$ be two Hamiltonian G -spaces. Show that $(M_1 \times M_2, \omega_1 \oplus \omega_2, G, \mu_1 \oplus \mu_2)$ is also a Hamiltonian G -space, where the action of G on $M_1 \times M_2$ is diagonal and the new moment $\mu_1 \oplus \mu_2: M_1 \times M_2 \rightarrow \mathfrak{g}^*$ is given by $(\mu_1 \oplus \mu_2)(x, y) \doteq \mu_1(x) + \mu_2(y)$.

We have seen that comoment maps are Poisson maps. It turns out that moment maps are also Poisson maps.

Exercise 40

For a Hamiltonian G -space (M, ω, G, μ) , show that the moment map $\mu: M \rightarrow \mathfrak{g}^*$ is a Poisson map, that is,

$$\{f \circ \mu, g \circ \mu\}_\omega = \{f, g\}_{\mathfrak{g}^*} \circ \mu$$

holds for all $f, g \in \mathcal{C}^\infty(\mathfrak{g}^*)$. Here, $\{\cdot, \cdot\}_{\mathfrak{g}^*}$ is the standard Lie-Poisson structure on $\mathfrak{g}^*$, as seen in Exercise 30 (p. 47). **Hint:** It suffices to show it holds for $f = \widehat{X}$ and $g = \widehat{Y}$, where $X, Y \in \mathfrak{g}$ and $\widehat{X}$ and $\widehat{Y}$ are the natural images of X and Y in $\mathfrak{g}^{**}$ (hence maps $\mathfrak{g}^* \rightarrow \mathbb{R}$). Use that $\mu_c^X = \widehat{X} \circ \mu$.

To further explore the properties of moment maps, let's introduce a useful little device: if (M, ω, G, μ) is a Hamiltonian G -space, for each $x \in M$ let $\theta_x: \mathfrak{g} \rightarrow T_x^*M$ be given by $\theta_x(X) = \omega_x(X_x^\#, \cdot)$. In other words, we have that

$$\theta_x(X)(v) = \omega_x(X_x^\#, v) = d(\mu_c^X)_x(v) = d\mu_x(v)X,$$

so θ_x is the dual of the derivative $d\mu_x: T_xM \rightarrow \mathfrak{g}^*$. Thus, simple linear algebra gives us the:

Proposition 8

Let (M, ω, G, μ) be a Hamiltonian G -space. Then $\mu: M \rightarrow \mathfrak{g}^*$ is a submersion at $x \in M$ if and only if the stabilizer G_x is discrete.

Proof: Just note that $\text{Ran}(d\mu_x) = \text{Ann}(\ker \theta_x) = \text{Ann}(\mathfrak{g}_x)$, since ω_x is non-degenerate and we use item (b) of Exercise 33 (p. 51). So $d\mu_x$ is surjective if and only if $\mathfrak{g}_x = 0$. The conclusion follows because we assume that G is connected. $\square$

The next natural thing to do would be to analyze the kernel of $d\mu_x$.

Proposition 9

Let (M, ω, G, μ) be a Hamiltonian G -space. Then for each $x \in M$, we have that $\ker d\mu_x$ and $T_x(G \cdot x)$ are ω_x -orthogonal.

Proof: Recall that $T_x(G \cdot x) = \{X_x^\# \mid X \in \mathfrak{g}\}$ (this is a general fact about Lie group actions on smooth manifolds), so it suffices to note that if $X \in \mathfrak{g}$ and $v \in \ker d\mu_x$, we have $\omega_x(X_x^\#, v) = d\mu_x(v)X = 0$, because $d\mu_x(v)$ is the zero linear functional. $\square$

Let's leave no stone unturned:

Proposition 10

Let (M, ω, G, μ) be a Hamiltonian G -space. Then the moment map $\mu: M \rightarrow \mathfrak{g}^*$ is G -equivariant, where in $\mathfrak{g}^*$ we consider the **coadjoint action** $G \curvearrowright \mathfrak{g}^*$ given by $g \cdot p \doteq p \circ \text{Ad}(g^{-1})$. Namely, we have that

$$\mu(g \cdot x) = \mu(x) \circ \text{Ad}(g^{-1})$$

for all $x \in M$ and $g \in G$.

Proof: Since G is connected, we may proceed infinitesimally, and just prove that

$$d\mu_x(X_M^\#|_x) = X_{\mathfrak{g}^*}^\#|_{\mu(x)},$$

by showing that both sides yield the same result when tested against an arbitrary $Y \in \mathfrak{g}$. On one hand, we have that

$$\begin{aligned} d\mu_x(X_M^\#|_x)Y &= \theta_x(Y)(X_M^\#|_x) = \omega_x(Y_M^\#|_x, X_M^\#|_x) \\ &= \{\mu_c^Y, \mu_c^X\}_\omega(x) = \mu_c^{[Y, X]}(x) \\ &= \mu(x)([Y, X]). \end{aligned}$$

On the other hand, we may take an arbitrary element $p \in \mathfrak{g}^*$, the 1-parameter subgroup $\gamma_X: \mathbb{R} \rightarrow G$ given by $\gamma_X(t) = \exp(tX)$, and compute

$$\begin{aligned} X_{\mathfrak{g}^*}^\#|_p &= \frac{d}{dt} \Big|_{t=0} \gamma_X(t) \cdot p = \frac{d}{dt} \Big|_{t=0} p \circ \text{Ad}(\gamma_X(t)^{-1}) \\ &= \frac{d}{dt} \Big|_{t=0} p \circ \text{Ad}(\gamma_{-X}(t)) = p \circ \frac{d}{dt} \Big|_{t=0} \text{Ad}(\gamma_{-X}(t)) \\ &= p \circ \text{ad}(-X) = p([\cdot, X]), \end{aligned}$$

concluding the proof. $\square$

Exercise 41

Let G be a Lie group. Show that the coadjoint action $G \curvearrowright \mathfrak{g}^*$ is a Poisson action, when we equip $\mathfrak{g}^*$ with its standard Lie-Poisson structure. Namely, show that for all $f_1, f_2 \in \mathcal{C}^\infty(\mathfrak{g}^*)$ and $g \in G$, we have $\{f_1 \circ g, f_2 \circ g\}_{\mathfrak{g}^*} = \{f_1, f_2\}_{\mathfrak{g}^*} \circ g$.

Hint: Again it suffices to show that the conclusion holds for $f = \widehat{X}$ and $g = \widehat{Y}$, where $X, Y \in \mathfrak{g}$. Verify that $\widehat{X} \circ g = \widehat{\text{Ad}(g^{-1})X}$.

Remark. For dimension reasons, a Lie algebra need not carry a symplectic form. It turns out that the **coadjoint orbits** carry natural symplectic forms — they’re called the **Kirillov-Kostant-Souriau forms** (KKS forms, for short). The coadjoint action restricted to each orbit is Hamiltonian, and the moment map is given by the inclusion into $\mathfrak{g}^*$.

Let’s conclude this section with one of the most powerful theorems in Mathematical Physics.

Theorem 17 (Noether)

Let (M, ω, G, μ) be a Hamiltonian G -space, and $H \in \mathcal{C}^\infty(M)$ be G -invariant. Then μ is constant along the integral curves of the Hamiltonian field X_H .

Proof: Since H is G -invariant, its derivative kills all action fields. So, let $\gamma: I \rightarrow M$ be an integral curve of X_H . For any $X \in \mathfrak{g}$, compute:

$$\begin{aligned} (\mu \circ \gamma)'(t)X &= d\mu_{\gamma(t)}(\gamma'(t))X = \theta_{\gamma(t)}(X)(\gamma'(t)) \\ &= \omega_{\gamma(t)}(X_{\gamma(t)}^\#, \gamma'(t)) = \omega_{\gamma(t)}(X_{\gamma(t)}^\#, X_H|_{\gamma(t)}) \\ &= -dH_{\gamma(t)}(X_{\gamma(t)}^\#) = 0. \end{aligned}$$

□

The fact that a powerful and celebrated theorem like this has such an easy proof should not be cause for concern. This Hamiltonian formulation essentially sweeps the dirt under the rug, as applying this theorem to some mechanical system requires not only recognizing the underlying symplectic structure, but also recognizing a convenient group G of symmetries acting on the configuration or phase space. It goes by the slogan “**symmetries generate conservation laws**”.

As our first example of how to apply this, let’s put together everything we have learned so far.

Example 32 (Killing fields)

Let (Q, g) be a Riemannian manifold, and $X \in \mathfrak{X}(Q)$ be a **Killing field** (that is, each stage of the flow of X is an isometry of Q or, equivalently, the covariant differential ∇X is skew-adjoint, where ∇ is the Levi-Civita connection of (Q, g)). Let’s assume, for simplicity, that X is complete^a, so we obtain an isometric action $\mathbb{R} \curvearrowright (Q, g)$ given by its flow. This action lifts to an action $\mathbb{R} \curvearrowright (T^*Q, \omega_{\text{can}})$, which is Hamiltonian by item (3) of Example 30 (p. 57), with moment map $\mu: T^*Q \rightarrow \mathbb{R}^*$ given by

$$\mu(x, p)a = \lambda_{(x,p)}(a_{T^*Q}^\#|_{(x,p)}) = p(a_Q^\#|_x) = p(aX_x) = ap(X_x).$$

The Hamiltonian $H: T^*Q \rightarrow \mathbb{R}$ given by $H(x, p) = g_x(p, p)/2$ is $\mathbb{R}$ -invariant, and

we know that if $(x, p): I \rightarrow T^*Q$ is an integral curve of X_H , then $x: I \rightarrow Q$ is a geodesic, by Example 19 (p. 27). Noether's conclusion is that if $X \in \mathfrak{X}(Q)$ is a Killing field and $x: I \rightarrow Q$ is a geodesic, then the function $t \mapsto g_{x(t)}(X_{x(t)}, \dot{x}(t))$ is constant, since if $p \in T_x^*Q$ corresponds to $v \in T_xQ$, then $p(X_x)$ reads $g_x(X_x, v)$.

^aThis assumption can be dropped.

Exercise 42

Let (Q, g) be a Riemannian manifold, $A \in \Omega^1(Q)$ be a magnetic potential, and $X \in \mathfrak{X}(Q)$ be a Killing field such that $\mathcal{L}_X A = 0$. Show that if $\gamma: I \rightarrow Q$ is a magnetic geodesic (in the sense of Exercise 19, p. 28), then there is $c \in \mathbb{R}$ such that

$$g_{\gamma(t)}(\dot{\gamma}(t), X_{\gamma(t)}) = A_{\gamma(t)}(X_{\gamma(t)}) + c, \quad \text{for all } t \in I,$$

and conclude that

$$B_{\gamma(t)}(\dot{\gamma}(t), X_{\gamma(t)}) = (\nabla_{\dot{\gamma}(t)} A)(X_{\gamma(t)}) + A_{\gamma(t)}\left(\frac{DX}{dt}(t)\right), \quad \text{for all } t \in I.$$

To what extent is this constant c gauge invariant? **Hint:** You can assume that X is complete, for simplicity (which allows you to mimic what we just did on the previous example). See the remark after Example 30 (p. 57), and justify it.

Example 33

Consider again the Hamiltonian $SO(3)$ -space $(\mathbb{R}^6, \omega_6, SO(3), \times)$ seen in Example 31 (p. 59), and the Hamiltonian $H: \mathbb{R}^6 \rightarrow \mathbb{R}$ given by

$$H(x, y) = \frac{\|y\|^2}{2} + V(x),$$

where $V: \mathbb{R}^3 \rightarrow \mathbb{R}$ is a radial function. Then H is clearly $SO(3)$ -invariant. But integral curves of X_H have the form $(\gamma, \dot{\gamma}): I \rightarrow \mathbb{R}^6$, with $\gamma: I \rightarrow \mathbb{R}^3$ a solution to Newton's force equation $\ddot{\gamma}(t) = F(\gamma(t))$, where $F(x) \doteq -\nabla V(x)$ is the gradient of V . Noether's theorem now says that if $\gamma: I \rightarrow \mathbb{R}^3$ is a solution to Newton's force equation, the angular momentum $t \mapsto \gamma(t) \times \dot{\gamma}(t)$ is constant.

Exercise 43 (Challenge #4 - The Lagrangian World)

Let Q be a smooth manifold and $L: TQ \rightarrow \mathbb{R}$ be a **Lagrangian function**^a. For each $(x, v) \in T^*Q$, define the **fiber derivative** $\mathbb{F}L: TQ \rightarrow T^*Q$ by

$$\mathbb{F}L(x, v)w \doteq \left. \frac{d}{dt} \right|_{t=0} L(x, v + tw).$$

Say that L is **hyperregular** if $\mathbb{F}L$ is a diffeomorphism. If $G \curvearrowright Q$ is an action and L is G -invariant, show that for each $X \in \mathfrak{g}$, the **Noether charge** $\mathcal{J}^X: TQ \rightarrow \mathbb{R}$ given by $\mathcal{J}^X(x, v) \doteq \mathbb{F}L(x, v)X_x^\#$ is constant along critical points of the action functional

$$\mathcal{A}_L[x] = \int_a^b L(x(t), \dot{x}(t)) dt.$$

Hints: The lifted action $G \curvearrowright (TQ, (\mathbb{F}L)^*\omega_{\text{can}})$ is Hamiltonian, the moment map is the Noether charge, and integral curves of the Hamiltonian field of L computed with $(\mathbb{F}L)^*\omega_{\text{can}}$ project to critical points of the action functional of L . Alternatively, use that L is hyperregular and send everything to $(T^*Q, \omega_{\text{can}})$ with $\mathbb{F}L$ and that Hamilton's equations for the Legendre transform of L are equivalent to the Euler-Lagrange equations for L . Note that the conclusion still holds if L is not assumed to be hyperregular. See [38] for more details.

^aThe name only means that the domain is TQ , just like when we say that a Hamiltonian on Q is a function defined on T^*Q .

3.4 Marsden-Weinstein reduction

Let's revisit the issue with quotients: quotients of symplectic manifolds under symplectic actions do not need to be symplectic, for dimension reasons. But, again, what exactly is the problem caused by the dimension of the space being odd? Since derivatives commute with pull-backs, and the pull-back map induced by a surjective submersion is injective, whatever 2-form survives in the quotient will be closed. But in odd dimensions, it will be degenerate. Think of the linear setting:

Exercise 44

Let (V, Ω) be a symplectic vector space, and $S \subseteq V$ be a subspace.

- (a) If S is isotropic, Ω survives in the quotient S^Ω/S as a symplectic form.
- (b) If S is coisotropic, Ω survives in the quotient S/S^Ω as a symplectic form.
- (c) If S is any subspace, Ω survives in the quotient $S/(S \cap S^\Omega)$ as a symplectic form.

On the above exercise, obviously items (a) and (b) follow from (c), but it still instructive to do them. The point here is that $S \cap S^\Omega$ is the **radical** of $\Omega|_{S \times S}$, namely, the kernel of the map $S \rightarrow S^*$ induced by Ω , and quotienting out the radical leads to non-degeneracy in the quotient. Related to dimensions of kernels, are dimensions of ranges.

Definition 22

Let M be a smooth manifold and $\omega \in \Omega^2(M)$. The **rank** of ω at $x \in M$ is the dimension of the range of the map $T_x M \rightarrow T_x^* M$ induced by ω_x .

Note that the rank function is lower semicontinuous and, in particular, we see that symplectic forms have full rank. So, consider the situation where ω has constant rank, even if not full.

Definition 23

Let M be a smooth manifold and $\omega \in \Omega^2(M)$ have constant rank. The **radical distribution** of ω is the distribution $\text{rad}(\omega) \hookrightarrow TM$ given by

$$\text{rad}(\omega)_x \doteq \text{rad}(\omega_x) = \{v \in T_x M \mid \omega_x(v, \cdot) = 0\}.$$

The first thing (probably) anyone does when they come across a distribution of subspaces, is to ask themselves when is it integrable. So, one checks for involutivity directly: if the manifold M has a constant rank $\omega \in \Omega^2(M)$ and $X, Y \in \mathfrak{X}(M)$ are tangent to $\text{rad}(\omega)$, we may use Cartan's homotopy formula together with the identity mentioned in the hint for Exercise 25 (p. 41), and $\iota_X \omega = \iota_Y \omega = 0$, to compute:

$$\begin{aligned} \iota_{[X,Y]} \omega &= \mathcal{L}_X(\iota_Y \omega) - \iota_X(\mathcal{L}_Y \omega) \\ &= \mathcal{L}_X(\iota_Y \omega) - \iota_X(\iota_Y(d\omega) + d(\iota_Y \omega)) \\ &= -d\omega(X, Y). \end{aligned}$$

Proposition 11

Let M be a smooth manifold and $\omega \in \Omega^2(M)$ have constant rank. Then $\text{rad}(\omega)$ is integrable if and only if $d\omega$ annihilates $\text{rad}(\omega)$ — which in particular happens when ω is closed.

This motivates the following definition:

Definition 24

Let M be a smooth manifold. A 2-form $\omega \in \Omega^2(M)$ is called **pre-symplectic** if it is closed and has constant rank. We'll say that the pair (M, ω) is a **pre-symplectic manifold**. The **radical foliation** $\mathcal{F}_\omega$ of (M, ω) consists of the leaves of $\text{rad}(\omega)$ and, we say that $\mathcal{F}_\omega$ is **simple** if the quotient leaf space $M/\mathcal{F}_\omega$ has the structure of a smooth manifold for which the projection $\pi_\omega: M \rightarrow M/\mathcal{F}_\omega$ is a surjective submersion.

Example 34

- (1) Let (M, ω) be a symplectic manifold. Then it is obviously pre-symplectic, $\text{rad}(\omega)$ is trivial, and $\mathcal{F}_\omega$ consists of points, so $M/\mathcal{F}_\omega$ is just M itself.
- (2) Consider the vector space $\mathbb{R}^{2n+r} = \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R}^r$ and define $\omega_{2n,r} \in \Omega^2(\mathbb{R}^{2n+r})$ by assigning to each tangent space to $\mathbb{R}^{2n+r}$ the skew-symmetric bilinear map $\Omega_{2n,r}: \mathbb{R}^{2n+r} \times \mathbb{R}^{2n+r} \rightarrow \mathbb{R}$ given by

$$\Omega_{2n,r}((x, y, z), (x', y', z')) \doteq \langle x, y' \rangle - \langle x', y \rangle,$$

where $\langle \cdot, \cdot \rangle$ is the standard Euclidean inner product in $\mathbb{R}^n$. Then $\omega_{2n,r}$ is pre-symplectic, we have that $\text{rad}(\omega_{2n,r})_{(x,y,z)} \cong \mathbb{R}^r$, and the leaf of $\mathcal{F}_{\omega_{2n,r}}$ passing through (x, y, z) is $\{(x, y)\} \times \mathbb{R}^r$. The quotient $\mathbb{R}^{2n+r}/\mathcal{F}_{\omega_{2n,r}}$ is just $(\mathbb{R}^{2n}, \omega_{2n})$.

Exercise 45

Let (V, Ω) be a pre-symplectic vector space. Show that V admits a basis

$$\mathcal{B} = (e_1, \dots, e_n, f_1, \dots, f_n, h_1, \dots, h_r)$$

such that for all $i, j = 1, \dots, n$ and $k, \ell = 1, \dots, r$ we have

$$\Omega(e_i, e_j) = \Omega(f_i, f_j) = \Omega(e_i, h_k) = \Omega(f_j, h_k) = \Omega(h_k, h_\ell) = 0 \quad \text{and} \quad \Omega(e_i, f_j) = \delta_{ij}.$$

Or, in other words, the matrix of Ω relative to $\mathcal{B}$ is $-J_{2n} \oplus 0_r$. The number r is intrinsic to (V, Ω) , identify it. You can call this $\mathcal{B}$ a **pre-Darboux basis**.

The fact that in the two previous examples the leaf spaces turned out to be symplectic was not a coincidence.

Proposition 12

Let (M, ω) be a pre-symplectic manifold with simple radical distribution. Then $M/\mathcal{F}_\omega$ has a unique symplectic structure $\tilde{\omega} \in \Omega^2(M/\mathcal{F}_\omega)$ such that $(\pi_\omega)^*\tilde{\omega} = \omega$.

Proof: Given any point $x \in M$, denote by $[x]$ the leaf of $\mathcal{F}_\omega$ passing through x . If $\pi_\omega: M \rightarrow M/\mathcal{F}_\omega$ is the quotient projection, as before, consider now the derivative $d(\pi_\omega)_x: T_x M \rightarrow T_{[x]}(M/\mathcal{F}_\omega)$. By definition of the leaf space, we have that its kernel is $\ker d(\pi_\omega)_x = T_x[x] = \text{rad}(\omega_x)$. Thus, we get an isomorphism

$$T_{[x]}(M/\mathcal{F}_\omega) \cong T_x M / \text{rad}(\omega_x),$$

and so ω_x passes to the quotient as a symplectic linear form $\tilde{\omega}_x$ on $T_{[x]}(M/\mathcal{F}_\omega)$. This defines a smooth $\tilde{\omega} \in \Omega^2(M/\mathcal{F}_\omega)$, which is non-degenerate and satisfies $(\pi_\omega)^*\tilde{\omega} = \omega$ by construction, and is closed since

$$0 = d\omega = d((\pi_\omega)^*\tilde{\omega}) = (\pi_\omega)^*(d\tilde{\omega}) \implies d\tilde{\omega} = 0,$$

as π_ω^* is injective. The condition $(\pi_\omega)^*\tilde{\omega} = \omega$ forces such $\tilde{\omega}$ to be unique. $\square$

Another way to produce symplectic quotients, apart from dealing with pre-symplectic objects, is to explore Hamiltonian actions further. Let (M, ω, G, μ) be a Hamiltonian G -space, and take a regular value $p \in \mathfrak{g}^*$ of $\mu: M \rightarrow \mathfrak{g}^*$, so that the inverse image $\mu^{-1}(p)$ is an embedded submanifold of M . Since we're always assuming that G is connected, μ is G -equivariant by Proposition 10 (p. 61). Now, the stabilizer G_p acts on $\mu^{-1}(p)$, for if $x \in \mu^{-1}(p)$, we have

$$\mu(g \cdot x) = \mu(x) \circ \text{Ad}(g^{-1}) = p \circ \text{Ad}(g^{-1}) = p,$$

so $g \cdot x \in \mu^{-1}(p)$ as well. Write $\iota_p: \mu^{-1}(p) \hookrightarrow M$ for the inclusion. The pull-back $(\iota_p)^*\omega$ is surely closed, but as you might expect, not necessarily non-degenerate. All we have seen so far would come full circle if $(\iota_p)^*\omega$ were pre-symplectic. But it is.

Lemma 4

Let (M, ω, G, μ) be a Hamiltonian G -space and $p \in \mathfrak{g}^*$ be a regular value of μ . Then for all $x \in \mu^{-1}(p)$, we have that

$$\text{rad}((\iota_p)^*\omega)_x = T_x(G_p \cdot x),$$

so that $\dim \text{rad}((\iota_p)^*\omega)_x = \dim G_p$ is independent of x , and thus $(\mu^{-1}(p), (\iota_p)^*\omega)$ is pre-symplectic.

Proof: Let's use the definition of radical with the result of Proposition 9 (p. 61) to compute

$$\begin{aligned} \text{rad}((\iota_p)^*\omega)_x &= T_x(\mu^{-1}(p)) \cap (T_x(\mu^{-1}(p)))^{\omega_x} \\ &= \ker d\mu_x \cap ((T_x(G \cdot x))^{\omega_x})^{\omega_x} \\ &= \ker d\mu_x \cap T_x(G \cdot x) \\ &= T_x(G_p \cdot x), \end{aligned}$$

as required. The very last step requires a quick explanation: if $v \in T_x(G \cdot x)$, then $v = (X_M^\#)_x$ for some $X \in \mathfrak{g}$ and, with this in place, $v \in \ker d\mu_x$ if and only if $(X_g^\#)_p = 0$ by Proposition 10 (p. 61) — which is equivalent to having $X \in \mathfrak{g}_p$ by item (b) of Exercise 33 (p. 51), meaning that $v \in T_x(G_p \cdot x)$. $\square$

Theorem 18 (Marsden-Weinstein, Meyer)

Let (M, ω, G, μ) be a Hamiltonian G -space and $p \in \mathfrak{g}^*$ be a regular value for μ . Assume that the action $G_p \curvearrowright \mu^{-1}(p)$ is free and proper. Then the quotient $\mu^{-1}(p)/G_p$ has a unique symplectic form $\omega_\circ$ characterized by the relation $\pi^*\omega_\circ = (\iota_p)^*\omega$, where $\pi: \mu^{-1}(p) \rightarrow \mu^{-1}(p)/G_p$ is the quotient projection. We call $(\mu^{-1}(p)/G_p, \omega_\circ)$ the **reduction of (M, ω) to level p** .

Proof: The space $\mu^{-1}(p)/G_p$ is a smooth manifold because the action is assumed to be free and proper. Now the 2-form $(i_p)^*\omega$ in $\mu^{-1}(p)$ is pre-symplectic and the leaves of radical foliation $\mathcal{F}_{(i_p)^*\omega}$ are precisely the G_p -orbits. We are done, by Proposition 12 (p. 66).

$$\begin{array}{ccc}
 & & (M, \omega) \\
 & \nearrow \iota_p & \\
 (\mu^{-1}(p), (i_p)^*\omega) & & \\
 & \searrow \pi & \\
 & & (\mu^{-1}(p)/G_p, \omega_\circ)
 \end{array}$$

□

Remark. When the Lie group G is abelian, the adjoint representation is trivial, so we have that $G = G_p$ for all $p \in \mathfrak{g}^*$. And, for $p = 0$, we have that $G_0 = G$. In this case, $\mu^{-1}(0) \rightarrow \mu^{-1}(0)/G$ turns out to be a principal G -bundle.

Just like what happened with Noether's theorem, we again see a strong theorem with a very simple proof. In practice, one wants to explicitly describe the **reduced form** $\omega_\circ$. Let's see how one can usually try to do this in practice.

Example 35

- (1) Let (M, ω) be a symplectic manifold, and $H: M \rightarrow \mathbb{R}$ be a smooth function for which the field X_H is complete. We have seen that $(M, \omega, \mathbb{R}, \mu)$ is a Hamiltonian $\mathbb{R}$ -space, where $\mu = H$ and the action is given by Hamiltonian flows. So, for each $e \in \mathbb{R}$, each quotient $\Sigma_e/\mathbb{R}$ of an energy level which turns out to be a manifold automatically inherits a symplectic structure $\omega_\circ$. Explicitly, if $L \in \Sigma_e/\mathbb{R}$, and $\tilde{v}, \tilde{w} \in T_L(\Sigma_e/\mathbb{R})$, we have

$$(\omega_\circ)_L(\tilde{v}, \tilde{w}) = \omega_x(v, w),$$

where $x \in L$ and $v, w \in T_x \Sigma_e$ are such that $d\pi_x(v) = \tilde{v}$, and similarly for $\tilde{w}$ — where $\pi: \Sigma_e \rightarrow \Sigma_e/\mathbb{R}$ is the quotient projection. And as a quick sanity check, do note that $\dim(\Sigma_e/\mathbb{R}) = \dim M - 2$ is even.

- (2) Consider the circular Hamiltonian action of S^1 on $\mathbb{C}^{n+1} \cong \mathbb{R}^{2n+2}$, equipped with ω_{2n+2} — we have seen that the moment map is $\mu: \mathbb{C}^{n+1} \rightarrow (i\mathbb{R})^*$ given by

$$\mu(z) = -\frac{\|z\|^2}{2}.$$

In particular, $\mu^{-1}(-1/2) = S^{2n+1}$. Hence the quotient $\mathbb{CP}^n = S^{2n+1}/S^1$ inherits a symplectic form $\omega_{\text{FS}} \in \Omega^2(\mathbb{CP}^n)$ — called the **Fubini-Study form**. There is something slightly deeper going on here, in the following way: S^{2n+1} inherits a Riemannian metric from $\mathbb{R}^{2n+2}$ and the S^1 -action consists of isometries, so there is a unique Riemannian metric g_{FS} (called the **Fubini-Study metric**)

on the quotient $\mathbb{S}^{2n+1}/\mathbb{S}^1$ for which the quotient projection $\mathbb{S}^{2n+1} \rightarrow \mathbb{S}^{2n+1}/\mathbb{S}^1$ is a **Riemannian submersion** (see [32] for more on those). With the natural complex structure J that $\mathbb{CP}^n$ has, the relation $g_{\text{FS}} = \omega_{\text{FS}}(\cdot, J\cdot)$ holds, making $\mathbb{CP}^n$ a Kähler manifold.

- (3) Let $(M, -d\lambda)$ be an exact symplectic manifold, and $G \curvearrowright M$ be a Lie group action preserving λ – hence automatically Hamiltonian by item (3) of Example 30 (p. 57). Assume that $0 \in \mathfrak{g}^*$ is a regular value of the moment map μ and that we are under the conditions of Theorem 18 (p. 67). In this case, the reduction of $(M, -d\lambda)$ to level 0 is called the **Marsden-Weinstein reduction** of $(M, -d\lambda)$ and it is denoted by $M//G$. Let's show that the reduced symplectic structure $(-d\lambda)_\circ$ on $M//G$ is also exact. Write $\iota: \mu^{-1}(0) \hookrightarrow M$ for the inclusion, $\pi: \mu^{-1}(0) \rightarrow M//G$ for the projection, and consider $\iota^*\lambda \in \Omega^1(\mu^{-1}(0))$. Clearly $\iota^*\lambda$ is G -invariant, but we also have that if $x \in \mu^{-1}(0)$ and $v \in \ker d\mu_x$ satisfies $d\pi_x(v) = 0$, then $(\iota^*\lambda)_x(v) = 0$, since $x \in \mu^{-1}(0)$ says that λ_x annihilates $T_x(G \cdot x) = \ker d\pi_x$. Hence there is $\lambda_\circ \in \Omega^1(M//G)$ such that $\pi^*(\lambda_\circ) = \iota^*\lambda$. Once this is in place, we have that

$$\pi^*((d\lambda)_\circ) = \iota^*(d\lambda) = d(\iota^*\lambda) = d(\pi^*\lambda_\circ) = \pi^*(d\lambda_\circ),$$

and since π being a surjective submersion implies that π^* is injective, it follows that $(-d\lambda)_\circ = -d\lambda_\circ$ is exact.

- (4) Let (M, ω, G, μ) be a Hamiltonian G -space under the conditions of Theorem 18 (p. 67), $p \in \mathfrak{g}^*$ be a regular value of the moment map μ , and $H: M \rightarrow \mathbb{R}$ be a G -invariant Hamiltonian. By G -invariance of H , we know that the restriction $H|_{\mu^{-1}(p)}: \mu^{-1}(p) \rightarrow \mathbb{R}$ passes to the quotient as a function $H_p: \mu^{-1}(p)/G_p$, namely, satisfying $H_p \circ \pi = H|_{\mu^{-1}(p)}$, where $\pi: \mu^{-1}(p) \rightarrow \mu^{-1}(p)/G_p$ is the quotient projection. But there's more to it. First, we claim that the Hamiltonian field X_H is tangent to $\mu^{-1}(p)$ along its points. More precisely, if $x \in \mu^{-1}(p)$ and γ is the integral curve of H starting at x , we have that

$$d\mu_x(X_H|_x) = \left. \frac{d}{dt} \right|_{t=0} \mu(\gamma(t)) = 0$$

by Noether's theorem, so $X_H|_x \in T_x\mu^{-1}(p)$. Next, G -invariance of H and ω also gives us that $d(L_g)_x(X_H|_x) = X_H|_{g \cdot x}$, since for each $v \in T_xM$ and $g \in G$ we know that $H \circ L_g = H$ implies $dH_{g \cdot x} \circ d(L_g)_x = dH_x$, so evaluating at v gives

$$\omega_x(X_H|_x, v) = \omega_{g \cdot x}(X_H|_{g \cdot x}, d(L_g)_x(v)) = \omega_x(d(L_{g^{-1}})_{g \cdot x} X_H|_{g \cdot x}, v).$$

Thus $X_H|_x = d(L_{g^{-1}})_{g \cdot x} X_H|_{g \cdot x}$ implies that $d(L_g)_x(X_H|_x) = X_H|_{g \cdot x}$, as claimed. Hence there is a vector field $\tilde{X} \in \mathfrak{X}(\mu^{-1}(p)/G_p)$ such that $d\pi \circ X_H = \tilde{X} \circ \pi$, and we claim that this field $\tilde{X}$ is nothing more than X_{H_p} , computed using the reduced symplectic form $\omega_\circ$. Let's check this, identifying ω with its pull-back

and H with its restriction, both to $\mu^{-1}(p)$. Again, we'll use that π^* is injective, so that

$$\begin{aligned}\pi^* \iota_{\tilde{X}} \omega_\circ &= \iota_{X_H} (\pi^* \omega_\circ) = \iota_{X_H} \omega \\ &= dH = d(H_p \circ \pi) \\ &= dH_p \circ d\pi = \pi^* (dH_p) \\ &= \pi^* \iota_{X_{H_p}} \omega_\circ,\end{aligned}$$

meaning that $\iota_{\tilde{X}} \omega_\circ = \iota_{X_{H_p}} \omega_\circ$ finally yields $\tilde{X} = X_{H_p}$ due to non-degeneracy of $\omega_\circ$.

Exercise 46 (Passing differential forms to action quotients)

Let's clarify one part of the argument used in item (3) of the previous example. Assume that M is a smooth manifold, $G \curvearrowright M$ is a free and proper Lie group action, with quotient projection $\pi: M \rightarrow M/G$, and $\alpha \in \Omega^k(M)$ is:

- G -invariant, and;
- **G -horizontal**, in the sense that for all $x \in M$ and $v_1, \dots, v_k \in T_x M$, we have that $\alpha_x(v_1, \dots, v_k) = 0$ whenever there is some i such that $v_i \in \ker d\pi_x$.

Show that there is a unique $\tilde{\alpha} \in \Omega^k(M/G)$ such that $\pi^* \tilde{\alpha} = \alpha$.

Hint: there is only one choice of definition for $\tilde{\alpha}$, but you need to check that the expression you're forced to consider is indeed well-defined; do the case $k = 1$ first to get intuition, and then telescope a certain difference for the general case.

Remark: note that requiring α to be G -horizontal is to be expected, as we have the isomorphism $T_{G \cdot x}(M/G) \cong T_x M / \ker d\pi_x$ — G -invariance alone is not strong enough to ensure α survives in M/G .

In particular, reducing Hamiltonians on phase spaces is a useful technique for reducing the number of degrees of freedom of a mechanical system.

Example 36 (Mechanics on the plane)

Consider, in $\mathbb{R}^2$, the motion of a particle with mass $m > 0$ under the action of a potential function $V: \mathbb{R}^2 \rightarrow \mathbb{R}$. Such motion is described by the integral curves of the Hamiltonian field of the total energy $H: T^*\mathbb{R}^2 \cong \mathbb{R}^4 \rightarrow \mathbb{R}$ given by

$$H(x, y, p_x, p_y) = \frac{1}{2m}(p_x^2 + p_y^2) + V(x, y).$$

Assume that V has rotational symmetry, that is, that V is invariant under the action $S^1 \curvearrowright \mathbb{R}^2$ given by rotation. The action $S^1 \curvearrowright \mathbb{R}^4$ given by cotangent lifts

is Hamiltonian with moment map $\mu: \mathbb{R}^4 \rightarrow \mathbb{R}$ given by

$$\mu(x, y, p_x, p_y) = -yp_x + xp_y.$$

Indeed, by item (3) of Example 30 (p. 57), and identifying $\text{Lie}(\mathbb{S}^1) = i\mathbb{R} \cong \mathbb{R} \cong \mathbb{R}^*$ by setting $\mu(x, y, p_x, q_y) \doteq \mu(x, y, p_x, p_y)(1)$, we just have that

$$\mu(x, y, p_x, p_y) = (p_x dx + p_y dy)|_{(x,y)}(-y\partial_x|_{(x,y)} + x\partial_y|_{(x,y)}) = -yp_x + xp_y,$$

as claimed. It's easy to see that every $a \in \mathbb{R}$, $a \neq 0$, is a regular value for μ . Using polar coordinates $x = r \cos \theta$ and $y = r \sin \theta$, a straightforward computation shows that

$$p_x = p_r \cos \theta - \frac{p_\theta \sin \theta}{r} \quad \text{and} \quad p_y = p_r \sin \theta + \frac{p_\theta \cos \theta}{r},$$

so that

$$H(r, \theta, p_r, p_\theta) = \frac{1}{2m} \left(p_r^2 + \frac{p_\theta^2}{r^2} \right) + V(r) \quad \text{and} \quad \mu(r, \theta, p_r, p_\theta) = p_\theta.$$

By Noether's theorem, the angular momentum p_θ is constant along the path the particle follows. Given $a \in \mathbb{R}$, $a \neq 0$, the level set $\mu^{-1}(a)$ consists of all possible initial conditions for the particle for which the angular momentum is constant and equal to a . Only the coordinates r and p_r survive in the quotient $\mu^{-1}(a)/\mathbb{S}^1$, and the reduced Hamiltonian is $H_a: \mu^{-1}(a)/\mathbb{S}^1 \rightarrow \mathbb{R}$ given by

$$H_a(r, p_r) = \frac{p_r^2}{2m} + V_{\text{eff}}(r),$$

where $V_{\text{eff}}: \mu^{-1}(a)/\mathbb{S}^1 \rightarrow \mathbb{R}$ given by

$$V_{\text{eff}}(r) = \frac{a^2}{2mr^2} + V(r)$$

is the **effective potential**.

Remark. Interesting dynamics are also induced by very complicated potentials. One example is the **Henon-Heiles potential** $V_{\text{HH},\lambda}: \mathbb{R}^2 \rightarrow \mathbb{R}$, given by

$$V_{\text{HH},\lambda}(x, y) = \frac{1}{2m}(x^2 + y^2) + \lambda \left(x^2 y - \frac{y^3}{3} \right).$$

Moving through energy levels, one even has chaotic behavior. It's a very frequent illustrative example in two-dimensional dynamics, usually studied with numerical techniques.

Exercise 47 (Mechanics on the sphere)

Consider, in the sphere S^2 , the motion of a particle with mass $m > 0$ under the action of a potential function $V: S^2 \rightarrow \mathbb{R}$. Such motion is described by the integral curves of the Hamiltonian field of the total energy $H: T^*S^2 \rightarrow \mathbb{R}$, given by

$$H(x, y, z, p_x, p_y, p_z) = \frac{1}{2m}(p_x^2 + p_y^2 + p_z^2) + V(x, y, z).$$

Of course, we have (by unwittingly identifying $TS^2 \cong T^*S^2$) that

$$T^*S^2 = \{(x, y, z, p_x, p_y, p_z) \in T^*\mathbb{R}^3 \mid x^2 + y^2 + z^2 = 1 \text{ and } xp_x + yp_y + zp_z = 0\}.$$

Consider the circular action $S^1 \curvearrowright S^2$, as in item (2) of Example 26 (p. 49), and assume that V is S^1 -invariant — which is to say that V only depends on z . Take spherical coordinates

$$x = \cos \theta \cos \phi, \quad y = \sin \theta \cos \phi, \quad \text{and} \quad z = \sin \phi,$$

where θ ranges from 0 to 2π and ϕ from $-\pi/2$ to $\pi/2$.

(a) Show that

$$p_x = -p_\theta \frac{\sin \theta}{\cos \phi} - p_\phi \cos \theta \sin \phi, \quad p_y = p_\theta \frac{\cos \theta}{\cos \phi} - p_\phi \sin \theta \sin \phi,$$

$$\text{and } p_z = p_\phi \cos \phi.$$

(b) Conclude that

$$H(\theta, \phi, p_\theta, p_\phi) = \frac{1}{2m} \left(\frac{p_\theta^2}{\cos^2 \phi} + p_\phi^2 \right) + V(\phi) \quad \text{and} \quad \mu(\theta, \phi, p_\theta, p_\phi) = p_\theta.$$

By Noether, p_θ is again a conserved quantity.

(c) Show that for each regular value $a \in \mathbb{R}$ of μ , the reduced Hamiltonian and the effective potential $H_a, V_{\text{eff}}: \mu^{-1}(a)/S^1 \rightarrow \mathbb{R}$ are given by and related via

$$H_a(\phi, p_\phi) = \frac{p_\phi^2}{2m} + V_{\text{eff}}(\phi) \quad \text{and} \quad V_{\text{eff}}(\phi) = \frac{a^2}{2m \cos^2 \phi} + V(\phi).$$

Exercise 48 (Mechanics on the cylinder)

Consider, in the sphere $S^1 \times \mathbb{R}$, the motion of a particle with mass $m > 0$ under the action of a potential function $V: S^1 \times \mathbb{R} \rightarrow \mathbb{R}$. Again, such motion will be described by the integral curves of the Hamiltonian field of the total energy

$H: T^*(\mathbb{S}^1 \times \mathbb{R}) \rightarrow \mathbb{R}$, given by

$$H(x, y, z, p_x, p_y, p_z) = \frac{1}{2m}(p_x^2 + p_y^2 + p_z^2) + V(x, y, z),$$

and we may write (again with the aid of the metric) that

$$T^*(\mathbb{S}^1 \times \mathbb{R}) = \{(x, y, z, p_x, p_y, p_z) \in T^*\mathbb{R}^3 \mid x^2 + y^2 = 1 \text{ and } xp_x + yp_y = 0\}.$$

Take cylindrical coordinates $x = r \cos \theta$, $y = r \sin \theta$, $z = z$.

- (a) Show that $p_\theta = -p_x \sin \theta + p_y \cos \theta$, and conclude that in such coordinates, the Hamiltonian is given by

$$H(\theta, z, p_\theta, p_z) = \frac{1}{2m}(p_\theta^2 + p_z^2) + V(\theta, z).$$

Note we did not impose conditions on V , nor considered symmetries yet.

- (b) Consider the action $\mathbb{S}^1 \curvearrowright \mathbb{S}^1 \times \mathbb{R}$ given by rotation and assume that the potential V is $\mathbb{S}^1$ -invariant, which is to say that it depends only on z . Show that the moment map $\mu: T^*(\mathbb{S}^1 \times \mathbb{R}) \rightarrow \mathbb{R}$ is given by $\mu(\theta, z, p_\theta, p_z) = p_\theta$. Conclude that for each regular value $a \in \mathbb{R}$ of μ , we have

$$H_a(z, p_z) = \frac{p_z^2}{2m} + V_{\text{eff}}(z), \quad \text{with} \quad V_{\text{eff}}(z) = \frac{a^2}{2m} + V(z)$$

on the quotient $\mu^{-1}(a)/\mathbb{S}^1$.

- (c) Consider the action $\mathbb{R} \curvearrowright \mathbb{S}^1 \times \mathbb{R}$ given by translations and assume that the potential V is $\mathbb{R}$ -invariant, which is to say it depends only on θ . Show that the moment map $\mu: T^*(\mathbb{S}^1 \times \mathbb{R}) \rightarrow \mathbb{R}$ is given by $\mu(\theta, z, p_\theta, p_z) = p_z$. Conclude that for each regular value $a \in \mathbb{R}$ of μ , we have

$$H_a(\theta, p_\theta) = \frac{p_\theta^2}{2m} + V_{\text{eff}}(\theta), \quad \text{with} \quad V_{\text{eff}}(\theta) = \frac{a^2}{2m} + V(\theta)$$

on the quotient $\mu^{-1}(a)/\mathbb{R}$.

We'll conclude the discussion with one last "fake" example:

Example 37

Let's see that what would be a linear Marsden-Weinstein quotient is in fact trivial. Namely, let (V, Ω) be a symplectic vector space, and let ω be the constant symplectic form, which assigns to each tangent space $T_x V \cong V$, $\omega_x \doteq \Omega$ itself. We have the obvious evaluation action $\text{Sp}(V, \Omega) \curvearrowright V$. Recall that the Lie algebra

$$\mathfrak{sp}(V, \Omega) = \{\psi \in \mathfrak{gl}(V) \mid \Omega(\psi(x), y) + \Omega(x, \psi(y)) = 0, \text{ for all } x, y \in V\}$$

consists of the so-called Hamiltonian operators. Important examples come as the velocities of curves of symplectic transvections, namely, $\dot{\tau}_v$ given by

$$\dot{\tau}_v(x) = \frac{d}{d\lambda} \Big|_{\lambda=0} \tau_{\lambda,v}(x) = \frac{d}{d\lambda} \Big|_{\lambda=0} (x + \lambda \Omega(x, v)v) = \Omega(x, v)v,$$

for any non-zero $v \in V$. Now, given $\psi \in \mathfrak{sp}(V, \Omega)$, we have $(\psi^\#)_x = \psi(x)$ because the original action is linear. Given any $(x, v) \in TV$, we have the relation $\omega_x((\psi^\#)_x, v) = \Omega(\psi(x), v)$, so that $\mu: V \rightarrow \mathfrak{sp}(V, \Omega)^*$ given by

$$\mu_c^\psi(x) = \frac{1}{2} \Omega(\psi(x), x)$$

does satisfy $d(\mu_c^\psi)_x = \omega_x((\psi^\#)_x, \cdot)$, using that ψ is Ω -skew. If $x \in \mu^{-1}(0)$, then $\mu_c^{\dot{\tau}_v}(x) = 0$ leads to $\Omega(x, v)^2 = 0$ for all $v \in V$, so $x = 0$ from non-degeneracy of Ω . But here a disaster happens: μ_c is not a comoment map, but instead is an anti-Poisson map. On one hand we have

$$\{\mu_c^{\psi_1}, \mu_c^{\psi_2}\}_\omega(x) = \omega_x((\psi_1^\#)_x, (\psi_2^\#)_x) = \Omega(\psi_1(x), \psi_2(x))$$

and on the other hand

$$\begin{aligned} \mu_c^{[\psi_1, \psi_2]}(x) &= \frac{1}{2} \Omega([\psi_1, \psi_2](x), x) \\ &= \frac{1}{2} (\Omega(\psi_1 \psi_2(x), x) - \Omega(\psi_2 \psi_1(x), x)) \\ &= -\Omega(\psi_1(x), \psi_2(x)). \end{aligned}$$

For more on symplectic reductions, see [13] and [26].

Where to go from here?

Now you have the basic Symplectic Geometry kit needed to move on with your mathematical life. Here are some topics you can try to explore next (perhaps in no particular order).

Symplectic Topology: We have seen that if an even-dimensional compact manifold is to admit a symplectic form, then all even degree de Rham cohomology spaces must be non-trivial, and the manifold has to be orientable — but those are not the only obstructions ($S^2 \times S^4$ is one possible witness). This raises the question of whether there is a complete obstruction. In dimension 2, the complete criterion is just orientability. In dimension 4, things already get much more complicated, and conditions must be imposed on the **Seiberg-Witten invariants** of the manifold (generally speaking, topology of 4-manifolds is a very complicated subject). In higher dimensions, some information is certainly encoded on the cohomology ring of the classifying space $B\mathrm{Sp}_{2n}(\mathbb{R})$ (via characteristic classes), but it seems not much is known about it. On the other hand, every symplectic manifold carries an almost-complex structure, which is compatible with ω , but not integrable in general — and there are also complex and almost-complex manifolds which carry no symplectic structure. In fact, given any group described with finitely many generators and relations is the fundamental group of a four-dimensional compact symplectic manifold. More on these relations between symplectic and complex geometry leads us to the next topic.

Some references: [3], [18], [28], [37], [39].

Kähler Geometry: It is the intersection of Riemannian geometry, complex geometry, and symplectic geometry. In the linear setting, we can say that a **Kähler vector space** is a quadruple (V, Ω, J, g) , where Ω is a symplectic structure, J is an almost complex structure which is also a symplectomorphism, and $g = \Omega(\cdot, J\cdot)$ is an inner product on V . We'll say that (Ω, J, g) is a **compatible triple** for V . A **Kähler basis** is a basis $\mathcal{B} = (e_1, \dots, e_n, Je_1, \dots, Je_n)$ which is both Darboux and orthonormal. If $S \subseteq V$ is a subspace, then $J[S^\Omega] = S^\perp$ holds. On the smooth setting, we say that a **Kähler manifold** is a quadruple (M, ω, J, g) where (ω, J, g) is a field of compatible triples on the tangent spaces to M , with ω closed and J integrable. This turns out to be equivalent to requiring a M to be a complex manifold equipped with a **hermitian metric** (i.e., a Riemannian metric for which $J^*g = g$) such that $\nabla J = 0$, where ∇ is the Levi-Civita connection. One can talk about things such as Lagrangian submanifolds, minimal and totally geodesic submanifolds, complex submanifolds, holomorphic curvature, and much more. Being complex manifolds, one can talk about complex differential forms of some bi-degree (k, ℓ) on any Kähler manifold, and talk about the **Dolbeault cohomology** spaces $H_{\bar{\partial}}^{k,\ell}(M)$ — in particular, the symplectic form ω itself defines a Dolbeault class $[\omega] \in H_{\bar{\partial}}^{1,1}(M)$. Kähler manifolds are also thoroughly studied in Algebraic Geometry, appearing very prominently as submanifolds of complex projective spaces.

Some references: [5], [19], [31], [45], etc.

Geometric Quantization: A **quantization procedure** consists in passing from a classical description of a physical system, to a quantum description. For example, if Q is the configuration space for a classical mechanical system and T^*Q is the phase space of positions and momenta, classical observables (by which we mean “physical quantities”) are simply functions $f: T^*Q \rightarrow \mathbb{R}$, which take a state (x, p) and return the value $f(x, p)$ of the physical quantity represented by f when the system is at state (x, p) (for example, the energy of the system when it is at said state). In the quantum setting, measurements of physical quantities — now understood as quantum observables — only give us statistical distributions of possible values instead of a determined value. The spectrum of such values is encoded as the mathematical spectrum of a symmetric operator on a complex Hilbert space, which will now play the role of the quantum state space. Namely, symmetric operators have real eigenvalues, and this is perhaps the first motivation for the use of functional analysis tools in the study of quantum mechanics. However, is it possible that two distinct classical descriptions give rise to the same quantum description under quantization, and so each such classical description is called a **classical limit**. In any case, T^*Q carries its standard symplectic structure ω_{can} , and one might wonder what this becomes after quantization. Since symplectomorphisms should naturally correspond to unitary operators (because both preserve the structure of whatever space they’ll act on) and the Poisson bracket of two classical observables should correspond to the commutator of the corresponding quantum observables, one can formulate the quantization problem in terms of category theory, by trying to find a functor from the core of symplectic category to the core of the category of Hilbert spaces. It turns out that imposing even weak assumptions on the spaces considered (motivated by the behavior of Schrödinger’s equation) prohibits the existence of such a functor (this is called **Van Hove’s theorem**). The next keyword, then, is **pre-quantization**. Again, this has more than one possible formulation. One can say that a prequantization of a symplectic manifold (M, ω) with integral class $[\omega]$ is a (complex) Hermitian line bundle $(\mathcal{L}, \langle \cdot, \cdot \rangle) \rightarrow M$ with a $U(1)$ -connection $\nabla: \mathfrak{X}(M) \times \Gamma(\mathcal{L}) \rightarrow \Gamma(\mathcal{L})$ parallelizing $\langle \cdot, \cdot \rangle$ and satisfying $R^\nabla = \omega$ (in the sense that for all $X, Y \in \mathfrak{X}(M)$ and $\psi \in \Gamma(\mathcal{L})$, the relation $R^\nabla(X, Y)\psi = \omega(X, Y)\psi$ holds) — taking the associated Hilbert space to be the space of (suitably normalized) L^2 -sections of $\mathcal{L}$ (to be thought of as wavefunctions), one obtains a (pre)quantum description satisfying most of the Dirac’s axioms for quantization procedures.

Some references: [2], [16], [40], [43], etc.

Contact Geometry: A **contact manifold** is a pair (M, Σ) , where $\Sigma \hookrightarrow TM$ is a non-integrable distribution of hyperplanes — called a **contact distribution**. Up to a non-zero functional multiple, choosing Σ amounts to choosing a non-vanishing 1-form $\alpha \in \Omega^1(M)$ such that $\Sigma = \ker \alpha$ and $d\alpha_x$ is non-degenerate on Σ_x , for all $x \in M$ — then α is called a **contact form**, and one often writes (M, α) instead of (M, Σ) . Darboux’s theorem implies that the dimension of a contact manifold must be odd, and then non-degenerability can be rephrased as $\alpha \wedge (d\alpha)^{\wedge n} \neq 0$. The **Reeb field** of (M, α) is the unique vector field $\xi \in \mathfrak{X}(M)$ satisfying $\alpha(\xi) = 1$

and $d\alpha(\xi, \cdot) = 0$ (the latter condition does not contradict non-degenerability of $d\alpha$ on Σ because ξ is transverse to Σ , not tangent). This field helps control the geometry of (M, α) , as $\mathcal{L}_\xi \alpha = 0$ by Cartan's homotopy formula. A diffeomorphism between contact manifolds preserving the contact forms is called a **contactomorphism**. One factory of examples comes from symplectic geometry: if $(M, -d\lambda)$ is an exact symplectic manifold, then $(M \times \mathbb{R}, \pi^*\lambda + dt)$ is a contact manifold, where $\pi: M \times \mathbb{R} \rightarrow M$ and $t: M \times \mathbb{R} \rightarrow \mathbb{R}$ are the projections. On the other hand, if (M, α) is a contact manifold, then $(M \times \mathbb{R}, d(e^t \alpha))$ is a symplectic manifold. The contact analogues of Lagrangian submanifolds are called **Legendrian submanifolds**. In summary, contact geometry is the natural odd-dimensional analogue of symplectic geometry, with applications in differential equations, control theory, geometric quantization, among others.

Some references: [10], [26], [29], etc.

Symplectic Connections: Given a symplectic manifold (M, ω) , a **symplectic connection** is a torsionfree connection ∇ which parallelizes ω . If S is any tensor field of type $(1, 2)$, and we define a connection ∇' by $\nabla'_X Y = \nabla_X Y + S_X Y$, then a very short computation shows that ∇' is symplectic if and only if the trilinear map $(X, Y, Z) \mapsto \omega(S_X Y, Z)$ is fully symmetric; since the difference of any two connections is a tensor, we conclude that the space of symplectic connections on (M, ω) is an affine space whose translation space is $\Gamma(T^*M^{\odot 3})$. This is very different from what happens in Riemannian geometry, where a metric uniquely determines its Levi-Civita connection. It is not possible to make a “canonical” choice of symplectic connection without having extra structure on the manifold (e.g., being Kähler, pseudo-Kähler, or a symmetric symplectic space). Generally, the Ricci endomorphism r^∇ of a symplectic connection (characterized by $\text{Ric}^\nabla(X, Y) = \omega(X, r^\nabla(Y))$) carries relevant geometric information (even though what would be the “scalar curvature” vanishes as ω is skew), and one usually looks for the so-called **preferred connections**, satisfying the cyclic identity $(\nabla_X r^\nabla)(Y, Z) + (\nabla_Y r^\nabla)(Z, X) + (\nabla_Z r^\nabla)(X, Y) = 0$. Such connections also have a variational characterization, and this entire theory has some links with geometric quantization. Symplectic connections are also useful in the study of **symplectic Lie groups**, that is, Lie groups equipped with left-invariant symplectic forms.

Some references: [6], [7], [9], [12], etc.

Mirror Symmetry: Briefly, a **Calabi-Yau manifold** is a compact and Ricci-flat Kähler manifold with vanishing first Chern class. Such manifolds are considered in **String Theory**, where point-like particles are treated as one-dimensional strings, and spacetimes have extra “hidden” dimensions. **Mirror symmetry** essentially discusses when two given Calabi-Yau manifolds yield the “same” physical theory, even though they might have different geometries. This is a relatively recent area of research in mathematical physics (from roughly around the 90's). Important tools and questions indeed come from symplectic geometry. Part of the challenge seems to be making the aforementioned equivalence mathematically precise. One such attempt is called **Homological mirror symmetry** (also known as **Kontsevich's conjecture**) — it discusses the so-called **Floer homology**: one

defines the Floer cochain groups as certain modules generated by transversal intersections of Lagrangian submanifolds of a given symplectic manifold, dualizes, and computes the homology of the resulting chain complex. The **Fukaya category** of a symplectic manifold has Lagrangian submanifolds as objects, and chain groups as morphisms. Understanding all of this requires a deep knowledge not only of symplectic geometry, but also of algebraic geometry, algebraic topology, category theory and, to some extent, actual Physics.

Some references: [3], [4], [22], etc.

Toric manifolds and Delzant polytopes: A **toric manifold** is simply a Hamiltonian $\mathbb{T}^k$ -space for which the action is effective. The **Atiyah-Guillemin-Sternberg convexity theorem** says that the set of fixed points of the action on a compact and connected toric manifold is a finite union of connected symplectic submanifolds, the moment map is constant on each submanifold, and the image of the moment map equals the convex hull of those values (inside the dual Lie algebra $\mathfrak{t}^* \cong \mathbb{R}^k$ of $\mathbb{T}^k$). A **Delzant polytope** is a polytope in $\mathbb{R}^k$ such that for which each vertex is the intersection of exactly n edges, and for each vertex we can find vectors $v_1, \dots, v_k \in \mathbb{Z}^k$ along those edges which form a basis for $\mathbb{Z}^k$. **Delzant's theorem** says that there is a bijective correspondence between toric manifolds and Delzant polytopes — namely, the correspondence takes a toric manifold to the image of its moment map. So, given a toric manifold, the associated Delzant polytope actually encodes all of its geometry.

Some references: [13], [28].

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